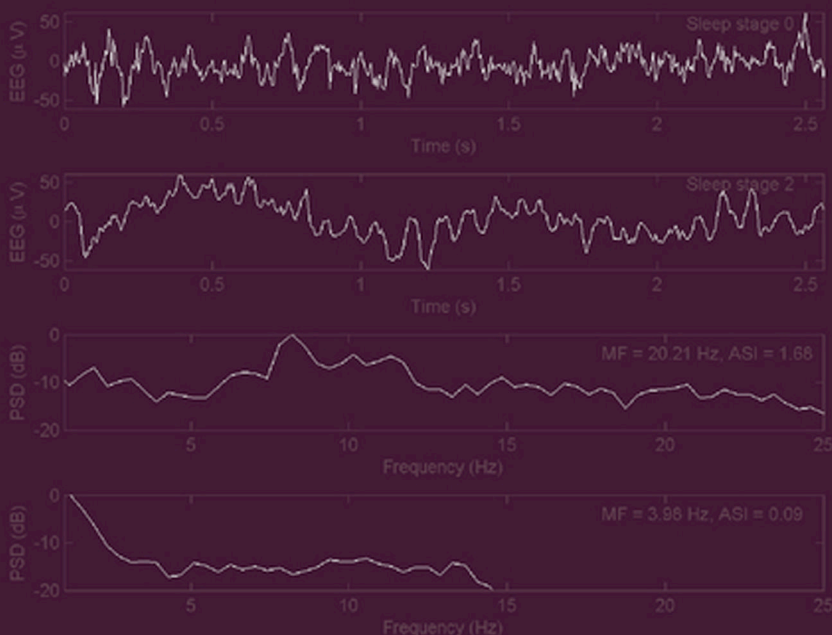


Biomedical Signal Analysis

Second Edition



RANGARAJ M. RANGAYYAN



IEEE Press Series in Biomedical Engineering
Metin Akay, Series Editor



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DEDICATION

Mátr dévô bhava

Pitr dévô bhava

Áchárya dévô bhava

Look upon your mother as your God

Look upon your father as your God

Look upon your teacher as your God

— from the sacred Vedic hymns of the *Taittireeya Upanishad* of India.

*This book is dedicated to the fond memory of
my mother Srimati Padma Srinivasan Rangayyan
and my father Sri Srinivasan Mandayam Rangayyan,
and to all of my teachers,
in particular, Professor Ivaturi Surya Narayana Murthy.*

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PREFACE

The first edition of this book has been received very well around the world. Professors at several universities across North America, Europe, Asia, and other regions of the world are using the book as a textbook. A low-cost paperback edition for selected regions of the world and a Russian edition have been published. I have received several messages and comments from many students, professors, and researchers via mail and at conferences with positive feedback about the book. I am grateful to IEEE and Wiley for publishing and promoting the book and to the many users of the book for their support and feedback.

I have myself used the book to teach my course ENEL 563 Biomedical Signal Analysis at the University of Calgary. In addition to positive responses, I have received suggestions from students and professors on revising the book to provide additional examples and including advanced topics and discussions on recent developments in the book. I also made notes identifying parts of the book that could be improved for clarity, augmented with details for improved comprehension, and expanded with additional examples for better illustrations of application. I have also identified a few new developments, novel applications, and advanced techniques for inclusion in the second edition to make the book more interesting and appealing to a wider readership.

New Material in the Second Edition

In view of the success of the first edition, I have not made any major change in the organization and style of the book. Notwithstanding a tighter format to reduce white space and control the total number of pages, the second edition of the book remains similar to the first edition in terms of organization and style of presentation. New material has been inserted into the same chapters as before, thereby expanding the book. The new topics have been chosen with care not only to fit with the structure and organization of the book but also to provide additional support material and advanced topics that can be assimilated and appreciated in a first course or an advanced study of the subject area.

Some of the substantial and important additions made to the book deal with the following topics:

- analysis of the variation of parameters of the electromyogram with force;
- illustrations of the electroencephalogram with application to sleep analysis and prediction of epileptic seizures;
- details on the theory of linear systems and numerical examples related to convolution;
- details on the z -transform and the Fourier transform along with additional examples of Fourier spectra and spectral analysis of biomedical signals;
- details on linear filters and their characteristics, such as the impulse response, transfer function, and pole–zero diagrams;
- description and demonstration of nonlinear order-statistic filters;
- derivation of the matched filter;
- derivations related to the complex cepstrum;
- details on random processes and their properties;
- wavelets and the wavelet transform with biomedical applications;
- fractal analysis with biomedical applications;
- time-frequency distributions and analysis of nonstationary signals with biomedical applications;
- principal component analysis, independent component analysis, and blind source separation with biomedical applications;
- monitoring of sleep apnea;
- analysis of various types of bioacoustic signals that could bear diagnostic information; and

- methods for pattern analysis and classification with illustrations of application to biomedical signals.

Discussions related to the topics listed above are spread throughout the book with several new references added to assist the reader in further studies. Many more problems and projects have been added at the ends of the chapters.

The first edition of the book (2002) has 516 pages (plus xxxv pages of front matter) with nine chapters, 538 numbered equations (with many more equations not numbered but as parts of procedures), 232 numbered figures (many with multiple subfigures), and 265 references. The second edition (2015) has 672 pages (plus xliii pages of front matter) in a more compact layout than the first edition, with nine chapters, 814 numbered equations (and many more equations not numbered but as parts of procedures), 370 numbered figures (many with multiple subfigures), and 505 references. The discussions on some of the new topics added were kept brief in order to control the size of the book; regardless, the second edition is approximately 50% larger than the first edition in several aspects.

Intended Audience

As with the first edition, the second edition is directed at engineering students in their final (senior) year of undergraduate studies or in the first year of their graduate studies. Electrical Engineering students with a good background in signals and systems will be well prepared for the material in the book. Students in other engineering disciplines, or in computer science, physics, mathematics, or geosciences, should also be able to appreciate the material in the book. A course on digital signal processing or digital filters would form a useful link to the material in the present book, but a capable student without this background should be able to gain a basic understanding of the subject matter. The introductory materials on systems, filters, and transforms added in the second edition should assist the reader without formal training on the same topics. Practicing engineers, computer scientists, information technologists, medical physicists, and data-processing specialists working in diverse areas such as telecommunications, seismic and geophysical applications, biomedical applications, and hospital information systems may find the book useful in their quest to learn advanced techniques for signal analysis. They could draw inspiration from other applications of signal processing or analysis, and satisfy their curiosity regarding computer applications in medicine and computer-aided medical diagnosis.

Teaching and Learning Plan

The book starts with an illustrated introduction to biomedical signals in Chapter 1. Chapter 2 continues the introduction, with emphasis on the analysis of multiple channels of correlated signals.

Chapter 3 deals exclusively with filtering of signals for removal of artifacts as an important step before signal analysis. The basic properties of systems and transforms

as well as signal processing techniques are reviewed and described where required. The chapter is written as a mix of theory and application so as to facilitate easy comprehension of the basics of signals, systems, and transforms [1–4]. The emphasis is on the application of filters to particular problems in biomedical signal analysis. A large number of illustrations are included to provide a visual impression of the problem and the effectiveness of the various filtering methods described.

Chapter 4 presents techniques that are particularly useful in the detection of events in biomedical signals. Analysis of waveshape and waveform complexity of events and components of signals is the focus of Chapter 5. Techniques for frequency-domain characterization of biomedical signals and systems are presented in Chapter 6. A number of diverse examples are provided in all of the chapters. Attention is directed to the characteristics of the problems that are encountered when analyzing and interpreting biomedical signals, rather than to any specific diagnostic application with particular signals.

The material in the book up to and including Chapter 6 provides more than adequate material for a one-semester (13-week) course at the senior (fourth-year) engineering level. My own teaching experience indicates that this material will require about 38 hours of lectures. It would be desirable to augment the lectures with about 12 hours of tutorials (problem-solving sessions) and 10 laboratory sessions.

Modeling biomedical signal-generating processes and systems for parametric representation and analysis is the subject of Chapter 7. Chapter 8 deals with the analysis of nonstationary and multicomponent signals. The topics in these chapters are of higher mathematical complexity than suitable for undergraduate courses. Some sections may be selected and included in a first course on biomedical signal analysis if there is particular interest in these topics. Otherwise, the two chapters could be left for self-study by those in need of the techniques, or included in an advanced course.

Chapter 9 presents the final aspect of biomedical signal analysis, and provides an introduction to pattern classification and diagnostic decision making. Although this topic is advanced in nature and could form a graduate-level course on its own, the material is introduced so as to draw the entire exercise of biomedical signal analysis to its concluding stage of diagnostic decision. It is recommended that a few sections from this chapter be included even in a first course on biomedical signal analysis so as to give the students a flavor of the end result.

The topic of data compression has been deliberately left out of the book. Advanced topics such as time-frequency distributions, wavelet-based analysis, independent component analysis, and fractal analysis have been introduced briefly in the second edition, but require further detailed study for completeness. Adaptive filters and nonstationary signal analysis techniques are introduced in the book, but deserve more attention, depth, and breadth. The references provided should assist the interested reader in obtaining further advanced material.

Each chapter includes a number of study questions and problems to facilitate preparation for tests and examinations. A number of laboratory exercises are also provided at the end of each chapter, which could be used to formulate hands-on exercises with real-life signals. Data files related to the problems and exercises at the end of each chapter are available at the site

<http://people.ucalgary.ca/~ranga/enel563>

MATLAB[®] programs to read the data are also provided in some cases.

It is strongly recommended that the first one or two laboratory sessions in the course include visits to a local hospital, health sciences center, or clinical laboratory to view and experience procedures related to biomedical signal acquisition and analysis in a practical (clinical) setting. Signals acquired from fellow students and professors could form interesting and motivating material for laboratory exercises, and may be used to supplement the data files provided. A few workshops by physiologists, neuroscientists, and cardiologists should also be included in the course so as to provide the students with a nonengineering perspective on the subject.

Practical experience with real-life signals is a key element in understanding and appreciating biomedical signal analysis. This aspect could be difficult and frustrating at times, but provides professional satisfaction and educational fun!

RANGARAJ MANDAYAM RANGAYYAN

Calgary, Alberta, Canada

February, 2015

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As always, I am grateful to my family — my wife Mayura, my daughter Vidya, and my son Adarsh — for their love and support.

I hope that this book will assist those who seek to enrich their lives and those of others with the exciting field of biomedical signal analysis.

RANGARAJ MANDAYAM RANGAYYAN

*Calgary, Alberta, Canada
February, 2015*

PREFACE: FIRST EDITION

Background and Motivation

The establishment of the clinical electrocardiograph (ECG) by the Dutch physician Willem Einthoven in 1903 marked the beginning of a new era in medical diagnostic techniques, including the entry of electronics into health care. Since then, electronics, and subsequently computers, have become integral components of biomedical signal analysis systems, performing a variety of tasks from data acquisition and pre-processing for removal of artifacts to feature extraction and interpretation. Electronic instrumentation and computers have been applied to investigate a host of biological and physiological systems and phenomena, such as the electrical activity of the cardiovascular system, the brain, the neuromuscular system, and the gastric system; pressure variations in the cardiovascular system; sound and vibration signals from the cardiovascular, the musculoskeletal, and the respiratory systems; and magnetic fields of the brain, to name a few.

The primary step in investigations of physiological systems requires the development of appropriate sensors and instrumentation to transduce the phenomenon of interest into a measurable electrical signal. The next step of analysis of the signals, however, is not always an easy task for a physician or life-sciences specialist. The clinically relevant information in the signal is often masked by noise and interference, and the signal features may not be readily comprehensible by the visual or

auditory systems of a human observer. Heart sounds, for example, have most of their energy at or below the threshold of auditory perception of most humans; the interference patterns of a surface electromyographic signal are too complex to permit visual analysis. Some repetitious or attention-demanding tasks, such as on-line monitoring of the ECG of a critically ill patient with cardiac rhythm problems, could be uninteresting and tiring for a human observer. Furthermore, the variability present in a given type of signal from one subject to another, and the interobserver variability inherent in subjective analysis performed by physicians or analysts make consistent understanding or evaluation of any phenomenon difficult, if not impossible. These factors created the need not only for improved instrumentation, but also for the development of methods for objective analysis via signal processing algorithms implemented in electronic hardware or on computers.

Processing of biomedical signals, until a few years ago, was mainly directed toward filtering for removal of noise and power-line interference; spectral analysis to understand the frequency characteristics of signals; and modeling for feature representation and parameterization. Recent trends have been toward quantitative or objective analysis of physiological systems and phenomena via signal analysis. The field of biomedical signal analysis has advanced to the stage of practical application of signal processing and pattern analysis techniques for efficient and improved non-invasive diagnosis, on-line monitoring of critically ill patients, and rehabilitation and sensory aids for the handicapped. Techniques developed by engineers are gaining wider acceptance by practicing clinicians, and the role of engineering in diagnosis and treatment is gaining much-deserved respect.

The major strength in the application of computers in biomedical signal analysis lies in the potential use of signal processing and modeling techniques for quantitative or objective analysis. Analysis of signals by human observers is almost always accompanied by perceptual limitations, interpersonal variations, errors caused by fatigue, errors caused by the very low rate of incidence of a certain sign of abnormality, environmental distractions, and so on. The interpretation of a signal by an expert bears the weight of the experience and expertise of the analyst; however, such analysis is almost always subjective. Computer analysis of biomedical signals, if performed with the appropriate logic, has the potential to add objective strength to the interpretation of the expert. It thus becomes possible to improve the diagnostic confidence or accuracy of even an expert with many years of experience. This approach to improved health care could be labeled as *computer-aided diagnosis*.

Developing an algorithm for biomedical signal analysis, however, is not an easy task; quite often, it might not even be a straightforward process. The engineer or computer analyst is often bewildered by the variability of features in biomedical signals and systems, which is far higher than that encountered in physical systems or observations. Benign diseases often mimic the features of malignant diseases; malignancies may exhibit a characteristic pattern, which, however, is not always guaranteed to appear. Handling all of the possibilities and degrees of freedom in a biomedical system is a major challenge in most applications. Techniques proven to work well with a certain system or set of signals may not work in another seemingly similar situation.

The Problem solving Approach

The approach I have taken in presenting material in this book is primarily that of development of algorithms for problem solving. Engineers are often said to be (with admiration, I believe) problem solvers. However, the development of a problem statement and gaining of a good understanding of the problem could require a significant amount of preparatory work. I have selected a logical series of problems, from the many case studies I have encountered in my research work, for presentation in the book. Each chapter deals with a certain type of a problem with biomedical signals. Each chapter begins with a statement of the problem, followed immediately with a few illustrations of the problem with real-life case studies and the associated signals. Signal processing, modeling, or analysis techniques are then presented, starting with relatively simple “textbook” methods, followed by more sophisticated research approaches directed at the specific problem. Each chapter concludes with one or more applications to significant and practical problems. The book is illustrated copiously with real-life biomedical signals and their derivatives.

The methods presented in the book are at a fairly high level of technical sophistication. A good background in signal and system analysis [1, 3, 4] as well as probability, random variables, and stochastic processes [5–10] is required in order to follow the procedures and analysis. Familiarity with systems theory and transforms such as the Laplace and Fourier, the latter in both continuous and discrete versions, will be assumed. We will not be getting into details of the transducers and instrumentation techniques essential for biomedical signal acquisition [11–14]; instead, we will be studying the problems present in the signals after they have been acquired, concentrating on how to solve the problems. Concurrent or prior study of the physiological phenomena associated with the signals of specific interest, with a clinical textbook, is strongly recommended.

Intended Readership

The book is directed at engineering students in their final year of undergraduate studies or in their graduate studies. Electrical Engineering students with a rich background in signals and systems [1, 3, 4] will be well prepared for the material in the book. Students in other engineering disciplines, or in computer science, physics, mathematics, or geophysics should also be able to appreciate the material in the book. A course on digital signal processing or digital filters [15] would form a useful link, but a capable student without this topic may not face much difficulty.

Practicing engineers, computer scientists, information technologists, medical physicists, and data-processing specialists working in diverse areas such as telecommunications, seismic and geophysical applications, biomedical applications, and hospital information systems may find the book useful in their quest to learn advanced techniques for signal analysis. They could draw inspiration from other applications of signal processing or analysis, and satisfy their curiosity regarding computer applications in medicine and computer-aided medical diagnosis.

Teaching and Learning Plan

The book starts with an illustrated introduction to biomedical signals in Chapter 1. Chapter 2 continues the introduction, but with emphasis on the analysis of multiple channels of related signals. This part of the book may be skipped in the teaching plan for a course if the students have had a previous course on biomedical signals and instrumentation. In such a case, the chapters should be studied as review material in order to get oriented toward the examples to follow in the book.

Chapter 3 deals exclusively with filtering for removal of artifacts as an important precursive step before signal analysis. Basic properties of systems and transforms as well as signal processing techniques are reviewed and described as and when required. The chapter is written so as to facilitate easy comprehension by those who have had a basic course on signals, systems, and transforms [1, 3, 4]. The emphasis is on the application to particular problems in biomedical signal analysis, and not on the techniques themselves. A large number of illustrations are included to provide a visual impression of the problem and the effectiveness of the various filtering methods described.

Chapter 4 presents techniques particularly useful in the detection of events in biomedical signals. Analysis of waveshape and waveform complexity of events and components of signals is the focus of Chapter 5. Techniques for frequency-domain characterization of biomedical signals and systems are presented in Chapter 6. A number of diverse examples are provided in these chapters. Attention is directed to the characteristics of the problems one faces in analyzing and interpreting biomedical signals, rather than to any specific diagnostic application with particular signals.

The material in the book up to and including Chapter 6 will provide more than adequate material for a one-semester (13-week) course at the senior (fourth-year) engineering level. My own teaching experience indicates that this material will require about 36 hours of lectures, augmented with about 12 hours of tutorials (problem-solving sessions) and 10 laboratory sessions.

Modeling biomedical signal-generating processes and systems for parametric representation and analysis is the subject of Chapter 7. Chapter 8 deals with the analysis of nonstationary signals. The topics in these chapters are of higher mathematical complexity than suitable for undergraduate courses. Some sections may be selected and included in a first course on biomedical signal analysis if there is particular interest in these topics. Otherwise, the two chapters could be left for self-study by those in need of the techniques, or included in an advanced course.

Chapter 9 presents the final aspect of biomedical signal analysis, and provides an introduction to pattern classification and diagnostic decision. Although this topic is advanced in nature and could form a graduate-level course on its own, the material is introduced so as to draw the entire exercise of biomedical signal analysis to its concluding stage of diagnostic decision. It is recommended that a few sections from this chapter be included even in a first course on biomedical signal analysis so as to give the students a flavor of the end result.

The topic of data compression has deliberately been left out of the book. Advanced topics such as nonlinear dynamics, time-frequency distributions, wavelet-

based analysis, chaos, and fractals are not covered in the book. Adaptive filters and nonstationary signal analysis techniques are introduced in the book, but deserve more attention, depth, and breadth. These topics will form the subjects of a follow-up book that I intend to write.

Each chapter includes a number of study questions and problems to facilitate preparation for tests and examinations. A number of laboratory exercises are also provided at the end of each chapter, which could be used to formulate hands-on exercises with real-life signals. Data files related to the problems and exercises at the end of each chapter are available at the site (revised for 2015)

<http://people.ucalgary.ca/~ranga/enel563>

MATLAB[®] programs to read the data are also provided where required.

It is strongly recommended that the first one or two laboratory sessions in the course be visits to a local hospital, health sciences center, or clinical laboratory to view biomedical signal acquisition and analysis in a practical (clinical) setting. Signals acquired from fellow students and professors could form interesting and motivating material for laboratory exercises, and should be used to supplement the data files provided. A few workshops by physiologists, neuroscientists, and cardiologists should also be included in the course so as to provide the students with a nonengineering perspective on the subject.

Practical experience with real-life signals is a key element in understanding and appreciating biomedical signal analysis. This aspect could be difficult and frustrating at times, but provides professional satisfaction and educational fun!

RANGARAJ MANDAYAM RANGAYYAN

*Calgary, Alberta, Canada
September, 2001*

ACKNOWLEDGMENTS: FIRST EDITION

To write a book on my favorite subject of biomedical signal analysis has been a long-cherished ambition of mine. Writing this book has been a major task with many facets: challenging, yet yielding more knowledge; tiring, yet stimulating the thirst to understand and appreciate more; difficult, yet satisfying when a part was brought to a certain stage of completion.

A number of very important personalities have shaped me and my educational background. My mother, Srimati Padma Srinivasan Rangayyan, and my father, Sri Srinivasan Mandayam Rangayyan, encouraged me to keep striving to gain higher levels of education and to set and achieve higher goals all the time. I have been very fortunate to have been taught and guided by a number of dedicated teachers, the most important of them being Professor Ivaturi Surya Narayana Murthy, my Ph.D. supervisor, who introduced me to the topic of this book at the Indian Institute of Science, Bangalore, Karnataka, India. It is with great respect and admiration that I dedicate this book as a humble offering to their spirits.

My basic education was imparted by many influential teachers at Saint Joseph's Convent, Saint Joseph's Indian High School, and Saint Joseph's College in Mandya and Bangalore, Karnataka, India. My engineering education was provided by the People's Education Society College of Engineering, Mandya, affiliated with the University of Mysore. I express my gratitude to all of my teachers.

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The book has benefited significantly from illustrations and text provided by a number of researchers worldwide, as identified in the references and permissions cited. I thank them all for enriching the book with their gifts of knowledge and kindness. I thank Bert Unterberger for drafting some of the illustrations in the book.

The research projects that have provided me with the background and experience essential in order to write the material in this book have been supported by many agencies. I thank the Natural Sciences and Engineering Research Council of Canada, the Alberta Heritage Foundation for Medical Research, the Alberta Breast Cancer Foundation, the Arthritis Society of Canada, the Nickle Family Foundation of Calgary, Control Data Corporation, the University of Calgary, the University of Manitoba, and the Indian Institute of Science for supporting my research projects.

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erous support from the Department of Electrical and Computer Engineering and the Faculty of Engineering at the University of Calgary in terms of supplies, services, and relief from other duties.

My association with the IEEE Engineering in Medicine and Biology Society (EMBS) in many positions has benefited me considerably in numerous ways. In particular, the period as an Associate Editor of the *IEEE Transactions on Biomedical Engineering* was very rewarding, as it provided me with a wonderful opportunity to work with many leading researchers and authors of scientific articles. I thank IEEE EMBS for lending professional support to my career on many fronts. I am grateful to the IEEE Press, in particular, Metin Akay, Series Editor, IEEE Press Series in Biomedical Engineering, for inviting me to write this book.

Writing this book has been a monumental task, often draining me of all of my energy. The infinite source of inspiration and recharging of my energy has been my family — my wife Mayura, my daughter Vidya, and my son Adarsh. While supporting me with their love and affection, they have had to bear the loss of my time and effort at home. I express my sincere gratitude to my family for their love and support, and record their contribution toward the preparation of this book.

It is my humble hope that this book will assist those who seek to enrich their lives and those of others with the wonderful powers of biomedical signal analysis. Electrical and Computer Engineering is indeed a great field in the service of humanity!

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September, 2001

ABOUT THE AUTHOR

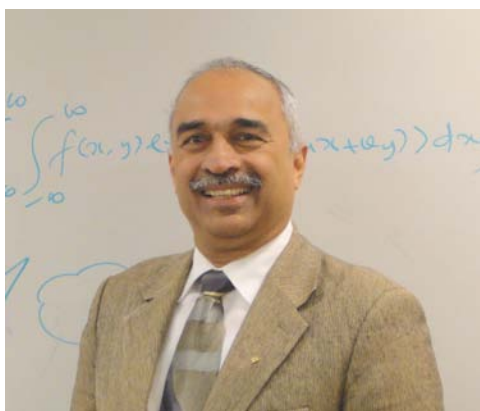


Photo by Faraz Oloumi.

Rangaraj (Raj) Mandayam Rangayyan was born in Mysore, Karnataka, India, on 21 July 1955. He received the Bachelor of Engineering degree in Electronics and Communication in 1976 from the University of Mysore at the People's Education Society College of Engineering, Mandya, Karnataka, India, and the Ph.D.

degree in Electrical Engineering from the Indian Institute of Science, Bangalore, Karnataka, India, in 1980. He was with the University of Manitoba, Winnipeg, Manitoba, Canada, from 1981 to 1984.

He is, at present, a Professor with the Department of Electrical and Computer Engineering as well as an Adjunct Professor of Surgery and Radiology at the University of Calgary, Calgary, Alberta, Canada. His research interests are in the areas of digital signal and image processing, biomedical signal analysis, medical imaging and image analysis, and computer vision. His research projects have been focused on mammographic image enhancement and analysis for computer-aided diagnosis of breast cancer; region-based image processing; knee-joint vibration signal analysis for noninvasive diagnosis of articular cartilage pathology; and analysis of textured images by cepstral filtering and sonification. He has published more than 150 papers in journals and 250 papers in proceedings of conferences. He has lectured extensively in many countries, including India, Canada, United States, Brazil, Argentina, Uruguay, Chile, United Kingdom, The Netherlands, France, Spain, Italy, Finland, Russia, Romania, Croatia, Egypt, Malaysia, Thailand, China, and Japan. He has collaborated with many research groups in India, Brazil, Italy, Spain, France, Romania, and China.

He was an Associate Editor of the *IEEE Transactions on Biomedical Engineering* from 1989 to 1996; the Program Chair and Editor of the Proceedings of the IEEE Western Canada Exhibition and Conference on “Telecommunication for Health Care: Telemetry, Teleradiology, and Telemedicine,” July 1990, Calgary, Alberta, Canada; the Canadian Regional Representative to the Administrative Committee of the IEEE Engineering in Medicine and Biology Society (EMBS), 1990–1993; a Member of the Scientific Program Committee and Editorial Board, International Symposium on Computerized Tomography, Novosibirsk, Siberia, Russia, August 1993; the Program Chair and Coeditor of the *Proceedings of the 15th Annual International Conference of the IEEE EMBS*, October 1993, San Diego, CA; and Program Cochair, 20th Annual International Conference of the IEEE EMBS, Hong Kong, October 1998.

He has been awarded the Killam Resident Fellowship thrice in support of his book-writing projects. He has been recognized with the 1997 and 2001 Research Excellence Awards of the Department of Electrical and Computer Engineering, the 1997 Research Award of the Faculty of Engineering, and by appointment as “University Professor” (2003–2013), at the University of Calgary. He is the author of two textbooks: *Biomedical Signal Analysis* (IEEE/ Wiley, 2002/2015) and *Biomedical Image Analysis* (CRC, 2005). He has coauthored and coedited several other books, including *Color Image Processing with Biomedical Applications* (SPIE, 2011). He has been elected as Fellow, IEEE (2001); Fellow, Engineering Institute of Canada (2002); Fellow, American Institute for Medical and Biological Engineering (2003); Fellow, SPIE (2003); Fellow, Society for Imaging Informatics in Medicine (2007); Fellow, Canadian Medical and Biological Engineering Society (2007); and Fellow, Canadian Academy of Engineering (2009). He has been recognized with the IEEE Third Millennium Medal (2000), the Distinguished Alumni award of the PES College of Engineering (2012), and the Outstanding Engineer Medal by IEEE Canada (2013).

SYMBOLS AND ABBREVIATIONS

Note: Boldfaced letters represent the vectorial or matrix form of the variable indicated by the corresponding italicized letters. Variables or symbols used within limited contexts are not listed: they are described within their contexts. The mathematical symbols listed may stand for other entities or variables in different applications; only the common associations are listed for ready reference.

a_k	autoregressive model or filter coefficients
au	arbitrary units
$aV\{F, L, R\}$	augmented ECG leads
Ag	silver
$AgCl$	silver chloride
A_z	area under the ROC curve
ACF	autocorrelation function
ADC	analog-to-digital converter
AHI	apnea–hypopnea index
AI	aortic insufficiency
AM	amplitude modulation
ANC	adaptive noise cancellation
ANN	artificial neural network
ANS	autonomic nervous system
AO	aorta, aortic (valve or pressure)

AP	action potential
AR	interval between atrial activity and the corresponding QRS
AR	autoregressive (model or filter)
ARMA	autoregressive, moving-average (model or filter)
AS	aortic stenosis
ASD	atrial septal defect
AV	atrioventricular
A2	aortic component of the second heart sound
b	bit
b_l	moving-average model or filter coefficients
bpm	beats per minute
BCG	ballistocardiogram
BP	blood pressure
BSS	blind source separation
C	covariance matrix
Ca	calcium
C_i	the i^{th} class in a pattern classification problem
Cl	chlorine
C_{xy}	covariance between x and y
CA	constant area
CAD	computer-aided diagnosis
CCF	cross-correlation function
CD	compact disk
CNS	central nervous system
CO	carbon monoxide
CP	carotid pulse
CPAP	continuous positive airway pressure
CSA	central sleep apnea
CSD	cross-spectral density, cross-spectrum
CV	coefficient of variation
CWT	continuous wavelet transform
$diag$	diagonal of a matrix
D	dicrotic notch in the carotid pulse
DAC	digital-to-analog converter
DC	direct current; zero frequency
DFT	discrete Fourier transform
DM	diastolic murmur
DW	dicrotic wave in the carotid pulse
DWT	discrete wavelet transform
$e(n), E(\omega)$	model or estimation error
ECG	electrocardiogram, electrocardiography
ECoG	electrocorticogram
EEG	electroencephalogram
EGG	electrogastrogram
EM	electromagnetic

EMD	empirical mode decomposition
EMG	electromyogram
ENG	electroneurogram
EOG	electrooculogram
EP	energy parameter
ERP	event-related potential
ESP	energy spread parameter
E_x	total energy of the signal x
$E[\]$	statistical expectation operator
f	frequency variable, usually in Hertz
fBm	fractional Brownian motion
f_c	cutoff frequency (usually at -3 dB) of a filter in Hertz
f_s	sampling frequency in Hertz
FD	fractal dimension
FDI	first dorsal interosseus
FF	form factor
FFT	fast Fourier transform
FIR	finite impulse response (filter)
FLDA	Fisher linear discriminant analysis
FM	frequency modulation
FN	false negative
FNF	false-negative fraction
FP	false positive
FP	frequency parameter
FPF	false-positive fraction
FSP	frequency spread parameter
FT	Fourier transform
GLR	generalized likelihood ratio
GTFD	generalized time-frequency distribution
h	hour
$h(t), h(n)$	impulse response of a filter
H	entropy
H	Hurst coefficient
H	Hermitian (complex-conjugate) matrix transposition
Hg	mercury
$H(s), H(z)$	transfer function of a filter
$H(s)$	Laplace transform of $h(t)$
$H(z)$	z -transform of $h(n)$
$H(\omega)$	frequency response of a filter
$H(\omega)$	Fourier transform of $h(t)$
HR	heart rate
HRV	heart rate variability
HSS	hypertrophic subaortic stenosis
Hz	Hertz
i	index of a series or discrete-time signal

ICA	independent component analysis
IFT	inverse Fourier transform
IIR	infinite impulse response (filter)
IMF	instantaneous mean frequency
IMF	intrinsic mode function
IPI	interpulse interval
j	index of a series or discrete-time signal
j	$\sqrt{-1}$
JM	Jeffries–Matusita
k -NN	k nearest neighbors
K	kurtosis
K	potassium
KLD	Kullback–Leibler distance or divergence
KLT	Karhunen–Loève transform
ln	natural logarithm (base e)
L_{ij}	loss function in pattern classification
LA	left atrium
LED	light emitting diode
LHS	left-hand side
LMS	least mean squares
LOO	leave one out
LP	linear prediction (model)
LSI	linear shift-invariant
LTI	linear time-invariant
LV	left ventricle
m	mean
m	mean vector of a pattern class
mA	milliamperes
min	minute
mm	millimeter
ms	millisecond
mV	millivolt
M	number of samples
MA	moving average
MCI	muscle-contraction interference
MI	mitral insufficiency
MMSE	minimum mean-squared error
MP	Matching Pursuit
MPC	minimum-phase correspondent
MPTFD	Matching Pursuit time-frequency distribution
MR	mitral regurgitation
MS	mitral stenosis
MS	mean-squared (value)
MSE	mean-squared error
MU	motor unit

MUAP	motor-unit action potential
MVC	maximal voluntary contraction
nA	nanoamperes
N	number of samples
N	filter order
Na	sodium
NPV	negative predictive value
p_k	pole of a model
$p(x)$	probability density function of the random variable x
$p(x C_i)$	likelihood function of class C_i or state-conditional PDF of x
ppm	pulses per minute
pps	pulses per second
Pa	Pascal
OAE	otoacoustic emission
OLS	orthogonal least-squares
OMPTFD	optimized Matching Pursuit time-frequency distribution
OSA	obstructive sleep apnea
P	atrial contraction wave in the ECG
P	percussion wave in the carotid pulse
P	model order or number of poles
$P(x)$	probability of the event x
$P(C_i x)$	posterior probability that the observation x is from class C_i
PA	predictive area
PCA	principal component analysis
PCG	phonocardiogram
PDA	patent ductus arteriosus
PDF	probability density function
PFP	patellofemoral pulse trains or signals
PI	pulmonary insufficiency
PLP	posterior leaflet prolapse
PNS	parasympathetic nervous system
PPC	physiological patellofemoral crepitus
PPV	positive predictive value
PQ	isoelectric segment in the ECG before ventricular contraction
PS	pulmonary stenosis
PSA	power spectral analysis
PSD	power spectral density, power spectrum
PSG	polysomnography
PVC	premature ventricular contraction
P2	pulmonary component of the second heart sound
Q	model order or number of zeros
QRS	ventricular contraction wave in the ECG
QRSTA	area under the QRS and T waves
$r, \mathbf{r}$	reference input to an adaptive filter
$r_j(\mathbf{x})$	average risk or loss in pattern classification

RA	right atrium
RBF	radial basis function
RBFN	radial basis function network
REM	rapid eye movement
RHS	right-hand side
RF	radio-frequency
RLS	recursive least-squares
RLSL	recursive least-squares lattice
RMS	root mean squared (value)
ROC	receiver operating characteristics
ROC	region of convergence
RR	interval between two successive QRS waves in an ECG
RSPWVD	reassigned smoothed pseudo Wigner–Ville distribution
RV	right ventricle
s	second
s	Laplace-domain variable
sec	second (in figures from other sources)
sgn	signum (sign)
S	skewness
$S(\omega), S(k)$	auto- or cross-spectral density; power spectral density
SA	sinoatrial
SaO_2	level of oxyhemoglobin in blood
SD	standard deviation
SEM	spectral error measure
SEP	somatosensory evoked potential
SL	signal length
SM	systolic murmur
SMUAP	single-motor-unit action potential
SNR	signal-to-noise ratio
SNS	sympathetic nervous system
SpO_2	level of oxyhemoglobin in blood
SPWVD	smoothed pseudo Wigner–Ville distribution
ST	isoelectric segment in the ECG during ventricular contraction
STFT	short-time Fourier transform
SWVD	smoothed Wigner–Ville distribution
S1	first heart sound
S2	second heart sound
S3	third heart sound
S4	fourth heart sound
S^+	sensitivity of a test
S^-	specificity of a test
t	time variable
T	ventricular relaxation wave in the ECG
T	tidal wave in the carotid pulse
T	sampling interval

T	as a superscript: vector or matrix transposition
T^+	positive test result
T^-	negative test result
TCR	turns count rate
TF	time-frequency
TFD	time-frequency distribution
Th	threshold
TI	tricuspid insufficiency
TN	true negative
TNF	true-negative fraction
TP	true positive
TPF	true-positive fraction
Tr	trace of a matrix (sum of the diagonal entries)
TS	tricuspid stenosis
TSE	total squared error
TV	television
V	Volt
V1 – V6	chest leads for ECG
VAG	vibroarthrogram
VCG	vectorcardiography
VLf	very low frequency
VMG	vibromyogram
VSD	ventricular septal defect
w	filter tap weight; weighting function
$\mathbf{w}$	filter weight vector
WP	wavelet packet
WT	wavelet transform
WVD	Wigner–Ville distribution
$x(t), x(n)$	a signal in the time domain; usually denotes input
$\mathbf{x}$	vectorial representation of the signal $x(n)$
$\mathbf{x}$	a feature vector in pattern classification
$X(f), X(\omega)$	Fourier transform of $x(t)$
$X(k)$	discrete Fourier transform of $x(n)$
$X(z)$	z -transform of $x(n)$
$X(\tau, \omega)$	short-time Fourier transform or time-frequency distribution of $x(t)$
$y(t), y(n)$	a signal in the time domain; usually denotes output
$\mathbf{y}$	vectorial representation of the signal $y(n)$
$Y(f), Y(\omega)$	Fourier transform of $y(t)$
$Y(k)$	discrete Fourier transform of $y(n)$
$Y(z)$	z -transform of $y(n)$
z	the z -transform variable
z^{-1}	unit delay operator in discrete-time systems
z_l	zeros of a system
$\mathbf{z}$	a prototype feature vector in pattern classification
ZCR	zero-crossing rate

ZT	the z -transform
1D	one-dimensional
2D	two-dimensional
3D	three-dimensional
I, II, III	limb leads for ECG
α	an EEG wave
β	an EEG wave
β	spectral component
γ	an EEG wave
γ_i	reflection coefficient
γ_{xy}	correlation coefficient between x and y
Γ_{xy}	coherence between x and y
δ	an EEG wave
δ	Dirac delta (impulse) function
ε	total squared error
η	a random variable or noise process
θ	an angle
θ	a threshold
θ	an EEG wave
θ, Θ	cross-correlation function
λ	forgetting factor in the RLS filter
μ	the mean (average) of a random variable
μ	a rhythmic wave in the EEG
μ	step size in the LMS filter
μV	microvolt
μm	micrometer
μs	microsecond
ρ	correlation coefficient
σ	the real part of the Laplace variable s (Neper frequency)
σ	the standard deviation of a random variable
σ^2	the variance of a random variable
τ	a time interval, delay, or shift
ϕ, Φ	autocorrelation
ω	frequency variable in radians per second
Ω	frequency variable in radians per second
*	when in-line: convolution
*	as a superscript: complex conjugation
$\bar{}$	average or normalized version of the variable
$\hat{}$	complex cepstrum of the signal, if a function of time
$\hat{}$	complex logarithm of the signal, if a function of frequency
$\sim$	estimate of the variable under the symbol
$'$, $''$	first and second derivatives of the preceding function
$\forall$	for all
$\in$	belongs to or is in (the set)
$ $	absolute value or magnitude of

$\angle$	argument of, angle of
$\lceil \rceil$	ceiling
$\lfloor \rfloor$	floor
$\langle, \rangle$ or $\cdot$	dot (inner) product

CHAPTER 1

INTRODUCTION TO BIOMEDICAL SIGNALS

1.1 The Nature of Biomedical Signals

Living organisms are made up of many component *systems* — the human body, for example, includes the nervous system, the cardiovascular system, and the musculoskeletal system, among others. Each system is made up of several subsystems that carry on many *physiological processes*. For example, the cardiac system performs the important task of rhythmic pumping of blood throughout the body to facilitate the delivery of nutrients, as well as pumping blood through the pulmonary system for oxygenation of the blood itself.

Physiological processes are complex phenomena, including nervous or hormonal stimulation and control; inputs and outputs that could be in the form of physical material, neurotransmitters, or information; and action that could be mechanical, electrical, or biochemical. Figure 1.1 shows a schematic representation of a generic physiological system to support the present discussion. Most physiological processes are accompanied by or manifest themselves as *signals* that reflect their nature and activities. Such signals could be of many types, including biochemical in the form of hormones and neurotransmitters, electrical in the form of potential or current, and physical in the form of pressure or temperature.

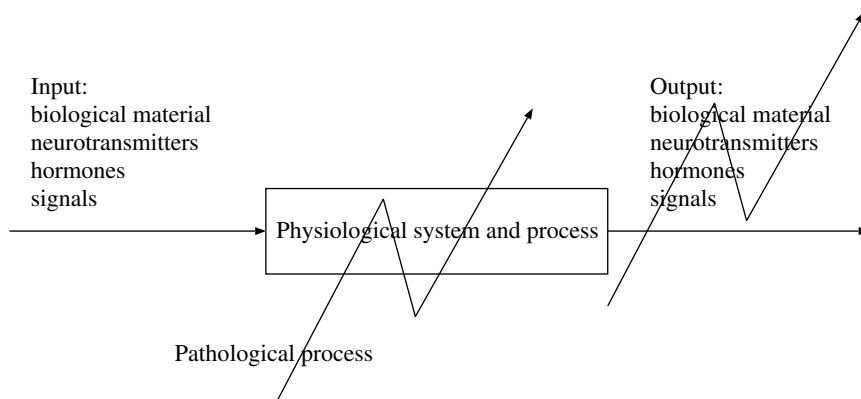


Figure 1.1 Schematic representation of a generic physiological system with various types of possible inputs and outputs. The effect of a pathological process is depicted by the zigzag line across the system and the list of possible outputs.

Diseases or defects in a biological system cause alterations in its normal physiological processes, leading to *pathological processes* that affect the performance, health, and general well-being of the system. A pathological process is typically associated with signals that are different in some aspects from the corresponding normal signals. If we possess a good understanding of a system of interest, it becomes possible to observe the corresponding signals and assess the state of the system. The task is not difficult when the signal is simple and appears at the outer surface of the body. For example, most infections cause a rise in the temperature of the body, which may be sensed easily, albeit in a relative and *qualitative* manner, via the palm of one's hand. Objective or *quantitative* measurement of temperature requires an instrument, such as a thermometer.

A single measurement x of temperature is a *scalar* and represents the thermal state of the body at a *particular or single instant of time* t (and at a particular position). If we record the temperature continuously in some form, such as a strip-chart record or on a magnetic tape, we obtain a *signal as a function of time*; such a signal may be expressed in *continuous-time* or *analog* form as $x(t)$. When the temperature is measured at *discrete* points of time, it may be expressed in *discrete-time* form as $x(nT)$ or $x(n)$, where n is the index or measurement sample number of the array of values, and T represents the uniform interval between the time instants of measurement. A discrete-time signal that can take amplitude values only from a limited list of *quantized* levels is called a *digital* signal; the distinction between discrete-time and digital signals is often ignored.

In intensive-care monitoring, the tympanic (eardrum) temperature may sometimes be measured using an infrared sensor. Occasionally, when catheters are being used for other purposes, a temperature sensor may also be introduced into an artery or the heart to measure the *core* temperature of the body. It then becomes possible to obtain a continuous measurement of temperature, although only a few samples taken

at intervals of a few minutes may be stored for subsequent analysis. Figure 1.2 illustrates representations of temperature measurements as a scalar, an array, and a signal plotted as a function of time. It is obvious that the graphical representation facilitates easier and faster comprehension of trends in the temperature than the numerical format. Long-term recordings of temperature can facilitate the analysis of temperature-regulation mechanisms [16, 17].

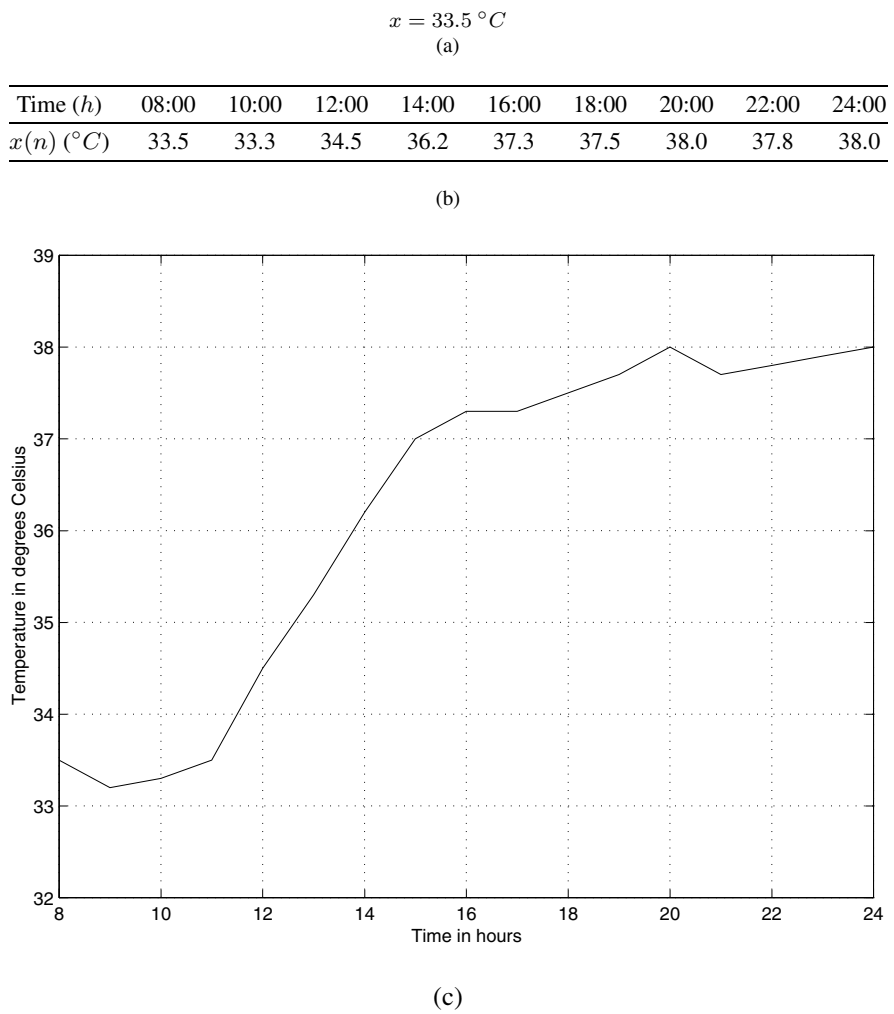


Figure 1.2 Measurements of the temperature of a patient presented as (a) a scalar with one temperature measurement x at an unspecified instant of time, (b) an array $x(n)$ made up of several measurements at different instants of time, and (c) a plot of the signal $x(n)$ or $x(t)$. The horizontal axis of the plot represents time in *hours*; the vertical axis gives temperature in *degrees Celsius*. Data courtesy of Foothills Hospital, Calgary.

Let us now consider another basic measurement in health care and monitoring: blood pressure (BP). Each measurement consists of two values — the systolic pressure and the diastolic pressure. BP is measured in millimeters of mercury (*mm of Hg*) in clinical practice, although the international standard unit for pressure is the *Pascal*, with $1 \text{ Pa} = 0.0075 \text{ mm of Hg}$. A single BP measurement could thus be viewed as a *vector* $\mathbf{x} = [x_1, x_2]^T$ with two components: x_1 indicating the systolic pressure and x_2 indicating the diastolic pressure. When BP is measured at a few instants of time, we obtain an array of vectors $\mathbf{x}(n)$. In intensive-care monitoring and surgical procedures, a pressure transducer may sometimes be inserted into an artery (along with other intraarterial or intravenous devices). It then becomes possible to obtain the arterial systolic and diastolic BP as a continuous-time recording, although the values may be transferred to a computer and stored only at sampled instants of time that are several seconds or minutes apart. The signal may then be expressed as a function of time $\mathbf{x}(t)$. Figure 1.3 shows BP measurements as a single two-component vector, as an array of vectors, and as a plot as a function of time. It is clear that the plot as a function of time facilitates rapid observation of trends in the pressure.

1.2 Examples of Biomedical Signals

The preceding example of body temperature as a signal is a simple example of a *biomedical signal*. Regardless of its simplicity, we can appreciate its importance and value in the assessment of the well-being of a child with a fever or that of a critically ill patient in a hospital. The origins and nature of a few other biomedical signals of various types are described in the following sections, with brief indications of their usefulness in diagnosis. Further detailed discussions on some of the signals are provided in the context of their analysis for various purposes in the chapters that follow.

This book is limited to the analysis of commonly encountered biomedical signals expressed as functions of time. Signals encountered in molecular biology and nanobioscience are not included. The topics of biomedical imaging and the analysis of biomedical images [18, 19] are not considered.

1.2.1 The action potential of a cardiac myocyte

The action potential is the basic component of all bioelectrical signals. It provides information on the nature of physiological activity at the single-cell level.

The action potential is the electrical signal that accompanies the mechanical contraction of a single muscle cell when stimulated by an electrical current (neural or external) [11, 20–26]. The action potential is caused by the flow of sodium (Na^+), potassium (K^+), chloride (Cl^-), and other ions across the cell membrane.

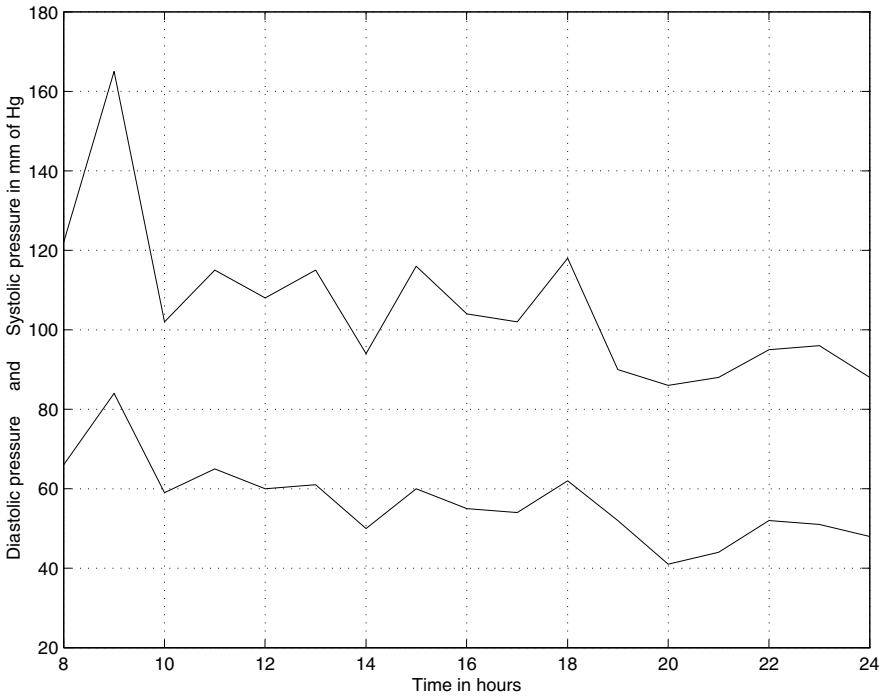
Action potentials are also associated with signals and messages transmitted in the nervous system with no accompanying contraction. Hodgkin and Huxley [20, 21] conducted pioneering work on recording action potentials from a nerve fiber.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \text{Systolic} \\ \text{Diastolic} \end{bmatrix} = \begin{bmatrix} 122 \\ 66 \end{bmatrix}$$

(a)

Time (h)	08:00	10:00	12:00	14:00	16:00	18:00	20:00	22:00	24:00
Systolic	122	102	108	94	104	118	86	95	88
Diastolic	66	59	60	50	55	62	41	52	48

(b)



(c)

Figure 1.3 Measurements of the BP of a patient presented as (a) a single pair or vector of systolic and diastolic measurements $\mathbf{x}$ in *mm of Hg* at an unspecified instant of time, (b) an array $\mathbf{x}(n)$ made up of several measurements at different instants of time, and (c) a signal $\mathbf{x}(t)$ or $\mathbf{x}(n)$. Note the use of boldface $\mathbf{x}$ to indicate that each measurement is a vector with two components. The horizontal axis of the plot represents time in *hours*; the vertical axis gives the systolic pressure (upper trace) and the diastolic pressure (lower trace) in *mm of Hg*. Data courtesy of Foothills Hospital, Calgary.

Recording an action potential requires the isolation of a single cell, and microelectrodes with tips of the order of a few micrometers to stimulate the cell and record the response [11].

Resting potential: Nerve and muscle cells are encased in a semipermeable membrane that permits selected substances to pass through while others are kept out. Body fluids surrounding cells are conductive solutions containing charged atoms known as ions. Figure 1.4 gives a schematic illustration of a cell and its characteristics.

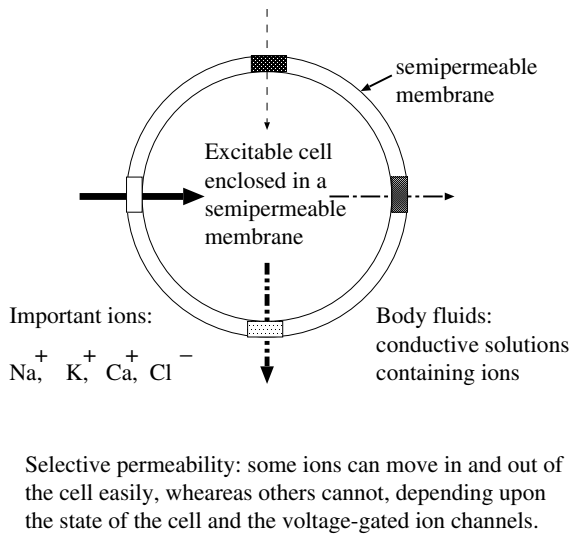


Figure 1.4 Schematic representation of a cell and its characteristics. The parts of the cell membrane with different shades and the corresponding arrows of different types and thickness represent, in a schematic manner, the variable permeability of the membrane to different ions.

In their resting state, the membranes of excitable cells readily permit the entry of K^+ and Cl^- ions, but effectively block the entry of Na^+ ions (the permeability for K^+ is 50–100 times that for Na^+). Various ions seek to establish a balance between the inside and the outside of a cell according to charge and concentration. The inability of Na^+ to penetrate a cell membrane results in the following [22]:

- Na^+ concentration inside the cell is far less than that outside.
- The outside of the cell is more positive than the inside of the cell.
- To balance the charge, additional K^+ ions enter the cell, causing higher K^+ concentration inside the cell than outside.
- Charge balance cannot be reached due to differences in membrane permeability for the various ions.

- A state of equilibrium is established with a potential difference, with the inside of the cell being negative with respect to the outside.

Figure 1.5 shows a schematic representation of a cell in its resting state. A cell in its resting state is said to be *polarized*. Most cells maintain a *resting potential* of the order of -60 to -100 mV until some disturbance or stimulus upsets the equilibrium.

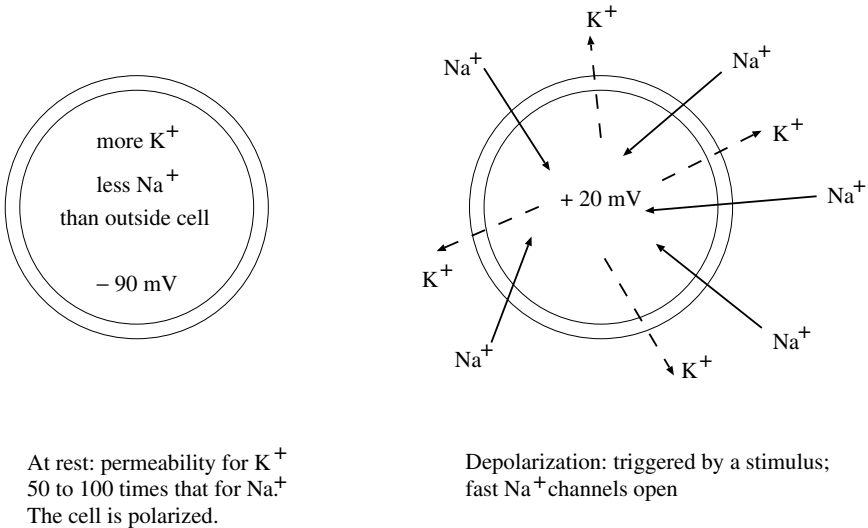


Figure 1.5 Schematic representation of a cell in its resting or polarized state (left) and the process of depolarization of the cell due to a stimulus (right).

Depolarization: When a cell is excited by ionic currents or an external stimulus, the membrane changes its characteristics and begins to allow Na^+ ions to enter the cell. This movement of Na^+ ions constitutes an ionic current, which further reduces the membrane barrier to Na^+ ions. This leads to an avalanche effect: Na^+ ions rush into the cell. K^+ ions try to leave the cell as they were in higher concentration inside the cell in the preceding resting state, but cannot move as fast as the Na^+ ions. The net result is that the inside of the cell becomes positive with respect to the outside due to an imbalance of K^+ ions. A new state of equilibrium is reached after the rush of Na^+ ions stops. This change represents the beginning of the *action potential*, with a peak value of about $+20$ mV for most cells. An excited cell displaying an action potential is said to be *depolarized*; the process is called *depolarization*. Figure 1.5 shows a schematic representation of a cell undergoing the process of depolarization.

Repolarization: After a certain period of being in the depolarized state the cell becomes polarized again and returns to its resting potential via a process known as *repolarization*. Repolarization occurs by processes that are analogous to those of depolarization, except that instead of Na^+ ions, the principal ions involved in repolarization are K^+ ions [24]. Membrane depolarization, while increasing the permeability for Na^+ ions, also increases the permeability of the membrane for K^+

ions via a specific class of ion channels known as voltage-dependent K^+ channels. Although this may appear to be paradoxical at first glance, the key to the mechanism for repolarization lies in the time-dependence and voltage-dependence of the membrane permeability changes for K^+ ions compared with those for Na^+ ions. The permeability changes for K^+ during depolarization occur considerably more slowly than those for Na^+ ions; hence, the initial depolarization is caused by an inrush of Na^+ ions. However, the membrane permeability changes for Na^+ spontaneously decrease near the peak of the depolarization process, whereas those for K^+ ions are beginning to increase. Hence, during repolarization, the predominant membrane permeability is for K^+ ions. Because K^+ concentration is much higher inside the cell than outside, there is a net efflux of K^+ from the cell, which makes the inside more negative, thereby effecting repolarization back to the resting potential.

It should be noted that the voltage-dependent K^+ permeability change is due to a distinctly different class of ion channels than those that are responsible for setting the resting potential. A mechanism known as the $Na^+ - K^+$ pump extrudes Na^+ ions in exchange for transporting K^+ ions back into the cell. However, this transport mechanism carries very little current in comparison with ion channels, and therefore makes a minor contribution to the repolarization process. The $Na^+ - K^+$ pump is essential for resetting the $Na^+ - K^+$ balance of the cell, but the process occurs on a longer time scale than the duration of an action potential.

Nerve and muscle cells repolarize rapidly, with an action potential duration of about 1 – 5 ms. Heart muscle cells repolarize slowly, with an action potential duration of 150 – 300 ms.

The action potential is always the same for a given cell, regardless of the method of excitation or the intensity of the stimulus beyond a threshold: This is known as the *all-or-none* or all-or-nothing phenomenon. After an action potential, there is a period during which a cell cannot respond to any new stimulus, known as the *absolute refractory period* (about 1 ms in nerve cells [27]). This is followed by a *relative refractory period* (3 – 5 ms in nerve cells), when another action potential may be triggered by a much stronger stimulus than in the normal situation [27]. Figure 1.6 shows the various phases or intervals of an action potential.

Figure 1.7 shows action potentials recorded from individual rabbit ventricular and atrial myocytes (muscle cells) [24]. Figure 1.8 shows a ventricular myocyte in its relaxed and fully contracted states. The tissues were first incubated in digestive enzymes, principally collagenase, and then dispersed into single cells using gentle mechanical agitation. The recording electrodes were glass patch pipettes; a whole-cell, current-clamp recording configuration was used to obtain the action potentials. The cells were stimulated at low rates (once per 8 s); this is far less than physiological rates. Moreover, the cells were maintained at 20 °C, rather than at body temperature. Nevertheless, the major features of the action potentials shown are similar to those recorded under physiological conditions.

The resting membrane potential of the cells (from 0 to 20 ms in the plots in Figure 1.7) is about -83 mV. A square pulse of current, 3 ms in duration and 1 nA in amplitude, was passed through the recording electrode and across the cell membrane, causing the cell to depolarize rapidly. The ventricular myocyte exhibits

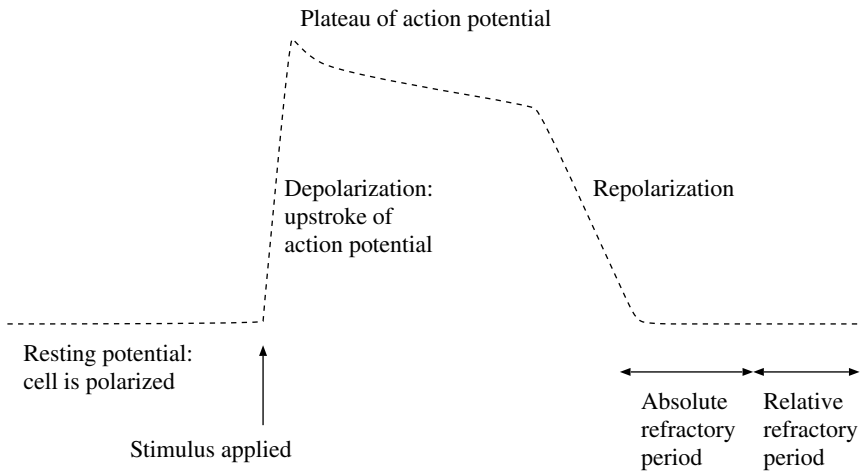


Figure 1.6 Illustration of the various phases or intervals of an action potential. Some specifications of the absolute refractory period include the duration of the action potential itself.

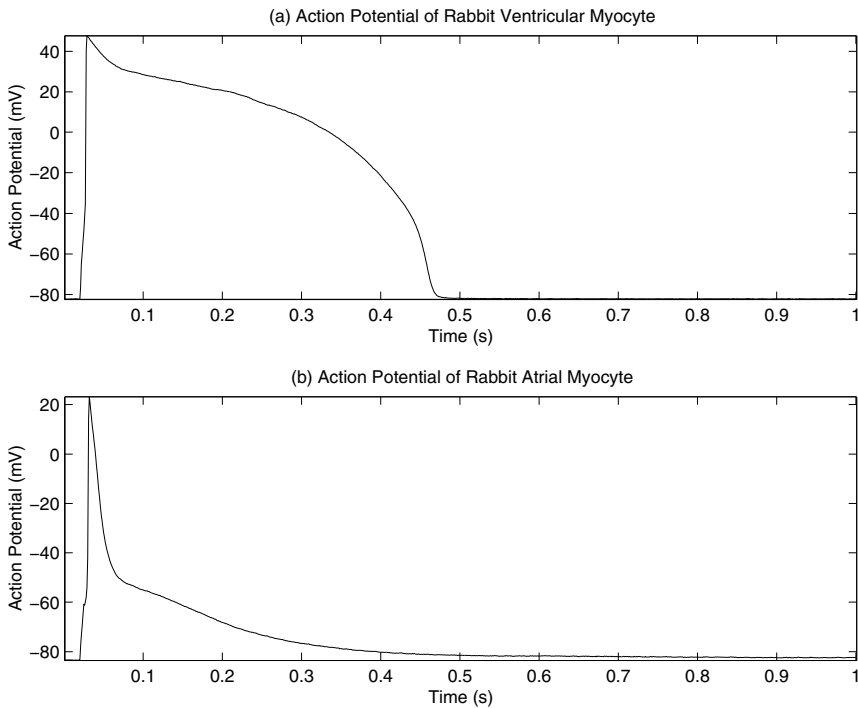
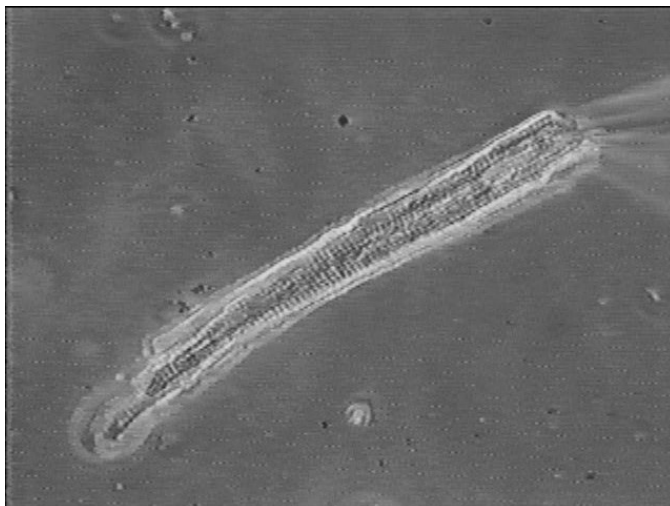


Figure 1.7 Action potentials of rabbit ventricular and atrial myocytes. Data courtesy of R. Clark, Department of Physiology and Biophysics, University of Calgary.



(a)



(b)

Figure 1.8 A single ventricular myocyte (of a rabbit) in its (a) relaxed and (b) fully contracted states. The length of the myocyte is approximately $25\ \mu\text{m}$. The tip of the glass pipette, faintly visible at the upper-right end of the myocyte, is approximately $2\ \mu\text{m}$ wide. Images courtesy of R. Clark, Department of Physiology and Biophysics, University of Calgary.

a depolarized potential of about $+40\text{ mV}$; it then slowly declines back to the resting potential level over an interval of about 500 ms . The initial rapid depolarization of the atrial cell is similar to that of the ventricular cell, but does not overshoot zero membrane potential as much as the ventricular action potential; repolarization occurs much more quickly than the case for the ventricular cell.

Figure 1.9 shows a schematic representation of a cell as a system. When a stimulus is applied, the cell provides a response or output. Depending upon the nature of the cell, such as a nerve cell or a muscle cell, the output could be an action potential, a twitch, or contraction (change in length). When several muscle cells are stimulated, the net result is the development of force.

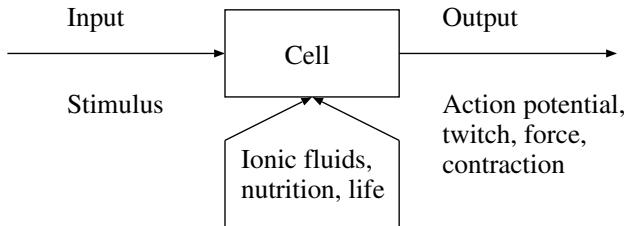


Figure 1.9 Schematic representation of a cell as a system. Upon receiving an input of a stimulus, the cell provides a response that could cause an action potential, a twitch, contraction, or force.

1.2.2 The action potential of a neuron

In studies of the central nervous system (CNS), it is desirable to record the action potentials of isolated neurons *in situ*. Hodgkin and Huxley [20, 21] conducted pioneering studies on recording action potentials from the giant axon of the squid; they proposed mathematical and electrical circuit models for the generation of action potentials.

Drake et al. [28] described the design and performance of a multisite microprobe system to record isolated and interrelated neuronal activity *in vivo*. Figure 1.10 shows a sample recording obtained from a rat's brain (cortex) using the microprobe. It was observed that the waveforms and durations of neuronal action potentials recorded with the microprobe were similar to those measured using single-needle microelectrodes [29]. Note that the durations of the neuronal action potentials are much shorter than those of cardiac myocytes. The study demonstrated the possibility of observing simultaneous spatiotemporal patterns of extracellular single-unit action potentials, which could be used to investigate the columnar organization and layering of cortical tissue. Furthermore, it was indicated that the microprobe could be useful for exploring the characteristics of extracellular fields around an active neuron.

Propagation of an action potential: An action potential propagates along a muscle fiber or an unmyelinated nerve fiber as follows [30]. Once initiated by a stimulus,

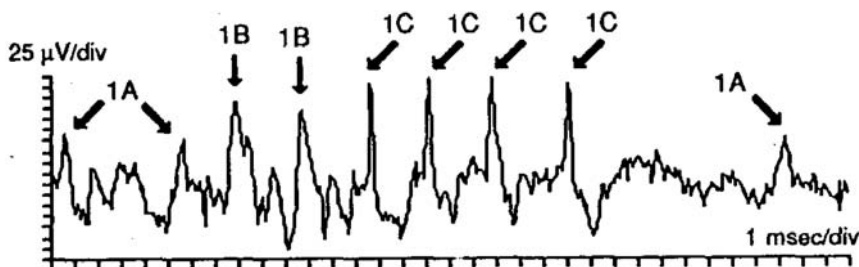


Figure 1.10 Neuronal action potentials recorded from a rat's brain using a multisite microprobe system. The action potentials marked 1A have low amplitude, just above the baseline neural noise level. The action potentials marked 1B possess broader waveforms and could possibly be composed of two superimposed action potentials. The action potentials marked 1C are clearly isolated, have the same amplitude, and are likely from the same cell. Reproduced with permission from K.L. Drake, K.D. Wise, J. Farraye, D.J. Anderson, and S.L. BeMent, Performance of planar multisite microprobes in recording extracellular single-unit intracortical activity, *IEEE Transactions on Biomedical Engineering*, 35(9):719–732, 1988. ©IEEE.

the action potential propagates along the whole length of a fiber without decrease in amplitude by progressive depolarization of the membrane. Current flows from a depolarized region through the intracellular fluid to adjacent inactive regions, thereby depolarizing them. Current also flows through the extracellular fluids, through the depolarized membrane, and back into the intracellular space, completing the local circuit. The energy to maintain conduction is supplied by the fiber itself.

Myelinated nerve fibers are covered by an insulating sheath of *myelin*. The sheath is interrupted every few millimeters by spaces known as the *nodes of Ranvier*, where the fiber is exposed to the interstitial fluid. Sites of excitation and changes of membrane permeability exist only at the nodes, and current flows by jumping from one node to the next in a process known as *saltatory conduction*.

1.2.3 The electroneurogram (ENG)

The ENG is an electrical signal observed as a stimulus and the associated nerve action potential propagate over the length of a nerve. It may be used to measure the velocity of propagation (or conduction velocity) of a stimulus or action potential in a nerve [11, 31, 32]. ENGs may be recorded using concentric needle electrodes or silver–silver-chloride electrodes ($Ag - AgCl$) at the surface of the body.

The conduction velocity in a peripheral nerve may be measured by stimulating a motor nerve and measuring the related activity at two points that are a known distance apart along its course. In order to minimize muscle contraction and other undesired effects, the experimental limb is held in a relaxed posture and a strong but short stimulus is applied in the form of a pulse of about 100 V amplitude and 100–300 μs duration [11, 31, 32]. The difference in the latencies of the ENGs recorded over

the associated muscle gives the conduction time. Knowing the separation distance between the stimulating and recording sites, it is possible to determine the conduction velocity in the nerve [11, 31, 32]. ENG's have amplitudes of the order of $10 \mu V$ and are susceptible to power-line interference and instrumentation noise.

Figure 1.11 illustrates the ENG's recorded in a study of nerve conduction velocity. The stimulus was applied to the ulnar nerve near the wrist. The ENG's were recorded at the wrist (marked "Wrist" in the figure), just below the elbow (BELbow), and just above the elbow (AEIbow) using surface electrodes, amplified with a gain of 2,000, and filtered to the bandwidth $10 - 10,000 \text{ Hz}$. The three traces in the figure indicate increasing latencies with respect to the stimulus time point, which is at the left margin of the plots. The responses shown in the figure are normal, indicate a BELbow–Wrist latency of 3.23 ms , and result in a nerve conduction velocity of 64.9 m/s .

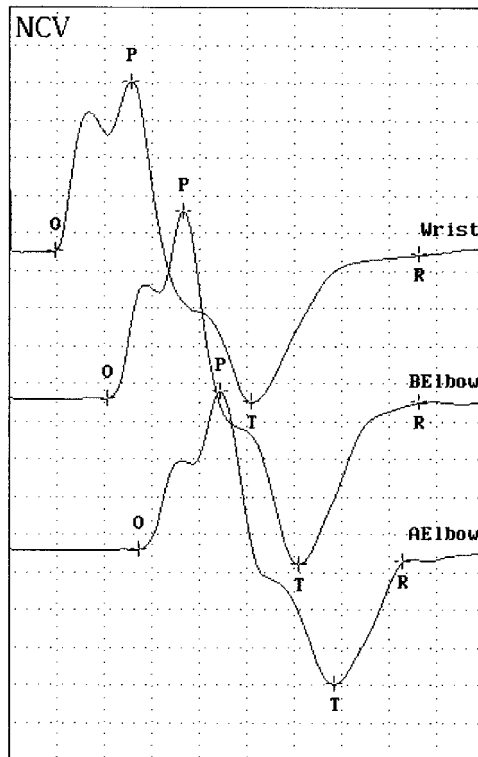


Figure 1.11 Nerve conduction velocity measurement via electrical stimulation of the ulnar nerve. The grid boxes represent 3 ms in width and $2 \mu V$ in height. AEIbow: above the elbow. BELbow: below the elbow. O: onset. P: Peak. T: trough. R: recovery of baseline. Courtesy of M. Wilson and C. Adams, Alberta Children's Hospital, Calgary.

Typical values of propagation rate or nerve conduction velocity are [11, 30, 33]:

- $45 - 70 \text{ m/s}$ in nerve fibers;

- $0.2 - 0.4 \text{ m/s}$ in heart muscle;
- $0.03 - 0.05 \text{ m/s}$ in time-delay fibers between the atria and ventricles.

Neural diseases may cause a decrease in conduction velocity.

1.2.4 The electromyogram (EMG)

Skeletal muscle fibers are considered to be twitch fibers because they produce a mechanical twitch response for a single stimulus and generate a propagated action potential. Skeletal muscles are made up of collections of *motor units* (MUs), each of which consists of an anterior horn cell (or motoneuron or motor neuron), its axon, and all muscle fibers innervated by that axon [34–36]. A motor unit is the smallest muscle unit that can be activated by volitional effort. The constituent fibers of a motor unit are activated synchronously. Component fibers of a motor unit extend lengthwise in loose bundles along the muscle. In cross-section, the fibers of a given motor unit are interspersed with the fibers of other motor units [11, 30, 37]. Figure 1.12 shows a schematic representation of two motor units. Figure 1.13 (top panel) illustrates a motor unit in schematic form along with related models [37]. When stimulated by a neural signal, each motor unit contracts and causes an electrical signal that is the summation of the action potentials of all of its constituent cells. This is known as the *single-motor-unit action potential* (SMUAP, or simply MUAP) and may be recorded using needle electrodes inserted into the muscle region of interest.

Muscles for gross movement or large force have hundreds of muscle fibers per motor unit; muscles for precise or fine movement have fewer fibers per motor unit. The number of muscle fibers per motor nerve fiber (or motor unit) is known as the *innervation ratio*. The mechanical output (contraction) of a muscle is the net result of stimulation and contraction of several of its motor units.

Various experiments have been conducted to estimate the innervation ratios of several muscles in animals and human beings, *in vivo* and using autopsy samples, with some variations in the methods and definitions used. Goodgold and Eberstein [30] and Kimura [27] provide tables including the following examples. The platysma (a large sheet-like muscle spanning parts of the pectoral muscle, deltoid, clavicle, and neck) has 1,826 large nerve fibers controlling 27,100 muscle fibers in 1,096 motor units, leading to an estimate of about 25 muscle fibers per motor unit. The first dorsal interosseus (FDI) muscle (on the back of the palm of the hand and the index finger) has 199 large nerve fibers and 40,500 muscle fibers in 119 motor units, with about 340 muscle fibers per motor unit. The medial gastrocnemius (calf muscle of the leg) has 965 large nerve fibers and 1,120,000 muscle fibers in 579 motor units, with about 1,934 muscle fibers per motor unit.

Laryngeal muscles have been estimated to have only two or three muscle fibers per motor unit [30]. Gath and Stålberg [38] used a multielectrode probe for *in situ* measurement of the innervation ratios of human muscles; they estimated the number of muscle fibers per motor unit to be 72 in the brachial biceps, 70 in the deltoid, and 124 in the tibialis anterior. Brown and Harvey [35] studied the muscles of cats'

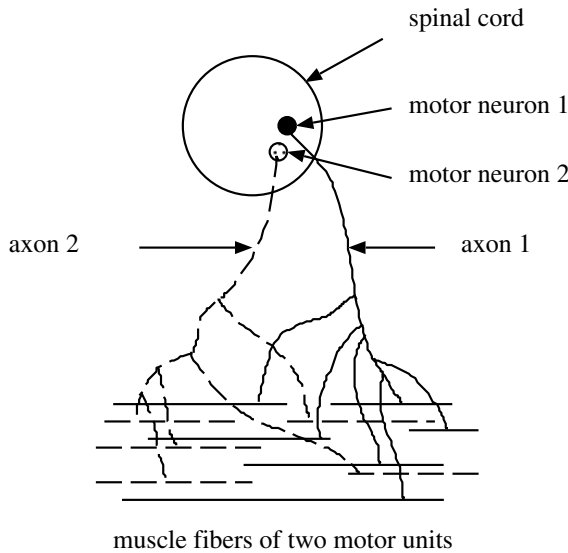


Figure 1.12 Schematic representation of two motor units, one in solid line and the other in dashed line.

eyes and noted that not more than 10 muscle fibers are supplied by a single nerve fiber in the extrinsic ocular muscles. See Goodgold and Eberstein [30], Kimura [27], Buchthal and Schmalbruch [34], and Brown et al. [36] for related discussions.

Normal SMUAPs are usually biphasic or triphasic, 3 – 15 ms in duration, 100 – 300 μV in amplitude, and appear with frequency in the range of 6 – 30/s [11, 30]. Schematic representations of monophasic, biphasic, and triphasic waveforms are shown in Figure 1.14. The shape of a recorded SMUAP depends upon the type of the needle electrode used, its position with respect to the active motor unit, and the projection of the electrical field of the activity on to the electrodes. Figure 1.15 illustrates simultaneous recordings of the activities of a few motor units from three channels of needle electrodes [39]. Although the SMUAPs are biphasic or triphasic, the same SMUAP displays variable shape from one channel to another. (*Note:* The action potentials in Figure 1.7 are monophasic; the first two SMUAPs in Channel 1 in Figure 1.15 are biphasic, and the third SMUAP in the same signal is triphasic.)

The shape of SMUAPs is affected by disease. Figure 1.16 illustrates SMUAP trains of a normal subject and those of patients with neuropathy and myopathy. Neuropathy causes slow conduction and/or desynchronized activation of the fibers within a motor unit, and a polyphasic SMUAP with an amplitude that is larger than normal. The same motor unit may be observed to fire at higher rates than normal before more motor units are recruited. Myopathy involves loss of muscle fibers in motor units, with the neurons presumably intact. *Splintering* of SMUAPs occurs due to asynchrony in activation as a result of patchy destruction of fibers (such as the case

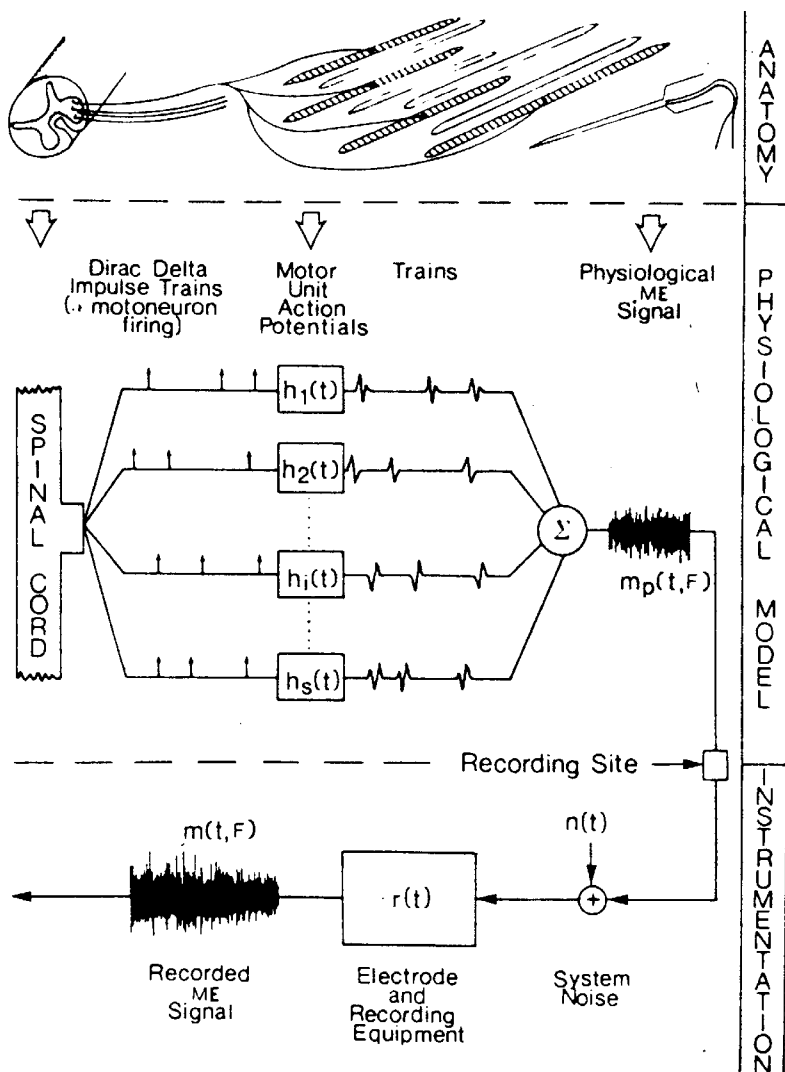


Figure 1.13 Schematic representation of a motor unit and model for the generation of EMG signals. Top panel: A motor unit includes an anterior horn cell or motor neuron (illustrated in a cross-section of the spinal cord), an axon, and several connected muscle fibers. The hatched fibers belong to one motor unit; the nonhatched fibers belong to other motor units. A needle electrode is also illustrated. Middle panel: The firing pattern of each motor neuron is represented by an impulse train. Each system $h_i(t)$ shown represents a motor unit that is activated and generates a train of SMUAPs. The net EMG is the sum of several SMUAP trains. Bottom panel: Effects of instrumentation on the EMG signal acquired. The observed EMG is a function of time t and muscular force produced F . ME: myoelectric. Reproduced with permission from C.J. de Luca, Physiology and mathematics of myoelectric signals, *IEEE Transactions on Biomedical Engineering*, 26:313–325, 1979. ©IEEE.

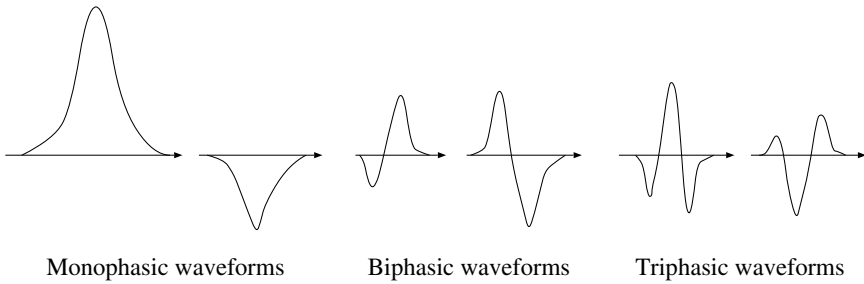


Figure 1.14 Schematic representations of monophasic, biphasic, and triphasic waveforms.

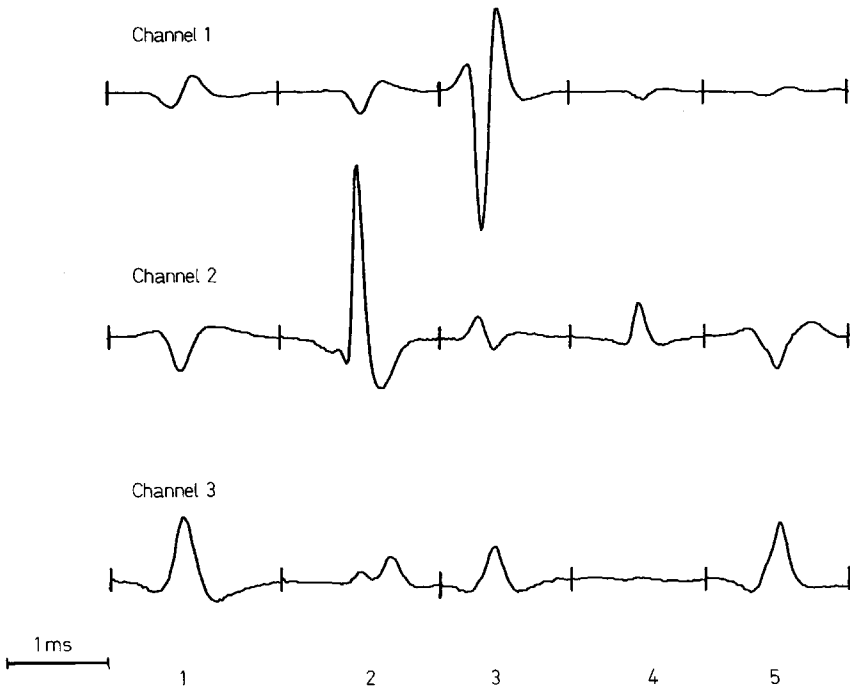


Figure 1.15 SMUAP trains recorded simultaneously from three channels of needle electrodes. Observe the different shapes of the same SMUAPs projected on to the axes of the three channels. Three different motor units are active over the duration of the signals illustrated. Reproduced with permission from B. Mambrito and C.J. de Luca, Acquisition and decomposition of the EMG signal, in *Progress in Clinical Neurophysiology*, Volume 10: *Computer-aided Electromyography*, Editor: J.E. Desmedt, pp 52–72, 1983. ©S. Karger AG, Basel, Switzerland.

in muscular dystrophy), leading to polyphasic SMUAPs. More motor units may be observed to be recruited at low levels of effort.

Gradation of muscular contraction: Muscular contraction levels are controlled in two ways:

- *Spatial recruitment*, by activating new motor units with increasing effort; and
- *Temporal recruitment*, by increasing the frequency of discharge (firing rate) of each motor unit with increasing effort.

Figure 1.17 gives a schematic illustration of spatiotemporal recruitment of motor units and the related EMG signals. Motor units are activated at different times and at different frequencies causing asynchronous contraction. The twitches of individual motor units sum and fuse to form tetanic contraction and increased force. Weak volitional effort causes motor units to fire at about 5 – 15 *pps* (pulses per second). As greater tension is developed, an *interference pattern* EMG is obtained, with the constituent and active motor units firing in the range of 25 – 50 *pps*. Grouping of MUAPs has been observed as fatigue develops, leading to decreased high-frequency content and increased amplitude in the EMG [37].

Spatiotemporal summation of the MUAPs of all of the active motor units gives rise to the EMG of the muscle; see Figure 1.17. EMG signals recorded using surface electrodes are complex signals including interference patterns of several MUAP trains and are difficult to analyze. An EMG signal indicates the level of activity of a muscle, and may be used to diagnose neuromuscular diseases such as neuropathy and myopathy.

Figure 1.18 illustrates an EMG signal recorded from the crural diaphragm of a dog using fine-wire electrodes sewn in-line with the muscle fibers and placed 10 *mm* apart [40]. The signal represents one period of breathing (inhalation being the active part as far as the muscle and EMG are concerned). It is seen that the overall level of activity in the signal increases during the initial phase of inhalation. Figure 1.19 shows the early parts of the same signal on an expanded time scale. SMUAPs are seen at the beginning stages of contraction, followed by increasingly complex interference patterns of several MUAPs.

Figure 1.20 shows an EMG signal recorded using surface electrodes placed on the forearm of a subject. In this experiment, designed to study the variation of the EMG signal with respect to the force exerted by a muscle, the subject performed a series of contractions using a gripping device equipped with a force transducer. (The output of the transducer was not calibrated in units of force.) The subject was instructed to squeeze the gripping device to the maximum extent possible; the corresponding output was noted as the level of maximal voluntary contraction (MVC). The subject was then asked to relax the muscle, and squeeze the device again in five steps, between approximately 20% and 100% MVC, each contraction lasting for about 2 – 3 *s*, with resting periods of about 2 – 3 *s* between each contraction. The EMG and force signals were lowpass filtered with the cutoff frequency at 1 *kHz*, highpass filtered with the cutoff frequency at 10 *Hz*, and recorded at the sampling rate of 2 *kHz* per channel. In the plot shown in Figure 1.20, the force signal has been normalized such that

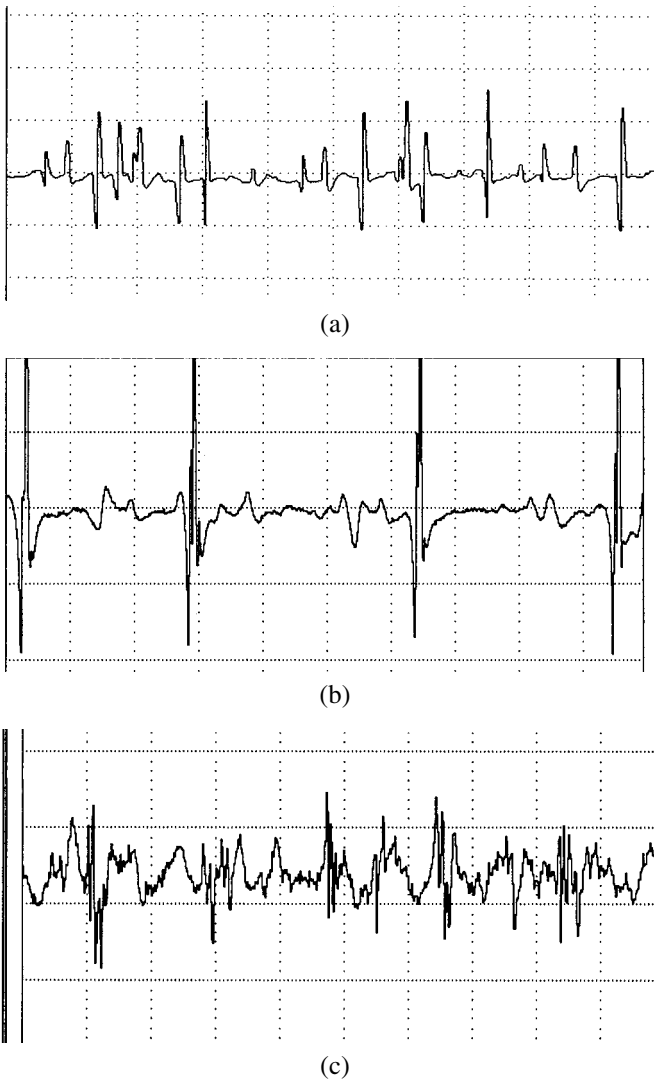
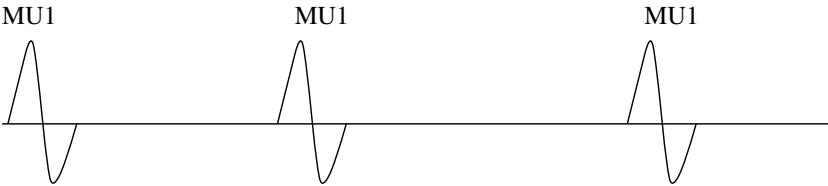
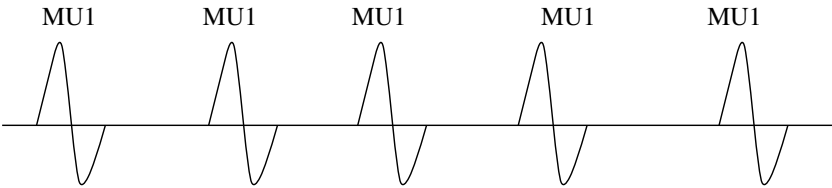


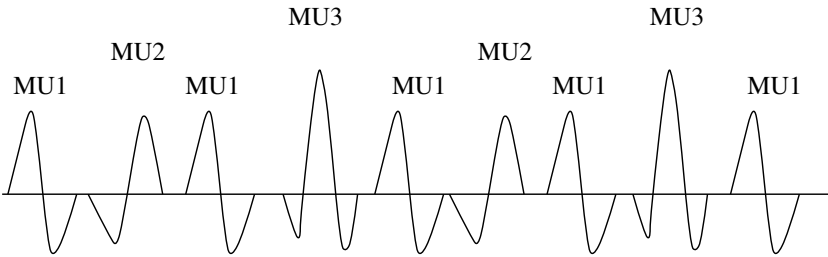
Figure 1.16 Examples of SMUAP trains. (a) From the right deltoid of a normal subject, male, 11 years; the SMUAPs are mostly biphasic, with duration in the range 3 – 5 *ms*. (b) From the deltoid of a six-month-old male patient with brachial plexus injury (neuropathy); the SMUAPs are polyphasic and large in amplitude (800 μV), and the same motor unit is firing at a relatively high rate at low-to-medium levels of effort. (c) From the right biceps of a 17-year-old male patient with myopathy; the SMUAPs are polyphasic and indicate early recruitment of more motor units at a low level of effort. The signals were recorded with gauge 20 needle electrodes. The width of each grid box represents a duration of 20 *ms*; its height represents an amplitude of 200 μV . Courtesy of M. Wilson and C. Adams, Alberta Children's Hospital, Calgary.



(a) At the beginning with low effort, only motor unit MU1 is firing at a low rate.



(b) At a slightly higher level of effort, with temporal recruitment, the firing rate of MU1 is increased. No other motor unit has been recruited yet.



(c) At an even higher level of effort, with spatial recruitment, new motor units MU2 and MU3 have been brought into action. MU1 continues to fire at the same rate as in (b).

Figure 1.17 Schematic representation of spatiotemporal recruitment of motor units and the resulting EMG signals. To keep the illustration simple, it is assumed that the MUAPs do not overlap.

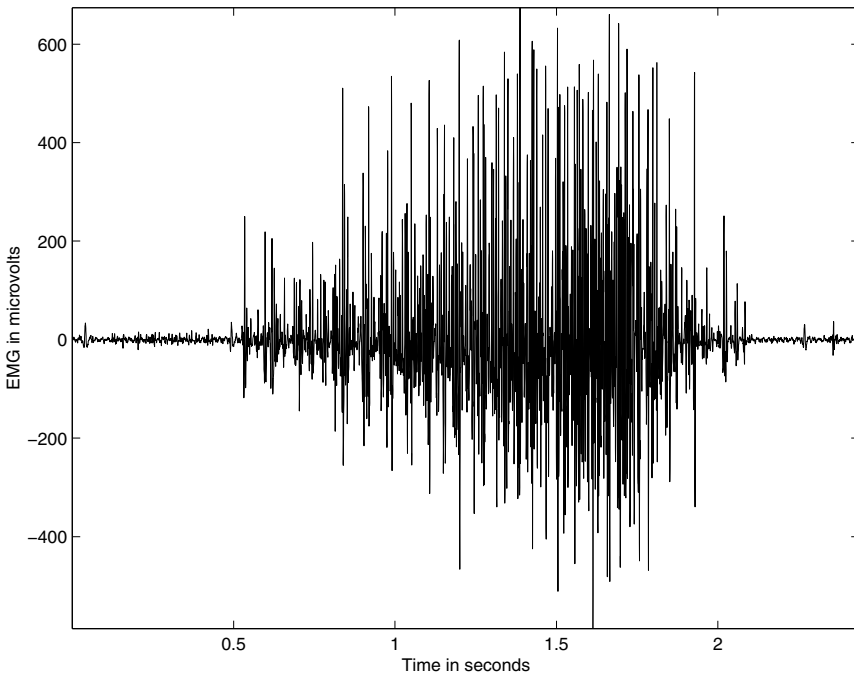


Figure 1.18 EMG signal recorded from the crural diaphragm muscle of a dog using implanted fine-wire electrodes; see also Figure 1.19. Data courtesy of R.S. Platt and P.A. Easton, Department of Clinical Neurosciences, University of Calgary.

the minimum is zero and the maximum is 100; thus, the force exerted is expressed in %MVC. It is evident that the dynamic range (peak-to-peak swing) and power of the EMG signal increase as the level of force exerted increases. Figure 1.21 shows as expanded view of the period 10 – 12 s of the force and EMG signals, where the increasing trend of the EMG signal with increasing force is clearly seen.

Signal processing techniques for the analysis of EMG signals are described in Sections 5.2.4, 5.6, 5.9, 5.10, 5.11, 5.13.4, 7.2.1, and 7.3. See Nikolic and Krarup [41] for the description of methods for decomposition of EMG signals into their constituent MUAPs and firing patterns for quantitative analysis. EMG signals are useful in the control of prosthetic devices [42, 43].

1.2.5 The electrocardiogram (ECG)

The ECG is the electrical manifestation of the contractile activity of the heart and can be recorded fairly easily with surface electrodes on the limbs or chest. The ECG is the most commonly known, recognized, and used biomedical signal. Original investigations on recording of the human ECG were conducted by Waller [44] and Einthoven [45] in the late 1800s and the early 1900s.

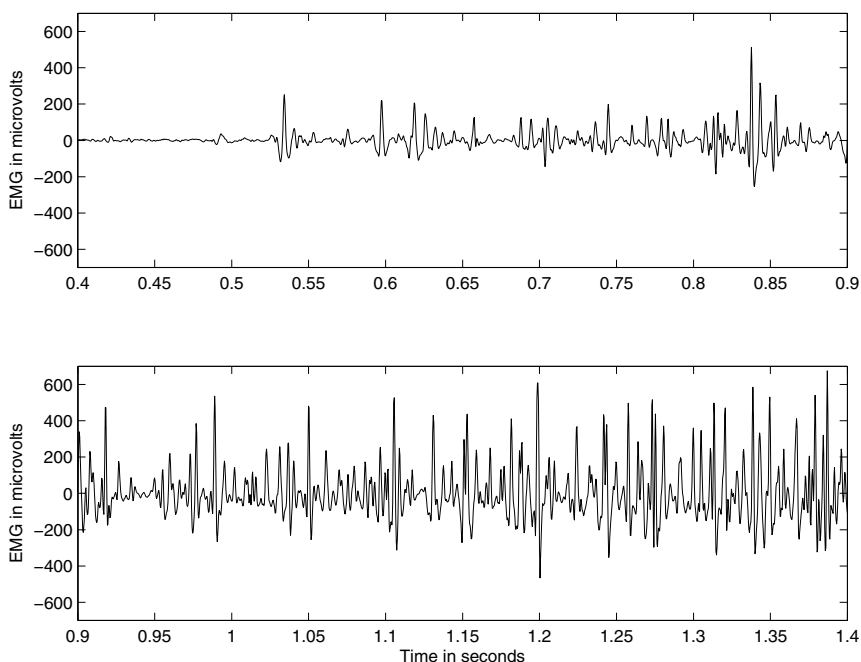


Figure 1.19 The initial part of the EMG signal in Figure 1.18 shown on an expanded time scale. Observe the SMUAPs at the initial stages of contraction, followed by increasingly complex interference patterns of several MUAPs. Data courtesy of R.S. Platt and P.A. Easton, Department of Clinical Neurosciences, University of Calgary.

The rhythm of the heart in terms of beats per minute (*bpm*) may be easily estimated by counting the readily identifiable waves. More important is the fact that the ECG waveshape is altered by cardiovascular diseases and abnormalities such as myocardial ischemia and infarction, ventricular hypertrophy, and conduction problems.

The heart: The heart is a four-chambered pump with two atria for collection of blood and two ventricles for pumping out of blood. Figure 1.22 shows a schematic representation of the four chambers and the major vessels connecting to the heart. The resting or filling phase of a cardiac chamber is called *diastole*; the contracting or pumping phase is called *systole*.

The right atrium (or auricle) collects deoxygenated blood from the superior and inferior vena cavae. During atrial contraction, blood is passed from the right atrium to the right ventricle through the tricuspid valve. During ventricular systole, the deoxygenated blood in the right ventricle is pumped out through the pulmonary valve to the lungs for oxygenation.

The left atrium receives oxygenated blood from the lungs, which is passed on during atrial contraction to the left ventricle via the mitral valve. The left ventricle is the largest and most important cardiac chamber. The left ventricle contracts the

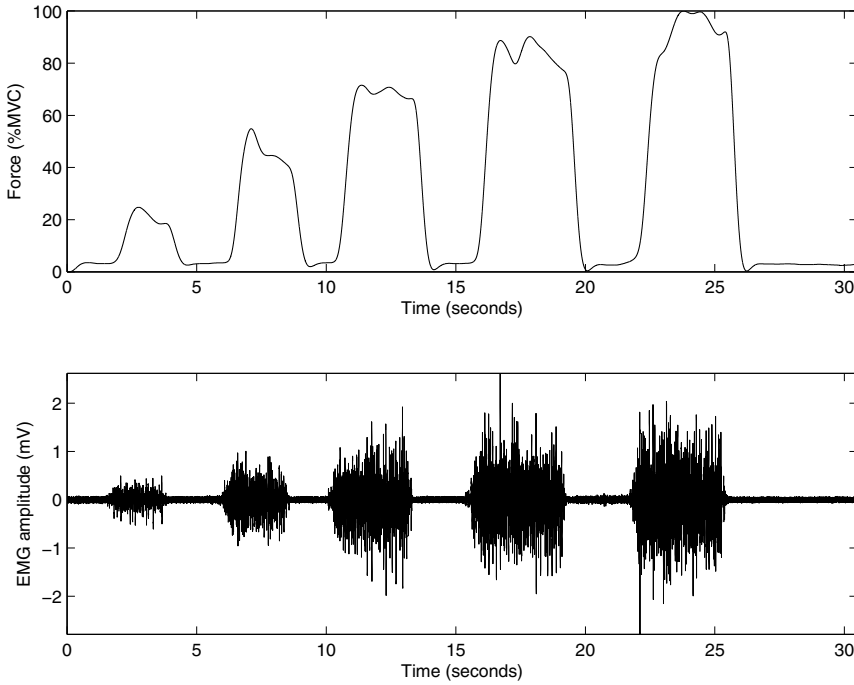


Figure 1.20 Force signal (upper plot) and the EMG signal (lower plot) recorded from the forearm muscle of a subject using surface electrodes; see also Figure 1.21. Data courtesy of Shantanu Banik.

strongest among the cardiac chambers, as it has to pump out the oxygenated blood through the aortic valve and the aorta against the pressure of the rest of the vascular system of the body. Due to the higher level of importance of contraction of the ventricles, the terms systole and diastole are applied to the ventricles by default.

The heart rate (HR) or cardiac rhythm is controlled by specialized pacemaker cells that form the sinoatrial (SA) node located at the junction of the superior vena cava and the right atrium [33]. The firing rate of the SA node is controlled by the autonomic nervous system (ANS) leading to the delivery of the neurotransmitters acetylcholine (for vagal stimulation, causing a reduction in heart rate) or epinephrine (for sympathetic stimulation, causing an increase in the heart rate). The normal (resting) heart rate is about 70 bpm . The heart rate is lower during sleep, but abnormally low heart rates below 60 bpm during activity could indicate a disorder called *bradycardia*. The instantaneous heart rate could reach values as high as 200 bpm during vigorous exercise or athletic activity; a high resting heart rate could be due to illness, disease, or cardiac abnormalities, and is termed *tachycardia*.

The electrical system of the heart: Coordinated electrical events and a specialized conduction system intrinsic and unique to the heart play major roles in the rhythmic contractile activity of the heart. The SA node is the basic, natural cardiac

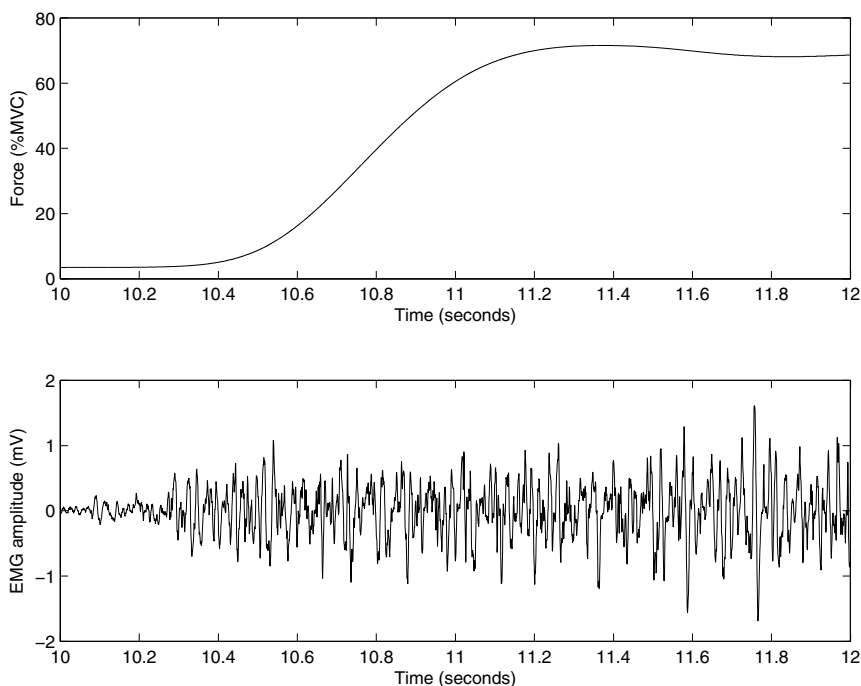


Figure 1.21 Expanded view of the part of the EMG (lower plot) and force signals (upper plot) in Figure 1.20 over the period 10 – 12 s. Observe the increasing levels of the range and power of the EMG signal at the initial stages of contraction. Data courtesy of Shantanu Banik.

pacemaker that triggers its own train of action potentials. The action potential of the SA node propagates through the rest of the heart, causing a particular pattern of excitation and contraction (see Figure 1.23). A schematic ECG signal is shown in Figure 1.24 with labels showing the names and durations of the various component waves. Also shown in the figure are representations of the action potentials of the atrial and ventricular myocytes. The sequence of events and waves in a cardiac cycle is as follows [33]:

1. The SA node fires.
2. Electrical activity is propagated through the atrial musculature at comparatively low rates, causing slow-moving depolarization (contraction) of the atria. This results in the P wave in the ECG (see Figures 1.24 and 1.25). Due to the slow contraction of the atria and their small size, the P wave is a slow, low-amplitude wave, with an amplitude of about $0.1 - 0.2 \text{ mV}$ and a duration of about $60 - 80 \text{ ms}$.
3. The excitation wave faces a propagation delay at the atrioventricular (AV) node, which results in a normally isoelectric segment of about $60 - 80 \text{ ms}$ after the P

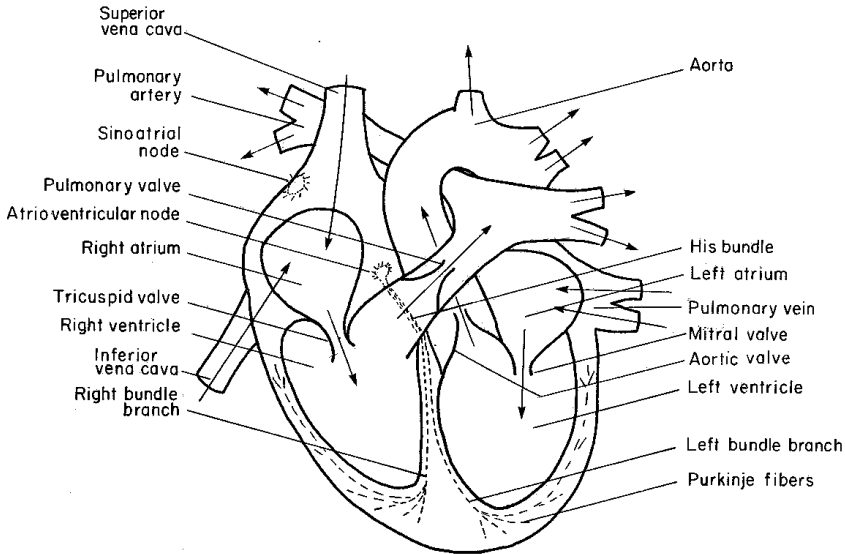


Figure 1.22 Schematic representation of the chambers, valves, vessels, and conduction system of the heart.

wave in the ECG, known as the PQ segment. The pause assists in the completion of the transfer of blood from the atria to the ventricles.

4. The AV node fires.
5. The His bundle, the bundle branches, and the Purkinje system of specialized conduction fibers propagate the stimulus to the ventricles at a high rate.
6. The wave of stimulus spreads rapidly from the apex of the heart upwards, causing rapid depolarization (contraction) of the ventricles. This results in the QRS wave of the ECG — a sharp biphasic or triphasic wave of about 1 mV amplitude and 80 ms duration (see Figure 1.25).
7. The plateau portion of the action potential causes a normally isoelectric segment of about $100\text{--}120\text{ ms}$ after the QRS, known as the ST segment; see Figure 1.24. This is because ventricular muscle cells possess a relatively long action potential duration of $300\text{--}350\text{ ms}$ (see Figure 1.7).
8. Repolarization (relaxation) of the ventricles causes the slow T wave, with an amplitude of $0.1\text{--}0.3\text{ mV}$ and duration of $120\text{--}160\text{ ms}$ (see Figure 1.25).

A summary of the various events described above, represented as parts of a cardiac cycle in relation to an ECG signal, is given in Figure 1.26.

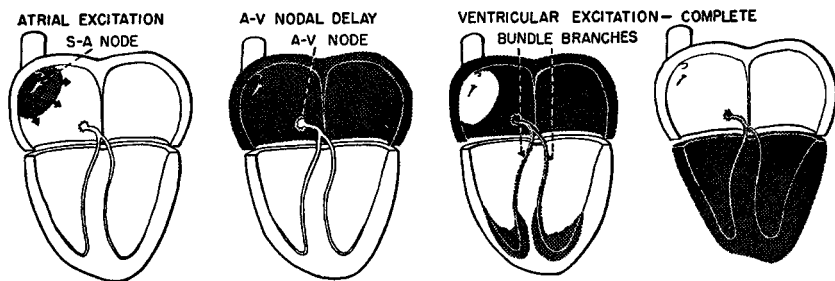


Figure 1.23 Propagation of the excitation pulse through the heart. Reproduced with permission from R.F. Rushmer, *Cardiovascular Dynamics*, 4th edition, ©W.B. Saunders, Philadelphia, PA, 1976.

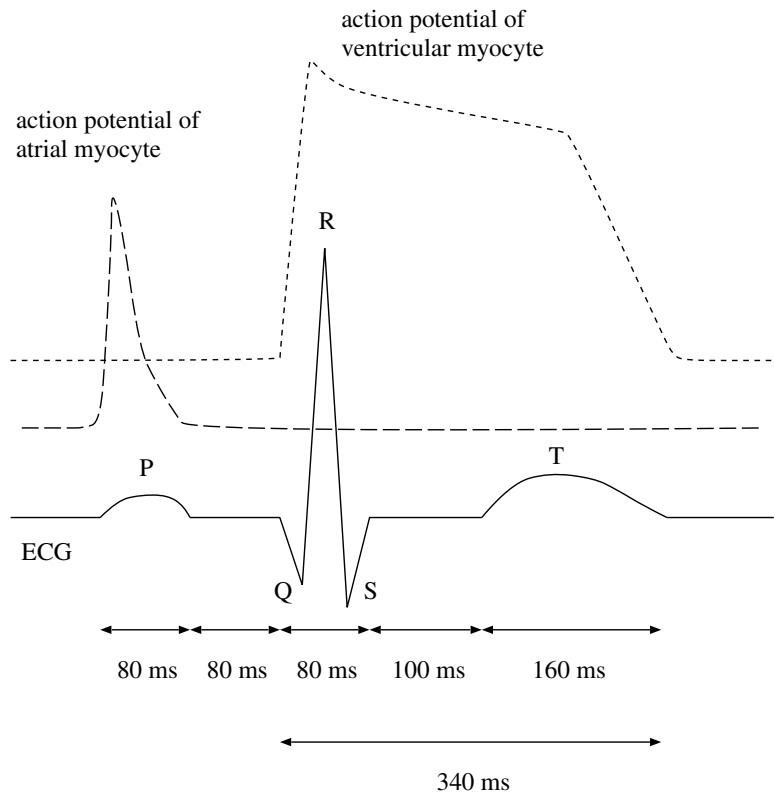


Figure 1.24 Schematic representations of an ECG signal and the action potentials of atrial and ventricular myocytes. See also Figure 1.7.

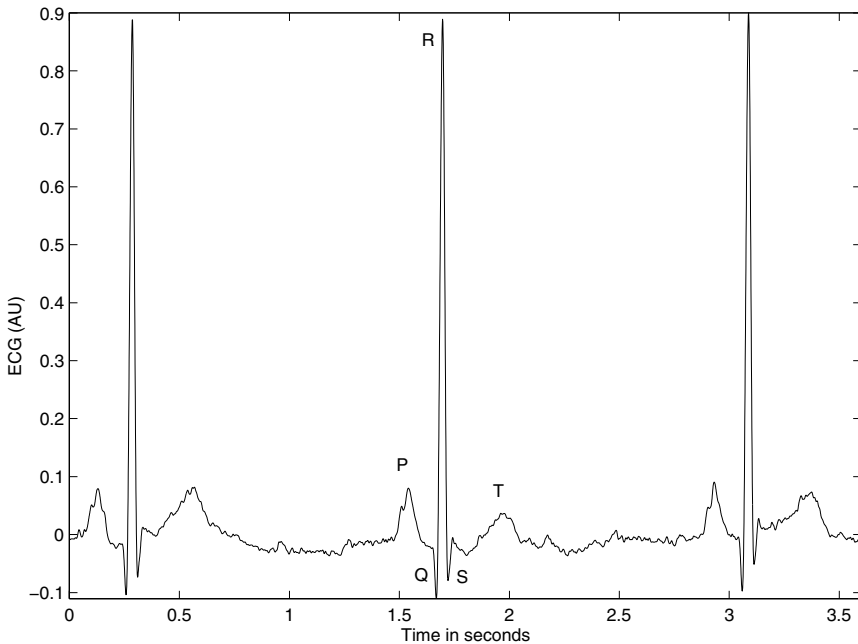


Figure 1.25 A typical ECG signal (male subject of age 24 years). (*Note:* Signal values may not be calibrated, that is, specified in actual units, in some situations. Sometimes, the calibration details may be lost in the acquisition and processing steps. As is the case in this plot, signal values in plots in this book are in arbitrary units (AU) or normalized units unless specified.)

Any disturbance in the regular rhythmic activity of the heart is called *arrhythmia*. Cardiac arrhythmia may be caused by irregular firing patterns from the SA node, or by abnormal and additional pacing activity from other parts of the heart. Many parts of the heart possess inherent rhythmicity and pacemaker properties; for example, the SA node, the AV node, the Purkinje fibers, atrial tissue, and ventricular tissue. If the SA node is inactive or depressed, any one of the above tissues may take over the role of the pacemaker or introduce *ectopic* beats. Different types of abnormal rhythm (arrhythmia) result from variations in the site and frequency of impulse formation. Premature ventricular contractions (PVCs) caused by ectopic foci on the ventricles upset the regular rhythm and may lead to ventricular dissociation and fibrillation — a state of disorganized contraction of the ventricles independent of the atria — resulting in no effective pumping of blood and possibly death. The waveshapes of PVCs are usually substantially different from those of the normal beats of the same subject due to the different conduction paths of the ectopic impulses and the associated abnormal contraction events. Figure 1.27 shows an ECG signal with a few normal beats and two PVCs. (See Figure 9.9 for an illustration of ventricular

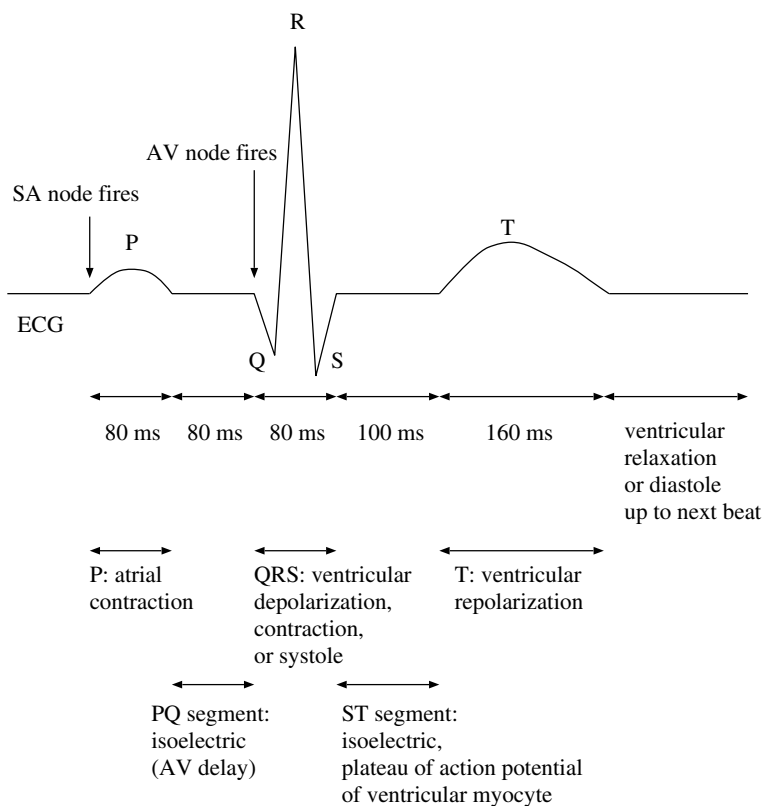


Figure 1.26 Summary of the parts and waves of a cardiac cycle as seen in an ECG signal.

bigeminy, where every second pulse from the SA node is replaced by a PVC with a full compensatory pause.)

The QRS waveshape is affected by conduction disorders; for example, bundle-branch block causes a widened and possibly jagged QRS. Figure 1.28 shows the ECG signal of a patient with right bundle-branch block. Observe the wider-than-normal QRS complex, which also displays a waveshape that is significantly different from the normal QRS waves. Ventricular hypertrophy (enlargement) could also cause a wider-than-normal QRS.

The ST segment, which is normally isoelectric (flat and in line with the PQ segment) may be elevated or depressed due to myocardial ischemia (reduced blood supply to a part of the heart muscles due to a block in the coronary arteries) or due to myocardial infarction (dead myocardial tissue incapable of contraction due to total lack of blood supply). Many other diseases cause specific changes in the ECG waveshape: the ECG is an important signal that is useful in heart-rate (rhythm) monitoring and the diagnosis of cardiovascular diseases.

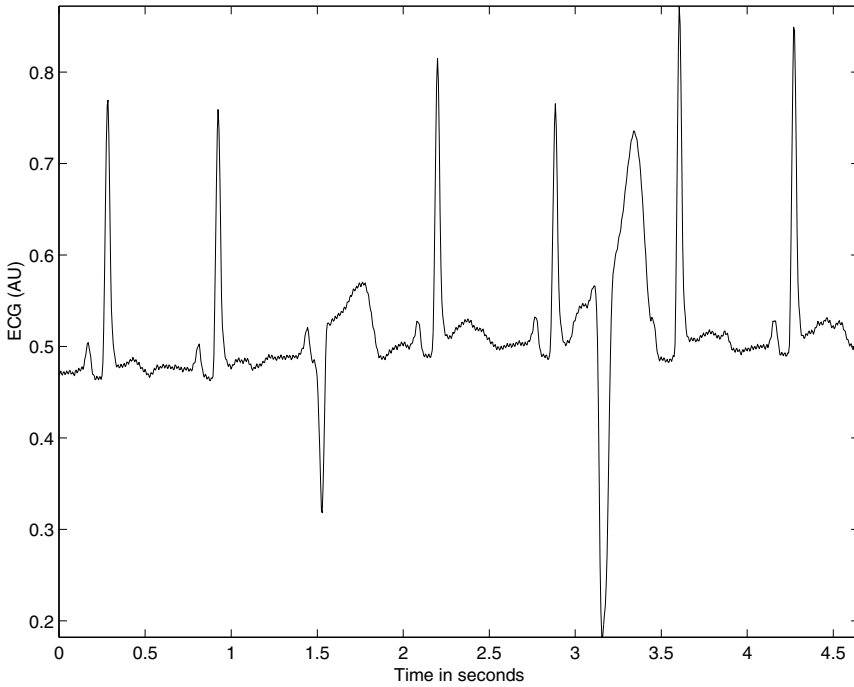


Figure 1.27 ECG signal with PVCs. The third and sixth beats are PVCs. The first PVC has blocked the normal beat that would have appeared at about the same time instant, but the second PVC has not blocked any normal beat triggered by the SA node. Data courtesy of G. Groves and J. Tyberg, Department of Physiology and Biophysics, University of Calgary.

ECG signal acquisition: In clinical practice, the standard 12-lead ECG is obtained using four limb leads and chest leads in six positions [33, 46]. The right leg is used to place the reference (ground) electrode. The left arm, right arm, and left leg are used to obtain leads I, II, and III. The illustration in Figure 1.29 shows the limb leads used to acquire the commonly used lead II ECG.

A combined reference known as *Wilson's central terminal* is formed by combining the left arm, right arm, and left leg leads; it is used as the reference for chest leads. The *augmented* limb leads known as aVR, aVL, and aVF (aV for the augmented lead, R for the right arm, L for the left arm, and F for the left leg) are obtained by using the exploring electrode on the limb indicated by the name of the lead, with the reference being Wilson's central terminal without the exploring limb lead.

Figure 1.30 shows the directions of the axes formed by the six limb leads. The hypothetical equilateral triangle formed by leads I, II, and III is known as *Einthoven's triangle*. The center of the triangle represents Wilson's central terminal. Schematically, the heart is assumed to be placed at the center of the triangle. The six leads measure projections of the three-dimensional (3D) cardiac electrical vector on to the axes illustrated in Figure 1.30. The six axes sample the $0^\circ - 180^\circ$ range in steps of

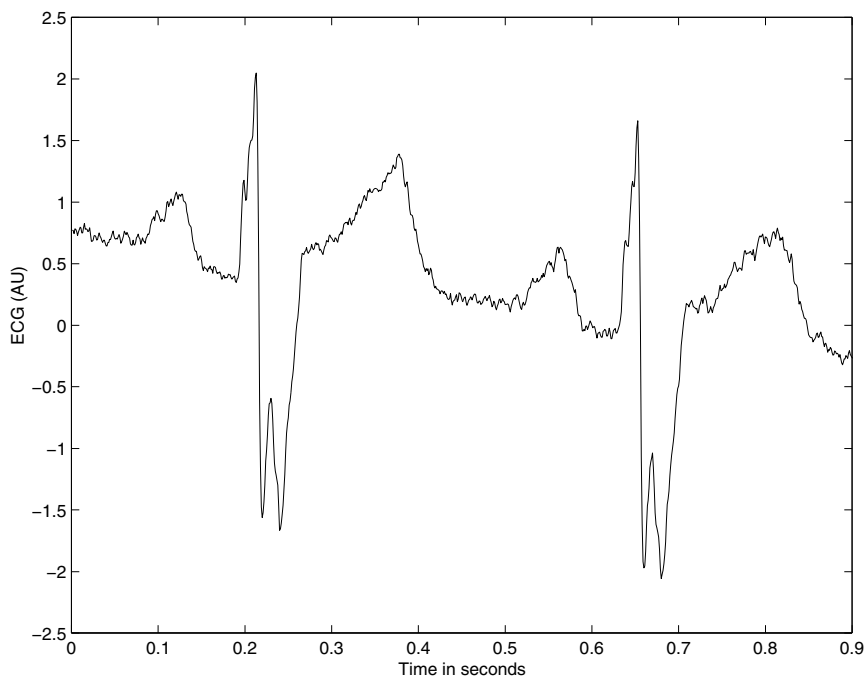


Figure 1.28 ECG signal of a patient with right bundle-branch block and hypertrophy (male patient of age 3 months). The QRS complex is wider than normal, and displays an abnormal, jagged waveform due to desynchronized contraction of the ventricles. (The signal also has a baseline drift, which has not been corrected.)

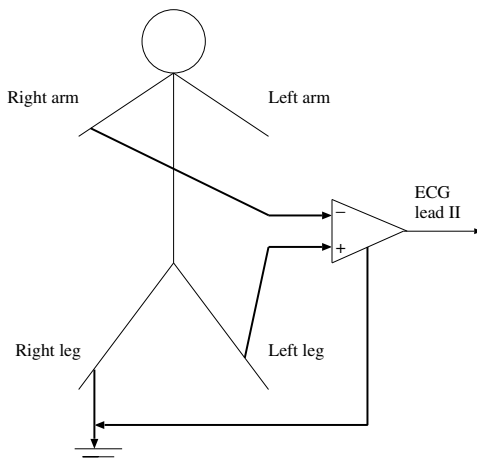


Figure 1.29 Limb leads used to acquire the commonly used lead II ECG. *Note:* The labeling of the left or right side refers to the corresponding side of the patient or subject, as in medical convention, and not the side of the reader.

approximately 30° . The projections facilitate viewing and analysis of the electrical activity of the heart from different perspectives in the frontal plane.

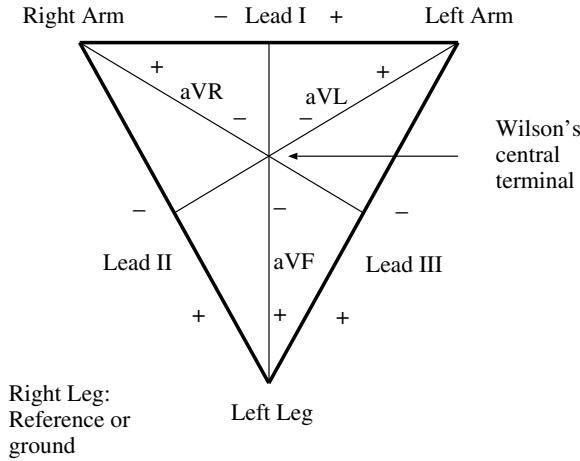


Figure 1.30 Einthoven's triangle and the axes of the six ECG leads formed by using four limb leads.

The six chest leads (written as V1–V6) are obtained from six standardized positions on the chest [33] with Wilson's central terminal as the reference. The positions for placement of the precordial (chest) leads are indicated in Figure 1.31. The V1 and V2 leads are placed at the fourth intercostal space just to the right and left of the sternum, respectively. V4 is recorded at the fifth intercostal space at the left mid-clavicular line. The V3 lead is placed half-way between the V2 and V4 leads. The V5 and V6 leads are located at the same level as the V4 lead, but at the anterior axillary line and the midaxillary line, respectively. The six chest leads permit viewing the cardiac electrical vector from different orientations in a cross-sectional plane: V5 and V6 are most sensitive to left-ventricular activity; V3 and V4 depict septal activity best; V1 and V2 reflect well activity in the right-half of the heart.

In spite of being redundant, the 12-lead system serves as the basis of the standard clinical ECG. Clinical ECG interpretation is mainly empirical. A compact and efficient system has been proposed for *vectorcardiography* or VCG [33,47], where loops inscribed by the 3D cardiac electrical vector in three mutually orthogonal planes, namely, the frontal, horizontal, and sagittal planes, are plotted and analyzed. Regardless, the 12-lead scalar ECG is the most commonly used procedure in clinical practice.

Because the external ECG is a projection of the internal 3D cardiac electrical vector, the external ECG recordings are not unique. Some of the lead interrelationships are [33, 46]:

- $II = I + III$,
- $aVL = (I - III) / 2$,

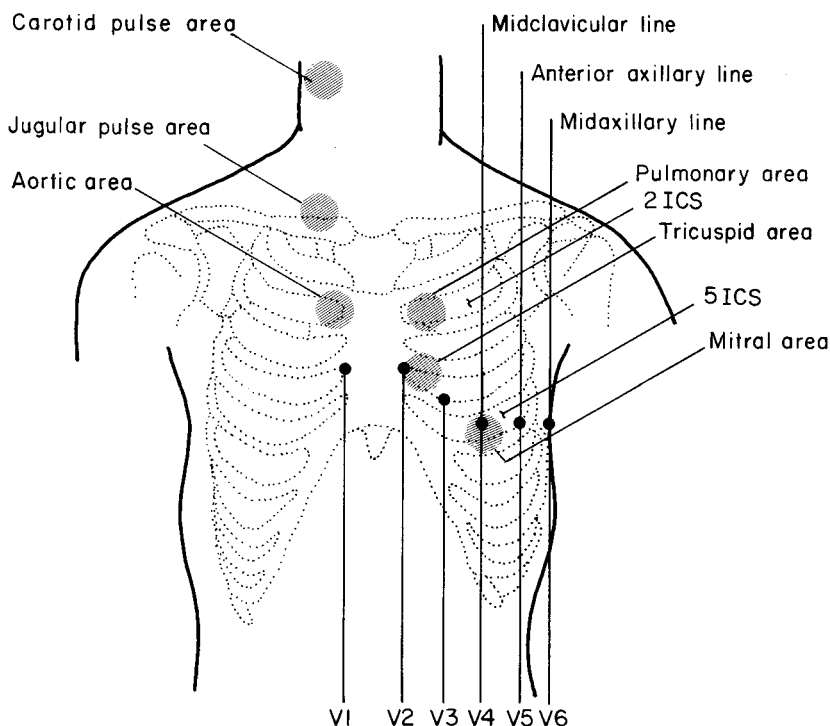


Figure 1.31 Positions for placement of the precordial (chest) leads V1–V6 for ECG, auscultation areas for heart sounds, and pulse transducer positions for the carotid and jugular pulse signals. ICS: intercostal space.

which are depicted in Figures 1.32 and 1.33.

Some of the important features of the standard clinical ECG are:

- A rectangular calibration pulse of 1 mV amplitude and 200 ms duration is applied to produce a pulse of 1 cm height on the plot.
- The typical recording speed used is 25 mm/s , resulting in a graphical scale of 0.04 s/mm or 40 ms/mm . Then, the width of the calibration pulse is 5 mm .
- The ECG signal peak value is normally about 1 mV .
- The amplifier gain used is 1,000.
- Clinical ECG is usually filtered to a bandwidth of about $0.05 - 100\text{ Hz}$, with a recommended sampling rate of 500 Hz for diagnostic ECG. Distortions in the shape of the calibration pulse may indicate improper filter settings or a poor signal acquisition system.

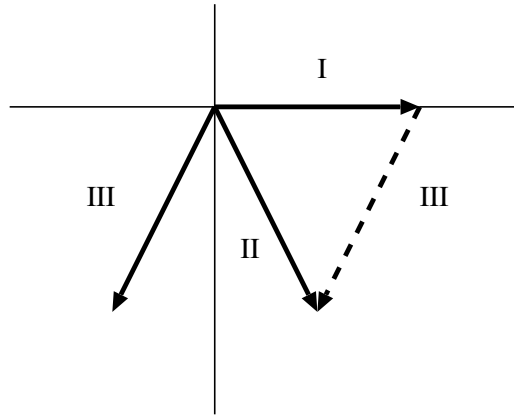


Figure 1.32 Vectorial relations between ECG leads I, II, and III. See also Figure 1.30.

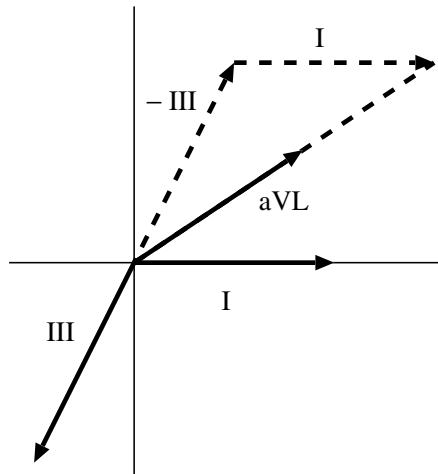


Figure 1.33 Vectorial relations between ECG leads I, II, and III. See also Figure 1.30.

- ECG for heart-rate monitoring could use a reduced bandwidth $0.5 - 50 \text{ Hz}$ and a lower sampling rate 100 Hz .
- High-resolution ECG requires a greater bandwidth of $0.05 - 500 \text{ Hz}$.

Some of the features mentioned above are related to recording of the ECG on paper and may not be relevant to digital or computer recording of the ECG.

For details of other systems of ECG leads and electrodes, see Rushmer [33], Friedman [48], Goldberger [47], Malmivuo and Plonsey [49], Webster [11], and Draper et al. [50].

Figure 1.34 shows the 12-lead ECG of a normal male adult. The system used to obtain the illustration records three channels at a time: leads I, II, III; aVR, aVL, aVF; V1, V2, V3; and V4, V5, V6 are recorded in the three available channels simultaneously. Other systems may record one channel at a time. Observe the changing shape of the ECG waves from one lead to another. A well-trained cardiologist is able to deduce the 3D orientation of the cardiac electrical vector by analyzing the wave-shapes in the six limb leads. Cardiac defects, if any, may be localized by analyzing the waveshapes in the six chest leads.

Figure 1.35 shows the 12-lead ECG of a patient with right bundle-branch block with secondary repolarization changes. The increased QRS width and distortions in the QRS shape indicate the effects of asynchronous activation of the ventricles due to the bundle-branch block.

Signal processing techniques to filter ECG signals are presented in Sections 3.3, 3.5, 3.6, 3.8, and 3.12. Detection of ECG waveforms is discussed in Sections 4.2.1, 4.3.2, 4.7, and 4.9. Analysis of ECG waveform shape and classification of beats are dealt with in Sections 5.2.1, 5.2.2, 5.2.3, 5.4, 5.7, 5.8, 9.2.1, and 9.11. Analysis of heart-rate variability (HRV) is described in Sections 7.2.2, 7.8, and 8.11. For reviews of computer applications in ECG analysis, see Jenkins [51, 52] and Cox et al. [53].

1.2.6 The electroencephalogram (EEG)

The EEG (popularly known as *brain waves*) represents the electrical activity of the brain [54–56]. A few important aspects of the organization of the brain are as follows. The main parts of the brain are the cerebrum, the cerebellum, the brain stem (including the midbrain, pons medulla, and the reticular formation), and the thalamus (between the midbrain and the hemispheres). Figure 1.36 gives a schematic representation of the various parts and functional areas of the human brain. The cerebrum is divided into two hemispheres, separated by a longitudinal fissure across which there is a large connective band of fibers known as the corpus callosum. The outer surface of the cerebral hemispheres, known as the cerebral cortex, is composed of neurons (gray matter) in convoluted patterns, and separated into regions by fissures (sulci). Beneath the cortex lie nerve fibers that lead to other parts of the brain and the body (white matter).

Cortical potentials are generated due to excitatory and inhibitory postsynaptic potentials developed by cell bodies and dendrites of pyramidal neurons. Physiological control processes, thought processes, and external stimuli generate signals in the corresponding parts of the brain that may be recorded at the scalp using surface electrodes. The scalp EEG is an average of the multifarious activities of many small zones of the cortical surface beneath the electrode.

In clinical practice, several channels of the EEG are recorded simultaneously from various locations on the scalp for comparative analysis of activities in different regions of the brain. The International Federation of Societies for Electroencephalography and Clinical Neurophysiology recommended the 10 – 20 system of electrode placement for clinical EEG recording [54], which is schematically illustrated in Figure 1.37. The name 10 – 20 indicates the fact that the electrodes along the midline are

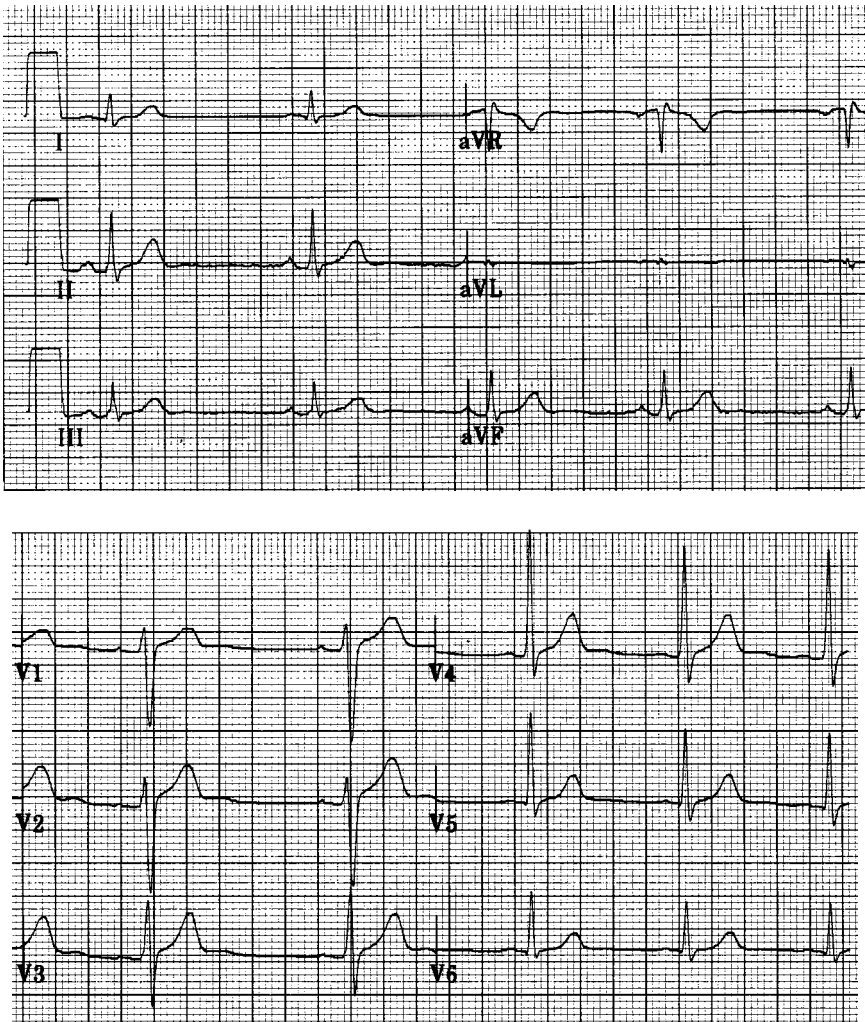


Figure 1.34 Standard 12-lead ECG of a normal male adult. Signal courtesy of E. Gedamu and L.B. Mitchell, Foothills Hospital, Calgary.

placed at 10%, 20%, 20%, 20%, 20%, and 10% of the total nasion–inion distance; the other series of electrodes are also placed at similar fractional distances of the corresponding reference distances [54]. The interelectrode distances are equal along any anteroposterior or transverse line, and electrode positioning is symmetrical. EEG signals may be used to study the nervous system, to monitor sleep stages [57,58], for biofeedback and control, for brain–computer interfacing, and for detection or diagnosis of epilepsy (seizure) [59–61].

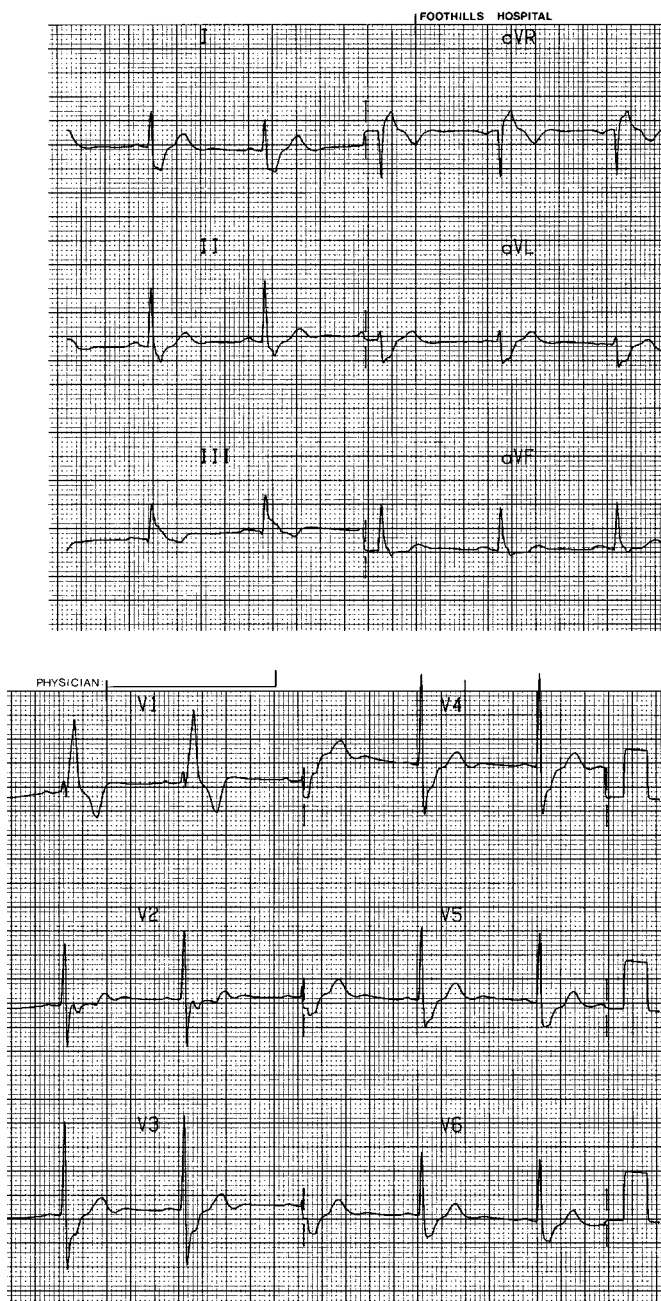


Figure 1.35 Standard 12-lead ECG of a patient with right bundle-branch block. Signal courtesy of L.B. Mitchell, Foothills Hospital, Calgary.

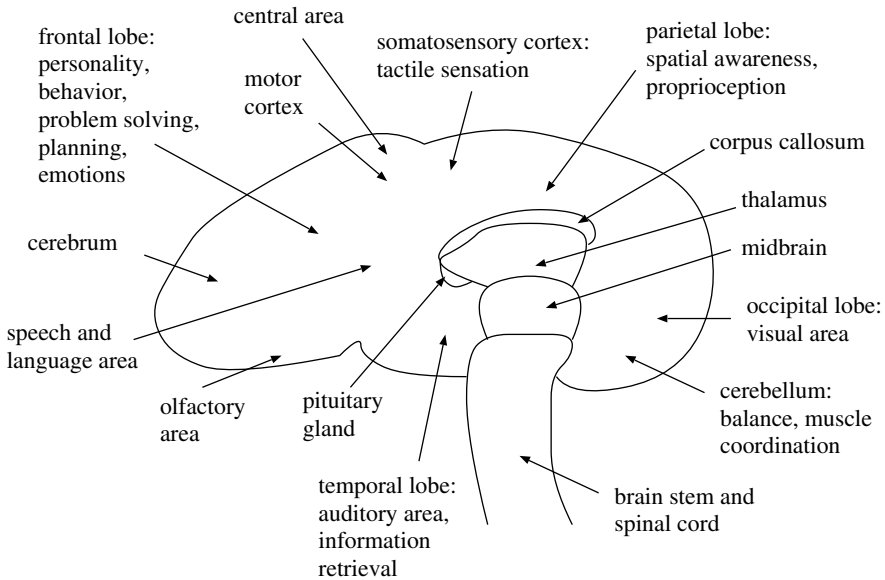


Figure 1.36 Schematic diagram showing the various parts and functional areas of the human brain.

Typical EEG instrumentation settings used are lowpass filtering at 75 Hz , and recording at $100\text{ }\mu\text{V}/\text{cm}$ and $30\text{ mm}/\text{s}$ for 10 – 20 minutes over 8 – 16 simultaneous channels. Monitoring of sleep EEG and detection of transients related to epileptic seizures may require multichannel EEG acquisition over several hours. Special EEG techniques include the use of needle electrodes and nasopharyngeal electrodes, recording the electrocorticogram (ECoG) from an exposed part of the cortex, and the use of intracerebral electrodes. Evocative techniques for recording the EEG include initial recording at rest (eyes open, eyes closed), hyperventilation (after breathing at 20 respirations per minute for 2 – 4 minutes), photic stimulation (with 1 – 50 flashes of light per second), auditory stimulation with loud clicks, sleep (different stages), and pharmaceuticals or drugs.

EEG signals exhibit several patterns of rhythmic or periodic activity. (*Note:* The term *rhythm* stands for different phenomena or events in the ECG and the EEG.) The commonly used terms for EEG frequency (f) bands are:

- Delta (δ): $0.5 \leq f < 4\text{ Hz}$;
- Theta (θ): $4 \leq f < 8\text{ Hz}$;
- Alpha (α): $8 \leq f \leq 13\text{ Hz}$; and
- Beta (β): $f > 13\text{ Hz}$.

Figure 1.38 illustrates traces of EEG signals with the rhythms listed above.

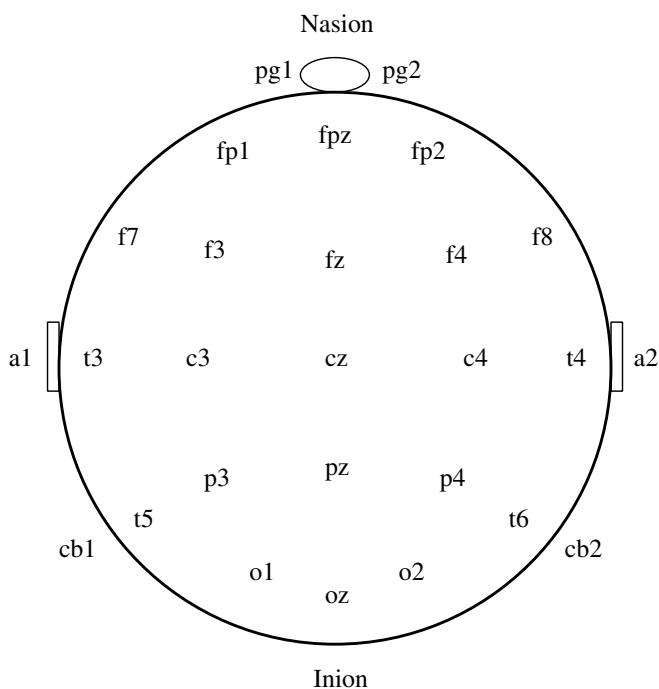


Figure 1.37 The 10 – 20 system of electrode placement for EEG recording [54]. Notes regarding channel labels: pg, nasopharyngeal; a, auricular (earlobes); fp, prefrontal; f, frontal; p, parietal; c, central; o, occipital; t, temporal; cb, cerebellar; z, midline; odd numbers on the left, even numbers on the right of the subject.

EEG rhythms are associated with various physiological and mental processes [55, 56]. The alpha rhythm is the principal resting rhythm of the brain; it is common in wakeful, resting adults, especially in the occipital area, with bilateral synchrony. Auditory and mental arithmetic tasks with the eyes closed lead to strong alpha waves, which are suppressed when the eyes are opened (that is, by a visual stimulus); see Figure 1.38 (e) [54].

In addition to the commonly studied rhythms mentioned above, the gamma rhythm is defined as activity in the range 30 – 80 Hz . The gamma rhythm is considered to be related to responses induced by various types of sensory input or stimuli, active sensory processes involving attention, and short-term memory processes [62]; see also Mantini et al. [63] and Rennie et al. [64] for related discussions.

The alpha wave is replaced by slower rhythms at various stages of sleep. Theta waves appear at the beginning stages of sleep; delta waves appear at deep-sleep stages. High-frequency beta waves appear as background activity in tense and anxious subjects. The depression or absence of the normal (expected) rhythm in a certain state of the subject could indicate abnormality. The presence of delta or theta (slow) waves in a wakeful adult may be considered to be abnormal. Focal brain injury

and tumors lead to abnormal slow waves in the corresponding regions. Unilateral depression (left–right asymmetry) of a rhythm could indicate disturbances in cortical pathways. Spikes and sharp waves could indicate the presence of epileptogenic regions in the corresponding parts of the brain.

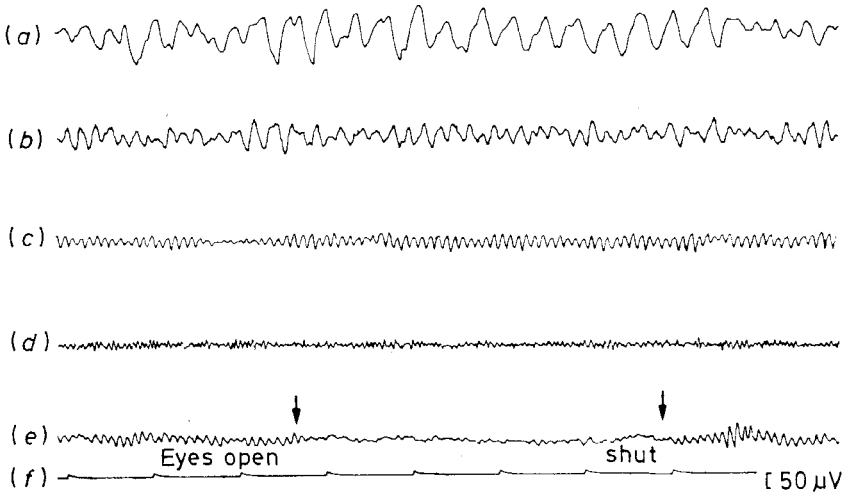


Figure 1.38 From top to bottom: (a) delta rhythm; (b) theta rhythm; (c) alpha rhythm; (d) beta rhythm; (e) blocking of the alpha rhythm by eye opening; (f) 1 s time markers and 50 μV marker. Reproduced with permission from R. Cooper, J.W. Osselton, and J.C. Shaw, *EEG Technology*, 3rd Edition, 1980. ©Butterworth Heinemann Publishers, a division of Reed Educational & Professional Publishing Ltd., Oxford, UK.

Figure 1.39 shows an example of eight channels of the EEG recorded simultaneously from the scalp of a subject. All channels display high levels of alpha activity. Figure 1.40 shows 10 channels of the EEG of a subject with spike-and-wave complexes. Observe the distinctly different waveshape and sharpness of the spikes in Figure 1.40 as compared to the smooth waves in Figure 1.39.

EEG signals also include spikes, transients, and other waves and patterns associated with various disorders of the nervous system (see Figure 4.1 and Section 4.2.4). Figure 1.41 shows a 21-channel record of a patient with a seizure starting at about the 50-s mark [65]. The signal is characterized by a recruiting theta rhythm at about 5 Hz in the channels labeled as T2, F8, T4, and T6. Artifacts are evident in the signal due to muscle activity (in T3, C3, and C4) and blinking of the eye (in Fp1 and Fp2). Increased amounts of relatively high-frequency activity are seen in several channels after the 50-s mark related to the seizure.

Detection of events and rhythms in EEG signals is discussed in Sections 4.4, 4.5, and 4.6. Spectral analysis of EEG signals is dealt with in Sections 6.3.3, 6.7, and 7.5.2. Adaptive segmentation of EEG signals is described in Section 8.2.2, 8.5, and 8.9.

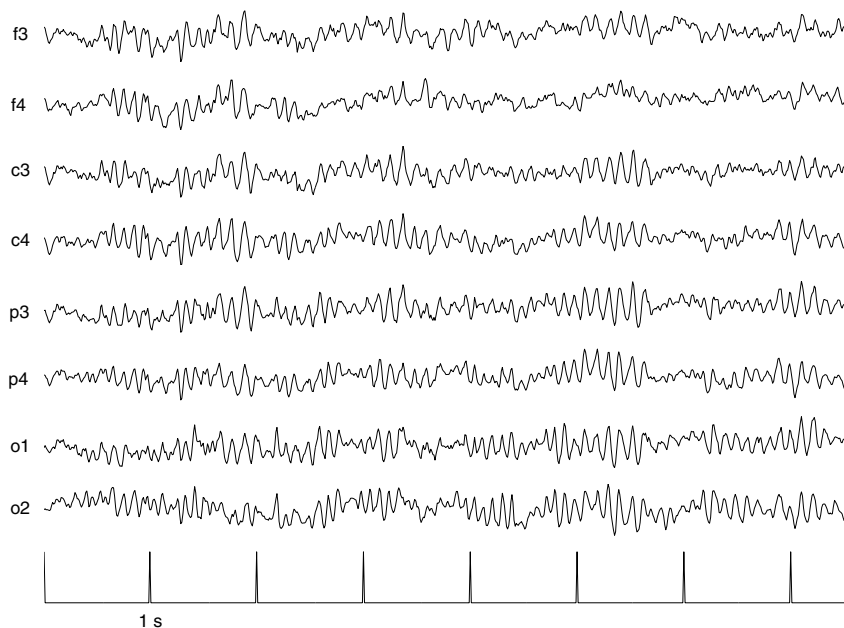


Figure 1.39 Eight channels of the EEG of a subject displaying alpha rhythm. See Figure 1.37 for details regarding channel labels. Data courtesy of Y. Mizuno-Matsumoto, Osaka University Medical School, Osaka, Japan.

1.2.7 Event related potentials (ERPs)

The term *event-related potential* is more general than and preferred to the term *evoked potential*, and includes the ENG or the EEG in response to light, sound, electrical, or other external stimuli. Short-latency ERPs are predominantly dependent upon the physical characteristics of the stimulus, whereas longer-latency ERPs are predominantly influenced by the conditions of presentation of the stimuli.

Somatosensory evoked potentials (SEPs) are useful for noninvasive evaluation of the nervous system from a peripheral receptor to the cerebral cortex. Median nerve short-latency SEPs are obtained by placing stimulating electrodes about 2 – 3 cm apart over the median nerve at the wrist with electrical stimulation at 5 – 10 pps, each stimulus pulse being of duration less than 0.5 ms with an amplitude of about 100 V (producing a visible thumb twitch). The SEPs are recorded from the surface of the scalp. The latency, duration, and amplitude of the response are measured.

ERPs and SEPs are weak signals typically buried in ongoing activity of the associated systems. Examples of ERPs are provided in Figures 3.2 and 3.41. Improvement of signal-to-noise ratio (SNR) is usually achieved by synchronized averaging and filtering. See Sections 3.5.1 and 3.11 for examples of ERPs and related signal processing methods.

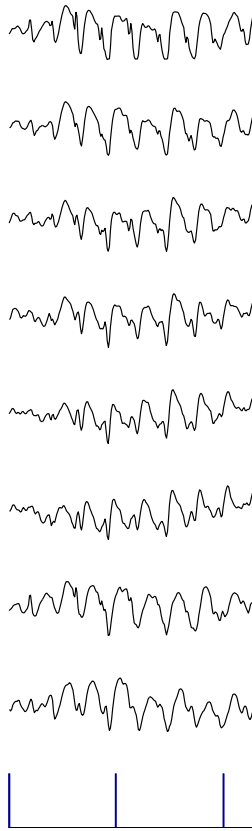


Figure 1.40 Ten channels of the EEG of a subject displaying spike-and-wave complexes. The channels shown are, from top to bottom: c3, c4, p3, p4, o1, o2, t3, t4, and time (1 s per mark). See Figure 1.37 for details regarding channel labels. Data courtesy of Y. Mizuno-Matsumoto, Osaka University Medical School, Osaka, Japan.

1.2.8 The electrogastrogram (EGG)

The electrical activity of the stomach consists of rhythmic waves of depolarization and repolarization of its constituent smooth muscle cells [66–68]. The activity originates in the midcorpus of the stomach, with intervals of about 20 s in humans. The waves of activity are always present and are not directly associated with contractions; they are related to the spatial and temporal organization of gastric contractions.

External (cutaneous) electrodes can record the signal known as the electrogastrogram (EGG). Chen et al. [69] used the following procedures to record cutaneous EGG signals. With the subject in the supine position and remaining motionless, the stomach was localized by using a 5 MHz ultrasound transducer array, and the orientation of the distal stomach was marked on the abdominal surface. Three active electrodes were placed on the abdomen along the antral axis of the stomach with an

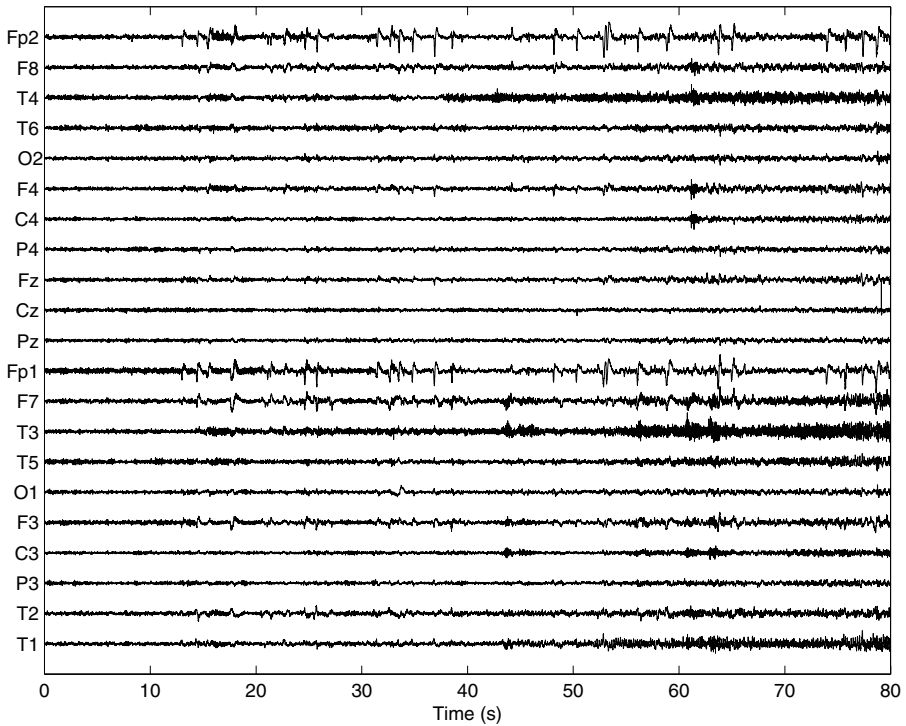


Figure 1.41 21 channels of the EEG of a subject displaying seizure activity. Data courtesy of M. De Vos, Katholieke Universiteit Leuven, Leuven, Belgium. T1 and T2 are sphenoidal electrodes.

interelectrode spacing of 3.5 cm . A common reference electrode was placed 6 cm away in the upper-right quadrant. Three bipolar signals were obtained from the three active electrodes in relation to the common reference electrode. The signals were amplified and filtered to the bandwidth of $0.02 - 0.3\text{ Hz}$ with 6 dB/octave transition bands, and sampled at 2 Hz .

The surface EGG is believed to reflect the overall electrical activity of the stomach, including the electrical control activity and the electrical response activity. Chen et al. [69] indicated that gastric dysrhythmia or arrhythmia may be detected via analysis of the EGG. Other researchers suggest that the diagnostic potential of the signal has not yet been established [66, 67]. Accurate and reliable measurement of the electrical activity of the stomach requires implantation of electrodes within the stomach [70], which limits its practical applicability.

1.2.9 The phonocardiogram (PCG)

The heart sound signal is perhaps the most traditional biomedical signal, as indicated by the fact that the stethoscope is the primary instrument carried and used by physi-

cians. The PCG is a vibration or sound signal related to the contractile activity of the cardiohemic system (the heart and blood together) [33, 71–76]; it represents a recording of the heart sound signal. Recording of the PCG signal requires a transducer to convert the vibration or sound signal into an electronic signal: Microphones, pressure transducers, or accelerometers may be placed on the chest surface for this purpose. The normal heart sounds provide an indication of the general state of the heart in terms of rhythm and contractility. Cardiovascular diseases and defects cause changes to the heart sounds or additional sounds and murmurs that could be useful in their diagnosis.

The genesis of heart sounds: The externally recorded heart sounds are not caused by valve leaflet movements *per se*, as believed earlier, but by vibrations of the whole cardiovascular system triggered by pressure gradients [33]. The cardiohemic system may be compared to a fluid-filled balloon, which, when stimulated at any location, vibrates as a whole. Externally, however, heart sound components are best heard at certain locations on the chest individually, and this localization has led to the concept of *secondary sources* on the chest related to the well-known auscultatory areas: the mitral, aortic, pulmonary, and tricuspid areas [33]. The standard auscultatory areas are indicated in Figure 1.31. The mitral area is near the apex of the heart. The aortic area is to the right of the sternum, in the second intercostal space. The tricuspid area is in the fourth intercostal space on either side of the sternum. The pulmonary area lies at the left parasternal line in the second or third intercostal space [33].

A normal cardiac cycle contains two major sounds — the first heart sound (S1) and the second heart sound (S2). Figure 1.42 shows a normal PCG signal, along with the ECG and carotid pulse tracings. S1 occurs at the onset of ventricular contraction, and corresponds in timing to the QRS complex in the ECG signal.

The initial vibrations in S1 occur when the first myocardial contractions in the ventricles move blood toward the atria, sealing the AV (mitral and tricuspid) valves (see Figure 1.43). The second component of S1 begins with abrupt tension of the closed AV valves, decelerating the blood. At this stage, known as isovolumic contraction, all four of the cardiac valves are closed. Next, the semilunar (aortic and pulmonary) valves open and the blood is ejected out of the ventricles. The third component of S1 may be caused by oscillation of blood between the root of the aorta and the ventricular walls. This is followed by the fourth component of S1, which may be due to vibrations caused by turbulence in the ejected blood flowing rapidly through the ascending aorta and the pulmonary artery.

Following the systolic pause in the PCG of a normal cardiac cycle, the second sound S2 is caused by the closure of the semilunar valves. While the primary vibrations occur in the arteries due to deceleration of blood, the ventricles and atria also vibrate, due to transmission of vibrations through the blood, the valves, and the valve rings. S2 has two components, one due to closure of the aortic valve (A2) and another due to closure of the pulmonary valve (P2). The aortic valve normally closes before the pulmonary valve, and hence A2 precedes P2 by a few milliseconds. The A2–P2 gap is widened in normal subjects during inspiration. The pulmonary impedance is lower during inspiration as compared to the condition during expiration. The com-

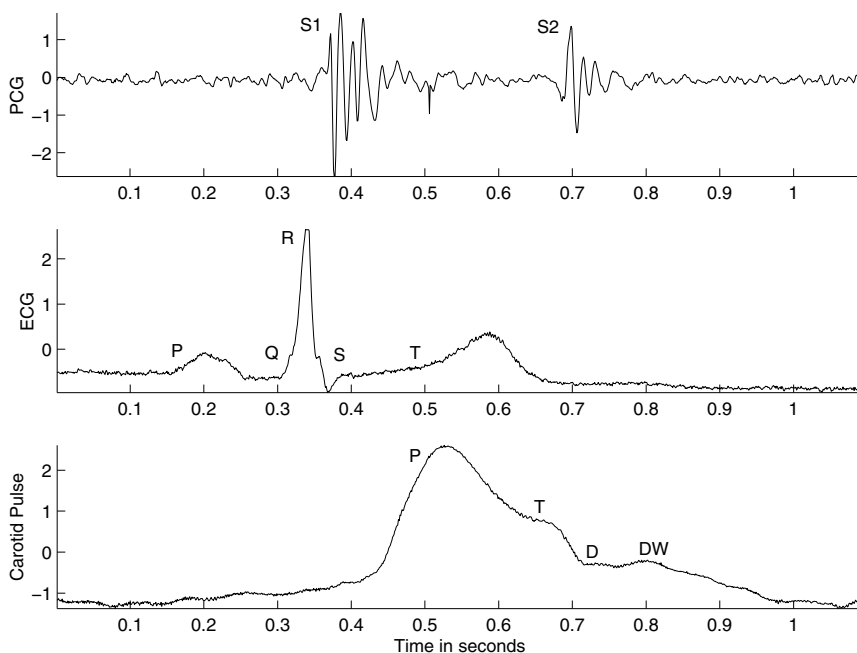


Figure 1.42 Three-channel simultaneous record of the PCG, ECG, and carotid pulse signals of a normal male adult.

pliance of the pulmonary vessels is increased during inspiration because they are dilated due to negative thoracic pressure. These conditions cause P2 to be delayed during inspiration as compared to the situation during expiration. (*Note:* The PCG signal in Figure 1.42 does not show the A2 and P2 components separately.) A wide gap is sustained during expiration in the case of pulmonary hypertension. Bundle-branch block could cause the A2–P2 gap to widen, or may even reverse the order of occurrence of A2 and P2.

In some cases, a third heart sound (S3) may be heard, corresponding to sudden termination of the ventricular rapid-filling phase. Because the ventricles are filled with blood and their walls are relaxed during this part of diastole, the vibrations of S3 are of very low frequency. In late diastole, a fourth heart sound (S4) may be heard sometimes, caused by atrial contractions displacing blood into the distended ventricles. In addition to these sounds, valvular clicks and snaps are occasionally heard.

Heart murmurs: The intervals between S1 and S2, and S2 and S1 of the next cycle (corresponding to ventricular systole and diastole, respectively), are normally silent. Murmurs, which are caused by certain cardiovascular defects and diseases, may occur in these intervals. Murmurs are high-frequency noise-like sounds that arise when the velocity of blood becomes high as it flows through an irregularity, such as a constriction, an orifice, or a baffle. Typical conditions in the cardiovascular

A. COMPONENTS OF FIRST HEART SOUND

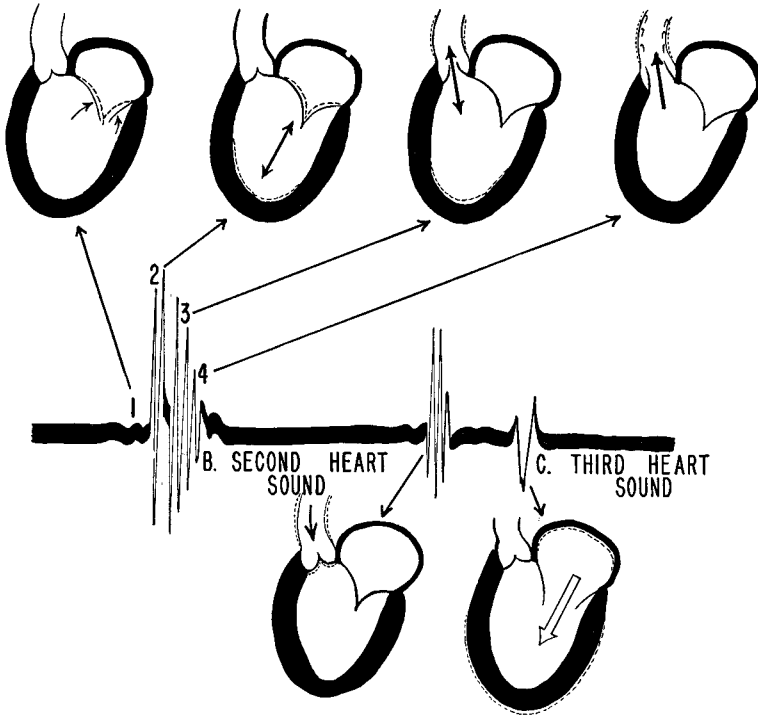


Figure 1.43 Schematic representation of the genesis of heart sounds. Only the left portion of the heart is illustrated as it is the major source of the heart sounds. The corresponding events in the right portion also contribute to the sounds. The atria do not directly contribute much to the heart sounds. Reproduced with permission from R.F. Rushmer, *Cardiovascular Dynamics*, 4th edition, ©W.B. Saunders, Philadelphia, PA, 1976.

system that cause turbulence in blood flow are valvular stenosis and insufficiency. A valve is said to be stenosed when, due to the deposition of calcium or other reasons, the valve leaflets are stiffened and do not open completely, thereby causing an obstruction or baffle in the path of the blood being ejected. A valve is said to be insufficient when it cannot close effectively and causes reverse leakage or regurgitation of blood through a narrow opening.

Systolic murmurs are caused by conditions such as ventricular septal defect (a hole in the septum or wall between the left ventricle and the right ventricle), aortic stenosis, pulmonary stenosis, mitral insufficiency, and tricuspid insufficiency. Semilunar valvular stenosis (aortic stenosis and pulmonary stenosis) causes an obstruction in the path of blood being ejected during systole. AV valvular insufficiency (mitral insufficiency and tricuspid insufficiency) causes regurgitation of blood to the atria during ventricular contraction.

Diastolic murmurs are caused by conditions such as aortic insufficiency, pulmonary insufficiency, mitral stenosis, and tricuspid stenosis. Other conditions causing murmurs are atrial septal defect and patent ductus arteriosus (an abnormal connection or shunt between the aorta and the pulmonary artery), as well as certain physiological or functional conditions that result in increased cardiac output or blood velocity.

Various features of heart sounds and murmurs, such as intensity, frequency content, and timing, are affected by many physical and physiological factors, including the recording site on the thorax, intervening thoracic structures, ventricular contractility, positions of the cardiac valves at the onset of systole, the degree of the defect present, the heart rate, and blood velocity. For example, S1 is loud and delayed in mitral stenosis; right bundle-branch block causes wide splitting of S2; left bundle-branch block results in reversed splitting of S2; acute myocardial infarction causes a pathologic S3; and severe mitral regurgitation leads to an increased S4 [71–75]. Although murmurs are noise-like events, their features aid in distinguishing between different causes. For example, aortic stenosis causes a diamond-shaped midsystolic murmur, whereas mitral stenosis causes a decrescendo–crescendo type of diastolic–presystolic murmur. Figure 1.44 illustrates the PCG, ECG, and carotid pulse signals of a patient with aortic stenosis; the PCG displays the typical diamond-shaped murmur in systole.

Recording of PCG signals: PCG signals are normally recorded using piezoelectric contact sensors that are sensitive to displacement or acceleration at the skin surface. The PCG signals illustrated in this section were obtained using a Hewlett Packard HP21050A transducer, which has a nominal bandwidth of $0.05 - 1,000\text{ Hz}$. The carotid pulse signals shown in this section were recorded using the HP21281A pulse transducer, which has a nominal bandwidth of $0 - 100\text{ Hz}$. PCG recording is normally performed in a quiet room, with the patient in the supine position with the head resting on a pillow. The PCG transducer is placed firmly at the desired position on the chest using a suction ring and/or a rubber strap.

Use of the ECG and carotid pulse signals in the analysis of PCG signals is described in Sections 2.2.1, 2.2.2, and 2.3. Segmentation of the PCG based on events detected in the ECG and carotid pulse signals is discussed in Section 4.10. A particular type of synchronized averaging to detect A2 in S2 is the topic of Section 4.11. Spectral analysis of the PCG and its applications are presented in Sections 6.2.1, 6.3.5, 6.5, and 7.10. Parametric modeling and detection of S1 and S2 are described in Sections 7.5.2 and 7.9. Modeling of sound generation in stenosed coronary arteries is discussed in Section 7.7.2. Adaptive segmentation of PCG signals with no other reference signal is considered in Section 8.10.

1.2.10 The carotid pulse

The carotid pulse is a pressure signal recorded over the carotid artery as it passes near the surface of the body at the neck. It provides a pulse signal indicating the variations in arterial BP and volume with each heart beat. Because of the proximity of the recording site to the heart, the carotid pulse signal closely resembles the morphology

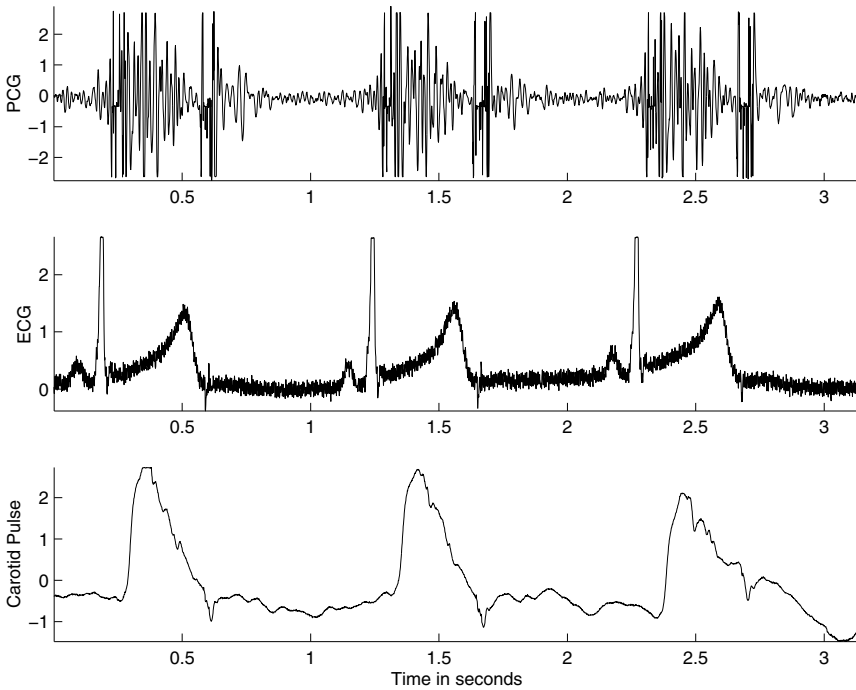


Figure 1.44 Three-channel simultaneous record of the PCG, ECG, and carotid pulse signals of a patient (female, 11 years) with aortic stenosis. Note the presence of the typical diamond-shaped systolic murmur and the split nature of S2 in the PCG.

of the pressure signal at the root of the aorta; however, it cannot be used to measure absolute pressure [72]. The carotid pulse is a useful adjunct to the PCG and can assist in the identification of S2 and its components.

The carotid pulse rises abruptly with the ejection of blood from the left ventricle to the aorta, reaching a peak called the percussion wave (marked as P in Figure 1.42). This is followed by a plateau or a secondary wave known as the tidal wave (marked as T in Figure 1.42), caused by a reflected pulse returning from the upper body. Next, closure of the aortic valve causes a notch known as the dicrotic notch (marked as D in Figure 1.42). The dicrotic notch may be followed by the dicrotic wave (marked as DW in Figure 1.42) due to a reflected pulse from the lower body [72]. The carotid pulse trace is affected by valvular defects such as mitral insufficiency and aortic stenosis [72]. In the case of aortic stenosis, obstruction of blood flow across the aortic valve leads to prolongation of ejection, which, in turn, causes an abnormally slow initial upstroke of the carotid pulse and longer ejection time [72]. In the case of mitral insufficiency, the dicrotic wave is accentuated [72]. Regardless of some benefits provided by the carotid pulse in the diagnosis of cardiovascular defects and diseases, it is not commonly used in current clinical practice.

The carotid pulse signals shown in this section were recorded using the HP21281A pulse transducer, which has a nominal bandwidth of $0 - 100\text{ Hz}$. The carotid pulse signal is usually recorded with the PCG and ECG signals. Placement of the carotid pulse transducer requires careful selection of a location on the neck as close to the carotid artery as possible, where the pulse is felt the strongest, usually by a trained technician (see Figure 1.31).

Details on intervals that may be measured from the carotid pulse and their use in segmenting the PCG are presented in Sections 2.2.2 and 2.3. Signal processing techniques for the detection of the dicrotic notch are described in Section 4.3.3. Use of the dicrotic notch for segmentation of PCG signals is explored in Sections 4.10 and 4.11. Use of the carotid pulse for averaging of PCG spectra in systole and diastole is described in Section 6.3.5.

1.2.11 Signals from catheter tip sensors

For specific and detailed monitoring of cardiac function, sensors placed on catheter tips may be inserted into the cardiac chambers. It then becomes possible to acquire several signals such as left ventricular pressure, right atrial pressure, aortic pressure, and intracardiac sounds [74, 75]. While these signals provide valuable and accurate information, the procedures are invasive and are associated with certain risks.

Figures 1.45 and 1.46 illustrate multichannel aortic, left ventricular, and right ventricular pressure recordings from a dog using catheter-tip sensors. The ECG signal is also shown. Observe in Figure 1.45 that the right ventricular and left ventricular pressures increase exactly at the instant of each QRS complex. The aortic pressure peaks slightly after the corresponding increase in the left ventricular pressure. The notch (incisura) in the aortic pressure signal is due to closure of the aortic valve. (The same notch propagates through the vascular system and appears as the dicrotic notch in the carotid pulse signal.) The left ventricular pressure range ($10 - 110\text{ mm of Hg}$) is much larger than the right ventricular pressure range ($5 - 25\text{ mm of Hg}$). The aortic pressure range is limited to the vascular BP range of $80 - 120\text{ mm of Hg}$.

The signals in Figure 1.46 display the effects of PVCs. Observe the depressed ST segment in the ECG signal in the figure, likely due to myocardial ischemia. (It should be noted that the PQ and ST segments of the ECG signal in Figure 1.45 are isoelectric, even though the displayed values indicate a nonzero level. On the other hand, in the ECG in Figure 1.46, the ST segment stays below the corresponding isoelectric PQ segment.) The ECG complexes appearing just after the 2 s and 3 s markers are PVCs arising from different ectopic foci, as evidenced by their markedly different waveforms. Although the PVCs cause a less-than-normal increase in the left ventricular pressure, they do not cause a rise in the aortic pressure, as no blood is effectively pumped out of the left ventricle during the ectopic beats.

1.2.12 The speech signal

Human beings are social creatures by nature and have an innate need to communicate. Humans are endowed with the most sophisticated vocal system in nature.

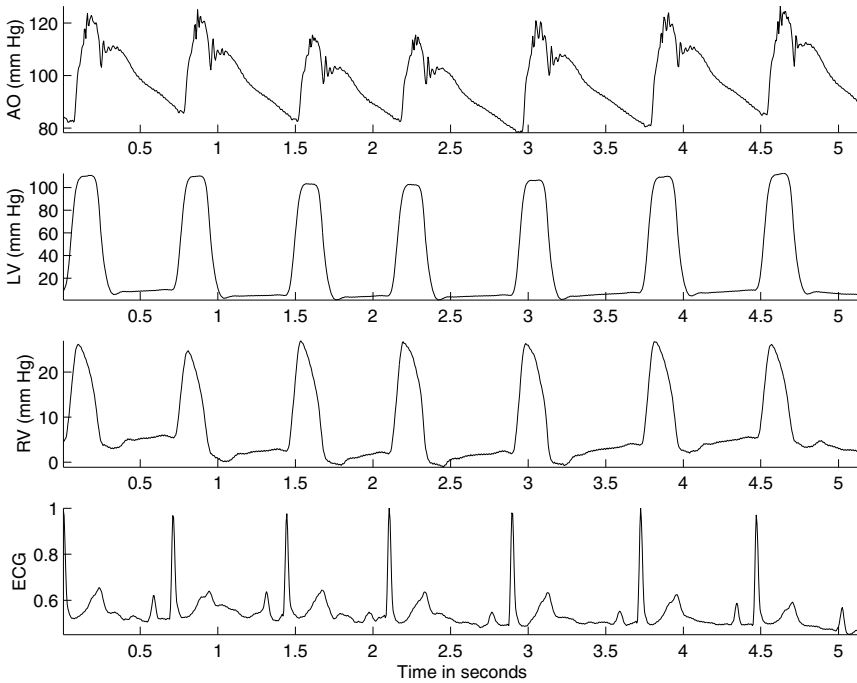


Figure 1.45 Normal ECG and intracardiac pressure signals from a dog. The pressure signals shown are AO, near the aortic valve; LV, in the left ventricle; and RV, in the right ventricle. Data courtesy of R. Sas and J. Tyberg, Department of Physiology and Biophysics, University of Calgary.

The speech signal is an important signal, although it is more commonly considered as a communication signal than a biomedical signal. However, the speech signal can serve as a diagnostic signal when speech and vocal-tract disorders need to be investigated [77, 78].

Speech sounds are produced by transmitting puffs of air from the lungs through the vocal tract (as well as the nasal tract for certain sounds) [79]. Figure 1.47 shows a schematic diagram of the anatomy of the vocal tract; Figure 1.48 gives a schematic representation of the parts of the speech production system. The vocal tract starts at the vocal cords or glottis in the throat and ends at the lips and the nostrils. The shape of the vocal tract is varied to produce different types of sound units or *phonemes* which, when concatenated, form speech. In essence, the vocal tract acts as a filter that modulates the spectral characteristics of the input puffs of air. It is evident that the system is dynamic; therefore, the filter and the speech signal produced have time-varying characteristics, that is, they are nonstationary (see Section 3.2.4).

Speech sounds may be classified as voiced, unvoiced, and plosive sounds [79]. Voiced sounds involve the participation of the glottis: air is forced through the vocal cords held at a certain tension. The vocal cords vibrate and the result is a series of

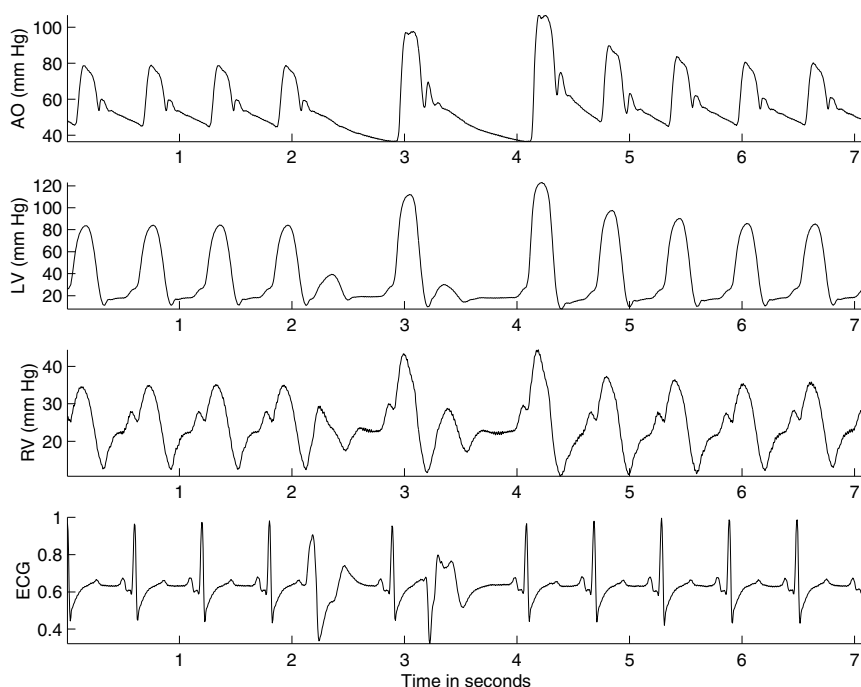


Figure 1.46 ECG and intracardiac pressure signals from a dog with PVCs. The pressure signals shown are AO, near the aortic valve; LV, in the left ventricle; and RV, in the right ventricle. Data courtesy of R. Sas and J. Tyberg, Department of Physiology and Biophysics, University of Calgary.

quasiperiodic pulses of air which is passed through the vocal tract; see Figure 1.49. The input to the vocal tract may be treated as a train of impulses or pulses that is almost periodic. The vocal tract acts as a filter: Upon convolution with the impulse response of the vocal tract, which is held steady in a certain configuration for the duration of the voiced sound desired, a quasiperiodic signal is produced with a characteristic waveshape that is repeated. All vowels are voiced sounds. Figure 1.50 shows the speech signal of the word “safety” spoken by a male. Figure 1.51 shows, in the upper trace, a portion of the signal corresponding to the phoneme /E/ (the letter “a” in the word). The quasiperiodic nature of the signal is evident. Features of interest in voiced signals are the pitch (average interval between the repetitions of the vocal-tract impulse response or basic wavelet) and the resonance or formant frequencies of the vocal-tract system.

An unvoiced sound (or fricative) is produced by forcing a steady stream of air through a narrow opening or constriction formed at a specific position along the vocal tract. The vocal cords do not vibrate for such sounds; see Figure 1.49. The result is a turbulent signal that appears like random noise. The input to the vocal tract is a broadband random signal, which is filtered by the vocal tract to yield the

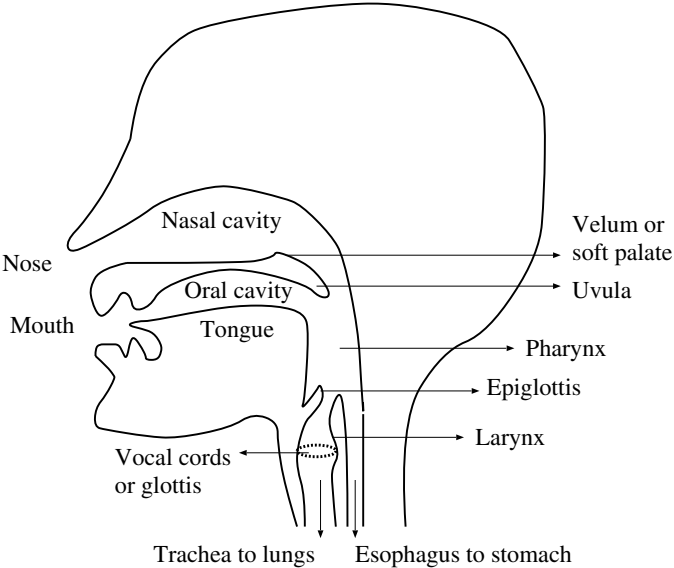


Figure 1.47 Schematic diagram of anatomy of the vocal tract.

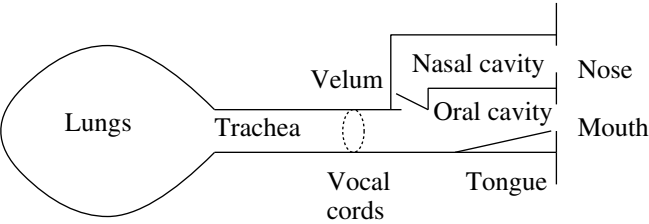


Figure 1.48 Schematic representation of the speech production system.

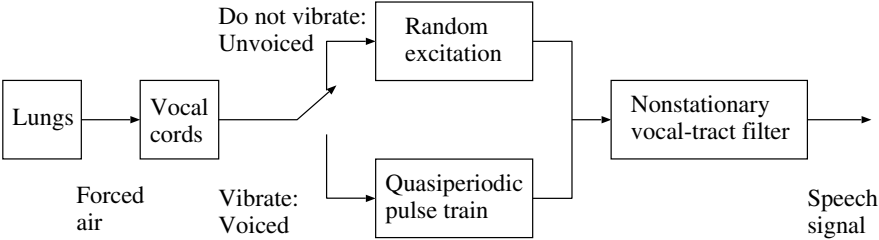


Figure 1.49 Schematic representation of the production of voiced and unvoiced speech.

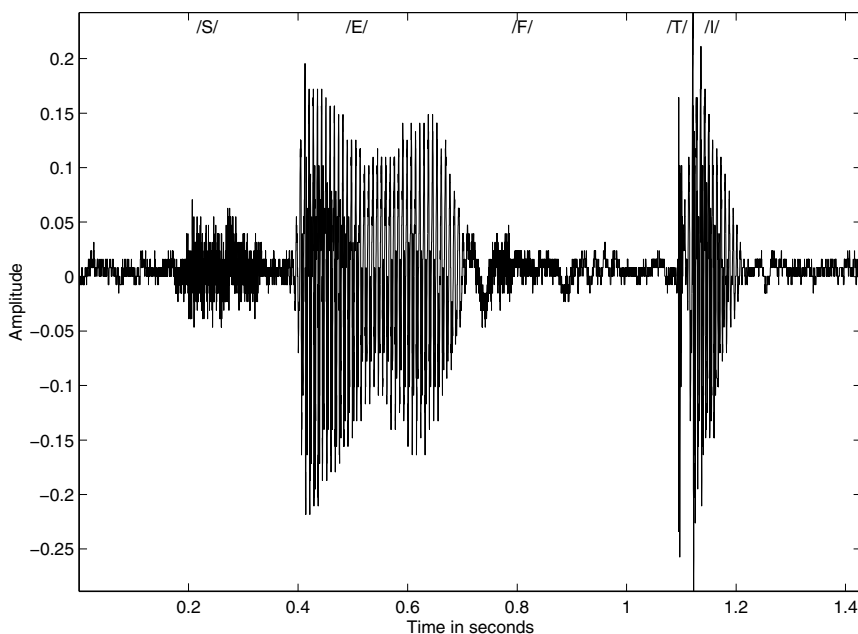


Figure 1.50 Speech signal of the word “safety” uttered by a male speaker. Approximate time intervals of the various phonemes in the word are /S/: 0.2 – 0.35 s; /E/: 0.4 – 0.7 s; /F/: 0.75 – 0.95 s; /T/: transient at 1.1 s; /I/: 1.1 – 1.2 s. Background noise is also seen in the signal before the beginning and after the termination of the speech segment, as well as during the stop interval before the plosive /T/.

desired sound. Fricatives are unvoiced sounds, as they do not involve any activity (vibration) of the vocal cords. The phonemes /S/, /SH/, /Z/, and /F/ are examples of fricatives. The lower trace in Figure 1.51 shows a portion of the signal corresponding to the phoneme /S/ in the word “safety.” The signal has no identifiable structure and appears to be random (see also Figures 3.1, 3.3, and 3.4, as well as Section 3.2.4). The transfer function of the vocal tract, as evidenced by the spectrum of the signal itself, would be of interest in analyzing a fricative.

Plosives, also known as stops, involve complete closure of the vocal tract, followed by an abrupt release of built-up pressure. The phonemes /P/, /T/, /K/, and /D/ are examples of plosives. The sudden burst of activity at about 1.1 s in Figure 1.50 illustrates the plosive nature of /T/. Plosives are difficult to characterize as they are transients; their properties are affected by the preceding phoneme as well. For more details on the speech signal, see Rabiner and Schafer [79].

Parkinson’s disease, which causes tremor, rigidity, and loss of muscle control, is also known to affect speech. The changes in speech caused by the disease include reduced loudness, increased vocal tremor, and breath-related noise. Vocal impairment caused by the disease are labeled as dysphonia, which refers to the inability to pro-

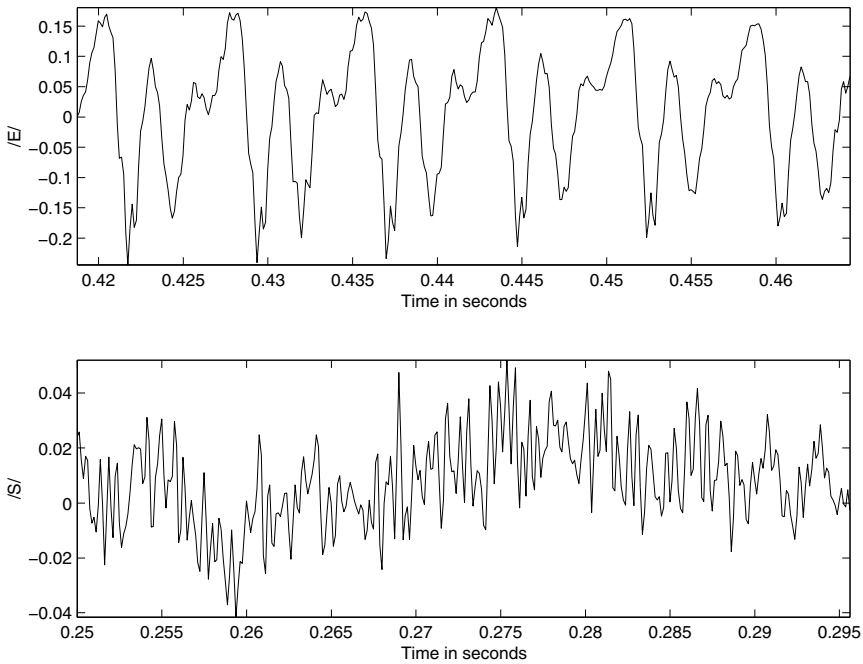


Figure 1.51 Segments of the signal in Figure 1.50 on an expanded scale to illustrate the quasiperiodic nature of the voiced sound /E/ in the upper trace and the almost random nature of the fricative /S/ in the lower trace.

duce normal vocal sounds, and dysarthria, which relates to difficulty in pronouncing words [78]. Tsanas et al. [78] describe several techniques for the analysis of speech signals for the classification of Parkinson's disease. Maciel et al. [80] describe techniques used to derive diagnostically useful parameters from speech signals. In the case of voiced speech segments, some of such measures represent temporal or spectral variations over time, perturbation of pitch, perturbation of frequency, and jitter. Whereas most methods for the analysis of speech abnormalities use signals of sustained vowels, Umapathy et al. [81] proposed methods to decompose continuous speech signals and extract features for classification of speech pathology. See also Umapathy and Krishnan [82], Ghoraani et al. [83], and Arias-Londoño et al. [84] for additional related material.

Signal processing techniques for extraction of the vocal-tract response from voiced speech signals are described in Section 4.8.3. Frequency-domain characteristics of speech signals are illustrated in Sections 7.6.3 and 8.4.1.

1.2.13 The vibromyogram (VMG)

The VMG is the direct mechanical manifestation of contraction of a skeletal muscle and is a vibration signal that accompanies the EMG. The signal has also been named the sound-, acoustic-, or phono-myogram. Muscle sounds or vibrations are related to variations in the dimensions (contraction) of the constituent muscle fibers (see Figure 1.8) and may be recorded using contact microphones or accelerometers (such as the Dytran 3115A accelerometer, Dytran, Chatsworth, CA) placed on the muscle surface [85, 86]. The frequency and intensity of the VMG have been shown to vary in direct proportion to the contraction level [85, 86]. The VMG, along with the EMG, may be useful in studies related to neuromuscular control, muscle contraction, athletic training, and biofeedback. VMG signal analysis, however, is not as well established or common as EMG analysis.

Simultaneous analysis of the VMG and EMG signals is discussed in Section 2.2.6. Adaptive cancellation of the VMG from knee-joint vibration signals is the topic of Sections 3.9.2, 3.9.3, and 3.14. Analysis of muscle contraction using the VMG is described in Section 5.11.

1.2.14 The vibroarthrogram (VAG)

The knee joint: As illustrated in Figure 1.52, the knee joint is formed between the femur, the patella, and the tibia. The knee joint is the largest articulation in the human body that can effectively move from 0° extension to 135° flexion, together with 20° to 30° rotation of the flexed leg on the femoral condyles. The joint has four important features: (1) a joint cavity, (2) articular cartilage, (3) a synovial membrane, and (4) a fibrous capsule [87, 88]. The knee joint is known as a synovial joint, as it contains a lubricating substance called the synovial fluid. The patella (knee cap), a sesamoid bone, protects the joint, and is precisely aligned to slide in the groove (trochlea) of the femur during leg movement. The knee joint is made up of three compartments: (1) the patellofemoral, (2) the lateral tibiofemoral, and (3) the medial tibiofemoral compartments. The patellofemoral compartment is classified as a synovial gliding joint and the tibiofemoral as a synovial hinge joint [89]. The anterior and posterior cruciate ligaments as well as the lateral and medial ligaments bind the femur and tibia together, give support to the knee joint, and limit movement of the joint. The various muscles around the joint help in the movement of the joint and contribute to its stability.

The knee derives its physiological movement and its typical rolling–gliding mechanism of flexion and extension from its six degrees of freedom: three in translation and three in rotation. The translations of the knee take place on the anterior–posterior, medial–lateral, and proximal–distal axes. The rotational motion consists of flexion–extension, internal–external rotation, and abduction–adduction.

Although the tibial plateaus are the main load-bearing structures in the knee, the cartilage, menisci, and ligaments also bear loads. The patella aids knee extension by lengthening the lever arm of the quadriceps muscle throughout the entire range of motion, and allows a better distribution of compressive stresses on the femur [90].

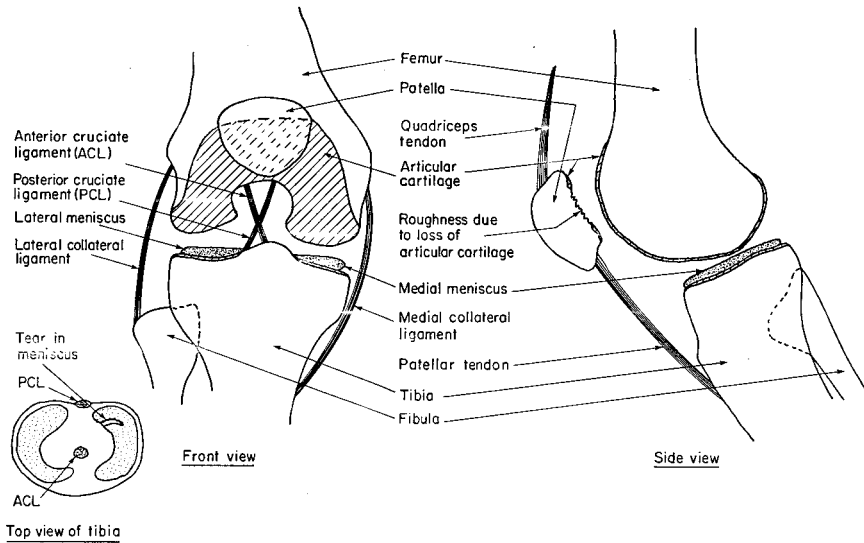


Figure 1.52 Front and side views of the knee joint (the two views are not mutually orthogonal). The inset shows the top view of the tibia with the menisci.

Articular cartilage: Two types of cartilage are present in the knee joint: the *articular cartilage*, which covers the ends of bones, and the wedge-shaped fibrocartilaginous structure called the *menisci*, located between the femur and the tibia [91]. The shock-absorbing menisci are composed of the medial meniscus and the lateral meniscus, which are two crescent-shaped plates of fibrocartilage that lie on the articular surface of the tibia.

The articular surfaces of the knee joint are the large curved condyles of the femur, the flattened condyles (medial and lateral plateaus) of the tibia, and the facets of the patella. There are three types of articulation: an intermediate articulation between the patella and the femur, as well as lateral and medial articulation between the femur and the tibia. The articular surfaces are covered by cartilage, like all the major joints of the body. Cartilage is vital to joint function because it protects the underlying bone during movement. Loss of cartilage function leads to pain, decreased mobility, and, in some instances, deformity and instability.

Knee-joint disorders: The knee is the most commonly injured joint in the body. Arthritic degeneration of injured knees is a well-known phenomenon and is known to result from a variety of traumatic causes. Damage to the stabilizing ligaments of the knee or to the shock-absorbing fibrocartilaginous pads (the menisci) are two of the most common causes of deterioration of knee-joint surfaces. Impact trauma to the articular cartilage surfaces themselves could lead to surface deterioration and secondary osteoarthritis.

Nontraumatic conditions of the knee joint include the extremely common idiopathic condition known as chondromalacia patella (soft cartilage of the patella), in

which articular cartilage softens, fibrillates, and sheds off the undersurface of the patella. Similarly, the meniscal fibrocartilage of the knee can apparently soften, which could possibly lead to degenerative tears and secondary changes in the regional hyaline surfaces.

Knee-joint sounds: Considerable noise is often associated with degeneration of knee-joint surfaces. The VAG is the vibration signal recorded from a joint during movement (articulation) of the joint. Normal joint surfaces are smooth and produce little or no sound, whereas joints affected by osteoarthritis and other degenerative diseases may have suffered cartilage loss and produce palpable and audible grinding vibration and sounds. Detection of knee-joint problems via the analysis of VAG signals could help avoid unnecessary exploratory surgery and also aid better selection of patients who would benefit from surgery [92–98]. The VAG signal, however, is not yet well understood, and it is a difficult signal to analyze due to its complex nonstationary characteristics.

Further details on the VAG signal are provided in Sections 2.2.7, 3.3.6, 5.12.1, and 8.2.3. Modeling of a specific type of VAG signal known as patellofemoral crepitus is presented in Sections 7.2.4, 7.3, and 7.7.3. Adaptive filtering of the VAG signal to remove muscle-contraction interference is described in Sections 3.9.2, 3.9.3, and 3.14. Adaptive segmentation of VAG signals into quasistationary segments is illustrated in Sections 8.6.1 and 8.6.2. The role of VAG signal analysis in the detection of articular cartilage pathology is discussed in Sections 5.12, 6.6, 8.14, and 9.12.

1.2.15 Otoacoustic emission (OAE) signals

The OAE signal represents the acoustic energy emitted by the cochlea either spontaneously or in response to an acoustic stimulus. The existence of this signal indicates that the cochlea not only receives sound but also produces acoustic energy [99]. The OAE signal could provide objective information on the micromechanical activity of the preneural or sensory components of the cochlea that are distal to the nerve-fiber endings. Analysis of the OAE signal could lead to improved noninvasive investigative techniques to study the auditory system. The signal may also assist in screening of hearing function and in the diagnosis of hearing impairment.

1.2.16 Bioacoustic signals

Several systems and parts of the human body produce sounds and vibrations in various bands of frequencies under both normal physiological and pathological conditions. In the preceding sections, we have studied the PCG, VMG, VAG, and OAE signals. A few other bioacoustic signals that have been studied by several researchers are breathing, tracheal, lung, and chest sounds [100–108]; snoring sounds [109, 110]; swallowing sounds [111]; gastrointestinal or bowel sounds [112]; sounds of the shoulder, temporomandibular, and hip joints [113]; and crying sounds of infants [114]. Kompis et al. [102] describe methods for acoustic imaging of the chest using multiple transducers. Kaniusas et al. [115] proposed techniques for integrated analysis of sound signals related to the cardiac system, respiration, and snoring.

1.3 Objectives of Biomedical Signal Analysis

The representation of biomedical signals in electronic form facilitates computer processing and analysis of the data. Figure 1.53 illustrates the typical steps and processes involved in computer-aided diagnosis (CAD) and therapy based upon biomedical signal analysis.

Some of the major objectives of biomedical instrumentation and signal analysis [11–14, 22] are the following:

- *Information gathering* — measurement of phenomena to interpret a system.
- *Diagnosis* — detection of malfunction, pathology, or abnormality.
- *Monitoring* — obtaining continuous or periodic information about a system.
- *Therapy and control* — modification of the behavior of a system based upon the outcome of the activities listed above to ensure a specific result.
- *Evaluation* — objective analysis to determine the ability to meet functional requirements, obtain proof of performance, perform quality control, or quantify the effect of treatment.

Signal acquisition procedures may be categorized as being invasive or noninvasive, and active or passive.

Invasive versus noninvasive procedures: Invasive procedures involve the placement of transducers or other devices inside the body, such as needle electrodes to record MUAPs, or insertion of catheter-tip sensors into the heart via a major artery or vein to record intracardiac signals. Noninvasive procedures are desirable in order to minimize risk to the subject. Recording of the ECG using limb or chest electrodes, the EMG with surface electrodes, or the PCG with microphones or accelerometers placed on the chest are noninvasive procedures.

Note that making measurements or imaging with X rays or ultrasound may be classified as invasive procedures, as they involve penetration of the body with externally administered radiation, even though the radiation is invisible and there is no visible puncturing or invasion of the body.

Active versus passive procedures: Active data acquisition procedures require external stimuli to be applied to the subject, or require the subject to perform a certain activity to stimulate the system of interest in order to elicit the desired response or signal. For example, recording an EMG signal requires contraction of the muscle of interest, such as clenching a fist; recording the VAG signal from the knee requires flexing of the leg over a certain joint angle range; and recording visual ERP signals requires the delivery of flashes of light to the subject. While these stimuli may appear to be innocuous, they do carry risks in certain situations for some subjects: Flexing the knee beyond a certain angle may cause pain for some subjects; and strobe lights may trigger epileptic seizures in some subjects. The investigator should be aware of such risks, factor them in a *risk–benefit analysis*, and be prepared to manage adverse reactions.

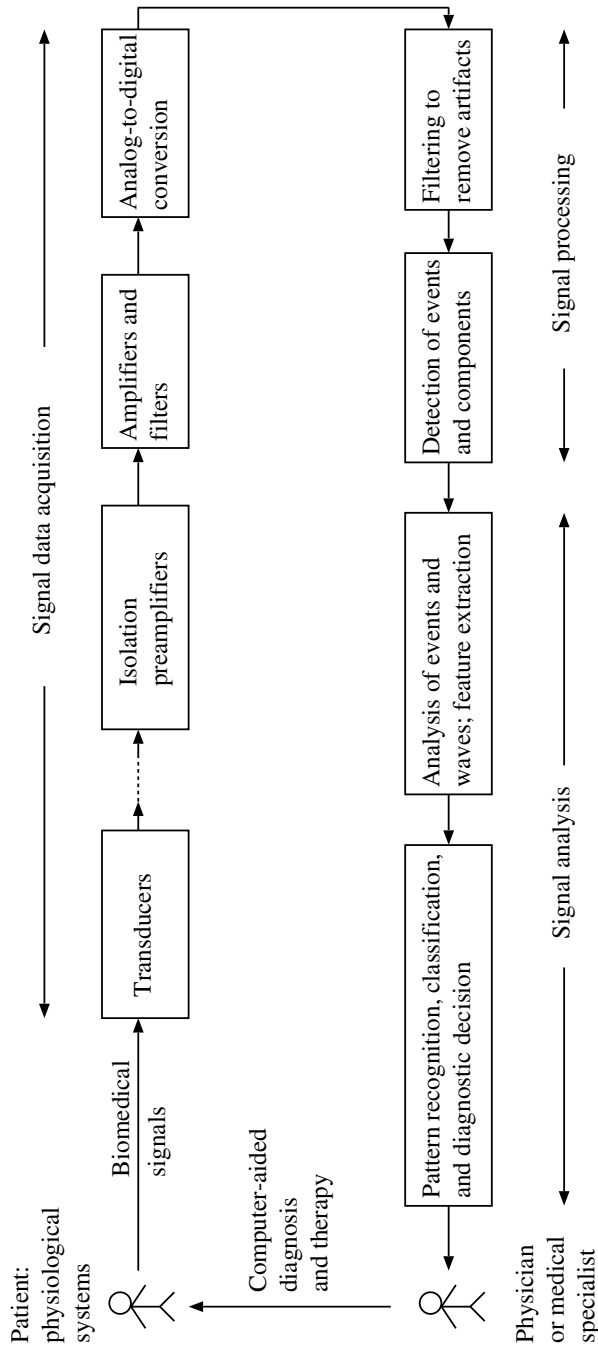


Figure 1.53 Computer-aided diagnosis and therapy based upon biomedical signal analysis.

Passive procedures do not require the subject to perform any activity. Recording of the ECG using limb or chest electrodes, the EEG during sleep using scalp-surface electrodes, or the PCG with microphones or accelerometers placed on the chest are passive procedures, but require contact between the subject and the instruments. Note that although the procedure is passive, the system of interest is active under its own natural control in these procedures. Acquiring an image of a subject with reflected natural light (with no flash from the camera) or with the natural infrared (thermal) emission could be categorized as a passive and noncontact procedure.

Most organizations require ethical approval by specialized committees for experimental procedures involving human or animal subjects, with the aim of minimizing the risk and discomfort to the subject and maximizing the benefits to both the subjects and the investigator.

The human–instrument system: Some of the components of a *human–instrument system* [11–14, 22] are the following:

- *The subject or patient:* It is important always to bear in mind that the main purpose of biomedical instrumentation and signal analysis is to provide a certain benefit to the subject or patient. All systems and procedures should be designed so as not to cause undue inconvenience to the subject and not to cause any harm or danger. In applying invasive or risky procedures, it is extremely important to perform a risk–benefit analysis and determine if the anticipated benefits of the procedure are worth placing the subject at the risks involved.
- *Stimulus or procedure of activity:* Application of stimuli to the subject in active procedures requires instruments such as strobe light generators, sound generators, and electrical pulse generators. Passive procedures require a standardized protocol of the desired activity to ensure repeatability and consistency of the experiment and the results.
- *Transducers:* electrodes, sensors.
- *Signal conditioning equipment:* amplifiers, filters.
- *Display equipment:* oscilloscopes, strip-chart or paper recorders, computer monitors, printers.
- *Recording, data processing, and transmission equipment:* analog instrumentation tape recorders, analog-to-digital converters (ADCs), digital-to-analog converters (DACs), digital tapes, compact disks (CDs), diskettes, computers, telemedicine systems.
- *Control devices:* power supply stabilizers and isolation equipment, patient intervention systems.

The science of measurement of physiological variables and parameters is known as *biometrics*. Some of the aspects to be considered in the design, specification, or use of biomedical instruments [11–14, 22] are the following:

- *Isolation of the subject or patient* — Safety is of paramount importance so that the subject is not placed at the risk of electrocution.
- *Dynamic range of operation* — The dynamic range is defined as the minimum to the maximum values of the signal or parameter being measured.
- *Sensitivity* — This can be represented by the smallest signal variation measurable and determines the resolution of the system.
- *Linearity* — Linear transfer characteristics are desired over at least a portion of the range of operation. Any nonlinearity present may need to be corrected at later stages of signal processing.
- *Hysteresis* — A lag in measurement due to the direction of variation of the entity being measured indicates hysteresis, which may add a bias to the measurement.
- *Frequency response* — This represents the variation of sensitivity with frequency. Most systems encountered in practice exhibit a lowpass behavior, that is, the sensitivity of the system decreases as the frequency of the input signal increases. Signal restoration techniques may be required to compensate reduced high-frequency sensitivity.
- *Stability* — An unstable system could preclude repeatability and consistency of measurements.
- *SNR* — Artifacts due to power-line interference, grounding problems, thermal noise, and other sources could compromise the quality of the signal being acquired. A good understanding of the signal-degrading phenomena present in the system is necessary in order to design appropriate filtering and correction procedures.
- *Accuracy* — The accuracy of a measurement could be affected by errors due to component tolerance; movement or mechanical errors; drift due to changes in temperature, humidity, or pressure; reading errors due to parallax; and zeroing or calibration errors.

The scientific and technological aspects of biomedical signal acquisition have advanced substantially over the past few decades. Systems are now available to acquire a few dozen to a few hundred channels of signals; wearable and portable systems are also available to facilitate wireless signal data acquisition from untethered and mobile subjects engaged in various activities [116–119]. Junnila et al. [120] developed a ballistocardiographic (BCG) chair, designed to look like a normal office chair, with a lightweight and flexible electromechanical film sensor to record the vibration signal related to cardiac activity along with the ECG and an impedance cardiogram. Guerrero et al. [121] used multiple pressure sensors integrated into a bed to obtain BCG signals. The envelope of the BCG signal was used to derive information related to respiration and detect sleep-related breathing disorders. Peltokangas et al. [122] describe a system with eight embroidered textile electrodes attached to a bed sheet

to measure multiple channels of the ECG and facilitate monitoring of patients during sleep. Mikhelson et al. [123] reported on a system with a millimeter-wave sensor, two cameras, and a pan/tilt base not only to detect and track a subject but also measure the subject's chest displacement and the heart rate in a noncontact manner.

1.4 Difficulties in Biomedical Signal Acquisition and Analysis

In spite of the long history of biomedical instrumentation and its extensive use in health care and research, many practical difficulties are encountered in biomedical signal acquisition, processing, and analysis [11–14, 22]. The characteristics of the problems, and hence their potential solutions, are unique to each type of signal. Particular attention should be paid to the following issues.

Accessibility of the variables to measurement: Most of the systems and organs of interest, such as the cardiovascular system and the brain, are located well within the body in protective enclosures (for good reasons!). While the ECG may be recorded using limb electrodes, the signal so acquired is only a projection of the true 3D cardiac electrical vector on to the axis of the electrodes. Such a signal may be sufficient for rhythm monitoring, but could be inadequate for more specific analysis of the cardiac system such as atrial electrical activity. Accessing the atrial electrical activity at the source requires the insertion of an electrode close to the atrial surface or within the atria.

Similarly, measurement of BP using a pressure cuff over an arm gives an estimate of the brachial arterial pressure. Detailed study of pressure variations within the cardiac chambers or arteries over a cardiac cycle would require the insertion of catheters with pressure sensors into the heart. Such invasive procedures provide access to the desired signals at their sources and often provide clear and useful signals, but carry high risks.

The surface EMG includes the interference pattern of the activities of several motor units even at low levels of muscular contraction. Acquisition of SMUAPs requires access to the specific muscle layer or unit of interest by insertion of fine-wire or needle electrodes. The procedure carries risks of infection and damage to muscle fibers, and it causes pain to the subject during muscular activity.

An investigator should assess the system and variables of interest carefully, and determine the minimal level of intervention absolutely essential to the data acquisition procedure. A trade-off between the integrity and quality of the information acquired versus the pain and risks to the subject may be needed.

Variability of the signal source: It is evident from the preceding sections that the various systems that comprise the human body are dynamic with several variables or degrees of freedom. Biomedical signals represent the dynamic activity of physiological systems and the states of their constituent variables. The nature of the processes or the variables could be deterministic or random (stochastic); a special case is that of periodicity or quasiperiodicity.

A normal ECG exhibits a regular rhythm with a readily identifiable waveshape (the QRS complex) in each period; under such conditions, the signal may be referred

to as a deterministic and periodic signal. However, the cardiovascular system of a heart patient may not stay in a given state over substantial periods, and the waveshape and rhythm may vary over time.

The surface EMG is the summation of the MUAPs of the motor units that are active at the given instant of time. Depending upon the level of contraction desired (at the volition of the subject), the number of active motor units varies, increasing with increasing effort. Furthermore, the firing intervals and the firing rate of each motor unit also vary in response to the level of contraction desired, and exhibit stochastic properties. While the individual MUAPs possess readily identifiable and simple monophasic, biphasic, or triphasic waveshapes, the interference pattern of several motor units firing at different rates will appear as an almost random signal with no visually recognizable waves or waveshapes.

The dynamic nature of biological systems causes most signals to exhibit stochastic and nonstationary behavior. This means that signal statistics such as mean, variance, and spectral density change with time. For this reason, signals from a dynamic system should be analyzed over extended periods of time including various possible states of the system, and the results should be placed in the context of the corresponding states.

Interrelationships and interactions among physiological systems: The various systems that compose the human body are not mutually independent; rather, they are interrelated and interact in various ways. Some of the interactive phenomena are compensation, feedback, cause-and-effect, collateral effects, loading, and take-over of function of a disabled system or part by another system or part. For example, the second heart sound exhibits a split during active inspiration in normal subjects due to reduced intrathoracic pressure and decreased venous return to the left side of the heart [72] (but not during expiration); this is due to normal physiological processes. However, the second heart sound is split in both inspiration and expiration due to delayed right ventricular contraction in right bundle-branch block, pulmonary valvular stenosis or insufficiency, and other conditions [72]. Ignoring this interrelationship could lead to misinterpretation of the signal. See Chapter 2 for more detailed discussions on this topic.

Effect of the instrumentation or procedure on the system: The placement of transducers on and connecting a system to instruments could affect the performance or alter the behavior of the system, and could cause spurious variations in the parameters being investigated. The experimental procedure or activity required to elicit the signal may lead to certain effects that could alter signal characteristics. This aspect may not always be obvious unless careful attention is paid. For example, the placement of a relatively heavy accelerometer may affect the vibration characteristics of a muscle and compromise the integrity of the vibration or sound signal being measured. Fatigue may set in after a few repetitions of an experimental procedure, and subsequent measurements may not be indicative of the true behavior of the system; the system may need some rest between procedures or their repetitions.

Physiological artifacts and interference: One of the prerequisites for obtaining a good ECG signal is for the subject to remain relaxed and still with no movement. Coughing, tensing of muscles, and movement of the limbs cause the corresponding

EMG to appear as an undesired artifact. In the absence of any movement by the subject, the only muscular activity in the body would be that of the heart. When chest leads are used, even normal breathing could cause the associated EMG of the chest muscles to interfere with the desired ECG. It should also be noted that breathing causes beat-to-beat variations in the RR interval, which should not be mistaken to be sinus arrhythmia (see Section 2.2.4 for more details and a related illustration). An effective solution would be to record the signal with the subject holding his or her breath for a few seconds. This simple solution is not applicable in long-term monitoring of critically ill patients or in recording the ECG of infants; signal processing procedures would then be required to remove the artifacts.

A unique situation is that of acquiring the ECG of a fetus through surface electrodes placed over the expectant mother's abdomen: the maternal ECG appears as an interference in this situation. No volitional or external control is possible or desirable to prevent the artifact in this situation, which calls for adaptive cancellation techniques using multiple channels of various signals [124]; see Chapter 3.

Another example of physiological interference or cross-talk is that of muscle-contraction interference (MCI) in the recording of the knee-joint VAG signal [125]. The rectus femoris muscle is active (contracting) during the swinging movement of the leg required to elicit the joint vibration signal. The VMG of the muscle is propagated to the knee and appears as an interference. Swinging the leg mechanically using a mechanical actuator is a possible solution; however, this represents an unnatural situation and may cause other sound or vibration artifacts from the machine. Adaptive filtering using multichannel vibration signals from various points is a feasible solution [125]; see Chapter 3.

Energy limitations: Most biomedical signals are generated at microvolt or millivolt levels at their sources. Recording such signals requires sensitive transducers and instrumentation with low noise levels. The connectors and cables need to be shielded, in order to obviate pickup of ambient electromagnetic (EM) signals. Some applications may require transducers with integrated amplifiers and signal conditioners so that the signal leaving the subject at the transducer is much stronger than ambient sources of potential interference. A Faraday cage or shield, constructed with a wire mesh or foil of a conducting material to enclose the entire data acquisition setup including the subject and all devices, may be used to prevent external EM waves from contaminating the signals being recorded.

When external stimuli are required to elicit a certain response from a system, the level of the stimulus is constrained due to safety factors and physiological limitations. Electrical stimuli to record the ENG need to be limited in voltage level so as to not cause local burns or interfere with the electrical control signals of the cardiac or nervous systems. Auditory and visual stimuli are constrained by the lower thresholds of detectability and upper thresholds related to frequency response, saturation, or pain.

Patient safety: Protection of the subject or patient from electrical shock or radiation hazards is an unquestionable requirement of paramount importance. The relative levels of any other risks involved should be assessed when a choice is available between various procedures and should be analyzed against their relative benefits.

Patient safety concerns may preclude the use of a procedure that may yield better signals or results than others, or require modifications to a procedure that may lead to inferior signals. Further signal processing steps would then become essential in order to improve signal quality or otherwise compensate for the initial loss.

In consideration of the various difficulties encountered in the acquisition of biomedical signals, several institutions and agencies have supported efforts to develop public databases of annotated signals. The PhysioNet [126–128] is one such resource that is useful in education and research activities related to biomedical signals.

1.5 Why Use CAD?

Physicians, cardiologists, neuroscientists, and health-care technologists are highly trained and skilled practitioners. Why then would we want to use computers or electronic instrumentation for the analysis of biomedical signals? The following points provide some arguments in favor of the application of computers to process and analyze biomedical signals.

- Humans are highly skilled and fast in the analysis of visual patterns and waveforms, but are slow in arithmetic operations with large numbers of values. The ECG of a single cardiac cycle (heart beat) could have up to 200 numerical values; the corresponding PCG up to 2,000. If signals need to be processed to remove noise or extract a parameter, it would not be practical for a person to perform such computation. Computers can perform millions of arithmetic operations per second. It should be noted, however, that recognition of waveforms and images using mathematical procedures typically requires huge numbers of operations that could lead to slow responses in such tasks from low-level computers.
- Humans could be affected by fatigue, boredom, and environmental factors, and are susceptible to committing errors. Long-term monitoring of signals, for example, the heart rate and ECG of a critically ill patient, by a human observer watching an oscilloscope or computer display is neither economical nor feasible. A human observer could be distracted by other events in the surrounding areas and may miss short episodes or transients in the signal. Computers, being inanimate but mathematically accurate and consistent machines, can be designed to perform computationally specific and repetitive tasks.
- Analysis by humans is usually subjective and qualitative. When comparative analysis is required between the signal of a subject and another or a standard pattern, a human observer would typically provide a qualitative response. For example, if the QRS width of the ECG is of interest, a human observer may remark that the QRS of the subject is wider than the reference or normal. More specific or objective comparison to the accuracy of the order of a few milliseconds would require the use of electronic instrumentation or a computer. Derivations of quantitative or numerical features from signals with large numbers of samples would certainly demand the use of computers.

- Analysis by humans is subject to interobserver as well as intraobserver variations (with time). Given that most analyses performed by humans are based upon qualitative judgment, they are liable to vary with time for a given observer, or from one observer to another. The former could also be due to lack of diligence or due to inconsistent application of knowledge, and the latter due to variations in training and level of understanding. Computers can apply a given procedure repeatedly and whenever recalled in a consistent manner. It is further possible to encode the knowledge (to be more specific, the logic) of many experts into a single computational procedure, and thereby enable a computer with the collective intelligence of several human experts in the area of interest.
- Most biomedical signals vary slowly over time (that is, they are lowpass signals), with their bandwidth limited to a few tens to a few thousand Hz. Typical sampling rates for digital processing of biomedical signals range from 100 Hz to 10 – 20 kHz. Sampling rates as above facilitate *on-line*, *real-time* analysis of biomedical signals with even low-end computers. The term “on-line” indicates that the patient or subject is connected to the system or computer that is acquiring and analyzing signals. The term “real-time analysis” may be used to indicate the processing of each sample of a signal before the next sample arrives, or the processing of an epoch or episode such as an ECG beat before the next one is received in its entirety in a buffer. Monitoring the heart rate of critically ill patients demands on-line and real-time ECG analysis. However, some applications do not require on-line, real-time analysis: For example, processing a VAG signal to diagnose cartilage degeneration, or analyzing a long-term ECG record obtained over several hours using an ambulatory system, usually does not demand immediate attention and results. In such cases, computers could be used for *off-line* analysis of prerecorded signals with sophisticated signal processing and time-consuming modeling techniques. The speed required for real-time processing and the computational complexities of modeling techniques in the case of off-line applications would both rule out the possibility of performance of the tasks by humans.

One of the important points to note in the above discussion is that *quantitative analysis* becomes possible by the application of computers to biomedical signals. The logic of medical or clinical diagnosis via signal analysis could then be *objectively* encoded and *consistently* applied in routine or repetitive tasks. However, it should be emphasized at this stage that the end-goal of biomedical signal analysis should be seen as computer-*aided* diagnosis and not automated diagnosis. A physician or medical specialist typically uses a significant amount of information in addition to signals and measurements, including the general physical appearance and mental state of the patient, family history, and socioeconomic factors affecting the patient, many of which are not amenable to quantification and logical rule-based processes. Biomedical signals are, at best, indirect indicators of the state of the patient; most cases lack a direct or unique signal–pathology relationship [53]. The results of signal analysis need to be integrated with other clinical signs, symptoms, and information by a physician. Above all, the *intuition* of the specialist plays an

important role in arriving at the final diagnosis. For these reasons, and keeping in mind the realms of practice of various licensed and regulated professions, liability, and legal factors, the final diagnostic decision is best left to the physician or medical specialist. It is expected that quantitative and objective analysis facilitated by the application of computers to biomedical signal analysis will lead to a more accurate diagnostic decision by the physician.

On the importance of quantitative analysis:

“When you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind: it may be the beginning of knowledge, but you have scarcely, in your thoughts, advanced to the stage of *science*.”

— Lord Kelvin (*William Thomson, 1824–1907*) [129]

On assumptions made in quantitative analysis:

“Things do not in general run around with their measure stamped on them like the capacity of a freight car; it requires a certain amount of investigation to discover what their measures are ... What most experimenters take for granted before they begin their experiments is infinitely more interesting than any results to which their experiments lead.”

— Norbert Wiener (*1894–1964*)

1.6 Remarks

We have taken a general look at the nature of biomedical signals in this chapter and have seen a few signals illustrated for the purpose of gaining familiarity with their typical appearance and features. Specific details of the characteristics of the signals and their processing or analysis are provided in the subsequent chapters.

We have also stated the objectives of biomedical instrumentation and signal analysis. Some practical difficulties that arise in biomedical signal investigation were discussed in order to draw attention to the relevant practical issues. The suitability and desirability of the application of computers for biomedical signal analysis were discussed, with emphasis on objective and quantitative analysis toward the end-goal of CAD. The remaining chapters provide descriptions of specific signal processing techniques and applications.

1.7 Study Questions and Problems

Note: Some of the questions may require background preparation with other sources on the ECG (for example, Rushmer [33]), the EMG (for example, Goodgold and Eberstein [30]), and biomedical instrumentation (for example, Webster [11]).

1. Give two reasons to justify the use of electronic instruments and computers in medicine.
2. State any two objectives of using biomedical instrumentation and signal analysis.

3. Distinguish between open-loop and closed-loop monitoring of a patient.
4. List three common types or sources of artifact in a biomedical instrument.
5. A nerve cell has an action potential of duration 10 *ms* including the refractory period. What is the maximum rate (in pulses per second) at which this cell can transmit electrical activity?
6. Consider a myocardial cell with an action potential of duration 300 *ms* including its refractory period. What is the maximum rate at which this cell can be activated (fired) into contraction?
7. Distinguish between spatial and temporal recruitment of motor units to obtain increasing levels of muscular activity.
8. Consider three motor units with action potentials (SMUAPs) that are of different biphasic and triphasic shapes. Consider the initial stages of contraction of the related muscle. Draw three plots of the net EMG of the three motor units for increasing levels of contraction with the spatial and temporal recruitment phenomena invoked individually and in combination. Assume low levels of contraction and that the SMUAPs do not overlap.
9. Draw a typical ECG waveform over one cardiac cycle indicating the important component waves, their typical durations, and the typical intervals between them. Label each wave or interval with the corresponding cardiac event or activity.
10. Draw the waveform corresponding to two cycles of a typical ECG signal and indicate the following waves and periods: (a) the P, QRS, and T waves; (b) the RR interval; (c) atrial contraction; (d) atrial relaxation; (e) ventricular contraction; and (f) ventricular relaxation.
11. Explain why the P and T waves are low-frequency signals whereas the QRS complex is a high-frequency signal. Include diagrams of action potentials and an ECG waveform in your reasoning.
12. Explain the reasons for widening of the QRS complex in the case of certain cardiac diseases.
13. Give two examples that call for the use of electronic instruments and/or computers in ECG analysis.
14. A heart patient has a regular SA node pulse (firing) pattern and an irregular ectopic focus. Over a period of 10 *s*, the SA node was observed to fire regularly at $t = 0, 1, 2, 3, 4, 5, 6, 7, 8,$ and 9 s . The ectopic focus was observed to fire at $t = 1.3, 2.8, 6.08,$ and 7.25 s . Draw two impulse sequences corresponding to the firing patterns of the SA node and the ectopic focus. Draw a schematic waveform of the resulting ECG of the patient. Explain the source of each beat (SA node or ectopic focus) and give reasons.
15. A patient has ventricular bigeminy, where every second pulse from the SA node is replaced by a premature ventricular ectopic beat with a full compensatory pause. (See Figure 9.9 for an illustration of bigeminy.) The firing rate of the SA node is regular at 80 beats a minute, and each ectopic beat precedes the blocked SA node pulse by 100 *ms*.
(a) Draw a schematic trace of the ECG for 10 beats, marking the time scale in detail.
(b) Draw a histogram of the RR intervals for the ECG trace. (c) What is the average RR interval computed over the 10 beats?
16. Draw a typical PCG (heart sound signal) waveform over one cardiac cycle indicating the important component waves, their typical durations, and the typical intervals between them. Label each wave or interval with the corresponding cardiac event or activity.

17. Give two examples that require the application of electronic instruments and/or computers in EEG analysis.
18. Distinguish between ECG rhythms and EEG rhythms. Sketch one example of each.
19. What are the causes and characteristics of ventricular ectopic beats (PVCs)? Draw a sample ECG signal including five normal beats and two PVCs. Label parts of the signal indicating atrial contraction, ventricular contraction, and the premature nature of PVCs.
20. Describe two cardiac abnormalities that cause changes in the shape of the QRS complex. Draw the ECG waveforms for the two cases and compare their features with those of a normal ECG waveform.
21. What is the ENG? Describe an experimental procedure to acquire an ENG. Describe the typical result obtained from ENGs and a potential application.
22. Explain the cardiac events that cause the second heart sound (S2). Identify the valves and their actions associated with S2. Explain how the dicrotic notch in the carotid pulse is related to S2.
23. Draw Einthoven's triangle and indicate the axes representing the six leads of the ECG obtained using the four limb leads. Mark the positions of the four limb leads in your diagram.
24. Draw a schematic representation of an ECG signal over one cardiac cycle. Label intervals related to atrial contraction and ventricular contraction. Overlay the ECG signal with schematic representations of the action potentials of atrial and ventricular myocytes. Explain the relationships between the various parts of the action potentials and the ECG signal.
25. Draw a schematic representation of Einthoven's triangle showing the directions (polarities) of leads I, II, and III of the ECG signal. Rearrange the vectors and derive the relationship between the three leads I, II, and III using vectorial arithmetic.
26. Consider the contraction of a muscle at a low level of effort. Suppose that two motor units are active. Assume that one of the motor units has a biphasic action potential and is firing at the rate of 10 *pps*. Assume that the other motor unit has a triphasic action potential and is firing at the rate of 12 *pps*. For a duration of 0.5 *s*, draw a schematic sketch of the action potential trains for the two motor units individually and separately. Draw a schematic sketch of the overall or combined EMG signal of the muscle. (Assume that the action potentials do not overlap.) Mark the time axis clearly in your drawings.
27. Draw a schematic diagram of a motor unit including labels (names) for each part. Explain the generation of the EMG signal, including the following: (a) single motor unit action potential, (b) innervation ratio, (c) temporal recruitment, and (d) spatial recruitment. Include sketches of at least two sample EMG signals in your discussion.
28. Describe the events in the cardiac system that cause (a) the components of the first heart sound (S1), and (b) the components of the second heart sound (S2). Describe an abnormal condition that could cause (c) a systolic murmur (SM), and (d) a diastolic murmur (DM). Draw a schematic representation of the PCG signal and the corresponding ECG signal over two cardiac cycles and label the PCG signal with the parts related to S1, S2, SM, and DM.
29. (a) Draw a schematic graphical representation of a normal ECG signal over one cardiac cycle. Identify all the waves and their typical durations. Identify the isoelectric parts of the signal. (b) Draw an ECG signal with a wide and jagged QRS complex. (c) Draw an

ECG signal with an elevated ST segment. (d) Draw an ECG signal with a depressed ST segment.

30. Draw a schematic sketch of a normal ECG signal as well as the corresponding PCG and carotid pulse signals over one cardiac cycle. Identify the following waves in the signals: P, QRS, T, S1, S2, and the dicrotic notch. Indicate the temporal relationships between the waves mentioned. Label your figure with the relationships between the waves listed above and the following events in the cardiac cycle: atrial contraction, ventricular contraction, ventricular relaxation, and closure of the aortic valve. Label the intervals where systolic and diastolic murmurs could appear.

1.8 Laboratory Exercises and Projects

1. Visit an ECG, EMG, or EEG laboratory in your local hospital or health sciences center. View a demonstration of the acquisition of a few biomedical signals. Request a specialist in a related field to explain how he or she would interpret the signals. Volunteer to be the experimental subject and experience first-hand a biomedical signal acquisition procedure!
2. Set up an ECG acquisition system and study the effects of the following conditions or actions on the quality and nature of the signal: loose electrodes; lack of electrode gel; the subject holding his/her breath or breathing freely during the recording procedure; and the subject coughing, talking, or squirming during signal recording. Record noise-free and noisy ECG signals under various conditions for use in exercises on filtering.
3. Using a stethoscope, listen to your own heart sounds and those of your friends. Examine the variability of the sounds with the site of auscultation. Study the effects of heavy breathing and speaking by the subject as you are listening to the heart sound signal.
4. Record speech signals of vowels (/A/, /I/, /U/, /E/, /O/), diphthongs (/EI/, /OU/), fricatives (/S/, /F/), and plosives (/T/, /P/), as well as words with all three types of sounds (for example, safety, explosive, hearty, heightened, house). You may be able to perform this experiment with the microphone on your computer. Study the waveform and characteristics of each signal. Use the signals in exercises on filtering, segmentation, and spectral analysis.
5. Using surface electrodes placed on the forearm, record EMG signals at various levels of force exerted by clenching the fist using a device with a force transducer. Provide for rest between repetitions of the experiment to avoid the development of fatigue in the muscle. Plot and study variations in the signal's characteristics with the level of force.

CHAPTER 2

ANALYSIS OF CONCURRENT, COUPLED, AND CORRELATED PROCESSES

The human body is a complex integration of a number of biological systems with several ongoing physiological, functional, and possibly pathological processes. Most biological processes within a body are not independent of one another; rather, they are mutually correlated and bound together by physical or physiological control and communication phenomena. Analyzing any single process without due attention to others that are concurrent, coupled, or correlated with the process may provide only partial information and pose difficulties in the comprehension of the process. Then, the problem is, how do we recognize the existence of concurrent, coupled, and correlated phenomena? How do we obtain the corresponding signals and identify the correlated features? Unfortunately, there is no simple or universal solution or rule to apply to this problem.

Ideally, an investigator should explore the system or process of interest from all possible angles and use multidisciplinary approaches to identify several potential sources of information. The signals so obtained may be electrical, mechanical, biochemical, or physical, among the many possibilities, and may exhibit interrelationships confounded by peculiarities of transduction, time delays, multipath transmission or reflection, waveform distortions, and filtering effects that may need to be accounted for in their simultaneous analysis. Events or waves in signals of interest

may be nonspecific and difficult to identify and analyze. How could we exploit the concurrency, coupling, and correlation present between processes or related signals to gain better understanding of the system or systems of interest?

2.1 Problem Statement

Determine the correspondences, correlation, and interrelationships present between concurrent signals related to a common underlying physiological system or process, and identify their potential applications.

The statement above represents, of necessity at this stage of the discussion, a vague and generic problem. The case studies and applications presented in the following sections provide a few illustrative examples dealing with specific systems and problems. Signal processing techniques for the various tasks identified in the case studies are developed in the chapters that follow. Note that the examples cover a diverse range of systems, processes, and signals. The specific problem of your interest will very likely not be directly related to any of the case studies presented here. It is expected that a study of the examples provided will expand the scope of your analytical skills and lead to improved solutions for your specific case.

2.2 Illustration of the Problem with Case Studies

2.2.1 The ECG and the PCG

A clinical ECG record typically includes 12 channels of sequentially or simultaneously recorded signals, and can be used on its own to diagnose many cardiac diseases. This is mainly due to (a) the simple and readily identifiable waveforms in the ECG, and the innumerable studies that have firmly established clinical ECG as a standard procedure. The PCG, on the other hand, is a more complex signal. PCG waveforms cannot be visually analyzed except for the identification of gross features such as the presence of murmurs, time delays as in a split S₂, and envelopes of murmurs. An advantage with the PCG is that it may be listened to; auscultation of heart sounds is more commonly performed than visual analysis of the PCG signal. However, objective analysis of the PCG requires the identification of components, such as S₁ and S₂, and subsequent analysis tailored to the nature of the components.

Given a run of a PCG signal over several cardiac cycles, visual identification of S₁ and S₂ is possible if there are no murmurs between the sounds, and if the heart rate is low such that the S₂ – S₁ (of the next beat) interval is longer than the S₁ – S₂ interval (as expected in normal situations). At high heart rates and with the presence of murmurs or premature beats, identification of S₁ and S₂ could be difficult.

Problem: *Identify the beginning of S₁ in a PCG signal and extract the heart sound signal over one cardiac cycle.*

Solution: The ECG and PCG are concurrent phenomena, with the noticeable difference that the former is electrical while the latter is mechanical (sound or vibra-

tion). It is customary to record the ECG with the PCG; see Figures 1.42 and 1.44 for examples.

The QRS wave in the ECG is directly related to ventricular contraction, as the summation of the action potentials of ventricular muscle cells (see Section 1.2.5). As the ventricles contract, the tension in the chordae tendineae and the pressure of retrograde flow of blood toward the atria seal the AV valves shut, thereby causing the initial vibrations of S1 [33] (see Section 1.2.9). Thus, S1 begins immediately after the QRS complex. Given the nonspecific nature of vibration signals and the various possibilities in the transmission of the heart sounds to the recording site on the chest, detection of S1 on its own is a difficult problem.

As shown in Sections 3.5.1, 4.3.1, and 4.3.2, detection of the QRS is fairly easy, given that the QRS is the sharpest wave in the ECG over a cardiac cycle; in fact, the P and T waves may be almost negligible in many ECG records. Thus, the QRS complex in the ECG is a reliable indicator of the beginning of S1 and may be used to segment a PCG record into individual cardiac cycles: from the beginning of one QRS (and thereby S1) to the beginning of the next QRS and S1. This method may be applied visually or via signal processing techniques: The former requires no further explanation but is expanded upon in Section 2.3; the latter is dealt with in Section 4.10.

2.2.2 The PCG and the carotid pulse

Identification of the diastolic segment of the PCG may be required in some applications in cardiovascular diagnosis [130]. Ventricular systole ends with the closure of the aortic and pulmonary valves, indicated by the aortic (A2) and pulmonary (P2) components of S2 (see Section 1.2.9). The end of contraction is also indicated by the T wave in the ECG, and S2 appears slightly after the end of the T wave (see Figure 1.42). S2 may be taken to be the end of systole and the beginning of ventricular relaxation or diastole. (*Note:* Shaver et al. [74] and Reddy et al. [75] have included S2 in the part of their article on systolic sounds.) However, as in the case of S1, S2 is also a nonspecific vibrational wave that cannot be readily identified (even visually), especially when murmurs are present.

Given the temporal relationship between the T wave and S2, it may appear that the former may be used to identify the latter. This, however, may not always be possible in practice, as the T wave is often a low-amplitude and smooth wave and is sometimes not recorded at all (see the normal beats in Figure 1.27). ST segment elevation (as in Figure 1.27) or depression (as in Figure 1.46) may make even visual identification of the T wave difficult. Thus, the T wave is not a reliable indicator to use for identification of S2.

Problem: *Identify the beginning of S2 in a PCG signal.*

Solution: Given the inadequacy of the T wave as an indicator of diastole, we need to explore other possible sources of information. Closure of the aortic valve is accompanied by deceleration and reversal of blood flow in the aorta. This causes a sudden drop in the pressure within the aorta, which is already on a downward slope due to the end of systolic activity. The sudden change in pressure causes an *incisura*

or notch in the aortic pressure wave (see Figures 1.45 and 1.46). The aortic pressure signal may be obtained using catheter-tip sensors [74, 75], but the procedure would be invasive. Fortunately, the notch is transmitted through the arterial system and may be observed in the carotid pulse (see Section 1.2.10) recorded at the neck.

The dicrotic notch (D) in the carotid pulse signal bears a delay with respect to the corresponding notch in the aortic pressure signal, but has the advantage of being accessible in a noninvasive manner. (Similar events occur in the pulmonary artery, but provide no externally observable effects.) See Figures 1.42 and 1.44 for examples of three-channel PCG – ECG – carotid pulse recordings that illustrate the D – S2 – T relationships. The dicrotic notch may be used as a reliable indicator of the end of systole or beginning of diastole that may be obtained in a noninvasive manner. The average S2 – D delay has been found to be 42.6 ms with a standard deviation (*SD*) of 5 ms [131] (see also Tavel [72]), which should be subtracted from the dicrotic notch position to obtain the beginning of S2.

Signal processing techniques for the detection of the dicrotic notch and segmentation of the PCG are described in Sections 4.3.3, 4.10, and 4.11.

2.2.3 The ECG and the atrial electrogram

Most studies on the ECG and the PCG pay more attention to ventricular activity than to atrial activity — and, even then, more to left ventricular activity than to the right. Rhythm analysis is commonly performed using QRS complexes to obtain interbeat intervals known as RR intervals. Such analysis neglects atrial activity.

Recollect that the AV node introduces a delay between atrial contraction initiated by the SA node impulse and the consequent ventricular contraction. This delay plays a major role in the coordinated contraction of the atria and the ventricles. Certain pathological conditions may disrupt this coordination and may even cause AV dissociation [33]. It then becomes necessary to study atrial activity independent of ventricular activity and establish their association, or lack thereof. Thus, the interval between the P wave and the QRS (termed the PR interval) would be a valuable adjunct to the RR interval in rhythm analysis. Unfortunately, the atria, being relatively small chambers with weak contractile activity, cause a small and smooth P wave in the external ECG. Quite often, the P wave may not be recorded or seen in the external ECG; see, for example, leads I and V3 – V6 in Figure 1.34.

Problem: *Obtain an indicator of atrial contraction to measure the PR interval.*

Solution: One of the reasons for the lack of specificity of the P wave is the effect of transmission from the atria to the external recording sites. An obvious solution would be to insert electrodes into one of the atria via a catheter and record the signal at the source. This would, of course, constitute an invasive procedure. Jenkins et al. [132, 133] and Jenkins [51, 52] proposed a unique and interesting procedure to obtain a strong and clear signal of atrial activity: they developed a pill electrode that could be swallowed and lowered through the esophagus to a position close to the left atrium (the bipolar electrode pill being held suspended by wires about 35 cm from the lips). The procedure may or may not be termed invasive, although an object is inserted into the body (and removed after the procedure), as the action required

is that of normal swallowing of a tablet-like object. The gain required to obtain a good atrial signal was 2 – 5 times that used in ECG amplifiers. With a 5 – 100 Hz bandpass filter, Jenkins et al. obtained an SNR of 10.

Figure 2.1 shows recordings from a normal subject of the atrial electrogram from the pill electrode and an external ECG lead. Atrial contraction is clearly indicated by a sharp spike in the atrial electrogram. Measurement of the PR interval (or the AR interval, as called by Jenkins et al.) now becomes an easy task, with identification of the spike in the atrial electrogram (the “A” wave, as labeled by Jenkins et al.) being easier than identification of the QRS in the ECG.

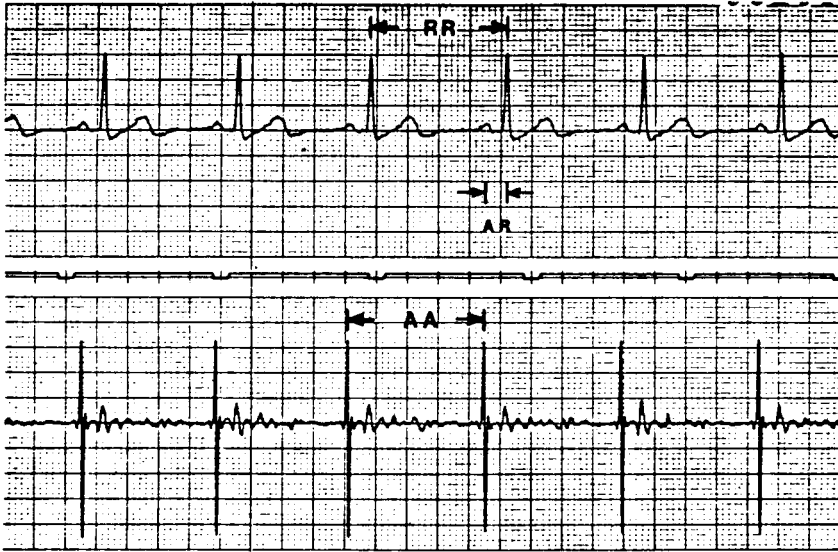


Figure 2.1 Pill-electrode recording of the atrial electrogram (lower tracing) and the external ECG (upper tracing) of a normal subject. The pulse train between the two signals indicates intervals of 1 s . Reproduced with permission from J.M. Jenkins, D. Wu, and R. Arzbacher, Computer diagnosis of abnormal cardiac rhythms employing a new P-wave detector for interval measurement, *Computers and Biomedical Research*, 11:17–33, 1978. ©Academic Press.

Figure 2.2 shows the atrial electrogram and external ECG of a subject with ectopic beats. The PVCs have no immediately preceding atrial activity. The first PVC has blocked the conduction of the atrial activity occurring immediately after, resulting in a compensatory pause before the following normal beat. The second PVC has not blocked the subsequent atrial wave, but has caused a longer-than-normal AV delay and an aberrant conduction path, which explains the different waveshape of the consequent beat. The third PVC has not affected the timing of the following SA-node-initiated pulse, but has caused a change in waveshape in the resulting QRS-T by altering the conduction path [51, 52, 132, 133].

Jenkins et al. developed a four-digit code for each beat, as illustrated in Figure 2.2. The first digit was coded as

0: abnormal waveshape or

1: normal waveshape,

as determined by a correlation coefficient computed between the beat being processed and a normal template (see Sections 3.5.1, 4.4.2, 5.4.1, and 5.8). The remaining three digits encoded the nature of the RR, AR, and AA intervals, respectively, as

0: short,

1: normal, or

2: long.

The absence of a preceding A wave related to the beat being analyzed was indicated by the code $\times$ in the fourth digit (in which case the AR interval is longer than the RR interval). Figure 2.2 shows the code for each beat. Based upon the code for each beat, Jenkins et al. were able to develop a computerized method to detect a wide variety of arrhythmia.

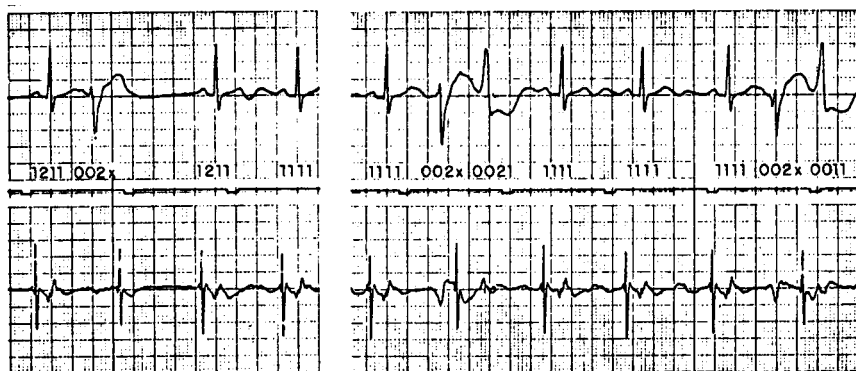


Figure 2.2 Atrial electrogram (lower tracing) and the external ECG (upper tracing) of a subject with ectopic beats. The pulse train between the two signals indicates intervals of 1 s. Reproduced with permission from J.M. Jenkins, D. Wu, and R. Arzbaecher, Computer diagnosis of abnormal cardiac rhythms employing a new P-wave detector for interval measurement, *Computers and Biomedical Research*, 11:17–33, 1978. ©Academic Press.

2.2.4 Cardiorespiratory interaction

The heart rate is affected by normal breathing due to the coupling and interaction existing between the cardiac and respiratory systems [134–139]. Figure 2.3 shows two recordings of the ECG of a normal subject taken a few minutes apart, the first with the subject breathing normally and the second with the subject holding his breath. Baroreceptors in the aorta detect changes in the aortic transmural pressure associated with variations in the intrapleural pressure with respiration. A decrease in the intrapleural pressure during inspiration causes the vagus nerve activity to be impeded, which causes an increase in the heart rate during inspiration. Therefore, as

a subject carries on breathing normally, there could be substantial variations in the heart rate and RR intervals.

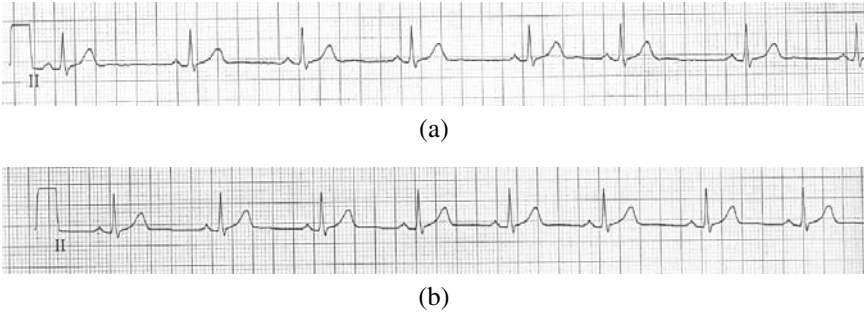


Figure 2.3 ECG signal of a subject (a) with the subject breathing normally, and (b) with the subject holding breath. Signal courtesy of E. Gedamu and L.B. Mitchell, Foothills Hospital, Calgary.

Breathing also affects the transmission of the heart sounds from the cardiac chambers to the chest surface. Durand et al. [140] recorded intracardiac and chest-surface PCG signals and derived the dynamic transfer function of the heart – thorax acoustic system in dogs. Analysis of the synchronization and coupling within the cardiorespiratory system could require sophisticated analysis of several signals acquired simultaneously from the cardiac and respiratory systems [141]. A few techniques for the analysis of heart-rate variability (HRV) based upon RR interval data are described in Sections 7.2.2, 7.8, and 8.11.

2.2.5 The importance of HRV

Even under resting and apparently steady conditions, the intervals between heart beats and the heart rate are not constant. Variability of the RR interval and heart rate is a normal and healthy physiological phenomenon. Reduced HRV in patients following acute myocardial infarction has been observed to be related to poor prognosis [142, 143]. Malik and Camm [142] noted that HRV is the single most important predictor of patients at high risk of sudden death and ventricular arrhythmia.

As noted in Section 1.2.5, cardiac rhythm is controlled by the ANS. Section 2.2.4 presented a discussion on the effects of normal breathing on the heart rate due to the coupling and interaction between the cardiac and respiratory systems. The ANS is divided into the sympathetic nervous system (SNS) and the parasympathetic nervous system (PNS). The SNS and PNS do not work in opposition to one another, but interact in a complex and dynamic manner, with the interactions modulated by secondary messengers [144]. The PNS can inhibit sympathetic nerve activity, and sympathetic activation can inhibit parasympathetic activation [144]. Vagal tone or tonic parasympathetic activation is stronger than the sympathetic tone at rest. The parasympathetic

influence on the heart rate is mediated by the release of acetylcholine by the vagus nerve; sympathetic stimulation is mediated by the release of epinephrine. HRV has been noted as a noninvasive signature of the interaction and balance between the effects of the PNS and SNS on the heart [145]; see Kamath et al. [146] for a recent compilation of several works on this topic.

Olshansky et al. [144] studied the role of the PNS in heart failure and made the following observations. The resting rate of a normal heart is governed by a parasympathetic mechanism. The resting heart rate, which is an indicator of vagus nerve function, can predict mortality. Increase in the parasympathetic component of HRV and higher vagus nerve activity result in slower heart rate and better outcome in terms of cardiac health. In cases of heart failure, the regulation of heart rate by parasympathetic activation is poor. High resting heart rate typically leads to adverse outcomes. They also noted that the high-frequency components of HRV are associated with the vagus nerve and the parasympathetic effect, that the low-frequency components are due to sympathetic and parasympathetic activation, and, furthermore, that parasympathetic activation and its physiological effects are attenuated in cases of heart failure. Olshansky et al. observed that the electrophysiological benefits of parasympathetic activation include antiinflammatory effects, reduced heart rate, increased HRV, and direct antiarrhythmic effects. Whereas high levels of sympathetic activity are associated with poor prognosis, a high level of parasympathetic activation could provide protection. Based on these observations, Olshansky et al. indicated that direct or indirect vagus nerve stimulation could have beneficial effects on clinical outcomes.

See Malik et al. [147] for recommendations on measurement, physiological interpretation, and clinical use of HRV. See Sections 7.2.2, 7.8, and 8.11 for discussions on methods for the analysis of HRV.

2.2.6 The EMG and VMG

The EMG signal has been studied extensively and the relationship between EMG signal parameters and muscle contraction level has been established [30, 37]. It is known that the root mean-squared (*RMS*) and mean frequency values of the EMG increase with increasing muscle contraction until fatigue sets in, at which point both values begin to decrease. In this situation, while the muscle output measured is mechanical contraction (using force or strain transducers), the signal analyzed is electrical in character. A direct mechanical signal related to basic muscle-fiber or motor-unit phenomena may be desired in some situations.

Problem: *Obtain a mechanical signal that is a direct indicator of muscle-fiber or motor-unit activity to study muscle contraction and force development.*

Solution: The VMG, as introduced in Section 1.2.13, is a vibration signal measured from a contracting muscle. The signal is a direct manifestation of the contraction of muscle fibers and, as such, represents mechanical activity at the muscle-fiber or motor-unit level. The VMG is the mechanical counterpart and contemporary of the EMG. Although no direct relationship has been established between the force outputs of individual motor units and the net force output of the muscle, it has been

shown that the *RMS* and mean frequency parameters of the VMG signal increase with muscle force output, in patterns that parallel those of the EMG. Thus, the VMG may be used to quantify muscular contraction [85].

Given the simplicity and noninvasive nature of EMG and VMG measurement, simultaneous analysis of the two signals is an attractive and viable application. Such techniques may find use in biofeedback and rehabilitation [86]. Figure 2.4 shows simultaneous EMG – VMG recordings at two levels of contraction of the rectus femoris muscle [86]. Both signals are interference patterns of several active motor units even at low levels of muscle effort and cannot be analyzed visually. However, a general increase in the power levels of the signals from the lower effort to the higher effort case may be observed. Signal processing techniques for simultaneous EMG – VMG studies are described in Section 5.11.

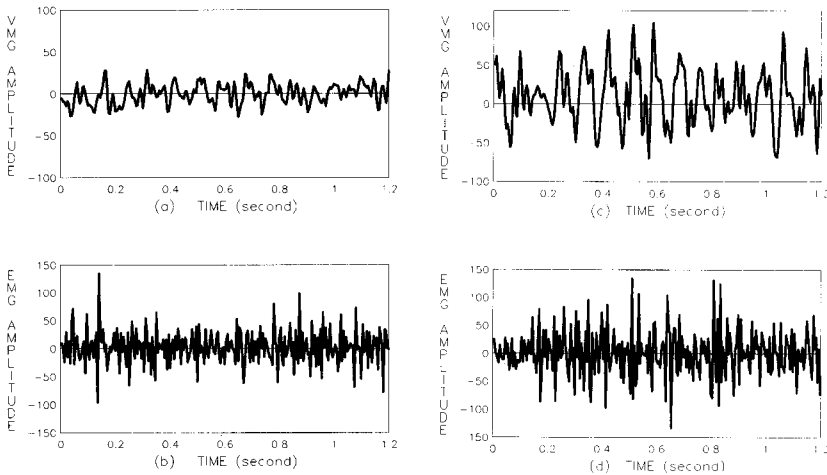


Figure 2.4 Simultaneous EMG – VMG records at two levels of contraction of the rectus femoris muscle. (a) VMG at 40% of the maximal voluntary contraction (MVC) level. (b) EMG at 40% MVC. (c) VMG at 60% MVC. (d) EMG at 60% MVC. Reproduced with permission from Y.T. Zhang, C.B. Frank, R.M. Rangayyan, and G.D. Bell, Relationships of the vibromyogram to the surface electromyogram of the human rectus femoris muscle during voluntary isometric contraction, *Journal of Rehabilitation Research and Development*, 33(4): 395–403, 1996. ©Department of Veterans Affairs.

2.2.7 The knee joint and muscle vibration signals

We saw in Section 1.2.14 that the vibration (VAG) signals produced by the knee joint during active swinging movement of the leg may bear diagnostic information. However, the VMG associated with the rectus femoris muscle that must necessarily be active during extension of the leg could appear as an interference and corrupt the VAG signal [125].

Problem: *Suggest an approach to remove muscle-contraction interference from the knee-joint vibration signal.*

Solution: The VMG interference signal gets transmitted from the source muscle location to the VAG recording position at the skin surface over the patella (knee cap) through the intervening muscles and bones (see Figure 3.11 and Section 3.3.6). Although the interference signal has been found to be of very low frequency (around 10 Hz), the frequency content of the signal varies with muscular effort and knee-joint angle. The rectus femoris muscle and the knee-joint systems are coupled dynamic systems with vibration characteristics that vary with activity level, and hence over time; thus, simple highpass or bandpass filtering of the VAG signal is not an appropriate solution.

An approach to solve the problem would be to record the VMG signal at the rectus femoris at the same time as the VAG signal of interest is acquired from the patella position. Adaptive filtering and noise cancellation techniques [124, 125, 148] could then be applied, with the VAG signal as the primary input and the VMG signal as the reference input. Assuming that the VMG signal that arrives at the patella is strongly correlated with the VMG signal at the rectus femoris and not correlated with the VAG signal of interest, the adaptive filter should remove the interference and estimate the desired VAG signal. Details of adaptive filters are provided in Sections 3.9 and 3.14.

2.3 Application: Segmentation of the PCG into Systolic and Diastolic Parts

Problem: *Show how the ECG and carotid pulse signals may be used to break a PCG signal into its systolic and diastolic parts.*

Solution: A cardiac cycle may be divided into two important parts based upon ventricular activity: systole and diastole. The systolic part starts with S1 and ends at the beginning of S2; it includes any systolic murmur that may be present in the signal. The diastolic part starts with S2, and it ends just before the beginning of the S1 of the next cardiac cycle. (The aortic and pulmonary valves close slightly before the A2 and P2 components of S2. Therefore, systole may be considered to have ended just before S2. Although Shaver et al. [74] and Reddy et al. [75] have included S2 in the part of their article on systolic sounds, we shall include S2 in the diastolic part of the PCG.) The diastolic part includes any diastolic murmur that may be present in the signal; it might also include S3 and S4, if present, as well as AV valve opening snaps, if any.

We saw in Section 2.2.1 that the QRS complex in the ECG may be used as a reliable marker of the beginning of S1. We also saw, in Section 2.2.2, that the dicrotic notch in the carotid pulse may be used to locate the beginning of S2. Thus, if we have both the ECG and carotid pulse signals along with the PCG, it becomes possible to break the PCG into its systolic and diastolic parts.

Figure 2.5 shows three-channel PCG – ECG – carotid pulse signals of a subject with systolic murmur due to aortic stenosis (the same as in Figure 1.44), with the systolic and diastolic parts of the PCG marked in relation to the QRS and D events.

The demarcation was performed by visual inspection of the signals in this example. Signal processing techniques to detect the QRS and D waves are presented in Section 4.3. Adaptive filtering techniques to break the PCG into stationary segments without the use of any other reference signal are described in Section 8.10.

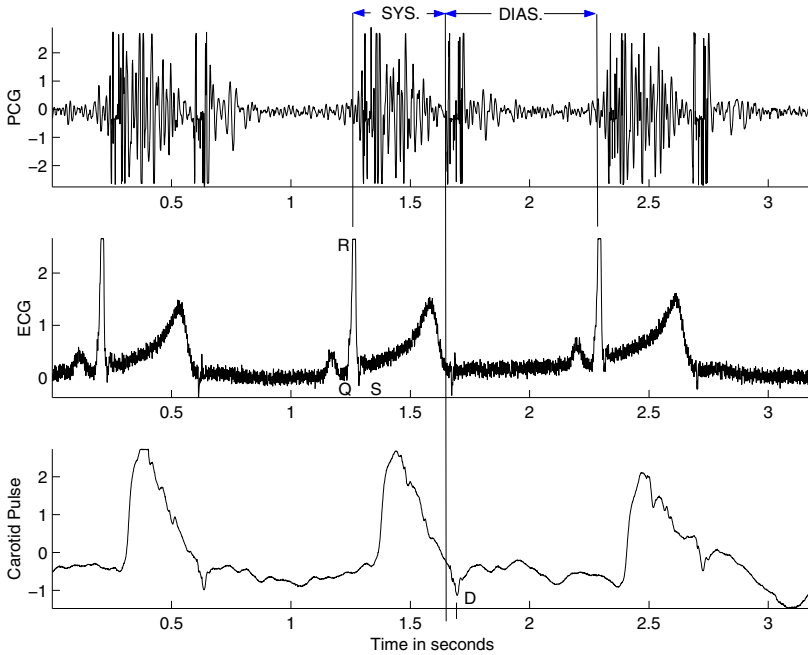


Figure 2.5 Demarcation of the systolic (SYS.) and diastolic (DIAS.) parts of the PCG signal in Figure 1.44 by using the ECG and carotid pulse as reference signals. The QRS complex and the dicrotic notch D are marked on the ECG and carotid pulse signals, respectively.

2.4 Application: Diagnosis and Monitoring of Sleep Apnea

Problem: *Propose approaches based on biomedical signal analysis to detect sleep apnea. Analyze the problem taking into consideration the various physiological systems that are either part of the problem or are affected by the resulting condition.*

Solution: The term “apnea” indicates a condition in which one stops breathing for several seconds, usually during sleep [127, 149–152]. The term “hypopnea” indicates a condition when airflow is diminished. According to Chesson et al. [153], the general condition of disordered breathing during sleep includes the conditions of apnea, defined as cessation or near cessation of respiration for a minimum period of 10 s; hypopnea, defined as a reduction in airflow for a minimum period of 10 s; and episodes of increased respiratory effort due to partial upper-airway obstruction. See Alshaer et al. [105] and Bradley and Floras [154] for additional conditions that de-

fine apnea and hypopnea. Disordered breathing during sleep causes fragmentation of sleep and lack of adequate rest. The total number of episodes of apnea and hypopnea per hour of sleep is defined as the apnea–hypopnea index (AHI). Whereas a patient affected by sleep apnea may stop breathing 10 – 100 times per hour in episodes of duration 10 – 30 s each, the diagnosis of sleep apnea is typically based on an AHI threshold of 10 to 15 [154].

Causes of sleep apnea include the lack of neural input from the CNS to the diaphragm to cause contraction and breathing, known as central sleep apnea (CSA), or collapse of the upper airway, referred to as obstructive sleep apnea (OSA). CSA involves disruption of breathing control where breathing does not match the metabolic requirements. In contrast, OSA is predominantly an anatomical disorder in which the upper airway does not remain open. During sleep, when muscle tone is reduced or absent, the airway collapses and the individual can no longer breathe sufficiently to maintain normal blood gases; as a result, the level of oxygen drops and that of carbon dioxide rises. OSA is the most common type of sleep apnea.

A common symptom of OSA is snoring, a respiratory noise caused by airflow through a partially obstructed airway. An episode of apnea is typically followed by arousal, allowing breathing to resume and blood gases to return to normal levels; there could be movements of the body or the limbs during such occasions of arousal. Frequent and numerous such arousals reduce the quality of sleep and the amount of rest obtained. Chronic reduction of oxygen levels and arousals increase SNS activity, and could cause numerous complications, including daytime sleepiness, hypertension, heart failure, depression, cardiac arrhythmia, and stroke. An effective treatment for OSA is continuous positive airway pressure (CPAP) applied via a face or nasal mask; the pressure applied inflates the upper airway and prevents it from collapsing.

Sleep apnea leads to decreased levels of oxyhemoglobin in the blood. Oxygen transported by blood is bound to hemoglobin in red blood cells. The absorption of different wavelengths of light by hemoglobin changes when it is bound to oxygen; thus, oxyhemoglobin appears red and deoxyhemoglobin appears blue. A finger-tip sensor with a light emitting diode (LED) and a photo sensor, known as a finger-tip pulse oximeter, or a CO -oximeter is commonly used to estimate the level of oxyhemoglobin. Using such a device, the absorption levels for red and infrared light are compared to estimate the proportion of oxyhemoglobin to deoxyhemoglobin in the blood. The measure known as SpO_2 is defined as the ratio of the amount of oxyhemoglobin to deoxyhemoglobin; SaO_2 is defined as the ratio of the amount of arterial-blood oxyhemoglobin to the sum of all types of hemoglobin.

One approach to detect sleep apnea is to measure oronasal airflow by using a pneumotachometer; however, due to the inconvenient nature of this approach, indirect indicators of airflow through the nose and mouth may be obtained by using an oronasal thermistor or by nasal pressure measurement [105, 106]. Respiratory inductance plethysmography is an indirect way of assessing airflow via movement of thoracic or abdominal belts; the signal obtained is proportional to volumetric displacement of the lungs [105, 106, 155].

2.4.1 Monitoring of sleep apnea by polysomnography

Polysomnography (PSG) [153, 156, 157], which involves multichannel recording of several biomedical signals and parameters, is the current standard for the evaluation of sleep-related problems, including sleep apnea. Some of the signals and parameters measured are respiratory effort, airflow, oxygenation, sleep state, EMG of the submental muscle (beneath the chin), EMG of the legs, the electrooculogram (EOG), EEG, ECG, and snoring sound. PSG requires the subject to sleep overnight in a laboratory.

Figure 2.6 shows eight channels selected from a 14-channel PSG data set of a patient with OSA [105]. Periods of diminished airflow to the lungs and consequently low SaO_2 levels are evident. Also seen are related intervals of increased submental EMG activity, snoring, and increased thoracic and/or abdominal activity related to recovery of respiration following episodes of apnea. (Artifact in the submental EMG channel has caused baseline movement; filters may be used to remove such artifact.) SaO_2 values are seen to increase following periods of increased ventilation, but fall again as OSA causes reduced airflow to the lungs. The leg EMG channel shows activity related to leg movement during periods of recovery from episodes of apnea, but is contaminated with the ECG.

2.4.2 Home monitoring of sleep apnea

Although PSG is a comprehensive and accurate method for diagnosing sleep apnea, it is expensive, inconvenient, and often not available. To address these difficulties, practical home-monitoring systems have been developed and are commercially available for the diagnosis and follow-up of sleep apnea. One such system [150] captures the following signals to diagnose patients suspected of having sleep apnea: SpO_2 using a finger-tip pulse oximeter; heart rate, also from the pulse oximeter; pulse amplitude; nasal airflow (through measurement of pressure with a nasal cannula); snoring sound signal using a microphone attached to the throat; and body position using an accelerometer. Additional signals that may be acquired are respiratory airflow (with a pneumotachograph, when CPAP is used); mask pressure (when CPAP is used); respiratory movements; EMG of the legs (for the diagnosis of periodic leg movement); and EMG of the masseter (jaw muscle, for the diagnosis of bruxism).

Figure 2.7 shows a segment of four channels of signals (duration = 4 min) from a home apnea monitor for a subject with mild sleep apnea. Concurrent episodes of increasing heart rate and snoring sound are seen in relation to the two brief episodes of apnea (periods of no or low variations in the nasal pressure signal just before the markers for 32 and 33 min).

Figure 2.8 shows a segment of four channels of signals (of duration 5 min) from an apnea monitor for a subject with severe apnea. Several episodes of substantial hemoglobin desaturation are seen in the record. It is evident that periods of apnea are directly associated with episodes of snoring, reduced SpO_2 , and increased heart rate. The clinical report of the study indicated that the patterns of airflow and oxygen saturation signals seen in the figure are compatible with an obstructive condition.

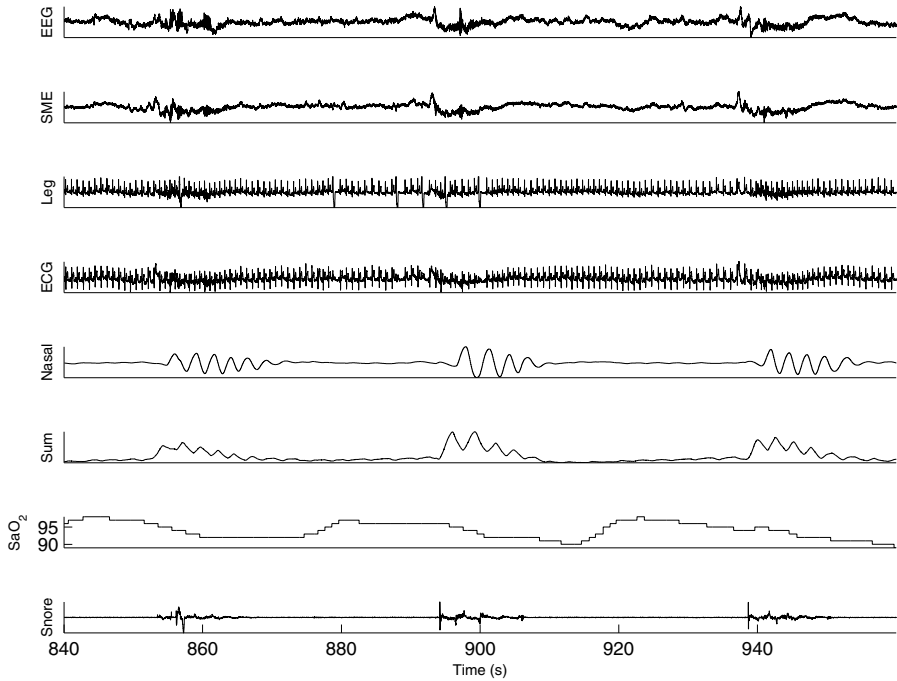


Figure 2.6 Top to bottom: EEG (F4), submental EMG (SME), leg EMG, ECG, airflow (nasal pressure cannula), sum of thoracic and abdominal activity (sum, from respiratory inductance plethysmography), SaO_2 %, and snoring sound signals selected from a 14-channel PSG record of a patient with OSA. Amplitude information has been removed from all channels except SaO_2 % to reduce clutter. Data obtained from Hisham Alshaer and T. Douglas Bradley, Sleep Research Laboratory of the University Health Network Toronto Rehabilitation Institute, Toronto, Ontario, Canada, with permission [105, 106].

See Koley and Dey [158] for the description of a system to detect apnea and hypopnea using the SpO_2 signal from a finger-tip oximeter.

2.4.3 Multivariate and multiorgan analysis

Bianchi et al. [159] proposed a multivariate and multiorgan approach for the analysis of cardiorespiratory variability signals with application to the analysis of sleep and sleep-related disorders. This approach emphasizes that the ANS influences several organs and systems, including the cardiovascular, respiratory, and endocrine-metabolic systems; it also indicates a direct connection to the central and peripheral nervous systems. Bianchi et al. [159] conducted PSG studies including the following signals: three EEG leads (C3–A2, C4–A1, and O2–A1), two EOG leads, three EMG leads (chin, right tibia, and left tibia), one ECG lead, nasal and oral airflow using thermistors, and thoracic and abdominal respiration with piezoelectric belts.

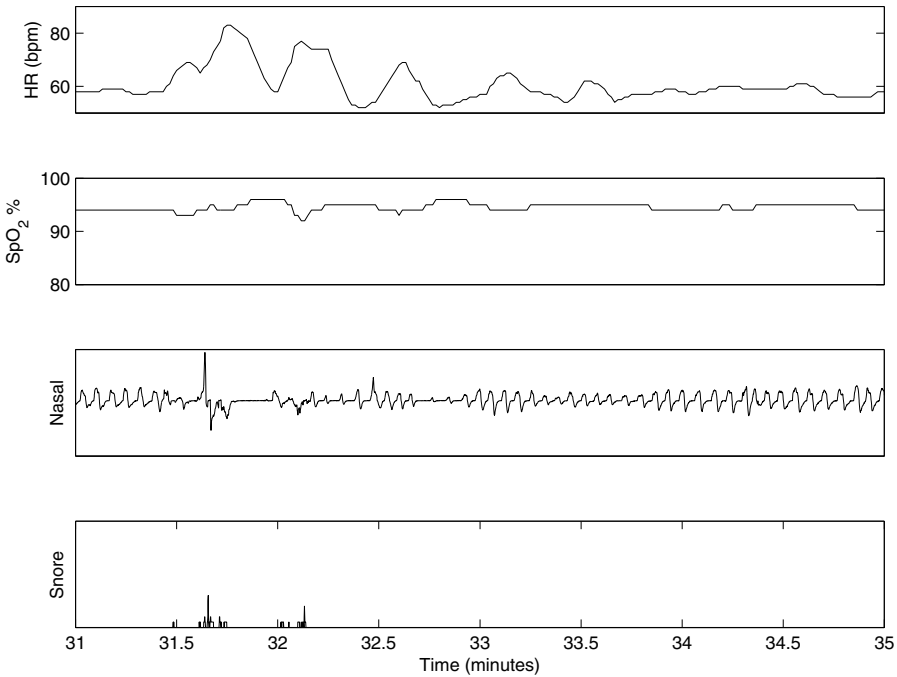


Figure 2.7 Top to bottom: Heart rate, SpO_2 , nasal pressure, and snoring sound signals from a home apnea monitoring record of a moderately symptomatic subject. Amplitude information has been removed from the nasal pressure and snoring sound channels to reduce clutter. Data courtesy of R. Platt, SagaTech Electronics Inc., Calgary, Alberta, Canada, sagatech.ca.

An HRV signal was obtained from the ECG as a sequence of RR intervals. The results obtained indicated that respiratory activation and tachycardia precede periodic leg movement, which was taken to correlate with sympathetic preactivation. Bianchi et al. observed that autonomic changes could be the first manifestation of central activity involving other peripheral systems, and precede related changes in the EEG and EMG. They also observed that HRV may start a few cardiac beats before short arousals, an arousal being considered as a homeostatic mechanism to recover cardiorespiratory functionality. The results were taken to indicate the existence of strong coupling between the ANS and CNS during sleep, and that a mismatch in the coupling strength may lead to pathological situations. It was noted that therapeutic strategies should not be targeted to a single organ or system but should take into consideration the interactions among various systems.

Cerutti [160] presents compelling arguments for an integrated approach for analysis of signals arising from coupled and correlated biological or physiological systems. Figure 2.9 shows four traces of signals including the EEG, EMG from the tibia, RR intervals, and respiration from a patient with myoclonus (involuntary twitching or jerking of a muscle) during sleep. Repeated jerking of the leg, known as the restless

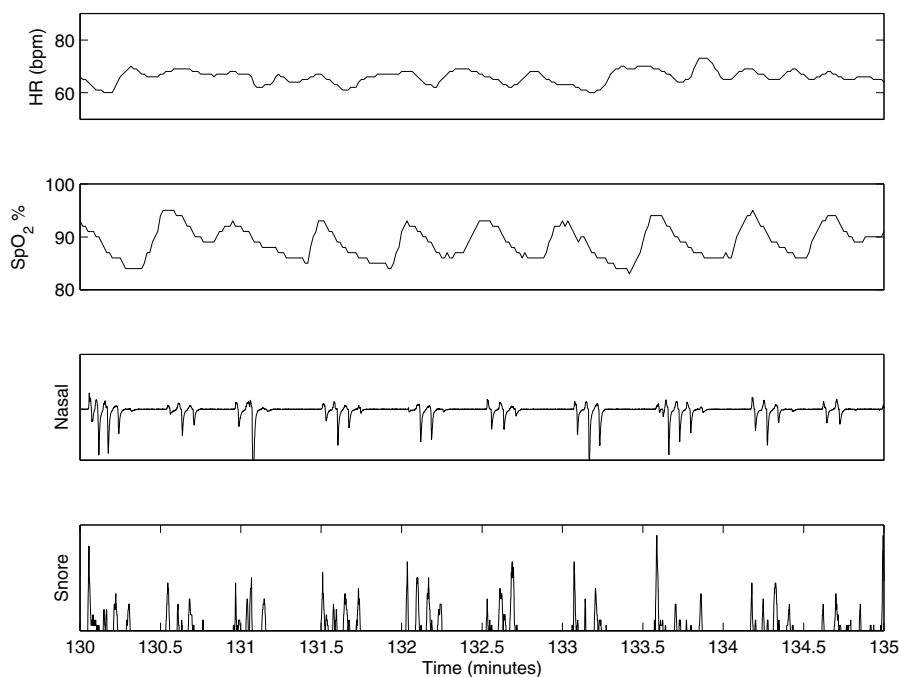


Figure 2.8 Top to bottom: Heart rate, SpO_2 , nasal pressure, and snoring sound signals from a home apnea monitoring record of a subject with severe sleep apnea. Amplitude information has been removed from the nasal pressure and snoring sound channels to reduce clutter. Data courtesy of R. Platt, SataTech Electronics Inc., Calgary, Alberta, Canada, sagatech.ca.

leg syndrome, is associated with sleep disruption and apnea, among several other disorders. The signals in Figure 2.9 demonstrate synchronization between arousal events in the EEG, spikes of activity related to myoclonus in the tibial EMG, increased heart rate (diminished RR intervals), and decreased respiratory activity. (See Somers et al. [161] for a discussion on the autonomic and hemodynamic responses to OSA and high sympathetic nerve activity in patients with OSA.)

In order to simplify the process of monitoring sleep apnea, several researchers have studied various secondary markers of apnea and hypopnea. Contrary to the multivariate approach described above, it might be desirable to be able to detect apnea by using only one or a small number of signals obtained by minimally intrusive ways. It is well known that the heart rate varies with respiration [134]; see Section 2.2.4 and Figure 2.3. Based on this knowledge, researchers have attempted to derive information related to respiration from HRV signals (see Sections 7.2.2, 7.8, and 8.11 for discussions on HRV). Episodes of apnea have been observed to be accompanied by cyclical variations in RR intervals of ECG signals, demonstrating bradycardia during apnea followed by tachycardia upon return to normal respiration. de Chazal et al. [162] proposed methods to derive a surrogate signal related to res-

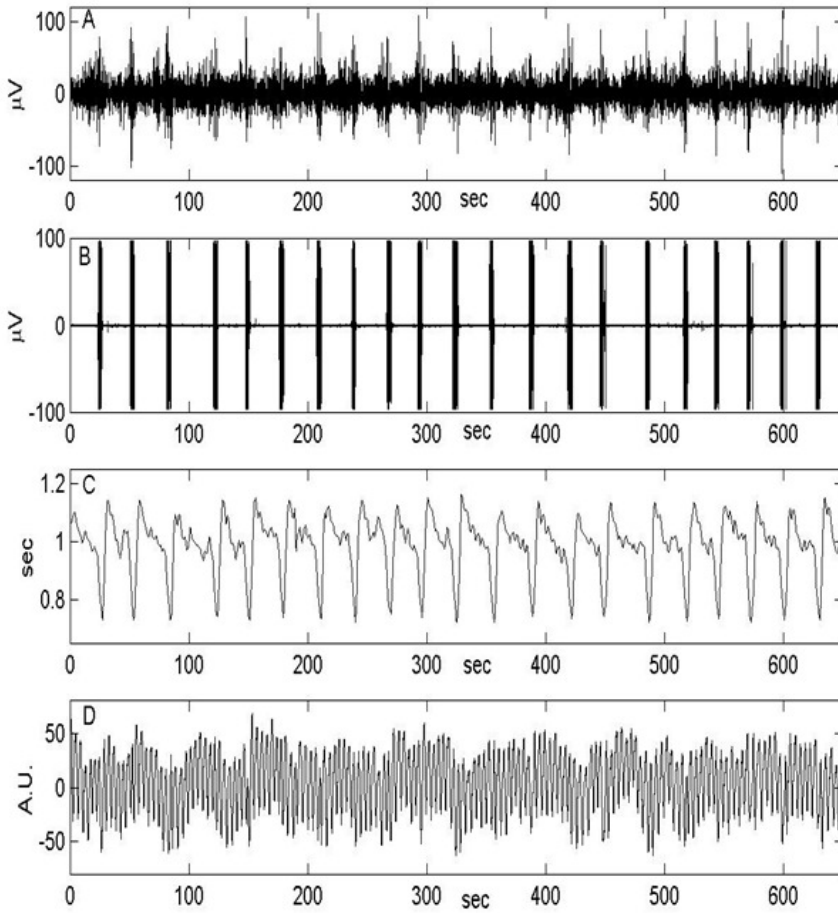


Figure 2.9 Top to bottom: EEG, EMG from the tibia, RR intervals, and respiration signal from a patient with myoclonus (involuntary twitching or jerking of a muscle) during sleep. Reproduced with permission from S. Cerutti, “Methods of biomedical signal processing: multiparametric and multidisciplinary integration toward a better comprehension of pathophysiological mechanisms,” pp 3–31, Chapter 1 in *Advanced Methods of Biomedical Signal Processing*, Edited by S. Cerutti and C. Marchesi, IEEE and Wiley, New York, NY, 2011. ©IEEE.

piration from a single-lead ECG signal. Several features were derived from the RR intervals of the ECG signal and the surrogate respiration signal, including the mean, *SD*, correlation coefficients, measures of variation, and spectral density. Features based only on RR intervals provided a minute-by-minute rate of over 85% in the identification of episodes of apnea. Bsoul et al. [149] developed a real-time sleep apnea monitor using single-lead ECG. Measures were extracted from an HRV signal formulated using RR intervals and from a surrogate respiration signal derived from one-minute segments of the ECG signal for the detection of sleep apnea. High sensitivity of up to 96% was obtained.

Madhav et al. [163] proposed methods to derive respiratory activity from signals such as the ECG and the photoplethysmogram using modeling techniques. Arunachalam and Brown [164] proposed a real-time algorithm to estimate and remove baseline wander and obtain a surrogate respiration signal from an ECG signal. The respiration signal was estimated from the amplitude modulation of R waves in the ECG caused by breathing. See Khandoker et al. [165] for related studies.

Mendez et al. [166] proposed a method for the detection of OSA based on the ECG recorded during sleep. Several parameters were derived from the ECG signal using time-variant modeling techniques. One of the features giving good results in the detection of OSA was the coherence between the RR interval data and the area under the QRS wave. This result was considered to suggest that respiration not only causes respiratory sinus arrhythmia, but also modifies the ECG waveshape. The coherence feature was considered to be a direct representation of the relation between the respiratory aspects of OSA and their synchronization with the heart rate.

Patangay et al. [167] proposed methods for the detection of apnea based on features of ECG as well as PCG signals. It was observed that the decrease in intrathoracic pressure during OSA causes a decrease in the left-ventricular pressure; this, in turn, results in increasingly stronger left-ventricular contraction. As a result, the PCG signal during OSA, in particular the S1 amplitude, has a crescendo-like change in amplitude. A composite feature vector was prepared using subband decomposition of S1 amplitudes and RR intervals. A sensitivity of 85.5% and a specificity of 92.2% were obtained in the detection of episodes of OSA.

Alshaer et al. [105, 106] developed a specially designed face frame with a microphone to record breath sounds. They proposed single-channel signal processing techniques to detect sleep-disordered breathing (CSA or OSA) via characterization of the envelope and spectral properties of the breath sounds. Using a diagnostic limit of $AHI \geq 10$ based on PSG, the overall accuracy of the breath-sound-based method was about 88%, with an overall correlation of AHI with PSG of 94%.

Notwithstanding the benefits of detecting apnea with a single signal, it should be noted that each such approach may present its own limitations and may not provide the desired degree of robustness or dependability in a clinical application. Similar to the acceptance of redundancy in the 12-lead system of ECG in clinical practice for the sake of the accompanying robustness, it may be desirable to use multiple signals from several systems that may demonstrate different effects or manifestations of apnea. Although substantial numbers of research studies are still being conducted on various issues related to OSA and CSA, as reviewed briefly in the present section,

systems and methods to detect and treat apnea are now commercially available [150, 168].

2.5 Remarks

This chapter has introduced the notion of using multiple channels of biomedical signals to obtain information on concurrent, coupled, and correlated phenomena with the aim of obtaining an improved understanding of physiological systems or obtaining reference signals for various purposes. The main point to note is that physiological systems are complex systems with multiple variables and outputs that should be studied from various approaches in order to gain multifaceted information. See Cerutti [160] and Baselli et al. [169] for discussions on several techniques and parametric models for the analysis of interactions between biomedical signals and systems.

Some of the problems described in the present chapter have been stated in fairly general terms due to the introductory nature of the chapter. The subsequent chapters present more illustrations of specific problems and applications of the notions gained from this chapter. A number of examples are provided in the chapters that follow to illustrate the use of multiple channels of signals to obtain clinically useful information.

2.6 Study Questions and Problems

1. A patient has ventricular bigeminy: Every second pulse from the SA node is replaced by a PVC with a full compensatory pause. (See Figure 9.9 for an illustration of bigeminy.) The firing rate of the SA node is regular at 80 beats a minute, and each ectopic beat precedes the blocked SA-node pulse by 100 *ms*.

Draw a schematic three-channel representation of the ECG, the atrial electrogram (or the SA node's firing pattern), and the firing pattern of the ectopic focus for 10 beats, marking the time scale in detail. Identify the correspondences and relationships between the activities in the three channels.

2. Draw schematic representations of the ECG, PCG, and carotid pulse signals. Label all waves in the three signals. Identify their common relationships to events in the cardiac cycle.

2.7 Laboratory Exercises and Projects

(Note: The following projects require access to a physiological signal recording laboratory.)

1. Using a multichannel biomedical signal acquisition system, obtain simultaneous recordings of an ECG channel and a signal related to respiration (temperature, airflow, or pressure in the nostril). Study the variations in the RR interval with inspiration and expiration. Repeat the experiment with the subject holding his/her breath during the signal acquisition period.

2. Obtain simultaneous recordings of an ECG lead, the PCG, the carotid pulse, and the pulse at the wrist. Study the temporal correspondences (and delays) between events in the various channels.
3. Record an ECG lead and PCG signals from two or three auscultation areas (mitral, aortic, pulmonary, tricuspid, and apex: see Figure 1.31) simultaneously. Study the variations in the intensities and characteristics of S1 and S2 and their components in the PCGs from the various recording sites.

CHAPTER 3

FILTERING FOR REMOVAL OF ARTIFACTS

Most biomedical signals appear as weak signals in an environment that is teeming with many other signals of various origins. Any signal other than that of interest could be termed as an interference, artifact, or simply *noise*. The sources of noise could be physiological, the instrumentation used, or the environment of the experiment.

This chapter starts with an introduction to the nature of the artifacts that are commonly encountered in biomedical signals. Several illustrations of signals corrupted by various types of artifacts are provided. Details of the design of filters, spanning a broad range of approaches, from linear time-domain and frequency-domain fixed filters to the optimal Wiener filter to adaptive filters, are then described. The chapter concludes with demonstrations of application of the filters described to ECG, ERP, and VAG signals.

(*Note:* A good background in signal and system analysis [1–4] as well as in probability, random variables, and stochastic processes [5–10] is required to follow the procedures and analysis described in this chapter. Familiarity with systems theory and transforms such as the Laplace transform, the Fourier transform in both the continuous and discrete form, and the z -transform is assumed.)

3.1 Problem Statement

Noise is omnipresent! The problems caused by artifacts in biomedical signals are vast in scope and variety; their potential for degrading the performance of the most sophisticated signal processing algorithms is high. The enormity of the problem of noise removal and its importance are reflected by the size of this chapter and its placement as the first one on signal processing techniques. Let us start with a generic statement of the problem and investigate its nature:

Analyze the various types of artifacts that corrupt biomedical signals and explore filtering techniques to remove them without degrading the signal of interest.

During a procedure to acquire the ECG signal, if the subject coughs or squirms, the EMG associated with such activity will create an interference or artifact. In adult patients, such physiological interference may be minimized by strict instructions and self-control; this solution may, however, not be applicable to infants, children, and severely ill patients. An intriguing example of physiological interference is that of the expectant mother's ECG appearing along with that of the fetus, with the latter being of interest. No external control is feasible or desirable in this case, and the investigator is forced to develop innovative solutions to extract the signal of interest.

Due to the weak levels of most biomedical signals at their source, high amplification factors of several hundred to several thousand may be required. Electronic noise in the instrumentation amplifiers also gets amplified along with the desired signal. While it is possible to reduce the thermal component of the noise by cooling the devices to low temperatures, this step may not be practical in most applications; the cost could also be prohibitive. Low-noise power supplies and specialized electronic amplifiers with high input impedance, high common-mode rejection ratio, and high power-supply rejection ratio are desirable for the acquisition of biomedical signals [11].

Our environment is filled with EM waves, both natural and man-made. EM waves broadcast by radio and television (TV) stations and those radiated by fluorescent lighting devices, computer monitors, and other systems used in the laboratory or work environment are picked up by cables, devices, and connectors. The 50 Hz or 60 Hz power-supply waveform is notorious for the many ways in which it can get mixed with and corrupt the signal of interest. Such interference may be termed as being due to the environment of the experiment. Simple EM shielding of cables and grounding of the chassis of equipment reduce EM and power-supply interference in most cases. Experiments dealing with weak signals such as ERPs and EEGs may require a wire-mesh-shielded (Faraday) cage to contain the subject and the instruments.

The ECG is a relatively strong signal with a readily identifiable waveform. Most types of interference that affect ECG signals may be removed by bandpass filters. Other signals of less recognizable waveforms and broader bandwidths may not be amenable to simple filtering procedures. In the case of signals such as ERPs or SEPs, the noise levels could be much higher than the signal levels, rendering the latter

unrecognizable in a single recording. It is important to gain a good understanding of the noise processes involved before one attempts to filter or preprocess a signal.

3.2 Random, Structured, and Physiological Noise

Several types of artifacts or noise could corrupt biomedical signals in various ways. The following sections provide descriptions of various types of noise and methods to characterize them.

3.2.1 Random noise

A *deterministic signal* is one whose value at a given instant of time may be computed using a closed-form mathematical function of time, or predicted from a knowledge of a few past values of the signal. A signal that does not meet this condition may be labeled as a *nondeterministic signal* or a random signal. The term *random noise* refers to an interference that arises from a random process, such as thermal noise in electronic devices.

Test for randomness: Random signals are generally expected to display more excursions about a certain reference level within a specified interval than signals that are predictable. Kendall [170] and Challis and Kitney [171] recommend a test for randomness based upon the number of peaks or troughs in the signal. A peak or a trough is defined by a set of three consecutive samples of the signal, with the central sample being either the maximum or minimum, respectively. As the direction of excursion of the signal changes at peaks and troughs, such points are collectively known as *turning points*. At a turning point, the sign of the first-order difference (derivative) at the current sample of the signal is not equal to that at the preceding sample. Given a signal of N samples, the signal may be labeled as being random if the number of turning points is greater than the threshold $\frac{2}{3}(N - 2)$ [170, 171]. In the case of a signal of varying characteristics, that is, a nonstationary signal, the test would have to be conducted using a running window of N samples. The width of the window should be chosen by taking into consideration the shortest duration over which the signal may remain in a given state. The method as above was used by Mintchev et al. [70] to study the dynamics of the level of randomness in EGG signals.

Figure 3.1 illustrates the variation in the number of turning points in a moving window of 50 *ms* (400 samples with the sampling frequency $f_s = 8 \text{ kHz}$) for the speech signal of the word “safety.” The limit on the number of turning points for randomness for $N = 400$ according to the threshold mentioned above is 265. It is seen from the figure that the test indicates that the signal is random for the fricatives /S/ (over the interval of 0.2 – 0.4 *s*, approximately) and /F/ (0.7 – 0.9 *s*), and not random for the remaining portions, as expected. (See also Section 1.2.12 and Figures 1.50 and 1.51.)

Statistical analysis of random processes: A random process is characterized by its probability density function (PDF) representing the probabilities of occurrence of

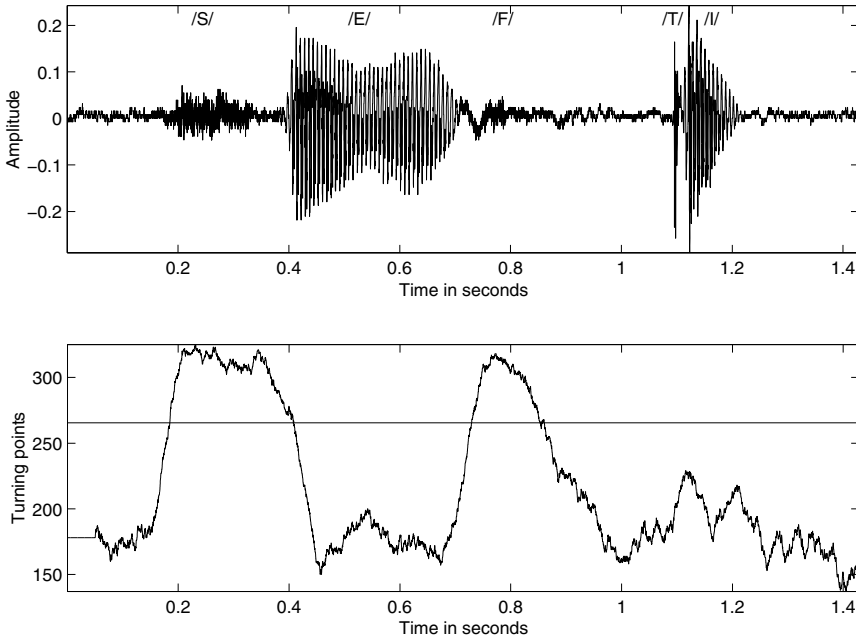


Figure 3.1 Top: Speech signal of the word “safety” uttered by a male speaker. Bottom: Count of turning points in a moving window of 50 ms (400 samples with $f_s = 8$ kHz). The threshold for randomness for $N = 400$ is 265.

all possible values of the related random variable. (See Papoulis [5] and Bendat and Piersol [6] for background material on probability, random variables, and stochastic processes.) Consider a random process η that is characterized by the PDF $p_\eta(\eta)$. The mean μ_η of the random process η is given by the first-order moment of the PDF, defined as

$$\mu_\eta = E[\eta] = \int_{-\infty}^{\infty} \eta p_\eta(\eta) d\eta, \quad (3.1)$$

where $E[\]$ represents the *statistical expectation operator*. It is common to assume the mean of a random noise process to be zero.

The mean-squared (*MS*) value of the random process η is given by the second-order moment of the PDF, defined as

$$E[\eta^2] = \int_{-\infty}^{\infty} \eta^2 p_\eta(\eta) d\eta. \quad (3.2)$$

The variance σ_η^2 of the process is defined as the second central moment:

$$\sigma_\eta^2 = E[(\eta - \mu_\eta)^2] = \int_{-\infty}^{\infty} (\eta - \mu_\eta)^2 p_\eta(\eta) d\eta. \quad (3.3)$$

The square root of the variance gives the *SD* σ_η of the process. Note that $\sigma_\eta^2 = E[\eta^2] - \mu_\eta^2$. If the mean is zero, it follows that $\sigma_\eta^2 = E[\eta^2]$, that is, the variance and the *MS* values are the same.

The coefficient of variation (*CV*) of a process is defined as the ratio σ/μ . This dimensionless measure places the variability of a process in the context of its mean and is useful in the analysis of the variability of processes with widely different means; however, as $\mu \rightarrow 0$, $CV \rightarrow \infty$. *CV* is applicable in the analysis of nonnegative quantities.

A few other measures used to characterize random processes are skewness, kurtosis, and entropy. Skewness is defined as a normalized version of the third central moment, given by

$$S_\eta = \frac{1}{\sigma_\eta^3} \int_{-\infty}^{\infty} (\eta - \mu_\eta)^3 p_\eta(\eta) d\eta. \quad (3.4)$$

Skewness characterizes the lack of symmetry of a PDF. Symmetric PDFs, such as the Gaussian, have a skewness value of zero. A process with higher probabilities of occurrence of larger values of the associated variable has a PDF with a negative skewness. On the other hand, a process with higher probabilities of occurrence of smaller values of the associated variable has a PDF with a positive skewness.

Kurtosis is defined as a normalized version of the fourth central moment, given by

$$K_\eta = \frac{1}{\sigma_\eta^4} \int_{-\infty}^{\infty} (\eta - \mu_\eta)^4 p_\eta(\eta) d\eta. \quad (3.5)$$

Kurtosis characterizes the presence of a long tail in the PDF. The Gaussian PDF has a kurtosis value of 3. The value $K' = K - 3$ (referred to as kurtosis excess) is used to represent the difference in the kurtosis of a PDF with respect to that of a Gaussian PDF. A positive K' indicates a PDF with a strong peak near the mean that declines with a heavy tail. A PDF that is nearly flat has a negative K' value.

The entropy of a random process is a statistical measure of the information conveyed by the process [18, 172, 173]. It is also a measure of the extent of disorder in the process. The most commonly used measure of entropy of a PDF is defined as

$$H_\eta = - \int_{-\infty}^{\infty} p_\eta(\eta) \log_2[p_\eta(\eta)] d\eta. \quad (3.6)$$

The unit of entropy, defined as above, is *bits* (*b*). The entropy of a process is at its maximum when all associated values or events occur with equal probability, that is, the PDF is uniform.

The definitions of the moments of a random process given above refer to a continuous variable and the situation when its PDF is known. In a practical situation, when we have only a certain number, N , of the values of the process observed as $\eta(n)$, $n = 0, 1, 2, \dots, N-1$, and no knowledge of its PDF, we could estimate the statistical measures from the given samples as

$$\mu_\eta = \frac{1}{N} \sum_{n=0}^{N-1} \eta(n), \quad (3.7)$$

$$MS_{\eta} = \frac{1}{N} \sum_{n=0}^{N-1} [\eta(n)]^2, \quad (3.8)$$

$$RMS_{\eta} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} [\eta(n)]^2}, \quad (3.9)$$

and

$$\sigma_{\eta} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} [\eta(n) - \mu_{\eta}]^2}. \quad (3.10)$$

[Some authors use the notation (nT) , $T = 1/f_s$ being the sampling interval, where f_s is the sampling frequency, to denote the index of a sampled signal; in this book we shall use just (n) , the sample number.]

In the case of a process with a finite number L of discrete (or quantized) values η_l , $l = 0, 1, 2, \dots, L-1$, the entropy is defined as

$$H_{\eta} = - \sum_{l=0}^{L-1} p_{\eta}(\eta_l) \log_2[p_{\eta}(\eta_l)], \quad (3.11)$$

where $p_{\eta}(\eta_l)$ is the probability of occurrence of the l^{th} quantized value of η . Each probability value in the equation given above may be estimated from a large number of observations of the values of the process. This form of entropy is known as Shannon entropy.

When the values of a random process η form a time series or a function of time, we have a random signal (or a stochastic process) $\eta(t)$. The statistical measures described above then have physical meanings: The mean represents the DC component, the MS value represents the average power, and the RMS value gives the average noise magnitude or level. The measures are useful in calculating the SNR , which is commonly defined as the ratio of the peak-to-peak amplitude range of the signal to the RMS value of the noise, or as the ratio of the average power of the signal to that of the noise.

Observe the use of the same symbol η to represent the random variable, the random process, and the random signal as a function of time. The subscript of the PDF or the statistical parameter derived indicates the random process of concern. The context of the discussion or expression should make the meaning of the symbol clear.

A biomedical signal of interest $x(t)$ may also, for the sake of generality, be considered to be a realization of a random process x . For example, although a normal heart sound signal is heard as the same comforting *lub-dub* sound over every cycle, the corresponding PCG vibration waveforms are not precisely the same from one cycle to another. The PCG signal may be represented as a random process exhibiting certain characteristics *on the average*.

When a (random) signal $x(t)$ is observed in an environment with random noise, the measured signal $y(t)$ may be treated as a realization of another random process

y . In most cases, the noise is additive, and the observed signal is expressed as

$$y(t) = x(t) + \eta(t). \quad (3.12)$$

Each of the random processes x and y is characterized by its own PDF $p_x(x)$ and $p_y(y)$, respectively.

In most practical applications, the random processes representing a signal of interest and the noise affecting the signal may be assumed to be *statistically independent processes*. Two random processes x and η are said to be statistically independent if their joint PDF $p_{x,\eta}(x, \eta)$ is equal to the product of their individual PDFs given as $p_x(x) p_\eta(\eta)$.

The first-order moment and second-order central moment of the signals $x(t)$ and $y(t)$ in Equation 3.12 are related as

$$E[y] = \mu_y = \mu_x + \mu_\eta; \quad (3.13)$$

if $\mu_\eta = 0$, then $\mu_y = \mu_x$; and

$$E[(y - \mu_y)^2] = \sigma_y^2 = \sigma_x^2 + \sigma_\eta^2, \quad (3.14)$$

where μ represents the mean and σ^2 represents the variance of the random process indicated by the subscript.

Ensemble averages: When the PDFs of the random processes of concern are not known, it is common to approximate the statistical expectation operation by averages computed using a collection or *ensemble* of sample observations of the random process. Such averages are known as *ensemble averages*. Suppose we have M observations of the random process x as functions of time: $x_1(t), x_2(t), \dots, x_M(t)$. We may estimate the mean of the process at a particular instant of time t_1 as

$$\mu_x(t_1) = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k=1}^M x_k(t_1). \quad (3.15)$$

Figure 3.2 illustrates ten sample acquisitions of flash visual ERPs (see also Figure 3.41). The vertical lines at $t = t_1$ and $t = t_2 = t_1 + \tau$ represent the ensemble averaging process at two different instants of time.

The autocorrelation function (ACF) $\phi_{xx}(t_1, t_1 + \tau)$ of a random process x that is a time series is given by

$$\begin{aligned} \phi_{xx}(t_1, t_1 + \tau) &= E[x(t_1) x(t_1 + \tau)] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(t_1) x(t_1 + \tau) p_{x_1, x_2}(x_1, x_2) dx_1 dx_2, \end{aligned} \quad (3.16)$$

where x_1 and x_2 represent the random variables corresponding to the processes $x(t_1)$ and $x(t_1 + \tau)$, respectively, and $p_{x_1, x_2}(x_1, x_2)$ is the joint PDF of the two processes. The ACF may be estimated as

$$\phi_{xx}(t_1, t_1 + \tau) = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k=1}^M x_k(t_1) x_k(t_1 + \tau), \quad (3.17)$$

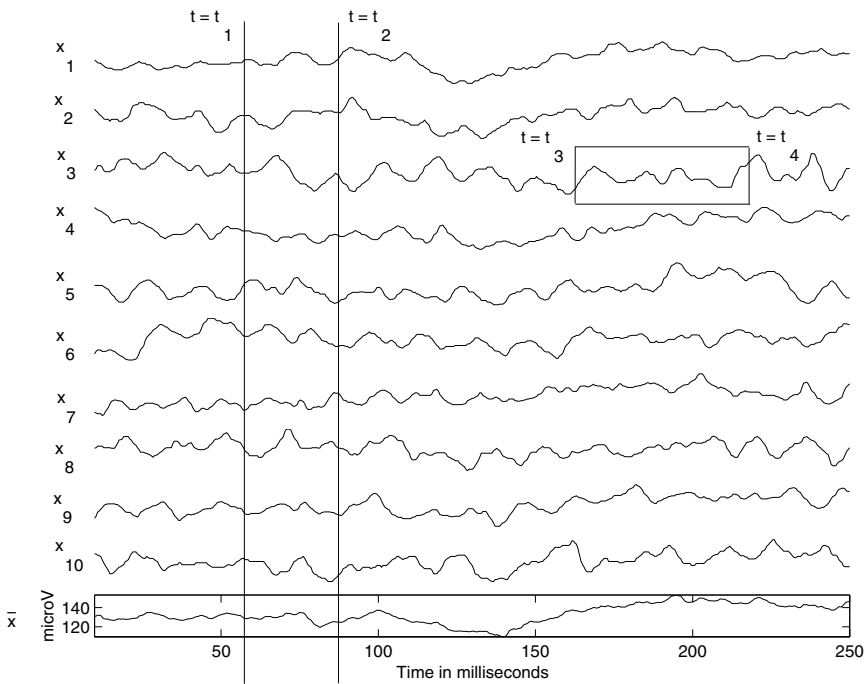


Figure 3.2 Ten sample acquisitions (x_1 to x_{10}) of individual flash visual ERPs from the occipital midline (oz) position of a normal adult male (the author of this book!). The earlobes were used to form the reference lead (a1a2), and the left forehead was used as the reference (see Figure 1.37). The signals may be treated as ten realizations of a random process in the form of time series or signals. The vertical lines at $t = t_1$ and $t = t_2 = t_1 + \tau$ represent the ensemble averaging process at two different instants of time. The last plot (framed) gives the ensemble average or prototype $\bar{x}(t)$ of the ten individual signals. The horizontal box superimposed on the third trace represents the process of computing temporal statistics over the duration $t = t_3$ to $t = t_4$ of the sample ERP $x_3(t)$. See also Figure 3.41. Data courtesy of L. Alfaro and H. Darwish, Alberta Children's Hospital, Calgary.

where τ is the delay parameter. If the signals are complex, one of the functions in the expression above should be conjugated; in this book, we shall deal with physiological signals that are always real. The two vertical lines at $t = t_1$ and $t = t_2 = t_1 + \tau$ in Figure 3.2 represent the ensemble averaging process to compute $\phi_{xx}(t_1, t_2)$. The ACF indicates how the values of a signal at a particular instant of time are statistically related to (or have characteristics in common with) the values of the same signal at another instant of time.

When dealing with random processes that are observed as functions of time (or stochastic processes), it becomes possible to compute ensemble averages at every

point of time. Then, we obtain an averaged function of time $\bar{x}(t)$ as

$$\bar{x}(t) = \mu_x(t) = \frac{1}{M} \sum_{k=1}^M x_k(t) \quad (3.18)$$

for all time t . The signal $\bar{x}(t)$ may be used to represent the random process x as a prototype; see the last trace (framed) in Figure 3.2.

Time averages: When we have a sample observation of a random process $x_k(t)$ as a function of time, it is possible to compute *time averages* or *temporal statistics* by integrating along the time axis, such as

$$\mu_x(k) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x_k(t) dt. \quad (3.19)$$

The integral would be replaced by a summation in the case of sampled or discrete-time signals. The time-averaged ACF $\phi_{xx}(\tau, k)$ is given by

$$\phi_{xx}(\tau, k) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x_k(t) x_k(t + \tau) dt. \quad (3.20)$$

(See Section 6.3 for details on estimation of the ACF of finite-length data sequences.) The horizontal box superimposed on the third trace in Figure 3.2 represents the process of computing temporal statistics over the duration $t = t_3$ to $t = t_4$ of the sample ERP $x_3(t)$ selected from the ensemble of ERPs illustrated in the figure.

Random noise may thus be characterized in terms of ensemble and/or temporal statistics. The mean does not play an important role: it is usually assumed to be zero, or may be subtracted out if it is not zero. The ACF plays an important role in the characterization of random processes. The Fourier transform of the ACF is the power spectral density (PSD) function, which is useful in spectral analysis and filter design.

Covariance and cross-correlation: When two random processes x and y need to be compared, we could compute the covariance between them as

$$C_{xy} = E[(x - \mu_x)(y - \mu_y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y) p_{x,y}(x, y) dx dy, \quad (3.21)$$

where $p_{x,y}(x, y)$ is the joint PDF of the two processes. The covariance parameter may be normalized to get the correlation coefficient, defined as

$$\rho_{xy} = \frac{C_{xy}}{\sigma_x \sigma_y}, \quad (3.22)$$

with $-1 \leq \rho_{xy} \leq +1$. A high covariance indicates that the two processes have similar statistical variability or behavior. The processes x and y are said to be uncorrelated if $\rho_{xy} = 0$. Two processes that are statistically independent are also uncorrelated; the converse of this property is, in general, not true.

When dealing with random processes x and y that are functions of time, the cross-correlation function (CCF) between them is defined as

$$\theta_{xy}(t_1, t_1 + \tau) = E[x(t_1)y(t_1 + \tau)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(t_1) y(t_1 + \tau) p_{x,y}(x, y) dx dy. \quad (3.23)$$

Correlation functions are useful in analyzing the nature of variability and spectral bandwidth of signals, as well as for detection of events by template matching.

3.2.2 Structured noise

Power-line interference at 50 Hz or 60 Hz is an example of structured noise: The typical waveform of the interference is known in advance. It should, however, be noted that the phase of the interfering waveform will not usually be known. Furthermore, the interfering waveform may not be an exact sinusoid; this is indicated by the presence of harmonics of the fundamental 50 Hz or 60 Hz component. Analysis of the power spectrum of a given noisy signal can reveal the presence of structured or periodic noise in the form of peaks or spikes at the fundamental frequency and its harmonics.

3.2.3 Physiological interference

The human body is a complex conglomeration of several systems and processes. Several physiological processes could be active at a given instant of time, each one producing many signals of different types. A patient or experimental subject may not be able to exercise control on all physiological processes and systems. The appearance of signals from systems or processes other than those of interest may be termed as physiological interference; several examples are listed below.

- EMG related to coughing, breathing, or squirming affecting the ECG
- EGG interfering with precordial ECG
- Maternal ECG getting added to the fetal ECG of interest
- ECG interfering with the EEG
- Ongoing EEG in ERPs and SEPs
- Breath, lung, or bowel sounds contaminating the heart sounds (PCG)
- Heart sounds getting mixed with breath or lung sounds
- Muscle sound (VMG) interference in joint sounds (VAG)
- Needle-insertion activity appearing at the beginning of a needle-EMG recording

Physiological interference may not be characterized by any specific waveform or spectral content, and is typically dynamic and nonstationary (varying with the level

of the activity of relevance and hence with time; see the next section for a discussion on stationarity). Thus, simple linear bandpass filters will usually not be effective in removing physiological interference.

3.2.4 Stationary, nonstationary, and cyclostationary processes

We saw in Section 3.2.1 that random processes may be characterized in terms of their ensemble and/or temporal statistics. A random process is said to be *stationary in the strict sense* or *strongly stationary* if its statistics are not affected by a shift in the origin of time. In practice, only first-order and second-order averages are used. A random process is said to be *weakly stationary* or *stationary in the wide sense* if its mean is a constant and its ACF depends only upon the difference (or shift) in time. Then, from Equations 3.15 and 3.17, we have $\mu_x(t_1) = \mu_x$ and $\phi_{xx}(t_1, t_1 + \tau) = \phi_{xx}(\tau)$. The ACF is now a function of the delay or shift parameter τ only; the PSD of the process does not vary with time.

A stationary process is said to be *ergodic* if the temporal statistics computed are independent of the sample observed; that is, the same result is obtained for any sample observation $x_k(t)$. The time averages in Equations 3.19 and 3.20 are then independent of k : $\mu_x(k) = \mu_x$ and $\phi_{xx}(\tau, k) = \phi_{xx}(\tau)$. All ensemble statistics may be replaced by temporal statistics when analyzing ergodic processes. Ergodic processes are an important type of stationary random processes because their statistics may be computed from a single observation as a function of time. The use of ensemble and temporal averages for noise filtering is illustrated in Sections 3.5.1 and 3.5.2, respectively.

Signals or processes that do not meet the conditions described above may be, in general, called *nonstationary processes*. A nonstationary process possesses statistics that vary with time. It is readily seen in Figure 1.28 (see also Figure 3.6) that the mean level (baseline) of the signal is varying over the duration of the signal. Therefore, the signal is nonstationary in the mean, a first-order statistical measure. Figure 3.3 illustrates the variance of the speech signal of the word “safety” computed in a moving window of 50 ms (400 samples with $f_s = 8$ kHz). Because the variance changes substantially from one portion of the signal to another, it should be concluded that the signal is nonstationary in its second-order statistics (variance, *SD*, or *RMS*). While the speech signal is stationary in the mean, this is not an important characteristic as the mean is typically removed from speech signals. (A DC signal bears no information related to vibration or sound.)

Note that variance displays a behavior that is almost the opposite of that of turning points in Figure 3.1. Variance is sensitive to changes in amplitude, with large swings about the mean leading to large variance values. The procedure to detect turning points examines the presence of peaks and troughs with no consideration of their relative amplitudes; the low-amplitude ranges of the fricatives in the signal have resulted in low variance values, even though their counts of turning points are high.

Most biomedical systems are dynamic and produce nonstationary signals (for example, EMG, EEG, VMG, PCG, VAG, and speech signals). However, a physical or physiological system has limitations in the rate at which it can change its character-

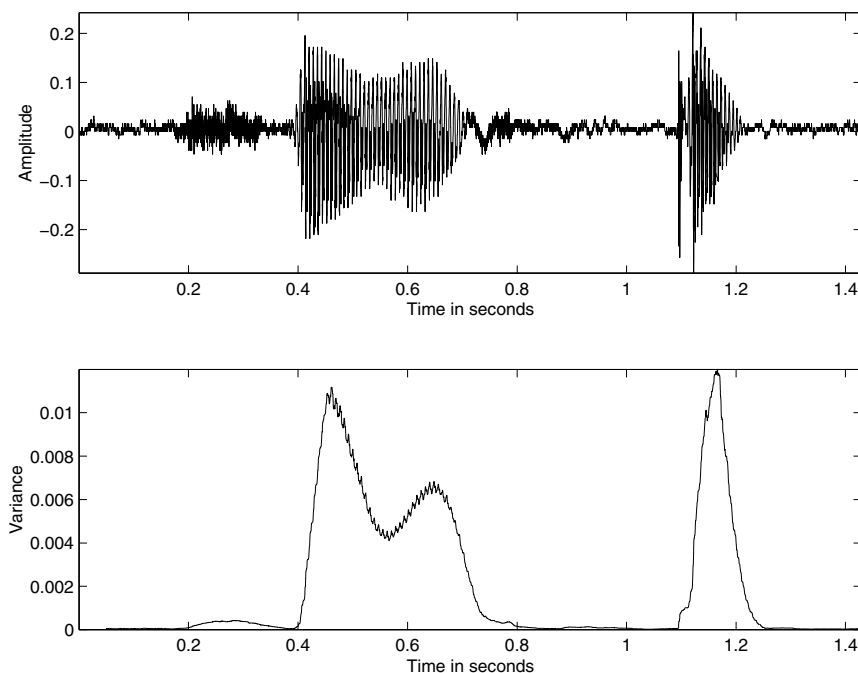


Figure 3.3 Top: Speech signal of the word “safety” uttered by a male speaker. Bottom: Variance computed in a moving window of 50 ms (400 samples with $f_s = 8\text{ kHz}$).

istics. This limitation facilitates breaking a signal into segments of short duration (typically a few tens of milliseconds), over which the statistics of interest are not varying, or may be assumed to remain the same. The signal is then referred to as a *quasistationary process*; the approach is known as *short-time analysis*. Figure 3.4 illustrates the spectrogram of the speech signal of the word “safety.” The spectrogram was computed by computing an array of magnitude spectra of segments of the signal of duration 64 ms ; an overlap of 32 ms was permitted between successive segments. It is evident that the spectral characteristics of the signal vary over its duration: The fricatives demonstrate more high-frequency content than the vowels and also lack formant (resonance) structure. The signal is, therefore, nonstationary in terms of its PSD; because the PSD is related to the ACF, the signal is also nonstationary in the second-order statistical measure of the ACF. However, it can be observed that the PSD shows the same or similar spectral content over the duration of each of the phonemes present in the signal. Thus, over such intervals, the signal may be considered to be quasistationary.

Further discussion and examples of techniques of this nature are presented in Sections 8.4.1 and 8.5. Adaptive signal processing techniques may also be designed to detect changes in certain statistical measures of an observed signal; the signal may then be broken into quasistationary segments of variable duration that meet the

specified conditions of stationarity. Methods for analysis of nonstationary signals are discussed in Chapter 8. Adaptive segmentation of the EEG, VAG, and PCG signals is discussed in Sections 8.5, 8.6, 8.9, and 8.10.

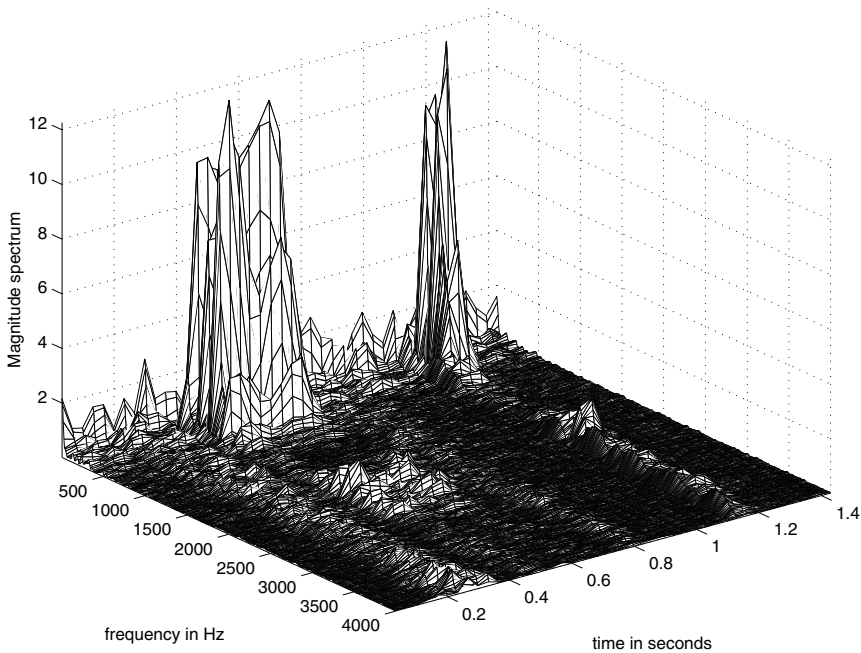


Figure 3.4 Spectrogram of the speech signal of the word “safety” uttered by a male speaker. (The signal is also illustrated in Figures 1.50, 3.1, and 3.3.) Each function or plot along a line parallel to the frequency axis represents the magnitude spectrum of the signal in a moving window of duration 64 ms (512 samples with $f_s = 8$ kHz), with the window advance interval being 32 ms. The spectrogram is plotted on a linear scale to display better the major differences between the voiced and unvoiced sounds.

Certain systems, such as the cardiac system, normally perform rhythmic operations. The resulting signal, such as the ECG, PCG, or carotid pulse, is then almost periodic and may be referred to as a *cyclostationary signal*. The statistics of the PCG signal vary within the duration of a cardiac cycle, especially when murmurs are present, but repeat themselves at regular intervals. The cyclic repetition of the process facilitates ensemble averaging, using epochs or events extracted from an observation of the signal over many cycles (which is, strictly speaking, a single function of time). Exploitation of the cyclic nature of the ECG signal for synchronized averaging to reduce noise is illustrated in Section 3.5.1. Application of the same concept to estimate the envelopes of PCG signals is described in Section 5.5.2. Further extensions of the approach to extract A2 from S2 in PCG signals are demonstrated in Section 4.11. Procedures to estimate the PSDs of PCG segments in systole and diastole are presented in Section 6.3.5.

3.3 Illustration of the Problem with Case Studies

The following case studies present several examples of various types of interference in biomedical signals of different origins. The aim of this section is to gain familiarity with the various possibilities of interference and their general characteristics. Filtering techniques to remove various types of interference are described in sections to follow.

3.3.1 Noise in event related potentials

An ERP is a signal obtained in response to a stimulus. The response is usually of small amplitude (of the order of $10 \mu V$), and it is submerged in ambient EEG activity and noise that could be larger than the ERP. The waveform of a single response may be barely recognizable against the background activity. Figure 3.2 shows ten individual flash visual ERP signals. The signals were recorded at the occipital mid-line position, with the left and right earlobes combined to form the reference lead. The left forehead was used as the reference. The ERP signals are buried in ongoing EEG and power-line (60 Hz) interference, and they cannot be analyzed using the individual acquisitions shown in the figure.

3.3.2 High frequency noise in the ECG

Figure 3.5 shows a segment of an ECG signal with high-frequency noise. The noise could be due to the instrumentation amplifiers, the recording system, pickup of ambient EM signals by the cables, and other sources. The signal illustrated has also been corrupted by power-line interference at 60 Hz and its harmonics, which may also be considered as a part of high-frequency noise relative to the low-frequency nature of the ECG signal.

3.3.3 Motion artifact in the ECG

Low-frequency artifacts and baseline drift may be caused in chest-lead ECG signals by coughing or breathing with large movement of the chest, or when an arm or leg is moved in the case of limb-lead ECG acquisition. The ECG is a common source of artifact in chest-lead ECG. Poor contact and polarization of the electrodes may also cause low-frequency artifacts. Baseline drift may sometimes be caused by variations in temperature and bias in the instrumentation and amplifiers as well. Figure 3.6 shows an ECG signal with low-frequency artifact. Baseline drift makes analysis of isoelectricity of the ST segment difficult. A large baseline drift may cause the positive or negative peaks in the ECG to be clipped by the amplifiers or the ADC.

3.3.4 Power line interference in ECG signals

The most commonly encountered periodic artifact in biomedical signals is the power-line interference at 50 Hz or 60 Hz . If the power-line waveform is not a pure sinu-

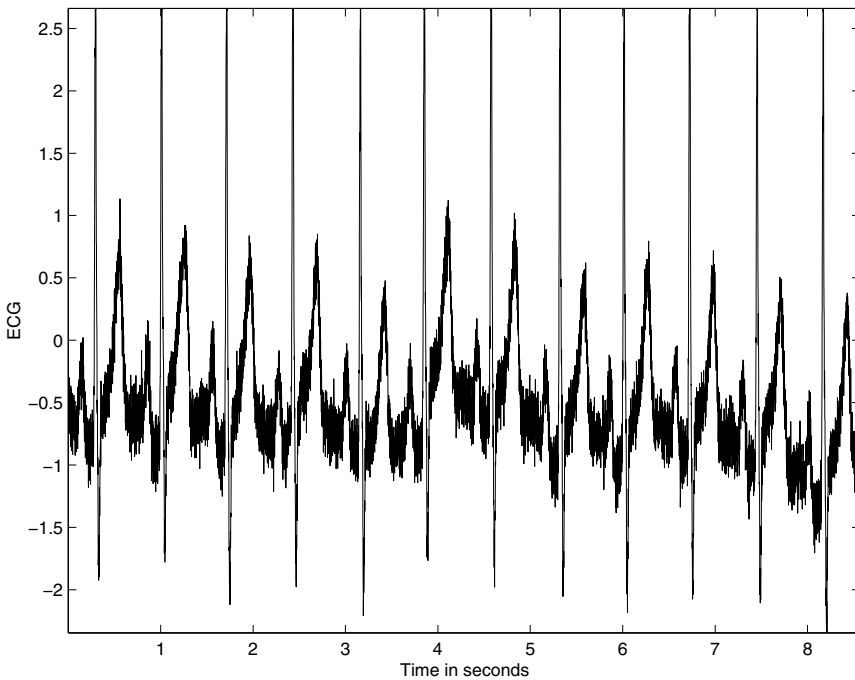


Figure 3.5 ECG signal with high-frequency noise.

soid due to distortions, harmonics of the fundamental frequency could also appear. Harmonics will also appear if the interference is a periodic waveform that is not a sinusoid (such as rectangular pulses).

Power-line interference may be difficult to detect visually in signals having non-specific waveforms such as the PCG or EMG; however, the interference is easily visible if present on well-defined signal waveforms such as the ECG or carotid pulse signals. In either case, the power spectrum of the signal should provide a clear indication of the presence of power-line interference as an impulse or spike at 50 Hz or 60 Hz ; harmonics, if present, will appear as additional spikes at integral multiples of the fundamental frequency.

Figure 3.7 shows a segment of an ECG signal with 60 Hz interference. Observe the regular or periodic structure of the interference, which rides on top of the ECG waves. Figure 3.8 shows the power spectrum of the signal. The periodic interference is clearly displayed not only as a spike at its fundamental frequency of 60 Hz , but also as spikes at 180 Hz and 300 Hz , which represent the third and fifth harmonics, respectively. (The recommended sampling rate for ECG signals is 500 Hz ; the higher rate of $1,000\text{ Hz}$ was used in this case because the ECG was recorded as a reference signal with the PCG. The larger bandwidth also permits better illustration of artifacts and filtering.)

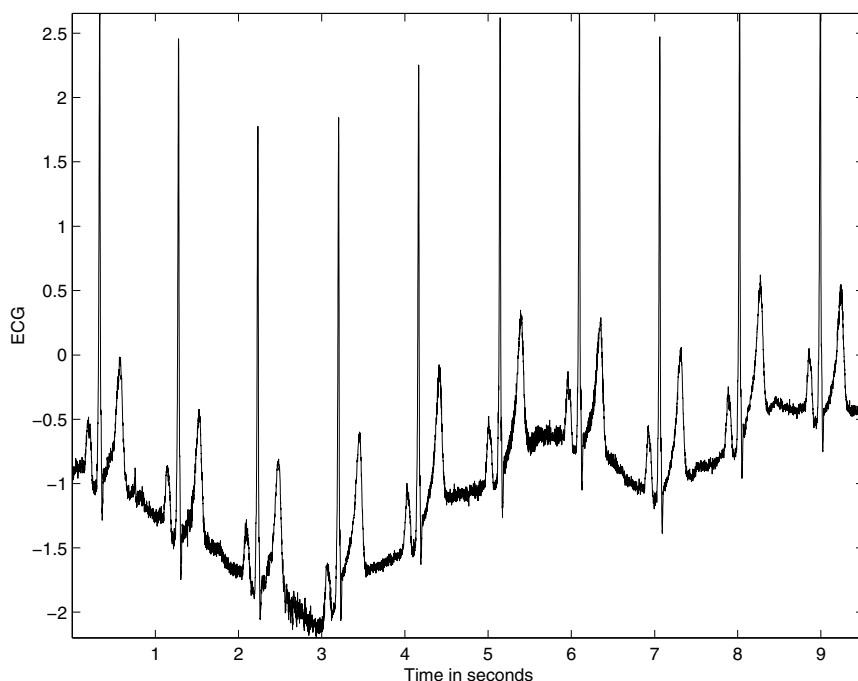


Figure 3.6 ECG signal with low-frequency artifact.

The bandwidth of interest of the ECG signal, which is usually in the range $0.05 - 100\text{ Hz}$, includes the 60 Hz component; hence simple lowpass filtering will not be appropriate for removal of power-line interference. Lowpass filtering of the ECG to a bandwidth lower than 60 Hz could smooth and blur the QRS complex as well as affect the PQ and ST segments. The ideal solution would be to remove the 60 Hz component without sacrificing any other component.

3.3.5 Maternal interference in fetal ECG

Figure 3.9 shows an ECG signal recorded from the abdomen of a pregnant woman. Shown also is a simultaneously recorded ECG from the woman's chest. Comparing the two, we see that the abdominal ECG demonstrates multiple peaks (QRS complexes) corresponding to the maternal ECG (occurring at the same time instants as the QRS complexes in the chest lead) as well as several others at weaker levels and a higher repetition rate. The QRS complexes that are not of the expectant mother represent the ECG of the fetus. Observe that the QRS complex shapes of the maternal ECG from the chest and abdominal leads have different shapes due to the projection of the cardiac electrical vector on to different axes. Given that the two signals being combined have almost the same bandwidth, how would we be able to separate them and obtain the fetal ECG of interest?

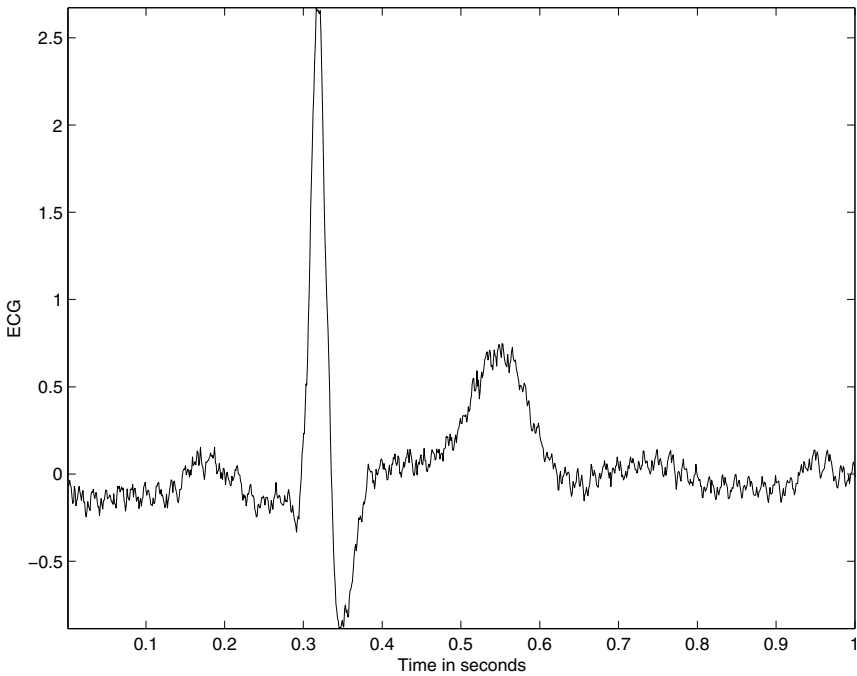


Figure 3.7 ECG signal with power-line (60 Hz) interference.

3.3.6 Muscle contraction interference in VAG signals

Figure 3.10 shows the recording setup used by Zhang et al. [125] to study the possibility of VMG signals appearing as muscle-contraction interference in VAG signals. The left-hand column in Figure 3.11 shows VMG signals recorded using accelerometers placed at the distal rectus femoris (thigh), midpatella (knee cap), tibial tuberosity, and midtibial shaft positions of a subject during isometric contraction of the rectus femoris muscle (with no leg or knee movement). The right-hand column of the figure shows vibration signals recorded at the same positions using the same accelerometers, but during isotonic contraction (swinging movement of the leg). The top signal (a) in the right-hand column indicates the VMG signal generated at the rectus femoris during acquisition of the VAG signals; parts (b)–(d) of the right-hand column show the VAG signals.

VAG signals are difficult to analyze because they have no predefined or recognizable waveforms; it is even more difficult to identify any noise or interference that may be present in VAG signals. The signals shown in Figure 3.11 indicate that a transformed version of the VMG could get added to the VAG, especially during extension of the leg when the rectus femoris muscle is active [the second halves of the VAG signals in parts (b)–(d) of the right-hand column]. The left-hand column of VMG signals in Figure 3.11 illustrates that the VMG generated at the distal rec-

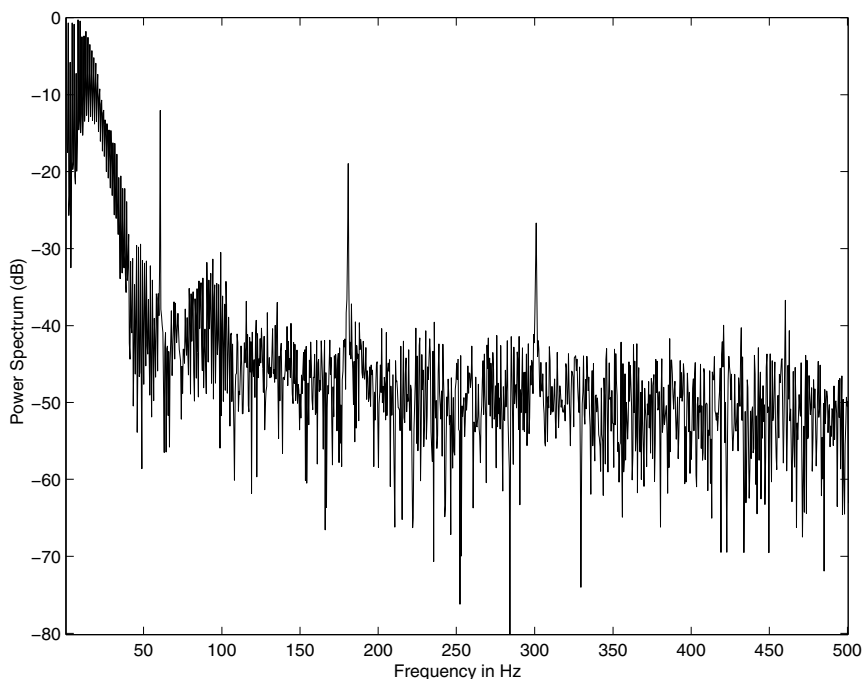


Figure 3.8 Power spectrum of the ECG signal in Figure 3.7 with power-line interference. The spectrum illustrates peaks at the fundamental frequency of 60 Hz as well as the third and fifth harmonics at 180 Hz and 300 Hz , respectively.

tus femoris gets transmitted well down the leg and appears at the other recording positions. It may be observed from the VAG signals in the right-hand column that vibration signals comparable to the VMG are present in the VAG channels (b)–(d) during extension (second halves) but are not as prominent in flexion (first halves). Interestingly, the knee-joint “crepitus” and click signals that appear in the first half of the VAG signal at the midpatella position [right (b)] have been transmitted downward along the leg to the tibial tuberosity [right (c)] and midtibial shaft [right (d)] positions farther down the leg, presumably along the tibia, but not upwards to the distal rectus femoris position [right (a)].

It should also be noted that the VAG signal cannot be expected to be the same during the extension and flexion parts of a swing cycle: Extension causes more stress or force per unit area on the patellofemoral joint than flexion. Furthermore, the VAG and VMG signals are nonstationary: Characteristics of the VAG vary with the quality of the cartilage surfaces that come into contact at different joint angles, while the VMG varies in accordance with the level of contraction of the muscles involved. To make the problem even more difficult, the bandwidths of the two signals overlap in the range of about 0 – 100 Hz . These factors make removal of the VMG or muscle-contraction interference from VAG signals a challenge.

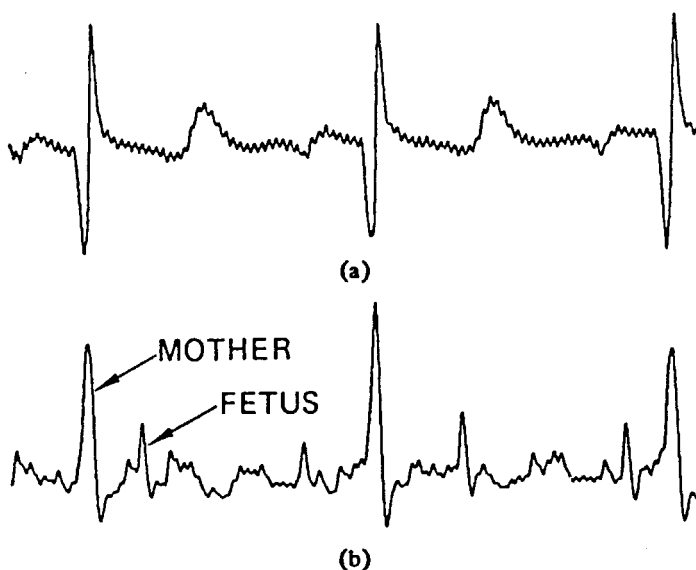


Figure 3.9 ECG signals of a pregnant woman from abdominal and chest leads: (a) chest-lead ECG and (b) abdominal-lead ECG; the former presents the maternal ECG whereas the latter is a combination of the maternal and fetal ECG signals. (See also Figure 3.101.) Reproduced with permission from B. Widrow, J.R. Glover, Jr., J.M. McCool, J. Kaunitz, C.S. Williams, R.H. Hearn, J.R. Zeidler, E. Dong, Jr., and R.C. Goodlin, Adaptive noise cancelling: Principles and applications, *Proceedings of the IEEE*, 63(12):1692–1716, 1975. ©IEEE.

3.3.7 Potential solutions to the problem

Now that we have gained an understanding of a few sources of artifacts in biomedical signals and their nature, we are prepared to look at specific problems and develop effective filtering techniques to solve them. The following sections present studies of artifacts of various types and demonstrate increasingly complex signal processing techniques to remove them. The problem statement at the beginning of each section defines the nature of the problem in as general terms as possible, sets the terms and conditions, and defines the scope of the investigation to follow. The solution proposed provides the details of an appropriate filtering technique. Each solution is demonstrated with an illustration of its application. Further examples of application of the techniques studied are provided at the end of the chapter. Comparative evaluation of filtering techniques is also provided where applicable. Examples of both success and failure of filtering methods are presented to facilitate learning.

A practical problem encountered by an investigator in the field may not precisely match a specific problem considered in this chapter. However, it is expected that the knowledge of several techniques and an appreciation of the results of their ap-

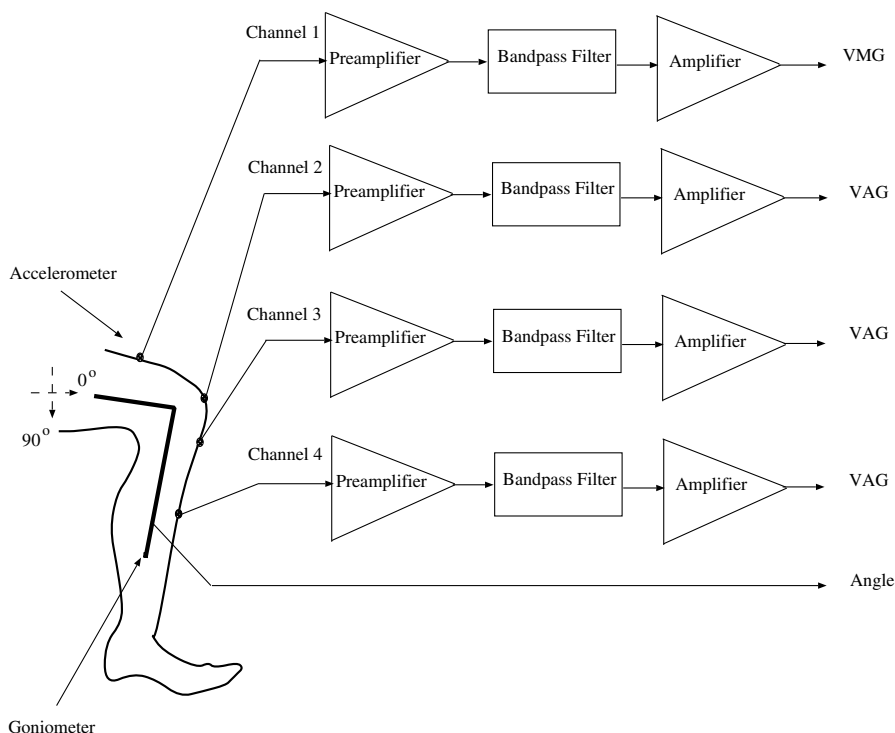


Figure 3.10 Experimental setup to measure VMG and VAG signals at various positions along the leg [125].

plication gained from this chapter will help in designing innovative and appropriate solutions to new problems.

3.4 Fundamental Concepts of Filtering

A filter is a signal processing system, algorithm, or method, realized in hardware or software, that is used to modify a given signal in a particular manner. Quite often, a filtering operation is performed on a signal to remove undesired components that are referred to as noise or artifacts. Regardless of the nature of the operations performed or their effects, filters may be categorized as

- linear or nonlinear,
- stationary or nonstationary,
- fixed (time-invariant) or adaptive (time-variant),
- active or passive, or

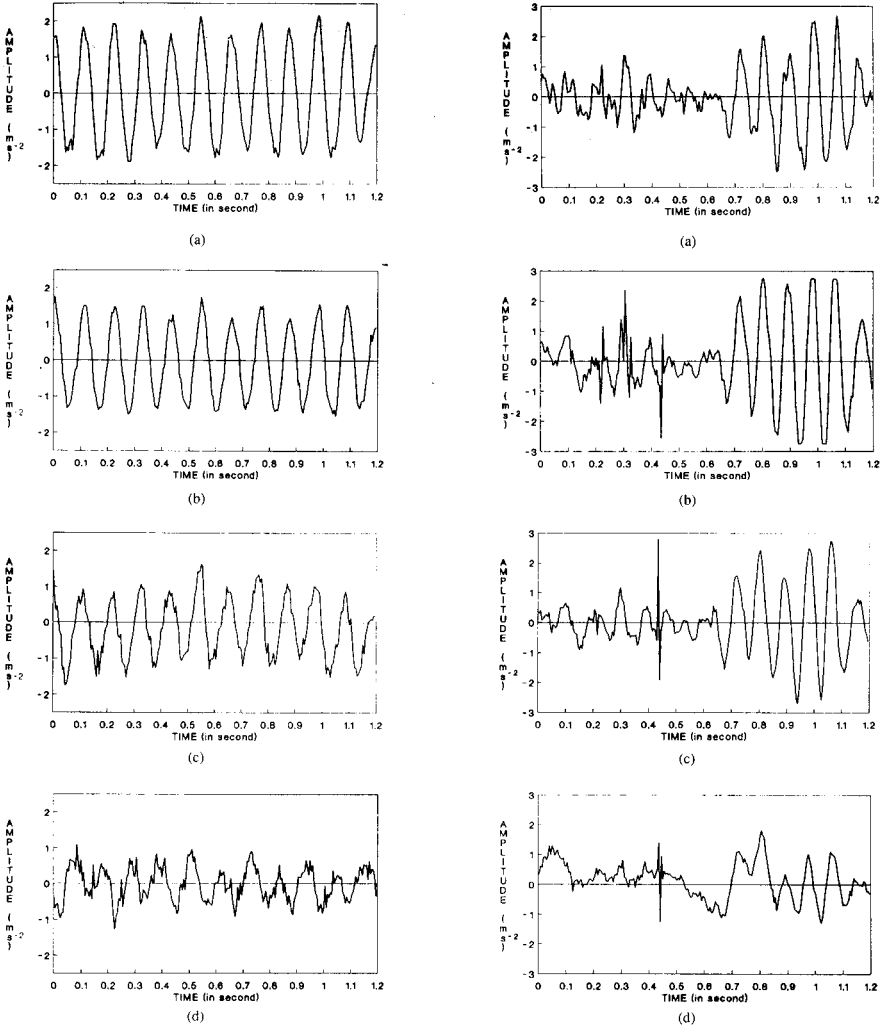


Figure 3.11 Left-hand column: VMG signals recorded simultaneously at (top-to-bottom) (a) the distal rectus femoris, (b) midpatella, (c) tibial tuberosity, and (d) midtibial shaft positions during isometric contraction (no leg or knee movement). Right-hand column: Vibration signals recorded simultaneously at the same positions as above during isotonic contraction (swinging movement of the leg). Observe the muscle-contraction interference appearing in the extension parts (second halves) of each of the VAG signals [plots (b)–(d)] in the right-hand column [125]. The recording setup is shown in Figure 3.10. Reproduced with permission from Y.T. Zhang, R.M. Rangayyan, C.B. Frank, and G.D. Bell, Adaptive cancellation of muscle-contraction interference from knee joint vibration signals, *IEEE Transactions on Biomedical Engineering*, 41(2):181–191, 1994. ©IEEE.

- statistical or deterministic.

In this chapter, we shall study several filters that span almost all of the categories mentioned above. Before studying specific filters, it would be beneficial to remind ourselves of a few fundamental notions associated with filters and their characteristics, as presented in the following sections.

3.4.1 Linear shift invariant filters

Linear time-invariant (LTI) or shift-invariant (LSI) filters form an important category of filters that have a long and well-established history in the field of signal processing [1–3, 15]. The following presentation summarizes important notions and characteristics of LSI filters and filtering operations. Both continuous-time and discrete-time notations are used, as appropriate or convenient. The proofs of the various properties stated are left as exercises for the reader. Readers unfamiliar with the material are referred to Lathi [1, 2], Oppenheim et al. [3], and Oppenheim and Schaffer [15].

A fundamental characteristic of an LSI system is its impulse response, which is the output of the system when the input is a Dirac delta or impulse function. Before looking at the details of the impulse response, let us review a few important definitions related to the delta or impulse function. In continuous time, the Dirac delta function is defined as [1–3, 15]

$$\delta(t) = \begin{cases} \text{undefined} & \text{at } t = 0, \\ 0 & \text{otherwise,} \end{cases} \quad (3.24)$$

and

$$\int_{t=-\infty}^{\infty} \delta(t) dt = 1. \quad (3.25)$$

The delta function may be visualized as the limit of a function whose integral over the full extent of time or an independent variable is maintained equal to unity while its duration or extent is compressed toward zero. Figure 3.12 demonstrates this definition graphically using a rectangular pulse function. Another example of this phenomenon is given by

$$\delta(t) = \frac{1}{2} \lim_{a \rightarrow 0} a|t|^{(a-1)}. \quad (3.26)$$

Figure 3.13 illustrates three plots of the function defined above for $a = 0.8, 0.4$, and 0.2 . The emergence of the delta function as $a \rightarrow 0$ is evident.

The delta function is the derivative of the unit step function $u(t)$, which is defined as

$$u(t) = \begin{cases} 1 & \text{for } t > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (3.27)$$

The delta function is also defined in terms of its action within an integral over a certain interval $[T_1, T_2]$:

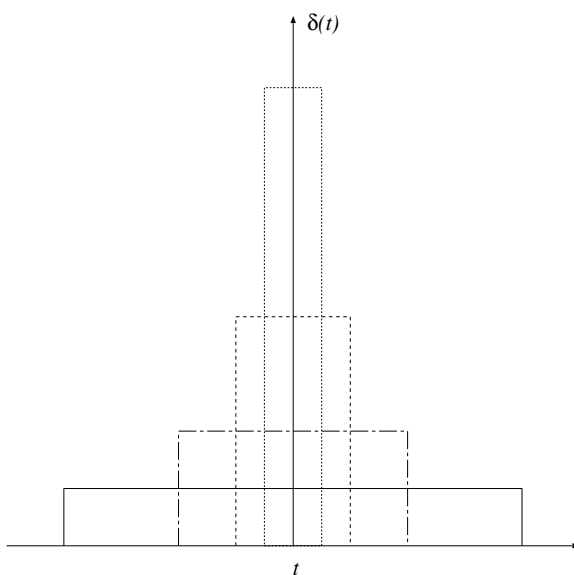


Figure 3.12 Schematic representation of the delta function as a limit of a rectangular pulse function. The time duration of the pulse is reduced while maintaining unit area under the function.

$$\int_{T_1}^{T_2} x(t) \delta(t - t_o) dt = \begin{cases} x(t_o) & \text{if } T_1 < t_o < T_2, \\ 0 & \text{otherwise,} \end{cases} \quad (3.28)$$

where $x(t)$ is a function that is continuous at t_o . This is known as the *sifting property* of the delta function, because the value of the function $x(t)$ at the location t_o of the delta function is sifted or selected from all of its values. The expression may be extended to all t as

$$x(t) = \int_{\alpha=-\infty}^{\infty} x(\alpha) \delta(t - \alpha) d\alpha, \quad (3.29)$$

where α is a temporary variable, which may also be interpreted as resolving the arbitrary signal $x(t)$ into a weighted combination of mutually orthogonal delta functions.

Let us consider a continuous-time signal, $x(t)$, that is being processed by an LSI system, as shown schematically in Figure 3.14. An LSI system is completely characterized or specified by its impulse response, $h(t)$, which is the output of the system when the input is a delta function. The output of the system, $y(t)$, is given by the convolution of the input, $x(t)$, with the impulse response, $h(t)$, defined as

$$y(t) = \int_{\tau=-\infty}^{\infty} x(\tau) h(t - \tau) d\tau, \quad (3.30)$$

where τ is a temporary variable of integration. An equivalent result is given by

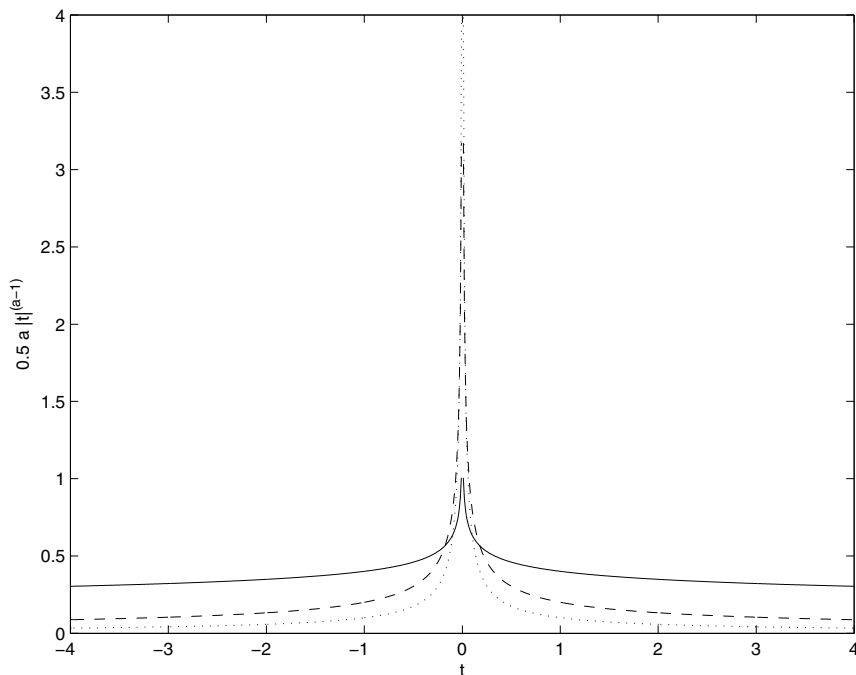


Figure 3.13 The delta function as the limit of $0.5 a |t|^{(a-1)}$ as $a \rightarrow 0$. The function is plotted for $a = 0.8$ (solid line), $a = 0.4$ (dashed line), and $a = 0.2$ (dotted line).

$$y(t) = \int_{\tau=-\infty}^{\infty} h(\tau) x(t - \tau) d\tau. \quad (3.31)$$

The convolution operation given above is *linear convolution*; another version of convolution known as periodic or circular convolution is defined in Equation 3.85. When a causal system or causality is considered, the lower limit of the integrals given above for convolution may be changed to zero and the upper limit changed to t , the current instant of time, resulting in

$$y(t) = \int_{\tau=0}^t x(\tau) h(t - \tau) d\tau \quad (3.32)$$

or

$$y(t) = \int_{\tau=0}^t h(\tau) x(t - \tau) d\tau. \quad (3.33)$$

The discrete-time unit impulse function or delta function is defined as [1–3]

$$\delta(n) = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{otherwise.} \end{cases} \quad (3.34)$$

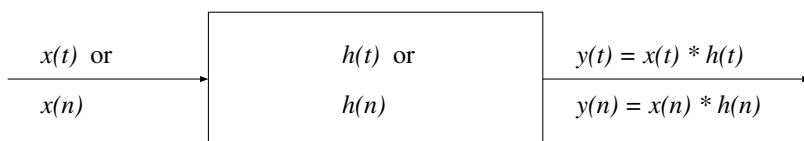


Figure 3.14 A schematic representation of a continuous-time or discrete-time LSI filter.

Figure 3.15 illustrates a few versions of the discrete-time delta function.

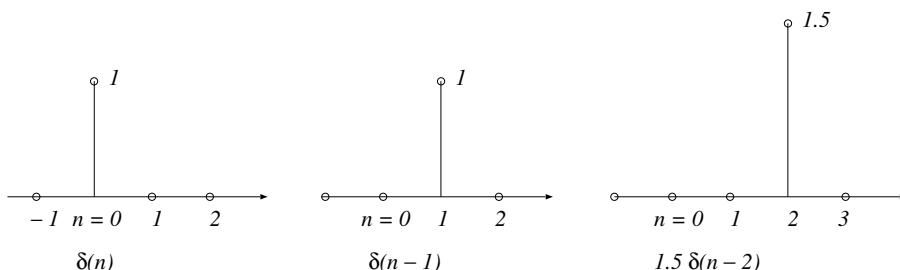


Figure 3.15 A schematic representation of the discrete-time unit impulse function or delta function as well as shifted and scaled versions of the same.

The discrete-time unit step function is defined as

$$u(n) = \begin{cases} 1 & \text{for } n \geq 0, \\ 0 & \text{otherwise.} \end{cases} \quad (3.35)$$

Figure 3.16 illustrates a few versions of the discrete-time unit step function.

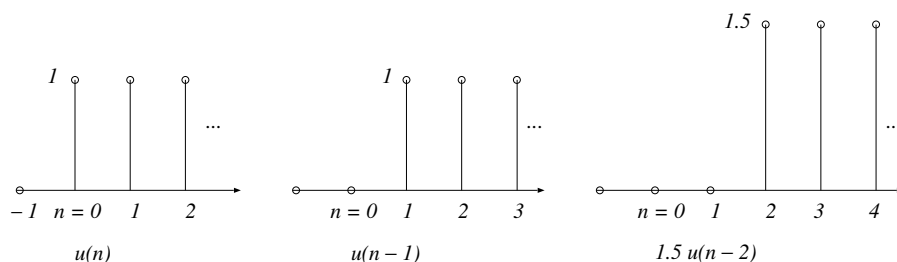


Figure 3.16 A schematic representation of the discrete-time unit step function as well as shifted and scaled versions of the same.

A discrete-time LSI system is shown schematically in Figure 3.17, displaying its impulse response, $h(n)$. The output, $y(n)$, of the system is given by the linear convolution of the input, $x(n)$, with the impulse response, $h(n)$, as

$$y(n) = \sum_{k=0}^n x(k) h(n-k), \quad (3.36)$$

where k is a temporary variable of summation. An equivalent result is given by

$$y(n) = \sum_{k=0}^n h(k) x(n-k). \quad (3.37)$$

In the two equations given above, causality has been assumed.

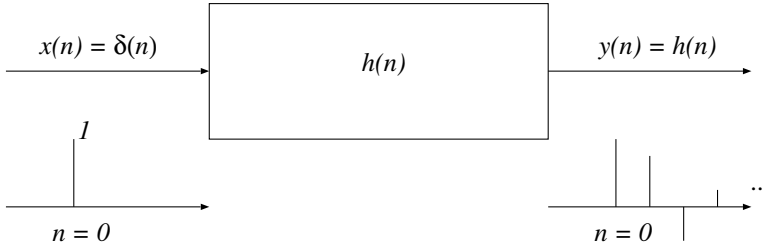


Figure 3.17 A schematic representation of the impulse response, $h(n)$, of a discrete-time LSI filter.

To understand convolution, let us consider the discrete-time version in Equation 3.36. In addition to the change of the independent (time) variable from n to k , there are two important points to note:

- $h(-k)$ represents a reversal in time of the function or signal $h(k)$; and
- $h(n-k)$ represents a shift of the reversed signal $h(-k)$ by n samples.

Multiplication of $h(n-k)$ by $x(k)$ can be viewed as scaling. The summation represents accumulation of the results or integration of $x(k) h(n-k)$ over the interval $k = 0$, the origin of time, to n , the present instant of time. A simple illustration of shifting or adding a delay in time, as well as scaling, is shown in Figure 3.15. A numerical illustration of shifting a signal is shown in Figure 3.18. An illustration of reversing as well as shifting a signal is shown in Figure 3.19.

The linear convolution of two discrete-time signals is illustrated numerically in Figure 3.20. To facilitate understanding of the procedure, we could expand and modify Equation 3.36 as follows:

$$y(n) = \sum_{k=0}^n x(k) h(n-k),$$

$$y(0) = \sum_{k=0}^0 x(k) h(0-k)$$

$n:$	0	1	2	3	4	5	6	7
$x(n):$	4	5	3	1				
$x(n-1):$		4	5	3	1			
$x(n-2):$			4	5	3	1		
$x(n-3):$				4	5	3	1	

Figure 3.18 A numerical illustration of shifted versions of signal. Blank spaces indicate samples of the signal that are undefined or zero in value. This type of shifting is known as linear shifting. See Figure 3.37 for an example of circular or periodic shifting.

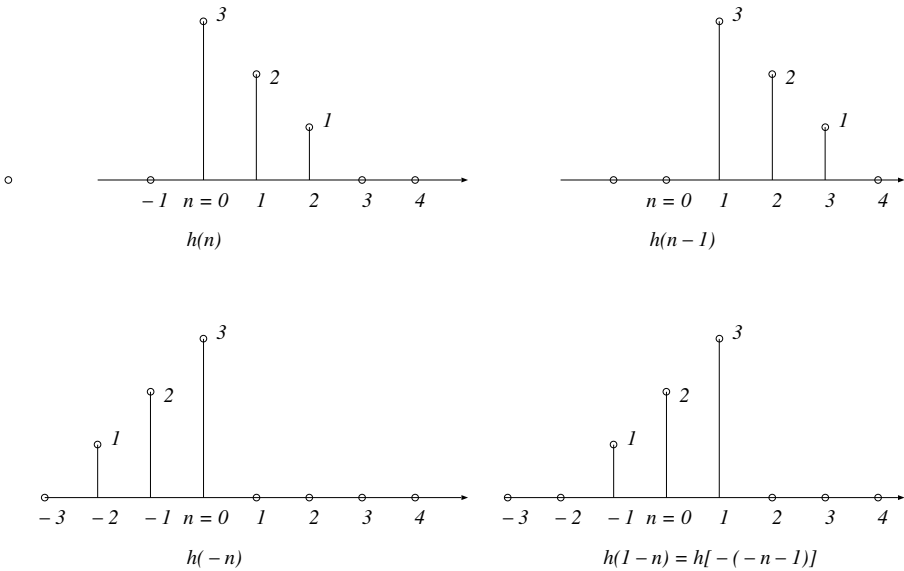


Figure 3.19 A schematic illustration of reversing and shifting of a signal.

$$\begin{aligned}
 &= x(0)h(0). \\
 y(1) &= \sum_{k=0}^1 x(k) h(1-k) \\
 &= x(0)h(1) + x(1)h(0). \\
 y(2) &= \sum_{k=0}^2 x(k) h(2-k) \\
 &= x(0)h(2) + x(1)h(1) + x(2)h(0).
 \end{aligned}$$

$$\begin{aligned}
y(3) &= \sum_{k=0}^3 x(k) h(3-k) \\
&= x(0)h(3) + x(1)h(2) + x(2)h(1) + x(3)h(0). \quad (3.38)
\end{aligned}$$

The procedure is stopped and the output ceases to exist (or has only zero values) when the shift exceeds a certain amount such that $x(k)$ and $h(n-k)$ do not overlap in time any more. It is evident that the linear convolution of two discrete-time signals with durations of N_1 and N_2 samples leads to a result with the duration of N_1+N_2-1 samples.

$$\begin{array}{rcccccccc}
n: & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
x(n): & 4 & 1 & 3 & 1 & & & & \\
h(n): & 3 & 2 & 1 & & & & & \\
k: & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
x(k): & 4 & 1 & 3 & 1 & 0 & 0 & 0 & 0 \\
\\
h(0-k): & 1 & 2 & 3 & & & & & \\
h(1-k): & & 1 & 2 & 3 & & & & \\
h(2-k): & & & 1 & 2 & 3 & & & \\
h(3-k): & & & & 1 & 2 & 3 & & \\
h(4-k): & & & & & 1 & 2 & 3 & \\
h(5-k): & & & & & & 1 & 2 & 3 \\
h(6-k): & & & & & & & 1 & 2 & 3 \\
y(n): & 12 & 11 & 15 & 10 & 5 & 1 & 0 & 0 \\
n: & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7
\end{array}$$

Figure 3.20 A numerical illustration of the convolution of two discrete-time signals. The notations used in the figure agree with Equation 3.36. For each shift, the corresponding output sample value is obtained by multiplying each pair of samples of $x(k)$ and $h(n-k)$ that overlap (exist at the same instant of time k) and then adding the results.

Taking a different approach, we could expand and modify Equation 3.36 as follows:

$$\begin{aligned}
y(n) &= \sum_{k=0}^n x(k) h(n-k), \\
&= x(0)h(n) + x(1)h(n-1) + x(2)h(n-2) + x(3)h(n-3) + \dots \quad (3.39)
\end{aligned}$$

This result may be interpreted as the sum of several delayed and weighted versions of the impulse response of the system, the weights being provided by the samples of the input signal. Just as the system produces the impulse response, $h(n)$, when the input is $\delta(n)$, when the system is triggered by the input $x(0)$ at $n = 0$, it produces the output $x(0)h(n)$, which lasts for the duration of $h(n)$. When the input is $x(1)$, occurring at $n = 1$, the output is $x(1)h(n - 1)$, recognizing the fact that the system is causal and the corresponding output can only start from $n = 1$. Note that the part of the output signal that started at $n = 0$ will, in general, continue past $n = 1$, and hence the parts of the output that start at $n = 0$ and $n = 1$ will overlap. The process continues, as illustrated in Figure 3.21.

n :	0	1	2	3	4	5	6	7
$x(n)$:	4	1	3	1				
$h(n)$:	3	2	1					
$x(0)h(n-0)$:	12	8	4	0	0	0	0	0
$x(1)h(n-1)$:	0	3	2	1	0	0	0	0
$x(2)h(n-2)$:	0	0	9	6	3	0	0	0
$x(3)h(n-3)$:	0	0	0	3	2	1	0	0
$y(n)$:	12	11	15	10	5	1	0	0
n :	0	1	2	3	4	5	6	7

Figure 3.21 A numerical illustration of the convolution of two discrete-time signals. The output, $y(n)$, is obtained by adding the corresponding values in the four rows labeled as $x(0)h(n-0)$ through $x(3)h(n-3)$. The notations used in the figure agree with Equation 3.39. Compare this procedure with that shown in Figure 3.20. Although the view is different, the end result is the same.

Illustrations of application: To illustrate the effects of random noise being added to a deterministic signal, the following signal was created:

$$x(t) = 5 \sin(2\pi 2t) + 2 \cos(2\pi 3t), \quad (3.40)$$

using a sampling frequency of 2 kHz . Random noise with a Gaussian PDF was simulated and added to the signal x for an effective SNR of 10 dB . The original signal, the noise samples generated, and the noisy signal are shown in Figure 3.22. The distinction between the deterministic nature of the original signal and the random nature of the noise are clearly evident. The histogram of a realization of the noise process used is shown in Figure 3.23, which approximates a Gaussian PDF with zero mean.

The noisy signal was filtered by computing the mean of each sample and the preceding 10 samples. The equation of the corresponding filter could be expressed as

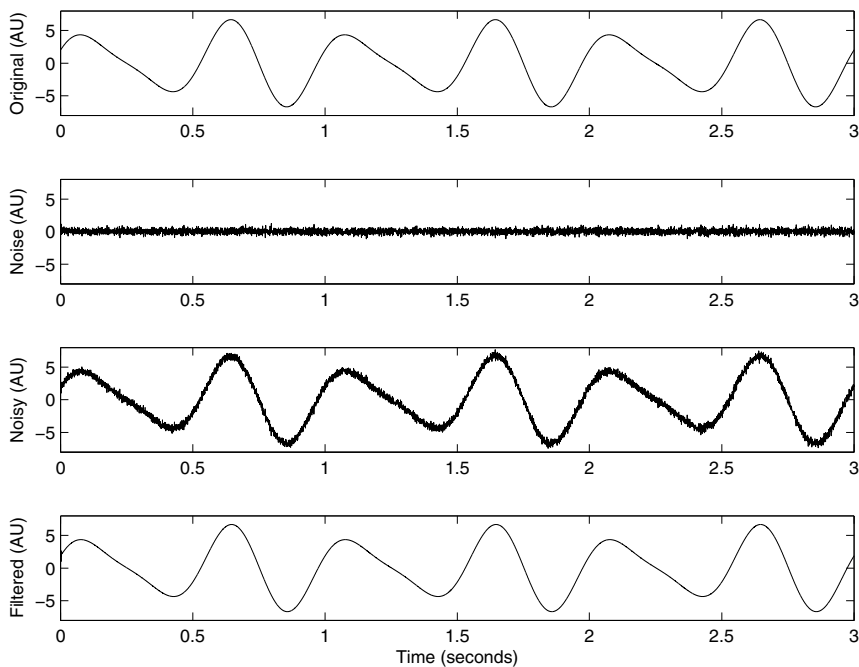


Figure 3.22 Top to bottom: original signal $x(t)$ as in Equation 3.40; Gaussian-distributed random noise; noisy signal; result of filtering using the mean of the signal values in a sliding window of width 11 samples or 5.5 ms . See Figure 3.23 for the histogram of the noise process.

$$y(n) = \frac{1}{11} \sum_{k=0}^{10} x(n-k), \quad (3.41)$$

for $n = 10, 11, \dots, N-1$, where N is the number of samples in the signal. A filter as above is known as a moving-average (MA) filter: average values of the input signal are computed in a moving temporal window and used to define the output signal. (See Figure 3.2 for a related illustration.) Note that the operation as defined in Equation 3.41 cannot be started until the first 11 samples are available in the input data stream. The operation may also be viewed as a convolution of $x(n)$ with the impulse response of the filter, as in Equation 3.37, with $h(k) = 1/11$, $k = 0, 1, \dots, 10$. Due to the random nature of the noise process (with the mean of the process being zero), the mean of a number of samples of the process tends to zero as the number of samples used increases. Therefore, the values of the filtered result, also shown in Figure 3.22, approach the corresponding values in the original signal. Whereas a small number of samples used in computing the mean would not suppress much of the noise, a large number of samples could overly smooth the signal and remove fine details.

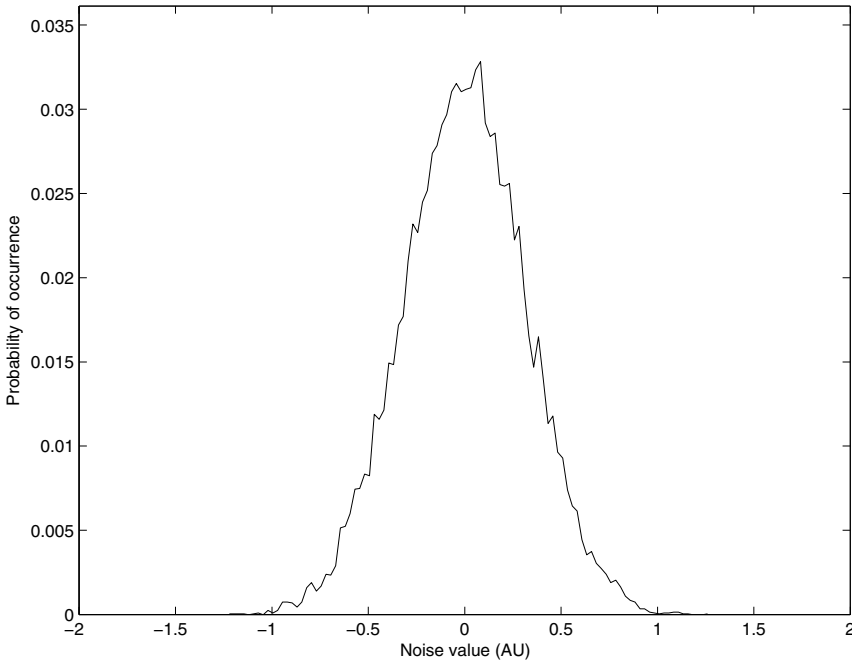


Figure 3.23 Histogram of a realization of the noise process used in the example in Figure 3.22.

Figure 3.24 shows another version of a noisy signal generated using the same processes as in the preceding example. The noisy signal was filtered with a filter having its impulse response as a linearly decreasing function or a ramp of duration 0.25 s , defined in the continuous-time notation as

$$h(t) = 10(0.25 - t), \quad 0 \leq t \leq 0.25\text{ s}, \quad (3.42)$$

with the sampling frequency of 2 kHz . The output was divided by the sum of all of the values of $h(n)$. The result is a weighted average of the corresponding values of the input signal. Figure 3.25 shows the filter function in Equation 3.42 superimposed on the noisy signal being filtered. The impulse response of the filter has been reversed in time, as required in Equation 3.36 for convolution, and placed at $t = 2.3\text{ s}$. The output at $t = 2.3\text{ s}$ is given by the area under the product of the signal and the impulse response; equivalently, it is given by the sum of the products of all of the overlapping samples of the signal and the impulse response. It is seen that the filtered output is smooth and free of noise; however, some of the minor details in the original signal have been suppressed by the lengthy filter. Furthermore, the delay introduced by the filter is clearly seen by comparing the peaks in the original signal with the corresponding peaks in the filtered output.

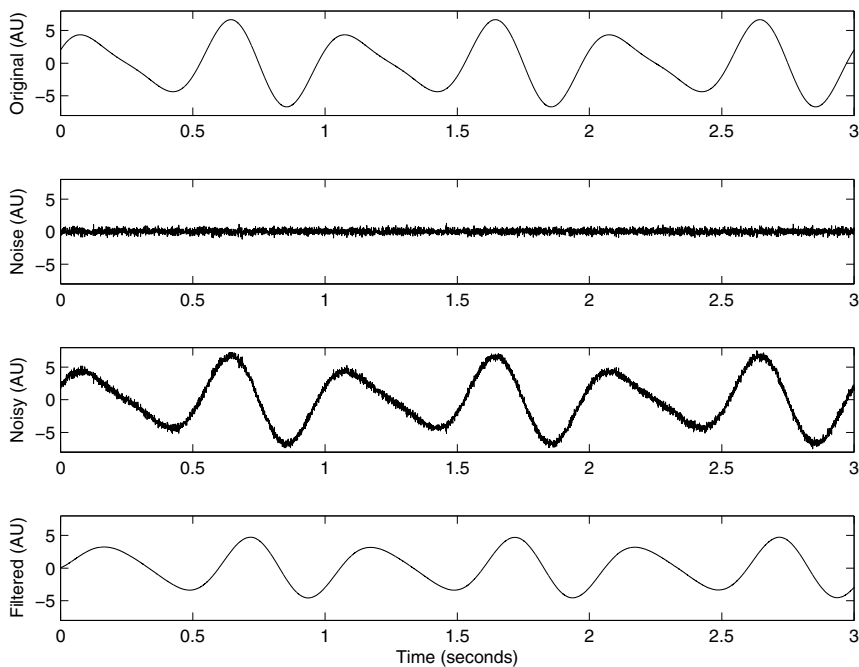


Figure 3.24 Top to bottom: original signal $x(t)$ as in Equation 3.40; Gaussian-distributed random noise; noisy signal; result of filtering using the ramp function in Equation 3.42. See Figure 3.23 for the histogram of the noise process. See also Figure 3.25.

LSI systems in series or parallel: When a number of LSI systems are used in series or in parallel, their effects may be combined into a single equivalent system or operation. Figure 3.26 shows two LSI systems in series (cascade). The first system, with the impulse response $h_1(n)$, operates upon the input, $x(n)$, to give the output as

$$s(n) = x(n) * h_1(n). \quad (3.43)$$

The second system, with the impulse response $h_2(n)$, operates upon $s(n)$ to produce the output

$$\begin{aligned} y(n) &= s(n) * h_2(n) \\ &= x(n) * h_1(n) * h_2(n) \\ &= x(n) * h(n), \end{aligned} \quad (3.44)$$

where

$$h(n) = h_1(n) * h_2(n) \quad (3.45)$$

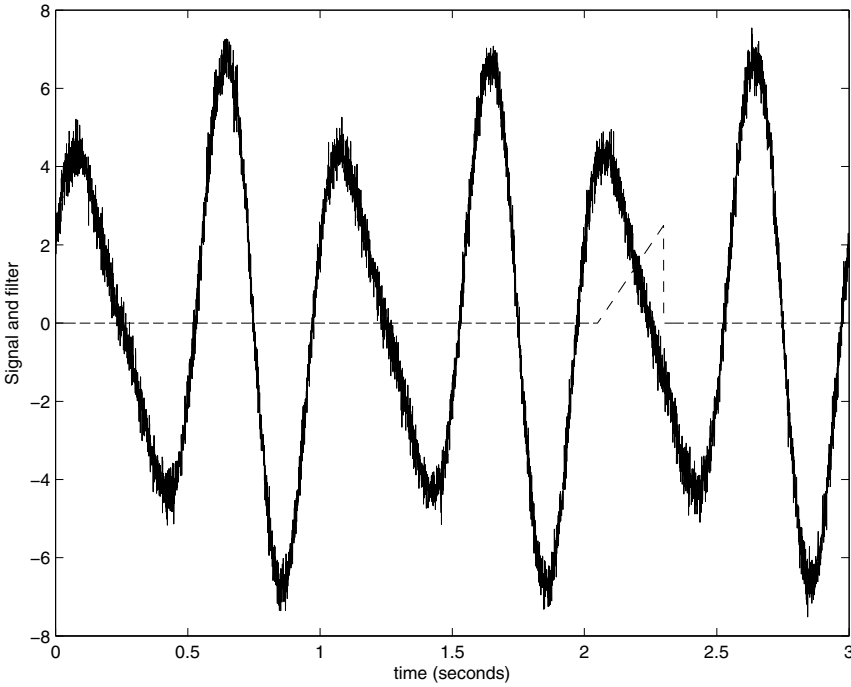


Figure 3.25 The linear ramp filter in Equation 3.42 is shown superimposed (in dashed line) on the noisy signal in the example shown in Figure 3.24. The impulse response of the filter has been reversed in time and placed at $t = 2.3$ s. The output at $t = 2.3$ s is given by the area under the product of the signal and the impulse response; equivalently, it is given by the sum of the products of all samples of the signal and the impulse response that overlap.

is the impulse response of the combination of the two LSI systems in series.

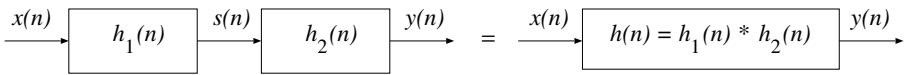


Figure 3.26 Two LSI systems in series and the equivalent system.

Figure 3.27 shows two LSI systems in parallel. For the first system, we have the output as

$$s_1(n) = x(n) * h_1(n). \quad (3.46)$$

Similarly, the second system produces the output

$$s_2(n) = x(n) * h_2(n). \quad (3.47)$$

The combined result is

$$\begin{aligned}
y(n) &= s_1(n) + s_2(n) \\
&= x(n) * h_1(n) + x(n) * h_2(n) \\
&= x(n) * [h_1(n) + h_2(n)] \\
&= x(n) * h(n),
\end{aligned} \tag{3.48}$$

where

$$h(n) = h_1(n) + h_2(n) \tag{3.49}$$

is the impulse response of the combination of the two LSI systems in parallel. Relationships such as above are useful in the analysis of sophisticated signal processing systems designed by combining several LSI systems.

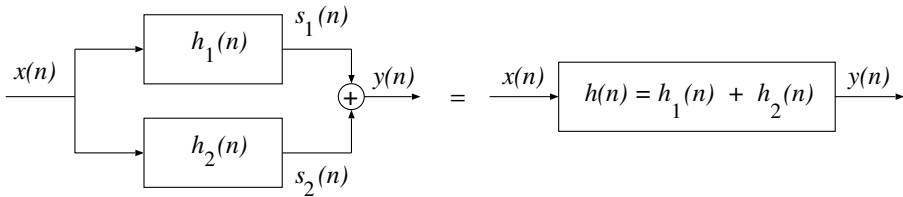


Figure 3.27 Two LSI systems in parallel and the equivalent system.

3.4.2 Transform domain analysis of signals and systems

Quite often, it is convenient to analyze the behavior, characteristics, and performance of an LTI or LSI system in a transform domain. The commonly used transforms for the analysis of continuous-time systems and signals are the Laplace and Fourier transforms [1–3]. The Laplace transform, $H(s)$, of the impulse response, $h(t)$, of an LTI system is defined as

$$H(s) = \int_{-\infty}^{\infty} h(t) \exp(-st) dt, \tag{3.50}$$

where $s = \sigma + j\omega$ is a complex variable representing the transform domain. Here, $\omega = 2\pi f$ represents the frequency variable in radians per second, with f being the frequency variable in Hz ; the unit for σ is neper or Napier. If the signal $h(t)$ is causal and of finite duration, existing only over the interval of time $[0, T]$, the limits of the integral could be changed as

$$H(s) = \int_0^T h(t) \exp(-st) dt. \tag{3.51}$$

The function $H(s)$ is known as the system transfer function, or simply the transfer function of the filter.

Figure 3.28 shows a schematic representation of the s -plane. When $H(s)$ is evaluated on the imaginary axis in the s -plane, with $s = j\omega$, we get the frequency response of the system as

$$H(\omega) = H(s)|_{s=j\omega} = \int_0^T h(t) \exp(-j\omega t) dt, \quad (3.52)$$

which is the Fourier transform of the impulse response, $h(t)$. [Note: Some authors express the function above as $H(e^{j\omega})$ or $H(j\omega)$.] In general, $H(\omega)$ is a complex quantity even when $h(t)$ is a real-valued signal. The magnitude response of the system or filter is given by $|H(\omega)|$ and the phase response is given by $\angle H(\omega)$.

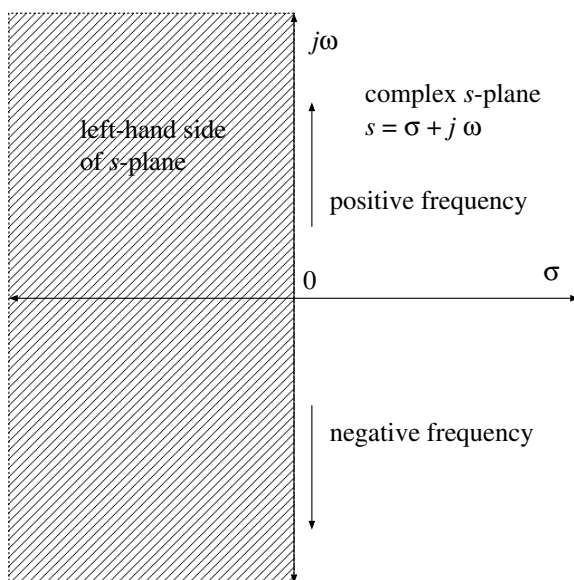


Figure 3.28 Schematic representation of the s -plane or Laplace transform domain.

It is common to express the transfer function, $H(s)$, of an LTI system as a ratio of two polynomials in s . The roots of the polynomial in the numerator give the zeros of the system whereas the roots of the polynomial in the denominator give the poles of the system. For a stable LTI system, all of the poles should be located on the left-hand side (LHS) of the s -plane with $\sigma < 0$. For a real-valued impulse response or signal, poles and zeros should occur in complex-conjugate pairs or with real values. The pole-zero plot of a system is adequate to derive its impulse response, transfer function, and output (for a given input signal) except for a scale factor or gain.

The Laplace transform is a linear and reversible transform. An important property of the Laplace transform when dealing with LTI systems is that the convolution of two signals in the time domain is converted to the product of their individual Laplace transforms in the s -domain, expressed as

$$\begin{aligned}
&\text{if } y(t) = x(t) * h(t), \\
&\text{then } Y(s) = X(s) H(s), \\
&\quad Y(\omega) = X(\omega) H(\omega).
\end{aligned} \tag{3.53}$$

Figure 3.29 shows the input–output relationship for an LTI system in the s -domain.

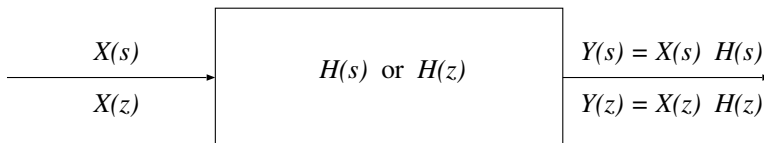


Figure 3.29 Input–output relationship for an LTI system in the s -domain or an LSI system in the z -domain.

It follows that, when two LTI systems are in parallel, as in Figure 3.27, we have the output as $Y(s) = [H_1(s) + H_2(s)]X(s)$, with the equivalent transfer function of $H(s) = H_1(s) + H_2(s)$. When two LTI systems are in series, as in Figure 3.26, we have the output as $Y(s) = H_1(s) H_2(s) X(s)$, with the equivalent transfer function of $H(s) = H_1(s) H_2(s)$. For further details regarding the properties of the Laplace transform, the inverse Laplace transform, and examples of signals and their transforms, refer to Lathi [1, 2] and Oppenheim et al. [3].

The discrete-time counterpart of the Laplace transform is the z -transform, which is useful for the analysis of discrete-time LSI systems and signals. The z -transform of a signal $x(n)$ is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}, \tag{3.54}$$

where z is a complex variable. Lathi [1, 2] and Oppenheim et al. [3] provide detailed discussions on the properties of the z -transform, its region of convergence (ROC), and the inverse z -transform.

If we consider the z -transform of the impulse response, $h(n)$, of a causal, LSI, finite-impulse response (FIR) system, with $h(n)$ existing only for $n = 0, 1, 2, \dots, N-1$, then we have

$$H(z) = \sum_{n=0}^{N-1} h(n) z^{-n}, \tag{3.55}$$

which represents the transfer function of the system.

An important property of the z -transform when dealing with LSI systems is that the convolution of two signals in the time domain is converted to the product of their individual z -transforms, expressed as

$$\begin{aligned} \text{if } y(n) &= x(n) * h(n), \\ \text{then } Y(z) &= X(z) H(z). \end{aligned} \quad (3.56)$$

Figure 3.29 shows the input–output relationship for an LSI system in the z -domain. The relationship expressed above may be derived in detail as follows:

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{\infty} y(n) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x(k) h(n-k) z^{-n} \\ &= \sum_{k=-\infty}^{\infty} x(k) \sum_{n=-\infty}^{\infty} h(n-k) z^{-n}. \end{aligned} \quad (3.57)$$

Here, the definition of convolution relationship $y(n) = x(n) * h(n)$ has been expanded as in Equation 3.36. The range of summation has been indicated as $[-\infty, \infty]$ for the sake of generality and ease of manipulation of the equations. In the last step above, the order of the two summation operations has been switched and the terms have been rearranged. Now, let $m = n - k$. Then, $n = m + k$, and we have

$$\begin{aligned} Y(z) &= \sum_{k=-\infty}^{\infty} x(k) \sum_{m=-\infty}^{\infty} h(m) z^{-(m+k)} \\ &= \sum_{k=-\infty}^{\infty} x(k) \sum_{m=-\infty}^{\infty} h(m) z^{-m} z^{-k} \\ &= \sum_{k=-\infty}^{\infty} x(k) z^{-k} \sum_{m=-\infty}^{\infty} h(m) z^{-m} \\ &= X(z) H(z). \end{aligned} \quad (3.58)$$

The effects of combining two LSI systems in series or in parallel, as seen in the z -domain, are shown in Figures 3.30 and 3.31. For the combination in series, we have

$$S(z) = X(z) H_1(z) \quad (3.59)$$

and

$$\begin{aligned} Y(z) &= S(z) H_2(z) \\ &= X(z) H_1(z) H_2(z) \\ &= X(z) H(z), \end{aligned} \quad (3.60)$$

where

$$H(z) = H_1(z) H_2(z). \quad (3.61)$$

For the combination in parallel, we have

$$S_1(z) = X(z) H_1(z), \quad (3.62)$$

$$S_2(z) = X(z) H_2(z), \quad (3.63)$$

and

$$\begin{aligned} Y(z) &= S_1(z) + S_2(z) \\ &= X(z) H_1(z) + X(z) H_2(z) \\ &= X(z) [H_1(z) + H_2(z)] \\ &= X(z) H(z), \end{aligned} \quad (3.64)$$

where

$$H(z) = H_1(z) + H_2(z). \quad (3.65)$$

Relationships such as these are useful in the analysis of sophisticated signal processing systems that are designed by combining several LSI systems.

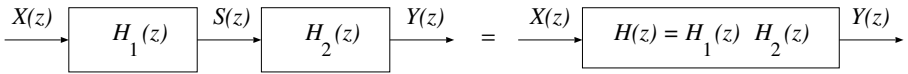


Figure 3.30 Two LSI systems in series and the equivalent system in the z -domain.

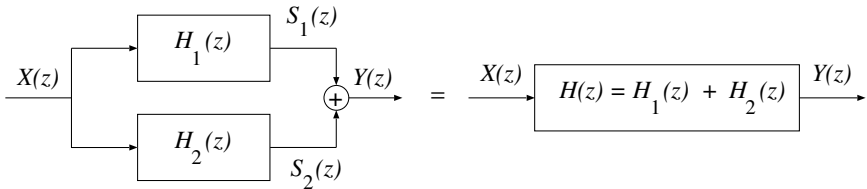


Figure 3.31 Two LSI systems in parallel and the equivalent system in the z -domain.

A few of the important properties of and examples related to the z -transform are as follows [1–3]:

- The z -transform of $\delta(n)$ is 1. The ROC is all z .
- The z -transform of the unit step function, $u(n)$, is $\frac{1}{1-z^{-1}}$. The ROC is $|z| > 1$.

- The z -transform of $a^n u(n)$ is $\frac{1}{1-az^{-1}}$. The ROC is $|z| > |a|$.
- The z -transform of a causal signal $x(n)$ shifted by k samples, that is, $x(n-k)$, is $z^{-k} X(z)$, where $X(z)$ is the z -transform of $x(n)$.
- If $y(n) = x(n) * h(n)$, then $Y(z) = X(z) H(z)$.

Figure 3.32 illustrates the relationship between the Laplace domain and the z -domain with $z = \exp(sT)$, where $T = 1/f_s$ is the sampling interval and f_s is the sampling frequency. (The maximum frequency permitted in the signal without aliasing errors is $f_m = f_s/2$; this is also referred to as the folding frequency or the Nyquist frequency.) The entire LHS of the s -plane is mapped to the region within the unit circle in the z -plane. Thus, all of the poles of a stable system should be located within the unit circle in the z -plane. See Oppenheim and Schaffer [174] for a detailed discussion on this relationship. The bilinear transformation is another method of mapping from the s -plane to the z -plane [1–3].

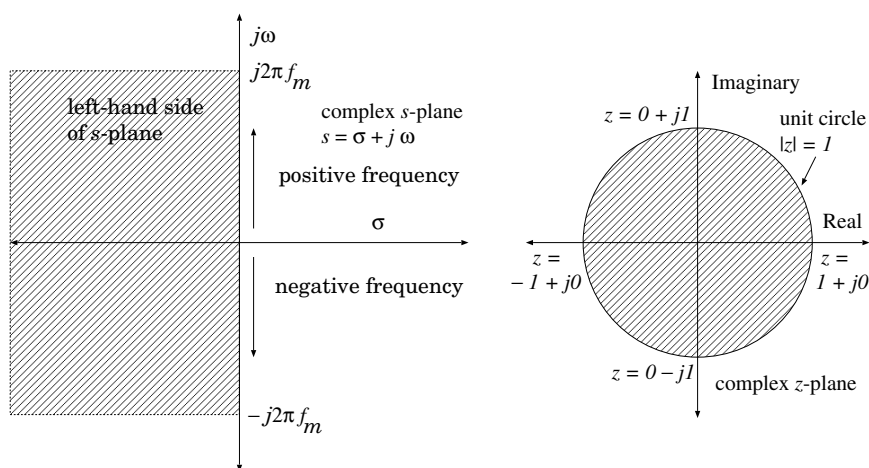
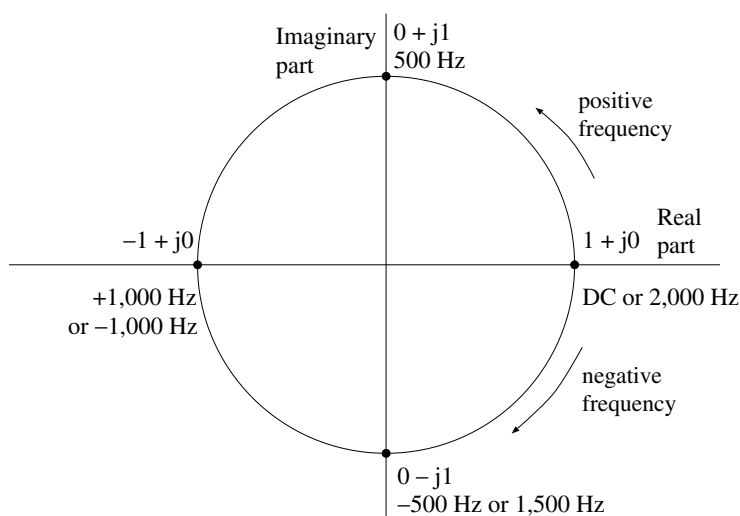
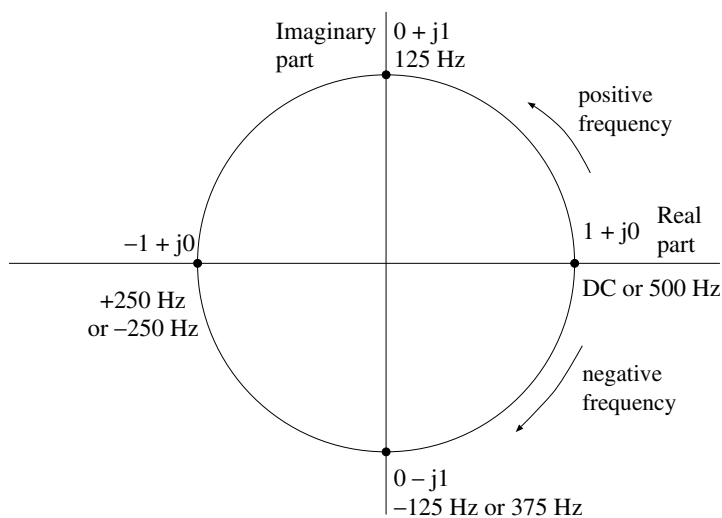


Figure 3.32 Transformation from the Laplace domain to the z -domain with $z = \exp(sT)$. $f_m = f_s/2$.

Figure 3.33 provides interpretation of the frequency variable around the unit circle in the z -domain for two different sampling frequencies, $f_s = 2,000 \text{ Hz}$ and $f_s = 500 \text{ Hz}$. The unit circle is given by $z = \exp(j\omega T)$, and represents the frequency axis in a circular and periodic manner instead of the linear vertical axis in the s -plane. (The same symbol ω is being used in the present discussion for the frequency variable in both cases of continuous-time and discrete-time analysis. Different symbols will be used for the two cases when the two variables need to be distinguished.) The notions of a limited bandwidth of $\pm f_s/2$ or $[0, f_s]$ and the introduction of aliasing errors if the Nyquist sampling rate [1–3] is not satisfied are demonstrated by the circular and periodic nature of the frequency axis.



(a) unit circle in the z -plane with $f_s = 2,000$ Hz



(b) unit circle in the z -plane with $f_s = 500$ Hz

Figure 3.33 Interpretation of the frequency variable with uniform sampling along the unit circle in the z -domain for two different sampling frequencies.

Just as the Fourier transform may be seen as the Laplace transform evaluated on the imaginary axis with $s = j\omega$, evaluation of the z -transform on the unit circle in the z -plane gives us the Fourier transform, as

$$\begin{aligned} X(\omega) &= \sum_{n=0}^{N-1} x(n) z^{-n} \Big|_{z=\exp(j\omega T)} \\ &= \sum_{n=0}^{N-1} x(n) \exp(-j\omega nT). \end{aligned} \quad (3.66)$$

Using $\omega = 2\pi f$ and $T = 1/f_s$, we have the argument of the exp function above as $-j\omega nT = -j2\pi n f/f_s$. We may now consider the ratio f/f_s to represent a normalized frequency variable in the range $[0, 1]$, with zero corresponding to DC and unity corresponding to f_s . The result of multiplication of this ratio with 2π may be seen as division of the range $[0, 2\pi]$ into the frequency axis spanning the range $[0, f_s]$. Thus, the variable T may be dropped.

3.4.3 The pole–zero plot

Consider a signal processing system or filter with the transfer function specified as a ratio of two polynomials in z or a rational function of z :

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^N b_k z^{-k}}{1 + \sum_{k=1}^M a_k z^{-k}}. \quad (3.67)$$

The equivalent time-domain difference equation of the filter is

$$y(n) = \sum_{k=0}^N b_k x(n-k) - \sum_{k=1}^M a_k y(n-k). \quad (3.68)$$

The equations given above do not include a gain factor that could be specified separately.

The polynomials in the numerator and denominator of Equation 3.67 may be solved for their roots. Let the roots of the numerator be z_k , $k = 1, 2, \dots, N$, which are the zeros of the transfer function $H(z)$. Let the roots of the denominator be p_k , $k = 1, 2, \dots, M$, which are the poles of the transfer function $H(z)$. A plot of the roots of $H(z)$ in the z -plane is the pole–zero plot of the system; see Figure 3.34. The transfer function may be expressed in terms of the poles and zeros as

$$H(z) = \frac{\prod_{k=1}^N (1 - z_k z^{-1})}{\prod_{k=1}^M (1 - p_k z^{-1})}, \quad (3.69)$$

which may be modified to

$$H(z) = z^{(M-N)} \frac{\prod_{k=1}^N (z - z_k)}{\prod_{k=1}^M (z - p_k)}. \quad (3.70)$$

The term $(z - z_k)$ represents the vector from an arbitrary point z to the zero z_k . Similarly, the term $(z - p_k)$ represents the vector from an arbitrary point z to the pole p_k .

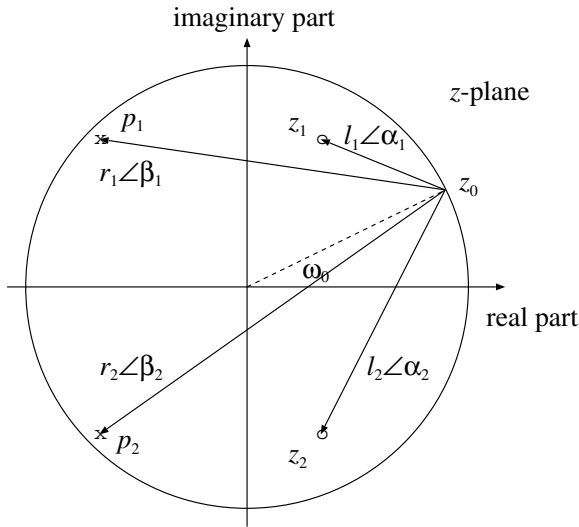


Figure 3.34 Derivation of the frequency response of a system from its pole-zero plot.

Now, consider the evaluation of $H(z)$ for a point z_0 on the unit circle in the z -plane, as shown in Figure 3.34. This gives the frequency response of the system for the corresponding radian frequency ω_0 , as

$$H(\omega_0)|_{z=z_0} = z_0^{(M-N)} \frac{\prod_{k=1}^N (z_0 - z_k)}{\prod_{k=1}^M (z_0 - p_k)}. \quad (3.71)$$

Following the illustration in Figure 3.34, let us represent the vector from z_0 to the zero z_1 as $(z_0 - z_1) = l_1 \angle \alpha_1$, the vector from z_0 to the pole p_1 as $(z_0 - p_1) = r_1 \angle \beta_1$, and apply the same procedure to the remaining poles and zeros. Then, we have

$$|H(\omega_0)| = \frac{\prod_{k=1}^N l_k}{\prod_{k=1}^M r_k} \quad (3.72)$$

and

$$\angle H(\omega_0) = \angle z_0^{(M-N)} + \sum_{k=1}^N \alpha_k - \sum_{k=1}^M \beta_k. \quad (3.73)$$

Thus, the magnitude of the response of a system at a particular frequency is given by the ratio of the product of the distances from the corresponding frequency point in the z -plane to all of the zeros of the system to the product of the distances from the same

frequency point to all of the poles of the system (except for a gain factor). Similarly, the phase response is given by the difference between the sum of the angles of the vectors from the frequency point to all of the zeros of the system and the sum of the angles of the vectors from the same frequency point to all of the poles of the system (except for any additional linear phase factor).

It is evident from the equations and the related discussions given above that, as the point z_0 approaches a zero of the system, one of the distances l_k will become small, leading to a low value of the response. If a zero is present on the unit circle, the corresponding l_k will be zero, resulting in a zero-valued response at the related frequency. Thus, a zero on the unit circle is associated with a spectral null of the system. As the frequency variable crosses from one side of a zero on the unit circle to the other side, there will be an associated 180° change in the phase response. On the contrary, as the point z_0 approaches a pole of the system that is close to the unit circle, the corresponding distance r_k will become small, leading to a high value of the response at the related frequency. Thus, a pole close to the unit circle is associated with a spectral peak or resonance of the system. In this manner, the pole-zero plot of a system may be inspected to elicit information related to its frequency response in an immediate, albeit qualitative, manner.

3.4.4 The discrete Fourier transform

If we consider evaluation of the Fourier transform for only certain sampled values of the frequency, we could span the range $[0, 2\pi]$ with K samples, with even steps of $2\pi/K$. This is equivalent to spanning the range $[0, 1]$ of the normalized frequency with K samples. The discrete Fourier transform (DFT) could then be expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n) \exp\left(-j\frac{2\pi}{K}nk\right), \quad (3.74)$$

with $k = 0, 1, 2, \dots, K-1$. It can be shown that, when the given signal has only N nonzero samples, we need only N samples of the DFT evenly spaced over the unit circle in the z -plane [15] for exact recovery of the original signal from the DFT. Then, we have

$$X(k) = \sum_{n=0}^{N-1} x(n) \exp\left(-j\frac{2\pi}{N}nk\right), \quad (3.75)$$

for $k = 0, 1, 2, \dots, N-1$, and

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) \exp\left(+j\frac{2\pi}{N}nk\right), \quad (3.76)$$

for $n = 0, 1, 2, \dots, N-1$, as the forward and inverse DFT relationships, respectively.

If we define a complex variable

$$W_N = \exp\left(-j\frac{2\pi}{N}\right), \quad (3.77)$$

which can also be seen as a vector or a phasor, we can write the DFT relationship as

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk}, \quad (3.78)$$

for $k = 0, 1, 2, \dots, N-1$. Considering a simple case with $N = 8$, Figure 3.35 shows the vectors (or phasors) representing the $N = 8$ roots of unity, with W_8^k , $k = 0, 1, 2, \dots, 7$, where $W_8 = \exp\left(-j\frac{2\pi}{8}\right)$. Given the relationship

$$W_N^{nk} = \exp\left(-j\frac{2\pi}{N}nk\right) = \cos\left(\frac{2\pi}{N}nk\right) - j \sin\left(\frac{2\pi}{N}nk\right), \quad (3.79)$$

the sinusoidal nature of the basis functions used in the DFT becomes clear. Figure 3.36 shows stem plots of the sin functions as above, for $k = 0, 1, 2, \dots, 7$. For the sake of illustration, each signal was created with 64 samples, with $n = 0, 1, 2, \dots, 63$.

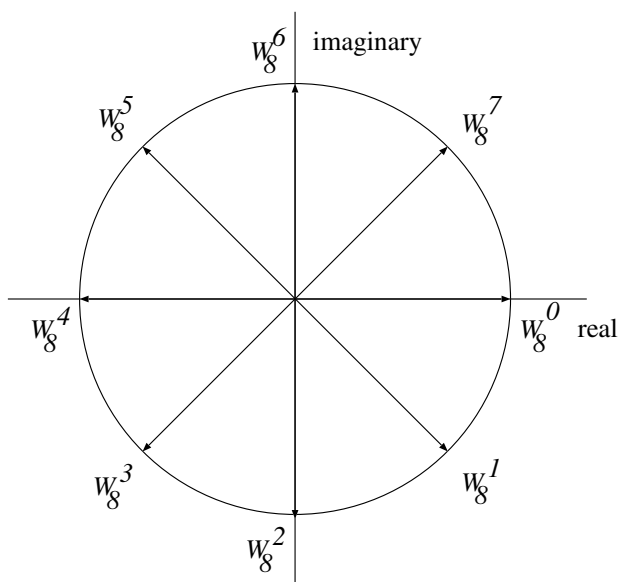


Figure 3.35 Vectors (or phasors) representing the $N = 8$ roots of unity, with W_8^k , $k = 0, 1, 2, \dots, 7$, where $W_8 = \exp\left(-j\frac{2\pi}{8}\right)$. Based upon a similar figure by Hall [175].

Let us consider the DFT relationship again, with the basis functions written in terms of the sin and cos functions as follows:

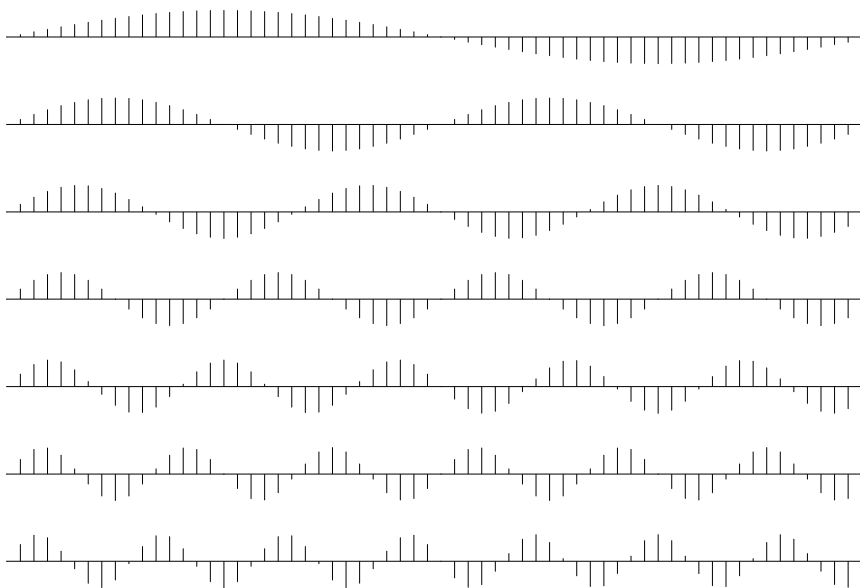


Figure 3.36 Sinusoidal basis functions used in the DFT. The eight plots (top to bottom) show the function $\sin(\frac{2\pi}{N}nk)$, for $k = 0, 1, 2, \dots, 7$, with $N = 64$. Each sinusoidal function (except for $k = 0$) varies over the range $[-1, 1]$. The axis labels have been suppressed for improved visualization.

$$\begin{aligned}
 X(k) &= \sum_{n=0}^{N-1} x(n) \exp\left(-j\frac{2\pi}{N}nk\right) \\
 &= \sum_{n=0}^{N-1} x(n) \left\{ \cos\left(\frac{2\pi}{N}nk\right) - j \sin\left(\frac{2\pi}{N}nk\right) \right\} \\
 &= \sum_{n=0}^{N-1} x(n) \cos\left(\frac{2\pi}{N}nk\right) - j \sum_{n=0}^{N-1} x(n) \sin\left(\frac{2\pi}{N}nk\right). \quad (3.80)
 \end{aligned}$$

It is evident that the real part of $X(k)$ is given by the dot product of the given signal, $x(n)$, with the k^{th} cos basis function, $\cos(\frac{2\pi}{N}nk)$. Similarly, the imaginary part of $X(k)$ is given by the dot product with the corresponding sin function. The dot product represents the projection of one vector or series of values on to another and indicates the extent of commonality between them. Thus, it is clear that the DFT coefficients indicate the amount or strength of each sinusoid present in the given signal. This is the essence of the analysis of a signal in the frequency domain.

Along similar lines, we can view the inverse DFT relationship as a regeneration or synthesis of the original signal as a weighted combination of sinusoids of various frequencies, the weights being the corresponding DFT coefficients, as follows:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) \cos\left(\frac{2\pi}{N}nk\right) + j \frac{1}{N} \sum_{k=0}^{N-1} X(k) \sin\left(\frac{2\pi}{N}nk\right), \quad (3.81)$$

for $n = 0, 1, 2, \dots, N - 1$.

The periodic nature of the frequency variable and the range of the basis functions used in the DFT relationships leads to an important property: The results of the DFT and inverse DFT operations are periodic signals. Even when we start with a discrete-time signal of finite duration that is not periodic, its spectrum obtained via the DFT is a periodic signal; furthermore, the result of application of the inverse DFT to the spectrum so obtained (and further manipulated as desired) is a periodic discrete-time signal. Unless proper attention is paid to this property, errors comparable to aliasing due to inadequate sampling of a continuous-time signal arise in the final result. Of particular interest are implications of the following property:

$$\begin{aligned} \text{if } y(n) &= x(n) * h(n), \\ \text{then } Y(k) &= X(k) H(k). \end{aligned} \quad (3.82)$$

The symmetry and periodicity of the basis functions used in the DFT can be exploited to reduce the computational requirements for the DFT, which has led to the creation of several versions of the fast Fourier transform (FFT) [15]. The two properties are expressed as follows:

$$W_N^{-nk} = (W_N^{nk})^* \quad (3.83)$$

and

$$W_N^{nk} = W_N^{n(k+N)} = W_N^{(n+N)k}. \quad (3.84)$$

The advantages provided by FFT algorithms are so substantial that it is usually computationally less expensive to use the second line of Equation 3.82 to realize convolution than the original definition of convolution itself. The algorithm to achieve convolution via the DFT is as follows:

1. Compute the DFTs, $X(k)$ and $H(k)$, of the two signals given, $x(n)$ and $h(n)$.
2. Multiply the two DFTs, sample by sample, to get the DFT of the output, $Y(k) = X(k) H(k)$.
3. Compute the inverse DFT of $Y(k)$. The output signal, $y(n)$, is given by the real part of the result.

However, it should be noted that the convolution achieved in this manner is circular or periodic convolution; the results, in general, will not be the same as those

provided by linear convolution. This arises due to the fact that the results of DFT and inverse DFT operations are periodic entities. When we are interested in computing the output of an LSI system, we need to perform linear convolution. To understand the differences between the two types of convolution, let us examine the following examples in detail.

Figure 3.37 shows a numerical illustration of periodic or circular shift applied to a signal. Comparing this figure with Figure 3.18 facilitates understanding the differences between circular and linear shifting.

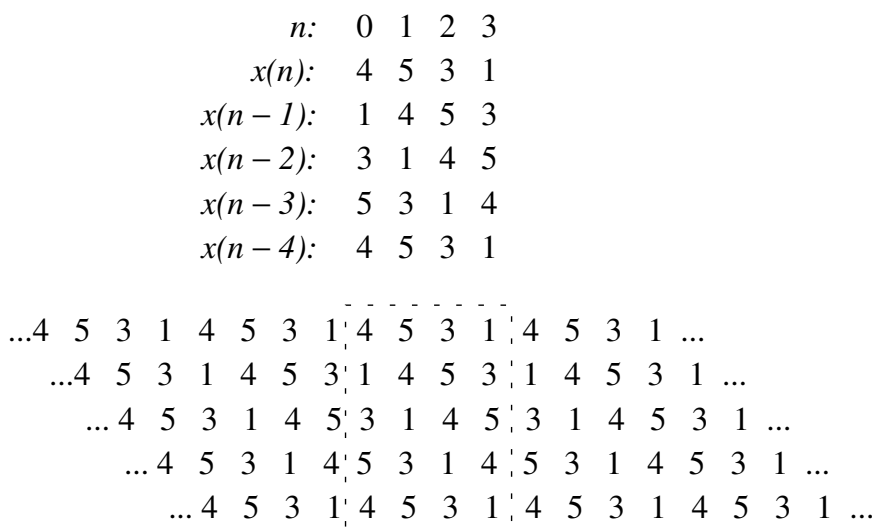


Figure 3.37 A numerical illustration of circularly shifted versions of a periodic signal. The window shown in dashed lines on the lower set of signals over four cycles represents one period. Shifted versions of the signal are shown for one period in the upper part of the figure. The circular nature of the shift is evident in the upper illustration, even if the shift in the lower illustration is considered to be linear. This type of shifting is known as circular or periodic shifting. A sample that gets shifted out of the observation window on the right reenters the window from the left. See Figure 3.18 for an example of linear shifting.

In view of circular shifting of periodic signals, we can express the convolution of two periodic signals, having the same period of N samples, as

$$y_p(n) = \sum_{k=0}^{N-1} x_p(k) h_p[(n-k) \bmod N], \quad (3.85)$$

where the subscript p indicates that the signals are periodic, and $[(n-k) \bmod N]$ returns values within the range $[0, N-1]$, by adding or subtracting N , as required. The result is also periodic, having the same period of N samples. Figure 3.38 shows a numerical example of circular or periodic convolution of two periodic discrete-time signals.

n :	0	1	2	3
$x(n)$:	4	1	3	1
$h(n)$:	4	3	2	1
k :	0	1	2	3
$x(k)$:	4	1	3	1
$h(0-k)$:	4	1	2	3
$h(1-k)$:	3	4	1	2
$h(2-k)$:	2	3	4	1
$h(3-k)$:	1	2	3	4
$y(n)$:	26	21	24	19
n :	0	1	2	3

Figure 3.38 A numerical illustration of the circular or periodic convolution of two periodic discrete-time signals.

Given two signals, $x(n)$, $n = 0, 1, 2, \dots, N_1$, and $h(n)$, $n = 0, 1, 2, \dots, N_2$, we know that the result of linear convolution of the two signals should have $N_1 + N_2 - 1$ samples. The application of periodic convolution in place of linear convolution leads to erroneous results; compare the numerical example in Figure 3.39 with that in Figure 3.20.

n :	0	1	2	3
$x(n)$:	4	1	3	1
$h(n)$:	3	2	1	0
k :	0	1	2	3
$x(k)$:	4	1	3	1
$h(0-k)$:	3	0	1	2
$h(1-k)$:	2	3	0	1
$h(2-k)$:	1	2	3	0
$h(3-k)$:	0	1	2	3
$y(n)$:	17	12	15	10
n :	0	1	2	3

Figure 3.39 A numerical illustration of the undesired effect of periodic convolution of two discrete-time signals when linear convolution is desired. Compare this with the illustration of linear convolution of the same signals in Figure 3.20.

In order to gain the computational benefits provided by the FFT, the DFT approach to convolution may be modified by padding two given nonperiodic signals of unequal duration with zeros so as to have the same extended duration as that of the desired result, and then considering them to represent one period each of their periodic extensions. The algorithm to achieve linear convolution via the DFT is as follows:

1. Extend the two signals given, $x(n)$ and $h(n)$, to have N samples each, with $N \geq N_1 + N_2 - 1$, by appending or padding with zeros.
2. Compute the DFTs, $X(k)$ and $H(k)$, of the two extended signals, for $k = 0, 1, 2, \dots, N - 1$.
3. Multiply the two DFTs, sample by sample, to get the DFT of the output, $Y(k) = X(k) H(k)$, $k = 0, 1, 2, \dots, N - 1$.
4. Compute the inverse DFT of $Y(k)$. The output signal, $y(n)$, for $n = 0, 1, 2, \dots, N_1 + N_2 - 1$, is given by the real part of the result.

Note that the sample-by-sample product given as $Y(k) = X(k) H(k)$ cannot be computed if $X(k)$ and $H(k)$ are of different lengths (or periods). The final result $y(n)$ will, in general be complex, and may have extra values for $N_1 + N_2 - 1 < n < N$ that are not of interest. Figure 3.40 shows a numerical example of circular or periodic convolution of two nonperiodic discrete-time signals. By comparing the illustration in Figure 3.40 with the related case of linear convolution of the same signals in Figure 3.20, it is evident that the final results are the same.

3.4.5 Properties of the Fourier transform

The Fourier transform is the most commonly used transform to study the frequency-domain characteristics of signals [1–3, 15, 174]. This is mainly because the Fourier transform uses sinusoidal functions as its basis functions. Projections are computed of the given signal $x(t)$ on to the complex exponential basis function of frequency ω radians/s, given by $\exp(j\omega t) = \cos(\omega t) + j \sin(\omega t)$, as

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt, \quad (3.86)$$

or in the frequency variable f in Hz as

$$X(f) = \int_{-\infty}^{\infty} x(t) \exp(-j2\pi ft) dt. \quad (3.87)$$

[The complex exponential function is conjugated in computing the projection. In some fields, the forward Fourier transform is defined with $\exp(+j\omega t)$ in the integral.] Equations 3.86 and 3.87 represent *analysis* of the signal $x(t)$ with reference to the complex exponential basis functions. The lower limit of the integral will be 0 if the

$n:$	0	1	2	3	4	5
$x(n):$	4	1	3	1	0	0
$h(n):$	3	2	1	0	0	0
$k:$	0	1	2	3	4	5
$x(k):$	4	1	3	1	0	0
$h(0-k):$	3	0	0	0	1	2
$h(1-k):$	2	3	0	0	0	1
$h(2-k):$	1	2	3	0	0	0
$h(3-k):$	0	1	2	3	0	0
$h(4-k):$	0	0	1	2	3	0
$h(5-k):$	0	0	0	1	2	3
$y(n):$	12	11	15	10	5	1
$n:$	0	1	2	3	4	5

Figure 3.40 A numerical illustration of the use of periodic convolution of two discrete-time and nonperiodic signals to achieve linear convolution by adequate zero padding. Compare this with the illustration of linear convolution of the same signals in Figure 3.20. With the expected number of samples in the result being six, the two input signals, with four and three samples each, have been padded with zeros to the same length of six samples. The effective period used in periodic convolution is six samples.

signal is causal; the upper limit will be equal to the duration of the signal in the case of a finite-duration signal. The value of $X(\omega)$ or $X(f)$ at each frequency of interest $\omega = 2\pi f$ represents the amount or strength of the corresponding cosine and sine functions present in the signal $x(t)$. Note that, in general, $X(\omega)$ is complex for real signals, and includes the magnitude and phase of the corresponding complex exponential.

The inverse transformation, representing *synthesis* of the signal $x(t)$ as a weighted combination of the complex exponential basis functions, is given as

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \exp(j\omega t) d\omega = \int_{-\infty}^{\infty} X(f) \exp(j2\pi ft) df. \quad (3.88)$$

The second version of the equation given above with the frequency variable f in Hz may be more convenient in some situations than the first one with ω in *radians/s*, due to the absence of the $\frac{1}{2\pi}$ factor. [If the forward Fourier transform is defined with $\exp(+j\omega t)$, the inverse Fourier transform will have $\exp(-j\omega t)$ in the integral.]

In the case of a discrete-time signal $x(n)$, we may still compute the Fourier transform with a continuous frequency variable ω as

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) \exp(-j\omega n), \quad (3.89)$$

with the normalized-frequency range $0 \leq \omega \leq 2\pi$ (equivalent to $0 \leq f \leq 1$). The lower limit of the summation will be 0 if the signal is causal. The upper limit of the summation will be equal to the index $(N - 1)$ of the last sample in the case of a finite-duration signal with N samples. Various relationships between the Laplace transform, the Fourier transform, and the z -transform, as well as sampling of the z -transform to obtain the DFT, are described in Section 3.4.2.

Some of the important properties of the DFT and their implications are summarized below [1–3, 15, 174].

- A signal $x(n)$ and its DFT $X(k)$ are both periodic sequences.
- If a signal $x(n)$ has N samples, its DFT $X(k)$ must be computed with at least N samples equally spaced over the normalized frequency range $0 \leq \omega \leq 2\pi$ (or, equivalently, around the unit circle in the z -plane) for complete representation and determination of $X(\omega)$, and hence exact reconstruction of $x(n)$ via the inverse DFT of $X(k)$. One may use more than N samples to compute $X(k)$ in order to employ an FFT algorithm with $L = 2^M \geq N$ samples, where M is an integer, or to obtain $X(\omega)$ with finer frequency sampling than $2\pi/N$.
- The DFT is linear: the DFT of $ax(n) + by(n)$ is $aX(k) + bY(k)$, where $X(k)$ and $Y(k)$ are the DFTs of $x(n)$ and $y(n)$, respectively.
- The DFT of $x(n - n_o)$ is $\exp(-j\frac{2\pi}{N}kn_o)X(k)$, where $X(k)$ is the DFT of $x(n)$. A time shift leads to a linear component being added to the phase of the original signal. As all sequences in DFT relationships are periodic, the shift operation should be defined as a circular or periodic shift. If at least n_o zeros are present or are padded at the end of the signal before the shift operation, a circular shift will be equivalent to a linear shift.
- The DFT of $x(n) * h(n)$ is $X(k)H(k)$, where $X(k)$ and $H(k)$ are the DFTs of $x(n)$ and $h(n)$, respectively. The inverse DFT of $X(k)H(k)$ is $x(n) * h(n)$. Similarly, $x(n)h(n)$ and $X(k) * H(k)$ form a DFT pair. Convolution in one domain is equivalent to multiplication in the other. It is necessary for all of the signals in the relationships mentioned above to have the same number of samples N .

As all sequences in DFT relationships are periodic, the convolution operations in the relationships mentioned here are *periodic convolution* and not linear convolution. Note that circular or periodic convolution is defined for periodic signals having the same period, and that the result will also be periodic with the same period as that of the individual input signals.

The result of linear convolution of two signals $x(n)$ and $h(n)$ with different durations N_x and N_h samples, respectively, will have a duration of $N_x + N_h - 1$ samples. If linear convolution is desired via the inverse DFT of $X(k)H(k)$, the DFTs must be computed with $L \geq N_x + N_h - 1$ samples. The individual signals should be padded with zeros at the end to make their effective durations equal

for the sake of DFT computation and multiplication. All signals and their DFTs are then periodic with the augmented period of L samples.

- The DFT of a real signal $x(n)$ possesses conjugate symmetry, that is, $X(-k) = X^*(k)$. As a consequence, the real part and the magnitude of $X(k)$ are even-symmetric sequences, whereas the imaginary part and the phase of $X(k)$ are odd-symmetric sequences.
- According to Parseval's theorem, the total energy of the signal must remain the same before and after Fourier transformation. We then have the following equalities:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega, \quad (3.90)$$

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2.$$

Because the integral of $|X(\omega)|^2$ over all ω or the sum of $|X(k)|^2$ over all k represents the total energy of the signal (or average power, if the quantity is divided by the duration of the signal), $|X(\omega)|^2$ and $|X(k)|^2$ represent the spread or *density* of the power of the signal along the frequency axis.

Note: A signal or function $x(n)$ possesses even symmetry if $x(-n) = x(n)$ or odd symmetry if $x(-n) = -x(n)$. An arbitrary signal $x(n)$ may be expressed as a combination of a part with even symmetry, $x_e(n)$, and a part with odd symmetry, $x_o(n)$, with the even part given by

$$x_e(n) = \frac{1}{2}[x(n) + x(-n)] \quad (3.91)$$

and the odd part given by

$$x_o(n) = \frac{1}{2}[x(n) - x(-n)]. \quad (3.92)$$

Then, we have

$$x(n) = x_e(n) + x_o(n). \quad (3.93)$$

See Figure 3.8 for an illustration of the PSD of the ECG signal in Figure 3.7 computed via the FFT. Several examples of Fourier spectra and PSDs are given in the sections that follow.

The remaining sections in the present chapter provide details of application of the various concepts described in the present section to filtering of biomedical signals for several purposes.

3.5 Time domain Filters

Certain types of noise may be filtered directly in the time domain using signal processing techniques or digital filters. An advantage of time-domain filtering is that spectral characterization of the signal and noise may not be required (at least in a direct manner). Time-domain processing may also be faster in most cases than frequency-domain filtering.

3.5.1 Synchronized averaging

Problem: *Propose a time-domain technique to remove random noise given the possibility of acquiring multiple realizations of the signal or event of interest.*

Solution: Linear filters fail to perform well or are not applicable when the signal and noise spectra overlap. Synchronized signal averaging can separate a repetitive signal from noise without distorting the signal [46, 171]. ERP or SEP epochs may be obtained a number of times by repeated application of the stimulus; they may then be averaged by using the stimulus as a trigger for aligning the epochs. ECG signals may be filtered by detecting the QRS complexes and using their positions to align the waveforms for synchronized averaging. If the noise is random with zero mean and is uncorrelated with the signal, averaging will improve the SNR .

Let $y_k(n)$ represent one realization of a signal, with $k = 1, 2, \dots, M$ representing the ensemble index, and $n = 0, 1, 2, \dots, N - 1$ representing the time-sample index. Here, M is the number of copies (events, epochs, observations, or realizations) of the signal available, and N is the number of time samples in each copy of the signal (event). We may express the observed signal as

$$y_k(n) = x_k(n) + \eta_k(n), \quad (3.94)$$

where $x_k(n)$ represents the original uncorrupted signal and $\eta_k(n)$ represents the noise in the k^{th} observed signal. Now, if for each instant of time n we add the M copies of the signal, we get

$$\sum_{k=1}^M y_k(n) = \sum_{k=1}^M x_k(n) + \sum_{k=1}^M \eta_k(n); \quad n = 0, 1, 2, \dots, N - 1. \quad (3.95)$$

If the repetitions of the signal are identical and aligned, $\sum_{k=1}^M x_k(n) = Mx(n)$. If the noise is random and has zero mean and variance σ_η^2 , $\sum_{k=1}^M \eta_k(n)$ will tend to zero as M increases, with a variance of $M\sigma_\eta^2$. The RMS value of the noise in the summed signal is $\sqrt{M}\sigma_\eta$. Thus, the SNR of the signal will increase by a factor of $\frac{M}{\sqrt{M}}$ or $\sqrt{M}$. The larger the number of epochs or realizations that are averaged, the better will be the SNR of the result. Note that synchronized averaging is a type of ensemble averaging.

An algorithmic description of synchronized averaging is as follows:

1. Obtain a number of realizations of the signal or event of interest.

2. Determine a reference point for each realization of the signal. This is directly given by the trigger if the signal is obtained by external stimulation (such as ERPs or SEPs), or may be obtained by detecting the repetitive events in the signal if it is quasiperiodic (such as the QRS complex in the ECG or S1 and S2 in the PCG).
3. Extract parts of the signal corresponding to the events and add them to a buffer. Note that it is possible for the various parts to be of different durations. Alignment of the copies at the trigger point is important; the tail ends of all parts may not be aligned.
4. Divide the result in the buffer by the number of events added.

Illustrations of application: Figure 3.41 illustrates two single-flash ERPs in the upper two traces. The results of averaging over 10 and 20 flashes are shown in the third and fourth plots, respectively, in the same figure. The averaging process has facilitated identification of the first positivity and the preceding and succeeding troughs (marked on the fourth trace) with certainty; the corresponding features are not reliably seen in the single acquisitions (see also the single-flash ERPs in Figure 3.2). Visual ERPs are analyzed in terms of the latencies of the first major peak or positivity, labeled as P120 due to the fact that the normal expected latency for adults is 120 *ms*; the trough or negativity before P120, labeled as N80; and the trough following P120, labeled as N145. The N80, P120, and N145 latencies measured from the averaged signal in Trace 4 of Figure 3.41 are 85.7, 100.7, and 117 *ms*, respectively, which are considered to be within the normal range for adults.

The upper trace in Figure 3.42 illustrates a noisy ECG signal over several beats. In order to obtain trigger points, a sample QRS complex of 86 *ms* duration (86 samples at a sampling rate of 1,000 *Hz*) was extracted from the first beat in the signal and used as a template. Template matching was performed using a normalized correlation coefficient defined as [171]

$$\gamma_{xy}(k) = \frac{\sum_{n=0}^{N-1} [x(n) - \bar{x}][y(k - N + 1 + n) - \overline{y_k}]}{\sqrt{\sum_{n=0}^{N-1} [x(n) - \bar{x}]^2 \sum_{n=0}^{N-1} [y(k - N + 1 + n) - \overline{y_k}]^2}}, \quad (3.96)$$

where

$$\overline{y_k} = \frac{1}{N} \sum_{n=0}^{N-1} y(k - N + 1 + n),$$

that is, $\overline{y_k}$ is the average of the part of the signal y being used in the template matching procedure. The operation given above is causal, and it is valid from $k = N$ to the last sample of the signal y . x is the template, and $\bar{x}$ is the average value of x . k is the time index of the signal y at which the template is placed. {Jenkins et al. [132] used a measure similar to $\gamma_{xy}(k)$ but without subtraction of the mean and without the shift parameter k to match segmented ECG cycles with a template.} The lower trace in Figure 3.42 shows $\gamma_{xy}(k)$, where it is seen that the cross-correlation result peaks to values near unity at the locations of the QRS complexes in the signal. Averaging

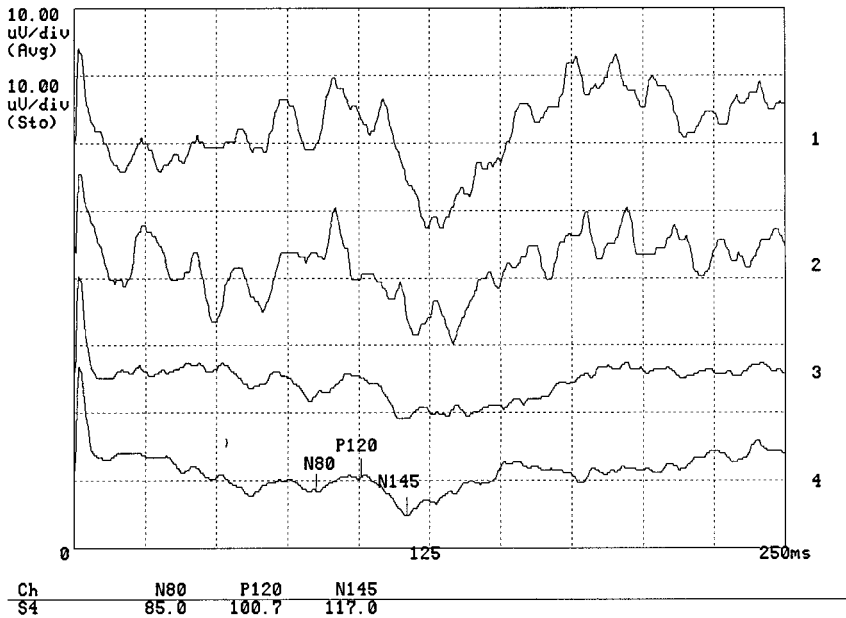


Figure 3.41 Traces 1 and 2: Two sample acquisitions of individual flash visual ERPs from the occipital midline (oz) position of a normal adult male. The earlobes were used to form the ground lead (a1a2), and the left forehead was used as the reference (see Figure 1.37). Trace 3: Average of 10 ERPs. Trace 4: Average of 20 ERPs. The latencies of interest have been labeled on Trace 4 by an EEG technologist as N80, 85.0 ms; P120, 100.7 ms; and N145, 117.0 ms. The height of each grid division (box) is 10 μV and the width is 25 ms. See also Figure 3.2. Data courtesy of L. Alfaro and H. Darwish, Alberta Children's Hospital, Calgary.

inherent in the cross-correlation formula (over N samples) has reduced the effect of noise on template matching.

By choosing an appropriate threshold, it becomes possible to obtain a trigger point to extract the QRS complex locations in the ECG signal. (*Note:* The QRS template matches with the P and T waves with cross-correlation values of about 0.5; wide QRS complexes may yield high cross-correlation values with tall P and T waves. The threshold has to be chosen so as to detect only the QRS complexes.) A threshold of 0.9 was applied to $\gamma_{xy}(k)$, and the QRS positions of all of the 12 beats in the signal were detected.

Figure 3.43 illustrates two ECG cycles extracted using the trigger points obtained by thresholding the cross-correlation function, as well as the result of averaging the first 11 cycles in the signal. It is seen that the noise has been effectively suppressed by synchronized averaging. The low-level baseline variation and power-line interference present in the signal have caused minor artifacts in the result, which are negligible in this illustration.

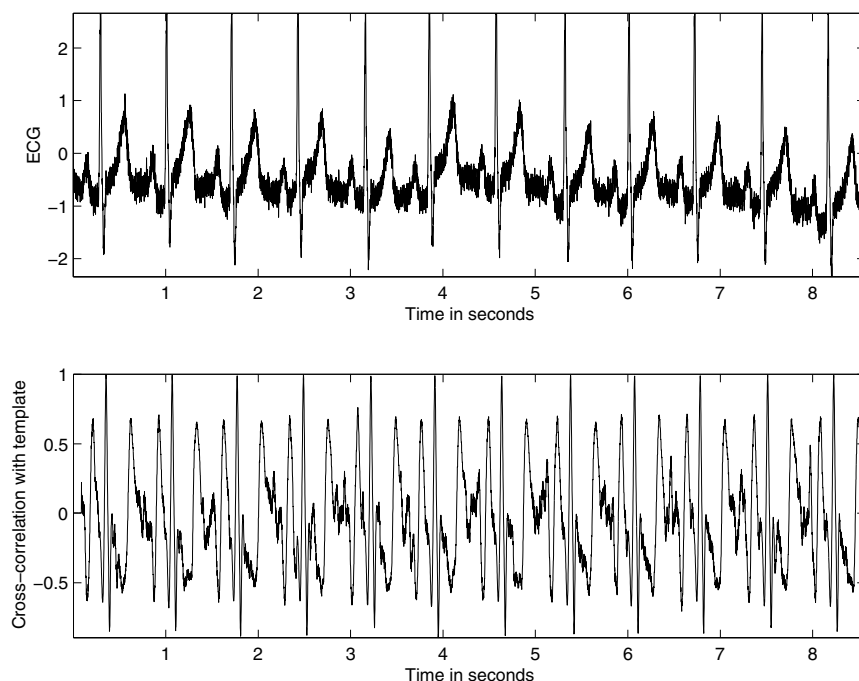


Figure 3.42 An ECG signal with noise (upper trace) and the result of cross-correlation (lower trace) with the QRS template selected from the first cycle. The cross-correlation coefficient is normalized to the range $[-1, 1]$.

The most important requirement in synchronized averaging is indicated by the first word in the name of the process: The realizations of the signal that are added for averaging *must be synchronized or aligned* such that the repetitive part of the signal appears at exactly the same instant in each realization of the signal. If this condition is not met, the waveform of the event in the signal will be blurred or smudged along the time axis.

A major advantage of synchronized averaging is that no frequency-domain filtering is performed — either explicitly or implicitly. No spectral content of the signal is lost as is the case with frequency-domain (lowpass) filters or other time-domain filters such as moving-window averaging filters.

Structured noise such as power-line interference may be suppressed by synchronized averaging if the phase of the interference in each realization is different. To facilitate this feature, the repetition rate of the stimulus should be set so that it is not directly related to the power-line frequency (for example, the flashes used to acquire the averaged ERPs in Figure 3.41 were delivered at 2.1 pps). Physiological interference such as background EEG in ERPs and SEPs may also be suppressed by synchronized averaging, as such activity may bear no interrelationship from one

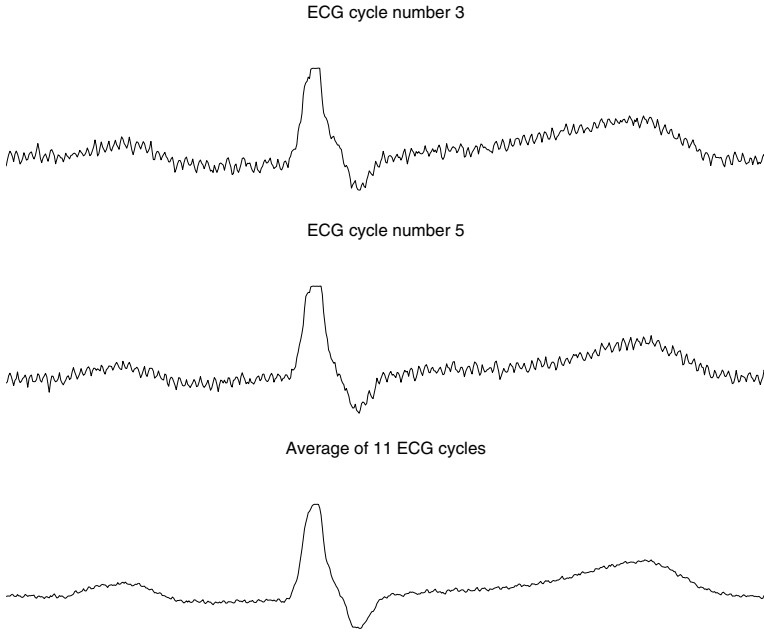


Figure 3.43 Upper two traces: two cycles of the ECG extracted from the signal in Figure 3.42. Bottom trace: the result of synchronized averaging of 11 cycles from the same ECG signal.

epoch of the desired signal to another. See Sections 3.11, 4.11, 5.5.2, and 6.3.5 for further applications and discussions on synchronized averaging.

3.5.2 MA filters

Problem: *Propose a time-domain technique to remove random noise given only one realization of the signal or event of interest.*

Solution: When an ensemble of several realizations of an event is not available, synchronized averaging will not be possible. We are then forced to consider temporal averaging for noise removal, with the assumption that the processes involved are ergodic, that is, temporal statistics may be used instead of ensemble statistics. As temporal statistics are computed using a few samples of the signal along the time axis and the temporal window of samples is moved to obtain the output at various points of time, such a filtering procedure is called a moving-window averaging filter in general; the term MA filter is commonly used. See Figures 3.22, 3.24, and 3.25 as well as the related discussions.

The general form of the difference equation of an MA filter is

$$y(n) = \sum_{k=0}^N b_k x(n - k), \quad (3.97)$$

where x and y are the input and output of the filter, respectively. The b_k values are the filter coefficients or tap weights, $k = 0, 1, 2, \dots, N$, where N is the order of the filter. The effect of division by the number of samples used ($N + 1$) is included in the values of the filter coefficients. The signal-flow diagram of a generic MA filter is shown in Figure 3.44.

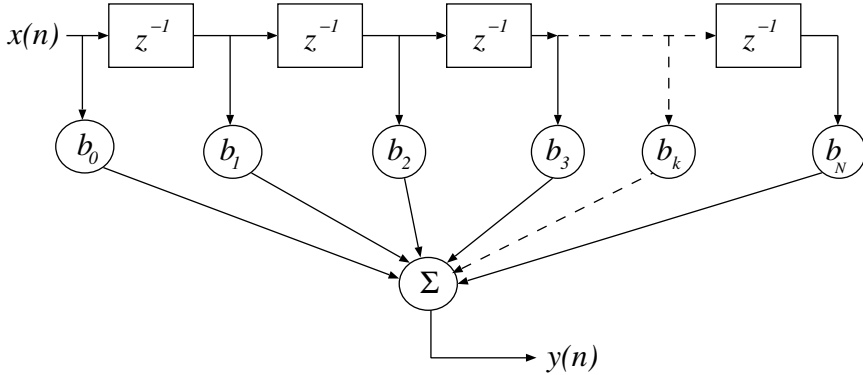


Figure 3.44 Signal-flow diagram of an MA filter of order N . Each block with the symbol z^{-1} represents a delay of one sample, and it serves as a memory unit for the corresponding signal sample value.

Applying the z -transform, we get the transfer function $H(z)$ of the filter as

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^N b_k z^{-k} = b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_N z^{-N}, \quad (3.98)$$

where $X(z)$ and $Y(z)$ are the z -transforms of $x(n)$ and $y(n)$, respectively.

A simple MA filter for filtering noise is the von Hann (also known as Hann or Hanning) filter [46], given by

$$y(n) = \frac{1}{4}[x(n) + 2x(n-1) + x(n-2)]. \quad (3.99)$$

The signal-flow diagram of the Hann filter is shown in Figure 3.45. The impulse response of the filter is obtained by letting $x(n) = \delta(n)$, resulting in

$$h(n) = \frac{1}{4}[\delta(n) + 2\delta(n-1) + \delta(n-2)]; \quad (3.100)$$

see Figure 3.46.

Applying the z -transform to Equation 3.99, we get

$$Y(z) = \frac{1}{4}[X(z) + 2z^{-1}X(z) + z^{-2}X(z)], \quad (3.101)$$

which leads to the derivation of the transfer function of the Hann filter as

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{4}[1 + 2z^{-1} + z^{-2}]. \quad (3.102)$$

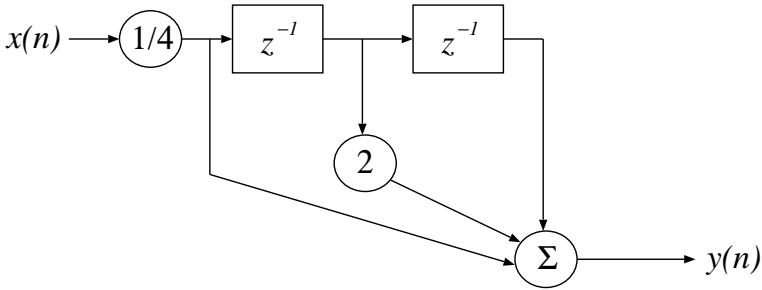


Figure 3.45 Signal-flow diagram of the Hann filter.

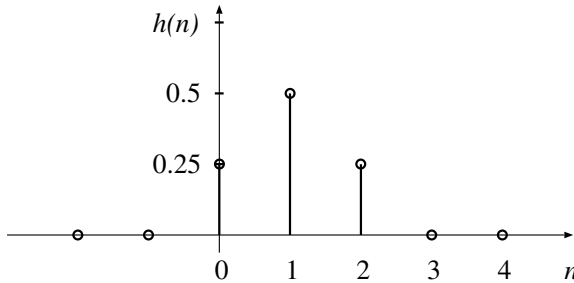


Figure 3.46 Impulse response of the Hann filter.

The transfer function has a double-zero at $z = -1$.

An MA filter is an FIR filter with the following attributes and advantages:

- The impulse response $h(k)$ has a finite number of terms: $h(k) = b_k$, $k = 0, 1, 2, \dots, N$.
- An FIR filter may be realized nonrecursively (with no feedback).
- The output depends only on the present input sample and a few past input samples.
- The filter is defined by a set of tap weights of the delay stages, as illustrated in Figure 3.44.
- The filter transfer function has no poles except at $z = 0$: the filter is inherently stable.
- The filter has linear phase if the series of tap weights is symmetric or antisymmetric.

(Note: A recursive filter uses previous values of the output to compute the current output value.)

The frequency response of a filter is obtained by substituting $z = e^{j\omega T}$ in the expression for $H(z)$, where T is the sampling interval in seconds and ω is the radian frequency ($\omega = 2\pi f$, where f is the frequency in Hz). Note that we may set $T = 1$ and deal with normalized frequency in the range $0 \leq \omega \leq 2\pi$ or $0 \leq f \leq 1$; then, $f = 1$ or $\omega = 2\pi$ represents the sampling frequency, with lower frequency values being represented as a normalized fraction of the sampling frequency.

The frequency response of the Hann filter is obtained as

$$H(\omega) = H(z)|_{z=\exp(j\omega)} = \frac{1}{4}[1 + 2e^{-j\omega} + e^{-j2\omega}]. \quad (3.103)$$

This expression can be simplified as follows:

$$\begin{aligned} H(\omega) &= \frac{1}{4}[1 + 2e^{-j\omega} + e^{-j2\omega}] \\ &= \frac{1}{4}e^{-j\omega}[e^{+j\omega} + 2 + e^{-j\omega}] \\ &= \frac{1}{4}e^{-j\omega}[2\cos(\omega) + 2] \\ &= \frac{1}{2}[1 + \cos(\omega)]e^{-j\omega}. \end{aligned} \quad (3.104)$$

Note that $e^{\pm j\omega} = \cos(\omega) \pm j\sin(\omega)$, $e^{+j\omega} + e^{-j\omega} = 2\cos(\omega)$, and $e^{+j\omega} - e^{-j\omega} = 2j\sin(\omega)$. The magnitude and phase of the frequency response are

$$|H(\omega)| = \left| \frac{1}{2}[1 + \cos(\omega)] \right| \quad (3.105)$$

and

$$\angle H(\omega) = -\omega. \quad (3.106)$$

The magnitude and phase responses of the Hann filter are plotted in Figure 3.47. It is clear that the filter is a lowpass filter with linear phase.

Note that, although we started with a description of the Hann filter in the time domain, subsequent analysis of the filter was performed in the frequency domain using the z -transform and the frequency response. System analysis is easier to perform in the z domain in terms of the poles and zeros of the transfer function and in the frequency domain in terms of the magnitude and phase responses. The magnitude and phase responses assist in understanding the effect of the filter on the frequency components of the signal (and noise).

It is seen from the magnitude response of the Hann filter (Figure 3.47) that components beyond about 20% of the sampling frequency of 1,000 Hz are reduced in amplitude by more than 3 dB , that is, to less than one-half of their levels in the input. High-frequency components beyond 40% of the sampling frequency are suppressed to less than 20 dB below their input levels. The filter will perform adequate filtering of ECG signals sampled at 200 Hz , with the gain being lower than -20 dB beyond 80 Hz . However, if the signal is sampled at 1,000 Hz (as in the present example),

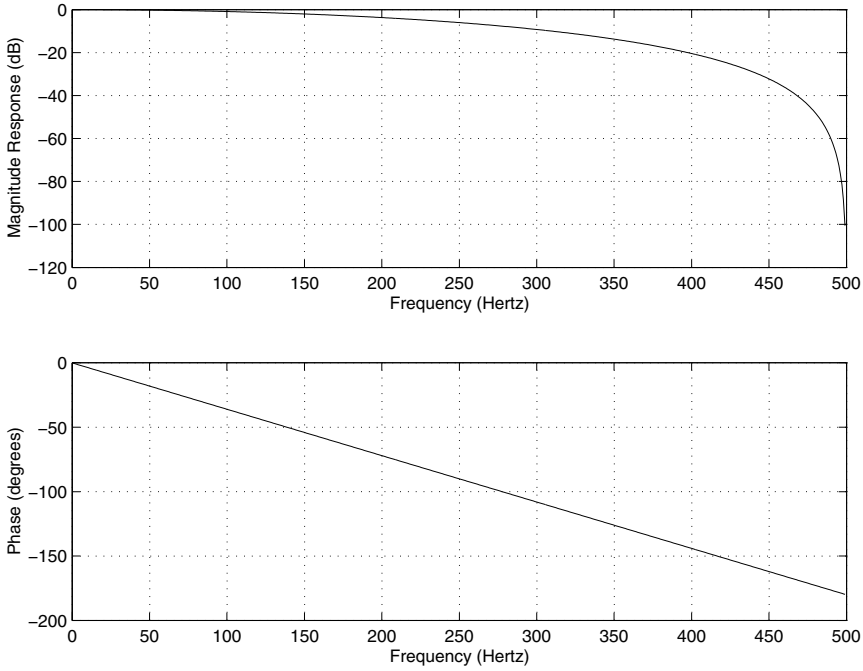


Figure 3.47 Magnitude and phase responses of the Hann (smoothing) filter.

the gain remains above -20 dB for frequencies up to 400 Hz; such a lowpass filter may not be adequate for filtering ECG signals, but may be appropriate for other signals, such as the PCG and the EMG.

Increased smoothing may be achieved by averaging signal samples over longer time windows, at the expense of increased filter delay. If the signal samples over a window of eight samples are averaged, we get the 8-point MA filter with the output defined as

$$\begin{aligned}
 y(n) &= \frac{1}{8} \sum_{k=0}^7 x(n-k) \\
 &= \frac{1}{8} [x(n) + x(n-1) + x(n-2) + x(n-3) + x(n-4) \\
 &\quad + x(n-5) + x(n-6) + x(n-7)].
 \end{aligned} \tag{3.107}$$

The impulse response of the filter is

$$\begin{aligned}
 h(n) &= \frac{1}{8} [\delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3) + \delta(n-4) \\
 &\quad + \delta(n-5) + \delta(n-6) + \delta(n-7)].
 \end{aligned} \tag{3.108}$$

The transfer function of the filter is

$$H(z) = \frac{1}{8} \sum_{k=0}^7 z^{-k}, \quad (3.109)$$

and the frequency response is given by

$$\begin{aligned} H(\omega) &= \frac{1}{8} \sum_{k=0}^7 \exp(-j\omega k) \\ &= \frac{1}{8} \{1 + \exp(-j4\omega) \\ &\quad \times [1 + 2 \cos(\omega) + 2 \cos(2\omega) + 2 \cos(3\omega)]\}. \end{aligned} \quad (3.110)$$

The frequency response of the 8-point MA filter is shown in Figure 3.48; the pole-zero plot of the filter is depicted in Figure 3.49. It is seen that the filter has zeros at $\frac{f_s}{8} = 125 \text{ Hz}$, $\frac{f_s}{4} = 250 \text{ Hz}$, $\frac{3f_s}{8} = 375 \text{ Hz}$, and $\frac{f_s}{2} = 500 \text{ Hz}$. Comparing the frequency response of the 8-point MA filter with that of the Hann filter in Figure 3.47, we see that the former provides increased attenuation in the range $90 - 400 \text{ Hz}$ over the latter. Note that the attenuation provided by the filter after about 100 Hz is nonuniform, which may not be desirable in certain applications. Furthermore, the phase response of the filter is not linear, although it is piecewise linear.

Relationship of MA filtering to integration: Disregarding the $\frac{1}{8}$ scale factor for a moment, the operation in Equation 3.107 may be interpreted as the summation or integration of the signal over the duration $n - 7$ to n . A comparable integration of a continuous-time signal $x(t)$ over the interval $t - \tau$ to t is expressed as

$$y(t) = \int_{t-\tau}^t x(t) dt. \quad (3.111)$$

The general definition of the integral of a signal is

$$y(t) = \int_{-\infty}^t x(t) dt, \quad (3.112)$$

or, if the signal is causal,

$$y(t) = \int_0^t x(t) dt. \quad (3.113)$$

The Fourier transforms of the signals in the relationship above are related as [1–3]

$$Y(\omega) = \frac{1}{j\omega} X(\omega) + \pi X(0)\delta(\omega). \quad (3.114)$$

The frequency response of the integration operator is

$$H(\omega) = \frac{1}{j\omega}, \quad (3.115)$$

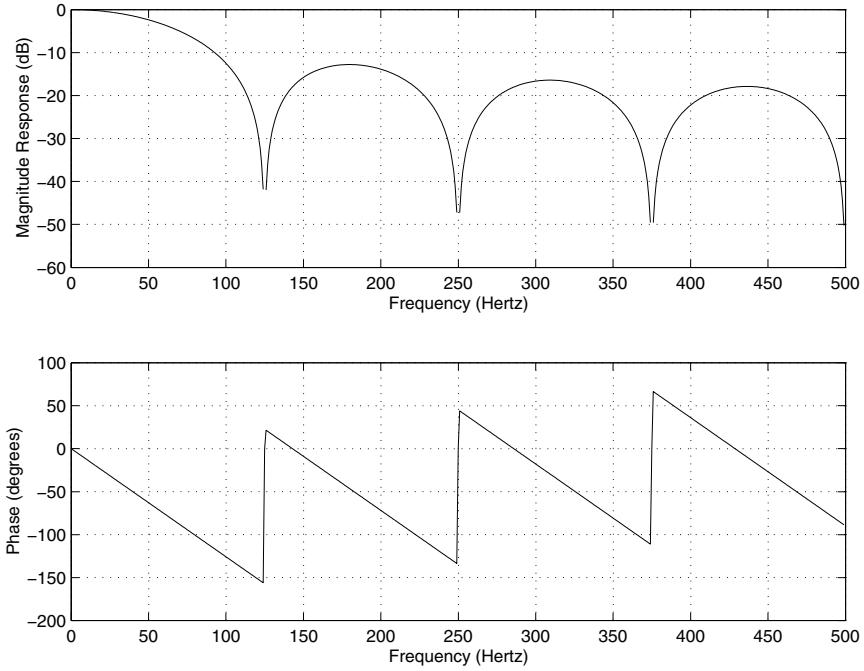


Figure 3.48 Magnitude and phase responses of the 8-point MA (smoothing) filter.

with the magnitude response

$$|H(\omega)| = \left| \frac{1}{\omega} \right| \quad (3.116)$$

and phase response

$$\angle H(\omega) = -\frac{\pi}{2}. \quad (3.117)$$

It is seen from the frequency response that the gain of the filter reduces (nonlinearly) as the frequency is increased; therefore, the corresponding filter has lowpass characteristics.

Integration or accumulation of a discrete-time signal for all samples up to the present sample results in the transfer function $H(z) = \frac{1}{1-z^{-1}} [1-3]$. Such an operation is seldom used in practice. Instead, a moving-window sum is computed, such as the 8-point MA filter in Equation 3.107. It follows from Equation 3.107 that

$$\begin{aligned} y(n-1) &= \frac{1}{8}[x(n-1) + x(n-2) + x(n-3) + x(n-4) + x(n-5) \\ &\quad + x(n-6) + x(n-7) + x(n-8)]. \end{aligned} \quad (3.118)$$

By combining Equations 3.107 and 3.118, we get the relationship

$$y(n) = y(n-1) + \frac{1}{8}x(n) - \frac{1}{8}x(n-8). \quad (3.119)$$

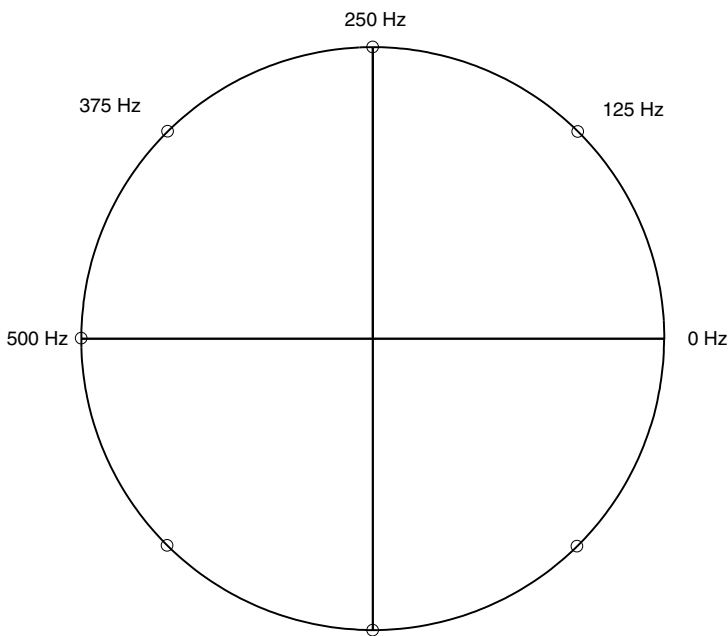


Figure 3.49 Pole-zero plot of the 8-point MA (smoothing) filter.

The recursive form as above clearly depicts the integration aspect of the filter. The transfer function of this expression is easily derived to be

$$H(z) = \frac{1}{8} \left[\frac{1 - z^{-8}}{1 - z^{-1}} \right]. \quad (3.120)$$

The frequency response of the filter is given by

$$H(\omega) = \frac{1}{8} \left[\frac{1 - e^{-j8\omega}}{1 - e^{-j\omega}} \right] = \frac{1}{8} e^{-j\frac{7}{2}\omega} \left[\frac{\sin(4\omega)}{\sin(\frac{\omega}{2})} \right], \quad (3.121)$$

which is equivalent to that in Equation 3.110. Summation over a limited discrete-time window results in a frequency response having sinc-type characteristics, as illustrated in Figure 3.48. See Tompkins [46] for a discussion on other types of integrators.

Illustration of application: Figure 3.50 shows a segment of an ECG signal with high-frequency noise. Figure 3.51 shows the result of filtering the signal with the 8-point MA filter described above. Although the noise level has been reduced, some noise is still present in the result. This is due to the fact that the attenuation of the simple 8-point MA filter is not more than -20 dB at most frequencies (except near the zeros of the filter).

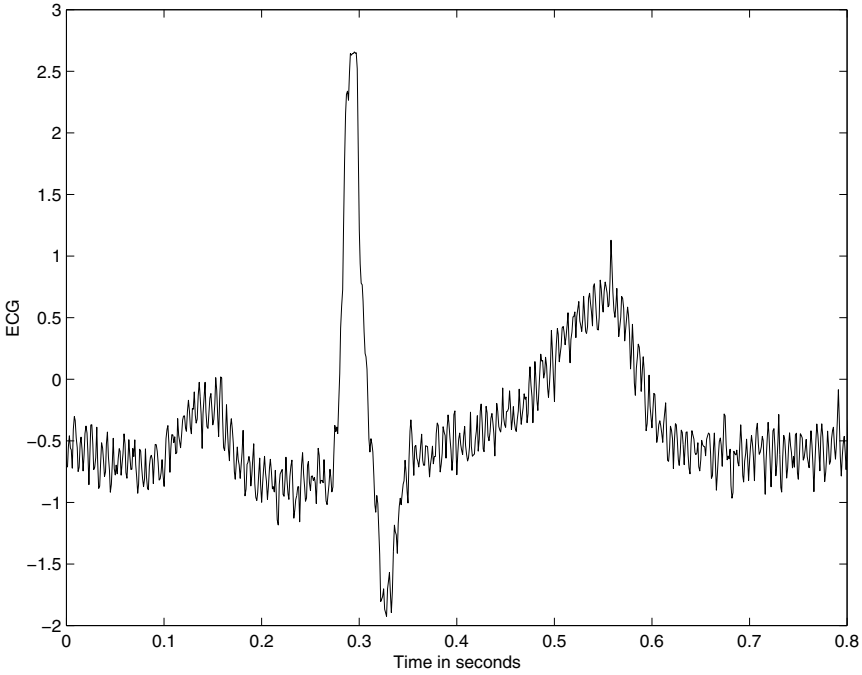


Figure 3.50 ECG signal with high-frequency noise; $f_s = 1,000 \text{ Hz}$.

3.5.3 Derivative based operators to remove low frequency artifacts

Problem: *Develop a time-domain technique to remove baseline drift in the ECG signal.*

Solution: The derivative operator in the time domain removes the parts of the input that are constant (the output is zero). Large changes in the input lead to high values in the output of the derivative operator. Improved understanding of the derivative operation may be obtained by studying its transform in the frequency domain.

The ideal $\frac{d}{dt}$ operator in the time domain results in multiplication of the Fourier transform of the original signal by $j\omega = j2\pi f$ in the frequency domain. If $X(f)$ represents the Fourier transform of the signal $x(t)$, then the Fourier transform of $\frac{dx}{dt}$ is $j2\pi f X(f)$ or $j\omega X(\omega)$. The frequency response of the operation is $H(\omega) = j\omega$. It is seen that the magnitude of the frequency response increases linearly with frequency, starting with $H(\omega) = 0$ at $\omega = 0$. Thus, the DC component is removed by the derivative operator, and higher frequencies receive linearly increasing gain: the operation represents a highpass filter. The derivative operator may be used to remove DC and suppress low-frequency components (and boost high-frequency components).

It follows readily that the second-order derivative operator $\frac{d^2}{dt^2}$ has the frequency response $H(\omega) = (j\omega)(j\omega) = -\omega^2$, with a quadratic increase in gain for higher

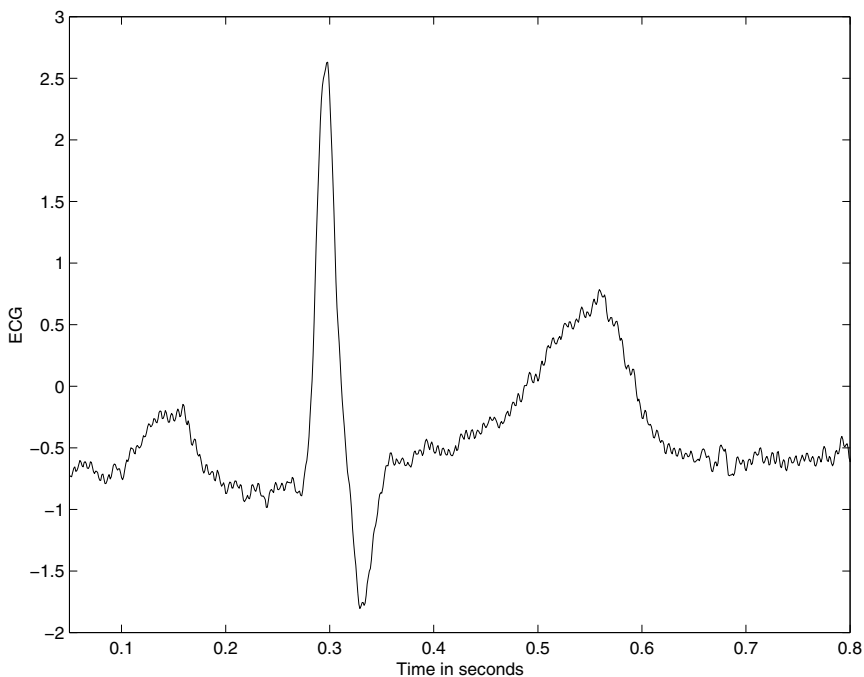


Figure 3.51 The ECG signal with high-frequency noise in Figure 3.50 after filtering by the 8-point MA filter shown in Figure 3.48 and Equation 3.107.

frequencies. The second-order derivative operator may be used to obtain even higher gain for higher frequencies than the first-order derivative operator; the former may be realized as a cascade of two of the latter.

In digital signal processing, the basic derivative is given by the first-order difference operator [46]

$$y(n) = \frac{1}{T} [x(n) - x(n-1)]. \quad (3.122)$$

The scale factor including the sampling interval T is required in order to obtain the rate of change of the signal with respect to the true time. The transfer function of the operator is

$$H(z) = \frac{1}{T} (1 - z^{-1}). \quad (3.123)$$

The filter has a zero at $z = 1$, the DC point.

The frequency response of the operator is

$$H(\omega) = \frac{1}{T} [1 - \exp(-j\omega)] = \frac{1}{T} \exp\left(-j\frac{\omega}{2}\right) \left[2j \sin\left(\frac{\omega}{2}\right)\right], \quad (3.124)$$

which leads to

$$|H(\omega)| = \frac{2}{T} \left| \sin\left(\frac{\omega}{2}\right) \right| \quad (3.125)$$

and

$$\angle H(\omega) = \frac{\pi}{2} - \frac{\omega}{2}. \quad (3.126)$$

The magnitude and phase responses of the first-order difference operator are plotted in Figure 3.52. The gain of the filter increases for higher frequencies up to the folding frequency $f_s/2$. The gain may be taken to approximate that of the ideal derivative operator, that is, $|\omega|$, for low values of ω . Any high-frequency noise present in the signal will be amplified significantly: therefore, the result could be noisy.

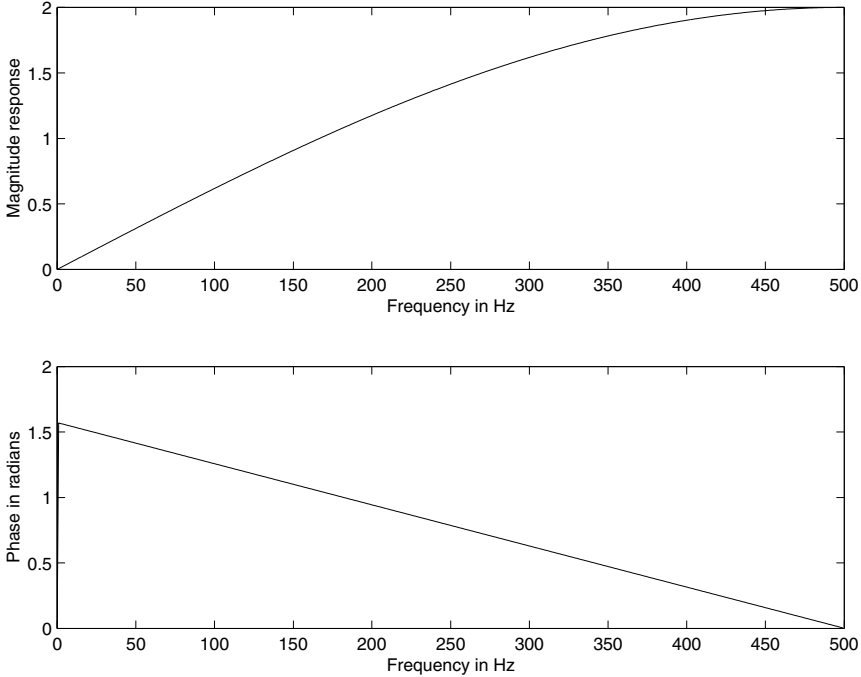


Figure 3.52 Magnitude and phase responses of the first-order difference operator in Equation 3.122. The magnitude response is shown on a linear scale in order to illustrate better its proportionality to frequency.

The noise-amplification problem with the first-order difference operator in Equation 3.122 may be controlled by taking the average of two successive output values:

$$\begin{aligned} y_3(n) &= \frac{1}{2} [y(n) + y(n-1)] \\ &= \frac{1}{2T} \{[x(n) - x(n-1)] + [x(n-1) - x(n-2)]\} \\ &= \frac{1}{2T} [x(n) - x(n-2)]. \end{aligned} \quad (3.127)$$

The transfer function of the operator above, known as the three-point central difference [46], is

$$H(z) = \frac{1}{2T} (1 - z^{-2}) = \left[\frac{1}{T} (1 - z^{-1}) \right] \left[\frac{1}{2} (1 + z^{-1}) \right]. \quad (3.128)$$

Observe that the transfer function of the three-point central-difference operator is the product of the transfer functions of the simple first-order difference operator and a two-point MA filter. The three-point central-difference operation may, therefore, be performed by the simple first-order difference operator and a two-point MA filter in series (cascade).

The magnitude and phase responses of the three-point central-difference operator are given by

$$|H(\omega)| = \frac{1}{T} |\sin(\omega)| \quad (3.129)$$

and

$$\angle H(\omega) = \frac{\pi}{2} - \omega, \quad (3.130)$$

and are plotted in Figure 3.53. The transfer function has zeros at $z = 1$ and $z = -1$, with the latter pulling the gain at the folding frequency ($f_s/2$) to zero: The operator is a bandpass filter. Although the operator does not have the noise-amplification problem of the first-order difference operator, the approximation of the $\frac{d}{dt}$ operation is poor after about $f_s/10$ [46].

Illustration of application: Figures 3.54 and 3.55 show the results of filtering the ECG signal with low-frequency noise shown in Figure 3.6, using the first-order difference and three-point central-difference operators, respectively. It is seen that the baseline drift has been removed, with the latter being less noisy than the former. However, it is obvious that the highpass and high-frequency emphasis effects inherent in both operators have removed the slow P and T waves, and have altered the QRS complexes to such an extent as to make the resulting waveforms look unlike ECG signals. (We shall see in Section 4.3 that, although the derivative operators are not useful in the present application, they are useful in detecting the QRS complex and the dicrotic notch.)

Problem: *How can we improve the performance of the basic first-order difference operator as a filter to remove low-frequency noise or baseline wander without distorting the QRS complex?*

Solution: The drawback of the first-order difference and the three-point central-difference operators lies in the fact that their magnitude responses remain low for a significant range of frequencies well beyond the band related to baseline wander. The zero of the first-order difference operator at $z = 1$ is desired in order to reject the DC component and low frequencies. However, we would like to maintain the levels of the components present in the signal beyond about $0.5 Hz$, that is, we would like the gain of the filter to be close to unity after about $0.5 Hz$.

The gain of a filter at specific frequencies may be boosted by placing poles at the related locations around the unit circle in the z -plane. For the sake of stability of the

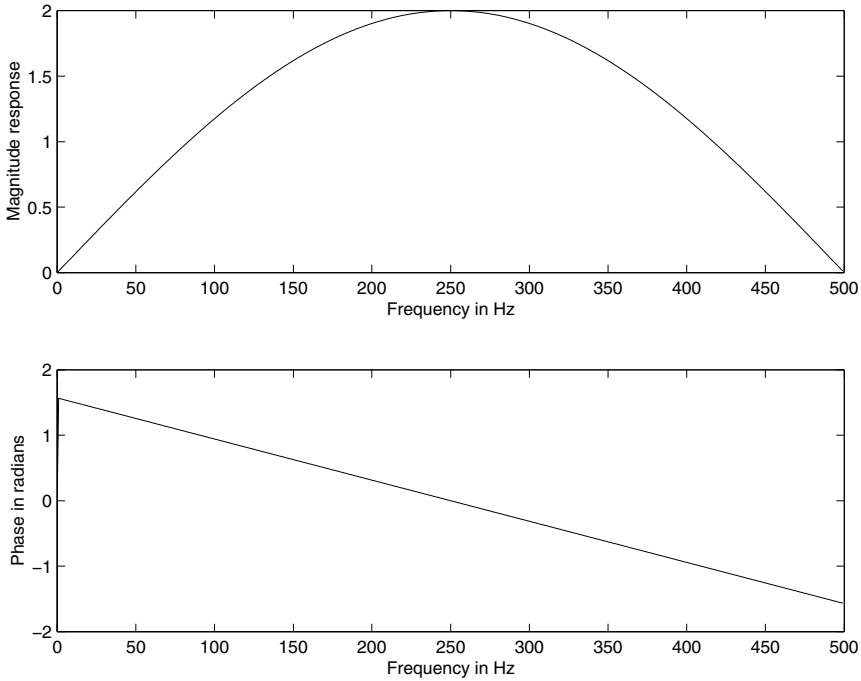


Figure 3.53 Magnitude and phase responses of the three-point central-difference operator in Equation 3.127. The magnitude response is shown on a linear scale.

filter, the poles should be placed within the unit circle. Since we are interested in maintaining a high gain at low frequencies, we could place a pole on the real axis (zero frequency), for example, at $z = 0.995$ [176]. The transfer function of the modified first-order difference filter is then

$$H(z) = \frac{1}{T} \left[\frac{1 - z^{-1}}{1 - 0.995 z^{-1}} \right] \quad (3.131)$$

or, equivalently,

$$H(z) = \frac{1}{T} \left[\frac{z - 1}{z - 0.995} \right]. \quad (3.132)$$

The time-domain input–output relationship is given as

$$y(n) = \frac{1}{T} [x(n) - x(n - 1)] + 0.995 y(n - 1). \quad (3.133)$$

Two equivalent signal-flow diagrams of the filter are shown in Figure 3.56. (*Note:* The filter is no longer an FIR filter; details on infinite impulse response or IIR filters are presented in Section 3.6.1.)

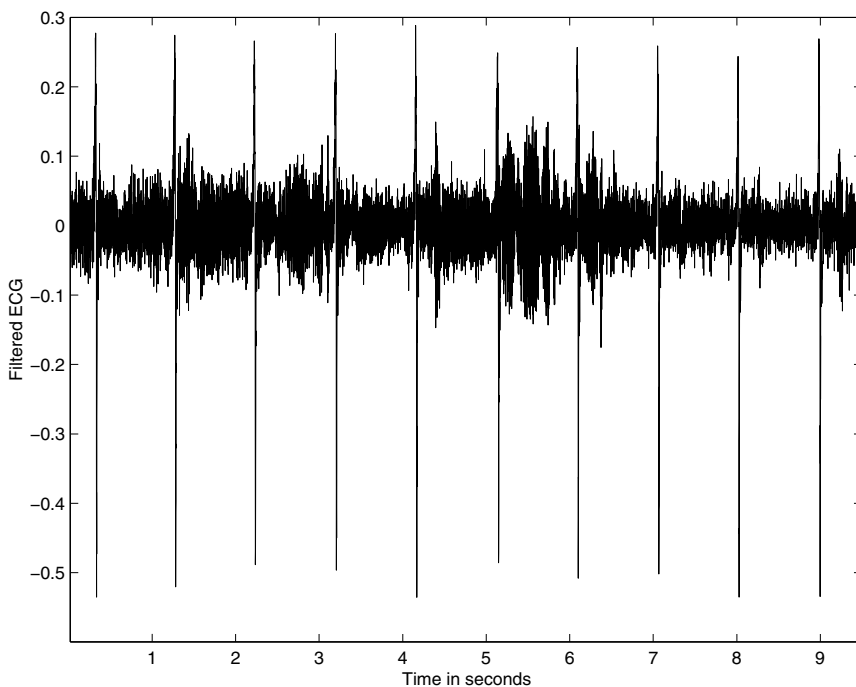


Figure 3.54 Result of filtering the ECG signal with low-frequency noise shown in Figure 3.6 using the first-order difference operator in Equation 3.122.

The form of $H(z)$ in Equation 3.132 in terms of z helps in understanding the graphical method for the evaluation of the frequency response of discrete-time filters [1–3, 46] shown in Figure 3.34. The frequency response of a system is obtained by evaluating its transfer function at various points on the unit circle in the z -plane, that is, by letting $z = \exp(j\omega)$ and evaluating $H(z)$ for various values of the frequency variable ω of interest. The numerator in Equation 3.132 expresses the vectorial distance between a specified point in the z -plane and the zero at $z = 1$; the denominator gives the distance to the pole at $z = 0.995$. The magnitude transfer function of a system for a particular value of z is given by the product of the distances from the corresponding point in the z -plane to all of the zeros of the system's transfer function, divided by the product of the distances to all of its poles. The phase response is given by the sum of the angles of the vectors joining the point to all of the zeros, minus the sum of the angles to all of the poles [1–3, 46]. It is obvious that the magnitude response of the filter in Equations 3.131 and 3.132 is zero at $z = 1$, due to the presence of a zero at that point. Furthermore, for values of z away from $z = 1$, the distances to the zero at $z = 1$ and the pole at $z = 0.995$ will be almost equal; therefore, the gain of the filter will be close to unity for frequencies greater than about 5 Hz. The magnitude and phase responses of the filter shown in Figure 3.57 confirm these observations: The filter is a highpass filter with nonlinear phase.

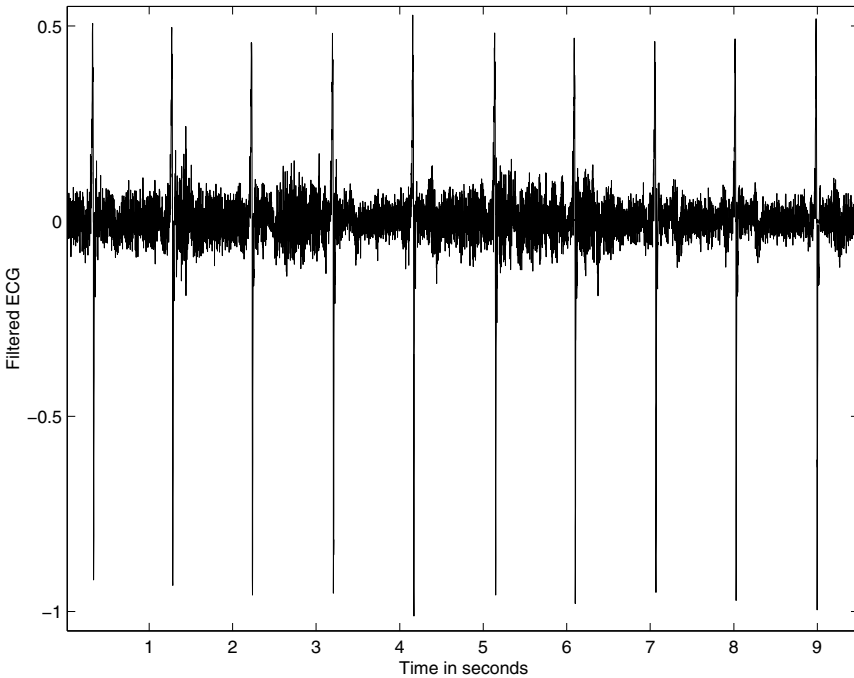


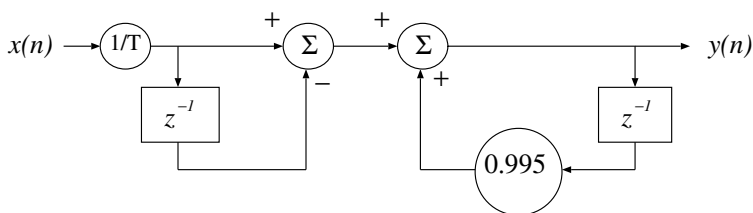
Figure 3.55 Result of filtering the ECG signal with low-frequency noise shown in Figure 3.6 using the three-point central-difference operator in Equation 3.127.

The result of application of the filter in Equation 3.133 to the ECG signal with low-frequency noise shown in Figure 3.6 is displayed in Figure 3.58. It is evident that the low-frequency baseline artifact has been removed without any significant distortion of the ECG waveforms, as compared with the results of differentiation in Figures 3.54 and 3.55. Close inspection, however, reveals that the S wave has been enhanced (made deeper) and that negative undershoots have been introduced after the P and T waves. Removal of the low-frequency baseline artifact has been achieved at the cost of distortion of the ECG waves due to the use of a derivative-based filter and its nonlinear phase response.

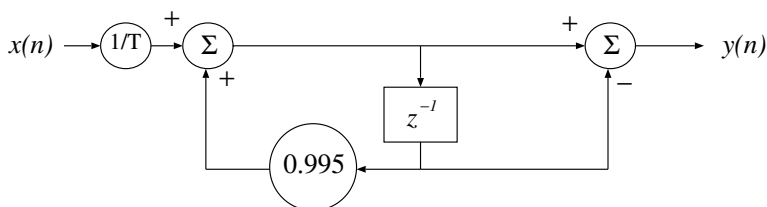
3.5.4 Various specifications of a filter

Although we started the present section with the design of filters in the time domain, we used several ways to specify the parameters and analyze the characteristics of filters that are not limited to the time domain. Some of the several possible specifications of a filter are listed below.

- Difference equation.
- Signal-flow diagram.



(a)



(b)

Figure 3.56 Two equivalent signal-flow diagrams of the filter to remove low-frequency noise or baseline wander. The form in (a) uses two delays, whereas that in (b) uses only one delay.

- Tap-weight or filter coefficients.
- Impulse response, $h(n)$.
- Transfer function, $H(z)$.
- Frequency response, $H(\omega)$, including its magnitude and phase parts.
- Pole-zero diagram and a gain factor.

The items listed above are interrelated, and any one of them may be used to derive any of the others, as seen in the several illustrations provided in the present and previous sections; the following sections provide more examples.

3.6 Frequency domain Filters

The filters described in the previous section perform relatively simple operations in the time domain; although their frequency-domain characteristics were explored, the operators were not specifically designed to possess any particular frequency response at the outset. The frequency response of the MA filter, in particular, is not attractive: the attenuation in the stopband is not high and is not uniform, with the gain falling below -20 dB only around the zeros of the transfer function.

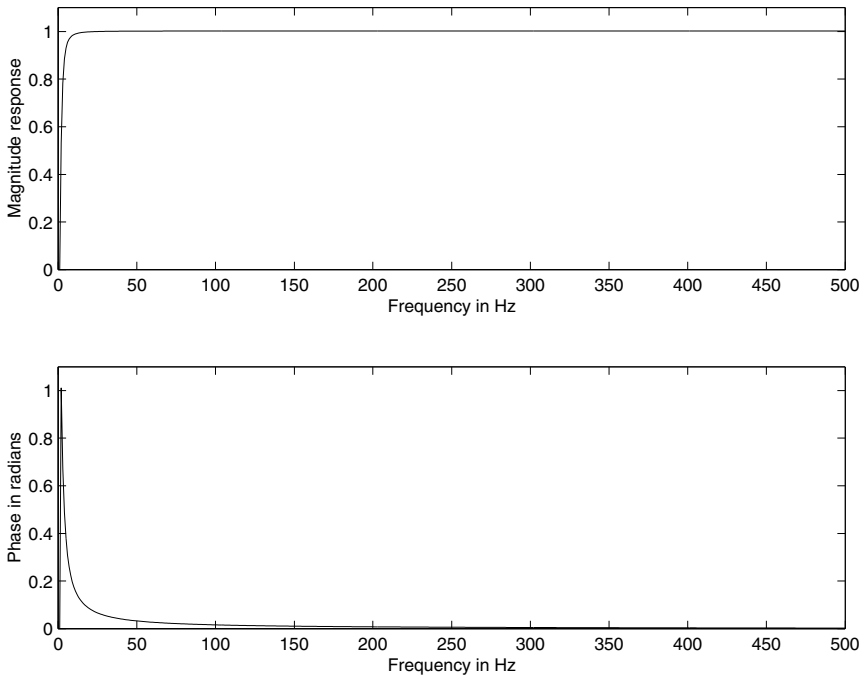


Figure 3.57 Normalized magnitude and phase responses of the filter to remove baseline wander as in Equation 3.131. The magnitude response is shown on a linear scale.

Filters may be designed in the frequency domain to provide specific lowpass, highpass, bandpass, or band-reject (notch) characteristics. Frequency-domain filters may be implemented in software after obtaining the Fourier transform of the input signal, or converted into equivalent time-domain filters and applied directly upon the signal samples.

Many design procedures are available in the literature to design various types of filters: The most-commonly used designs are the Butterworth, Chebyshev, elliptic, and Bessel filters [15,40,177–181]. Since these filters have been well-established in the analog-filter domain, it is common to commence with an analog design and apply the bilinear transformation to obtain a digital filter in the z -domain. It is also common to design a lowpass filter with the desired passband, transition, and stopband characteristics on a normalized-frequency axis, and then transform it to the desired lowpass, highpass, bandpass, or band-reject characteristics [15,177]. Frequency-domain filters may also be specified directly in terms of the values of the desired frequency response at certain frequency samples only, and then transformed into the equivalent time-domain filter coefficients via the inverse Fourier transform.

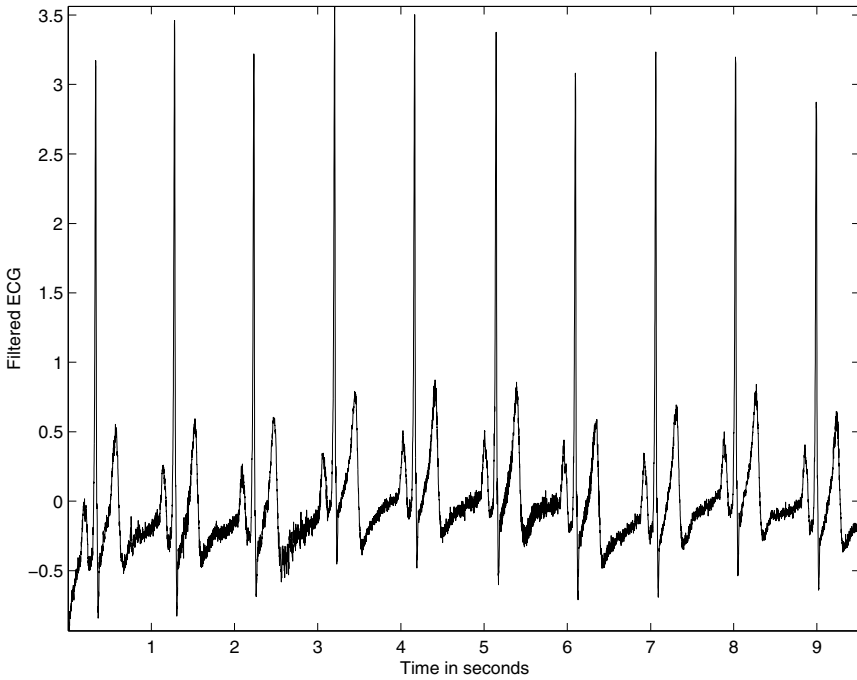


Figure 3.58 Result of processing the ECG signal with low-frequency noise shown in Figure 3.6 using the filter to remove baseline wander as in Equation 3.133. (Compare with the results in Figures 3.54 and 3.55.)

3.6.1 Removal of high frequency noise: Butterworth lowpass filters

Problem: Design a frequency-domain filter to remove high-frequency noise with minimal loss of signal components in the specified passband.

Solution: The Butterworth filter is a commonly used frequency-domain filter due to its simplicity and the property of a maximally flat magnitude response in the passband. For a Butterworth lowpass filter of order N , the first $2N - 1$ derivatives of the squared magnitude response are zero at $\Omega = 0$, where Ω represents the analog radian frequency. The Butterworth filter response is monotonic in the passband as well as in the stopband.

The basic Butterworth lowpass filter function is given as [15, 174]

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{j\Omega}{j\Omega_c}\right)^{2N}}, \quad (3.134)$$

where H_a is the frequency response of the analog filter and Ω_c is the cutoff frequency (in *radians/s*). A Butterworth filter is completely specified by its cutoff frequency Ω_c and order N . As the order N increases, the filter response becomes more

flat in the passband, and the transition to the stopband becomes faster or sharper. $|H_a(j\Omega_c)|^2 = \frac{1}{2}$ for all N .

Changing to the Laplace variable s , we get

$$H_a(s)H_a(-s) = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}}. \quad (3.135)$$

The poles of the squared transfer function are located with equal spacing around a circle of radius Ω_c in the s -plane, distributed symmetrically on either side of the imaginary axis $s = j\Omega$. No pole will lie on the imaginary axis itself; poles will appear on the real axis for odd N . The angular spacing between the poles is $\frac{\pi}{N}$. If $H_a(s)H_a(-s)$ has a pole at $s = s_p$, it will have a pole at $s = -s_p$ as well. Furthermore, for the filter coefficients to be real, complex poles must appear in conjugate pairs. In order to obtain a stable and causal filter, we need to form $H_a(s)$ with only the N poles on the LHS of the s -plane. The pole positions in the s -plane are given by

$$s_k = \Omega_c \exp \left[j\pi \left(\frac{1}{2} + \frac{(2k-1)}{2N} \right) \right], \quad (3.136)$$

$k = 1, 2, \dots, 2N$ [177].

Once the pole positions are obtained in the s -plane, they may be combined to obtain the transfer function in the analog Laplace domain as

$$H_a(s) = \frac{G}{(s-p_1)(s-p_2)(s-p_3)\cdots(s-p_N)}, \quad (3.137)$$

where $p_k, k = 1, 2, \dots, N$, are the N poles of the transfer function in the left-half of the s -plane, and G is a gain factor specified as needed or calculated to normalize the gain at DC ($s = 0$) to be unity.

If we use the bilinear transformation

$$s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right], \quad (3.138)$$

the Butterworth circle in the s -plane maps to a circle in the z -plane with its real-axis intercepts at $z = \frac{2-\Omega_c T}{2+\Omega_c T}$ and $z = \frac{2+\Omega_c T}{2-\Omega_c T}$. The poles at $s = s_p$ and $s = -s_p$ in the s -plane map to the locations $z = z_p$ and $z = 1/z_p$, respectively. The poles in the z -plane are not uniformly spaced around the transformed Butterworth circle. For stability, all poles of $H(z)$ must lie within the unit circle in the z -plane.

Consider the unit circle in the z -plane given by $z = e^{j\omega}$. For points on the unit circle, we have

$$s = \sigma + j\Omega = \frac{2}{T} \left(\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} \right) = \frac{2j}{T} \tan \left(\frac{\omega}{2} \right). \quad (3.139)$$

For the unit circle, $\sigma = 0$; therefore, we get the relationships between the continuous-time (s -domain) frequency variable Ω and the discrete-time (z -domain) frequency variable ω as

$$\Omega = \frac{2}{T} \tan \left(\frac{\omega}{2} \right) \quad (3.140)$$

and

$$\omega = 2 \tan^{-1} \left(\frac{\Omega T}{2} \right). \quad (3.141)$$

This is a nonlinear relationship that warps the frequency values as they are mapped from the imaginary (vertical) axis in the s -plane to the unit circle in the z -plane (or viceversa), and it should be taken into account in specifying cutoff frequencies.

The transfer function $H_a(s)$ may be mapped to the z -domain by applying the bilinear transformation, that is, by substituting $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$ in Equation 3.137. The transfer function $H(z)$ may then be simplified to the form

$$H(z) = \frac{G' (1 + z^{-1})^N}{\sum_{k=0}^N a_k z^{-k}}, \quad (3.142)$$

where $a_k, k = 0, 1, 2, \dots, N$, are the filter coefficients or tap weights (with $a_0 = 1$), and G' is the gain factor [usually calculated so as to obtain $|H(z)| = 1$ at DC, that is, at $z = 1$]. Observe that the filter has N zeros at $z = -1$ due to the use of the bilinear transformation. The filter is now in the familiar form of an IIR filter. Two forms of realization of a generic IIR filter are illustrated as signal-flow diagrams in Figures 3.59 and 3.60: The former represents a direct realization using $2N$ delays and $2N + 1$ multipliers (with $a_0 = 1$), whereas the latter uses only N delays and $2N + 1$ multipliers.

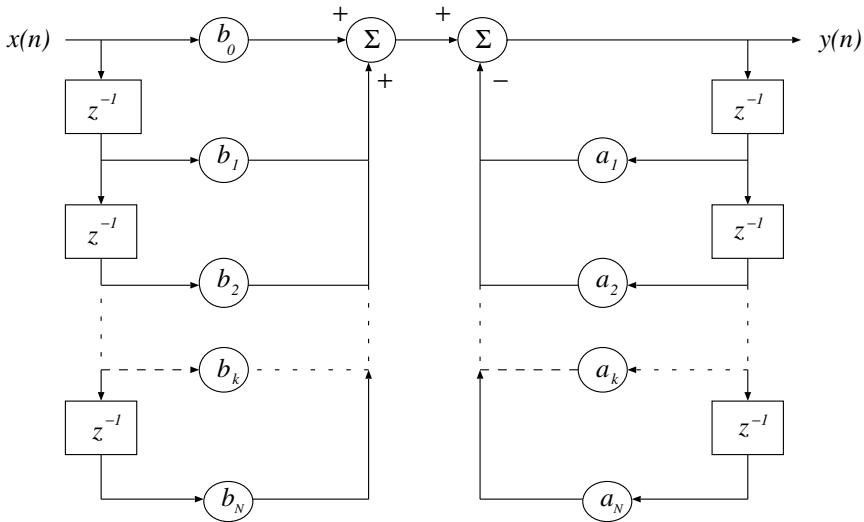


Figure 3.59 Signal-flow diagram of a direct realization of a generic IIR filter. This form uses $2N$ delays and $2N + 1$ multipliers for a filter of order N .

A time-domain representation of the filter will be required if the filter is to be applied to data samples directly in the time domain. From the filter transfer function

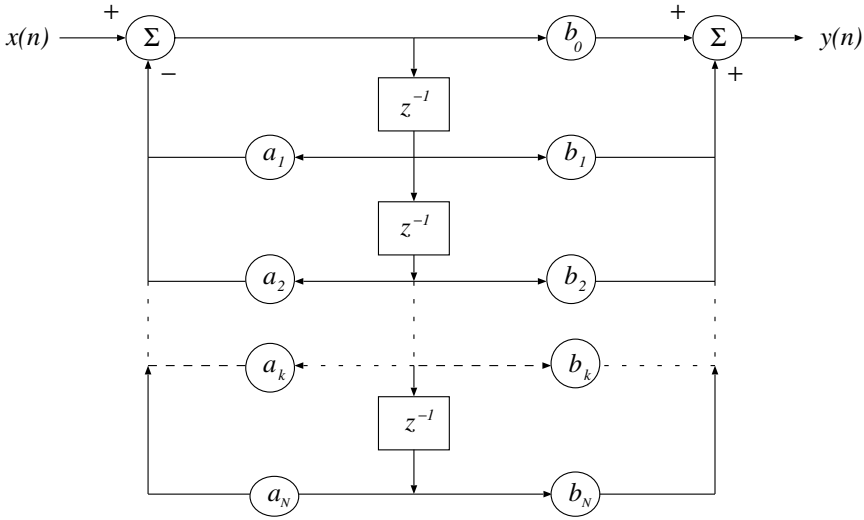


Figure 3.60 Signal-flow diagram of a realization of an IIR filter that uses only N delays and $(2N + 1)$ multipliers for a filter of order N .

$H(z)$ in Equation 3.142, it becomes easy to represent the filter in the time domain with the difference equation

$$y(n) = \sum_{k=0}^N b_k x(n-k) - \sum_{k=1}^N a_k y(n-k). \quad (3.143)$$

The coefficients b_k are given by the coefficients of the expansion of $G'(1 + z^{-1})^N$. (The MATLAB[®] [182] command *butter* provides Butterworth filters.)

It is also possible to directly specify the Butterworth filter as

$$|H(\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}}, \quad (3.144)$$

with ω normalized to the range $(0, 2\pi)$ for sampled or discrete-time signals; in such a case, the equation is valid only for the range $(0, \pi)$, with the function in the range $(\pi, 2\pi)$ being a reflection of that over $(0, \pi)$. The cutoff frequency ω_c should be specified in the range $(0, \pi)$.

If the DFT is used to compute the Fourier transforms of the signals being filtered, Equation 3.144 may be modified to

$$|H(k)|^2 = \frac{1}{1 + \left(\frac{k}{k_c}\right)^{2N}}, \quad (3.145)$$

where k is the index of the DFT array standing for discretized frequency. With K being the number of points in the DFT array, k_c is the array index corresponding to

the cutoff frequency ω_c (that is, $k_c = \lceil K \frac{\omega_c}{\omega_s} \rceil$). The equation given above is valid for $k = 0, 1, 2, \dots, \frac{K}{2}$, with the second half over $(\frac{K}{2} + 1, K - 1)$ being a reflection of the first half [that is, $H(k) = H(K - k)$, $k = \frac{K}{2} + 1, \dots, K - 1$]. Note that the DFT includes two unique values: the DC component in $H(0)$ and the folding-frequency component in $H(\frac{K}{2})$. The variable k in the filter equation could also be used to represent normalized frequency in the range $(0, 1)$, with unity standing for the sampling frequency, 0.5 standing for the maximum frequency present in the sampled signal (that is, the folding frequency), and k_c being specified in the range $(0, 0.5)$. (Note: MATLAB[®] normalizes one-half of the sampling frequency to unity; the maximum normalized frequency present in the sampled signal is then unity. MATLAB[®] and a few programming languages do not allow an array index to be zero: In such a case, the indices mentioned above must be incremented by one.)

One could compute the DFT of the given signal, multiply the result by $|H(k)|$, and compute the inverse DFT to obtain the filtered signal. The advantage of this procedure is that no phase change is involved: The filter is a strictly magnitude-only transfer function. The time-domain implementation described earlier will include a phase response which may not be desired. However, time-domain implementation will be required in on-line signal processing applications.

Butterworth lowpass filter design example: In order to design a Butterworth lowpass filter, we need to specify two parameters: ω_c and N . The two parameters may be specified based on a knowledge of the characteristics of the filter as well as those of the signal and noise. It is also possible to specify the required minimum gain at a certain frequency in the passband and the required minimum attenuation at another frequency in the stopband. The two values may then be used with Equation 3.134 to obtain two equations in the two unknowns ω_c and N , which may be solved to derive the filter parameters [174].

Given the 3 dB cutoff frequency f_c and order N , the procedure to design a Butterworth lowpass filter is as follows:

1. Convert the specified 3 dB cutoff frequency f_c to radians in the normalized range $(0, 2\pi)$ as $\omega_c = \frac{f_c}{f_s} 2\pi$. Then, $T = 1$. Prewarp the cutoff frequency ω_c by using Equation 3.140 and obtain Ω_c .
2. Derive the positions of the poles of the filter in the s -plane as given by Equation 3.136.
3. Form the transfer function $H_a(s)$ of the Butterworth lowpass filter in the Laplace domain by using the poles in the left-half plane only as given by Equation 3.137.
4. Apply the bilinear transformation as per Equation 3.138 and obtain the transfer function of the filter $H(z)$ in the z -domain as in Equation 3.142.
5. Convert the filter to the series of coefficients b_k and a_k as in Equation 3.143.

Let us now design a Butterworth lowpass filter with $f_c = 40 \text{ Hz}$, $f_s = 200 \text{ Hz}$, and $N = 4$. We have $\omega_c = \frac{40}{200} 2\pi = 0.4\pi \text{ radians/s}$. The prewarped s -domain cutoff frequency is $\Omega_c = \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = 1.453085 \text{ radians/s}$.

The poles of $H_a(s)H_a(-s)$ are placed around a circle of radius 1.453085 *radians* / *s* with an angular separation of $\frac{\pi}{N} = \frac{\pi}{4}$ *radians*. The poles of interest are located at angles $\frac{5}{8}\pi$ and $\frac{7}{8}\pi$ and the corresponding conjugate positions. Figure 3.61 shows the positions of the poles of $H_a(s)H_a(-s)$ in the Laplace plane. The coordinates of the poles of interest are $(-0.556072 \pm j 1.342475)$ and $(-1.342475 \pm j 0.556072)$. The transfer function of the filter is

$$H_a(s) = \frac{4.458247}{(s^2 + 1.112143s + 2.111456)(s^2 + 2.684951s + 2.111456)}. \quad (3.146)$$

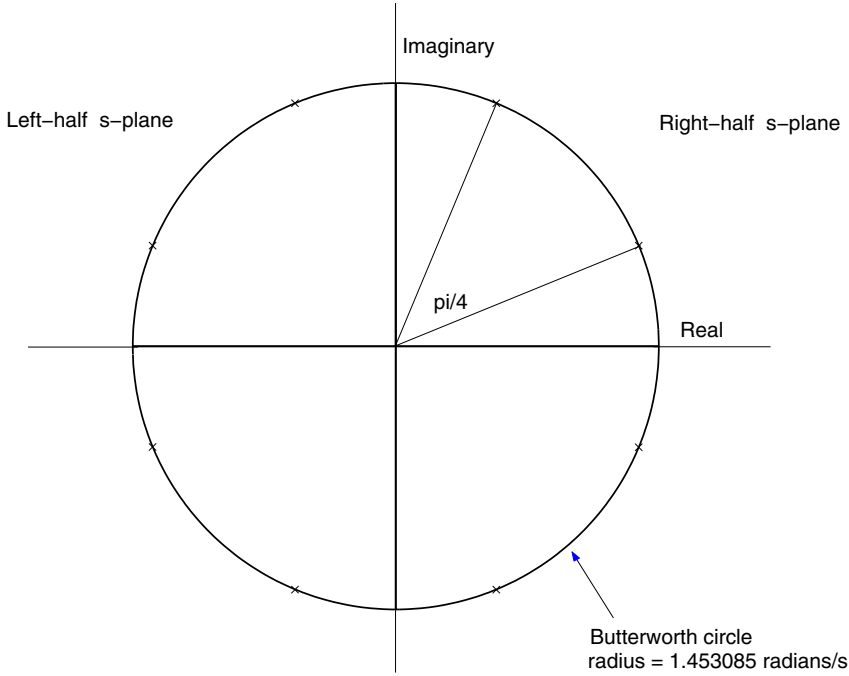


Figure 3.61 Pole positions in the s -plane of the squared magnitude response of the Butterworth lowpass filter with $f_c = 40$ *Hz*, $f_s = 200$ *Hz*, and $N = 4$.

Applying the bilinear transformation, we get

$$H(z) = \frac{0.046583(1 + z^{-1})^4}{(1 - 0.447765z^{-1} + 0.460815z^{-2})(1 - 0.328976z^{-1} + 0.064588z^{-2})}. \quad (3.147)$$

This filter has four poles at $(0.223882 \pm j 0.640852)$ and $(0.164488 \pm j 0.193730)$, and four zeros at $-1 + j 0$. The b_k coefficients of the filter as in Equation 3.143 are $\{0.0465829, 0.186332, 0.279497, 0.186332, 0.046583\}$, and the a_k coefficients are $\{1, -0.776740, 0.672706, -0.180517, 0.029763\}$. The pole-zero plot and the frequency response of the filter are given in Figures 3.62 and 3.63, respectively. The

frequency response displays the expected monotonic decrease in gain and -3 dB power point or 0.707 gain at 40 Hz.

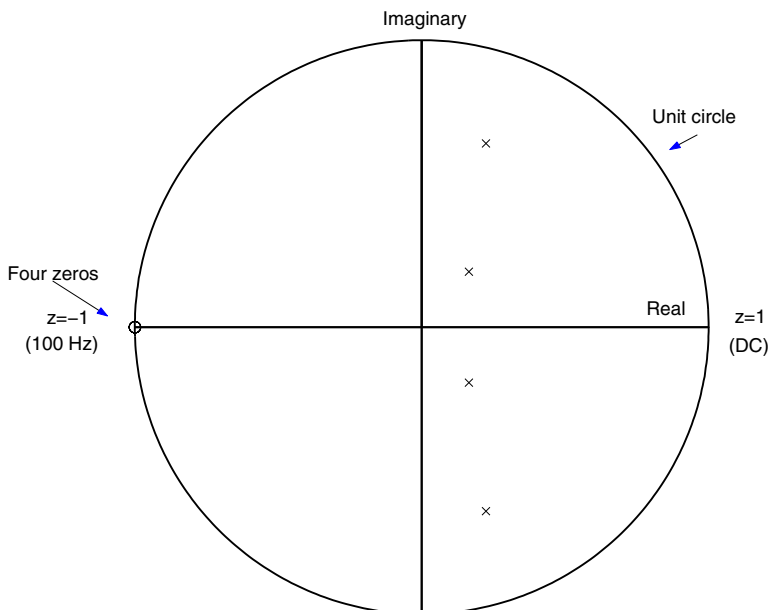


Figure 3.62 Positions of the poles and zeros in the z -plane of the Butterworth lowpass filter with $f_c = 40$ Hz, $f_s = 200$ Hz, and $N = 4$.

Figure 3.64 compares the magnitude responses of three Butterworth lowpass filters with $f_c = 40$ Hz and $f_s = 200$ Hz, with the order increasing from $N = 4$ (dotted) to $N = 8$ (dashed) to $N = 12$ (solid). All three filters have their half-power points (gain = 0.707) at 40 Hz, but the transition band becomes sharper as the order N is increased.

The Butterworth design is popular because of its simplicity, a monotonically decreasing magnitude response, and a maximally flat magnitude response in the passband. Its main disadvantages are a slow (or wide) transition from the passband to the stopband, and a nonlinear phase response. The nonlinear phase may be corrected by passing the filter output again through the same filter but after a reversal in time [178]. This process, however, leads to a magnitude response that is the square of that provided by the initial filter design. The squaring effect may be compensated for in the initial design; however, the approach cannot be applied in real time. The elliptic filter design provides a sharp transition band at the expense of ripples in the passband and the stopband. The Bessel design provides a group delay that is maximally flat at DC, and a phase response that approximates a linear response. Details on the design of Bessel, Chebyshev, elliptic, and other filters may be found in other sources on filter design [15, 40, 177–181].

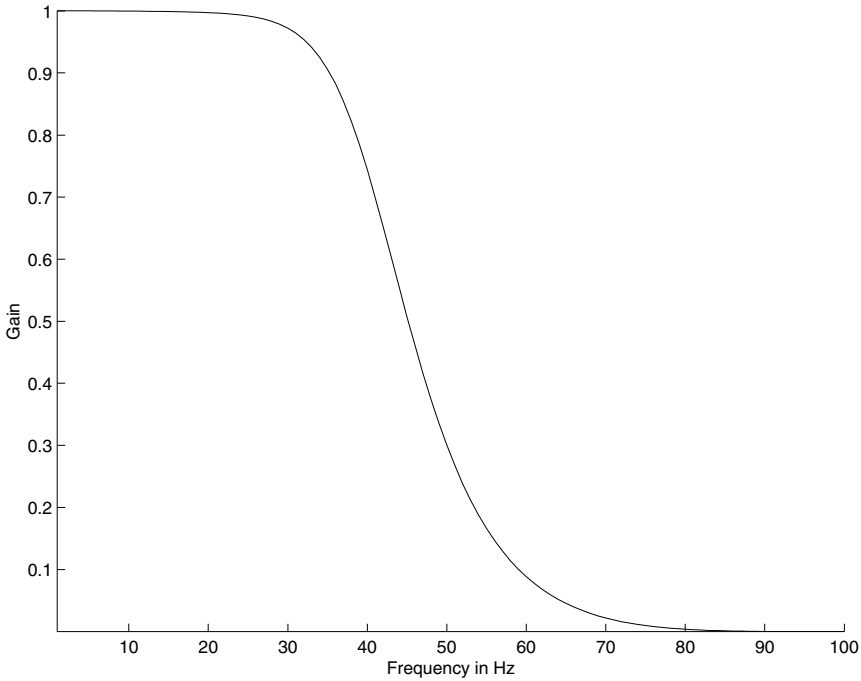


Figure 3.63 Magnitude response of the Butterworth lowpass filter with $f_c = 40 \text{ Hz}$, $f_s = 200 \text{ Hz}$, and $N = 4$.

Illustrations of application: The upper trace in Figure 3.65 illustrates a carotid pulse signal with high-frequency noise. The lower trace in the same figure shows the result of processing with a Butterworth lowpass filter ($f_c = 40 \text{ Hz}$, $f_s = 200 \text{ Hz}$, and $N = 12$). The high-frequency noise has been effectively reduced.

Figure 3.66 shows the result of filtering the noisy ECG signal shown in Figure 3.50 with an eighth-order Butterworth lowpass filter as in Equations 3.144 and 3.145 and a cutoff frequency of 70 Hz . The frequency response $|H(\omega)|$ of the filter is shown in Figure 3.67. It is evident that the high-frequency noise has been suppressed by the filter.

3.6.2 Removal of low frequency noise: Butterworth highpass filters

Problem: Design a frequency-domain filter to remove low-frequency noise with minimal loss of signal components in the passband.

Solution: Highpass filters may be designed on their own, or obtained by transforming a normalized prototype lowpass filter [174, 177]. The latter approach is easier since lowpass filter prototypes with various characteristics are readily available, as are the transformations required to derive highpass, bandpass, and band-

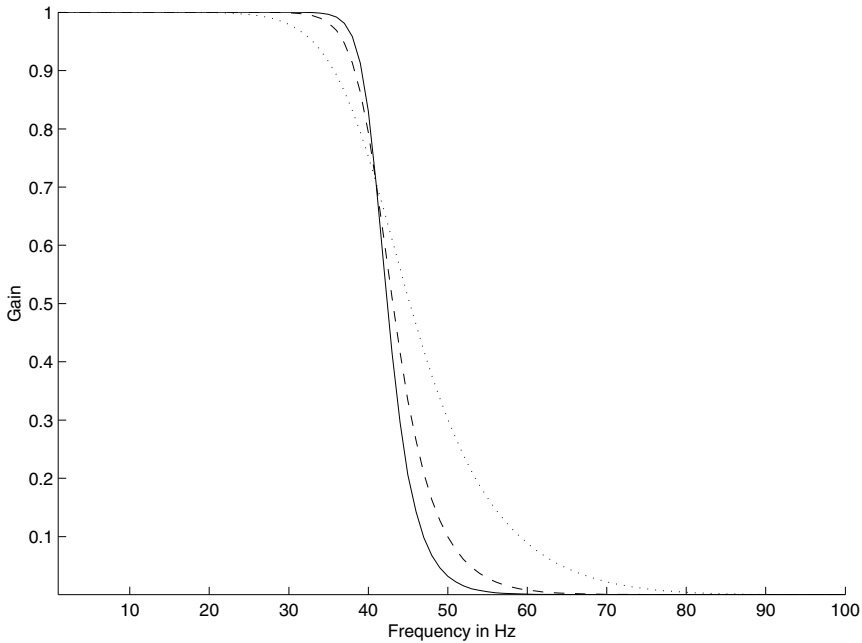


Figure 3.64 Magnitude responses of three Butterworth lowpass filters with $f_c = 40 \text{ Hz}$, $f_s = 200 \text{ Hz}$, and variable order: $N = 4$ (dotted), $N = 8$ (dashed), and $N = 12$ (solid).

stop filters [174, 177]. [MATLAB[®] provides highpass filters with the command `butter(N,  $f_c$ , 'high')`.]

As in the case of the Butterworth lowpass filter in Equation 3.145, the Butterworth highpass filter may be specified directly in the discrete-frequency domain as

$$|H(k)|^2 = \frac{1}{1 + \left(\frac{k_c}{k}\right)^{2N}}. \quad (3.148)$$

Illustration of application: Figure 3.6 shows a segment of an ECG signal with low-frequency noise appearing in the form of a wandering baseline (baseline drift). Figure 3.68 shows the result of filtering the signal with an eighth-order Butterworth highpass filter as in Equation 3.148 and a cutoff frequency of 2 Hz . The frequency response of the filter is shown in Figure 3.69. While the low-frequency artifact has been removed by the filter, it should be noted that the high-frequency noise present in the signal has not been affected.

Observe that the filtered result retains the characteristics of the QRS complex, unlike the case with the derivative-based time-domain filters (compare Figure 3.68 with Figures 3.54 and 3.55.) This advantage is due to the fact that the Butterworth highpass filter that was used has a gain of almost unity over the frequency range of $3 - 100 \text{ Hz}$; the derivative-based filters severely attenuate these components and hence distort the QRS complex. However, it should be observed that the filter has

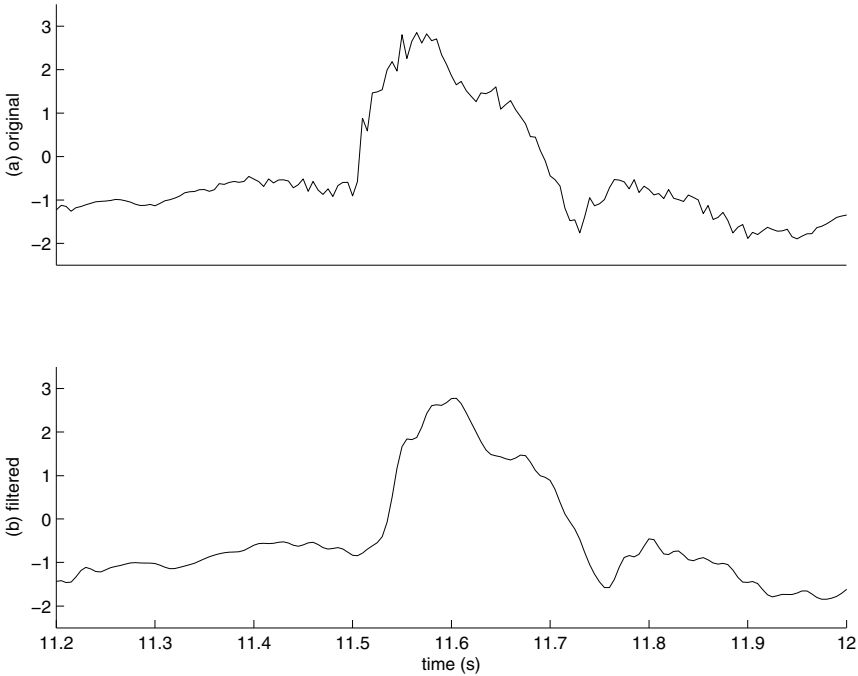


Figure 3.65 Upper trace: a carotid pulse signal with high-frequency noise. Lower trace: result of filtering using a Butterworth lowpass filter with $f_c = 40 \text{ Hz}$, $f_s = 200 \text{ Hz}$, and $N = 12$. The filtering operation was performed in the time domain using the MATLAB[®] *filter* command.

distorted the P and T waves to some extent. The result in Figure 3.68 compares well with that in Figure 3.58, obtained using the simpler IIR filter in Equation 3.133. (Compare the frequency responses in Figures 3.69, 3.52, 3.53, and 3.57.)

3.6.3 Removal of periodic artifacts: Notch and comb filters

Problem: Design a frequency-domain filter to remove periodic artifacts, such as power-line interference.

Solution: The simplest method to remove periodic artifacts is to compute the Fourier transform of the signal, delete the undesired component(s) from the spectrum, and then compute the inverse Fourier transform. The undesired components could be set to zero, or better, to the average level of the signal components over a few frequency samples around the component that is to be removed; the former method will remove the noise components as well as the signal components at the frequencies of concern, whereas the latter assumes that the signal spectrum is smooth in the affected regions.

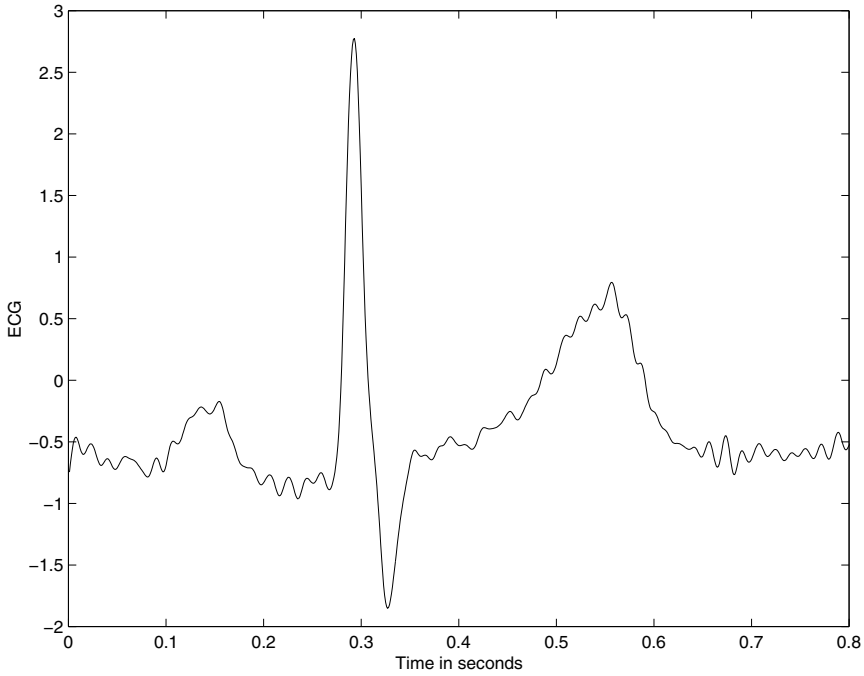


Figure 3.66 Result of frequency-domain filtering of the noisy ECG signal in Figure 3.50 with an eighth-order Butterworth lowpass filter with cutoff frequency = 70 Hz.

Periodic interference may also be removed by notch filters with zeros on the unit circle in the z -domain at the specific frequencies to be rejected. If f_o is the interference frequency, the angles of the (complex conjugate) zeros required will be $\pm \frac{f_o}{f_s}(2\pi)$; the radius of the zeros will be unity. If harmonics are also present, multiple zeros will be required at $\pm \frac{n f_o}{f_s}(2\pi)$, n representing the orders of all of the harmonics present. The angles of the zeros are limited to the range $(-\pi, \pi)$. The filter is then called a “comb” filter. In some situations, higher-order harmonics beyond $\frac{f_s}{2}$ may appear at aliased locations (see Figures 3.8 and 3.100); zeros may then be placed at such frequencies as well.

Notch filter design example: Consider a signal with power-line interference at $f_o = 60$ Hz and sampling rate of $f_s = 1,000$ Hz (see Figures 3.7 and 3.8). The notch filter is then required to have zeros at $\omega_o = \pm \frac{f_o}{f_s}(2\pi) = \pm 0.377$ radians = $\pm 21.6^\circ$. The locations of the zeros are given by $\cos(\omega_o) \pm j \sin(\omega_o)$ or $z_1 = 0.92977 + j0.36812$ and $z_2 = 0.92977 - j0.36812$. The transfer function is

$$H(z) = (1 - z^{-1} z_1)(1 - z^{-1} z_2) = 1 - 1.85955z^{-1} + z^{-2}. \quad (3.149)$$

Substituting $z = 1$ in the expression above, we get the DC response or gain as $H(1) = 1 - 1.85955 + 1 = 0.14045$. Therefore, if the gain at DC is required to be unity, $H(z)$ as above should be divided by 0.14045.

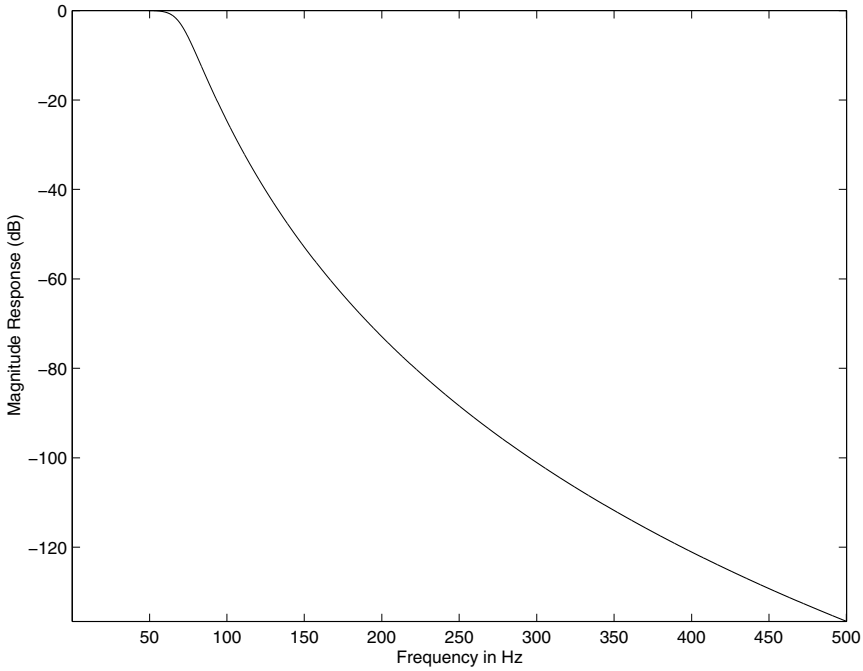


Figure 3.67 Frequency response of the eighth-order Butterworth lowpass filter with cutoff frequency $f_c = 70 \text{ Hz}$ and $f_s = 1,000 \text{ Hz}$.

Figure 3.70 shows a plot of the zeros of the notch filter in the z -plane. Figure 3.71 shows the magnitude and phase responses of the notch filter. Observe that the filter attenuates not only the 60 Hz component but also a band of frequencies around 60 Hz . The sharpness of the notch may be improved by placing a few poles near or symmetrically around the zeros and inside the unit circle [1, 2, 176]. Note also that the gain of the filter is at its maximum at $f_s/2$; additional lowpass filtering in the case of application to ECG signals could be used to reduce the gain at frequencies beyond about 80 Hz .

Comb filter design example: Let us consider the presence of a periodic artifact with the fundamental frequency of 60 Hz and odd harmonics at 180 Hz , 300 Hz , and 420 Hz . Let $f_s = 1,000 \text{ Hz}$, and assume the absence of any aliasing error. Zeros are then desired at 60 Hz , 180 Hz , 300 Hz , and 420 Hz , which translate to $\pm 21.6^\circ$, $\pm 64.8^\circ$, $\pm 108^\circ$, and $\pm 151.2^\circ$, with 360° corresponding to $1,000 \text{ Hz}$. The coordinates of the zeros are $0.92977 \pm j0.36812$, $0.42578 \pm j0.90483$, $-0.30902 \pm j0.95106$, and $-0.87631 \pm j0.48175$. The transfer function of the filter is

$$\begin{aligned}
 H(z) &= G (1 - 1.85955z^{-1} + z^{-2})(1 - 0.85156z^{-1} + z^{-2}) \\
 &\times (1 + 0.61803z^{-1} + z^{-2})(1 + 1.75261z^{-1} + z^{-2}), \quad (3.150)
 \end{aligned}$$

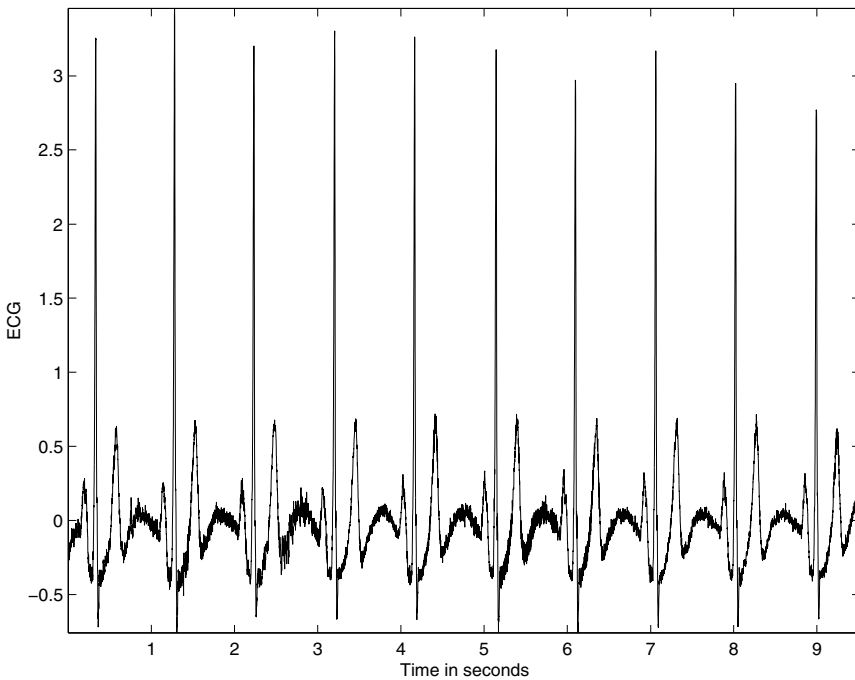


Figure 3.68 Result of frequency-domain filtering of the ECG signal with low-frequency noise in Figure 3.6 with an eighth-order Butterworth highpass filter with cutoff frequency = 2 Hz. (Compare with the results in Figures 3.54, 3.55, and 3.58.)

where G is the desired gain or scaling factor. With G computed so as to set the gain at DC to be unity, the filter transfer function becomes

$$\begin{aligned}
 H(z) = & 0.6310 - 0.2149z^{-1} + 0.1512z^{-2} - 0.1288z^{-3} + 0.1227z^{-4} \\
 & - 0.1288z^{-5} + 0.1512z^{-6} - 0.2149z^{-7} + 0.6310z^{-8}. \quad (3.151)
 \end{aligned}$$

A plot of the locations of the zeros in the z -plane is shown in Figure 3.72. The frequency response of the comb filter is shown in Figure 3.73. Observe the low gain at not only the notch frequencies but also in the adjacent regions.

Illustrations of application: Figure 3.74 shows an ECG signal with power-line interference at $f_o = 60$ Hz. Figure 3.75 shows the result of applying the notch filter in Equation 3.149 to the signal. The 60 Hz interference has been effectively removed, with no perceptible distortion of the ECG waveform.

Figure 3.76 shows an ECG signal with noise including power-line interference at 60 Hz and $f_s = 200$ Hz. A notch filter was designed with two zeros at 60 Hz with radius equal to unity. The coordinates of the zeros in the z -plane are $-0.3090 \pm j 0.9511$. The frequency response of the notch filter is shown in Figure 3.77 (dashed line). The result of filtering the noisy signal in Figure 3.76 is shown in Figure 3.78, which indicates effective removal of the interference.

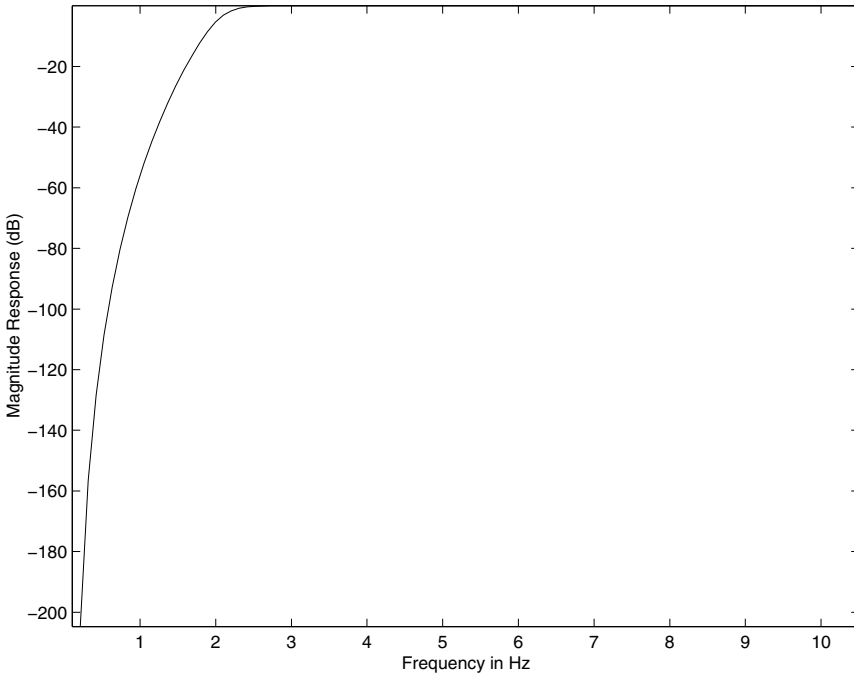


Figure 3.69 Frequency response of an eighth-order Butterworth highpass filter with cutoff frequency = 2 Hz. $f_s = 1,000$ Hz. The frequency response is shown on an expanded scale for the range 0 – 10 Hz only.

In order to improve the notch filter, specifically, to make the notch in the frequency response narrower or sharper, four poles were incorporated in the filter. The frequencies of the poles were set at 59.95 Hz and 60.05 Hz; the radii of the poles were set at 0.4 to place them well within the unit circle in the z -plane. The coordinates of the poles in the z -plane are $-0.1230 \pm j 0.3806$ and $-0.1242 \pm j 0.3802$. The frequency response of the modified notch filter is shown in Figure 3.77 (solid line). It is evident that the notch is sharper or narrower for the modified filter as compared to the response of the filter with only two zeros; however, the gain increases for higher frequencies. As a result, the output of the filter, shown in Figure 3.79, has more noise than the previous result in Figure 3.78.

An illustration of the application of the comb filter is provided in Section 3.12.

3.7 Order statistic filters

The class of filters based on order statistics includes several nonlinear filters that are useful in filtering different types of noise in signals [18, 183, 184]. The first step in order-statistic filtering is to arrange in rank order, usually from the minimum

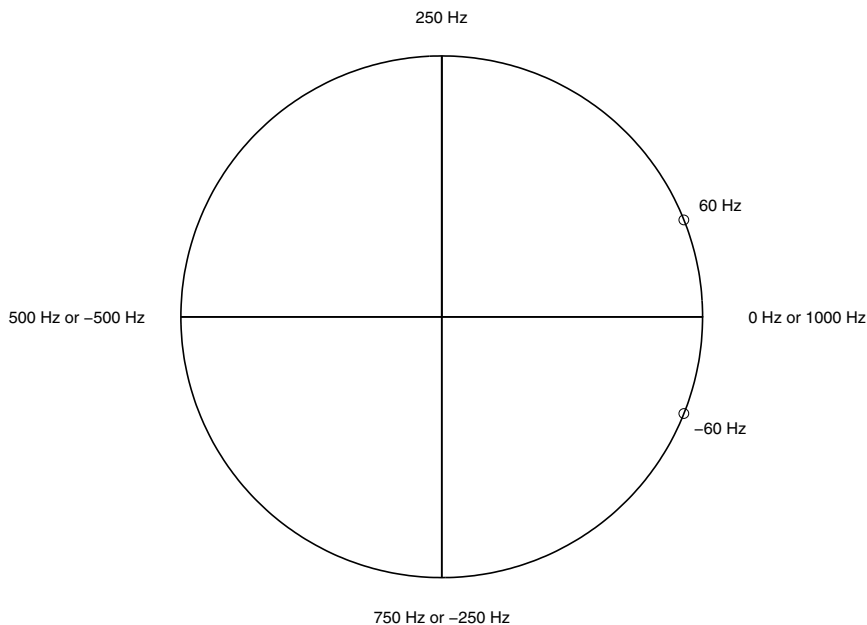


Figure 3.70 Zeros of the notch filter (in the z -domain) to remove 60 Hz interference, the sampling frequency being 1,000 Hz .

to the maximum, the values of the signal in a moving window positioned at the current sample being processed. Then, the i^{th} entry in the list is the output of the i^{th} order-statistic filter. A few commonly used order-statistic filters are defined in the following list.

- *Min filter*: the first entry in the rank-ordered list, useful in removing high-valued impulsive noise.
- *Max filter*: the last entry in the rank-ordered list, useful in removing low-valued impulsive noise.
- *Min/Max filter*: sequential application of the Min and Max filters, useful in removing impulsive noise of both of the types mentioned above.
- *Median filter*: the entry in the middle of the list. The median filter is the most popular and commonly used filter among the order-statistic filters.
- *α -trimmed mean filter*: the mean of a reduced or trimmed list, where the first $\alpha \times 100\%$ and the last $\alpha \times 100\%$ of the entries in the original list are removed, with $0 \leq \alpha < 0.5$. Outliers, which are samples with values substantially different from the rest of the samples in the list, are rejected by the trimming process. A value close to but less than 0.5 for α leads to the rejection of the entire list

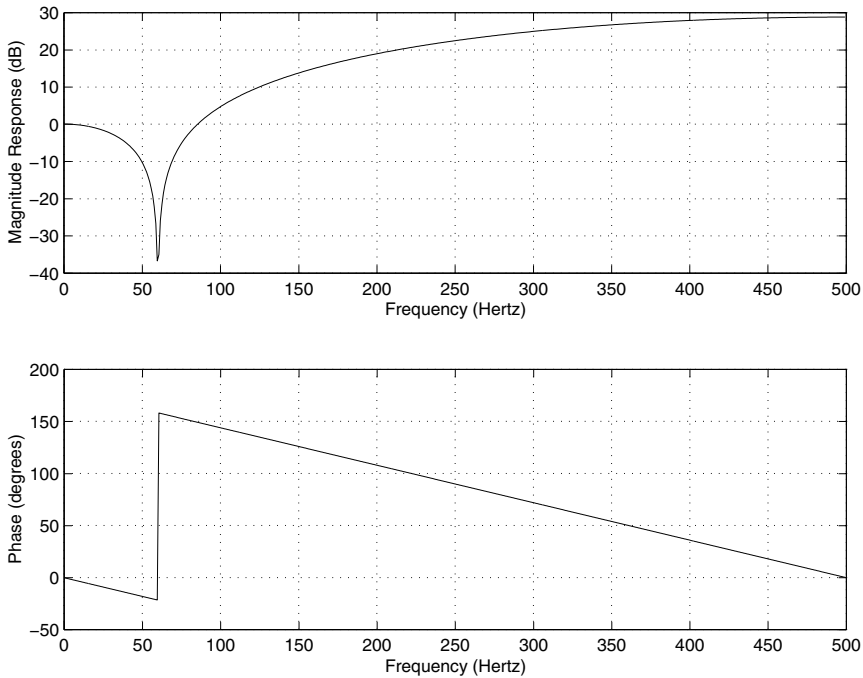


Figure 3.71 Magnitude and phase responses of the 60 Hz notch filter with zeros as shown in Figure 3.70. $f_s = 1,000$ Hz.

except the median or a few values close to the median; the output is then close to or equal to that of the median filter. The mean of the trimmed list provides a compromise between the generic mean and median filters.

- *L-filters*: a weighted combination of all of the elements in the rank-ordered list. Appropriate weights can provide outputs equivalent to those of all of the filters listed above, and facilitate the design of several nonlinear filters based on order statistics.

The methods described above may be used to realize several types of linear, nonlinear, nonstationary, and adaptive filters for the removal of various kinds of noise. It should be noted that nonlinear filters are not amenable to analysis using the Fourier transform.

Illustrations of application: Figure 3.80 (a) shows a synthesized test signal with a rectangular pulse; part (b) of the same figure shows the test signal contaminated with simulated impulsive (shot) noise. The results of filtering the noisy signal using the mean and median with filter length $M = 3$ samples are shown in parts (c) and (d) of Figure 3.80, respectively. The mean filter has blurred the edges of the pulse, which is undesirable; it has also created variations or details within the pulse that are artifacts. Furthermore, the result of the mean filter has retained the noise spikes

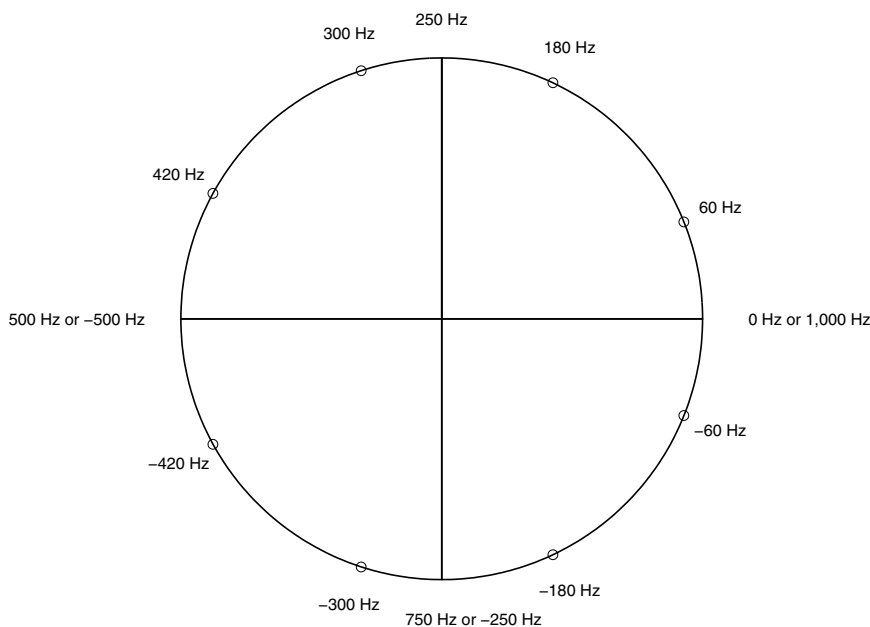


Figure 3.72 Zeros of the comb filter (in the z -domain) to remove 60 Hz interference with odd harmonics; the sampling frequency is 1,000 Hz .

present near the beginning and end of the signal, albeit reduced in amplitude. The median filter has completely eliminated the noise without distorting the pulse.

Figure 3.81 (a) shows another synthesized test signal with two rectangular pulses, the first one having a width of only two samples. Part (b) of the same figure shows the test signal degraded with uniformly distributed noise. The results of filtering the noisy signal using the mean and median with filter length $M = 5$ samples are shown in parts (c) and (d) of Figure 3.81, respectively. The mean filter has reduced the noise, but has also blurred the edges of the pulses; furthermore, the amplitude of the first short pulse has been reduced. Whereas the median filter has removed the noise to some extent without distorting the edges of the long pulse, it has obliterated the short pulse. The examples demonstrate the need to choose an appropriate type and length of the filter in accordance with the nature and strength of the noise as well as the signal.

Figure 3.82 shows a noisy ECG signal ($f_s = 1,000$ Hz) and the results of filtering with the mean and median using a causal sliding window including nine samples. Both of the filters have substantially reduced the noise without causing any noticeable distortion.

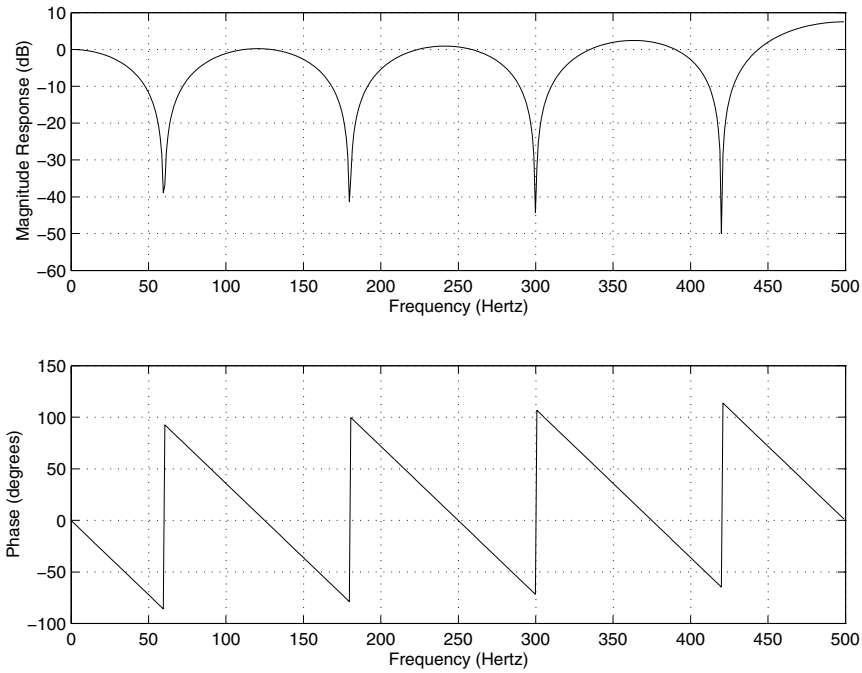


Figure 3.73 Magnitude and phase responses of the comb filter with zeros as shown in Figure 3.72.

3.8 Optimal Filtering: The Wiener Filter

The filters described in the preceding sections can take into account only limited information about the temporal or spectral characteristics of the signal and noise processes. They are often labeled as *ad hoc* filters: One may have to try several filter parameters and settle upon the filter that appears to provide a usable result. The output is not guaranteed to be the best achievable result: It is not optimized in any sense.

Problem: *Design an optimal filter to remove noise from a signal, given that the signal and noise processes are independent, stationary, random processes. You may assume that the “desired” or ideal characteristics of the uncorrupted signal are known. The noise characteristics may also be assumed to be known.*

Solution: Wiener filter theory provides for *optimal* filtering by taking into account the statistical characteristics of the signal and noise processes. The filter parameters are *optimized* with reference to a *performance criterion*. The output is guaranteed to be the best achievable result under the conditions imposed and the information provided. The Wiener filter is a powerful tool that changed traditional approaches to signal processing.

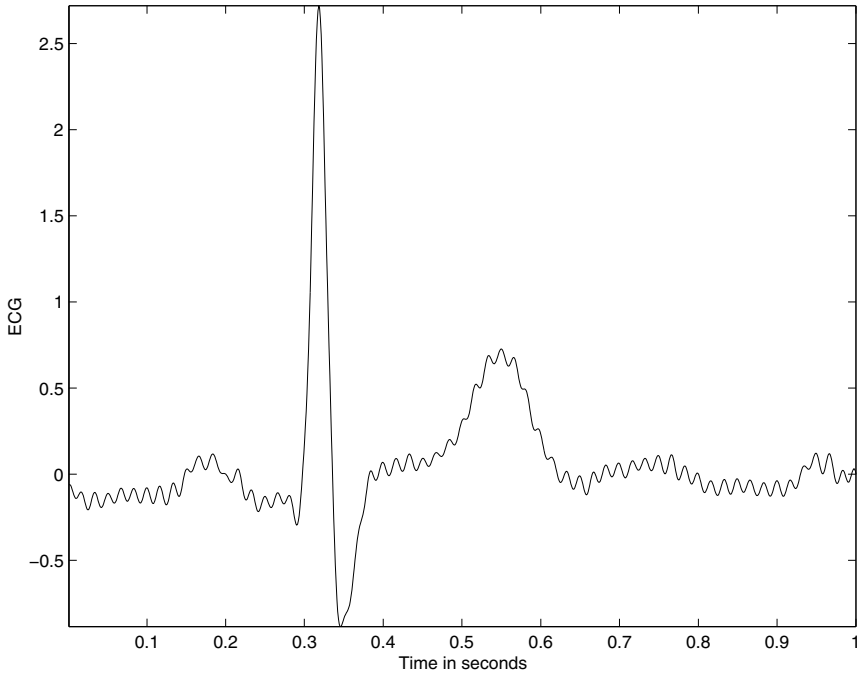


Figure 3.74 ECG signal with 60 Hz interference.

Considering the application of filtering a biomedical signal to remove noise, let us limit ourselves to a single-input, single-output, FIR filter with real input signal values and real coefficients. Figure 3.83 shows the general signal-flow diagram of a transversal filter with coefficients or tap weights w_i , $i = 0, 1, 2, \dots, M - 1$, input $x(n)$, and output $\tilde{d}(n)$ [148]. The output is usually considered to be an estimate of some “desired” signal $d(n)$ that represents the ideal, uncorrupted signal, and is, therefore, indicated as $\tilde{d}(n)$. If we assume, for the moment, that the desired signal is available, we could compute the *estimation error* between the output and the desired signal as

$$e(n) = d(n) - \tilde{d}(n). \quad (3.152)$$

Since $\tilde{d}(n)$ is the output of a linear FIR filter, it can be expressed as the convolution of the input $x(n)$ with the tap-weight sequence w_i (which is also the impulse response of the filter) as

$$\tilde{d}(n) = \sum_{k=0}^{M-1} w_k x(n-k). \quad (3.153)$$

For easier handling of the optimization procedures, the tap-weight sequence may be written as an $M \times 1$ *tap-weight vector*

$$\mathbf{w} = [w_0, w_1, w_2, \dots, w_{M-1}]^T, \quad (3.154)$$

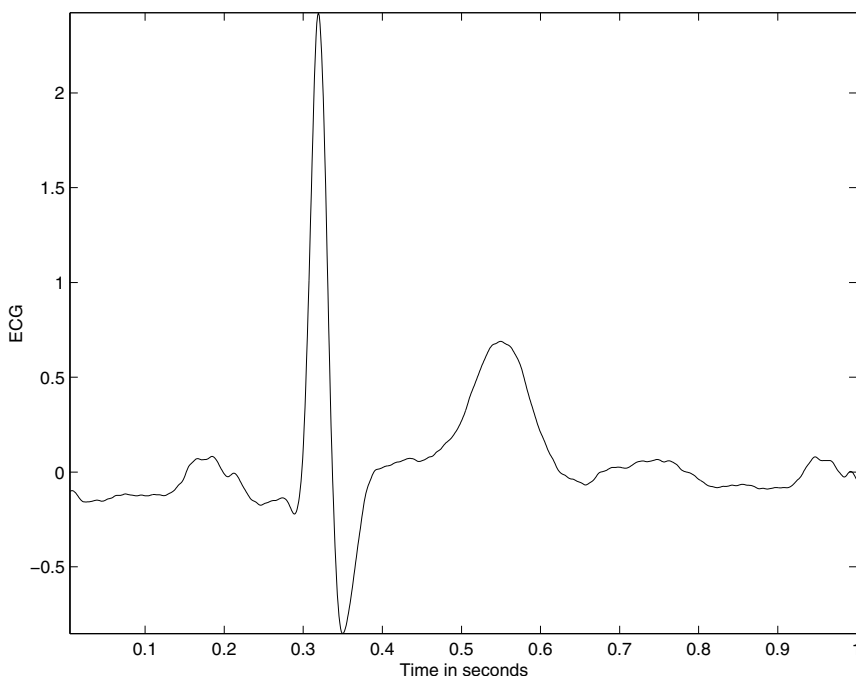


Figure 3.75 The ECG signal in Figure 3.74 after filtering with the 60 Hz notch filter shown in Figures 3.70 and 3.71.

where the boldfaced character $\mathbf{w}$ represents a vector and the superscript T indicates vector transposition. As the tap weights are combined with M values of the input in the convolution expression, we could also write the M input values as an $M \times 1$ vector:

$$\mathbf{x}(n) = [x(n), x(n-1), \dots, x(n-M+1)]^T. \quad (3.155)$$

Note that the vector $\mathbf{x}(n)$ varies with time: At a given instant n , the vector contains the current input sample $x(n)$ and the preceding $(M-1)$ input samples from $x(n-1)$ to $x(n-M+1)$. The convolution expression in Equation 3.153 may now be written in a simpler form as the inner or dot product of the vectors $\mathbf{w}$ and $\mathbf{x}(n)$:

$$\tilde{d}(n) = \mathbf{w}^T \mathbf{x}(n) = \mathbf{x}^T(n) \mathbf{w} = \langle \mathbf{x}, \mathbf{w} \rangle. \quad (3.156)$$

The estimation error is then given by

$$e(n) = d(n) - \mathbf{w}^T \mathbf{x}(n). \quad (3.157)$$

Wiener filter theory estimates the tap-weight sequence that minimizes the MS value of the estimation error; the output could then be called the *minimum mean-squared error* (MMSE) estimate of the desired response, the filter then being an

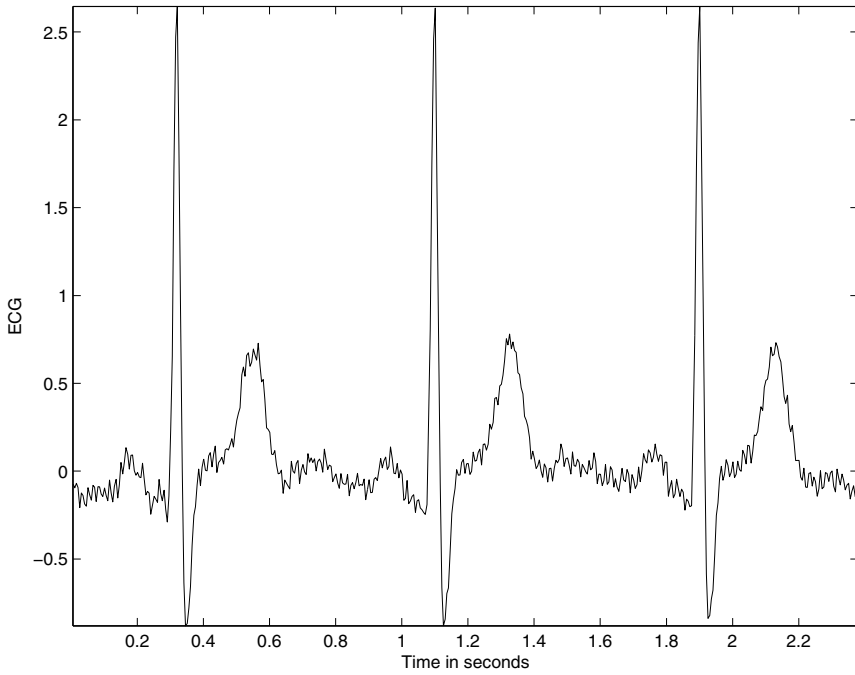


Figure 3.76 An ECG signal with power-line interference at 60 Hz.

optimal filter. The mean-squared error (MSE) is used to define the performance criterion as

$$\begin{aligned}
 J(\mathbf{w}) &= E[e^2(n)] \\
 &= E[\{d(n) - \mathbf{w}^T \mathbf{x}(n)\} \{d(n) - \mathbf{x}^T(n) \mathbf{w}\}] \\
 &= E[d^2(n)] - \mathbf{w}^T E[\mathbf{x}(n)d(n)] - E[d(n)\mathbf{x}^T(n)]\mathbf{w} \\
 &\quad + \mathbf{w}^T E[\mathbf{x}(n)\mathbf{x}^T(n)]\mathbf{w}.
 \end{aligned} \tag{3.158}$$

Note that the expectation operator is not applicable to $\mathbf{w}$ as it is not a random variable.

Under the assumption that the input vector $\mathbf{x}(n)$ and the desired response $d(n)$ are jointly stationary, the statistical expectation expressions in the equation given above have the following interpretations [148]:

- $E[d^2(n)]$ is the variance of $d(n)$, written as σ_d^2 , with the further assumption that the mean of $d(n)$ is zero.
- $E[\mathbf{x}(n)d(n)]$ is the cross-correlation between the input vector $\mathbf{x}(n)$ and the desired response $d(n)$, which is an $M \times 1$ vector:

$$\mathbf{\Theta} = E[\mathbf{x}(n)d(n)]. \tag{3.159}$$

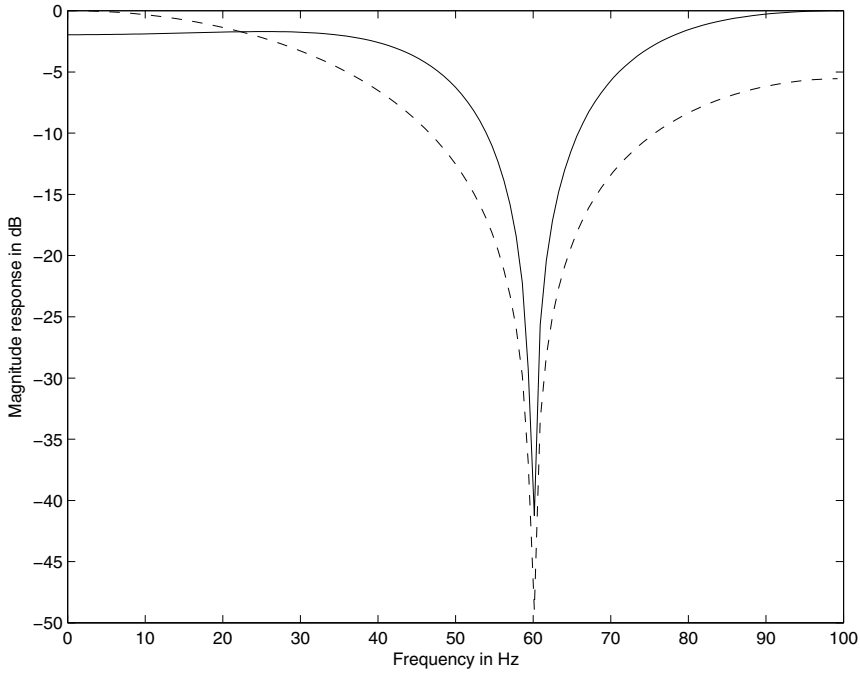


Figure 3.77 Dashed line: frequency response (magnitude, in dB) of a notch filter with two zeros at 60 Hz ; $f_s = 200\text{ Hz}$. Solid line: response of the filter with the inclusion of four poles; see the text for details.

Note that $\Theta = [\theta(0), \theta(-1), \dots, \theta(1-M)]^T$, where

$$\theta(-k) = E[x(n-k)d(n)], \quad k = 0, 1, 2, \dots, M-1. \quad (3.160)$$

- $E[d(n)\mathbf{x}^T(n)]$ is the transpose of $E[\mathbf{x}(n)d(n)]$; therefore,

$$\Theta^T = E[d(n)\mathbf{x}^T(n)]. \quad (3.161)$$

- $E[\mathbf{x}(n)\mathbf{x}^T(n)]$ represents the autocorrelation of the input vector $\mathbf{x}(n)$ computed as the outer product of the vector with itself, written as

$$\Phi = E[\mathbf{x}(n)\mathbf{x}^T(n)] \quad (3.162)$$

or in its full $M \times M$ matrix form as

$$\Phi = \begin{bmatrix} \phi(0) & \phi(1) & \cdots & \phi(M-1) \\ \phi(-1) & \phi(0) & \cdots & \phi(M-2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(-M+1) & \phi(-M+2) & \cdots & \phi(0) \end{bmatrix} \quad (3.163)$$

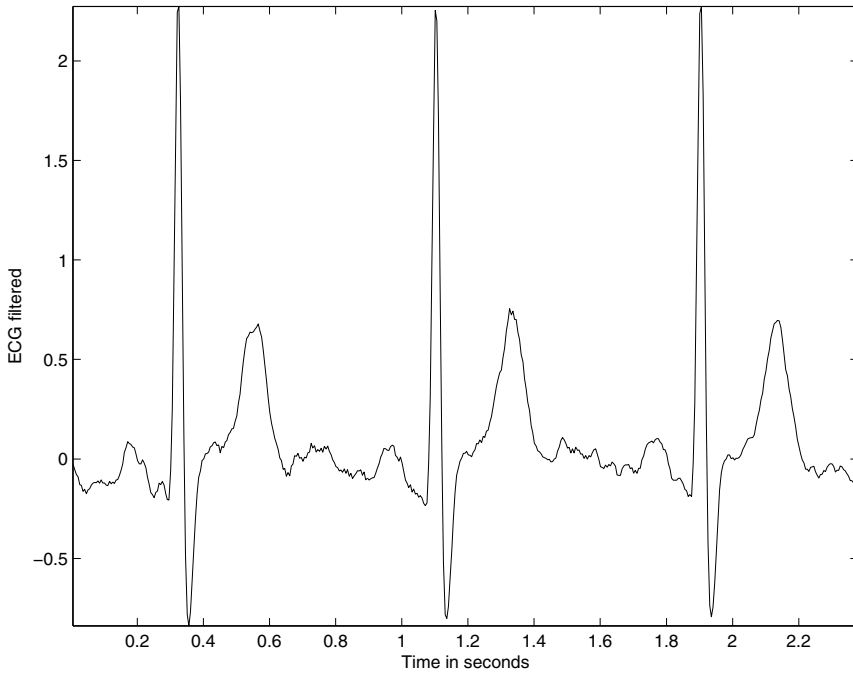


Figure 3.78 The result of filtering the noisy signal in Figure 3.76 using the notch filter with two zeros at 60 Hz.

with the element in row k and column i given by

$$\phi(i - k) = E[x(n - k)x(n - i)], \quad (3.164)$$

with the property that $\phi(i - k) = \phi(k - i)$. (Note: $\phi = \phi_{xx}$.) With the assumption of wide-sense stationarity, the $M \times M$ matrix Φ is completely specified by M values of the autocorrelation $\phi(0), \phi(1), \dots, \phi(M - 1)$ for lags $0, 1, \dots, M - 1$.

With the interpretations as listed above, the MSE expression in Equation 3.158 is simplified to

$$J(\mathbf{w}) = \sigma_d^2 - \mathbf{w}^T \Theta - \Theta^T \mathbf{w} + \mathbf{w}^T \Phi \mathbf{w}. \quad (3.165)$$

This expression indicates that the MSE is a second-order function of the tap-weight vector $\mathbf{w}$. To determine the optimal tap-weight vector, denoted by $\mathbf{w}_o$, we could differentiate $J(\mathbf{w})$ with respect to $\mathbf{w}$, set it to zero, and solve the resulting equation. To perform this differentiation, we should note the following derivatives:

$$\begin{aligned} \frac{d}{d\mathbf{w}}(\Theta^T \mathbf{w}) &= \Theta, \\ \frac{d}{d\mathbf{w}}(\mathbf{w}^T \Theta) &= \Theta, \end{aligned}$$

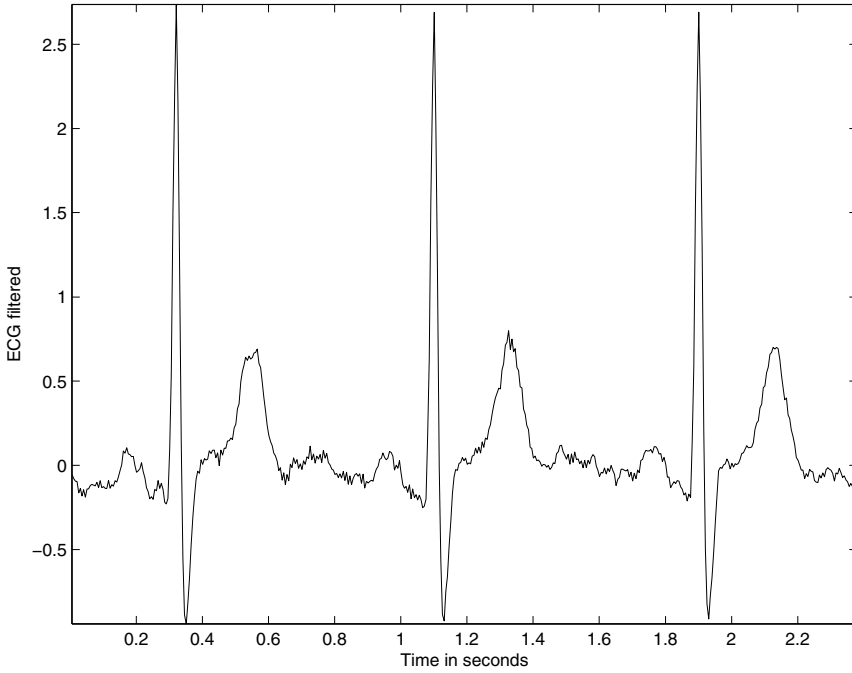


Figure 3.79 The result of filtering the noisy signal in Figure 3.76 using the notch filter with two zeros and four poles. Compare with the result in Figure 3.78.

$$\frac{d}{d\mathbf{w}}(\mathbf{w}^T \Phi \mathbf{w}) = 2\Phi \mathbf{w}.$$

Now, we obtain the derivative of $J(\mathbf{w})$ with respect to $\mathbf{w}$ as

$$\frac{dJ(\mathbf{w})}{d\mathbf{w}} = -2\Theta + 2\Phi \mathbf{w}. \quad (3.166)$$

Setting this expression to zero, we obtain the condition for the optimal filter as

$$\Phi \mathbf{w}_o = \Theta. \quad (3.167)$$

This equation is known as the *Wiener–Hopf* equation. It is also known as the *normal equation* as it can be shown that [148], for the optimal filter, each element of the input vector $\mathbf{x}(n)$ and the estimation error $e(n)$ are mutually orthogonal and, furthermore, that the filter output $\hat{d}(n)$ and the error $e(n)$ are mutually orthogonal (that is, the expectation of their products is zero). The optimal filter is obtained as

$$\mathbf{w}_o = \Phi^{-1} \Theta. \quad (3.168)$$

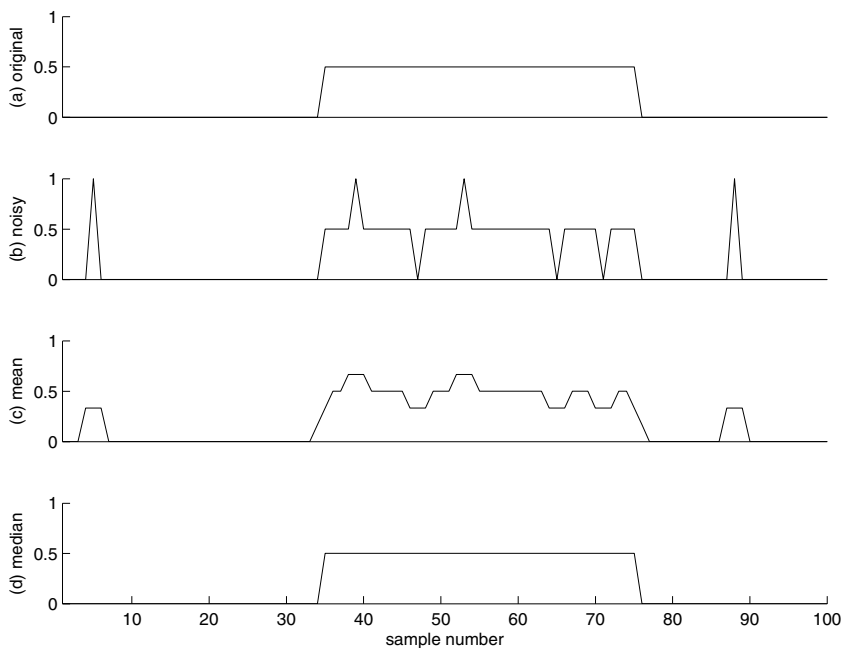


Figure 3.80 (a) A synthesized test signal with a rectangular pulse. (b) Test signal contaminated with simulated impulsive or shot noise. Result of filtering the noisy signal using (c) the mean and (d) the median with a sliding window having $M = 3$ samples.

In expanded form, we have the Wiener–Hopf equation as

$$\begin{bmatrix} \phi(0) & \phi(1) & \cdots & \phi(M-1) \\ \phi(-1) & \phi(0) & \cdots & \phi(M-2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(-M+1) & \phi(-M+2) & \cdots & \phi(0) \end{bmatrix} \begin{bmatrix} w_{o0} \\ w_{o1} \\ \vdots \\ w_{o(M-1)} \end{bmatrix} = \begin{bmatrix} \theta(0) \\ \theta(-1) \\ \vdots \\ \theta(1-M) \end{bmatrix} \quad (3.169)$$

or as

$$\sum_{i=0}^{M-1} w_{oi} \phi(i-k) = \theta(-k), \quad k = 0, 1, 2, \dots, M-1. \quad (3.170)$$

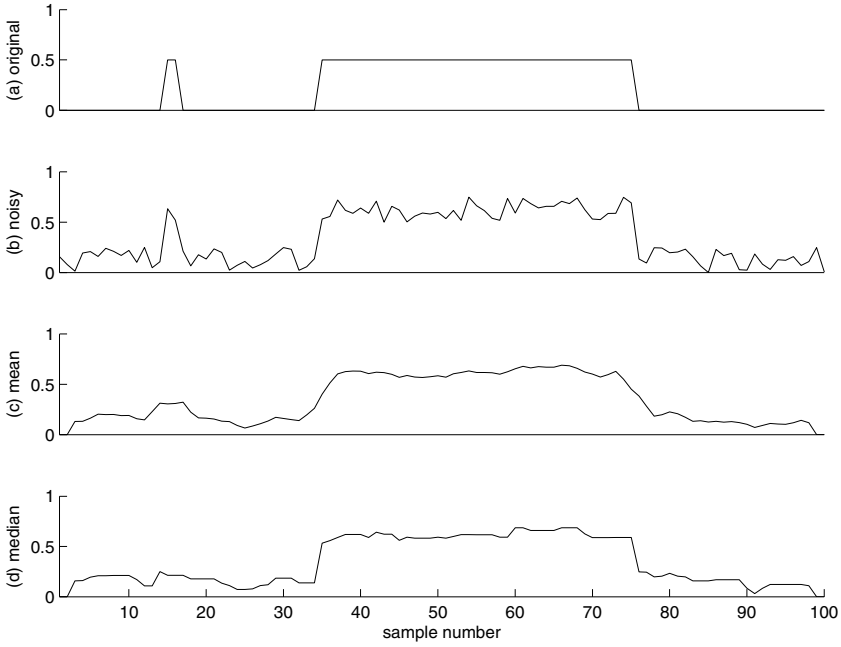


Figure 3.81 (a) A synthesized test signal with two rectangular pulses. (b) Degraded signal with uniformly distributed noise. Result of filtering the degraded signal using (c) the mean and (d) the median operation with a sliding window having $M = 5$ samples.

The minimum MSE is given by

$$J_{\min} = \sigma_d^2 - \Theta^T \Phi^{-1} \Theta. \quad (3.171)$$

Given the condition that the signals involved are stationary, we have $\phi(i - k) = \phi(k - i)$ and $\theta(-k) = \theta(k)$, that is, the functions ϕ and θ are even-symmetric. Then, we may write Equation 3.170 as

$$\sum_{i=0}^{M-1} w_{oi} \phi(k - i) = \theta(k), \quad k = 0, 1, 2, \dots, M - 1. \quad (3.172)$$

Thus, we have the convolution relationship

$$w_{ok} * \phi(k) = \theta(k). \quad (3.173)$$

Applying the Fourier transform to the equation given above, we get

$$W(\omega) S_{xx}(\omega) = S_{xd}(\omega), \quad (3.174)$$

which may be modified to obtain the Wiener filter frequency response $W(\omega)$ as

$$W(\omega) = \frac{S_{xd}(\omega)}{S_{xx}(\omega)}, \quad (3.175)$$

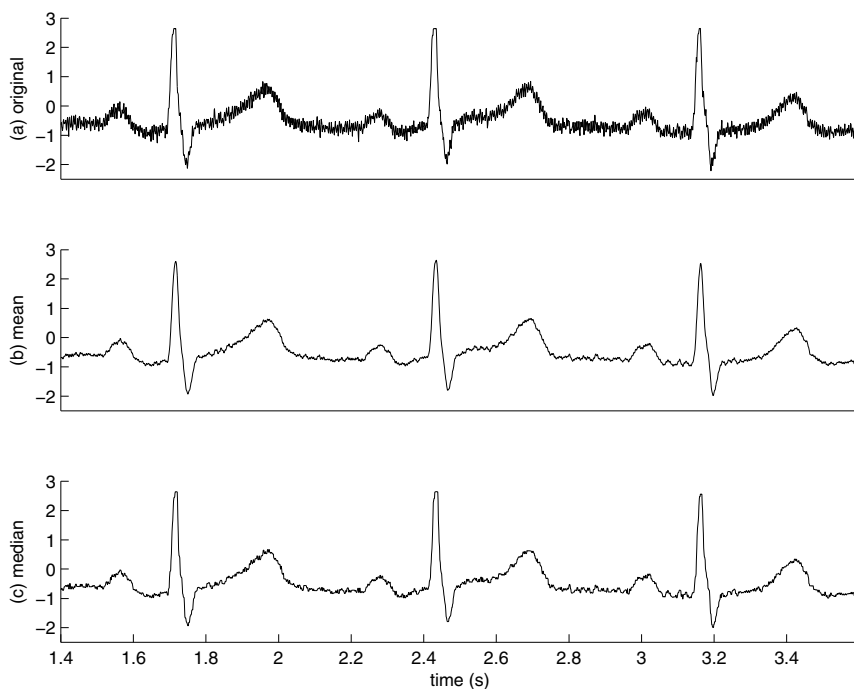


Figure 3.82 (a) An ECG signal with noise. Result of filtering the ECG signal using (b) the mean and (c) the median operation with a sliding window having $M = 9$ samples.

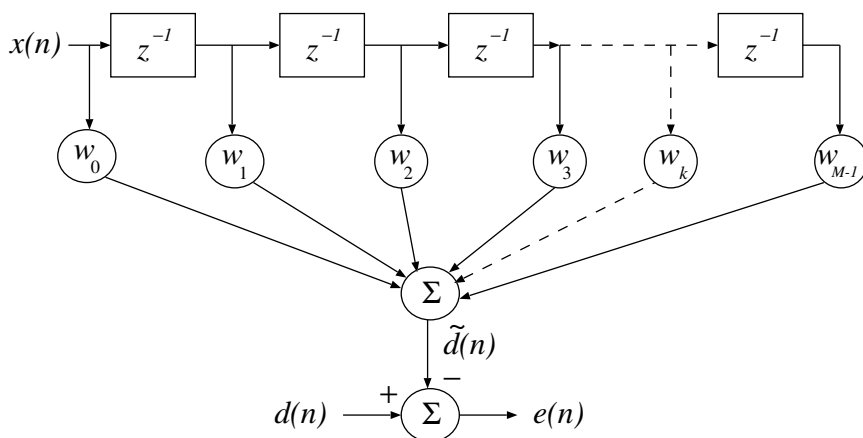


Figure 3.83 Signal-flow diagram of the Wiener filter.

where $S_{xx}(\omega)$ is the PSD of the input signal and $S_{xd}(\omega)$ is the cross-spectral density (CSD) between the input signal and the desired signal.

Note that derivation of the optimal filter requires specific knowledge about the input and the desired response in the form of the autocorrelation Φ of the input $x(n)$ and the cross-correlation Θ between the input $x(n)$ and the desired response $d(n)$. In practice, although the desired response $d(n)$ may not be known, it should be possible to obtain an estimate of its temporal or spectral statistics, which may be used to estimate Θ . Proper estimation of the statistical entities mentioned above requires a large number of samples of the corresponding signals.

{Note: Haykin [148] allows all the entities involved to be complex. Vector transposition T is then Hermitian or complex-conjugate transposition H . Products of two entities require one to be conjugated: for example, $e^2(n)$ is obtained as $e(n)e^*(n)$, and Equation 3.153 will have w_k^* in place of w_k . Furthermore, $\frac{d}{d\mathbf{w}}(\Theta^H \mathbf{w}) = \mathbf{0}$ and $\frac{d}{d\mathbf{w}}(\mathbf{w}^H \Theta) = 2\Theta$. The final Wiener–Hopf equation, however, simplifies to the same as above in Equation 3.170.}

Let us now consider the problem of removing noise from a corrupted input signal. For this case, let the input $x(n)$ contain a mixture of the desired (original) signal $d(n)$ and noise $\eta(n)$, that is,

$$x(n) = d(n) + \eta(n). \quad (3.176)$$

Using the vector notation as before, we have

$$\mathbf{x}(n) = \mathbf{d}(n) + \boldsymbol{\eta}(n), \quad (3.177)$$

where $\boldsymbol{\eta}(n)$ is the vectorial representation of the noise function $\eta(n)$. The autocorrelation matrix of the input is given by

$$\Phi = E[\mathbf{x}(n)\mathbf{x}^T(n)] = E[\{\mathbf{d}(n) + \boldsymbol{\eta}(n)\}\{\mathbf{d}(n) + \boldsymbol{\eta}(n)\}^T]. \quad (3.178)$$

If we assume that the noise process is statistically independent of the signal process and that at least one of the processes has a mean value of zero, we have

$$E[\mathbf{d}(n)\boldsymbol{\eta}^T(n)] = E[\boldsymbol{\eta}^T(n)\mathbf{d}(n)] = \mathbf{0}. \quad (3.179)$$

(This result is valid also if the two processes are mutually orthogonal.) Then,

$$\Phi = E[\mathbf{d}(n)\mathbf{d}^T(n)] + E[\boldsymbol{\eta}(n)\boldsymbol{\eta}^T(n)] = \Phi_d + \Phi_\eta, \quad (3.180)$$

where Φ_d and Φ_η are the $M \times M$ autocorrelation matrices of the signal and noise, respectively. Furthermore,

$$\Theta = E[\mathbf{x}(n)d(n)] = E[\{\mathbf{d}(n) + \boldsymbol{\eta}(n)\}d(n)] = E[\mathbf{d}(n)d(n)] = \Phi_{1d}, \quad (3.181)$$

where Φ_{1d} is an $M \times 1$ autocorrelation vector of the desired signal. The optimal Wiener filter is then given by

$$\mathbf{w}_o = (\Phi_d + \Phi_\eta)^{-1}\Phi_{1d}. \quad (3.182)$$

The frequency response of the Wiener filter may be obtained by modifying Equation 3.175 by taking into account the spectral relationships

$$S_{xx}(\omega) = S_d(\omega) + S_\eta(\omega) \quad (3.183)$$

and

$$S_{xd}(\omega) = S_d(\omega), \quad (3.184)$$

which leads to

$$W(\omega) = \frac{S_d(\omega)}{S_d(\omega) + S_\eta(\omega)} = \frac{1}{1 + \frac{S_\eta(\omega)}{S_d(\omega)}}, \quad (3.185)$$

where $S_d(\omega)$ and $S_\eta(\omega)$ are the PSDs of the desired signal and the noise process, respectively. Note that designing the optimal filter requires knowledge of the PSDs of the desired signal and the noise process (or models thereof).

From Equation 3.185, the following important properties of the Wiener filter are evident: $W(\omega) = 0$ wherever $S_d(\omega) = 0$, $W(\omega) = 1$ wherever $S_\eta(\omega) = 0$, and $W(\omega)$ decreases as $S_\eta(\omega)$ increases. Therefore, the Wiener filter has the following characteristics:

- frequency components that are not present in the input (that is, have a value of zero) will not be modified or restored,
- the input signal's frequency components are passed with unit gain at frequencies where noise does not exist, and
- the gain of the Wiener filter decreases as the SNR , expressed as a function of frequency, decreases.

Illustrations of application: The upper trace in Figure 3.84 shows one ECG cycle extracted from the signal with noise in Figure 3.5. A piecewise linear model of the desired version of the signal was created by concatenating linear segments to provide P, QRS, and T waves with amplitudes, durations, and intervals similar to those in the given noisy signal. The baseline of the model was set to zero. The noise-free model is shown in the middle trace of Figure 3.84. The log PSDs of the given noisy signal and the noise-free model, the latter being $S_d(\omega)$ in Equation 3.185, are shown in the upper two plots of Figure 3.85.

The T-P intervals between successive cardiac cycles in an ECG (the interbeat intervals) may be taken to represent the isoelectric baseline. Then, any activity present in these intervals constitutes noise. Four T-P intervals were selected from the noisy signal in Figure 3.5, and their Fourier power spectra were averaged to derive the noise PSD $S_\eta(\omega)$ required in the Wiener filter (Equation 3.185). The estimated log PSD of the noise is shown in the third trace of Figure 3.85. Observe the relatively high levels of power in the noise PSD above 100 Hz as compared to the PSDs of the original noisy signal and the model. Observe also the peaks in the original and noise PSDs near 180 Hz , 300 Hz , and 420 Hz , representing the third, fifth, and seventh harmonics of 60 Hz , respectively; the peak at 460 Hz is an aliased version

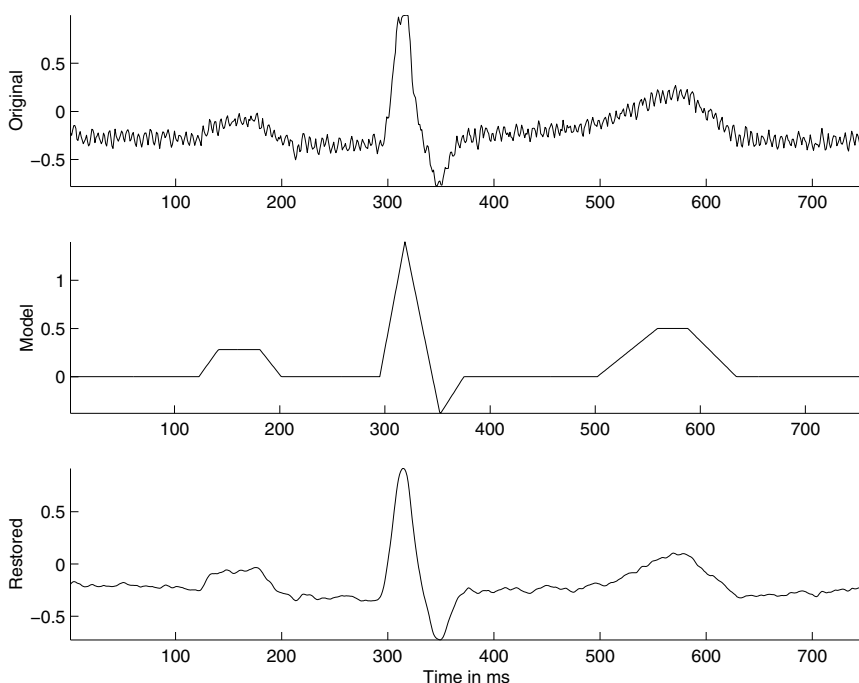


Figure 3.84 From top to bottom: one cycle of the noisy ECG signal in Figure 3.5 (labeled as Original); a piecewise linear model of the desired noise-free signal (Model); and the output of the Wiener filter (Restored).

of the ninth harmonic at 540 Hz . The 60 Hz component itself appears to have been suppressed by a notch filter in the signal acquisition system. (See Sections 3.3.4 and 3.6.3 for details.)

The Wiener filter frequency response was derived as in Equation 3.185, and is shown in the fourth plot in Figure 3.85. Observe the low gain of the filter near 180 Hz , 300 Hz , 420 Hz , and 460 Hz corresponding to the peaks in the noise spectrum. As indicated by Equation 3.185, the Wiener filter gain is inversely related to the noise PSD and directly related to the signal PSD. The result of application of the Wiener filter to the given signal is shown in the third trace of Figure 3.84. It is evident that almost all of the noise has been effectively removed by the filter. This is also confirmed by comparing the first and last plots in Figure 3.85: The spectrum of the restored signal has values below -50 dB for frequencies greater than 100 Hz .

The most important point to observe here is that the filter was derived with models of the noise and signal processes (PSDs), which were obtained from the given signal itself in the present application. No cutoff frequency was required to be specified in designing the Wiener filter, whereas the Butterworth filter requires the specification of a cutoff frequency and filter order.

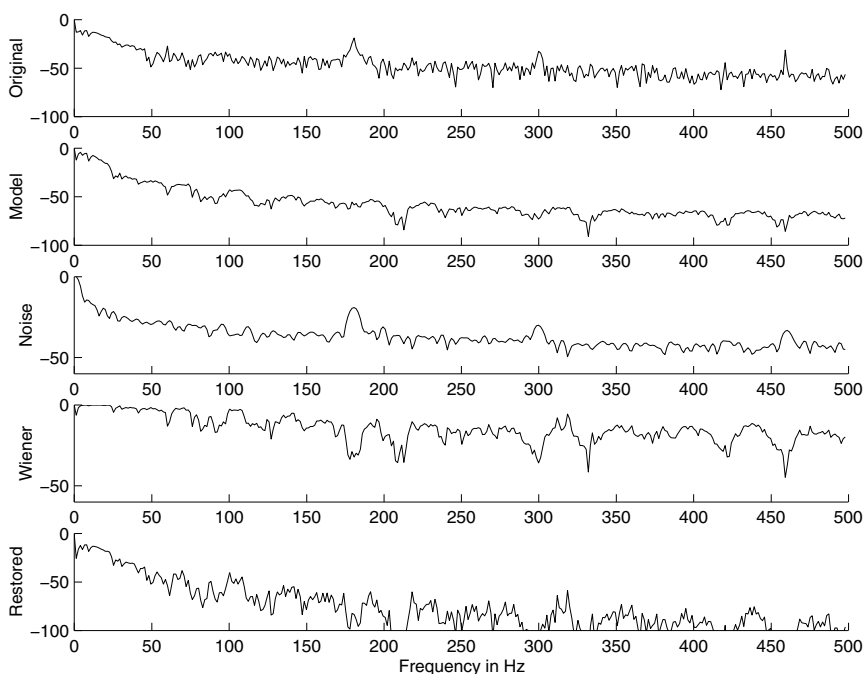


Figure 3.85 From top to bottom: log PSD (in dB) of the given noisy signal (labeled as Original); log PSD of the noise-free model (Model); estimated log PSD of the noise process (Noise); log frequency response of the Wiener filter (Wiener); and log PSD of the filter output (Restored). The scaling of the vertical axis is not the same for all PSDs in the figure.

In another experiment, a noise-free ECG signal was obtained from a subject under controlled conditions; a one-second-long segment of this signal was selected to derive the desired signal PSD (model) for use in the design of the Wiener filter. Subsequently, a noisy ECG was recorded from the same subject by including artifacts related to motion and contraction of the limbs. The sampling frequency used was 200 Hz . Ten segments of noise were selected from the noisy ECG in the T-P intervals and their average PSD was computed. The impulse response of the Wiener filter was derived by taking the inverse Fourier transform of the frequency response derived according to Equation 3.185.

Figure 3.86 shows one-second-long traces of the model ECG, the noisy ECG, and the filtered result. Also shown in the same figure are the ACFs of the model ECG and noise, as well as the impulse response of the Wiener filter. The impulse response of the Wiener filter, which possesses even symmetry due to the real-valued nature of the related frequency response, was truncated by selecting only 10 samples on either side of its maximal value (which occurs at zero time but appears at 0.5 s in Figure 3.86 due to the shift applied for causality), resulting in the filter order of $M = 21$. The coefficients obtained as above were used to filter the noisy signal in the time domain

(over a duration of about two minutes). It is evident from the last two traces of Figure 3.86 that the noise has been effectively suppressed without any noticeable degradation of the ECG wave components. Figure 3.87 shows longer traces of the noisy and filtered ECG signals. Close inspection of the results indicates that, while relatively high-frequency noise has been suppressed, noise within the bandwidth of the ECG signal remains in the output. Furthermore, low-frequency artifacts remain for the same reason, along with the fact that the short segments of noise used to model the noise PSD do not effectively incorporate the low-frequency artifacts in the model. The PSDs and frequency response of the Wiener filter shown in Figure 3.88 confirm the observations made above for the present example as well as those in the previous illustration.

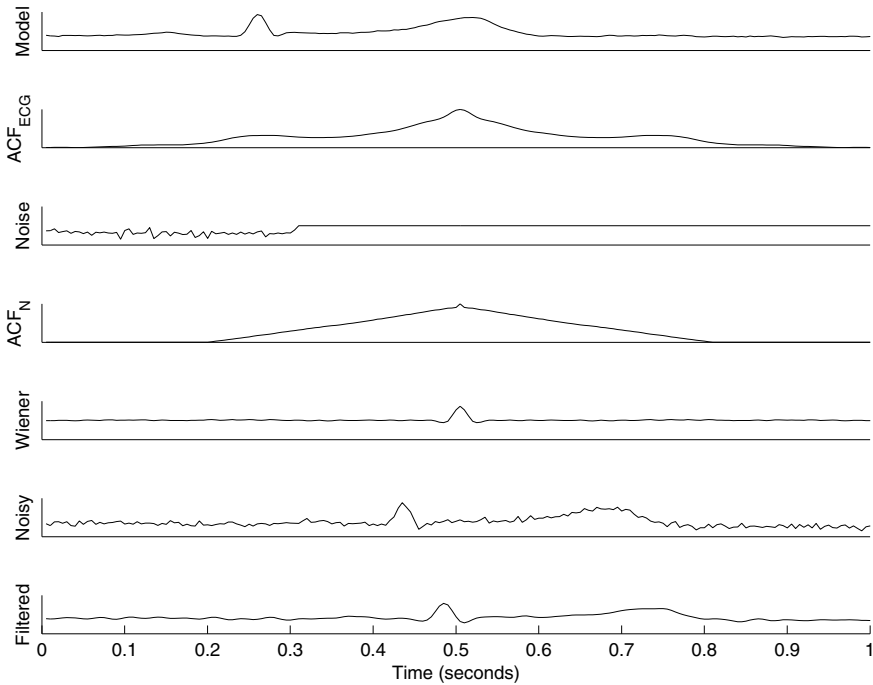


Figure 3.86 From top to bottom: one cycle of the noise-free ECG of a subject (labeled as Model); the ACF of the noise-free ECG; a sample segment of noise from a noisy ECG of the same subject (the actual noise segment has a duration of 0.3 s but has been padded with zeros to the same length as the ECG, which is 1 s); the ACF of the noise (ACF_N) obtained using 10 segments; the impulse response of the Wiener filter (shifted for causality); a segment of the noisy ECG to be filtered; and the corresponding filtered result. The amplitude labels have been suppressed to prevent clutter. ECG signal data courtesy of Emily Marasco and Matthew LaRocque, University of Calgary. See also Figure 3.87.

Most signal acquisition systems should permit the measurement of at least the variance or power level of the noise present. A uniform (white) PSD model may

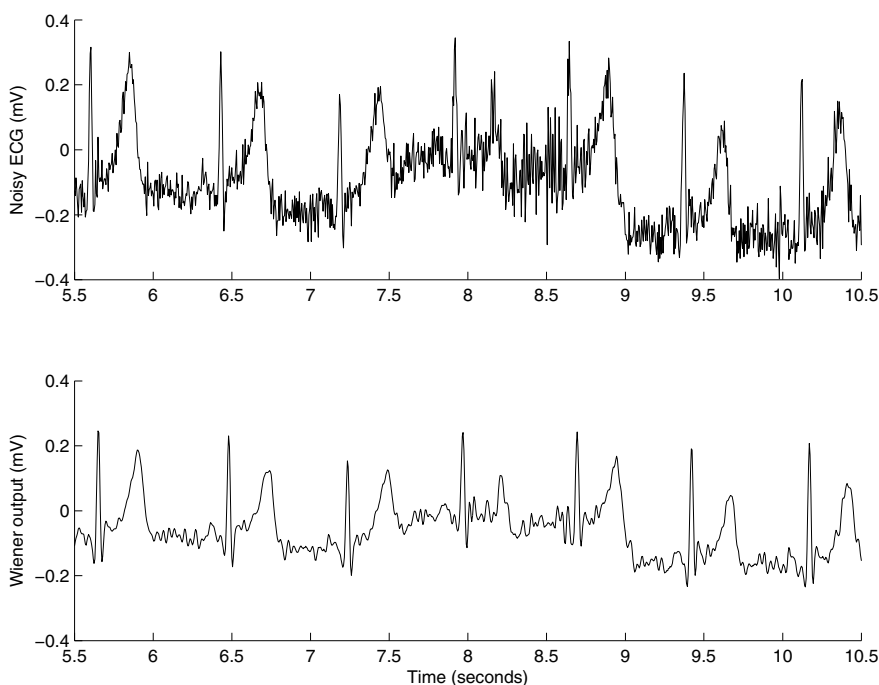


Figure 3.87 From top to bottom: a segment of the noisy ECG to be filtered and the corresponding filtered result. ECG signal data courtesy of Emily Marasco and Matthew LaRocque, University of Calgary. See also Figure 3.86.

then be easily derived. Models of the ideal signal and the noise processes may also be created using parametric Gaussian or Laplacian models either in the time domain (ACF) or directly in the frequency domain (PSD). However, it should be observed that the noise PSDs in Figures 3.85 and 3.88 are not flat or white, and that the noise ACF in Figure 3.86 is not an impulse. These characteristics indicate that the noise process is not random but has some underlying structure.

3.9 Adaptive Filters for Removal of Interference

Filters with fixed characteristics (tap weights or coefficients), as seen in the preceding sections, are suitable when the characteristics of the signal and noise (random or structured) are stationary and known. Design of frequency-domain filters requires detailed knowledge of the spectral contents of the signal and noise. Such filters are not applicable when the characteristics of the signal and/or noise vary with time, that is, when they are nonstationary. They are also not suitable when the spectral contents of the signal and the interference overlap significantly.

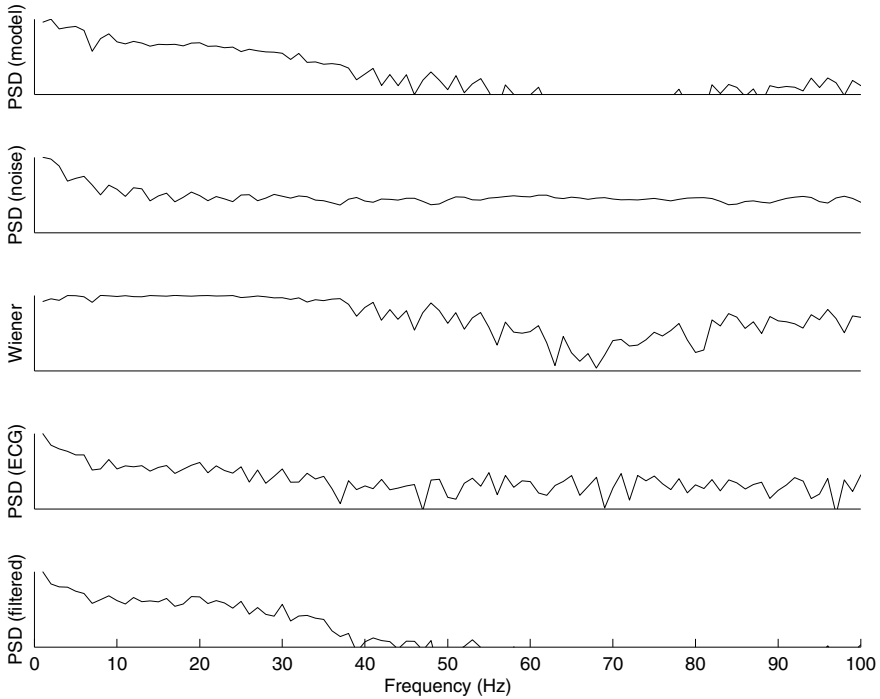


Figure 3.88 From top to bottom: log PSD (in dB) of the noise-free ECG (model); log PSD of the noise obtained using 10 segments; log frequency response of the Wiener filter; log PSD of the noisy ECG; and log PSD of the filtered result. The display of all PSDs is limited to the range $[-50, 0]$ dB . $f_s = 200$ Hz . The PSDs of ECG signals were obtained using only one cardiac cycle of each as shown in Figure 3.86.

Consider the situation when two ECG signals such as those of a fetus and the expectant mother, or two vibration signals such as the VAG and the VMG, arrive at the recording site and get added in some proportion. The spectra of the signals in the mixture span the same or similar frequency ranges, and hence fixed filtering cannot separate them. In the case of the VAG/VMG mixture, it is also possible for the spectra of the signals to vary from one point in time to another, due to changes in the characteristics of the cartilage surfaces causing the VAG signal, and due to the effect of variations in the recruitment of the motor units of a muscle on the VMG signal. Such a situation calls for the use of a filter that can learn and adapt to the characteristics of the interference, estimate the interfering signal, and remove it from the mixture to obtain the desired signal. This requires the filter to adjust automatically its impulse response (and hence its frequency response) as the characteristics of the signal and/or noise vary. (Several methods for the analysis of nonstationary and multicomponent signals are presented in Chapter 8.)

Problem: *Design an optimal filter to remove a nonstationary interference from a nonstationary signal. An additional channel of information related to the interference is available for use. The filter should continuously adapt to the changing characteristics of the signal and interference.*

Solution: We need to address two different concerns in this problem:

1. The filter should be *adaptive*; the tap-weight vector of the filter will then vary with time. The principles of the adaptive filter, also known as the adaptive noise canceler (ANC), are explained in Section 3.9.1.
2. The filter should be *optimal*. Two well-established methods for optimization of the adaptive filter are presented in Sections 3.9.2 and 3.9.3.

Illustrations of the application of the methods are presented at the end of Sections 3.9.2 and 3.9.3 as well as at the end of the chapter, in Sections 3.13 and 3.14.

3.9.1 The adaptive noise canceler

Figure 3.89 shows a generic block diagram of an adaptive filter or ANC [124, 185]. The “primary input” to the filter $x(n)$ is a mixture of the signal of interest $v(n)$ and the “primary noise” $m(n)$:

$$x(n) = v(n) + m(n). \quad (3.186)$$

$x(n)$ is the primary observed signal; it is desired that the interference or noise $m(n)$ be estimated and removed from $x(n)$ in order to obtain the signal of interest $v(n)$. (The notation and terminology used in the present section, while being different from those in the preceding sections, are commonly used in the literature on adaptive filters.) It is assumed that $v(n)$ and $m(n)$ are uncorrelated. Adaptive filtering requires a second input, known as the “reference input” $r(n)$, that is uncorrelated with the signal of interest $v(n)$ but closely related to or correlated with the interference or noise $m(n)$ in some manner that need not be known. The ANC filters or modifies the reference input $r(n)$ to obtain a signal $y(n)$ that is as close to the noise $m(n)$ as possible. $y(n)$ is then subtracted from the primary input to estimate the desired signal:

$$\tilde{v}(n) = e(n) = x(n) - y(n). \quad (3.187)$$

Let us now analyze the function of the filter. Let us assume that the signal of interest $v(n)$, the primary noise $m(n)$, the reference input $r(n)$, and the primary noise estimate $y(n)$ are statistically stationary and have zero means. (*Note:* The requirement of stationarity is removed later when the expectations are computed in moving windows.) We have already stated that $v(n)$ is uncorrelated with $m(n)$ and $r(n)$, and that $r(n)$ is correlated with $m(n)$. The output of the ANC is

$$\begin{aligned} e(n) &= x(n) - y(n) \\ &= v(n) + m(n) - y(n), \end{aligned} \quad (3.188)$$

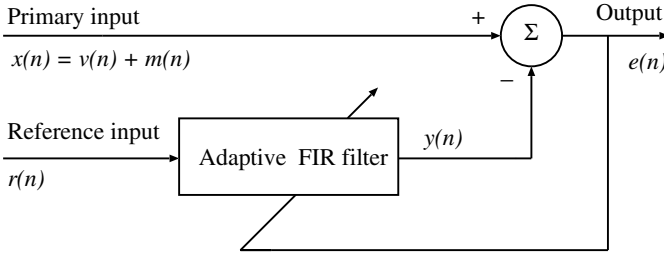


Figure 3.89 Block diagram of a generic ANC or adaptive filter.

where $y(n) = \tilde{m}(n)$ is the estimate of the primary noise obtained at the output of the adaptive FIR filter. By taking the square and expectation (statistical average) of both sides of Equation 3.188, we obtain

$$E[e^2(n)] = E[v^2(n)] + E[\{m(n) - y(n)\}^2] + 2E[v(n)\{m(n) - y(n)\}]. \quad (3.189)$$

Since $v(n)$ is uncorrelated with $m(n)$ and $y(n)$ and all of them have zero means, we have

$$E[v(n)\{m(n) - y(n)\}] = E[v(n)]E[m(n) - y(n)] = 0. \quad (3.190)$$

Equation 3.189 can be rewritten as

$$E[e^2(n)] = E[v^2(n)] + E[\{m(n) - y(n)\}^2]. \quad (3.191)$$

With reference to Figure 3.89, note that the output $e(n)$ is used (fed back) to control the adaptive FIR filter. In ANC applications, the objective is to obtain an output $e(n)$ that is a least-squares fit to the desired signal $v(n)$. This is achieved by feeding the output back to the adaptive FIR filter and adjusting the filter to minimize the total system output power. The system output serves as the error signal for the adaptive process.

The signal power $E[v^2(n)]$ will be unaffected as the filter is adjusted to minimize $E[e^2(n)]$; accordingly, the minimum output power is

$$\min E[e^2(n)] = E[v^2(n)] + \min E[\{m(n) - y(n)\}^2]. \quad (3.192)$$

As the filter is adjusted so that $E[e^2(n)]$ is minimized, $E[\{m(n) - y(n)\}^2]$ is also minimized. Thus, the filter output $y(n)$ is the MMSE estimate of the primary noise $m(n)$. Moreover, when $E[\{m(n) - y(n)\}^2]$ is minimized, $E[\{e(n) - v(n)\}^2]$ is also minimized, since from Equation 3.188, we have

$$e(n) - v(n) = m(n) - y(n). \quad (3.193)$$

Adjusting or adapting the filter to minimize the total output power is, therefore, equivalent to causing the output $e(n)$ to be the MMSE estimate of the signal of interest $v(n)$ for the given structure and adjustability of the adaptive FIR filter and for the given reference input.

The output $e(n)$ will contain the signal of interest $v(n)$ and some noise. From Equation 3.193, the output noise is given by $e(n) - v(n) = \tilde{v}(n) - v(n) = m(n) - y(n)$. Because minimizing $E[e^2(n)]$ minimizes $E[\{m(n) - y(n)\}^2]$, *minimizing the total output power minimizes the output noise power*. Since the signal component $v(n)$ in the output remains unaffected, *minimizing the total output power maximizes the output SNR*.

From Equation 3.191, we have the condition that the output power is minimum when $E[e^2(n)] = E[v^2(n)]$. When this condition is achieved, $E[\{m(n) - y(n)\}^2] = 0$. We then have $y(n) = m(n)$ and $e(n) = v(n)$; that is, the output is a perfect and noise-free estimate of the desired signal.

Optimization of the filter may be performed by expressing the error in terms of the tap-weight vector and applying the procedure of choice. The output $y(n)$ of the adaptive filter (see Figure 3.89) in response to its input $r(n)$ is given by

$$y(n) = \sum_{k=0}^{M-1} w_k r(n-k), \quad (3.194)$$

where w_k , $k = 0, 1, 2, \dots, M-1$, are the tap weights, and M is the order of the filter. The estimation error $e(n)$ or the output of the ANC system is

$$e(n) = x(n) - y(n). \quad (3.195)$$

For the sake of notational simplicity, let us define the tap-weight vector at time n as

$$\mathbf{w}(n) = [w_0(n), w_1(n), \dots, w_{M-1}(n)]^T. \quad (3.196)$$

Similarly, the input vector at each time instant n may be defined as the M -dimensional vector

$$\mathbf{r}(n) = [r(n), r(n-1), \dots, r(n-M+1)]^T. \quad (3.197)$$

Then, the estimation error $e(n)$ given in Equation 3.195 may be rewritten as

$$e(n) = x(n) - \mathbf{w}^T(n)\mathbf{r}(n). \quad (3.198)$$

It is worth noting that the derivations made above required no knowledge about the processes behind $v(n)$, $m(n)$, and $r(n)$ or their interrelationships, other than the assumptions of statistical independence between $v(n)$ and $m(n)$ and some form of correlation between $m(n)$ and $r(n)$. The arguments can be extended to situations where the primary and reference inputs contain additive random noise processes that are mutually uncorrelated and also uncorrelated with $v(n)$, $m(n)$, and $r(n)$. The procedures may also be extended to cases where $m(n)$ and $r(n)$ are deterministic or structured rather than stochastic processes, such as power-line interference, an ECG, or a VMG signal [124].

Several methods are available to maximize the output SNR; two such methods based on the least-mean-squares (LMS) and the recursive least-squares (RLS) approaches are described in the following sections.

3.9.2 The least mean squares adaptive filter

The purpose of adaptive filtering algorithms is to adjust the tap-weight vector to minimize the MSE. By squaring the expression for the estimation error $e(n)$ given in Equation 3.198, we get

$$e^2(n) = x^2(n) - 2x(n)\mathbf{r}^T(n)\mathbf{w}(n) + \mathbf{w}^T(n)\mathbf{r}(n)\mathbf{r}^T(n)\mathbf{w}(n). \quad (3.199)$$

The squared error is a second-order (quadratic) function of the tap-weight vector (and the inputs), and may be depicted as a concave hyperparaboloidal (bowl-like) surface that is never negative. The aim of the filter optimization procedure would be to reach the bottom of the bowl-like function. Gradient-based methods may be used for this purpose.

By taking the expected values of the entities in Equation 3.199 and taking the derivative with respect to the tap-weight vector, we may derive the Wiener–Hopf equation for the present application. The LMS algorithm takes a simpler approach by assuming the square of the instantaneous error as in Equation 3.199 to stand for an estimate of the MSE [124]. The LMS algorithm is based on the method of steepest descent, where the new tap-weight vector $\mathbf{w}(n+1)$ is given by the present tap-weight vector $\mathbf{w}(n)$ plus a correction proportional to the negative of the gradient $\nabla e^2(n)$ of the squared error:

$$\mathbf{w}(n+1) = \mathbf{w}(n) - \mu \nabla e^2(n). \quad (3.200)$$

The parameter μ controls the stability and rate of convergence of the algorithm: the larger the value of μ , the larger the gradient of the error that is introduced and the faster the convergence of the algorithm, and viceversa.

The LMS algorithm approximates $\nabla e^2(n)$ by the derivative of the squared error in Equation 3.199 with respect to the tap-weight vector as

$$\widetilde{\nabla e^2}(n) = -2x(n)\mathbf{r}(n) + 2\{\mathbf{w}^T(n)\mathbf{r}(n)\}\mathbf{r}(n) = -2e(n)\mathbf{r}(n). \quad (3.201)$$

Using this estimate of the gradient in Equation 3.200, we get

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu e(n) \mathbf{r}(n). \quad (3.202)$$

This expression is known as the Widrow–Hoff LMS algorithm.

The advantages of the LMS algorithm lie in its simplicity and ease of implementation: Although the method is based on the MSE and gradient-based optimization, the filter expression itself is free of differentiation, squaring, or averaging. It has been shown that the expected value of the tap-weight vector provided by the LMS algorithm converges to the optimal Wiener solution when the input vectors are uncorrelated over time [124, 186]. The procedure may be started with an arbitrary tap-weight vector; it will converge in the mean and remain stable as long as μ is greater than zero but less than the reciprocal of the largest eigenvalue of the autocorrelation matrix of the reference input [124].

Illustration of application: Zhang et al. [125] used a two-stage adaptive LMS filter to cancel muscle-contraction interference from VAG signals. The first stage

was used to remove the measurement noise in the accelerometers and associated amplifiers, and the second stage was designed to cancel the muscle signal.

Zhang et al. [125] also proposed a procedure for optimization of the step size μ by using an *RMS*-error-based misadjustment factor and a time-varying estimate of the input signal power, among other entities. The LMS algorithm was implemented as

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu(n) e(n) \mathbf{r}(n). \quad (3.203)$$

The step size μ was treated as a variable, its value being determined dynamically as

$$\mu(n) = \frac{\mu}{(M+1) \bar{x}^2(n) [\alpha, r(n), \bar{x}^2(n-1)]}, \quad (3.204)$$

with $0 < \mu < 1$. The notation $\bar{x}^2(n) [\alpha, r(n), \bar{x}^2(n-1)]$ indicates that the updated value of $\bar{x}^2(n)$ is a function of $\alpha, r(n)$, and $\bar{x}^2(n-1)$. A forgetting factor α was introduced in the adaptation process, with $0 \leq \alpha \ll 1$; this feature was expected to overcome problems caused by high levels of nonstationarity in the signal. $\bar{x}^2(n)$ is a time-varying estimate of the input signal power, computed as $\bar{x}^2(n) = \alpha r^2(n) + (1 - \alpha) \bar{x}^2(n-1)$.

The filtered versions of the VAG signals recorded from the midpatella and the tibial tuberosity positions, as shown in Figure 3.11 [traces (b) and (c), right-hand column], are shown in Figure 3.90. The muscle-contraction signal recorded at the distal rectus femoris position was used as the reference input [Figure 3.11, right-hand column, trace (a)]. It is seen that the low-frequency muscle-contraction artifact has been successfully removed from the VAG signals (compare the second half of each signal in Figure 3.90 with the corresponding part in Figure 3.11).

3.9.3 The RLS adaptive filter

When the input process of an adaptive system is quasistationary, the best steady-state performance results from slow adaptation. However, when the input statistics are time-variant (nonstationary), the best performance is obtained by a compromise between fast adaptation (necessary to track variations in the input process) and slow adaptation (necessary to limit the noise in the adaptive process). The LMS adaptation algorithm is a simple and efficient approach for ANC; however, it is not appropriate for fast-varying signals due to its slow convergence and due to the difficulty in selecting the correct value for the step size μ . An alternative approach based on the exact minimization of the least-squares criterion is the RLS method [148, 187]. The RLS algorithm has been widely used in real-time system identification and noise cancellation because of its fast convergence, which is about an order of magnitude higher than that of the LMS method. (The derivation of the RLS filter in this section has been adapted from Sesay [187] and Krishnan [185] with permission.)

An important feature of the RLS algorithm is that it utilizes information contained in the input data and extends it back to the instant of time when the algorithm was initiated [148]. Given the least-squares estimate of the tap-weight vector of the filter at time $n-1$, the updated estimate of the vector at time n is computed upon the arrival of new data.

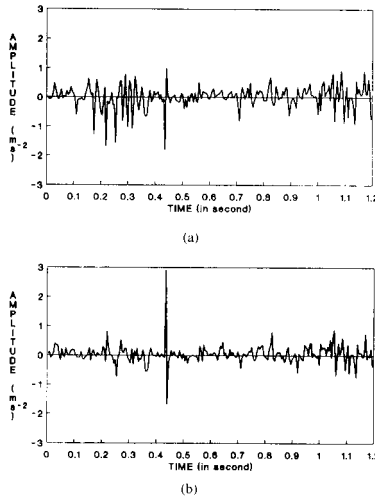


Figure 3.90 LMS-filtered versions of the VAG signals recorded from the midpatella and the tibial tuberosity positions, as shown in Figure 3.11 [traces (b) and (c), right-hand column]. The muscle-contraction signal recorded at the distal rectus femoris position was used as the reference input [Figure 3.11, right-hand column, trace (a)]. The recording setup is shown in Figure 3.10. Reproduced with permission from Y.T Zhang, R.M. Rangayyan, C.B. Frank, and G.D. Bell, Adaptive cancellation of muscle-contraction interference from knee joint vibration signals, *IEEE Transactions on Biomedical Engineering*, 41(2):181–191, 1994. ©IEEE.

In the derivation of the RLS algorithm, the *performance criterion* or *objective function* $\xi(n)$ to be minimized in the sense of least squares is defined as

$$\xi(n) = \sum_{i=1}^n \lambda^{n-i} |e(i)|^2, \quad (3.205)$$

where $1 \leq i \leq n$ is the observation interval, $e(i)$ is the estimation error as defined in Equation 3.198, and λ is a weighting factor (also known as the *forgetting factor*) with $0 < \lambda \leq 1$. The values of $\lambda^{n-i} < 1$ give more “weight” to the more recent error values. Such weighting is desired in the case of nonstationary data, where changes in the signal statistics make the inclusion of past data less appropriate. The inverse of $(1 - \lambda)$ is a measure of the memory of the algorithm.

The Wiener–Hopf equation is the necessary and sufficient condition [148] for minimizing the performance index in the least-squares sense and for obtaining the optimal values of the tap weights, and may be derived in a manner similar to that presented in Section 3.8 for the Wiener filter. The normal equation to be solved in the RLS procedure is

$$\Phi(n) \tilde{\mathbf{w}}(n) = \Theta(n), \quad (3.206)$$

where $\tilde{\mathbf{w}}(n)$ is the optimal tap-weight vector for which the performance index is at its minimum, $\Phi(n)$ is an $M \times M$ time-averaged (and weighted) autocorrelation

matrix of the reference input $\mathbf{r}(i)$, defined as

$$\Phi(n) = \sum_{i=1}^n \lambda^{n-i} \mathbf{r}(i) \mathbf{r}^T(i), \quad (3.207)$$

and $\Theta(n)$ is an $M \times 1$ time-averaged (and weighted) cross-correlation matrix between the reference input $\mathbf{r}(i)$ and the primary input $x(i)$, defined as

$$\Theta(n) = \sum_{i=1}^n \lambda^{n-i} \mathbf{r}(i) x(i). \quad (3.208)$$

The general scheme of the RLS filter is illustrated in Figure 3.91.

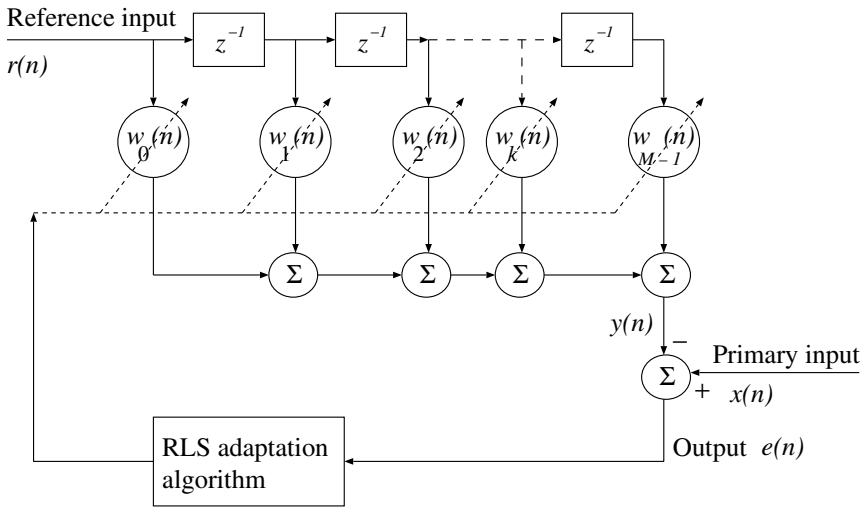


Figure 3.91 General structure of the adaptive RLS filter.

Because of the difficulty in solving the normal equation for the optimal tap-weight vector, recursive techniques need to be considered. In order to obtain a recursive solution, we could isolate the term corresponding to $i = n$ from the rest of the summation on the right-hand side (RHS) of Equation 3.207 and obtain

$$\Phi(n) = \lambda \left[\sum_{i=1}^{n-1} \lambda^{n-i-1} \mathbf{r}(i) \mathbf{r}^T(i) \right] + \mathbf{r}(n) \mathbf{r}^T(n). \quad (3.209)$$

According to the definition in Equation 3.207, the expression inside the square brackets on the RHS of Equation 3.209 equals the time-averaged and weighted autocorrelation matrix $\Phi(n-1)$. Hence, Equation 3.209 can be rewritten as a recursive expression, given by

$$\Phi(n) = \lambda \Phi(n-1) + \mathbf{r}(n) \mathbf{r}^T(n). \quad (3.210)$$

Similarly, Equation 3.208 can be written as the recursive equation

$$\Theta(n) = \lambda \Theta(n-1) + \mathbf{r}(n)x(n). \quad (3.211)$$

To compute the least-squares estimate $\tilde{\mathbf{w}}(n)$ for the tap-weight vector in accordance with Equation 3.206, we have to determine the inverse of the correlation matrix $\Phi(n)$. In practice, such an operation is time-consuming (particularly if M is large). To reduce the computational requirements, a matrix inversion lemma known as the “ABCD lemma” could be used (a similar form of the lemma can be found in Haykin [148]). According to the ABCD lemma, given matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$,

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{DA}^{-1}\mathbf{B} + \mathbf{C}^{-1})^{-1}\mathbf{DA}^{-1}. \quad (3.212)$$

The matrices $\mathbf{A}$, $\mathbf{C}$, $(\mathbf{A} + \mathbf{BCD})$, and $(\mathbf{DA}^{-1}\mathbf{B} + \mathbf{C}^{-1})$ are assumed to be invertible. With the correlation matrix $\Phi(n)$ assumed to be positive definite and therefore nonsingular, we may apply the matrix inversion lemma to Equation 3.210 by assigning

$$\begin{aligned} \mathbf{A} &= \lambda \Phi(n-1), \\ \mathbf{B} &= \mathbf{r}(n), \\ \mathbf{C} &= 1, \\ \mathbf{D} &= \mathbf{r}^T(n). \end{aligned}$$

We then have

$$\begin{aligned} \Phi^{-1}(n) &= \lambda^{-1}\Phi^{-1}(n-1) \\ &\quad - \lambda^{-1}\Phi^{-1}(n-1)\mathbf{r}(n) \left[\lambda^{-1}\mathbf{r}^T(n)\Phi^{-1}(n-1)\mathbf{r}(n) + 1 \right]^{-1} \\ &\quad \times \lambda^{-1}\mathbf{r}^T(n)\Phi^{-1}(n-1). \end{aligned} \quad (3.213)$$

Since the expression inside the brackets of the equation given above is a scalar, the equation can be rewritten as

$$\Phi^{-1}(n) = \lambda^{-1}\Phi^{-1}(n-1) - \frac{\lambda^{-2}\Phi^{-1}(n-1)\mathbf{r}(n)\mathbf{r}^T(n)\Phi^{-1}(n-1)}{1 + \lambda^{-1}\mathbf{r}^T(n)\Phi^{-1}(n-1)\mathbf{r}(n)}. \quad (3.214)$$

For convenience of notation, let

$$\mathbf{P}(n) = \Phi^{-1}(n), \quad (3.215)$$

with $\mathbf{P}(0) = \delta^{-1}\mathbf{I}$, where δ is a small constant and $\mathbf{I}$ is the identity matrix. Furthermore, let

$$\mathbf{k}(n) = \frac{\lambda^{-1}\mathbf{P}(n-1)\mathbf{r}(n)}{1 + \lambda^{-1}\mathbf{r}^T(n)\mathbf{P}(n-1)\mathbf{r}(n)}. \quad (3.216)$$

$\mathbf{k}(n)$ is analogous to the *Kalman gain vector* in Kalman filter theory [148]. Equation 3.214 may then be rewritten in a simpler form as

$$\mathbf{P}(n) = \lambda^{-1}\mathbf{P}(n-1) - \lambda^{-1}\mathbf{k}(n)\mathbf{r}^T(n)\mathbf{P}(n-1). \quad (3.217)$$

By multiplying both sides of Equation 3.216 by the denominator on its RHS, we get

$$\mathbf{k}(n) [1 + \lambda^{-1} \mathbf{r}^T(n) \mathbf{P}(n-1) \mathbf{r}(n)] = \lambda^{-1} \mathbf{P}(n-1) \mathbf{r}(n), \quad (3.218)$$

or

$$\mathbf{k}(n) = [\lambda^{-1} \mathbf{P}(n-1) - \lambda^{-1} \mathbf{k}(n) \mathbf{r}^T(n) \mathbf{P}(n-1)] \mathbf{r}(n). \quad (3.219)$$

Comparing the expression inside the brackets on the RHS of the equation given above with Equation 3.217, we have

$$\mathbf{k}(n) = \mathbf{P}(n) \mathbf{r}(n). \quad (3.220)$$

$\mathbf{P}(n)$ and $\mathbf{k}(n)$ have the dimensions $M \times M$ and $M \times 1$, respectively.

By using Equations 3.206, 3.211, and 3.215, a recursive equation for updating the least-squares estimate $\tilde{\mathbf{w}}(n)$ of the tap-weight vector can be obtained as

$$\begin{aligned} \tilde{\mathbf{w}}(n) &= \Phi^{-1}(n) \Theta(n) \\ &= \mathbf{P}(n) \Theta(n) \\ &= \lambda \mathbf{P}(n) \Theta(n-1) + \mathbf{P}(n) \mathbf{r}(n) x(n). \end{aligned} \quad (3.221)$$

Substituting Equation 3.217 for $\mathbf{P}(n)$ in the first term of Equation 3.221, we get

$$\begin{aligned} \tilde{\mathbf{w}}(n) &= \mathbf{P}(n-1) \Theta(n-1) - \mathbf{k}(n) \mathbf{r}^T(n) \mathbf{P}(n-1) \Theta(n-1) \\ &+ \mathbf{P}(n) \mathbf{r}(n) x(n) \\ &= \Phi^{-1}(n-1) \Theta(n-1) - \mathbf{k}(n) \mathbf{r}^T(n) \Phi^{-1}(n-1) \Theta(n-1) \\ &+ \mathbf{P}(n) \mathbf{r}(n) x(n) \\ &= \tilde{\mathbf{w}}(n-1) - \mathbf{k}(n) \mathbf{r}^T(n) \tilde{\mathbf{w}}(n-1) + \mathbf{P}(n) \mathbf{r}(n) x(n). \end{aligned} \quad (3.222)$$

Finally, from Equation 3.220, using the fact that $\mathbf{P}(n) \mathbf{r}(n)$ equals the gain vector $\mathbf{k}(n)$, the equation given above can be rewritten as

$$\begin{aligned} \tilde{\mathbf{w}}(n) &= \tilde{\mathbf{w}}(n-1) - \mathbf{k}(n) [x(n) - \mathbf{r}^T(n) \tilde{\mathbf{w}}(n-1)] \\ &= \tilde{\mathbf{w}}(n-1) + \mathbf{k}(n) \alpha(n), \end{aligned} \quad (3.223)$$

where $\tilde{\mathbf{w}}(0) = \mathbf{0}$, and

$$\begin{aligned} \alpha(n) &= x(n) - \mathbf{r}^T(n) \tilde{\mathbf{w}}(n-1) \\ &= x(n) - \tilde{\mathbf{w}}^T(n-1) \mathbf{r}(n). \end{aligned} \quad (3.224)$$

The quantity $\alpha(n)$ is often referred to as the *a priori error*, reflecting the fact that it is the error obtained using the “old” filter (that is, the filter before being updated with the new data at the n^{th} time instant). It is evident that, in the case of ANC applications, $\alpha(n)$ will be the estimated signal of interest $\tilde{v}(n)$ after the filter has converged, that is,

$$\alpha(n) = \tilde{v}(n) = x(n) - \tilde{\mathbf{w}}^T(n-1) \mathbf{r}(n). \quad (3.225)$$

Furthermore, after convergence, the primary noise estimate, that is, the output of the adaptive filter $y(n)$, can be written as

$$y(n) = \tilde{m}(n) = \tilde{\mathbf{w}}^T(n-1)\mathbf{r}(n). \quad (3.226)$$

By substituting Equations 3.188 and 3.226 in Equation 3.225, we get

$$\begin{aligned} \tilde{v}(n) &= v(n) + m(n) - \tilde{m}(n) \\ &= v(n) + m(n) - \tilde{\mathbf{w}}^T(n-1)\mathbf{r}(n) \\ &= x(n) - \tilde{\mathbf{w}}^T(n-1)\mathbf{r}(n). \end{aligned} \quad (3.227)$$

Equation 3.223 gives a recursive relationship for obtaining the optimal values of the tap weights, which, in turn, provide the least-squares estimate $\tilde{v}(n)$ of the signal of interest $v(n)$ as in Equation 3.227.

Illustration of application: Figure 3.92 shows plots of the VAG signal of a normal subject [trace (a)] and a simultaneously recorded channel of muscle-contraction interference [labeled as MCI, trace (b)]. The characteristics of the vibration signals in this example are different from those of the signals in Figure 3.11, due to a different recording protocol in terms of speed and range of swinging motion of the leg [185]. The results of adaptive filtering of the VAG signal with the muscle-contraction interference channel as the reference are also shown in Figure 3.92: Trace (c) shows the result of LMS filtering, and trace (d) shows that of RLS filtering. A single-stage LMS filter with variable step size $\mu(n)$ as in Equation 3.204 was used; no attempt was made to remove instrumentation noise. The LMS filter used $M = 7$, $\mu = 0.05$, and a forgetting factor $\alpha = 0.98$; other values resulted in poor results. The RLS filter used $M = 7$ and $\lambda = 0.98$.

The relatively low-frequency muscle-contraction interference has been removed better by the RLS filter than by the LMS filter; the latter failed to track the nonstationarities in the interference and has caused additional artifacts in the result. The spectrograms of the primary, reference, and RLS-filtered signals are shown in Figures 3.93, 3.94, and 3.95, respectively. (The logarithmic scale is used to display better the minor differences between the spectrograms.) It is seen that the predominantly low-frequency artifact, indicated by the high energy levels at low frequencies for the entire duration in the spectrograms in Figures 3.93 and 3.94, has been removed by the RLS filter.

3.10 Selecting an Appropriate Filter

We have so far examined five approaches to remove noise and interference: (1) synchronized or ensemble averaging of multiple realizations or copies of a signal, (2) MA filtering, (3) frequency-domain filtering, (4) optimal (Wiener) filtering, and (5) adaptive filtering. The first two approaches work directly with the signal in the time domain. Frequency-domain (fixed) filtering is performed on the spectrum of the signal. Note that the impulse response of a filter designed in the frequency domain could be used to implement the filter in the time domain as an IIR or FIR filter. Furthermore,

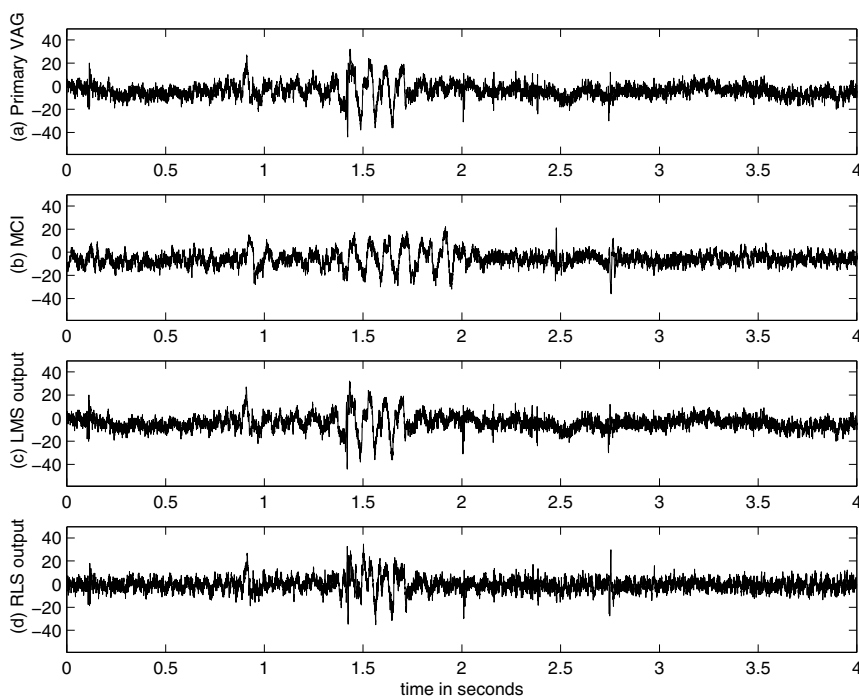


Figure 3.92 (a) VAG signal of a normal subject. (b) Muscle-contraction interference (MCI). (c) Result of LMS filtering. (d) Result of RLS filtering. The recording setup is shown in Figure 3.10.

time-domain filters may be analyzed in the frequency domain via their transfer function or frequency response to understand better their characteristics and effects on the input signal. The Wiener filter may be implemented either in the time domain as a transversal filter or in the frequency domain. Adaptive filters work directly on the signal in the time domain, but dynamically alter their characteristics in response to changes in the interference; their frequency response varies from one point in time to another.

What are the guiding principles to determine which of these filters is the best for a given application? The following points should assist in making this decision.

Synchronized or ensemble averaging is possible when:

- The signal is statistically stationary, (quasi)periodic, or cyclostationary.
- Multiple realizations or recordings of the signal of interest are available.
- A trigger point or time marker is available, or can be derived to extract and align the realizations of the signal.

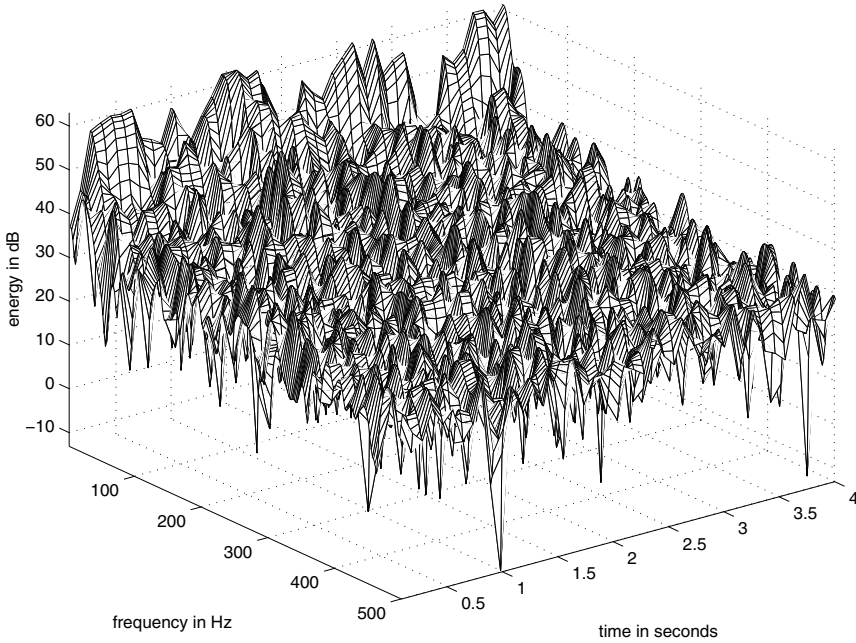


Figure 3.93 Spectrogram of the VAG signal in Figure 3.92 (a). A Hann window of length 256 samples (128 *ms*) was used; an overlap of 32 samples (16 *ms*) was allowed between adjacent segments.

- The noise is a stationary random process that is uncorrelated with the signal and has a zero mean (or a known mean).

Temporal MA filtering is suitable when:

- The signal is statistically stationary at least over the duration of the moving window.
- The noise is a zero-mean random process that is stationary at least over the duration of the moving window and is independent of the signal.
- The signal is a relatively slow (low-frequency) phenomenon.
- Fast, on-line, real-time filtering is desired.

Frequency-domain fixed filtering is applicable when:

- The signal is statistically stationary.
- The noise is a stationary random process that is statistically independent of the signal.

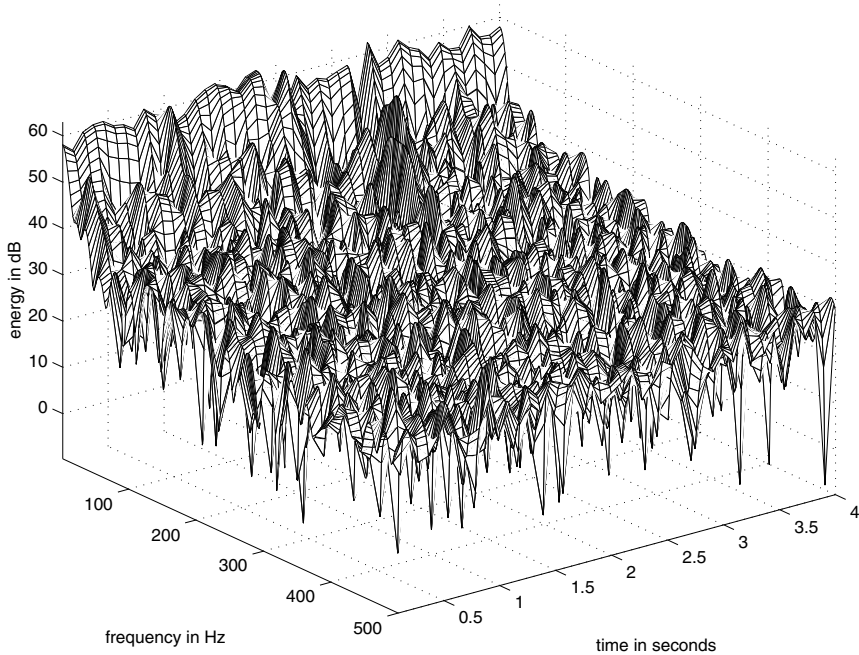


Figure 3.94 Spectrogram of the muscle-contraction interference signal in Figure 3.92 (b). A Hann window of length 256 samples (128 *ms*) was used; an overlap of 32 samples (16 *ms*) was allowed between adjacent segments.

- The signal spectrum is limited in bandwidth compared to that of the noise (or viceversa).
- Loss of information in the spectral band removed by the filter does not seriously affect the signal.
- On-line, real-time filtering is not required (if implemented in the spectral domain via the Fourier transform).

The optimal Wiener filter can be designed if:

- The signal is statistically stationary.
- The noise is a stationary random process that is statistically independent of the signal.
- Specific details (or models) are available regarding the ACFs or the PSDs of the signal and noise.

Adaptive filtering is called for and possible when:

- The noise or interference is not stationary and not necessarily a random process.

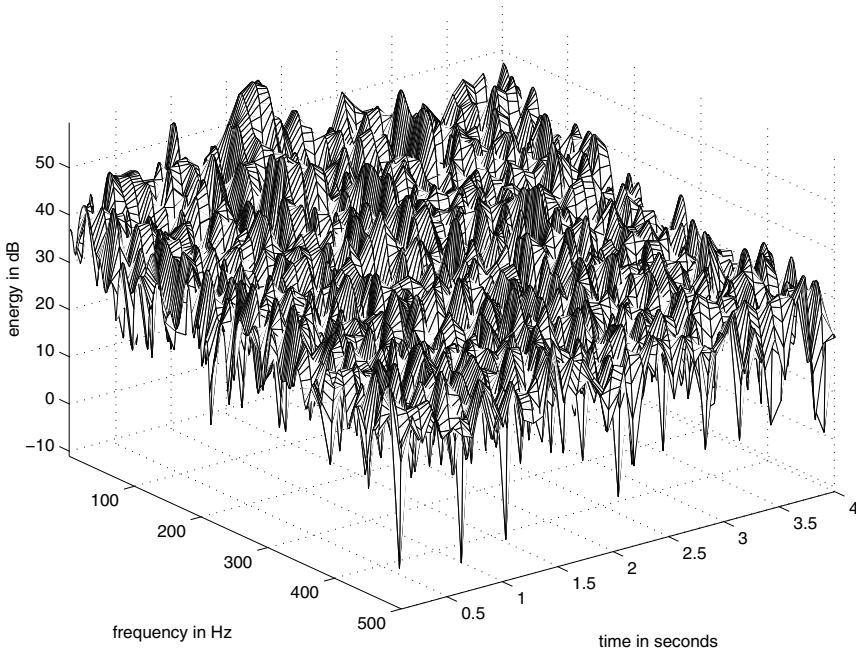


Figure 3.95 Spectrogram of the RLS-filtered VAG signal in Figure 3.92 (d). A Hann window of length 256 samples (128 ms) was used; an overlap of 32 samples (16 ms) was allowed between adjacent segments.

- The noise is uncorrelated with the signal.
- No information is available about the spectral characteristics of the signal and noise, which may also overlap significantly.
- A second source or recording site is available to obtain a reference signal that is strongly correlated with the noise but uncorrelated with the signal.

It is worth noting that an adaptive filter acts as a fixed filter when the signal and noise are stationary. An adaptive filter can also act as a notch or a comb filter when the interference is periodic. It should be noted that all of the filters mentioned above are applicable only when the noise is additive. Techniques such as homomorphic filtering (see Section 4.8) may be used as preprocessing steps if signals combined with operations other than addition need to be separated.

3.11 Application: Removal of Artifacts in ERP Signals

Problem: *Propose a method to improve the SNR of ERP signals. Suggest methods to measure the improvement in the results obtained.*

Solution: Kamath et al. [188] applied synchronized averaging to improve the SNR of cortical evoked potentials or ERPs related to electrical and mechanical stimulation of the esophagus. We have seen the details of ensemble averaging or synchronized averaging in Section 3.5.1; examples of application of synchronized averaging are shown in Figure 3.2 and 3.41. Although improvement in SNR was obtained in some experiments conducted by Kamath et al. [188], they also observed that habituation took place as the number of stimuli was increased beyond a certain limit, and that the use of the ERPs obtained after habituation in averaging led to a reduction in the SNR .

Let $y_k(n)$ represent one realization or observation of an ERP, with $k = 1, 2, \dots, M$ representing the ensemble index, and $n = 0, 1, 2, \dots, N - 1$ representing the time-sample index. Here, M is the number of realizations of the ERP available, and N is the number of the time samples in each signal. We may express the observed signal as

$$y_k(n) = x_k(n) + \eta_k(n), \quad (3.228)$$

where $x_k(n)$ represents the original uncorrupted signal and $\eta_k(n)$ represents the noise in the k^{th} realization of the observed signal. The result of ensemble averaging or synchronized averaging is obtained, for each instant of time n , by averaging the M observations of the ERP as

$$\begin{aligned} \bar{y}(n) &= \frac{1}{M} \sum_{k=1}^M y_k(n) \\ &= \frac{1}{M} \sum_{k=1}^M x_k(n) + \frac{1}{M} \sum_{k=1}^M \eta_k(n); \quad n = 0, 1, 2, \dots, N - 1. \end{aligned} \quad (3.229)$$

Figure 3.96 shows superimposed plots of all of the $M = 24$ ERPs available from the experiments conducted by Kamath et al. [188]. Whereas the overall trend of the ERPs is visible, the extent of the noise and inherent variations present in the ERPs can also be observed. Figure 3.97 shows the result of synchronized averaging of all of the 24 ERPs available. The result demonstrates the benefits of averaging in terms of reduced noise and clear depiction of the trends in the ERP signal.

To study the effects of habituation, various sets of the 24 available ERPs were averaged. Figure 3.98 shows three such results, each being the average of eight ERPs. Whereas the average of the first eight ERPs for $k = 1, 2, \dots, 8$ (solid line) appears to be a good result, the result for the next set of eight ERPs for $k = 9, 10, \dots, 16$ (dashed line) appears to be noisy. Furthermore, the average of the last set of eight ERPs for $k = 17, 18, \dots, 24$ (dotted line) is not similar to the other results of averaging at all, indicating that habituation (or fatigue) could have resulted in the last eight ERPs being substantially different from the earlier responses.

In Section 3.5.1, we saw that synchronized averaging can, theoretically, improve the SNR by the factor of $\sqrt{M}$. In a practical application, the actual improvement

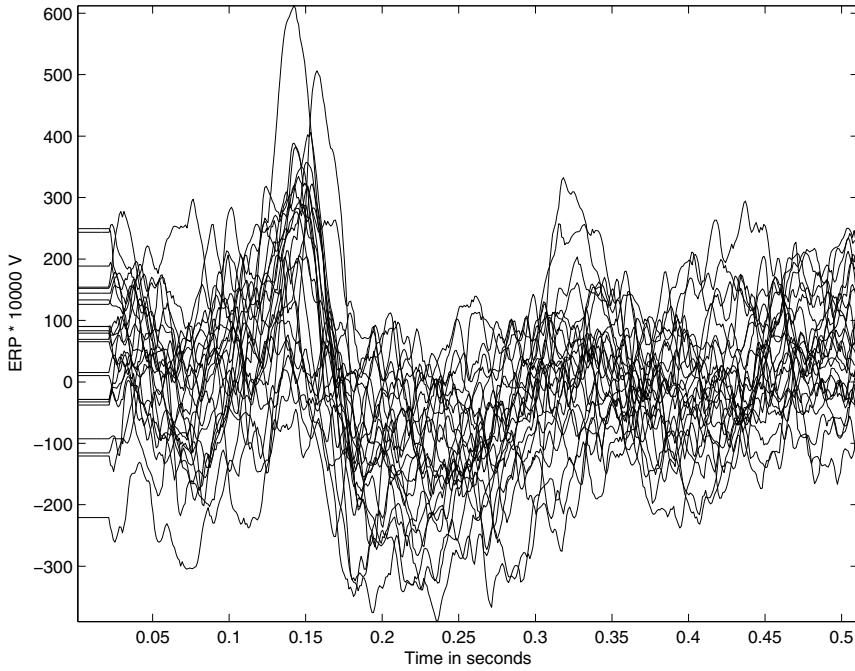


Figure 3.96 Superimposed plots of 24 cortical ERPs related to electrical stimulation of the esophagus. Data courtesy of M.V. Kamath, McMaster University, Hamilton, ON, Canada.

obtained in the SNR may be different. Kamath et al. [188] estimated the SNR by first deriving the power of the noise in the output as

$$\sigma_{\eta}^2 = \frac{1}{NT(M-1)} \sum_{k=1}^M \sum_{n=0}^{N-1} [y_k(n) - \bar{y}(n)]^2. \quad (3.230)$$

Here, $T = 0.001$ s is the sampling interval. The signal power in the output was derived as

$$\sigma_{\bar{y}}^2 = \frac{1}{NT} \left\{ \sum_{n=0}^{N-1} [\bar{y}(n)]^2 \right\} - \frac{\sigma_{\eta}^2}{M}. \quad (3.231)$$

Then, SNR was estimated as

$$SNR = \frac{\sigma_{\bar{y}}^2}{\sigma_{\eta}^2}. \quad (3.232)$$

Kamath et al. [188] also computed the Euclidean distance between the original ERP signals and the averaged signal obtained as

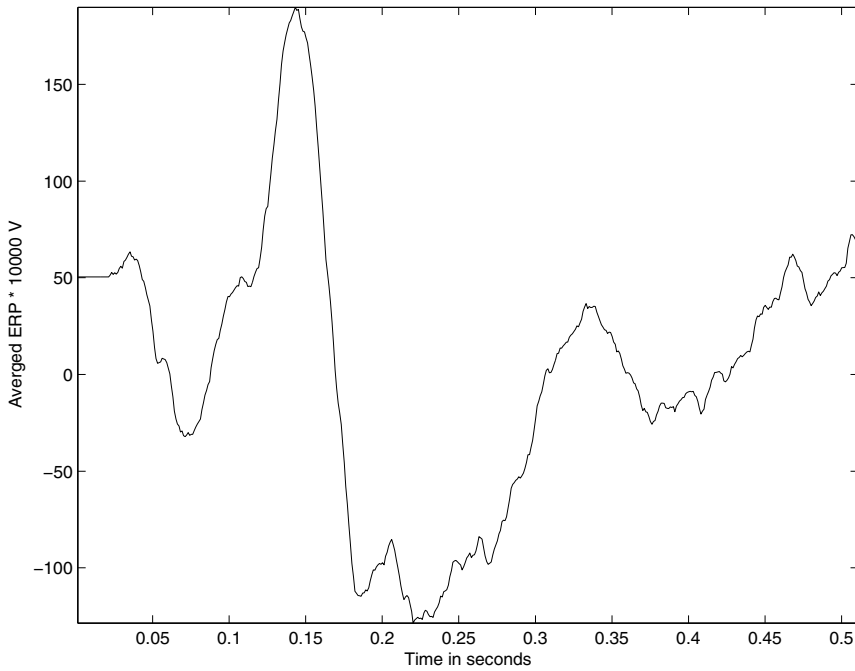


Figure 3.97 Result of synchronized averaging of all of the 24 ERPs shown in Figure 3.96.

$$D = \frac{1}{M} \sum_{k=1}^M \sqrt{\sum_{n=0}^{N-1} [y_k(n) - \bar{y}(n)]^2}. \quad (3.233)$$

The *SNR* for the average of all of the 24 ERPs, shown in Figure 3.97, was found to be 0.37. The *SNRs* for the three results of averaging eight ERPs, shown in Figure 3.98, were computed to be 0.73, 0.65, and 0.06, respectively, for the first, second, and third sets of eight ERPs. These results confirm, in a quantitative manner, the subjective observations made in the preceding paragraphs: repeated stimulation of a biological system could lead to habituation and fatigue, which could cause the response to be substantially different. Although, based on theoretical considerations, one would expect the averaging of more observations of a certain process to lead to better results with higher *SNR*, the results of the present study demonstrate otherwise. A judicious design of the experiment is essential, by taking into account various beneficial as well as adverse effects of the methods used.

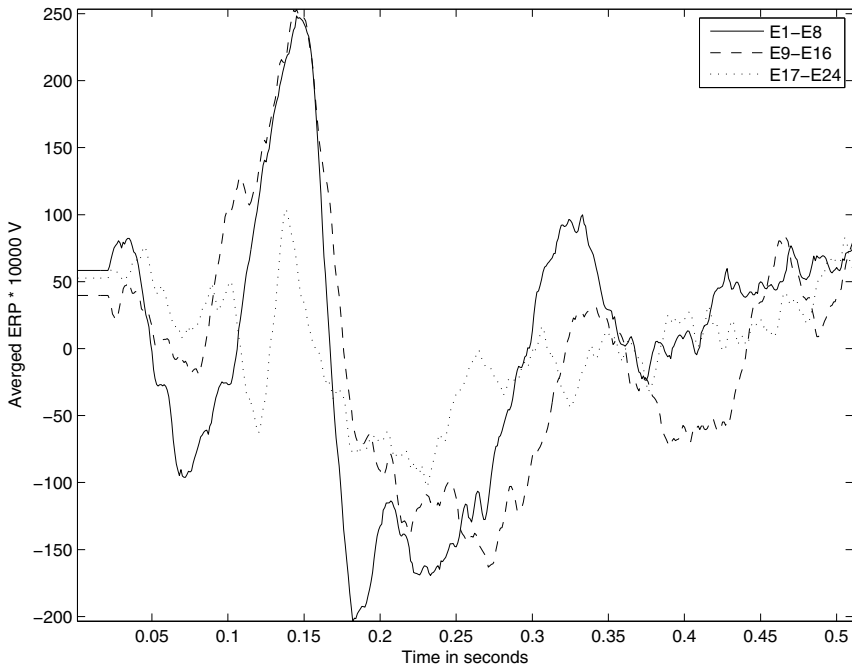


Figure 3.98 Results of synchronized averaging of selected sets of the ERPs shown in Figure 3.96. The three plots shown were obtained by averaging ERPs in groups of eight, for $k = 1, 2, \dots, 8$ (solid line); $k = 9, 10, \dots, 16$ (dashed line); and $k = 17, 18, \dots, 24$ (dotted line).

3.12 Application: Removal of Artifacts in the ECG

Problem: Figure 3.99 (top trace) shows an ECG signal with a combination of baseline drift, high-frequency noise, and power-line interference. Design filters to remove the artifacts.

Solution: The power spectrum of the given signal is shown in the topmost plot in Figure 3.100. Observe the relatively high amount of spectral energy present near DC, from 100 Hz to 500 Hz, and at the power-line frequency and its harmonics located at 60 Hz, 180 Hz, 300 Hz, and 420 Hz. The fundamental component at 60 Hz is lower than the third, fifth, and seventh harmonics, perhaps due to a notch filter included in the signal acquisition system, which has not been effective.

A Butterworth lowpass filter with order $N = 8$ and $f_c = 70$ Hz (see Section 3.6.1 and Equation 3.145), a Butterworth highpass filter of order $N = 8$ and $f_c = 2$ Hz (see Section 3.6.2 and Equation 3.148), and a comb filter with zeros at 60 Hz, 180 Hz, 300 Hz, and 420 Hz (see Section 3.6.3 and Equation 3.151) were applied in series to the signal. The signal spectrum displays the presence of further harmonics (ninth and eleventh) of the power-line interference at 540 Hz and 660 Hz

that have been aliased to the peaks apparent at 460 Hz and 340 Hz , respectively. However, the comb filter in the present example was not designed to remove these components. The lowpass and highpass filters were applied in the frequency domain to the Fourier transform of the signal using the form indicated by Equations 3.145 and 3.148. The comb filter was applied in the time domain using the coefficients in Equation 3.151.

The combined frequency response of the filters is shown in the middle plot in Figure 3.100. The spectrum of the ECG signal after the application of all three filters is shown in the bottom plot in Figure 3.100. The filtered signal spectrum has no appreciable energy beyond about 100 Hz , and it displays significant attenuation at 60 Hz .

The outputs after the lowpass filter, the highpass filter, and the comb filter are shown in Figure 3.99. Observe that the baseline drift is present in the output of the lowpass filter, and that the power-line interference is present in the outputs of the lowpass and highpass filters. The final trace is free of all three types of interference. Note, however, that the highpass filter has introduced a noticeable distortion (undershoot) in the P and T waves.



Figure 3.99 ECG signal with a combination of artifacts and its filtered versions. The duration of the signal is 10.7 s .

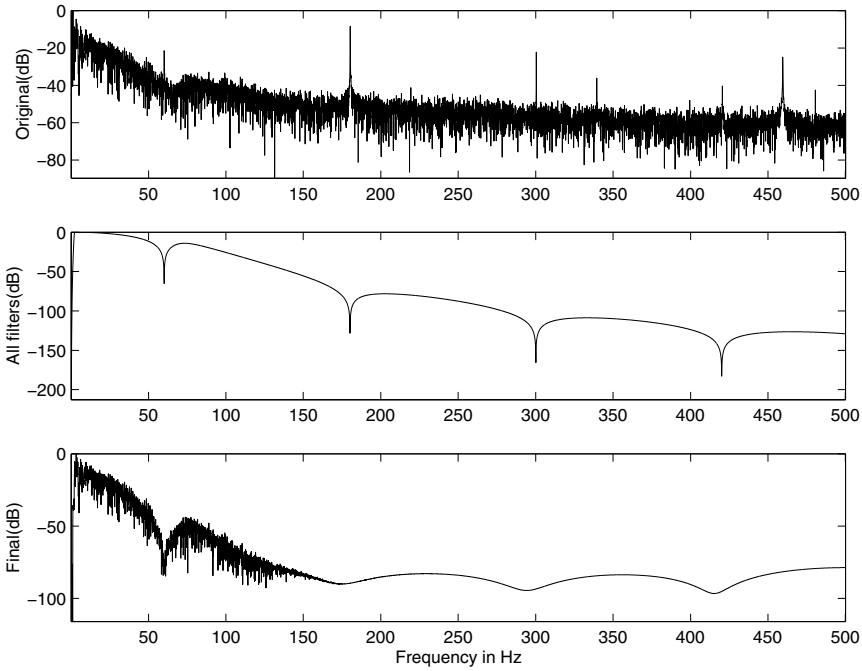


Figure 3.100 Top and bottom plots: Power spectra of the ECG signals in the top and bottom traces of Figure 3.99. Middle plot: Frequency response of the combination of lowpass, highpass, and comb filters. The cutoff frequency of the highpass filter is 2 Hz ; the highpass portion of the frequency response is not clearly seen in the plot.

3.13 Application: Adaptive Cancellation of the Maternal ECG to Obtain the Fetal ECG

Problem: *Propose an adaptive noise cancellation filter to remove the maternal ECG signal from the abdominal-lead ECG shown in Figure 3.9 to obtain the fetal ECG. Chest-lead ECG signals of the expectant mother may be used for reference.*

Solution: Widrow et al. [124] described a multiple-reference ANC for removal of the maternal ECG in order to obtain the fetal ECG. The combined ECG was obtained from a single abdominal lead, whereas the maternal ECG was obtained via four chest leads. The model was designed to permit the treatment of not only multiple sources of interference, but also of components of the desired signal present in the reference inputs, and further to consider the presence of uncorrelated noise components in the reference inputs. It should be noted that the maternal cardiac vector is projected on to the axes of different ECG leads in different ways, and hence the characteristics of the maternal ECG in the abdominal lead would be different from those of the chest-lead ECG signals used as reference inputs.

Each filter channel used by Widrow et al. [124] had 32 taps and a delay of 129 *ms*. The signals were prefiltered to the bandwidth 3 – 35 *Hz* and a sampling rate of 256 *Hz* was used. The optimal Wiener filter used (see Section 3.8) included transfer functions and cross-spectral vectors between the input source and each reference input. Further extension of the method to more general multiple-source, multiple-reference noise cancelling problems was also discussed by Widrow et al.

The result of cancellation of the maternal ECG from the abdominal lead ECG signal in Figure 3.9 is shown in Figure 3.101. Comparing the two figures, it is seen that the filter output has successfully extracted the fetal ECG and suppressed the maternal ECG. See Widrow et al. [124] for details; see also Ferrara and Widrow [189] and Zarzoso and Nandi [190] for additional related discussion.

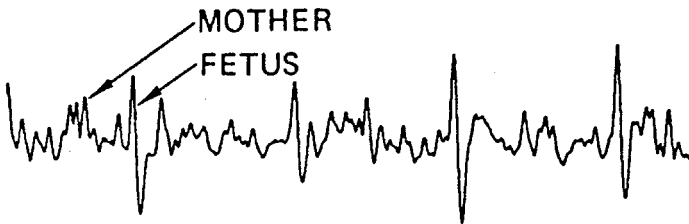


Figure 3.101 Result of adaptive cancellation of the maternal chest ECG from the abdominal ECG in Figure 3.9. The QRS complexes extracted correspond to the fetal ECG. Reproduced with permission from B. Widrow, J.R. Glover, Jr., J.M. McCool, J. Kaunitz, C.S. Williams, R.H. Hearn, J.R. Zeidler, E. Dong, Jr., and R.C. Goodlin, Adaptive noise cancelling: Principles and applications, *Proceedings of the IEEE*, 63(12):1692–1716, 1975. ©IEEE.

3.14 Application: Adaptive Cancellation of Muscle contraction Interference in VAG Signals

Problem: Study the applicability of adaptive noise cancellation filters to remove the muscle-contraction interference caused by the rectus femoris in the VAG signal recorded at the patella.

Solution: Rangayyan et al. [191] conducted a study on the impact of cancellation of muscle-contraction interference on modeling and classification of VAG signals and further classification of the filtered signals as normal or abnormal. Both the LMS (see Section 3.9.2) and the RLS (see Section 3.9.3) methods were investigated, and the RLS method was chosen for its more efficient tracking of nonstationarities in the input and reference signals.

Figure 3.102 shows plots of the VAG signal of a subject with chondromalacia patella of grade II [trace (a)] and a simultaneously recorded channel of muscle-contraction interference [labeled as MCI, trace (b)]. The results of adaptive filtering of the VAG signal with the muscle-contraction interference channel as the reference are also shown in Figure 3.102: Trace (c) shows the result of LMS filtering and trace

(d) shows that of RLS filtering. A single-stage LMS filter with variable step size $\mu(n)$ as in Equation 3.204 was used, with $M = 7$, $\mu = 0.05$, and $\alpha = 0.98$. The RLS filter used $M = 7$ and $\lambda = 0.98$.

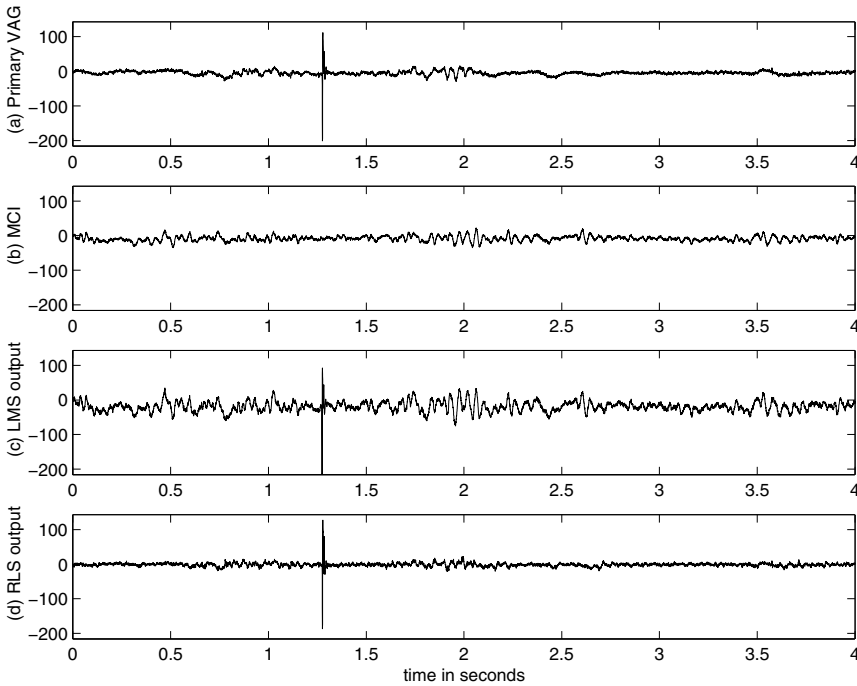


Figure 3.102 Top to bottom: (a) VAG signal of a subject with chondromalacia patella of grade II; (b) muscle-contraction interference (MCI); (c) result of LMS filtering; and (d) result of RLS filtering. The recording setup is shown in Figure 3.10.

As in the example in Figure 3.92, it is seen that the muscle-contraction interference has been removed by the RLS filter; however, the LMS filter failed to perform well, due to its limited capabilities in tracking the nonstationarities in the interference. The spectrograms of the primary, reference, and RLS-filtered signals are shown in Figures 3.103, 3.104, and 3.105, respectively. (The logarithmic scale is used to display better the minor differences between the spectrograms.) It is seen that the frequency components of the muscle-contraction interference have been suppressed by the RLS filter.

The primary (original) and filtered VAG signals of 53 subjects were adaptively segmented and modeled in the study of Rangayyan et al. [191] (see Chapter 8). The segment boundaries were observed to be markedly different for the primary and the filtered VAG signals. Parameters extracted from the filtered VAG signals were expected to provide higher discriminant power in pattern classification when compared to the same parameters of the unfiltered or primary VAG signals. However, classification experiments indicated otherwise: The filtered signals gave a lower classification

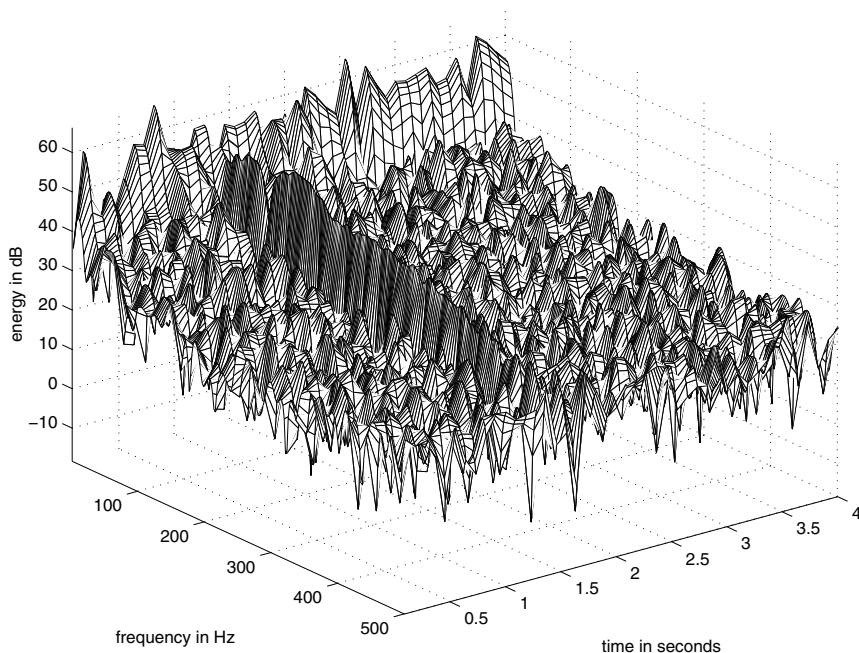


Figure 3.103 Spectrogram of the original VAG signal in Figure 3.102 (a). A Hann window of length 256 samples (128 *ms*) was used; an overlap of 32 samples (16 *ms*) was allowed between adjacent segments.

accuracy by almost 10%. It was reasoned that, after removal of the predominantly low-frequency muscle-contraction interference, the transient VAG signals of clinical interest were not modeled well by the prediction-based methods. It was concluded that the adaptive filtering procedure used was not an appropriate preprocessing step before signal modeling for pattern classification. However, it was noted that cancellation of muscle-contraction interference may be a desirable step before auditory or spectral analysis of VAG signals.

3.15 Remarks

We have investigated problems posed by artifact, noise, and interference of various forms in the acquisition and analysis of several biomedical signals. Random noise, structured interference, and physiological interference have been identified and analyzed separately. Attention has been drawn to the different characteristics of various types of noise, such as frequency content and nonstationarity. Fixed, optimal, and adaptive filters were developed in the time and frequency domains for several applications, and guidelines were drawn to assist in choosing the appropriate filter for various types of artifacts. Advanced methods for adaptive denoising based on

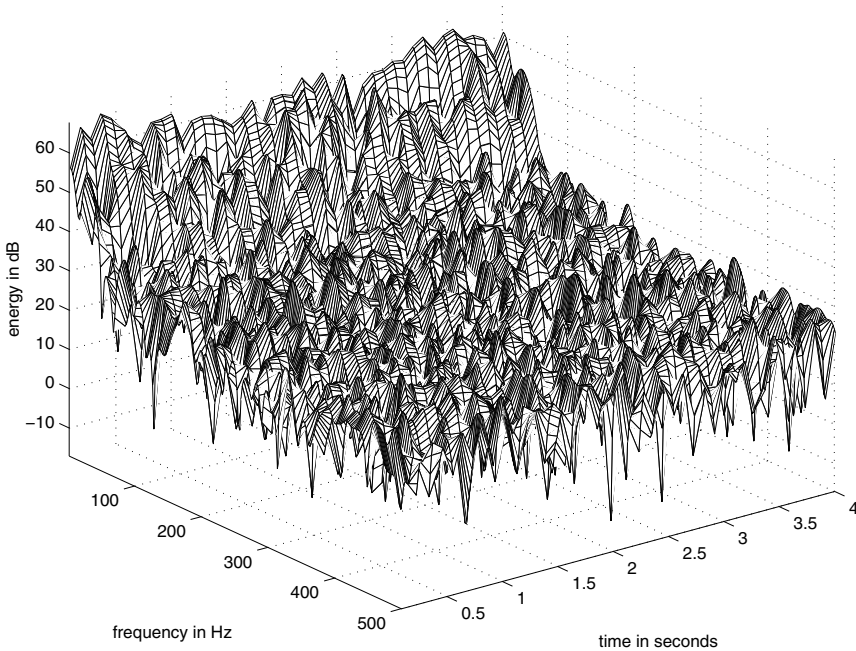


Figure 3.104 Spectrogram of the muscle-contraction interference signal in Figure 3.102 (b). A Hann window of length 256 samples (128 *ms*) was used; an overlap of 32 samples (16 *ms*) was allowed between adjacent segments.

wavelet and time-frequency decomposition methods have not been discussed in this chapter, but are described by Krishnan and Rangayyan [192] for filtering VAG signals; see Chapter 8 for related material. Another category of filters that has not been considered in this chapter is that of morphological filters [193, 194], which include nonlinear statistics-based operations and could be formulated under certain conditions to include linear filter operations as well.

It is important to observe that each practical problem needs to be studied carefully to determine the type and characteristics of the artifact present; the nature of the signal and its relationship to, or interaction with, the artifact; and the effect of the filter being considered on the desired signal or features computed from the filtered result. Different filters may be suitable for different subsequent steps of signal analysis. It is unlikely that a single filter will address all of the problems and the requirements in a wide variety of practical situations and applications. Regardless of one's expertise in filters, it should be remembered that *prevention is better than cure*: most filters, while removing an artifact, may introduce another. Attempts should be made at the outset to acquire artifact-free signals to the extent possible.

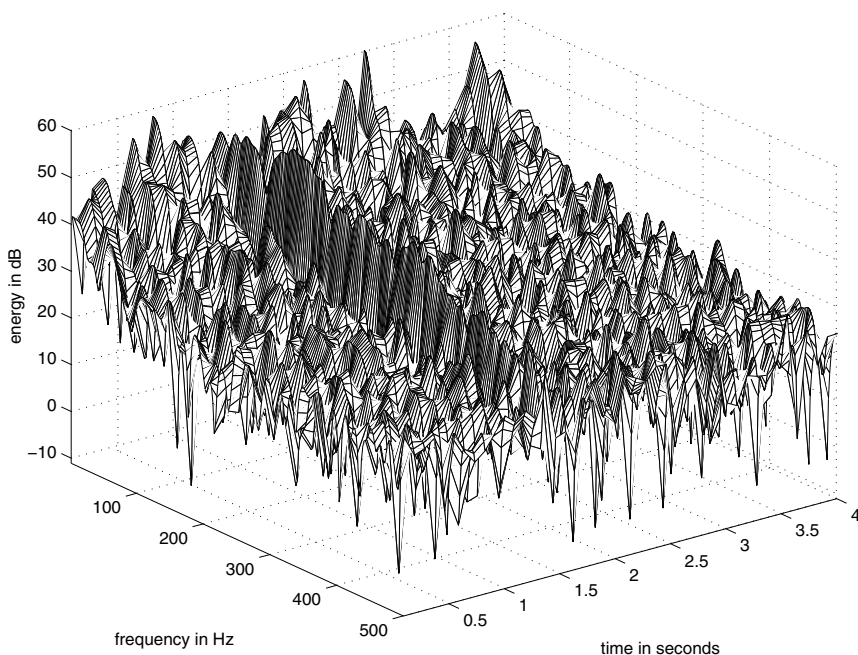


Figure 3.105 Spectrogram of the RLS-filtered VAG signal in Figure 3.102 (d). A Hann window of length 256 samples (128 *ms*) was used; an overlap of 32 samples (16 *ms*) was allowed between adjacent segments.

3.16 Study Questions and Problems

Note: Some of the questions deal with the fundamentals of signals and systems, and may require background preparation with other sources such as Lathi [1,2] or Oppenheim et al. [3]. Such problems are included for the sake of recollection of the related concepts.

1. What are the potential sources of instrumentation and physiological artifacts in recording the PCG signal? Propose nonelectronic methods to prevent or suppress the latter type of artifacts.
2. List four potential sources of instrumentation and physiological artifacts in recording the ECG signal. Describe methods to prevent or remove each artifact. Identify the possible undesired effects of your procedures on the ECG signal.
3. Identify at least three potential sources of physiological artifacts in recording the EEG signal.
4. In recording the EEG in a clinical laboratory, some channels were found to contain the ECG as an artifact. Will simple lowpass or bandpass filtering help in removing the artifact? Why (not)? Propose a scheme to remove the artifact.
5. A biomedical signal is bandpass filtered to the range $0 - 150 \text{ Hz}$. Assume the filter to be ideal, and assume any distribution of spectral energy over the bandwidth of the signal.

- (a) What is the minimum frequency at which the signal should be sampled in order to avoid aliasing errors? (b) A researcher samples the signal at 500 Hz . Draw a schematic representation of the spectrum of the sampled signal. (c) Another researcher samples the signal at 200 Hz . Draw a schematic representation of the spectrum of the sampled signal. Explain the differences between case (b) and case (c).
- Distinguish between ensemble averages and temporal (time) averages. Identify applications of first-order and second-order averages of both types in EEG analysis.
 - Explain how one may apply ensemble averaging and temporal (time) averaging procedures to process ECG signals. Identify applications of first-order and second-order averages of both types in ECG analysis.
 - Explain how you would apply synchronized averaging to remove noise in (a) ECG signals, (b) event-related (or evoked) potentials, (c) heart sound (PCG) signals, and (d) EMG signals. In each case, explain (i) how you will obtain the information required for synchronization of the signals epochs or episodes; (ii) sources of artifacts and how you will deal with them; (iii) limitations and practical difficulties; and (iv) potential for success of the method.
 - Draw a typical ECG waveform over one cardiac cycle indicating the important component waves. How is the waveform affected by passage through the following: (a) A lowpass filter with a cutoff frequency of 40 Hz ? (b) A highpass filter with a cutoff frequency of 5 Hz ? Draw schematic representations of the expected outputs and explain their characteristics.
 - What is the z -transform of a signal whose samples are given in the series $\{4, 3, 2, 1, 0, -1, 0, 1, 0\}$? (The first sample represents zero time in all the signal sample arrays given in the problems, unless stated otherwise.)
 - A digital filter is used to process a signal at a sampling rate of $2,000\text{ Hz}$. (a) Draw the unit circle in the complex z -plane and identify the frequencies corresponding to the points $z = (1 + j0)$, $z = (0 + j1)$, $z = (-1 + j0)$, and $z = (0 - j1)$, as well as the point $z = (1 + j0)$ again as approached in the counterclockwise direction. (b) What are the frequencies corresponding to these same points if the sampling rate is 500 Hz ?
 - What is the transfer function of an LSI system whose impulse response is given by the series $\{2, 1, 0, 0, -1, 0, 1, 0\}$ for $n = 0, 1, 2, \dots, 7$?
 - The impulse response of a digital filter is $\{1, -2, 1\}$. What will be the response of the filter to the unit step?
 - The impulse response of a filter is $\{3, -2, 2\}$. What will be the response of the filter to the input $\{6, 4, 2, 1\}$?
 - The transfer function of a filter is $H(z) = z^{-1} - 3z^{-2} + 2z^{-4} - z^{-6}$. What is the difference equation relating the output to the input? What is the impulse response of the filter?
 - The impulse response of a filter is given by the series of values $\{3, 2, 1, 0, -1, 0, 0, 1\}$. What is its transfer function?
 - The impulse response of a filter is specified by the series of sample values $\{3, 1, -1, 1\}$. (a) What will be the response of the filter to the input whose sample values are $\{4, 4, 2, 1\}$? (b) Is the filter response obtained by linear convolution or circular convolution of the input with the impulse response? (c) What will be the response with the type of convolution other than the one you indicated as the answer to the question above? (d) How

- would you implement convolution of the two signals listed above using the DFT? Which type of convolution will this procedure provide? How would you get the other type of convolution for the signals in this problem via the DFT-based procedure?
18. A biomedical signal is expected to be band-limited to 100 Hz, with significant components of interest up to 80 Hz. However, the signal is contaminated with a periodic artifact with a fundamental frequency of 60 Hz and significant third and fifth harmonics. A researcher samples the signal at 200 Hz without prefiltering the signal. Draw a schematic representation of the spectrum of the signal and indicate the artifactual components. Label the frequency axis clearly in Hz. What kind of a filter would you recommend to remove the artifact?
 19. A biomedical signal sampled at 500 Hz was found to have a significant amount of 60 Hz interference. (a) Design a notch filter with two zeros to remove the interference. (b) What is the effect of the filter if a signal sampled at 100 Hz is applied as the input?
 20. Two filters with transfer functions $H_1(z) = \frac{1}{3}(1 + z^{-1} + z^{-2})$ and $H_2(z) = 1 - z^{-1}$ are cascaded. (a) What is the transfer function of the complete system? (b) What is its impulse response? (c) What is its gain at DC and at the folding frequency (that is, $f_s/2$)?
 21. A filter has the transfer function $H(z) = (1 + 2z^{-1} + z^{-2})/(1 - z^{-2})$. (a) Write the difference equation relating the output to the input. (b) Draw the signal-flow diagram of a realization of the filter. (c) Draw its pole-zero diagram.
 22. A digital filter has zeros at $0.5 \pm j0.5$ and poles at $-0.6 \pm j0.3$. (a) Derive the transfer function of the filter. (b) Derive the time-domain difference equation (input-output relationship) of the filter. (c) If the filter is used at the sampling frequency of 1,000 Hz, what are the frequencies at which the gain of the filter is at its maximum and minimum?
 23. Two filters with the transfer functions $H_1(z) = \frac{1}{2T}(1 - z^{-2})$ and $H_2(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$ are cascaded. (a) What is the transfer function of the complete system? (b) Draw its pole-zero diagram. (c) Write the difference equation relating the output to the input. (d) Draw the signal-flow diagram of a realization of the filter. (e) Compute the first six values of the impulse response of the filter. (f) The filter is used to process a signal sampled at 1,000 Hz. What is its gain at 0, 250, and 500 Hz?
 24. A filter is described by the difference equation $y(n) = y(n-1) + \frac{1}{4}x(n) - \frac{1}{4}x(n-4)$. (a) What is its transfer function? (b) Draw the signal-flow diagram of a realization of the filter. (c) Draw its pole-zero diagram.
 25. Under what conditions will synchronized averaging fail to reduce noise?
 26. A signal sampled at the rate of 100 Hz has the samples $\{0, 10, 0, -5, 0\}$ in mV. The signal is passed through a filter described by the transfer function $H(z) = \frac{1}{T}(1 - z^{-1})$. What will be the output sequence? Plot the output and indicate the amplitude and time scales in detail with appropriate units.
 27. A signal sampled at the rate of 100 Hz has the samples $\{0, 10, 0, -5, 0\}$ in mV. It is supposed to be processed by a differentiator with the difference equation $y(n) = \frac{1}{T}[x(n) - x(n-1)]$ and then squared. By mistake the squaring operation is performed before the differentiation. What will be the output sequence? Plot the outputs for both cases and indicate the amplitude and time scales in detail with appropriate units. Explain the differences between the two results.
 28. A certain signal analysis technique requires the following operations in order: (a) differentiation, (b) squaring, and (c) lowpass filtering with a filter $H(\omega)$. Considering a

- generic signal $x(t)$ as the input, write the time-domain and frequency-domain expressions for the output of each stage. Will changing the order of the operations change the final result? Why (not)?
29. A signal sampled at the rate of 100 Hz has the samples $\{0, 10, 0, -5, 0\}$ in mV. The signal is processed by a differentiator with the difference equation $y(n) = \frac{1}{T}[x(n) - x(n-1)]$, and then filtered with a 4-point MA filter. (a) Derive the transfer function and frequency response of each filter and the combined system. (b) Derive the values of the signal samples at each stage. (c) Does it matter which filter is placed first? Why (not)? (d) Plot the output and indicate the amplitude and time scales in detail with appropriate units.
 30. Distinguish between ensemble averages and temporal (time) averages. Identify potential applications of first-order and second-order averages of both types in heart sound (PCG) analysis. Explain how you would obtain a trigger for synchronization.
 31. Is the heart sound signal (PCG) a stationary signal or not? Provide your answer in the context of one full cardiac cycle and give reasons. If you say that the PCG signal is nonstationary, identify parts (segments) that could possibly be stationary, considering the possibility of murmurs in both systole and diastole.
 32. A signal $x(t)$ is transmitted through a channel. The received signal $y(t)$ is a scaled, shifted, and noisy version of $x(t)$ given as $y(t) = \alpha x(t - t_0) + \eta(t)$ where α is a scale factor, t_0 is the time delay, and $\eta(t)$ is noise. Assume that the noise process has zero mean and is statistically independent of the signal process, and that all processes are stationary. Derive expressions for the PSD of $y(t)$ in terms of the PSDs of x and η .
 33. A signal $x(n)$ that is observed in an experiment is modeled as a noisy version of a desired signal $d(n)$ as $x(n) = d(n) + \eta(n)$. The noise process η is a zero-mean, unit-variance random process with uncorrelated samples [“white” noise, with ACF $\phi_\eta(\tau) = \delta(\tau)$] that is statistically independent of the signal process d . The ACF $\phi_d(\tau)$ of d is given by the sequence $\{1.0, 0.6, 0.2\}$, for $\tau = 0, 1, 2$, respectively. Prepare the Wiener–Hopf equation and derive the coefficients of the optimal Wiener filter.
 34. Draw the waveform of a typical normal ECG over one cardiac cycle. Label the names of the component waves and give their typical durations and intervals. Draw a version of the ECG waveform including power-line artifact at 60 Hz. Draw another version of the ECG waveform including high-frequency noise. Describe potential causes of the two types of artifact and methods to prevent or remove each artifact.
 35. A researcher is designing a Wiener filter and is in need of help in obtaining the autocorrelation matrix, defined as $\Phi = E[\mathbf{x}\mathbf{x}^T]$. Help the researcher by computing the autocorrelation matrix for the sample vector $\mathbf{x} = [1 \ 2 \ 2 \ 1]^T$. Explain the meaning of the operator $E[\]$ and describe how you would implement the operation in practice.
 36. A researcher uses a combination of two digital filters in cascade (series). The output of the first filter is the first derivative or difference of the input. The output of the second filter is the average of the current input sample and the preceding input sample. (a) Give the input–output relationship in the time domain (difference equation) for each filter. (b) Derive the transfer function for each filter. (c) Derive the impulse response of the complete system. (d) Derive the transfer function of the complete system. (e) Does it matter which filter is placed first? Explain. (f) Compute the gain of the complete system at 0, $f_s/4$, and $f_s/2$, where f_s is the sampling frequency.
 37. A noisy signal $x(n)$ is expressed in vector notation as $\mathbf{x}(n) = \mathbf{d}(n) + \boldsymbol{\eta}(n)$, where $\mathbf{d}(n)$ is the original (ideal) signal and $\boldsymbol{\eta}(n)$ is random noise that is statistically independent of

- the signal. All signals are assumed to be second-order stationary. Derive an expression for the ACF matrix of $\mathbf{x}$ in terms of the ACF matrices of $\mathbf{d}$ and $\boldsymbol{\eta}$. Explain each step of your derivation.
38. A random signal $x(t)$ is characterized by its PDF $p_x(x)$. Write the basic definition of the mean-squared value of x based upon its PDF. Given an observation of the signal in terms of its samples, expressed as $x(n)$, $n = 0, 1, 2, \dots, N-1$, give a formula to compute the mean-squared value of x (without the PDF). Give a formula to obtain the *RMS* value of x . If x represents an electrical signal, what do the mean-squared and *RMS* values represent?
 39. For a discrete-time signal $x(n)$, write the basic definition of the z -transform. The signal $x(n)$ is processed by an LSI system with the impulse response $h(n)$. Write the complete mathematical expression for the output $y(n)$. Showing all steps, derive the relationship between the z -transforms of $x(n)$, $h(n)$, and $y(n)$.
 40. A digital filter is specified by the difference equation $y(n) = x(n) - x(n-2)$, where $x(n)$ is the input and $y(n)$ is the output. Derive the transfer function of the filter. Derive the magnitude and phase parts of the frequency response of the filter. Let the sampling frequency be normalized to unity. Plot the magnitude and phase responses. Indicate the values of the functions at normalized frequency values of 0, 0.25, and 0.5. Explain the nature and effects of the filter.
 41. A researcher is designing an experiment to record visual ERPs. Provide advice to the researcher on the following: (a) Identify a potential source of artifact in the form of random noise. Propose a strategy or method to prevent or remove the artifact. (b) Identify a potential source of structured noise. Propose a strategy or method to prevent or remove the artifact. (c) Identify a potential source of physiological interference. Propose a strategy or method to prevent or remove the artifact. No equation is required for your answer to this question.
 42. Two LSI filters are specified in terms of their impulse responses as $\delta(n) - \delta(n-1)$ and $\delta(n) + 2\delta(n-1) + \delta(n-2)$. A researcher prepares a new filter by connecting the two filters described above in series. (a) Derive the transfer function of each filter. (b) Derive the transfer function of the combined filter. (c) Derive the impulse response of the combined filter. (d) Does it matter which filter is placed first? Explain and justify your answer. (e) Draw the signal-flow diagram of the combined filter. (f) What is the gain of the combined filter at DC, $f_s/4$, and $f_s/2$, where f_s is the sampling frequency? From these values, give an interpretation of the nature of the combined filter.
 43. A signal of interest $x(t)$ is affected by additive noise $\eta(t)$ and is observed as $y(t) = x(t) + \eta(t)$. It is assumed that the processes are random, and that the noise process is statistically independent of the signal process. It is also assumed that the mean of η is zero. Write the basic definition of the expected value (mean) of the random process x based upon its PDF. Derive the relationship between the mean of y and the statistics of x and η . Derive the relationship between the variance of y and the statistics of x and η . Interpret and explain your results.
 44. A researcher obtains a set of ERP signals $x_k(n)$, $k = 1, 2, \dots, M$, and $n = 0, 1, 2, \dots, N-1$, where $x_k(n)$ is the k^{th} signal, M is the number of signals recorded, and N is the number of samples in each signal. Help the researcher with the following: (a) Give a step-by-step procedure (algorithm) to obtain the synchronized average of the ERPs. (b) State the conditions and requirements for synchronized averaging to be applicable and yield good results. (c) Give an equation to define the synchronized or ensemble

- average of the ERPs. (d) Give an equation to define the temporal average of one of the ERPs over a specified window of time. (e) Give an equation to obtain the average Euclidean distance between the result of synchronized averaging and the M ERPs available. Explain the meaning of the Euclidean distance in this context.
45. A researcher is designing an experiment to record ENGs. Provide advice to the researcher on the following: (a) Identify a potential source of artifact in the form of random noise. Propose a strategy or method to prevent or remove the artifact. (b) Identify a potential source of structured noise. Propose a strategy or method to prevent or remove the artifact. (c) Identify a potential source of physiological interference. Propose a strategy or method to prevent or remove the artifact. No equation is required for your answer to this question.
 46. You are given two signals, $x(n)$ and $y(n)$, each with N samples, $n = 0, 1, 2, \dots, N-1$. No information is available regarding the PDFs of the related processes. Give an equation to compute the normalized correlation coefficient between the two signals. Explain the meaning and purpose of each part of your equation.
 47. A filter is specified with the transfer function $H(z) = \frac{1}{T} \left[\frac{1-z^{-1}}{1-0.95z^{-1}} \right]$. (a) Derive an expression for the input–output relationship of the filter in the time domain. (b) Draw a signal-flow diagram of the filter using only one delay element. (c) Draw the pole–zero plot of the system. (d) What is the gain of the system at zero frequency and at one-half of the sampling frequency?
 48. You are given a set of M signals, $x_k(n)$, $k = 1, 2, \dots, M$, each with N samples, $n = 0, 1, 2, \dots, N-1$. The index n represents sampled time and the index k represents the k^{th} signal in the set. (a) Explain the differences between ensemble averages and temporal averages. Give equations to define the following: (b) the ensemble mean at an instant of time $n = n_1$; (c) the temporal mean of the k^{th} signal computed over the period $n = n_1$ to $n = n_2$; and (d) the average signal or template, $\bar{x}(n)$, $n = 0, 1, 2, \dots, N-1$, computed over the set of signals.
 49. You are given two processes that generate two signals, x and y , with the PDFs $p_x(x)$ and $p_y(y)$, respectively. The joint PDF between the two processes is $p_{x,y}(x,y)$. (a) Write an expression to define the mean of the signal x . (b) Write an expression to define the variance of the signal x . (c) Write an expression to define the covariance between the two signals x and y . (d) Write an expression to define the normalized correlation coefficient between the two signals. (e) Give interpretations of the variance and normalized correlation coefficient and indicate practical use or applications of these measures.
 50. You are given an ensemble of M signals, $y_k(n)$, $k = 1, 2, \dots, M$, with each signal having N samples, indexed as $n = 0, 1, 2, \dots, N-1$. Write mathematical equations or expressions for the following: (a) The ensemble or synchronized average, $s(n)$, of the M signals. (b) The total power of the difference, error, or noise between the result of averaging, $s(n)$, and all of the given signals, $y_k(n)$, $k = 1, 2, \dots, M$. (c) The total power of the result, $s(n)$. (d) The SNR of the result.
 51. You are given two signals, $x(n)$ and $y(n)$, $n = 0, 1, 2, \dots, N$. Write a mathematical equation or expression to derive the normalized correlation coefficient between the two signals. Suppose that $x(n) = u(n) - u(n-4)$, where $u(n)$ is the unit step function, and $y(n) = x(n-8)$. Sketch the signals $x(n)$ and $y(n)$. Compute the correlation coefficient (without normalization) between the two signals and plot the result. Explain the relationships between the signals and the result.

52. A digital filter is specified by the difference equation $y(n) = \frac{1}{4} \sum_{k=0}^{k=3} x(n-k)$, where $x(n)$ is the input and $y(n)$ is the output. (a) Give an equation for and draw a sketch of the impulse response of the filter. (b) Draw the signal-flow diagram of the filter. (c) Derive the transfer function of the filter. (d) Derive the magnitude and phase parts of the frequency response of the filter. (e) What is the gain of the filter at 0 Hz and $f_s/2\text{ Hz}$, where f_s is the sampling frequency? (f) Explain the nature and effects of the filter.
53. Considering the acquisition of the ECG signal, identify and describe one potential source or cause of each of the following types of artifact: (a) high-frequency noise, (b) periodic artifact, and (c) a physiological artifact. In each case, explain how the artifact is caused, how the artifact gets combined with the ECG, and how you would remove or prevent the artifact.
54. Write the equations to define (a) the convolution of two signals, and (b) the Fourier transform of a signal. Prove that the Fourier transform of the convolution of two signals is equal to the product of the Fourier transforms of the two individual signals. Show and explain all steps. You may use continuous-time or discrete-time notation.
55. Write an equation to define the cross-correlation between two signals. Explain the computational procedures required to obtain the cross-correlation. Derive an expression for the Fourier transform of the cross-correlation of two signals in terms of the Fourier transforms of the individual signals. Show all steps. If you use any property of the Fourier transform, give its proof.
56. A biomedical engineer working in a neurophysiology laboratory is frustrated by the appearance of the ECG as an artifact in the EEG of a patient. You are hired to help the engineer. (a) Compare the typical amplitude ranges and frequency bandwidths of the two signals. (b) Would you recommend the use of a fixed lowpass, highpass, or bandpass filter to remove the artifact? If yes, give the essential characteristics of the filter that you recommend. If not, explain your reasons. (c) The engineer has heard about adaptive filters (ANCs). Draw a schematic (block) diagram of an ANC. In the context of the problem mentioned above, explain what the primary input should be; how and from where on the patient you would obtain an appropriate reference input; and how the various inputs and outputs of the system relate to one another. Explain the basic assumptions made in the design and application of the filter. You do not have to derive any equation in your answer to this problem.
57. A discrete-time signal $x(n)$ is passed through an LSI filter with the impulse response $h(n)$. Write the full expression that gives the output $y(n)$ in terms of $x(n)$ and $h(n)$. Starting with the basic definition of the z -transform, derive the relationship between the z -transforms of $y(n)$, $x(n)$, and $h(n)$.
58. (a) A filter is specified to have a zero in the z domain at $z = 1$. (i) Derive the transfer function $H_1(z)$ of the filter. (ii) Derive the impulse response $h_1(n)$ of the filter. (iii) Is this a lowpass, highpass, bandpass, or band-reject filter?
 (b) Another filter is specified to have a double-zero (or two zeros) at $z = -1$. (i) Derive the transfer function $H_2(z)$ of the filter. (ii) Derive the impulse response $h_2(n)$ of the filter. (iii) Is this a lowpass, highpass, bandpass, or band-reject filter?
 (c) A researcher uses the two filters $H_1(z)$ and $H_2(z)$ as above in cascade (series). (i) Derive the transfer function $H(z)$ of the combined filter. (ii) Derive the impulse response $h(n)$ of the combined filter. (iii) Draw the pole-zero plot for the combined

- filter. (iv) Draw the signal-flow diagram for the combined filter. (v) Is this a lowpass, highpass, bandpass, or band-reject filter?
59. A biomedical signal is sampled at 400 Hz with no aliasing error. The signal is known to contain power-line artifact at the fundamental frequency of 50 Hz and all of its harmonics up to and including the maximum frequency present in the signal. Draw the unit circle in the z -domain and show the positions of the zeros of a comb filter to remove the artifact. Mark the angle and frequency of each zero. (You do not have to derive the transfer function of the filter.)
 60. Two discrete-time filters are specified in terms of their impulse responses as $h_1(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3)$ and $h_2(n) = \delta(n) - \delta(n-1)$. The two filters are used in series to filter a signal. Derive and plot the impulse response of the combined filter. Derive the transfer function and frequency response of the combined filter.
 61. In the optimization procedure for the derivation of the Wiener filter, the coefficients of the filter are expressed as the vector $\mathbf{w} = [w_0, w_1, w_2, \dots, w_{M-1}]^T$, where M is the order of the filter and T indicates the transpose. The current input sample $x(n)$ and $M-1$ previous input samples are expressed in another vector as $\mathbf{x}(n) = [x(n), x(n-1), \dots, x(n-M+1)]^T$. (a) Write the full expression of convolution to define the output of the filter in terms of the input signal and the impulse response of the filter. Write the equivalent expression using the vectors as defined above and explain how the two methods lead to the same result. (b) Explain the difference between $\mathbf{x}(n) \mathbf{x}^T(n)$ and $\mathbf{x}^T(n) \mathbf{x}(n)$. (c) What does $E[\mathbf{x}(n) \mathbf{x}^T(n)]$ represent? Write a mathematical expression to give the detailed relationship between an element in the result and the input sample values.
 62. A signal $x(n)$ is given in terms of its samples as $\{4, 3, 2, 1, 2, 4, 2, 1\}$, for $n = 0, 1, 2, \dots, 7$. The signal is processed using an LSI filter with the impulse response, $h(n)$, having the sampled values $\{1, 2, 1\}$, for $n = 0, 1, 2$. Give the procedure to compute the output of the filter. Derive the output of the filter showing all steps.
 63. Draw a block diagram of the ANC (adaptive filter). State the expected relationships between the inputs to the filter. State the conditions that must be met for optimal functioning of the adaptive filter. Give equations for the output of the filter using both the summation form and the vectorial form for convolution of signals. Explain the composition of the vectors in the vectorial form of the equation.
 64. A filter is specified in terms of its pole-zero plot as follows: a zero at $z = 1$ and a zero at $z = -1$. (a) Derive the transfer function of the filter. (b) Derive the difference equation and draw a signal-flow diagram of the filter. (c) Derive and plot the impulse response of the filter. (d) Derive the magnitude and phase of the frequency response of the filter. (e) Draw a sketch of the magnitude of the frequency response of the filter. Assuming the sampling rate to be 200 Hz , label the frequency axis in Hz . (f) Derive the gain at 0 Hz and 100 Hz and explain the nature of the filter.
 65. Draw a schematic sketch of a speech signal including segments of voiced and unvoiced speech. Explain their characteristics. Explain the test for randomness. Indicate the result you would expect if you were to apply the test for randomness to your speech signal example.
 66. A biomedical signal sampled at 240 Hz contains power-line interference at 60 Hz . Design a notch filter to remove the artifact. Draw the unit circle in the complex z domain. Indicate the values of z and the frequency in Hz at the intersections of the circle with

the axes. Mark the locations of the poles and/or zeros in your filter design. Derive the transfer function and the impulse response of the filter. Show all steps.

67. A researcher is interested in recording vibration signals from the knee joint during swinging movement of the leg. However, muscle vibration signals from the thigh muscle were observed to contaminate the knee-joint vibration signal. Provide recommendations to the researcher on how a filter for ANC may be designed to reduce the muscle artifact. Give a schematic block diagram of the ANC filter. You do not need to derive the mathematical procedures for the filter. Indicate where and which signals need to be provided as input to the ANC filter, and where the filtered output is to be obtained. Which part of the ANC filter has time-varying characteristics? Write an equation to describe the input–output relationship of this part.
68. Compare the schematic representations of a cell in Figure 1.9 and an LSI system in Figures 3.14 and 3.17. Draw analogies between their inputs, outputs, and characteristics.

3.17 Laboratory Exercises and Projects

Note: Data files related to the exercises are available at the site

<http://people.ucalgary.ca/~ranga/enel563>

1. The data file `ecg2x60.dat` contains an ECG signal, sampled at 200 Hz, with a significant amount of 60 Hz power-line artifact. (See also the file `ecg2x60.m`.) (a) Design a notch filter with two zeros to remove the artifact and implement it in MATLAB[®]. (b) Add two poles at the same frequencies as those of the zeros, but with a radius that is less than unity. Study the effect of the poles on the output of the filter as their radius is varied.
2. A noisy ECG signal is provided in the file `ecg_hfn.dat`. (See also the file `ecg_hfn.m`.) The sampling rate of this signal is 1,000 Hz.

Develop a MATLAB[®] program to perform synchronized averaging as described in Section 3.5.1. Select a QRS complex from the signal for use as the template and use a suitable threshold on the cross-correlation function in Equation 3.96 for beat detection. Plot the resulting averaged QRS complex. Ensure that the averaged result covers one full cardiac cycle. Plot a sample ECG cycle from the noisy signal for comparison.

Select the QRS complex from a different beat for use as the template and repeat the experiment. Observe the results when the threshold on the cross-correlation function is low (for example, 0.4) or high (for example, 0.95) and comment.

3. Filter the noisy ECG signal in the file `ecg_hfn.dat` (see also the file `ecg_hfn.m`; $f_s = 1,000$ Hz) using four different Butterworth lowpass filters (individually) realized with MATLAB[®] with the following characteristics: (a) Order 2, cutoff frequency 10 Hz. (b) Order 8, cutoff frequency 20 Hz. (c) Order 8, cutoff frequency 40 Hz. (d) Order 8, cutoff frequency 70 Hz. Use “help butter” and “help filter” in MATLAB[®] to get details about the Butterworth filter.

Compare the results obtained using each of the four Butterworth filters (individually) with those obtained by synchronized averaging, and comment upon the improvement or distortion in the outputs. Relate your discussions to specific characteristics observed in plots of the signals.

4. The ECG signal in the file `ecg_lfn.dat`, with $f_s = 1,000 \text{ Hz}$, has a wandering baseline (low-frequency artifact). (See also the file `ecg_lfn.m`.) Filter the signal with the derivative-based filters described in Section 3.5.3 and study the results. Study the effect of variation of the position of the pole in the filter in Equation 3.131 on the signal.
5. Filter the signal in the file `ecg_lfn.dat` ($f_s = 1,000 \text{ Hz}$) using Butterworth highpass filters with orders 2 to 8 and cutoff frequencies 0.5 to 5 Hz. (See also the file `ecg_lfn.m`.) Study the efficacy of the filters in removing the baseline artifact and the effect on the ECG waveform itself. Determine the best compromise acceptable.
6. Design a Wiener filter to remove the artifacts in the ECG signal in the file `ecg_hfn.dat`. (See also the file `ecg_hfn.m`.) The equation of the desired filter is given in Equation 3.185. The required model PSDs may be obtained as follows:

Create a piecewise linear model of the desired version of the signal by concatenating linear segments to provide P, QRS, and T waves with amplitudes, durations, and intervals similar to those in the given noisy ECG signal. Compute the PSD of the model signal.

Select a few segments from the given ECG signal that are expected to be isoelectric (for example, the T–P intervals). Compute their PSDs and obtain their average. The selected noise segments should have zero mean or have the mean subtracted out.

Compare the results of the Wiener filter with those obtained by synchronized averaging and lowpass filtering.

CHAPTER 4

DETECTION OF EVENTS

Biomedical signals carry signatures of physiological events. The part of a signal related to a specific event of interest is often referred to as an *epoch*. Analysis of a signal for monitoring or diagnosis requires the identification of epochs and investigation of the corresponding events. Once an event has been identified, the corresponding waveform may be segmented and analyzed in terms of its amplitude, waveshape (morphology), time duration, intervals between events, energy distribution, frequency content, and other characteristics. Detection of events is, therefore, an important step in biomedical signal analysis.

4.1 Problem Statement

A generic problem statement applicable to the theme of this chapter may be formulated as follows:

Given a biomedical signal, identify discrete signal epochs and correlate them with events in the related physiological processes.

In the sections to follow, we shall first study a few examples of epochs in different biomedical signals, with the aim of understanding the nature of the related physio-

logical events. Such an understanding will help in the subsequent development of signal processing techniques to emphasize, detect, and analyze epochs.

4.2 Illustration of the Problem with Case Studies

The following sections provide illustrations of several events in biomedical signals. The aim of the illustrations is to develop an appreciation of the nature of signal events. A good understanding of signal events will help in designing appropriate signal processing techniques for their detection.

4.2.1 The P, QRS, and T waves in the ECG

As shown in Section 1.2.5, a cardiac cycle is represented in a period of the repetitive ECG signal as the series of waves labeled P, QRS, and T. If we view the cardiac cycle as a series of events, we have the following epochs in an ECG waveform:

- **The P wave:** Contraction of the atria is triggered by the SA-node impulse. The atria do not possess any specialized conduction system as the ventricles do; as such, contraction of the atrial muscles takes place in a slow squeezing manner, with the excitation stimulus being propagated by the muscle cells themselves. For this reason, the P wave is a slow waveform, with a duration of about 80 *ms*. The P wave amplitude is much smaller (about 0.1 – 0.2 *mV*) than that of the QRS because the atria are smaller than the ventricles. The P wave is the epoch related to the event of atrial contraction. (Atrial relaxation does not produce any distinct waveform in the ECG as it is overshadowed by the following QRS wave.)
- **The PQ segment:** The AV node provides a delay to facilitate completion of atrial contraction and transfer of blood to the ventricles before ventricular contraction is initiated. The resulting PQ segment, of about 80 *ms* duration, is thus a “nonevent”; however, it is important in recognizing the baseline as the interval is usually isoelectric.
- **The QRS wave:** The specialized system of Purkinje fibers stimulate contraction of ventricular muscles in a rapid sequence from the apex upwards. The almost-simultaneous contraction of the entire ventricular musculature results in a sharp and tall QRS complex of about 1 *mV* amplitude and 80 – 100 *ms* duration. The event of ventricular contraction is represented by the QRS epoch.
- **The ST segment:** The normally flat (isoelectric) ST segment is related to the plateau in the action potential of the left ventricular muscle cells (see Figures 1.7 and 1.24). The duration of the plateau in the action potential is about 200 *ms*; the ST segment duration is usually about 100 – 120 *ms*. As in the case of the PQ segment, the ST segment may also be called a nonevent. However, myocardial ischemia or infarction could change the action potentials of a portion of

the left ventricular musculature and cause the ST segment to be depressed (see Figure 1.46) or elevated. The PQ segment serves as a useful reference when the isoelectric nature of the ST segment needs to be verified.

- **The T wave:** The T wave appears in a normal ECG signal as a discrete wave separated from the QRS by an isoelectric ST segment. However, it relates to the last phase of the action potential of ventricular muscle cells, when the potential returns from the plateau of the depolarized state to the resting potential through the process of repolarization [33]. The T wave is commonly referred to as the wave corresponding to ventricular relaxation. While this is correct, it should be noted that relaxation through repolarization is simply the final phase of contraction: Contraction and relaxation are indicated by the upstroke and downstroke of the same action potential. For this reason, the T wave may be said to relate to a nonspecific event.

The T wave is elusive, being low in amplitude ($0.1 - 0.3 \text{ mV}$) and being a slow wave extending over $120 - 160 \text{ ms}$. It is almost absent in many ECG recordings. Rather than attempt to detect the often obscure T wave, one may extract a segment of the ECG $80 - 360 \text{ ms}$ from the beginning of the QRS and use it to represent the ST segment and the T wave.

4.2.2 The first and second heart sounds

We observed in Section 1.2.9 that the normal cardiac cycle manifests as a series of the first and second heart sounds — S1 and S2. Murmurs and additional sounds may appear in the presence of cardiovascular diseases or defects. We shall concentrate on S1, S2, and murmurs only.

- **The first heart sound S1:** S1 reflects a sequence of events related to ventricular contraction — closure of the atrioventricular valves, isovolumic contraction, opening of the semilunar valves, and ejection of the blood from the ventricles [33]. The epoch of S1 is directly related to the event of ventricular contraction.
- **The second heart sound S2:** S2 is related to the end of ventricular contraction, signified by closure of the aortic and pulmonary valves. As we observed in the case of the T wave, the end of ventricular contraction cannot be referred to as a specific event per se. However, in the case of S2, we do have the specific events of closure of the aortic and pulmonary valves to relate to, as indicated by the corresponding A2 and P2 components of S2. Unfortunately, separate identification of A2 and P2 is confounded by the fact that they usually overlap in normal signals. If A2 and P2 are separated due to a cardiovascular disorder, simultaneous multisite PCG recordings will be required to identify each component definitively as they may be reversed in order (see Tavel [72] and Rushmer [33]).
- **Murmurs:** Murmurs, if present, could be viewed as specific events. For example, the systolic murmur of aortic stenosis relates to the event of turbulent

ejection of blood from the left ventricle through a restricted aortic valve opening. The diastolic murmur in the case of aortic insufficiency corresponds to the event of regurgitation of blood from the aorta back into the left ventricle through a leaky aortic valve.

4.2.3 The dicrotic notch in the carotid pulse

As we saw in Sections 1.2.10 and 1.2.11, closure of the aortic valve causes a sudden drop in aortic pressure that is already on a downward slope at the end of ventricular systole. The dicrotic notch inscribed in the carotid pulse is a delayed, upstream manifestation of the incisura in the aortic pressure wave. The dicrotic notch is a specific signature on the relatively nondescript carotid pulse signal, and may be taken as an epoch related to the event of aortic valve closure (albeit with a time delay); the same event also signifies the end of ventricular systole and ejection as well as the beginning of S2 and diastole.

4.2.4 EEG rhythms, waves, and transients

We have already studied a few basic characteristics of the EEG in Section 1.2.6, and noted the nature of the α , β , δ , and θ waves. We shall now consider a few events and transients that occur in EEG signals [54–56, 195–197]. Figure 4.1 shows typical manifestations of the activities described below [54].

- **K-complex:** This is a transient complex waveform with slow waves, sometimes associated with sharp components, and often followed by 14 Hz waves. It occurs spontaneously or in response to a sudden stimulus during sleep, with an amplitude of about 200 μV .
- **Lambda waves:** These are monophasic, positive, sharp waves that occur in the occipital location with an amplitude of less than 50 μV . They are related to eye movement and are associated with visual exploration.
- **Mu rhythm:** This rhythm appears as a group of waves in the frequency range of 7–11 Hz with an arcade or comb shape in the central location. The mu rhythm usually has an amplitude of less than 50 μV , and is blocked or attenuated by contralateral movement, thought of movement, readiness to move, or tactile stimulation.
- **Spike:** A spike is defined as a transient with a pointed peak, having a duration in the range of 20–30 ms .
- **Sharp wave:** A sharp wave is also a transient with a pointed peak, but with a longer duration than a spike, in the range of 70–200 ms .
- **Spike-and-wave rhythm:** A sequence of surface-negative slow waves in the frequency range of 2.5–3.5 Hz and having a spike associated with each wave is referred to as a spike-and-wave rhythm. There could be several spikes of

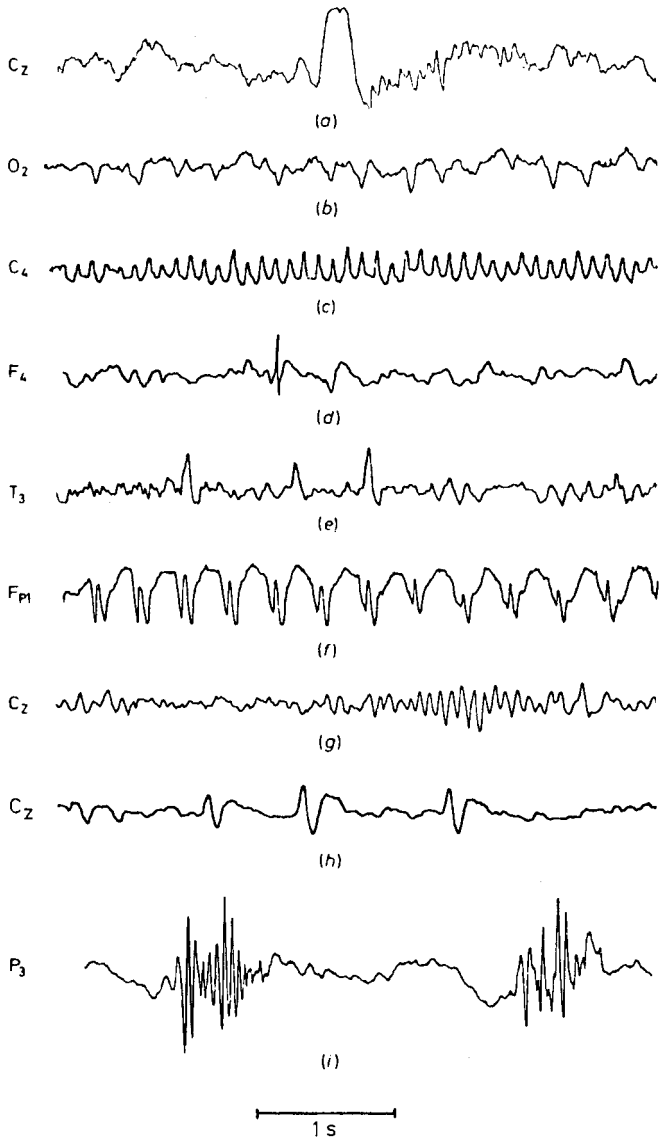


Figure 4.1 Top to bottom: (a) the K-complex; (b) the lambda wave; (c) the mu rhythm; (d) a spike; (e) sharp waves; (f) spike-and-wave complexes; (g) a sleep spindle; (h) vertex sharp waves; and (i) polyspike discharges. The horizontal bar at the bottom indicates a duration of 1 s; the vertical bars at the right indicate 100 μV . Reproduced with permission from R. Cooper, J.W. Osselson, and J.C. Shaw, *EEG Technology*, 3rd edition, 1980. ©Butterworth Heinemann Publishers, a division of Reed Educational & Professional Publishing Ltd., Oxford, UK.

amplitude up to $1,000 \mu V$ in each complex, in which case the rhythm is called a polyspike-and-wave complex.

- **Sleep spindle:** This is an episodic rhythm at about 14 Hz and $50 \mu V$, occurring maximally over the frontocentral regions during certain stages of sleep. A spindle is defined, in general, as a short sequence of monomorphic waves having a fusiform appearance [55].
- **Vertex sharp transient or V-wave:** This wave represents a sharp potential that is maximal at the vertex at about $300 \mu V$ and is negative in relation to the EEG in other areas. It occurs spontaneously during sleep or in response to a sensory stimulus during sleep or wakefulness.

In addition to the above, the term “burst” is used to indicate a phenomenon composed of two or more waves that are different from the principal (background) activity in terms of amplitude, frequency, or waveform. A burst is abrupt and has a relatively short duration [55].

An EEG record is described in terms of [54]

- the most persistent rhythm (for example, α);
- the presence of other rhythmic features, such as δ , θ , or β ;
- discrete features of relatively long duration, such as an episode of spike-and-wave activity;
- discrete features of relatively short duration, such as isolated spikes or sharp waves;
- the activity remaining when all the previous features have been described, referred to as background activity; and
- artifacts, if any, giving rise to ambiguity in interpretation.

Each of the EEG waves or activities is described in chronological sequence in terms of amplitude; frequency, in the case of rhythmic features; waveform, in the case of both rhythmic and transient features; location or spatial distribution; incidence or temporal variability; the presence or absence of right–left symmetry in the location of activity; and responsiveness to stimuli, such as eye opening and closure. The EEG record at rest is first described as above; effects of evocative techniques are then specified in the same terms. Behavioral changes, such as the subject becoming drowsy or falling asleep, are also noted [54].

The EEG signals in Figure 1.39 demonstrate the presence of the α rhythm in all the channels. The EEG signals in Figure 1.40 depict spike-and-wave complexes in almost all the channels.

4.3 Detection of Events and Waves

We shall now see how the knowledge that we have gained so far of several biomedical signal events may be applied to develop signal processing techniques for their detection. Each of the following sections deals with the problem of detection of a specific type of event. The techniques described should find applications in the detection of other events of comparable characteristics.

4.3.1 Derivative based methods for QRS detection

Problem: *Develop signal processing techniques to facilitate detection of the QRS complex, given that it is the sharpest wave in an ECG cycle.*

Solution 1: We noted in Section 1.2.5 that the QRS complex has the largest slope (rate of change of voltage) in a cardiac cycle by virtue of the rapid conduction and depolarization characteristics of the ventricles. As the rate of change is given by the derivative operator, the $\frac{d}{dt}$ operation would be the most logical starting point in an attempt to develop an algorithm to detect the QRS complex.

We saw in Section 3.5.3 that the derivative operator enhances the QRS, although the resulting wave does not bear any resemblance to a typical QRS complex. Observe in Figures 3.54 and 3.55 that the slow P and T waves have been suppressed by the derivative operators, while the output is the highest at the QRS. However, given the noisy nature of the results of the derivative-based operators, it is also evident that significant smoothing will be required before further processing can take place.

Balda et al. [198] proposed a derivative-based algorithm for QRS detection, which was further studied and evaluated by Ahlstrom and Tompkins [199], Friesen et al. [200], and Tompkins [46]. The algorithm progresses as follows. In a manner similar to Equation 3.127, the smoothed three-point first derivative $y_0(n)$ of the given signal $x(n)$ is approximated as

$$y_0(n) = |x(n) - x(n-2)|. \quad (4.1)$$

The second derivative is approximated as

$$y_1(n) = |x(n) - 2x(n-2) + x(n-4)|. \quad (4.2)$$

The two results are weighted and combined to obtain

$$y_2(n) = 1.3y_0(n) + 1.1y_1(n). \quad (4.3)$$

The result $y_2(n)$ is scanned with a threshold of 1.0. Whenever the threshold is crossed, the subsequent eight samples are also tested against the same threshold. If at least six of the eight points pass the threshold test, the segment of eight samples is taken to be a part of a QRS complex. The procedure results in a pulse with its width proportional to that of the QRS complex; however, the method is sensitive to noise.

Illustration of application: Figure 4.2 illustrates, in the topmost trace, two cycles of a filtered version of the ECG signal shown in Figure 3.5. The signal was filtered

with an eighth-order Butterworth lowpass filter with $f_c = 90 \text{ Hz}$, down-sampled by a factor of five, and filtered with a notch filter with $f_o = 60 \text{ Hz}$. The effective sampling rate is 200 Hz . The signal was normalized by dividing by its maximum value.

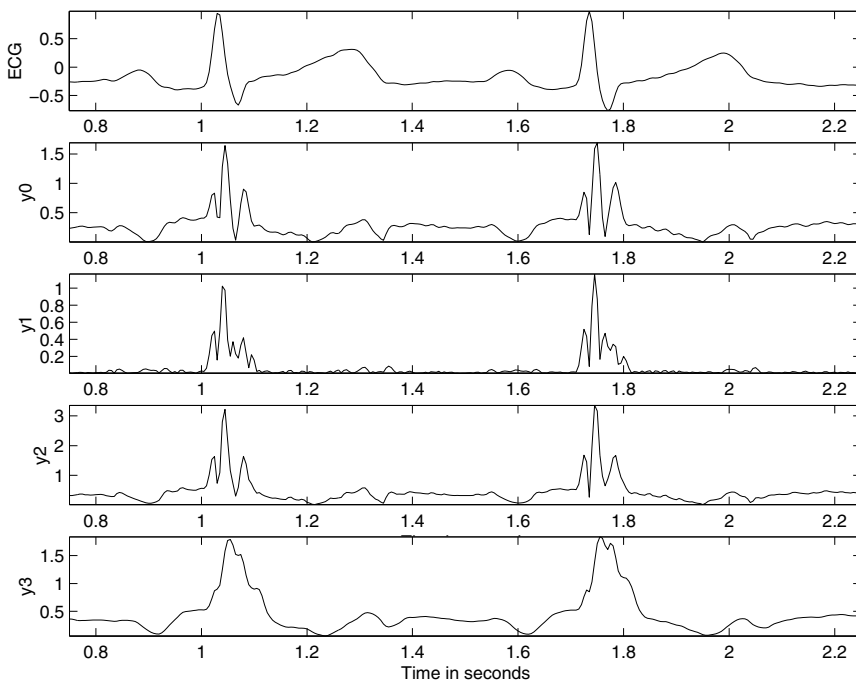


Figure 4.2 From top to bottom: two cycles of a filtered version of the ECG signal shown in Figure 3.5; output $y_0(n)$ of the first-derivative-based operator in Equation 4.1; output $y_1(n)$ of the second-derivative-based operator in Equation 4.2; the combined result $y_2(n)$ from Equation 4.3; and the result $y_3(n)$ of passing $y_2(n)$ through the 8-point MA filter in Equation 3.107.

The second and third plots in Figure 4.2 show the derivatives $y_0(n)$ and $y_1(n)$, respectively; the fourth plot illustrates the combined result $y_2(n)$. Observe the relatively high values in the derivative-based results at the QRS locations; the outputs are low or negligible at the P and T wave locations, in spite of the fact that the original signal possesses an unusually sharp and tall T wave. It is also seen that the results have multiple peaks over the duration of the QRS wave, due to the fact that the QRS complex includes three major swings: Q–R, R–S, and S–ST baseline in the present example (an additional PQ baseline–Q swing may also be present in other ECG signals).

The last plot in Figure 4.2 shows the smoothed result $y_3(n)$ obtained by passing $y_2(n)$ through the 8-point MA filter in Equation 3.107. We now have a single pulse with amplitude greater than 1.0 over the duration of the corresponding QRS complex.

A simple peak-searching algorithm may be used to detect each ECG beat. The net delay introduced by the filters should be subtracted from the detected peak location in order to obtain the corresponding QRS location.

Note that peak searching cannot be performed directly on an ECG signal: The QRS might not always be the highest wave in a cardiac cycle, and artifacts may easily upset the search procedure. Observe also that the ECG signal in the present illustration was filtered to a restricted bandwidth of 90 *Hz* before the derivatives were computed and that it is free of baseline drift.

Solution 2: Murthy and Rangaraj [201] proposed a QRS detection algorithm based upon a weighted and squared first-derivative operator and an MA filter. In this method, a filtered-derivative operator was defined as

$$g_1(n) = \sum_{i=1}^N |x(n-i+1) - x(n-i)|^2 (N-i+1), \quad (4.4)$$

where $x(n)$ is the ECG signal, and N is the width of a window within which first-order differences are computed, squared, and weighted by the factor $(N-i+1)$. The weighting factor decreases linearly from the current difference to the difference N samples earlier in time and provides a smoothing effect. Further smoothing of the result was performed by an MA filter over M points to obtain

$$g(n) = \frac{1}{M} \sum_{j=0}^{M-1} g_1(n-j). \quad (4.5)$$

With the sampling rate of 100 *Hz*, the filter window widths were set as $M = N = 8$ samples. The algorithm provides a single peak for each QRS complex and suppresses P and T waves.

Searching for the peak in a processed signal such as $g(n)$ may be accomplished by a simple peak-searching algorithm as follows:

1. Scan a portion of the signal $g(n)$ that may be expected to contain a peak and determine the maximum value $g_{\max}$, or let $g_{\max}$ be the maximum of $g(n)$ over its entire available duration.
2. Define a threshold as a fraction of the maximum, for example, $Th = 0.5 g_{\max}$.
3. For all $g(n) > Th$, select those samples for which the corresponding $g(n)$ values are greater than a certain predefined number M of preceding and succeeding samples of $g(n)$, that is,

$$\begin{aligned} \{p\} = \{n \mid & [g(n) > Th] \text{ AND} \\ & [g(n) > g(n-i), i = 1, 2, \dots, M] \text{ AND} \\ & [g(n) > g(n+i), i = 1, 2, \dots, M]\}. \end{aligned} \quad (4.6)$$

The set $\{p\}$ defined as above contains the indices of the peaks in $g(n)$.

Additional conditions may be imposed to reject peaks due to artifacts, such as a minimum interval between two adjacent peaks. A more elaborate peak-searching algorithm is described in Section 4.3.2.

Illustration of application: Figure 4.3 illustrates, in the topmost trace, two cycles of a filtered version of the ECG signal shown in Figure 3.5. The signal was filtered with an eighth-order Butterworth lowpass filter with $f_c = 40\text{ Hz}$ and was down-sampled by a factor of ten. The effective sampling rate is 100 Hz to match the parameters used by Murthy and Rangaraj [201]. The signal was normalized by dividing by its maximum value.

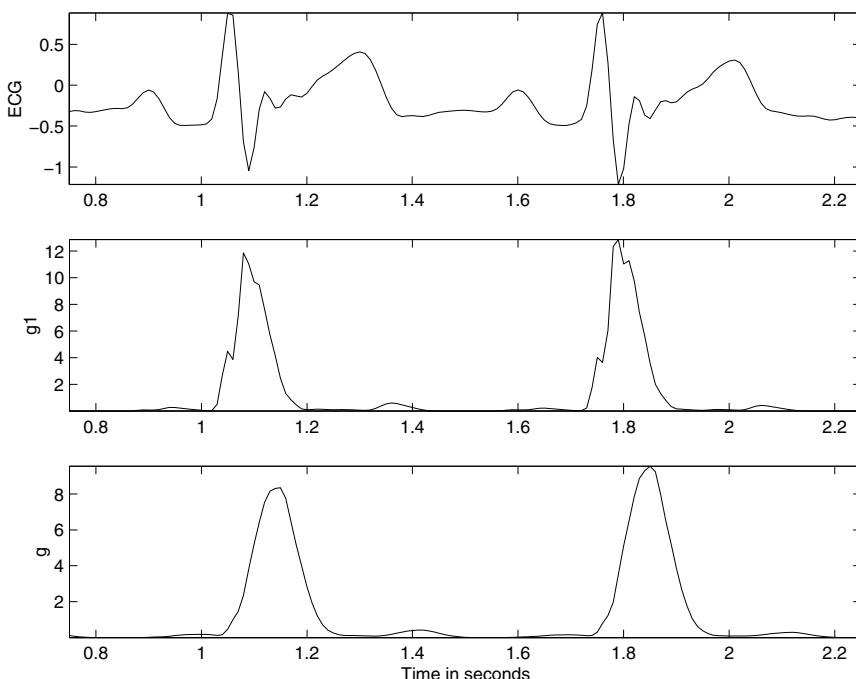


Figure 4.3 From top to bottom: two cycles of a filtered version of the ECG signal shown in Figure 3.5; output $g_1(n)$ of the weighted and squared first-derivative operator in Equation 4.4; output $g(n)$ of the smoothing filter in Equation 4.5.

The second and third plots in Figure 4.3 show the outputs of the derivative-based operator and the smoothing filter. Observe that the final output contains a single, smooth peak for each QRS, and that the P and T waves produce no significant output. A simple peak-searching algorithm may be used to detect and segment each beat [201]. The net delay introduced by the filter used should be subtracted from the detected peak location in order to obtain the corresponding QRS location.

4.3.2 The Pan–Tompkins algorithm for QRS detection

Problem: Propose an algorithm to detect QRS complexes in an ongoing ECG signal.

Solution: Pan and Tompkins [202] (see also Tompkins [46]) proposed a real-time QRS detection algorithm based on analysis of the slope, amplitude, and width of QRS complexes. The algorithm includes a series of filters and methods that perform lowpass, highpass, derivative, squaring, integration, adaptive thresholding, and search procedures. Figure 4.4 illustrates the steps of the algorithm in schematic form.

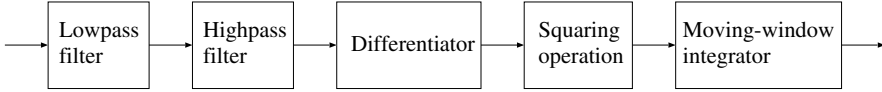


Figure 4.4 Block diagram of the Pan–Tompkins algorithm for QRS detection.

Lowpass filter: The recursive lowpass filter used in the Pan–Tompkins algorithm has integers as its coefficients to reduce computational complexity, with the transfer function defined as

$$H(z) = \frac{1}{32} \frac{(1 - z^{-6})^2}{(1 - z^{-1})^2}. \quad (4.7)$$

(See also Equations 3.119 and 3.120.) The output $y(n)$ is related to the input $x(n)$ as

$$y(n) = 2y(n-1) - y(n-2) + \frac{1}{32} [x(n) - 2x(n-6) + x(n-12)]. \quad (4.8)$$

With the sampling rate being 200 Hz, the filter has a low cutoff frequency of $f_c = 11$ Hz and introduces a delay of 5 samples or 25 ms. The filter provides an attenuation greater than 35 dB at 60 Hz and effectively suppresses power-line interference, if present.

Highpass filter: The highpass filter used in the algorithm is implemented as an allpass filter minus a lowpass filter. The lowpass component has the transfer function

$$H_{lp}(z) = \frac{(1 - z^{-32})}{(1 - z^{-1})}; \quad (4.9)$$

the input–output relationship is

$$y(n) = y(n-1) + x(n) - x(n-32). \quad (4.10)$$

The transfer function $H_{hp}(z)$ of the highpass filter is specified as

$$H_{hp}(z) = z^{-16} - \frac{1}{32} H_{lp}(z). \quad (4.11)$$

Equivalently, the output $p(n)$ of the highpass filter is given by the difference equation

$$p(n) = x(n-16) - \frac{1}{32} [y(n-1) + x(n) - x(n-32)], \quad (4.12)$$

with $x(n)$ and $y(n)$ being related as in Equation 4.10. If all of the operations from Equation 4.9 to Equation 4.12 are combined, the input–output relationship of the overall highpass filter is

$$p(n) = p(n-1) - \frac{1}{32}x(n) + x(n-16) - x(n-17) + \frac{1}{32}x(n-32). \quad (4.13)$$

The highpass filter has a cutoff frequency of 5 Hz and introduces a delay of 80 ms for $f_s = 200$ Hz.

Derivative operator: The derivative operation used by Pan and Tompkins is specified as

$$y(n) = \frac{1}{8} [2x(n) + x(n-1) - x(n-3) - 2x(n-4)] \quad (4.14)$$

and approximates the ideal $\frac{d}{dt}$ operator up to 30 Hz. The derivative procedure suppresses the low-frequency components of the P and T waves, and provides a large gain to the high-frequency components arising from the high slopes of the QRS complex. (See Section 3.5.3 for details on the properties of derivative-based filters.)

Figure 4.5 shows the magnitude of the frequency response for each of the filters in the Pan–Tompkins algorithm for QRS detection described to this point. Observe the response of the highpass filter that was derived from a lowpass filter. The combined procedure has a bandpass nature that is evident from the final response shown in the figure.

Squaring: The squaring operation makes the result positive and emphasizes large differences resulting from QRS complexes; the small differences arising from P and T waves are suppressed. The high-frequency components in the signal related to the QRS complex are further enhanced.

Integration: As observed in Section 4.3.1, the output of a derivative-based operation will exhibit multiple peaks within the duration of a single QRS complex. The Pan–Tompkins algorithm performs smoothing of the output of the preceding operations through a moving-window integration filter as

$$y(n) = \frac{1}{N} \{x[n - (N-1)] + x[n - (N-2)] + \cdots + x(n)\}. \quad (4.15)$$

The choice of the window width N is to be made with the following considerations: too large a value will result in the outputs due to the QRS and T waves being merged, whereas too small a value could yield multiple peaks for a single QRS. A window width of $N = 30$ samples was found to be suitable for $f_s = 200$ Hz. Figure 4.6 illustrates the effect of the window width on the output of the integrator and its relationship to the QRS width. (See Section 3.5.2 for details on the properties of MA and integrating filters.)

Adaptive thresholding: The thresholding procedure in the Pan–Tompkins algorithm adapts to changes in the ECG signal by computing running estimates of signal and noise peaks. A peak is said to be detected whenever the final output $y(n)$ changes direction within a specified time interval. In the following discussion, $SPKI$ represents the peak level that the algorithm has learned to correspond to QRS peaks, and $NPKI$ represents the peak level related to non-QRS events (such as noise, EMG,

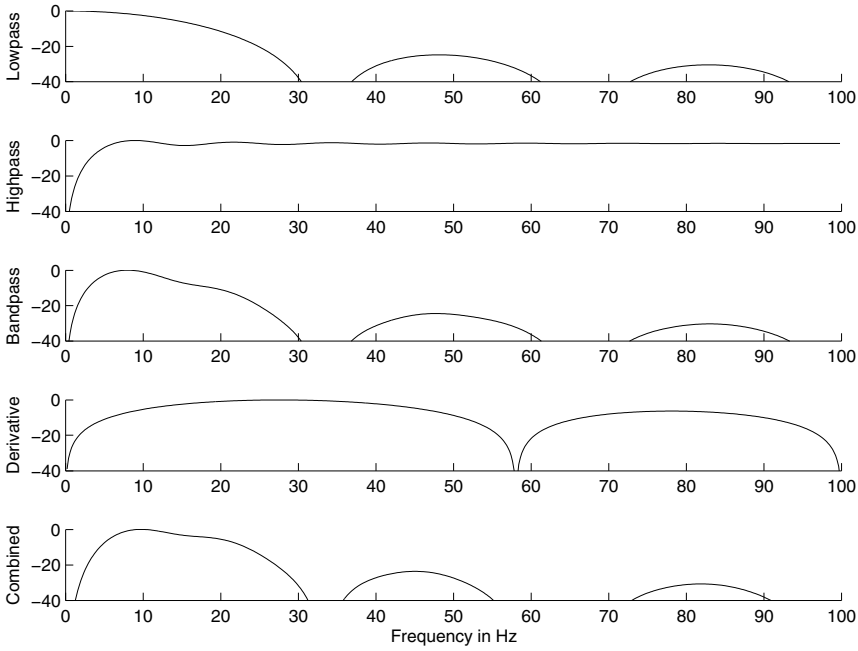


Figure 4.5 Frequency response (magnitude, in *dB*) of the filters used in the Pan–Tompkins algorithm for QRS detection. Top to bottom: the initial lowpass filter, the highpass filter, the bandpass filter resulting from the combination of the two filters, the derivative operator, and the combination of the bandpass filter and the derivative operator. The bandpass nature of the combined procedure is evident from the final response.

and various artifacts). *THRESHOLD I1* and *THRESHOLD I2* are two thresholds used to categorize peaks detected as signal (QRS) or noise.

Every new peak detected is categorized as a signal peak or a noise peak. If a peak exceeds *THRESHOLD I1* during the first step of analysis, it is classified as a QRS (signal) peak. If the search-back technique (described in the next paragraph) is used, the peak should be above *THRESHOLD I2* to be called a QRS. The peak levels and thresholds are updated after each peak is detected and classified as

$$\begin{aligned} SPKI &= 0.125 PEAKI + 0.875 SPKI \quad \text{if } PEAKI \text{ is a signal peak;} \\ NPKI &= 0.125 PEAKI + 0.875 NPKI \quad \text{if } PEAKI \text{ is a noise peak;} \end{aligned} \quad (4.16)$$

$$\begin{aligned} THRESHOLD I1 &= NPKI + 0.25 (SPKI - NPKI); \\ THRESHOLD I2 &= 0.5 THRESHOLD I1. \end{aligned} \quad (4.17)$$

The updating formula for *SPKI* is changed to

$$SPKI = 0.25 PEAKI + 0.75 SPKI \quad (4.18)$$

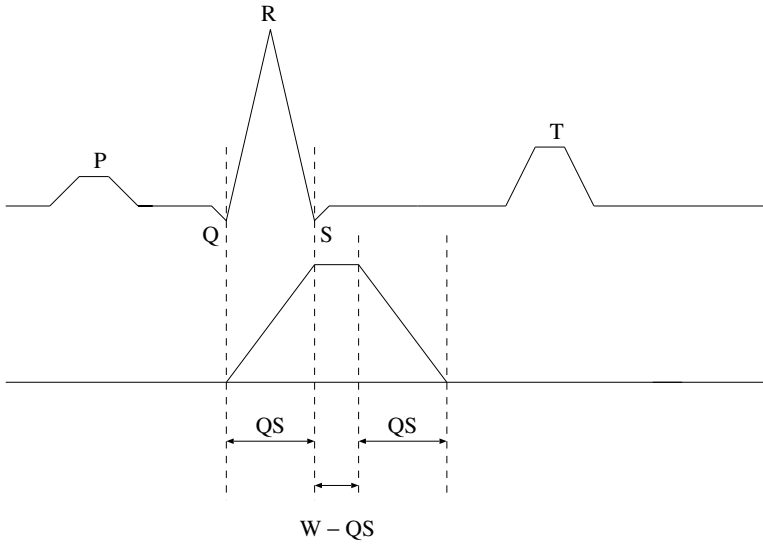


Figure 4.6 The relationship of a QRS complex to the moving-window integrator output. Upper plot: Schematic ECG signal. Lower plot: Schematic representation of the output of the moving-window integrator. QS: QRS complex width. W: width of the integrator window, given as N/f_s s. It is assumed that the QRS complex will provide a result in the form of a rectangular pulse of width QS before the integrator; then, the output of the integrator is a trapezoid as shown, with a total width of $W+QS$, assuming that $W > QS$. Adapted from Tompkins [46].

if a QRS is detected in the search-back procedure using *THRESHOLD I2*.

Search-back procedure: The Pan–Tompkins algorithm maintains two averages of *RR* intervals: *RR AVERAGE1* is the average of the eight most-recent beats, and *RR AVERAGE2* is the average of the eight most-recent beats having *RR* intervals within the range specified by *RR LOW LIMIT* = $0.92 \times \text{RR AVERAGE2}$ and *RR HIGH LIMIT* = $1.16 \times \text{RR AVERAGE2}$. Whenever a QRS is not detected for a certain interval specified as *RR MISSED LIMIT* = $1.66 \times \text{RR AVERAGE2}$, the QRS is taken to be the peak between the established thresholds applied in the search-back procedure.

The Pan–Tompkins algorithm performed with a low error rate of 0.68%, or 33 beats per hour on a database of about 116,000 beats obtained from 24-hour records of the ECGs of 48 patients (see Tompkins [46] for details).

If N_B QRS complexes or beats are detected in an ECG signal over a duration of T s, we have

$$HR = 60 \frac{N_B}{T} \quad (4.19)$$

in *bpm* (on the average). If the average RR interval measured over several beats is RR_a s, we have

$$HR = \frac{60}{RR_a} \quad (4.20)$$

in *bpm* (on the average). The equation given above may also be used with the beat-to-beat RR interval instead of RR_a to obtain the instantaneous heart rate. In a practical ECG monitoring system, it would be desirable to derive the average heart rate and update its display a few times per minute, such as every 10 s.

Illustration of application: Figure 4.7 illustrates, in the topmost trace, the same ECG signal as in Figure 4.2. The Pan–Tompkins algorithm as above was implemented. The outputs of the various stages of the algorithm are illustrated in sequence in the same figure. The observations to be made are similar to those in the preceding section on the derivative-based method. The derivative operator suppresses the P and T waves and provides a large output at the QRS locations. The squaring operation preferentially enhances large values and boosts high-frequency components. The result still possesses multiple peaks for each QRS and hence needs to be smoothed. The final output of the integrator is a single smooth pulse for each QRS. Observe the shift between the actual QRS location and the pulse output due to the cumulative delays of the various filters. The thresholding and search procedures and their results are not illustrated. More examples of QRS detection are presented in Sections 4.9 and 4.10.

4.3.3 Detection of the dicrotic notch

Problem: *Propose a method to detect the dicrotic notch in the carotid pulse signal.*

Solution: Lehner and Rangayyan [131] proposed a method for detection of the dicrotic notch that used the least-squares estimate of the second derivative $p(n)$ of the carotid pulse signal $y(n)$, defined as

$$p(n) = 2y(n-2) - y(n-1) - 2y(n) - y(n+1) + 2y(n+2). \quad (4.21)$$

Observe that this expression is noncausal; it may be made causal by adding a delay of two samples.

The second derivative was used due to the fact that the dicrotic notch appears as a short wave riding on the downward slope of the carotid pulse signal (see also Starmer et al. [203]). A first-derivative operation would give an almost-constant output for the downward slope. The second-derivative operation removes the effect of the downward slope and enhances the notch itself. The result was squared and smoothed to obtain

$$s(n) = \sum_{k=1}^M p^2(n-k+1)w(k), \quad (4.22)$$

where $w(k) = (M-k+1)$, $k = 1, 2, \dots, M$, is a linear weighting function, and $M = 16$ samples for $f_s = 256$ Hz.

The method yields two peaks for each period of the carotid pulse signal. The first peak in the result represents the onset of the carotid upstroke. The second peak that

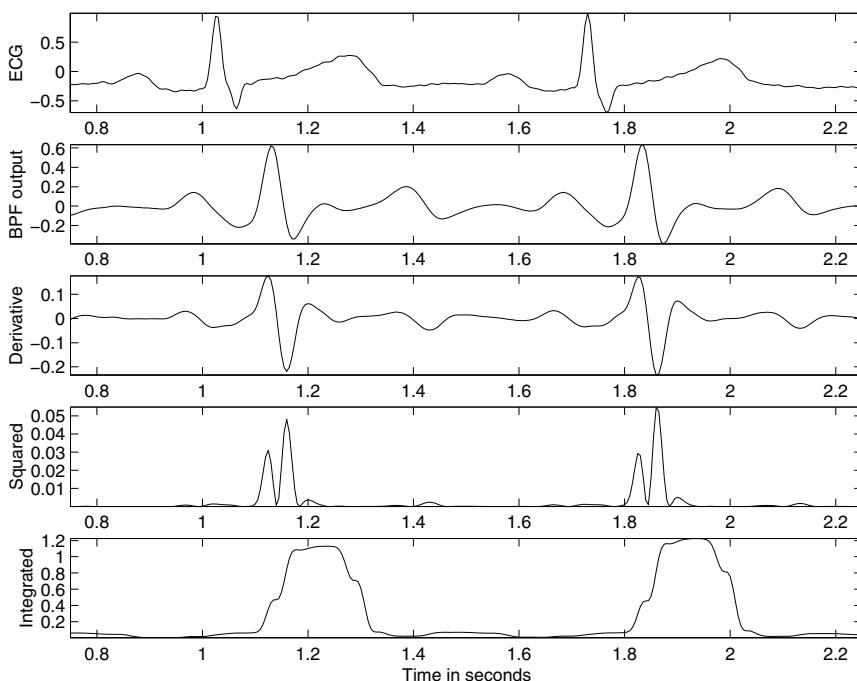


Figure 4.7 Results of the Pan–Tompkins algorithm. From top to bottom: two cycles of a filtered version of the ECG signal shown in Figure 3.5 (the same as that in Figure 4.2); output of the bandpass filter (BPF, a combination of lowpass and highpass filters); output of the derivative-based operator; the result of squaring; and $100\times$ the result of the final integrator.

appears in the result within a cardiac cycle is related to the diastolic notch. To locate the diastolic notch, the local minimum in the carotid pulse within a $\pm 20\text{ ms}$ interval of the second peak was used.

Illustration of application: The upper plot in Figure 4.8 illustrates two cycles of a carotid pulse signal. The signal was lowpass filtered at 100 Hz and sampled at 250 Hz . The result of application of the Lehner and Rangayyan method to the signal is shown in the lower plot. It is evident that the second derivative has successfully accentuated the diastolic notch. A simple peak-searching algorithm may be used to detect the first and second peaks in the result. The diastolic notch may then be located by searching for the minimum in the carotid pulse signal within a $\pm 20\text{ ms}$ interval around the second peak location.

Observe that the result illustrated in Figure 4.8 may benefit from further smoothing by increasing the window width M in Equation 4.22. The window width needs to be chosen in accordance with the characteristics of the signal on hand as well as the lowpass filter and sampling rate used. Further illustration of the detection of the diastolic notch is provided in Section 4.10.

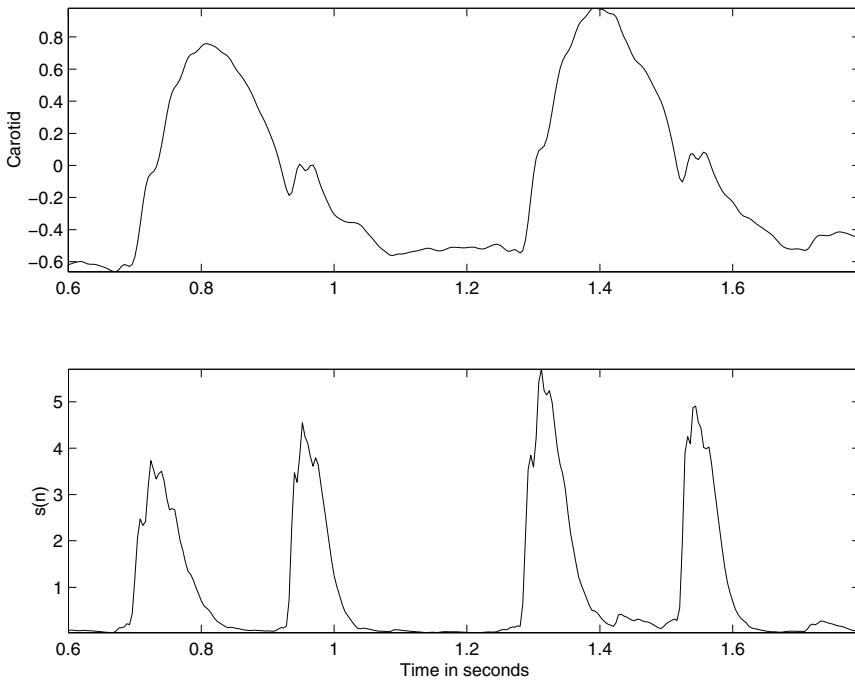


Figure 4.8 Two cycles of a carotid pulse signal and the result of the Lehner and Rangayyan method for detection of the diastolic notch.

4.4 Correlation Analysis of EEG Rhythms

EEG signals are usually acquired simultaneously over multiple channels. Event detection in and epoch analysis of EEG signals becomes more complicated than the problems we have seen with the single-channel ECG and carotid pulse signals, due to the need to detect similar or related events across multiple channels. Autocorrelation and cross-correlation techniques in both the time and frequency domains serve such needs.

4.4.1 Detection of EEG rhythms

Problem: *Propose a method to detect the presence of the α rhythm in an EEG channel. How would you extend the method to detect the presence of the same rhythm simultaneously in two EEG channels?*

Solution: Two signals may be compared to detect the characteristics present in common between them via their dot product (also known as the inner or scalar prod-

uct), defined as

$$x \cdot y = \langle x, y \rangle = \sum_{n=0}^{N-1} x(n) y(n), \quad (4.23)$$

where the signals $x(n)$ and $y(n)$ have N samples each. The dot product represents the projection of one signal on to the other, with each signal being viewed as an N -dimensional vector. The dot product may be normalized by the geometric mean of the energies of the two signals to obtain a correlation coefficient as [132]

$$\gamma_{xy} = \frac{\sum_{n=0}^{N-1} x(n) y(n)}{\left[\sum_{n=0}^{N-1} x^2(n) \sum_{n=0}^{N-1} y^2(n) \right]^{1/2}}. \quad (4.24)$$

The means of the signals may be subtracted out, if desired, as in Equation 3.96.

In the case of two continuous-time signals $x(t)$ and $y(t)$, the projection of one signal on to the other is defined as

$$\theta_{xy} = \int_{-\infty}^{\infty} x(t) y(t) dt. \quad (4.25)$$

When a shift or time delay may be present in the occurrence of the epoch of interest in the two signals being compared, it becomes necessary to introduce a time-shift parameter to compute the projection for every possible position of overlap. The shift parameter facilitates searching one signal for the occurrence of an event matching that in the other signal at any time instant within the available duration of the signals. The CCF between two signals for a shift or delay of τ seconds or k samples may be obtained as

$$\theta_{xy}(\tau) = \int_{-\infty}^{\infty} x(t) y(t + \tau) dt, \quad (4.26)$$

or

$$\theta_{xy}(k) = \sum_n x(n) y(n + k). \quad (4.27)$$

The range of summation in the latter case needs to be limited to the range of the available overlapped data. A scale factor, depending upon the number of data samples used, needs to be introduced to obtain the true CCF, but is neglected here (see Section 6.3). An extended version of the correlation coefficient γ_{xy} in Equation 4.24, to include time shift, is provided in Equation 3.96.

When the ACF or the CCF is computed for various shifts, a question arises about the data samples in one of the signal segments beyond the duration of the other. We may add zeros to one of the signals and increase its length by the maximum shift of interest, or we may use the true data samples from the original signal record, if available. The latter method was used wherever possible in the following illustrations.

In the case of random signals, we need to take the expectation or sample average of the outer product of the vectors formed by the available samples of the signals. Let $\mathbf{x}(n) = [x(n), x(n-1), \dots, x(n-N+1)]^T$ and $\mathbf{y}(n) = [y(n), y(n-1), \dots, y(n-N+1)]^T$ represent the N -dimensional vectorial form of the two signals $x(n)$ and

$y(n)$ with the most-recent N samples being available in each signal at the time instant n . If $\mathbf{x}(n)$ and $\mathbf{y}(n)$ are sample observations of random processes, their CCF is defined as

$$\Theta_{xy} = E[\mathbf{x}(n) \mathbf{y}^T(n)], \quad (4.28)$$

in a manner similar to Equations 3.162 and 3.163. The outer product, which is an $N \times N$ matrix, provides the cross-terms that include all possible delays (shifts) within the duration of the given signals.

All of the equations given above may be modified to obtain the ACF by replacing the second signal y with the first signal x . The signal x is then compared with itself.

The ACF displays peaks at intervals corresponding to the period (and integral multiples thereof) of any periodic or repetitive pattern present in the signal. This property facilitates the detection of rhythms in signals such as the EEG: The presence of the α rhythm would be indicated by a peak in the neighborhood of 0.1 s . The ACF of most signals decays and reaches negligible values after delays of a few milliseconds, except for periodic signals of infinite or indefinite duration for which the ACF will also exhibit periodic peaks. The ACF will also exhibit multiple peaks when the same event repeats itself at regular or irregular intervals. One may need to compute the ACF only up to certain delay limits depending upon the expected characteristics of the signal being analyzed.

The CCF displays peaks at the period of any periodic pattern present in *both* of the signals being analyzed. The CCF may, therefore, be used to detect rhythms present in common between two signals, for example, between two channels of the EEG. When one of the functions being used to compute the CCF is a template representing an event, such as an ECG cycle as in the illustration in Section 3.5.1 or an EEG spike-and-wave complex as in Section 4.4.2, the procedure is known as *template matching*.

Illustration of application: Figure 4.9 shows, in the upper trace, the ACF of a segment of the p4 channel of the EEG in Figure 1.39 over the time interval 4.67 – 5.81 s . The ACF displays peaks at time delays of 0.11 s and its integral multiples. The inverse of the delay of the first peak corresponds to 9 Hz , which is within the α rhythm range. (The PSD in the lower trace of Figure 4.9 and the others to follow are described in Section 4.5.) It is, therefore, obvious that the signal segment analyzed contains the α rhythm. A simple peak-searching algorithm may be applied to the ACF to detect the presence of peaks at specific delays of interest or over the entire range of the ACF.

To contrast with the preceding example, the upper trace of Figure 4.10 shows the ACF of the 4.2 – 4.96 s segment of the f3 channel of the EEG in Figure 1.39. The ACF shows no peak in the 0.08 – 1.25 s region, indicating absence of the α rhythm in the segment analyzed.

Figures 4.11, 4.12, and 4.13 illustrate the CCF results comparing the following portions of the EEG signal shown in Figure 1.39 in order: the p3 and p4 channels over the duration 4.72 – 5.71 s when both channels exhibit the α rhythm; the o2 and c4 channels over the duration 5.71 – 6.78 s when the former has the α rhythm but not the latter channel; and the f3 and f4 channels over the duration 4.13 – 4.96 s when

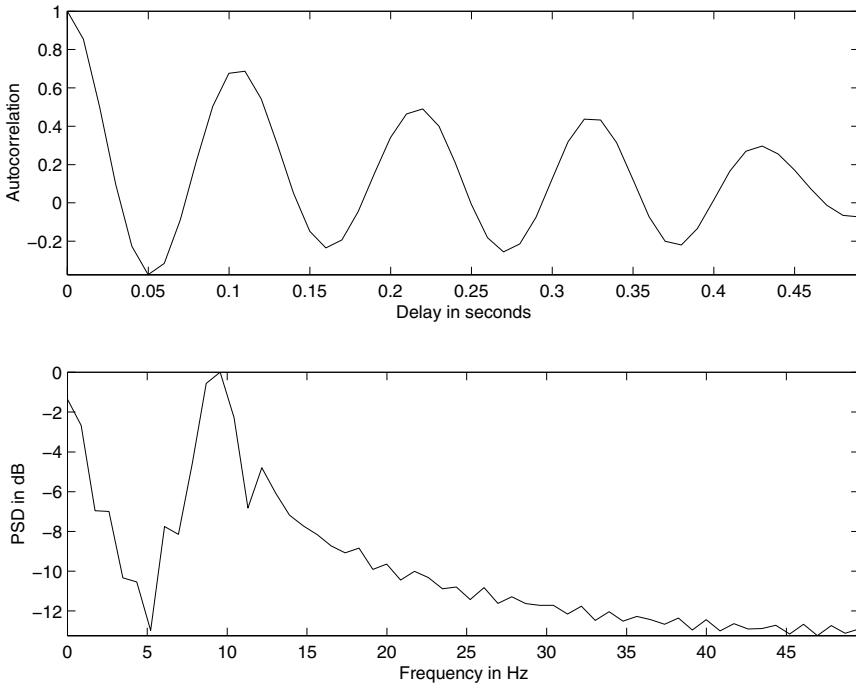


Figure 4.9 Upper trace: ACF of the 4.67 – 5.81 s portion of the p4 channel of the EEG signal shown in Figure 1.39. Lower trace: The PSD of the signal segment in *dB*, given by the Fourier transform of the ACF.

neither channel has α activity. The relative strengths of the peaks in the α range, as described earlier, agree with the joint presence, singular presence, or absence of the α rhythm in the various segments (channels) analyzed.

It should be noted that the segments of EEG signals used in the illustrations in the present section are of short duration. See Section 6.7 for an example of analysis of longer EEG records.

4.4.2 Template matching for EEG spike and wave detection

We have seen the use of template matching for the extraction of ECG cycles for use in synchronized averaging in Section 3.5.1. We shall now consider another application of template matching.

Problem: *Propose a method to detect spike-and-wave complexes in an EEG signal. You may assume that a sample segment of a spike-and-wave complex is available.*

Solution: A spike-and-wave complex is a well-defined event in an EEG signal. The complex is composed of a sharp spike followed by a wave with a frequency of about 3 Hz; the wave may contain a half period or a full period of an almost-

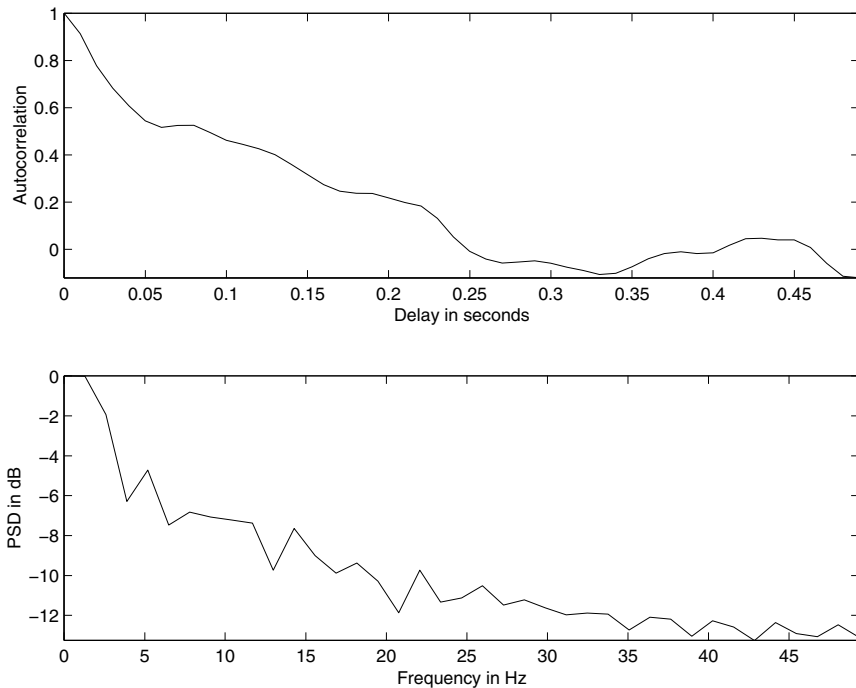


Figure 4.10 Upper trace: ACF of the 4.2 – 4.96 s portion of the f3 channel of the EEG signal shown in Figure 1.39. Lower trace: The PSD of the signal segment in *dB*.

sinusoidal pattern. One may, therefore, extract an epoch of a spike-and-wave complex from an EEG channel and use it for template matching with the same formula as in Equation 3.96 (see also Barlow [196]). The template may be correlated with the same channel from which it was extracted to detect similar events that appear at a later time, or with another channel to search for similar events. A simple threshold on the result should yield the time instants where the events appear.

Illustration of application: The c3 channel of the EEG signal in Figure 1.40 is shown in the upper trace of Figure 4.14. The spike-and-wave complex between 0.60 s and 0.82 s in the signal was selected for use as the template, and template matching was performed with the same channel signal using the formula in Equation 3.96. The result in the lower trace of Figure 4.14 demonstrates strong and clear peaks at each occurrence of the spike-and-wave complex in the EEG signal. The peaks in the result occur at the same instants of time as the corresponding spike-and-wave complexes.

Figure 4.15 shows the f3 channel of the EEG signal in Figure 1.40, along with the result of template matching, using the same template that was used in the previous example from channel c3. The result shows that the f3 channel also has spike-and-wave complexes that match the template.

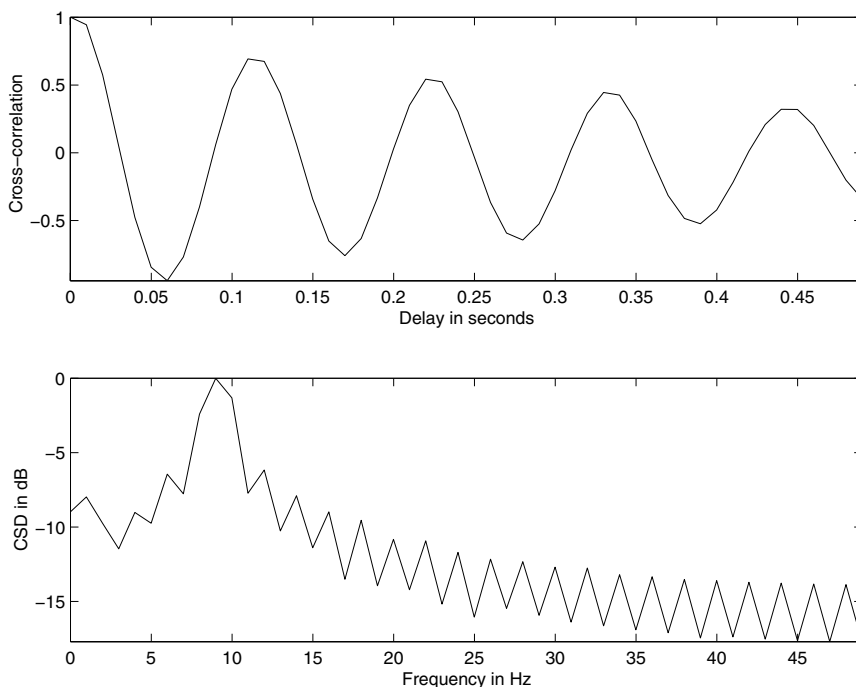


Figure 4.11 Upper trace: CCF between the 4.72 – 5.71 s portions of the p3 and p4 channels of the EEG signal shown in Figure 1.39. Lower trace: The CSD of the signal segments in *dB*, computed as the Fourier transform of the CCF.

4.4.3 Detection of EEG rhythms related to seizure

Yadav et al. [59] noted that the recurring nature of seizures of a given patient could be used to design patient-specific templates for the recognition of seizures. They noted that, in most patients, a few types of seizures tend to occur repeatedly, with related similar, though not identical, patterns in the EEG. Figure 4.16 illustrates the temporal evolution of a seizure event on a single-channel EEG, including varying patterns of piecewise stationary rhythms. (Note that the seizure rhythms are substantially different from the α and other rhythms seen in the normal EEG.) Yadav et al. [59] suggested that it is possible to train a patient-specific seizure detector using previously identified templates, and they proposed a model-based patient-specific method using statistically optimal null filters for automatic detection of seizures in intracranial EEG. See Qu and Gotman [204] and Grewal and Gotman [60] for additional details on detection of seizures.

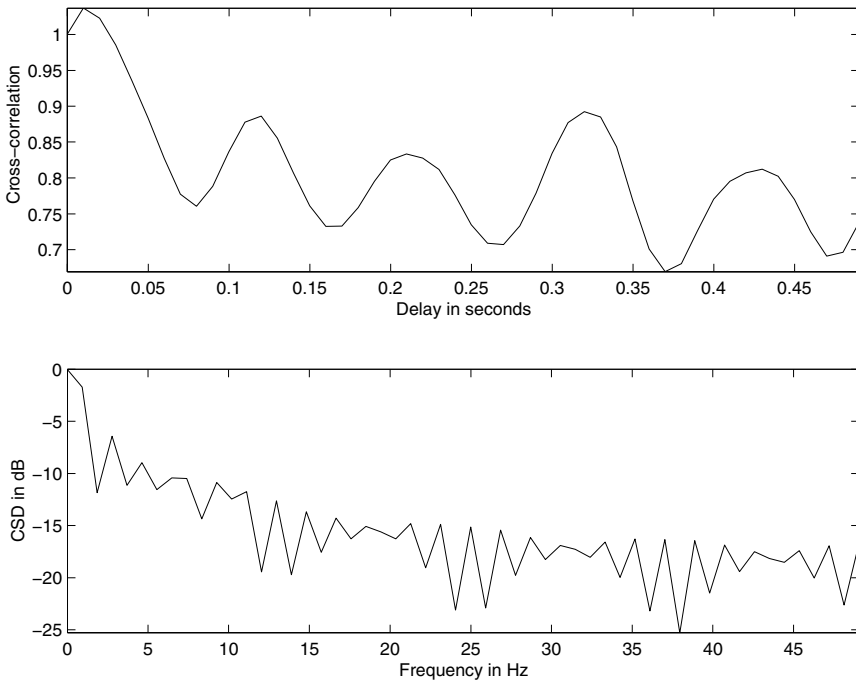


Figure 4.12 Upper trace: CCF between the 5.71 – 6.78 s portions of the o2 and c4 channels of the EEG signal shown in Figure 1.39. Lower trace: The CSD of the signal segments in *dB*.

4.5 Cross spectral Techniques

The multiple peaks that arise in the ACF or CCF may cause confusion in the detection of rhythms; the analyst may be required to discount peaks that appear at integral multiples of the delay corresponding to a fundamental frequency and other delays. The Fourier-domain equivalents of the ACF or the CCF permit easier and more intuitive analysis in the frequency domain than in the time domain. The notion of rhythms would be easier to associate with frequencies in *cps* or *Hz* than with the corresponding inversely related periods (see also the introductory section of Chapter 6).

4.5.1 Coherence analysis of EEG channels

Problem: Describe a frequency-domain approach to study the presence of rhythms in multiple channels of an EEG signal.

Solution: The Fourier-domain equivalents of the ACF and CCF are the PSD (also known as the autospectrum) and the cross-spectrum (or cross-spectral density — CSD), respectively. The PSD $S_{xx}(f)$ of a signal is related to its ACF via the Fourier

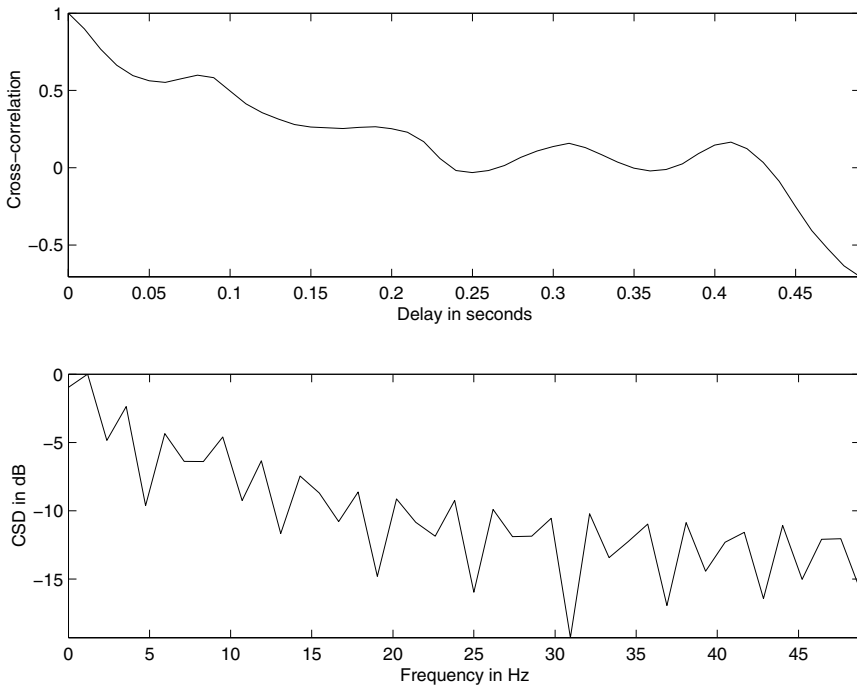


Figure 4.13 Upper trace: CCF between the 4.13 – 4.96 s portions of the f3 and f4 channels of the EEG signal shown in Figure 1.39. Lower trace: The CSD of the signal segments in *dB*.

transform:

$$S_{xx}(f) = FT[\phi_{xx}(\tau)] = X(f)X^*(f) = |X(f)|^2, \quad (4.29)$$

where $FT[\]$ indicates the Fourier transform. The Fourier transform of the CCF between two signals gives the CSD:

$$S_{xy}(f) = FT[\theta_{xy}(\tau)] = X(f)Y^*(f). \quad (4.30)$$

(For the sake of simplicity, the double-symbol subscripts xx and yy may be replaced by their singular versions, or dropped entirely when not relevant in subsequent discussions.)

The PSD displays peaks at frequencies corresponding to periodic activities in the signal. This property facilitates the detection of rhythms in signals such as the EEG: The presence of the α rhythm would be indicated by a peak or multiple peaks in the neighborhood of 8 – 13 Hz. The PSD may also be studied to locate the presence of activity spread over specific bands of frequencies, such as formants in the speech signal or murmurs in the PCG.

The CSD exhibits peaks at frequencies that are present in both of the signals being compared. The CSD may be used to detect rhythms present in common between two channels of the EEG.

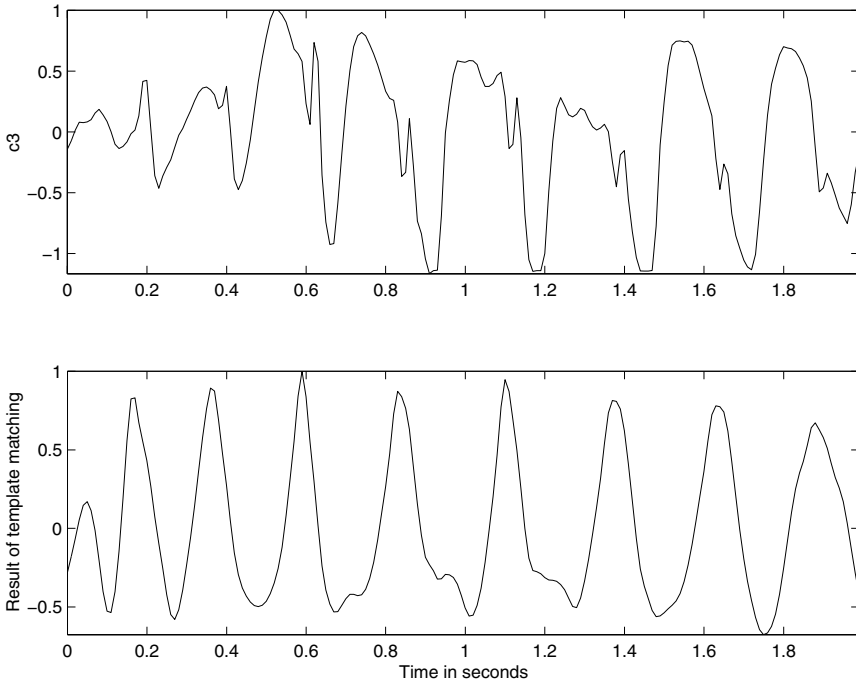


Figure 4.14 Upper trace: the c3 channel of the EEG signal shown in Figure 1.40. Lower trace: result of template matching. The spike-and-wave complex between 0.60 s and 0.82 s in the signal was used as the template.

The normalized magnitude of the *coherence spectrum* of two signals is given by [6, 54]

$$\Gamma_{xy}(f) = \left[\frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)} \right]^{1/2}. \quad (4.31)$$

If this expression is computed for two individual signals directly, the magnitude of the result will be equal to unity for all f , which is incorrect. In order to evaluate the magnitude coherence spectrum as defined in Equation 4.31, each spectral density function involved — S_{xy} , S_{xx} , and S_{yy} — must be estimated using an averaging procedure applied to several observations of the processes generating the signals [6, 205, 206]. See Section 6.3 for procedures for the estimation of spectral density functions.

The phase of the coherence spectrum is given by $\psi_{xy}(f) = \angle S_{xy}(f)$, which represents the average phase difference (related to the time delay) between frequency components in the two signals.

Rosenberg et al. [207] applied coherence to identify functional coupling between neuronal spike trains. They observed that frequency-domain analysis using coher-

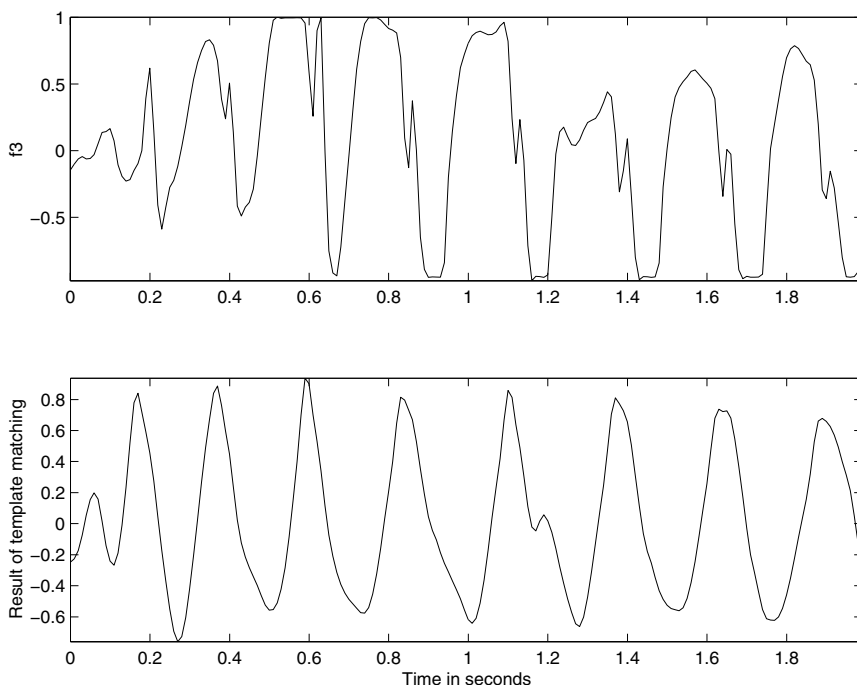


Figure 4.15 Upper trace: the f3 channel of the EEG signal shown in Figure 1.40. Lower trace: result of template matching. The spike-and-wave complex between 0.60 s and 0.82 s in the c3 channel (see Figure 4.14) was used as the template.

ence as a measure of association can lead to new insight and understanding of interactions between spike trains. Amjad et al. [208] described several methods to estimate the coherence function and its application to the analysis of SMUAP discharges and physiological tremor; see also Farmer et al. [209] and Farmer [210]. Johnston et al. [211] used coherence analysis to study the correlation between neural inputs to motor units innervating different muscles. See Section 6.3 for procedures for the estimation of spectral density functions.

Illustration of application: The coherence between EEG signals recorded from different positions on the scalp depends upon the structural connectivity or functional coupling between the corresponding parts of the brain. Investigations into the neurophysiology of seizure discharges and behavior attributable to disorganization of cerebral function may be facilitated by coherence analysis [54]. The symmetry, or lack thereof, between two EEG channels on the left and right sides of the same position (for example, c3 and c4) may be analyzed via the CSD or the coherence function.

The lower traces in Figures 4.9 and 4.10 illustrate the PSDs of EEG segments with and without the α rhythm, respectively. The former shows a strong and clear peak at about 9 Hz, indicating the presence of the α rhythm. Observe that the

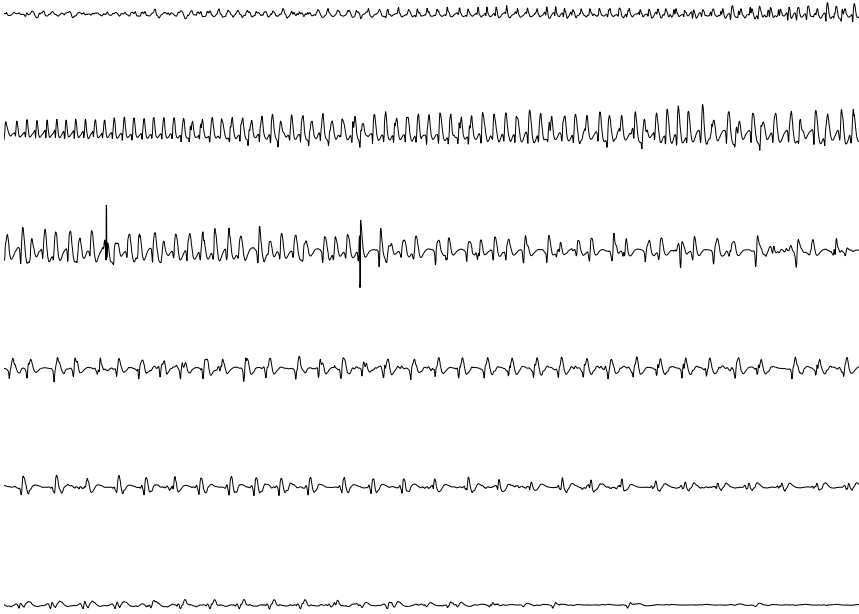


Figure 4.16 The evolution of various rhythms related to a seizure event as seen in an intracranial EEG signal. Each row represents a duration of 15 s and is a continuation of the preceding row. The minimum to maximum range of the signal shown is about 120 μV . Data courtesy of R. Agarwal [59].

PSD displays a single peak although the corresponding ACF has multiple peaks at two, three, and four times the delay corresponding to the fundamental period of the α wave in the signal. The PSD in Figure 4.10 exhibits no peak in the α range, indicating the absence of the α rhythm in the signal.

The lower traces in Figures 4.11, 4.12, and 4.13 illustrate the CSDs corresponding to the CCFs in the respective upper traces. Once again, it is easier to deduce the common presence of strong α activity between channels p3 and p4 from the CSD rather than the CCF in Figure 4.11. The single peak at 9 Hz in the CSD is more easily interpretable than the multiple peaks in the corresponding CCF. The CSD in Figure 4.12 lacks a clear peak in the α range, even though the corresponding CCF shows a peak at about 0.1 s, albeit less significant than that in Figure 4.11. The results agree with the fact that one channel has α activity while the other does not. Finally, the CSD in Figure 4.13 is clearly lacking a peak in the α range; the two signal segments have no α activity. Further methods for the analysis of α activity are presented in Sections 6.3.3, 6.7, and 7.5.2.

4.6 The Matched Filter

When a sample observation or template of a typical version of a signal event is available, it becomes possible to design a filter that is *matched* to the characteristics of the event and maximizes the *SNR* of the output. If a signal that contains repetitions of the event with almost the same characteristics is passed through the *matched filter*, the output should provide peaks at the time instants of occurrence of the event. Matched filters are commonly used for the detection of signals of known characteristics that are buried in noise [212, 213]. They are designed to perform a correlation operation between the input signal and the signal template, and hence are also known as *correlation filters*.

4.6.1 Derivation of the transfer function of the matched filter

In order to derive the transfer function, $H(\omega)$, of the matched filter [212], let the signal $x(t)$ be the input to the matched filter. The Fourier transform of $x(t)$ is

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt. \quad (4.32)$$

The output of the matched filter, $y(t)$, is given by the inverse Fourier transform of $Y(\omega) = X(\omega)H(\omega)$, as follows:

$$\begin{aligned} y(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) H(\omega) \exp(+j\omega t) d\omega \\ &= \int_{-\infty}^{\infty} X(f) H(f) \exp(+j2\pi ft) df. \end{aligned} \quad (4.33)$$

In the second expression of the equation given above, the frequency variable has been changed from ω in radians per second to f in Hz .

Consider the presence of white noise at the input, with the PSD

$$S_{\eta i}(f) = \frac{P_{\eta i}}{2}, \quad (4.34)$$

where $P_{\eta i}$ is the average noise power at the input. Then, the noise PSD at the output is

$$S_{\eta o}(f) = \frac{P_{\eta i}}{2} |H(f)|^2. \quad (4.35)$$

The average output noise power is

$$P_{\eta o} = \frac{P_{\eta i}}{2} \int_{-\infty}^{\infty} |H(f)|^2 df. \quad (4.36)$$

The *RMS* value of the noise in the absence of any signal is $\sqrt{P_{\eta o}}$.

Letting $t = t_0$ in Equation 4.33, the magnitude of the instantaneous output signal at $t = t_0$ is

$$M_y = |y(t_0)| = \left| \int_{-\infty}^{\infty} X(f) H(f) \exp(+j 2\pi f t_0) df \right|. \quad (4.37)$$

Thus, the SNR at the output is $\frac{M_y}{\sqrt{P_{\eta o}}}$.

To derive the optimal transfer function of the matched filter, we could maximize the SNR , which is equivalent to maximizing the expression

$$\frac{M_y^2}{P_{\eta o}} = \frac{\text{instantaneous peak power of signal}}{\text{noise mean power}}, \quad (4.38)$$

which represents peak-power SNR [212].

For a given signal $x(t)$, the total energy is a constant, given by

$$E_x = \int_{-\infty}^{\infty} x^2(t) dt = \int_{-\infty}^{\infty} |X(f)|^2 df. \quad (4.39)$$

Let us consider the following ratio:

$$\frac{M_y^2}{E_x P_{\eta o}} = \frac{\left| \int_{-\infty}^{\infty} H(f) X(f) \exp(+j 2\pi f t_0) df \right|^2}{\frac{P_{\eta i}}{2} \int_{-\infty}^{\infty} |H(f)|^2 df \int_{-\infty}^{\infty} |X(f)|^2 df}. \quad (4.40)$$

The quantity E_x is a constant for a given input signal; hence, maximizing the expression in Equation 4.40 is equivalent to maximizing the expression in Equation 4.38.

In order to determine the condition for maximization of the expression in Equation 4.40, recall Schwarz's inequality for two arbitrary complex functions $A(f)$ and $B(f)$:

$$\left| \int_{-\infty}^{\infty} A(f) B(f) df \right|^2 \leq \left[\int_{-\infty}^{\infty} |A(f)|^2 df \right] \left[\int_{-\infty}^{\infty} |B(f)|^2 df \right]. \quad (4.41)$$

For any two real functions $a(t)$ and $b(t)$, the corresponding inequality is

$$\left[\int_{-\infty}^{\infty} a(t) b(t) dt \right]^2 \leq \left[\int_{-\infty}^{\infty} a^2(t) dt \right] \left[\int_{-\infty}^{\infty} b^2(t) dt \right]. \quad (4.42)$$

For any two vectors $\mathbf{a}$ and $\mathbf{b}$, Schwarz's inequality states that

$$|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}| |\mathbf{b}|, \quad (4.43)$$

and

$$|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|. \quad (4.44)$$

In the inequalities stated above, equality is achieved if $\mathbf{a} = K \mathbf{b}$, that is, $\mathbf{a}$ and $\mathbf{b}$ are collinear; if $a(t) = K b(t)$; or if $A(f) = K B^*(f)$, where K is a real constant.

The inequality in Equation 4.41 can be applied to Equation 4.40 by considering $A(f) = H(f)$ and $B(f) = X(f) \exp(+j 2\pi f t_0)$. Then, we have

$$\frac{P_{\eta i} M_y^2}{2 E_x P_{\eta o}} \leq 1, \quad (4.45)$$

because

$$\left| \int_{-\infty}^{\infty} X(f) H(f) \exp(+j 2\pi f t_0) df \right|^2 \leq \int_{-\infty}^{\infty} |X(f)|^2 df \int_{-\infty}^{\infty} |H(f)|^2 df. \quad (4.46)$$

The LHS of Equation 4.46 represents the instantaneous output power, M_y^2 , evaluated in the Fourier domain. Thus, the ratio in Equation 4.40 is maximized when equality is achieved, that is, $A(f) = K B^*(f)$ with the functions $A(f)$ and $B(f)$ as explained above, leading to the condition

$$\begin{aligned} H(f) &= K [X(f) \exp(+j 2\pi f t_0)]^* \\ &= K X^*(f) \exp(-j 2\pi f t_0), \end{aligned} \quad (4.47)$$

which leads to maximal peak output SNR .

Taking the inverse Fourier transform of the last expression given above, we have

$$h(t) = K x[-(t - t_0)]. \quad (4.48)$$

Therefore, the impulse response of the matched filter is a scaled, reversed, and shifted version of the signal of interest.

The output of the matched filter is then obtained by the following steps:

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} X(f) H(f) \exp(+j 2\pi f t) df \\ &= \int_{-\infty}^{\infty} X(f) K X^*(f) \exp(-j 2\pi f t_0) \exp(+j 2\pi f t) df \\ &= K \int_{-\infty}^{\infty} |X(f)|^2 \exp[+j 2\pi f (t - t_0)] df \\ &= K \phi_x(t - t_0). \end{aligned} \quad (4.49)$$

The last relationship listed above is based on the property that the Fourier transform of the ACF is equal to the PSD of the signal. [Note: The Fourier transform of $x(-t)$ is $X^*(f)$, and that of $x(t - \tau)$ is $\exp(-j 2\pi f \tau) X(f)$.]

It is seen from the derivations given above that, when the impulse response of the matched filter is related to the desired signal as in Equation 4.48, the output is maximized and is equal to a scaled and delayed version of the ACF of the signal.

Illustration of application: Figure 4.17 shows a signal $x(n)$ composed of three events. The signal may be defined in terms of impulses as

$$\begin{aligned} x(n) &= 3\delta(n-5) + 2\delta(n-6) + \delta(n-7) \\ &+ 1.5\delta(n-16) + \delta(n-17) + 0.5\delta(n-18) \\ &+ 0.75\delta(n-26) + 0.5\delta(n-27) + 0.25\delta(n-28). \end{aligned} \quad (4.50)$$

From the plot of the signal in Figure 4.17, it is seen that $x(n)$ is composed of three occurrences of a basic signal $g(n)$ that may be defined as

$$g(n) = 3\delta(n) + 2\delta(n-1) + \delta(n-2), \quad (4.51)$$

with the composite signal given by

$$x(n) = g(n-5) + 0.5g(n-16) + 0.25g(n-26). \quad (4.52)$$

If we consider $g(n)$ as the basic pattern to be detected by using a matched filter, we need the impulse response of the filter to be $h(n) = K g(-n + n_0)$, where K is a scale factor or gain and n_0 is a delay to obtain a causal $h(n)$. In the present case, we need $n_0 = 2$ samples. Let $K = 1$. Then, we have

$$h(n) = \delta(n) + 2\delta(n-1) + 3\delta(n-2), \quad (4.53)$$

which is plotted in the trace in the middle of Figure 4.17. The trace at the bottom in the same figure shows the output of the matched filter. It is evident that the output possesses peaks at the locations of occurrences of $g(n)$ in the input signal, with a delay of 2 samples introduced by the filtering process. The values of the peaks are in proportion to the amplitude scale factors of the repeated occurrences of the basic pattern. It can be verified that the output values for the samples 5–9 represent the ACF of $g(n)$.

4.6.2 Detection of EEG spike and wave complexes

Problem: Design a matched filter to detect spike-and-wave complexes in an EEG signal. A reference spike-and-wave complex is available.

Solution: Let $x(t)$ be the given reference signal, representing an ideal observation of the event of interest. Let $X(f)$ be the Fourier transform of $x(t)$. Consider passing $x(t)$ through an LSI filter whose impulse response is $h(t)$; the transfer function of the filter is $H(f) = FT[h(t)]$. The output is given by $y(t) = x(t) * h(t)$ or $Y(f) = X(f)H(f)$.

Based on the derivation of the matched filter in Section 4.6.1, it is evident that the output energy is maximized when

$$H(f) = K X^*(f) \exp(-j2\pi f t_0), \quad (4.54)$$

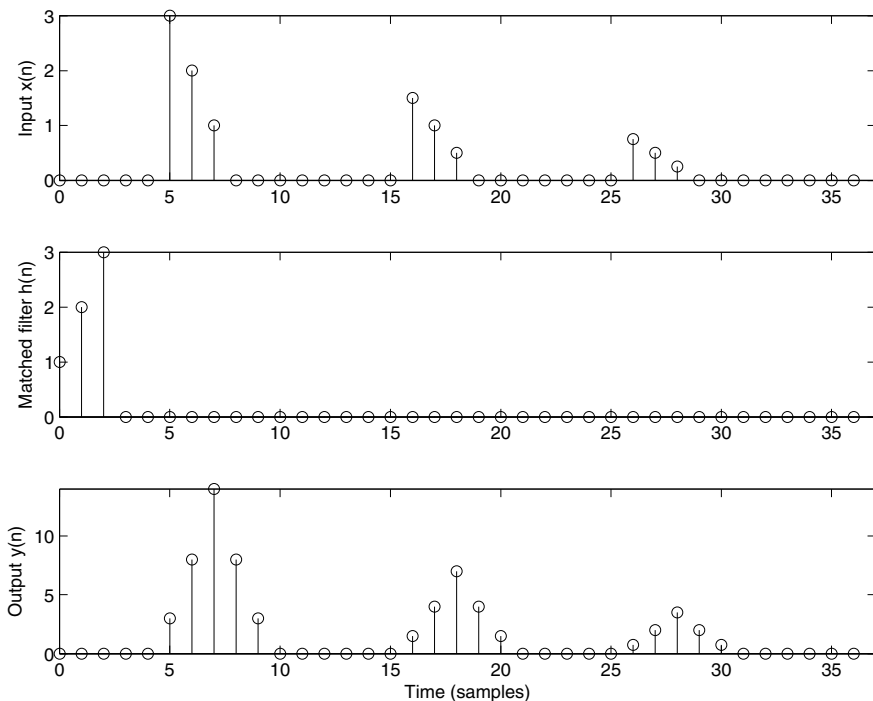


Figure 4.17 Top: A test signal with three events. Middle: Impulse response of the matched filter. Bottom: Output of the matched filter with peaks at the locations of occurrence of the basic signal pattern.

where K is a scale factor and t_0 is a time instant or delay [212]. This corresponds to the impulse response being

$$h(t) = Kx(t_0 - t). \quad (4.55)$$

Thus, the transfer function of the matched filter is proportional to the complex conjugate of the Fourier transform of the signal event to be detected. In the time domain, the impulse response is simply a *reversed* or *reflected* version of the reference signal that is scaled and delayed. A suitable delay will have to be added to make the filter causal, as determined by the duration of the reference signal.

If the derivation of the impulse response of the matched filter is based upon Equation 4.54 and implemented using the DFT, with the number of samples in the DFT equal to the number of samples N in the reference signal or template x , due to the periodicity of the DFT, the time delay or shift provided inherently will be $(N - 1)$. This is short by one sample; the desired shift is N samples to make the reversed signal causal.

Because the impulse response is a reversed version of $x(t)$, the convolution operation performed by the matched filter is equivalent to correlation: The output is then

equal to the cross-correlation between the input and the reference signal. When a portion of an input signal that is different from $x(t)$ matches the reference signal, the output approximates the ACF ϕ_x of the reference signal at the corresponding time delay. The corresponding frequency domain result is

$$Y(f) = X(f)H(f) = X(f)X^*(f) = S_x(f), \quad (4.56)$$

which is the PSD of the reference signal (ignoring the time delay and scale factors). The output is, therefore, maximum at the time instant of occurrence of an approximation to the reference signal. (See also Barlow [196] for related discussion.)

Illustration of application: To facilitate comparison with template matching, the spike-and-wave complex between 0.60 s and 0.82 s in the c3 channel of the EEG in Figure 1.40 was used as the reference signal to derive the matched filter. Figure 4.18 shows the extracted reference signal in the upper trace. The lower trace in the same figure shows the impulse response of the matched filter, which is simply a time-reversed version of the reference signal. The matched filter was implemented as an FIR filter using the MATLAB[®] *filter* command.

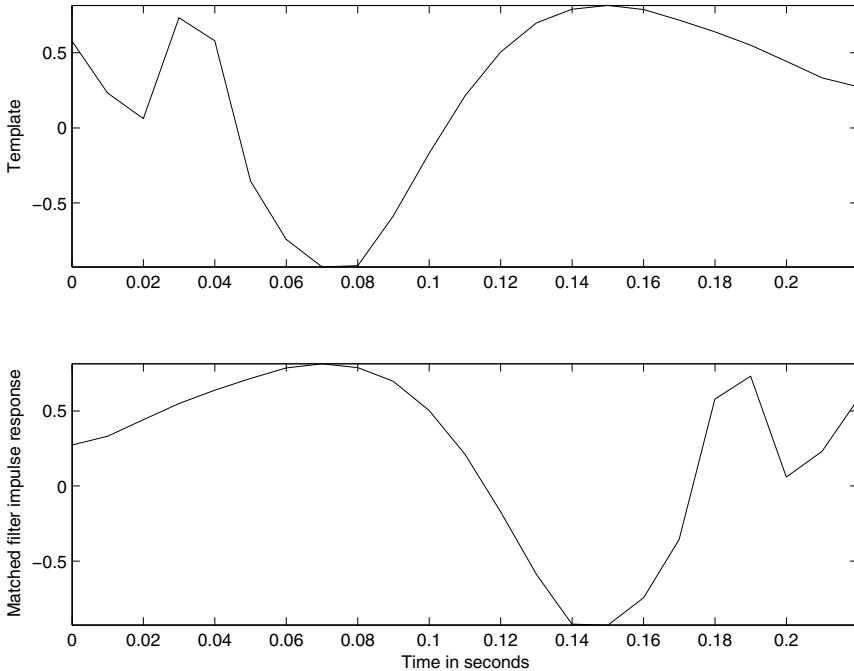


Figure 4.18 Upper trace: The spike-and-wave complex between 0.60 s and 0.82 s in the c3 channel of the EEG signal shown in Figure 1.40. Lower trace: Impulse response of the matched filter derived from the signal segment in the upper trace. Observe that the latter is a time-reversed version of the former.

Figures 4.19 and 4.20 show the outputs of the matched filter applied to the c3 and f3 channels of the EEG in Figure 1.40, respectively. The upper trace in each plot shows the signal, and the lower trace shows the output of the matched filter. It is evident that the matched filter provides a large output for each spike-and-wave complex. Comparing the outputs of the matched filter in Figures 4.19 and 4.20 with those of template matching in Figure 4.14 and 4.15, respectively, we observe that they are similar, with the exception that the results of the matched filter peak with a delay of 0.22 s after the corresponding spike-and-wave complex. The delay corresponds to the duration of the impulse response of the filter. The values differ due to different normalization in Equation 3.96. (*Note:* MATLAB[®] provides the command *filtfilt* for zero-phase forward and reverse digital filtering; this method is not considered in the book.)

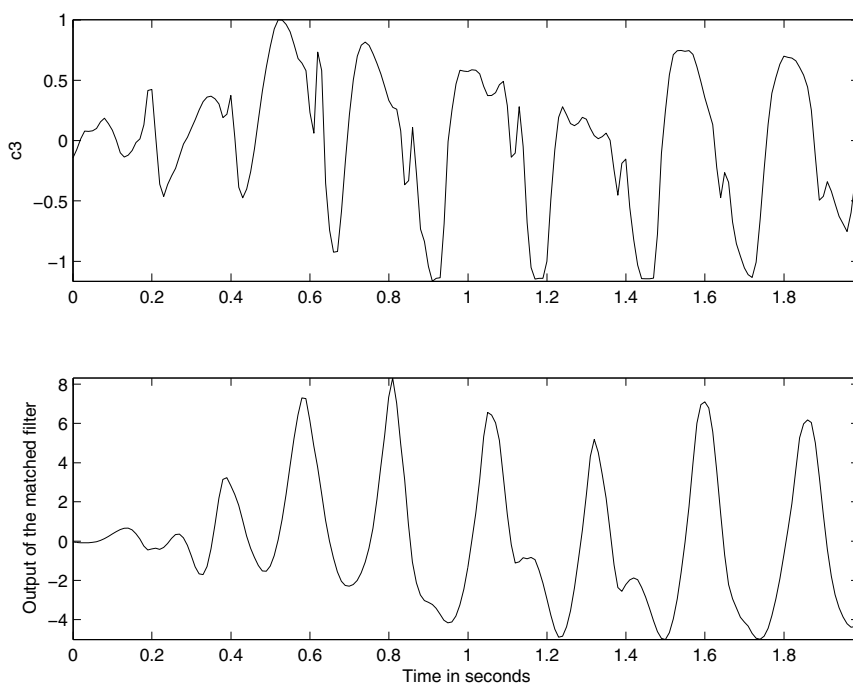


Figure 4.19 Upper trace: The c3 channel of the EEG signal shown in Figure 1.40, used as input to the matched filter in Figure 4.18. Lower trace: Output of the matched filter. See also Figure 4.14.

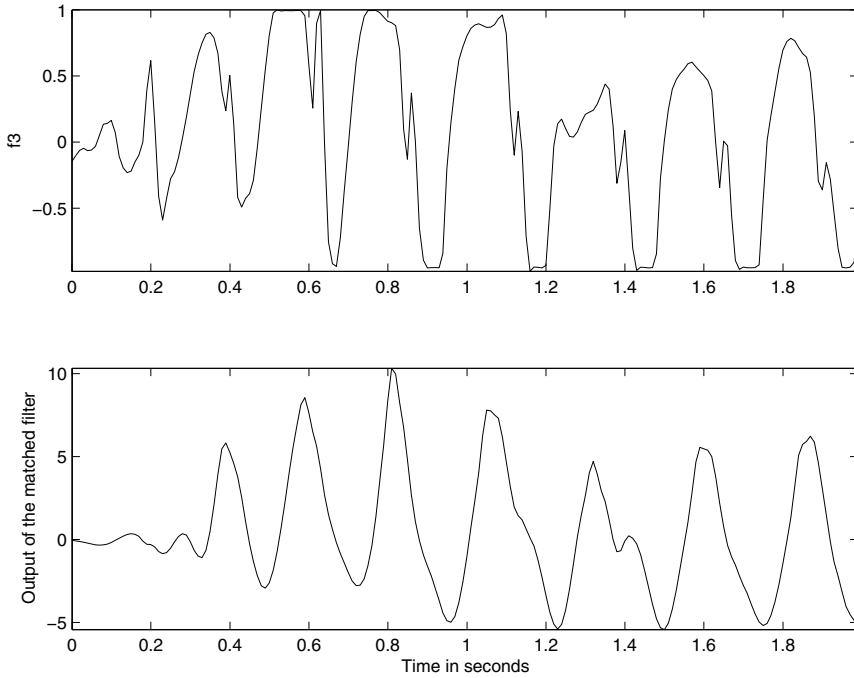


Figure 4.20 Upper trace: The $f3$ channel of the EEG signal shown in Figure 1.40, used as input to the matched filter in Figure 4.18. Lower trace: Output of the matched filter. See also Figure 4.15.

4.7 Detection of the P Wave in the ECG

Detection of the P wave in the ECG is difficult, because the P wave typically has a low amplitude, has an ill-defined and variable shape, and could be placed in a background of noise of varying size and origin.

Problem: *Propose an algorithm to detect the P wave in the ECG signal.*

Solution 1: In the method proposed by Hengeveld and van Bommel [214], VCG signals are processed as follows:

1. The QRS is detected, deleted, and replaced with the baseline. The baseline is determined by analyzing a few samples preceding the QRS complex.
2. The resulting signal is bandpass filtered with -3 dB points at 3 Hz and 11 Hz.
3. The end of the preceding T wave is estimated with reference to the current QRS by using $QT_{\max} = \frac{2}{9}RR + 250$ ms, where RR is the interval between two successive QRS complexes.
4. The maximum and minimum values are found in all three VCG leads in the search interval from the preceding T wave to the current QRS.

5. The signal is rectified and thresholded at 50% and 75% of the maximum to obtain a ternary (three-level) signal.
6. The cross-correlation of the result is computed with a ternary template derived in a manner similar to the procedure in the previous step from a representative set of P waves.
7. The peak in the cross-correlation corresponds to the P location in the original ECG.

The algorithm overcomes the dominance of the QRS complex by first detecting the QRS and then deleting it. Observe that the cross-correlation is computed not with an original P wave, which we have noted could be obscure and variable, but with a ternary wave derived from the P wave. The ternary wave represents a simplified template of the P wave.

Figure 4.21 illustrates the results of the various stages of the P-finding algorithm of Hengeveld and van Bommel [214]. Observe that the original ECG signal shown in part (a) of the figure has a P wave that is hardly discernible. The processed versions of the signal after deleting the QRS, filtering, and rectification are shown in parts (b), (c), and (d). The ternary version in part (e) shows that the P wave has been converted into two pulses corresponding to its upstroke and return parts. The result of cross-correlation with the template in part (f) is shown in part (g). A simple peak-picking algorithm with search limits may be used to detect the peak in the result, and hence determine the P wave position.

Note that the result in part (d) has other waves preceding those related to the P wave. An appropriate search interval should be used so as to disregard the unwanted components.

Solution 2: Gritzali et al. [215] proposed a common approach to detect the QRS, T, and P waves in multichannel ECG signals based upon a transformation they labeled as the “length” transformation. Given a collection of ECG signals from N simultaneous channels $x_1(t), x_2(t), \dots, x_N(t)$, the length transformation was defined as

$$L(N, w, t) = \int_t^{t+w} \sqrt{\sum_{j=1}^N \left(\frac{dx_j}{dt} \right)^2} dt, \quad (4.57)$$

where w is the width of the time window over which the integration is performed. In essence, the procedure computes the total squared derivative of the signals across the various channels available, and integrates the summed quantity over a moving time window. The advantage of applying the derivative-based operator across multiple channels of an ECG signal is that P and T waves may be detected even when they well-defined in only one or a few of the channels used.

In the procedure for waveform detection proposed by Gritzali et al., the QRS is first detected by applying a threshold to $L(N, w, t)$, with w set equal to the average QRS width. The onset (start) and offset (end) points of the QRS are represented by a pulse waveform, as indicated in Figure 4.22. The QRS complexes in the signals are then replaced by the isoelectric baseline of the signals, the procedure is repeated

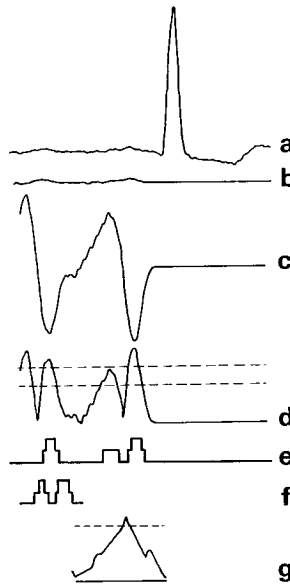


Figure 4.21 Illustration of the results at various stages of the Hengeveld and van Bommel method for P wave detection. From top to bottom: (a) the original ECG signal; (b) after replacement of the QRS with the baseline; (c) after bandpass filtering; (d) after rectification, with the dashed lines indicating the thresholds; (e) the thresholded ternary signal; (f) the ternary P wave template; and (g) result of cross-correlation between the signals in (e) and (f). Reproduced with permission from S.J. Hengeveld and J.H. van Bommel, *Computer detection of P waves*, *Computers and Biomedical Research*, 9:125–132, 1976. ©Academic Press.

with w set equal to the average T duration, and the T waves are detected. The same steps are repeated to detect the P waves. Figure 4.22 illustrates the detection of the QRS, T, and P waves in a three-channel ECG signal. Gritzali et al. also proposed a procedure based upon correlation analysis and least-squares modeling to determine the thresholds required.

4.8 Homomorphic Filtering and the Complex Cepstrum

In Chapter 3, we studied linear filters designed to separate signals that were added together. The question asked was the following: Given $y(t) = x(t) + \eta(t)$, how can one extract $x(t)$ only? Given that the Fourier transform is linear, we know that the Fourier transforms of the signals are also combined in an additive manner: $Y(\omega) = X(\omega) + \eta(\omega)$. Therefore, a linear filter will facilitate the separation of $X(\omega)$ and $\eta(\omega)$, with the assumption that they have substantial portions of their energy in different frequency bands.

Suppose now that we are presented with a signal that contains the product of two signals, for example, $y(t) = x(t)p(t)$. From the multiplication or convolution

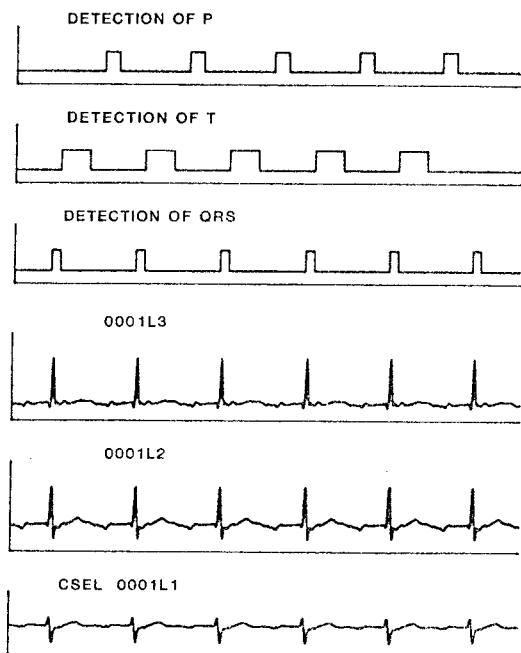


Figure 4.22 Detection of the P, QRS, and T waves in a three-channel ECG signal using the length transformation. The lower three traces show the three ECG channels. The upper three traces indicate the onset and end of the P, QRS, and T waves detected by the procedure in the form of pulse trains. The first P and the last T waves have not been processed. CSEL: CSE library. L1/L2/L3: three ECG leads. Reproduced with permission from F. Gritzali, G. Frangakis, and G. Papakonstantinou, Detection of the P and T waves in an ECG, *Computers and Biomedical Research*, 22:83–91, 1989. ©Academic Press. See Willems et al. [216,217] for details on the ECG database used by Gritzali et al.

property of the Fourier transform we have $Y(\omega) = X(\omega) * P(\omega)$, where $*$ represents convolution in the frequency domain. How would we be able to separate $x(t)$ from $p(t)$?

Furthermore, suppose we have $y(t) = x(t) * h(t)$, where $*$ stands for convolution as in the case of the passage of the glottal pulse train or random excitation $x(t)$ through the vocal-tract system with the impulse response $h(t)$. The Fourier transforms of the signals are related as $Y(\omega) = X(\omega) H(\omega)$. How would we attempt to separate $x(t)$ and $h(t)$?

4.8.1 Generalized linear filtering

Given that linear filters are well established and understood, it is attractive to consider extending their application to signals that have been combined by operations other than addition, especially by multiplication and convolution as indicated in the

preceding paragraphs. An interesting possibility to achieve this is via conversion of the operation combining the signals into addition by one or more transformations. Under the assumption that the transformed signals occupy different portions of the transformed space, linear filters may be applied to separate them. The inverses of the transformations used initially would then take us back to the original space of the signals. This approach was proposed in a series of papers by Bogert et al. [218], Oppenheim et al. [219], and Oppenheim and Schafer [220, 221]. Because the procedure extends the application of linear filters to multiplied and convolved signals, it has been referred to as *generalized linear filtering*. Furthermore, as the operations can be represented by algebraically linear transformations between the input and output vector spaces, they have been called *homomorphic systems*.

As a simple illustration of a homomorphic system for multiplied signals, consider again the signal

$$y(t) = x(t) p(t). \quad (4.58)$$

Given the goal of converting the multiplication operation to addition, it is evident that a simple logarithmic transformation is appropriate:

$$\log[y(t)] = \log[x(t) p(t)] = \log[x(t)] + \log[p(t)]; \quad x(t) \neq 0, p(t) \neq 0 \forall t. \quad (4.59)$$

The logarithms of the two signals are now combined in an additive manner. Taking the Fourier transform, we get

$$Y_l(\omega) = X_l(\omega) + P_l(\omega), \quad (4.60)$$

where the subscript l indicates that the Fourier transform has been applied to a log-transformed version of the signal.

Assuming that the logarithmic transformation has not affected the separability of the Fourier components of the two signals $x(t)$ and $p(t)$, a linear filter (such as a lowpass or a highpass filter) may now be applied to $Y_l(\omega)$ to separate them. An inverse Fourier transform will yield the filtered signal in the time domain. An exponential operation will complete the reversal procedure (if required).

Figure 4.23 illustrates the operations involved in a multiplicative homomorphic system (or filter). The symbol at the input or output of each block indicates the operation that combines the signal components at the corresponding step. A system of this type is useful in image processing, where an image may be treated as the product of an illumination function and a transmittance or reflectance function. The homomorphic filter facilitates separation of the illumination function and correction for nonuniform lighting. The method has been used to achieve simultaneous dynamic range compression and contrast enhancement [18, 174, 219, 222].

4.8.2 Homomorphic deconvolution

Problem: Propose a homomorphic filter to separate two signals that have been combined through the convolution operation.

Solution: Consider the case expressed by the relation

$$y(t) = x(t) * h(t). \quad (4.61)$$

As in the case of the multiplicative homomorphic system, our goal is to convert the convolution operation to addition. From the convolution property of the Fourier transform, we know that

$$Y(\omega) = X(\omega) H(\omega). \quad (4.62)$$

Thus, application of the Fourier transform converts convolution to multiplication. Now, it is readily seen that the multiplicative homomorphic system may be applied to convert the multiplication to addition. Taking the complex logarithm of $Y(\omega)$, we have

$$\log[Y(\omega)] = \log[X(\omega)] + \log[H(\omega)]; \quad X(\omega) \neq 0, H(\omega) \neq 0 \forall \omega. \quad (4.63)$$

[Note: $\log_e[X(\omega)] = \hat{X}(\omega) = \log_e[|X(\omega)| \angle X(\omega)] = \log_e[|X(\omega)|] + j \angle X(\omega)$, where $|X(\omega)|$ and $\angle X(\omega)$ are the magnitude and phase spectra of $x(t)$.]

A linear filter may now be used to separate the transformed components of x and h , with the assumption as before that they are separable in the transformed space. A series of the inverses of the transformations applied initially will take us back to the original domain.

While the discussion here has been in terms of application of the Fourier transform, the general formulation of the homomorphic filter by Oppenheim and Schaffer [174] is in terms of the z -transform. However, the Fourier transform is equivalent to the z -transform evaluated on the unit circle in the z -plane, and the Fourier transform is more commonly used in signal processing than the z -transform.

Figure 4.24 gives a block diagram of the steps involved in a homomorphic filter for convolved signals. The path formed by the first three blocks (in the top row) transforms the convolution operation at the input to addition. The third block with the inverse Fourier transform is used to move back to a pseudo time domain. The last three blocks (in the bottom row) perform the reverse transformation, converting addition to convolution. The filter in between deals with (transformed) signals that are combined by simple addition. Further details of these operations and results are presented in the following section.

4.8.3 Extraction of the vocal tract response

Problem: *Design a homomorphic filter to extract the basic wavelet corresponding to the vocal-tract response from a voiced speech signal.*

Solution: We noted in Section 1.2.12 that voiced speech is generated by excitation of the vocal tract, while it is held in a particular form, with a glottal waveform that may be approximated as a series of pulses. The voiced speech signal may, therefore, be expressed in discrete-time terms as $y(n) = x(n) * h(n)$, where $y(n)$ is the speech signal, $x(n)$ is the glottal waveform (excitation sequence), and $h(n)$ is the impulse response of the vocal tract (basic wavelet). The $*$ symbol represents convolution, with the assumption that the vocal-tract filter may be approximated by an LSI filter. We may, therefore, use the homomorphic filter for convolved signals as introduced in the preceding section to separate $h(n)$ and $x(n)$.

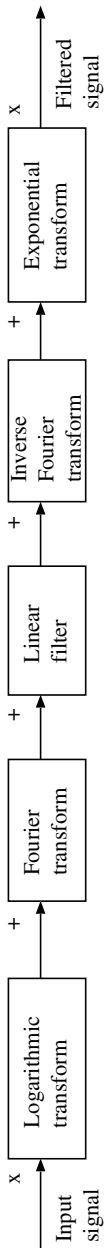


Figure 4.23 Operations involved in a multiplicative homomorphic system or filter. The symbol at the input or output of each block indicates the operation that combines the signal components at the corresponding step.

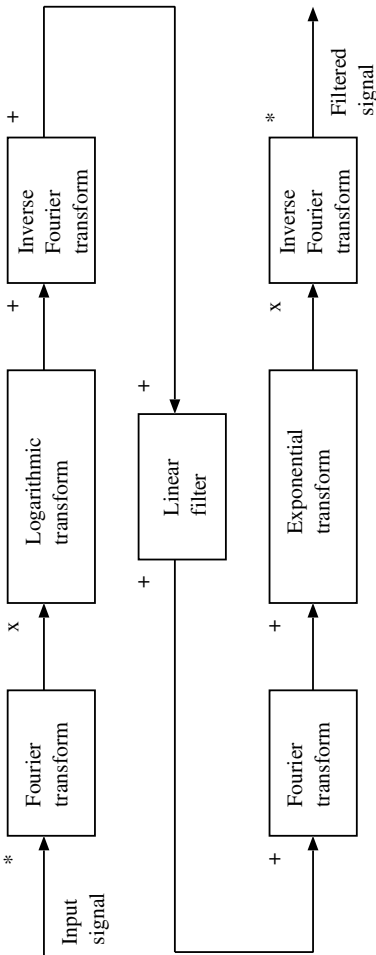


Figure 4.24 Operations involved in a homomorphic filter for convolved signals. The symbol at the input or output of each block indicates the operation that combines the signal components at the corresponding step.

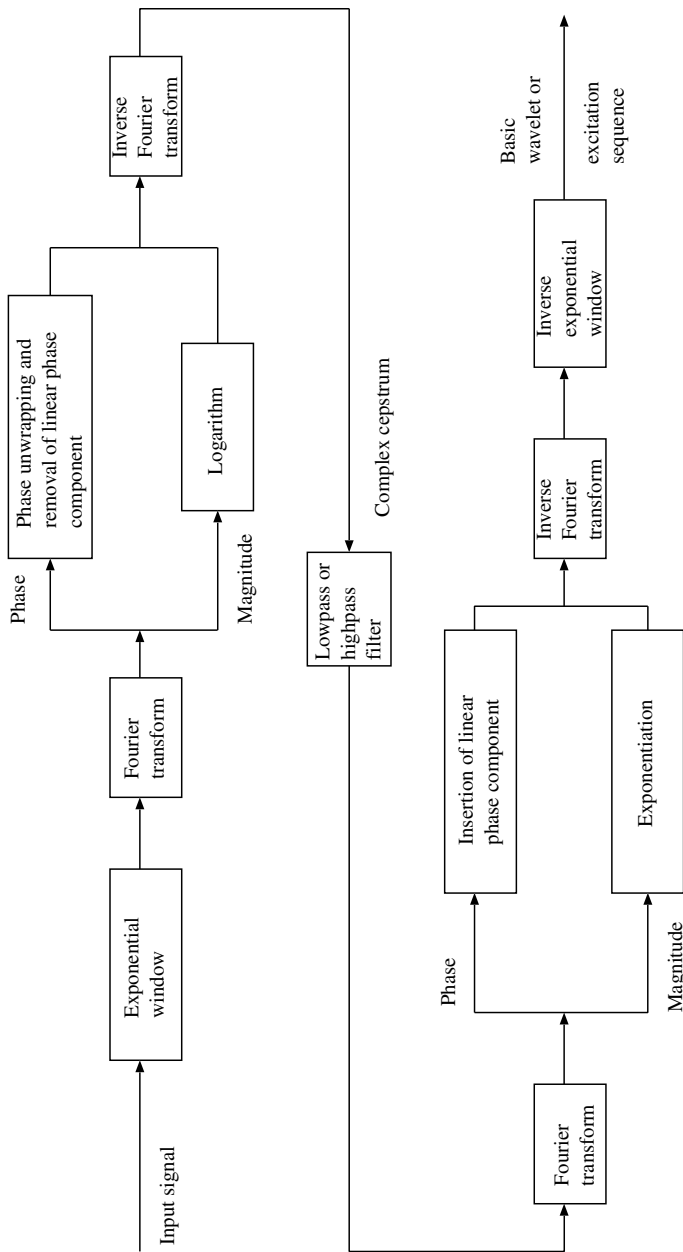


Figure 4.25 Detailed block diagram of the steps involved in deconvolution of signals using the complex cepstrum.

The glottal excitation sequence may be further expressed as $x(n) = p(n) * g(n)$, where $p(n)$ is a train of ideal impulses (Dirac delta functions) and $g(n)$ is a smoothing function, to indicate that the physical vocal-cord system cannot produce ideal impulses but rather pulses of finite duration and slope [174]. This aspect will be neglected in our discussions.

Practical application of the homomorphic filter is not simple. Figure 4.25 gives a detailed block diagram of the procedure [174, 223]. Some of the details and practical techniques are explained in the following paragraphs.

The complex cepstrum: The formal definition of the complex cepstrum states that it is the inverse z -transform of the complex logarithm of the z -transform of the input signal [174, 223]. (The name “cepstrum” was derived by transposing the syllables of the word “spectrum”; other transposed terms [174, 218, 223] are less commonly used.) If $y(n)$ is the input signal and $Y(z)$ is its z -transform, the complex cepstrum $\hat{y}(n)$ is defined as

$$\hat{y}(n) = \frac{1}{2\pi j} \oint \log[Y(z)] z^{n-1} dz. \quad (4.64)$$

The contour integral performs the inverse z -transform, and should be evaluated within an annular region in the complex z -plane where $\hat{Y}(z) = \log[Y(z)]$ is single-valued and analytic [3, 174]. The unit of n in $\hat{y}(n)$, that is, in the cepstral domain, is referred to as *quefrency*, a term obtained by switching the syllables in the term frequency; it is also referred to as pseudo time [218, 221].

Given $y(n) = x(n) * h(n)$, it follows that

$$\hat{Y}(z) = \hat{X}(z) + \hat{H}(z) \quad \text{or} \quad \hat{Y}(\omega) = \hat{X}(\omega) + \hat{H}(\omega), \quad (4.65)$$

and further that the complex cepstra of the signals are related as

$$\hat{y}(n) = \hat{x}(n) + \hat{h}(n). \quad (4.66)$$

Here, the $\hat{}$ symbol over a function of z or ω indicates the complex logarithm of the corresponding function of z or ω , whereas the same symbol over a function of time (n) indicates the complex cepstrum of the corresponding signal. It should be noted that, if the original signal $y(n)$ is real, its complex cepstrum $\hat{y}(n)$ is *real*; the prefix *complex* is used to indicate the fact that the preceding z and logarithmic transformations are computed as complex functions. Furthermore, it should be noted that the complex cepstrum is a function of time.

An important consideration in the evaluation of the complex logarithm of $Y(z)$ or $Y(\omega)$ relates to the phase of the signal. The phase spectrum computed as its principal value in the range $[0, 2\pi]$, given by $\arctan \left[\frac{\text{imaginary}\{Y(\omega)\}}{\text{real}\{Y(\omega)\}} \right]$, will almost always have discontinuities that will conflict with the requirements of the inverse z -transformation or inverse Fourier transform to follow later. Thus, $Y(\omega)$ needs to be separated into its magnitude and phase components, the logarithmic operation applied to the magnitude, the phase corrected to be continuous by adding correction factors of $\pm 2\pi$ at discontinuities larger than π , and the two components combined

again before the subsequent inverse transformation. Correcting the phase spectrum as above is referred to as *phase unwrapping* [174, 223]. It has been shown that a linear phase term, if present in the spectrum of the input signal, may cause rapidly decaying oscillations in the complex cepstrum [223]. It is advisable to remove the linear phase term, if present, in the phase-unwrapping step. The linear phase term may be added to the filtered result (as a time shift) if necessary.

The complex cepstrum of exponential sequences: Exponential sequences are signals that have a rational z -transform, that is, their z -transforms may be expressed as ratios of polynomials in z [174]. Such signals are effectively represented as weighted sums of exponentials in the time domain. Consider a signal $x(n)$ whose z -transform is expressed as

$$X(z) = A z^r \frac{\prod_{k=1}^{M_I} (1 - a_k z^{-1}) \prod_{k=1}^{M_O} (1 - b_k z)}{\prod_{k=1}^{N_I} (1 - c_k z^{-1}) \prod_{k=1}^{N_O} (1 - d_k z)}, \quad (4.67)$$

with $|a_k|, |b_k|, |c_k|, |d_k| < 1$. The function $X(z)$ possesses M_I zeros that are located inside the unit circle, given by a_k , $k = 1, 2, \dots, M_I$; M_O zeros outside the unit circle, given by $1/b_k$, $k = 1, 2, \dots, M_O$; N_I poles inside the unit circle, given by c_k , $k = 1, 2, \dots, N_I$; and N_O poles outside the unit circle, given by $1/d_k$, $k = 1, 2, \dots, N_O$.

In the process of computing the complex cepstrum, we get [174]

$$\begin{aligned} \hat{X}(z) &= \log[X(z)] \\ &= \log[A] + \log[z^r] \\ &\quad + \sum_{k=1}^{M_I} \log(1 - a_k z^{-1}) + \sum_{k=1}^{M_O} \log(1 - b_k z) \\ &\quad - \sum_{k=1}^{N_I} \log(1 - c_k z^{-1}) - \sum_{k=1}^{N_O} \log(1 - d_k z). \end{aligned} \quad (4.68)$$

For real sequences, A is real. If $A > 0$, $\hat{x}(0) = \log[A]$.

The factor z^r indicates a delay or advance by r samples, depending upon the sign of r .

Recall the power series

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad \text{if } |x| < 1. \quad (4.69)$$

Thus, we have

$$\log(1 - \alpha z^{-1}) = - \sum_{n=1}^{\infty} \frac{\alpha^n}{n} z^{-n}, \quad \text{if } |z| > |\alpha|, \quad (4.70)$$

and

$$\log(1 - \beta z) = - \sum_{n=1}^{\infty} \frac{\beta^n}{n} z^n, \quad \text{if } |z| < |\beta^{-1}|. \quad (4.71)$$

Expanding all of the log terms in Equation 4.68 into their equivalent power series and taking the inverse z -transform, we get [174]

$$\hat{x}(n) = \begin{cases} \log |A| & \text{for } n = 0; \\ - \sum_{k=1}^{M_I} \frac{a_k^n}{n} + \sum_{k=1}^{N_I} \frac{c_k^n}{n} & \text{for } n > 0; \\ \sum_{k=1}^{M_O} \frac{b_k^{-n}}{n} - \sum_{k=1}^{N_O} \frac{d_k^{-n}}{n} & \text{for } n < 0. \end{cases} \quad (4.72)$$

From the steps shown above, a few properties of the complex cepstrum that are evident and important are as follows [174]:

- The complex cepstrum, $\hat{x}(n)$, decays at least as fast as $1/n$. We have

$$|\hat{x}(n)| < K \left| \frac{\alpha^n}{n} \right|, \quad -\infty < n < \infty, \quad (4.73)$$

where $\alpha = \max(|a_k|, |b_k|, |c_k|, |d_k|)$, and K is a constant.

- $\hat{x}(n)$ will be of infinite duration even if $x(n)$ is of finite duration, and exists for $-\infty < n < \infty$, in general.
- If $x(n)$ is a minimum-phase signal, all of its poles and zeros are inside the unit circle in the z -plane. Then, $\hat{x}(n) = 0$ for $n < 0$; that is, the complex cepstrum is causal.
- If $x(n)$ is a maximum-phase signal, it has no poles or zeros inside the unit circle in the z -plane. Then, $\hat{x}(n) = 0$ for $n > 0$; that is, the complex cepstrum is anticausal.

Limiting ourselves to causal signals of finite energy, we need not consider the presence of poles on or outside the unit circle in the z -plane. However, the z -transform of a finite-energy signal may have zeros outside the unit circle. Such a composite signal may be separated into its minimum-phase component and maximum-phase component by extracting the causal part ($n > 0$) and anticausal part ($n < 0$), respectively, of its complex cepstrum, followed by the inverse procedures. The composite signal is equal to the convolution of its minimum-phase component and maximum-phase component. (See also Section 5.4.2.)

Effect of echoes or repetitions of a wavelet: Let us consider a simplified signal $y(n) = x(n) * h(n)$, where

$$x(n) = \delta(n) + a \delta(n - n_0), \quad (4.74)$$

with a and n_0 being two constants. (The sampling interval T is ignored, or assumed to be normalized to unity in this example.) The signal may also be expressed as

$$y(n) = h(n) + a h(n - n_0). \quad (4.75)$$

The signal thus has two occurrences of the basic wavelet $h(n)$ at $n = 0$ and $n = n_0$. The coefficient a indicates the magnitude of the second appearance of the basic wavelet (called an *echo* in seismic applications), and n_0 indicates its delay (pitch in the case of a voiced speech signal). The topmost plot in Figure 4.26 shows a synthesized signal with a wavelet and an echo at one-half of the amplitude (that is, $a = 0.5$) of the first wavelet arriving at $n_0 = 0.01125$ s.

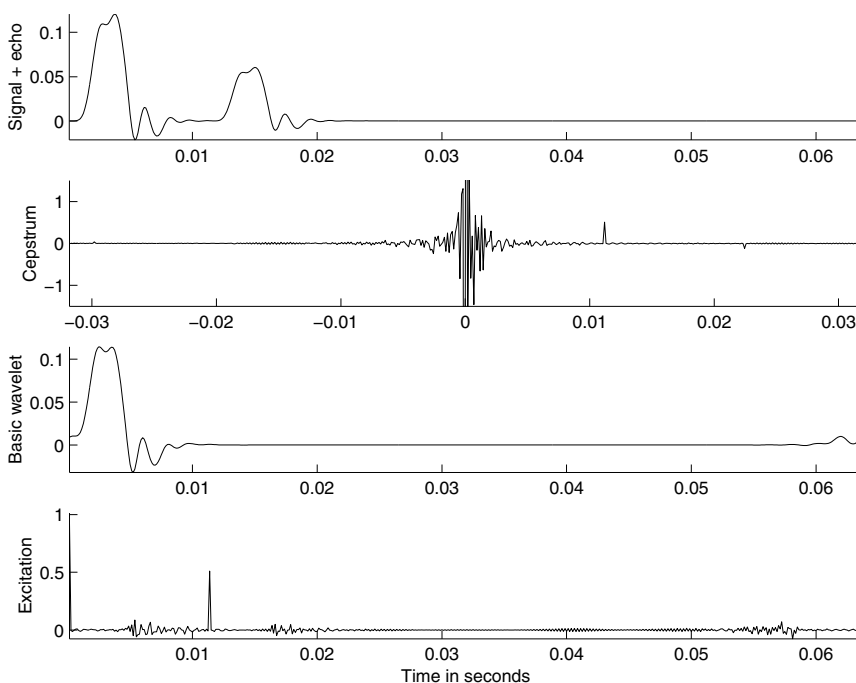


Figure 4.26 From top to bottom: a composite signal with a wavelet and an echo; the complex cepstrum of the signal (the amplitude axis has been stretched to make the peaks at the echo time and its multiples more readily visible; values outside the range ± 1.5 have been clipped); the basic wavelet extracted by shortpass filtering the cepstrum; and the excitation sequence extracted by longpass filtering the cepstrum.

Taking the z -transform of the signal in Equation 4.75, we have

$$Y(z) = (1 + az^{-n_0})H(z). \quad (4.76)$$

If the z -transform is evaluated on the unit circle, we get the Fourier-transform-based expression

$$Y(\omega) = [1 + a \exp(-j\omega n_0)]H(\omega). \quad (4.77)$$

Taking the logarithm, we have

$$\hat{Y}(\omega) = \hat{H}(\omega) + \log[1 + a \exp(-j\omega n_0)]. \quad (4.78)$$

If $a < 1$, the log term may be expanded in a power series, to get

$$\hat{Y}(\omega) = \hat{H}(\omega) + a \exp(-j\omega n_0) - \frac{a^2}{2} \exp(-2j\omega n_0) + \frac{a^3}{3} \exp(-3j\omega n_0) - \dots \quad (4.79)$$

Taking the inverse Fourier transform, we get

$$\hat{y}(n) = \hat{h}(n) + a \delta(n - n_0) - \frac{a^2}{2} \delta(n - 2n_0) + \frac{a^3}{3} \delta(n - 3n_0) - \dots \quad (4.80)$$

The derivation given above shows that the complex cepstrum of a signal with a basic wavelet and an echo is equal to the complex cepstrum of the basic wavelet plus a series of impulses at the echo delay and integral multiples thereof [174, 223]. The amplitudes of the impulses are proportional to the echo amplitude (the factor a) and decay for the higher-order repetitions (if $a < 1$). It may be readily seen that, if the signal has multiple echoes or repetitions of a basic wavelet, the cepstrum will possess multiple impulse trains, with an impulse at the arrival time of each wavelet and integral multiples thereof. In the case of a voiced speech signal, the location of the first peak will give the pitch. The second plot in Figure 4.26 shows the complex cepstrum of the signal in the first plot of the same figure. It is seen that the cepstrum has a peak at 0.01125 s, the echo arrival time; a smaller (negative) peak is also seen at twice the echo arrival time.

Under the assumption that the complex cepstrum of the basic wavelet decays to negligible values before the first impulse $a \delta(n - n_0)$ related to the echo, $\hat{h}(n)$ may be extracted from the complex cepstrum $\hat{y}(n)$ of the composite signal by a window that has unit value for $|n| < n_c$, n_c being the cutoff point. (This filter is referred to as a “shortpass” filter as the cepstrum is a function of pseudo time [218, 221]; it may also be called a lowpass filter.) The inverse procedures will yield $h(n)$. The remaining portion of the cepstrum (obtained by “longpass” or highpass filtering) will give $\hat{x}(n)$, which upon application of the inverse procedures will yield $x(n)$. The third and fourth plots in Figure 4.26 show the basic wavelet $h(n)$ and the excitation sequence $x(n)$ extracted by filtering the cepstrum with the cutoff point at $n_c = 0.005$ s.

In the case where $a \geq 1$, it can be shown that the complex cepstrum will have a train of impulses on its negative time axis, that is, at $(n + kn_0)$, $k = 1, 2, \dots$ [174, 223]. An appropriate exponential weighting sequence may be used to achieve the condition $a < 1$, in which case the impulse train will appear on the positive axis of the cepstrum. If the weighted signal satisfies the minimum-phase condition, the cepstrum will be causal.

The power cepstrum: Several variants of the cepstrum have been proposed in the literature; Childers et al. [223] provided a review of the related techniques. One variant that is commonly used is the *real cepstrum* or the *power cepstrum*, which is defined as the square of the inverse z -transform of the logarithm of the squared magnitude of the z -transform of the given signal. In practice, the z -transform in the

definition stated is replaced by the DFT. The power cepstrum has the computational advantage of not requiring phase unwrapping, but does not facilitate separation of the components of the signal.

By evaluating the inverse z -transform on the unit circle in the z -plane, the power cepstrum $\hat{y}_p(n)$ of a signal $y(n)$ may be defined as

$$\hat{y}_p(n) = \left\{ \frac{1}{2\pi j} \oint \log |Y(z)|^2 z^{n-1} dz \right\}^2. \quad (4.81)$$

If, as before, we consider $y(n) = x(n) * h(n)$, we have $|Y(z)|^2 = |X(z)|^2 |H(z)|^2$, and it follows that $\log |Y(z)|^2 = \log |X(z)|^2 + \log |H(z)|^2$. Applying the inverse z -transform to this relationship, we get

$$\hat{y}_p(n) = \hat{x}_p(n) + \hat{h}_p(n), \quad (4.82)$$

where $\hat{h}_p(n)$ is the power cepstrum of the basic wavelet and $\hat{x}_p(n)$ is the power cepstrum of the excitation signal. Note that, in the equation given above, the cross-product term was neglected; the cross-term will be zero if the two component power cepstra occupy nonoverlapping quefrequency ranges. The final squaring operation in Equation 4.81 is omitted in some definitions of the power cepstrum; in such a case, the cross-term does not arise, and Equation 4.82 is valid.

The power cepstrum does not retain the phase information of the original signal. However, it is useful in the identification of the presence of echoes in the signal, and in the estimation of their arrival times. The power cepstrum is related to the complex cepstrum as [223]

$$\hat{y}_p(n) = [\hat{y}(n) + \hat{y}(-n)]^2. \quad (4.83)$$

Let us again consider the situation of a signal with two occurrences of a basic wavelet $h(n)$ at $n = 0$ and $n = n_0$ as in Equations 4.74 and 4.75. Then [223],

$$|Y(z)|^2 = |H(z)|^2 |1 + az^{-n_0}|^2. \quad (4.84)$$

By taking the logarithm of both sides of the equation and substituting $z = \exp(j\omega)$, we get

$$\begin{aligned} \log |Y(\omega)|^2 &= \log |H(\omega)|^2 + \log[1 + a^2 + 2a \cos(\omega n_0)] \\ &= \log |H(\omega)|^2 + \log(1 + a^2) \\ &\quad + \log \left(1 + \frac{2a}{1 + a^2} \cos(\omega n_0) \right). \end{aligned} \quad (4.85)$$

It is now seen that the logarithm of the PSD of the signal has sinusoidal components (ripples) due to the presence of an echo. The amplitudes and frequencies of the ripples are related to the amplitude a of the echo and its time delay n_0 .

Illustration of application: A voiced speech signal $y(n)$ is the result of convolution of a slowly varying vocal-tract response $h(n)$ with a relatively fast-varying glottal pulse train $x(n)$, expressed as $y(n) = x(n) * h(n)$. Under these conditions,

it may be demonstrated that the contributions of $h(n)$ to the complex cepstrum $\hat{y}(n)$ will be limited to low values of n within the range of the pitch period of the speech signal [174]. The complex cepstrum $\hat{y}(n)$ will possess impulses at the pitch period and integral multiples thereof. Therefore, a filter that selects a portion of the complex cepstrum for low values of n , followed by the inverse transformations, will yield an estimate of $h(n)$.

When the repetitions of the basic wavelet have magnitudes almost equal to (or even greater than) that of the first wavelet in the given signal record, the contributions of the pulse-train component to the complex cepstrum may not decay rapidly and may cause aliasing artifacts when the cepstrum is computed over a finite duration. A similar situation is caused when the delay between the occurrences of the multiple versions of the basic wavelet is a significant portion of the duration of the given signal. The problem may be ameliorated by applying an exponential weight α^n to the data sequence, with $\alpha < 1$. Childers et al. [223] recommended values of α in the range $[0.96, 0.99]$, depending upon the signal characteristics as listed above. Furthermore, they recommend appending or padding zeros to the input signal to facilitate computation of the cepstrum to a longer duration than the signal in order to avoid aliasing errors and ambiguities in time-delay estimates. (See Figure 4.25 for an illustration of the various steps in homomorphic filtering of convolved signals.)

Figure 4.27 illustrates a segment of a voiced speech signal (obtained by filtering the signal shown in Figure 1.51) and the basic wavelet extracted by shortpass filtering of its complex cepstrum with $n_c = 0.003125$ s. The signal was padded with zeros to twice its duration; exponential weighting with $\alpha = 0.99$ was used. It is seen that the basic vocal-tract response wavelet has been successfully extracted. Extraction of the vocal-tract response facilitates spectral analysis without the effect of the quasiperiodic repetitions in the speech signal as indicated by Equation 4.85.

The fourth trace in Figure 4.27 shows the glottal (excitation) waveform extracted by longpass filtering of the cepstrum with the same parameters as in the preceding paragraph. The result shows impulses at the time of arrival of each wavelet in the composite speech signal. The peaks are decreasing in amplitude due to the use of exponential weighting (with $\alpha = 0.99$) prior to computation of the cepstrum. Inverse exponential weighting can restore the pulses to their original levels; however, the artifact at the end of the excitation signal gets amplified to much higher levels than the desired pulses due to progressively higher values of α^{-n} for large n . Hence, the inverse weighting operation was not applied in the present illustration. Regardless, the result indicates that pitch information may also be recovered by homomorphic filtering of voiced speech signals.

4.9 Application: ECG Rhythm Analysis

Problem: Describe a method to measure the heart rate and average RR interval from an ECG signal.

Solution: Algorithms for QRS detection such as the Pan–Tompkins method described in Section 4.3.2 are useful for ECG rhythm analysis or heart-rate monitoring.

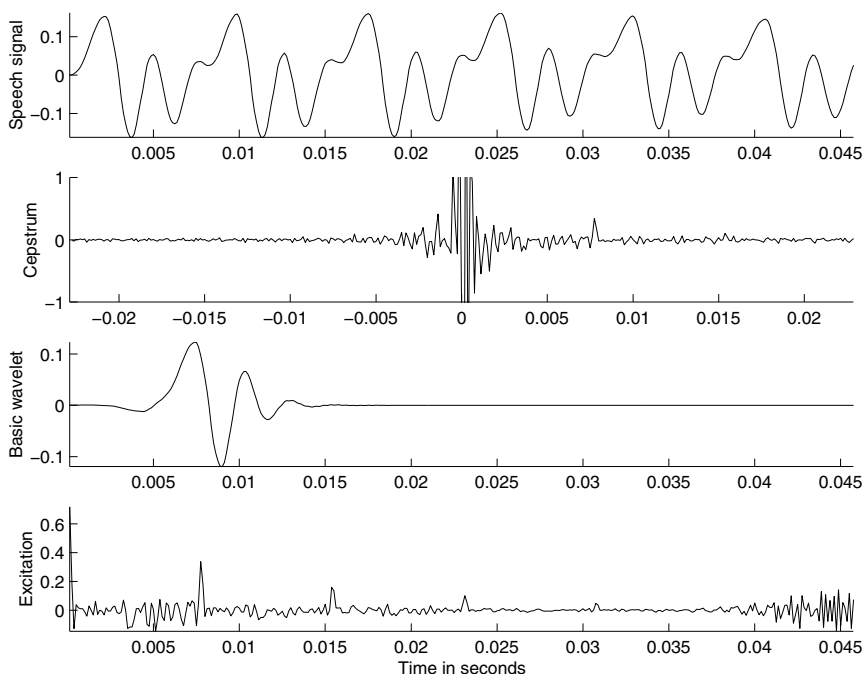


Figure 4.27 From top to bottom: a segment of a voiced speech signal over six pitch periods (extracted from the signal shown in Figure 1.51 and lowpass filtered); the complex cepstrum of the signal (the amplitude axis has been stretched to make the peaks at the echo time and its multiples more readily visible; values outside the range ± 1.0 have been clipped); the (shifted) basic wavelet extracted by shortpass filtering the cepstrum; and the excitation sequence extracted by longpass filtering the cepstrum.

The output of the final smoothing filter could be subjected to a peak-searching algorithm to obtain a time marker for each QRS or ECG beat. The search procedure proposed by Pan and Tompkins was explained in Section 4.3.2. The intervals between two such consecutive markers gives the *RR* interval, which could be averaged over a number of beats to obtain a good estimate of the interbeat interval. The heart rate may be computed in *bpm* as 60 divided by the average *RR* interval in seconds. The heart rate may also be obtained by counting the number of beats detected over a certain period, for example, 10 s, and multiplying the result with the required factor (6 in the present case) to get the number of beats over one minute.

The upper plot in Figure 4.28 shows a filtered version of the noisy ECG signal shown in Figure 3.5. The noisy signal was filtered with an eighth-order Butterworth lowpass filter with a cutoff frequency of 90 Hz, and the signal was down-sampled by a factor of five to an effective sampling rate of 200 Hz. The lower plot shows the output of the Pan–Tompkins method. The Pan–Tompkins result was normalized by dividing by its maximum over the data record available (as the present example was

computed off-line). A fixed threshold of 0.1 and a blanking interval of 250 *ms* was used in a simple search procedure, which was successful in detecting all the beats in the signal. (The blanking interval indicates the period over which threshold checking is suspended once the threshold has been crossed.) The average *RR* interval was computed as 716 *ms*, leading to an effective heart rate of 84 *bpm*.

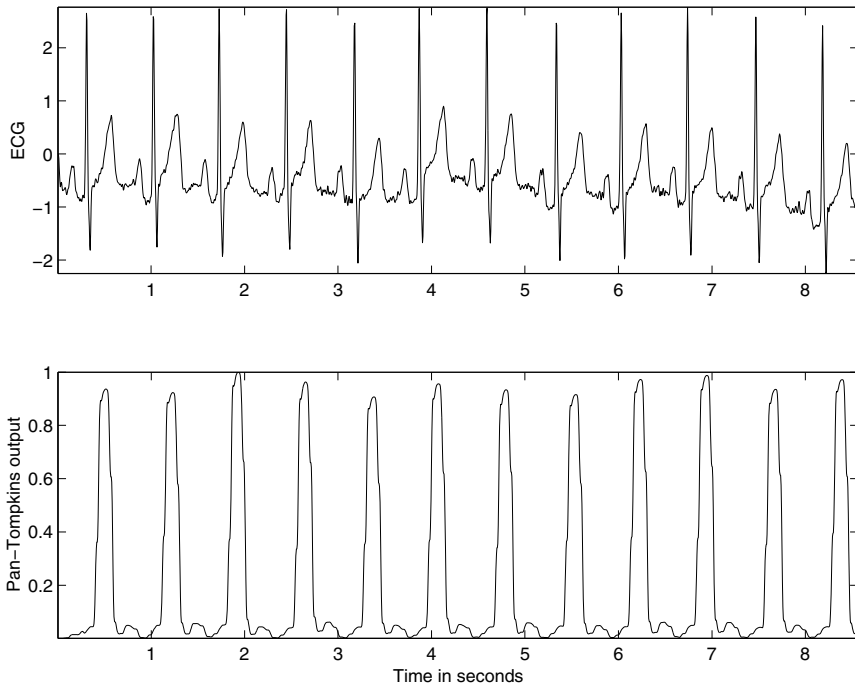


Figure 4.28 Results of the Pan–Tompkins algorithm. Top: lowpass-filtered version of the ECG signal shown in Figure 3.5. Bottom: normalized result of the final integrator.

Results at the various stages of the Pan–Tompkins algorithm for a noisy ECG signal sampled at 200 *Hz* are shown in Figure 4.29. The bandpass filter has efficiently removed the low-frequency artifact in the signal. The final output has two peaks that are much larger than the others: one at the beginning of the signal due to filtering artifacts, and one at about 7.5 *s* due to an artifact in the signal. Furthermore, the output has two peaks for the beat with an artifact at 7.5 *s*. The simple peak-searching procedure as explained in the previous paragraph was applied, which resulted in the detection of 46 beats: one more than the 45 present in the signal due to the artifact at about 7.5 *s*. The average *RR* interval was computed to be 446.6 *ms*, leading to an effective heart rate of 137 *bpm*.

The illustration demonstrates the need for prefiltering a given ECG signal to remove artifacts and the need to apply an adaptive threshold to the output of the Pan–Tompkins algorithm for QRS detection. It is readily seen that direct thresholding of

the original ECG signal will not be successful in detecting all of the QRS complexes in the signal.

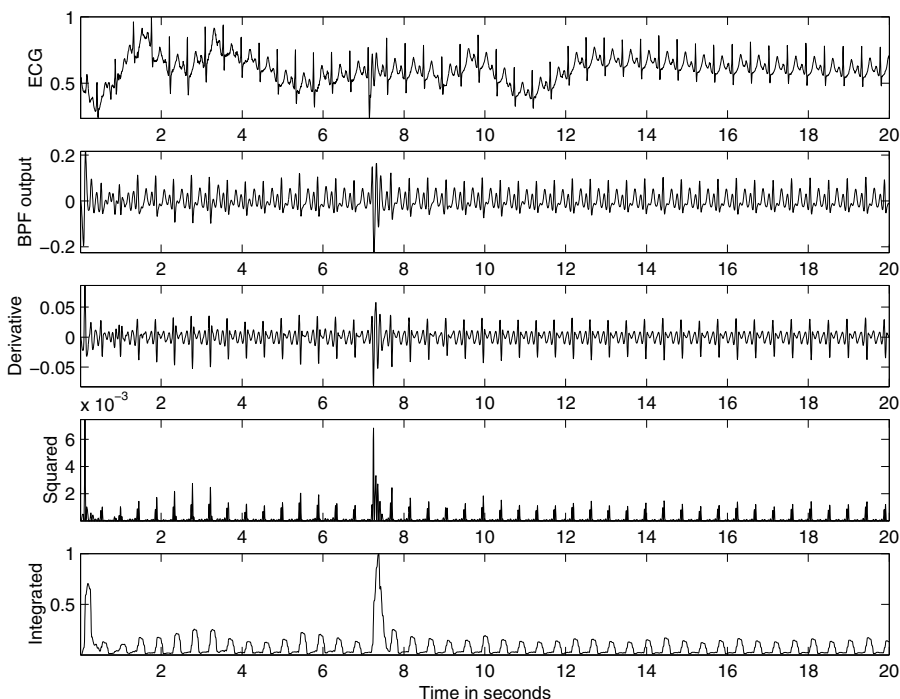


Figure 4.29 Results of the Pan–Tompkins algorithm with a noisy ECG signal. From top to bottom: ECG signal sampled at 200 Hz ; output of the bandpass filter (BPF); output of the derivative-based operator; the result of squaring; and normalized result of the final integrator.

4.10 Application: Identification of Heart Sounds

Problem: Outline a signal processing algorithm to identify $S1$ and $S2$ in a PCG signal, and further segment the PCG signal into its systolic and diastolic parts. The ECG and carotid pulse signals are available for reference.

Solution: We saw in Section 2.3 how the ECG and carotid pulse signals could be used to demarcate the onset of $S1$ and $S2$ in the PCG; the procedure, however, was not based upon signal processing but upon visual analysis of the signals. We have, in the present chapter, studied signal processing techniques to detect the QRS complex in the ECG and the diastolic notch in the carotid pulse signal. We may, therefore, use these methods to transfer the timing information from the ECG and carotid pulse signals to the PCG signal. In order to perform this task, we need to recognize a few timing relationships between the signals [72, 131].

The beginning of S1 may be taken to be the same instant as the beginning of the QRS. The QRS itself may be detected using one of the three methods described in the present chapter, such as the Pan–Tompkins method.

Detection of the beginning of S2 is more involved. Let the heart rate be HR bpm. The preejection period PEP is defined as the interval from the beginning of the QRS to the onset of the corresponding carotid upstroke. The rate-corrected PEP is defined as $PEPC = PEP + 0.4HR$, with the periods in ms . $PEPC$ is in the range of 131 ± 13 ms for normal adults [72].

The ejection time ET is the interval from the onset of the carotid upstroke to the dicrotic notch. The rate-corrected ejection time in ms is $ETC = ET + 1.6HR$, and is in the ranges of 395 ± 13 ms for normal adult males and 415 ± 11 ms for normal adult females.

Using $PEPC_{\max} = 144$ ms and $HR_{\min} = 60$ bpm, we get $PEP_{\max} = 120$ ms . With $HR_{\min} = 60$ bpm and $ETC_{\max} = 425$ ms , we get $ET_{\max} = 329$ ms . Based on the parameters defined in the preceding sentences, the maximum interval between the QRS and the dicrotic notch can be approximated to be 380 ms . The procedure proposed by Lehner and Rangayyan [131] for detection of the dicrotic notch recommends searching the output of the derivative-based method described in Section 4.3.3 in a 500 ms interval after the QRS. After the dicrotic notch is detected, we need to subtract the time delay (S2–D) between the beginning of S2 and the dicrotic notch to get the time instant where S2 begins. Lehner and Rangayyan [131] measured the average S2–D delay over the PCG and carotid pulse signals of 60 pediatric patients to be 42.6 ms , with $SD = 5.0$ ms .

The following procedure may be used to segment a PCG signal into its systolic and diastolic parts.

1. Use the Pan–Tompkins method described in Section 4.3.2 to locate the QRS complexes in the ECG.
2. Identify one period of the PCG as the interval between two successive QRS locations. Note that the delay introduced by the filters used in the Pan–Tompkins method needs to be subtracted from the detected peak locations to obtain the starting points of the QRS complexes.
3. Use the Lehner and Rangayyan method described in Section 4.3.3 to detect the dicrotic notch in the carotid pulse signal.
4. Let the standardized S2–D delay be its mean $+2 \times SD$ as reported by Lehner and Rangayyan [131], that is, 52.6 ms . Subtract the standardized S2–D delay from the detected dicrotic notch location to obtain the onset of S2.
5. Use the S1–S2 interval to obtain the systolic part of the PCG cycle.
6. Use the interval between the S2 point and the next detected S1 to obtain the diastolic part of the PCG cycle.

Figures 4.30 and 4.31 illustrate the results of application of the procedure described above to the PCG, ECG, and carotid pulse signals of a normal subject and

a patient with a split S2, systolic ejection murmur, and opening snap of the mitral valve. (Clinical diagnosis indicated the possibility of ventricular septal defect, pulmonary stenosis, or pulmonary hypertension for the 14-month-old female patient with murmur.) The peak positions detected in the output of the Pan–Tompkins method (the third trace in each figure) and the output of the Lehner and Rangayyan method (the fifth trace) have been marked with the * symbol. A threshold of 0.75 times the maximum value was used to detect the peaks in the output of the Pan–Tompkins method, with a blanking interval of 250 *ms*.

The QRS and dicrotic notch positions have been marked on the ECG and carotid pulse traces with the triangle and diamond symbols, respectively. Finally, the S1 and S2 positions are marked on the PCG trace with triangles and diamonds, respectively. The filter delays and timing relationships between the three channels of signals described previously have been accounted for in the process of marking the events. Note how the results of event detection in the ECG and carotid pulse signals have been transferred to locate the corresponding events in the PCG. Lehner and Rangayyan [131] used a similar procedure to break PCG signals into systolic and diastolic segments; the segments were then analyzed separately in the time and frequency domains. (See also Sections 6.3.5 and 7.9.)

4.11 Application: Detection of the Aortic Component of the Second Heart Sound

Heart sounds are preferentially transmitted to different locations on the chest. The aortic and pulmonary components A2 and P2 of S2 are best heard at the aortic area (to the right of the sternum, in the second right intercostal space) and the pulmonary area (left parasternal line in the third left intercostal space), respectively (see Figure 1.31). A2 is caused by the closure of the aortic valve at the end of systole, and is usually louder than P2 at all locations on the chest. Earlier theories on the genesis of heart sounds attributed the sounds to the opening and closing actions of the valve leaflets *per se*. The more commonly accepted theory at the present time is that described by Rushmer [33]; see Section 1.2.9.

The relative timing of A2 and P2 depends upon the pressure differences across the corresponding valves in the left and right ventricular circulatory systems. In a normal individual, the timing of P2 with reference to A2 varies with respiration; the timing of A2 itself is independent of respiration. The pulmonary pressure (intrathoracic pressure) is decreased during inspiration, leading to a delayed closure of the pulmonary valve and hence an increased (audible and visible) gap between A2 and P2 [33, 72, 73]. The gap is closed and A2 and P2 overlap during expiration in normal individuals. A2 and P2 have individual durations of about 50 *ms*. The normal inspiratory gap between A2 and P2 is of the order of 30 – 40 *ms*, although splits as long as 100 *ms* have been recorded [72].

A split in S2 longer than 40 *ms* during sustained expiration is considered to be abnormal [72]. Complete right bundle-branch block could cause delayed activation of the right ventricle and, therefore, delayed pulmonary valve closure, a delayed P2,

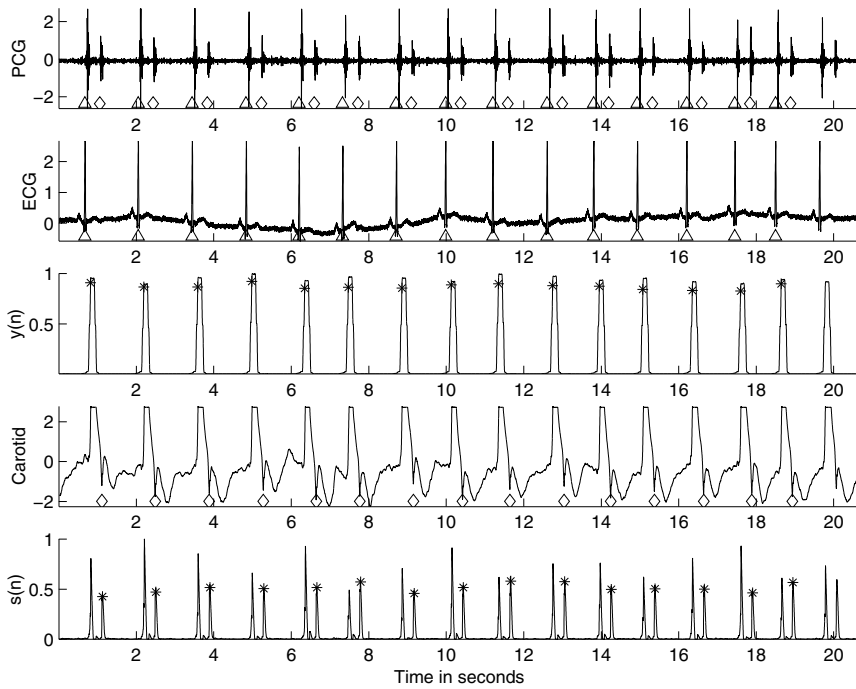


Figure 4.30 Results of segmentation of a PCG signal into systolic and diastolic parts using the ECG and carotid pulse signals for reference. From top to bottom: the PCG signal of a normal subject (male, 23 years); the ECG signal; $y(n)$, the output of the Pan–Tompkins method for detection of the QRS after normalization to the range (0, 1); the carotid pulse signal; $s(n)$, the output of the Lehner and Rangayyan method for detection of the dicotic notch, normalized to the range (0, 1). The peaks detected in the outputs of the two methods have been identified with * marks. The QRS and the dicotic notch positions have been marked with the triangle and diamond symbols, respectively. The S1 and S2 positions are marked on the PCG trace with triangles and diamonds, respectively. The last cardiac cycle was not processed.

and hence a widely split S2. Some of the other conditions that cause a wide split in S2 are atrial septal defect, ventricular septal defect, and pulmonary stenosis. Left bundle-branch block leads to delayed left ventricular contraction and aortic valve closure (with reference to the right ventricle and the pulmonary valve), causing A2 to appear after P2, and *reversed splitting* of the two components. Some of the other conditions that could cause reversed splitting of S2 are aortic insufficiency and abnormally early pulmonary valve closure. It is thus seen that identification of A2 and P2 and their temporal relationships could assist in the diagnosis of several cardiovascular defects and diseases.

MacCanon et al. [224] conducted experiments on a dog for direct detection and timing of aortic valve closure. They developed a catheter with an electrical contacting device that could be placed at the aortic valve to detect the exact moment of

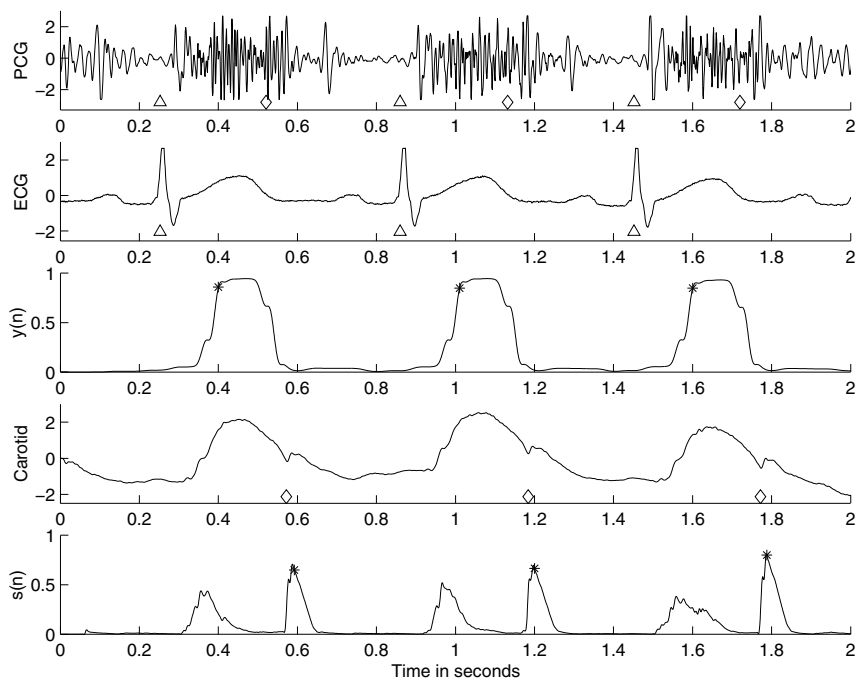


Figure 4.31 Results of segmentation of a PCG signal into systolic and diastolic parts using the ECG and carotid pulse signals for reference. From top to bottom: the PCG signal of a patient (female, 14 months) with a split S2, systolic ejection murmur, and opening snap of the mitral valve; the ECG signal; $y(n)$, the output of the Pan–Tompkins method for detection of the QRS; the carotid pulse signal; $s(n)$, the output of the Lehner and Rangayyan method for detection of the dicotic notch. The peaks detected in the outputs of the two methods have been identified with * marks. The QRS and the dicotic notch positions have been marked with the triangle and diamond symbols, respectively. The S1 and S2 positions are marked on the PCG trace with triangles and diamonds, respectively.

closure of the aortic valve. They also measured the aortic pressure and the PCG at the third left intercostal space. It was demonstrated that the aortic valve closes at least 5 – 13 *ms* before the incisura appears in the aortic pressure wave (see Figures 1.45 and 1.46 for illustrations of aortic pressure waves recorded from a dog). The conclusion reached was that S2 is caused not by the collision of the valve leaflets themselves, but due to the rebound of the aortic blood column and walls after valve closure. MacCanon et al. also hypothesized that the relative high-frequency characteristics of the incisura and S2 result from elastic recoil of the aortic wall and valve in reaction to the distention by the rebounding aortic blood column.

Stein et al. [225] and Stein and Sabbah [226] conducted experiments in which intracardiac and intraarterial sounds were recorded and analyzed. Their experiments indicated that S2 begins *after* the aortic valve closes. They argued that the intensity of

S2 depends upon, among other factors, the distensibility of the aortic and pulmonary valves; hemodynamic factors that cause the valves to distend and vibrate; viscosity of the blood and its ability to inhibit diastolic valve motion; and the configuration of the aorta, the pulmonary artery, and the ventricles. It was demonstrated that the pulmonary valve, due to its larger surface area than that of the aortic valve, is more distensible and hence produces a larger sound than the aortic valve even for the same pressure gradient across the valve. In the case of pulmonary hypertension, it was argued that the pulmonary valve would distend further at a higher speed: the rate of development of the diastolic pressure gradient across the closed pulmonary valve would be higher than in normal cases.

Problem: *Given that S2 is made up of an aortic component A2 and a pulmonary component P2 with variable temporal relationships, propose a method to detect only A2.*

Solution: We have seen in the preceding section how the dicrotic notch in the carotid pulse signal may be used to detect the beginning of S2. The technique is based upon the direct relationship between aortic valve closure and the aortic incisura, and consequently the dicrotic notch, as explained above. Now, if we were to detect and segment S2 over several cardiac cycles and several respiratory cycles, we could perform synchronized averaging of S2. A2 should appear at the same instant in every S2 segment and should be strengthened by the synchronized averaging process. P2, on the other hand, would appear at different times, and should be cancelled out (suppressed) by the averaging process.

Figure 4.32 shows segments of duration 300 ms containing S2 segmented from nine successive cardiac cycles of the PCG of a patient with atrial septal defect. The PCG signal was segmented using the ECG and carotid pulse signals for reference in a method similar to that illustrated in Figures 4.30 and 4.31. The PCG signal was recorded at the second left intercostal space, which is closer to the pulmonary area than to the aortic area. The nine S2 segments clearly show the fixed timing of A2 and the variable timing of P2. The last plot is the average of S2 segments extracted from 21 successive cardiac cycles. The averaged signal displays A2 very well, while P2 has been suppressed.

The detection of A2 could have been better had the PCG been recorded at the aortic area, where A2 would be stronger than P2. Once A2 is detected, it could be subtracted from each S2 record to obtain individual estimates of P2. Sarkady et al. [227], Baranek et al. [228], and Durand et al. [229] proposed averaging techniques as above with or without envelope detection (but without the use of the carotid pulse); the methods were called aligned ensemble averaging to detect wavelets or coherent detection and averaging.

Nigam and Priemer [230] proposed a method to extract the A2 and P2 components from S2 via blind source separation (BSS) techniques. Xu et al. [231, 232] used time-frequency representations of transient nonlinear chirp signals to model A2 and P2 and derive the A2–P2 splitting interval; they showed that the splitting interval is strongly correlated with the pulmonary arterial pressure measured by catheterization.

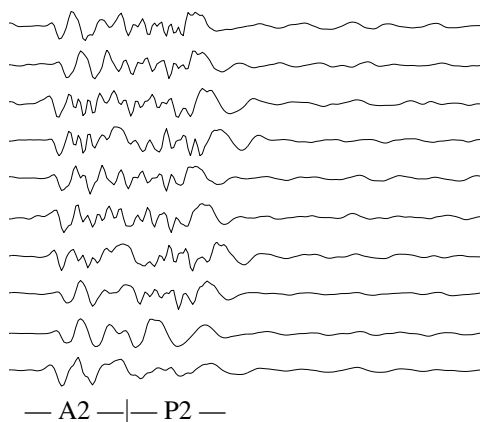


Figure 4.32 Synchronized averaging of S2 to detect A2 and suppress P2. The figure displays nine consecutive segments of S2 (duration = 300 *ms*) of a patient (female, 7 years) with atrial septal defect leading to a variable split in S2. The trace at the bottom is the average of S2 over 21 consecutive beats. The expected time intervals of A2 and P2 are indicated below the last trace.

4.12 Remarks

We have now established links between the characteristics of certain epochs in a number of biomedical signals and the corresponding physiological or pathological events in the biomedical systems of concern. We have seen how derivative-based operators may be applied to detect QRS complexes in the ECG signal as well as the dicrotic notch in the carotid pulse signal. The utility of correlation and spectral density functions in the detection of rhythms and events in EEG signals was also demonstrated. We have studied how signals with repetitions of a certain event or wavelet, such as a voiced speech signal, may be analyzed using the complex cepstrum and homomorphic filtering. Finally, we also saw how events detected in one signal may be used to locate the corresponding or related events in other signals: the task of detecting S1 and S2 in the PCG was made simpler by using the ECG and carotid pulse signals, where the QRS wave and the dicrotic notch can be detected more readily than the heart sounds themselves.

Event detection is an important step that is required before we may attempt to analyze the corresponding waves or wavelets in more detail. After a specific wave of interest has been detected, isolated, and extracted, methods targeted to the expected characteristics of the event may be applied for specifically focused analysis of the corresponding physiological or pathological event. Analysis of the event is then not hindered or obscured by other events or artifacts in the acquired signal.

4.13 Study Questions and Problems

1. Prove that the ACF $\phi_{xx}(\tau)$ of any stationary function $x(t)$ is maximum at $\tau = 0$. *Hint:* Start with $E[\{x(t + \tau) \pm x(t)\}^2] \geq 0$.
2. For a stationary process x , prove that the ACF is even symmetric, that is, $\phi_{xx}(\tau) = \phi_{xx}(-\tau)$. You may use the expectation or time-average definition of the ACF.
3. Starting with the continuous time-average definition of the ACF, prove that the Fourier transform of the ACF is the PSD of the signal.
4. What are the Fourier-domain equivalents of the ACF and the CCF? Describe their common features and differences. List their applications in biomedical signal analysis.
5. A signal $x(t)$ is transmitted through a channel. The received signal $y(t)$ is a scaled, shifted, and noisy version of $x(t)$ given as $y(t) = \alpha x(t - t_0) + \eta(t)$ where α is a scale factor, t_0 is the time delay, and $\eta(t)$ is noise. Assume that the noise process has zero mean and is statistically independent of the signal process, and that all processes are stationary. Derive expressions for the mean and the ACF of $y(t)$ in terms of the statistics of x and η .
6. Derive an expression for the ACF of the signal $x(t) = \sin(\omega_0 t)$. Use the time-average definition of the ACF. From the ACF, derive an expression for the PSD of the signal. Show all steps.
7. A rhythmic episode of a theta wave in an EEG signal is approximated by a researcher to be a sine wave of frequency 5 Hz. The signal is sampled at 100 Hz. Draw a schematic representation of the ACF of the episode for delays up to 0.5 s. Label the time axis in samples and in seconds. Draw a schematic representation of the PSD of the episode. Label the frequency axis in Hz.
8. The values of a signal sampled at 100 Hz are given by the series $\{0, 0, 0, 0, 10, 10, 10, 0, 0, 0, 0, 0, 5, 5, 5, 0, 0, 0, 0, -3, -3, -3, 0, 0, 0\}$. An investigator performs template matching with the pattern $\{0, 5, 5, 5, 0\}$. The first sample in each array stands for zero time. Plot the output of the template matching operation and interpret the result. Label the time axis in seconds.
9. A biphasic signal $x(n)$ is represented by the series of samples $x(n) = \{0, 1, 2, 1, 0, -1, -2, -1, 0\}$ for $n = 0, 1, 2, \dots, 8$. (a) Draw a plot of $x(n)$. (b) Compose a signal $y(n)$ defined as $y(n) = 3x(n) + 2x(n - 12) - x(n - 24)$. Draw a plot of $y(n)$. (c) Design a matched filter to detect the presence of $x(n)$ in $y(n)$. Explain how the impulse response $h(n)$ and the frequency response $H(\omega)$ of the filter are related to $x(n)$. Plot $h(n)$. (d) Compute the output of the filter and plot it. Interpret the output of the filter.
10. A researcher uses the derivative operator (filter) specified as $w(n) = x(n) - x(n - 1)$, where $x(n)$ is the input and $w(n)$ is the output. The result is then passed through the MA filter $y(n) = \frac{1}{3}[w(n) + w(n - 1) + w(n - 2)]$, where $y(n)$ is the final output desired. (a) Derive the transfer functions (in the z -domain) of the two filters individually as well as that of the combination. (b) Does it matter which of the two filters is placed first? Why (not)? (c) Derive the impulse response of each filter and that of the combination. Plot the three signals. (d) The signal described by the samples $\{0, 0, \dots, 0, 6, 6, 6, 6, 6, 6, 6, 6, 0, 0, \dots\}$ is applied to the system. Derive the values of the final output signal. Explain the effect of the operations on the signal.

11. You have been hired to develop two software packages for the analysis of 10-channel EEG signals for the following purposes: (a) Detection of the presence of the alpha rhythm: (i) in any one single channel and (ii) jointly in a pair of left-right channels. (b) Detection of spike-and-wave complexes of a prespecified shape in any channel.
Design two signal processing packages to address the two problems mentioned above. For each package, provide the following details: (i) A schematic block diagram representing the various algorithms or signal processing steps that you recommend. Include at least three distinct and nontrivial procedures in your design. (ii) Explain each block or part of your package. Describe the reason or logic behind your recommendation. (iii) Write at least two nontrivial equations related to your procedures. (iv) Provide graphical illustrations representing a typical EEG signal as it is processed by each block of your package.
12. The template of a signal is specified by the series of samples $\{2, 3, -1\}$ for $n = 0, 1, 2$. (a) Design a matched filter to detect the signal. Give the impulse response and transfer function of the filter. (b) The signal specified by the series of samples $\{1, -1, -2, 4, 6, -2, 1, -1, 0\}$ is applied to the matched filter. Compute the output and explain its characteristics.
13. Describe the significance of the P wave in the analysis of ECG signals. Describe a method for the detection of P waves in an ECG signal. Explain the purpose and reasoning behind each step of your algorithm. Give at least one nontrivial equation representing a procedure in your algorithm. Draw schematic sketches representing a sample input signal and the corresponding output at each stage of your method.
14. Draw a block diagram representing the various steps in the Pan-Tompkins method to detect QRS complexes in ECG signals. Explain the purpose and nature of each step in the procedure, including detection of the peaks in the output corresponding to the QRS complexes. No equations are required in your answer to this question.
Draw a schematic sketch of a noisy ECG signal including three cardiac cycles with high-frequency and low-frequency noise, and illustrate how it is modified by each step. Explain how the result could be used to measure the heart rate of a patient.
15. A researcher in signal processing seeks your help to design filters to obtain a smoothed estimate of the first derivative (difference) of a signal digitized at the sampling rate of 200 Hz. Help the researcher with the following: (a) Give the difference equation and impulse response of a filter to compute the first derivative (difference) of the signal. (b) Give the transfer function of the derivative filter. Derive the magnitude and phase parts of the frequency response of the filter. (c) Draw a sketch of the magnitude of the frequency response of the derivative filter. Explain the nature of the filter by deriving the gain at 0 Hz and 100 Hz. (d) Give the difference equation and impulse response of a filter to compute the average of the current input sample and the previous input sample (that is, an MA filter of length 2). (e) Give the transfer function of the MA filter. Derive the magnitude and phase parts of the frequency response of the filter. (f) Draw a sketch of the magnitude of the frequency response of the MA filter. Explain the nature of the filter by deriving the gain at 0 Hz and 100 Hz. (g) Derive the impulse response and the difference equation of a combined filter with the derivative and MA filters in series. (h) Derive the transfer function and plot the pole-zero diagram of the combined filter. (i) Derive the magnitude and phase parts of the frequency response of the combined filter. Draw a sketch of the magnitude of the frequency response of the filter. Explain the nature of the filter by deriving the gain at 0 Hz and 100 Hz. (j) Draw a signal-flow diagram of the combined filter.

16. Describe a procedure for the detection of QRS complexes in an ECG signal. Give a schematic block diagram of the important steps in the procedure. Include at least four steps. Give one suitable equation for each step of your algorithm either in the discrete-time domain as a difference equation or in the z -domain as a transfer function. Sketch an ECG signal with at least two beats and show the corresponding output at the various steps of your procedure. Explain the effect of each step of the procedure on the signal.

4.14 Laboratory Exercises and Projects

Note: Data files related to the exercises are available at the site

<http://people.ucalgary.ca/~ranga/enel563>

1. Implement the Pan–Tompkins method for QRS detection. You may employ a simple threshold-based method to detect QRS complexes as the procedure will be run off-line. Apply the procedure to the signals in the files ECG3.dat, ECG4.dat, ECG5.dat, and ECG6.dat, sampled at a rate of 200 Hz (see the file ECGS.m). Compute the averaged heart rate and QRS width for each record. Verify your results by measuring the parameters visually from plots of the signals.
2. The files eeg1-xx.dat (where xx indicates the channel name) give eight simultaneously recorded channels of EEG signals with the alpha rhythm. (You may read the signals using the MATLAB[®] program in the file eeg1.m.) The sampling rate is 100 Hz per channel. Cut out a portion of a signal with a clear presence of the alpha rhythm for use as a template or reference signal. Perform cross-correlation of the template with running (short-time) windows of various channels and study the use of the results for the detection of the presence of the alpha rhythm.
3. The files eeg2-xx.dat (where xx indicates the channel name) give ten simultaneously recorded channels of EEG signals with spike-and-wave complexes. (You may read the signals using the MATLAB[®] program in the file eeg2.m.) The sampling rate is 100 Hz per channel. Cut out one spike-and-wave complex from any EEG channel and use it as a template. Perform template matching by cross-correlation or by designing a matched filter. Apply the procedure to the same channel from which the template was selected as well as to other channels. Study the results and explain how they may be used to detect spike-and-wave complexes.
4. The files pec1.dat, pec33.dat, and pec52.dat give three-channel recordings of the PCG, ECG, and carotid pulse signals (sampled at 1,000 Hz; you may read the signals using the program in the file plotpec.m). The signals in pec1.dat (adult male) and pec52.dat (male subject, 23 years) are normal; the PCG signal in pec33.dat has systolic murmur, and is of a patient suspected to have pulmonary stenosis, ventricular septal defect, and pulmonary hypertension (female, 14 months).

Apply the Pan–Tompkins method for QRS detection to the ECG channel and the Lehner and Rangayyan method to detect the dicrotic notch in the carotid pulse channel. Extrapolate the timing information from the ECG and carotid pulse channels to detect the onset of S1 and S2 in the PCG channel. What are the corrections required to compensate the delays between the corresponding events in the three channels?

CHAPTER 5

ANALYSIS OF WAVESHAPE AND WAVEFORM COMPLEXITY

Certain biomedical signals, such as the ECG and the carotid pulse, have simple waveshapes (although the QRS wave is often referred to as a *complex*!). The readily identifiable signatures of the ECG and the carotid pulse are modified by abnormal events and pathological processes. Hence, analysis of waveshapes could be useful in the diagnosis of various diseases.

Signals such as the EMG and the PCG do not have waveshapes that may be identified easily. EMG signals are complex interference patterns of innumerable SMUAPs. PCG signals represent vibration waves that do not possess specific waveshapes. Regardless, even the complexity of the waveforms in some signals, such as the EMG and the PCG, does vary in relation to physiological and pathological phenomena. Analyzing the waveform complexity of such signals may assist in gaining an understanding of the processes they reflect.

5.1 Problem Statement

Explain how waveshapes and waveform complexity in biomedical signals relate to the characteristics of the underlying physiological and pathological phenomena. Propose techniques to parameterize and analyze the signal features you identify.

As in the preceding chapters, the problem statement given above is generic and represents the theme of the present chapter. The following section presents illustrations of the problem with case studies that provide more specific definitions of the problem with a few signals of interest. The remaining sections of the chapter describe techniques to address the stated problems. It should be noted that, although signal analysis techniques are proposed in the context of specific signals and applications, they may find applications in other fields where signals with comparable characteristics and behavior are encountered.

5.2 Illustration of the Problem with Case Studies

5.2.1 The QRS complex in the case of bundle branch block

We saw in Section 1.2.5 that the His bundle and its branches conduct the cardiac excitation pulse from the AV node to the ventricles. A block in one of the bundle branches causes asynchrony between the contraction of the left and the right ventricles. This, in turn, causes a staggered summation of the action potentials of the myocytes of the left and the right ventricles over a longer-than-normal duration. The result is a longer and possibly jagged QRS complex, as illustrated by the ECG of a patient with right bundle-branch block in Figure 1.28.

5.2.2 The effect of myocardial ischemia and infarction on QRS wave shape

The occlusion of a coronary artery, or a branch thereof, due to the deposition of fat, calcium, and other substances, results in reduced blood supply to a portion of the cardiac musculature. The part of the myocardium served by the affected artery then suffers from ischemia, that is, lack of blood supply. Complete blockage of an artery leads to myocardial infarction when the affected tissues die. The deceased myocytes cannot contract any more and no longer produce action potentials.

The action potential of an under-nourished ventricular myocyte reflects altered repolarization characteristics: The action potential is of smaller amplitude and shorter duration [11, 233] than the normal case. The result of the summation of the action potentials of all of the active ventricular myocytes will then be different from the normal QRS complex. The primary change reflected in the ECG is a modified ST segment that is either elevated or depressed, depending upon the lead used and the position of the affected region; the T wave may also be inverted. Chronic myocardial infarction causes a return to a normal ST segment and a pronounced Q wave [33].

5.2.3 Ectopic beats

Ectopic beats are generated by cardiac tissues that possess abnormal pacing capabilities. Ectopic beats originating from focal points on the atria could cause altered P waveshapes due to different paths of propagation of the excitation pulse and hence different activation sequences of atrial muscle fibers. However, the QRS complex of atrial ectopic beats will appear normal because the conduction of the excitation past the AV node would be normal.

Ectopic beats originating on the ventricles (that are necessarily premature beats, that is, PVCs) typically possess bizarre waveshapes due to widely differing paths of conduction and excitation of the ventricular muscle fibers. Figure 1.27 illustrates an ECG signal with PVCs. PVCs typically lack a preceding P wave; however, an ectopic beat triggered during the normal AV node delay will demonstrate a normal preceding P wave. PVCs triggered by ectopic foci close to the AV node may possess near-normal QRS shape as the path of conduction may be almost the same as in the case of a normal impulse from the AV node. On the other hand, beats triggered by ectopic foci near the apex of the heart could take widely different paths of propagation, resulting in substantially different QRS waveshapes. In addition to waveshape, the preceding and succeeding *RR* intervals play important roles in determining the nature of ectopic beats.

5.2.4 Complexity of the EMG interference pattern

We saw in Section 1.2.4 that motor units are recruited by two mechanisms — spatial and temporal recruitment — in order to produce increasing levels of contraction and muscular force output. As more and more motor units are brought into action and their individual firing rates increase (within certain limits), the SMUAPs of the active motor units overlap and produce a complex interference pattern. Figures 1.18 and 1.19 illustrate an EMG signal obtained from the crural diaphragm of a dog during one normal breath cycle. The increasing complexity of the waveform with deeper levels of inspiration is clearly seen in the expanded plot in Figure 1.19.

Although a surface EMG interference pattern is typically too complex for visual analysis, the general increase in the level of activity (“busy-ness”) may be readily seen. It used to be common practice in EMG laboratories to feed EMG signals to an amplified speaker: low levels of activity when the SMUAPs are not overlapping (that is, separated in time) result in discrete “firing” type of sounds; increasing levels of contraction result in increased “chatter” in the sound produced. EMG signals may be analyzed to derive parameters of waveform complexity that increase with increasing muscular contraction, and hence to obtain correlates to mechanical activity that are derived from its electrical manifestation.

5.2.5 PCG intensity patterns

Although the vibration waves in a PCG signal may not be amenable to direct visual analysis, the general intensity pattern of the signal over a cardiac cycle may be

readily appreciated either by auscultation or visual inspection. Certain cardiovascular diseases and defects alter the relative intensity patterns of S1 and S2, cause additional sounds or murmurs, and/or split S2 into two distinct components, as described in Section 1.2.9. While many cardiovascular diseases and defects may cause systolic murmurs, the intensity pattern or envelope of the murmur could assist in arriving at a specific diagnosis. It should also be noted that definitive diagnosis based on the PCG would usually require comparative analysis of PCG signals from a few positions on the chest. Figures 1.42, 1.44, 4.30, and 4.31 illustrate PCG signals of a normal subject and patients with systolic murmur, split S2, and opening snap of the mitral valve. The differences in the overall intensity patterns of the signals are obvious. However, signal processing techniques are desirable to convert the signals into positive-valued envelopes that could be treated as distributions of signal energy over time. Such a transformation permits the treatment of signal intensity patterns as PDFs, which lends to the computation of various statistical measures and moments.

5.3 Analysis of ERPs

The most important parameter extracted from a visual ERP is the timing or latency of the first major positivity; since the average of this latency is about 120 *ms* for normal adults, it is referred to as P120 (see Figure 3.41). The latencies of the troughs before and after P120, called N80 and N145, respectively, are also of interest. The amplitudes of the ERP at the corresponding instants are of lesser importance. Delays in the latencies that are well beyond the normal range could indicate problems in the visual system. Asymmetries in the latencies of the left and right parts of the visual system could also be indicative of disorders.

The lowest trace in Figure 3.41 is an averaged flash visual ERP recorded from a normal adult male subject. The signal has been labeled to indicate the N80, P120, and N145 points; the corresponding actual latencies for the subject are 85, 100.7, and 117 *ms*, respectively.

Auditory ERPs are weaker and more complex than visual ERPs, requiring averaging over several hundred or a few thousand stimuli. Auditory ERPs are analyzed for the latencies and amplitudes of several peaks and troughs. Clinical ERP analysis is usually performed manually, since there is no pressing need for signal processing techniques beyond synchronized averaging.

5.4 Morphological Analysis of ECG Waves

The waveshape of an ECG cycle could be changed by many different abnormalities, including myocardial ischemia or infarction, bundle-branch block, and ectopic beats. It is not possible to propose a single analysis technique that can assist in categorizing all possible abnormal causes of change in waveshape. The following sections address a few illustrative cases.

5.4.1 Correlation coefficient

Problem: *Propose a general index to indicate altered QRS waveshape. You are given a normal QRS template.*

Solution: Jenkins et al. [132] applied the correlation coefficient γ_{xy} as defined in Equation 4.24 to classify ECG cycles as normal beats or beats with abnormal morphology. A normal beat was used as a template to compute γ_{xy} for each detected beat. They found that most normal beats possessed γ_{xy} values above 0.9, and that PVCs and beats with abnormal shape had considerably lower values. A threshold of 0.9 was used to assign a code to each beat as 0 for abnormal or 1 for normal in terms of waveshape. Figure 2.2 shows an ECG signal with five abnormal beats that have the first symbol in the 4-symbol code shown for each beat as 0, indicating an abnormal shape due to generation by an ectopic focus or due to aberrant conduction of a pulse generated by the SA node. The normal beats have the first symbol of the code as 1, indicating high correlation with the normal template. See Section 5.8 for related discussion.

5.4.2 The minimum phase correspondent and signal length

The normal ECG signal contains epochs of activity where the signal's energy is concentrated. Discounting the usually low-amplitude P and T waves, most of the energy of a normal ECG signal is concentrated within the interval of about 80 ms that is spanned by the QRS complex. The normally isoelectric PQ, ST, and TP segments contain no energy as the signal amplitude is typically zero over the corresponding intervals. We have observed that certain abnormal conditions cause the QRS to widen or the ST segment to possess a nonzero value. In such a case, it could be said that the energy of the signal is being spread over a longer duration. Let us now consider how we may capture this information, and investigate if it may be used for waveshape analysis.

Problem: *Investigate the effect of the distribution of energy over the time axis on a signal's characteristics. Propose measures to parameterize the effects and study their use in the classification of ECG beats.*

Solution: A signal $x(t)$ may be seen as a distribution of the amplitude of a certain variable over the time axis. The square of the signal, that is, $x^2(t)$, may be interpreted as the instantaneous energy of the signal-generating process. The function $x^2(t)$, $0 \leq t \leq T$, may then be viewed as an energy distribution or density function, with the observation that the total energy of the signal is given by

$$E_x = \int_0^T x^2(t) dt. \quad (5.1)$$

Such a representation facilitates the definition of moments of the energy distribution, leading to a centroidal time

$$t_{\bar{x}} = \frac{\int_0^T t x^2(t) dt}{\int_0^T x^2(t) dt}, \quad (5.2)$$

and dispersion of energy about the centroidal time

$$\sigma_{t_x}^2 = \frac{\int_0^T (t - t_{\bar{x}})^2 x^2(t) dt}{\int_0^T x^2(t) dt}. \quad (5.3)$$

Observe the similarity between the equations given above and Equations 3.1 and 3.3: The normalized function

$$p_x(t) = \frac{x^2(t)}{\int_0^T x^2(t) dt} \quad (5.4)$$

is now treated as a PDF. Other moments may also be defined to characterize and study the distribution of $x^2(t)$ over the time axis. The preceding equations have been stated in continuous time for the sake of generality; they are valid for discrete-time signals, with a simple change of t to n and $\int dt$ to $\sum_n$.

Minimum-phase signals: The distribution of the energy of a signal over its duration is related to its amplitude spectrum and, more importantly, to its phase spectrum. The notion of minimum phase is useful in analyzing related signal characteristics. The minimum-phase property of signals may be explained in both the time and frequency domains [174, 201, 234–238].

In the time domain, a signal $x(n)$ is a minimum-phase signal if both the signal and its inverse $x_i(n)$ are one-sided (that is, completely causal or anticausal) signals with finite energy, that is, $\sum_{n=0}^{\infty} x^2(n) < \infty$ and $\sum_{n=0}^{\infty} x_i^2(n) < \infty$. [Note: The inverse of a signal is defined such that $x(n) * x_i(n) = \delta(n)$; equivalently, we have $X_i(z) = \frac{1}{X(z)}$.]

Some of the important properties of a minimum-phase signal are:

- For a given amplitude spectrum there exists one and only one minimum-phase signal.
- Of all finite-energy, one-sided signals with identical amplitude spectra, the energy of the minimum-phase signal is optimally concentrated toward the origin, and the signal has the smallest phase lag and phase-lag derivative at each frequency.
- The z -transform of a minimum-phase signal has all of its poles and zeros inside the unit circle in the z -plane.
- The complex cepstrum of a minimum-phase signal is causal (see also Section 4.8.3).

The extreme example of a minimum-phase signal is the delta function $\delta(t)$, which has all of its energy concentrated at $t = 0$. The magnitude spectrum of the delta function is real and equal to unity for all frequencies; the phase lag at every frequency is zero.

Minimum-phase and maximum-phase components: A signal $x(n)$ that does not satisfy the minimum-phase condition, referred to as a composite signal or a

mixed-phase signal, may be split into its minimum-phase component and maximum-phase component by filtering its complex cepstrum $\hat{x}(n)$ [174, 223, 239]. To obtain the minimum-phase component, the causal part of the complex cepstrum (see Section 4.8.3) is chosen as follows:

$$\hat{x}_{\min}(n) = \begin{cases} 0, & n < 0, \\ 0.5 \hat{x}(n), & n = 0, \\ \hat{x}(n), & n > 0. \end{cases} \quad (5.5)$$

Application of the inverse procedures yields the minimum-phase component $x_{\min}(n)$. Similarly, the maximum-phase component is obtained by application of the inverse procedures to the anticausal part of the cepstrum, selected as

$$\hat{x}_{\max}(n) = \begin{cases} \hat{x}(n), & n < 0, \\ 0.5 \hat{x}(n), & n = 0, \\ 0, & n > 0. \end{cases} \quad (5.6)$$

The minimum-phase and maximum-phase components of a signal satisfy the following relationships:

$$\hat{x}(n) = \hat{x}_{\min}(n) + \hat{x}_{\max}(n) \quad (5.7)$$

and

$$x(n) = x_{\min}(n) * x_{\max}(n). \quad (5.8)$$

The minimum-phase correspondent: A mixed-phase signal may be converted to a minimum-phase signal that has the same spectral magnitude as the original signal by filtering the complex cepstrum of the original signal as

$$\hat{x}_{\text{MPC}}(n) = \begin{cases} 0, & n < 0, \\ \hat{x}(n), & n = 0, \\ \hat{x}(n) + \hat{x}(-n), & n > 0, \end{cases} \quad (5.9)$$

and applying the inverse procedures [174, 223, 239]. The result is known as the *minimum-phase correspondent* (MPC) of the original signal [201]. The MPC possesses optimal concentration of energy near the origin under the constraint imposed by the specified magnitude spectrum (of the original mixed-phase signal).

Observe that $\hat{x}_{\text{MPC}}(n)$ is equal to twice the even part of $\hat{x}(n)$ for $n > 0$. This leads to a simpler procedure to compute the MPC, as follows. Let us assume $\hat{X}(z) = \log X(z)$ to be analytic over the unit circle in the z -plane. We can write $\hat{X}(\omega) = \hat{X}_R(\omega) + j\hat{X}_I(\omega)$, where the subscripts R and I indicate the real and imaginary parts, respectively. $\hat{X}_R(\omega)$ and $\hat{X}_I(\omega)$ are the log-magnitude and phase spectra of $x(n)$, respectively. Now, the inverse Fourier transform of $\hat{X}_R(\omega)$ is equal to the even part of $\hat{x}(n)$, defined as $\hat{x}_e(n) = [\hat{x}(n) + \hat{x}(-n)]/2$. Thus, we have

$$\hat{x}_{\text{MPC}}(n) = \begin{cases} 0, & n < 0, \\ \hat{x}_e(n), & n = 0, \\ 2\hat{x}_e(n), & n > 0. \end{cases} \quad (5.10)$$

This result means that we do not need to compute the complex cepstrum, which requires the unwrapped phase spectrum of the signal, but need only to compute a *real cepstrum* using the log-magnitude spectrum. Furthermore, given that the PSD is the Fourier transform of the ACF, we have $\log\{\text{FT}[\phi_{xx}(n)]\} = 2\hat{X}_R(\omega)$. It follows that, in the cepstral domain, $\hat{\phi}_{xx}(n) = 2\hat{x}_e(n)$, and therefore [239]

$$\hat{x}_{\text{MPC}}(n) = \begin{cases} 0, & n < 0, \\ 0.5 \hat{\phi}_{xx}(n), & n = 0, \\ \hat{\phi}_{xx}(n), & n > 0, \end{cases} \quad (5.11)$$

where $\hat{\phi}_{xx}(n)$ is the cepstrum of the ACF $\phi_{xx}(n)$ of $x(n)$.

Signal length: The notion of signal length (SL), as introduced by Berkhout [235], is different from signal duration. The duration of a signal is the extent of time over which the signal exists, that is, has nonzero values (neglecting periods within the total duration of the signal where the signal's amplitude could be zero). SL relates to how the energy of a signal is distributed over its duration. SL depends upon both the magnitude and phase spectra of the signal. For one-sided signals, minimum SL implies minimum phase; the converse is also true [235].

The general definition of the SL of a signal $x(n)$ is given as [235]

$$SL = \frac{\sum_{n=0}^{N-1} w(n) x^2(n)}{\sum_{n=0}^{N-1} x^2(n)}, \quad (5.12)$$

where $w(n)$ is a nondecreasing, positive weighting function with $w(0) = 0$. The definition of $w(n)$ depends upon the application and the desired characteristics of SL . It is readily seen that samples of the signal away from the origin $n = 0$ receive progressively heavier weights given by $w(n)$. The definition of SL as above may be viewed as a normalized moment of $x^2(n)$. If $w(n) = n$, we get the centroidal time instant of $x^2(n)$ as in Equation 5.2.

For a given amplitude spectrum and hence total energy, the minimum-phase signal has its energy optimally concentrated near the origin. Therefore, the minimum-phase signal will have the lowest SL of all signals with the specified amplitude spectrum. Signals with increasing phase lag have their energy spread over a longer time duration and will have larger SL due to the increased weighting by $w(n)$.

Illustration of application: The QRS-T wave is the result of the spatiotemporal summation of the action potentials of ventricular myocytes. The duration of normal QRS-T waves is in the range of 350 – 400 *ms*, with the QRS itself limited to about 80 *ms* due to rapid and coordinated depolarization of the ventricular muscle fibers via the Purkinje fibers. However, PVCs, in general, have QRS-T complexes that are wider than normal, that is, they have their energy distributed over longer time spans within their total duration. This is due to different and possibly slower and disorganized excitation sequences triggering the ventricular muscle fibers: Ectopic triggers may not get conducted through the Purkinje system, and may be conducted through the ventricular muscle fibers themselves. Furthermore, PVCs do not, in general, display separate QRS and T waves; that is, they lack an isoelectric ST segment.

Regardless of the distinctions described above, normal ECG beats and PVCs have similar amplitude spectra, indicating that the difference between the signals may lie in their phase. SL depends upon both the amplitude spectrum and the phase spectrum of the given signal, and it parameterizes the distribution of energy over the duration of the signal. Based upon the arguments above, Murthy and Rangaraj [201] proposed the application of SL to classify ECG beats as normal or ectopic (or PVC, along with the use of the RR interval to indicate prematurity). Furthermore, to overcome ambiguities in the determination of the onset of each beat, they computed the SL of the MPC of the ECG signals (segmented so as to include the P, QRS, and T waves of each cycle). Use of the MPC resulted in a “rearrangement” of the signal such that the dominant QRS wave appeared at the origin in the MPC.

Figure 5.1 illustrates a normal ECG signal and three PVCs of a patient with multiple ectopic foci generating PVCs of widely differing shapes [201]. The figure also illustrates the corresponding MPCs and lists the SL values of all of the signals. The SL values of the MPCs of the abnormal waves are higher than the SL of the MPC of the normal signal (see the right-hand column of signals in Figure 5.1). The SL values of the original PVCs do not exhibit such a separation from the SL of the normal signal (see the left-hand column of signals in Figure 5.1). Ambiguities due to the presence of baseline segments of variable lengths at the beginning of the signals have been overcome by the use of the MPCs. In each case, the MPCs have the most-dominant wave at the origin, reflecting a rearrangement of energy in order to meet the minimum-phase criteria.

Figure 5.2 shows plots of the RR intervals and SL values computed using the original ECG signals and their MPCs for several beats of the same patient whose representative ECG waveforms are illustrated in Figure 5.1 [201]. The SL values of the normal signals and the ectopic beats exhibit a significant overlap in the range 28 – 35 [plot (a) in Figure 5.2]. However, the SL values of the MPCs of the PVCs are higher than those of the normal beats, which facilitates their classification [plot (b) in Figure 5.2].

Murthy and Rangaraj [201] applied their QRS detection method (described in Section 4.3.1) to ECG signals of two patients with ectopic beats, and used the SL of MPC to classify the beats with a linear discriminant function (described in Section 9.4.1). They analyzed 208 beats of the first patient (whose signals are illustrated in Figures 5.1 and 5.2): 132 out of 155 normals and 48 out of 53 PVCs were correctly classified; one beat was missed by the QRS detection algorithm. Misclassification of normal beats as PVCs was attributed to wider-than-normal QRS complexes and depressed ST segments in some of the normal beats of the patient (see Figure 5.2). The signal of the second patient included 89 normals and 18 PVCs, all of which were detected and classified correctly. It was observed that computation of the MPC was not required in the case of the second patient: The SL values of the original signals provided adequate separation between the normal and ectopic beats. The segments of normal ECG cycles used by Murthy and Rangaraj included the P wave; better results could be obtained by using only the QRS and T waves since most PVCs do not include a distinct P wave and essentially correspond to the QRS and T waves in a normal ECG signal.

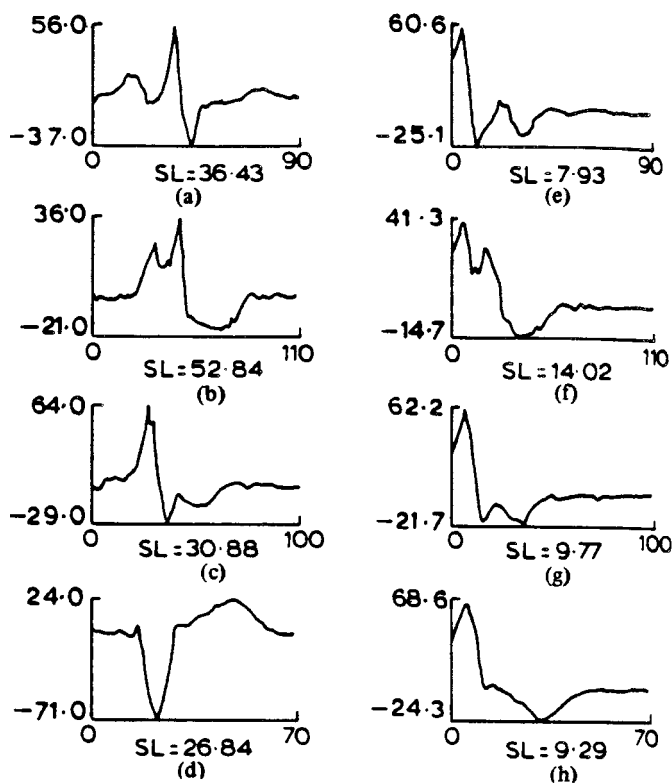
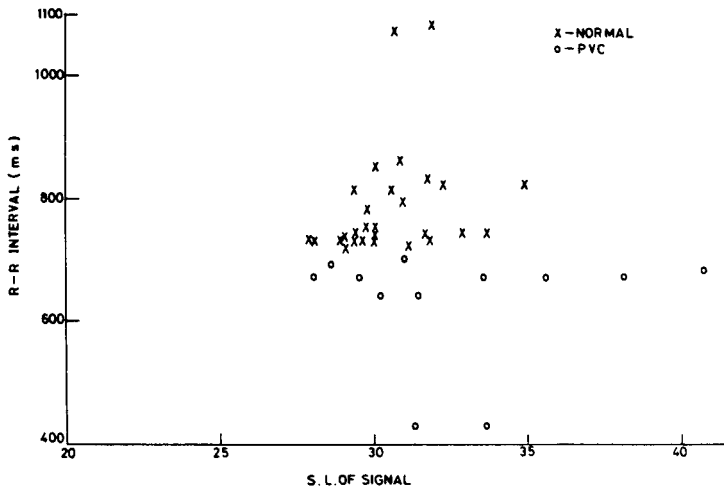
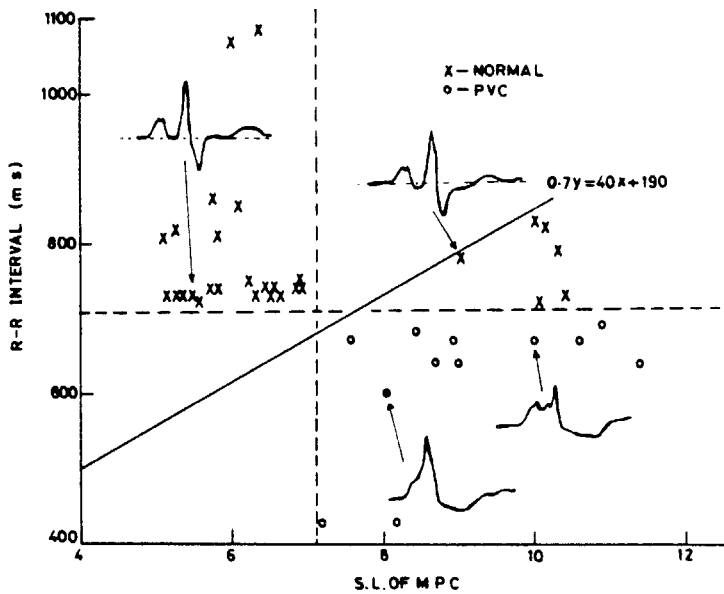


Figure 5.1 (a) A normal ECG beat and (b)–(d) three ectopic beats (PVCs) of a patient with multiple ectopic foci. (e)–(h) MPCs of the signals in (a)–(d). The SL values of the signals are also indicated [201]. Note that the abscissa is labeled in samples, with a sampling interval of 10 ms. The ordinate is not calibrated. The signals have different durations and amplitudes although plotted to the same size. Reproduced with permission from I.S.N. Murthy and M.R. Rangaraj, New concepts for PVC detection, *IEEE Transactions on Biomedical Engineering*, 26(7):409–416, 1979. ©IEEE.

It should be noted that the QRS width may be increased by other abnormal conditions such as bundle-branch block; the definition of SL as above would lead to higher SL for wider-than-normal QRS complexes. Furthermore, ST segment elevation or depression would be interpreted as the presence of energy in the corresponding time interval in the computation of SL . Abnormally large T waves could also lead to SL values that are larger than those for normal signals. More sophisticated logic and other parameters in addition to SL could be used to rule out these possibilities and affirm the classification of a beat as an ectopic beat.



(a)



(b)

Figure 5.2 (a) Plot of RR and SL values of several beats of a patient with multiple ectopic foci (as in Figure 5.1). (b) Same as (a) but with the SL of the MPCs of the signals. A few representative ECG cycles are illustrated. The linear discriminant (decision) function used to classify the beats is also shown [201]. Reproduced with permission from I.S.N. Murthy and M.R. Rangaraj, New concepts for PVC detection, *IEEE Transactions on Biomedical Engineering*, 26(7):409–416, 1979. ©IEEE.

5.4.3 ECG waveform analysis

Measures such as the correlation coefficient and SL described in the preceding sections provide general parameters that could assist in comparing waveforms. The representation, however, is in terms of gross features, and many different waveforms could possess the same or similar feature values. Detailed analysis of ECG waveforms will require the use of several features or measurements for accurate categorization of various QRS complex shapes and correlation with cardiovascular diseases. Since the ECG waveform depends upon the lead system used, sets of features may have to be derived for multiple-lead ECGs, including as many as 12 leads that are commonly used in clinical practice.

The steps required for ECG waveform analysis may be expressed as [53]:

1. Detection of ECG waves, primarily the QRS complex, and possibly the P and T waves.
2. Delimitation of wave boundaries, including the P, QRS, and T waves.
3. Measurement of interwave intervals, such as RR, PQ, QT, ST, QQ, and PP intervals.
4. Characterization of the morphology (shape) of the waves.

The last step given above may be achieved using parameters such as the correlation coefficient and SL as described earlier, or via detailed measurements of the peaks of the P, Q, R, S, and T waves (some of which could be negative); the durations of the P, Q, R, S, QRS, and T waves; and the interwave intervals defined above [53]. The nature of the PQ and ST segments, in terms of their being isoelectric or not (in case of the latter, as being positive, negative, elevated, or depressed), should also be documented. However, a large number of such features would make the development of further pattern classification rules difficult.

Cox et al. [53] and Nolle [240] proposed four measures to characterize QRS complexes, as illustrated in Figure 5.3 and defined as follows:

1. *Duration* — the duration or width of the QRS complex.
2. *Height* — the maximum amplitude minus the minimum amplitude of the QRS complex.
3. *Offset* — the positive or negative vertical distance from the midpoint of the baseline to the center of the QRS complex. The baseline is defined as the straight line connecting the temporal boundary points of the QRS complex. The center is defined as the midpoint between the highest and lowest bounds in amplitude of the QRS complex.
4. *Area* — the area under the QRS waveform rectified with respect to a straight line through the midpoint of the baseline.

Since the measures are independent of time, they are not sensitive to the preceding procedures for the detection of fiducial markers.

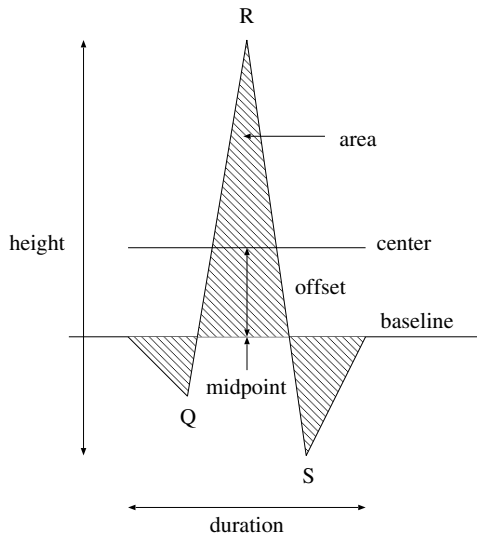


Figure 5.3 ECG waveform features used by Cox et al. [53] and Nolle [240].

The measures of Cox et al. [53] and Nolle [240] were used to develop a system for arrhythmia monitoring, known as “Argus” for Arrhythmia Guard System, for use in coronary-care units. Figure 5.4 shows the grouping of more than 200 QRS complexes of a patient with multifocal PVCs into 16 dynamic families by Argus using the four features defined above [53]. The families labeled 00, 01, 02, 04, 06, and 10 were classified as normal beats by Argus (163 beats which were all classified as normals by a cardiologist; 91% of the normals were correctly labeled by Argus). PVCs of different shapes from more than two ectopic foci form the remaining families, with some of them having shapes close to those of the patient’s normal beats. Of the 52 beats in the remaining families, 96% were labeled as PVCs by the cardiologist; Argus labeled 85% of them as PVCs, 13% as not PVCs, and 2% as borderline beats [240]. Some noteworthy points of one of the clinical tests of Argus with over 50,000 beats are as follows: 85% of 45,364 normal beats detected and classified correctly, with 0.04% beats missed; 78% of 4,010 PVCs detected and classified correctly, with 5.3% beats missed; and 38 normals (less than 0.1% of the beats) falsely labeled as PVCs.

For recent developments in the analysis and classification of ECG signals, see Ye et al. [241] and Ince et al. [242].

5.5 Envelope Extraction and Analysis

Signals with complex patterns such as the EMG and PCG may not permit direct analysis of their waveshape. In such cases, the intricate high-frequency variations may not be of interest; rather, the general trends in the level of the overall activity might

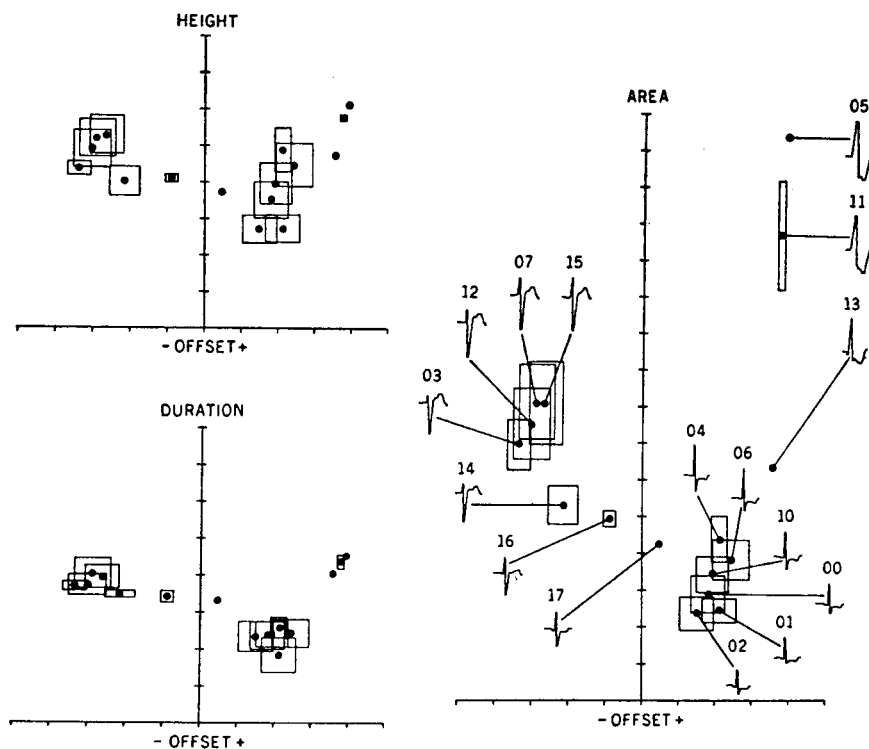


Figure 5.4 Use of four features to catalog QRS complexes into one of 16 dynamic families of similar complexes enclosed by four-dimensional boxes. The waveforms of typical members of each family are shown in the area-versus-offset feature plane. The family numbers displayed are in the octal (base eight) system. The families labeled 00, 01, 02, 04, 06, and 10 were classified as normal beats, with the others being PVCs or borderline beats. Reproduced with permission from J.R. Cox, Jr., F.M. Nolle, and R.M. Arthur, Digital analysis of the electroencephalogram, the blood pressure wave, and the electrocardiogram, *Proceedings of the IEEE*, 60(10):1137–1164, 1972. ©IEEE.

convey useful information. Considering, for example, the EMG in Figure 1.18, observe that the general signal level increases with the level of activity (breathing). As another example, the PCG in the case of aortic stenosis, as illustrated in Figure 1.44, demonstrates a diamond-shaped systolic murmur: The *envelope* of the overall signal carries important information. Let us, therefore, consider the problem of extraction of the envelope of a seemingly complex signal.

Problem: Formulate algorithms to extract the envelope of an EMG or PCG signal to facilitate analysis of trends in the level of activity or energy in the signal.

Solution: The first step required in order to derive the envelope of a signal with positive and negative deflections is to obtain the absolute value of the signal at each

time instant, that is, perform full-wave rectification. This procedure will create abrupt discontinuities at time instants when the original signal values change sign, that is, at zero-crossings. The discontinuities create high-frequency components of significant magnitude. This calls for the application of a lowpass filter with a relatively low bandwidth in the range of 0 – 10 or 0 – 50 Hz to obtain smooth envelopes of EMG and PCG signals. An MA filter may be used to perform lowpass filtering, leading to the basic definition of a time-averaged envelope as

$$y(t) = \frac{1}{T_a} \int_{t-T_a}^t |x(t)| dt, \quad (5.13)$$

where T_a is the duration of the MA filtering window.

In a procedure similar in principle to that described above, Lehner and Rangayyan [131] applied a weighted MA filter to the squared PCG signal to obtain a smoothed energy distribution curve $E(n)$ as

$$E(n) = \sum_{k=1}^M x^2(n - k + 1) w(k), \quad (5.14)$$

where $x(n)$ is the PCG signal, $w(k) = M - k + 1$, and $M = 32$ with the signal sampled at 1,024 Hz . Observe that the difference between energy and power is simply a division by the time interval being considered, which may be treated as a scale factor or ignored.

The envelope represents the total averaged activity (such as electrical or acoustic) within the averaging window. An improved filter such as a Bessel filter [40] may be required if a smooth envelope is desired. The filter should strike a balance between the need to smooth discontinuities in the rectified signal and the requirement to maintain good sensitivity to represent relevant changes in signal level or amplitude. This procedure is known as envelope detection or amplitude demodulation. A few related procedures and techniques are described in the following sections.

5.5.1 Amplitude demodulation

Amplitude modulation (AM) of signals for radio transmission involves multiplication of the signal $x(t)$ to be transmitted by the carrier $\cos(\omega_c t)$, where ω_c is the carrier frequency [1–3]. The AM signal is given as $y(t) = x(t) \cos(\omega_c t)$. If the exact carrier wave used at the transmitting end were available at the receiving end as well (including the phase), *synchronous demodulation* becomes possible by multiplying the received signal $y(t)$ with the carrier. We then have the demodulated signal as

$$x_d(t) = y(t) \cos(\omega_c t) = x(t) \cos^2(\omega_c t) = \frac{1}{2}x(t) + \frac{1}{2}x(t) \cos(2\omega_c t). \quad (5.15)$$

The AM component at $2\omega_c$ may be removed by a lowpass filter, which will leave us with the desired signal $x(t)$.

If $x(t)$ is always positive, or a DC bias is added to meet this requirement, it becomes readily apparent that the envelope of the AM signal is equal to $x(t)$. A simple *asynchronous demodulation* procedure that does not require the carrier then becomes feasible: We only need to follow the envelope of $y(t)$. Given that the carrier frequency ω_c is far greater than the maximum frequency present in $x(t)$, the positive envelope of $y(t)$ may be extracted by performing half-wave rectification. A lowpass filter with an appropriate time constant to “fill the gaps” between the peaks of the carrier wave will provide a good estimate of $x(t)$. The difference between the use of a full-wave rectifier or a half-wave rectifier (that is, the larger gaps between the peaks of the carrier wave available after the latter type of rectification) can be smoothed over by increasing the time constant of the filter. The main differences between various envelope detectors lie in the way the rectification operation is performed and in the lowpass filter used [1, 3].

In a related procedure known as complex demodulation, a given arbitrary signal is demodulated to derive the time-varying amplitude and phase characteristics of the signal for each frequency (band) of interest [243–245]. In this approach, an arbitrary signal $x(t)$ is expressed as

$$x(t) = a(t) \cos[\omega_o t + \psi(t)] + x_r(t), \quad (5.16)$$

where ω_o is the frequency of interest; $a(t)$ and $\psi(t)$ are the time-varying amplitude and phase of the component at ω_o , respectively; and $x_r(t)$ is the remainder of the signal $x(t)$ after the component at ω_o has been removed. It is assumed that $a(t)$ and $\psi(t)$ vary slowly in relation to the frequencies of interest. The signal $x(t)$ may be equivalently expressed in terms of complex exponentials as

$$x(t) = \frac{1}{2} a(t) [\exp\{j[\omega_o t + \psi(t)]\} + \exp\{-j[\omega_o t + \psi(t)]\}] + x_r(t). \quad (5.17)$$

In the procedure of complex demodulation, the signal is shifted in frequency by $-\omega_o$ via multiplication with $2 \exp(-j\omega_o t)$, to obtain the result $y(t)$ as

$$\begin{aligned} y(t) &= 2x(t) \exp(-j\omega_o t) \\ &= a(t) \exp[j\psi(t)] + a(t) \exp\{-j[2\omega_o t + \psi(t)]\} + 2x_r(t) \exp(-j\omega_o t). \end{aligned} \quad (5.18)$$

The second term in the expression above is centered at $2\omega_o$, whereas the third term is centered at ω_o ; only the first term is placed at DC. Therefore, a lowpass filter may be used to extract the first term, to obtain the final result $y_o(t)$ as

$$y_o(t) \approx a(t) \exp[j\psi(t)]. \quad (5.19)$$

The desired entities may then be extracted as $a(t) \approx |y_o(t)|$ and $\psi(t) \approx \angle y_o(t)$.

The frequency resolution of the method depends upon the bandwidth of the lowpass filter used. The procedure may be repeated at every frequency (band) of interest. The result may be interpreted as the envelope of the signal for the specified frequency (band). The method was applied for the analysis of HRV by Shin et al. [243] and for the analysis of heart rate and arterial pressure variability by Hayano et al. [244].

In applying envelope detection to biomedical signals such as the PCG and the EMG, it should be noted that there is no underlying RF carrier wave in the signal: The envelope rides on relatively high-frequency acoustic or electrical activity that has a composite spectrum. The difference in frequency content between the envelope and the “carrier activity” will not be comparable to that in AM. Regardless, we could expect at least a ten-fold difference in frequency content: The envelope of an EMG or PCG signal may have an extremely limited bandwidth of 0 – 20 *Hz*, whereas the underlying signal has components up to at least 200 *Hz*, if not to 1,000 *Hz*. Application of envelope detection to the analysis of EMG related to respiration is illustrated in Section 5.10.

5.5.2 Synchronized averaging of PCG envelopes

The ECG and PCG form a good signal pair for synchronized averaging: The latter could be averaged over several cardiac cycles using the former as the trigger. However, the PCG is not amenable to direct synchronized averaging as the vibration waves may interfere in a destructive manner and cancel themselves out. Karpman et al. [246] proposed to first rectify the PCG signal, smooth the result using a low-pass filter, and then perform synchronized averaging of the nonnegative envelopes so obtained using the ECG as the trigger. The PCG envelopes were averaged over up to 128 cardiac cycles to get repeatable averaged envelopes. It should be noted that while synchronized averaging can reduce the effects of noise, breathing, coughing, and other types of artifacts, it can also smudge the time boundaries of cardiac events if the heart rate is not constant during the averaging procedure.

Figure 5.5 illustrates the envelopes obtained for a normal case and seven cases of systolic murmur due to various cardiovascular diseases and defects. The typical diamond-shaped envelope in the case of aortic stenosis results in an envelope shaped like an isosceles triangle due to rectification. Mitral regurgitation results in a rectangular holosystolic (spanning the entire systolic period) murmur envelope.

Karpman et al. analyzed 400 cases of systolic murmurs due to six types of diseases and defects, and obtained an accuracy of 89% via envelope analysis. They proposed a decision tree to classify systolic murmurs based upon the shape of a given murmur’s envelope as well as its relation to S1 and S2, which is illustrated in Figure 5.6.

5.5.3 The envelopogram

Sarkady et al. [227] proposed a Fourier-domain algorithm to obtain envelopes of PCG signals. They defined the *envelopogram estimate* as the magnitude of the analytic signal $y(t)$ formed using the PCG, $x(t)$, and its Hilbert transform, $x_H(t)$, as

$$y(t) = x(t) + jx_H(t). \quad (5.20)$$

(*Note:* An analytic function is a complex function of time having a Fourier transform that vanishes for negative frequencies [6, 174].) The Hilbert transform of a signal is

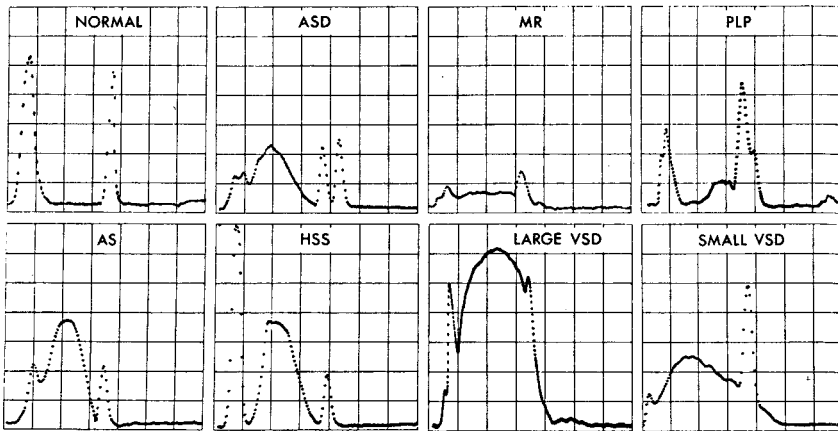


Figure 5.5 Averaged envelopes of the PCG signals of a normal subject and patients with systolic murmur due to aortic stenosis (AS), atrial septal defect (ASD), hypertrophic subaortic stenosis (HSS), rheumatic mitral regurgitation (MR), ventricular septal defect (VSD), and mitral regurgitation with posterior leaflet prolapse (PLP). Reproduced with permission from L. Karpman, J. Cage, C. Hill, A.D. Forbes, V. Karpman, and K. Cohn, Sound envelope averaging and the differential diagnosis of systolic murmurs, *American Heart Journal*, 90(5):600–606, 1975. ©American Heart Association.

defined as the convolution of the signal with $\frac{1}{\pi t}$, that is,

$$x_H(t) = \int_{-\infty}^{\infty} \frac{x(\tau)}{\pi(t - \tau)} d\tau. \quad (5.21)$$

The Fourier transform of $\frac{1}{\pi t}$ is $-j \operatorname{sgn}(\omega)$, where

$$\operatorname{sgn}(\omega) = \begin{cases} -1 & \omega < 0 \\ 0 & \omega = 0 \\ 1 & \omega > 0 \end{cases}, \quad (5.22)$$

is the signum function. Then, we have $Y(\omega) = X(\omega)[1 + \operatorname{sgn}(\omega)]$. $Y(\omega)$ is a one-sided or single-sideband function of ω containing only positive-frequency terms.

Based upon the definitions and properties described above, Sarkady et al. [227] proposed the following algorithm to obtain the envelopogram estimate:

1. Compute the DFT of the PCG signal.
2. Set the negative-frequency terms to zero; that is, $X(k) = 0$ for $\frac{N}{2} + 2 \leq k \leq N$, with the DFT indexed $1 \leq k \leq N$.
3. Multiply the positive-frequency terms, that is, $X(k)$ for $2 \leq k \leq \frac{N}{2} + 1$, by 2; the DC term $X(1)$ remains unchanged.

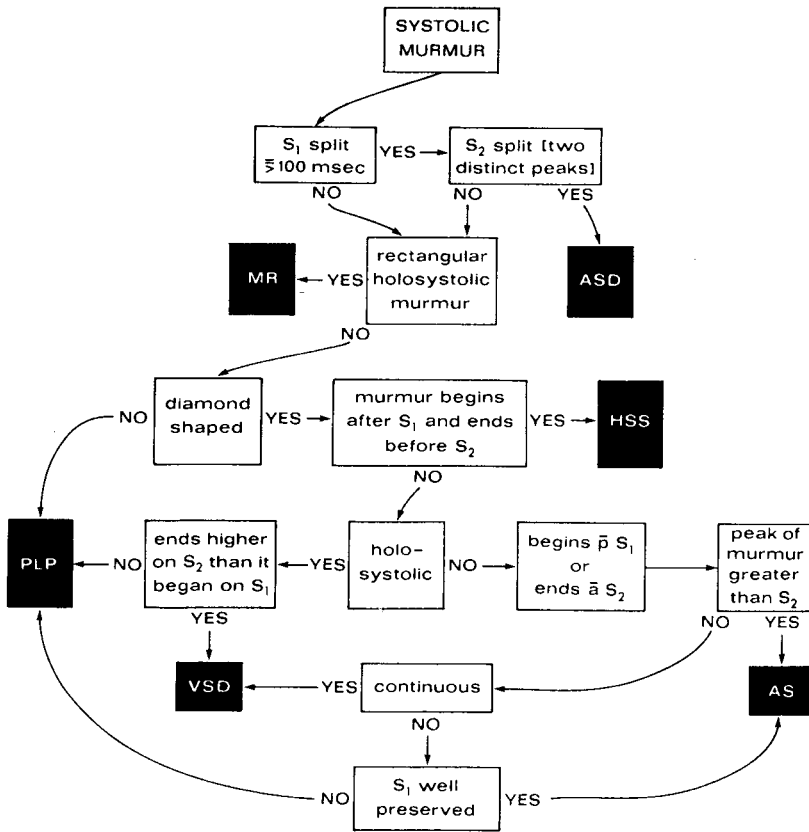


Figure 5.6 Decision tree to classify systolic murmurs based upon envelope analysis. For details on the abbreviations used, refer to the caption of Figure 5.5. $\bar{p}S_1$, after S_1 ; $\bar{a}S_2$, before S_2 . Reproduced with permission from L. Karpman, J. Cage, C. Hill, A.D. Forbes, V. Karpman, and K. Cohn, Sound envelope averaging and the differential diagnosis of systolic murmurs, *American Heart Journal*, 90(5):600–606, 1975. ©American Heart Association.

4. Compute the inverse DFT of the result.
5. Obtain an estimate of the envelopogram as the magnitude of the result.

The procedure described above, labeled also as complex demodulation by Sarkady et al., yields a high-resolution envelope of the input signal. Envelopograms and PSDs computed from PCG signals over single cardiac cycles tend to be noisy and are affected by respiration and muscle noise. Sarkady et al. recommended synchronized averaging of both envelopograms and PSDs of PCGs over several cycles. A similar method was used by Baranek et al. [228] to obtain the envelopes of PCG signals for the detection of the aortic component A2 of S2.

See Guerrero et al. [121] for illustrations of the use of the Hilbert transform to obtain envelopes of BCG signals and their application to detect sleep-related breathing disorders.

Illustration of application: The topmost plots in Figures 5.7 and 5.8 show one cycle each of the PCG signals of a normal subject and of a patient with systolic murmur, split S2, and opening snap of the mitral valve. The PCG signals were segmented by using the Pan–Tompkins method to detect the QRS complexes in the ECG signal, as illustrated in Figures 4.30 and 4.31 for the same signals. The envelopograms of the PCG cycles illustrated and the averaged envelopograms (over 16 beats for the normal case and 26 beats for the case with murmur) obtained using the method of Sarkady et al. [227] are shown in the second and third plots of Figures 5.7 and 5.8, respectively. Observe that while a split S2 is visible in the individual signal and envelopogram illustrated in Figure 5.7, the split is not clearly seen in the averaged envelopogram and envelope, due to variations related to breathing over the duration of the signal and averaging.

Furthermore, based upon the method of Karpman et al. [246], the averaged envelopes were computed by taking the absolute value of the signal over each cardiac cycle, smoothing with a Butterworth lowpass filter with $N = 8$ and $f_c = 50$ Hz, and synchronized averaging. The last plots in Figures 5.7 and 5.8 show the averaged envelopes. (The Butterworth filter has introduced a small delay in the envelope.) The averaged envelopograms and averaged envelopes for the normal case display the envelopes of S1 and S2; the individual components of S1 and S2 have been smoothed over and merged in the averaged results. The averaged envelopograms and averaged envelopes for the case with murmur clearly demonstrate the envelopes of S1, the systolic murmur, the split S2, and the opening snap of the mitral valve.

5.6 Analysis of Activity

Problem: *Propose measures of waveform complexity or activity that may be used to analyze the extent of variability in signals such as the PCG and EMG.*

Solution: The samples of a given EMG or PCG signal may, for the sake of generality, be treated as a random variable x . Then, the variance $\sigma_x^2 = E[(x - \mu_x)^2]$ represents an averaged measure of the variability or *activity* of the signal about its mean. If the signal has zero mean, or is preprocessed to meet the same condition, we have $\sigma_x^2 = E[x^2]$; that is, the variance is equal to the average power of the signal. Taking the square root, we get the *SD* of the signal as equal to its *RMS* value. Thus, the *RMS* value could be used as an indicator of the level of activity about the mean of the signal. A much simpler indicator of activity is the number of zero-crossings within a specified interval; the zero-crossing rate (*ZCR*) increases as the high-frequency content of the signal increases. A few measures related to the concepts introduced above are described in the following sections, with illustrations of application.

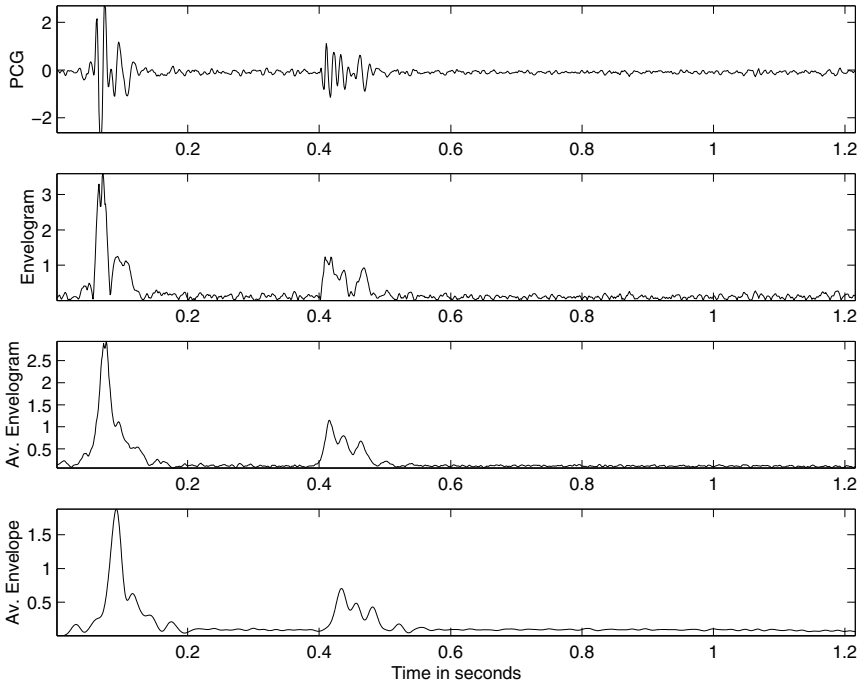


Figure 5.7 Top to bottom: PCG signal of a normal subject (male, 23 years); envelopegram estimate of the signal shown; averaged envelopegram over 16 cardiac cycles; averaged envelope over 16 cardiac cycles. The PCG signal starts with S1. See Figure 4.30 for an illustration of segmentation of the same signal. Av. stands for average.

5.6.1 The *RMS* value

The *RMS* value of a signal $x(n)$ over its total duration of N samples is given by

$$RMS = \left[\frac{1}{N} \sum_{n=0}^{N-1} x^2(n) \right]^{\frac{1}{2}}. \quad (5.23)$$

This global measure of signal level (related to power), however, is not useful for the analysis of trends in nonstationary signals. A running estimate of the *RMS* value of the signal computed over a causal window of M samples, defined as

$$RMS(n) = \left[\frac{1}{M} \sum_{k=0}^{M-1} x^2(n-k) \right]^{\frac{1}{2}}, \quad (5.24)$$

could serve as a useful indicator of the average power of the signal as a function of time. The duration of the window M needs to be chosen in accordance with the bandwidth of the signal, with $M \ll N$. Such an approach for computing

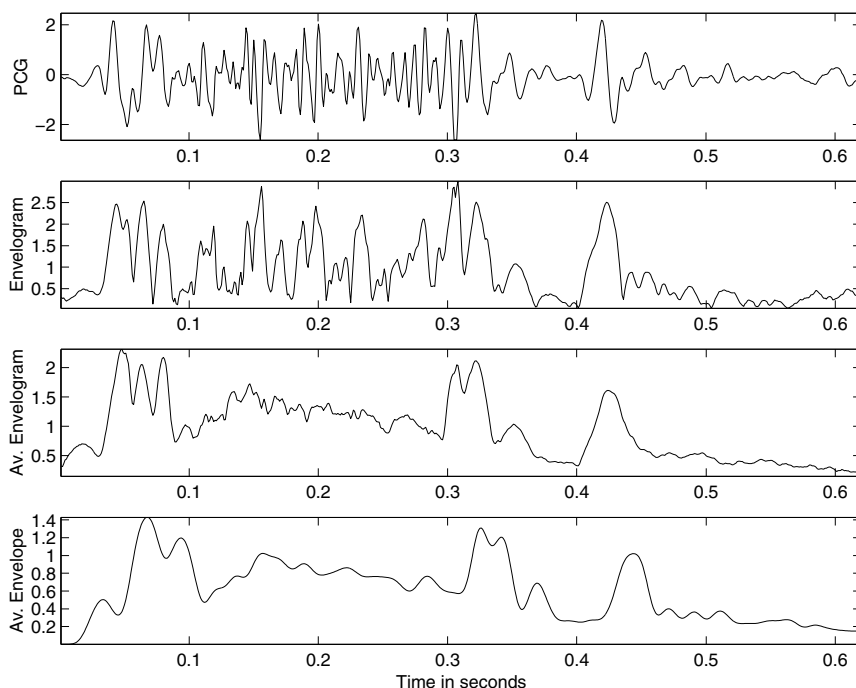


Figure 5.8 Top to bottom: PCG signal of a patient (female, 14 months) with systolic murmur (approximately $0.1 - 0.28$ s), split S2 ($0.28 - 0.38$ s), and opening snap of the mitral valve ($0.38 - 0.43$ s); envelopogram estimate of the signal shown; averaged envelopogram over 26 cardiac cycles; averaged envelope over 26 cardiac cycles. The PCG signal starts with S1. Av. stands for average. See Figure 4.31 for an illustration of segmentation of the same signal.

running parameters of signals falls under the general scheme of *short-time analysis* of nonstationary signals [79].

Gerbarg et al. [247, 248] derived power-versus-time curves of PCG signals by computing the average power in contiguous segments of duration 10 ms, and they used the curves to identify systolic and diastolic segments of the signals. They noted that within an interval of 10 s of a PCG signal, at least one diastolic segment would be longer than the corresponding systolic segment, and that all systolic segments in the interval would have approximately the same duration. Innocent (physiological) systolic murmurs in children were observed to be limited to the first and middle thirds of the systolic interval between S1 and S2, whereas pathological systolic murmurs due to mitral regurgitation were noted to be holosystolic. Based upon these observations, Gerbarg et al. computed ratios of the mean power of the last third of systole to the mean power of systole and also to a certain “standard” noise level. A ratio was also computed of the mean energy of systole to the mean energy of the PCG over the complete cardiac cycle. Agreement in the range of $78 - 91\%$ was

obtained between computer classification based upon the three ratios defined above and clinical diagnosis of mitral regurgitation in different groups of subjects.

Use of the *RMS* value for the analysis of EMG and VMG signals, and thereby analysis of muscular activity, is illustrated in Section 5.11.

5.6.2 Zero crossing rate

An intuitive indication of the “busy-ness” of a signal is provided by the number of times it crosses the zero-activity line or some other reference level. *ZCR* is defined as the number of times the signal crosses the reference within a specified interval. However, *ZCR* could be easily affected by DC bias, baseline wander, and low-frequency artifacts. For these reasons, it would be advisable to measure the *ZCR* of the derivative of the signal, which would be similar to the definition of turning points in the test for randomness described in Section 3.2.1. Saltzberg and Burch [249] discuss the relationship between *ZCR* and moments of PSDs, and their application to EEG analysis.

In spite of its simplicity, *ZCR* has been used in practical applications such as speech signal analysis to perform speech-versus-silence decision and to discriminate between voiced and unvoiced sounds [79] (see also Figure 3.1), and PCG analysis for the detection of murmurs. Jacobs et al. [250] used *ZCR* to perform normal-versus-abnormal classification of PCG signals using the ECG as a trigger, and obtained correct classification rates of 95% for normals (58/61) and 94% for abnormals (77/82). They indicated a decision limit of 20 zero-crossings in a cardiac cycle. Yokoi et al. [251] proposed a screening system based upon measurements of the maximum amplitude and *ZCR* in 8 *ms* segments of PCG signals (sampled at 2 *kHz*). They obtained correct classification rates of 98% with 4,809 normal subjects and 76% with 1,217 patients with murmurs.

5.6.3 Turns count

Willison [252] proposed to analyze the level of activity in EMG signals by determining the number of spikes occurring in the interference pattern (see also Goodgold and Eberstein [30], Fuglsang-Frederiksen and Månsson [253], and Dowling et al. [254]). Instead of counting zero-crossings, Willison’s method investigates the significance of every change in phase (direction or slope) of the EMG signal called a *turn*. Turns greater than 100 μV are counted, with the threshold selected so as to avoid counting insignificant fluctuations due to noise. The method is similar to counting turning points as in the test for randomness described in Section 3.2.1, but is expected to be robust in the presence of noise due to the threshold imposed. The method is not directly sensitive to SMUAPs, but significant phase changes caused by superimposed SMUAPs are counted. Willison [252] found that EMG signals of subjects with myopathy possessed higher turns counts than those of normal subjects at comparable levels of volitional effort.

Illustration of application: The topmost plot in Figure 5.9 illustrates the EMG signal over two breath cycles from the crural diaphragm of a dog recorded via im-

planted fine-wire electrodes [40]. The subsequent plots illustrate, in top-to-bottom order, the short-time *RMS* values, the turns count by Willison's procedure, and the smoothed envelope of the signal. The *RMS* and turns count values were computed using a causal moving window of duration 70 *ms* (210 samples). The window duration needs to be chosen to strike a balance between the extent of smoothing desired in the turns count series and the accuracy in reflecting the nonstationary nature of the signal; in the present example, nonstationarity is related to the increasing level of EMG activity with inspiration. The envelope was obtained by taking the absolute value of the signal (equivalent to full-wave rectification) followed by a Butterworth lowpass filter of order $N = 8$ and cutoff frequency $f_c = 8$ *Hz*. It is seen that all three of the derived features demonstrate the expected increasing trend with the level of contraction (breath), and can serve as correlates or indicators of muscle contraction and the concomitant EMG complexity. The results may be further smoothed (lowpass filtered) if desired.

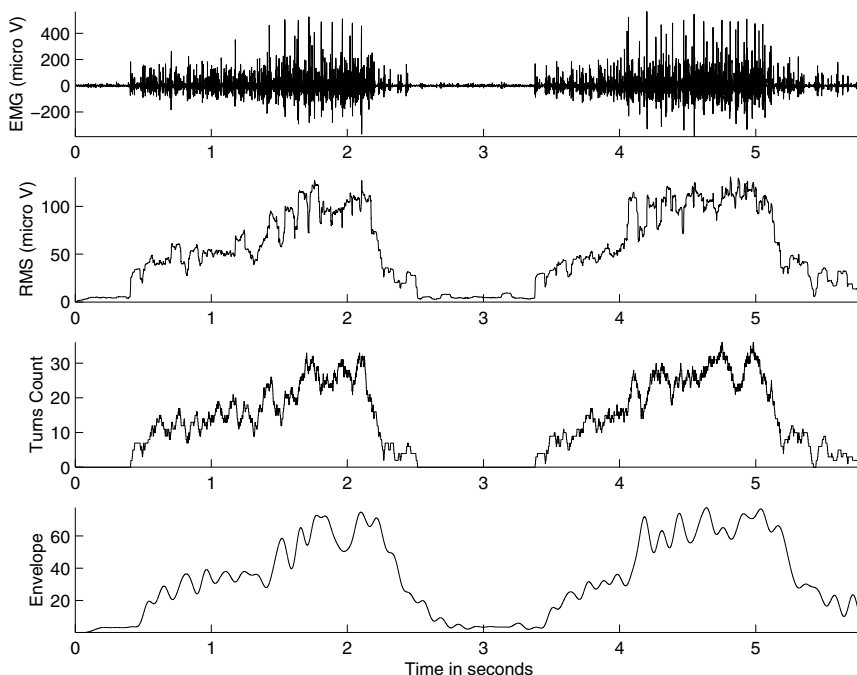


Figure 5.9 Top to bottom: EMG signal over two breath cycles from the crural diaphragm of a dog recorded via implanted fine-wire electrodes; short-time *RMS* values; turns count using Willison's procedure; and smoothed envelope of the signal. The *RMS* and turns count values were computed using a causal moving window of 70 *ms* duration. EMG signal courtesy of R.S. Platt and P.A. Easton, Department of Clinical Neurosciences, University of Calgary.

Figure 5.10 illustrates one 70 *ms* segment of the EMG signal in Figure 5.9 with the boundary points of the significant turns as detected by Willison's proce-

ture marked by the ‘*’ symbol. The procedure was implemented by first computing the derivative of the EMG signal and detecting points of change in its sign. A turn was marked wherever the EMG signal differed by at least $100 \mu V$ between successive points of sign change in the derivative. Observe from Figure 5.10 that the EMG signal need not cross the zero line to cause a turns count, and that zero-crossings with voltage swings of less than $100 \mu V$ are not counted as turns.

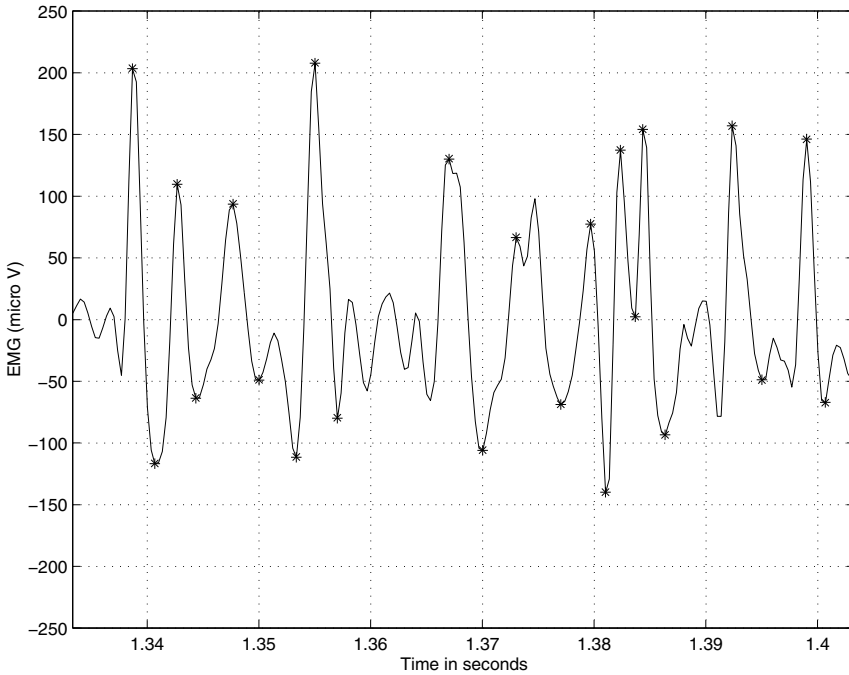


Figure 5.10 Illustration of the detection of turns in a 70 ms window of the EMG signal in Figure 5.9. The segments of the signal between pairs of ‘*’ marks have been identified as significant turns.

5.6.4 Form factor

Based upon the notion of variance as a measure of signal activity, Hjorth [255–257] (see also Cooper et al. [54]) proposed a method for the analysis of EEG waves. In this method, short-time segments of duration 1 s or longer are analyzed and three parameters are computed. The first parameter is called *activity* and is simply the variance σ_x^2 of the signal segment $x(n)$. The second parameter, called *mobility* M_x , is computed as the square root of the ratio of the activity of the first derivative of the

signal to the activity of the (original) signal:

$$M_x = \left[\frac{\sigma_{x'}^2}{\sigma_x^2} \right]^{\frac{1}{2}} = \frac{\sigma_{x'}}{\sigma_x}, \quad (5.25)$$

where x' stands for the first derivative of x . The third parameter, called *complexity* or the *form factor* (FF), is defined as the ratio of the mobility of the first derivative of the signal to the mobility of the signal itself:

$$FF = \frac{M_{x'}}{M_x} = \frac{\sigma_{x''}/\sigma_{x'}}{\sigma_{x'}/\sigma_x}, \quad (5.26)$$

where x'' stands for the second derivative of the signal. The complexity of a sinusoidal wave is unity; other waveforms have complexity values increasing with the extent of variations present in them.

Hjorth [256, 257] described the mathematical relationships between the activity, mobility, complexity, and PSD of a signal, and applied them to model EEG signal generation. Binnie et al. [258, 259] described the application of FF and spectrum analysis to EEG analysis for the detection of epilepsy. However, because the computation of FF is based upon the first and second derivatives of the signal and their variances, the measure is sensitive to noise. A complex and relatively wideband signal such as the EMG is not amenable to analysis via FF . Application of FF to discriminate between normal and ectopic ECG beats is illustrated in Section 5.7.

We have explored a few measures to characterize waveform complexity in this section. Many authors have proposed several other diverse measures and interpretations of waveform or system complexity in the literature, examples of which include features based upon nonlinear dynamics and the correlation dimension [260], and the embedding dimension of time-varying dynamic systems [261].

5.7 Application: Parameterization of Normal and Ectopic ECG Beats

Problem: *Develop a parameter to discriminate between normal ECG waveforms and ectopic beats (PVCs).*

Solution: We have observed several times that ectopic beats, due to the abnormal propagation paths of the associated excitation pulses, typically possess waveforms that are significantly different from those of the normal QRS waveforms of the same subject. More often than not, ectopic beats have bizarre and complex waveshapes. The form factor FF described in Section 5.6.4 parameterizes the notion of waveform complexity, providing a value that increases with complexity. Therefore, FF appears to be a suitable measure to discriminate between normal and ectopic beats. Note that the RR interval by itself cannot indicate ectopic beats, as the RR interval could vary due to sinus arrhythmia and conduction problems, as well as due to HRV.

Figure 5.11 displays a segment of the ECG of a patient with ectopic beats; the segment illustrates the initiation of an episode of *ventricular bigeminy*, where every normal beat is followed by an ectopic beat [33]. The ECG of the patient was

processed using the Pan–Tompkins algorithm for QRS detection (see Section 4.3.2). QRS marker points were detected using a simple threshold applied to the output of the Pan–Tompkins algorithm. Each beat was segmented at points 160 *ms* before and 240 *ms* after the detected marker point; the diamond and circle symbols on the ECG in Figure 5.11 indicate the starting and ending points of the corresponding beats. The *FF* value was computed for each segmented beat. The *RR* interval (in *ms*) and the *FF* value are shown for each beat in Figure 5.11. It can be readily seen that the *FF* values for the PVCs are higher than those for the normal beats.

Note from Figure 5.11 that the *RR* intervals for the PVCs are lower than those for the normal beats, and that the normal beats that follow the PVCs have higher-than-normal *RR* intervals due to the compensatory pause. Pattern classification of the ECG beats in this example as normal or PVCs using *RR* and *FF* is described in Section 9.11.

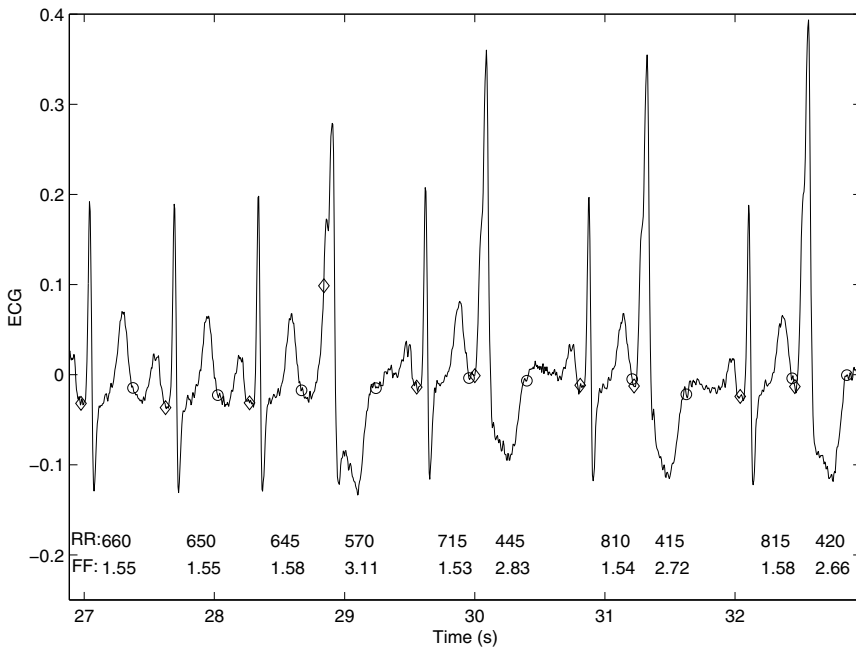


Figure 5.11 Segment of the ECG of a patient (male, 65 years) with ectopic beats. The diamond and circle symbols indicate the starting and ending points, respectively, of each QRS-T wave obtained using the Pan–Tompkins algorithm. The *RR* interval (in *ms*) and the *FF* value are shown for each beat.

5.8 Application: Analysis of Exercise ECG

Problem: Develop an algorithm to analyze changes in the *ST* segment of the ECG during exercise.

Solution: Hsia et al. [262] developed a method to analyze changes in the ST segment of the ECG signal while the subject performed physical exercises. The analysis was performed as part of a radionuclide ventriculography (gated blood-pool imaging) procedure. In this procedure, nuclear medicine images are obtained of the left ventricle before and after the patient performs exercises on a treadmill or bicycle ergometer. Images are obtained at different phases of the cardiac cycle by gating the radionuclide (gamma ray) emission data with reference to the ECG; image data for each phase are averaged over several cardiac cycles [18]. Analysis of exercise ECG is complicated due to baseline artifacts caused by the effects of respiration, skin resistance changes due to perspiration, and soft tissue movement affecting electrode contact. Detection of changes in the ST segment in the presence of such artifacts poses a major challenge.

One of the main parameters used by Hsia et al. is related to the correlation coefficient as defined in Equation 3.96. The measure, however, is affected by baseline variations. To address this, a modified correlation coefficient was defined as

$$\gamma_{xy} = \frac{\sum_{n=0}^{N-1} [x(n)][y(n) - \Delta]}{\sqrt{\sum_{n=0}^{N-1} [x(n)]^2 \sum_{n=0}^{N-1} [y(n) - \Delta]^2}}. \quad (5.27)$$

Here, $x(n)$ is the template, $y(n)$ is a segment of the ECG signal being analyzed, Δ is a baseline correction factor defined as the difference between the baseline of $y(n)$ and the baseline of $x(n)$, and N is the duration (number of samples) of the template and the signal segment being analyzed. The template was generated by averaging up to 20 QRS complexes that met a specified RR interval constraint.

Hsia et al. proposed a method to establish the baseline of each ECG beat by searching for the PQ segment by backtracking from the detected R point (trigger for gating the image data). The region of three consecutive samples with the minimum change (maximum flatness) preceding the QRS was taken to represent the baseline of the beat. (*Note:* The PQ segment is almost always isoelectric, whereas the ST segment is variable in the presence of cardiac diseases.) The search procedure also established the width of the QRS complex to be used in template matching (N in Equation 5.27). Beats with $\gamma_{xy} < 0.85$ were considered to be abnormal. The baseline correction factor in Equation 5.27 provided the robustness required.

Groups of 16 successive normal beats were aligned and averaged to obtain a representative waveform. The ST segment level was computed as the difference between a reference ST point and the isoelectric level of the current averaged beat. The averaging procedure included a condition to reject beats with abnormal morphology, such as PVCs. The ST reference point was defined as $R + 64 \text{ ms} + \max(4, \frac{200 - HR}{16}) \times 4 \text{ ms}$ or $S + 44 \text{ ms} + \max(4, \frac{200 - HR}{16}) \times 4 \text{ ms}$, where R or S indicates the position of the R or S wave of the present beat in ms , and HR is in bpm . Significant ST level differences were reported by the algorithm. (Elevation or depression of the ST segment by more than 0.1 mV with reference to the baseline is considered to be significant.) Furthermore, the slope of the ST segment was computed by using two samples before and two samples after the detected ST point as described above (a duration of 16 ms , with the sampling rate of 250 Hz).

In addition to the analysis of the ST segment, the method of Hsia et al. performed rhythm analysis, identified PVCs and other abnormal beats, and assisted in the rejection of radionuclide emission data related to abnormal beats from the imaging procedure. The combined use of nuclear medicine imaging and ECG analysis can improve the accuracy of the diagnosis of myocardial ischemia.

5.9 Application: Quantitative Analysis of the EMG in Relation to Force Exerted

Problem: *Propose methods for parametric analysis of the variations in the EMG signal with respect to the force exerted by a muscle.*

Solution: We have seen several examples demonstrating increasing levels of power and complexity of the EMG signal with increasing muscular activity or force exerted; see, for example, Figures 1.18, 1.19, 1.20, and 5.9. We have also studied how such variations in a signal may be represented quantitatively using measures such as the *RMS* value, *ZCR*, and turns count. Let us now explore the relationships between these entities in a formal and mathematical manner.

Figure 5.12 shows the ECG and force signals as in Figure 1.20, but with automatic delineation of the portions of the force signal corresponding to the five intervals of muscular contraction. To perform this task, starting from the first sample, the point where the force signal increased beyond 10% MVC was identified. Then, the next point where the signal dropped below 10% MVC was identified. This process was repeated until the end of the signal. To refine the definition of each interval of contraction, a threshold was defined as 0.7 times the maximum level of contraction within the interval. Then, the smaller extent of each interval previously identified, within which the force remained above the threshold, was detected. The final intervals detected by this procedure are labeled in Figure 5.12 with ‘*’ marks.

Within each interval of the force signal identified as above as well as the corresponding interval in the EMG signal, the average force exerted, the *RMS* value, *ZCR*, and the turns count divided by the time duration of the interval (referred to as the turns count rate or *TCR*) were computed. The threshold to detect significant turns was set at 100 μV .

Figure 5.13 demonstrates the important difference between zero-crossings and turns. The cumulative number of zero-crossings detected keeps increasing with time even during the intervals when no muscular force is exerted, due to noisy variations in the EMG signal during the periods of rest. The threshold to detect significant turns, which was set at 100 μV in this experiment, disregards the small turns associated with the background variations in the EMG signal, as well as small variations during periods of contraction. Thus, the cumulative number of significant turns detected is seen to increase only during the periods of contraction and not during the periods of rest. Therefore, we may expect *TCR* to be a better indicator of muscular force exerted than *ZCR*.

In order to analyze the variation of the EMG signal with force in a quantitative manner, a linear fit was computed to represent the variation of each EMG parameter

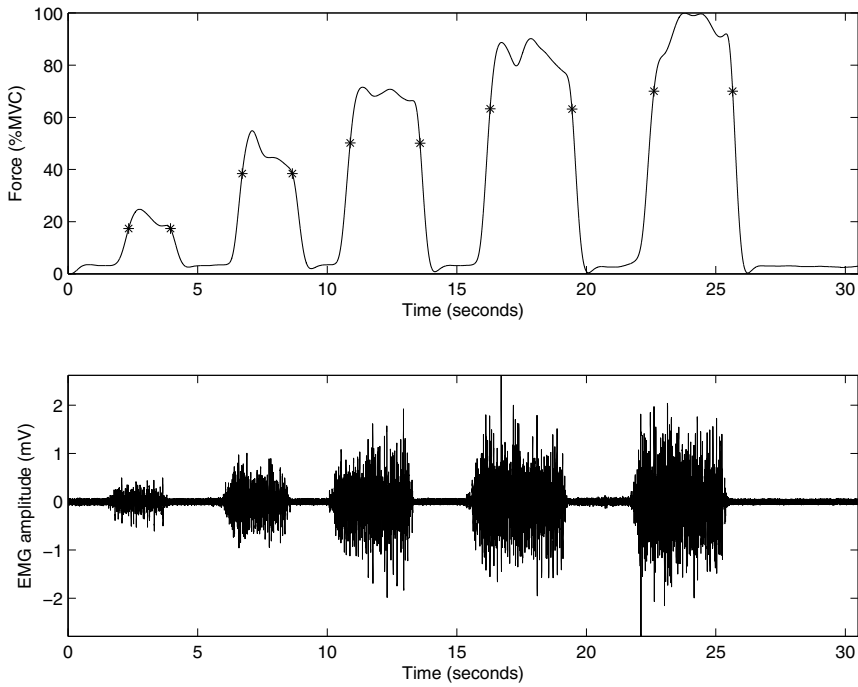


Figure 5.12 Force and EMG signals recorded from the forearm muscle of a subject using surface electrodes. The portions of muscular contraction have been automatically identified in the force signal and delimited by the “*” marks.

versus force. The linear models (straight-line fits) obtained are shown in the plots in Figures 5.14, 5.15, and 5.16 for the *RMS*, *ZCR*, and *TCR* values, respectively. To obtain a measure of fit between each parameter and force, the correlation coefficient was computed as

$$r^2 = \frac{\left[\sum_{n=1}^{n=N} x(n)y(n) - N\bar{x}\bar{y} \right]^2}{\left[\sum_{n=1}^{n=N} x^2(n) - N\bar{x}^2 \right] \left[\sum_{n=1}^{n=N} y^2(n) - N\bar{y}^2 \right]}, \quad (5.28)$$

where N is the number of samples of x or y representing the variables *RMS*, *ZCR*, *TCR*, or force ($N = 5$ in the present study); $\bar{x}$ is the mean of x . The values of r^2 for *RMS*, *ZCR*, and *TCR* are given in the captions of Figures 5.14, 5.15, and 5.16, respectively. The value of *ZCR* shows limited variation and low correlation with force. It is evident that *RMS* and *TCR* have a high degree of correlation and linear behavior with respect to force.

The study described above is limited due to the use of a single EMG signal and a simple method to delimit the intervals of muscular contraction. It is seen in Figure 5.12 that the muscular force exerted is not constant within each interval. For

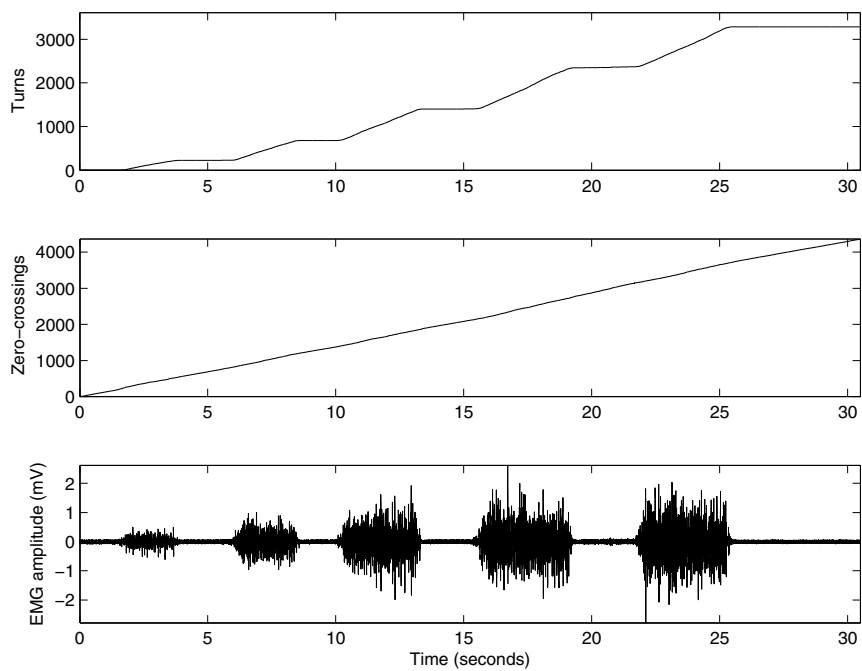


Figure 5.13 Bottom to top: EMG signal, cumulative count of zero-crossings, and cumulative number of significant turns. See also Figure 5.12.

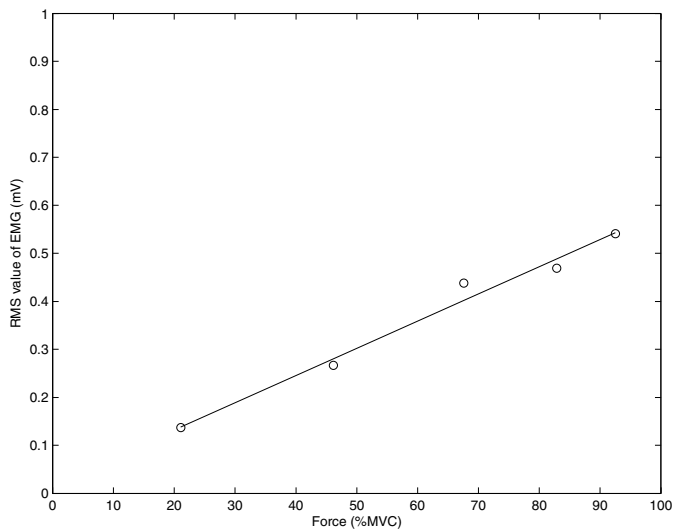


Figure 5.14 Variation of the *RMS* value of the EMG signal in Figure 5.12 with the average muscular force exerted in each of the five periods of contraction. The ‘o’ marks indicate the measured values and the straight line indicates the computed linear model. $r^2 = 0.98$.

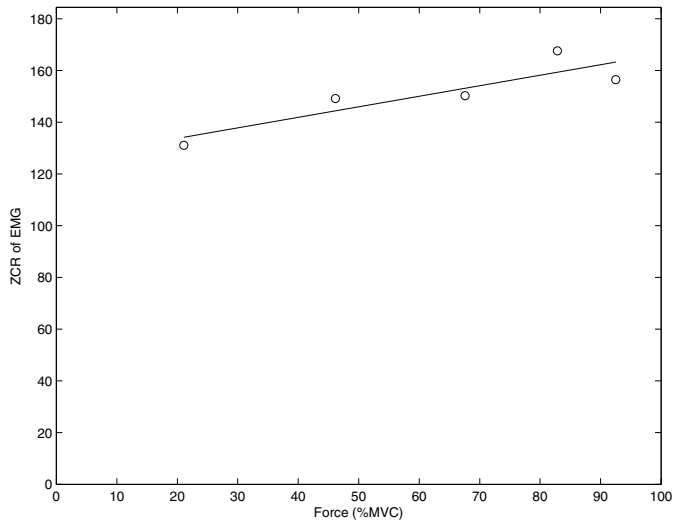


Figure 5.15 Variation of the ZCR value of the EMG signal in Figure 5.12 with the average muscular force exerted in each of the five periods of contraction. The ‘o’ marks indicate the measured values and the straight line indicates the computed linear model. $r^2 = 0.78$.

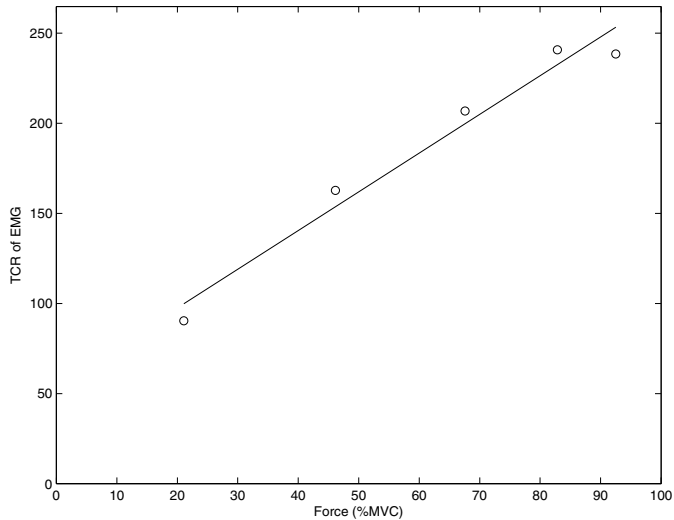


Figure 5.16 Variation of the TCR value of the EMG signal in Figure 5.12 with the average muscular force exerted in each of the five periods of contraction. The ‘o’ marks indicate the measured values and the straight line indicates the computed linear model. $r^2 = 0.97$.

robust analysis, it is necessary to repeat the experiment several times with many subjects and analyze a large number of samples of the parameters. See Section 5.11 for further related discussion and illustration of the relationships between EMG and force.

5.10 Application: Analysis of Respiration

Problem: *Propose a method to relate EMG activity to airflow during respiration.*

Solution: Platt et al. [40] recorded EMG signals from the parasternal intercostal and crural diaphragm muscles of dogs. One EMG signal was obtained from a pair of electrodes mounted at a fixed distance of 2 mm placed between fibers in the third left parasternal intercostal muscle about 2 cm from the edge of the sternum. The crural diaphragm EMG was obtained via fine-wire electrodes sewn in-line with the muscle fibers and placed 10 mm apart. During the signal acquisition experiment, the dog breathed through a snout mask, and a pneumotachograph was used to measure airflow. Figures 1.18, 1.19, and 5.9 show samples of the crural EMG signal.

Although the EMG signal is commonly used in many physiological studies including analysis of respiration, the intricate variations in the signal are often not of interest. A measure of the total or integrated electrical activity, ideally reflecting the global activity in the pool of active motor units of the muscle, would serve the purposes of most analyses [40]. As the EMG signal is nonstationary, short-time measures are called for. The smoothed envelope of the EMG signal is commonly used under these circumstances.

Platt et al. observed that the filters commonly used for smoothing rectified EMG signals had poor high-frequency attenuation, resulting in noisy envelopes. They proposed a modified Bessel filter for application to the EMG signal after full-wave rectification; the filter severely attenuated frequencies beyond 20 Hz with gain < -70 dB, and yielded EMG envelopes that were much smoother than those given by other filters.

The EMG envelopes derived by Platt et al. agreed very well with the inspiratory airflow pattern. Figure 5.17 shows plots of the parasternal intercostal EMG signal over two breath cycles, the corresponding filtered envelope, and the airflow pattern. Figure 5.18 shows the correlation between the filtered EMG envelope amplitude and the airflow in liters per second. It is evident that the envelope extracted by this method is an excellent correlate of inspiratory airflow.

5.11 Application: Electrical and Mechanical Correlates of Muscular Contraction

Problem: *Derive parameters from the electrical and mechanical manifestations of muscular activity that correlate with the level of contraction or force produced.*

Solution: Zhang et al. [85, 86] studied the usefulness of simultaneously recorded EMG and VMG signals in the analysis of muscular force produced by contraction.

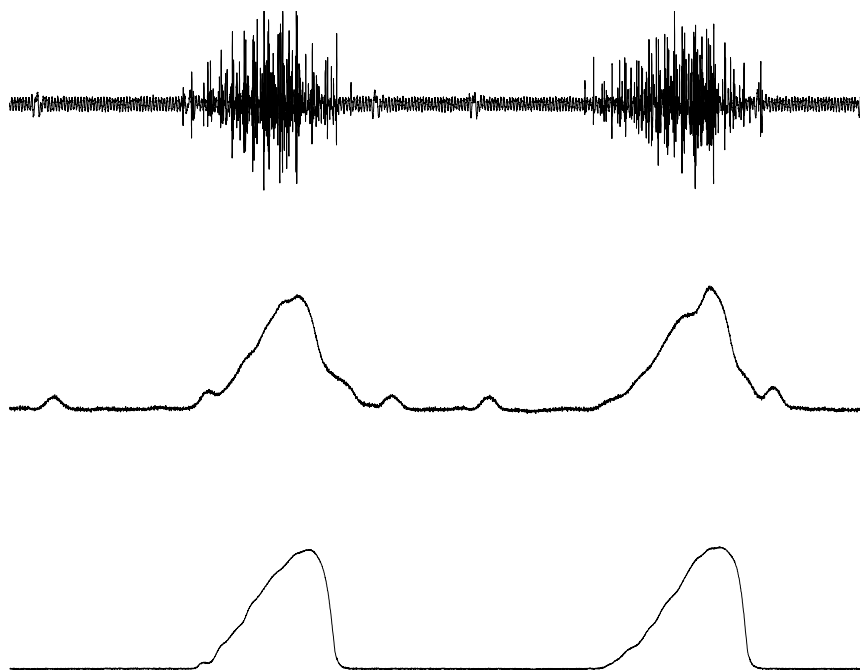


Figure 5.17 Top to bottom: EMG signal over two breath cycles from the parasternal intercostal muscle of a dog recorded via implanted electrodes; EMG envelope obtained with the modified Bessel filter with a time constant of 100 ms ; and inspiratory airflow. The duration of the signals plotted is 5 s . The several minor peaks appearing in the envelope are related to the ECG which appears as an artifact in the EMG signal. Data courtesy of R.S. Platt and P.A. Easton, Department of Clinical Neurosciences, University of Calgary [40].

In their experimental procedure, the subjects performed isometric contraction (that is, with no movement of the associated leg) of the rectus femoris (thigh) muscle to different levels of torque with a Cybex II dynamometer. Four levels of contraction were performed from 20% to 80% of the MVC level of the individual subject. The experiments were performed at three knee-joint angles of 30° , 60° , and 90° . Each contraction was held for a duration of about 6 s , and the subjects rested in between experiments to prevent the development of muscle fatigue. The VMG signal was recorded using a Dytran 3115a accelerometer, and surface EMG signals were recorded using disposable $\text{Ag} - \text{AgCl}$ electrodes. The VMG signals were filtered to the bandwidth $3 - 100\text{ Hz}$ and the EMG signals were filtered to $10 - 300\text{ Hz}$. The VMG and EMG signals were sampled at 250 Hz and $1,000\text{ Hz}$, respectively. Figure 2.4 illustrates sample recordings of the VMG and EMG signals at two levels of contraction.

RMS values were computed for each contraction level over a duration of 5 s . Figure 5.19 shows the variation of the *RMS* values of the EMG and VMG signals

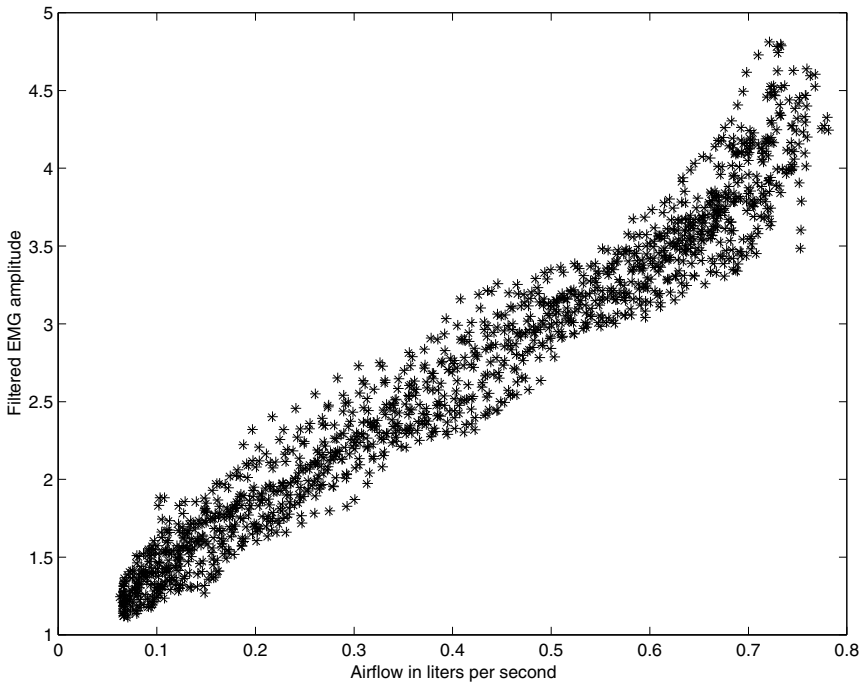


Figure 5.18 Correlation between EMG amplitude obtained from the Bessel-filtered envelope versus inspiratory airflow. The EMG envelope was filtered using a modified Bessel filter with a time constant of 100 ms. Data courtesy of R.S. Platt and P.A. Easton, Department of Clinical Neurosciences, University of Calgary [40].

acquired at a knee-joint angle of 60° and averaged over four subjects. The almost-linear trends of the *RMS* values of both the signals with muscular contraction indicate the usefulness of the derived parameter in the analysis of muscular activity. It should, however, be noted that the relationship between *RMS* values and contraction may not follow the same (linear) pattern for different muscles. Figure 5.20 shows the *RMS*-versus-%MVC relationships for three muscles: The relationship is linear for the FDI, whereas it is nonlinear for the biceps and deltoid muscles [263].

5.12 Application: Statistical Analysis of VAG Signals

Problem: *Knee-joint VAG signals possess different levels of waveform complexity depending upon the state of the cartilage covering the joint surfaces. Explore the use of statistical parameters that characterize variability for parametric representation and classification of VAG signals.*

Solution: As described in Section 1.2.14, a normal knee joint has smooth cartilage surfaces and produces almost no sound or vibration; see also Section 8.2.3.

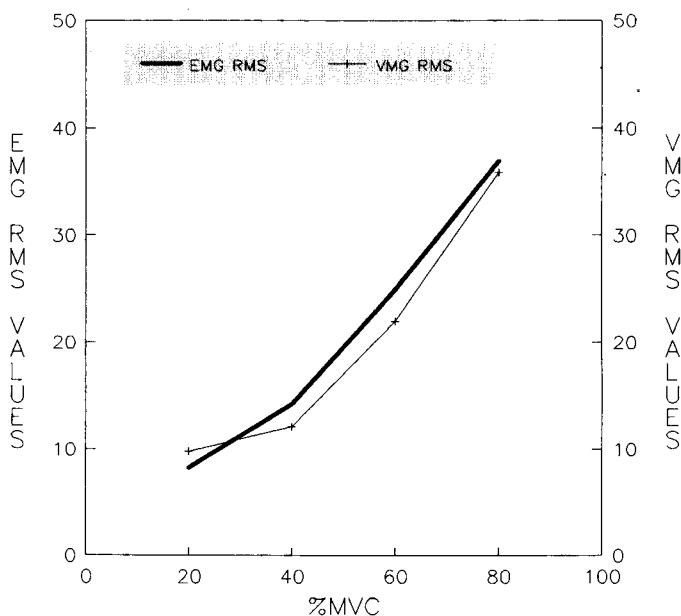


Figure 5.19 *RMS values of the VMG and EMG signals for four levels of contraction of the rectus femoris muscle at 60° knee-joint angle averaged over four subjects. Reproduced with permission from Y.T. Zhang, C.B. Frank, R.M. Rangayyan, and G.D. Bell, Relationships of the vibromyogram to the surface electromyogram of the human rectus femoris muscle during voluntary isometric contraction, *Journal of Rehabilitation Research and Development*, 33(4): 395–403, 1996. ©Department of Veterans Affairs.*

Regardless, when a VAG signal is recorded from a normal knee joint, a signal with some variations is obtained. Under pathological conditions, the cartilage surfaces are eroded and could generate additional sounds. We could expect the statistics of the variability of normal and abnormal VAG signals to demonstrate differences due to the different nature of the underlying causes. Rangayyan and Wu [264–266] explored the use of several statistical parameters for screening VAG signals. The related background and methods are described in the following paragraphs.

5.12.1 Acquisition of knee joint VAG signals

The data set used in the studies of Rangayyan and Wu [264–266] consists of 89 signals, with 51 from normal volunteers (22 male, 29 female, age 28 ± 9.5 years) and 38 from subjects with knee-joint pathology (20 male, 18 female, age 35 ± 13.8 years). The normal condition of volunteers was established by clinical examination and history. The abnormal signals were collected from symptomatic patients scheduled to undergo arthroscopy independent of the VAG studies. Informed consent was ob-

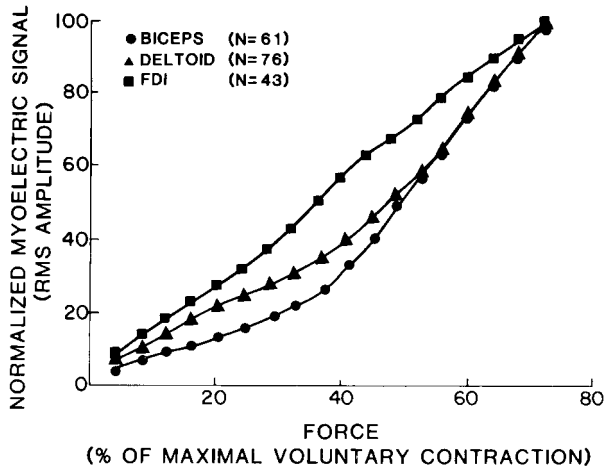


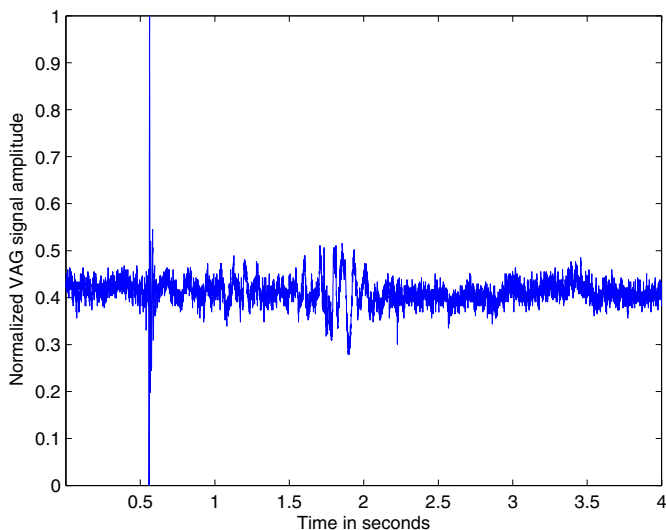
Figure 5.20 EMG *RMS* value versus level of muscle contraction expressed as a percentage of MVC for each subject. The relationship is displayed for three muscles. FDI: first dorsal interosseus. N: number of muscles (subjects) in the study. Reproduced with permission from J.H. Lawrence and C.J. de Luca, Myoelectric signal versus force relationship in different human muscles, *Journal of Applied Physiology*, 54(6):1653–1659, 1983. ©American Physiological Society.

tained from each subject. The experimental protocol was approved by the Conjoint Health Research Ethics Board of the University of Calgary.

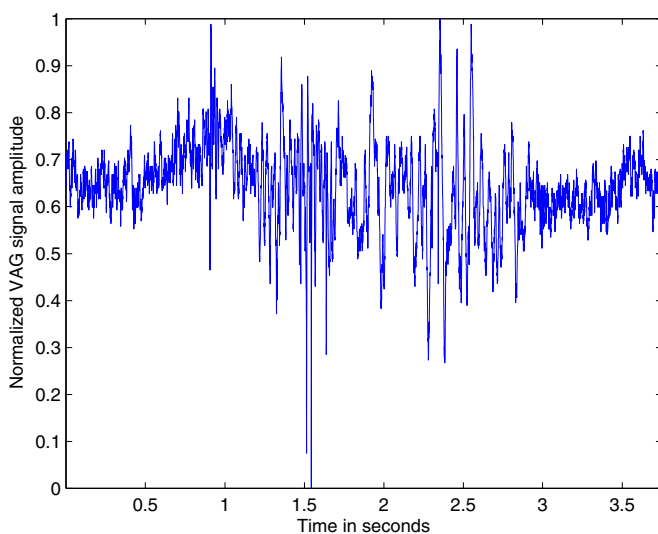
Each subject sat on a rigid table in a relaxed position with the leg to be tested being freely suspended in air. The VAG signal was recorded by an accelerometer (model 3115a, Dytran, Chatsworth, CA), placed at the midpatella position of the knee, as the subject swung the leg over an approximate angle range of 135° (approximately full flexion) to 0° (full extension) and back to 135° in 4 s [96]. The first half of each VAG signal corresponds, approximately, to extension, and the second half corresponds to flexion of the leg. The VAG signals were prefiltered (10 Hz to 1 kHz) and digitized at the sampling rate of 2 kHz.

The abnormal cases in the data set include chondromalacia of different grades at the patella, meniscal tear, tibial chondromalacia, and anterior cruciate ligament injuries, as confirmed during arthroscopic examination. Due to the inadequacy of the data available for classification of the signals into various types or stages of pathology, studies using this data set have been limited to screening, that is, classification of the signals and the corresponding knee joints as normal or abnormal.

Figure 5.21 shows examples of normal and abnormal VAG signals. Each signal was normalized to the amplitude range $[0, 1]$. The abnormal signal exhibits a higher degree of overall variation, activity, or complexity than the normal signal.



(a)



(b)

Figure 5.21 VAG signal of (a) a normal subject and (b) a subject with knee-joint pathology. Reproduced from R.M. Rangayyan and Y.F. Wu, Screening of knee-joint vibroarthrographic signals using probability density functions estimated with Parzen windows, *Biomedical Signal Processing & Control*, January 2010, 5(1):53–58, with permission from Elsevier. ©Elsevier.

5.12.2 Estimation of the PDFs of VAG signals

In the work of Rangayyan and Wu [266], models of the PDFs of VAG signals were derived using the Parzen-window approach [267, 268] to represent their basic statistical characteristics. A histogram was computed by combining all of the normal signals into one group. The histogram, denoted by $h_n(x_l)$, with x_l , $l = 0, 1, 2, \dots, L - 1$, represents the L bins used to quantize the range of the values of the signal x . Rangayyan and Wu used $L = 100$ bins to represent the normalized range of $[0, 1]$ for the VAG signal values. A large value for L could result in several bins with negligible or zero counts, whereas a small value could cause diminished differences between the histograms for normal and abnormal signals. The value of $L = 100$ was selected based on an analysis of the results with several values of L . Similarly, a histogram $h_a(x_l)$ was obtained by pooling together all of the abnormal signals. Each histogram was normalized by dividing by the total number of samples in the population to have unit area under the normalized histogram. A Gaussian fit was then obtained for each of the two histograms, denoted as $g_n(x_l)$ and $g_a(x_l)$.

The Parzen-window approach to obtain a nonparametric estimate of the PDF from a collection of samples was applied as follows [267, 268]. Consider the situation where we have a set of independent samples, $Z = \{z_1, z_2, \dots, z_K\}$, with an unknown underlying PDF $p(z)$. A nonparametric estimate of $p(z)$ from Z is provided by the function [267, 268]

$$\hat{p}(z) = \frac{1}{K} \sum_{k=1}^K \kappa(z - z_k), \quad (5.29)$$

where κ is a window or kernel function that integrates to unity. Rangayyan and Wu [266] used

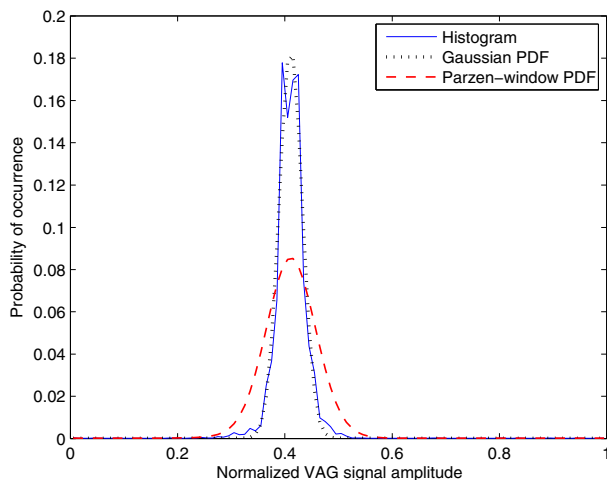
$$\kappa(z - z_k) = \frac{1}{\sigma_P \sqrt{2\pi}} \exp \left[-\frac{(z - z_k)^2}{2\sigma_P^2} \right]. \quad (5.30)$$

The Parzen-window PDF was estimated using the quantized values of the signals, x_l , $l = 0, 1, 2, \dots, L - 1$, with $L = 100$ levels, in the normalized range $[0, 1]$. Experiments were conducted with the value of the parameter σ_P in Equation 5.30 varying over the range $[0.01, 0.1]$ in steps of 0.01; the final value was set equal to 0.04.

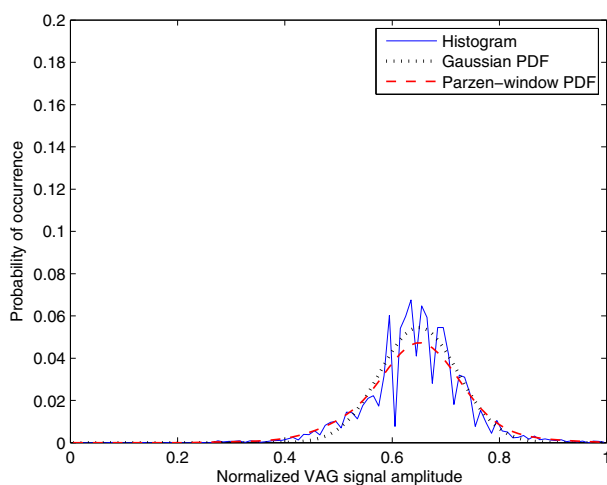
Figure 5.22 shows the Parzen-window estimates of the PDFs of the normal and abnormal VAG signals shown in Figure 5.21. Figure 5.23 shows the Parzen-window estimates of the PDFs of the 51 normal and 38 abnormal VAG signals in the data set used. The amplitude of the VAG signals has been normalized to the range $[0, 1]$. The figures also show the normalized histogram and the Gaussian fit for each case.

The following observations were made from the PDFs of the normal and abnormal VAG signals:

- In general, the Parzen-window PDFs are closer to the normalized histograms of the VAG signals than the corresponding Gaussian models. The closeness of the fit depends upon the value of σ_P used: The smaller the value of σ_P , the better

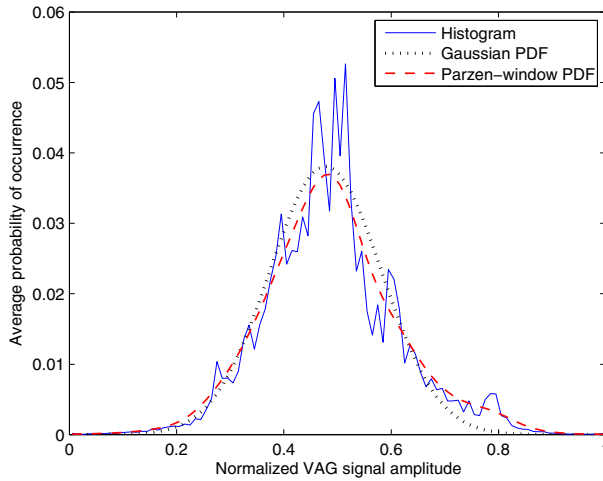


(a)

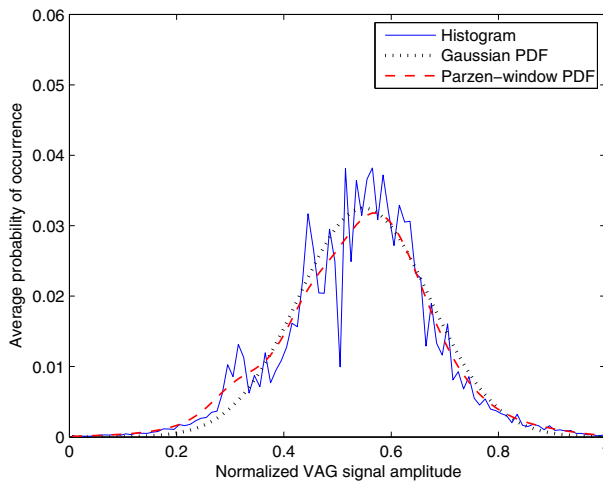


(b)

Figure 5.22 Nonparametric Parzen-window estimates of the PDFs of the VAG signals in Figure 5.21: (a) of a normal subject and (b) of a patient with knee-joint pathology. The amplitude has been normalized to the range $[0, 1]$. The figure also shows the normalized histogram and the Gaussian fit for each case. The parameters of the Gaussian fit are (a) mean $= 0.4107$, $SD = 0.0216$; and (b) mean $= 0.6499$, $SD = 0.0730$. Reproduced from R.M. Rangayyan and Y.F. Wu, Screening of knee-joint vibroarthrographic signals using probability density functions estimated with Parzen windows, *Biomedical Signal Processing & Control*, January 2010, 5(1):53–58, with permission from Elsevier. ©Elsevier.



(a)



(b)

Figure 5.23 Nonparametric Parzen-window estimates of the PDFs of VAG signals: (a) derived from VAG signals of 51 normal volunteers and (b) derived from VAG signals of 38 subjects with knee-joint pathology. The amplitude has been normalized to the range $[0, 1]$. The figure also shows the normalized histogram and the Gaussian fit for each case. The parameters of the Gaussian fit are (a) mean = 0.4778, $SD = 0.1047$; and (b) mean = 0.5495, $SD = 0.1227$. Reproduced from R.M. Rangayyan and Y.F. Wu, Screening of knee-joint vibroarthrographic signals using probability density functions estimated with Parzen windows, *Biomedical Signal Processing & Control*, January 2010, 5(1):53–58, with permission from Elsevier. ©Elsevier.

the fit. However, a larger value of σ_P is desirable to obtain smooth model PDFs that do not include irrelevant details of the histograms of the signals. A balance needs to be achieved between these two requirements.

- The PDF models for the abnormal signals indicate that the abnormal signals have higher probabilities of occurrence of higher values within the normalized range $[0, 1]$.
- Based on the differences between the PDF models for the normal and abnormal signals, it should be possible to classify VAG signals using parameters derived from their PDFs.
- The apparent differences between the PDF models suggest the existence of different underlying signal-generation processes or statistical models for normal and abnormal VAG signals.

The following section provides details on the derivation of parameters from PDFs for classification of VAG signals.

5.12.3 Screening of VAG signals using statistical parameters

VAG signals related to various types of knee-joint pathology have been observed to possess a larger extent of variability over the duration of a swing cycle of the leg than normal VAG signals [94, 95, 264–266, 269–271]. To characterize this nature of VAG signals, Moussavi et al. [94] used the variance of the means of the segments of a given VAG signal; the signals were segmented adaptively using the RLS algorithm. Rangayyan and Wu explored the use of several statistical parameters [272], including FF (see Section 5.6.4), skewness, kurtosis, and entropy [264] as well as an adaptive turns count and the variance of the MS value [265] (see Section 3.2.1) for screening of VAG signals. Other methods that have been proposed for the analysis of VAG signals include AR modeling [95, 96], cepstral coefficients [96], time-frequency distributions (TFDs) [273], and wavelet packet decomposition [270]; see Wu et al. [271] for a review.

As described in Section 3.2.1, various statistical measures may be computed based upon the moments of the PDF of a given signal, $p_x(x_l)$, with $x_l, l = 0, 1, 2, \dots, L - 1$, representing the L bins used to span the range of the values of the signal x . The k^{th} central moment of the PDF $p_x(x_l)$ is defined as

$$m_k = \sum_{l=0}^{L-1} (x_l - \mu)^k p_x(x_l), \quad (5.31)$$

where μ is the mean value, given by

$$\mu = \sum_{l=0}^{L-1} x_l p_x(x_l). \quad (5.32)$$

See Section 3.2.1 for the definition of several related statistical parameters.

The Kullback–Leibler distance or divergence (KLD) between two PDFs $p_1(x_l)$ and $p_2(x_l)$ is defined as [267]

$$KLD(p_1, p_2) = \sum_{l=0}^{L-1} p_2(x_l) \ln \left[\frac{p_2(x_l)}{p_1(x_l)} \right]. \quad (5.33)$$

Rangayyan and Wu [266] computed the KLD between the PDF estimated for the signal to be classified and the PDF models for the normal and abnormal VAG signals. In obtaining the PDF models for the normal and abnormal signals with the limited data set available, the leave-one-out (LOO) method was used: the signal to be classified was left out of the procedure to obtain the Parzen-window model for the corresponding class. In this manner, the signal being tested does not contribute to the training process.

Using only the KLD feature obtained with the Parzen-window models, a normal-versus-abnormal classification accuracy of 73% was obtained with the data set of 89 VAG signals used. The sensitivity and specificity of classification (screening) were calculated to be 68.42% and 76.47%, respectively. The use of KLD with the Gaussian models resulted in poorer performance, with overall accuracy of 69.7%. Note that the Gaussian model does not facilitate the characterization of asymmetric or skewed PDFs.

Pattern classification using the set of parameters (KLD, K, H, μ, σ) led to an overall accuracy of 77.5%, sensitivity of 71.1%, and specificity of 82.4% using radial basis functions (RBF) with LOO. Other combinations of the parameters gave poorer results. Mu et al. [274] obtained better classification performance using some of the features described above but with advanced methods including a genetic algorithm for feature selection and the strict two-surface proximal classifier.

Given the nonstationary nature of VAG signals, it would be advantageous to compute some of the parameters described above using segments of VAG signals instead of their full duration [264, 265]. See Sections 6.6, 8.14, 9.4.2, 9.8.1, and 9.12 for descriptions of other methods for the analysis of VAG signals.

5.13 Application: Fractal Analysis of the EMG in Relation to Force

Problem: *Fractals exhibit patterns of various levels of complexity. Can fractal analysis be applied to characterize the variations in an EMG signal with respect to the force exerted by a muscle?*

Solution: Several biomedical systems and signals exhibit varying levels of complexity in their characteristics and patterns that vary across different states of health and disease. The fractal dimension (FD) can be used to represent such complexity in a quantitative manner. Various methods are available to derive estimates of FD from a given signal, as described in the following sections.

5.13.1 Fractals in nature

The term “fractal” was coined by Mandelbrot [275] to represent patterns that possess self-similarity at several scales or levels of magnification [276–283]. Fractal geometry has received considerable attention as a mathematical tool to model randomness in nature. Self-similarity refers to the property of an object that has a substructure that resembles its superstructure: Under several levels of magnification, a fractal appears to have the same form. A common example of a self-similar fractal in nature is a fern that has the same pattern repeated at multiple scales in its leaves; broccoli and cauliflower also exhibit fractal patterns. On the other hand, a Euclidean form, such as a square or a triangle, does not retain its original form when magnified. The Cantor bar, the Koch curve, the Sierpinski triangle, and the Hilbert curve are well-known fractals [276, 282].

A fractal is typically formed using a recursive procedure, whereas a Euclidean shape is defined by an algebraic formula. Fractals possess irregular and complex forms that cannot be described adequately using simple Euclidean geometry or shapes. It is readily understood that the dimension of a straight line is unity, that of a square is two, and that of a cube is three. However, a VAG signal that is a 1D function of time, but occupies the 2D space in a more complicated manner than a straight line, may be considered to possess a fractional dimension that is between unity and two. Fractal analysis could be used to model and analyze apparently complex or complicated patterns. However, natural objects may not possess self-similarity at all scales. A real-life pattern does not possess infinite levels of detail or granularity. Furthermore, a natural pattern, when magnified, may appear similar but not exactly identical to its original unmagnified form. This characteristic is referred to as statistical self-similarity. A natural fractal-like object may be viewed as an approximate fractal with statistical self-similarity over a finite range of scales. Fractal analysis is a part of a new set of advanced techniques for nonlinear and dynamic analysis of biomedical signals [284–286].

5.13.2 Fractal dimension

Consider a self-similar pattern that exhibits a number of self-similar parts represented by the variable a , at the reduction factor of $1/s$. The reduction factor is related to the measurement scale. The self-similarity dimension (D) is defined in a power-law relationship as [276]

$$a = \frac{1}{s^D}. \quad (5.34)$$

Applying the log and solving for D , we get

$$D = \frac{\log(a)}{\log(1/s)}. \quad (5.35)$$

The slope of a plot of the log of the number of self-similar parts, $\log(a)$, versus the log of the reduction factor, $\log(1/s)$, can provide an estimate of D . A straight-line

approximation could be fitted to estimate the slope. Two popular methods to estimate FD are the ruler method and the box-counting method [276], which are described in the following paragraphs.

The ruler method: Using rulers of different length, the total length of a pattern or signal can be estimated to different levels of accuracy. If a large ruler is used, the small details in the given pattern are ignored. As smaller and smaller rulers are used, finer details get measured. The measured or estimated length increases and improves in accuracy as the size of the ruler decreases. An estimate of FD is obtained from the slope of a straight-line fit to a plot of the log of the measured length versus the log of the measuring unit or ruler size.

Let u be the length measured with a ruler of size s . The precision of measurement is represented by $1/s$. A fractal is expected to satisfy the power law [276]

$$u = c \frac{1}{s^d}, \quad (5.36)$$

where c is a constant of proportionality and $FD = 1 + d$. Applying the log, we have

$$\log(u) = \log(c) + d \log(1/s). \quad (5.37)$$

The slope of a plot of $\log(u)$ versus $\log(1/s)$ provides an estimate of FD as $FD = 1 + d$.

If we let $u = ns$, where n is the number of times the ruler is used to measure the length u with the ruler of size s , then

$$\log(n) = \log(c) + (1 + d) \log(1/s). \quad (5.38)$$

The slope of a plot of $\log(n)$ versus $\log(1/s)$ directly provides an estimate of FD .

The box-counting method: The box-counting method [276, 282, 287–290] involves partitioning the pattern or signal space into square boxes of equal size, and counting the number of boxes that contain a part of the signal. The process is repeated by partitioning the signal space into smaller and smaller squares. The log of the number of boxes counted is plotted against the log of the magnification index for each stage of partitioning. The slope of the straight line fitted to the plot gives an estimate of the FD .

Higuchi's method: Higuchi [291] proposed an algorithm to estimate the FD of a signal by computing measures of the length of the signal using multiple versions of the signal reconstructed for varying measurement intervals. Given a signal $x(n)$, $n = 1, 2, \dots, N$, the method creates new signals as [291, 292]

$$x_k(m) = x(m), x(m+k), x(m+2k), \dots, x\left(m + \left\lfloor \frac{N-m}{k} \right\rfloor k\right), \quad (5.39)$$

for various values of k , where k and m are integers with $m = 1, 2, \dots, k$. Here, m represents the initial point and k is the interval between the points of $x_k(m)$. The length of each derived signal is computed as

$$L(m, k) = \frac{1}{k} \frac{N-1}{k \lfloor \frac{N-m}{k} \rfloor} \sum_{i=1}^{\lfloor \frac{N-m}{k} \rfloor} |x(m+ik) - x[m+(i-1)k]| \quad (5.40)$$

and

$$L(k) = \frac{1}{k} \sum_{m=1}^k L(m, k). \quad (5.41)$$

The slope of a straight-line fit to a log-log plot of $L(k)$ against $1/k$ gives the FD of the original signal.

When applying the ruler or box-counting methods to a signal or pattern, it is important to select an appropriate range of the size of the ruler or the box [276, 282, 289]. The range of size needs to be related to the range of the values of the signal for both the independent variable (such as time) and the dependent variable (such as voltage). It could be convenient to normalize both ranges of a given signal to $[0, 1]$ and to define the ruler or box size in relation to the normalized variables.

For discussions on other methods of fractal analysis, see Section 6.6, Cabral and Rangayyan [282], and Banik et al. [293].

5.13.3 Fractal analysis of physiological signals

Fractal properties have been observed in several physiological structures and processes. Many anatomical structures have fractal-like appearance, such as the coronary arteries, venous branching patterns, bronchial trees, certain muscle fiber bundles, and the His-Purkinje network in the ventricles [294]. The branching pattern of the His-Purkinje network of conduction pathways provides an efficient way to distribute the depolarization stimulus to the ventricles. The electrogenesis of the QRS complex in the ECG has been modeled by using a fractal-like conduction system. The normal QRS complex has been shown to follow an inverse-power-law distribution of frequency content in the log-log scale [295]; this property has been noted as being consistent with depolarization of the myocardium by a self-similar branching network [294]. Studies of such branching networks used to depolarize a network of cells via computer modeling have shown that, after 10 generations of branching, it is possible to simulate realistic QRS complexes [296]. The frequency content of QRS complexes has been shown to be affected by changes in the geometry of the branching network.

Goldberger et al. [295] and Goldberger and West [297] showed that the rhythm of a healthy heart is not highly regular, but that it is a temporal fractal with a high degree of variability in the heart rate. The PSD of a time-series representation of the heart rate follows the inverse power law (see Section 6.6 for related discussion). It has been shown that, in the case of time series of heart rate, the loss of physiological complexity can lead to greater regularity. The phenomena associated with fractals,

self-similar scaling, $1/f$ noise, and inverse-power-law distributions offer interesting models as well as useful methods to characterize physiological processes and biomedical signals [297].

Li et al. [298] applied a method of fractal and wavelet-based spectral analysis to analyze EEG signals of rats. It was hypothesized that variations in FD -related parameters, such as the Hurst and spectral exponents, can be used to describe the dynamic characteristics of the brain in different states, in particular before seizures. They found that the method revealed characteristic signs of an approaching seizure, including the emergence of long-range correlation and decrease in the FD value. Liang et al. [299] applied the Hurst exponent, extracted from EEG recordings, as a measure of the effects of anesthesia on brain activity. The maximal overlap discrete wavelet transform (DWT) was used to suppress the effects of artifacts in the EEG and the scaling properties of the data in designated frequency bands were calculated prior to estimation of the Hurst exponent. It was observed that the Hurst exponent decreased (especially in low-frequency bands) when anesthesia deepened and that it is a useful measure for estimating the depth of anesthesia.

Shah et al. [300] studied acceleration signals obtained from finger joints of patients with calcium pyrophosphate deposition disease, rheumatoid arthritis, or spondyloarthropathy of the finger joint. The ruler method was used to estimate FD . The results showed that there were significant differences between the FD values of acceleration signals from patients in the three categories. Specifically, the FD values of acceleration signals from the patients with calcium pyrophosphate deposition disease (1.709 ± 0.097) were higher than those of patients with rheumatoid arthritis (1.6 ± 0.069) or spondyloarthropathy (1.569 ± 0.081).

There is increasing interest in nonlinear dynamical analysis of biomedical signals using several recently developed methods related to fractals, chaos, and nonlinear modeling [282, 284–286]; see Stam [301] for a detailed review of nonlinear dynamical analysis of the EEG and other signals. The recently developed methods of nonlinear dynamical analysis are expected to make it possible to study self-organization, pattern formation, and attractors of trajectories in the state space of nonlinear and complex systems that may not be captured by traditional linear methods.

For discussions on the application of fractal analysis to images of breast lesions in mammograms, see Rangayyan and Nguyen [302], Cabral and Rangayyan [282], and Banik et al. [293].

5.13.4 Fractal analysis of EMG signals

We have seen several examples of the increasing complexity of EMG signals with increasing force and studied many methods to capture such complexity in quantitative terms; see, for example, Figure 1.20 and Sections 5.9, 5.10, and 5.11. Figure 5.24 shows the variation of FD , obtained using Higuchi's method, for the EMG signal shown in Figure 1.20. The EMG signal was first filtered with a Butterworth lowpass filter of 6th order and cutoff frequency of 300 Hz. Segments of duration 1 s were cut for each level of contraction to estimate FD . It is evident that FD increases with the level of contraction (except for the last level of contraction) with high correlation.

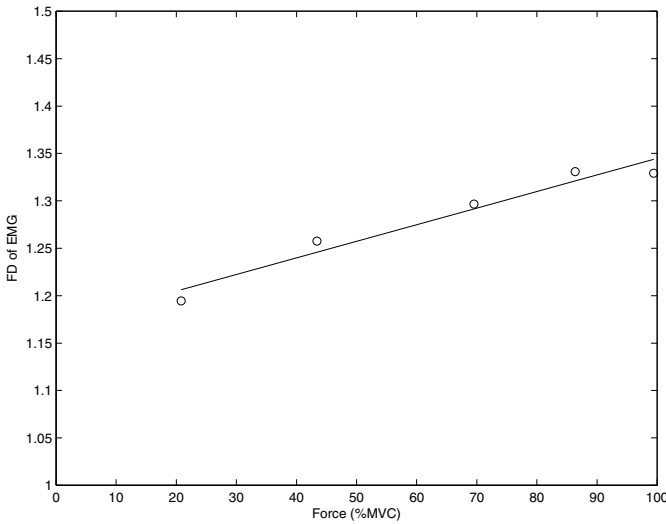


Figure 5.24 Variation of FD with the level of muscle contraction for the EMG signal shown in Figure 1.20; $r^2 = 0.95$. Figure courtesy of Faraz Oloumi.

Anmuth et al. [303] applied fractal analysis to surface EMG signals obtained from the FDI at various levels of isometric contraction. A correlation coefficient of 0.991 was obtained between pooled FD and %MVC data for 10 subjects.

Gitter and Czerniecki [304] studied the use of FD , estimated using the box-counting method, in the analysis of EMG signals. They obtained needle EMG signals at various levels of contraction of the biceps brachii muscle. FD values, averaged over three repetitions of each level of contraction, were obtained in the range [1.1, 1.4] and exhibited positive correlation with force. However, whereas the correlation was high with $r = 0.94$ in the range [10, 70] %MVC, it was low with $r = 0.35$ in the range [70, 90] %MVC.

Gupta et al. [305] studied the variation of the FD of surface EMG signals of the biceps brachii during flexion and extension of the arm at various speeds and with different loads placed on the palm. Using linear regression, they found FD to be correlated well to both the load ($r = 0.99$) and the rate of flexion-extension ($r = 0.98$).

Muscle fatigue is known to cause reductions in the mean and median frequency of the related EMG signals. Ravier et al. [306] applied the $1/f$ model (see Section 6.6.1) to analyze EMG signals related to muscle force and fatigue. They found that the right slope of the EMG spectrum (beyond the peak frequency in the range of 60 – 90 Hz) decreased with increasing levels of force, but there was no substantial effect of fatigue on the slope. However, Talebinejad et al. [307] applied multifractal detrended fluctuation analysis methods to obtain optimal Hurst exponents, which showed a high degree of correlation with the progress of fatigue; see also Talebinejad et al. [308, 309].

5.14 Remarks

We have now reached the stage in our study where we can derive parameters from segments of biomedical signals. In the present chapter, we focused our attention on characteristics that could be observed or derived in the time domain. The parameters considered were designed with the aim of discriminating between different types of waveshapes, or of representing change in waveform complexity through the course of a physiological or pathological process. We have seen how the various parameters explored in the present chapter can help in distinguishing between normal and ectopic ECG beats, and how certain measures can serve as quantities that are correlated with levels of physiological activity, such as respiration and muscular contraction.

It should be borne in mind that, in most practical applications, a single parameter or a couple of measures may not adequately serve the purposes of signal analysis or diagnostic decision making. A single parameter such as FF or SL may assist in distinguishing some types of PVCs from normal ECG beats; however, several cardiovascular diseases and defects may cause changes in the ECG signal that may lead to similar variations in the FF or SL values. A practical application would need to maintain a broad scope of analysis and use several parameters to detect various possible abnormalities. As always, an investigator should consider the possibility that a parameter observed to be useful in, for example, ECG analysis in the time domain, may serve the needs in the analysis of some other signal, such as the PCG or EMG, in a different domain.

5.15 Study Questions and Problems

1. Prove that the values of FF of a sinusoidal wave is equal to unity.
2. The following discrete-time signals are defined over the interval 0 to 10 s with the sampling frequency being 1 Hz: (a) $x_1(n) = u(n) - u(n-5)$. (b) $x_2(n) = 2u(n-3) - 2u(n-8)$. (c) $x_3(n) = u(n-2) - u(n-9)$. (d) $x_4(n) = u(n-2) - u(n-10)$. $u(n)$ is the discrete-time unit step function. The SL of a signal $x(n)$ is defined as

$$SL = \frac{\sum_{n=0}^{N-1} w(n) x^2(n)}{\sum_{n=0}^{N-1} x^2(n)},$$

where $w(n)$ is a nondecreasing weighting function, and N is the number of samples in the signal. Let $w(n) = n$, $n = 0, 1, 2, \dots, N-1$. Draw sketches of each signal with the weighting function $w(n)$ superimposed. Compute the SL values for the four signals given. Interpret your results and compare the characteristics of the four signals in terms of their SL values.

3. You are given a signal with the samples $\{0, 0, 2, 2, 3, -3, 2, 0, 0\}$ and a template with the samples $\{1, -1\}$. Perform the template matching operation and derive the sample values for the output. Provide an interpretation of the result.
4. Discuss the similarities and differences between the problems of (i) detection of spike transients in EEG signals, and (ii) the detection of QRS complexes in ECG signals.

5. You have been hired to develop a heart-rate monitor for use in a coronary-care unit. Design a system to accept a patient's ECG signal, filter it to remove artifacts and noise, sample the signal, measure the heart rate, and set off alarms as appropriate. Provide a block diagram of the system, with details of the signal processing steps to be performed in each block. Specify the important parameters for each processing step.
6. A needle EMG signal under low levels of muscle contraction was observed to contain a mixture of three trains of MUAPs. One of the trains contains quasiperiodic occurrences of a monophasic MUAP, the second contains occurrences of a biphasic MUAP, and the third contains occurrences of a triphasic MUAP. It was also observed that the MUAPs do not overlap in the EMG signal. Propose a signal analysis procedure to: (a) detect the occurrence (location in time) of each MUAP of each type individually and (b) determine the firing rate of each motor unit. Note that each MUAP needs to be detected and labeled as being one of monophasic, biphasic, or triphasic type.
Your solution should include: (i) plots of the EMG signal (make up one according to the description above) with labels for the components; (ii) plots of the signal at various stages of your procedure; (iii) equations for important steps of your signal analysis procedure; and (iv) statements describing the reason or logic behind each step you propose.
7. A researcher is attempting to develop a digital signal processing system for the acquisition and analysis of heart sound signals (PCG signals). Assist the researcher in addressing the following concerns and problems: (a) What are the typical bandwidths of normal PCG signals and those with murmurs? What is the recommended sampling frequency? (b) What are the sources of artifacts that one has to consider in recording PCG signals? Name one physiological source and one other source, and recommend techniques to limit or eliminate both. (c) How can one identify the locations of S1 and S2? Which other biomedical signals would you recommend for assistance in this problem? Draw schematic diagrams of the signals and identify the corresponding cardiac events and timing relationships. (d) Propose a technique to obtain the envelope of the PCG signal. List all steps of the method you propose and provide the required parameters. (e) Draw schematic PCG signals and their envelopes over one cardiac cycle for a normal case, a case with systolic murmur, and a case with diastolic murmur. Identify each event in each case.
8. You are given a database of SMUAPs containing several types of normal and abnormal patterns. Each signal record has one SMUAP. The patterns and features of interest are: (i) Monophasic SMUAPs. (ii) Biphasic SMUAPs. (iii) Triphasic SMUAPs. (iv) Polyphasic SMUAPs with more than three phases.
(a) Propose two parameters (computed features) to help in separating the four classes of SMUAPs. Give the required equations or procedures and explain their relationship to the signal characteristics described above. Describe conditions or preprocessing steps that are required in order for your methods to work well. (b) Draw a schematic plot of the feature-vector space and demarcate regions where you expect features of the four SMUAP types to lie. (c) State decision rules to classify the four SMUAP types using the two measures you propose.
9. Why is the ST segment of the ECG relevant in diagnosis? Recommend signal analysis techniques for the analysis of ST segment variations in clinical applications.
10. A researcher new to the field of biomedical signal analysis is assigned a project on the analysis of heart sounds and murmurs. The researcher is provided with a database of PCG signals of four types: (a) normal, (b) systolic murmur, (c) diastolic murmur, and (d) systolic and diastolic murmur.

The researcher surfs the internet and obtains two programs to compute the ZCR using a moving window of 20 ms and a filtered envelope of a given signal. The sampling rate used is 1,000 Hz . Although the programs generated graphs of the given signal, the ZCR , and the smoothed envelope, the researcher encountered difficulties in interpreting the results.

Assist the researcher by providing the following information: (i) For each type of PCG signal listed above, draw plots of a typical PCG signal, its ZCR as a function of time, and the averaged envelope. Explain the important expected characteristics of the results. (ii) Estimate the ranges of ZCR values (using a moving window of 20 ms) for typical normal heart sounds and murmurs. Explain your procedure and assumptions.

11. A researcher has obtained a system to record the EMG signal from a muscle that is directly involved in breathing (respiration). Help the researcher in understanding the nature of the signal and developing a signal processing system to analyze the signal as follows: (a) Draw a schematic representation of the EMG over two cycles of respiration, showing the ranges of inspiration and expiration. (b) Propose an algorithm to obtain the envelope of the EMG signal. Describe the nature and purpose of each step in your algorithm. Give an equation for each step, and plot the result of the operation on the EMG signal. (c) Propose a method to obtain a measure of activity (to indicate how 'busy' the signal is) as a function of time, in order to characterize the variation in the signal with breathing. Give an equation and describe the nature of the method. Plot the result of the operation on the EMG signal. (d) The peak values in the envelope and activity functions over a given cycle of respiration are expected to be correlated with the peak air flow to the lungs. Propose algorithms to detect the peak envelope and peak activity values. (e) For each procedure that you propose in items (b), (c), and (d) above, discuss potential artifacts that could mislead your procedure and indicate how you would prevent them.
12. A researcher is developing methods to extract features from ECG signals to distinguish between normal beats and PVCs. Draw typical waveforms of the two types of beats and explain the major differences between them in qualitative terms. Propose two quantitative measures or features to characterize the differences between normal beats and PVCs. Give an equation or a step-by-step algorithm to compute each feature.
13. A student new to the field of biomedical engineering approaches you seeking assistance to develop methods to analyze the relationship between the force produced by contracting a muscle and the corresponding EMG signal. Help the student by providing detailed responses to the following questions and requests:
 - (a) How can one acquire EMG signals corresponding to different levels of force? (i) How many electrodes are required? (ii) Where and how should the electrodes be placed? (iii) How can one get another signal or variable that is directly proportional to the force? (iv) What should be the settings for the bandwidth of the filters for the EMG signal? (v) What should be the settings for the bandwidth of the filters for the force signal? (vi) What should be the sampling frequency? (vii) Give a step-by-step experimental procedure to acquire the required signals. (viii) Draw a schematic diagram to illustrate the expected nature of the EMG and force signals.
 - (b) (i) What are the precautions to be taken in recording the signals? (ii) List one potential source each for random noise, structured noise, and physiological artifact that could corrupt the EMG signal. (iii) For each source of noise or artifact that you identify, recommend a procedure to prevent the same. (iv) For each source of noise or artifact that you identify, recommend a postacquisition procedure to remove the same.

- (c) (i) What kind of measures or parameters may one derive from an EMG signal so that the measures vary in proportion to muscular force? (ii) Give equations to define three parameters that are suitable for this purpose. For each parameter, give a step-by-step procedure to compute the parameter from the EMG signal. (iii) Explain the expected relationships between the three parameters that you recommend and muscular force. Draw a plot with the force on the abscissa (x -axis) and the proposed parameters on the ordinate (y -axis).
14. Given a sampled signal, $x(n)$, $n = 0, 1, 2, \dots, N - 1$, write an equation to compute the variance, σ_x^2 , of the signal. Give the definitions of the parameters *mobility* and *form factor*. Explain how the three parameters mentioned above may be computed and used to analyze the relationship between the force generated by muscular contraction and the related EMG signal.
 15. Explain the concept of an envelope of a signal. Give a step-by-step algorithm to compute an envelope of a signal. Include at least two equations for parts of the procedure. Draw a normal PCG signal over one cardiac signal and a corresponding envelope. Explain the relationships between the two and suggest a use of the envelope in a practical application.

5.16 Laboratory Exercises and Projects

Note: Data files related to the exercises are available at the site

<http://people.ucalgary.ca/~ranga/enel563>

1. The signal in the file `emg_dog2.dat` was recorded from the crural diaphragm of a dog using fine-wire electrodes sewn in-line with the muscle fibers and placed 10 mm apart. The signal represents two cycles of breathing, and has been sampled at 10 kHz. (See also the file `emg_dog2.m`.)

Write a MATLAB[®] program to perform full-wave rectification (absolute value) or half-wave rectification (threshold at zero, with the mean value of the signal being zero). Apply a lowpass Butterworth filter of order four and cutoff frequency in the range 5 to 20 Hz to the result. Analyze and evaluate the results with the two methods of rectification and at least two different lowpass cutoff frequencies. Compare the results with the envelope provided in the file `emg_dog2_env.dat`.

2. The *RMS* value of a signal within a specific duration is related to the average power level of the signal. Write a MATLAB[®] program to compute the *RMS* value at each instant for the EMG signal in the file `emg_dog2.dat` by using a causal short-time analysis window of duration in the range 50 – 150 ms. Use at least two different window durations and analyze the results. (See also the file `emg_dog2.m`.)
3. Develop a program to compute the turns count in causal moving windows of duration in the range 50 – 150 ms. Apply the method to the EMG signal in the file `emg_dog2.dat`. (See also the file `emg_dog2.m`.) Study the results for different thresholds in the range 0 – 200 μV . Compare the envelope, *RMS*, and turns count curves in terms of their usefulness as representatives of inspiratory airflow (data provided in the file `emg_dog2_flo.dat`).
4. The file `safety.wav` contains the speech signal for the word “safety” uttered by a male speaker, sampled at 8 kHz. (See also the file `safety.m`.) The signal has a significant

amount of background noise (as it was recorded in a computer laboratory). Develop procedures to derive short-time *RMS*, turns count, and *ZCR* in moving windows of duration in the range 10 – 100 *ms*. Study the variations in the parameters in relation to the voiced, unvoiced, and silence (background noise) portions of the signal.

What do you expect the results to be if the procedures are applied to the first derivative of the signal? Confirm your assertions or expectations by performing the study.

5. Develop a program to derive the envelopegram. Apply the procedure to the PCG signals in the files pec1.dat, pec33.dat, and pec52.dat. (See the file plotpec.m.)

Extend the procedure to average the envelopegrams over several cardiac cycles using the ECG as the trigger. How will you handle the variations in the duration (number of samples) of the signals from one beat to another?

6. The ECG signal in the file ecgpvc.dat contains a large number of PVCs, including episodes of bigeminy. (See the file ecgpvc.m.) Apply the Pan–Tompkins procedure to detect and segment each beat. Label each beat as normal or PVC by visual inspection. Record the number of beats missed, if any, by your detection procedure.

Compute the *RR* interval and the form factor *FF* for each beat. Use a duration of 80 samples (400 *ms*) spanning the QRS–T portion of each beat to compute *FF*. The P wave need not be considered in the present exercise.

Compute the mean and *SD* of the *FF* and *RR* values for the normal beats and the PVCs. Evaluate the variation of the two parameters between the two categories of beats.

7. Obtain EMG signals related to various levels of muscular activity or force. You may use the data files EMGforce.txt and EMGforce2.txt as well as the program EMGforce.m provided; the sampling rate is 2,000 *Hz* per channel. Normalize the force signal such that the minimum value is zero and the maximum value (corresponding to MVC) is 100. Filter the force signal to remove noise and artifacts. Plot the EMG signal (in *mV*) and normalized force (in %MVC) against the time axis.

Develop a program for automatic identification of portions (segments) corresponding to each level of contraction within which the force remains close to the corresponding peak values. For each segment of the EMG signal identified as above, compute four suitable parameters. Ensure that your list of EMG features includes the following: *RMS* value, form factor, and turns count. Compute also the average force (in %MVC) for each segment.

Plot the values of the various parameters versus force in %MVC. Label the axes with the appropriate units. Analyze the results in terms of statistical variation of the parameters in relation to force. Using the `polyfit` function in MATLAB[®], obtain a straight-line (linear) fit to represent the variation of each EMG parameter versus force. Use `polyval` to evaluate the values of the dependent variable given by the model for the available values of the independent variable. Superimpose the linear models (straight-line fits) obtained on the plots of the parameters in the preceding step. Analyze the results.

Compute the correlation coefficient, *r*, with *r*² given by the formula

$$r^2 = \frac{\left[\sum_{n=1}^{n=N} x(n)y(n) - N\bar{x}\bar{y} \right]^2}{\left[\sum_{n=1}^{n=N} x^2(n) - N\bar{x}^2 \right] \left[\sum_{n=1}^{n=N} y^2(n) - N\bar{y}^2 \right]},$$

where *N* is the number of samples of *x* or *y* representing each of the EMG parameters or force, and $\bar{x}$ is the mean of *x*. Using *r*, analyze the goodness of fit for each parameter

and discuss the appropriateness of the linear model. Tabulate the parameters of the linear model and r for each of the EMG parameters. Analyze the results and describe your findings. Perform all of the above steps with the two EMG signals mentioned above. Analyze and compare the results.

CHAPTER 6

FREQUENCY DOMAIN CHARACTERIZATION OF SIGNALS AND SYSTEMS

Many biomedical systems exhibit innate rhythms and periodicity that are more readily expressed and appreciated in terms of frequency than temporal measures. As a basic example, consider cardiac function: We express cardiac rhythm more conveniently in terms of *beats per minute* — a measure of the frequency of occurrence or the rate of repetition — than in terms of the duration of a beat or the interval between beats in seconds (the *RR* interval). A cardiac rhythm expressed as *72 bpm* is more easily understood than a statement of the corresponding *RR* interval as *0.833 s*. By the same token, the notion of an EEG rhythm is more easily conveyed by description in *cycles per second* in lay terms, or in *Hertz (Hz)* in technical terms. Even engineers would find a frequency-domain expression easier to appreciate than a time-domain description, such as an alpha rhythm having a frequency of *11.5 Hz* versus the equivalent period of *0.087 s*.

When the signal being studied is made up of discrete (that is, separate and distinct) events in time, such as the ECG or a train of SMUAPs, the basic rhythm or rate of activity present in the signal can be assessed directly in the time domain. On the other hand, signals such as the PCG display complex or complicated patterns in the time domain that do not facilitate ready appreciation of their frequency-domain char-

acteristics; furthermore, the time-domain waveforms may differ from one occurrence of the signal (one heart beat) to another.

The PCG provides an interesting example of a signal with multiple frequency-domain features: In addition to the beat-to-beat periodicity or rhythm, the heart sounds within a cardiac cycle exhibit *resonance*. Due to the multicompartmental nature of the cardiac system, we should expect heart sounds to possess multiple resonance frequencies; this leads to the need to describe the PCG not only in terms of a rhythm (the heart rate) or a single resonance frequency, but also a composite *spectrum* of several dominant or resonance frequencies. Furthermore, constrained flow of blood through an orifice such as a septal defect or across a stenosed valve acting as a baffle could lead to turbulence, resulting in *wideband noise*. In the case of noise-like murmurs, we would be able to identify neither rhythms nor resonance frequencies: The need arises to consider the distribution of the signal's energy or power over a wide band of frequencies, leading to the notion of the power *spectral density* function.

We have seen in Chapter 3 that it is often more convenient and meaningful to describe filters in terms of their transfer function or frequency response — $H(s)$, $H(z)$, $H(\omega)$, or $H(f)$ — than in terms of their impulse response $h(t)$ or the time-domain input–output relationship (difference equation). Furthermore, we saw in Section 4.4 that it is easier to interpret the PSDs of EEG waves than it is to interpret their theoretically equivalent ACFs. The Fourier and other transforms provide an invertible or reversible transformation from the time domain to the frequency domain (and vice-versa). Therefore, it may be argued that no new information is created by taking a given signal from the time domain to the frequency domain. However, the distribution of the energy or power of the signal in the frequency domain that is provided by the Fourier transform — the spectrum or PSD of the signal — facilitates better analysis and description of the frequency-domain characteristics of the signal. The PSD of a signal is not only useful in analyzing the signal, but also in designing amplifiers, filters, data-acquisition and transmission systems, and signal processing systems to treat the signal appropriately. We have seen in Section 3.8 that we need not only the signal PSD but also the noise PSD in order to be able to implement the optimal Wiener filter.

The treatment of biomedical signals as stochastic processes provides flexibility and a sense of generality in analysis, but imposes conditions and requirements in the estimation of their statistics including the ACF and PSD. In the present chapter, we shall investigate methods to estimate the PSD and frequency-domain parameters of biomedical signals and systems. We shall also study methods to derive spectral parameters that can characterize the given signal as well as the system that generated the signal. The motivation for the study, as always, shall be to distinguish between normal and abnormal signals or systems, and the potential use of the methods in diagnosis.

6.1 Problem Statement

Investigate the potential use of the Fourier spectrum and parameters derived thereof in the analysis of biomedical signals. Identify physiological and pathological processes that could modify the frequency content of the corresponding signals. Outline the signal processing tasks needed to perform spectral analysis of biomedical signals and systems.

As in the preceding chapters, the problem statement given above is generic and represents the theme of the present chapter. The various signal analysis techniques described and the examples used for illustration in the following sections will address the points raised in the problem statement, with attention to specific problems and techniques.

6.2 Illustration of the Problem with Case Studies

6.2.1 The effect of myocardial elasticity on heart sound spectra

The first and second heart sounds — S1 and S2 — are typically composed of low-frequency components; this is to be expected due to the fluid-filled and elastic nature of the cardiohemic system. Sakai et al. [310] processed recorded heart sound signals by using tunable bandpass filters (with a bandwidth of 20 Hz , tuned over the range 20 – 40 Hz to 400 – 420 Hz), and estimated the frequency distributions of S1 and S2. They found the heart sound spectra to be maximum in the 20 – 40 Hz band, that S1 had a tendency to demonstrate peaks at lower frequencies than those of S2, and that S2 exhibited a “gentle peaking” between 60 Hz and 220 Hz .

Gerbarg et al. [247, 248] developed a computer program to simulate a filter bank and obtained averaged power spectra of S1 and S2 of 1,000 adult males, 32 high-school children, and 75 patients in a hospital. The averaged PSDs of S1 and S2 obtained by them indicated peak power in the range 60 – 70 Hz , and relative power levels lower than -10 dB beyond 150 Hz . The PSD of S2 displayed slightly more high-frequency energy than that of S1.

Frome and Frederickson [311] applied the FFT to the analysis of first and second heart sounds. They described how segmented S1 and S2 data may be combined into a single complex signal and how a single FFT may be used to obtain the FFTs of the two signals. Computer data processing techniques were described to obtain smoothed, averaged periodograms (described in Section 6.3.1) of S1 and S2 separately.

Yoganathan et al. [312] applied the FFT for the analysis of S1 of 29 normal subjects. The FFT spectra of 250 ms windows containing S1 were averaged over 15 beats for each subject. It was found that the frequency spectrum of S1 contains peaks in a low-frequency range (10–50 Hz) and a medium-frequency range (50–140 Hz) [312]. In a related study [313], the spectrum of S2 was observed to contain peaks in low-frequency (10–80 Hz), medium-frequency (80–220 Hz), and high-frequency ranges (220–400 Hz). It has been suggested that the resonance peaks in the spectra

may be related to the elastic properties of the heart muscles and the dynamic events causing the various components of S1 and S2 (see Section 1.2.9).

Adolph et al. [314] used a dynamic spectrum analyzer to study the frequency content of S1 during the isovolumic contraction period. The center frequency of a filter with 20 Hz bandwidth was initially set to 30 Hz and then varied in 10 Hz increments up to 70 Hz. The outputs of the filters were averaged over the same (prerecorded) 10 consecutive beats. Finally, the ratios of the average peak voltage of the filtered outputs to that of the total S1 signal during the isovolumic contraction period were computed.

Adolph et al. hypothesized that the frequency content of S1 during the isovolumic contraction period should depend on the relative contributions of the mass and elasticity of the left ventricle. The mass of the left ventricle with its blood content remains constant during isovolumic contraction. Therefore, it was reasoned that the frequency content of S1 should decrease (that is, shift toward lower frequencies) in the case of diseases that reduce ventricular elasticity, such as myocardial infarction.

Figure 6.1 shows averaged S1 spectra for normal subjects and patients with acute or healed myocardial infarction; it is seen that the reduced elasticity due to myocardial infarction has reduced the relative content of power near 40 Hz. However, Adolph et al. also noted that an increase in ventricular mass as in the case of trained athletes, or a reduction in elasticity combined with an increase in the mass as in the case of cardiomyopathy, could also cause a similar shift in the frequency content of S1. Regardless, they found that frequency analysis of S1 was of value in differentiating acute pulmonary embolism from acute myocardial infarction. Clarke et al. [315] also found reduction in the spectral energy of S1 to be a common accompaniment of myocardial ischemia.

6.2.2 Frequency analysis of murmurs to diagnose valvular defects

As we noted in Section 1.2.9, cardiovascular valvular defects and diseases cause high-frequency, noise-like sounds known as murmurs. Murmurs are often the only indicators of the early stages of certain cardiovascular diseases; prompt diagnosis could prevent further deterioration of the condition and possible complications.

We noted in Section 5.6.2 that zero-crossing analysis in the time domain was applied to assist in the detection of murmurs by Jacobs et al. [250] and Yokoi et al. [251]. Although *ZCR* increases with the presence of higher-frequency components, it does not yield a direct measure of the frequency content or the spectrum of the signal.

Application of electronic signal filtering techniques to analyze the frequency content of heart sounds and murmurs was initiated as early as the 1950s. Geckeler et al. [316] and McKusick et al. [317, 318] studied the applicability of the sound spectrograph for the analysis of heart sounds and murmurs. The sound spectrograph was developed in the late 1940s by Bell Telephone Laboratories as a tool to produce what was labeled as *visible speech*. The spectrograph used a bandpass filter (or a bank of bandpass filters) to determine the power of the given signal in each frequency band of interest. The signal was usually recorded and played back repeatedly

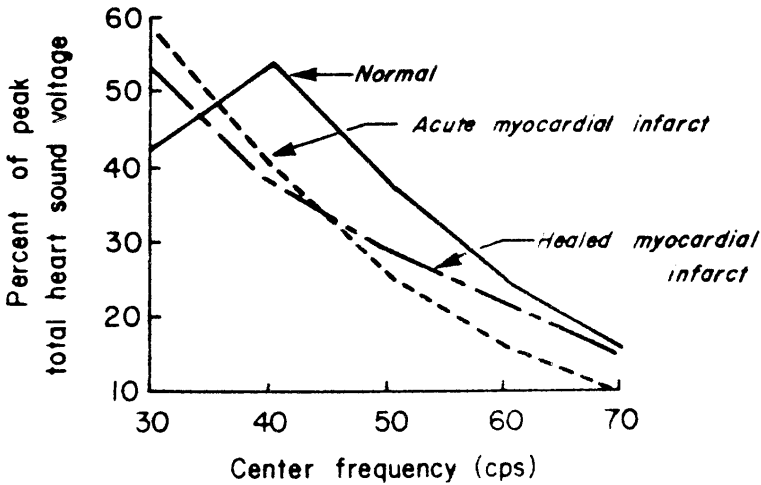


Figure 6.1 First heart sound spectra for normal, acute myocardial infarct, and healed myocardial infarct cases. The latter two cases exhibit an increased percentage of low-frequency components. Reproduced with permission from R.J. Adolph, J.F. Stephens, and K. Tanaka, The clinical value of frequency analysis of the first heart sound in myocardial infarction, *Circulation*, 41:1003–1014, 1970. ©American Heart Association.

as the center frequency of the bandpass filter was varied. The output was recorded on heat-sensitive or light-sensitive paper to produce a 2D distribution of frequency content of the signal at every instant of time as a gray-level image (essentially a TFD; see Section 8.4.1). Winer et al. [319] proposed isointensity contour plotting of the spectrogram instead of using variations in intensity (gray scale); they reported that, whereas normal heart sounds indicated the presence of regularity in the contours of equal intensity, abnormal sounds and murmurs produced irregular contour line structures with extensive “convolutions” and roughness. It was suggested that the cardiospectrograms (or spectral phonocardiography) could provide physiologic and pathologic information beyond that provided by auscultation, without suffering from the psychoacoustic impediments that affect human observers [33].

Yoshimura [320] used a tunable bandpass filter with low and high cutoff frequencies in the range $18 - 1,425\text{ Hz}$ to process recorded PCG signals. It was determined that the diastolic *rumble* of mitral stenosis occupied the range $20 - 200\text{ Hz}$, whereas the diastolic *blow* of aortic regurgitation spanned a much higher frequency range of $200 - 1,600\text{ Hz}$ (although the characteristic range was $400 - 800\text{ Hz}$).

Gerbarg et al. [247, 248] developed a computer program to simulate a filter bank and obtain power spectra of heart sounds and murmurs, with the aim of developing a system for screening to detect cardiovascular diseases. They argued that innocent (physiological) systolic murmur in children is limited to the first and middle thirds of the systolic interval between S1 and S2, whereas pathological systolic murmur due to mitral regurgitation is holosystolic. Therefore, they computed ratios of the

mean power of the *last third of systole* to the mean power of systole and also to a certain “standard” noise level. A ratio was also computed of the mean energy of systole to the mean energy of the PCG over the complete cardiac cycle. Gerbarg et al. obtained 78 – 91% agreement of their computer classification based upon the three ratios defined above with clinical diagnosis of mitral regurgitation in different groups of subjects. Although they did not claim that a fully automated program for the diagnosis of mitral regurgitation had been developed, they indicated that the feasibility of computer-based diagnosis had been established, and that simulation of human auscultation had been partially achieved.

The specific problem of detection of the murmur due to aortic insufficiency in the presence of the murmur due to mitral stenosis was considered by van Vollenhoven et al. [321]. Aortic insufficiency causes an early diastolic murmur (with a blowing or hissing quality) that is best heard in the aortic area (second right-intercostal space, just right of the sternum), whereas the middiastolic rumbling murmur of mitral stenosis is best heard at the apex. A tunable bandpass filter with 50 *Hz* bandwidth and center frequency tunable in steps of 50 *Hz* was used by van Vollenhoven et al. to study the frequency content in a 100 *ms* window during the diastolic phase of recorded PCG signals. They found that the murmur of mitral stenosis was limited in frequency content to less than 400 *Hz*, whereas the murmur in the case of aortic insufficiency combined with mitral stenosis had more high-frequency energy in the range 300 – 1,000 *Hz*.

Sarkady et al. [227] suggested synchronized averaging of the PSDs of PCG signals over several cardiac cycles computed using the FFT algorithm. Johnson et al. [322,323] studied FFT-based PSDs of the systolic murmur due to aortic stenosis. They computed the PSDs of systolic windows of duration 86, 170, and 341 *ms*, and averaged the results over 10 cardiac cycles. Johnson et al. hypothesized that higher murmur frequencies are generated as the severity of aortic stenosis increases. In their study of patients who underwent catheterization and cardiac fluoroscopy, the transvalvular systolic pressure gradient was measured during pull-back of the catheter from the left ventricle through the aortic valve and was found to be in the range 10 – 140 *mm of Hg*. Spectral power ratios (described in Section 6.4.2) were computed considering the band 25 – 75 *Hz* as the constant area (*CA*) related to normal sounds and the band 75 – 150 *Hz* as the predictive area (*PA*) related to murmurs.

Figure 6.2 illustrates the PSDs of four patients with aortic stenosis of different levels of severity. The PSDs in the figure are segmented into the *CA* and *PA* parts as described above; the transvalvular systolic pressure gradient (in *mm of Hg*) and the *PA/CA* spectral power ratio are also shown for each case. Johnson et al. found that the spectral power ratio increased linearly with the transvalvular systolic pressure gradient, and hence correlated well with the severity of aortic stenosis. The importance of recording the PCG in the aortic area, prefiltering the PCG to 25 – 1,500 *Hz*, and the selection of an appropriate systolic murmur window was discussed by Johnson et al. Although there were confounding factors, it was indicated that the noninvasive PCG-based technique could be useful in identifying the need for catheterization as well as follow-up of patients with aortic stenosis.

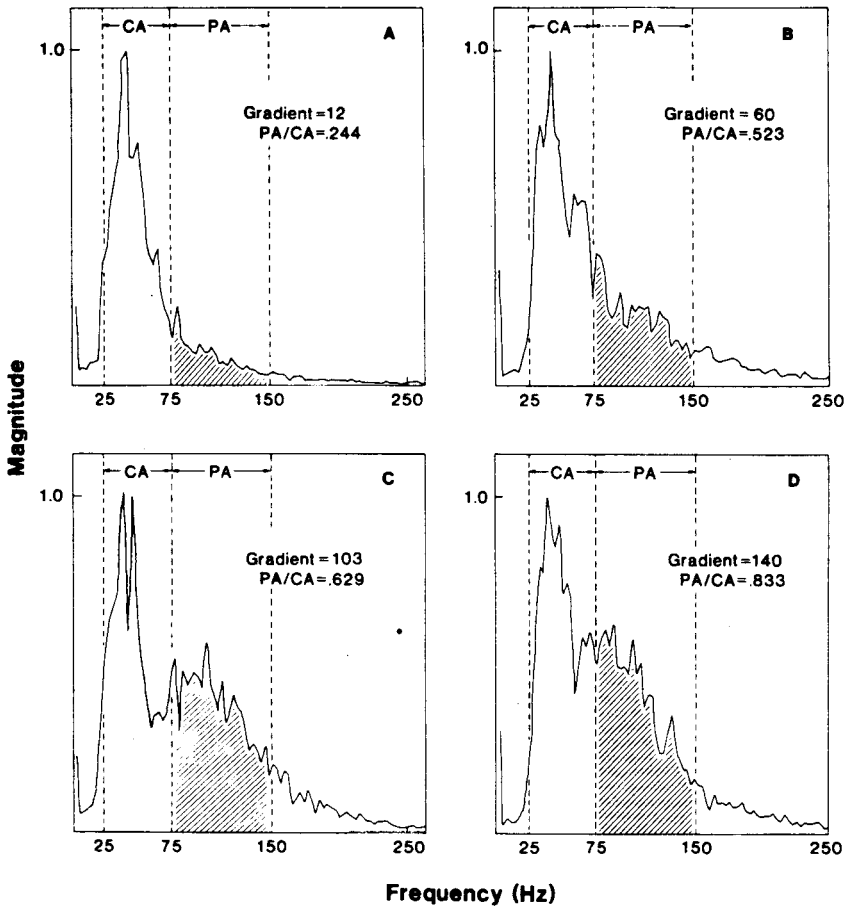


Figure 6.2 Averaged and normalized PSDs of four patients with aortic stenosis of different levels of severity. Each PSD is segmented into two parts: a constant area *CA* and a predictive area *PA*. The transvalvular systolic pressure gradient (measured via catheterization in *mm of Hg*) and the *PA/CA* spectral power ratio are shown for each case. Reproduced with permission from the American College of Cardiology, from G.R. Johnson, R.J. Adolph, and D.J. Campbell, Estimation of the severity of aortic valve stenosis by frequency analysis of the murmur, *Journal of the American College of Cardiology*, 1(5):1315–1323, 1983 ©Elsevier Science.

6.3 Estimation of the PSD

We encountered the ACF and CCF in Equations 3.17, 3.17, 3.20, and 4.27: The first equation cited here provides a general definition of the ACF as a statistical expectation or an integral over an indefinite duration; the last one treats the CCF as the projection of one signal onto another and neglects a scale factor that may be of no consequence in a given application. We shall now investigate more closely the procedures required to estimate the ACF, and hence the PSD, from finite-length signal records.

Let us consider a signal record of N samples: $x(n)$, $n = 0, 1, 2, \dots, N - 1$. In order to compute the time-averaged ACF $\phi_{xx}(m)$ for a delay of m samples, we need to form the product $x(n)x(n \pm m)$ and sum over the available range of data samples. The true ACF is given as $\phi_{xx}(m) = E[x(n)x(n+m)]$. Note that one of the copies of the signal entering the computation of the ACF should be conjugated if the signal is complex.

It is readily seen that we may sum from $n = 0$ to $n = N - 1$ when computing $\phi_{xx}(0)$ with $x(n)x(n) = x^2(n)$. However, when computing $\phi_{xx}(1)$ with $x(n)x(n+1)$, we can only sum from $n = 0$ to $n = N - 2$. As we apply a linear shift of m samples to one copy of the signal to compute $\phi_{xx}(\pm m)$, m samples of one of the copies of the signal drop out of the window of analysis indicated by the overlap between the two versions of the signal; see Figure 6.3. Therefore, only $N - |m|$ pairs of data samples are available to estimate the ACF for the delay of $\pm m$ samples. We then have a sample-mean estimate of the ACF given by

$$\phi_1(m) = \frac{1}{N - |m|} \sum_{n=0}^{N-|m|-1} x(n)x(n+m). \quad (6.1)$$

The subscript xx has been omitted in the equation given above; the subscript 1 indicates the use of one type of averaging scale factor in estimating the ACF. Oppenheim and Schaffer [174] have shown that $\phi_1(m)$ is a consistent estimate of $\phi_{xx}(m)$: It has zero bias and has a variance that tends to zero as $N \rightarrow \infty$. However, the variance of the estimate becomes exceptionally large as m approaches N : Very few nonzero pairs of samples are then available to compute the ACF, and the estimate is useless.

An alternative definition of the ACF ignores the lack of $|m|$ nonzero pairs of samples, and applies the same scale factor for all delays, leading to

$$\phi_2(m) = \frac{1}{N} \sum_{n=0}^{N-|m|-1} x(n)x(n+m). \quad (6.2)$$

Note that the upper limit of summation in the above expression could be stated as $N - 1$ with no effect on the result; the first or the last $|m|$ samples of $x(n)$ will not overlap with $x(n+m)$, and result in zero product terms. Oppenheim and Schaffer [174] have shown that $\phi_2(m)$ has a bias equal to $\frac{m}{N}\phi_{xx}(m)$: The bias tends to the actual value being estimated as m approaches N , although the variance is almost independent of m and tends to zero as $N \rightarrow \infty$. Regardless, both the ACF

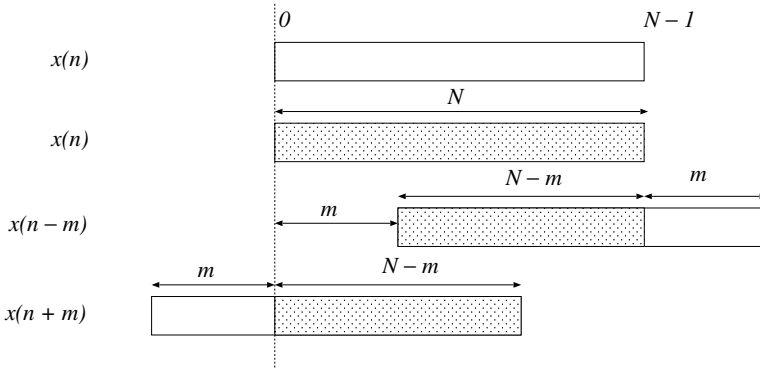


Figure 6.3 The effect of shifting in the computation of the ACF or CCF: as the shift increases, the number of available pairs of samples is reduced.

estimates are asymptotically unbiased [the bias of $\phi_2(m)$ tends to zero as $N \rightarrow \infty$], and they yield good estimates of the ACF as long as the number of samples N is large and $m \ll N$.

Note that the two ACF estimates $\phi_1(m)$ and $\phi_2(m)$ are interrelated as

$$\phi_2(m) = \frac{N - |m|}{N} \phi_1(m). \quad (6.3)$$

Thus, $\phi_2(m)$ is a scaled version of $\phi_1(m)$. However, since the scaling factor is a function of m , it is more commonly referred to as a *window*; more discussion on this interpretation is presented in Section 6.3.1. It should also be observed that the distinction between $\phi_1(m)$ and $\phi_2(m)$ is comparable to that between the unbiased and biased sample variance measures, where the division is by $N - 1$ or N , respectively, with N being the number of samples available.

See Bingham et al. [324], Welch [325], Kay and Marple [326], Thomson [327], Robinson [328], and Childers et al. [329] for discussions on several issues related to estimation of PSDs.

6.3.1 The periodogram

Since the PSD and the ACF are a Fourier transform pair, we may compute an estimate of the PSD as

$$S_2(\omega) = \sum_{m=-(N-1)}^{N-1} \phi_2(m) \exp(-j\omega m), \quad (6.4)$$

assuming that, indeed, the ACF is computed or available for $|m|$ up to $N - 1$. The Fourier transform of the signal $x(n)$, $n = 0, 1, 2, \dots, N - 1$, is given as

$$X(\omega) = \sum_{n=0}^{N-1} x(n) \exp(-j\omega n). \quad (6.5)$$

It can be shown that

$$S_2(\omega) = \frac{1}{N} |X(\omega)|^2. \quad (6.6)$$

The PSD estimate $S_2(\omega)$ is known as the *periodogram* of the signal $x(n)$ [174]. Oppenheim and Schaffer [174] have shown that $S_2(\omega)$ is a biased estimate of the PSD, with

$$E[S_2(\omega)] = \sum_{m=-(N-1)}^{N-1} \frac{N-|m|}{N} \phi_{xx}(m) \exp(-j\omega m). \quad (6.7)$$

If we consider the Fourier transform of $\phi_1(m)$, we get a different estimate of the PSD as

$$S_1(\omega) = \sum_{m=-(N-1)}^{N-1} \phi_1(m) \exp(-j\omega m), \quad (6.8)$$

with the expected value [174]

$$E[S_1(\omega)] = \sum_{m=-(N-1)}^{N-1} \phi_{xx}(m) \exp(-j\omega m). \quad (6.9)$$

Because of the finite limits of the summation, $S_1(\omega)$ is a biased estimate of the PSD.

The two estimates $S_2(\omega)$ and $S_1(\omega)$ may be seen as the Fourier transforms of windowed ACFs, with the window functions being a triangular function, known as the Bartlett window, $w_B(m)$, in the first case and a rectangular window, $w_R(m)$, in the second case:

$$w_B(m) = \begin{cases} \frac{N-|m|}{N}, & |m| < N, \\ 0, & \text{otherwise,} \end{cases} \quad (6.10)$$

$$w_R(m) = \begin{cases} 1, & |m| < N, \\ 0, & \text{otherwise.} \end{cases} \quad (6.11)$$

Note that the windows defined above have a (nonzero) duration of $(2N-1)$ samples.

Since the ACF is multiplied with the window function, the PSD is convolved with the Fourier transform of the window function, leading to spectral leakage and loss of resolution (more details on windows follow in Section 6.3.3). The Fourier transforms of the Bartlett and rectangular windows are, respectively [174],

$$W_B(\omega) = \frac{1}{N} \left[\frac{\sin(\omega N/2)}{\sin(\omega/2)} \right]^2 \quad (6.12)$$

and

$$W_R(\omega) = \frac{\sin[\omega(2N-1)/2]}{\sin(\omega/2)}. \quad (6.13)$$

Oppenheim and Schaffer [174] have shown that the periodogram has a variance that does not approach zero as $N \rightarrow \infty$; instead, the variance of the periodogram is of the order of σ_x^4 regardless of N . Thus, the periodogram is not a consistent estimate of the PSD.

6.3.2 The need for averaging

A common approach to reduce the variance of an estimate is to average over a number of independent estimates or observations. We have seen in Section 3.5.1 how the variance of the noise in noisy signals may be reduced by synchronized averaging over a number of observations of the corrupted signal. In a similar vein, a number of periodograms may be computed over multiple observations of a signal and averaged to obtain a better estimate of the PSD. It is necessary for the process to be stationary, at least during the period over which the periodograms are computed and averaged.

Problem: *How can we obtain an averaged periodogram when we are given only one signal record of finite duration?*

Solution: Oppenheim and Schaffer [174] have described the following procedure, attributed to Bartlett, to average periodograms of segments of the given signal record:

1. Divide the given data sequence $x(n)$, $n = 0, 1, 2, \dots, N - 1$, into K segments of M samples each. We then have the segments given by

$$x_i(n) = x[n + (i - 1)M], \quad 0 \leq n \leq M - 1, \quad 1 \leq i \leq K. \quad (6.14)$$

2. Compute the periodogram of each segment as

$$S_i(\omega) = \frac{1}{M} \left| \sum_{n=0}^{M-1} x_i(n) \exp(-j\omega n) \right|^2, \quad 1 \leq i \leq K. \quad (6.15)$$

The Fourier transform in the equation given above is evaluated as a DFT (using the FFT algorithm) in practice.

3. If the ACF $\phi_{xx}(m)$ is negligible for $|m| > M$, the periodograms of the K segments of duration M samples each may be assumed to be mutually independent. Then, the Bartlett estimate $S_B(\omega)$ of the PSD is obtained as the sample mean of the K independent observations of the periodogram:

$$S_B(\omega) = \frac{1}{K} \sum_{i=1}^K S_i(\omega). \quad (6.16)$$

Oppenheim and Schaffer [174] have shown that the expected value of the Bartlett estimate $S_B(\omega)$ is the convolution of the true PSD $S_{xx}(\omega)$ with the Fourier transform of the Bartlett window given in Equation 6.12 (with N replaced by M). The convolution relationship indicates the bias in the estimate, and has the effect of spectral smearing and leakage; the bias may, therefore, be interpreted as a loss in resolution. Although $S_B(\omega)$ is a biased estimate, its variance tends to zero as the number of segments K increases. Therefore, it is a consistent estimate.

When we have a (stationary) signal of fixed duration of N samples, we will face limitations on the number of segments K that we may obtain. While the variance of the estimate decreases as K is increased, it should be recognized that there is a

concomitant decrease in the number of samples M per segment. As M decreases, the main lobe of the Fourier transform of the Bartlett window (see Equation 6.12) widens; frequency resolution is lost because the estimate is the convolution of the true PSD with the window's frequency response. An illustration of application of the Bartlett procedure is provided at the end of Section 6.3.3.

Cyclostationary signals such as the PCG offer a unique and interesting approach to synchronized averaging of periodograms over a number of cycles, without the trade-off between the reduction of variance and the loss of resolution imposed by segmentation as described above. This is presented as an illustration of application in Section 6.3.5.

6.3.3 The use of windows: Spectral resolution and leakage

The Bartlett procedure may be viewed as an ensemble averaging approach to reduce the variance (which may be interpreted as noise) of the periodogram. Another approach to obtain a smooth spectrum is to convolve the periodogram $S(\omega)$ with a filter or smoothing function $W(\omega)$ in the frequency domain (similar to the use of an MA filter in the time domain). The smoothed estimate $S_s(\omega)$ is given by

$$S_s(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} S(\nu) W(\omega - \nu) d\nu, \quad (6.17)$$

where ν is a temporary variable for integration.

As the PSD is a nonnegative function, the smoothing function $W(\omega)$ should satisfy $W(\omega) \geq 0$, $-\pi \leq \omega \leq \pi$. The Fourier transform of the Bartlett window $W_B(\omega)$ meets this requirement. Oppenheim and Schaffer [174] have shown that the variance of the smoothed periodogram is reduced approximately by the factor

$$\frac{1}{N} \sum_{m=-(M-1)}^{M-1} w^2(m) = \frac{1}{2\pi N} \int_{-\pi}^{\pi} W^2(\omega) d\omega, \quad (6.18)$$

with reference to the variance of the original periodogram; here, N is the total number of samples in the signal and $(2M - 1)$ is the number of samples in the smoothing window function. A rectangular window offers a variance-reduction factor of approximately $\frac{2M}{N}$, whereas the factor for the Bartlett window is $\frac{2M}{3N}$ [174]. It should be noted that smoothing of the spectrum (reduction of variance) is achieved at the price of loss of frequency resolution.

Since the periodogram is the Fourier transform of the ACF estimate $\phi(m)$, the convolution operation in the frequency domain in Equation 6.17 is equivalent to multiplying $\phi(m)$ with $w(m)$, the inverse Fourier transform of $W(\omega)$. This result suggests that the same PSD estimate as $S_s(\omega)$ may be obtained by applying a window to the ACF estimate and then taking the Fourier transform of the result. As the ACF is an even function, the window should also be even.

Based upon the arguments outlined above, Welch [325] (see also Oppenheim and Schaffer [174]) proposed a method to average modified periodograms. In Welch's

procedure, the given signal is segmented as in the Bartlett procedure, but a window is applied directly to the original signal segments before Fourier transformation. The periodograms of the windowed segments are defined as

$$S_{W_i}(\omega) = \frac{1}{ME_w} \left| \sum_{n=0}^{M-1} x_i(n)w(n) \exp(-j\omega n) \right|^2, \quad i = 1, 2, \dots, K, \quad (6.19)$$

where E_w is the average power of the window given by

$$E_w = \frac{1}{M} \sum_{n=0}^{M-1} w^2(n). \quad (6.20)$$

Note that the duration of the window defined above is M samples. The Welch PSD estimate $S_W(\omega)$ is obtained by averaging the modified periodograms as

$$S_W(\omega) = \frac{1}{K} \sum_{i=1}^K S_{W_i}(\omega). \quad (6.21)$$

Welch [325] showed that, if the segments are not overlapping, the variance of the averaged modified periodogram is inversely proportional to K , the number of segments used. Welch also suggested that the segments may be allowed to overlap, in which case the modified periodograms are not mutually independent. The spectral window that is effectively convolved with the PSD in the frequency domain is proportional to the squared magnitude of the Fourier transform of the time-domain data window applied. Therefore, no matter which type of a data window is used, the spectral smoothing function is nonnegative, thereby guaranteeing that the PSD estimate will be nonnegative as well.

Some of the commonly used data windows are defined below [174, 330]; the windows are of length N samples and causal, defined for $0 \leq n \leq N-1$.

Rectangular:

$$w(n) = 1. \quad (6.22)$$

Bartlett (triangular):

$$w(n) = \begin{cases} \frac{2n}{N-1}, & 0 \leq n \leq \frac{N-1}{2}, \\ 2 - \frac{2n}{N-1}, & \frac{N-1}{2} \leq n \leq N-1. \end{cases} \quad (6.23)$$

Hamming:

$$w(n) = 0.54 - 0.46 \cos \left(\frac{2\pi n}{N-1} \right). \quad (6.24)$$

von Hann (also known as Hann or Hanning):

$$w(n) = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi n}{N-1} \right) \right]. \quad (6.25)$$

Blackman:

$$w(n) = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right). \quad (6.26)$$

Figure 6.4 illustrates the rectangular, Bartlett, Hann, and Hamming windows with $N = 256$ samples. A Hann window with $N = 128$ samples is also illustrated (centered with reference to the longer-duration windows).

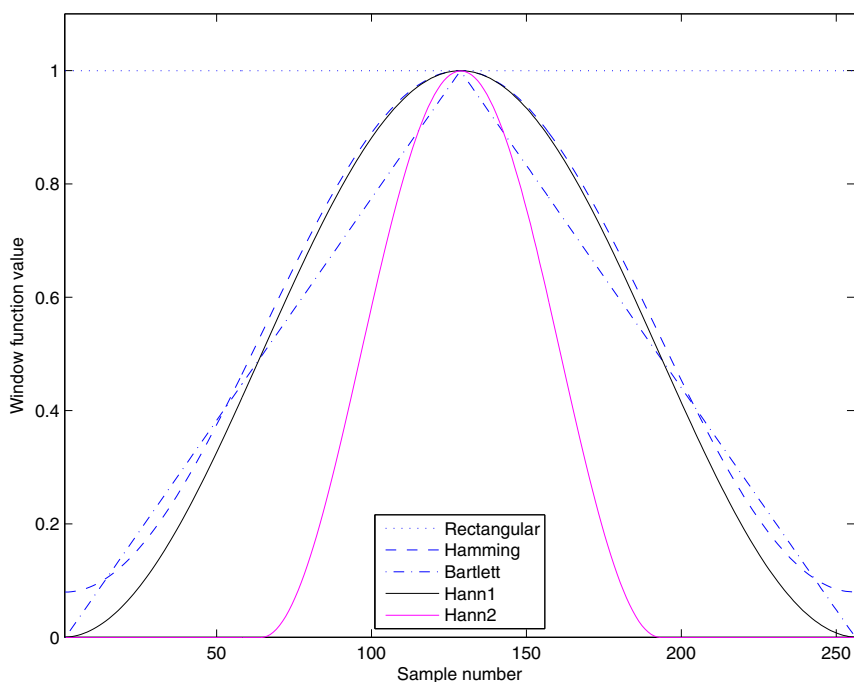


Figure 6.4 Commonly used window functions: rectangular, Bartlett, Hamming, and Hann windows with $N = 256$ (Hann1), and Hann window with $N = 128$ samples (Hann2). All windows are centered at the 128th sample.

Use of the tapered windows (all of the above, except the rectangular window) provides the advantage that the ends of the given signal are reduced to zero (with the further exception of the Hamming window, for which the end-values are not zero but 0.08). This feature means that there are no discontinuities in the periodic version of the signal encountered in DFT-based procedures. All of the window functions listed above are symmetric (even) functions and, therefore, have a linear phase (or a real spectrum with zero phase if the window is centered at the origin).

Figures 6.5 to 6.9 illustrate the log-magnitude frequency responses of the window functions shown in Figure 6.4. The frequency responses were computed after padding the windows to a total duration of $L = 2,048$ samples for FFT computation.

The plots are on an expanded scale over the limited normalized-frequency range of $(0, 0.1)$ in order to illustrate clearly the characteristics of the main lobe and the side lobes. The plot for the frequency response of the rectangular window has been plotted to a different scale as compared to the rest. The discontinuities in the frequency responses of the rectangular and Bartlett windows in Figures 6.5 and 6.6 are due to the log of the zeros of the responses being $-\infty$.

The rectangular window has the narrowest main lobe of width $\frac{4\pi}{N}$; the main lobe is wider at $\frac{8\pi}{N}$ for the Bartlett, Hann, and Hamming windows; it is the widest at $\frac{12\pi}{N}$ for the Blackman window [174]. A reduction in window width will lead to an increase in the main-lobe width, as illustrated by the frequency responses of the two Hann windows in Figures 6.8 and 6.9. Note that the wider the main lobe, the greater the spectral smoothing, and hence the loss of spectral resolution is more severe.

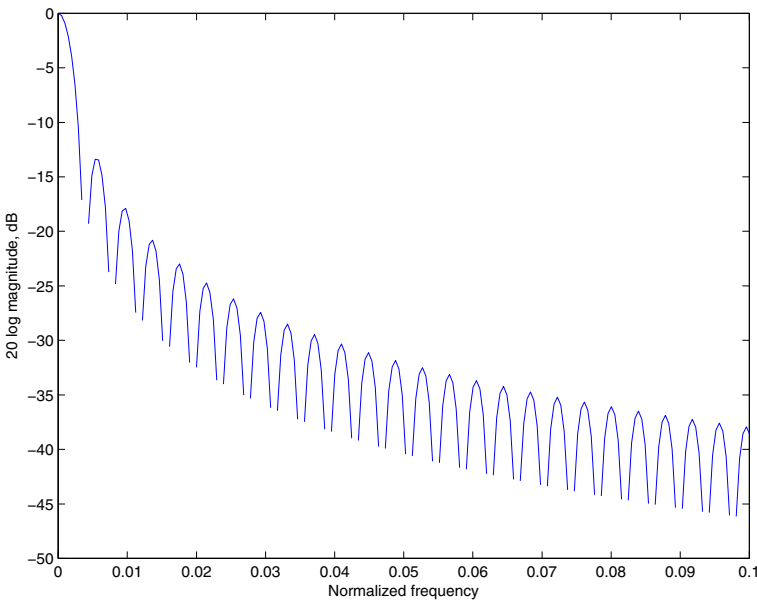


Figure 6.5 Log-magnitude frequency response of the rectangular window illustrated in Figure 6.4. The window width is $N = 256$ samples.

The rectangular window has the highest peak side-lobe levels of all of the windows listed at -13 dB, with the Bartlett, Hann, Hamming, and Blackman windows having their peak side-lobe levels at -25 dB, -31 dB, -41 dB, and -57 dB, respectively [174]. Higher levels of the side lobes will cause increased spectral leakage (weighted summation of spectral components with significant weights over a wide range of frequencies due to convolution in the frequency domain), resulting in a more distorted spectrum. Note that reduction of leakage through the use of the tapered windows comes at the price of increased main-lobe width and, therefore, more severe loss of spectral resolution (more smoothing).

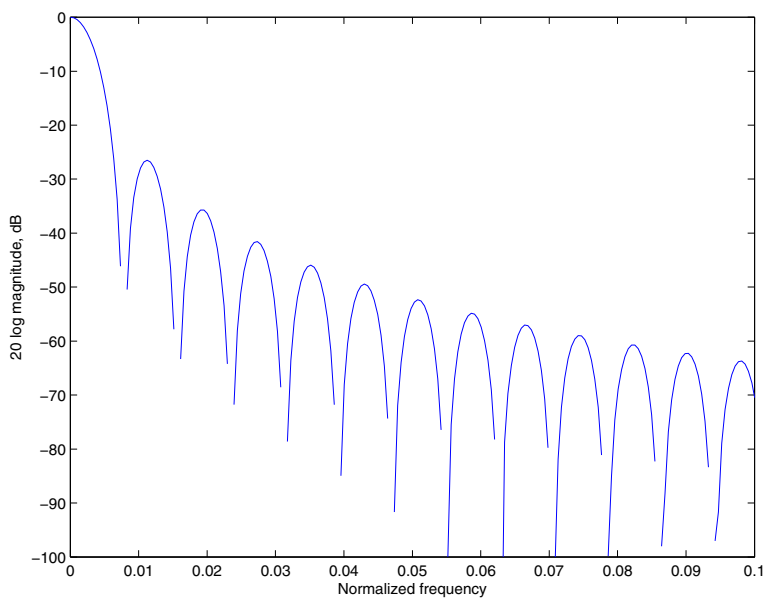


Figure 6.6 Log-magnitude frequency response of the Bartlett window illustrated in Figure 6.4. The window width is $N = 256$ samples.

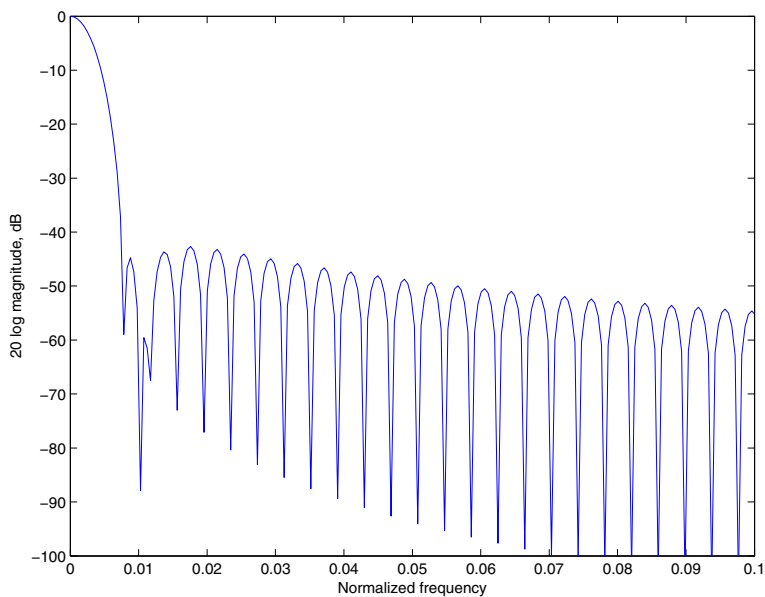


Figure 6.7 Log-magnitude frequency response of the Hamming window illustrated in Figure 6.4. The window width is $N = 256$ samples.

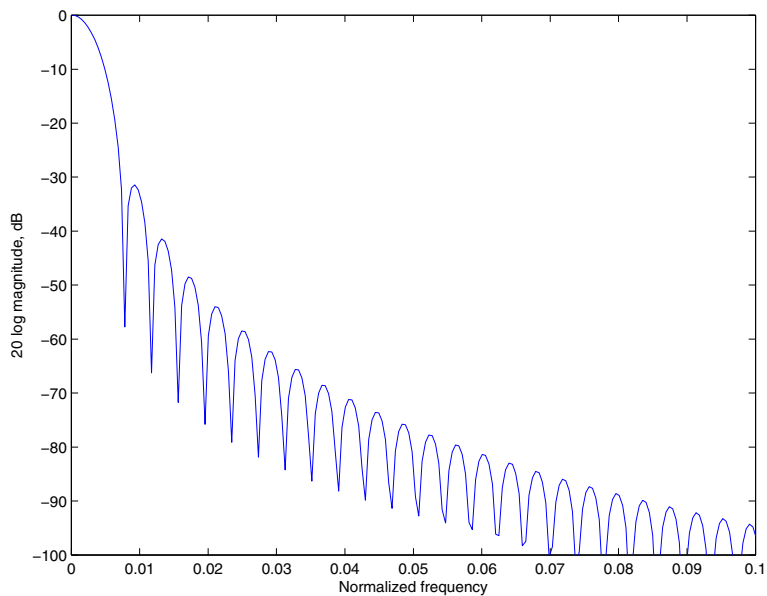


Figure 6.8 Log-magnitude frequency response of the Hann1 window illustrated in Figure 6.4. The window width is $N = 256$ samples.

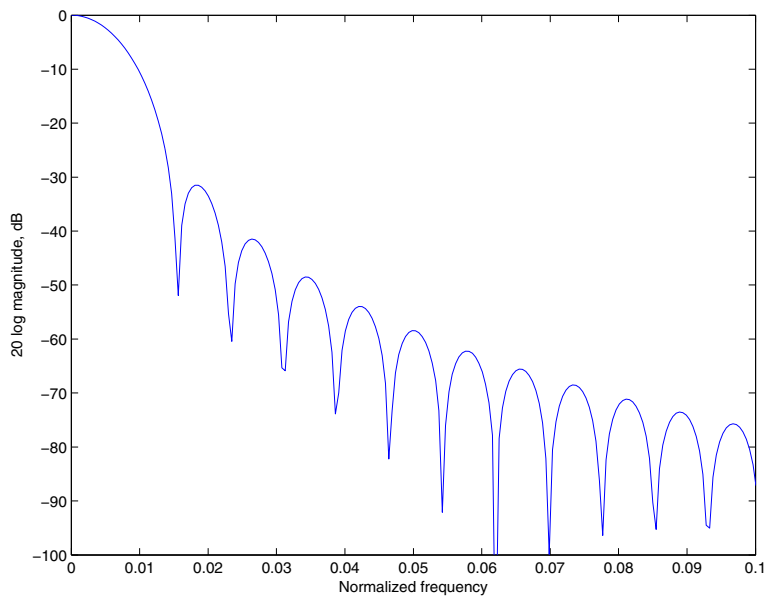


Figure 6.9 Log-magnitude frequency response of the Hann2 window illustrated in Figure 6.4. The window width is $N = 128$ samples.

Illustrations of application: The Welch method of windowing signal segments and averaging their PSDs was applied to the o2 channel of the EEG signal illustrated in Figure 1.39. The number of samples in the signal is $N = 750$, with the sampling frequency being $f_s = 100 \text{ Hz}$. Note that the specific EEG signal record may be assumed to be stationary over its relatively short duration of 7.5 s . The dominant activity in the signal is the alpha rhythm, which appears throughout the duration of the signal record.

The PSD of the entire signal was first computed using no window (that is, the rectangular window was applied implicitly); the FFT array was computed with $L = 1,024$ samples. The top trace in Figure 6.10 illustrates the PSD of the signal.

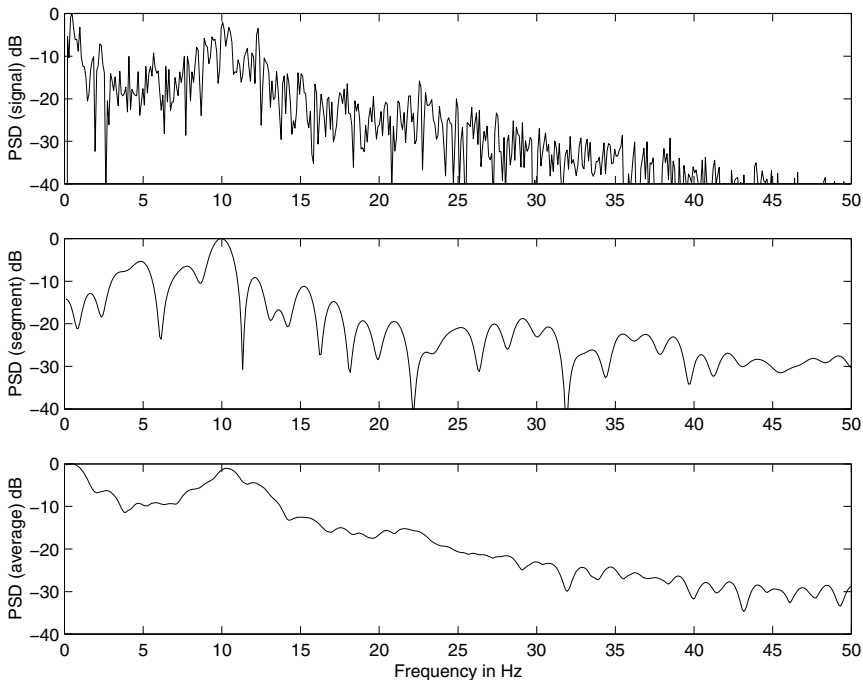


Figure 6.10 Bartlett PSD estimate of the o2 channel of the EEG signal in Figure 1.39. Top trace: PSD of the entire signal. Middle trace: PSD of the 11th segment. Bottom trace: Averaged PSD using $K = 11$ segments of the signal. The rectangular window was (implicitly) used in all cases. Number of samples in the entire signal: $N = 750$. Number of samples in each segment: $M = 64$. All FFT arrays were computed with $L = 1,024$ samples. Sampling frequency $f_s = 100 \text{ Hz}$. See also Figure 6.11.

For the first averaged periodogram procedure, the EEG signal was segmented with $M = 64$ samples each, with implicit use of the rectangular window (equivalent to the Bartlett method). A total of $K = 11$ segments were obtained. Each segment was padded with zeros to a length of $L = 1,024$ for the sake of FFT computation. The PSDs of the segments were then averaged, followed by normalization and logarithm-

mic transformation. The second and third plots in Figure 6.10 illustrate the PSD of a sample segment (the 11th segment) and the averaged PSD (the Bartlett estimate), respectively. It is seen that the averaged PSD (third trace) provides a smooth spectral estimate with a clearly dominant peak at approximately 10 Hz, representing the alpha rhythm present in the signal. The PSD of the individual segment (middle trace) displays many peaks and valleys that are possibly spurious and not significant, and have been suppressed or smoothed by the averaging process, as evident in the third plot in the figure. The single PSD computed from the entire signal (top trace) exhibits numerous variations that may not be relevant and could confound visual or automated analysis. (*Note:* Direct comparison of the PSDs is possible since they have the same number of samples, that is, the same frequency sampling rate.)

Figure 6.11 illustrates a second set of PSDs similar to that in Figure 6.10, but with the use of the Hann window in the Welch procedure. The effect of the Hann window is not significant in the case of the PSD of the entire signal (top trace), as the window length is reasonably large ($N = 750$). However, the Hann window has clearly smoothed the multiple (possibly spurious) peaks and valleys in the PSD of the segment illustrated in the middle trace. The wider main lobe of the Hann window's frequency response has caused a more severe loss of frequency resolution (smoothing) than the rectangular window in the case of the corresponding PSD in Figure 6.10. Finally, the averaged PSD in the lowest trace of Figure 6.11 clearly illustrates the benefit of the Hann window in the significantly reduced power levels beyond 30 Hz. The lower side-lobe levels of the Hann window have resulted in less spectral leakage than that in the case of the rectangular window as illustrated by the corresponding PSD in Figure 6.10. The price paid, however, is evidenced by the wider peak in the averaged PSD with the Hann window, which spans the range 5 – 15 Hz at the -10 dB level. The two distinct peaks at about 10 Hz and 12 Hz that are evident in the top traces of Figures 6.10 and 6.11 as well as in the smoothed PSD in the bottom trace of Figure 6.10 are no longer seen separately in the bottom trace of Figure 6.11. Regardless, the averaged PSD with the Hann window appears to be smoother and more amenable to analysis than the corresponding result with the rectangular window.

It should be noted that the segments of EEG signals used in the illustrations in the present section are of short duration. See Section 6.7 for an example of analysis of longer EEG records.

6.3.4 Estimation of the ACF

Good estimates of the ACF are required in applications such as the design of the optimal Wiener filter and estimation of the statistics of stochastic processes. Once a PSD estimate has been obtained by a method such as the Bartlett or Welch procedures, we may take the inverse Fourier transform of the result and use the result as an estimate of the ACF. We may also fit a smooth curve or a parametric model (such as an exponential or a Laplacian function) to the PSD or to the equivalent ACF model.

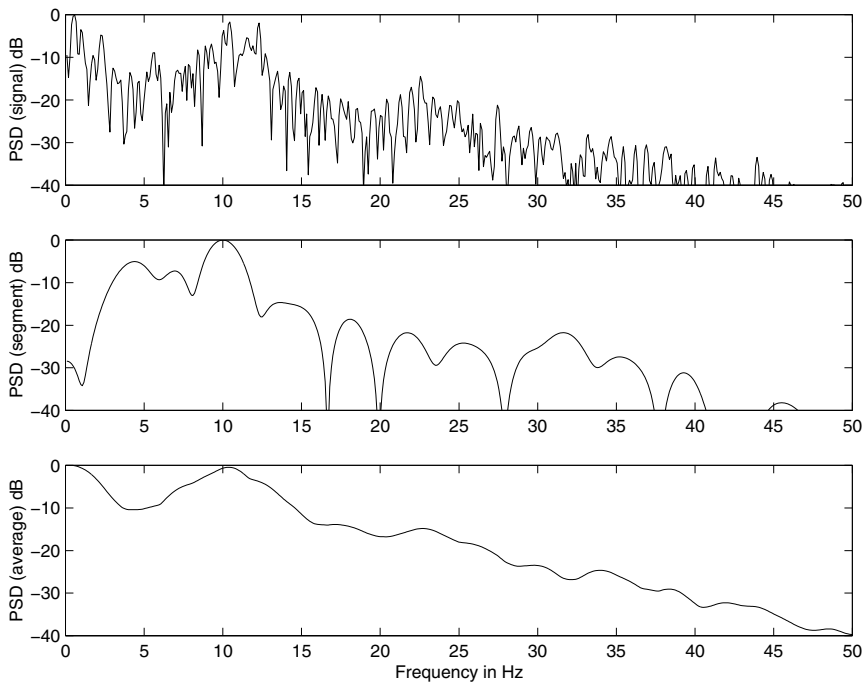


Figure 6.11 Welch PSD estimate of the o2 channel of the EEG signal in Figure 1.39. Top trace: PSD of the entire signal. Middle trace: PSD of the 11th segment. Bottom trace: Averaged PSD using $K = 11$ segments of the signal. The Hann window was used in all cases. Number of samples in the entire signal and the size of the Hann window used in computing the PSD of the entire signal: $N = 750$. Number of samples in each segment and the size of the Hann window used in the averaged periodogram method: $M = 64$. All FFT arrays were computed with $L = 1,024$ samples. Sampling frequency $f_s = 100$ Hz. See also Figure 6.10.

Let us consider again the expression

$$\phi_2(m) = \frac{1}{N} \sum_{n=0}^{N-|m|-1} x(n)x(n+m). \quad (6.27)$$

As the ACF is an even function, we need to compute it only for positive m . It is evident that the ACF estimate is simply the result of linear convolution of $x(n)$ with $x(-n)$ (with the scale factor $\frac{1}{N}$). If the DFT of $x(n)$ is $X(k)$, the DFT of $x(-n)$ is $X^*(k)$. Since convolution in the time domain is multiplication in the frequency domain, we could compute the DFT $X(k)$ of $x(n)$, obtain $X(k)X^*(k) = |X(k)|^2$, and take its inverse DFT. However, the DFT procedure provides circular convolution and not linear convolution. Therefore, we need to pad $x(n)$ with at least $M - 1$ zeros, where M is the largest lag for which the ACF is desired. The DFT must then be computed with at least $L = N + M - 1$ samples, where N is the number of

samples in the original signal. If this requirement is built into the periodogram or averaged periodogram procedure, the inverse DFT of the final PSD estimate may be used as an estimate of the ACF [with the scale factor $\frac{1}{N}$, or division by $\phi(0)$ to get the normalized ACF].

6.3.5 Synchronized averaging of PCG spectra

Every individual is familiar with the comforting *lub–dub* sounds of his or her heart beat; every prospective parent would have taken pleasure in listening to the throbbing heart of the yet-to-be-born baby. Use of the heart sounds is extremely common in clinical practice: the stethoscope is the most common sign and tool of a physician. Yet, behind this common signal lie many sophisticated and potentially complicating characteristics.

The PCG is a nonstationary signal due to the fact that the amount of blood in each cardiac chamber and the state of contraction of the muscles change continuously during each cardiac cycle. S2 usually has more high-frequency content than S1: The PSD of a normal PCG signal changes within about 300 *ms*. Valve opening or closing sounds, being of short duration of the order of 10 *ms*, are of a transient and high-frequency character. The presence of murmurs adds another dimension of nonstationarity, with frequency content well beyond that of the normal heart sounds: the PSD of an abnormal PCG could change every 100 *ms* or less. Individual epochs of S1, S2, valve snaps, and murmurs are of limited durations of the order of 10 – 300 *ms*. These aspects of the PCG preclude segmented averaging as recommended by the Bartlett or Welch procedures.

Over and above all of the factors mentioned in the preceding paragraph, the transmission characteristics of the chest wall change during breathing. (Living systems are dynamic!) The PCG signals recorded at various locations on the chest are also subject to different transmission-path effects. While adult subjects may cooperate in PCG signal acquisition by holding their breath or performing other maneuvers, these possibilities cannot be considered in the case of infants and young children in poor states of health. The PCG signal presents more challenges in acquisition and analysis than most of the other biomedical signals we have encountered [71].

Problem: *Propose a method to obtain averaged PSD estimates of the systolic and diastolic heart sounds.*

Solution: The cyclostationarity of the PCG lends itself to a unique approach to averaging PCG segments corresponding to the same phase of the cardiac cycle extracted from multiple beats. If the subject were to hold his/her breath during the period of acquisition of the PCG record, the chest-wall transmission characteristics will be stationary over the multiple cardiac cycles in the record. Therefore, we may segment S1, S2, or any portion of the cardiac cycle of interest from as many beats as are available, and then average their PSD estimates in a procedure similar to the Bartlett or Welch procedures. (*Note:* Direct averaging of the PCG signals themselves or of their complex Fourier transforms could lead to undesired cancellation of noise-like murmurs or asynchronous frequency components and their disappearance from

the result! Refer to Sections 4.11 and 6.5 for discussions on intentional cancellation of asynchronous components in the PCG via synchronized averaging.)

We saw in Sections 5.5.2 and 5.5.3 how the envelope or the envelopogram of the PCG may be averaged over several cardiac cycles. However, there was no need to segment parts of a cardiac cycle in envelope analysis: Nonstationarity of the signal within a cardiac cycle was not of concern. In the present application of PSD analysis, there is a need to segment the PCG further.

A procedure was described in Section 4.10 for segmentation of the systolic and diastolic parts of PCG signals based upon the detection of the QRS complex in the ECG and the detection of the dicrotic notch in the carotid pulse signal. Further segmentation of the systolic or diastolic parts into S1 and systolic murmur or S2 and diastolic murmur, respectively, would require more sophisticated methods, which are the topics of Chapter 8. For now, let us consider the task of obtaining averaged PSDs of the systolic and diastolic parts of a PCG signal.

Figure 6.12 shows the PCG signal over one cardiac cycle of a normal subject segmented using the procedure described in Section 4.10 and illustrated in Figure 4.30 (see also Figure 5.7). The periodograms of the systolic and diastolic parts of the PCG cycle illustrated are also shown in the figure. In order to obtain better PSD estimates, the periodogram of each systolic or diastolic segment was computed separately and averaged over 16 cardiac cycles. No data window was applied (the rectangular window was used, in effect); therefore, the procedure used is similar to the Bartlett procedure. Individual systolic or diastolic segments could be of different durations; for the present illustration, all periodograms were computed with the same number of samples, which was taken to be the maximum RR interval in the ECG record of the subject. The averaged systolic and diastolic PSD estimates are shown in Figure 6.12. (Averaging was performed using the nonnegative PSDs before conversion to dB .) The averaging procedure provides smoother estimates of the PSDs by removing beat-to-beat variations that are neither significant nor of interest. Spectral peaks may be clearly observed in the averaged periodograms and may be considered to be more reliable estimates of resonance than the peaks found in individual periodograms.

Figure 6.13 illustrates a PCG signal cycle as well as the individual and averaged systolic and diastolic PSD estimates for a patient with systolic murmur, split S2, and opening snap of the mitral valve (see also Figures 4.31 and 5.8). It is unlikely that the patient held her breath during data acquisition. The presence of increased high-frequency power in the range $120 - 250\text{ Hz}$ due to the systolic murmur is evident in the averaged systolic PSD. The diastolic PSDs are comparable to the corresponding normal diastolic PSDs in Figure 6.12.

6.4 Measures Derived from Power Spectral Density Functions

The Fourier spectrum or PSD provides us with a density function of signal amplitude, power, or energy versus frequency. We would typically have a large number of samples of the PSD over a wide frequency range, which may not lend itself to easy

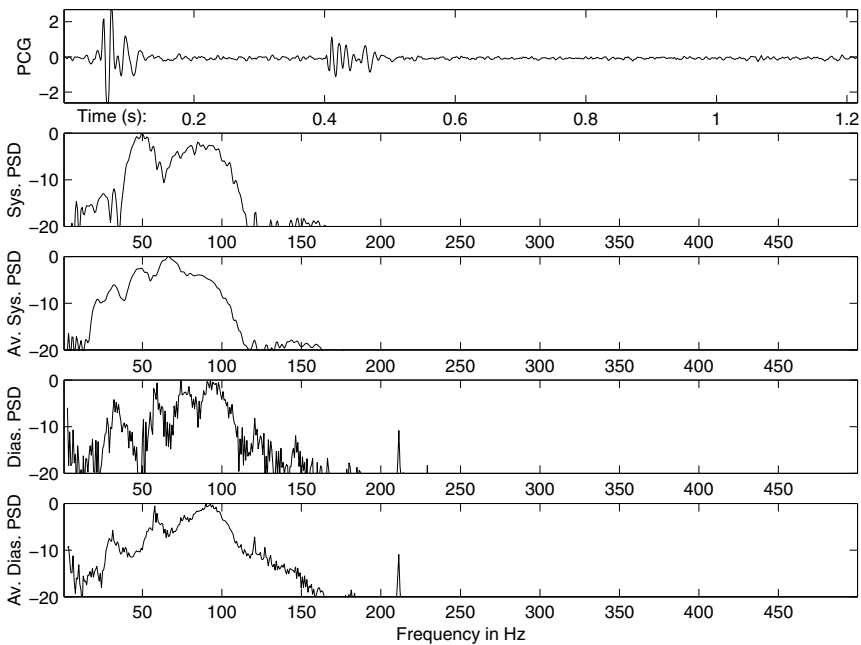


Figure 6.12 Top to bottom: A sample PCG signal over one cardiac cycle of a normal subject (male, 23 years; see also Figures 4.30 and 5.7); periodogram of the systolic portion of the signal (approximately 0 – 0.4 s); averaged periodogram of the systolic parts of 16 cardiac cycles segmented as illustrated in Figure 4.30; periodogram of the diastolic portion of the signal shown in the first plot (approximately 0.4 – 1.2 s); averaged periodogram of the diastolic parts of 16 cardiac cycles. The periodograms are on a log scale (dB). Av. = average. Sys. = systolic. Dias. = diastolic.

analysis. We may, of course, study the shape of the spectrum graphically and observe its general characteristics. Such an approach is often referred to as *nonparametric* spectral analysis. The spectral models described in Section 7.4 are characterized by a small number of parameters; hence, the procedure is known as *parametric* spectral analysis (or modeling).

Problem: *Derive parameters or measures from a Fourier spectrum or PSD that can help in the characterization of the spectral variations or features contained therein.*

Solution: Since the PSD is a nonnegative function as well as a density function, we may readily treat it as a PDF, and compute statistics using moments. We may also detect peaks corresponding to resonance, measure their bandwidth or quality factor, and derive measures of concentration of power in specific frequency bands of interest or concern. Although the PSD itself is nonparametric, we may derive several parameters that, while not completely representing the entire PSD, may facilitate the

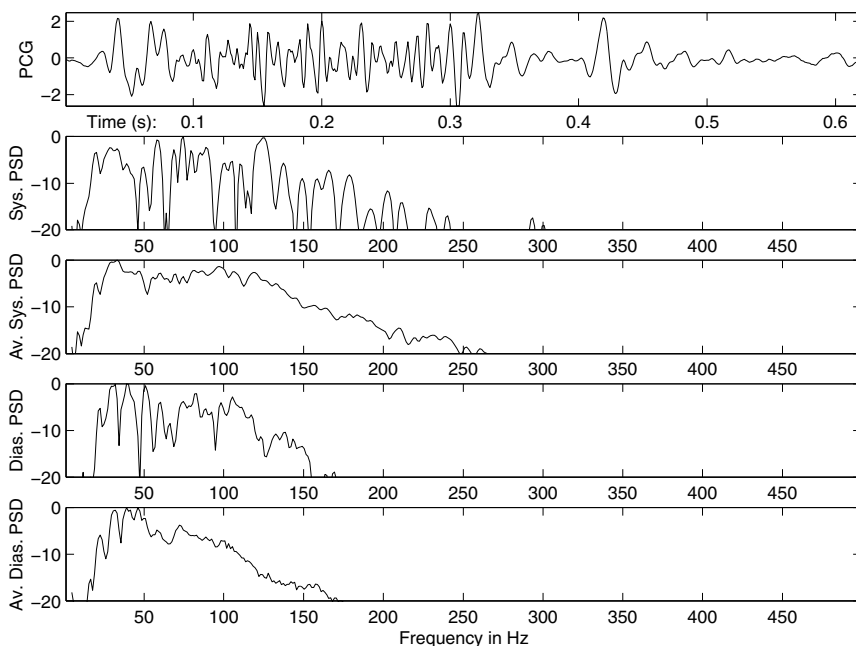


Figure 6.13 Top to bottom: A sample PCG signal over one cardiac cycle of a patient (female, 14 months; see also Figures 4.31 and 5.8) with systolic murmur, split S2, and opening snap of the mitral valve; periodogram of the systolic portion of the signal (approximately 0 – 0.28 s); averaged periodogram of the systolic parts of 26 cardiac cycles segmented as illustrated in Figure 4.31; periodogram of the diastolic portion of the signal shown in the first plot (approximately 0.28 – 0.62 s); averaged periodogram of the diastolic parts of 26 cardiac cycles. The periodograms are on a log scale (*dB*). Av. = average. Sys. = systolic. Dias. = diastolic.

identification of physiological and/or pathological phenomena. We shall investigate a few different approaches toward this end in the following sections.

6.4.1 Moments of PSD functions

As the area under the PSD curve represents the total signal power or energy which need not be unity, we have to normalize all moments by the total power or energy of the signal E_x , given by

$$\begin{aligned}
 E_x &= \sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2 \\
 &= \frac{1}{2\pi} \int_0^{2\pi} |X(\omega)|^2 d\omega = \int_{f_n=0}^1 |X(f_n)|^2 df_n. \quad (6.28)
 \end{aligned}$$

The frequency variable f_n above is normalized as $f_n = f/f_s$, such that $0 \leq f_n \leq 1$. Assuming that the PSD has been obtained using one of the methods described in the preceding sections, we may replace $|X(\cdot)|^2$ in the above expressions by $S_{xx}(\cdot)$.

In all definitions of the moments of PSDs in the present section,

$$S_{xx}(k) = \frac{1}{N} |X(k)|^2,$$

where $X(k)$ is the DFT of $x(n)$. $S_{xx}(k)$ represents the PSD of $x(n)$. Note the presence of the factor $\frac{1}{N}$ in the definition of the PSD. An additional factor of $\frac{1}{N}$ is required in the definitions of the moments of PSDs in this section, in order to scale the frequency index k to the appropriate frequency in Hz in relation to the sampling frequency f_s . Two versions of the moments are provided: one using the normalized frequency variable $0 \leq f_n \leq 1$ and the other using the DFT index $0 \leq k \leq N-1$.

As a measure of the concentration or spread of the signal power over its frequency range, we may compute the mean frequency f as the first-order moment

$$\overline{f_n} = \frac{2}{E_x} \int_{f_n=0}^{0.5} f_n S_{xx}(f_n) df_n \quad (6.29)$$

or as

$$\overline{f_{Hz}} = \frac{f_s}{N} \frac{2}{E_x} \sum_{k=0}^{N/2} k S_{xx}(k), \quad (6.30)$$

where $\overline{f_n}$ and $\overline{f_{Hz}}$ indicate the mean frequency in its normalized form and in Hz , respectively, and N is the number of samples in the DFT-based representation of the PSD. The upper limit of integration of 0.5 represents integration from DC to the maximum frequency present in the signal, which is one-half of the sampling frequency, the frequency variable having been normalized to the range $0 \leq f_n \leq 1$. Note that the integration or summation is performed over one-half period of the periodic function $S_{xx}(f_n)$ or $S_{xx}(k)$, which possesses even symmetry about one-half of the sampling frequency for real signals.

The median frequency f_{med} is defined as that frequency which splits the PSD in half:

$$\begin{aligned} f_{med} &= \frac{m}{N} f_s \text{ with the largest } m \text{ such that} \\ \frac{2}{E_x} \sum_{k=0}^m S_{xx}(k) &< \frac{1}{2}; \quad 0 \leq m \leq \frac{N}{2}. \end{aligned} \quad (6.31)$$

We may also compute higher-order statistics as follows:

- Variance f_{m2} as the second-order moment by using $(f_n - \overline{f_n})^2$ in place of f_n [the function of frequency that is multiplied with $S_{xx}(f_n)$] in Equation 6.29 or the equivalent expression in k in Equation 6.30; that is,

$$f_{m2} = \frac{2}{E_x} \int_{f_n=0}^{0.5} (f_n - \overline{f_n})^2 S_{xx}(f_n) df_n \quad (6.32)$$

or

$$f_{m2} = \left(\frac{f_s}{N}\right)^2 \frac{2}{E_x} \sum_{k=0}^{N/2} (k - \bar{k})^2 S_{xx}(k), \quad (6.33)$$

where $\bar{k} = N \overline{f_{Hz}} / f_s$ is the frequency sample index corresponding to $\overline{f_{Hz}}$.

▪ Skewness as

$$\text{skewness} = \frac{f_{m3}}{(f_{m2})^{3/2}}, \quad (6.34)$$

where the third-order moment f_{m3} is computed with $(f_n - \overline{f_n})^3$ in place of f_n in Equation 6.29, that is,

$$f_{m3} = \frac{2}{E_x} \int_{f_n=0}^{0.5} (f_n - \overline{f_n})^3 S_{xx}(f_n) df_n \quad (6.35)$$

or

$$f_{m3} = \left(\frac{f_s}{N}\right)^3 \frac{2}{E_x} \sum_{k=0}^{N/2} (k - \bar{k})^3 S_{xx}(k). \quad (6.36)$$

▪ Kurtosis as

$$\text{kurtosis} = \frac{f_{m4}}{(f_{m2})^2}, \quad (6.37)$$

where the fourth-order moment f_{m4} is computed with $(f_n - \overline{f_n})^4$ in place of f_n in Equation 6.29, that is,

$$f_{m4} = \frac{2}{E_x} \int_{f_n=0}^{0.5} (f_n - \overline{f_n})^4 S_{xx}(f_n) df_n \quad (6.38)$$

or

$$f_{m4} = \left(\frac{f_s}{N}\right)^4 \frac{2}{E_x} \sum_{k=0}^{N/2} (k - \bar{k})^4 S_{xx}(k). \quad (6.39)$$

In order to avoid notational clutter, the same symbols have been used for the two versions each of f_{m2} , f_{m3} , and f_{m4} ; it is understood that similar versions of the moments are used to compute the ratios for skewness and kurtosis.

The mean frequency is a useful measure of the concentration of signal power, and could indicate the resonance frequency in the case of unimodal distributions. However, a nearly uniform PSD could lead to one-half of the maximum frequency as the mean frequency, which by itself may not be a useful representation of the PSD. The presence of multiple resonance frequencies could also lead to a mean frequency that may not be a useful measure. Multimodal PSDs may be characterized better by a series of peak frequencies, along with measures of their relative levels and bandwidths or quality factors (described in the next section).

The square root of f_{m2} provides a measure of spectral spread (SD about the mean) and an indication of the bandwidth (but not at -3 dB) about the mean frequency.

The skewness is zero if the density function is symmetric about the mean frequency; otherwise, it indicates the extent of asymmetry of the distribution. Kurtosis indicates if the PSD is a long-tailed function.

Moments of PSDs may be useful in characterizing the general trends in the distribution of the power of a signal over its bandwidth. The higher-order moments are sensitive to noise or spurious variations in the PSD estimate, and they may not yield reliable measures if the PSD pattern is not simple or if the PSD estimate is poor (has a high variance). The reliability of moments may be improved by smoothing the PSD estimate, or by fitting a smooth parametric curve (such as a Gaussian, a Laplacian, or a spline) as a model of the PSD estimate and computing the moments of the model. Saltzberg and Burch [249] discuss the relationship between moments of PSDs and ZCR , along with their application to EEG analysis.

6.4.2 Spectral power ratios

The moments described in the preceding section provide general statistical characterization of the PSD treated as a PDF. In the case of analysis of biomedical signals, it may be more advantageous to define specific measures based upon *a priori* information or empirical knowledge about the signals, the systems, and the physiological or pathological processes of concern. For example, in the case of PCG analysis for the detection of murmurs, we could specifically investigate the presence of signal power in the frequency range beyond that of S1 and/or S2. If a specific type of pathology of interest is known to cause a shift in the frequency content within a certain band of frequencies, we may measure spectral power ratios over partitions of the band of interest. We have seen in Sections 6.2.1 and 6.2.2 how such measures have been used for the analysis of ventricular elasticity, diagnosis of myocardial infarction, and detection of murmurs.

The fraction of signal power in a frequency band of interest ($f_1 : f_2$) may be computed as

$$E_{(f_1:f_2)} = \frac{2}{E_x} \int_{f=f_1}^{f_2} |X(f)|^2 df = \frac{2}{NE_x} \sum_{k=k_1}^{k_2} |X(k)|^2, \quad (6.40)$$

where k_1 and k_2 are the DFT indices corresponding to f_1 and f_2 , respectively. Fractions of power as above may be computed for several bands of interest that may or may not span the entire signal bandwidth.

In a variation of the fractional-power measure defined above, Johnson et al. [322] compared the integral of the magnitude spectrum of the systolic murmurs due to aortic stenosis over the band 75 : 150 Hz to that over the band 25 : 75 Hz. They considered the higher-frequency band to represent the *predictive area* (PA) of the spectrum related to aortic stenosis and considered the lower-frequency band to represent a *constant area* (CA) that would be common to all systolic PCG signal

segments. The ratio of PA to CA was defined as

$$\frac{PA}{CA} = \frac{\int_{f=f_2}^{f_3} |X(f)| df}{\int_{f=f_1}^{f_2} |X(f)| df}, \quad (6.41)$$

with $f_1 = 25 \text{ Hz}$, $f_2 = 75 \text{ Hz}$, and $f_3 = 150 \text{ Hz}$. The $\frac{PA}{CA}$ ratio is provided for the PSDs of systolic murmurs of four patients with aortic stenosis in Figure 6.2; Johnson et al. showed that the ratio correlates well with the severity of aortic stenosis.

Binnie et al. [258, 259] described the application of spectral analysis to EEG for the detection of epilepsy. Their method was based upon partitioning or banding of the EEG spectrum into not only the traditional δ , θ , α , and β bands, but also into seven other nonuniform bands specified as 1 – 2, 2 – 4, 4 – 6, 6 – 8, 8 – 11, 11 – 14, and $> 14 \text{ Hz}$. Additional features related to FF (see Section 5.6.4) were also used. In a study with 275 patients with suspected epilepsy, 90% of the signals of the patients with pathology were classified as abnormal by their methods; conversely, 86% of the patients whose EEGs were classified as abnormal had confirmed pathology.

When analyzing a spectral peak, we may also compute the -3 dB bandwidth of the peak and, furthermore, compute its quality factor as the ratio of the peak frequency to the bandwidth. Such measures may be computed for not only the dominant peak, but several peaks at progressively lower levels of signal power. Essentially, each potential resonance peak is treated and characterized as a bandpass filter. Durand et al. [331] used such measures to characterize the PSDs of sounds produced by prosthetic heart valves (discussed in Section 6.5).

6.5 Application: Evaluation of Prosthetic Heart Valves

Efficient opening and closing actions of cardiac valves are of paramount importance for proper pumping of blood by the heart. When native valves fail, they may be replaced by mechanical prosthetic valves or by bioprosthetic valves extracted from pigs. Mechanical prosthetic valves are prone to sudden failure due to fracture of their components. Bioprosthetic valves fail gradually due to tissue degeneration and calcification, and have been observed to last 7 – 12 years [331]. Follow-up of the health of patients with prosthetic valves requires periodic, noninvasive assessment of the functional integrity of the valves.

Problem: *Deposition of calcium causes the normally pliant and elastic bioprosthetic valve leaflets to become stiff. Propose a method to assess the functional integrity of bioprosthetic valves.*

Solution: Based on the theory that valve opening and closure contribute directly to heart sounds, analysis of PCG components offers a noninvasive and passive approach to evaluation of prosthetic valves. The increased stiffness is expected to lead to higher-frequency components in the opening or closing sounds of the valve. Durand et al. [331] studied the spectra of the entire S1 signal segment to evaluate the sounds contributed by the closure of porcine (pig) bioprosthetic valves implanted in the mitral position in humans. They demonstrated that, whereas normal S1 spectra

were limited in bandwidth to about 100 Hz , degenerated bioprosthetic valves created significant spectral energy in the range 100 – 250 Hz . Figure 6.14 shows the power spectra of S1 in the case of a normal bioprosthetic valve and a degenerated bioprosthetic valve; increased high-frequency power is evident in the latter.

Durand et al. derived several parameters from S1 spectra and used them to discriminate normal from degenerated bioprosthetic valves. Some of the parameters used by them are the first and second dominant peak frequencies; the bandwidth and quality factor of the dominant peak; integrated mean area above -20 dB ; the highest frequency found at -3 dB ; total area and RMS value of the spectrum; area and RMS value in the 20 – 100 Hz , 100 – 200 Hz , and 200 – 300 Hz bands; and the median frequency. Normal-versus-degenerated valve classification accuracies as high as 98% were achieved.

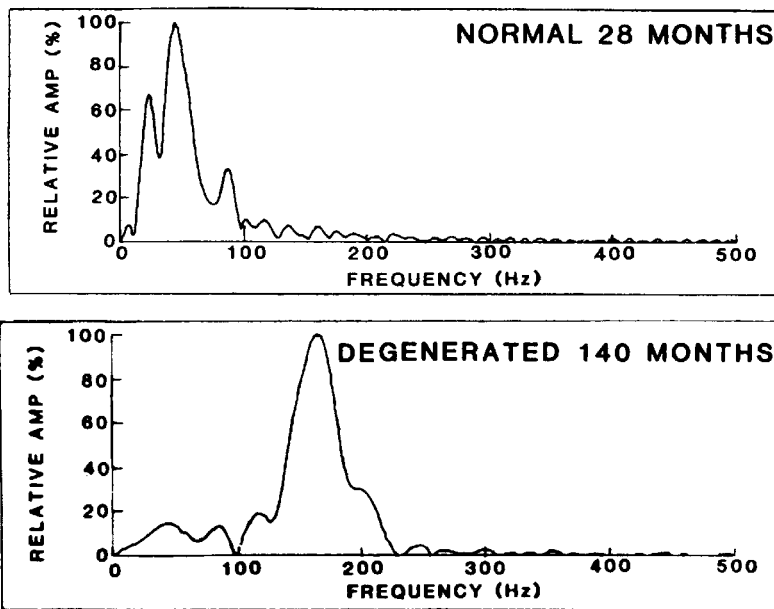


Figure 6.14 First heart sound spectra in the case of normal and degenerated porcine bioprosthetic valves implanted in the mitral position. Reproduced with permission from L.G. Durand, M. Blanchard, G. Cloutier, H.N. Sabbah, and P.D. Stein, Comparison of pattern recognition methods for computer-assisted classification of spectra of heart sounds in patients with a porcine bioprosthetic valve implanted in the mitral position, *IEEE Transactions on Biomedical Engineering*, 37(12):1121–1129, 1990 ©IEEE.

Durand et al. [229] also studied the sounds of bioprosthetic valves in the aortic position. They argued that the aortic and pulmonary components (A2 and P2, respectively) of S2, each lasting about 50 ms , are not temporally correlated during normal breathing. The two components of S2 are separated by 30 – 60 ms during inspiration, but get closer and could overlap during expiration. Furthermore, P2 is weaker

than A2 if the PCG is recorded in the aortic area. Thus, P2 may be suppressed and A2 strengthened by coherent detection and averaging of S2 over several cardiac and breath cycles; see Section 4.11. Durand et al. performed spectral analysis of A2 extracted as above for the purpose of evaluation of bioprosthetic valves in the aortic position. Among a selection of spectral analysis methods including the basic periodogram, Welch's averaged periodogram, all-pole modeling (see Section 7.5), and pole-zero modeling (see Section 7.6), they found the basic periodogram to provide the best compromise for estimating both the spectral distribution and the dominant frequency peaks of bioprosthetic valve sounds.

Cloutier et al. [332] studied the bias and variability of several diagnostic spectral parameters computed from simulated closing sounds of bioprosthetic valves in the mitral position. They found that the most-dominant spectral peak frequency and its quality factor were best estimated using an FFT-based PSD estimate with a rectangular window. However, the -3 dB bandwidth of the most-dominant spectral peak, the frequency of the second-dominant peak, and a few other parameters were best estimated by the Steiglitz-McBride method of pole-zero modeling (see Section 7.6.2). Some other parameters were best estimated by all-pole modeling using the covariance method (see Section 7.5). It was concluded that a single method would not provide the best estimates of all possible spectral parameters of interest.

6.6 Application: Fractal Analysis of VAG Signals

Problem: *Knee-joint VAG signals exhibit varying levels of waveform complexity depending upon the state of the cartilage covering the joint surfaces. Given that fractals demonstrate complex patterns depending upon certain parameters, can fractal analysis assist in parametric representation and classification of VAG signals?*

Solution: As explained in Section 1.2.14, normal knee joints with smooth cartilage surfaces produce little sound or vibration. However, when the cartilage surfaces are eroded or damaged due to pathological conditions, additional sounds could be generated. Rangayyan and Oloumi [333] and Rangayyan et al. [334] explored the use of FD as a parameter for screening of VAG signals. Given the nonstationary nature of VAG signals, multiple FD values were derived from the PSDs of segments of VAG signals. The related background and methods are described in the following paragraphs.

6.6.1 Fractals and the $1/f$ model

One of the several definitions of fractals is based on geometric structures that have self-similarity at different scales of length or size [275, 283, 335]; see Section 5.13. Fractals lack any single scale of length or size, and have a noninteger (fractional) dimension, referred to as FD . Fractal geometry has been used to represent and synthesize several natural forms such as leaves (especially ferns), trees, mountains, and clouds. Fractal analysis forms a part of advanced techniques for nonlinear analysis of biomedical signals [284–286].

Wiener's geometric model of physical Brownian motion has been used as the basis for algorithms to generate fractal signals or images. In this model, the unpredictable variation of a quantity, V , over time, t , is viewed as a noise process, $V(t)$. The PSD, $P_V(f)$, is used to estimate the power of fluctuations at a given frequency, f , and also of the variations over a time scale of the order of $1/f$.

A time-varying quantity, $V(t)$, with the best fitting line to its PSD, $P_V(f)$, varying as $1/f^\beta$ on a log-log scale, is referred to as $1/f$ noise; this is known as the inverse power law or the $1/f$ model. According to Voss [283], most natural phenomena under this model have β in the range of $[0.5, 1.5]$. The PSD of a noise process represented by Brownian motion or random walk varies as $1/f^2$; the power decreases quadratically with frequency. The trace of such a signal is a fractal curve. A direct relationship exists between the FD of the signal and the slope, β , of the best fitting line to its PSD on a log-log scale [283].

The fractional Brownian motion (fBm) model of Mandelbrot [275] has formed the basis of several mathematical models for computer generation of natural fractal scenery. Models based on fBm have been extended to 2D for the synthesis of Brownian surfaces and 3D to generate Brownian clouds [335].

Figure 6.15 shows examples of signals, $V_H(t)$, generated as functions of an arbitrary time variable, t , based on the fBm model. The scaling of the traces is characterized by the scaling parameter H , known as the Hurst coefficient, in the range $0 \leq H \leq 1$. A high value of H close to 1 results in a relatively smooth signal. A lower value of H produces a rougher trace. The variable H relates the changes in V , $\Delta V = V(t_2) - V(t_1)$, to differences in the time variable, $\Delta t = t_2 - t_1$, by the scaling law [283]

$$\Delta V \propto (\Delta t)^H. \quad (6.42)$$

Self-similar patterns repeat themselves at various levels of magnification. However, fBm traces repeat statistically only when t and V are magnified by different amounts. If t is magnified by a factor r as rt , the value of V is magnified by the factor r^H and becomes $r^H V$. Nonuniform scaling of this nature is known as self-affinity [275]. The zero-set of an fBm signal is the intersection of the signal with the horizontal axis. The zero-set is a set of disconnected points with a topological dimension of zero and fractal dimension, D_0 , given by [283]

$$D_0 = 1 - H. \quad (6.43)$$

The zero-set of a self-affine signal is self-similar; different estimates of D_0 will yield the same result. The FD of the signal is related to D_0 as

$$FD = D_0 + 1 \quad (6.44)$$

and to the scaling parameter, H , as

$$FD = 2 - H. \quad (6.45)$$

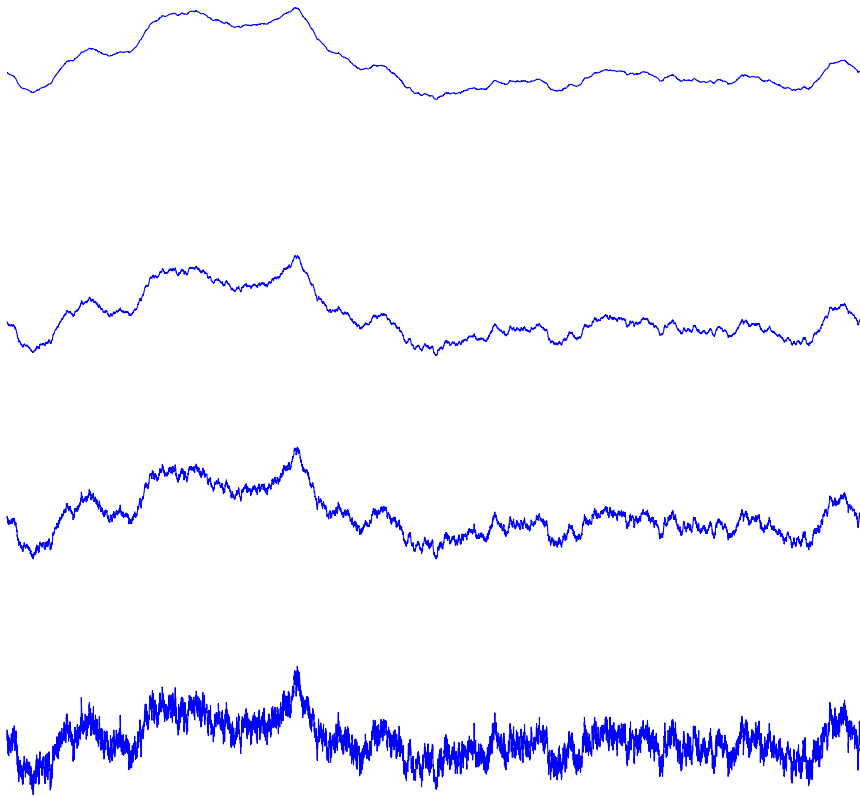


Figure 6.15 Examples of signals generated based on the fBm model for different values of H and FD . Top to bottom: $H = 0.9, 0.6, 0.4, 0.1$; model $FD = 1.1, 1.4, 1.6, 1.9$; estimated $FD = 1.103, 1.401, 1.600, 1.898$. Reproduced with permission from R.M. Rangayyan and F. Oloumi, Fractal analysis of knee-joint vibroarthrographic signals, Proceedings of the 10th IEEE International Conference on Information Technology and Applications in Biomedicine (ITAB 2010), Corfu, Greece, November 2010, pp 1–4. ©IEEE.

6.6.2 FD via power spectral analysis

The best method available to estimate the FD of a self-affine signal is power spectral analysis (PSA). As explained in Section 6.6.1, an fBm signal has a PSD that follows the $1/f^\beta$ model. A high value of β indicates a rapid decrease in the high-frequency content of the signal. A self-affine fBm function in an E -dimensional Euclidean space has its PSD $P_V(f) \propto 1/f^\beta$, with [283]

$$FD = E + 1 - H, \quad (6.46)$$

where

$$H = \frac{\beta - 1}{2}. \quad (6.47)$$

The FD , in terms of the spectral component, β , for a 1D signal with $E = 1$, is

$$FD = \frac{5 - \beta}{2}. \quad (6.48)$$

In the studies of Rangayyan and Oloumi [333] and Rangayyan et al. [334], to estimate β , a Hann window was first applied to the given VAG signal; then, the DFT of the windowed signal was calculated, and the squared magnitude of the result was used to obtain the PSD of the signal. The slope of the best fitting line to the log–log plot of PSD was then determined. See Chang et al. [336] for related discussions.

When applying the PSA method, it is important to specify an appropriate frequency range to obtain the linear fit. Low-frequency details related to any slow drift present in the signal as well as high-frequency content dominated by noise or artifacts in the signal should be disregarded in order to obtain accurate estimates of β and FD . In the works of Rangayyan and Oloumi [333] and Rangayyan et al. [334], when applying the PSA method to the synthesized fBm signals, PSD components spanning 1% of the total frequency range were removed at both the low and high ends of the positive frequency axis. For VAG signals, the frequency range to obtain the linear fit was varied between $[10, 300]$ Hz and $[10, 380]$ Hz to study the effect of the frequency range used on classification accuracy. This was based upon experimentation and analysis of the band of frequencies over which appreciable differences were observed between normal and abnormal VAG signals. (The VAG signals had been prefiltered to the range 10 Hz to 1 kHz at the time of data acquisition; therefore, there is no useful information below 10 Hz.)

6.6.3 Examples of synthesized fractal signals

Two types of methods are available to generate fBm signals with known FD : approximation by spatial methods and approximation by spectral synthesis. The most reliable method is spectral synthesis by means of inverse Fourier filtering [337]. If a_k is the k^{th} complex coefficient of the DFT, to obtain $P(f) \propto 1/f^\beta$, the condition is

$$E[|a_k|^2] \propto \frac{1}{k^\beta}, \quad (6.49)$$

where k denotes the frequency index corresponding to f . By randomly selecting coefficients that satisfy the stated condition and then taking their inverse DFT, we can obtain the corresponding signal in the time domain. In the works of Rangayyan and Oloumi [333] and Rangayyan et al. [334], a direct implementation of the algorithm given by Saupe [337] was used to generate 110 synthetic signals, 10 for each value of H , with $0 \leq H \leq 1$ in intervals of 0.1. Figure 6.15 illustrates four examples of the synthesized signals. It is evident from the illustration that the variability and complexity of the signals increase as H decreases (or as FD increases).

The estimated FD values for the four synthesized signals shown in Figure 6.15 are given in the caption of the figure. The RMS error between the known and estimated values of FD for the 110 synthesized signals was computed to be 0.0198, which indicates accurate estimation of FD by the PSA method. The PSA method performed well with the synthesized signals for FD in the range of $[1.1, 1.8]$, but overestimated FD values in the range $[1.9, 2.0]$.

Estimates of FD were also obtained for the synthesized fBm signals using the well-known and popular box-counting and ruler methods [275, 276, 282] (see Section 5.13); the two methods led to slightly poorer results, with higher RMS errors of 0.1387 and 0.2243, respectively. The results indicate that the PSA method is the best suited method for the estimation of FD of fBm and self-affine signals, among the three methods mentioned.

6.6.4 Fractal analysis of segments of VAG signals

The database used in the studies of Rangayyan and Oloumi [333] and Rangayyan et al. [334] consists of 89 VAG signals; see Section 5.12.1 for details. Examples of a normal signal and an abnormal signal are shown in Figure 6.16 and 6.17. Each signal was normalized to the amplitude range $[0, 1]$ and resampled to the length of 8,192 samples by linear interpolation.

It is known that the VAG signal is nonstationary and that it is appropriate to analyze segments of fixed or adaptive duration [96, 264–266, 269] or apply nonstationary signal processing techniques such as wavelets [270] and TFDs [273]. However, such methods increase the computational load. Furthermore, it is difficult, in practice, to associate parts of the knee joint affected by pathology to segments of VAG signals. The use of VAG signal segments of limited duration may lead to inaccuracies in the estimation of the PSD in low-frequency ranges. On the other hand, it has been observed that parts of VAG signals during certain portions of the swing cycle, especially extension, can provide features with higher discriminant capability than other parts [264–266, 269]. In the works of Rangayyan and Oloumi [333] and Rangayyan et al. [334], to facilitate separate analysis of parts of VAG signals during extension and flexion and to derive multiple features, the PSA method was applied not only to get a parameter for the full VAG signal in each case ($FD1$), but also to the first and second halves of the normalized duration ($FD1h$ and $FD2h$, corresponding to extension and flexion), as well as to four segments with each spanning one quarter of the total duration ($FD1q$, $FD2q$, $FD3q$, and $FD4q$).

Figures 6.18 and 6.19 show the PSDs and the linear fits derived to estimate the $FD1$ values of the normal and abnormal VAG signals shown in Figures 6.16 and 6.17, respectively. The estimated $FD1$ values for the two signals are given in the captions of Figures 6.18 and 6.19. The average and SD values of $FD1$ for the 51 normal signals were 1.8061 ± 0.2398 ; those for the 38 abnormal VAG signals were 1.6695 ± 0.2226 (using the frequency range $[10, 300]$ Hz). The discriminant capability of the FD values was assessed in terms of the area, A_z , under the receiver operating characteristic (ROC) curve by using ROCKIT [338]. The FD values obtained by the PSA method applied to the full VAG signals led to $A_z = 0.6872$.

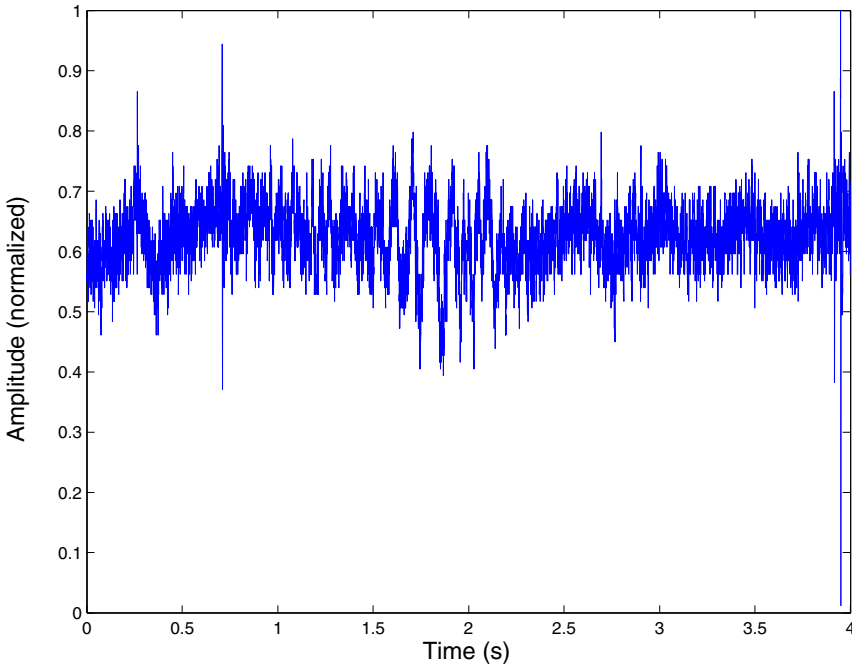


Figure 6.16 Example of a normal VAG signal. Reproduced with permission from R.M. Rangayyan and F. Oloumi, Fractal analysis of knee-joint vibroarthrographic signals, Proceedings of the 10th IEEE International Conference on Information Technology and Applications in Biomedicine (ITAB 2010), Corfu, Greece, November 2010, pp 1–4. ©IEEE.

The values of FD estimated for the first half (extension) and second half (flexion) of each signal led to poorer A_z values of 0.6133 and 0.5916, respectively. With the FD values derived using quarter portions of the VAG signals, the four parts, in order, led to $A_z = 0.6546, 0.7394, 0.5959$, and 0.7023 (using the frequency range $[10, 380]$ Hz). Based on the p -values obtained via the t -test, it was noted that the differences between the values of $FD1$, $FD2q$, and $FD4q$ were statistically highly significant ($p < 0.01$), whereas the differences in the $FD1q$ values for the normal and abnormal categories were statistically significant ($0.01 < p < 0.05$). Thus, the usefulness of some of the FD -based parameters in discriminating between normal and abnormal VAG signals was demonstrated.

The FD values for abnormal VAG signals were observed to be, on the average, lower than those for normal signals, which was interpreted as follows. The presence of defects in the articular surfaces of the knee joint due to loss or shedding of cartilage leads to structured vibration components that disrupt or replace the random, fBm-like signals associated with the normal friction-free cartilage surfaces. Although the VAG signals of some normal knees could possibly contain nonrandom components, such as clicks and crepitus, they would be fewer and occur less often than the grinding

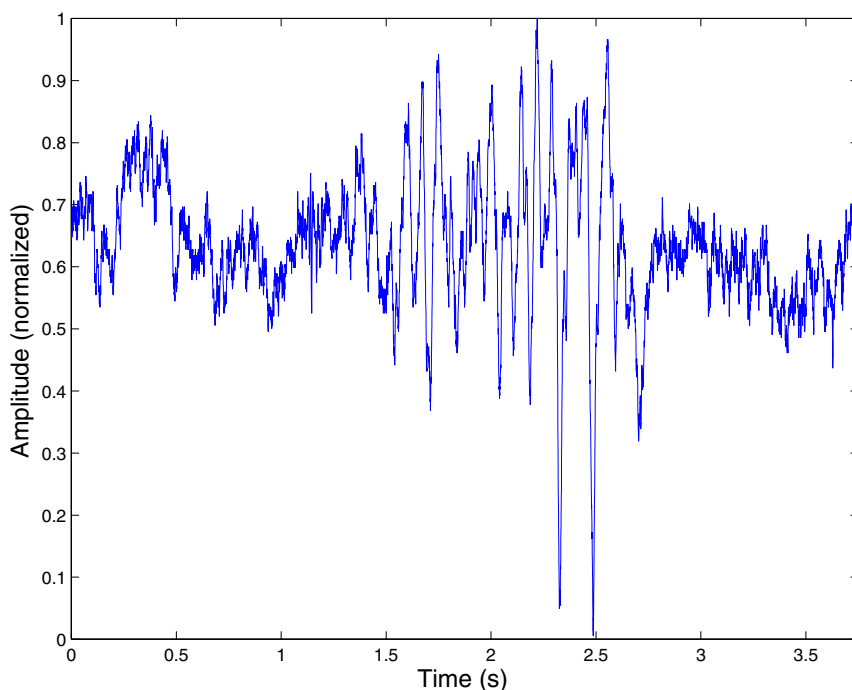


Figure 6.17 Example of an abnormal VAG signal. Reproduced with permission from R.M. Rangayyan and F. Oloumi, Fractal analysis of knee-joint vibroarthrographic signals, Proceedings of the 10th IEEE International Conference on Information Technology and Applications in Biomedicine (ITAB 2010), Corfu, Greece, November 2010, pp 1–4. ©IEEE.

noise components found in abnormal knee joints. The disruption of the fBm-like characteristics of normal VAG signals by pathological conditions may be expected to result in lower FD values for abnormal VAG signals via the PSA approach and the $1/f$ model. The range of the FD values of the abnormal VAG signals obtained in the study of Rangayyan et al. [334] agrees with the ranges of FD of acceleration signals obtained from finger joints of patients with calcium pyrophosphate deposition disease, rheumatoid arthritis, or spondyloarthropathy in the study of Shah et al. [300] (see Section 5.13.3).

The classification performance of the features derived via fractal analysis, with the highest A_z value of about 0.74 in ROC analysis, was noted to be comparable to the performance provided by several features obtained by predictive modeling, cepstral analysis, measures of variability of power, turns count, probabilistic models, wavelets, and TFDs [96, 264–266, 270, 273]. Selected combinations of FD with some of the other features mentioned above led to better results with A_z values in the range [0.92, 0.96]. See Rangayyan et al. [334] for further details and additional results with various feature selection and pattern classification procedures.

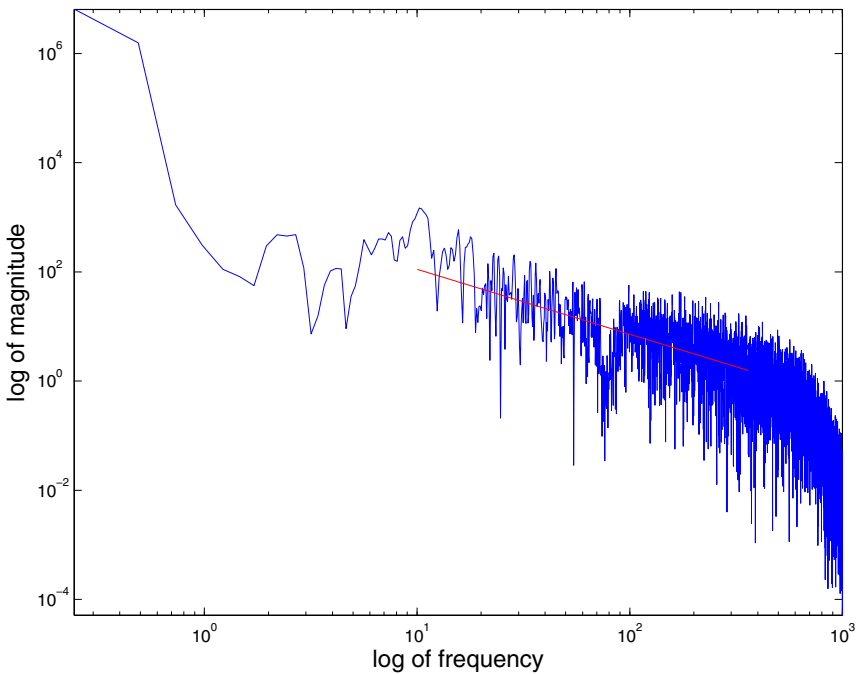


Figure 6.18 PSD of the normal VAG signal in Figure 6.16 with the straight-line fit to the range $[10, 360]$ Hz. $FD1 = 1.905$. Reproduced with permission from R.M. Rangayyan and F. Oloumi, Fractal analysis of knee-joint vibroarthrographic signals, Proceedings of the 10th IEEE International Conference on Information Technology and Applications in Biomedicine (ITAB 2010), Corfu, Greece, November 2010, pp 1–4. ©IEEE.

6.7 Application: Spectral Analysis of EEG Signals

Problem: It is known that the EEG signal exhibits shifts towards lower frequencies as the subject goes into deeper stages of sleep [57, 58]. Propose methods to obtain quantitative measures to characterize the spectral content of EEG signals and relate them to stages of sleep.

Solution: As shown in Section 1.2.6, EEG signals exhibit rhythms in certain frequency bands. A subject with eyes closed and awake will typically demonstrate alpha waves in the EEG. As the subject goes to sleep, the EEG rhythms are expected to shift to lower bands of theta and delta rhythms (see Figure 1.38). Episodic EEG activities related to rapid eye movement (REM) and sleep spindles (see Figure 4.1) could contribute power in frequency bands that are higher than the theta and delta bands. Overnight PSG studies commonly include several channels of the EEG signal that are used to establish the stage of sleep by either manual (visual) analysis or computer-aided analysis [57]. The amount of low-frequency power in the delta and

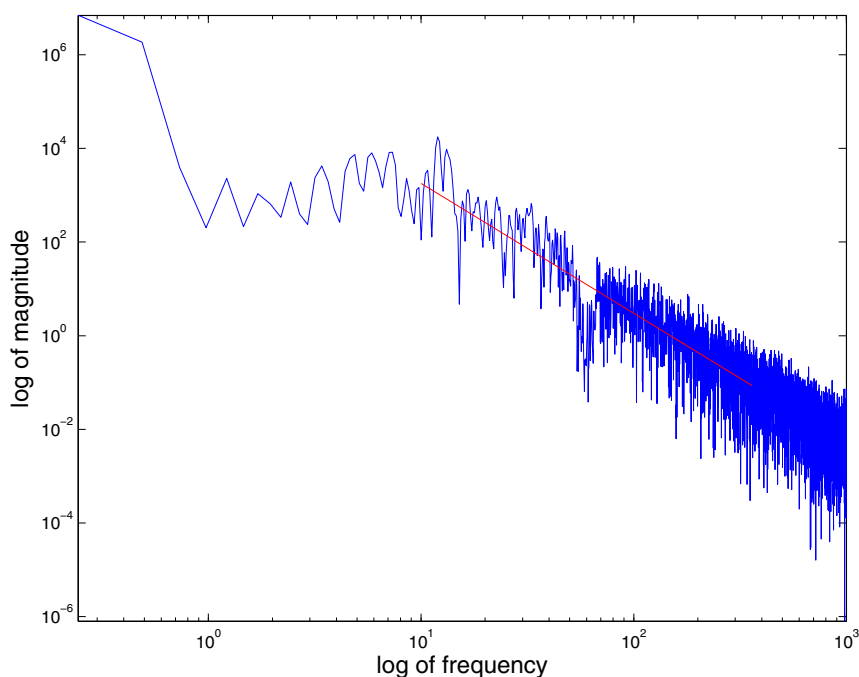


Figure 6.19 PSD of the abnormal VAG signal in Figure 6.17 with the straight-line fit to the range $[10, 360]$ Hz. $FD1 = 1.113$. Reproduced with permission from R.M. Rangayyan and F. Oloumi, Fractal analysis of knee-joint vibroarthrographic signals, Proceedings of the 10th IEEE International Conference on Information Technology and Applications in Biomedicine (ITAB 2010), Corfu, Greece, November 2010, pp 1–4. ©IEEE.

theta bands in relation to the power in the alpha band is a useful feature in sleep staging [57].

Figure 6.20 illustrates three traces of the C3–A2 EEG channel of a subject during sleep in stages 0, 1, and 2. (Stage 0 indicates that the subject is awake.) The EEG signal was analyzed by an expert who assigned sleep stages to 5,988-sample segments, corresponding to intervals of 29.94 s with $f_s = 200$ Hz [57]. Regardless of the noise present in the signals, it is evident that, in general, the signals shift to increasing power in lower frequencies as the sleep stage progresses from 0 to 1 to 2.

In order to obtain estimates of the PSDs of the EEG signals, the Welch method of windowing signal segments and averaging their PSDs was applied to 5,988-sample segments of the EEG signal. Nonoverlapping segments of length 512 samples were obtained and the Hann window was applied before computing the DFT. The magnitude spectra of 10 segments were averaged. Figure 6.21 illustrates the average log-magnitude PSDs of three EEG segments corresponding to sleep stages 0, 1, and 2 (top to bottom). Sample traces of the same EEG segments with durations of 512 samples are shown in Figure 6.20. The PSDs clearly demonstrate a reduction in the

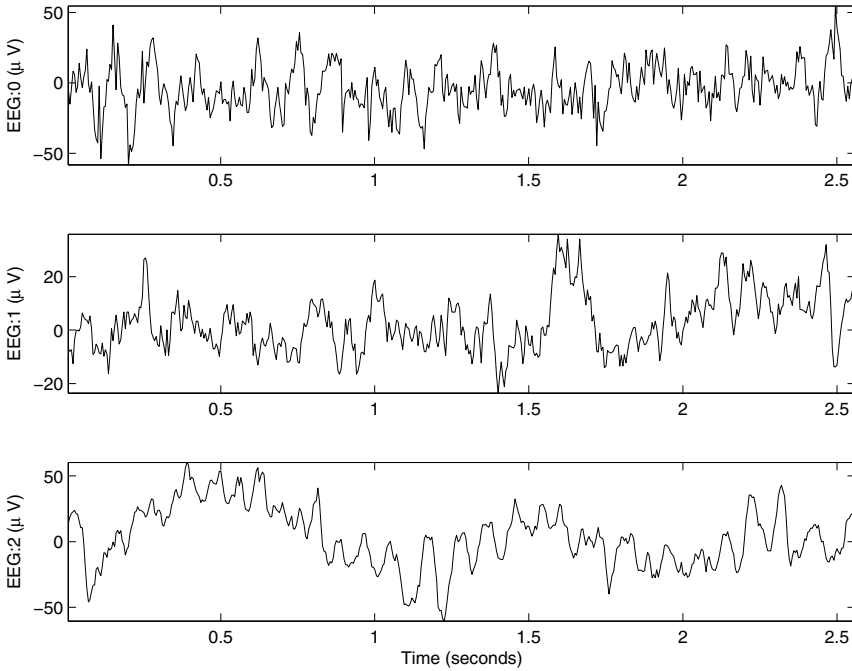


Figure 6.20 Top to bottom: Segments of the C3–A2 EEG channel of a subject during sleep in stages 0, 1, and 2. See also Figure 6.21. EEG data courtesy of R. Agarwal [57].

power of the EEG signals in the higher parts of the range of $0 - 25 \text{ Hz}$ plotted in the figure. The EEG signal for stage 0, which represents a wakeful state with the eyes closed (presumably), shows a clear peak at about 8 Hz corresponding to the alpha rhythm. No peak in the alpha range is seen in the remaining two PSDs, which also show a reduction in high-frequency power. Peaks at lower frequencies (in the delta and theta bands) are evident in the PSDs of the EEG for sleep stages 1 and 2.

The mean frequencies of the PSDs were computed according to Equation 6.30 over the band of $[0, 100] \text{ Hz}$. For the PSDs illustrated in Figure 6.21, the mean frequencies are 20.21, 10.09, and 3.98 Hz , respectively, indicating a clear downward trend over the sleep stages of 0, 1, and 2. Lowpass filtering the signals to remove noise and reducing the frequency range used to compute the mean frequency could lead to lower values.

Instead of computing a single mean frequency value over an entire PSD, one could also compute fractions of the total power in several frequency bands, as illustrated in Figure 6.2 and in the discussion in Section 6.4.2. One of the features used by Agarwal and Gotman [57] is the ratio of power in the alpha band (defined by them as 8.0 to 11 Hz) to the combined power in the delta (0.5 to 3.5 Hz) and theta (3.5 to 8.0 Hz) bands; this was referred to as the alpha-to-slow-wave index. The values of this index for the PSDs illustrated in Figure 6.21 are 1.68, 0.21, and 0.09,

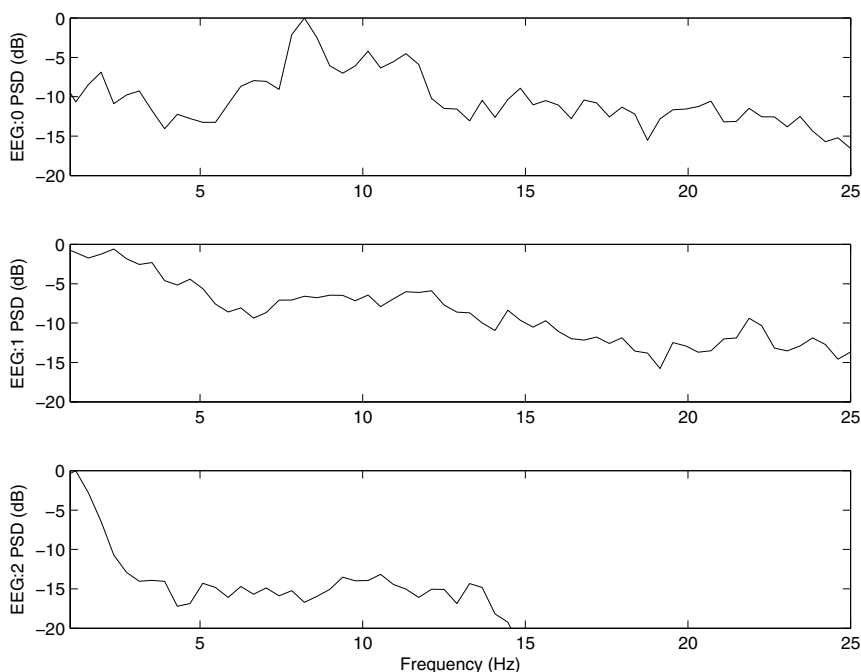


Figure 6.21 Top to bottom: Average PSDs of segments of the C3–A2 EEG channel of a subject during sleep in stages 0, 1, and 2. Figure 6.20 shows sample segments used in the derivation of the PSDs.

which indicate the expected reducing trend from stage 0 to 1 to 2, representative of the fact that alpha waves are replaced by slower theta and delta waves.

Figures 6.22 and 6.23 show cluster plots of the mean frequency and alpha-to-slow-wave index values for 702 segments of an overnight sleep EEG record with sleep stages of 0, 1, and 2. (Each segment has 5,988 samples, corresponding to an interval of 29.94 s with $f_s = 200$ Hz. The total EEG record has a duration of about 6 h and 50 min.) Each ‘×’ mark in the plots represents the value of the parameter and the corresponding sleep stage for one EEG segment. In general, the two parameters show trends of decreasing values as the sleep stage varies from 0 to 1 to 2. However, both parameters show substantial variance and overlap in their ranges for the three stages of sleep. It should be noted that EEG signals of a subject who is awake could contain substantial spectral variations. The presence of alpha waves in a stage-0 segment could cause the alpha-to-slow-wave index to be higher than that for a segment without alpha: This is indicated by the presence of multiple clusters of the index values for stage 0 in Figure 6.23.

Figure 6.24 illustrates a cluster plot of the mean frequency and alpha-to-slow-wave index values combined into a 2D vector for each EEG segment. The diamond marks for the feature vectors of stage 2 segments have formed a tight cluster with low

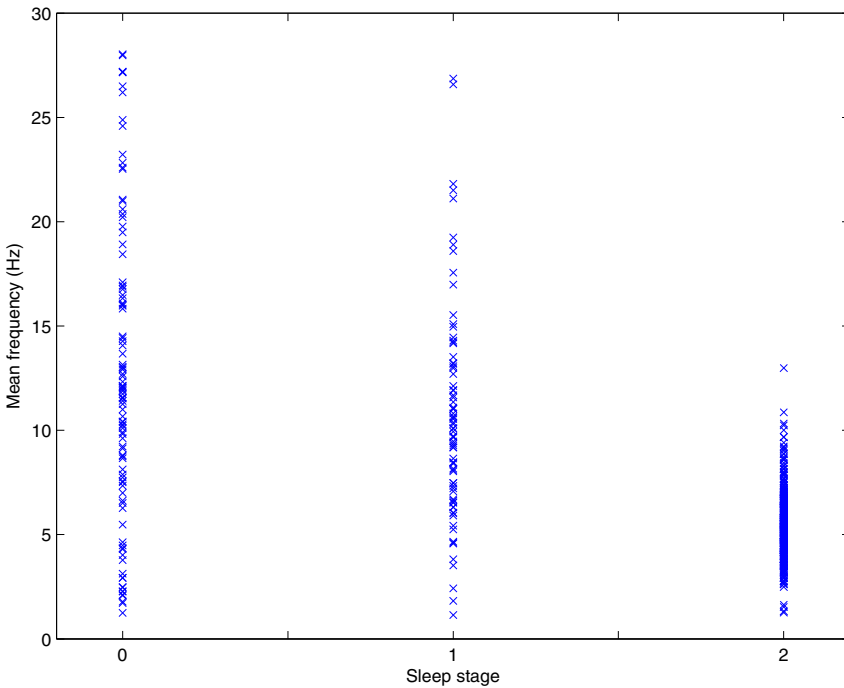


Figure 6.22 Cluster plot of mean frequency values for 702 EEG segments with sleep stages of 0, 1, and 2.

values for both features, as expected. However, the ‘o’ and ‘+’ marks representing the features for EEG segments corresponding to stages 0 and 1 overlap substantially. As mentioned in the preceding paragraph, stage-0 EEG segments could have a broad spread of the parameters depending upon the presence or absence of alpha; Figure 6.24 indicates the presence of two broad clusters for stage 0. Stage-1 EEG segments are also known to possess various types of waves and a broad spread of spectral features, as seen in Figure 6.24. These characteristics of EEG signals create difficulties in manual labeling of sleep stages and in automatic staging with a small number of parameters. The results shown in the present section indicate the need for additional features or measures to separate EEG segments of sleep stages 0 and 1.

EEG segments corresponding to sleep stages 3 and 4 were not present in the entire sleep record of the subject in the present case. Such segments could be expected to possess lower values for the two spectral parameters described in the present section.

Notwithstanding the results demonstrated in this section, it should be noted that staging of sleep requires several parameters derived from not only multiple channels of the EEG signal but also additional channels of the EOG and EMG signals; see Agarwal and Gotman [57] for descriptions of several measures for this application.

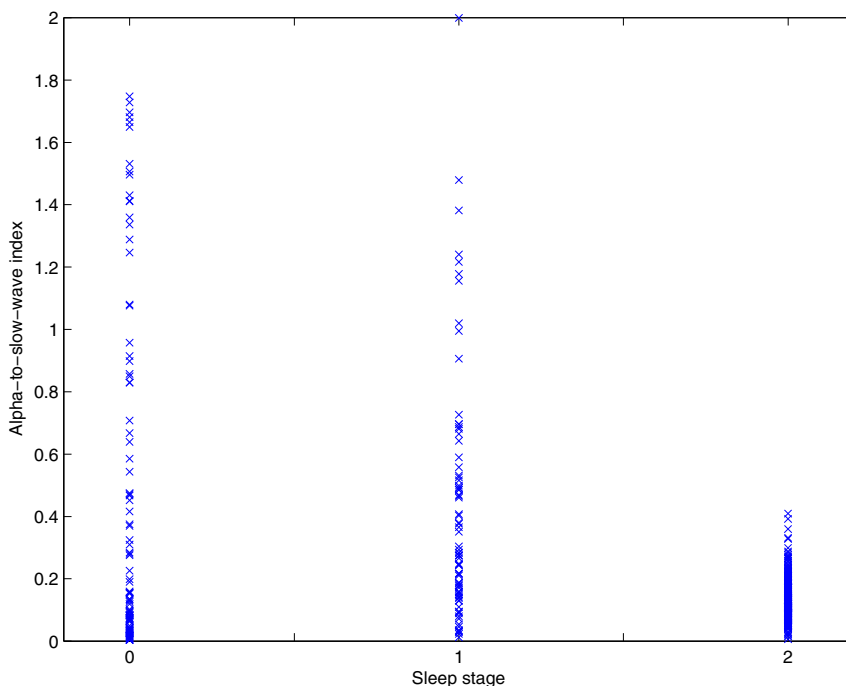


Figure 6.23 Cluster plot of the alpha-to-slow-wave index for 702 EEG segments with sleep stages of 0, 1, and 2.

6.8 Remarks

We have investigated the frequency-domain characteristics of a few biomedical signals and the corresponding physiological systems, with particular attention to the PCG and the cardiovascular system. Frequency-domain analysis via PSDs and parameters derived from PSDs can enable us to view the signal from a different perspective than the time domain. Certain signals such as the PCG and EEG may not lend themselves to easy interpretation in the time domain and, therefore, may benefit from a move to the frequency domain.

PSDs and their parameters facilitate investigation of the behavior of physiological systems in terms of rhythms, resonance, and parameters that could be related to the physical characteristics of anatomical entities (for example, the loss of elasticity of the myocardial muscles due to ischemia or infarction, the extent of aortic valvular stenosis, or the extent of calcification and stiffness of bioprosthetic valves). Pathological states may also be derived or simulated by modifying the spectral parameters or representations of the corresponding normal physiological states and signals.

It is worthwhile to pause at this stage of our study, and recognize the importance of the topics presented in the preceding chapters. A good understanding of the phy-

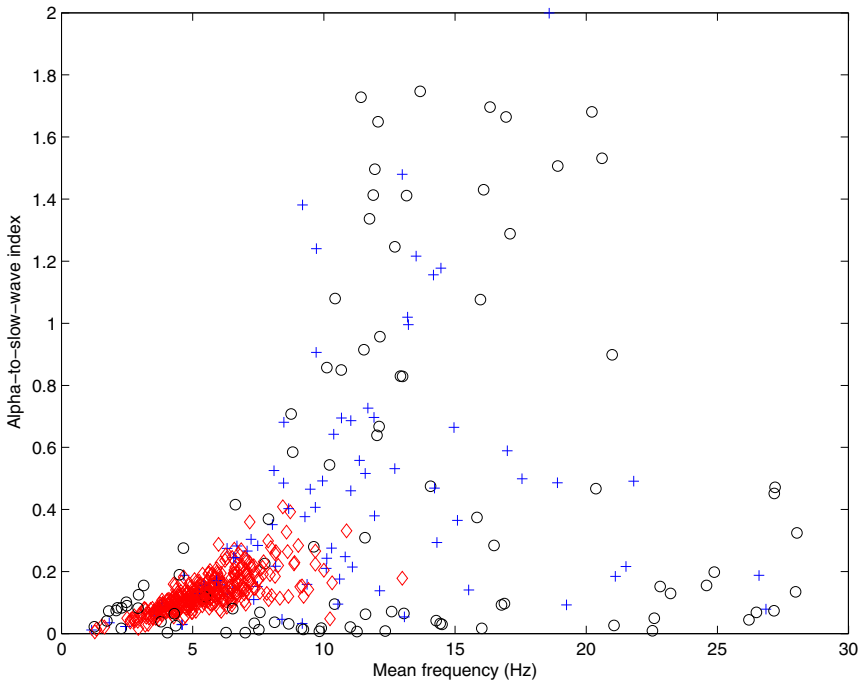


Figure 6.24 Cluster plot of the mean frequency and alpha-to-slow-wave index (ASI) for 702 EEG segments with sleep stages of 0 ('o' mark), 1 ('+' mark), and 2 (diamond mark).

biological systems that produce the biomedical signals we deal with, as well as of the pathological processes that alter their characteristics, is of paramount importance before we may process the signals. Preprocessing the signals to remove artifacts and detect events is essential before we may derive parameters to facilitate their analysis in the time and/or frequency domains. The design of biomedical signal analysis techniques requires a thorough understanding of the characteristics and properties of the biomedical systems behind the signals, in addition to detailed knowledge of mathematical principles, computer techniques, and digital signal processing algorithms.

6.9 Study Questions and Problems

1. The impulse response of a filter is specified by the series of sample values $\{3, 1, -1, 0\}$.
 - (a) What will be the response of the filter to the input whose sample values are $\{4, 4, 2, 1\}$?
 - (b) Is the filter response obtained by linear convolution or circular convolution of the input with the impulse response? (c) What will be the response with the type of convolution other than the one you indicated as the answer to the questions above? (d) How would you implement convolution of the two signals listed above using the FFT? Which type of convolution will this procedure provide? How would you get the other type of convolution for the signals in this problem via the FFT-based procedure?

2. A conjugate symmetric (even) signal $x_e(n)$ is defined as a signal with the property $x_e(n) = x_e^*(-n)$. A conjugate antisymmetric (odd) signal $x_o(n)$ is defined as a signal with the property $x_o(n) = -x_o^*(-n)$. An arbitrary signal $x(n)$ may be expressed as the sum of its conjugate symmetric and conjugate antisymmetric parts as $x(n) = x_e(n) + x_o(n)$, where $x_e(n) = \frac{1}{2}[x(n) + x^*(-n)]$ and $x_o(n) = \frac{1}{2}[x(n) - x^*(-n)]$. Prove that $FT[x_e(n)] = \text{real}[X(\omega)]$, and $FT[x_o(n)] = j\text{imag}[X(\omega)]$, where $FT[x(n)] = X(\omega)$, and FT stands for the Fourier transform [174].
3. A signal $x(t)$ is transmitted through a channel. The received signal $y(t)$ is a scaled, shifted, and noisy version of $x(t)$ given as $y(t) = \alpha x(t - t_0) + \eta(t)$ where α is a scale factor, t_0 is the time delay, and $\eta(t)$ is noise. Assume that the noise process has zero mean and is statistically independent of the signal process, and that all processes are stationary. Derive expressions for the PSD of $y(t)$ in terms of the PSDs of x and η [6, 212].
4. Consider a continuous-time sinusoidal signal of frequency 10 Hz. (a) Derive an analytical expression for the ACF of the signal. (b) Draw a schematic plot of the ACF, including detailed labeling of the time axis. (c) State the relationship of the PSD to the ACF. (d) Derive an analytical expression for the PSD of the given signal. (e) Draw a schematic plot of the PSD, including detailed labeling of the frequency axis.
5. Two real signals $x_1(n)$ and $x_2(n)$ are combined to form a complex signal defined as $y(n) = x_1(n) + jx_2(n)$. Derive a procedure to extract the DFTs $X_1(k)$ and $X_2(k)$ of $x_1(n)$ and $x_2(n)$, respectively, from the DFT $Y(k)$ of $y(n)$.
6. Distinguish between ensemble averages and temporal (time) averages. Identify applications of first-order and second-order averages of both types in PCG analysis.
7. Propose a procedure to process PCG signals to identify the possible presence of murmurs due to aortic stenosis.
8. Propose an algorithm to detect the presence of the alpha rhythm in an EEG signal. Propose an extension to the algorithm to detect the joint presence of the same rhythm in four simultaneously recorded EEG channels.
9. Give an equation to define the mean frequency (centroidal frequency) of a PSD function. Explain the role of each item of your equation. Explain how you would implement the procedure to obtain the mean frequency of a signal. Draw schematic sketches of two PSD functions and indicate their approximate mean frequencies. Explain the difference between the two examples.
10. Using continuous-time and continuous-frequency representation, the fraction of the energy of a signal $x(t)$ in the frequency band $[f_1 : f_2]$ Hz is given by

$$E_{f_1:f_2} = \frac{\int_{f_1}^{f_2} |X(f)|^2 df}{\int_0^\infty |X(f)|^2 df}.$$

Here, $X(f)$ is the Fourier transform of the signal $x(t)$ being processed. A researcher obtains a sampled version $x(n)$ of the signal $x(t)$ with the sampling frequency $f_s = 1,000$ Hz. The number of samples in the signal is $N = 1,900$. The researcher then pads the signal with zeros to increase the length to $M = 2,048$ samples and applies the FFT routine to obtain the DFT $X(k)$ of the signal. The researcher wants to compute the fractions of the energy of the signal in the frequency bands $[0 : 50]$ Hz, $[51 : 100]$ Hz, and $[101 : 400]$ Hz. Help the researcher by writing an algorithm to compute the three

- fractions. Ensure that you give the index of the DFT array for each frequency listed above. Sketch a schematic representation of the spectrum $X(k)$ over the $M = 2,048$ samples in the DFT array. Indicate all of the frequency bands required to compute the three fractions of energy, in terms of the frequency in Hz and the index of the DFT array.
11. A researcher in biomedical signal processing wishes to compute averaged estimates of the PSD for the systolic parts and diastolic parts of PCG signals. The related ECG and carotid pulse (CP) signals are available. The researcher also wants to derive quantitative measures from the averaged PSD. Help the researcher with the following: (a) Give a step-by-step algorithm to identify the beginning and end points of the systolic part and the diastolic part for each cardiac cycle in a given PCG signal. Explain how the ECG and CP signals may be used for this purpose. No equation is required for this part. (b) Give a step-by-step algorithm to compute the PSD of each systolic or diastolic segment and to derive the averaged PSD for systole and diastole over an entire PCG signal including several cardiac cycles. (c) Explain the notions of cyclostationary signals and synchronized averaging. Relate these concepts to the PCG, ECG, and CP signals and the procedure to obtain averaged PSDs as above. (d) Given the averaged PSD, $S(k)$, $k = 1, 2, \dots, N$, where N is the number of the samples used in the computation of the DFT, explain how the mean frequency of the PSD may be computed. Give an equation and explain each term or variable used.
 12. A student new to the area of biomedical signal processing wishes to develop methods to detect the presence of murmurs in PCG signals. The student wishes to derive quantitative measures from segments of the PCG signal to facilitate characterization of murmurs. Help the student with the following: (a) Explain the notion of waveform complexity. Give equations to compute the mean, the variance, and the form factor (FF) of a signal $x(n)$ with N samples. Explain how FF may be expected to differ between normal heart sounds and murmurs. (b) Give a step-by-step algorithm to compute the PSD of a signal. Include mathematical formulas to compute the DFT of the signal and to obtain the PSD from the DFT. (c) Give typical bandwidths of normal heart sounds and murmurs. Draw schematic diagrams of the PSDs of normal heart sounds and murmurs, and explain the differences between them. (d) Give a step-by-step procedure to compute the ratio of the power of a given signal in a particular band of frequencies $[f_1, f_2]$ to the total power of the signal. (e) Assume that a signal segment has $N = 500$ samples, that the DFT is computed using the same number of samples, and that the sampling rate is $1,000 Hz$. For the purpose of detecting murmurs in PCG signals, give a typical range, in Hz , for the band $[f_1, f_2]$ mentioned above. For the same range of frequencies, give the range of the DFT samples to be used to compute the ratio of power mentioned above. Explain how this measure could be used to detect murmurs and to distinguish murmurs from normal heart sounds.

6.10 Laboratory Exercises and Projects

Note: Data files related to the exercises are available at the site

<http://people.ucalgary.ca/~ranga/enel563>

1. Using MATLAB[®], prepare a signal that contains the sum of two cosine waves of equal amplitude at $40 Hz$ and $45 Hz$. Let the sampling rate be $1 kHz$. (a) Compute the power

spectrum of the signal with a rectangular window of duration 2 s . (b) Compute the power spectrum of the signal with a Hamming window of duration 2 s . (c) Compute the power spectrum of the signal with a rectangular window of duration 0.5 s . (d) Compute the power spectrum of the signal with a Hamming window of duration 0.5 s .

To obtain the power spectrum, you may take the FFT and square the absolute value of the result. Compare the spectra obtained in parts (a)–(d) and comment upon their similarities and/or differences. In order to visualize the differences clearly, use 2,048-point FFTs and plot the logarithm of the magnitude-squared spectra with an expanded scale from 0 to 100 Hz only. Be sure to label the frequency axis in Hz !

What should the ideal spectrum look like?

2. Two VAG signals are given in the files `vag1.dat` and `vag2.dat` (see also the file `vag.m`). The sampling rate is 2 kHz . Obtain and plot their PSDs using MATLAB[®]. Label the frequency axis in Hz !

Compute the mean frequency as the first moment of the PSD for each signal. Compute also the variance frequency (second central moment) of each PSD. What are the units of these parameters?

Compare the spectra and the parameters derived and give your evaluation of the frequency content of the signals.

3. The file `safety.wav` contains the speech signal for the word “safety” uttered by a male speaker, sampled at 8 kHz (see also the file `safety.m`). The signal has a significant amount of background noise (as it was recorded in a computer laboratory). Develop procedures to segment the signal into voiced, unvoiced, and silence (background noise) portions using short-time *RMS*, turns count, or *ZCR* measures. Compute the PSD for each segment that you obtain and study its characteristics.
4. The files `pec1.dat`, `pec33.dat`, and `pec52.dat` give three-channel recordings of the PCG, ECG, and carotid pulse signals (sampled at $1,000\text{ Hz}$; you may read the signals using the program in the file `plotpec.m`). The signals in `pec1.dat` and `pec52.dat` are normal; the PCG signal in `pec33.dat` has systolic murmur, and is of a patient suspected to have pulmonary stenosis, ventricular septal defect, and pulmonary hypertension.

Apply the Pan–Tompkins method for QRS detection to the ECG channel and the Lehner and Rangayyan method to detect the dicrotic notch in the carotid pulse channel. Extrapolate the timing information from the ECG and carotid pulse channels to segment the PCG signal into two parts: the systolic part from the onset of an S1 and to the onset of the following S2, and the diastolic part from the onset of an S2 to the onset of the following S1. Compute the PSD of each segment.

Extend the procedure to average the systolic and diastolic PSDs over several cardiac cycles. Compare the PSDs obtained for the three cases.

5. Compute the mean frequency and the ratio of the energy in the range $100 - 300\text{ Hz}$ to the total energy for each PSD derived in the previous problem. What can you infer from these measures?
6. Compute the PSDs of a few channels of the EEG in the file `eeg1-xx.dat` using Welch’s procedure (see also the file `eeg1.m`). Study the changes in the PSDs derived with variations in the window width, the number of segments averaged, and the type of the window used. Compare the results with the PSDs computed using the entire signal in each channel. Discuss the results in terms of the effects of the procedures and parameters on spectral resolution and leakage.

7. Compute the PSD of an ECG signal over a long duration including several cardiac cycles or beats. It would be good to select a segment with a steady heart rate (almost the same RR interval for each beat) and without any PVCs. Extract a segment over one cardiac cycle from the same signal and compute its PSD. Compare the two PSDs and explain the similarities and/or differences between them.

CHAPTER 7

MODELING BIOMEDICAL SIGNAL GENERATING PROCESSES AND SYSTEMS

Up to this point in the book, we have concentrated on processing and analysis of biomedical signals. The signals were treated in their own right as conveyors of diagnostic information. While it was emphasized that the design and application of signal analysis procedures require an understanding of the physiological and pathological processes and systems that generate the signals, no specific mathematical model was used to represent the genesis of the signals in the methods we studied in the preceding chapters.

We shall now consider the modeling approach, where an explicit mathematical model is used to represent the process or the system that generates the signal of interest. The parameters of the model are then investigated for use in signal analysis, pattern recognition, and decision making. As we shall see, the model parameters may also be related to the physical or physiological aspects of the related systems. The *parametric modeling* approach often leads to succinct and efficient representation of signals and systems. Regardless of the emphasis on modeling, the final aim of the methods described in this chapter is analysis of the signal of interest.

7.1 Problem Statement

Propose mathematical models to represent the generation of biomedical signals. Identify the possible relationships between the mathematical models and the physiological and pathological processes and systems that generate the signals. Explore the potential use of the model parameters in signal analysis, pattern recognition, and classification.

Given the diversity of the biomedical signals that we have already encountered and the many others that exist, a generic model cannot be expected to represent a large number of signals. Indeed, a very specific model is often required for each signal. Bioelectric signals such as the ECG and EMG may be modeled using the basic action potential or SMUAP as the building block. Sound and vibration signals such as the PCG and speech may be modeled using fluid-filled resonating chambers, turbulent flow across a baffle or through a constriction, vibrating pipes, and acoustic or vibrational excitation of a tract of variable shape. We shall investigate a few representative signals and models in the following sections, and then study a few modeling techniques that facilitate signal analysis based upon the parameters extracted.

7.2 Illustration of the Problem with Case Studies

7.2.1 Motor unit firing patterns

We saw in Section 1.2.4 that the surface EMG of an active skeletal muscle is the spatiotemporal summation of the action potentials of a large number of motor units that have been recruited into action (see Figure 1.13). If we consider the EMG of a single motor unit, we have a train of SMUAPs; the same basic wave (spike, pulse, or wavelet) is repeated in a quasiperiodic sequence. For the sake of generality, we may represent the intervals between the SMUAPs by a random variable: Although an overall periodicity exists and is represented by the firing rate in *pps*, the intervals between the pulses, known as the *interpulse interval* or IPI, may not precisely be the same from one SMUAP to another.

Agarwal and Gottlieb [339] modeled the single-motor-unit EMG as the convolution of a series of unit impulses or Dirac delta functions — known as a *point process* [340–343] — with the basic SMUAP wave. The SMUAP train $y(t)$ may then be modeled as the output of a linear system whose impulse response $h(t)$ is the SMUAP, and the input is a point process $x(t)$:

$$y(t) = \int_0^t h(t - \tau) x(\tau) d\tau. \quad (7.1)$$

Physiological conditions dictate that successive action potentials of the same motor unit cannot overlap: The interval between any two pulses should be greater than the SMUAP duration. In normal muscle activation, SMUAP durations are of the order of $3 - 15 \text{ ms}$ and motor unit firing rates are in the range $6 - 30 \text{ pps}$; the IPI is, therefore, in the range of about $30 - 170 \text{ ms}$, which is significantly higher than

the SMUAP duration. An SMUAP train, therefore, consists of discrete (distinct and separated) events or waves.

The model as above permits independent analysis of SMUAP waveshape and firing pattern: The two are, indeed, physiologically separate entities. The SMUAP waveshape depends upon the spatial arrangement and characteristics of the muscle fibers that constitute the motor unit, whereas the firing pattern is determined by the motor neuron that stimulates the muscle fibers. Statistics of the point process representing the IPI may be used to study the muscle activation process independently of the SMUAP waveshape. Details on point processes and their application to EMG modeling are presented in Section 7.3.

7.2.2 Cardiac rhythm

The ECG is a quasiperiodic signal that is also cyclostationary in the normal case (see Section 1.2.5). Each beat is triggered by a pulse from the SA node. The P wave is the combined result of the action potentials of the atrial muscle cells, while the QRS and T waves are formed by the spatiotemporal summation of the action potentials of the ventricular muscle cells.

In rhythm analysis, one is more interested in the timing of the beats than in their individual waveshape (with the exception of PVCs). Diseases that affect the SA node could disturb the normal rhythm, and lead to abnormal variability in the RR intervals. Disregarding the details of atrial and ventricular ECG waves, an ECG rhythm may be modeled by a point process representing the firing pattern of the SA node. Sinus arrhythmia and HRV may then be investigated by studying the distribution and statistics of the RR interval.

Figure 7.1 illustrates the representation of ECG complexes in terms of the instantaneous heart rate values defined as the inverse of the RR interval of each beat, in terms of a series of RR interval values, and as a train of delta functions at the SA node firing instants [137]. A discrete-time signal may be derived by sampling the signal in Figure 7.1 (b) at equidistant points; the result, however, may not be continuous or differentiable [137]. The signal in Figure 7.1 (c), known as the interval series, has values $I_k = t_k - t_{k-1}$, where the instants t_k represent the time instants at which the QRS complexes occur in the ECG signal. The I_k series is defined as a function of interval number and not of time, and hence may pose difficulties regarding interpretation in the frequency domain. Finally, the signal in Figure 7.1 (d) is defined as a train of Dirac delta functions $s(t) = \sum \delta(t - t_k)$. The series of impulses represents a point process that may be analyzed and interpreted with relative ease, as shown in Section 7.3. The last two representations may be used to analyze cardiac rhythm and HRV, as demonstrated in Section 7.8 (see also Section 8.11).

7.2.3 Formants and pitch in speech

Speech signals are formed by exciting the vocal tract with either a pulse train or a random signal produced at the glottis, and possibly their combination as well (see Section 1.2.12). The shape of the vocal tract is varied according to the nature of the

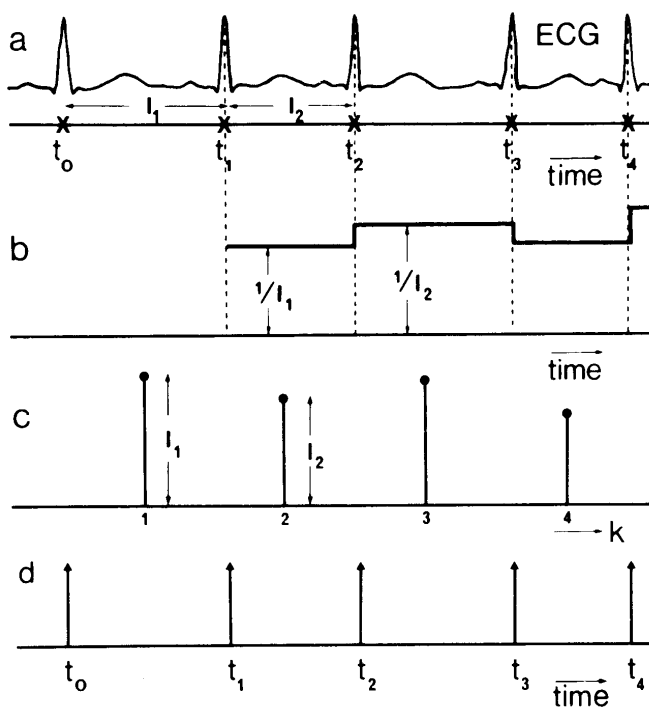


Figure 7.1 The train of ECG complexes in (a) is represented in terms of: (b) the instantaneous heart rate values defined as the inverse of the RR interval of each beat; (c) a series of RR interval values (known as the interval series); and (d) a train of delta functions at the SA node firing instants. Reproduced with permission from R.W. DeBoer, J.M. Karemaker, and J. Strackee, Comparing spectra of a series of point events particularly for heart rate variability studies, *IEEE Transactions on Biomedical Engineering*, 31(4):384–387, 1984. ©IEEE.

sound or phoneme to be produced; the system is, therefore, a time-variant system. We may model the output as the convolution of the (time-variant) impulse response of the vocal tract with the input glottal waveform. The input may be modeled by a random process for unvoiced speech and as a point process for voiced speech; see Figures 1.47, 1.48, 1.49, and 7.2. Clearly, the speech signal is a nonstationary signal; however, the signal may be considered to be quasistationary over short intervals of time during which the same phoneme is being produced.

Figure 7.2 illustrates the commonly used model for speech production [79]; see also Figures 1.47, 1.48, and 1.49. The speech signal may be modeled using the same convolutional relationship as in Equation 7.1, with the limitation that the expression is valid over durations of time when the vocal-tract shape is held fixed and the same glottal excitation is applied. Then, $h(t)$ represents the impulse response of the vocal-tract system (filter) for the time interval considered, and $x(t)$ represents the glottal waveform that is applied as input to the system. In the case of voiced speech, the IPI

statistics of the point-process input, in particular its mean, are related to the pitch. Furthermore, the frequency response of the filter $H(\omega)$ representing the vocal tract determines the spectral content of the speech signal: The dominant frequencies or peaks are known as formants in the case of voiced speech.

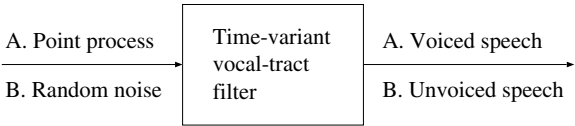


Figure 7.2 Model for production of speech, treating the vocal tract as a time-variant linear system. A point-process input generates quasiperiodic voiced speech, whereas a random-noise input generates unvoiced speech. See also Figures 1.47, 1.48, and 1.49.

Point processes are described in Section 7.3. Parametric spectral modeling and analysis techniques suitable for formant extraction are described in Sections 7.4, 7.5, and 7.6.

7.2.4 Patellofemoral crepitus

Among the various types of VAG signals produced by the knee joint (see Section 1.2.14), the most common is a signal known as physiological patellofemoral crepitus (PPC) [97, 344–347]. The PPC signal is a random sequence of vibrational pulses generated between the surfaces of the patella and the femur, typically observed during slow movement of the knee joint. The PPC signal may carry information on the state and lubrication of the knee joint. A mechanical model of the knee-joint surfaces that generate PPC, as proposed by Beverland et al. [346], is described in Section 7.7.3.

Zhang et al. [344] proposed a model for generation of the PPC signal based on point processes, similar to that for the SMUAP train described in Section 7.2.1. The effects of the repetition rate (or IPI) and the basic patellofemoral pulse (PFP) waveform on the spectrum of the PPC signal were analyzed separately. It was suggested that the model could represent the relationships between physiological parameters, such as the mean and SD of the IPI as well as the PFP waveshape, and parameters that could be measured from the PPC signal, such as its mean, RMS , and PSD-based features. Illustrations related to this application are provided at the end of Section 7.3.

7.3 Point Processes

Problem: *Formulate a mathematical model representing the generation of a train of SMUAPs and derive an expression for the PSD of the signal.*

Solution: In the model for EMG generation proposed by Agarwal and Gottlieb [339], a point process is used to represent the motor neuron firing sequence,

and the SMUAP train is modeled by the convolution integral as in Equation 7.1. The IPI is treated as a sequence of independent random variables with identical normal (Gaussian) PDFs.

Let the interval between the i^{th} SMUAP and the preceding one be τ_i , and let the origin be set at the instant of appearance of the first SMUAP at $i = 0$ with $\tau_0 = 0$. The time of arrival of the i^{th} SMUAP is then given by $t_i = \tau_1 + \tau_2 + \cdots + \tau_i$. The variable t_i is the sum of i independent random variables; note that $\tau_i > 0$. It is assumed that the mean μ and variance σ^2 of the random variable representing each IPI are the same. Then, the mean of t_i is $i\mu$, and its variance is $i\sigma^2$. Furthermore, t_i is also a random variable with the Gaussian PDF

$$p_{t_i}(t_i) = \frac{1}{\sqrt{2\pi i} \sigma} \exp \left[-\frac{(t_i - i\mu)^2}{2i\sigma^2} \right]. \quad (7.2)$$

If the SMUAP train has $N + 1$ SMUAPs labeled as $i = 0, 1, 2, \dots, N$, the motor neuron firing sequence is represented by the point process

$$x(t) = \sum_{i=0}^N \delta(t - t_i). \quad (7.3)$$

The Fourier transform of the point process is

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} \sum_{i=0}^N \delta(t - t_i) \exp(-j\omega t) dt \\ &= \sum_{i=0}^N \exp(-j\omega t_i). \end{aligned} \quad (7.4)$$

$X(\omega)$ is a function of the random variable t_i , which is, in turn, a function of i random variables $\tau_1, \tau_2, \dots, \tau_i$. Therefore, $X(\omega)$ is random. The ensemble average of $X(\omega)$ may be obtained by computing its expectation, taking into account the PDF of t_i , as follows [339]:

$$\overline{X}(\omega) = E[X(\omega)] = \sum_{i=0}^N E[\exp(-j\omega t_i)]. \quad (7.5)$$

$$E[\exp(-j\omega t_i)] = \int_{-\infty}^{\infty} \exp(-j\omega t_i) p_{t_i}(t_i) dt_i. \quad (7.6)$$

Using the expression for $p_{t_i}(t_i)$ in Equation 7.2, we get

$$E[\exp(-j\omega t_i)] = \frac{1}{\sqrt{2\pi i} \sigma} \int_{-\infty}^{\infty} \exp(-j\omega t_i) \exp \left[-\frac{(t_i - i\mu)^2}{2i\sigma^2} \right] dt_i. \quad (7.7)$$

Substituting $t_i - i\mu = r$, where r is a temporary variable, we get

$$E[\exp(-j\omega t_i)] = \frac{1}{\sqrt{2\pi i} \sigma} \exp(-j\omega i\mu) \int_{-\infty}^{\infty} \exp \left[-\frac{r^2}{2i\sigma^2} \right] \exp(-j\omega r) dr. \quad (7.8)$$

Given that the Fourier transform of $\exp(-\frac{t^2}{2\sigma^2})$ is $\sigma\sqrt{2\pi}\exp(-\frac{\sigma^2\omega^2}{2})$ [1], it follows that

$$E[\exp(-j\omega t_i)] = \exp(-j\omega i\mu) \exp\left[-\frac{i\sigma^2\omega^2}{2}\right]. \quad (7.9)$$

Finally, we have

$$\bar{X}(\omega) = \sum_{i=0}^N \exp(-j\omega i\mu) \exp\left[-\frac{i\sigma^2\omega^2}{2}\right]. \quad (7.10)$$

(In the equations in the present context, i is an index and $j = \sqrt{-1}$.) The ensemble-averaged Fourier transform of the SMUAP train is given by

$$\bar{Y}(\omega) = \bar{X}(\omega)H(\omega), \quad (7.11)$$

where $H(\omega)$ is the Fourier transform of an individual SMUAP. The Fourier transform of an SMUAP train is, therefore, a multiplicative combination of the Fourier transform of the point process representing the motor neuron firing sequence and the Fourier transform of an individual SMUAP.

Illustration of application to EMG: Figure 7.3 illustrates EMG signals synthesized using the point-process model as above using 1, 20, 40, and 60 motor units, all with the same biphasic SMUAP of 8 *ms* duration and IPI statistics $\mu = 50$ *ms* and $\sigma = 6.27$ *ms* [339]. It is seen that the EMG signal complexity increases as more motor units are activated. The interference patterns resemble real EMG signals but obscure the shape of the SMUAP used to generate the signals.

Figure 7.4 shows the magnitude spectra of two synthesized EMG signals, one with one active motor unit and the other with 15 active motor units, with biphasic SMUAP duration of 8 *ms*, $\mu = 20$ *ms*, and $\sigma = 4.36$ *ms* [339]. The smooth curve superimposed on the second spectrum in Figure 7.4 was derived from the mathematical model described in the preceding paragraphs. An important point to observe from the spectra is that the average magnitude spectrum of several identical motor units approaches the spectrum of a single MUAP. The spectral envelope of an SMUAP train or that of an interference pattern of several SMUAP trains with identical SMUAP waveshape is determined by the shape of an individual SMUAP.

Figure 7.5 shows the magnitude spectra of surface EMG signals recorded from the gastrocnemius-soleus muscle, averaged over 1, 5, and 15 signal records [339]. The spectra in Figures 7.4 and 7.5 demonstrate comparable features. If all of the motor units active in a composite EMG record were to have similar or identical MUAPs, the spectral envelope of the signal could provide information on the MUAP waveshape (via an inverse Fourier transform). As we have noted earlier in Section 1.2.4, MUAP shape could be useful in the diagnosis of neuromuscular diseases. In reality, however, many motor units of different MUAP shapes could be contributing to the EMG signal even at low levels of effort, and analysis as above may have limited applicability. Regardless, the point-process model provides an interesting approach to model EMG signals. The same model is applicable to the generation of voiced-speech signals, as illustrated in Figure 7.2. For other models to represent the characteristics of

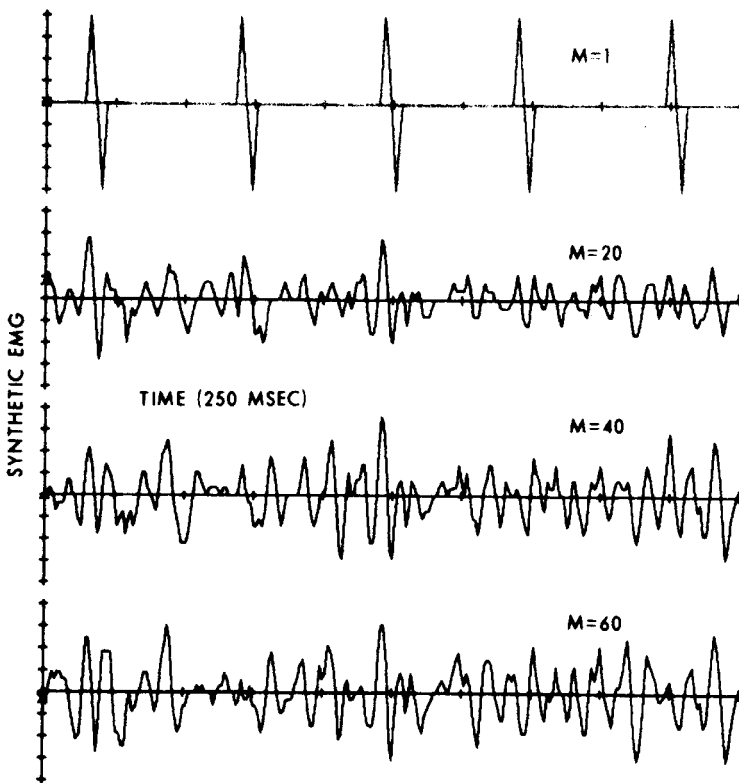


Figure 7.3 Synthesis of an SMUAP train and EMG interference pattern using the point-process model. Top to bottom: SMUAP train of a single motor unit, and interference patterns of the activities of 20, 40, and 60 motor units. SMUAP duration = 8 ms. IPI statistics $\mu = 50$ ms and $\sigma = 6.27$ ms. The duration of each signal is 250 ms. Reproduced with permission from G.C. Agarwal and G.L. Gottlieb, An analysis of the electromyogram by Fourier, simulation and experimental techniques, *IEEE Transactions on Biomedical Engineering*, 22(3):225–229, 1975. ©IEEE.

the EMG signal, refer to the papers by Parker et al. [348], Lindström and Magnusson [349], Zhang et al. [350], Parker and Scott [351], Shweddyk et al. [352], Person and Libkind [353], Person and Kudina [354], de Luca [37, 355], Lawrence and de Luca [263], and de Luca and van Dyk [356].

Illustration of application to PPC: Zhang et al. [344] proposed a point-process model to represent knee-joint PPC signals, which they called PFP trains or signals (see Section 7.2.4). Figure 7.6 illustrates the PSDs of two point processes simulated with mean repetition rate $\mu_r = 21$ pps and coefficient of variation $CV_r = \sigma_r / \mu_r = 0.1$ and 0.05, where σ_r is the SD of the repetition rate. A Gaussian distribution was used to model the IPI statistics. The spectra clearly show the most-dominant peak at the mean repetition rate of the point process, followed by smaller peaks at its

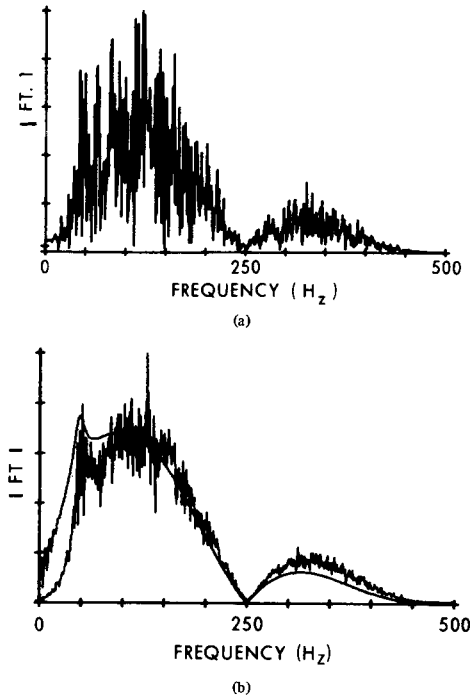


Figure 7.4 Magnitude spectra of synthesized EMG signals with (a) one motor unit, and (b) 15 motor units, with biphasic SMUAP duration of 8 ms , $\mu = 20\text{ ms}$, and $\sigma = 4.36\text{ ms}$. The smooth curve superimposed on the spectrum in (b) was derived from the point-process model with 10 SMUAPs. Reproduced with permission from G.C. Agarwal and G.L. Gottlieb, An analysis of the electromyogram by Fourier, simulation and experimental techniques, *IEEE Transactions on Biomedical Engineering*, 22(3):225–229, 1975. ©IEEE.

harmonics. The higher-order harmonics are better defined in the case with the lower CV_r ; in the limit, the PSD will be a periodic impulse train with all impulses of equal strength when the point process is exactly periodic ($\sigma_r = 0$, $CV_r = 0$).

Zhang et al. [344] simulated PFP trains for different IPI statistics using a sample PFP waveform from a real VAG signal recorded at the patella of a normal subject using an accelerometer. The duration of the PFP waveform was 21 ms , and the IPI statistics μ_r and CV_r were limited such that the PFP trains synthesized would have nonoverlapping PFP waveforms and resemble real PFP signals. Figures 7.7 and 7.8 illustrate the PSDs of synthesized PFP signals for different μ_r but with the same CV_r , and for the same μ_r but with different CV_r , respectively. The PSDs clearly illustrate the influence of IPI statistics on the spectral features of signals generated by point processes. Some important observations to be made are:

- The PSD envelope of the PFP train remains the same, regardless of the IPI statistics.

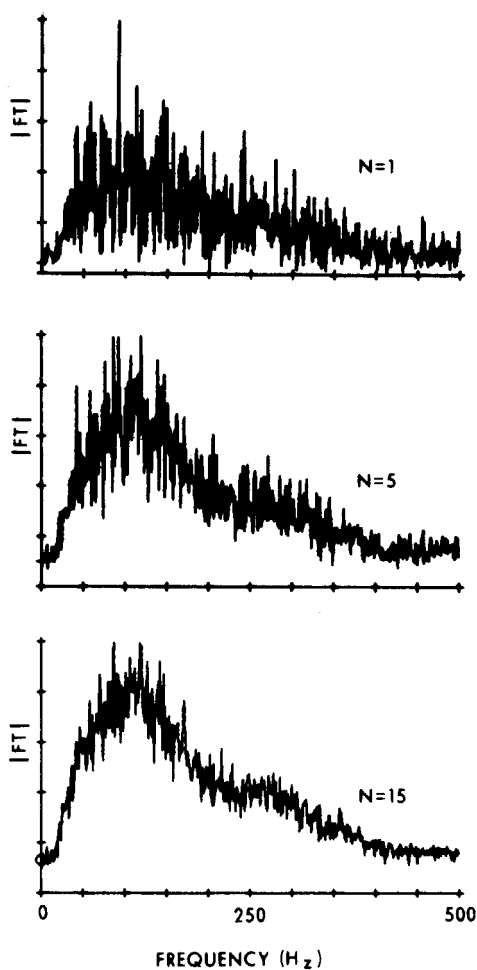


Figure 7.5 Magnitude spectra of surface EMG signals recorded from the gastrocnemius-soleus muscle, averaged over 1, 5, and 15 signal records. Reproduced with permission from G.C. Agarwal and G.L. Gottlieb, An analysis of the electromyogram by Fourier, simulation and experimental techniques, *IEEE Transactions on Biomedical Engineering*, 22(3):225–229, 1975. ©IEEE.

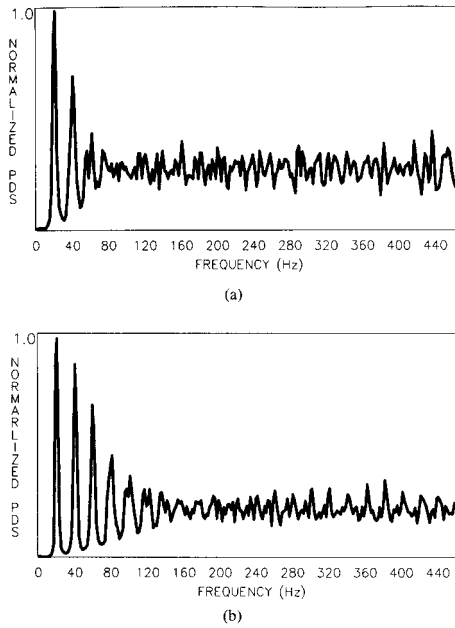


Figure 7.6 Normalized PSDs of synthesized point processes with (a) $\mu_r = 21$ pps and $CV_r = 0.1$, and (b) $\mu_r = 21$ pps and $CV_r = 0.05$. *Note:* PDS represents the power density spectrum, which is the same as the PSD. Reproduced with permission from Y.T. Zhang, C.B. Frank, R.M. Rangayyan, and G.D. Bell, Mathematical modelling and spectrum analysis of the physiological patello-femoral pulse train produced by slow knee movement, *IEEE Transactions on Biomedical Engineering*, 39(9):971–979, 1992. ©IEEE.

- The PSD envelope of the PFP train is determined by the PSD of an individual PFP waveform.
- The PSD envelope of the PFP train is modulated by a series of impulses with characteristics determined by the IPI statistics. The first impulse indicates the mean repetition rate.
- The point process has a highpass effect: low-frequency components of the PSD of the basic PFP are suppressed due to multiplication with the PSD of the point process.
- Physiological signals rarely exhibit precise periodicity. The CV_r value will be reasonably large, thereby limiting the effect of repetition to low frequencies in the PSD of the PFP train.

The observations made above are, in general, valid for all signals generated by point processes, including SMUAP trains and voiced-speech signals.

Zhang et al. [344] verified the point-process model for PFP signals by computing the IPI statistics and PSDs of real PFP signals recorded from normal subjects.

Figure 7.9 shows the IPI histograms computed from the PFP signals of two normal subjects. The IPI statistics computed for the two cases are $\mu_r = 25.2$ pps and $CV_r = 0.07$ for the first signal, and $\mu_r = 16.1$ pps and $CV_r = 0.25$ for the second signal. While the IPI histogram for the first signal appears to be close to a Gaussian distribution, the second is not. The PSDs of the two signals are shown in Figure 7.10. The PSDs of the real signals demonstrate features that are comparable to those observed from the PSDs of the synthesized signals and agree with the observations listed above. The envelopes of the two PSDs demonstrate minor variations: The basic PFP waveform in the two cases were not identical.

7.4 Parametric System Modeling

The importance of spectral analysis of biomedical signals is demonstrated in Chapter 6. The methods described are based on the computation and use of the Fourier spectrum; while this approach is, to begin with, nonparametric, a few parameters can be computed from Fourier spectra. The limitations of such an approach are also discussed in Chapter 6. We shall now study methods for parametric modeling and analysis that, although based on time-domain data and models at the outset, can facilitate parametric characterization of the spectral properties of signals and systems.

Problem: *Explore the possibility of parametric modeling of signal characteristics using the general linear system model.*

Solution: The difference equation that gives the output of a general discrete-time LSI system is

$$y(n) = - \sum_{k=1}^P a_k y(n-k) + G \sum_{l=0}^Q b_l x(n-l), \quad (7.12)$$

with $b_0 = 1$. (*Note:* The advantage of the negative sign before the summation with a_k will become apparent later in this section; some model formulations use a positive sign, which does not make any significant difference in the rest of the derivation.) The input to the system is $x(n)$; the output is $y(n)$; the parameters $b_l, l = 0, 1, 2, \dots, Q$, indicate how the present and Q past samples of the input are combined, in a linear manner, to generate the present output sample; the parameters $a_k, k = 1, 2, \dots, P$, indicate how the past P samples of the output are linearly combined (in a feedback loop) to produce the current output; G is a gain factor; and P and Q determine the order of the system. The summation over x represents the *moving-average* or MA part of the system; the summation over y represents the *autoregressive* or AR part of the system; the entire system may be viewed as a combined *autoregressive, moving-average* or ARMA system. The feedback part typically makes the impulse response of the system infinitely long; the system may then be viewed as an IIR filter (see Figures 3.59 and 3.60).

Equation 7.12 indicates that the output of the system is simply a linear combination of the present input sample, a few past input samples, and a few past output samples. The use of the past input and output samples in computing the present output sample represents the memory of the system. The model also indicates that the

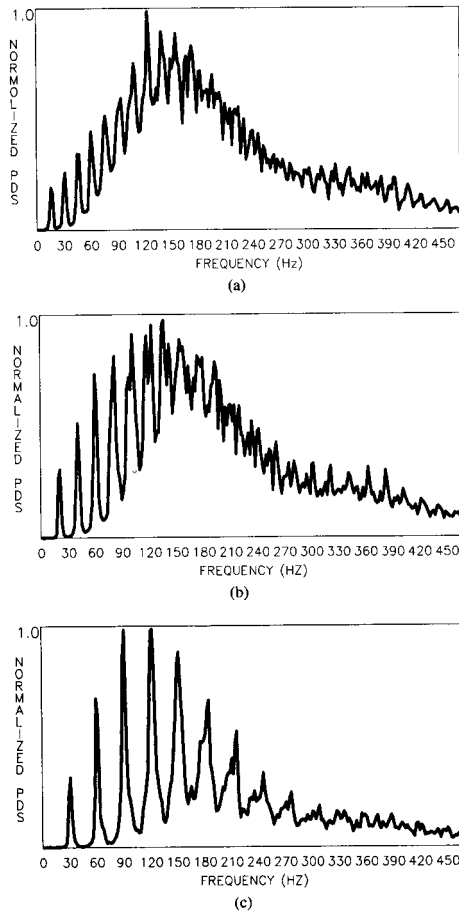


Figure 7.7 Normalized PSDs of synthesized PFP trains using a real PFP waveform with a duration of 21 ms, $CV_r = 0.05$, and (a) $\mu_r = 16$ pps, (b) $\mu_r = 21$ pps, and (c) $\mu_r = 31$ pps. *Note:* PDS represents the power density spectrum, which is the same as the PSD. Reproduced with permission from Y.T. Zhang, C.B. Frank, R.M. Rangayyan, and G.D. Bell, Mathematical modelling and spectrum analysis of the physiological patello-femoral pulse train produced by slow knee movement, *IEEE Transactions on Biomedical Engineering*, 39(9):971–979, 1992. ©IEEE.

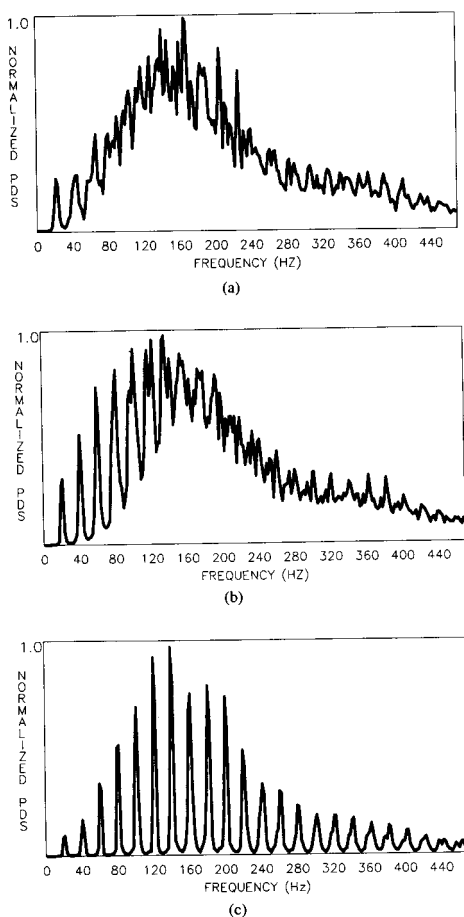


Figure 7.8 Normalized PSDs of synthesized PFP trains using a real PFP waveform with a duration of 21 ms, $\mu_r = 21$ pps, and (a) $CV_r = 0.1$, (b) $CV_r = 0.05$, and (c) $CV_r = 0.01$. *Note:* PDS represents the power density spectrum, which is the same as the PSD. Reproduced with permission from Y.T. Zhang, C.B. Frank, R.M. Rangayyan, and G.D. Bell, Mathematical modelling and spectrum analysis of the physiological patello-femoral pulse train produced by slow knee movement, *IEEE Transactions on Biomedical Engineering*, 39(9):971–979, 1992. ©IEEE.

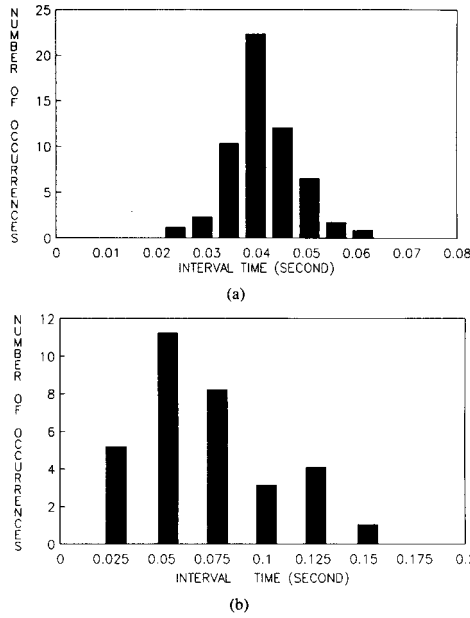


Figure 7.9 IPI histograms computed from real PFP trains recorded from two normal subjects. The statistics computed were (a) $\mu_r = 25.2$ pps and $CV_r = 0.07$, and (b) $\mu_r = 16.1$ pps and $CV_r = 0.25$. See also Figure 7.10. Reproduced with permission from Y.T. Zhang, C.B. Frank, R.M. Rangayyan, and G.D. Bell, Mathematical modelling and spectrum analysis of the physiological patello-femoral pulse train produced by slow knee movement, *IEEE Transactions on Biomedical Engineering*, 39(9):971–979, 1992. ©IEEE.

present output sample may be *predicted* as a *linear combination* of the present and a few past input samples, and a few past output samples. For this reason, the model is also known as the *linear prediction* or LP model [79, 148, 357, 358].

Applying the z -transform to Equation 7.12, we can obtain the transfer function of the system as

$$H(z) = \frac{Y(z)}{X(z)} = G \frac{1 + \sum_{l=1}^Q b_l z^{-l}}{1 + \sum_{k=1}^P a_k z^{-k}}. \quad (7.13)$$

(The advantage of the negative sign before the summation with a_k in Equation 7.12 is now apparent in the numerator – denominator symmetry of Equation 7.13.) The system is completely characterized by the parameters $a_k, k = 1, 2, \dots, P$; $b_l, l = 1, 2, \dots, Q$; and G . In most applications, the gain factor G is not important; the system is, therefore, completely characterized by the a and b parameters, with the exception of a gain factor. Furthermore, we may factorize the numerator and denominator polynomials in Equation 7.13 and express the transfer function as

$$H(z) = G \frac{\prod_{l=1}^Q (1 - z_l z^{-1})}{\prod_{k=1}^P (1 - p_k z^{-1})}, \quad (7.14)$$

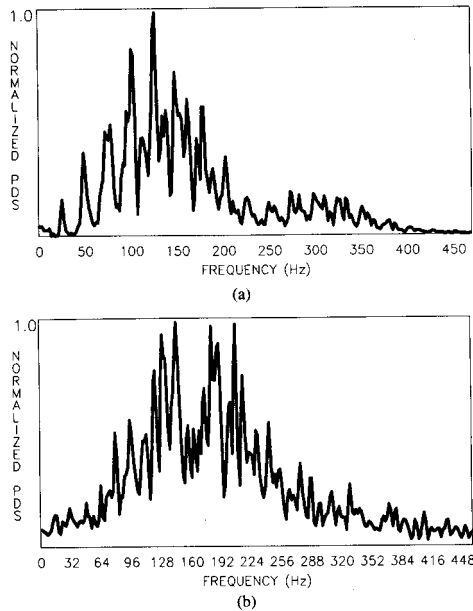


Figure 7.10 Normalized PSDs of the real PFP trains recorded from two normal subjects whose IPI histograms are shown in Figure 7.9. The IPI statistics of the two cases are (a) $\mu_r = 25.2$ pps and $CV_r = 0.07$, and (b) $\mu_r = 16.1$ pps and $CV_r = 0.25$. *Note:* PDS represents the power density spectrum, which is the same as the PSD. Reproduced with permission from Y.T. Zhang, C.B. Frank, R.M. Rangayyan, and G.D. Bell, Mathematical modelling and spectrum analysis of the physiological patello-femoral pulse train produced by slow knee movement, *IEEE Transactions on Biomedical Engineering*, 39(9):971–979, 1992. ©IEEE.

where $z_l, l = 1, 2, \dots, Q$, are the zeros of the system and $p_k, k = 1, 2, \dots, P$, are the poles of the system. The model may now be referred to as a *pole-zero model*. It is evident from Equation 7.14 that the system is completely characterized by its poles and zeros, but for a gain factor.

Equations 7.12, 7.13, and 7.14 demonstrate the applicability of the same conceptual model in the time and frequency domains. The a and b parameters are directly applicable in both the time and the frequency domains in expressing the input–output relationship or the system transfer function. The poles and zeros are more specific to the frequency domain, although the contribution of each pole or zero to the time-domain impulse response of the system may be derived directly from its coordinates in the z -plane [1].

Given a particular input signal $x(n)$ and the corresponding output of the system $y(n)$, we could derive their z -transforms $X(z)$ and $Y(z)$, and subsequently obtain the system transfer function $H(z)$ in some form. Difficulties arise at values of z for which $X(z) = 0$; as the system is linear and $Y(z) = H(z)X(z)$, we have $Y(z) = 0$ at such points as well. Then, $H(z)$ cannot be determined at the corresponding val-

ues of z . [The ideal test signal is the unit-impulse function $x(n) = \delta(n)$, for which $X(z) = 1$ for all z : The response of an LSI system to an impulse completely characterizes the system with the corresponding $y(n) = h(n)$ or its z -domain equivalent $H(z)$.] Methods to determine an AR or ARMA model for a given signal for which the corresponding input to the system is not known (or is assumed to be a point process or a random process) are described in the following sections.

7.5 Autoregressive or All pole Modeling

Problem: *How can we obtain an AR (or LP) model when the input to the system that caused the given signal as its output is unknown?*

Solution: In the AR or all-pole model [79, 357], the output is modeled as a linear combination of P past values of the output and the present input sample as

$$y(n) = - \sum_{k=1}^P a_k y(n-k) + G x(n). \quad (7.15)$$

(The discussion on AR modeling here closely follows that in Makhoul [357], with permission.) Some model formulations use a positive sign in place of the negative sign before the summation in the equation given above. It should be noted that the model in Equation 7.15 does not account for the presence of noise.

The all-pole transfer function corresponding to Equation 7.15 is

$$H(z) = \frac{G}{1 + \sum_{k=1}^P a_k z^{-k}}. \quad (7.16)$$

In the case of biomedical signals such as the EEG or the PCG, the input to the system is totally unknown. Then, we can only approximately predict the current sample of the output signal using its past values as

$$\tilde{y}(n) = - \sum_{k=1}^P a_k y(n-k), \quad (7.17)$$

where the $\sim$ symbol indicates that the predicted value is only approximate. The error in the predicted value (also known as the residual) is

$$e(n) = y(n) - \tilde{y}(n) = y(n) + \sum_{k=1}^P a_k y(n-k). \quad (7.18)$$

The general signal-flow diagram of the AR model viewed as a prediction or error filter is illustrated in Figure 7.11.

The least-squares method: In the least-squares method, the parameters a_k are obtained by minimizing the MSE with respect to all of the parameters. The procedure is similar to that used to derive the optimal Wiener filter (see Section 3.8 and Haykin [148]).

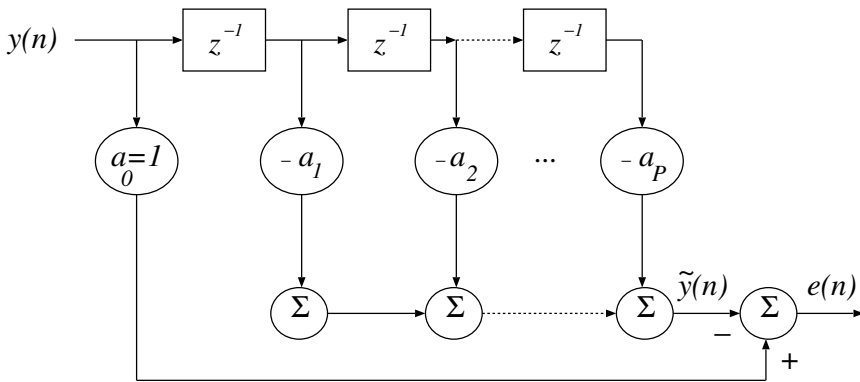


Figure 7.11 Signal-flow diagram of the AR model.

Given an observed signal $y(n)$, the following procedure is applicable for minimization of the MSE [357]. The total squared error (TSE) ε is given by

$$\varepsilon = \sum_n e^2(n) = \sum_n \left(y(n) + \sum_{k=1}^P a_k y(n-k) \right)^2. \quad (7.19)$$

(Note: The TSE is the same as the MSE except for a scale factor.) Although the range of the summation in Equation 7.19 is important, we may minimize ε without specifying the range for the time being. Minimization of ε is performed by applying the conditions

$$\frac{\partial \varepsilon}{\partial a_k} = 0, \quad 1 \leq k \leq P \quad (7.20)$$

to Equation 7.19, which yields

$$\sum_{k=1}^P a_k \sum_n y(n-k) y(n-i) = - \sum_n y(n) y(n-i), \quad 1 \leq i \leq P. \quad (7.21)$$

For a given signal $y(n)$, Equation 7.21 provides a set of P equations in the P unknowns $a_k, k = 1, 2, \dots, P$, known as the *normal equations*; the similarities between the normal equations here and those in the case of the Wiener filter (see Section 3.8) will become apparent later.

By expanding Equation 7.19 and using the relationship in Equation 7.21, the minimum TSE ε_P for the model of order P is obtained as

$$\varepsilon_P = \sum_n y^2(n) + \sum_{k=1}^P a_k \sum_n y(n) y(n-k). \quad (7.22)$$

The expression for TSE will be simplified in the following sections.

The autocorrelation method: If the range of summation in Equations 7.19 and 7.21 is specified to be $-\infty < n < \infty$, the error is minimized over an infinite duration, and we have

$$\phi_y(i) = \sum_{n=-\infty}^{\infty} y(n) y(n-i), \quad (7.23)$$

where $\phi_y(i)$ is the ACF of $y(n)$. In practice, the signal $y(n)$ will be available only over a finite interval, such as $0 \leq n \leq N-1$; the given signal may then be assumed to be zero outside this range and treated as a windowed version of the true signal, as we have seen in Section 6.3. Then, the ACF may be expressed as

$$\phi_y(i) = \sum_{n=i}^{N-1-i} y(n) y(n-i), \quad i \geq 0, \quad (7.24)$$

where the scale factor $\frac{1}{N}$ is omitted. (It will become apparent later that the scale factor is immaterial in the derivation of the model coefficients.) The normal equations then become

$$\sum_{k=1}^P a_k \phi_y(i-k) = -\phi_y(i), \quad 1 \leq i \leq P. \quad (7.25)$$

We now see that an AR model may be derived for a signal with the knowledge of only its ACF; the signal samples themselves are not required. It is also obvious now that the scale factor $\frac{1}{N}$ that was omitted in defining the ACF is of no consequence in deriving the model coefficients. It may be advantageous to use the normalized ACF values given as $\bar{\phi}_y(i) = \phi_y(i)/\phi_y(0)$, which have the property that $|\bar{\phi}_y(i)| \leq 1$.

The minimum TSE is given by

$$\varepsilon_P = \phi_y(0) + \sum_{k=1}^P a_k \phi_y(k). \quad (7.26)$$

Application to random signals: If the signal $y(n)$ is a sample of a random process, the error $e(n)$ is also a sample of a random process. We then have to use the expectation operation to obtain the *MSE* as follows:

$$\varepsilon = E[e^2(n)] = E \left[\left(y(n) + \sum_{k=1}^P a_k y(n-k) \right)^2 \right]. \quad (7.27)$$

Applying the condition for minimum error as in Equation 7.20, we get the normal equations as

$$\sum_{k=1}^P a_k E[y(n-k) y(n-i)] = -E[y(n) y(n-i)], \quad 1 \leq i \leq P. \quad (7.28)$$

The minimum MSE is

$$\varepsilon_P = E[y^2(n)] + \sum_{k=1}^P a_k E[y(n) y(n-k)]. \quad (7.29)$$

If the signal is a sample of a stationary random process, we have $E[y(n-k) y(n-i)] = \phi_y(i-k)$. This leads to the same normal equations as in Equation 7.25. If the process is ergodic, the ACF may be computed as a time average as in Equation 7.24.

If the signal is a sample of a nonstationary random process, $E[y(n-k) y(n-i)] = \phi_y(n-k, n-i)$; the ACF is a function of not only the shift but also time. We would then have to compute the model parameters for every instant of time n ; we then have a time-variant model. Methods for modeling and analysis of nonstationary signals are described in Chapter 8.

Computation of the gain factor G : Since we assumed earlier that the input to the system being modeled is unknown, the gain parameter G is not important. Regardless, the derivation of G demonstrates a few important points. Equation 7.18 may be rewritten as

$$y(n) = - \sum_{k=1}^P a_k y(n-k) + e(n). \quad (7.30)$$

Comparing this with Equation 7.15, we see that the only input signal $x(n)$ which can result in $y(n)$ at the output is given by the condition $Gx(n) = e(n)$. This condition indicates that the input signal is proportional to the error of prediction when the estimated model parameters are equal to the real system parameters. Regardless of the input, a condition that could be applied is that the energy of the output be equal to that of the signal $y(n)$ being modeled. Since the transfer function $H(z)$ is fixed, we then have the condition that the total energy of the input signal $Gx(n)$ be equal to the total energy of the error ε_P .

As illustrated in the model for speech generation in Figure 7.2, two types of input that are of interest are the impulse function and a random process that is stationary white noise. In the case when $x(n) = \delta(n)$, we have the impulse response $h(n)$ at the output, and

$$h(n) = - \sum_{k=1}^P a_k h(n-k) + G \delta(n). \quad (7.31)$$

Multiplying both sides of the expression above with $h(n-i)$ and summing over all n , we get expressions in terms of the ACF $\phi_h(i)$ of $h(n)$ as

$$\phi_h(i) = - \sum_{k=1}^P a_k \phi_h(i-k), \quad 1 \leq |i| \leq \infty \quad (7.32)$$

and

$$\phi_h(0) = - \sum_{k=1}^P a_k \phi_h(k) + G^2. \quad (7.33)$$

Due to the condition that the total energy of $h(n)$ be equal to that of $y(n)$, the condition $\phi_h(0) = \phi_y(0)$ must be satisfied. Comparing Equations 7.32 and 7.25, we then have

$$\phi_h(i) = \phi_y(i), \quad 0 \leq i \leq P. \quad (7.34)$$

Therefore, for a model of order P , the first $(P + 1)$ ACF terms of the impulse response $h(n)$ must be equal to the corresponding ACF terms of the signal $y(n)$ being modeled. It follows from Equations 7.33, 7.34, and 7.26 that

$$G^2 = \varepsilon_P = \phi_y(0) + \sum_{k=1}^P a_k \phi_y(k). \quad (7.35)$$

In the case when the input is a sequence of uncorrelated samples of a random process (white noise) with zero mean and unit variance, we could use the same procedure as for the impulse-input case, with the difference being that expectations are taken instead of summing over all n . {The conditions to be noted in this case are $E[x(n)] = 0$ and $E[x(n) x(n - i)] = \delta(i)$.} The same relations as above for the impulse-input case are obtained. The identical nature of the results for the two cases follows from the fact that the two types of input have identical ACFs and PSDs. These characteristics are relevant in the speech model shown in Figure 7.2.

Computation of the model parameters: For low orders of the model, Equation 7.25 may be solved directly. However, direct methods may not be feasible when P is large.

The normal equations in Equation 7.25 may be written in matrix form as

$$\begin{bmatrix} \phi_y(0) & \phi_y(1) & \cdots & \phi_y(P-1) \\ \phi_y(1) & \phi_y(0) & \cdots & \phi_y(P-2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_y(P-1) & \phi_y(P-2) & \cdots & \phi_y(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_P \end{bmatrix} = - \begin{bmatrix} \phi_y(1) \\ \phi_y(2) \\ \vdots \\ \phi_y(P) \end{bmatrix}. \quad (7.36)$$

For real signals, the $P \times P$ ACF matrix is symmetric and the elements along any diagonal are identical, that is, it is a Toeplitz matrix.

It is worth noting the following similarities and differences between the normal equations in the case of the Wiener filter as given in Equation 3.169 and those given above in Equation 7.36:

- The matrix on the LHS is the ACF of the input to the filter in the case of the Wiener filter, whereas it is the ACF of the output of the prediction filter in the present case.
- The filter vector on the LHS contains the coefficients of the filter being designed in both cases — the optimal Wiener filter or the optimal prediction filter.
- The vector on the RHS is the CCF between the input and the desired response in the case of the Wiener filter, whereas it is the ACF of the output of the prediction filter in the present case.

Haykin [148] provides a more detailed correspondence between the AR model and the Wiener filter.

A procedure known as Durbin's method [359, 360] or the Levinson–Durbin algorithm (see Makhoul [357], Rabiner and Schafer [79], or Haykin [148]) provides a recursive method to solve the normal equations in Equation 7.36. The procedure starts with a model order of 1; computes the model parameters, the error, and a secondary set of parameters known as the reflection coefficients; updates the model order and the parameters; and repeats the procedure until the model of the desired order is obtained. The Levinson–Durbin algorithm is summarized below.

Initialize model order $i = 0$ and error $\varepsilon_0 = \phi_y(0)$. Perform the following steps recursively for $i = 1, 2, \dots, P$.

1. Increment model order i and compute the i^{th} reflection coefficient γ_i as

$$\gamma_i = -\frac{1}{\varepsilon_{i-1}} \left[\phi_y(i) + \sum_{j=1}^{i-1} a_{i-1,j} \phi_y(i-j) \right], \quad (7.37)$$

where $a_{i-1,j}$ denotes the j^{th} model coefficient at iteration $(i-1)$; the iteration index is also the recursively updated model order.

2. Let $a_{i,i} = \gamma_i$.
3. Update the predictor coefficients as

$$a_{i,j} = a_{i-1,j} + \gamma_i a_{i-1,i-j}, \quad 1 \leq j \leq i-1. \quad (7.38)$$

4. Compute the error value as

$$\varepsilon_i = (1 - \gamma_i^2) \varepsilon_{i-1}. \quad (7.39)$$

The final model parameters are given as $a_k = a_{P,k}$, $1 \leq k \leq P$. The Levinson–Durbin algorithm computes the model parameters for all orders up to the desired order P . As the order of the model is increased, the TSE reduces, and hence we have $0 \leq \varepsilon_i \leq \varepsilon_{i-1}$. The reflection coefficients may also be used to test the stability of the model (filter) being designed: $|\gamma_i| < 1$, $i = 1, 2, \dots, P$, is the required condition for stability of the model of order P .

The covariance method: In deriving the autocorrelation method, the range of summation of the prediction error in Equations 7.19 and 7.21 was specified to be $-\infty < n < \infty$. If, instead, we specify the range of summation to be a finite interval, such as $0 \leq n \leq N-1$, we get

$$\sum_{k=1}^P a_k C(k, i) = -C(0, i), \quad 1 \leq i \leq P \quad (7.40)$$

instead of Equation 7.25 based upon the ACF, and the minimum TSE is given by

$$\varepsilon_P = C(0, 0) + \sum_{k=1}^P a_k C(0, k) \quad (7.41)$$

instead of Equation 7.26, where

$$C(i, k) = \sum_{n=0}^{N-1} y(n-i)y(n-k) \quad (7.42)$$

is the covariance of the signal $y(n)$ in the specified interval. The matrix formed by the covariance function is symmetric as $C(i, k) = C(k, i)$, similar to the ACF matrix in Equation 7.36; however, the elements along each diagonal will not be equal, as $C(i+1, k+1) = C(i, k) + y(-i-1)y(-k-1) - y(N-1-i)y(N-1-k)$. Computation of the covariance coefficients also requires $y(n)$ to be known for $-P \leq n \leq N-1$. The distinctions disappear as the specified interval of summation (error minimization) tends to infinity.

7.5.1 Spectral matching and parameterization

The AR model was derived in the preceding section based upon time-domain formulations in the autocorrelation and covariance methods. We shall now see that equivalent formulations can be derived in the frequency domain, which can lead to a different interpretation of the model. Applying the z -transform to Equation 7.18, we get

$$E(z) = \left[1 + \sum_{k=1}^P a_k z^{-k} \right] Y(z) = A(z)Y(z) \quad (7.43)$$

and

$$H(z) = \frac{G}{A(z)}, \quad (7.44)$$

where

$$A(z) = 1 + \sum_{k=1}^P a_k z^{-k}, \quad (7.45)$$

and $E(z)$ is the z -transform of $e(n)$. We can now view the error $e(n)$ as the result of passing the signal being modeled $y(n)$ through the filter $A(z)$, which may be considered to be an *inverse filter*. In the case of $y(n)$ being a deterministic signal, applying Parseval's theorem, the TSE to be minimized may be written as

$$\varepsilon = \sum_{n=-\infty}^{\infty} e^2(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |E(\omega)|^2 d\omega, \quad (7.46)$$

where $E(\omega)$ is obtained by evaluating $E(z)$ on the unit circle in the z -plane. Using $S_y(\omega)$ to represent the PSD of $y(n)$, we have

$$\varepsilon = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_y(\omega) |A(\omega)|^2 d\omega, \quad (7.47)$$

where $A(\omega)$ is the frequency response of the inverse filter and is given by evaluating $A(z)$ on the unit circle in the z -plane.

From Equations 7.16 and 7.44, we get

$$\tilde{S}_y(\omega) = |H(\omega)|^2 = \frac{G^2}{|A(\omega)|^2} = \frac{G^2}{\left|1 + \sum_{k=1}^P a_k \exp(-jk\omega)\right|^2}. \quad (7.48)$$

Here, $\tilde{S}_y(\omega)$ represents the PSD of the modeled signal $\tilde{y}(n)$ that is an approximation of $y(n)$ as in Equation 7.17. From Equation 7.43 we have

$$S_y(\omega) = \frac{|E(\omega)|^2}{|A(\omega)|^2}. \quad (7.49)$$

Now, $\tilde{S}_y(\omega)$ is the model's approximation of $S_y(\omega)$. Comparing Equations 7.48 and 7.49, we see that the error PSD $|E(\omega)|^2$ is modeled by a uniform (or “flat” or “white”) PSD equal to G^2 . For this reason, the filter $A(z)$ is also known as a “whitening” filter.

From Equations 7.48 and 7.49, we get the TSE as

$$\varepsilon = \frac{G^2}{2\pi} \int_{-\pi}^{\pi} \frac{S_y(\omega)}{\tilde{S}_y(\omega)} d\omega. \quad (7.50)$$

As the model is derived by minimizing the TSE ε , we see now that the model is effectively minimizing the integrated ratio of the signal PSD $S_y(\omega)$ to its approximation $\tilde{S}_y(\omega)$. Makhoul [357] describes the equivalence of the model in the following terms:

- As the model order $P \rightarrow \infty$, the TSE is minimized, that is, $\varepsilon_P \rightarrow 0$.
- For a model of order P , the first $(P+1)$ ACF values of its impulse response are equal to those of the signal being modeled. Increasing P increases the range of the delay parameter (time) over which the model ACF is equal to the signal ACF.
- Given that the PSD and the ACF are Fourier-transform pairs, the preceding point leads to the frequency-domain statement that increasing P leads to a better fit of $\tilde{S}_y(\omega)$ to $S_y(\omega)$. As $P \rightarrow \infty$, the model ACF and PSD become identical to the signal ACF and PSD, respectively. Thus, *any spectrum may be approximated by an all-pole model of an appropriate order* (see Section 7.5.2 for a discussion on the optimal model order).

Noting from Equation 7.35 that $G^2 = \varepsilon_P$, Equation 7.50 yields another important property of the model as

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{S_y(\omega)}{\tilde{S}_y(\omega)} d\omega = 1. \quad (7.51)$$

Equations 7.50 and 7.51 lead to the following spectral-matching properties of the AR model [357]:

- Due to the fact that the TSE is determined by the ratio of the true PSD to the model PSD, the spectral-matching process performs uniformly over the entire

frequency range irrespective of the spectral shape. (Had the error measure been dependent on the difference between the true PSD and the model PSD, the spectral match would have been better at higher-energy frequency coordinates than at lower-energy frequency coordinates.)

- $S_y(\omega)$ will be greater than $\tilde{S}_y(\omega)$ at some frequencies and lower at others, while satisfying Equation 7.51 on the whole; the contribution to the TSE is more significant when $S_y(\omega) > \tilde{S}_y(\omega)$ than in the opposite case. Thus, when the error is minimized, the fitting of $\tilde{S}_y(\omega)$ to $S_y(\omega)$ is better at frequencies where $S_y(\omega) > \tilde{S}_y(\omega)$. Thus, the model PSD fits better at the *peaks* of the signal PSD.
- The preceding point leads to another interpretation: The AR-model spectrum $\tilde{S}_y(\omega)$ is a good estimate of the *spectral envelope* of the signal PSD. This is particularly useful when modeling quasiperiodic signals such as voiced speech, PCG, and other signals that have strong peaks in their spectra representing harmonics, formants, or resonance. By following the envelope, the effects of repetition, that is, the effects of the point-process excitation function (see Section 7.3), are removed.

Since the model PSD is entirely specified by the model parameters (as in Equation 7.48), we now have a *parametric representation* of the PSD of the given signal (subject to the error in the model). The TSE may be related to the signal PSD as follows [357].

$$\hat{y}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \log[\tilde{S}_y(\omega)] d\omega \quad (7.52)$$

represents the zeroth coefficient of the (power or real) cepstrum (see Sections 4.8.3 and 5.4.2) of $\tilde{y}(n)$. Using the relationship in Equation 7.48, we get

$$\hat{y}(0) = \log G^2 - \frac{1}{2\pi} \int_{-\pi}^{\pi} \log |A(\omega)|^2 d\omega. \quad (7.53)$$

As all the roots (zeros) of $A(z)$ are inside the unit circle in the z -plane (for the AR model to be stable), the integral in the equation given above is zero [357]. We also have $G^2 = \varepsilon_P$. Therefore,

$$\varepsilon_P = \exp[\hat{y}(0)]. \quad (7.54)$$

The minimum of ε_P is reached as $P \rightarrow \infty$, and is given by

$$\varepsilon_{\min} = \varepsilon_{\infty} = \exp[\hat{y}(0)]. \quad (7.55)$$

This relationship means that the TSE ε_P is the geometric mean of the model PSD $\tilde{S}_y(\omega)$, which is always positive for a positive-definite PSD. The quantity ε_P represents that portion of the signal's information content that is not predictable by a model of order P .

7.5.2 Optimal model order

Given that the AR model performs better and better as the order P is increased, where do we stop? Makhoul [357] shows that if the given signal is the output of a P -pole system, then an AR model of order P would be the optimal model with the minimum error. But how would one find in practice if the given signal was indeed produced by a P -pole system?

One possibility to determine the optimal order for modeling a given signal is to follow the trend in the TSE as the model order P is increased. This is feasible in a recursive procedure such as the Levinson–Durbin algorithm, where models of all lower orders are computed in deriving a model of order P , and the error at each order is readily available. The procedure could be stopped when there is no significant reduction in the error as the model order is incremented.

Makhoul [357] describes the use of a normalized error measure $\bar{\varepsilon}_P$ defined as

$$\bar{\varepsilon}_P = \frac{\varepsilon_P}{\phi_y(0)} = \frac{\exp[\hat{y}(0)]}{\phi_y(0)}. \quad (7.56)$$

As the model order $P \rightarrow \infty$,

$$\bar{\varepsilon}_{\min} = \bar{\varepsilon}_{\infty} = \frac{\exp[\hat{y}(0)]}{\phi_y(0)}. \quad (7.57)$$

$\bar{\varepsilon}_{\min}$ is a monotonically decreasing function of P , with $\bar{\varepsilon}_0 = 1$ and $\bar{\varepsilon}_{\infty} = \bar{\varepsilon}_{\min}$; furthermore, it can be expressed as a function of the model PSD as

$$\bar{\varepsilon}_P = \frac{\exp\left[\frac{1}{2\pi} \int_{-\pi}^{\pi} \log \tilde{S}_y(\omega) d\omega\right]}{\frac{1}{2\pi} \int_{-\pi}^{\pi} \tilde{S}_y(\omega) d\omega}. \quad (7.58)$$

It is evident that $\bar{\varepsilon}_P$ depends only upon the shape of the model PSD and that $\bar{\varepsilon}_{\min}$ is determined solely by the signal PSD. The quantity $\bar{\varepsilon}_P$ may be viewed as the ratio of the geometric mean of the model PSD to its arithmetic mean, which is a measure of the spread of the PSD: The smaller the spread, the closer is the ratio to unity; the larger the spread, the closer is the ratio to zero. If the signal is the result of an all-pole system with P_0 poles, $\bar{\varepsilon}_P = \bar{\varepsilon}_{P_0}$ for $P \geq P_0$, that is, the curve remains flat. In practice, the incremental reduction in the normalized error may be checked with a condition such as

$$1 - \frac{\bar{\varepsilon}_{P+1}}{\bar{\varepsilon}_P} < \Delta, \quad (7.59)$$

where Δ is a small threshold. The optimal order may be considered to have been reached if the condition is satisfied for several consecutive model orders.

A measure based on an information-theoretic criterion proposed by Akaike [361] is expressed as [357]

$$I(P) = \log \bar{\varepsilon}_P + \frac{2P}{N_e}, \quad (7.60)$$

where N_e is the effective number of data points in the signal taking into account windowing (for example, $N_e = 0.4N$ for a Hamming window, where N is the

number of data samples). As P is increased, the first term in the equation given above decreases while the second term increases. Akaike's measure $I(P)$ may be computed up to the maximum order P of interest or the maximum that is feasible, and then the model of the order for which $I(P)$ is at its minimum could be taken as the optimal model.

Model parameters: The AR (all-pole) model $H(z)$ and its inverse $A(z)$ are uniquely characterized by any one set of the following sets of parameters [357]:

- The model parameters $a_k, k = 1, 2, \dots, P$. The series of a_k parameters is also equal to the impulse response of the inverse filter.
- The impulse response $h(n)$ of the AR model.
- The poles of $H(z)$, which are also the roots (zeros) of $A(z)$.
- The reflection coefficients $\gamma_i, i = 1, 2, \dots, P$.
- The ACF (or PSD) of the a_k coefficients.
- The ACF (or PSD) of $h(n)$.
- The cepstrum of a_k or $h(n)$.

With the inclusion of the gain factor G as required, all of the above sets have a total of $(P + 1)$ values and are equivalent in the sense that one set may be derived from another. Any particular set of parameters may be used, depending upon its relevance, interpretability, or relationship to the real-world system being modeled.

Illustration of application to EEG signals: Identification of the existence of rhythms of specific frequencies is an important aspect of EEG analysis. The direct relationship between the poles of an AR model and resonance frequencies makes this technique an attractive tool for the analysis of EEG signals.

Figure 7.12 shows the Fourier spectrum and AR-model spectra with $P = 6$ and $P = 10$ for the o1 channel of the EEG signal shown in Figure 1.39. The Fourier spectrum in the lowest trace of Figure 7.12 includes many spurious variations which make its interpretation difficult. On the other hand, the AR spectra indicate distinct peaks at about 10 Hz corresponding to an alpha rhythm; a peak at 10 Hz is clearly evident even with a low model order of $P = 6$ (the middle trace in Figure 7.12).

The poles of the AR model with order $P = 10$ are plotted in Figure 7.13. The dominant pole (closest to the unit circle in the z -plane) appears at 9.9 Hz, corresponding to the peak observed in the spectrum in the topmost plot in Figure 7.12. The radius of the dominant pole is $|z| = 0.95$; the other complex-conjugate pole pairs have $|z| \leq 0.76$. The model with $P = 6$ resulted in two complex-conjugate pole pairs and two real poles, with the dominant pair at 10.5 Hz with $|z| = 0.91$; the magnitude of the other pole pair was 0.74. A simple search for the dominant (complex) pole can provide an indication of the prevalent EEG rhythm with fairly low AR model orders.

Illustration of application to PCG signals: Application of AR modeling is an attractive possibility for the analysis of PCG signals due to the need to identify significant frequencies of resonance in the presence of multiple components, artifacts,

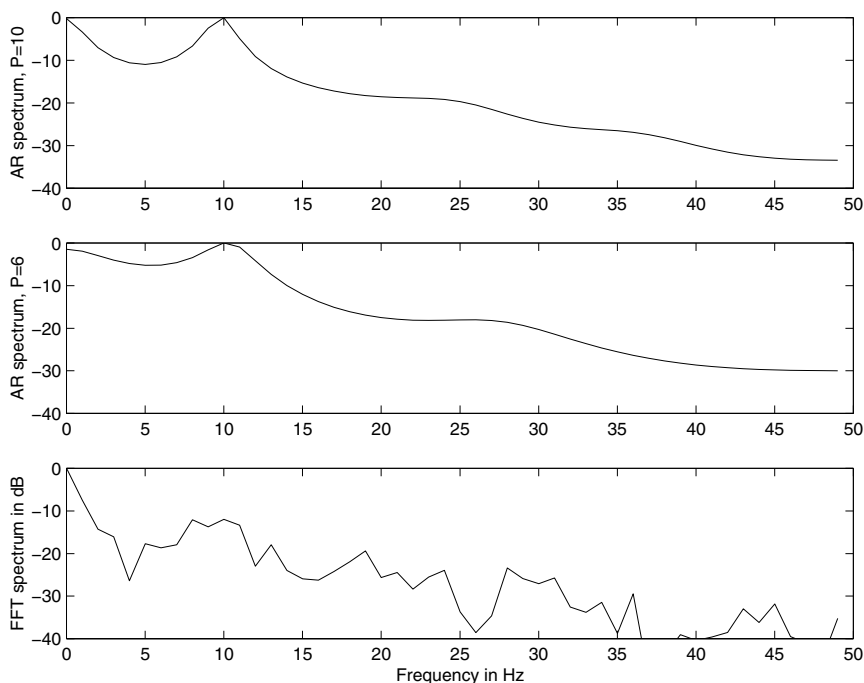


Figure 7.12 From bottom to top: Fourier and AR-model spectra with $P = 6$ and $P = 10$ for the o1 channel of the EEG signal shown in Figure 1.39.

and noise. Although the model coefficients themselves do not carry any physical correlates or significance, the poles may be related directly to the physical or physiological characteristics of hearts sounds and murmurs.

Figures 7.14 and 7.15 illustrate the Fourier spectrum of a segment containing one S1 and the subsequent systolic portion of the PCG signal of a normal subject, the AR-model spectra for order $P = 10$ and $P = 20$, and the poles of the model of order $P = 20$ (see also Figures 4.30, 5.7, and 6.12). Figures 7.16 and 7.17 illustrate the same items for a segment containing one S2 and the subsequent diastolic portion of the same subject. It is evident that the AR spectra follow the dominant peaks in the spectra of the original signals. The spectra for the models of order $P = 20$ provide closer fits than those for $P = 10$; peaks in the $P = 10$ spectra gloss over multiple peaks in the original spectra. Observe the presence of poles close to the unit circle in the z -plane at frequencies corresponding to the peaks in the spectra of the signals. The AR-model spectra are smoother and easier to interpret than the periodogram-based spectra illustrated in Figure 6.12 for the same subject. The spectra for the diastolic portion indicate more medium-frequency energy than those for the systolic portion, as expected. The model coefficients or poles provide a compact parametric representation of the signals and their spectra.

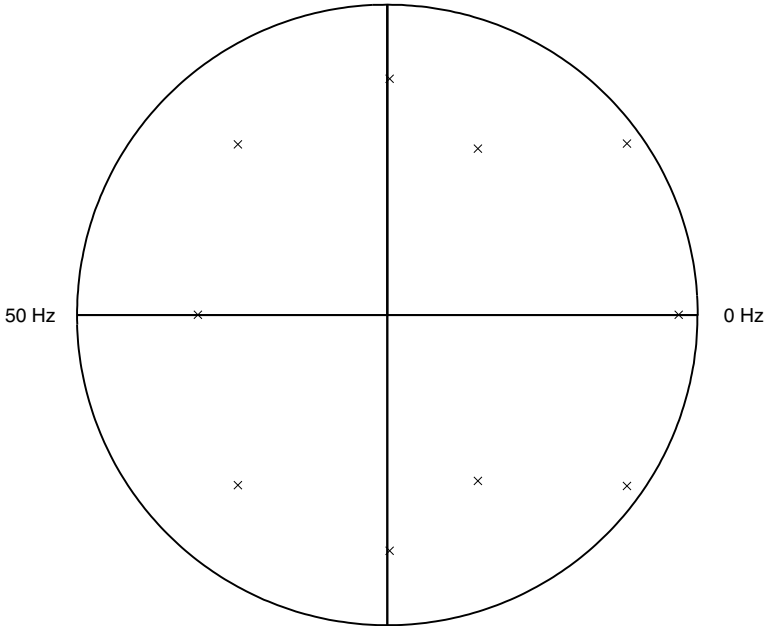


Figure 7.13 Poles of the AR model with $P = 10$ for the o1 channel of the EEG signal shown in Figure 1.39. See also Figure 7.12.

Figures 7.18 and 7.19 illustrate the Fourier spectrum of a segment containing one S1 and the subsequent systolic portion of the PCG signal of a subject with systolic murmur, split S2, and opening snap of the mitral valve (see also Figures 4.31, 5.8, and 6.13); the AR-model spectra for order $P = 10$ and $P = 20$; and the poles of the model of order $P = 20$. Figures 7.20 and 7.21 illustrate the same items for a segment containing one S2 and the subsequent diastolic portion of the same subject. The systolic murmur has given rise to more medium-frequency components than in the case of the normal subject in Figure 7.14. The AR-model spectra clearly indicate additional and stronger peaks at 150 Hz and 250 Hz , which are confirmed by poles close to the unit circle at the corresponding frequencies in Figure 7.19.

7.5.3 AR and cepstral coefficients

If the poles of $H(z)$ are inside the unit circle in the complex z -plane, from the theory of complex variables, $\ln H(z)$ can be expanded into a Laurent series as

$$\ln H(z) = \sum_{n=1}^{\infty} \hat{h}(n) z^{-n}. \quad (7.61)$$

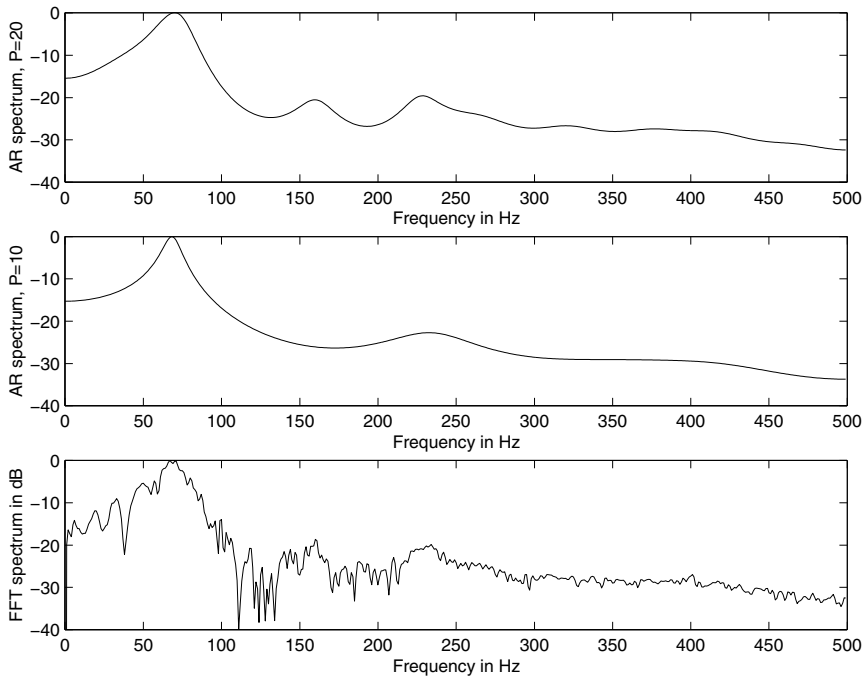


Figure 7.14 Bottom to top: Fourier spectrum of the systolic portion of the PCG of a normal subject (male, 23 years); AR-model spectrum with order $P = 10$; AR-model spectrum with order $P = 20$. (See also Figures 4.30, 5.7, and 6.12.)

Considering the definition of the complex cepstrum as the inverse z -transform of the logarithm of the z -transform of the signal, and the fact that the LHS of the equation given above represents the z -transform of $\hat{h}(n)$, it is clear that the coefficients of the series $\hat{h}(n)$ are the cepstral coefficients of $h(n)$. If $H(z)$ has been approximated by an AR model with coefficients a_k , $1 \leq k \leq P$, we have

$$\ln \left(\frac{1}{1 + \sum_{k=1}^P a_k z^{-k}} \right) = \sum_{n=1}^{\infty} \hat{h}(n) z^{-n}. \quad (7.62)$$

Differentiating both sides of the equation given above with respect to z^{-1} , we get

$$\frac{- \left(\sum_{k=1}^P k a_k z^{-k+1} \right)}{1 + \sum_{k=1}^P a_k z^{-k}} = \sum_{n=1}^{\infty} n \hat{h}(n) z^{-n+1}, \quad (7.63)$$

or

$$- \sum_{k=1}^P k a_k z^{-k+1} = \left(1 + \sum_{k=1}^P a_k z^{-k} \right) \sum_{n=1}^{\infty} n \hat{h}(n) z^{-n+1}. \quad (7.64)$$

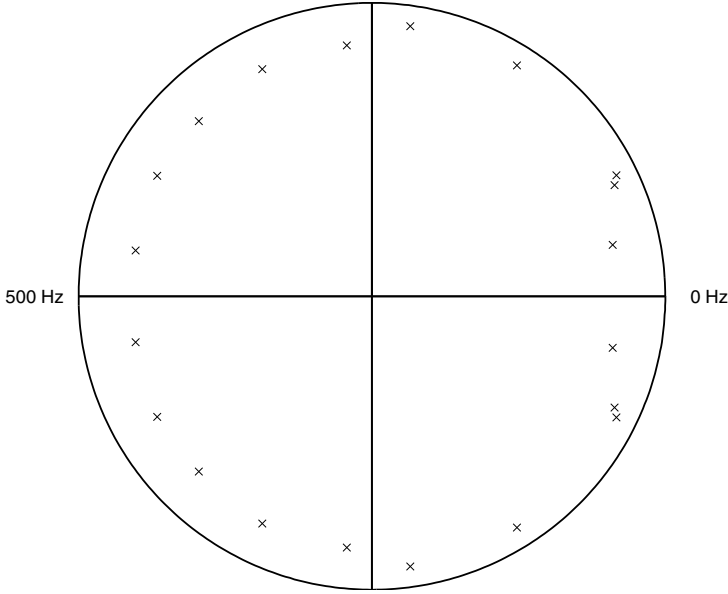


Figure 7.15 Poles of the AR model with order $P = 20$ of the systolic portion of the PCG of a normal subject. (See also Figure 7.14.)

By equating the constant term and the like powers of z^{-1} on both sides, the following relationship can be obtained [362]:

$$\begin{aligned}\hat{h}(1) &= -a_1, \\ \hat{h}(n) &= -a_n - \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) a_k \hat{h}(n-k), \quad 1 < n \leq P. \quad (7.65)\end{aligned}$$

As noted in Section 4.8.3, phase unwrapping could cause difficulties in estimating the cepstral coefficients using the inverse Fourier transform of the logarithm of the Fourier transform of a given signal [223]. Estimation of the cepstral coefficients using the AR coefficients has the advantage that it does not require phase unwrapping. Although the cepstral coefficients are deduced from the AR coefficients, it is expected that the nonlinear characteristics of the transformation could lead to an improvement in signal classification using the former than the latter set of coefficients. Cepstral coefficients have provided better classification than AR coefficients in speech [362], EMG [363], and VAG [96] signal analysis.

See Section 7.9 for a discussion on the application of AR modeling for the analysis of PCG signals. See Chisci et al. [61] for details on the application of AR modeling for the extraction of features from intracranial EEG signals and prediction of epileptic seizures.

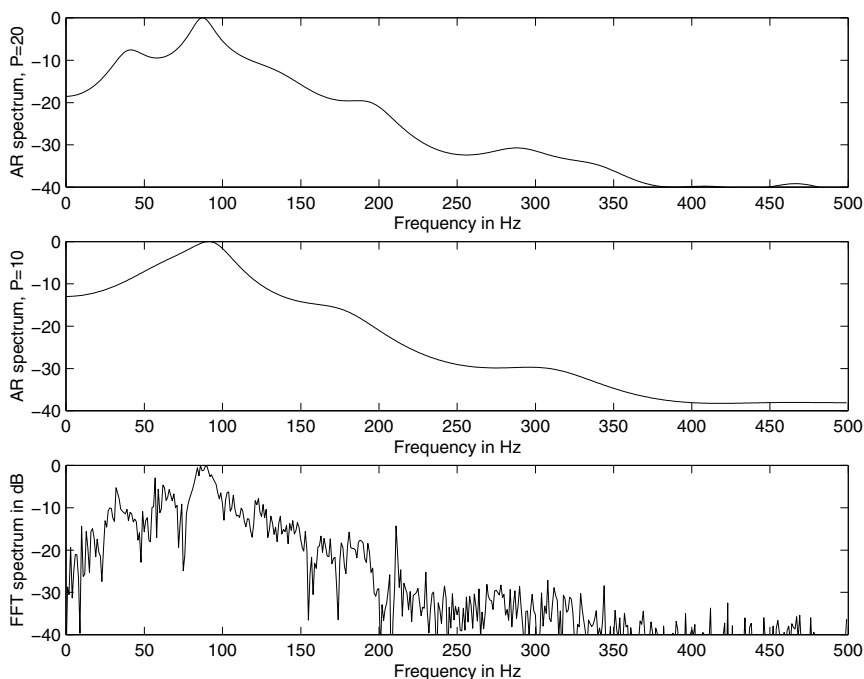


Figure 7.16 Bottom to top: Fourier spectrum of the diastolic portion of the PCG of a normal subject (male, 23 years); AR-model spectrum with order $P = 10$; AR-model spectrum with order $P = 20$. (See also Figures 4.30, 5.7, and 6.12.)

7.6 Pole zero Modeling

Although AR or all-pole modeling can provide good spectral models for any kind of spectra with appropriately high model orders, it has a few limitations. The AR model essentially follows the peaks in the PSD of the signal being modeled, and thus resonance characteristics are represented well. However, if the signal has spectral nulls or valleys (antiresonance), the AR-model spectrum will not provide a good fit in such spectral segments. Spectral zeros are important in modeling certain signals, such as nasal speech signals [364]. Furthermore, an all-pole model assumes the signal to be a minimum-phase signal or a maximum-phase signal, and does not allow mixed-phase signals [239].

The main conceptual difficulty posed by pole-zero modeling is that it is inherently nonunique, because a zero can be approximated by a large number of poles, and viceversa [357]. However, if the system being modeled has a number of influential zeros, the number of poles required for an all-pole model can become large. For these reasons, ARMA or pole-zero modeling [176, 239, 357, 364–366] is important in certain applications.

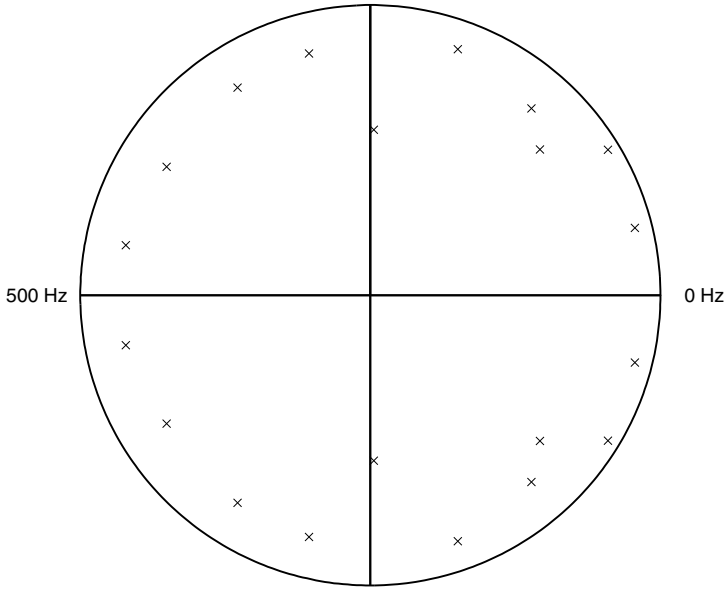


Figure 7.17 Poles of the AR model with order $P = 20$ of the diastolic portion of the PCG of a normal subject. (See also Figure 7.16.)

The ARMA normal equations: From the ARMA model represented by Equation 7.13, we can write the model PSD as [357]

$$\tilde{S}_y(\omega) = |H(\omega)|^2 = G^2 \frac{|B(\omega)|^2}{|A(\omega)|^2} = G^2 \frac{S_b(\omega)}{S_a(\omega)}, \quad (7.66)$$

where

$$S_a(\omega) = \left| 1 + \sum_{k=1}^P a_k \exp(-jk\omega) \right|^2 \quad (7.67)$$

and

$$S_b(\omega) = \left| 1 + \sum_{l=1}^Q b_l \exp(-jl\omega) \right|^2. \quad (7.68)$$

The total spectral-matching error is given by

$$\varepsilon = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_y(\omega) \frac{S_a(\omega)}{S_b(\omega)} d\omega, \quad (7.69)$$

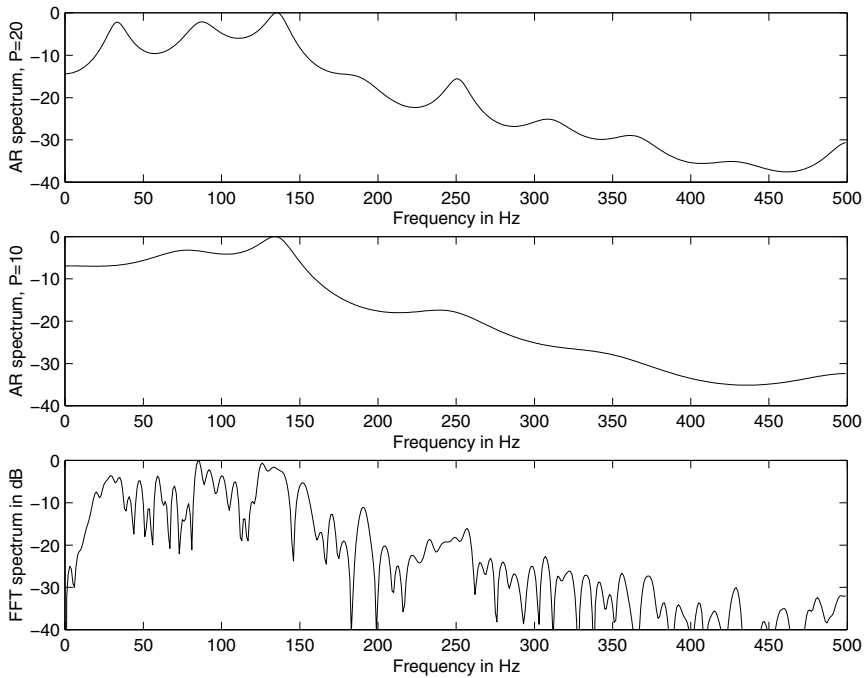


Figure 7.18 Bottom to top: Fourier spectrum of the systolic portion of the PCG of a subject (female, 14 months) with systolic murmur, split S2, and opening snap of the mitral valve; AR-model spectrum with order $P = 10$; AR-model spectrum with order $P = 20$. (See also Figures 4.31, 5.8, and 6.13.)

which may be viewed as the residual energy after passing the modeled signal through the inverse filter $\frac{A(z)}{B(z)}$. In order to obtain the optimal pole–zero model, we need to determine the coefficients a_k and b_l such that the error ε is minimized.

Before taking the derivatives of ε with respect to a_k and b_l , the following relationships are worth noting. Taking the partial derivative of $S_a(\omega)$ in Equation 7.67 with respect to a_i , we get

$$\frac{\partial S_a(\omega)}{\partial a_i} = 2 \sum_{k=0}^P a_k \cos[(i - k)\omega], \quad (7.70)$$

with $a_0 = 1$. Similarly, from Equation 7.68 we get

$$\frac{\partial S_b(\omega)}{\partial b_i} = 2 \sum_{l=0}^Q b_l \cos[(i - l)\omega]. \quad (7.71)$$

Let

$$\phi_{y\alpha\beta}(i) = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_y(\omega) \frac{[S_a(\omega)]^\beta}{[S_b(\omega)]^\alpha} \cos(i\omega) d\omega. \quad (7.72)$$

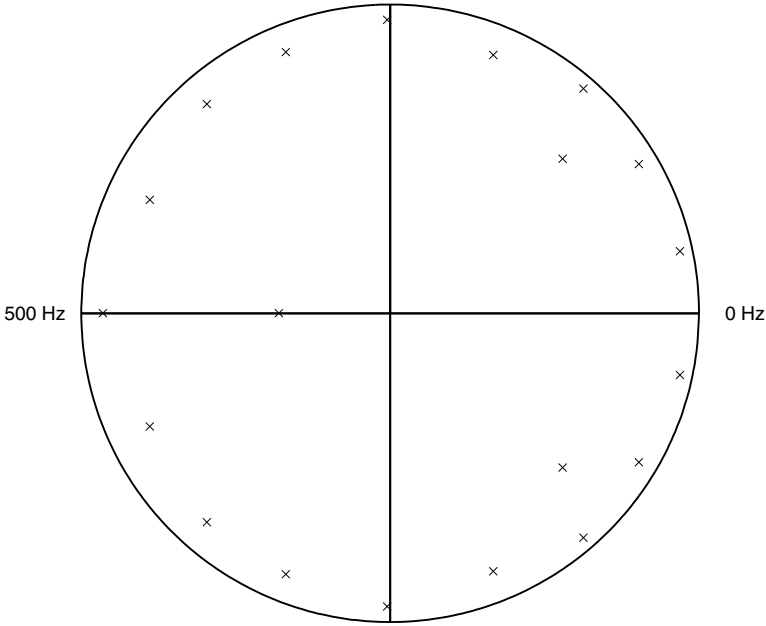


Figure 7.19 Poles of the AR model with order $P = 20$ of the systolic portion of the PCG of a subject with systolic murmur, split S2, and opening snap of the mitral valve. (See also Figure 7.18.)

$\phi_{y00}(i)$ is the inverse Fourier transform of $S_y(\omega)$ and hence simply $\phi_y(i)$.

Now, we can take the partial derivative of ε in Equation 7.69 with respect to a_i as

$$\begin{aligned}
 \frac{\partial \varepsilon}{\partial a_i} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{S_y(\omega)}{S_b(\omega)} \frac{\partial}{\partial a_i} S_a(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{S_y(\omega)}{S_b(\omega)} 2 \sum_{k=0}^P a_k \cos[(i-k)\omega] d\omega \\
 &= 2 \sum_{k=0}^P a_k \phi_{y10}(i-k), \quad 1 \leq i \leq P.
 \end{aligned} \tag{7.73}$$

In the same manner, we can obtain

$$\frac{\partial \varepsilon}{\partial b_i} = -2 \sum_{l=0}^Q b_l \phi_{y21}(i-l), \quad 1 \leq i \leq Q. \tag{7.74}$$

Because $\phi_{y10}(i-k)$ in Equation 7.73 is not a function of a_k , we obtain a set of linear equations by setting the final expression in Equation 7.73 to zero, which could

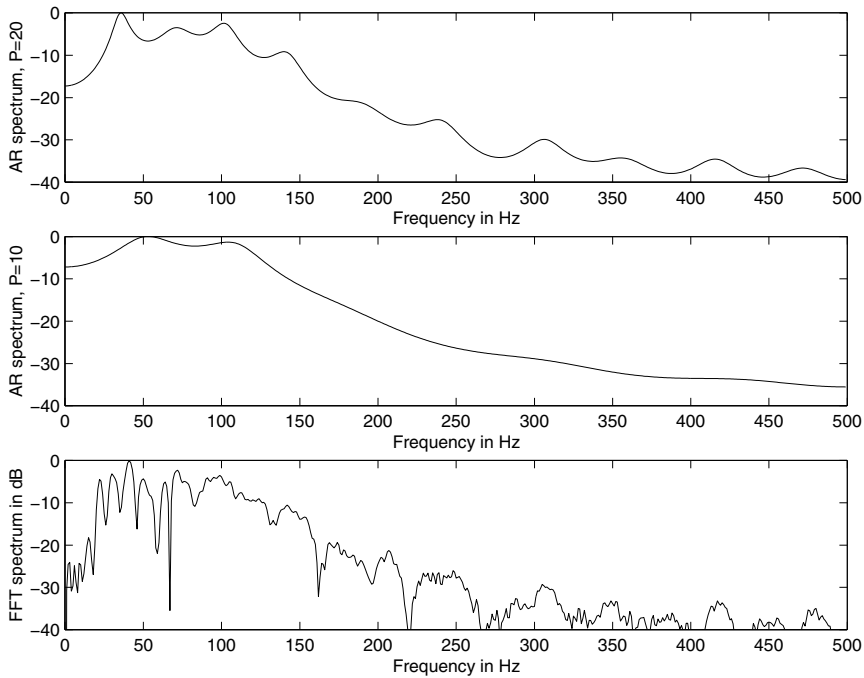


Figure 7.20 Bottom to top: Fourier spectrum of the diastolic portion of the PCG of a subject (female, 14 months) with systolic murmur, split S2, and opening snap of the mitral valve; AR-model spectrum with order $P = 10$; AR-model spectrum with order $P = 20$. (See also Figures 4.31, 5.8, and 6.13.)

be solved to obtain the a_k coefficients in a manner similar to the procedures used in AR modeling. However, $\phi_{y21}(i - l)$ in Equation 7.74 is a function of the b_l coefficients, which leads to a set of nonlinear equations that must be solved to obtain the b_l coefficients; the zeros of the model may then be derived from the b_l coefficients. Obtaining the ARMA model, therefore, requires solving P linear equations and Q nonlinear equations.

Iterative solution of the ARMA normal equations: Makhoul [357] describes the following iterative procedure to solve the $(P + Q)$ ARMA model normal equations based on the Newton–Raphson procedure:

Let $\mathbf{a} = [a_1, a_2, \dots, a_P]^T$, $\mathbf{b} = [b_1, b_2, \dots, b_Q]^T$, and $\mathbf{c} = [a_1, a_2, \dots, a_P, b_1, b_2, \dots, b_Q]^T$ represent the model coefficients to be derived in vectorial form. The vector at iteration $(m + 1)$ is derived from that at iteration m as

$$\mathbf{c}_{m+1} = \mathbf{c}_m - \mathbf{J}^{-1} \left. \frac{\partial \varepsilon}{\partial \mathbf{c}} \right|_{\mathbf{c}=\mathbf{c}_m}, \quad (7.75)$$

where $\mathbf{J}$ is the $(P + Q) \times (P + Q)$ symmetric Hessian matrix defined as $\mathbf{J} = \frac{\partial^2 \varepsilon}{\partial \mathbf{c} \partial \mathbf{c}^T}$. The vector $\mathbf{c}$ may be partitioned as $\mathbf{c}^T = [\mathbf{a}^T, \mathbf{b}^T]$, and the iterative procedure may

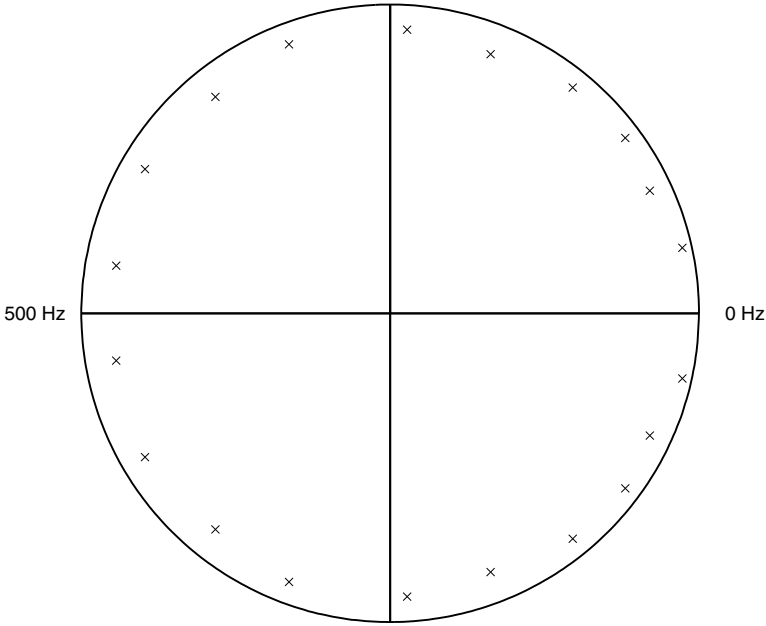


Figure 7.21 Poles of the AR model with order $P = 20$ of the diastolic portion of the PCG of a subject with systolic murmur, split S2, and opening snap of the mitral valve. (See also Figure 7.20.)

be expressed as

$$\begin{aligned}
 \begin{bmatrix} \mathbf{a}_{m+1} \\ \mathbf{b}_{m+1} \end{bmatrix} &= \begin{bmatrix} \mathbf{a}_m \\ \mathbf{b}_m \end{bmatrix} \\
 &- \begin{bmatrix} \frac{\partial^2 \varepsilon}{\partial \mathbf{a} \partial \mathbf{a}^T} & \frac{\partial^2 \varepsilon}{\partial \mathbf{a} \partial \mathbf{b}^T} \\ \frac{\partial^2 \varepsilon}{\partial \mathbf{b} \partial \mathbf{a}^T} & \frac{\partial^2 \varepsilon}{\partial \mathbf{b} \partial \mathbf{b}^T} \end{bmatrix}^{-1} \begin{bmatrix} \frac{\partial \varepsilon}{\partial \mathbf{a}} \\ \frac{\partial \varepsilon}{\partial \mathbf{b}} \end{bmatrix} \begin{bmatrix} \mathbf{a} = \mathbf{a}_m \\ \mathbf{b} = \mathbf{b}_m \end{bmatrix}. \quad (7.76)
 \end{aligned}$$

Equations 7.73 and 7.74 give the first-order partial derivatives required above. The second-order partial derivatives are given as follows [357]:

$$\frac{\partial^2 \varepsilon}{\partial a_i \partial a_j} = 2\phi_{y10}(i - j), \quad (7.77)$$

$$\frac{\partial^2 \varepsilon}{\partial a_i \partial b_j} = -2 \sum_{k=0}^P \sum_{l=0}^Q a_k b_l [\phi_{y20}(j + i - l - k) + \phi_{y20}(j - i - l + k)], \quad (7.78)$$

and

$$\frac{\partial^2 \varepsilon}{\partial b_i \partial b_j} = -2\phi_{y21}(i-j) + 4 \sum_{k=0}^P \sum_{l=0}^Q b_k b_l [\phi_{y31}(j+i-l-k) + \phi_{y31}(j-i-l+k)]. \quad (7.79)$$

(In the present context, i and j are both indices with integral values.) The iterative procedure works well if the initial estimate is close to the optimal model; otherwise, one of the noniterative methods described in the following sections may be considered.

7.6.1 Sequential estimation of poles and zeros

Given the difficulties with the nonlinear nature of direct pole-zero modeling, a few methods have been proposed to split the problem into two parts: Identify the poles first by AR modeling, and then treat the residual error in some manner to estimate the zeros [176, 239, 357, 364–366]. (*Note:* Several notational differences exist between the various references cited here. The following derivations use notations consistent with those used so far in the present chapter.)

Shanks' method: Let us consider a slightly modified version of Equation 7.13 as

$$H(z) = \frac{Y(z)}{X(z)} = \frac{B(z)}{A(z)} = \frac{1 + \sum_{l=1}^Q b_l z^{-l}}{1 + \sum_{k=1}^P a_k z^{-k}}, \quad (7.80)$$

where the gain factor G has been set to be unity: $G = 1$. The difference equation relating the output to the input is given by a small change to Equation 7.12 as

$$y(n) = - \sum_{k=1}^P a_k y(n-k) + \sum_{l=0}^Q b_l x(n-l). \quad (7.81)$$

The effect of the numerator and denominator polynomials in Equation 7.80 may be separated by considering $Y(z) = V(z)B(z)$, where $V(z) = \frac{X(z)}{A(z)}$. This leads to the all-zero or MA part of the system

$$y(n) = \sum_{l=0}^Q b_l v(n-l), \quad (7.82)$$

with $v(n)$ given by the all-pole or AR part of the model as

$$v(n) = - \sum_{k=1}^P a_k v(n-k) + x(n). \quad (7.83)$$

Let us consider the case of determining the a_k and b_l coefficients (equivalently, the poles and zeros) of the system $H(z)$ given its impulse response. Recollect that

$y(n) = h(n)$ when $x(n) = \delta(n)$; consequently, we have $X(z) = 1$, and $Y(z) = H(z)$. The impulse response of the system is given by

$$h(n) = - \sum_{k=1}^P a_k h(n-k) + \sum_{l=0}^Q b_l \delta(n-l), \quad (7.84)$$

which simplifies to

$$h(n) = - \sum_{k=1}^P a_k h(n-k), \quad n > Q. \quad (7.85)$$

The effect of the impulse input does not last beyond the number of zeros in the system: the system output is then perfectly predictable from the preceding P samples, and hence an AR or all-pole model is adequate to model $h(n)$ for $n > Q$. As a consequence, Equation 7.32 is modified to

$$\phi_h(i) = - \sum_{k=1}^P a_k \phi_h(i-k), \quad i > Q. \quad (7.86)$$

This system of equations may be solved by considering P equations with $Q+1 \leq i \leq Q+P$. Thus, the a_k coefficients, and hence the poles of the system, may be computed independently of the b_l coefficients or the zeros by restricting the AR error analysis to $n > Q$.

In a practical application, the error of prediction needs to be considered, as the model order P will not be known or some noise will be present in the estimation. Kopec et al. [364] recommend that the covariance method described in Section 7.5 be used to derive the AR model by considering the error of prediction as

$$e(n) = h(n) + \sum_{k=1}^P a_k h(n-k) \quad (7.87)$$

and minimizing the TSE defined as

$$\varepsilon = \sum_{n=Q+1}^{\infty} |e(n)|^2. \quad (7.88)$$

The first Q points are left out as they are not predictable with an all-pole model.

Let us assume that the AR modeling part has been successfully performed by the procedure described above. Let

$$\tilde{A}(z) = 1 + \sum_{k=1}^P \tilde{a}_k z^{-k} \quad (7.89)$$

represent the denominator polynomial of the system that has been estimated. The TSE in modeling $h(n)$ is given by

$$\varepsilon = \sum_{n=0}^{\infty} \left| h(n) - \sum_{l=0}^Q b_l \tilde{v}(n-l) \right|^2, \quad (7.90)$$

where $\tilde{v}(n)$ is the impulse response of the AR model derived, with $\tilde{V}(z) = \frac{1}{\tilde{A}(z)}$. Minimization of ε as above leads to the set of linear equations

$$\sum_{l=0}^Q b_l \phi_{\tilde{v}\tilde{v}}(l, j) = \phi_{h\tilde{v}}(0, j), \quad 0 \leq j \leq Q, \quad (7.91)$$

where

$$\phi_{h\tilde{v}}(l, j) = \sum_{n=0}^{\infty} h(n-l) \tilde{v}(n-j) \quad (7.92)$$

is the CCF between $h(n)$ and $\tilde{v}(n)$, and $\phi_{\tilde{v}\tilde{v}}$ is the ACF of $\tilde{v}(n)$.

The frequency-domain equivalents of the steps above may be analyzed as follows. The TSE is

$$\begin{aligned} \varepsilon &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| H(\omega) \tilde{A}(\omega) - \sum_{l=0}^Q b_l \exp(-jl\omega) \right|^2 |\tilde{V}(\omega)|^2 d\omega \quad (7.93) \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |E_h(\omega) - B(\omega)|^2 |\tilde{V}(\omega)|^2 d\omega, \end{aligned}$$

where $E_h(z) = H(z)\tilde{A}(z)$ is the AR-model error in the z -domain. [Recall that the Fourier spectrum of a signal is obtained by evaluating the corresponding function of z with $z = \exp(j\omega)$.] The method described above, which is originally attributed to Shanks [176] and has been described as above by Kopec et al. [364], therefore estimates the numerator polynomial of the model by fitting a polynomial (spectral function) to the z -transform of the error of the AR or all-pole model.

Makhoul [357] and Kopec et al. [364] suggest another method labeled as *inverse LP* modeling, where the inverse of the AR-model error $e_h^{-1}(n)$ given as the inverse z -transform of $E_h^{-1}(z)$ is subjected to all-pole modeling. The poles so obtained are the zeros of the original system being modeled.

7.6.2 Iterative system identification

Problem: *Given a noisy observation of the output of a linear system in response to a certain input, develop a method to estimate the numerator and denominator polynomials of a rational z -domain model of the system.*

Solution: In consideration of the difficulty in solving the nonlinear problem inherent in ARMA modeling or pole-zero estimation, Steiglitz and McBride [365] proposed an iterative procedure based upon an initial estimate of the denominator (AR) polynomial. Since their approach to system identification is slightly different from the LP approach we have been using so far in this chapter, it is appropriate to restate the problem.

The Steiglitz–McBride method: Figure 7.22 shows a block diagram illustrating the problem of system identification. The system is represented by its transfer function $H(z)$, input $x(n)$, output $y(n) = h(n) * x(n)$, and the noisy observation

$w(n) = y(n) + \eta(n)$, where $\eta(n)$ is a noise process that is statistically independent of the signals being considered. $H(z)$ is represented as a rational function of z , as

$$H(z) = \frac{Y(z)}{X(z)} = \frac{B(z)}{A(z)} = \frac{\sum_{l=0}^Q b_l z^{-l}}{\sum_{k=0}^P a_k z^{-k}}. \quad (7.94)$$

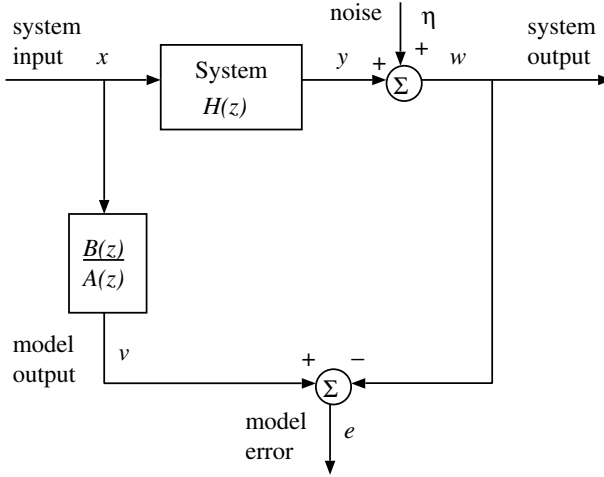


Figure 7.22 Schematic representation of system identification. Adapted from Steiglitz and McBride [365].

The error to be minimized may be written as [365]

$$\sum_{n=0}^{N-1} e^2(n) = \frac{1}{2\pi j} \oint \left| X(z) \frac{B(z)}{A(z)} - W(z) \right|^2 \frac{dz}{z}, \quad (7.95)$$

where the RHS represents the inverse z -transform of the function of z involved, $W(z)$ is the z -transform of $w(n)$, and N is the number of data samples available. The functions of z within the integral essentially compare the predicted model output with the observed output of the physical system.

As seen earlier, this approach leads to a nonlinear problem. The problem may be simplified (linearized) by taking the approach of separate identification of the numerator and denominator polynomials: the estimation problem illustrated in Figure 7.23 treats $A(z)$ and $B(z)$ as separate systems. The error to be minimized may then be written as [365]

$$\sum_{n=0}^{N-1} e^2(n) = \frac{1}{2\pi j} \oint |X(z)B(z) - W(z)A(z)|^2 \frac{dz}{z}. \quad (7.96)$$

(This approach was originally proposed by Kalman [367].)

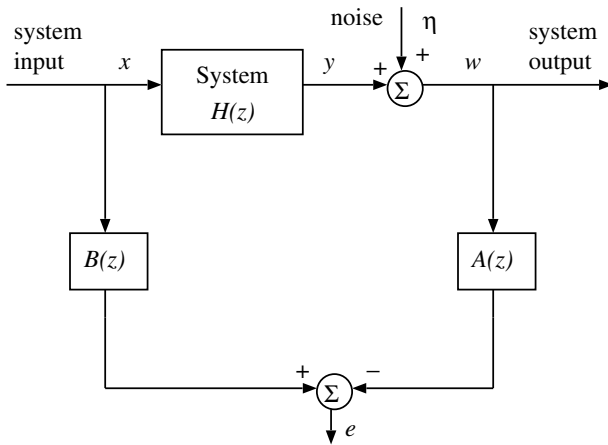


Figure 7.23 Schematic representation of system identification with separate estimation of $A(z)$ and $B(z)$. Adapted from Steiglitz and McBride [365].

The sample-model error is given by

$$e(n) = \sum_{l=0}^Q b_l x(n-l) - \sum_{k=1}^P a_k w(n-k) - w(n). \quad (7.97)$$

The model coefficients and the input–output data samples may be written in vectorial form as

$$\mathbf{c} = [b_0, b_1, \dots, b_Q, -a_1, -a_2, \dots, -a_P]^T \quad (7.98)$$

and

$$\mathbf{d}(n) = [x(n), x(n-1), \dots, x(n-Q), w(n-1), w(n-2), \dots, w(n-P)]^T, \quad (7.99)$$

with the vectors being of size $(P + Q + 1)$. Then, the error is given by

$$e(n) = \mathbf{d}^T(n) \mathbf{c} - w(n). \quad (7.100)$$

The condition for minimum TSE is given by

$$\frac{\partial}{\partial \mathbf{c}} \sum_{n=0}^{N-1} e^2(n) = 2 \sum_{n=0}^{N-1} \frac{\partial e(n)}{\partial \mathbf{c}} e(n) = 2 \sum_{n=0}^{N-1} \mathbf{d}(n) e(n) = 0. \quad (7.101)$$

Substitution of the expression for the error in Equation 7.100 in the condition stated above gives

$$\left(\sum_{n=0}^{N-1} \mathbf{d}(n) \mathbf{d}^T(n) \right) \mathbf{c} = \sum_{n=0}^{N-1} w(n) \mathbf{d}(n). \quad (7.102)$$

If we let

$$\Phi = \sum_{n=0}^{N-1} \mathbf{d}(n) \mathbf{d}^T(n) \quad (7.103)$$

represent the $(P + Q + 1) \times (P + Q + 1)$ correlation matrix of the combined string of input–output data samples $\mathbf{d}(n)$ and let

$$\Theta = \sum_{n=0}^{N-1} w(n) \mathbf{d}(n) \quad (7.104)$$

represent the correlation between the signal $w(n)$ and the data vector $\mathbf{d}(n)$ of size $(P + Q + 1)$, we get the solution to the estimation problem as

$$\mathbf{c} = \Phi^{-1} \Theta. \quad (7.105)$$

Although the vectors and matrices related to the filter coefficients and the signal correlation functions are defined in a different manner, the solution obtained above is comparable to that of the optimal Wiener filter (see Section 3.8 and Equation 3.168).

The limitation of the approach described above is that the error used has no physical significance. The separation of the numerator and denominator functions as in Figure 7.23, while simplifying the estimation problem, has led to a situation that is far from reality.

To improve upon the match between the real physical situation and the estimation problem, Steiglitz and McBride [365] proposed an iterative procedure which is schematically illustrated in Figure 7.24. The basic approach is to treat the system identified using the simplified procedure described above as an initial estimate, labeled as $A_1(z)$ and $B_1(z)$; filter the original signals $x(n)$ and $w(n)$ with the system $\frac{1}{A_1(z)}$; use the filtered signals to obtain new estimates $A_2(z)$ and $B_2(z)$; and iterate the procedure until convergence is achieved.

The error to be minimized may be written as [365]

$$\begin{aligned} \sum_{n=0}^{N-1} e^2(n) &= \frac{1}{2\pi j} \oint \left| X(z) \frac{B_i(z)}{A_{i-1}(z)} - W(z) \frac{A_i(z)}{A_{i-1}(z)} \right|^2 \frac{dz}{z} \quad (7.106) \\ &= \frac{1}{2\pi j} \oint \left| X(z) \frac{B_i(z)}{A_i(z)} - W(z) \right|^2 \left| \frac{A_i(z)}{A_{i-1}(z)} \right|^2 \frac{dz}{z}, \end{aligned}$$

with $A_0(z) = 1$. It is obvious that, upon convergence, when $A_i(z) = A_{i-1}(z)$, the minimization problem stated above reduces to the ideal (albeit nonlinear) situation expressed in Equation 7.95 and illustrated in Figure 7.22.

Steiglitz and McBride [365] suggest a modified iterative procedure to improve the estimate further, by imposing the condition that the partial derivatives of the true error criterion with respect to the model coefficients be equal to zero at convergence. The error of the true (ideal) model is given in the z -domain as (refer to Figure 7.22)

$$E(z) = X(z) \frac{B(z)}{A(z)} - W(z) = V(z) - W(z). \quad (7.107)$$

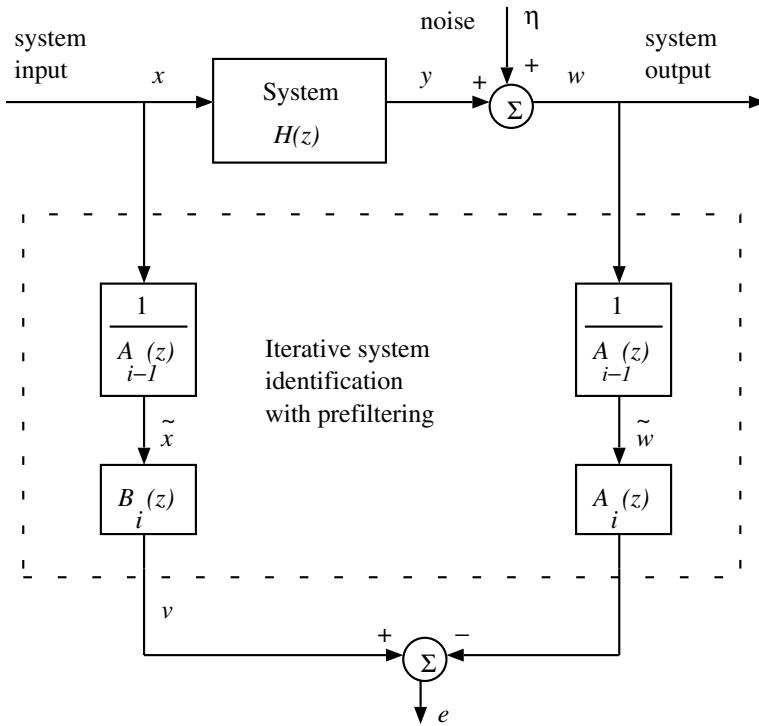


Figure 7.24 Schematic representation of system identification via iterative prefiltering. Adapted from Steiglitz and McBride [365].

$V(z)$ is the output of the model for the input $x(n)$. The derivatives of $E(z)$ with respect to the model coefficients are given by

$$\frac{\partial \varepsilon}{\partial a_i} = -\frac{X(z)B(z)}{A^2(z)} z^{-i} = -\frac{V(z)}{A(z)} z^{-i} = -\tilde{V}(z)z^{-i} \quad (7.108)$$

and

$$\frac{\partial \varepsilon}{\partial b_i} = \frac{X(z)}{A(z)} z^{-i} = \tilde{X}(z)z^{-i}, \quad (7.109)$$

where the superscript $\sim$ represents a filtered version of the corresponding signal, the filter transfer function being $\frac{1}{A(z)}$. A new data vector is defined as

$$\mathbf{d}_1(n) = [\tilde{x}(n), \tilde{x}(n-1), \dots, \tilde{x}(n-Q), \tilde{v}(n-1), \tilde{v}(n-2), \dots, \tilde{v}(n-P)]^T. \quad (7.110)$$

The error gradient in Equation 7.101 is modified to

$$\frac{\partial}{\partial \mathbf{c}} \sum_{n=0}^{N-1} e^2(n) = 2 \sum_{n=0}^{N-1} \frac{\partial e(n)}{\partial \mathbf{c}} e(n) = 2 \sum_{n=0}^{N-1} \mathbf{d}_1(n) e(n) \quad (7.111)$$

$$= 2 \sum_{n=0}^{N-1} [\mathbf{d}_1(n) \mathbf{d}_1^T(n) \mathbf{c} - w(n) \mathbf{d}_1(n)],$$

where the last equality is true only at convergence. The rest of the procedure remains the same as before, but with the correlation functions defined as

$$\Phi_1 = \sum_{n=0}^{N-1} \mathbf{d}_1(n) \mathbf{d}_1^T(n) \quad (7.112)$$

and

$$\Theta_1 = \sum_{n=0}^{N-1} w(n) \mathbf{d}_1(n). \quad (7.113)$$

Once the a_k and b_l coefficients are obtained, the related polynomials may be solved to obtain the poles and zeros of the system being modeled. Furthermore, the polynomials may be used to derive spectral models of the system or the signal of interest. Note that the procedures given above are applicable to the special case of system identification when the impulse response $h(n)$ is given: We just need to change $x(n) = \delta(n)$ and $X(z) = 1$. Steiglitz and McBride [365] did not provide any proof of convergence of their methods; however, it was indicated that the method performed successfully in many practical applications.

The Steiglitz–McBride method was applied to the modeling and classification of PCG signals by Joo et al. [368]. The first and second peak frequencies were detected from the model spectra and used to analyze porcine prosthetic valve function. Murthy and Prasad [369] applied the Steiglitz–McBride method to ECG signals. Pole–zero models derived from ECG strips including a few cardiac cycles were used to reconstruct and identify the ECG waveform over a single cycle, and also to reconstruct separately (that is, to segment) the P, QRS, and T waves.

7.6.3 Homomorphic prediction and modeling

Problem: *Given the relative ease of all-pole modeling, is it possible to convert the zeros of a system to poles?*

Solution: As mentioned earlier, an all-pole model assumes the signal being modeled to be a minimum-phase signal or a maximum-phase signal, and does not allow mixed-phase signals [239]. We have seen in Sections 4.8.3 and 5.4.2 that homomorphic filtering can facilitate the separation of the minimum-phase and maximum-phase components of a mixed-phase signal, and further facilitate the derivation of a minimum-phase version or correspondent (MPC) of a mixed-phase signal. Makhoul [357], Oppenheim et al. [239], and Kopec et al. [364] suggested methods to combine homomorphic filtering and LP into a procedure that has been labeled *homomorphic prediction* or *cepstral prediction*.

An intriguing property that arises in homomorphic prediction is that if a signal $x(n)$ has a rational z -transform, then $n\hat{x}(n)$ [where $\hat{x}(n)$ is the complex cepstrum of $x(n)$] has a rational z -transform whose poles correspond to the poles and zeros

of $x(n)$. The basic property of the z -transform we need to recollect here is that if $X(z)$ is the z -transform of $x(n)$, then the z -transform of $nx(n)$ is $-z \frac{dX(z)}{dz}$. Now, the complex cepstrum $\hat{x}(n)$ of $x(n)$ is defined as the inverse z -transform of $\hat{X}(z) = \log X(z)$. Therefore, we have

$$ZT[n\hat{x}(n)] = -z \frac{d\hat{X}(z)}{dz} = -z \frac{1}{X(z)} \frac{dX(z)}{dz}, \quad (7.114)$$

where $ZT[\cdot]$ represents the z -transform operator. If $X(z) = \frac{B(z)}{A(z)}$, we get

$$ZT[n\hat{x}(n)] = -z \frac{A(z)B'(z) - B(z)A'(z)}{A(z)B(z)}, \quad (7.115)$$

where the prime ' denotes the derivative of the associated function with respect to z . A general representation of a rational function of z (which represents an exponential signal in the z -domain) in terms of its poles and zeros is given by [239]

$$X(z) = A z^r \frac{\prod_{l=1}^{Q_i} (1 - z_{il} z^{-1}) \prod_{n=1}^{Q_o} (1 - z_{on} z)}{\prod_{k=1}^{P_i} (1 - p_{ik} z^{-1}) \prod_{m=1}^{P_o} (1 - p_{om} z)}, \quad (7.116)$$

with the magnitudes of all of the z_i, z_o, p_i , and p_o coefficients being less than unity (see also Sections 3.4.3, 4.8.3, and 5.4.2). The p_i and z_i values above give the P_i poles and Q_i zeros, respectively, of the system that are inside the unit circle in the z -plane; $\frac{1}{p_o}$ and $\frac{1}{z_o}$ give the P_o poles and Q_o zeros, respectively, that lie outside the unit circle. A causal and stable system will not have any poles outside the unit circle; regardless, the general representation given above permits the analysis and modeling of maximum-phase signals that are anticausal. Computation of the complex cepstrum requires the removal of any linear phase component that may be present, and hence we could impose the condition $r = 0$. We then have

$$\begin{aligned} \hat{X}(z) &= \log X(z) = \log A \\ &+ \sum_{l=1}^{Q_i} \log(1 - z_{il} z^{-1}) + \sum_{n=1}^{Q_o} \log(1 - z_{on} z) \\ &- \sum_{k=1}^{P_i} \log(1 - p_{ik} z^{-1}) - \sum_{m=1}^{P_o} \log(1 - p_{om} z), \end{aligned} \quad (7.117)$$

and, furthermore,

$$\begin{aligned} -z \frac{d\hat{X}(z)}{dz} &= - \sum_{l=1}^{Q_i} \frac{z_{il} z^{-1}}{(1 - z_{il} z^{-1})} + \sum_{n=1}^{Q_o} \frac{z_{on} z}{(1 - z_{on} z)} \\ &+ \sum_{k=1}^{P_i} \frac{p_{ik} z^{-1}}{(1 - p_{ik} z^{-1})} - \sum_{m=1}^{P_o} \frac{p_{om} z}{(1 - p_{om} z)}. \end{aligned} \quad (7.118)$$

From the expression given above, it is evident that $n\hat{x}(n)$ has simple (first-order) poles at every pole as well as every zero of $x(n)$. Therefore, we could apply an all-pole modeling procedure to $n\hat{x}(n)$, and then separate the poles so obtained into the desired poles and zeros of $x(n)$. An initial all-pole model of $x(n)$ can assist in the task of separating the poles from the zeros. Oppenheim et al. [239] show further that even if $X(z)$ is irrational, $n\hat{x}(n)$ has a rational z -transform with first-order poles corresponding to each irrational factor in $X(z)$.

Illustration of application to a synthetic speech signal: Figures 7.25 and 7.26 show examples of the application of several pole-zero and all-pole modeling techniques to a synthetic speech signal [239]. The impulse response of a synthetic system with two poles at 292 Hz and 3,500 Hz with bandwidth equal to 79 Hz and 100 Hz, respectively, and one zero at 2,000 Hz with bandwidth equal to 200 Hz is shown in Figure 7.25 (a), with its log-magnitude spectrum in Figure 7.26 (a). The formant or resonance structure of the signal is evident in the spectral peaks. (The sampling rate is 12 kHz.) Excitation of the system with a pulse train with the repetition rate of 120 Hz resulted in the signal in Figure 7.25 (b), whose spectrum is shown in Figure 7.26 (b); the spectrum clearly shows the effect of repetition of the basic wavelet in the series of waves that are superimposed on the basic spectrum of the wavelet. Application of homomorphic filtering to the signal in Figure 7.25 (b) provided an estimate of the basic wavelet as shown in Figure 7.25 (c), with the corresponding spectrum in Figure 7.26 (c).

The pole-zero modeling method of Shanks was applied to the result of homomorphic filtering in Figure 7.25 (c) with four poles and two zeros. The impulse response of the model and the corresponding spectrum are shown in Figure 7.25 (d) and Figure 7.26 (d), respectively. It is seen that the two peaks and the valley in the original spectrum are faithfully reproduced in the modeled spectrum. The frequencies of the poles (and their bandwidths) given by the model are 291 Hz (118 Hz) and 3,498 Hz (128 Hz), and those of the zero are 2,004 Hz (242 Hz), which compare well with those of the synthesized system listed in the preceding paragraph.

Application of the autocorrelation method of LP modeling with six poles to the original signal in Figure 7.25 (a) resulted in the model impulse response and spectrum illustrated in Figures 7.25 (e) and 7.26 (e). While the all-pole model spectrum has followed the spectral peaks well, it has failed to represent the valley or null related to the zero.

Illustration of application to a real speech signal: Figure 7.27 (a) shows the log-magnitude spectrum of a real speech signal (preemphasized) of the nasalized vowel /U/ in the word “moon” [364]. Part (b) of the same figure shows the spectrum after homomorphic filtering to remove the effects of repetition of the basic wavelet. Parts (c) and (d) show 10-pole, 6-zero model spectra obtained using Shanks’ method and inverse LP modeling, respectively. The spectra of the models have successfully followed the peaks and valleys in the signal spectrum.

Shanks’ method was applied to the minimum-phase and maximum-phase components of ECG signals obtained via homomorphic filtering by Murthy et al. [370]. Akay et al. [371] used ARMA techniques to model diastolic heart sounds for the

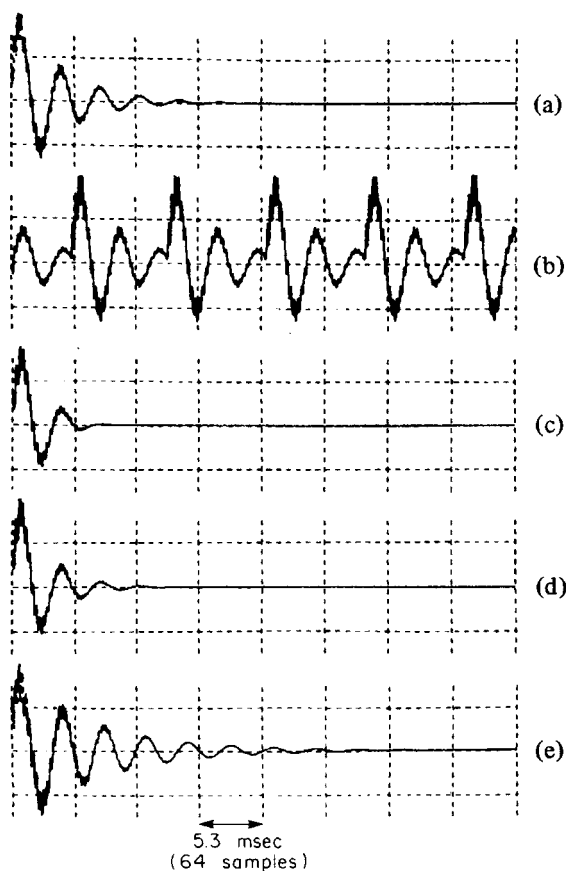


Figure 7.25 Time-domain signals: (a) impulse response of a 4-pole, 2-zero synthetic system; (b) synthesized voiced-speech signal obtained by triggering the system with an impulse train; (c) result of basic wavelet extraction via application of homomorphic filtering to the signal in (b); (d) impulse response of a 4-pole, 2-zero model of the signal in (c) obtained by Shanks' method; (e) impulse response of a 6-pole AR model. msec = ms. See also Figure 7.26. Reproduced with permission from A.V. Oppenheim, G.E. Kopec, and J.M. Tribolet, Signal analysis by homomorphic prediction, *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 24(4):327–332, 1976. ©IEEE.

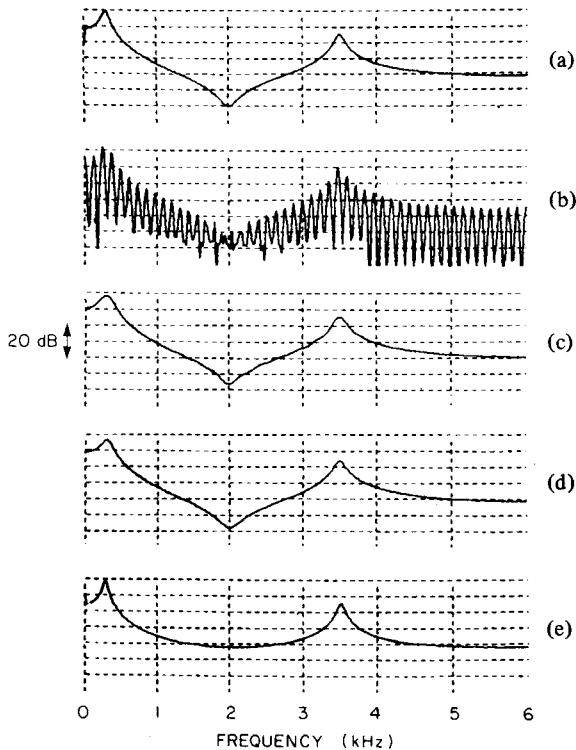


Figure 7.26 Log-magnitude spectra of the time-domain signals in Figure 7.25: (a) actual spectral response of the 4-pole, 2-zero synthetic system; (b) spectrum of the synthesized voiced-speech signal obtained by triggering the system with an impulse train; (c) spectrum of the basic wavelet extracted via application of homomorphic filtering to the signal corresponding to (b); (d) spectral response of a 4-pole, 2-zero model of the signal in (c) obtained by Shanks' method; (e) spectral response of a 6-pole AR model. Reproduced with permission from A.V. Oppenheim, G.E. Kopec, and J.M. Tribolet, *Signal analysis by homomorphic prediction*, *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 24(4):327–332, 1976. ©IEEE.

detection of coronary heart disease; however, only the dominant poles of the model were used in pattern analysis (see Section 7.10 for details of this application).

7.7 Electromechanical Models of Signal Generation

While purely mathematical models of signal generation, such as point processes and linear system models, provide the advantage of theoretical elegance and convenience, they may not be able to represent directly certain physical and physiological aspects of the systems that generate the signals. For example, the models described in the preceding sections cannot directly accommodate the physical dimensions of blood

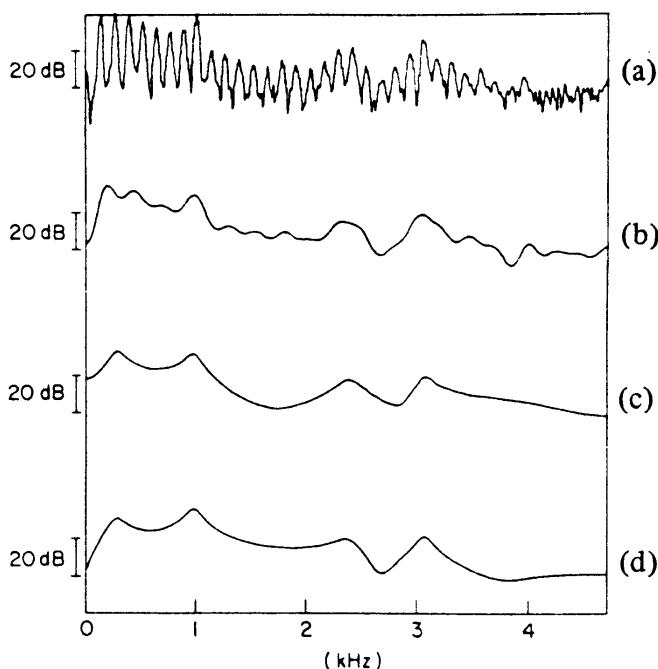


Figure 7.27 (a) Log-magnitude spectrum of the preemphasized, real speech signal of the nasalized vowel /U/ in the word “moon”; (b) spectrum after homomorphic filtering to remove the effects of repetition of the basic wavelet; (c) spectral response of a 10-pole, 6-zero model obtained by Shanks’ method; (d) spectral response of a 10-pole, 6-zero model obtained by inverse LP modeling. Reproduced with permission from G.E. Kopec, A.V. Oppenheim, and J.M. Tribolet, Speech analysis by homomorphic prediction, *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 25(1):40–49, 1977. ©IEEE.

vessels or valves, the loss in the compliance of a valve leaflet due to stenosis, or the lubrication (or the lack thereof) or friction between joint surfaces.

Sikarskie et al. [372] proposed a model to characterize aortic valve vibration for the analysis of its contribution to S2; in addition to mathematical relationships, they included physical factors, such as the valve forcing function, valve mass, and valve stiffness. It was shown that the amplitude and frequency of A2 depend strongly on the valve forcing function and valve stiffness. Valve mass was shown to have little effect on the amplitude and frequency of A2; blood density was shown to have no effect on the same parameters.

We shall now study three representative applications of electromechanical modeling, where mechanical models and their electrical counterparts are used to represent the generation and altered characteristics of sounds in the respiratory system, coronary arteries, and knee joints.

7.7.1 Modeling of respiratory sounds

Problem: *Given that parts of the respiratory system are composed of pipe-like segments, propose an electromechanical model to characterize the sounds produced due to respiration.*

Solution: Moussavi [108] proposed mechanical as well as electrical circuit models to characterize parts of the respiratory system. The models were used in the analysis of sounds related to breathing.

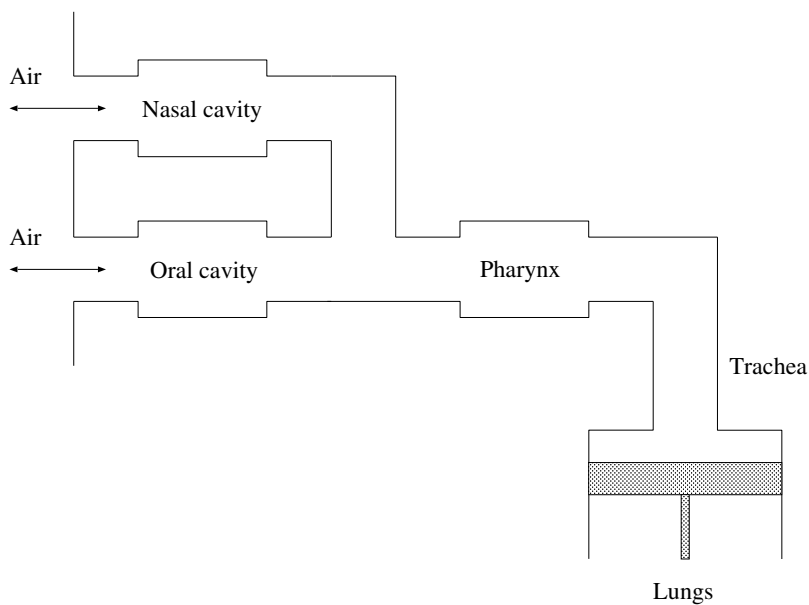
Regardless of the complex structure of the human vocal and respiratory tract, its main characteristics can be represented by mechanical models with tubes or pipes and their equivalent electrical circuit models [108, 373]; see also Section 1.2.12. A simple model of the vocal tract is given by a pipe closed at one end by the glottis and open at the other end at the lips. A pipe with length L has resonance frequencies given by $f_n = \frac{nv}{4L}$, $n = 1, 3, 5, \dots$, where n is the harmonic number and v is the velocity of air through the pipe. In order to take into account the variations of the respiratory tract along its length, the model could be modified to include multiple short segments of pipes or tubes with different diameter and length. Due to the mass of the air in each segment of the pipe model, the related inertance that opposes acceleration needs to be taken into account. Furthermore, the compliance related to the compressibility of air and the elasticity of the pipes' walls need to be represented. Overall, these considerations lead to a lossy transmission line or circuit model [108, 373]. Figure 7.28 shows a mechanical model and an electrical circuit model for a segment of the respiratory system given by Moussavi [108]; see also Figures 1.47 and 1.48 as well as Flanagan [373].

If a tube is smooth and its walls are hard, loss of energy occurs via viscous friction at the walls of the tube and heat conduction. The viscous loss is proportional to the square of particle velocity and the loss due to heat conduction is proportional to the square of the pressure [373]. Some of the similarities that can be drawn between mechanical and electrical models for the system described above are as follows [108, 373]: The sound pressure, P , is analogous to voltage, V ; the acoustic volume velocity, U , is comparable to current, I ; viscous loss can be represented by $I^2 R$ loss, where R is resistance; heat conduction loss can be related to V^2/R or $V^2 G$ loss, where G is conductance; inertance of air mass is analogous to inductance; and compliance can be modeled by capacitance.

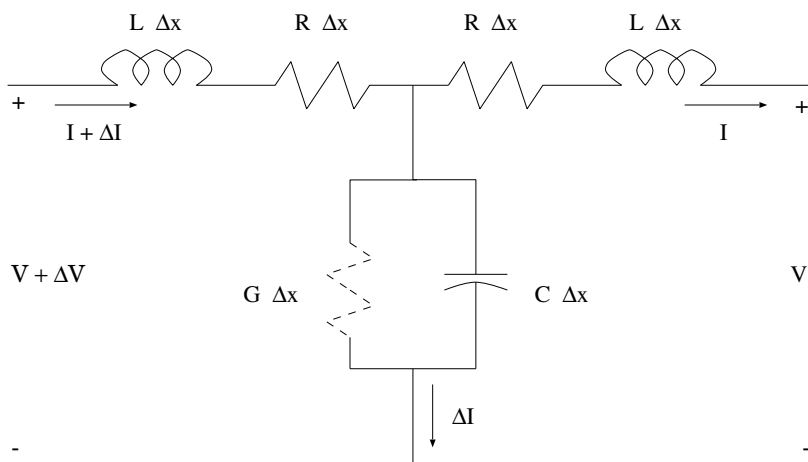
Moussavi [108] gives the following derivations for various parameters of the models; see also Flanagan [373]. Starting with Newton's second law, we have force $F = ma$, where m is the mass and a is the acceleration of an object. In a tube with cross-sectional area A and pressure P , we have $F = PA$; $m = \rho Al$, where ρ is the density of air and l is the length of the pipe; and $a = \frac{du}{dt}$, where u is the particle velocity and t is time. The volume velocity U is given by $U = Au$. Then, we have

$$PA = \rho Al \frac{du}{dt} = \rho l \frac{dU}{dt}, \quad (7.119)$$

which leads to



(a)



(b)

Figure 7.28 (a) Mechanical and (b) electrical circuit models of parts of the respiratory system. Adapted with permission from Z. Moussavi, *Fundamentals of Respiratory Sounds and Analysis*, Morgan & Claypool, San Francisco, CA, 2006, ©Morgan & Claypool.

$$P = \rho \frac{l}{A} \frac{dU}{dt}. \quad (7.120)$$

Comparing the equation given above with that for the voltage across an inductance L given by

$$V = L \frac{dI}{dt}, \quad (7.121)$$

we have the acoustic equivalent of inductance as

$$L_a = \rho \frac{l}{A}. \quad (7.122)$$

Now, consider the adiabatic gas law $PV_a^\eta = \text{a constant}$, where V_a is the volume of the gas under pressure P , and η is the adiabatic constant. Differentiating this equation with respect to time, we have

$$P \eta V_a^{\eta-1} \frac{dV_a}{dt} + V_a^\eta \frac{dP}{dt} = 0, \quad (7.123)$$

which leads to

$$\frac{1}{P} \frac{dP}{dt} = -\frac{\eta}{V_a} \frac{dV_a}{dt} = \frac{\eta}{V_a} U \quad (7.124)$$

and

$$U = \frac{V_a}{P \eta} \frac{dP}{dt}. \quad (7.125)$$

Comparing the last equation given above with the equation for the current through a capacitor C , expressed as

$$I = C \frac{dV}{dt}, \quad (7.126)$$

we have the acoustic equivalent of capacitance or compliance as

$$C_a = \frac{V_a}{P \eta}. \quad (7.127)$$

Acoustic resistance is given by

$$R_a = \frac{l S}{A^2} \sqrt{\frac{\omega \rho \mu}{2}}, \quad (7.128)$$

where ω is the frequency component of air flow, μ is the viscosity coefficient, and S is the circumference of the pipe, and acoustic conductance is given by

$$G_a = S l \frac{\eta - l}{\rho c^2} \sqrt{\frac{\lambda \omega}{2 c_p \rho}}, \quad (7.129)$$

where λ is the coefficient of heat conduction and c_p is the specific heat of air at constant pressure; see Flanagan [373] and Moussavi [108] for detailed derivations of these expressions.

Parameters derived as above may be used to model segments of tubes or pipes that can be combined to form more elaborate models of parts of the respiratory system. The effects of various segments could be analyzed as filtering of a certain input signal (such as an impulse, a pulse, or random noise) as it passes through the segments. Various respiratory diseases cause narrowing of parts of the respiratory tract as well changes in their smoothness, stiffness, and other mechanical properties. If such pathological changes can be included in the mechanical or electrical model, it becomes possible to analyze the related respiratory sounds and their characteristics. See Moussavi [108] for examples of sounds related to various diseases that affect the respiratory system and their characteristics.

7.7.2 Sound generation in coronary arteries

Problem: *Propose an electromechanical model to characterize the sounds produced due to blood flow in stenosed arteries.*

Solution: Blood vessels are normally flexible, elastic, and pliant, with smooth internal surfaces. When a segment of a blood vessel is hardened due to the deposition of calcium and other minerals, the segment becomes rigid. Furthermore, the development of plaque inside the vessel causes narrowing or constriction of the vessel, which impedes the flow of blood. The result is turbulent flow of blood, with accompanying high-frequency sounds.

Wang et al. [374, 375] proposed a sound-source model combining an incremental-network model of the left coronary artery tree with a transfer-function model describing the resonance characteristics of arterial chambers. The network model, illustrated in Figure 7.29, predicts flow in normal and stenosed arteries. It was noted that stenotic branches may require division into multiple segments in the model due to greater geometric variations. Furthermore, it was observed that a stenotic segment may exhibit poststenotic dilation as illustrated in Figure 7.30, due to increased pressure fluctuations caused by turbulence at the point of stenosis.

The resonance frequency of a segment of an artery depends upon the length and diameter of the segment, as well as upon the distal (away from the heart) hydraulic pressure loading the segment. The physical parameters required for the model were obtained from arteriograms of the patient being examined. The terminal resistances, labeled Z in Figure 7.29, represent loading of the resistive arteriolar beds, assumed to be directly related to the areas that the terminal branches serve.

Wang et al. related the network elements (resistance R , inertance or inductance L , and capacitance C) to the physical parameters of the artery segments as

$$R = 8\pi\nu \frac{l}{A^2}, \quad (7.130)$$

$$L = \rho \frac{l}{A}, \quad (7.131)$$

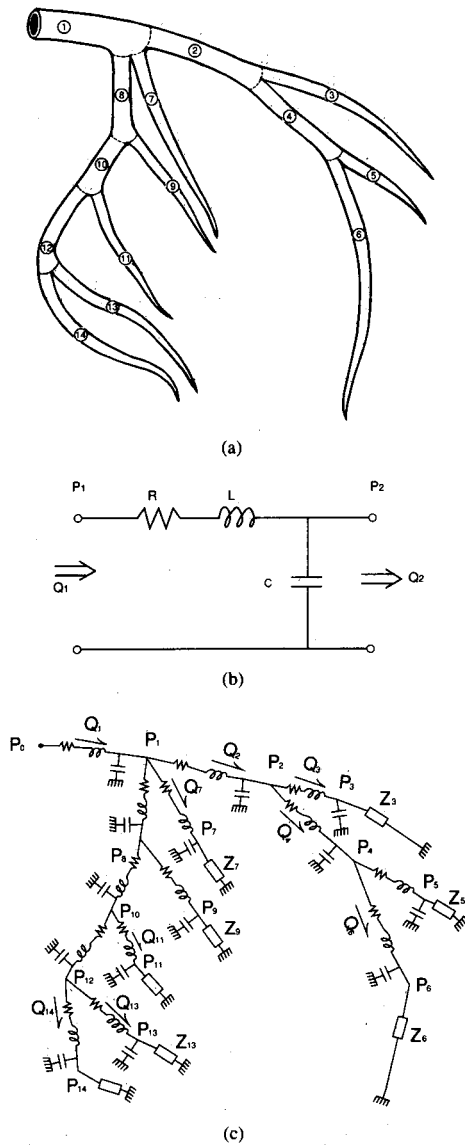
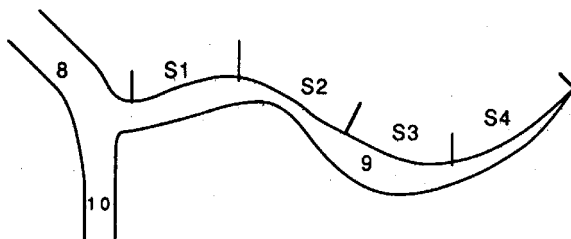


Figure 7.29 Electromechanical model of a coronary artery tree. (a) The left coronary artery tree is divided into 14 branches. (b) Circuit model of a segment. (c) Circuit model of the artery tree. Reproduced with permission from J.Z. Wang, B. Tie, W. Welkowitz, J.L. Semmlow, and J.B. Kostis, Modeling sound generation in stenosed coronary arteries, *IEEE Transactions on Biomedical Engineering*, 37(11):1087–1094, 1990, ©IEEE; and J.Z. Wang, B. Tie, W. Welkowitz, J. Kostis, and J. Semmlow, Incremental network analogue model of the coronary artery, *Medical & Biological Engineering & Computing*, 27:416–422, 1989. ©IFMBE.



S1: Proximal to Stenosis Segment
S2: Stenotic Segment
S3: Poststenotic Dilatation Segment
S4: Distal to Dilatation Segment

Figure 7.30 Hypothetical example of stenosis in coronary artery branch number 9. Reproduced with permission from J.Z. Wang, B. Tie, W. Welkowitz, J.L. Semmlow, and J.B. Kostis, Modeling sound generation in stenosed coronary arteries, *IEEE Transactions on Biomedical Engineering*, 37(11):1087–1094, 1990. ©IEEE.

and

$$C = A l h \frac{D}{E}, \quad (7.132)$$

where $\nu = 0.04 \text{ g cm}^{-1} \text{ s}$ is the viscosity of blood, $\rho = 1.0 \text{ g cm}^{-3}$ is the density of blood, $E = 2 \times 10^6 \text{ g cm}^{-1} \text{ s}^2$ is the Young's modulus of the blood vessel, D is the diameter of the segment, $A = \pi \frac{D^2}{4}$ is the cross-sectional area of the segment, $h \approx 0.08D$ is the wall thickness of the segment, and l is the length of the segment (see also Section 7.7.1). Wang et al. [375] remarked that, while the network elements may be assumed to be approximately constant during diastole, the assumption would not be valid during systole due to variations in the parameters of the segments.

In analyzing the artery–network model, voltage is analogous to pressure (P) and current is analogous to blood flow (Q). State-variable differential equations were used by Wang et al. [375] to derive the flow through the artery tree model for various pressure waveforms. It was hypothesized that turbulence at the point of stenosis would provide the excitation power, and that the stenotic segment and the dilated segment distal to the point of stenosis (see Figure 7.30) would act as resonance chambers.

Wang et al. [374] used the following relationships to compute the *RMS* pressure fluctuation (see also Fredberg [376]):

$$\langle P^2 \rangle_{\max} = 10^{-4} \rho u^2 f(x), \quad (7.133)$$

$$f(x) = 25.1 - 37.1x + 15.5x^2 - 0.08x^3 - 0.89x^4 + 0.12x^5, \quad (7.134)$$

and

$$x = 10^{-3} \frac{ud}{\nu} \left(\frac{D}{d} \right)^{0.75}, \quad (7.135)$$

where u is the blood velocity in the stenotic segment, and d is the diameter of the stenotic segment. The incremental network model was used to estimate the blood velocity in each segment.

The wideband spectrum of the sound associated with turbulent flow was modeled as (see also Fredberg [376])

$$S(f) = \frac{0.7 \frac{d}{U} \langle P^2 \rangle_{\max}}{1 + 0.5 \left[f \frac{d}{U} \right]^{\frac{10}{3}}}, \quad (7.136)$$

where U is the velocity of blood in a normal segment and f is frequency in Hz . Wang et al. used the function $S(f)$ given above as the source of excitation power to derive the response of their network model. It was observed that the model spectra indicated two resonance frequencies, the magnitude and frequency of which depended upon the geometry and loading of the segments. Wang et al. cautioned that the results of the model are sensitive to errors in the estimation of the required parameters from arteriograms or other sources.

Figure 7.31 illustrates the model spectra for segment number 12 of the artery tree model in Figure 7.29 with no stenosis and with stenosis of two grades. Narrowing of the segment with increasing stenosis is seen to shift the second peak in the spectrum to higher frequencies, while the magnitude and frequency of the first peak are both reduced. The results were confirmed by comparing the model spectra with spectra of signals recorded from a few patients with stenosed coronary arteries. Examples of spectral analysis of signals recorded from patients before and after angioplasty to correct for stenosis are presented in Section 7.10.

7.7.3 Sound generation in knee joints

Problem: *Develop a mechanical analog of the knee joint to model the generation of the pulse train related to PPC.*

Solution: Beverland et al. [346] studied the PPC signals produced during very slow movement of the leg (at about $4^\circ/s$). The signals were recorded by taping accelerometers to the skin above the upper pole and/or the lower pole of the patella. Reproducible series of bursts of vibration were recorded in their experiments. Figure 7.32 illustrates two channels of simultaneously recorded PPC signals from the upper and lower poles of the patella during extension and flexion of the leg. The signals display reversed similarity when extension versus flexion or upper-pole versus lower-pole recordings are compared.

Beverland et al. proposed a mechanical model to explain the generation of the PPC signals. The patella was considered to behave like a seesaw in the model, which was supported by the observation that a pivot point exists at the midpoint of the patella. The apparatus constructed, as illustrated in Figure 7.33, included a rubber

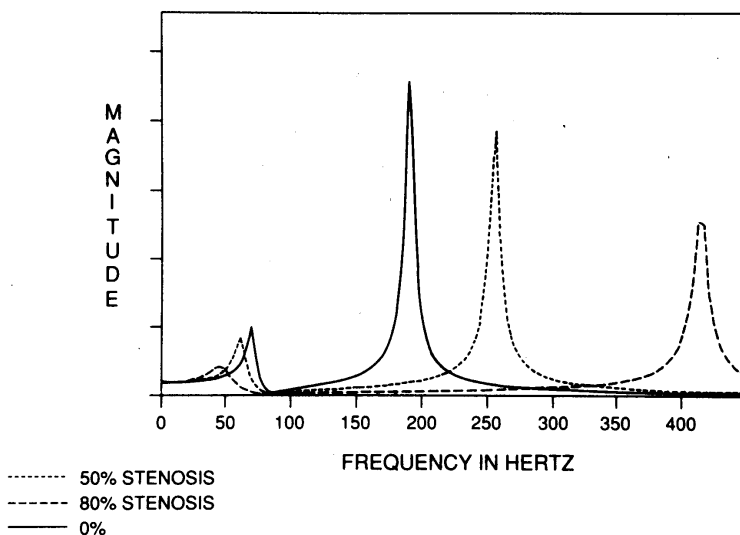


Figure 7.31 Shift in frequency components predicted by the transfer-function model for the case of stenosis in element number 12 in the model of the coronary artery in Figure 7.29. Reproduced with permission from J.Z. Wang, B. Tie, W. Welkowitz, J.L. Semmlow, and J.B. Kostis, Modeling sound generation in stenosed coronary arteries, *IEEE Transactions on Biomedical Engineering*, 37(11):1087–1094, 1990. ©IEEE.

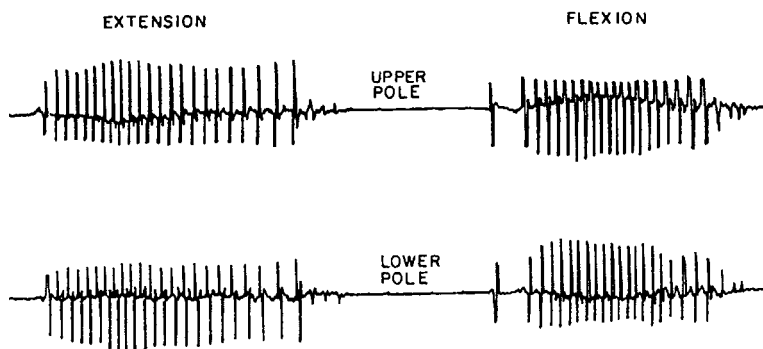


Figure 7.32 Simultaneously recorded PPC signals from the upper and lower poles of the patella during extension and flexion of the leg. The duration of the signal was not specified. Reproduced with permission from D.E. Beverland, W.G. Kernohan, G.F. McCoy, and R.A.B. Mollan, What is physiological patellofemoral crepitus?, *Proceedings of XIV International Conference on Medical and Biological Engineering and VII International Conference on Medical Physics*, pp 1249–1250, Espoo, Finland, 1985. ©IFMBE

wheel to represent the trochlear surface of the femur, on top of which was tensioned a rectangular piece of hardboard to represent the patella.

It was argued that, as the wheel in the model is slowly rotated clockwise (representing extension), it would initially stick to the overlying patella (hardboard) due to static friction. This would tend to impart an anticlockwise rotatory motion, as a rotating cogwheel would impart an opposite rotation to a cog in contact with it (as illustrated in the upper right-hand corner of Figure 7.33). The upper end of the patella would then move toward the wheel. A point would be reached where the static friction would be overcome, when the patella would slip and the rotation is suddenly reversed, with the upper pole jerking outward and the lower pole jerking inward. The actions would be the opposite to those described above in the case of flexion. The mechanical model was shown to generate signals similar to those recorded from subjects, thereby confirming the *stick-slip* frictional model for the generation of PPC signals.

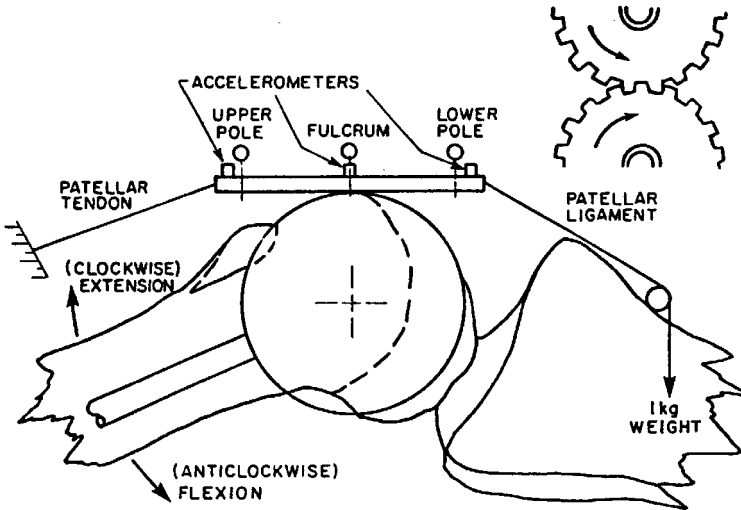


Figure 7.33 Apparatus to mimic the generation of PPC signals via a stick-slip frictional model. Reproduced with permission from D.E. Beverland, W.G. Kernohan, G.F. McCoy, and R.A.B. Mollan, What is physiological patellofemoral crepitus?, *Proceedings of XIV International Conference on Medical and Biological Engineering and VII International Conference on Medical Physics*, pp 1249–1250, Espoo, Finland, 1985. ©IFMBE

7.8 Application: Analysis of Heart rate Variability

Problem: Explore the applicability of Fourier spectral analysis methods to study HRV.

Solution: DeBoer et al. [137] applied Fourier analysis techniques to two types of measures derived from heart-rate data (see also Akselrod et al. [377]). They noted that the standard Fourier analysis methods cannot be applied directly to a series of point events. Therefore, they derived three types of signals from trains of ECG beats as illustrated in Figure 7.1.

The *interval spectrum* was derived by computing the Fourier spectrum of the interval series, normalized as $\tilde{I}_k = (I_k - \bar{I}) / \bar{I}$, where $\bar{I}$ is the mean interval length. The frequency axis was scaled by considering the time-domain data to be spaced at distances equal to the mean interval length $\bar{I}$, that is, the effective sampling frequency is $1 / \bar{I}$.

The *spectrum of counts* was derived by taking the Fourier transform of the impulse-train representation, derived from *RR* interval data as shown in Figure 7.1. The signal was normalized and scaled as $\tilde{s}(t) = \sum [\bar{I} \delta(t - t_k)] - N$, where N is the number of data samples, and the Fourier transform was computed. The spectra computed were smoothed with a 27-point rectangular window. DeBoer et al. demonstrated that the two spectra exhibit similar characteristics under certain conditions of slow or slight modulation of the data about the mean heart rate.

The *RR* interval data of a subject breathing freely and the two spectra derived from the data are shown in Figure 7.34. Three peaks are seen in both the spectra, which were explained as follows [137]:

- the effect of respiration at about 0.3 Hz ;
- the peak at 0.1 Hz related to 10 s waves seen in the BP; and
- a peak at a frequency lower than 0.1 Hz caused by the thermoregulatory system.

Figure 7.35 shows the *RR* interval data and spectra for a subject breathing at a fixed rate of 0.16 Hz . The spectra display well-defined peaks at both the average heart rate (1.06 Hz) and at the breathing rate, as well as their harmonics. The spectra clearly illustrate the effect of respiration on the heart rate, and may be used to analyze the coupling between the cardiovascular and respiratory systems (see Section 2.2.4).

Note that direct Fourier analysis of a stream of ECG signals will not provide the same information as above. The reduced representation (model) of the *RR* interval data, as illustrated in Figure 7.1, has permitted Fourier analysis of the heart rate and its relationship with respiration. The methods have application in studies on HRV [134–136, 138, 139, 145, 146].

As described in Section 2.4.3, Mendez et al. [166] proposed a method for the detection of OSA based on the ECG recorded during sleep. Features that gave good results in the detection of OSA included coherence between the *RR* interval data and the QRS area as well as measures of HRV.

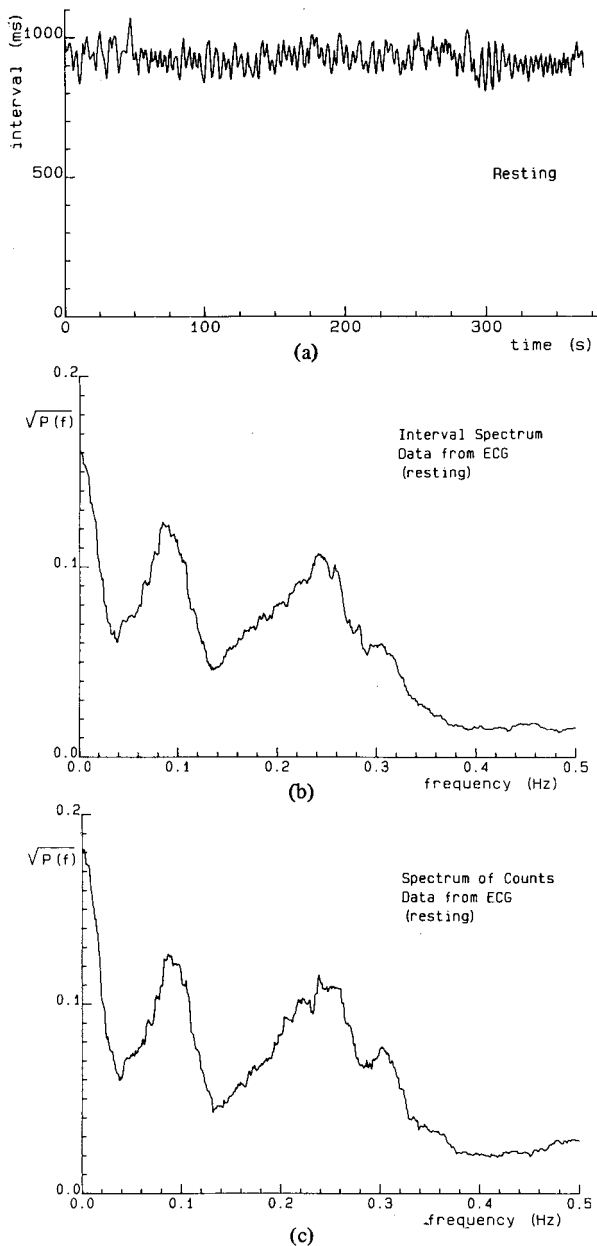


Figure 7.34 (a) 400 *RR* interval values from a healthy subject breathing freely. (b) Interval spectrum computed from a total of 940 intervals, including the 400 shown in (a) at the beginning. (c) Spectrum of counts. The spectra are shown for the range 0 – 0.5 *Hz* only. Reproduced with permission from R.W. DeBoer, J.M. Karemaker, and J. Strackee, Comparing spectra of a series of point events particularly for heart rate variability studies, *IEEE Transactions on Biomedical Engineering*, 31(4):384–387, 1984. ©IEEE.

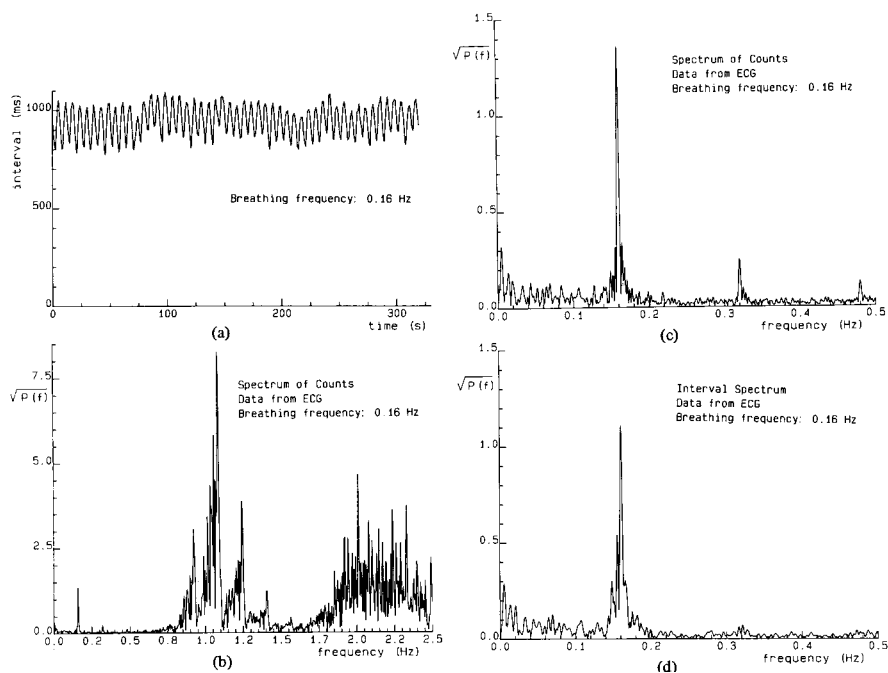


Figure 7.35 (a) 340 RR interval values from a healthy subject breathing at a fixed rate of 0.16 Hz . (b) Spectrum of counts for the range $0 - 2.5\text{ Hz}$. (c) Spectrum of counts for the range $0 - 0.5\text{ Hz}$. (d) Interval spectrum. Reproduced with permission from R.W. DeBoer, J.M. Karemaker, and J. Strackee, Comparing spectra of a series of point events particularly for heart rate variability studies, *IEEE Transactions on Biomedical Engineering*, 31(4):384–387, 1984. ©IEEE.

7.9 Application: Spectral Modeling and Analysis of PCG Signals

Iwata et al. [378, 379] applied AR modeling and parametric spectral analysis techniques to PCG signals for the detection of murmurs as well as the detection of the onset of S1 and S2. Their techniques included AR modeling, extraction of the dominant poles for pattern classification, and spectral tracking, which are explained in the following paragraphs.

Dominant poles: After the $a_k, k = 1, 2, \dots, P$, coefficients of an all-pole or AR model of order P have been computed, the polynomial $A(z)$ may be factorized and solved to obtain the locations of the poles $p_k, k = 1, 2, \dots, P$, of the system. The closer a pole is to the unit circle in the z -plane, the narrower its bandwidth and the stronger its contribution to the impulse response of the system. Poles that are close to the unit circle may be related to the resonance frequencies of the system, and used in system identification and pattern recognition.

In view of the nonstationary nature of the signal, Iwata et al. [378] computed a new model with order $P = 8$ for every window or frame of duration 25 *ms*, allowing an overlap of 12.5 *ms* between adjacent frames (with the sampling rate $f_s = 2 \text{ kHz}$). The frequency of a pole p_k was calculated as

$$f_k = \frac{\angle p_k}{2\pi} f_s, \quad (7.137)$$

and its bandwidth was calculated as

$$bw_k = \frac{\log |p_k|}{\pi} f_s. \quad (7.138)$$

Conditions based upon the difference in the spectral power estimate of the model from one frame to the next, and the existence of poles with $f_k < 300 \text{ Hz}$ with the minimal bandwidth for the model considered, were used to segment each PCG signal into four phases: S1, a systolic phase spanning the S1 – S2 interval, S2, and a diastolic phase spanning the interval from one S2 to the following S1. (See also Section 4.10.)

Figures 7.36 and 7.37 show the PCG signals, spectral contours, the spectral power estimate, and the dominant poles for a normal subject and a patient with murmur due to patent ductus arteriosus. Most of the dominant poles of the model for the normal subject are below 300 *Hz*. The model for the patient with murmur indicates many dominant poles above 300 *Hz*.

The mean and *SD* of the poles with $bw_k < 80 \text{ Hz}$ of the model of each PCG phase were computed and used for pattern classification. The five coefficients of a fourth-order polynomial fitted to the series of spectral power estimates of the models for each phase were also used as features. Twenty-six out of 29 design samples and 14 out of 19 test samples were correctly classified. However, the number of cases was low compared to the number of features used in most of the six categories present in the samples (normal, aortic insufficiency, mitral insufficiency, mitral stenosis, aortic stenosis combined with insufficiency, and atrial septal defect).

Spectral tracking: In another application of AR modeling for the analysis of PCG signals, Iwata et al. [379] proposed a spectral-tracking procedure based upon AR modeling to detect S1 and S2. PCG signals were recorded at the apex with a high-pass filter that, at 100 *Hz*, had a gain -40 dB below the peak gain at 300 *Hz* (labeled Ap-H). The signals were lowpass filtered with a gain of -20 dB at 1,000 *Hz* and sampled at 2 *kHz*. The AR model was computed with order $P = 8$ for frames of length 25 *ms*; the frame-advance interval was only 5 *ms*. The model PSD was computed as

$$\tilde{S}(\omega) = \frac{\sigma_r^2}{2\pi} \frac{1}{\sum_{k=0}^P \phi_a(k) \cos(\omega T)}, \quad (7.139)$$

where

$$\phi_a(k) = \sum_{j=0}^{P-k} a_j a_{j+k}, \quad (7.140)$$

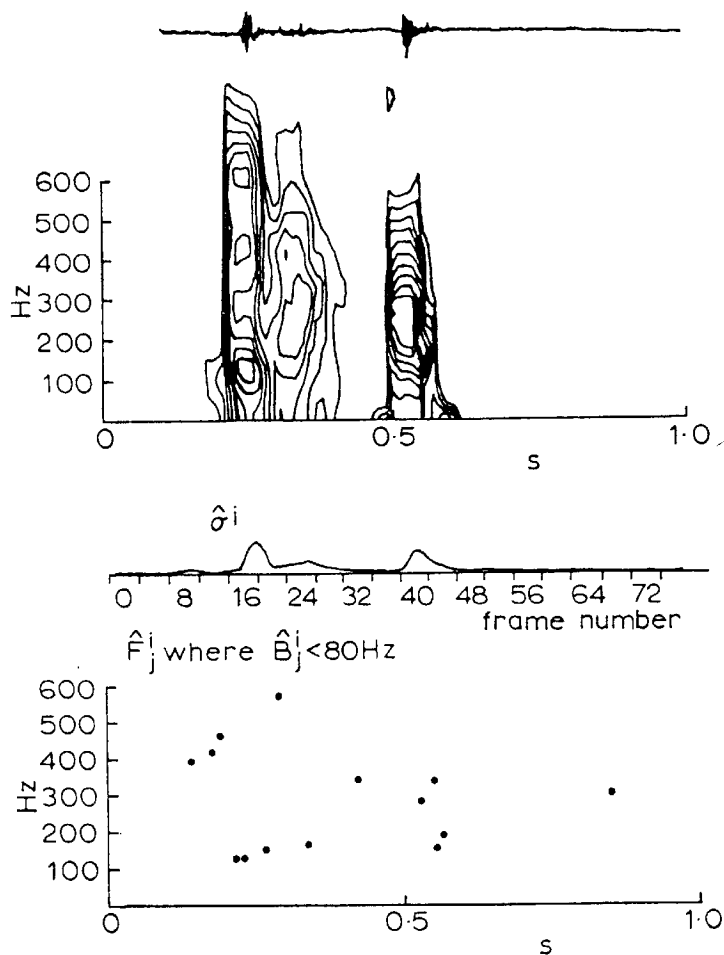


Figure 7.36 Illustration of feature extraction based upon all-pole modeling of a normal PCG signal. From top to bottom: PCG signal; model spectrum in the form of isointensity contours; model spectral power estimate $\hat{\sigma}^i$, where i refers to the frame number; the frequencies $\hat{F}_j^i$ of the dominant poles with bandwidth $\hat{B}_j^i < 80 \text{ Hz}$. Reproduced with permission from A. Iwata, N. Suzumara, and K. Ikegaya, Pattern classification of the phonocardiogram using linear prediction analysis, *Medical & Biological Engineering & Computing*, 15:407–412, 1977 ©IFMBE.

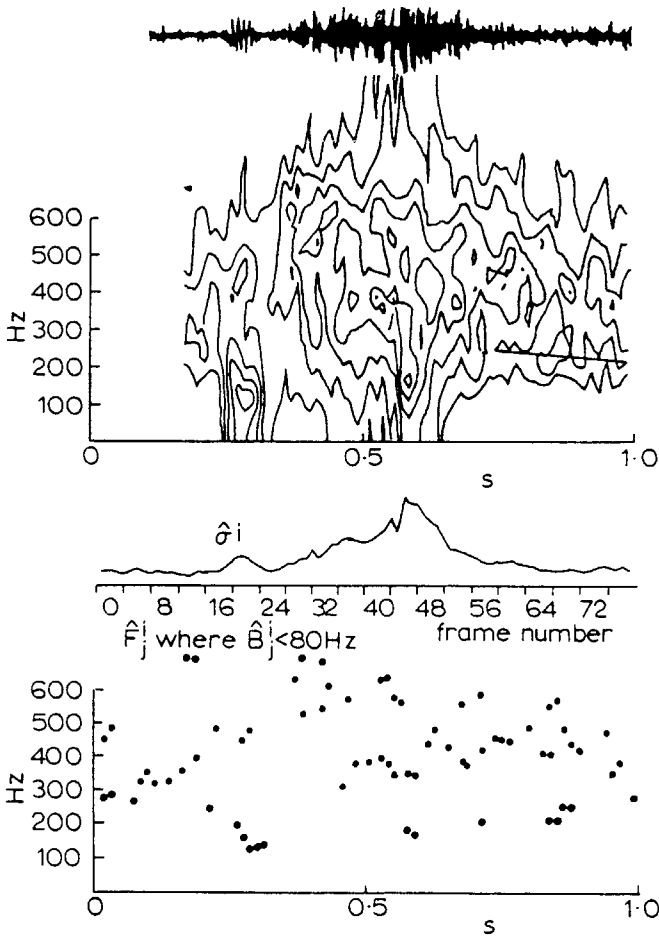


Figure 7.37 Illustration of feature extraction based upon all-pole modeling of the PCG signal of a patient with murmur due to patent ductus arteriosus. From top to bottom: PCG signal; model spectrum in the form of isointensity contours; model spectral power estimate $\hat{\sigma}^i$, where i refers to the frame number; the frequencies $\hat{F}_j^i$ of the dominant poles with bandwidth $\hat{B}_j^i < 80$ Hz. Reproduced with permission from A. Iwata, N. Suzumara, and K. Ikegaya, Pattern classification of the phonocardiogram using linear prediction analysis, *Medical & Biological Engineering & Computing*, 15:407–412, 1977 ©IFMBE.

with a_k being the AR-model coefficients, $P = 8$, $T = 0.5 \text{ ms}$, σ_r^2 being the model residual energy (error), and $a_0 = 1$.

Based upon a study of the spectra of 69 normal and abnormal PCG signals, Iwata et al. [379] found the mean peak frequency of S1 to be 127 Hz and that of S2 to be 170 Hz ; it should be noted that the PCG signals were highpass filtered (as described in the preceding paragraph) at the time of data acquisition. The model spectral power at 100 Hz was used as the tracking function to detect S1: The peak in the tracking function after the location t_R of the R wave in the ECG was taken to be the position of S1. The tracking function to detect S2 was based upon the spectral power at 150 Hz ; the peak in the interval $t_R + 0.25RR \leq t \leq t_R + 0.6RR$, where RR is the interbeat interval, was treated as the position of S2. The use of a normalized spectral density function based upon the AR-model coefficients but without the σ_r^2 factor in Equation 7.139 was recommended, in order to overcome problems due to the occurrence of murmurs close to S2.

Figure 7.38 illustrates the performance of the tracking procedure with a normal PCG signal. The peaks in the 100 Hz and 150 Hz spectral-tracking functions (lowest traces) coincide well with the timing instants of S1 and S2, respectively. Figure 7.39 illustrates the application of the tracking procedure to the PCG signal of a patient with mitral insufficiency. The systolic murmur completely fills the interval between S1 and S2, and no separation is seen between the sounds and the murmur. Whereas the 150 Hz spectral-tracking function labeled (b) in the figure does not demonstrate a clear peak related to S2, the normalized spectral-tracking function labeled (c) shows a clear peak corresponding to S2. The two additional PCG traces shown at the bottom of the figure (labeled Ap-L for the apex channel including more low-frequency components with a gain of -20 dB at 40 Hz , and 3L-H for a channel recorded at the third left intercostal space with the same bandwidth as the Ap-H signal) illustrate S2 more distinctly than the Ap-H signal, confirming the peak location of the spectral-tracking function labeled (c) in the figure.

7.10 Application: Detection of Coronary Artery Disease

The diastolic segment of a normal PCG signal after S2 is typically silent; in particular, the central portion of the diastolic segment after the possible occurrence of AV valve-opening snaps is silent. Akay et al. [130] conjectured that blood flow in the coronary arteries is at its maximum during middiastole, and further that the effects of coronary artery disease, such as occlusion or stenosis, could present high-frequency sounds in this period due to turbulent blood flow (see Section 7.7.2).

Akay et al. [130] studied the spectra of middiastolic segments of the PCGs, averaged over 20–30 beats, of normal subjects and patients with coronary artery disease confirmed by angiography. It was found that the PCG signals in the case of coronary artery disease exhibited greater portions of their energy above 300 Hz than the normal signals.

Figure 7.40 illustrates the AR-model spectra of two normal subjects and two patients with coronary artery disease. The signals related to coronary artery disease are

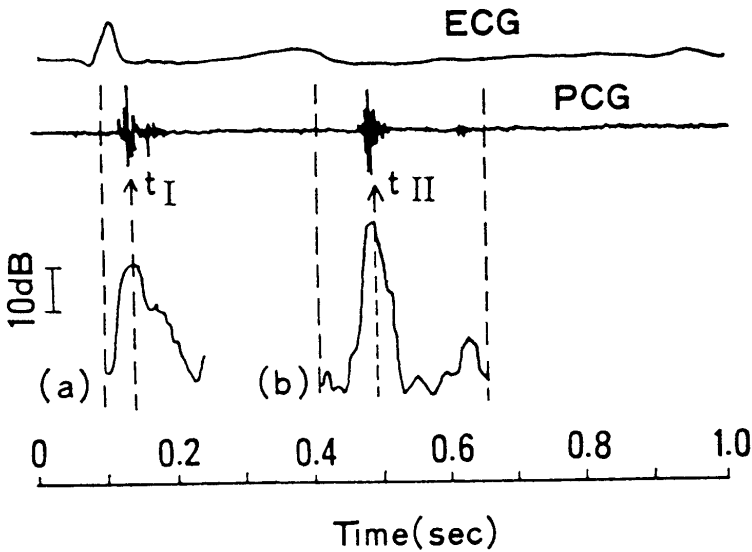


Figure 7.38 Illustration of the detection of S1 and S2 via spectral tracking based upon all-pole modeling of a normal PCG signal. From top to bottom: ECG signal; PCG signal; spectral-tracking functions at (a) 100 Hz for S1 and (b) 150 Hz for S2. The S1 and S2 locations detected are marked as t_I and t_{II} , respectively. Reproduced with permission from A. Iwata, N. Ishii, N. Suzumara, and K. Ikegaya, Algorithm for detecting the first and the second heart sounds by spectral tracking, *Medical & Biological Engineering & Computing*, 18:19–26, 1980 ©IFMBE.

seen to possess a high-frequency peak in the range 400 – 600 Hz that is not evident in the normal cases.

Akay et al. [380] further found that the high relative-power levels of resonance frequencies in the range of 400 – 600 Hz that were evident in patients with coronary artery disease were reduced after angioplasty. Figure 7.41 shows the spectra of the diastolic heart sounds of a patient before and after coronary artery occlusion was corrected by angioplasty. It may be readily observed that the high-frequency components that were present before surgery (“preang.”) are not present after the treatment (“postang.”). (The minimum-norm method of PSD estimation used by Akay et al. [380] — labeled as “MINORM” in the figure — is not discussed in this book.)

7.11 Remarks

In this chapter, we have studied how mathematical models may be derived to represent physiological processes that generate biomedical signals and, furthermore, how the models may be related to changes in signal characteristics due to functional and

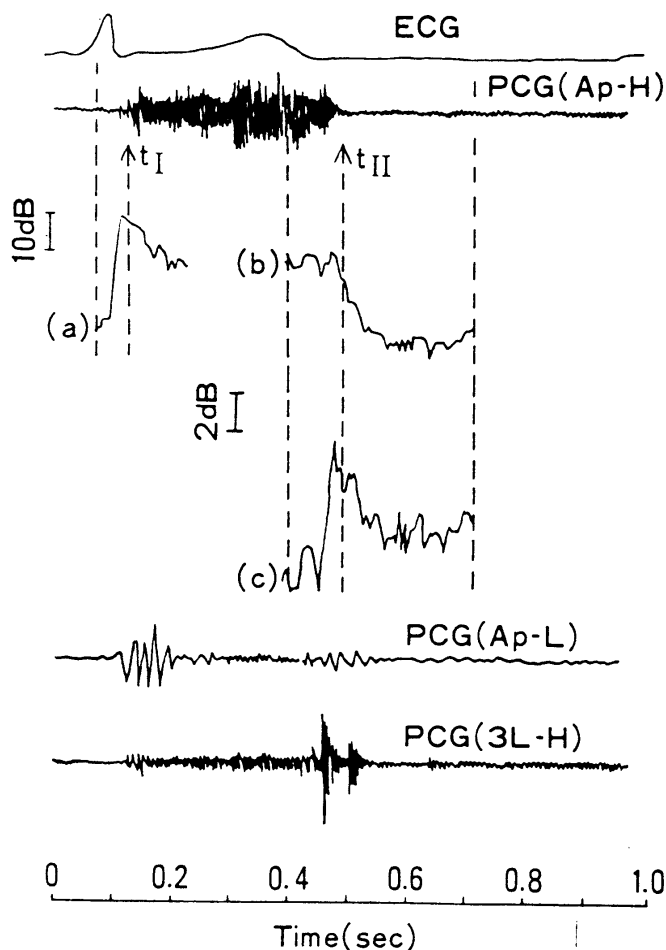


Figure 7.39 Illustration of the detection of S1 and S2 via spectral tracking based upon all-pole modeling of a PCG signal with systolic murmur due to mitral insufficiency. From top to bottom: ECG signal; PCG (Ap-H) signal; spectral-tracking functions at (a) 100 Hz for S1, (b) 150 Hz for S2, and (c) normalized spectral-tracking function at 150 Hz for S2; PCG (Ap-L) signal from the apex with more low-frequency components included; PCG (3L-H) signal from the third left intercostal space with the same filters as for Ap-H. The S1 and S2 locations detected are marked as t_I and t_{II} , respectively. Reproduced with permission from A. Iwata, N. Ishii, N. Suzumara, and K. Ikegaya, Algorithm for detecting the first and the second heart sounds by spectral tracking, *Medical & Biological Engineering & Computing*, 18:19–26, 1980 ©IFMBE.

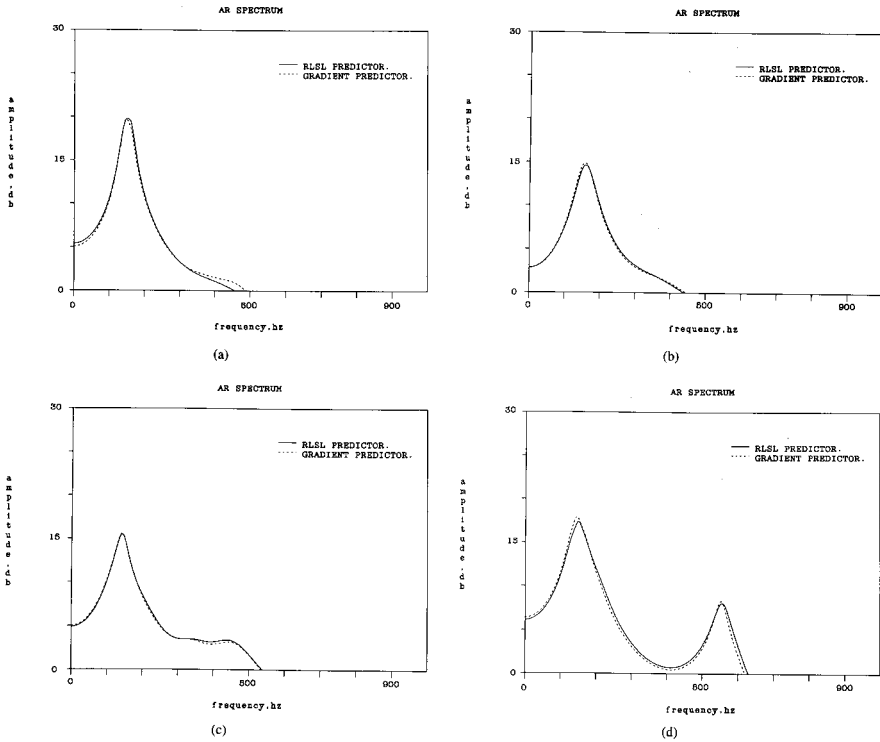


Figure 7.40 Diastolic heart sound spectra of (a, b) two normal subjects and (c, d) two patients with coronary artery disease. The method of estimating AR models identified in the figure as “RLSL” is described in Section 8.6.2; the gradient predictor method is not discussed in this book. Reproduced with permission from A.M. Akay, J.L. Semmlow, W. Welkowitz, M.D. Bauer, and J.B. Kostis, Detection of coronary occlusions using autoregressive modeling of diastolic heart sounds, *IEEE Transactions on Biomedical Engineering*, 37(4):366–373, 1990. ©IEEE.

pathological processes. The important point to note in the modeling approach is that the models provide a small number of *parameters* that characterize the signal and/or system of interest; the modeling approach is, therefore, useful in *parametric analysis* of signals and systems. As the number of parameters derived is usually much smaller than the number of signal samples, the modeling approach could also assist in data compression and compact representation of signals and related information.

Pole–zero models could be used to view physiological systems as control systems. Pathological states may be derived or simulated by modifying the parameters of the related models. Models of signals and systems are also useful in the design and control of prostheses.

A combination of mathematical modeling with electromechanical modeling can allow the inclusion of physical parameters, such as the diameter of a blood vessel,

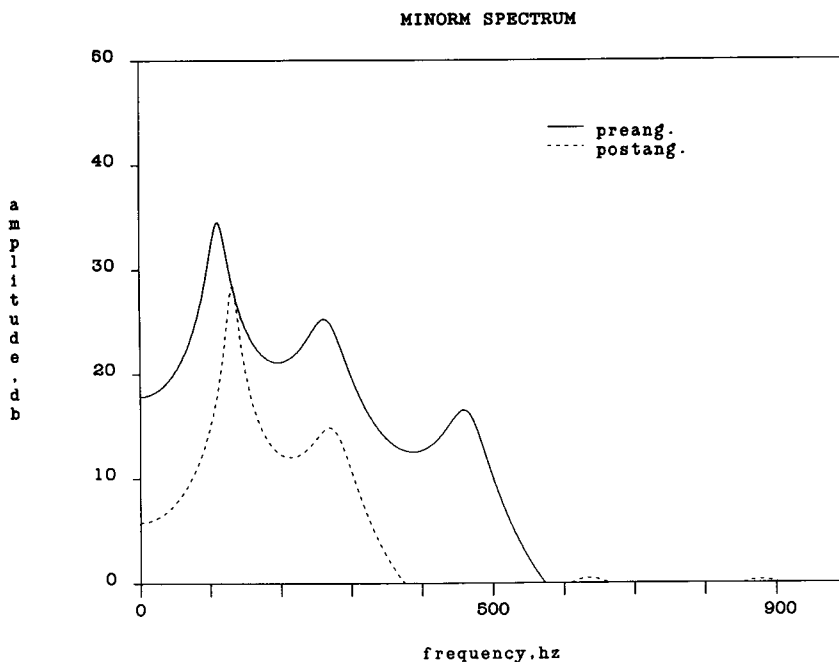


Figure 7.41 Diastolic heart sound spectra before (preang.) and after angioplasty (postang.) of a patient for whom coronary artery occlusion was corrected. (The minimum-norm method of PSD estimation used by Akay et al. [380] — labeled as “MINORM” in the figure — is not discussed in this book.) Reproduced with permission from A.M. Akay, J.L. Semmlow, W. Welkowitz, M.D. Bauer, and J.B. Kostis, Noninvasive detection of coronary stenoses before and after angioplasty using eigenvector methods, *IEEE Transactions on Biomedical Engineering*, 37(11):1095–1104, 1990. ©IEEE.

constriction due to plaque, stiffness due to stenosis, and friction coefficient. Although accurate estimation of such parameters for human subjects may not always be possible, the models could lead to better understanding of the related biomedical signals and systems.

7.12 Study Questions and Problems

1. Consider the LP model given by $\tilde{y}(n) = ay(n-1)$. Define the prediction error, and derive the optimal value for a by minimizing the TSE.
2. The AR-model coefficients of a signal are $a_0 = 1, a_1 = 1, a_2 = 0.5$. What is the transfer function of the model? Draw the pole-zero diagram of the model. What are the resonance frequencies of the system?
3. The AR-model coefficient vectors of a number of signals are made available to you. Propose two measures to compare the signals for (a) similarity, and (b) dissimilarity.

4. In AR modeling of signals, show why setting the derivative of the TSE with respect to any coefficient to zero will always lead to the minimum error (and not the maximum).
5. What type of a filter can convert the autocorrelation matrix of a signal to a diagonal matrix?
6. A biomedical signal is sampled at 500 Hz and subjected to AR modeling. The poles of the model are determined to be at $0.4 \pm j0.5$ and $-0.7 \pm j0.6$. (a) Derive the transfer function of the model. (b) Derive the difference equation in the time domain. (c) What are the resonance frequencies of the system that is producing the signal?
7. A model is described by the relationship $y(n) = x(n) + 0.5x(n-1) + 0.25x(n-2)$, where $x(n)$ is the input and $y(n)$ is the output. What is the type of this system among AR, MA, and ARMA systems? What is the model order? What is its transfer function? Draw the pole-zero diagram of the system. Comment upon the stability of the system.
8. A model is described by the relationship $y(n) = -0.5y(n-1) - y(n-2) + x(n) + 0.5x(n-1) - x(n-2)$, where $x(n)$ is the input and $y(n)$ is the output. What is the type of this system among AR, MA, and ARMA systems? What is the model order? What is its transfer function? Draw the pole-zero diagram of the system. Comment upon the stability of the system.

7.13 Laboratory Exercises and Projects

Note: Data files related to the exercises are available at the site

<http://people.ucalgary.ca/~ranga/enel563>

1. The file `safety.wav` contains the speech signal for the word “safety” uttered by a male speaker, sampled at 8 kHz (see the file `safety.m`). The signal has a significant amount of background noise (as it was recorded in a computer laboratory). Develop procedures to segment the signal into voiced, unvoiced, and silence (background noise) portions using short-time *RMS*, turns count, or *ZCR* measures.

Apply the AR modeling procedure to each segment; the command `lpc` in MATLAB[®] may be used for this purpose. Compute the model PSD for each segment. Compare the model PSD with the Fourier PSD for each segment. What are the advantages and disadvantages of the model-based PSD in the case of voiced and unvoiced sounds?

2. Derive the poles of the models you obtained in the preceding problem. Express each pole in terms of not only its z -plane coordinates but also its frequency and bandwidth. Study the variations in the pole positions as the type of the sound varies from one segment to the next over the duration of the signal.
3. The files `pec1.dat`, `pec33.dat`, and `pec52.dat` give three-channel recordings of the PCG, ECG, and carotid pulse signals (sampled at 1,000 Hz ; you may read the signals using the program in the file `plotpec.m`). The signals in `pec1.dat` and `pec52.dat` are normal; the PCG signal in `pec33.dat` has systolic murmur, and is of a patient suspected to have pulmonary stenosis, ventricular septal defect, and pulmonary hypertension.

Segment each signal into its systolic and diastolic parts. Apply the AR modeling procedure to each segment and derive the model PSD. Compare the result with the corresponding PSDs obtained using Welch’s procedure.

4. Derive the poles of the models you obtained in the preceding problem. Express each pole in terms of not only its z -plane coordinates but also its frequency and bandwidth. Study the variations in the pole positions from the systolic part to the diastolic part of each signal. What are the major differences between the pole plots for the normal cases and the case with murmur?
5. The files ECG3, ECG4, ECG5, and ECG6 contain ECG signals sampled at the rate of 200 Hz (see the file ECGS.m). Apply the Pan–Tompkins method for QRS detection to each signal. Create impulse sequences including a delta function at every QRS location for the four signals. Create also the interval series for each signal as illustrated in Figure 7.1. Compute the spectra corresponding to the two representations of cardiac rhythm and study their relationship to the heart rate and its variability in each case.

CHAPTER 8

ANALYSIS OF NONSTATIONARY AND MULTICOMPONENT SIGNALS

A stationary (or homogeneous) signal is one that possesses the same statistical measures for all time, or at least over the duration of observation. We have seen in the preceding chapters that most biomedical signals, being manifestations of dynamic systems and pathophysiological processes, are *nonstationary* (or heterogeneous): Figure 3.3 shows that the variance of the speech signal used as an example varies with time; Figure 3.4 shows that the spectrum or frequency content of the speech signal also varies considerably over its duration. Figures 6.12 and 6.13 show that the spectrum of a heart sound signal or PCG varies from systole to diastole and could vary in between the two events as well.

When the characteristics of a signal being studied vary considerably over the duration of interest, measures and transforms computed over the entire duration do not carry useful information: They gloss over the dynamics of the signal. A single PSD computed from a long EMG, PCG, VAG, or speech record is of no practical value. The PSD does not provide information on time localization of the frequency components of the signal. We addressed this concern in PCG signal analysis in Section 6.3.5 by segmenting the PCG into its systolic and diastolic parts by using the ECG and carotid pulse signals as timing references. But how would we be able to handle the situation when murmurs are present in systole and diastole, and when we

need to analyze the spectra of the murmurs without the contributions of S1 and S2? How could one perform segmentation of a PCG cycle into separate portions with S1, S2, and murmurs (if present)?

Furthermore, the EEG signal changes its nature in terms of rhythms, waves, transients, and spindles for which no independent references are available (see Section 1.2.6). In fact, the EEG represents a conglomeration of a number of mental and physiological processes going on in the brain at any given instant of time. The VAG signal has nonstationary characteristics related to the cartilage surfaces that come into contact depending upon the activity performed, and no other source of information can assist in identifying time instants when the signal properties change (see Section 1.2.14). Indeed, a VAG signal contains no specific events that may be identified as such, but is a concatenation of nonspecific vibrations (with, perhaps, the exception of clicks). Would we able to extend the application of the well-established signal analysis techniques that we have studied so far to such nonstationary signals?

In addition to being nonstationary, several biomedical signals possess multiple components from the same source as well as other sources that may be active at the same time. Cohen [381] defines a multicomponent signal as one that has delineated concentrations of power or energy in the time-frequency (TF) plane, with each such portion possibly related to a component of the signal; see Boashash [382] for related discussions. The speech signal is considered to be an example of a multicomponent signal: The formant structure of voiced speech indicates the presence of multiple resonances that relate to different components (see Section 1.2.12). Extending the same interpretation, we could consider EEG, PCG, and VAG signals to be multicomponent signals. In addition to the presence of multiple delineated components in the TF plane, biomedical signals could also possess multiple components arising from different sources that are active at the same time. A VAG signal could possibly contain simultaneous contributions from multiple sources of sound or vibration in the same knee joint due to different types of cartilage pathology (see Section 8.2.3). The ECG of an expectant mother recorded from the surface of her abdomen contains a mixture of the maternal as well as fetal ECGs (see Section 3.3.5). Such mixtures are also multicomponent signals with multiple sources. Given such composite signals that are also nonstationary, how can we separate them into their components for further detailed analysis?

8.1 Problem Statement

Develop methods to study the dynamic characteristics of nonstationary and multicomponent biomedical signals. Propose schemes to apply the well-established Fourier transform and AR-modeling techniques to analyze and parameterize nonstationary and multicomponent signals.

The case studies presented in the following section provide the motivation for the present study from the perspective of a few representative biomedical signals. Approaches to solve the stated problem are presented in the sections to follow. This chapter presents several techniques for segmentation-based analysis of nonstationary

signals. In addition, the chapter includes brief introductions to wavelets and joint TF analysis of nonstationary and multicomponent signals without segmentation.

8.2 Illustration of the Problem with Case Studies

8.2.1 Heart sounds and murmurs

We noted in Section 6.3.5 that the spectral contents of S1 and S2 are different due to the different states of contraction or relaxation of the ventricular muscles and the differences in their blood content during the corresponding cardiac phases. In the normal case, the QRS in the ECG signal and the dicrotic notch in the carotid pulse signal may be used to split the PCG into S1 and S2, and separate PSDs may be obtained for the signal parts as illustrated in Section 6.3.5. However, when a PCG signal contains murmurs in systole and/or diastole and possibly valve opening snaps (see Figure 6.13), it may be desirable to split the signal further.

Iwata et al. [378] applied AR modeling to PCG signals by breaking the signal into fixed segments of 25 *ms* duration (see Section 7.9). While this approach may be satisfactory, it raises questions on optimality. What should be the window duration? Is it necessary to break the intervals between S1 and S2 into multiple segments? Would it not be more efficient to compute a single AR model for the entire durations of each of S1, S2, systolic murmur, and diastolic murmur — that is, a total of only four models? It is conceivable that each model would be more accurate as all available signal samples would be used to estimate the required ACF if the signal were to be segmented adaptively as mentioned above.

8.2.2 EEG rhythms and waves

The scalp EEG represents a combination of the multifarious activities of many small zones of the cortical surface beneath each electrode. The signal changes its characteristics in relation to mental tasks, external stimuli, and physiological processes. As we have noted in Section 1.2.6 and observed in Figure 1.38, a visual stimulus blocks the alpha rhythm; slower waves become prominent as the subject goes to deeper stages of sleep (see Section 6.7); and patients with epilepsy may exhibit sharp spikes and trains of spike-and-wave complexes (see Sections 1.2.6 and 4.4.3). Description of an EEG record, as outlined in Sections 1.2.6, 4.2.4, and 4.4.3, requires the identification of several types of waves and rhythms. This suggests that the signal may first have to be broken into segments, each possessing certain properties that remain the same for the duration of the segment. Each segment may then be described in terms of its characteristic features.

8.2.3 Articular cartilage damage and knee joint vibrations

Movement of the knee joint consists of coupled translation and rotation. The configuration of the patella is such that some portion of the articular surface is in contact with

the femur throughout knee flexion and to almost full extension (see Section 1.2.14 and Figure 1.52). Goodfellow et al. [383] demonstrated that initial patellofemoral engagement occurs at approximately 20° of flexion involving both the medial and lateral facets. Figure 8.1 shows the patellar contact areas at different joint angles. As the knee is flexed, the patellofemoral contact area moves progressively upward, involving both the medial and lateral facets. At 90° of flexion, the band of contact engages the upper pole of the patella. The odd facet does not articulate with the lateral margin of the medial femoral condyle until about $120^\circ - 135^\circ$ of knee flexion.

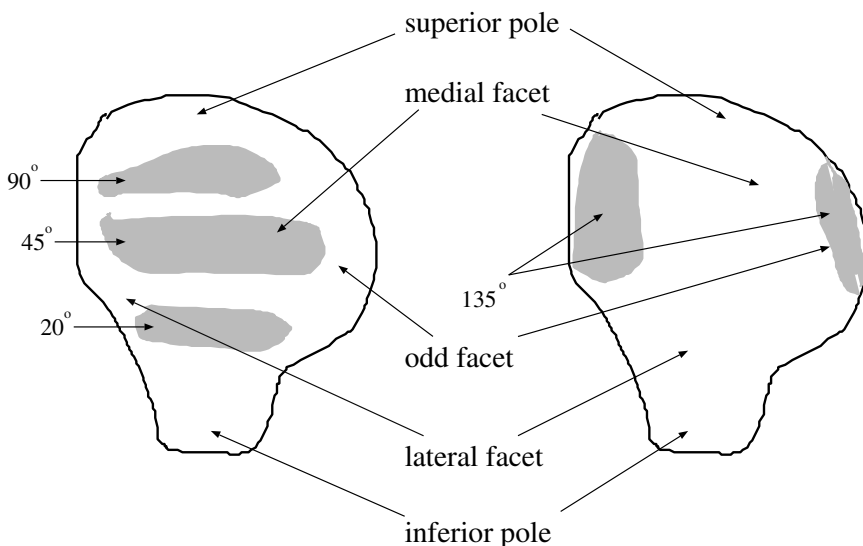


Figure 8.1 Contact areas of the patella with the femur during patellofemoral articulation. Adapted with permission from S. Krishnan, Adaptive Signal Processing Techniques for Analysis of Knee Joint Vibroarthrographic Signals, Ph.D. Thesis, University of Calgary, 1999.

Articular cartilage is composed of a solid matrix and synovial fluid [384]; it has no nerves, blood vessels, or lymphatics and is nourished by the synovial fluid covering its free surface. During articulation, friction between the bones is reduced as a result of the lubrication provided by the viscous synovial fluid [87,90]. The material properties of articular cartilage and cartilage thickness are variable not only from joint to joint but also within the same joint. In cases of abnormal cartilage, alterations in the matrix structure, such as increased hydration, disruption of the collagen fibrillar network, and disaggregation or loss of proteoglycans occur. As the compositional and biomechanical properties of abnormal articular cartilage continue to deteriorate, substance loss eventually occurs. This may be focal or diffuse, restricted to superficial fraying and fibrillation, or partial-thickness loss to full-thickness loss. In some cases, focal swelling or blistering of the cartilage may be seen before there is fraying of the articular surface [385].

Chondromalacia patella (soft cartilage of the patella) is a condition in which there is degeneration of patellar cartilage, often associated with anterior knee pain. Exposed subchondral bone and surface fibrillation of the articular cartilage are evident on the posterior patellar surface in chondromalacia patella [386]. Chondromalacia patella is usually graded in terms of the severity of the lesions [387,388] as follows:

- *Grade I:* Softening, cracking, and blistering, but no loss of articular cartilage.
- *Grade II:* Damage is moderate and there is some loss of cartilage.
- *Grade III:* Severe damage of fibrocartilage has occurred but bone is not exposed.
- *Grade IV:* The cartilage is eroded and the subchondral bone is exposed.

Osteoarthritis is a degenerative joint disease that involves specific changes to bone in addition to cartilage. In the late stages of osteoarthritis, there is full-thickness articular cartilage degeneration and exposed bone. Other structural changes include fibrous changes to the synovium, joint capsule thickening, and further alterations to the bone such as osteophyte formation [389]. Higher-grade chondromalacia may be categorized as osteoarthritis.

The menisci are subject to vertical compression, horizontal distraction, and rotary and shearing forces of varying degrees in the course of normal activities [390]. Advance of the aging process in both articular cartilage and fibrocartilage causes progressive liability to horizontal cleavage lesion [390].

The semiinvasive procedure of *arthroscopy* (fiber-optic inspection of joint surfaces, usually under general anesthesia) is often used for diagnosis of cartilage pathology. Through an arthroscope, the surgeon can usually see the patellofemoral joint, the femoral condyles, the tibial plateau (menisci), the anterior cruciate ligament, and the medial and lateral synovial spaces. Arthroscopy has emerged as the “gold standard” for relatively low-risk assessment of joint surfaces in order to determine the prognosis and treatment for a variety of conditions. Figure 8.2 shows the different stages of chondromalacia patella as viewed during arthroscopy.

Abnormal structures and surfaces in the knee joint are more likely to generate sound during extension and flexion movements than normal structures. Softened articular cartilage in chondromalacia patella — as well as cracks, fissures, or thickened areas in osteoarthritis — almost certainly increase the friction between the articular surfaces and are, therefore, likely to increase the sounds emitted during normal joint movement [92,391]. Injury to the menisci in the form of tearing causes irregularity in shape and disruption to normal joint movement, and may produce sharp clicking sounds during normal knee movement [92,97,392].

It is obvious from this discussion (and the related introduction in Section 1.2.14) that the VAG signal is a nonstationary and multicomponent signal. Different aspects of the articulating surfaces come into contact at different joint angles; their quality in terms of lubrication and functional integrity could vary from one position to another. Inspection of the VAG signals and their spectrograms illustrated in Sections 3.9.3 and 3.14 reveals that the nature of a VAG signal changes significantly over the duration of the signal. As no prior or independent information is available about changes

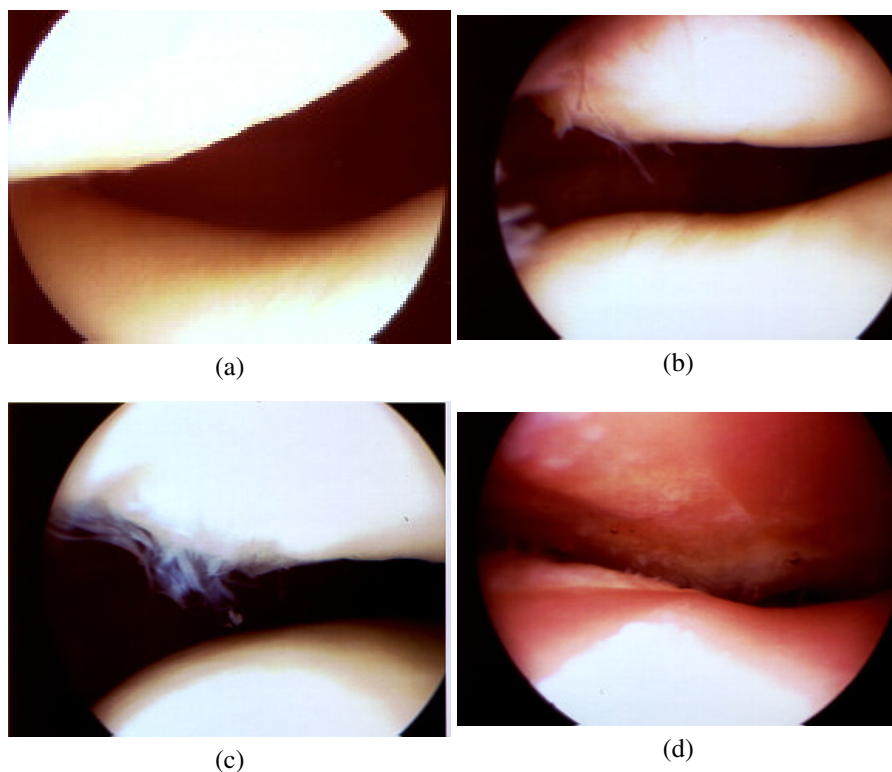


Figure 8.2 Arthroscopic views of the patellofemoral joint. (a) Normal cartilage surfaces. (b) Chondromalacia Grade II at the patella. (c) Chondromalacia Grade III at the patella. (d) Chondromalacia Grade IV at the patella and the femur; the bones are exposed. The under-surface of patella is at the top and the femoral condyle is at the bottom. Figure courtesy: G.D. Bell, Sport Medicine Centre, University of Calgary.

in the knee-joint structures that could lead to vibrations, adaptive segmentation of the VAG signal is desirable before it may be analyzed using the methods we have studied to this point in the present book. Illustrations of adaptive segmentation of VAG signals are provided in Sections 8.6.1 and 8.6.2. In addition, methods for analysis of VAG signals without segmentation are described in Section 8.14.

8.3 Time variant Systems

The linear system model represented by Equation 7.12 is a time-invariant system: The coefficients a_k and b_l of the system do not change with time, and consequently, the poles and zeros of the system stay fixed for all time. A nonstationary (or dynamic) system will possess coefficients that do vary with time: We saw in Sections 3.9.2

and 3.9.3 that the coefficient or tap-weight vectors of the adaptive LMS and RLS filters are expressed as functions of time. (*Note:* The Wiener filter described in Section 3.8, once optimized for a given set of signal and noise statistics, is a time-invariant filter.) Since the coefficients of an LMS or RLS filter vary with time, so do the transfer function and the frequency response of the filter. It follows that the impulse response of such a system also varies with time.

Let us consider an all-pole filter for the sake of simplicity; the filter characteristics are determined by the positions of the poles (except for a gain factor). If the poles are expressed in terms of their polar coordinates, their angles correspond to (resonance) frequencies and their radii are related to the associated bandwidths. We may, therefore, characterize time-variant or nonstationary systems and signals by describing their pole positions in the complex z -plane — or, equivalently, the related frequencies and bandwidths — as functions of time. A description of the variation or the modulation of the pole parameters over time can thus capture the nonstationary or dynamic nature of a time-variant system or signal. Variations in the gain factor also lead to nonstationarities in the signal produced by the system. Appel and v. Brandt [393, 394] describe the simulation of different types of nonstationary behavior of signals and systems.

In the general case of a nonstationary system that is an AR process, we may modify Equation 7.17 to indicate that the model coefficients are functions of time:

$$\tilde{y}(n) = - \sum_{k=1}^P a_k(n)y(n-k). \quad (8.1)$$

Methods related to the Kalman filter or the least-squares approach may be used to analyze such a system [18, 148, 395–397] (not considered in this book). Time-varying AR and ARMA modeling techniques have been applied to analyze the EEG [398], EGG [69], PCG [399], and HRV [400] signals; the application of time-varying analysis techniques to HRV signals is discussed in Section 8.11.

8.3.1 Characterization of nonstationary signals and dynamic systems

The output of a time-variant or dynamic system will be a nonstationary signal. The system may be characterized in terms of its time-variant model coefficients, transfer function, or related parameters derived thereof. Various short-time statistical measures computed over moving windows may be used to characterize a nonstationary signal; the measures may also be used to test for stationarity, or lack thereof, of a signal.

- **Mean:** The short-time mean represents the average or DC level of the signal in the analysis window. Variation of the mean from one window to another is usually an indication of the presence of a wandering baseline or low-frequency artifact, as in the case of the ECG signal in Figure 3.6. Clearly, the signal in Figure 3.6 is nonstationary in the mean. However, the mean is not an important measure in most signals, and it is typically blocked at the data-acquisition stage

via capacitive coupling and/or a highpass filter. Furthermore, since a DC level carries no sound or vibration information, its removal is of no consequence in the analysis of signals such as heart sounds, speech, VAG, and the VMG.

- **Variance:** Figure 3.3 illustrates the short-time variance for a speech signal. It is evident that the variance is high in regions of high signal variability (swings or excursions) about the mean, as in the case of the vowels or voiced-speech segments in the signal. The variance is low in the regions related to the fricatives or unvoiced-speech segments in the signal where the amplitude swing is small, in spite of their high-frequency nature. Since the mean of the signal is zero, the variance is equal to the MS value and represents the average power level in the corresponding signal windows. Although variations in the power level of speech signals may be useful in making voiced/ unvoiced/ silence decision, the parameter does not bear much information and provides a limited representation of the general statistical variability of signal characteristics. Regardless of the interpretation of the parameter, it is seen that the speech signal in Figure 3.3 is nonstationary in its variance (and the related measures of SD , MS , and RMS). From the discussion in Section 1.2.12, it is also clear that the vocal-tract system producing the speech signal is a dynamic system with time-varying configuration and filtering characteristics.
- **Measures of activity:** We have studied several measures of activity that indicate the “busy-ness” of the given signal, such as turning points, ZCR , and turns count, in Chapters 3 and 5. (The term “activity” has several connotations in the literature; see Chapter 5.) The short-time count of turning points is plotted in Figure 3.1 for a speech signal: It is evident that the signal is more active or busy in the periods related to the fricatives than those related to the vowels (a trend that is the opposite of that in the short-time variance of the same signal shown in Figure 3.3). The short-time turns count plot of the EMG signal in Figure 5.9 indicates the rising level of complexity of the signal with the level of breathing (inspiration). Although turning points, ZCR , and turns count are not among the traditional statistical measures derived from PDFs, they characterize signal variability and complexity in different ways. Both the examples cited above illustrate variation of the parameters measured over the duration of the corresponding signals: The signals are, therefore, nonstationary in terms of the number of turning points or the turns count.
- **ACF:** The ACF was defined in Section 3.2.1 in general as $\phi_{xx}(t_1, t_1 + \tau) = E[x(t_1)x(t_1 + \tau)]$. In Section 3.2.4, one of the conditions for (wide-sense or second-order) stationarity was defined as the ACF being independent of time, that is, $\phi_{xx}(t_1, t_1 + \tau) = \phi_{xx}(\tau)$. A nonstationary signal will not satisfy this condition and will have an ACF that varies with time. Since the ACF is based on the expectation of pairs of signal samples separated by a certain time difference or shift, it is a more general measure of signal variability than the variance and related measures. Note that the ACF for zero lag is the MS value of the signal.

One faces limitations in computing the ACF of short-time segments of a signal to investigate (non)stationarity: The shorter the analysis window, the shorter the maximum lag up to which the ACF may be estimated reliably. Regardless, the short-time ACF may be used to track nonstationarities in a signal. If the signal is the result of a dynamic AR system, the system parameters may be derived from the ACF (see Section 7.5).

- **PSD:** The PSD and ACF of a signal are interrelated by the Fourier transform. Therefore, a signal that is (non)stationary in its ACF is also (non)stationary in its PSD. However, the PSD is easier to interpret than the ACF, as we have seen in Chapter 6. The spectrogram of the speech signal in Figure 3.4 indicates significant variations in the short-time PSD of the signal: The speech signal is clearly nonstationary in its PSD (and ACF). The spectrograms of VAG signals in Sections 3.9.3 and 3.14 illustrate the nonstationary nature of VAG signals.
- **Higher-order statistics:** A major limitation of signal analysis using the ACF (or equivalently the PSD) is that the phase information is lost. The importance of phase in signals is discussed by Oppenheim and Lim [401]. Various conditions under which a signal may be reconstructed from its magnitude spectrum only or from its phase spectrum only are described by Hayes et al. [402] and Oppenheim and Lim [401]. Analysis based only upon the ACF cannot be applied to signals that are of mixed phase (that is, not minimum phase), that are the result of nonlinear systems, or that follow a PDF other than a Gaussian [403].

The general n^{th} -order moment of a random signal $x(t)$ at the instant t_1 is defined as [148, 403, 404]

$$m_x^n(t_1, t_1 + \tau_1, t_1 + \tau_2, \dots, t_1 + \tau_{n-1}) = E[x(t_1) x(t_1 + \tau_1) x(t_1 + \tau_2) \cdots x(t_1 + \tau_{n-1})], \quad (8.2)$$

where $\tau_1, \tau_2, \dots, \tau_{n-1}$ are various shifts. It is evident that the ACF is a special case of the above with $n = 2$, that is, the ACF is the second-order moment.

A set of parameters known as cumulants may be related to the moments as follows: The second-order and third-order cumulants are equal to the corresponding moments. The fourth-order cumulant is related to the fourth-order moment as [148, 403, 404]

$$\begin{aligned} c_x^4(t_1, t_1 + \tau_1, t_1 + \tau_2, t_1 + \tau_3) &= m_x^4(t_1, t_1 + \tau_1, t_1 + \tau_2, t_1 + \tau_3) \\ &- m_x^2(t_1, t_1 + \tau_1) m_x^2(t_1 + \tau_2, t_1 + \tau_3) \\ &- m_x^2(t_1, t_1 + \tau_2) m_x^2(t_1 + \tau_3, t_1 + \tau_1) \\ &- m_x^2(t_1, t_1 + \tau_3) m_x^2(t_1 + \tau_1, t_1 + \tau_2). \end{aligned} \quad (8.3)$$

The Fourier transforms of the cumulants provide the corresponding higher-order spectra or polyspectra (with as many frequency variables as the order minus

one). The Fourier transforms of the second-order, third-order, and fourth-order cumulants are known as the power spectrum (PSD), bispectrum, and trispectrum, respectively. A Gaussian process possesses only first-order and second-order statistics: Moments and spectra of order higher than two are zero. Higher-order moments, cumulants, and spectra may be used to characterize nonlinear, mixed-phase, and non-Gaussian signals [148, 403, 404]. Variations over time of such measures may be used to detect the related types of nonstationarity.

- **System parameters:** When a time-varying model of the system producing the signal is available in terms of its coefficients, such as $a_k(n)$ in Equation 8.1, we may follow or track changes in the coefficients over time. Significant changes in the model parameters indicate corresponding changes in the output signal.

The following sections present several examples of application of the concepts and notions listed above.

8.4 Fixed Segmentation

Given a nonstationary signal, the simplest approach to break it into quasistationary segments would be to consider small windows of fixed duration. Given a signal $x(i)$ for $i = 0, 1, 2, \dots, N-1$, we could consider a fixed segment duration of M samples, with $M \ll N$, and break the signal into $K = N/M$ parts as

$$x_k(n) = x[n + (k-1)M], \quad 0 \leq n \leq M-1, \quad 1 \leq k \leq K. \quad (8.4)$$

With the assumption that the signal does not change its characteristics to any significant extent within the duration corresponding to M samples (or $\frac{M}{f_s}$ s), each segment may be considered to be quasistationary.

Note that the segmentation here is similar to that in the Bartlett and Welch procedures described in Sections 6.3.2 and 6.3.3. However, we will not be averaging the spectra over the segments now, but will be treating them as separate entities. The signal processing techniques we have studied so far may then be applied to analyze each segment separately.

8.4.1 The short time Fourier transform

Once a given signal has been segmented into quasistationary parts $x_k(n)$ as in Equation 8.4, we may compute the Fourier spectrum for each segment as

$$X_k(\omega) = \sum_{n=0}^{M-1} x_k(n) \exp(-j\omega n). \quad (8.5)$$

The array of spectra $X_k(\omega)$ for $k = 1, 2, \dots, K$ will describe the time-varying spectral characteristics of the signal.

Segmentation of the given signal as above may be interpreted as the application of a moving window to the signal. The k^{th} segment $x_k(n)$ may be expressed as the multiplication of the signal $x(n)$ with a window function $w(n)$ positioned at the beginning of the segment as

$$x_k(n) = x(n) w[n - (k - 1)M], \quad 1 \leq k \leq K, \quad (8.6)$$

where

$$w(n) = \begin{cases} 1 & \text{for } 0 \leq n \leq M - 1, \\ 0 & \text{otherwise.} \end{cases} \quad (8.7)$$

Figure 8.3 (a) illustrates the PCG of a patient with systolic murmur and opening snap of the mitral valve, with a moving rectangular analysis window of duration 64 *ms* superimposed on the signal at three different instants of time. The duration of each window is 64 samples, equal to 64 *ms* with $f_s = 1 \text{ kHz}$. The three windows have been positioned approximately over the S1, systolic murmur, and S2 events in the signal. Figure 8.3 (b) shows the log PSDs of the signal segments extracted by the three analysis windows. It is seen that the PSDs differ significantly, with the second window displaying the largest amount of high-frequency power due to the murmur. The third window displays more medium-frequency content than the first. It is clear that the PCG signal is nonstationary in the PSD.

In general, the window may be positioned at any time instant m , and the resulting segment may be expressed as $x(n)w(n - m)$. We need to state how the window is moved or advanced from one segment to another; in the extreme situation, we may advance the window one sample at a time, in which case adjacent windows would have an overlap of $(M - 1)$ samples. We may then compute the Fourier transform of every segment as

$$X(m, \omega) = \sum_{n=0}^{M-1} [x(n) w(n - m)] \exp(-j\omega n). \quad (8.8)$$

In the case when both the time and frequency variables are continuous, we may write the expression above in a more readily understandable form as

$$X(\tau, \omega) = \int_{-\infty}^{\infty} [x(t) w(t - \tau)] \exp(-j\omega t) dt. \quad (8.9)$$

The spectrum is now expressed not only as a function of frequency ω , but also as a function of time τ . Although the limits of the integral have been stated as $(-\infty, \infty)$, the finite duration of the window placed at time τ performs segmentation of the signal as desired.

The spectral representation of the signal as a function of time in Equations 8.8 and 8.9 is known as a *time-frequency distribution* or TFD [405–408]. In practice, only the squared magnitudes of the expressions in Equations 8.8 and 8.9 are used for the display of TFDs. Because the Fourier transform is applied, in the procedure

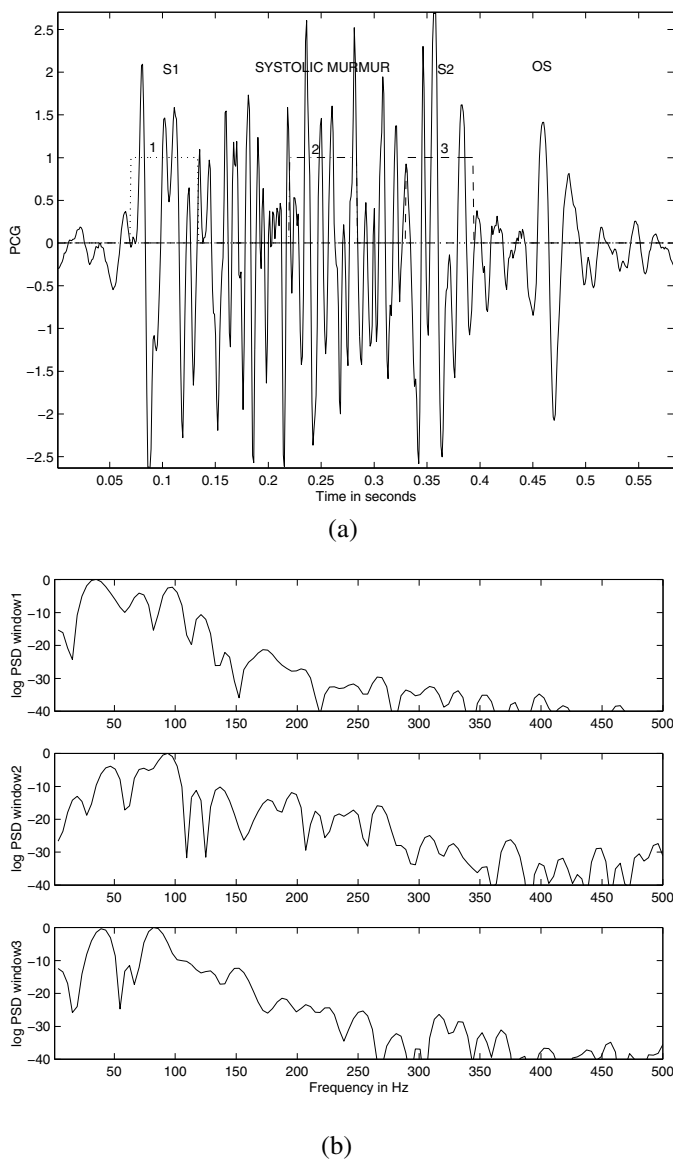


Figure 8.3 (a) PCG signal of a patient (female, 14 months) with systolic murmur and opening snap (OS) of the mitral valve. Three short-time analysis windows are superimposed, each one being a rectangular window of duration 64 ms. (b) Log PSDs of the three windowed signal segments. Each DFT was computed with zero-padding to a total length 256 samples. $f_s = 1$ kHz. See also Figures 6.13 and 8.4.

above, to short windows of the signal in time, the result is known as the *short-time Fourier transform* or STFT of the signal. The method of analysis of a nonstationary signal in short windows is, in general, known as *short-time analysis*. The magnitude of the STFT (squared and/or with the logarithmic operation if desired) is known as the *spectrogram* of the signal.

Figure 8.4 illustrates the spectrogram of the PCG signal of a patient with systolic murmur and opening snap of the mitral valve: The signal and the window parameters are the same as in Figure 8.3, but now the spectra are plotted for every window position with a displacement of 32 *ms*. The relatively high-frequency nature of the murmur as compared to S1 and S2 is clearly evident in the spectrogram.

We have previously encountered spectrograms of speech and VAG signals: Refer to Figure 3.4 and Sections 3.9.3 and 3.14. More examples of spectrograms are provided at the end of this section and later in this chapter.

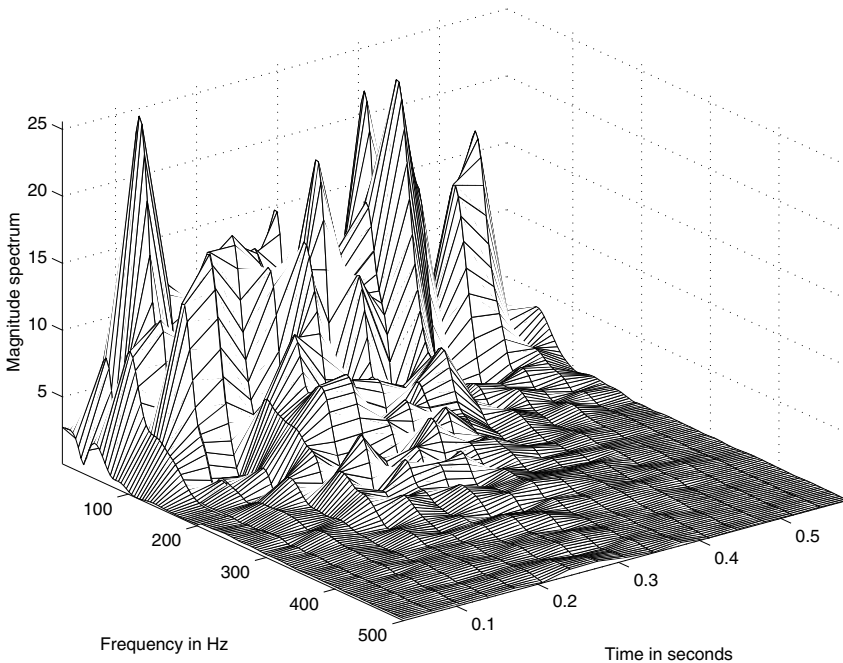


Figure 8.4 Spectrogram of the PCG signal of a patient (female, 14 months) with systolic murmur and opening snap of the mitral valve, computed with a moving short-time analysis window of duration 64 samples (64 *ms* with $f_s = 1$ *kHz*), with the window advance interval being 32 samples. Each DFT was computed with zero-padding to a total length 256 samples. $f_s = 1$ *kHz*. See also Figures 6.13 and 8.3.

8.4.2 Considerations in short time analysis

Short-time analysis of signals could be computationally expensive. In the case of the STFT, the Fourier transform has to be computed for each segment of the signal. In practice, there should be no need to compute the Fourier transform for every possible window position, that is, for every m in Equation 8.8. We could advance the analysis window by M samples, in which case adjacent windows will not overlap. It is common practice to advance the analysis window by $\frac{M}{2}$ samples, in which case adjacent windows will overlap for $\frac{M}{2}$ samples; some overlap is desirable in order to maintain continuity in the STFT or TFD computed.

An important question arises regarding the duration of the analysis window M to be used. The window should be short enough to ensure that the segment is stationary, but long enough to permit meaningful analysis and adequate representation of low-frequency components. We have seen in Section 6.4 that a short window possesses a wide main lobe in its frequency response. Since the given signal is multiplied in the time domain with the analysis window, the spectrum of the signal gets convolved with the spectral response of the window in the frequency domain. Convolution in the frequency domain with a function having a large main lobe leads to significant blurring and loss of spectral resolution.

The limitation imposed by the use of a window is related to the uncertainty principle or time–bandwidth product, expressed as [406]

$$\Delta t \times \Delta \omega \geq \frac{1}{2}, \quad (8.10)$$

where

$$(\Delta t)^2 = \int_{-\infty}^{\infty} (t - \bar{t})^2 |x(t)|^2 dt, \quad (8.11)$$

$$\bar{t} = \int_{-\infty}^{\infty} t |x(t)|^2 dt, \quad (8.12)$$

$$(\Delta \omega)^2 = \int_{-\infty}^{\infty} (\omega - \bar{\omega})^2 |X(\omega)|^2 d\omega, \quad (8.13)$$

$$\bar{\omega} = \int_{-\infty}^{\infty} \omega |X(\omega)|^2 d\omega, \quad (8.14)$$

and Δt and $\Delta \omega$ represent the time extent (duration) and frequency extent (bandwidth) of the signal $x(t)$ and its Fourier transform $X(\omega)$, respectively. The gist of the limitation stated above is that both a signal and its Fourier transform cannot simultaneously be made arbitrarily narrow. The effect of this limitation on the STFT and TFD-based analysis is that we cannot simultaneously obtain arbitrarily high resolution along both the time and frequency axes.

At the extremes, a continuous-time signal $x(t)$ provides infinite time resolution but no frequency resolution: The value of the signal is known at every instant of time t , but nothing is known about the frequency content of the signal. Conversely, the PSD $S_{xx}(f)$ provides infinite frequency resolution but no time resolution: The

overall strength of sinusoids at every frequency f present in the signal over all time t is known, but nothing is known about where exactly in time a given frequency component begins or ends. (The phase spectrum contains this information but cannot be readily interpreted and used for the purposes of this discussion.)

In the case of sampled signals and spectra, the sampling intervals Δt in the time domain and Δf in the frequency domain will be finite and will be limited by Heisenberg's inequality as stated above. Increasing the time resolution of the STFT by making the analysis window short in duration will compromise frequency resolution; on the other hand, increasing the window duration will lead to a loss in time resolution.

In general, the window function $w(n)$ included in Equation 8.8 need not be a rectangle: any of the window functions listed in Section 6.3.3 may be used. Once a window is chosen, the joint time and frequency resolution is the same over the entire TF plane.

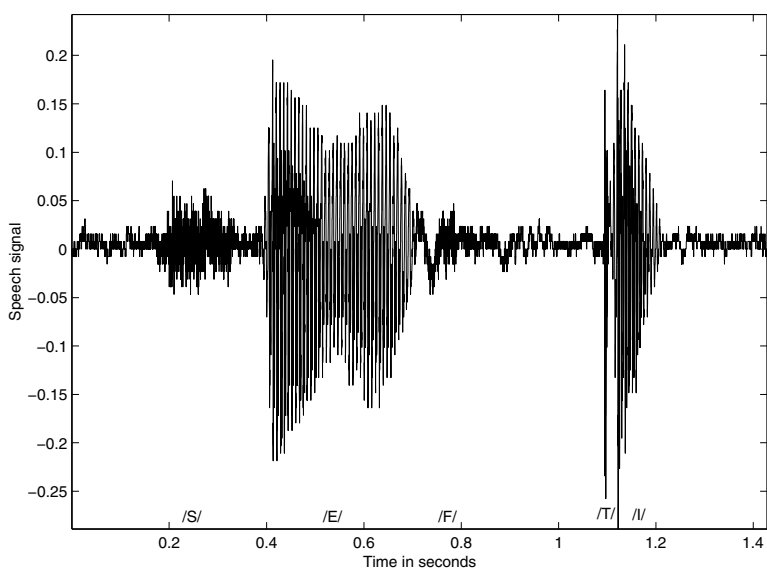
The STFT expression in Equation 8.8 indicated the placement of a causal analysis window beginning at the time instant of reference m in the argument of the STFT. It is also common practice to use a symmetrical noncausal window defined for $-\frac{M}{2} \leq n \leq \frac{M}{2}$, in which case the reference point of the analysis window would be the center of the window.

See Allen and Rabiner [409], Portnoff [410], and Rabiner and Schafer [79] for discussions on short-time Fourier analysis.

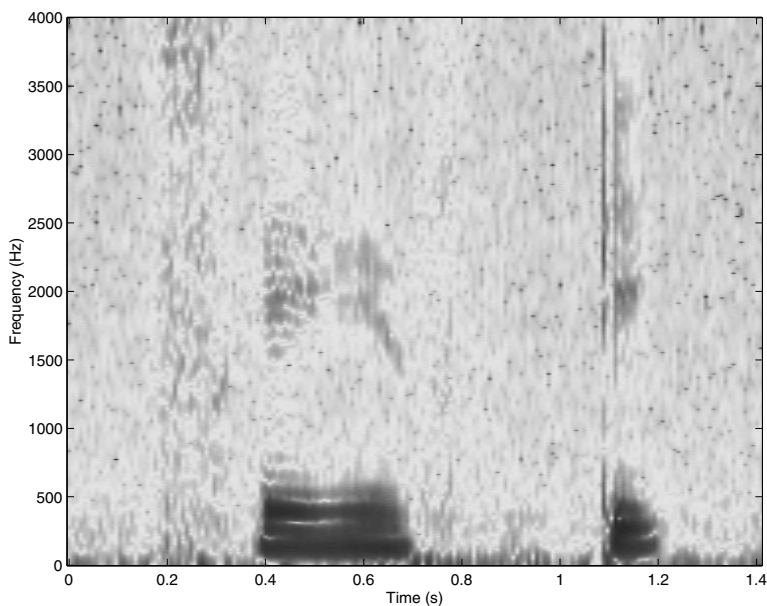
Illustration of application: Spectrograms of the speech signal in Figure 1.50 with different window parameters are provided in Figures 8.5 and 8.6. The spectrograms are shown here as gray-scale images, with the darkness at each point being proportional to the log PSD for the corresponding temporal analysis window position and frequency coordinate. It is evident that increasing the length of the analysis window provides better frequency resolution (the definition or clarity of the frequency components) while at the same time reducing the temporal resolution (that is, causing smearing in the temporal dimension). Decreasing the window length causes the reverse effects. The spectrogram in Figure 8.5 (b) with the analysis window duration being 16 ms clearly illustrates the high-frequency (broadband) nature of the fricatives; the transient and broadband nature of the plosive /T/ is also clearly shown. The same features are not clearly depicted by the spectrogram in Figure 8.6 (b) where the analysis window is fairly long (128 ms); however, the formant structure of the voiced-speech components (the vowels) is clearly depicted. The formant structure of the voiced-speech components is not clearly visible in the spectrogram in Figure 8.5 (b).

8.5 Adaptive Segmentation

One of the limitations of short-time analysis lies with the use of a fixed window duration. A signal may remain stationary for a certain duration of time much longer than the window duration chosen, and yet the signal would be broken into many segments over such a duration. Conversely, a signal may change its characteristics

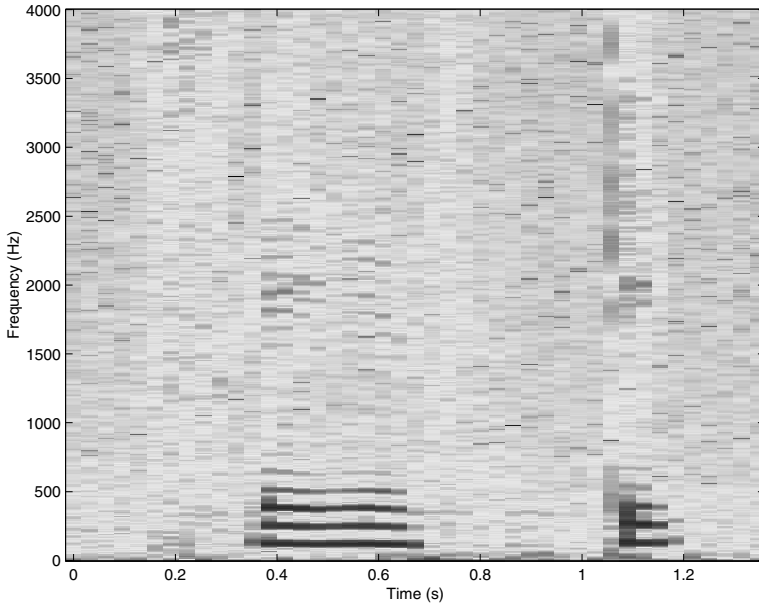


(a)

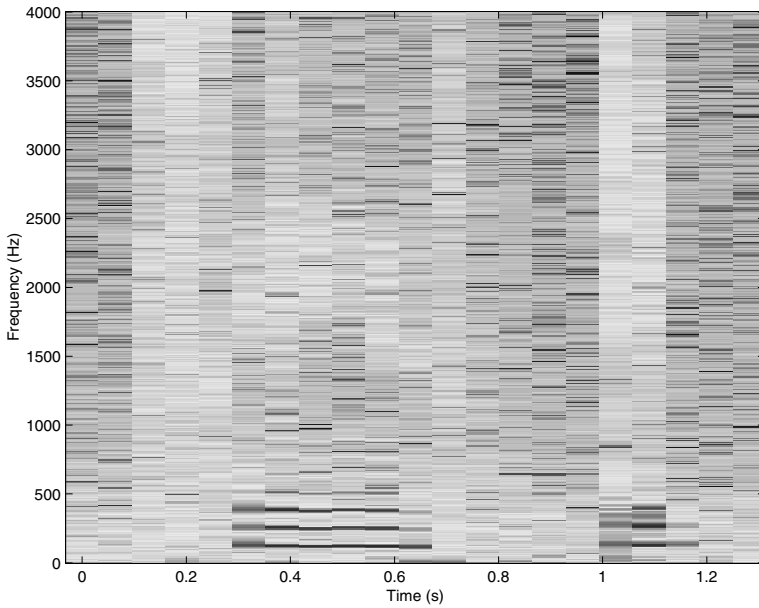


(b)

Figure 8.5 (a) Time-domain speech signal of the word “safety” uttered by a male speaker. (The signal is also illustrated in Figures 1.50, 3.1, and 3.3.) (b) Spectrogram (log PSD) of the signal computed with a moving short-time analysis window of duration 16 ms (128 samples with $f_s = 8\text{ kHz}$), with the window advance interval being 8 ms .



(a)



(b)

Figure 8.6 Spectrograms (log PSD) of the speech signal in Figure 8.5: (a) with a moving window of duration 64 *ms* (512 samples with $f_s = 8$ *kHz*) and window advance interval of 32 *ms*, and (b) with a moving window of duration 128 *ms* (1,024 samples) and window advance interval of 64 *ms*.

within the duration of the fixed window: Short-time analysis cannot guarantee stationarity of the signal over even the relatively short duration of the analysis window used. It would be desirable to adapt the analysis window to changes in the given signal, allowing the window to be as long as possible while the signal remains stationary, and to start a new window at the exact instant when the signal or the related system changes its characteristics.

Problem: *Propose adaptive methods to break a nonstationary signal into quasi-stationary segments of variable duration.*

Solution: We saw in Section 7.5 that a signal may be represented or modeled as a linear combination of a small number of past values of the signal, subject to a small error of prediction. It then follows that if a signal were to change its behavior, it would no longer be predictable from its preceding samples as they would correspond to the previous state of the time-variant system generating the nonstationary signal. Therefore, we could expect a large jump in the prediction error at instants of time when the signal changes in its characteristics. Furthermore, the AR-model parameters represent the system generating the signal and provide the poles of the system. If the system were to change in terms of the locations of its poles (related to its resonance frequencies), the same model would no longer hold: A new model would have to be initiated at such instants of change. This suggests that we could estimate AR models on a short-time basis and monitor the model parameters from segment to segment: A significant change in the model parameters would indicate a point of change in the signal. (We have seen in Section 7.9 how a similar approach was used by Iwata et al. [379] to detect S1 and S2 in PCGs.) Adjacent segments that have the same or similar model parameters could be concatenated to form longer segments in a subsequent stage of processing. As the AR model provides several parameters and may be interpreted in several ways (see Section 7.5.2), tracking the behavior of the model over a moving analysis window may be accomplished in many ways. The following sections provide the details of a few approaches for adaptive segmentation based upon the notions stated above.

8.5.1 Spectral error measure

Bodenstein and Praetorius [197] and Praetorius et al. [411] used the all-pole LP or AR model (see Section 7.5) for adaptive segmentation of EEG signals into quasistationary segments and also for further feature extraction. They made the following observations about the application of AR modeling to EEG signals:

- *Time domain:* The present value of the prediction error indicates the instantaneous degree of “unexpectedness” in the signal.
- *Autocorrelation domain:* The prediction error is decorrelated.
- *Spectral domain:* The prediction error being white noise, the AR model yields an all-pole representation of the signal spectrum, which is particularly suitable for the modeling of resonance.

These properties are useful for

- detection and elimination of transients;
- segmentation of the EEG into quasistationary segments; and
- feature extraction and pattern recognition (diagnosis).

Ferber [412] provided a description of nonstationarities in the EEG and suggested a few approaches to treat the same.

Analysis of spectral change: Let the PSD of the given nonstationary signal be $S(0, \omega)$ at zero time and $S(t, \omega)$ at time t . The *spectral error* of $S(t, \omega)$ with respect to $S(0, \omega)$ may be taken to be dependent upon the difference between the corresponding log PSDs — that is, to be proportional to $\{\log[S(t, \omega)] - \log[S(0, \omega)]\}$ or to be proportional to $\log \left[\frac{S(t, \omega)}{S(0, \omega)} \right]$. Consider the state when an AR model has been adapted to the signal spectrum $S(0, \omega)$ at zero time. If we pass the signal at time t through the AR model, the prediction error will have an instantaneous spectrum given by

$$S_e(\omega) = \frac{S(t, \omega)}{S(0, \omega)}, \quad (8.15)$$

which is similar to the spectral ratio in Equation 7.50. Thus, the problem of comparing two arbitrary PSDs of a nonstationary signal at two different instants of time may be expressed as testing $S_e(\omega)$ for deviation from a uniform PSD.

Let $a_R(k)$, $k = 1, 2, \dots, P$, represent the reference AR model. When the current signal $y(n)$ is passed through the filter represented by the AR model, we obtain the prediction error

$$e(n) = \sum_{k=0}^P a_R(k) y(n - k), \quad (8.16)$$

with the prediction model defined as in Equation 7.17, the prediction error as given in Equation 7.18, and $a_R(0) = 1$. The error indicates the deviation of the current signal from the previously computed model. Consider the integral

$$\varepsilon = \int_{-\infty}^{\infty} [1 - S_e(\omega)]^2 d\omega, \quad (8.17)$$

where $S_e(\omega)$ is the PSD of the prediction error. Ideally, when the AR model has been optimized for the signal on hand, the prediction error is expected to have a uniform PSD. However, if the signal is nonstationary, some changes would have occurred in the spectral characteristics of the signal, which would be reflected in the PSD of the error. If $\phi_e(k)$ is the ACF corresponding to $S_e(\omega)$, the latter is given by the Fourier transform of the former. However, since both functions are real and even, we have

$$S_e(\omega) = \phi_e(0) + 2 \sum_{k=1}^{\infty} \phi_e(k) \cos(2\pi\omega k). \quad (8.18)$$

Then,

$$\varepsilon = \int_{-\infty}^{\infty} \left[1 - \phi_e(0) - 2 \sum_{k=1}^{\infty} \phi_e(k) \cos(2\pi\omega k) \right]^2 d\omega. \quad (8.19)$$

Due to the orthonormality of the trigonometric functions, we get

$$\varepsilon = [1 - \phi_e(0)]^2 + 2 \sum_{k=1}^{\infty} \phi_e^2(k). \quad (8.20)$$

In practice, the summation may be performed up to some lag, M . Bodenstein and Praetorius [197] recommended normalization of the error measure by division by $\phi_e^2(0)$, leading to the *spectral error measure (SEM)*

$$SEM = \left[\frac{1}{\phi_e(0)} - 1 \right]^2 + 2 \sum_{k=1}^M \left[\frac{\phi_e(k)}{\phi_e(0)} \right]^2. \quad (8.21)$$

Here, the first term represents the change in the total power of the prediction error; the second term depends upon the change in spectral shape only. Note that the prediction error is expected to have a uniform (flat) PSD as long as the signal remains stationary with respect to the AR model designed. The *SEM* was shown to vary significantly in response to changes in the spectral characteristics of EEG signals and to be useful in breaking the signals into quasistationary parts. Figure 8.7 shows the general scheme of EEG segmentation by using the *SEM*.

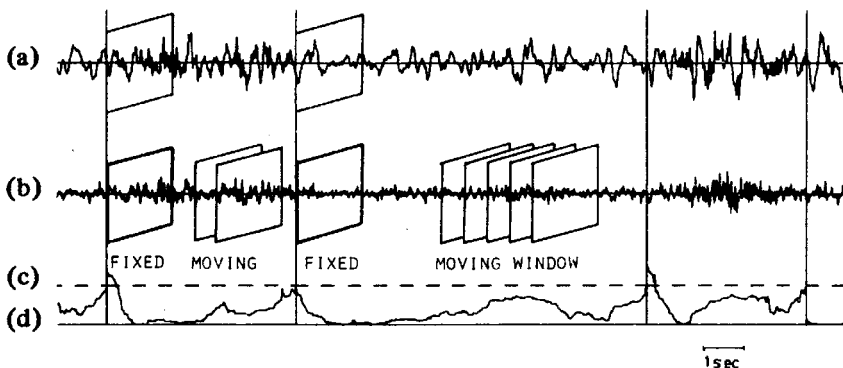


Figure 8.7 Adaptive segmentation of EEG signals via use of *SEM*. (a) Original EEG signal. The rectangular window at the beginning of each adaptive segment indicates the signal window to which the AR model has been optimized. (b) Prediction error. The initial ACF of the error is computed over the fixed window; the running ACF of the error is computed over the moving window. (c) Segmentation threshold. (d) *SEM*. The vertical lines represent the segmentation boundaries. Reproduced with permission from G. Bodenstein and H.M. Praetorius, Feature extraction from the electroencephalogram by adaptive segmentation, *Proceedings of the IEEE*, 65(5):642–652, 1977. ©IEEE.

Algorithm for adaptive segmentation [197]: Let $n = 0$ represent the starting point of analysis where the first reference or fixed analysis window is placed for each adaptive segment, as in Figure 8.7 (a). $(N + P)$ samples of the signal $y(n)$ should be available prior to the arbitrarily designated origin at $n = 0$, where $(2N + 1)$ is the size of the analysis window and P is the order of the AR model to be used.

1. Using the signal samples $y(-N)$ to $y(N)$, compute the signal ACF up to lag P .
2. Derive the corresponding AR model of order P .
3. Using the signal values $y(-N - P)$ to $y(n + N)$, compute the prediction error $e(-N)$ to $e(n + N)$ and compute the running short-time ACF $\phi_e(n, m)$ of the prediction error as

$$\phi_e(n, m) = \frac{1}{2N + 1} \sum_{k=-N}^{N-m} e(n + k) e(n + k + m). \quad (8.22)$$

Note that the ACF now has two indices: the first index n indicates the position of the short-time analysis window, and the second index m indicates the lag for which the ACF is computed.

4. Calculate $\phi_e(0, m)$ for $m = 0, 1, \dots, M$. This represents the fixed window at the beginning of each adaptive segment in Figure 8.7 (a).

Perform the following three steps for each data point:

5. Compute $\phi_e(n, m)$ for the moving window [see Figure 8.7 (b)] by the recursive relationship

$$\begin{aligned} \phi_e(n, m) &= \phi_e(n - 1, m) + e(n + N) e(n + N - m) \\ &\quad - e(n - N - 1) e(n - N - 1 - m). \end{aligned} \quad (8.23)$$

This represents the moving window in Figure 8.7 (b).

6. Compute the SEM at time n as

$$SEM(n) = \left[\frac{\phi_e(0, 0)}{\phi_e(n, 0)} - 1 \right]^2 + 2 \sum_{k=1}^M \left[\frac{\phi_e(n, k)}{\phi_e(n, 0)} \right]^2, \quad (8.24)$$

where $\phi_e(0, 0)$ accounts for the fact that the signal may have an arbitrary power level.

7. Test if $SEM(n) > Th_1$, where Th_1 is a threshold.

If the condition is not satisfied, increase n by 1 and return to Step 5.

If the condition is satisfied, a segment boundary has been detected at time n , as indicated by the vertical lines in Figure 8.7. Reset the procedure by the following step:

8. Shift the time axis by substituting $(n + k)$ with $(k - N)$ and start the procedure again with Step 1.

In the investigations of Bodenstein and Praetorius [197], SEM demonstrated sharp jumps as transients of duration less than 100 ms entered and left the moving

analysis window of duration 2 s ($2N + 1 = 101$ samples with $f_s = 50\text{ Hz}$). Such jumps could lead to inappropriate segmentation, especially with burst-suppression type EEG episodes as illustrated in Figure 8.8. To overcome this problem, it was suggested that the prediction error $e(n)$ be limited (clipped) by a threshold Th_2 as

$$e(n) = \begin{cases} e(n), & \text{if } |e(n)| < Th_2, \\ \text{sgn}[e(n)] Th_2, & \text{if } |e(n)| \geq Th_2. \end{cases} \quad (8.25)$$

The threshold Th_2 is shown by the dashed lines in Figure 8.8 (c). The *SEM* computed from the clipped $e(n)$ is shown in Figure 8.8 (d), which, when checked against the original threshold Th_1 , will yield the correct segmentation boundary. The signal reconstructed from the clipped prediction error is shown in Figure 8.8 (e), which shows that the clipping procedure has suppressed the effect of the transient without affecting the rest of the signal.

In spite of the clipping procedure as in Equation 8.25, it was indicated by Bodenstein and Praetorius [197] that the procedure was more sensitive than desired and caused false alarms. To further limit the effects of random fluctuations in the prediction error, a smoothed version $e_s(n)$ of the squared prediction error was computed as

$$e_s(n) = e^2(n-1) + 2e^2(n) + e^2(n+1) \quad (8.26)$$

for those samples of $e(n)$ that satisfied the condition $|e(n)| > Th_2$. Another threshold Th_3 was applied to $e_s(n)$, and the triplet $\{y(n-1), y(n), y(n+1)\}$ was considered to be a part of a transient only if $e_s(n) > Th_3$. The procedure of Bodenstein and Praetorius combines adaptive segmentation of EEG signals with transient detection as the two tasks are interrelated.

Illustration of application: Figure 8.9 shows the EEG signal of a child in sleep stage I, superimposed with 14 Hz spindles. The *SEM* and its components are also shown in the figure. The vertical lines indicate the segment boundaries detected. Bodenstein et al. [413] and Creutzfeldt et al. [414] describe further extension of the approach to computerized pattern classification of EEG signals including clustering of similar segments and labeling of the types of activity found in an EEG record.

The *SEM* method was applied by Tavathia et al. [93] for adaptive segmentation of VAG signals. It was indicated that each segment could be characterized by the frequency of the most-dominant pole obtained via AR modeling and the spectral power ratio $E_{40:120}$ as per Equation 6.40; however, no classification experiments were performed. More examples of application of the *SEM* technique are presented in Sections 8.5.4 and 8.9.

8.5.2 ACF distance

Michael and Houchin [415] proposed a method comparable to that of Bodenstein and Praetorius [197], but based upon a simpler scheme using the ACF. It should be noted that the AR-model coefficients are derived from the ACF and that the spectra used to compute *SEM* are related to the corresponding ACFs by the Fourier transform.

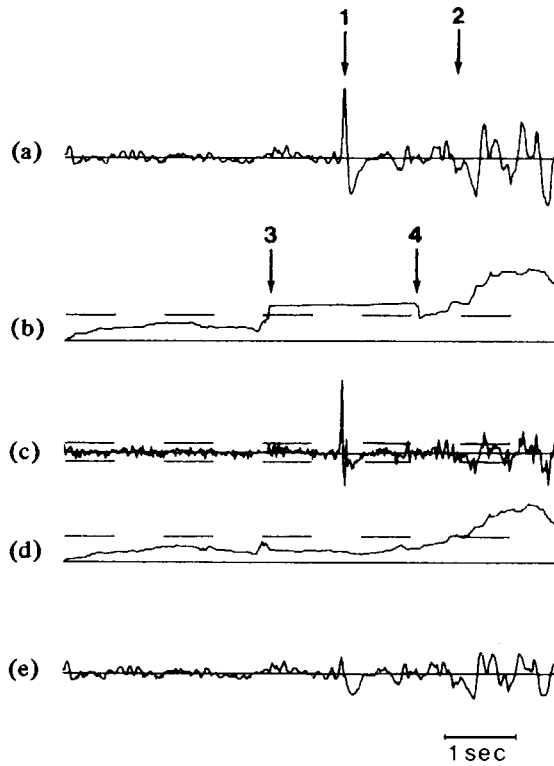


Figure 8.8 Elimination of transients by clipping the prediction error. (a) Original EEG signal of the burst-suppression type. The sharp wave marked by the arrow 1 is followed by the onset of a burst marked by the arrow 2. (b) *SEM* showing sudden jumps at points indicated by the arrows 3 and 4 as the sharp wave enters and leaves the analysis window. (c) Clipping of the prediction error with threshold Th_2 . (d) *SEM* after clipping the prediction error. The dashed line represents the threshold Th_1 . (e) Signal reconstructed from the clipped prediction error. Reproduced with permission from G. Bodenstein and H.M. Praetorius, Feature extraction from the electroencephalogram by adaptive segmentation, *Proceedings of the IEEE*, 65(5):642–652, 1977. ©IEEE.

However, direct use of the ACF removes the assumption made in AR modeling that the signal is the result of an AR process.

In the method of Michael and Houchin, the ACF is treated as a statistical measure of the given signal, and significant variations in the ACF are used to detect nonstationarity. A reference window is extracted at the beginning of each scan, and the given signal (EEG) is observed through a moving window. The duration of the window has to be chosen such that it is shorter than the shortest expected quasistationary segment of the given signal, but long enough to characterize the lowest frequency present. If the difference between the signal's statistics (ACF) in the moving win-

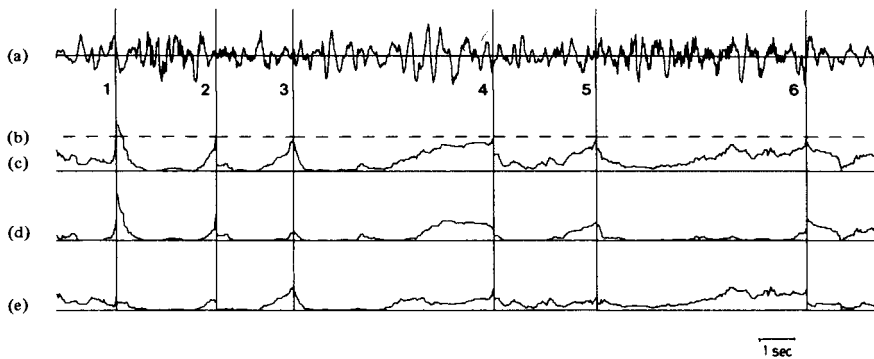


Figure 8.9 Use of *SEM* to segment an EEG signal. (a) Original EEG signal of a child in sleep stage I with superimposed 14 *Hz* spindles. (b) Segmentation threshold. (c) *SEM*. (d) Deviation in prediction error power. (e) Deviation in prediction error spectral shape. The vertical lines represent the segmentation boundaries. Reproduced with permission from G. Bodenstein and H.M. Praetorius, Feature extraction from the electroencephalogram by adaptive segmentation, *Proceedings of the IEEE*, 65(5):642–652, 1977. ©IEEE.

dow and the reference window is significant, a segment boundary is drawn, and the procedure is restarted.

Let $\phi_R(k)$ be the ACF of the reference window at the beginning of a new segmentation step, where k is the lag or delay. Let $\phi_T(n, k)$ be the ACF of the test window positioned at time instant n . Given that the ACF for zero lag is the power of the signal, Michael and Houchin computed a normalized power distance $d_P(n)$ between the ACFs as (see also Appel and v. Brandt [394])

$$d_P(n) = \frac{|\sqrt{\phi_T(n, 0)} - \sqrt{\phi_R(0)}|}{\min\{\sqrt{\phi_T(n, 0)}, \sqrt{\phi_R(0)}\}}. \quad (8.27)$$

A spectral distance $d_F(n)$ was computed using the ACF coefficients only up to lag q as

$$d_F(n) = \frac{\sum_{k=1}^q |\phi_T(n, k) - \phi_R(k)|}{0.5 + \sum_{k=1}^q \min\{\sqrt{\phi_T(n, k)}, \sqrt{\phi_R(k)}\}}. \quad (8.28)$$

The lag limit q was set as the lower value of the lags at which the ACFs changed from positive to negative values for the first time. The net ACF distance $d(n)$ was then computed as

$$d(n) = \frac{d_P(n)}{Th_P} + \frac{d_F(n)}{Th_F}, \quad (8.29)$$

where Th_P and Th_F are thresholds. The condition $d(n) > 1$ was considered to represent a significant change in the ACF, and used to mark a segment boundary.

Due to the use of a moving window of finite size, the true boundary or point of change in the signal characteristics will lie within the last test window before a segment boundary is triggered. Michael and Houchin used a linear interpolation procedure based upon the steepness of the ACF distance measure to correct for such a

displacement. Barlow et al. [416] provide illustrations of application of the method to clinical EEGs. Their work includes clustering of similar segments based upon mean amplitude and mean frequency measures, “dendrograms” to illustrate the clustering of segments [417], and labeling of the various states found in an EEG record. Illustrations of application of the ACF method are provided in Section 8.5.4.

8.5.3 The generalized likelihood ratio

Appel and v. Brandt [393] proposed the generalized likelihood ratio (GLR) method, which uses a reference window that is continuously grown as long as no new boundary is marked. The test window is a sliding window of constant duration as in the case of the *SEM* and ACF methods. Figure 8.10 illustrates the windows used. The advantage of the growing reference window is that it contains the maximum amount of information available from the beginning of the new segment to the current instant. Three different data sets are defined: the growing reference window, the sliding test window, and a pooled window formed by concatenating the two. Distance measures are then derived using AR-model prediction errors computed for the three data sets.

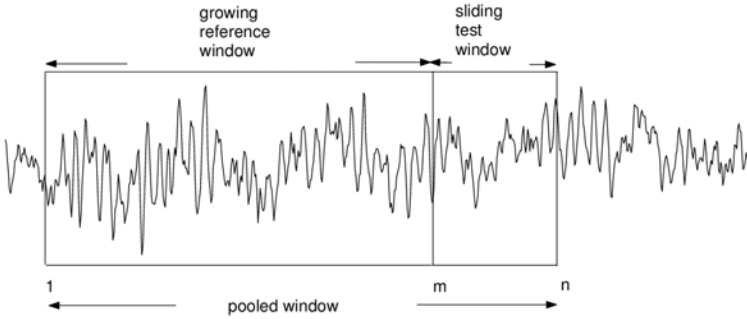


Figure 8.10 The growing reference window, the sliding test window, and the pooled window used in the GLR method for adaptive segmentation.

Let $\varepsilon(m : n)$ represent the prediction error energy (TSE ε as in Equation 7.19) within an arbitrary data set or window with boundaries m and n . The maximum log likelihood measure $H(m : n)$ for the window is defined as

$$H(m : n) = (n - m + 1) \ln \left[\frac{\varepsilon(m : n)}{(n - m + 1)} \right]. \quad (8.30)$$

Three measures are computed for the three data sets described above as $H(1 : m - 1)$ for the growing reference window, $H(m : n)$ for the test window, and $H(1 : n)$ for the composite or pooled window. Here, the reference window is denoted as commencing from the time instant or sample 1, m is the last sample of the growing reference window, and the current test window spans the duration from m to the current time instant n (see Figure 8.10). The GLR distance measure is defined as

$$d(n) = H(1 : n) - [H(1 : m - 1) + H(m : n)]. \quad (8.31)$$

Here, the first quantity represents the TSE if the test window is added to the growing reference window; the second quantity represents the TSE of the reference window grown so far; and the third quantity represents the TSE in modeling the test window itself. The measure $d(n)$ answers the question, “How much is the increase in the TSE if we add the test window to the growing reference window?”

Appel and v. Brandt [393] and Cohen [343] provide more details on the GLR. The GLR distance is a measure of the statistical similarity of the reference and test data sequences, with the assumption that their AR-model coefficients have a normal (Gaussian) distribution. The GLR distance is also a measure of the loss of information caused if no segment boundary is drawn at the position of the test window, that is, if it is assumed that the null hypothesis that the two sequences are similar is true.

Appel and v. Brandt [393] discuss issues related to the choice of the parameters involved in the GLR method, including the AR-model order, the test window length, and the threshold, on the GLR distance measure. The GLR method was also used by Willsky and Jones [418] to detect abrupt changes (sporadic anomalies and failures) in the variables of stochastic linear systems, as well as by Basseville and Benveniste [419] for segmentation of nonstationary signals (see also Cohen [343]). Illustrations of application of the GLR method are provided in Section 8.5.4.

8.5.4 Comparative analysis of the ACF, *SEM*, and GLR methods

Appel and v. Brandt [394] performed a comparative analysis of the performance of the ACF, *SEM*, and GLR methods of adaptive segmentation using synthesized signals as well as EEG signals. A simple two-pole system was used as the basis to simulate nonstationary signals. The gain, pole radius, and pole angle were individually varied back and forth between two sets of values. Several outputs of the dynamic system were computed with random signals (Gaussian-distributed white noise) as input. The signals were processed by the ACF, *SEM*, and GLR methods for adaptive segmentation. The variability of the segment boundaries detected for various realizations of the nonstationary (random) output signals for the same sequences of system parameters was analyzed.

Figure 8.11 shows the results related to variations in the angles of the poles, that is, in the resonance frequency of the system. The angle of the pole in the upper-half of the z -plane was changed from 20° to 40° and back at samples 200 and 400; the conjugate pole was also varied accordingly. The same changes were repeated at samples 700 and 800. The upper panel in the figure shows the pole positions and the related PSDs. The middle panel illustrates one sample of the 200 test signals generated: The higher-frequency characteristics of the signal related to the shifted pole positioned at 40° are evident over the intervals 200 – 400 and 700 – 800 samples. The lower panel illustrates the variability in the detected segment indices (dotted curve) and the estimated segment boundary positions (solid curves) for the three methods over 200 realizations of the test signals. (The true segment indices and boundaries are 1 : 200, 2 : 400, 3 : 700, and 4 : 800; ideally, the curves should exhibit steps at the points of change.) It is evident that the GLR method has provided the most consistent and accurate segmentation results, although at the price of increased computational load.

The *SEM* method has performed better than the ACF method, the latter showing the poorest results.

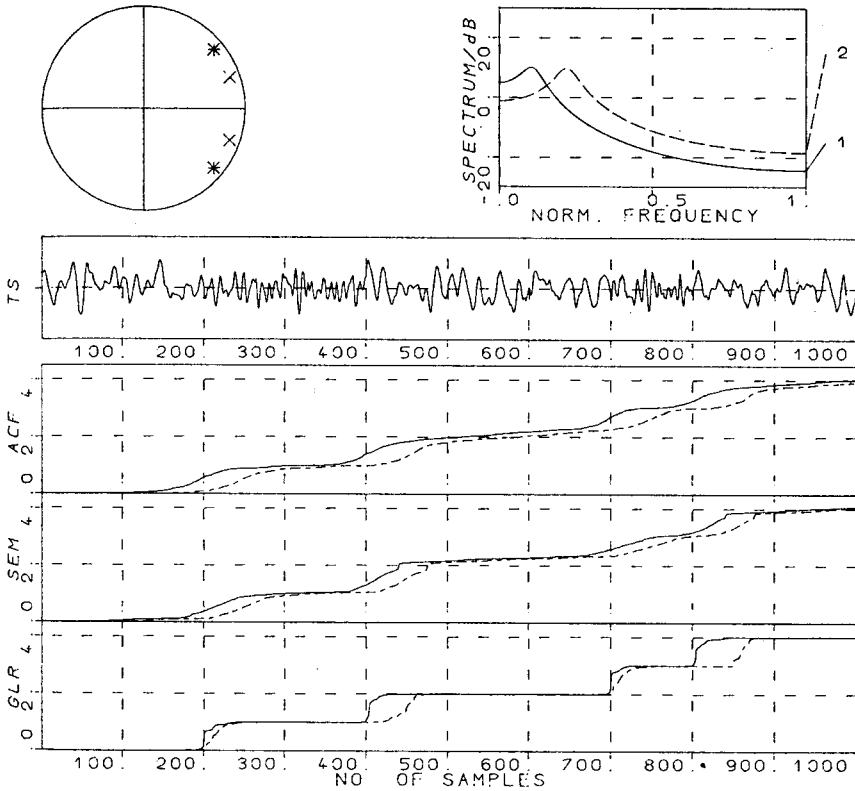


Figure 8.11 Comparative analysis of the ACF, *SEM*, and GLR methods for adaptive segmentation with the pole angle varied. Upper panel: pole positions and the related PSDs. *Note:* Norm. Frequency is normalized frequency such that the maximum frequency present in the sampled signal is unity. Middle panel: sample test signal; TS = time series. Lower panel: variability in the detected segment indices (dotted curve) and the estimated segment boundary positions (solid curves) for the three methods over 200 realizations of the test signals. See the text for more details. Reproduced with permission from U. Appel and A. v. Brandt, A comparative analysis of three sequential time series segmentation algorithms, *Signal Processing*, 6:45–60, 1984. ©Elsevier Science Publishers B.V. (North Holland).

Figure 8.12 shows the results related to variations in the distance of the poles from the origin, that is, in the bandwidth of the resonance frequency of the system. The distance of the poles from the origin was changed from 0.7 to 0.9 and back at samples 200 and 400. The same changes were repeated at samples 700 and 800. The PSDs display the increased prominence of the spectral peak when the poles are pushed toward the unit circle. The ACF method has not performed well in recognizing the

nonstationarities of this type in the test signals. The GLR method has performed better than the ACF method in segmentation.

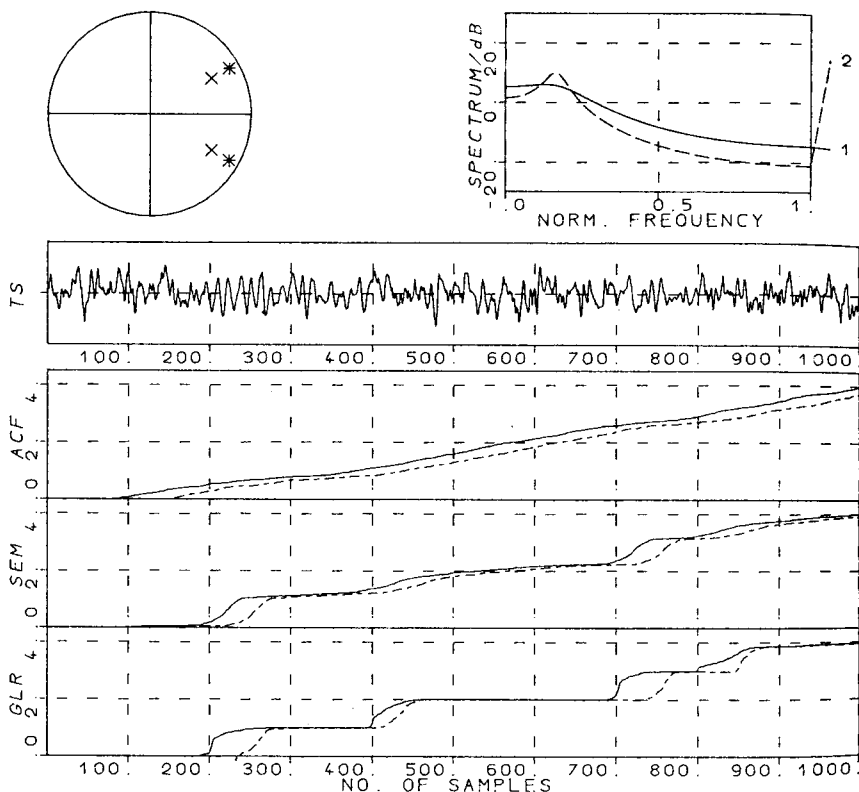


Figure 8.12 Comparative analysis of the ACF, *SEM*, and GLR methods for adaptive segmentation with the pole radius varied. Upper panel: pole positions and the related PSDs. *Note:* Norm. Frequency is normalized frequency such that the maximum frequency present in the sampled signal is unity. Middle panel: sample test signal; TS = time series. Lower panel: variability in the detected segment indices (dotted curve) and the estimated segment boundary positions (solid curves) for the three methods over 200 realizations of the test signals. See the text for more details. Reproduced with permission from U. Appel and A. v. Brandt, A comparative analysis of three sequential time series segmentation algorithms, *Signal Processing*, 6:45–60, 1984. ©Elsevier Science Publishers B.V. (North Holland).

Figure 8.13 shows the results of application of the three methods to an EEG signal. Although the exact locations where the signal changes its characteristics are not known for the EEG signal, the boundaries indicated by the GLR method appear to be the most accurate. It may be desirable in real-life applications to err on the side of superfluous segmentation; a subsequent clustering step could merge adjacent segments with similar model parameters.

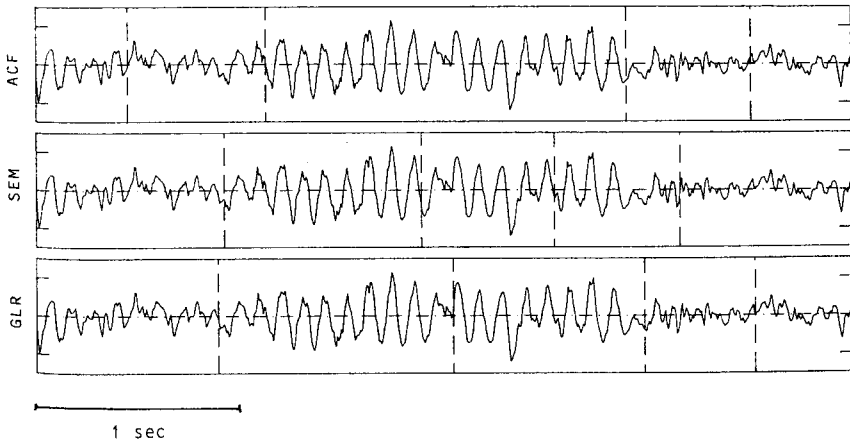


Figure 8.13 Comparative analysis of the ACF, *SEM*, and GLR methods for adaptive segmentation of an EEG signal. Reproduced with permission from U. Appel and A. v. Brandt, A comparative analysis of three sequential time series segmentation algorithms, *Signal Processing*, 6:45–60, 1984. ©Elsevier Science Publishers B.V. (North Holland).

8.6 Use of Adaptive Filters for Segmentation

We saw in Sections 3.9.2 and 3.9.3 that the coefficient (tap-weight) vectors of the adaptive LMS and RLS filters are expressed as functions of time. The filters adapt to changes in the statistics of the primary and reference signals. Could we, therefore, use the tap-weight vector $\mathbf{w}(n)$ to detect nonstationarities in a signal?

Problem: Investigate the potential use of the RLS adaptive filter for adaptive segmentation of nonstationary signals.

Solution: When we have only one signal to work with — the signal that is to be segmented — the question arises as to how we may provide two inputs, namely, the primary and reference signals, to the adaptive filter. If we assume that the signal to be segmented (applied at the primary input) was generated by an AR system, then we may provide the same signal with a delay as the reference input to the adaptive filter. The delay is to be set such that the reference input at a given instant of time is uncorrelated with the primary input; the delay may also be set on the basis of the order of the filter. (It is also possible to apply white noise at the reference input.) In essence, the adaptive filter then acts the role of an adaptive AR model. The filter tap-weight vector is continually adapted to changes in the statistics (ACF) of the input signal. The output represents the prediction error. Significant changes in the tap-weight vector or the prediction error may be used to mark points of prominent nonstationarities in the signal. Figure 8.14 shows a signal-flow diagram of the adaptive filter as described above; the filter structure is only slightly different from that in Figure 3.91.

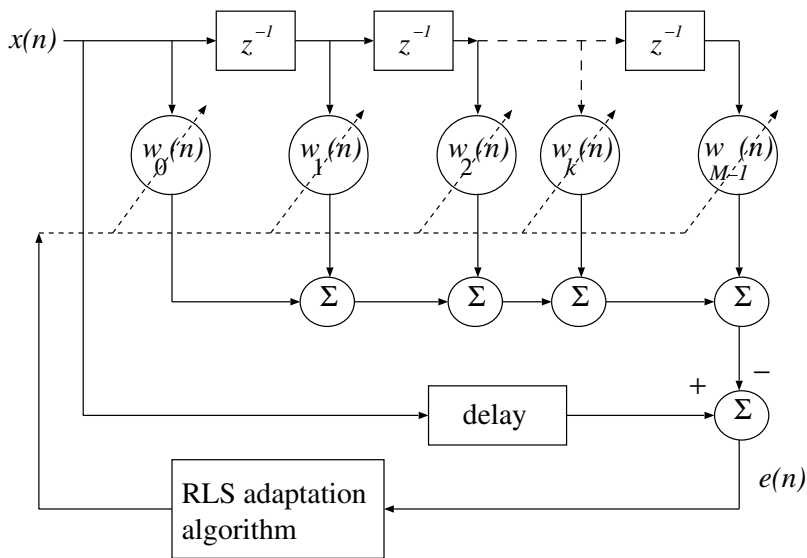


Figure 8.14 Adaptive RLS filter for segmentation of nonstationary signals. See also Figure 3.91.

8.6.1 Monitoring the RLS filter

The RLS filter as in Figure 8.14 attempts to predict the current signal sample from the available knowledge of the previous samples stored in the filter's memory units. If a large change occurs in the signal, the prediction error exhibits a correspondingly large value. In response, the adaptive filter's tap-weight vector is modified by the RLS algorithm.

Moussavi et al. [94] applied the RLS filter for segmentation of VAG signals. The order of the filter was set to be 5 in order to be low enough to detect transient changes and also to provide fast convergence. The forgetting factor was defined as $\lambda = 0.98$ so that the filter may be assumed to operate in an almost-stationary situation. The delay between the input and the reference input was set to be 7 samples (which corresponds to 3.5 ms with $f_s = 2 \text{ kHz}$).

The adaptive segmentation algorithm of Moussavi et al. is as follows:

1. Initialize the RLS algorithm.
2. Find the squared Euclidean distance between the current tap-weight vector $\mathbf{w}(n)$ and the preceding vector $\mathbf{w}(n-1)$ as

$$\Delta(n) = |\mathbf{w}(n) - \mathbf{w}(n-1)|^2. \quad (8.32)$$

3. After computing $\Delta(n)$ for all samples of the signal available (in off-line processing), compute the *SD* of the $\Delta(n)$ values. Define a threshold as three times the *SD*.

4. Label all samples n for which $\Delta(n)$ exceeds the threshold as primary segment boundaries.
5. Compute the primary segment lengths (durations) as the differences between successive primary segment boundaries. Reject all primary segment boundaries that result in segment duration less than a preset minimum (defined in the work of Moussavi et al. [94] as 120 samples or 60 *ms*, corresponding to a knee-joint angle range of approximately 4°).
6. Mark the remaining boundary points as the final segment boundaries.

The main advantage of the RLS method is that there are no explicit reference and test windows as in the case of the ACF, *SEM*, and GLR methods. The RLS method computes a new filter tap-weight vector at each sample of the incoming signal. The method was found to perform well in the detection of gradual changes as well as sudden variations in VAG signals.

Illustration of application: Figures 8.15 and 8.16 illustrate the segmentation of the VAG signals of a normal subject and a patient with arthroscopically confirmed cartilage pathology, respectively. The figures also illustrate the spectrograms of the two signals. While the segmentation of the abnormal signal in Figure 8.16 may appear to be superfluous at first sight, close inspection of the corresponding spectrogram indicates that the spectral characteristics of the signal do change within short intervals. It is evident that the RLS method has detected the different types of nonstationarity present in the signals. Moussavi et al. [94] tested the method with 46 VAG signals and observed that the segmentation boundaries agreed well with the nature of the joint sounds heard via auscultation with a stethoscope as well as with the spectral changes observed in the spectrograms of the signals.

8.6.2 The RLS lattice filter

In order to apply the RLS method for adaptive segmentation in a nonstationary environment, it is necessary to solve the least-squares problem recursively and rapidly. The *recursive least-squares lattice* (RLSL) algorithm is well suited for such purposes. Since the RLSL method uses a lattice filter, and is based upon forward and backward prediction and time-varying reflection coefficients, it is necessary to define some of the related procedures.

Forward and backward prediction: Let us rewrite Equation 7.17 related to LP or AR modeling as

$$\tilde{y}(n) = - \sum_{k=1}^M a_{M,k} y(n-k), \quad (8.33)$$

with the inclusion of the order of the model M as a subscript for the model coefficients a_k . In this procedure, M past samples of the signal, $y(n-1), y(n-2), \dots, y(n-M)$, are used in a linear combination to predict the current sample

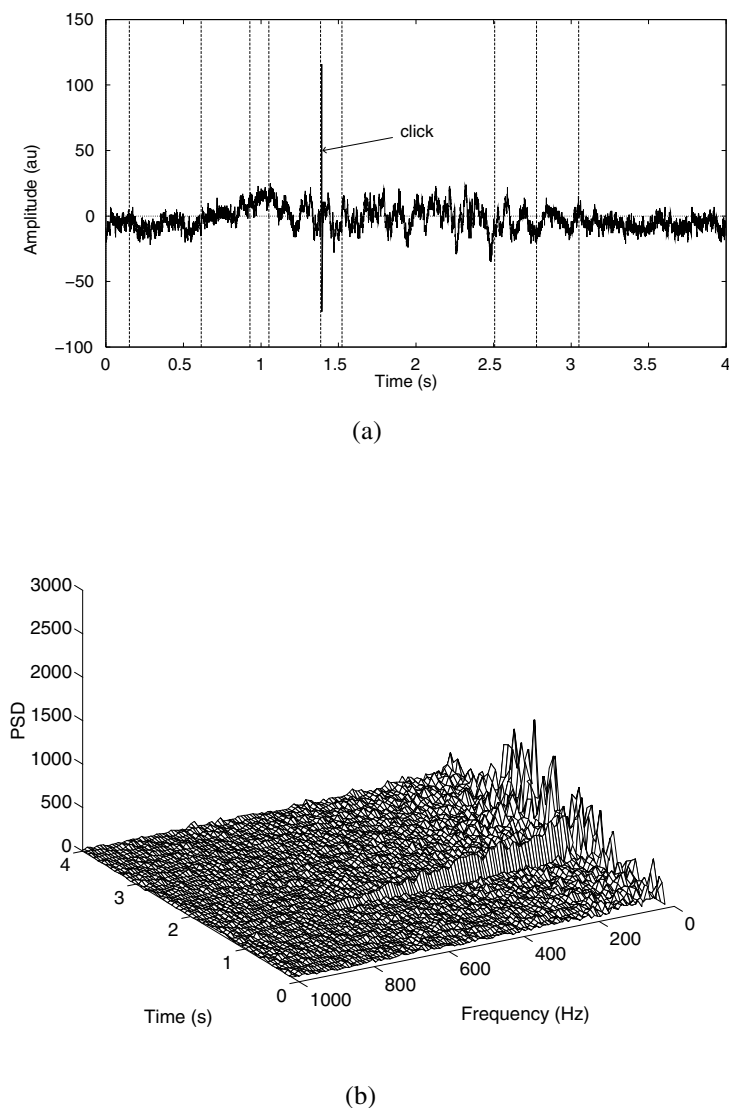
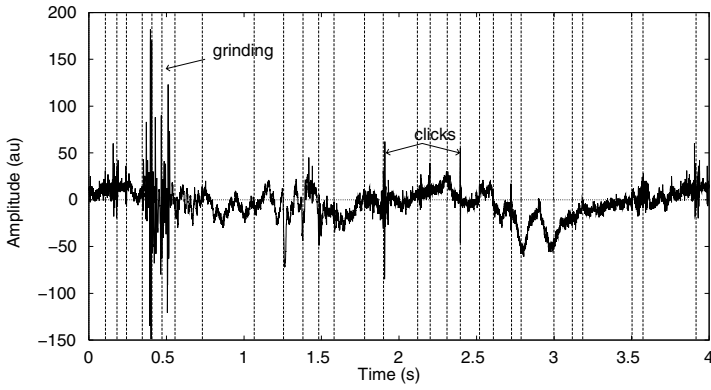
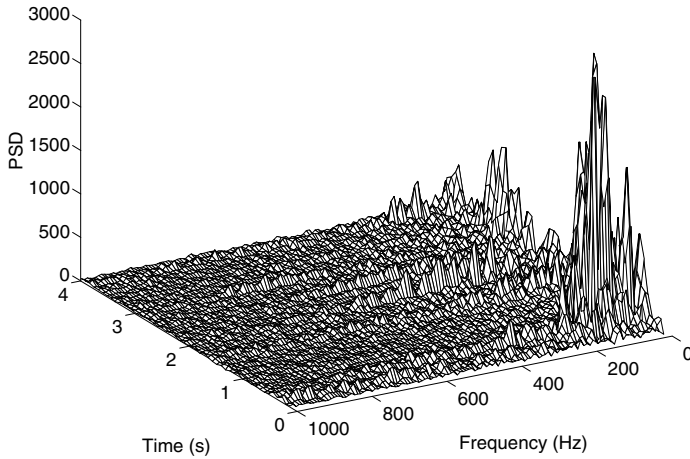


Figure 8.15 (a) Segmentation of the VAG signal of a normal subject using the RLS method. A click heard in auscultation of the knee joint is labeled. (b) Spectrogram (STFT) of the signal. Reproduced with permission from Z.M.K. Moussavi, R.M. Rangayyan, G.D. Bell, C.B. Frank, K.O. Ladly, and Y.T. Zhang, Screening of vibroarthrographic signals via adaptive segmentation and linear prediction modeling, *IEEE Transactions on Biomedical Engineering*, 43(1):15–23, 1996. ©IEEE.



(a)



(b)

Figure 8.16 (a) Segmentation of the VAG signal of a subject with cartilage pathology using the RLS method. Clicking and grinding sounds heard during auscultation of the knee joint are labeled. (b) Spectrogram (STFT) of the signal. Reproduced with permission from Z.M.K. Moussavi, R.M. Rangayyan, G.D. Bell, C.B. Frank, K.O. Ladly, and Y.T. Zhang, Screening of vibroarthrographic signals via adaptive segmentation and linear prediction modeling, *IEEE Transactions on Biomedical Engineering*, 43(1):15–23, 1996. ©IEEE.

$y(n)$ in the *forward* direction. The *forward prediction error* is

$$e_{M,f}(n) = y(n) - \tilde{y}(n) = \sum_{k=0}^M a_{M,k} y(n-k), \quad (8.34)$$

with $a_{M,0} = 1$. This equation is a restatement of Equation 7.18 with the inclusion of the order of the model M as a subscript for the error e as well as the subscript f to indicate that the prediction is being performed in the forward direction.

The term *backward prediction* refers to the estimation of $y(n-M)$ from the samples $y(n), y(n-1), \dots, y(n-M+1)$ as

$$\tilde{y}(n-M) = - \sum_{k=0}^{M-1} a_{M,k}^{\#} y(n-k), \quad (8.35)$$

where $a_{M,k}^{\#}$ are the backward prediction coefficients. Application of the least-squares method described in Section 7.5 for a stationary signal leads to the result

$$a_{M,k}^{\#} = a_{M,M-k}, \quad k = 0, 1, 2, \dots, M, \quad (8.36)$$

that is, the backward prediction coefficients are the same as the forward prediction coefficients, but in reverse order [148]. The *backward prediction error* is, therefore, given by

$$\begin{aligned} e_{M,b}(n) &= y(n-M) - \tilde{y}(n-M) \\ &= \sum_{k=0}^M a_{M,k}^{\#} y(n-k) = \sum_{k=0}^M a_{M,M-k} y(n-k). \end{aligned} \quad (8.37)$$

The Burg-lattice method: The Burg-lattice method [148] is based on minimizing the sum of the squared forward and backward prediction errors. Assuming that the input $y(n)$ is ergodic, the *performance index* ξ_m is given by

$$\xi_m = \sum_{n=m+1}^N [e_{m,f}^2(n) + e_{m,b}^2(n)], \quad (8.38)$$

where $e_{m,f}(n)$ is the forward prediction error and $e_{m,b}(n)$ is the backward prediction error, with the model order m being recursively updated as $m = 1, 2, \dots, M$. The length of the available block of data is N samples.

If we use the Levinson–Durbin method to estimate the forward prediction coefficients, we get (see Section 7.5 and Equation 7.38)

$$a_{m,k} = a_{m-1,k} + \gamma_m a_{m-1,m-k}, \quad (8.39)$$

where γ_m is the reflection coefficient for order m . Similarly, for the case of backward prediction, we get

$$a_{m,m-k} = a_{m-1,m-k} + \gamma_m a_{m-1,m}, \quad (8.40)$$

including the substitution $a_{m,k}^{\#} = a_{m,m-k}$.

Combining the relationships in Equations 8.34, 8.37, 8.39, and 8.40 leads to the lattice structure for computation of the forward and backward prediction errors, where the two prediction error series are interrelated recursively as [148]

$$e_{m,f}(n) = e_{m-1,f}(n) + \gamma_m e_{m-1,b}(n-1) \quad (8.41)$$

and

$$e_{m,b}(n) = e_{m-1,b}(n-1) + \gamma_m e_{m-1,f}(n). \quad (8.42)$$

(All coefficients are assumed to be real-valued in this derivation; Haykin [148] allows for all coefficients to be complex-valued.) Figure 8.17 illustrates a basic unit of the lattice structure that performs the recursive operations in Equations 8.41 and 8.42. The reflection coefficient γ_m may be chosen so as to minimize the performance index given in Equation 8.38, that is, by setting

$$\frac{\partial \xi_m}{\partial \gamma_m} = 2 \sum_{n=m+1}^N \left[e_{m,f}(n) \frac{\partial e_{m,f}(n)}{\partial \gamma_m} + e_{m,b}(n) \frac{\partial e_{m,b}(n)}{\partial \gamma_m} \right] = 0. \quad (8.43)$$

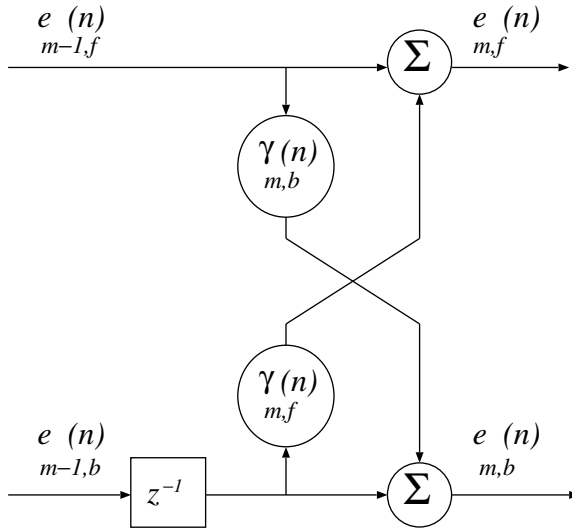


Figure 8.17 Basic unit of the lattice structure that performs the recursive operations in Equations 8.41 and 8.42 as well as the recursive operations in Equations 8.52 and 8.53. In the case of the former, due to the stationarity of the processes involved, the forward and backward reflection coefficients are the same and are independent of time (γ_m). Adapted with permission from S. Krishnan, Adaptive Signal Processing Techniques for Analysis of Knee Joint Vibroarthrographic Signals, Ph.D. Thesis, University of Calgary, 1999.

Partial differentiation of Equations 8.41 and 8.42 with respect to γ_m yields

$$\frac{\partial e_{m,f}(n)}{\partial \gamma_m} = e_{m-1,b}(n-1) \quad (8.44)$$

and

$$\frac{\partial e_{m,b}(n)}{\partial \gamma_m} = e_{m-1,f}(n). \quad (8.45)$$

Substituting the results above in Equation 8.43, we get

$$\sum_{n=m+1}^N [e_{m,f}(n) e_{m-1,b}(n-1) + e_{m,b}(n) e_{m-1,f}(n)] = 0. \quad (8.46)$$

Substituting Equations 8.41 and 8.42 in Equation 8.46, we get

$$\begin{aligned} \sum_{n=m+1}^N & [\{e_{m-1,f}(n) + \gamma_m e_{m-1,b}(n-1)\} e_{m-1,b}(n-1) \\ & + \{e_{m-1,b}(n-1) + \gamma_m e_{m-1,f}(n)\} e_{m-1,f}(n)] = 0. \end{aligned} \quad (8.47)$$

The reflection coefficients γ_m can then be calculated as

$$\gamma_m = -2 \frac{\sum_{n=m+1}^N e_{m-1,f}(n) e_{m-1,b}(n-1)}{\sum_{n=m+1}^N [e_{m-1,f}^2(n) + e_{m-1,b}^2(n-1)]}. \quad (8.48)$$

The magnitudes of the reflection coefficients are less than unity. The Burg formula always yields a minimum-phase design for the lattice predictor.

The prediction coefficients or the AR-model parameters can be computed from the reflection coefficients by using the relationship in Equation 8.39. The order m is updated recursively as $m = 1, 2, \dots, M$, with $a_{m,0} = 1$, and $a_{m-1,k} = 0$ for $k > m-1$. From Equation 8.39 and Figure 8.17, it can be observed that the AR coefficients can be computed for any model order by simply adding one or more lattice stages without affecting the earlier computations for lower orders. This is one of the main advantages of the Burg-lattice AR-modeling algorithm, especially in situations where the order of the system being modeled is not known in advance.

RLSL algorithm for adaptive segmentation: A general schematic representation of the RLSL filter structure is given in Figure 8.18. Two levels of updating are used in the RLSL algorithm:

1. *Order-update:* This involves updating the forward prediction error $e_{m,f}(n)$, the backward prediction error $e_{m,b}(n)$, the forward prediction error power $\varepsilon_{m,f}(n)$, and the backward prediction error power $\varepsilon_{m,b}(n)$. Here, m indicates the model order and n indicates the time instant.
2. *Time-update:* This involves time-updating of the parameters that ensure adaptation, including the forward reflection coefficients $\gamma_{m,f}(n)$ and backward reflection coefficients $\gamma_{m,b}(n)$. Note that, in the general nonstationary environment, $\gamma_{m,f}(n) \neq \gamma_{m,b}(n)$.

Order-updating and time-updating together enable the RLSL algorithm to achieve fast convergence and excellent tracking capability.

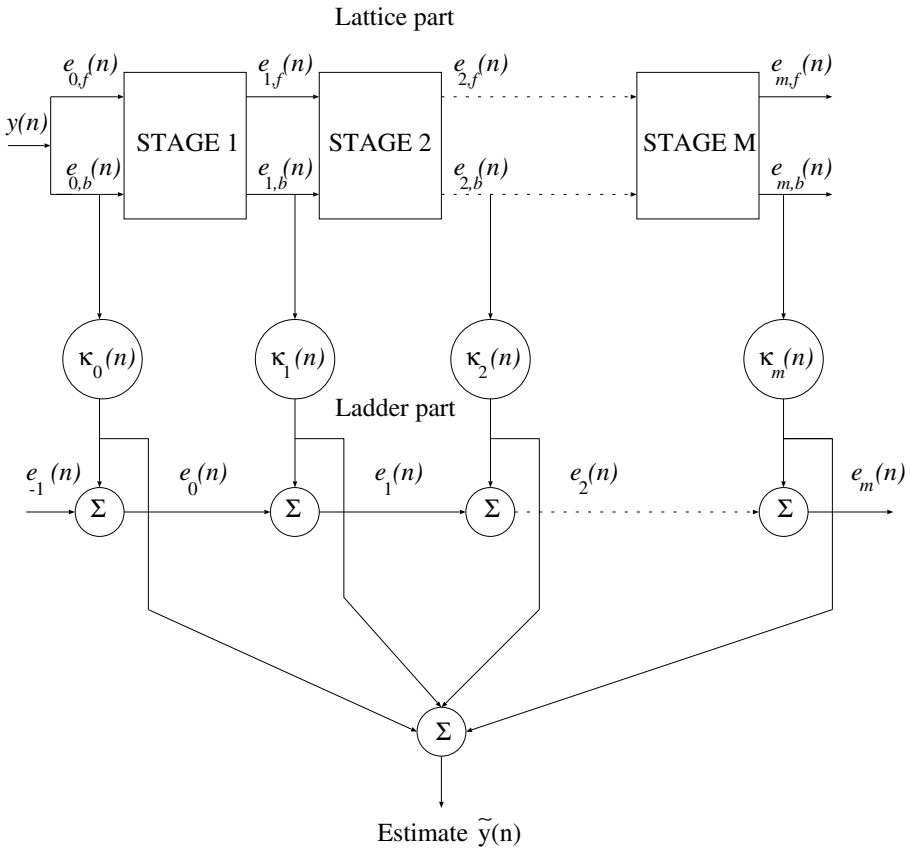


Figure 8.18 General schematic representation of the RLSL filter structure for adaptive segmentation of nonstationary signals. Adapted with permission from S. Krishnan, Adaptive Signal Processing Techniques for Analysis of Knee Joint Vibroarthrographic Signals, Ph.D. Thesis, University of Calgary, 1999.

The RLSL algorithm can be expressed in three stages [148, 185, 187]:

1. *Initialization of the algorithm and lattice for filter order M* : The parameters of the algorithm are initialized at $n = 0$ and for each order $m = 1, 2, \dots, M$ by setting the forward prediction error power $\varepsilon_{m-1,f}(0)$ and the backward prediction error power $\varepsilon_{m-1,b}(0)$ equal to a small positive constant; the forward reflection coefficients $\gamma_{m,f}(0) = 0$; the backward reflection coefficients $\gamma_{m,b}(0) = 0$; the conversion factor $\gamma_{0,c}(0) = 1$; and an auxiliary variable $\Delta_{m-1}(0) = 0$.

For each time instant $n \geq 1$, the following zeroth-order variables are generated: the forward prediction error $e_{0,f}(n)$ equal to the data input $y(n)$; the backward

prediction error $e_{0,b}(n) = y(n)$; $\varepsilon_{0,f}(n) = \varepsilon_{0,b}(n) = \lambda \varepsilon_{0,f}(n-1) + |y(n)|^2$, where λ is the forgetting factor, and $\gamma_{0,c}(n) = 1$.

The variables involved in joint process estimation, for each order $m = 0, 1, \dots, M$ at time $n = 0$, are initialized by setting the scalar $\rho_m(0) = 0$, and for each instant $n \geq 1$ the zeroth-order variable of *a priori* estimation error $e_0 = d(n)$, where $d(n)$ is the desired response of the system.

2. *Prediction part of the RLSL algorithm:* For $n = 1, 2, \dots, N_s$, where N_s is the number of signal samples available, the various order-updates are computed in the sequence $m = 1, 2, \dots, M$, where M is the final order of the least squares predictor, as follows:

$$\Delta_{m-1}(n) = \lambda \Delta_{m-1}(n-1) + \frac{e_{m-1,b}(n-1) e_{m-1,f}(n)}{\gamma_{m-1,c}(n-1)}, \quad (8.49)$$

where $\Delta_{m-1}(n)$ is the cross-correlation between the delayed backward prediction error $e_{m-1,b}(n-1)$ and the forward prediction error $e_{m-1,f}(n)$ of the lattice filter.

The forward reflection coefficient $\gamma_{m,f}(n)$ is then updated as

$$\gamma_{m,f}(n) = -\frac{\Delta_{m-1}(n)}{\varepsilon_{m-1,b}(n)}. \quad (8.50)$$

Similarly, the backward reflection coefficient is updated as

$$\gamma_{m,b}(n) = -\frac{\Delta_{m-1}(n)}{\varepsilon_{m-1,f}(n-1)}. \quad (8.51)$$

In general, $\varepsilon_{m-1,f}(n)$ and $\varepsilon_{m-1,b}(n-1)$ are unequal, so that in the RLSL algorithm, unlike in the Burg algorithm described earlier in this section, we have $\gamma_{m,f}(n) \neq \gamma_{m,b}(n)$.

From the lattice structure as described earlier in the context of Equations 8.41 and 8.42 and depicted in Figure 8.17, and noting that the reflection coefficients $\gamma_{m,f}(n)$ and $\gamma_{m,b}(n)$ are now different and time-variant parameters, we can write the order-update recursion of the forward prediction error as (see Figure 8.17)

$$e_{m,f}(n) = e_{m-1,f}(n) + \gamma_{m,f}(n) e_{m-1,b}(n-1), \quad (8.52)$$

and the order-update recursion of the backward prediction error as

$$e_{m,b}(n) = e_{m-1,b}(n-1) + \gamma_{m,b}(n) e_{m-1,f}(n). \quad (8.53)$$

The prediction error powers are updated as

$$\varepsilon_{m,f}(n) = \varepsilon_{m-1,f}(n) + \gamma_{m,f}(n) \Delta_{m-1}(n) \quad (8.54)$$

and

$$\varepsilon_{m,b}(n) = \varepsilon_{m-1,b}(n-1) + \gamma_{m,b}(n) \Delta_{m-1}(n). \quad (8.55)$$

The conversion factor $\gamma_{m,c}(n-1)$ is updated as

$$\gamma_{m,c}(n) = \gamma_{m-1,c}(n) - \frac{e_{m-1,b}^2(n)}{\varepsilon_{m-1,b}(n)}. \quad (8.56)$$

The equations in this step constitute the basic order-update recursions for the RLSL predictor. The recursions generate two sequences of prediction errors: the forward prediction error and the backward prediction error. The two error sequences play key roles in the recursive solution of the linear least-squares problem.

3. *Filtering part of the RLSL algorithm:* For $n = 1, 2, \dots, N_s$, the various order-updates are computed in the sequence $m = 0, 1, \dots, M$ as follows:

$$\rho_m(n) = \lambda \rho_m(n-1) + \frac{e_{m,b}(n)}{\gamma_{m,c}(n)} e_{m-1}(n). \quad (8.57)$$

The regression coefficients $\kappa_m(n)$ of the joint process estimator are defined in terms of the scalar $\rho_m(n)$ as

$$\kappa_m(n) = \frac{\rho_m(n)}{\varepsilon_{m,b}(n)}. \quad (8.58)$$

The order-update recursion of the *a posteriori* estimation error $e_m(n)$ is then given as

$$e_m(n) = e_{m-1}(n) - \kappa_m(n) e_{m,b}(n). \quad (8.59)$$

The dynamics of the input signal, that is, the statistical changes occurring in the signal, are reflected in the lattice filter parameters. Parameters such as the reflection coefficients (γ_f and γ_b) and the *MS* value of the estimation error {that is, $E[e_m^2(n)]$ } may therefore be used to monitor the statistical changes.

The conversion factor γ_c that appears in the algorithm can be used as a good statistical detection measure of the “unexpectedness” of the recent data samples. As long as the data belong to the same distribution, the variable γ_c will be near unity. If the recent data samples belong to a different distribution, γ_c will tend to fall from unity. This will cause the factor $\frac{1}{\gamma_c}$ appearing in the time-update formula (Equation 8.49) to be large, which leads to abrupt changes in the lattice parameters. The quantities γ_c , $\frac{1}{\gamma_c}$, or $\frac{1}{1-\gamma_c}$ may be used for fast tracking of changes in the input data and to test for segment boundaries in a nonstationary environment.

Illustration of application: The advantage in using the RLSL filter for segmentation of VAG signals is that the statistical changes in the signals are well reflected in the filter parameters, and hence segment boundaries can be detected by monitoring any one of the filter parameters, such as the MSE, conversion factor, or the reflection coefficients. Krishnan et al. [95] and Krishnan [185] used the conversion factor (γ_c)

to monitor statistical changes in VAG signals. In a stationary environment, γ_c starts with a low initial value and remains small during the early part of the initialization period. After a few iterations, γ_c begins to increase rapidly toward the final value of unity. In the case of nonstationary signals, such as VAG, γ_c will fall from its steady-state value of unity whenever a change occurs in the statistics of the signal. This can be used in segmenting VAG signals into quasistationary components. The segmentation procedure proposed by Krishnan et al. [95] and Krishnan [185] is summarized as follows:

1. The VAG signal is passed twice through the segmentation filter: The first pass is used to allow the filter to converge, and the second pass is used to test the γ_c value at each sample against a threshold value for the detection of segment boundaries.
2. Whenever γ_c at a particular sample during the second pass is less than the threshold, a primary segment boundary is marked.
3. If the difference between two successive primary segment boundaries is less than the minimum desired segment length (120 samples in the work of Krishnan et al.), the later of the two boundaries is deleted.

Figures 8.19 and 8.20 show the results of application of the RLSSL segmentation method to two VAG signals. Plots of $\gamma_c(n)$ are also included in the figures. It may be observed that the value of $\gamma_c(n)$ drops whenever there is a significant change in the characteristics of the signal. Whereas the direct application of a threshold on $\gamma_c(n)$ would result in superfluous segmentation, inclusion of the condition on the minimum segment length that is meaningful in the application is seen to provide practically useful segmentation. The number of segments was observed to be, on the average, eight segments per VAG signal. Signals of patients with cartilage pathology were observed to result in more segments than normal signals.

An advantage of the RLSSL method of adaptive segmentation is that a fixed threshold may be used; Krishnan et al. [95] found a fixed threshold value of 0.9985 to give good segmentation results with VAG signals. The adaptive segmentation procedure was found to provide segments that agreed well with manual segmentation based upon auscultation and/or arthroscopy. Adaptive analysis of VAG signals is described further in Section 9.12.

8.7 Wavelets and Time frequency Analysis

Biomedical signals are often mixtures of multiple components with time-varying properties; to complicate matters, such components may have varying durations and overlap one another. For example, PCG signals could include not only S1 and S2 but also murmurs and opening snaps of valves. Furthermore, S1 and S2 are themselves made up of multiple overlapping components, murmurs are noise-like high-frequency sounds, and opening snaps are transients. We have seen that VAG signals

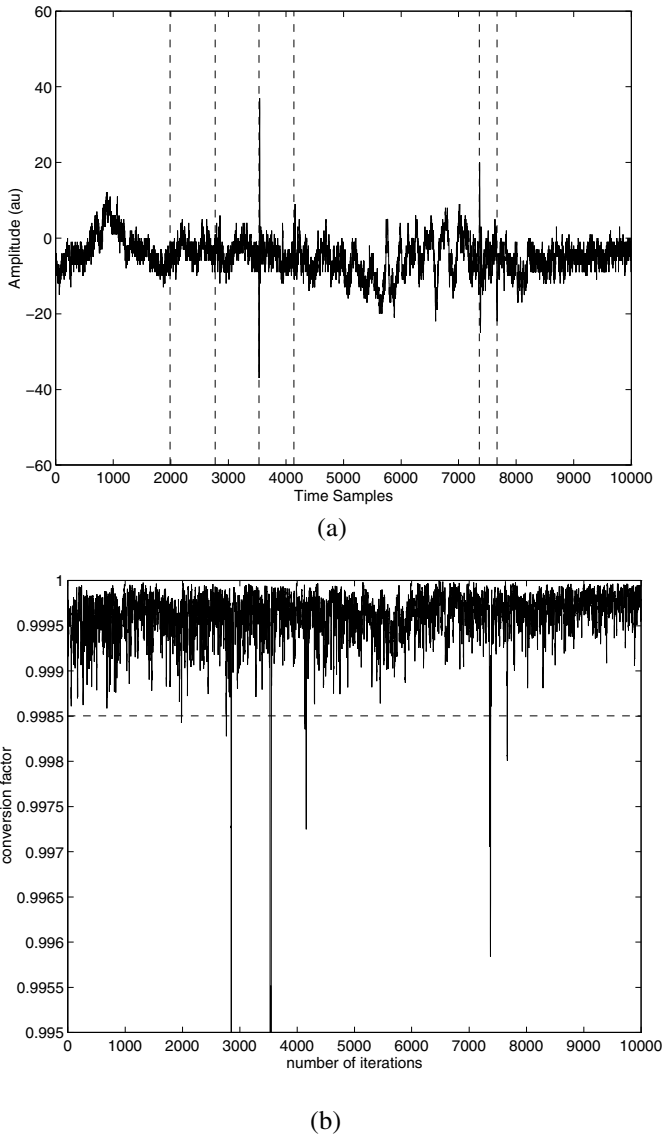


Figure 8.19 (a) VAG signal of a normal subject with the final segment boundaries given by the RLSL method shown by vertical dashed lines. (b) Plot of the conversion factor $\gamma_c(n)$; the horizontal dashed line represents the fixed threshold used to detect segment boundaries. The duration of the signal is 5 s, with $f_s = 2 \text{ kHz}$. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, C.B. Frank, and K.O. Ladly, Adaptive filtering, modelling, and classification of knee joint vibroarthrographic signals for non-invasive diagnosis of articular cartilage pathology, *Medical and Biological Engineering and Computing*, 35(6):677–684, 1997. ©IFMBE.

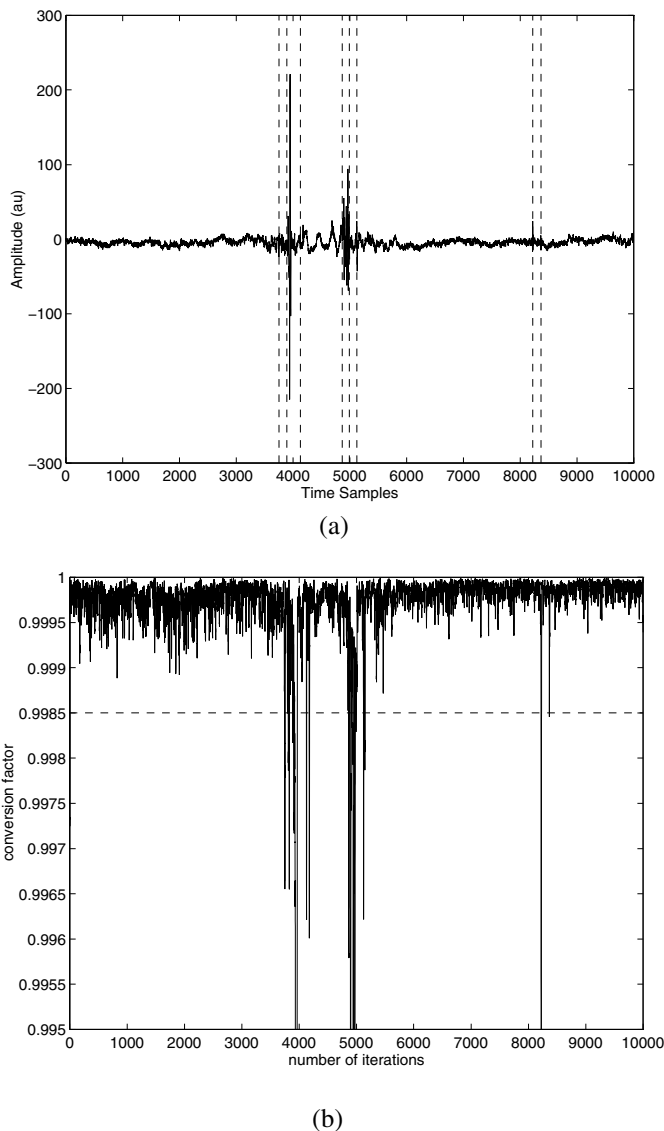


Figure 8.20 (a) VAG signal of a subject with cartilage pathology, with the final segment boundaries given by the RLSL method shown by vertical dashed lines. (b) Plot of the conversion factor $\gamma_c(n)$; the horizontal dashed line represents the fixed threshold used to detect segment boundaries. The duration of the signal is 5 s, with $f_s = 2 \text{ kHz}$. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, C.B. Frank, and K.O. Ladly, Adaptive filtering, modelling, and classification of knee joint vibroarthrographic signals for non-invasive diagnosis of articular cartilage pathology, *Medical and Biological Engineering and Computing*, 35(6):677–684, 1997. ©IFMBE.

could have multiple components related to various types and stages of deterioration of knee-joint surfaces causing sustained grinding sounds or transient clicks. When analyzing such signals, it would be desirable to obtain a joint distribution of their temporal and frequency-domain characteristics. The STFT and the various segmentation-based approaches described in the preceding sections of the present chapter are approaches that provide information of this nature, but impose several limitations.

A basic limitation of Fourier analysis is that the sinusoidal signals used as the basis functions are of infinite duration: The integral (see Equation 3.86) or summation (see Equation 3.75) used to compute the projection of the given signal on to each of the basis functions is defined over all time (or the entire duration of the given signal). Therefore, the properties of the given signal are averaged over its duration. Such global averaging provides good frequency resolution but no temporal resolution: The procedure obscures transients and components or events that exist over short durations of time. The STFT provides some improvement in temporal resolution by using short windows along the time axis but suffers from related limitations as described in Sections 8.4.1 and 8.4.2. Another limitation of Fourier analysis is that the basis function is limited to sinusoids; this limitation applies also to the STFT.

Problem: How can we overcome the limitations of Fourier analysis and study the time-varying characteristics of nonstationary and multicomponent signals with improved accuracy and flexibility? How can we tailor the tool or procedure of analysis to adapt to the particular nature of a given signal?

Solution: The answers to the questions stated above are provided by wavelets and methods for joint TF analysis that were developed over the past few decades and have gained substantial popularity in the analysis of biomedical signals [286, 406, 408, 420–429]. TFDs may be used for representation and analysis of nonstationary signals; furthermore, TF and other decomposition methods may be used for parameterization of signals and feature extraction for classification. The following sections provide brief descriptions of wavelet analysis, TFDs, and signal decomposition methods with illustrations of their application.

8.7.1 Approximation of a signal using wavelets

The analysis of a signal using a transform may be viewed as decomposition of the given signal in the form of a weighted combination of the basis functions that define the transform (see Equation 3.75). For example, in Fourier analysis, the given signal is expressed as a weighted combination of mutually orthonormal sinusoids (see Equation 3.76). If only some of the basis functions are retained and others are rejected (or given selective weighting), we obtain an approximation of the original signal, commonly referred to as a filtered version of the original signal. We will now generalize this concept and extend its definition and application.

In linear approximation, the given signal is projected over M mutually orthogonal basis functions that are predetermined or chosen a priori. A linear approximation of a signal $x(t)$ may be written as

$$\tilde{x}(t) = \sum_{m=0}^{M-1} \langle x, \psi_m \rangle \psi_m(t), \quad (8.60)$$

where $\langle x, \psi_m \rangle$ denotes the projection, inner product, or dot product of $x(t)$ with one of the M orthogonal basis vectors $\psi_m(t)$; see also Equation 3.80. An optimal linear approximation is provided by the Karhunen–Loève transform (KLT) [430]; see Section 8.8.1.

An adaptive approximation may be achieved by tailoring the M orthogonal basis functions to the properties of the given signal. The use of signal-adaptive basis functions leads to nonlinear decomposition, which may be expressed as

$$\tilde{x}(t) = \sum_{m \in B_M} \langle x, \psi_m \rangle \psi_m(t), \quad (8.61)$$

where B_M denotes a group of basis functions from a dictionary that provides the first M inner-product values $\langle x, \psi_m \rangle$ arranged in decreasing order. The functions in B_M are chosen as functions that correlate best with $x(t)$ and may be interpreted as the main features of $x(t)$. Instead of sinusoids as in the Fourier transform, one could define a set of wavelets that facilitate analysis of the characteristics of a certain type of signals (see Section 8.7.6 for examples of wavelets).

Filtering may be performed using wavelets by approximating the given signal with a small number of nonzero wavelet coefficients. If $w_m = \langle x, \psi_m \rangle$ denotes the wavelet coefficients, hard thresholding is performed as

$$\tilde{w}_m = \begin{cases} w_m & \text{if } |w_m| \geq T, \\ 0 & \text{if } |w_m| < T, \end{cases} \quad (8.62)$$

where T denotes a threshold value. Soft thresholding is given by

$$\tilde{w}_m = \begin{cases} \text{sgn}(w_m)(|w_m| - T) & \text{if } |w_m| \geq T, \\ 0 & \text{if } |w_m| < T. \end{cases} \quad (8.63)$$

The filtered signal (or the approximation) is given by

$$\tilde{x}(t) = \sum_{m \in B_M} \tilde{w}_m \psi_m(t). \quad (8.64)$$

The wavelet transform provides an approximation as above, where the basis functions or wavelets are obtained by dilating and translating a prototype function, $\psi(t)$, also known as the mother wavelet, written as

$$\psi_{\tau,s}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-\tau}{s}\right), \quad (8.65)$$

where s denotes the dilation or scale parameter and τ is the translation or shift parameter. The parameter τ represents the position on the time axis where the wavelet ψ is placed; the notion of scale represented by the parameter s is related to the concept of frequency or a band of frequencies. Formally, the continuous wavelet transform (CWT) is defined as

$$X(\tau, s) = \frac{1}{\sqrt{s}} \int_{-\infty}^{\infty} x(t) \psi^*\left(\frac{t-\tau}{s}\right) dt. \quad (8.66)$$

The integral in the CWT equation represents the projection of the given signal $x(t)$ on to the scaled wavelet with a shift so as to be positioned at $t = \tau$. The coefficient $X(\tau, s)$ resulting from the projection indicates the commonality between the two functions or the extent of the wavelet that is present in the signal being analyzed. The CWT equation also represents analysis of the signal using several wavelets so as to derive the corresponding projections or coefficients $X(\tau, s)$.

In practice, the CWT may be implemented with suitable discretization of the (τ, s) space. Typically, the time axis is discretized to the same sampling interval as that of the signal being processed. Due to substantial overlap between the shifted and scaled wavelets used, the CWT results in a redundant representation of the signal being analyzed. The (τ, s) space is referred to as the time–scale space and facilitates TF analysis if the relationship between the scaled wavelets and their spectral characteristics are taken into account (see Section 8.7.6).

Functions to be used as wavelets are required to possess certain properties. The property of finite energy is expressed as

$$\int_{-\infty}^{\infty} |\psi(t)|^2 dt < \infty. \quad (8.67)$$

The admissibility condition is stated as

$$C_\psi = \int_0^\infty \frac{|\Psi(f)|^2}{f} df < \infty, \quad (8.68)$$

where $\Psi(f)$ is the Fourier transform of $\psi(t)$ and C_ψ is known as the admissibility constant. If the wavelet is complex, its Fourier transform should be real and be zero for negative frequencies.

The inverse CWT is defined as

$$x(t) = \frac{1}{C_\psi} \int_{\tau=-\infty}^{\infty} \int_{s=0}^{\infty} X(\tau, s) \frac{1}{\sqrt{s}} \psi\left(\frac{t-\tau}{s}\right) d\tau \frac{ds}{s^2}. \quad (8.69)$$

The integral in the inverse CWT represents reconstruction or synthesis of the signal $x(t)$ using a weighted combination of scaled and shifted wavelets $\psi_{\tau,s}(\tau)$; each wavelet is weighted by the corresponding CWT coefficient $X(\tau, s)$.

The DWT is defined as

$$X(m, n) = \int_{-\infty}^{\infty} x(t) \psi_{m,n}^*(t) dt \quad (8.70)$$

with

$$\psi_{m,n}(t) = \frac{1}{\sqrt{s_0^m}} \psi \left(\frac{t - n \tau_0 s_0^m}{s_0^m} \right), \quad (8.71)$$

where the integers m and n control dilation and translation of the wavelet on a discrete grid; the parameters s_0 and τ_0 are the step parameters for dilation and translation, respectively. (In the equation given above, only the wavelet parameters have been shown with discrete variables; if the input signal has been sampled, the functions x and ψ need to be changed to discrete-time notation and the integral should be changed to summation.) The dyadic grid is obtained by setting $s_0 = 2$ and $\tau_0 = 1$; then, we have

$$\psi_{m,n}(t) = \frac{1}{\sqrt{2^m}} \psi (2^{-m}t - n). \quad (8.72)$$

The use of an appropriate grid for the DWT reduces the redundancy of representation as compared to the CWT. (The critically sampled DWT is a shift-variant transform; see Bradley [431] for related discussions.) The wavelet coefficients $X(m, n)$ obtained as above may be subjected to thresholding or selection procedures and an approximation of the signal may be reconstructed as

$$\tilde{x}(t) = \sum_m \sum_n \tilde{X}(m, n) \psi_{m,n}(t), \quad (8.73)$$

where $\tilde{\cdot}$ indicates the result of approximation or filtering and the range of summation depends upon the choices made and the application.

The wavelet approach permits simultaneous analysis of the local characteristics of a given nonstationary signal in both the temporal and spectral domains. The method facilitates the analysis of short-duration high-frequency signal components as well as long-duration low-frequency components. A multicomponent signal may be analyzed using multiple wavelets that match the expected characteristics of the various components so as to extract measures related to their properties in separate channels. Several types of wavelets and related procedures have been described in the literature [286, 406, 408, 420–429]; the presentation in this chapter is limited to a few concepts that can be related to the remaining topics in the chapter. The use of appropriate wavelets can assist in the identification and analysis of transient, aperiodic, multicomponent, and nonstationary signals.

The approach of selecting the best basis among a dictionary of bases by minimizing a cost function is known as the method of wavelet packets (WP) [432]. The WP approach uses a family of orthogonal bases that include different types of localized

TF functions (also known as TF atoms). The WP approach decomposes the given signal into TF atoms that are adapted to the TF structures expected or known to be present in the signal. The WP coefficients may then be used to obtain filtered or approximate versions of the signal; the coefficients may also be used in the form of a compact representation of the signal for further analysis or classification.

8.7.2 Signal decomposition using the Matching Pursuit algorithm

Matching Pursuit (MP) is a generic iterative signal approximation framework, whose properties depend on the type of dictionary of TF atoms used. In the MP algorithm [424], the given signal $x(t)$ is projected on to a dictionary of TF atoms obtained by scaling, translating, and modulating a window function $g(t)$ as

$$x(t) = \sum_{n=0}^{\infty} a_n g_{\gamma_n}(t), \quad (8.74)$$

where

$$g_{\gamma_n}(t) = \frac{1}{\sqrt{s_n}} g\left(\frac{t - \tau_n}{s_n}\right) \exp[j(2\pi f_n t + \phi_n)]. \quad (8.75)$$

Here, a_n is an expansion coefficient, the scale factor s_n is used to control the width of the window function, the parameter τ_n controls temporal placement, and the parameters f_n and ϕ_n are the frequency and phase of the exponential function, respectively. The subscript or index γ_n represents the set of parameters $(s_n, \tau_n, f_n, \phi_n)$. If the window is a Gaussian function, that is,

$$g(t) = 2^{\frac{1}{4}} \exp(-\pi t^2), \quad (8.76)$$

the TF atoms are known as Gabor atoms.

In the MP algorithm, the given signal is iteratively projected on to a dictionary of TF atoms. The first projection decomposes the signal into two parts:

$$x(t) = \langle x, g_{\gamma_0} \rangle g_{\gamma_0}(t) + R^1 x(t), \quad (8.77)$$

where $\langle x, g_{\gamma_0} \rangle$ denotes the inner product or projection of $x(t)$ on to the first TF atom $g_{\gamma_0}(t)$; the term $R^1 x(t)$ represents the residue after approximating $x(t)$ using $g_{\gamma_0}(t)$. The process of decomposition is continued by projecting the residue on to the subsequent functions in the dictionary, and after M iterations, we have

$$x(t) = \sum_{n=0}^{M-1} \langle R^n x, g_{\gamma_n} \rangle g_{\gamma_n}(t) + R^M x(t), \quad (8.78)$$

with $R^0 x(t) = x(t)$.

The iterative decomposition process may be stopped after using a prespecified number M of the TF atoms or until the energy of the residue $R^M x(t)$ decreases below a certain value. A decay parameter, defined as [424]

$$\lambda(m) = \sqrt{1 - \frac{\|R^m x\|^2}{\|R^{m-1} x\|^2}}, \quad (8.79)$$

may also be used; the process is continued until the decay parameter does not reduce substantially any further. The selected atoms or components may be considered to represent the coherent structures present in the signal with respect to the dictionary used. The residue may be assumed to be due to random noise. The approximate signal reconstructed using M selected (coherent) structures is given by

$$\tilde{x}(t) = \sum_{n=0}^{M-1} \langle R^n x, g_{\gamma_n} \rangle g_{\gamma_n}(t), \quad (8.80)$$

which may be taken to be a filtered version of the original signal. As in the case of wavelets, the coefficients $a_n = \langle R^n x, g_{\gamma_n} \rangle$, $n = 0, 1, 2, \dots, M-1$, may be used, along with the set of corresponding parameters, as a compact representation of the signal for further analysis or classification.

8.7.3 Empirical mode decomposition

Empirical mode decomposition (EMD) [433–437] is a decomposition procedure with which a multicomponent and nonstationary signal can be decomposed into a small number of simpler signals referred to as intrinsic mode functions (IMFs). IMFs may be analyzed using the Hilbert transform and other methods to derive TFDs [433]. The decomposition procedure is based on the local characteristics of the given signal; it is a data-driven algorithm that does not use predetermined basis functions.

An IMF is required to have the following properties [433, 435]:

- Over the duration of an IMF, the number of extrema and the number of zero-crossings must be equal or differ at most by one.
- At any location, the mean of the envelopes defined by the local maxima and the local minima is zero.

An IMF is not restricted to be a narrowband signal; it can be modulated in both amplitude and frequency and also be nonstationary [433].

IMFs are extracted level by level [433, 434]. First, high-frequency local oscillations riding on the corresponding low-frequency parts of the given signal are extracted. Then, the next level of high-frequency local oscillations in the residual of the signal are extracted. This iterative procedure is continued until no complete oscillation can be identified in the residual, which is considered to be the trend in the original signal; then, the final residual is a monotonic function.

In practice, EMD is implemented through a sifting process that detects and uses the local extrema. The standard EMD algorithm works as follows [433–437]:

1. Find the locations of all of the extrema in the given signal $x(n)$.
2. Interpolate between the minima to obtain the lower signal envelope, $x_{\min}(n)$; do the same with the maxima to obtain $x_{\max}(n)$. The cubic spline function is recommended for interpolation.
3. Compute the mean $x_m(n) = [x_{\min}(n) + x_{\max}(n)]/2$.
4. Subtract the mean obtained in the previous step from the signal to obtain the oscillatory mode signal $s(n) = x(n) - x_m(n)$.
5. If $s(n)$ meets the criteria for an IMF, define $c(n) = s(n)$ as an IMF; otherwise, set $x(n) = s(n)$ and repeat the process from Step 1.
6. Initially, the procedure given above results in the first IMF. To get the remaining IMFs, define the residual as $x(n) - c(n)$ and repeat the procedure.

If $c(n)$ is an IMF obtained as above, let

$$c_a(n) = c(n) + j c_H(n), \quad (8.81)$$

where $c_H(n)$ is the Hilbert transform of $c(n)$ and $c_a(n)$ is the related analytic signal. Let

$$c_a(n) = a(n) \exp[j \theta(n)], \quad (8.82)$$

with

$$a(n) = \sqrt{c^2(n) + c_H^2(n)} \quad (8.83)$$

and

$$\theta(n) = \arctan \left[\frac{c_H(n)}{c(n)} \right]. \quad (8.84)$$

Then, according to one of the definitions used, the instantaneous frequency $\omega(n)$ is given by the derivative of $\theta(n)$ with respect to time. Analysis as above can be performed on all IMFs and the results combined to perform TFD analysis of the original signal. See Section 5.5.1, Section 5.5.3, and Huang et al. [433] for related discussions.

The original signal may be expressed using the IMFs as

$$x(n) = \text{real} \left\{ \sum_i a_i(n) \exp[j \theta_i(n)] \right\}, \quad (8.85)$$

where the index i indicates the i^{th} IMF and the summation is over all IMFs. The expression provided above gives both the amplitude and the frequency of each component as functions of time; the residual component is neglected. With EMD and

expansion of the given signal into its IMFs, the amplitude and the frequency modulation in the original signal are separated: this facilitates TFD analysis.

Figure 8.21 shows an example of application of EMD to a VAG signal. The interpretation of IMFs is not straightforward because of the data-dependent nature of the components. In the example, the first three IMFs may be considered to be high-frequency or small-scale noise and not used in further analysis. The IMFs numbered 4 to 7 may be considered to contain most of the useful information. The remaining components may represent low-frequency or large-scale variations and trends that are not of interest.

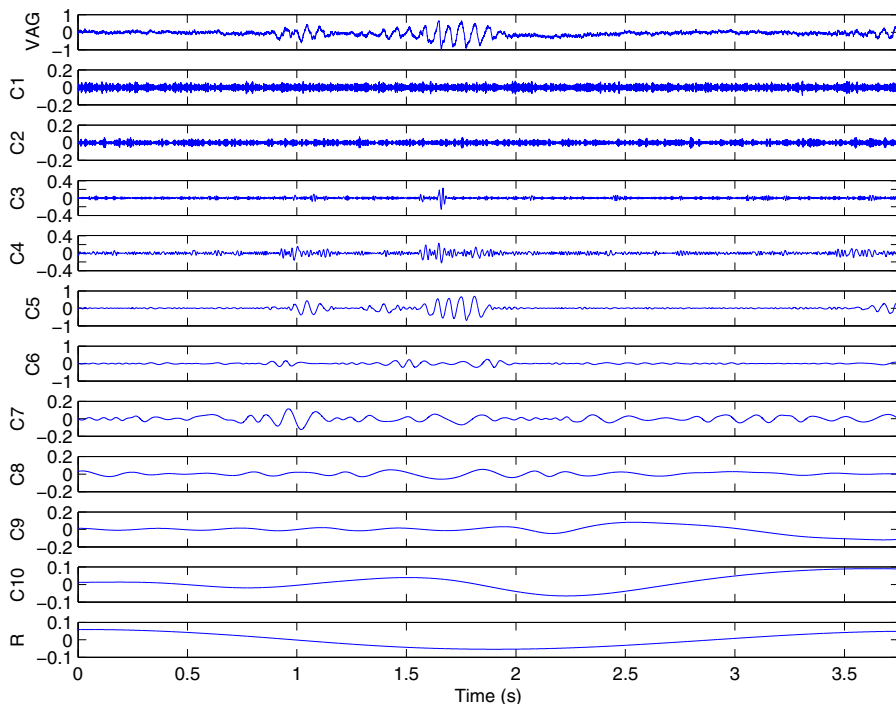


Figure 8.21 A VAG signal and 10 IMFs obtained via EMD. The last plot labeled “R” is the residual component. The number of IMFs was fixed at 10; the residual signal is not monotonic and the EMD procedure could be continued further. Figure courtesy of Yunfeng Wu, Xiamen University, China.

Wu et al. [438] applied a modified method known as ensemble EMD and detrended fluctuation analysis for removal of artifacts in knee-joint VAG signals. They found that EMD can improve the quality of VAG signals and that the results obtained were superior to those achieved via decomposition by the MP algorithm.

Echeverria et al. [437] applied EMD to analyze HRV data. The results were said to facilitate isolation of at least four components with overlapping and dynamic frequency excursions localized in the currently recognized spectral bands of autonomic

modulation. In addition, EMD was observed to permit the separation of the very-low-frequency (VLF) content as well.

Kaleem et al. [439] used EMD to analyze pathological speech signals. Instead of using sustained vowels, EMD was used to decompose randomly chosen portions of speech signals into IMFs, which were then analyzed to extract temporal and spectral features. Instantaneous features were used to capture discriminative information in signals that were suspected to be hidden at local time scales. Using a database consisting of continuous speech from 51 normal speakers and 161 speakers with speech pathology, a classification accuracy of 95.7% was obtained.

8.7.4 TFDs and their characteristics

If $x(t)$ is a signal and $\text{TFD}(t, \omega)$ is its TFD, the important criteria that the TFD is expected to satisfy are listed in the following paragraphs [406, 408].

Total energy:

$$\int \int \text{TFD}(t, \omega) dt d\omega = \int |x(t)|^2 dt = \int |X(\omega)|^2 d\omega, \quad (8.86)$$

where $X(\omega)$ is the Fourier transform of $x(t)$. This criterion indicates that, at particular values of t and ω , $\text{TFD}(t, \omega)$ gives the fractional energy of the signal; thus, $\text{TFD}(t, \omega)$ may be viewed as a joint TF density function.

Invariance: The TFD should be invariant to any linear shift in time and frequency; it is also expected that the TFD is scale-invariant.

Nonnegativity: $\text{TFD}(t, \omega) \geq 0$ for all t and ω . This criterion helps in treating the TFD as a density function.

Marginals: The instantaneous energy of the signal is given by

$$\int \text{TFD}(t, \omega) d\omega = |x(t)|^2. \quad (8.87)$$

Integration along the time axis gives the PSD of the signal as

$$\int \text{TFD}(t, \omega) dt = |X(\omega)|^2. \quad (8.88)$$

Expectation values:

$$E[g(t, \omega)] = \int \int g(t, \omega) \text{TFD}(t, \omega) dt d\omega. \quad (8.89)$$

The equation given above expresses a generalized moment of a TFD. The function $g(t, \omega)$ needs to be chosen according to the desired moment. The instantaneous mean frequency (IMF) is given by the time-varying first moment of the TFD along the frequency axis as

$$E_t[\omega] = \frac{1}{|x(t)|^2} \int \omega \text{TFD}(t, \omega) d\omega. \quad (8.90)$$

The group delay is given by the frequency-varying first moment of the TFD along the time axis as

$$E_{\omega} [t] = \frac{1}{|X(\omega)|^2} \int t \text{TFD}(t, \omega) dt. \quad (8.91)$$

See Boashash [382] for discussions on analysis of the instantaneous frequency of nonstationary signals.

Finite support: If $x(t)$ is zero at $t = t_1$, then $\text{TFD}(t_1, \omega)$ should be zero. Similarly, if $X(\omega)$ is zero at $\omega = \omega_1$, then $\text{TFD}(t, \omega_1)$ should be zero.

The simplest TFD is the spectrogram, which is obtained by taking the squared magnitude of the STFT defined in Equation 8.9.

The Wigner–Ville distribution (WVD) of a signal $x(t)$ is given by [406]

$$\text{WVD}(t, \omega) = \int_{-\infty}^{\infty} x(t + \tau/2) x^*(t - \tau/2) \exp(-j\omega\tau) d\tau. \quad (8.92)$$

A drawback of the WVD is that, in the case of multicomponent signals, cross-terms (referred to as “ghost frequencies”) are generated in the TFD that could lead to misinterpretation.

The Cohen’s class of generalized TFD (GTFD) is given by

$$\begin{aligned} \text{GTFD}(t, \omega) &= \frac{1}{4\pi^2} \int \int \int x(u + \tau/2) x^*(u - \tau/2) \Phi(\theta, \tau) \\ &\quad \times \exp(-j\theta t - j\tau\omega + j\theta u) du d\tau d\theta, \end{aligned} \quad (8.93)$$

where $\Phi(\theta, \tau)$ is the TF transformation kernel. The kernel acts as a lowpass filter and minimizes cross-terms. The selection of the kernel, which may be signal-independent or signal-dependent, determines the properties of the resulting distribution [405, 406].

Considering a signal-independent kernel, cross-terms may be minimized by 2D filtering of the WVD; the smoothed version of the WVD (SWVD) can be expressed as

$$\text{SWVD}(t, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi(u, \Omega) \text{WVD}(t - u, \omega - \Omega) du d\Omega, \quad (8.94)$$

where $\phi(u, \Omega)$ is a kernel function with lowpass-filter characteristics. The kernel function may be expressed as

$$\phi(u, \Omega) = g(u)H(\Omega), \quad (8.95)$$

where $g(u)$ and $H(\Omega)$ are smoothing functions or windows in the time and frequency domains, respectively. The smoothing windows are independent of the signal; Gaussian functions are commonly used as smoothing windows. The resulting

TFD, known as the smoothed pseudo-WVD (SPWVD) [440] is expressed as

$$\text{SPWVD}(t, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(u) H(\Omega) \text{WVD}(t-u, \omega-\Omega) du d\Omega. \quad (8.96)$$

Smoothing windows suppress cross-terms but smear localized components, leading to less accurate TF localization of signal components as compared to the WVD. A reassignment method was proposed by Auger and Flandrin [441] to improve TF localization in smoothed TFDs, such as SPWVDs. In the reassignment method, the window is moved from a point (t, ω) to the center of energy, $(\bar{t}, \bar{\omega})$, of the TFD. The center of energy is given by the group delay and the mean frequency computed as the mean values of the WVD:

$$\bar{t}(t, \omega) = t - \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u g(u) H(\Omega) \text{WVD}(t-u, \omega-\Omega) du d\Omega}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(u) H(\Omega) \text{WVD}(t-u, \omega-\Omega) du d\Omega}, \quad (8.97)$$

$$\bar{\omega}(t, \omega) = \omega - \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Omega g(u) H(\Omega) \text{WVD}(t-u, \omega-\Omega) du d\Omega}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(u) H(\Omega) \text{WVD}(t-u, \omega-\Omega) du d\Omega}. \quad (8.98)$$

The modified TFD, known as the reassigned SPWVD (or RSPWVD), can be expressed as

$$\begin{aligned} \text{RSPWVD}(t', \omega') &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \text{SPWVD}(t, \omega) \\ &\times \delta[t' - \bar{t}(t, \omega)] \delta[\omega' - \bar{\omega}(t, \omega)] dt d\omega. \end{aligned} \quad (8.99)$$

8.7.5 Decomposition based adaptive TFD

Krishnan et al. [273] presented the design of an adaptive TFD based on signal decomposition for the analysis of VAG signals; a schematic representation of their procedure is shown in Figure 8.22. In this method, the given signal is first decomposed into components of a specified mathematical representation; the MP algorithm described in Section 8.7.2 was used for this purpose. By knowing the components of the signal, the interaction between them can be established and used to remove or prevent cross-terms.

The TFD based on signal decomposition using the MP algorithm (or MPTFD) is obtained by taking the WVD of the TF atoms used to represent the signal as in Equation 8.80 and is given by [424]

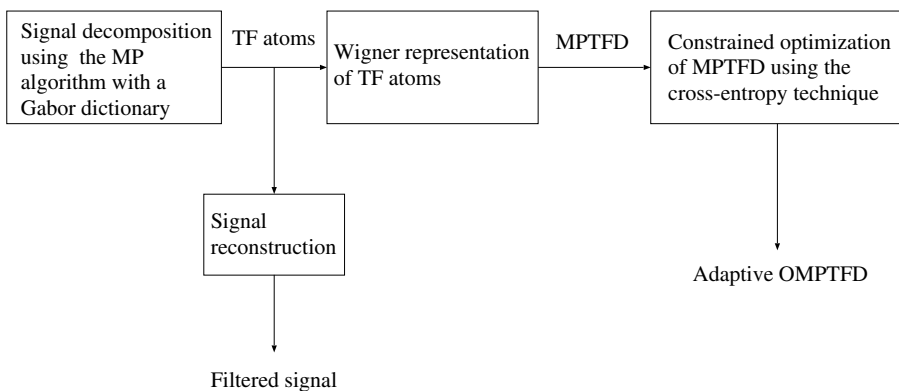


Figure 8.22 Schematic representation of the procedure to construct a decomposition-based adaptive TFD. Adapted with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, and C.B. Frank, Adaptive time-frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Transactions on Biomedical Engineering*, 47(6):773–783, June 2000. ©IEEE.

$$\begin{aligned}
 W(t, \omega) &= \sum_{n=0}^{M-1} |\langle R^n x, g_{\gamma_n} \rangle|^2 W g_{\gamma_n}(t, \omega) \\
 &+ \sum_{n=0}^{M-1} \sum_{\substack{m=0 \\ m \neq n}}^{M-1} \langle R^n x, g_{\gamma_n} \rangle \langle R^m x, g_{\gamma_m} \rangle^* W_{[g_{\gamma_n}, g_{\gamma_m}]}(t, \omega),
 \end{aligned} \tag{8.100}$$

where $W g_{\gamma_n}(t, \omega)$ is the WVD of the Gaussian window function. The double sum corresponds to the cross-terms of the WVD indicated by $W_{[g_{\gamma_n}, g_{\gamma_m}]}(t, \omega)$ and should be rejected in order to obtain a cross-term-free distribution of $x(t)$ in the TF plane. Thus, only the first term is retained, and the resulting TFD is given by

$$W'(t, \omega) = \sum_{n=0}^{M-1} |\langle R^n x, g_{\gamma_n} \rangle|^2 W g_{\gamma_n}(t, \omega). \tag{8.101}$$

Because the MPTFD is constructed using the WVD of TF atoms, most of the properties of the WVD are applicable to the MPTFD, except the marginals.

The MPTFD may be modified to satisfy the marginal requirements while preserving its other important characteristics. One way to optimize the MPTFD is by applying the cross-entropy minimization method [442–444]. Cross-entropy minimization is a general method of inference about an unknown PDF when there exists a prior estimate of the PDF and new information is available in the form of constraints on its expected values. The optimized MPTFD (or OMPTFD), denoted by $M(t, \omega)$, should satisfy the condition on the marginals, stated as

$$\int M(t, \omega) d\omega = |x(t)|^2 = m(t), \quad (8.102)$$

$$\int M(t, \omega) dt = |X(\omega)|^2 = m(\omega). \quad (8.103)$$

The two conditions given above may be treated as constraints for optimization. The desired $M(t, \omega)$ may be obtained from $W'(t, \omega)$, which is taken to be a prior estimate of the density, by minimizing the cross-entropy between them, given by

$$H(M, W') = \int \int M(t, \omega) \log \left(\frac{M(t, \omega)}{W'(t, \omega)} \right) dt d\omega. \quad (8.104)$$

Considering the marginals, the OMPTFD may be written as [443]

$$M(t, \omega) = W'(t, \omega) \exp\{-[\alpha_0(t) + \beta_0(\omega)]\}, \quad (8.105)$$

where α and β are Lagrange multipliers that may be determined using the constraints in an iterative algorithm as proposed by Loughlin et al. [444].

At the first iteration, let

$$M_1(t, \omega) = W'(t, \omega) \exp[-\alpha_0(t)]. \quad (8.106)$$

Imposing the constraint related to the time marginal given by Equation 8.102 on Equation 8.106, we obtain

$$\alpha_0(t) = \ln \left(\frac{m'(t)}{m(t)} \right), \quad (8.107)$$

where $m(t)$ is the desired time marginal and $m'(t)$ is the time marginal estimated from $W'(t, \omega)$. Then,

$$M_1(t, \omega) = W'(t, \omega) \frac{m(t)}{m'(t)}. \quad (8.108)$$

To have the desired frequency marginal $m(\omega)$, let

$$M_2(t, \omega) = M_1(t, \omega) \exp[-\beta_0(\omega)]. \quad (8.109)$$

By imposing the constraint on the frequency marginal given by Equation 8.103 on Equation 8.109, we obtain

$$\beta_0(\omega) = \ln \left(\frac{m'(\omega)}{m(\omega)} \right), \quad (8.110)$$

where $m(\omega)$ is the desired frequency marginal and $m'(\omega)$ is the frequency marginal estimated from $W'(t, \omega)$. Now, we have

$$M_2(t, \omega) = M_1(t, \omega) \frac{m(\omega)}{m'(\omega)}. \quad (8.111)$$

At this point, $M_2(t, \omega)$ may be altered and need not necessarily give the desired time marginal. Successive iteration could overcome this problem and modify the TFD to get closer to $M(t, \omega)$ [444]. This follows from the fact that the cross-entropy between the desired TFD and the estimated TFD decreases with the number of iterations [443, 444]. Because the iterative procedure is started with a positive distribution, $W'(t, \omega)$, the TFD at the n^{th} iteration, $M_n(t, \omega)$, is guaranteed to be a positive distribution.

8.7.6 Illustrations of application

Some of the commonly used wavelets are the Mexican hat (or sombrero), Morlet, Daubechies, and symmlet functions [286, 406, 408, 420–429]. The Mexican hat function is defined as the negative of the second derivative of a Gaussian, and is given by

$$\psi(t) = (1 - t^2) \exp(-0.5 t^2). \quad (8.112)$$

Figure 8.23 shows the wavelet defined above for three scales of 1, 2, and 4 (top to bottom), scaled according to Equation 8.72. It is evident that, as the scale value increases, the duration of the wavelet is increased and its frequency bandwidth is reduced; thus, there exists an inverse relationship between scale and frequency.

The Morlet function is defined as

$$\psi(t) = \frac{1}{\pi^{1/4}} [\exp(j \omega_0 t) - \exp(-0.5 \omega_0^2)] \exp(-0.5 t^2), \quad (8.113)$$

where ω_0 is the central frequency of the frequency response of the mother wavelet. The term $\exp(-0.5 \omega_0^2)$ is negligible for $\omega_0 > 5$ and may be ignored. Figures 8.24 and 8.25 show Morlet wavelets obtained using the real part of Equation 8.113 for three scales of 1, 2, and 4 and $\omega_0 = 2$ and 6 *radians/s*, respectively; the wavelets have been scaled according to Equation 8.72. The Morlet wavelets in Figure 8.24 are similar in appearance to the Mexican hat wavelets in Figure 8.23.

Figure 8.26 shows a noisy ECG signal over three cardiac cycles, sampled at 200 *Hz*. Figure 8.27 shows the STFT of the ECG signal using a window width of 32 samples, which was moved in steps of 16 samples. It is evident from the STFT that the spectrum of the signal changes substantially over its duration, with increased

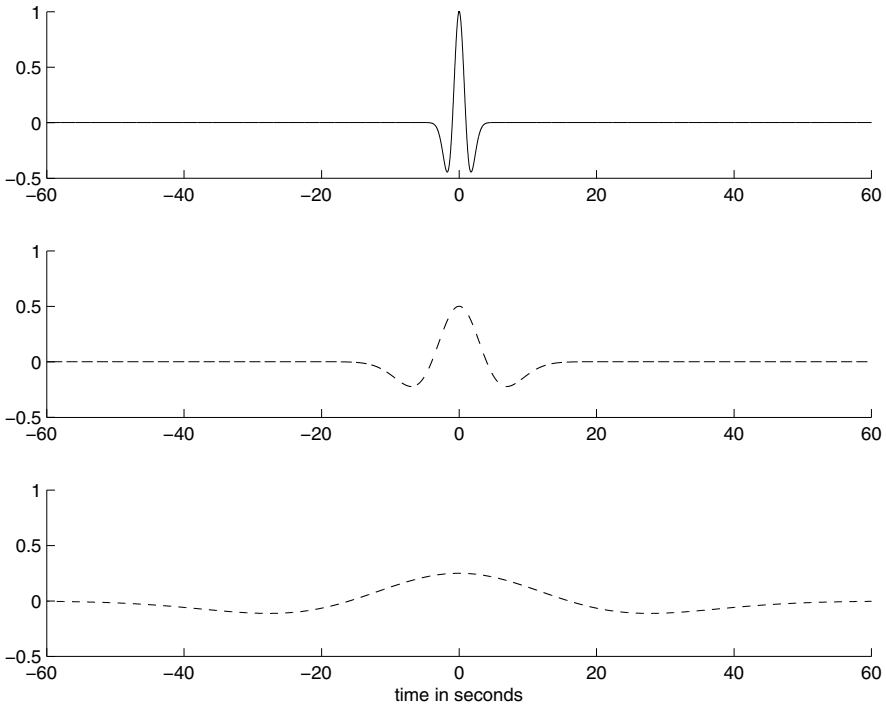


Figure 8.23 Top to bottom: The Mexican hat wavelet in Equation 8.112 for three scales of 1, 2, and 4. The wavelets have been scaled according to Equation 8.72.

high-frequency content during the intervals corresponding to the QRS complexes and predominantly low-frequency composition in periods related to the P and T waves.

In order to analyze the ECG signal using wavelets, wavelets were prepared as per Equation 8.113 (using only the real part) with $\omega_0 = 200 \text{ radians/s}$ and by changing the Gaussian term to $\exp(-5000 t^2)$. The parameters were selected so as to create wavelets that were comparable to typical QRS waves in ECG signals. Figure 8.28 shows the wavelets for four scales of 1, 2, 4, and 8. Figure 8.29 shows the ECG signal (top) and the results of the CWT for the four scales used (labeled correspondingly as wt1, wt2, wt4, and wt8 in the figure). In each case, the wavelet was translated by one sample per step. It is evident that the wavelets of smaller scale result in higher coefficients corresponding to the sharp QRS complexes, whereas the wavelets of larger scale provide larger coefficients at the locations of the slower T waves. One may, if desired, compute the CWT for several values of the scale parameter and compose the result for display as an image, known as the scalogram (see Figure 8.30).

The last plot in Figure 8.29 (labeled as wr) shows an approximation of the ECG signal reconstructed using only the coefficients for scales of 2, 4, and 8; that is, the result for the smallest scale of 1 was discarded. To obtain the reconstructed signal,

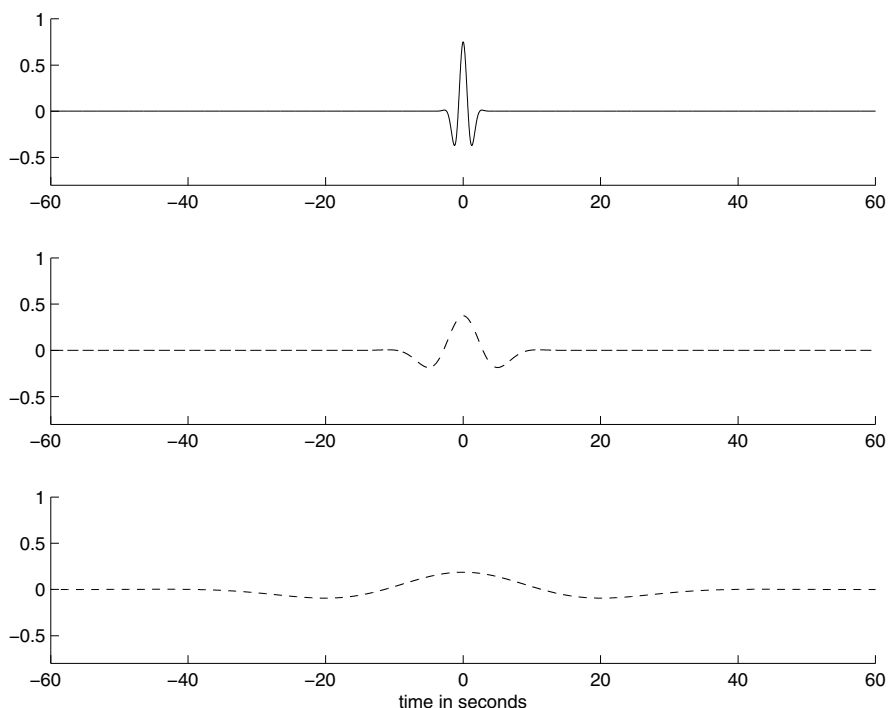


Figure 8.24 Top to bottom: The Morlet wavelet in Equation 8.113 for three scales of 1, 2, and 4, and $\omega_0 = 2$ radians/s. Only the real part has been used and the wavelets have been scaled according to Equation 8.72.

the wavelet transform for each scale was convolved with the corresponding wavelet according to Equation 8.69; however, due to the redundancy of the CWT, a reconstruction may be obtained by simply adding the results of the wavelet transform for the selected scales [428, 429]. The result of approximation is seen to retain well the P and QRS waves with most of the small-scale noise removed; however, the T waves have not been reproduced well. This is a simplified illustration of the CWT with only four scales used; in practice, one may have to use a large number of scales depending upon the nature of the signal and the desired result.

Representation and analysis of a given signal as shown above may be viewed as an application of matched filters, correlation filters, and multiscale filter banks. The CWT given in Equation 8.66 represents a convolution operation between the given signal and a wavelet. When a symmetric wavelet is used, convolution is the same as correlation. The CWT operation is then comparable to matched filtering (see Section 4.6.1). Thus, the CWT may be interpreted as seeking a match between a collection of wavelets and the given signal: the illustration in Figure 8.29 demonstrates this aspect of wavelet analysis.

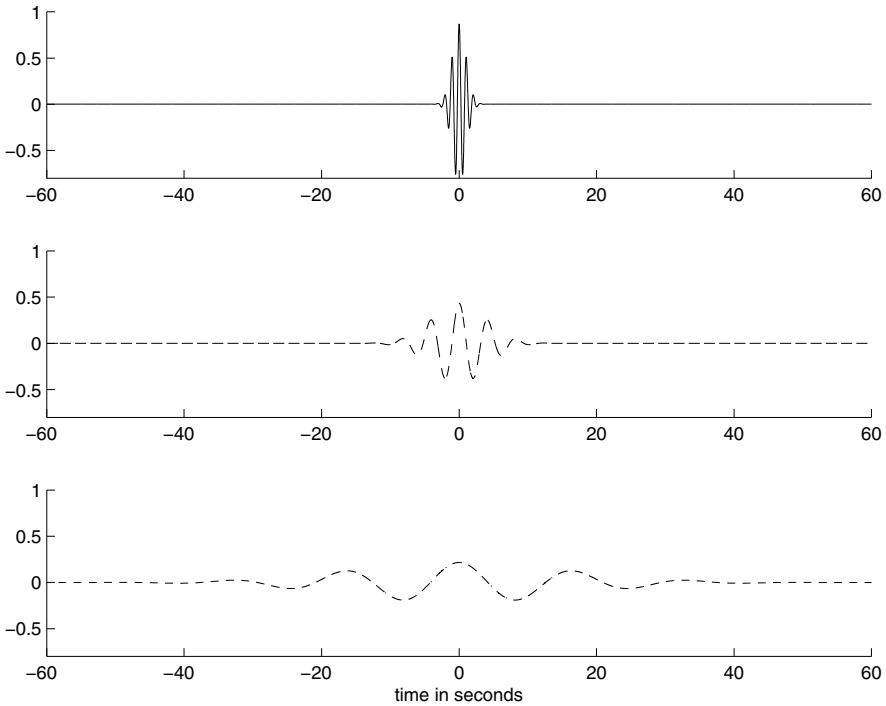


Figure 8.25 Top to bottom: The Morlet wavelet in Equation 8.113 for three scales of 1, 2, and 4, and $\omega_0 = 6$ radians/s. Only the real part has been used and the wavelets have been scaled according to Equation 8.72.

Further insight into the interpretation of wavelet analysis as an application of multiscale filter banks can be gained by examining the PSDs of the wavelets used. Figure 8.31 shows the normalized PSDs of the four wavelets in Figure 8.28; it is evident that the central frequency of the frequency response of the filter corresponding to each wavelet shifts towards lower values as the scale is increased. When several wavelets are used over a range of scales, a number of filtered versions of the given signal are obtained, as illustrated in Figure 8.29. The coefficients obtained using the CWT or DWT may also be used for further analysis of signals such as the ECG for various purposes, including detection of waves and classification [420,421,427].

In the work of Krishnan and Rangayyan [192], several methods were applied to remove noise in multicomponent and nonstationary synthetic signals as well as VAG signals. Gaussian random noise was added to the synthetic signals such that the resulting signals had SNR of 10 dB and 0 dB. The symmlet 4 wavelet [430] was used for wavelet-based filtering. A soft threshold ($T = 0.5$) was applied to the wavelet coefficients. In the case of the WP method, the best basis was selected by using the Schur concavity cost function [432], and the filtered version was obtained by applying the soft thresholding method ($T = 0.5$) to the WP coefficients. Gaussian

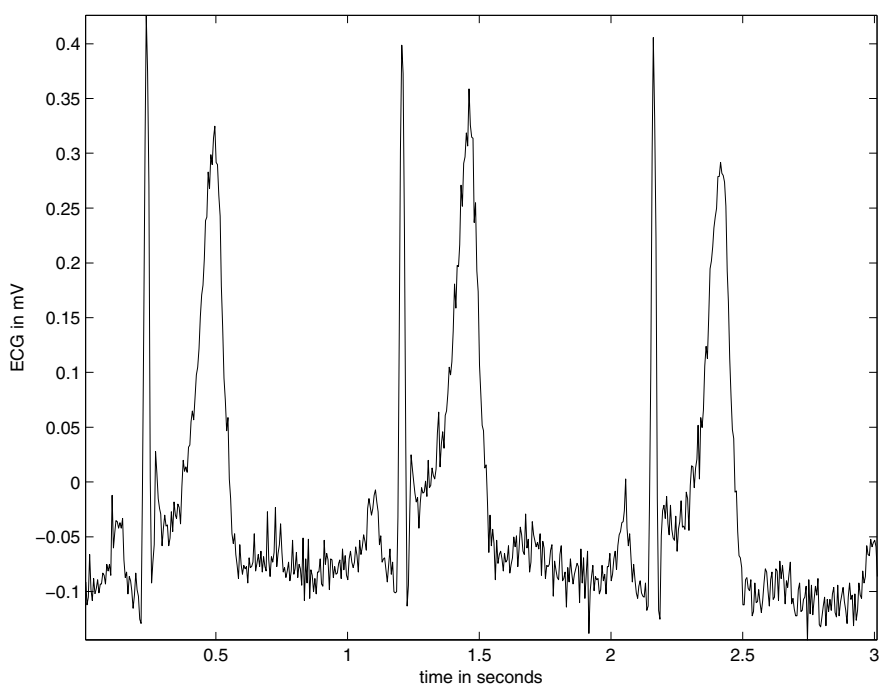


Figure 8.26 A noisy ECG signal over three cardiac cycles or beats.

functions were used for the MP method. The stopping criterion was based on the decay parameter as given by Equation 8.79.

Figure 8.32 (a) shows a synthetic signal and part (b) of the same figure shows the signal with noise added such that $SNR = 0$ dB. The filtered versions of the signal using the wavelet method, the WP method, and the MP method are shown in Figures 8.33 and 8.34. Visual comparison indicates that the MP result has preserved most of the important characteristics of the signal — in particular, the transient component.

The WP method may lead to better results if the threshold is selected in an optimal manner. The MP method locally optimizes the choice of the decomposition function for the signal residue at each stage. In the case of a multicomponent signal where different types of structures are located at different times but in the same frequency band, there may be no WP basis that is well adapted to all of them. MP is a translation-invariant method if a translation-invariant dictionary such as a Gabor dictionary is used. See Krishnan and Rangayyan [192] for more details and quantitative evaluation of the results.

The VAG signal of a subject with cartilage pathology is shown in Figure 8.35 (a), its filtered version using MP is shown in part (b) of the same figure, and the difference between the original signal and the filtered result is shown in Figure 8.36. From

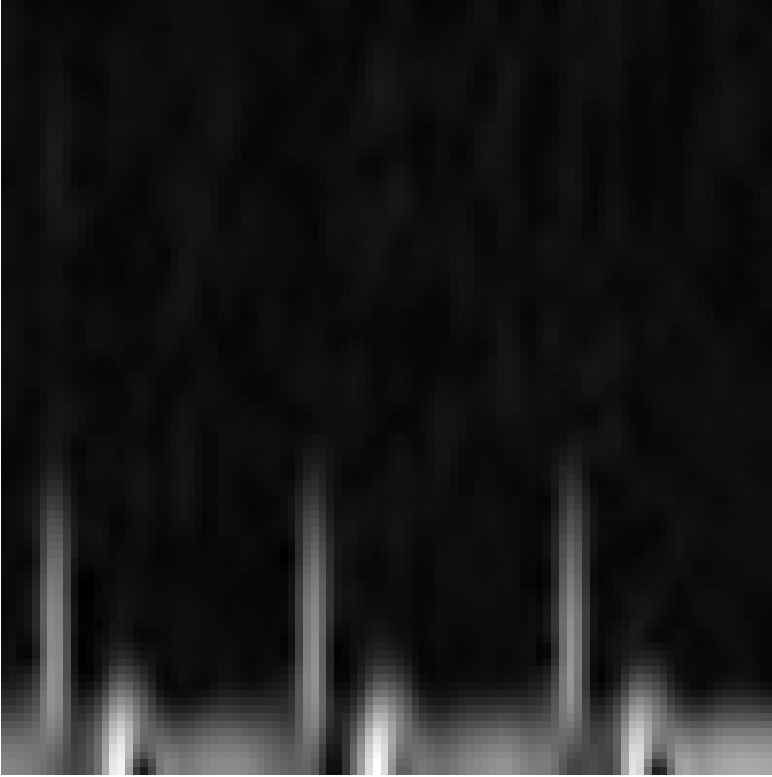


Figure 8.27 STFT of the ECG signal in Figure 8.26. The horizontal axis represents the time axis from 0 to 3 s (left to right) and the vertical axis represents frequency from 0 to 100 Hz (bottom to top).

Figure 8.36, it is evident that a substantial amount of random noise has been removed from the original signal.

The TFD of the original VAG signal in Figure 8.35 computed using the STFT is shown in Figure 8.37 (a); the TFD of the MP-filtered version is shown in part (b) of the same figure. The two TFDs were computed using the same STFT parameters. Tonal and FM components are seen more clearly in the TFD of the filtered signal than that of the original signal. For demonstrations of the use of MPTFDs for feature identification and classification of VAG signals, see Krishnan et al. [273] and Rangayyan and Krishnan [445]; see also Section 8.14. See Kovács et al. [446, 447] for illustrations of MP-based analysis of fetal PCG signals.

In the work of Krishnan [448], the synthetic signal shown in Figure 8.38 was prepared by combining overlapping chirp, impulse, and sinusoidal FM components; part (b) of the same figure shows the ideal TFD of the signal (known by design). Figure 8.39 shows a noisy version of the signal with $SNR = 10$ dB and its RSP-WVD. Even though the basic nature of the ideal TFD is visible in the RSPWVD, a

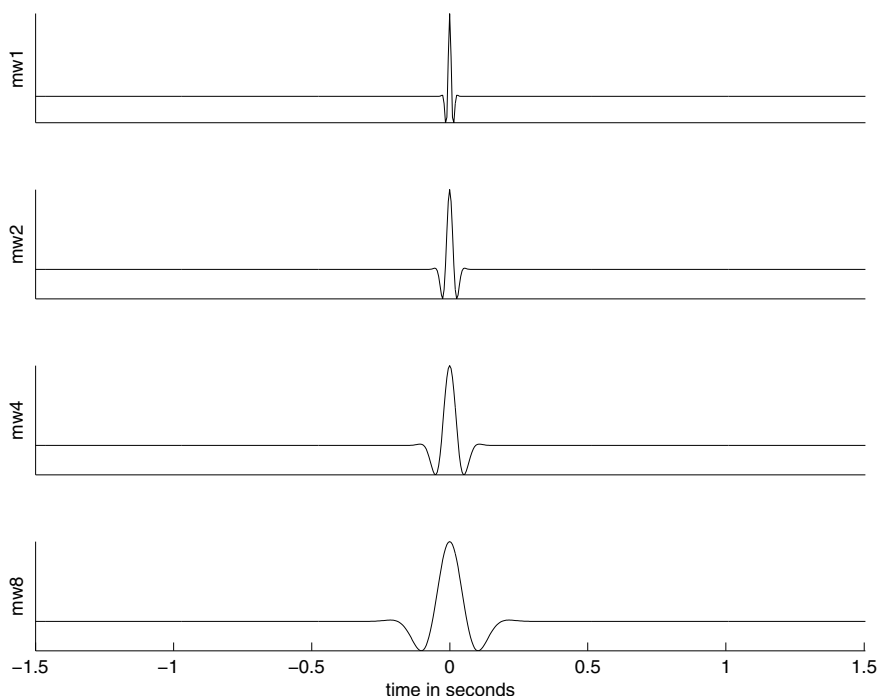


Figure 8.28 Wavelets to analyze the ECG signal in Figure 8.26. The wavelets were prepared using Equation 8.113 with $\omega_0 = 200$ radians/s and by changing the Gaussian term to $\exp(-5000 t^2)$, and are shown for four scales of 1, 2, 4, and 8. The amplitude details have been suppressed and all signals have been mapped to the full range of the ordinate available for clearer display.

large amount of interference is present. Figure 8.40 shows the MPTFD of the noisy signal, which clearly demonstrates the various components present in the signal and their TF characteristics without degrading interference.

See Section 8.14 for further illustrations and discussion of the application of TFD analysis to VAG signals. See Section 8.12 for a discussion on the application of the DWT to analyze EEG signals.

8.8 Separation of Mixtures of Signals

In studies of biomedical signals, it is common to acquire multiple channels of signals from several sources, some of which could provide mutually independent information while others could be correlated and cause redundancy. As shown in Chapter 2, studies of the cardiac, nervous, and neuromuscular systems are often conducted with multiple channels of signals, some of which could be inherently electrical while others could be basically of mechanical, acoustic, and other types, although all are

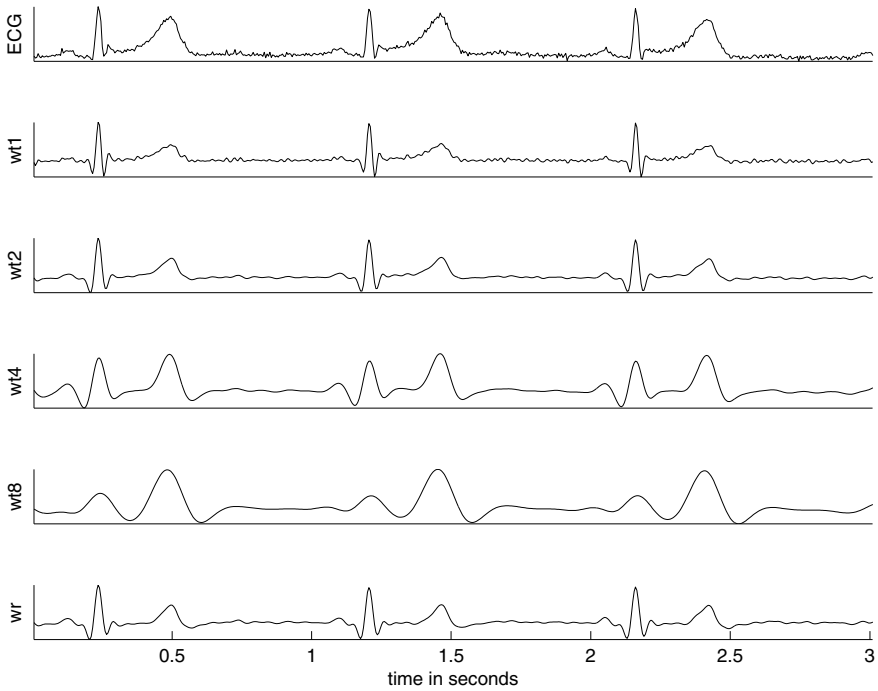


Figure 8.29 Results of analysis of the ECG signal in Figure 8.26 using the wavelets shown in Figure 8.28. Top to bottom: the original ECG signal; results of the CWT for the four scales used (wt1, wt2, wt4, and wt8); and approximation of the signal (wr) reconstructed using only wt2, wt4, and wt8. The amplitude details have been suppressed and all signals have been mapped to the full range of the ordinate available for clearer display.

acquired as electrical signals via suitable instrumentation. Regardless of the nature of the signals, cross-talk and cross-contamination can occur at their sources, along their paths of propagation, and at the sites of their acquisition. When analyzing such collections of multichannel signals, questions arise regarding their interrelationships, the potential for existence of correlation, the presence of redundancy in the data, and the quality of the information extracted. It would be desirable to remove redundant and correlated information for the sake of efficiency and extract information related to each source without contamination from others that are active at the same time. These considerations lead to the need to separate mixtures of signals into their components. Depending upon the assumptions made and the techniques used, one can extract uncorrelated components via principal component analysis (PCA) or statistically independent components via methods of independent component analysis (ICA) and BSS [65,449–463].

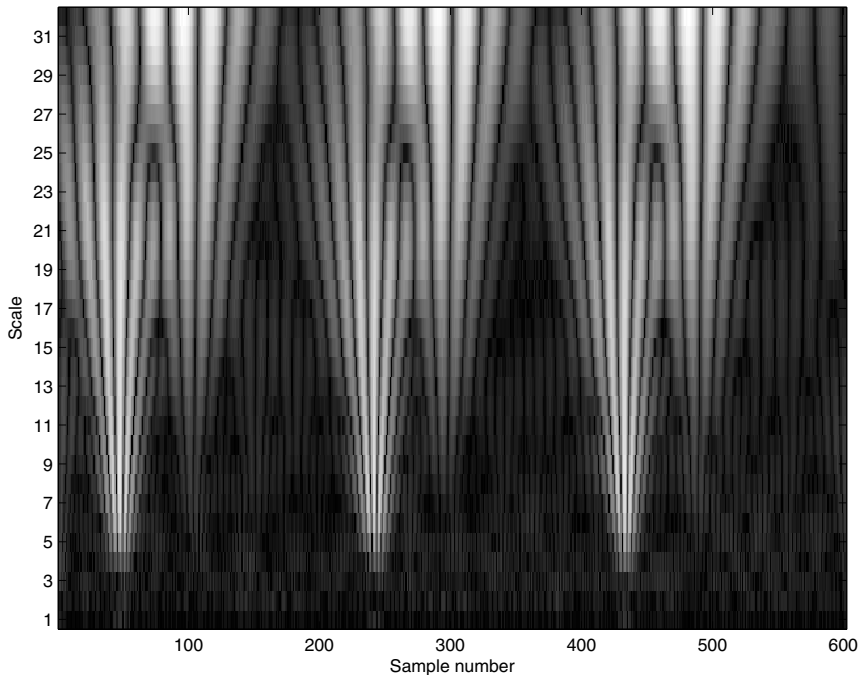


Figure 8.30 CWT of the ECG signal in Figure 8.26 using Morlet wavelets. The horizontal axis represents the time axis from 0 to 3 s sampled at 200 Hz and the vertical axis represents scale from 1 to 32.

8.8.1 Principal component analysis

In order to facilitate efficient representation of collections of signals, measurements, and features for pattern analysis and classification, it is desirable to prepare a compact set of the data with no redundancy or mutual correlation, and to keep the dimension of the result as small as possible. An approach to reduce the dimension of a data vector is PCA, which facilitates the selection of a compact set of uncorrelated components. The PCA method, which is related to the KLT [18, 175, 222], is described briefly in the following paragraphs.

Consider a set of signal samples, measurements, or features represented as a vector $\mathbf{y}$, of size $K \times 1$. The vector $\mathbf{y}$ may be represented without error by a deterministic linear transformation of the form

$$\mathbf{y} = \mathbf{W} \mathbf{x} = \sum_{k=1}^K x_k \mathbf{W}_k, \quad (8.114)$$

$$\mathbf{W} = [\mathbf{W}_1 \ \mathbf{W}_2 \ \cdots \ \mathbf{W}_K], \quad (8.115)$$

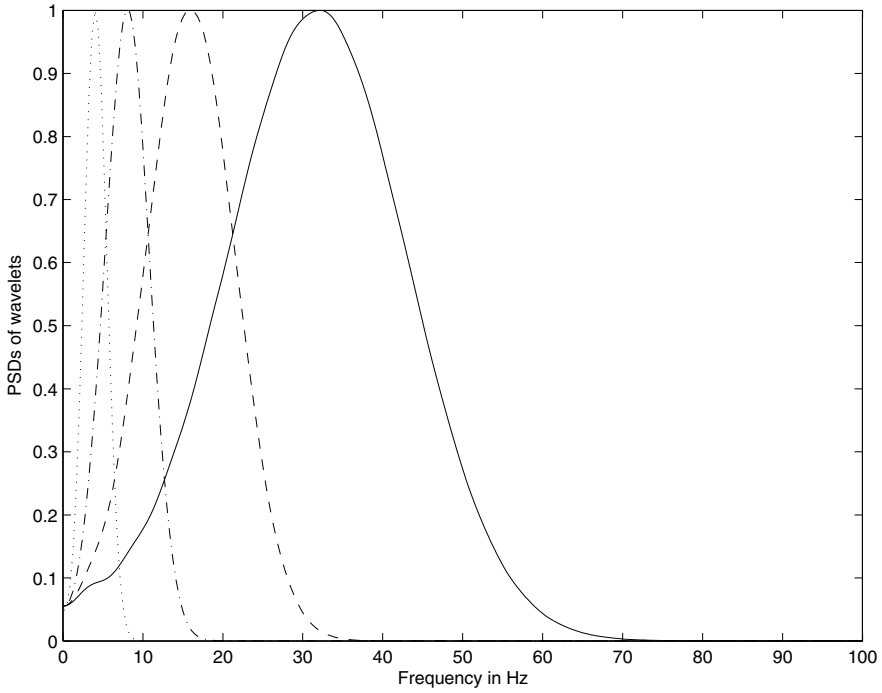


Figure 8.31 Normalized PSDs of the four wavelets in Figure 8.28. The shift towards lower values in the central frequency of the filter corresponding to each wavelet is evident as the scale is increased from 1 (solid line), to 2 (dashed line), 4 (dot–dash line), and 8 (dotted line).

where $|\mathbf{W}| \neq 0$, and $\mathbf{W}_k$ are $K \times 1$ column vectors that make up the $K \times K$ matrix $\mathbf{W}$. The matrix $\mathbf{W}$ should be formulated such that the vector $\mathbf{x}$, also of size $K \times 1$, facilitates a reduced and efficient representation of the original vector $\mathbf{y}$.

The matrix $\mathbf{W}$ may be viewed as being composed of K linearly independent column vectors in the K -dimensional space containing $\mathbf{y}$. Let $\mathbf{W}$ be orthonormal, that is,

$$\mathbf{W}_k^T \mathbf{W}_l = \begin{cases} 1, & k = l, \\ 0, & k \neq l. \end{cases} \quad (8.116)$$

Then,

$$\mathbf{W}^T \mathbf{W} = \mathbf{I} \text{ or } \mathbf{W}^{-1} = \mathbf{W}^T. \quad (8.117)$$

The column vectors of $\mathbf{W}$ form a set of orthonormal basis vectors of a linear transformation. The inverse relationship is given by

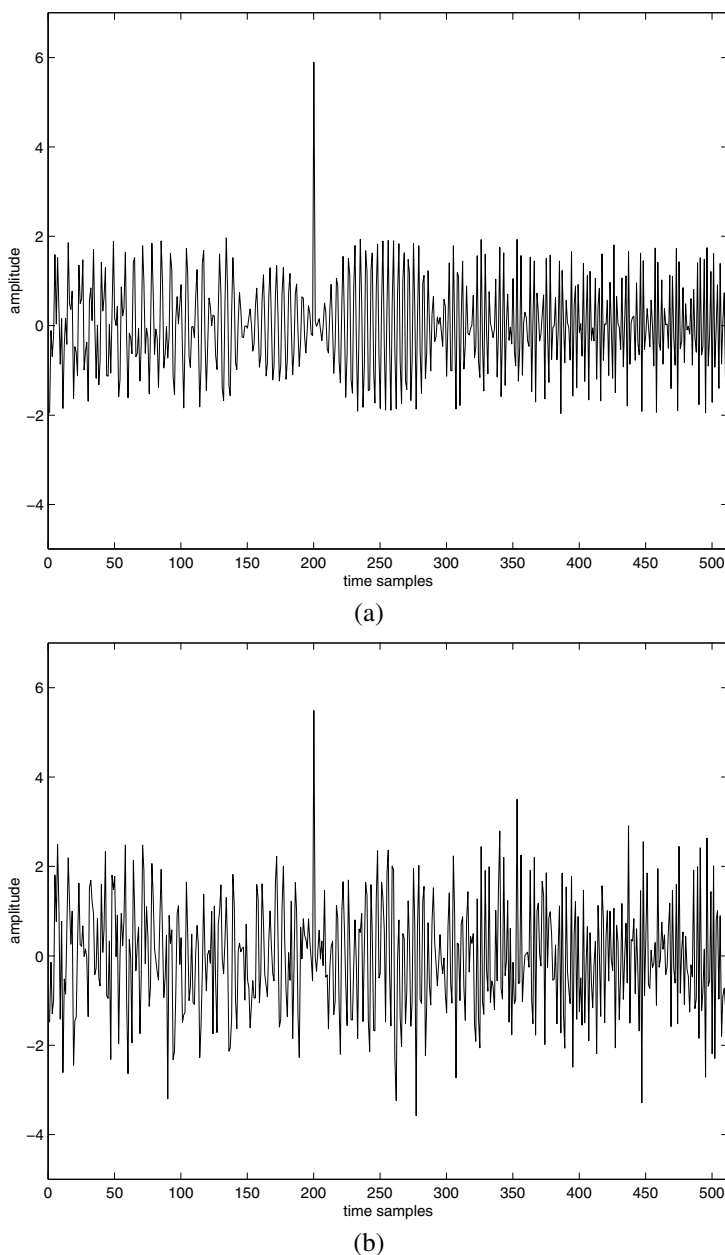


Figure 8.32 (a) Multicomponent and nonstationary synthetic signal with a linear frequency modulation (FM), a nonlinear FM, and a transient component. (b) The signal with noise ($SNR = 0$ dB). Reproduced from S. Krishnan and R. M. Rangayyan, Automatic de-noising of knee-joint vibration signals using adaptive time-frequency representations, *Medical and Biological Engineering and Computing*, 38(1):2–8, 2000, with kind permission from Springer Science+Business Media B.V. ©IFMBE and Springer.

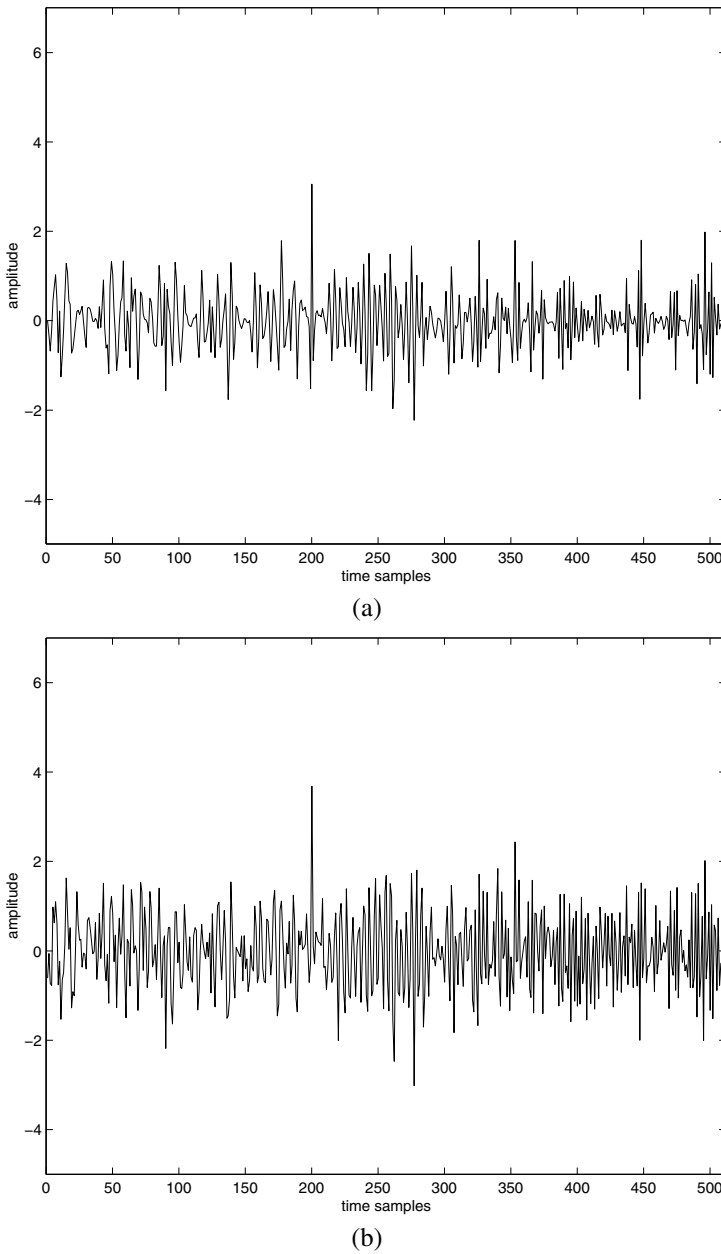


Figure 8.33 Filtered version of the noisy signal in Figure 8.32 (b) using (a) wavelets (symmlet 4) and (b) the WP method. Reproduced from S. Krishnan and R. M. Rangayyan, Automatic de-noising of knee-joint vibration signals using adaptive time-frequency representations, *Medical and Biological Engineering and Computing*, 38(1):2–8, 2000, with kind permission from Springer Science+Business Media B.V. ©IFMBE and Springer.

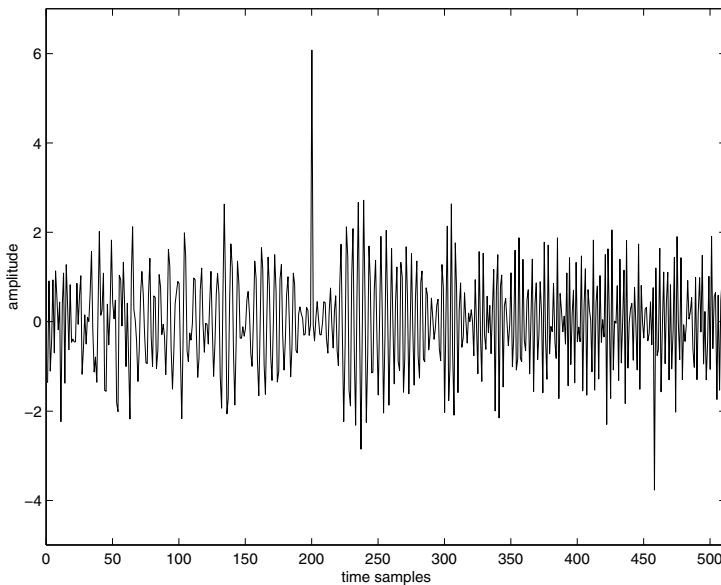


Figure 8.34 Filtered version of the noisy signal in Figure 8.32 (b) using the MP method. Reproduced from S. Krishnan and R. M. Rangayyan, Automatic de-noising of knee-joint vibration signals using adaptive time-frequency representations, *Medical and Biological Engineering and Computing*, 38(1):2–8, 2000, with kind permission from Springer Science+Business Media B.V. ©IFMBE and Springer.

$$\mathbf{x} = \mathbf{W}^{-1} \mathbf{y} = \sum_{k=1}^K y_k \mathbf{W}_k^{-1}. \quad (8.118)$$

In the representation formulated above, each component of $\mathbf{x}$ contributes to the representation of $\mathbf{y}$ and $\mathbf{W}_k^{-1}$ is the k^{th} $K \times 1$ column vector of the $K \times K$ matrix $\mathbf{W}^{-1}$. With the design of $\mathbf{W}$ as a linear and reversible transformation, $\mathbf{x}$ provides a complete or lossless representation of $\mathbf{y}$ if all of its K elements are used.

To achieve compact and efficient representation of the original vector $\mathbf{y}$ by extracting the most significant information contained, we may choose to use only $L < K$ components of $\mathbf{x}$. The omitted components of $\mathbf{x}$ may be replaced with other values $\tilde{x}_k$, $k = L + 1, L + 2, \dots, K$. We now have an approximation of $\mathbf{y}$, represented as

$$\tilde{\mathbf{y}} = \sum_{k=1}^L x_k \mathbf{W}_k + \sum_{k=L+1}^K \tilde{x}_k \mathbf{W}_k. \quad (8.119)$$

The error of approximation is

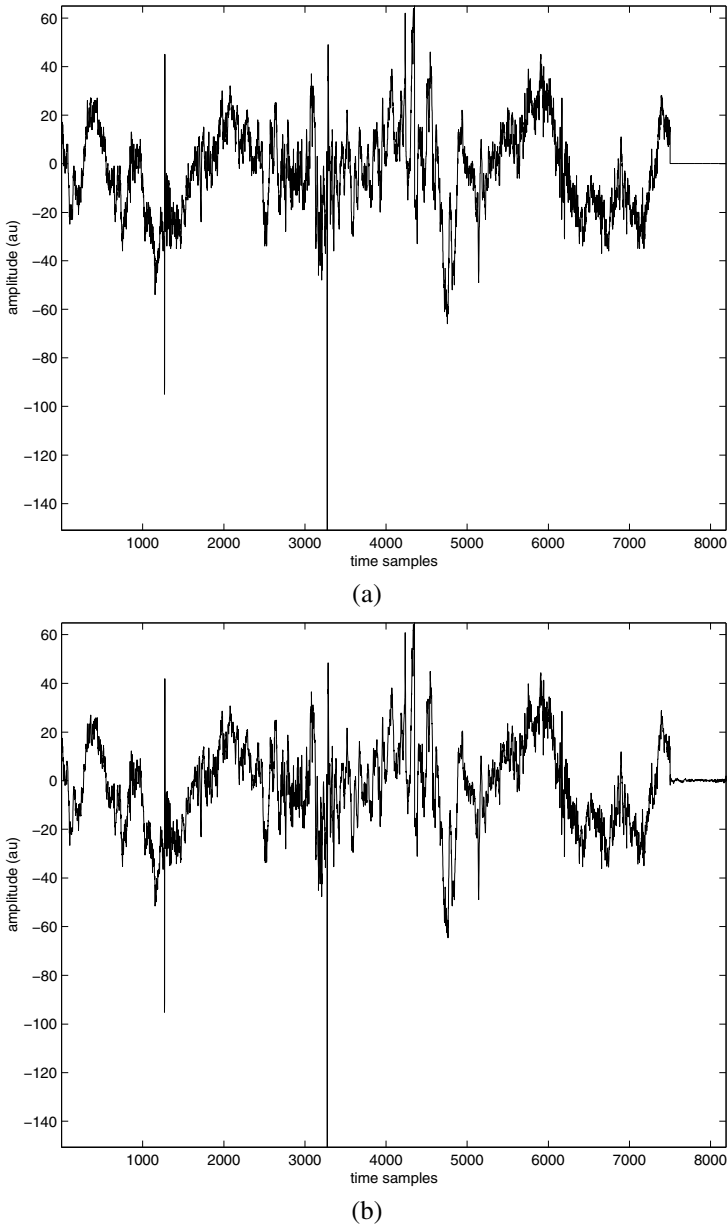


Figure 8.35 (a) VAG signal of a subject with cartilage pathology and (b) MP-filtered version of the signal. Reproduced from S. Krishnan and R. M. Rangayyan, Automatic de-noising of knee-joint vibration signals using adaptive time-frequency representations, *Medical and Biological Engineering and Computing*, 38(1):2–8, 2000, with kind permission from Springer Science+Business Media B.V. ©IFMBE and Springer.

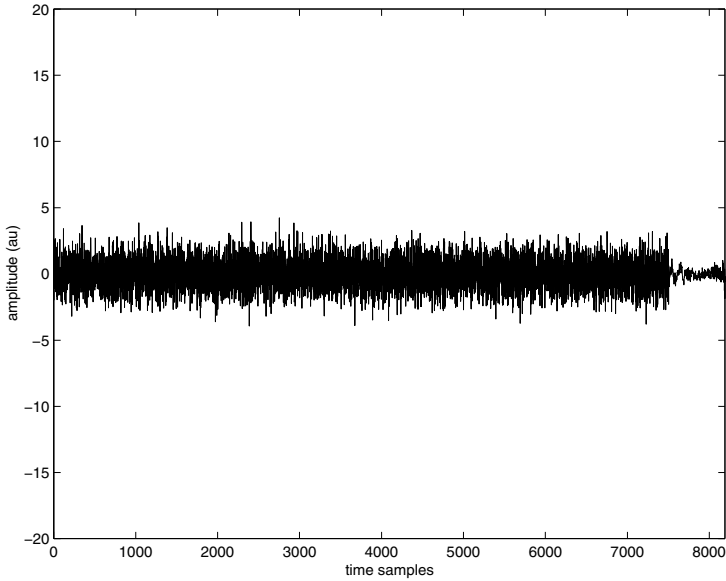


Figure 8.36 Difference between the original VAG signal in Figure 8.35 (a) and the MP-filtered version in Figure 8.35 (b). Reproduced from S. Krishnan and R. M. Rangayyan, Automatic de-noising of knee-joint vibration signals using adaptive time-frequency representations, *Medical and Biological Engineering and Computing*, 38(1):2–8, 2000, with kind permission from Springer Science+Business Media B.V. ©IFMBE and Springer.

$$\boldsymbol{\varepsilon} = \mathbf{y} - \tilde{\mathbf{y}} = \sum_{k=L+1}^K (x_k - \tilde{x}_k) \mathbf{W}_k. \quad (8.120)$$

The MSE is

$$\begin{aligned} \overline{\boldsymbol{\varepsilon}^2} &= E[\boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}] \\ &= E\left[\sum_{k=L+1}^K \sum_{l=L+1}^K (x_k - \tilde{x}_k)(x_l - \tilde{x}_l) \mathbf{W}_k^T \mathbf{W}_l \right] \\ &= \sum_{k=L+1}^K E[(x_k - \tilde{x}_k)^2]. \end{aligned} \quad (8.121)$$

The last step above is due to the orthonormality of $\mathbf{W}$.

Taking the derivative of the MSE with respect to $\tilde{x}_k$ and setting the result to zero, we get

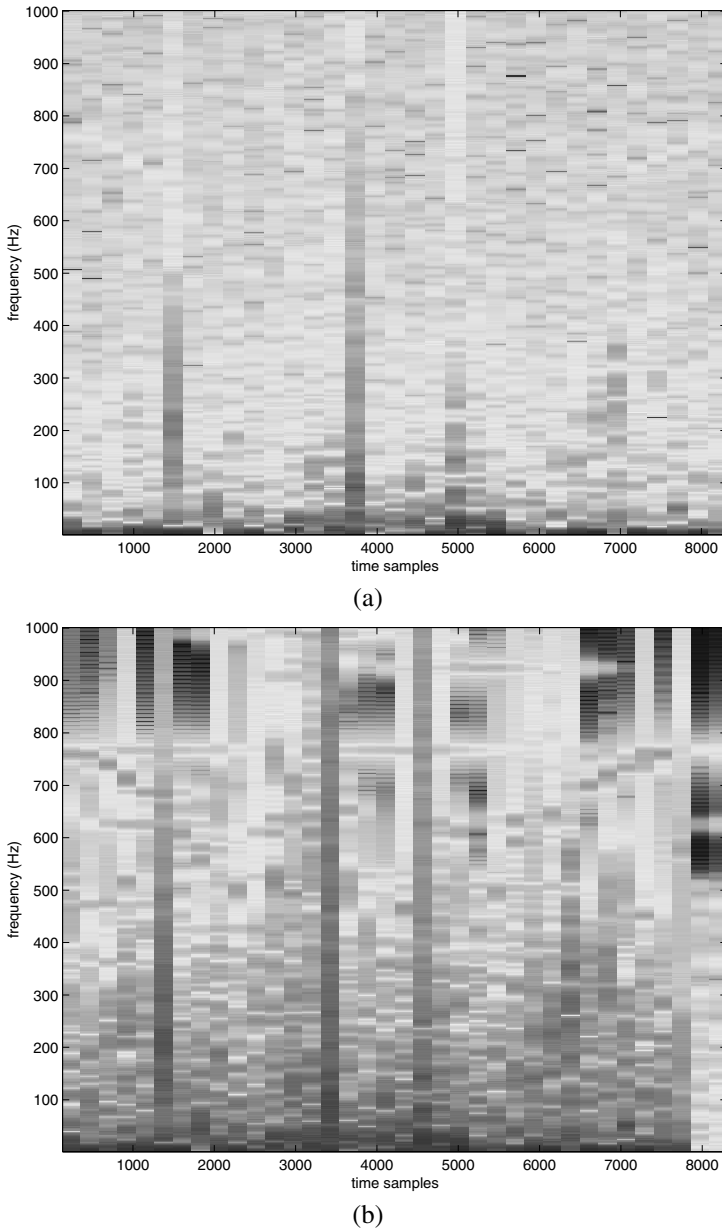


Figure 8.37 (a) TFD of the VAG signal in Figure 8.35 (a) and (b) the TFD of the MP-filtered VAG signal in Figure 8.35 (b) computed using the STFT. $f_s = 2,000 \text{ Hz}$. Reproduced from S. Krishnan and R. M. Rangayyan, Automatic de-noising of knee-joint vibration signals using adaptive time-frequency representations, *Medical and Biological Engineering and Computing*, 38(1):2–8, 2000, with kind permission from Springer Science+Business Media B.V. ©IFMBE and Springer.

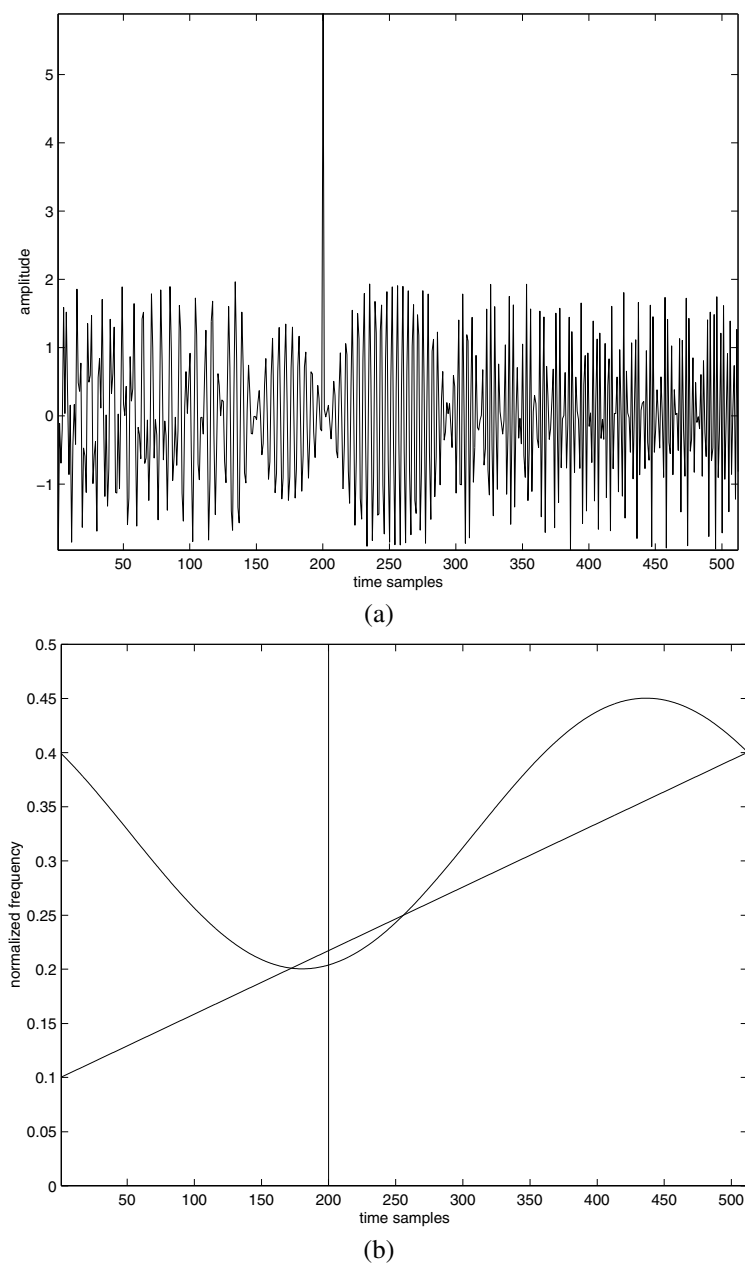


Figure 8.38 (a) Multicomponent and nonstationary synthetic signal consisting of overlapping chirp, impulse, and sinusoidal FM components and (b) its ideal TFD. Reproduced with permission from S. Krishnan, *Adaptive Signal Processing Techniques for Analysis of Knee Joint Vibroarthrographic Signals*, Ph.D. Thesis, University of Calgary, 1999.

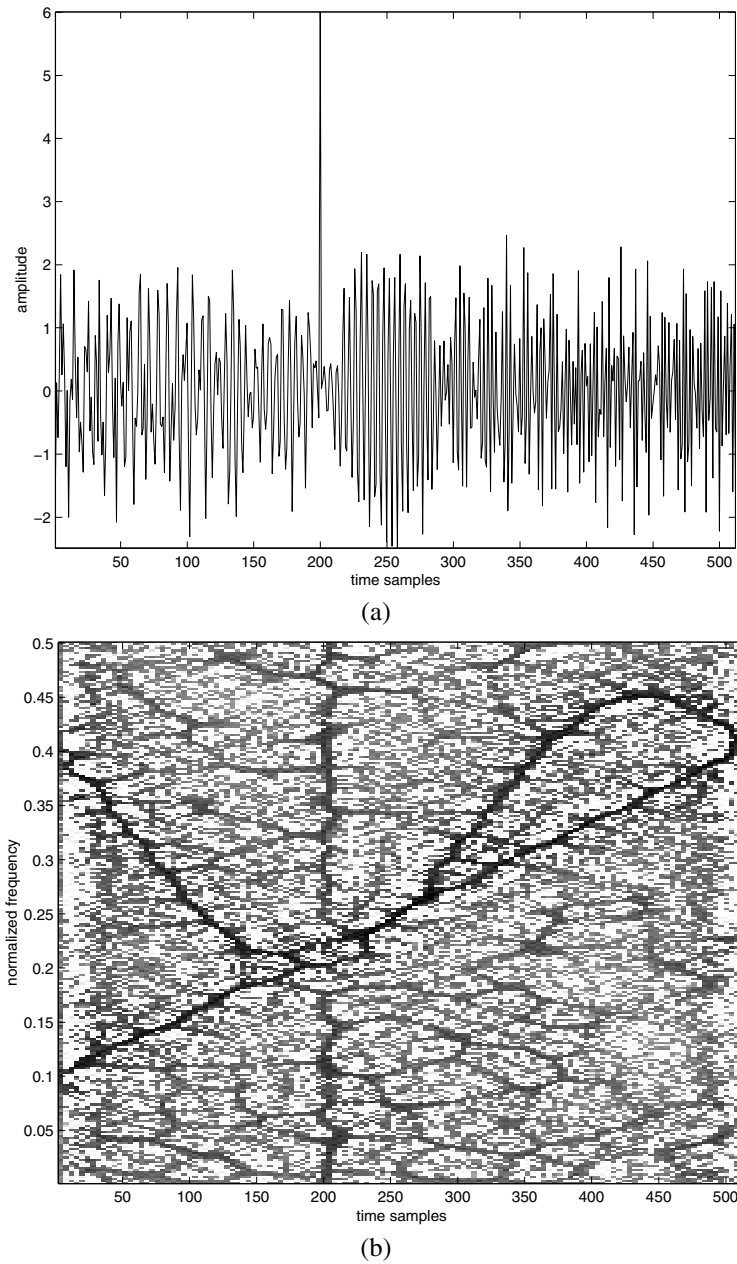


Figure 8.39 (a) The signal in Figure 8.38 (a) with random noise added such that $SNR = 10$ dB. (b) RSPWVD of the noisy signal. Reproduced with permission from S. Krishnan, Adaptive Signal Processing Techniques for Analysis of Knee Joint Vibroarthrographic Signals, Ph.D. Thesis, University of Calgary, 1999.

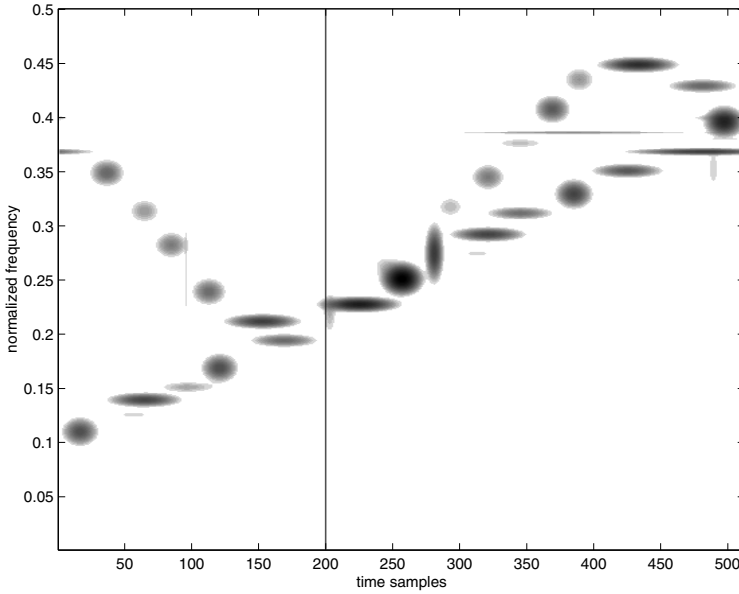


Figure 8.40 MPTFD of the noisy signal in Figure 8.39 (a). Reproduced with permission from S. Krishnan, Adaptive Signal Processing Techniques for Analysis of Knee Joint Vibroarthrographic Signals, Ph.D. Thesis, University of Calgary, 1999.

$$\frac{\partial \overline{\epsilon^2}}{\partial \tilde{x}_k} = -2 E[(x_k - \tilde{x}_k)] = 0. \quad (8.122)$$

The optimal MMSE choice for $\tilde{x}_k$ is, therefore,

$$\tilde{x}_k = E[x_k] = \bar{x}_k = \mathbf{W}_k^{-1} E[y_k], \quad k = L + 1, L + 2, \dots, K. \quad (8.123)$$

This result indicates that the omitted components are replaced by their means estimated for the given population of the vectors $\mathbf{y}$. The MMSE is given by

$$\begin{aligned} \overline{\epsilon^2}_{\min} &= \sum_{k=L+1}^K E[(x_k - \bar{x}_k)^2] \\ &= \sum_{k=L+1}^K E[(\mathbf{W}_k^{-1} y_k - \mathbf{W}_k^{-1} \bar{y}_k)(\mathbf{W}_k^{-1} y_k - \mathbf{W}_k^{-1} \bar{y}_k)^T] \\ &= \sum_{k=L+1}^K E[\mathbf{W}_k^{-1} (y_k - \bar{y}_k)(y_k - \bar{y}_k)^T \{\mathbf{W}_k^{-1}\}^T] \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=L+1}^K E[\mathbf{W}_k^{-1}(y_k - \bar{y}_k)(y_k - \bar{y}_k)^T \mathbf{W}_k] \\
&= \sum_{k=L+1}^K \mathbf{W}_k^T \boldsymbol{\sigma}_y \mathbf{W}_k,
\end{aligned} \tag{8.124}$$

where $\boldsymbol{\sigma}_y = E[(\mathbf{y} - \bar{\mathbf{y}})(\mathbf{y} - \bar{\mathbf{y}})^T]$ is the $K \times K$ covariance matrix of $\mathbf{y}$.

If the basis vectors $\mathbf{W}_k$ are selected as the eigenvectors of $\boldsymbol{\sigma}_y$, that is,

$$\boldsymbol{\sigma}_y \mathbf{W}_k = \lambda_k \mathbf{W}_k, \tag{8.125}$$

where

$$\lambda_k = \mathbf{W}_k^T \boldsymbol{\sigma}_y \mathbf{W}_k \tag{8.126}$$

are the corresponding eigenvalues, then we have

$$\bar{\varepsilon}_{\min}^2 = \sum_{k=L+1}^K \lambda_k. \tag{8.127}$$

Therefore, the MSE may be minimized by ordering the eigenvectors such that the corresponding eigenvalues are arranged in decreasing order, that is, $\lambda_1 > \lambda_2 > \dots > \lambda_K$. Then, if a component x_k of $\mathbf{x}$ is replaced by $\tilde{x}_k = \bar{x}_k$, the MSE increases by λ_k . By replacing the components of $\mathbf{x}$ corresponding to the eigenvalues at the lower end of the list by their respective mean values, the MSE is kept at its minimum for a chosen number of components L .

From the procedure and the properties described above, it is evident that the components of $\mathbf{x}$ are mutually uncorrelated:

$$\boldsymbol{\sigma}_x = \mathbf{W}^T \boldsymbol{\sigma}_y \mathbf{W} = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_K \end{bmatrix} = \boldsymbol{\Lambda}, \tag{8.128}$$

where $\boldsymbol{\Lambda}$ is a diagonal matrix with the eigenvalues λ_k placed along its diagonal. Because the eigenvalues λ_k are equal to the variances of x_k , a selection of the larger eigenvalues implies the selection of the transformed components with the higher variance or information content across the ensemble of the data vectors considered.

Illustration of application: In a study reported by Mesin et al. [457], surface EMG signals were recorded from the biceps brachii muscle. The electrodes were placed 5 mm apart and two channels of EMG were acquired from approximately the same group of muscle fibers. Therefore, the two EMG signals, shown in Figure 8.41, parts (a) and (b), are expected to be highly correlated. The correlation between the two EMG signals is demonstrated in part (c) of the figure, which also

shows the directions of the two principal components (eigenvectors). The two principal components of the EMG signals are shown in parts (d) and (e) of the figure; whereas the first component is close in appearance to the two EMG signals, the second component could be considered to be random noise. By design, the two principal components are uncorrelated. Due to the high degree of correlation between the two input signals, there is considerable redundancy present in them. PCA has extracted most of the power in the signals within the first principal component, which may be considered to be an adequate and redundancy-free representation of both input channels. In this manner, PCA can achieve data reduction and efficient representation of information extracted from multiple measurements.

PCA is commonly used to achieve reduction in the dimension of feature vectors to facilitate efficient pattern classification.

8.8.2 Independent component analysis

Let us consider the situation where we have an observation with K channels of signals, labeled as $y_1(n), y_2(n), \dots, y_K(n)$. The observation at one instant of time may be expressed as the $K \times 1$ vector $\mathbf{y}(n) = [y_1(n), y_2(n), \dots, y_K(n)]^T$. Suppose that the observed signals are being produced by L sources $x_1(n), x_2(n), \dots, x_L(n)$ that are mutually statistically independent. Let us also assume that we are able to gather more than the required number of observations to determine the sources, that is, the system is overdetermined, with $K \geq L$. Then, letting the $L \times 1$ vector $\mathbf{x}(n) = [x_1(n), x_2(n), \dots, x_L(n)]^T$ represent the values of the sources at the instant of time n , we have

$$\mathbf{y}(n) = \mathbf{M} \mathbf{x}(n) + \eta(n), \quad (8.129)$$

where η is random noise that is assumed to be independent of $\mathbf{y}$ and $\mathbf{x}$. This model relates the K observed values in $\mathbf{y}(n)$ at the instant of time n to the corresponding values of the L sources in $\mathbf{x}(n)$ through the $K \times L$ mixing matrix $\mathbf{M}$. The problem in ICA and BSS is to derive the values of $\mathbf{x}$ from the given observations $\mathbf{y}$ without knowing $\mathbf{M}$ (hence the term “blind” in BSS). We may assume that there is no noise in the observed data or let η be one of the sources contained in $\mathbf{x}$, and simplify Equation 8.129 to

$$\mathbf{y}(n) = \mathbf{M} \mathbf{x}(n). \quad (8.130)$$

PCA, as described in Section 8.8.1, facilitates the decomposition of a data vector into uncorrelated components. If the sources of the data or signal components have Gaussian PDFs, the property of being uncorrelated implies that the components are statistically independent; this, however, is not always true. PCA also does not assume any mixing matrix, and characterizes the redundancy in the input using its covariance matrix. Thus, PCA is based on second-order statistics.

ICA achieves separation of the sources by assuming them to be statistically independent and not only being uncorrelated as in PCA. Various measures of statistical

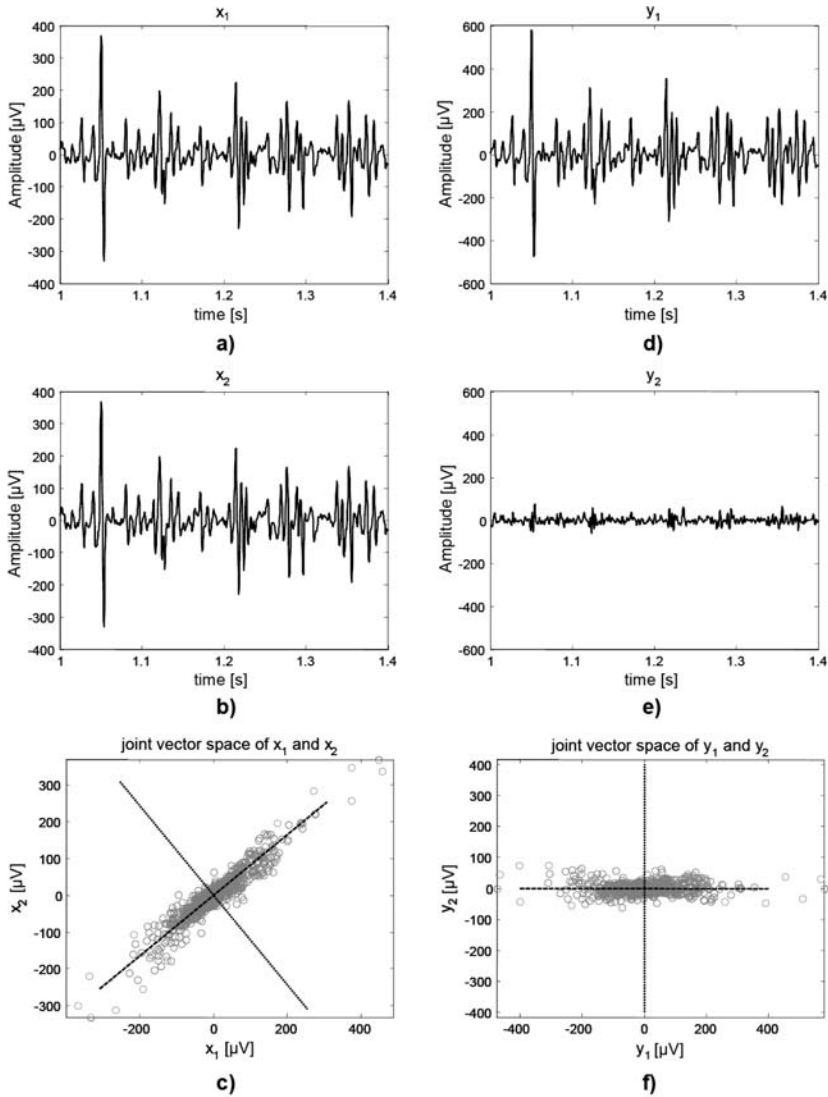


Figure 8.41 Illustration of PCA of correlated EMG signals. (a) and (b) Two channels of EMG signals from the biceps brachii muscle. (c) Directions of the principal components and demonstration of the correlation between the two EMG channels. (d) and (e) Principal components extracted via PCA. (f) Joint vector space of the principal components. Reproduced with permission from L. Mesin, A. Holobar, and R. Merletti, Blind source separation: Application to biomedical signals, Chapter 15 in *Advanced Methods of Biomedical Signal Processing*, IEEE and Wiley, 2011, Edited by S. Cerutti and C. Marchesi, pp. 379–409. ©IEEE.

independence can be used in optimization procedures to estimate a separation or “un-mixing” matrix $\mathbf{W}$ (of size $L \times K$) such that estimates of the original source signals can be obtained as

$$\tilde{\mathbf{x}}(n) = \mathbf{W} \mathbf{y}(n). \quad (8.131)$$

The central limit theorem [5] states that the PDF of a linear combination of several statistically independent random variables tends towards a Gaussian PDF regardless of the PDFs of the individual variables. Therefore, the mixed observed signal $\mathbf{y}$ in Equation 8.130 is likely to have a PDF that is close to a Gaussian regardless of the PDFs of the sources of the input signals $\mathbf{x}$. One approach to achieve BSS is to employ means to minimize the Gaussian characteristics of the reconstructed or estimated signals $\tilde{\mathbf{x}}$. The kurtosis excess (see Section 3.2.1) of a Gaussian PDF is zero: This property may be used to construct a function $F(\tilde{\mathbf{x}})$ that can be used in an iterative optimization procedure to estimate the separation matrix as [457]

$$\tilde{\mathbf{W}}_{(k+1)} = \tilde{\mathbf{W}}_k - \mu_k \nabla F(\tilde{\mathbf{x}}), \quad (8.132)$$

where k is the iteration number, μ is the learning rate, and ∇ is a gradient measure. This procedure is a gradient descent approach for optimization (see Section 3.9.2). Several other algorithms exist for ICA and BSS [65, 449–463]; some BSS methods use PCA as a preprocessing step.

Illustration of application: Zarzoso and Nandi [190] performed comparative analysis of BSS with adaptive filtering (see Section 3.9.1) for the purpose of extraction of the fetal ECG from maternal ECG recordings. Figure 8.42 shows the input ECG signals used in their work, including eight channels of the maternal ECG obtained using abdominal and thoracic surface electrodes. The result of BSS is shown in Figure 8.43. Although BSS separates the mixed input signals and provides independent components, the method, by itself, does not identify the various sources. However, knowing that the fetal heart rate is typically higher than the maternal heart rate, one can determine, by visual inspection, that the fifth and seventh traces from the top in Figure 8.43 represent the extracted fetal ECG over the interval of the recordings with 24 beats. Similarly, the first, second, and fourth traces can be taken to be representations of the maternal ECG (with 14 beats over the same interval) without interference from the fetal ECG. The third, sixth, and eighth traces do not show any QRS complexes with a regular rhythm and, hence, could be considered to be noise. If the derivation of the fetal heart rate is the goal of the recording, it can be easily achieved by using one of the extracted fetal ECG channels. The maternal heart rate may also be estimated from one of the extracted maternal ECG channels without interference from the fetal ECG.

It should be observed that BSS could lead to multiple representations of a single source in the resulting output channels. In the present example, the electrical activity of the maternal heart is shown in three output channels and that of the fetal heart is shown in two channels; the user is required to analyze the results and select appropriate channels for further analysis. (Indeed, all of the eight input channels demonstrate various views of the maternal ECG, with some of them including the fetal ECG as

well.) Zarzoso and Nandi [190] showed that BSS performs better than adaptive filtering in the extraction of fetal ECG from maternal ECG signals. See Callaerts et al. [464], Vanderschoot et al. [465], and De Lathauwer et al. [453] for other methods for the same application.

8.9 Application: Adaptive Segmentation of EEG Signals

Problem: *Propose a method for parametric representation of nonstationary EEG signals.*

Solution: Bodenstein and Praetorius [197] applied their adaptive segmentation procedure based upon the *SEM* (see Section 8.5.1) for representation and analysis of EEG signals with the following propositions.

1. An EEG signal consists of quasistationary segments upon which transients may be superimposed.
2. A segment is specified by its time of occurrence, duration, and PSD (represented by its AR-model coefficients). A transient is specified by its time of occurrence and a set of graph elements (or directly by its samples).
3. An EEG signal consists of a finite number of recurrent states.

It should be noted that, whereas the adaptive segments have variable length, each adaptive segment is represented by the same number of AR-model coefficients. The number of parameters is, therefore, independent of segment duration, which is convenient when pattern classification techniques are applied to the segments. Since the AR model is computed once at the beginning of each segment and some prediction error is permitted in the moving analysis window, the initial AR model may not adequately represent the entire adaptive segment. A new model may be computed using the signal samples over the entire duration of each adaptive segment. Instead, Bodenstein and Praetorius maintained the initial AR model of order P of each adaptive segment, and an additional *corrective predictor* of order M was derived for each adaptive segment using the ACF of the prediction error which is computed and readily available in the segmentation procedure. Each adaptive segment was then represented by the $(P + M)$ AR-model coefficients, the associated prediction error *RMS* values, and the segment length. The PSD of the segment may be derived from the two sets of AR-model coefficients.

With the EEG signals bandpass filtered to the range $1 - 25$ Hz and sampled at 50 Hz in the work of Bodenstein and Praetorius [197], the ACF window length was set to be 2 s with $2N + 1 = 101$ samples. Bodenstein and Praetorius used the rule of thumb that the AR-model order should be at least twice the number of expected resonances in the PSD of the signal. Short segments of EEG signals rarely demonstrate more than two spectral peaks, which suggests that an AR-model order of $P = 5$ should be adequate. Regardless, Bodenstein and Praetorius used $P = 8$, which met the Akaike criterion as well (see Section 7.5.2). The order of the ACF of the prediction error and the associated corrective predictor was set to a low value of

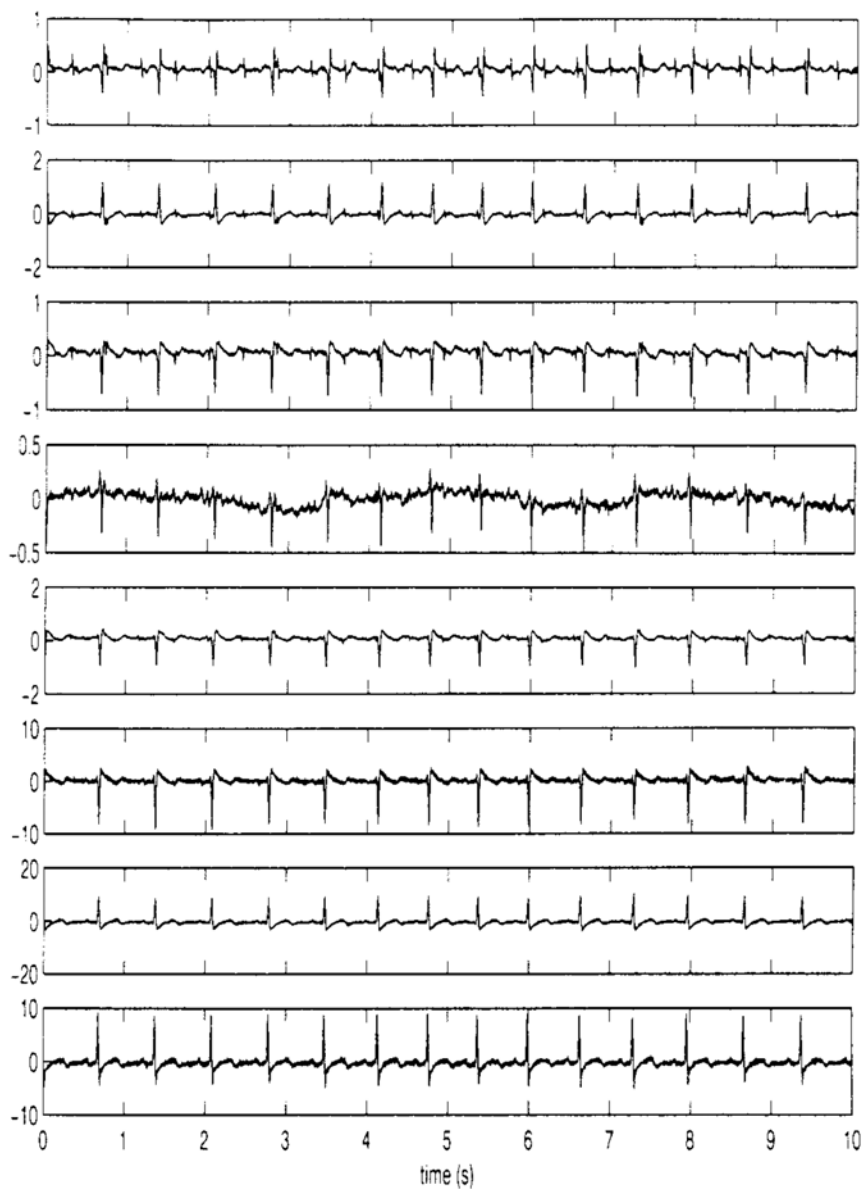


Figure 8.42 Eight channels of maternal ECG signals using abdominal and thoracic surface electrodes. Reproduced with permission from V. Zarzoso and A.K. Nandi, Noninvasive fetal electrocardiogram extraction: blind separation versus adaptive noise cancellation, *IEEE Transactions on Biomedical Engineering*, 48(1):12–18, 2001. ©IEEE.

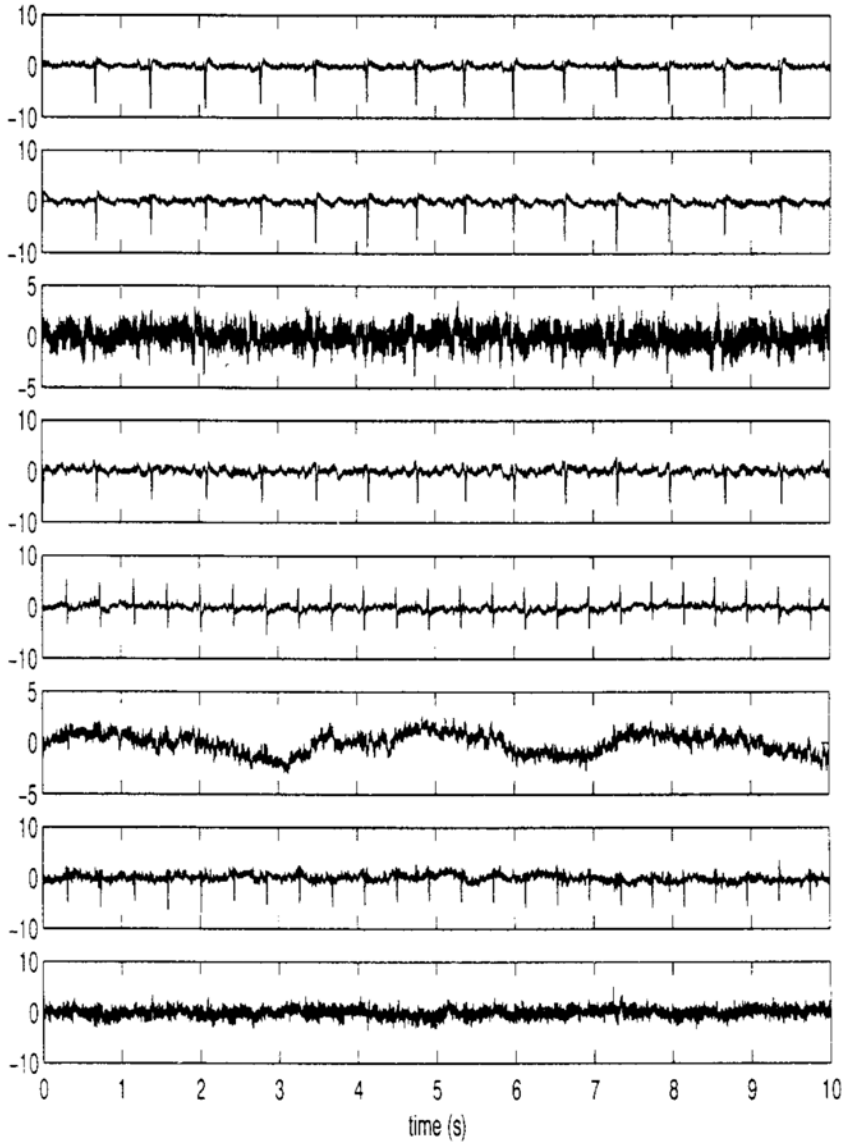


Figure 8.43 Results of BSS showing eight independent components. The fifth and seventh traces from the top show the extracted fetal ECG with 24 beats over the 10-s interval of the recordings. The first, second, and fourth traces show the maternal ECG with 14 beats over the same interval. The remaining traces represent noise. Reproduced with permission from V. Zarzoso and A.K. Nandi, Noninvasive fetal electrocardiogram extraction: blind separation versus adaptive noise cancellation, *IEEE Transactions on Biomedical Engineering*, 48(1):12–18, 2001. ©IEEE.

$M = 3$, allowing for one spectral peak (the error should ideally have a flat PSD). The thresholds were defined as $Th_1 = 0.5$ (empirical) and $Th_2 = 2.5\sigma$, where σ is the *RMS* value of the prediction error (see Section 8.5.1). The range of $20\sigma^2$ to $40\sigma^2$ was recommended for Th_3 . A transitional delay of 25 samples was allowed between each segmentation boundary and the starting point of the following fixed window to prevent the inclusion of the spectral components of one segment into the following segment.

Figure 8.44 shows a few examples of adaptive segmentation of EEG signals. A clustering procedure was included to remove spurious boundaries, some examples of which may be seen in Figure 8.44 (d); neighboring segments with similar parameters were merged in a subsequent step. Visual inspection of the results indicates that most of the adaptive segments are stationary (that is, they have the same appearance) over their durations. It is worth noting that the longest segment in Figure 8.44 (d) of duration 16 s or 800 samples is represented by just 12 parameters.

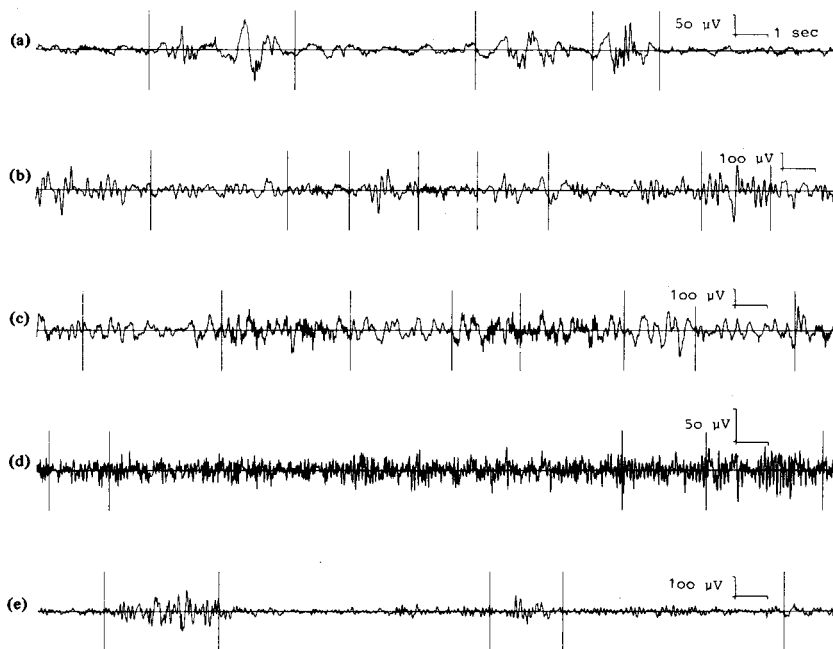


Figure 8.44 Examples of segmentation of EEG signals. (a) Newborn in non-REM sleep. REM = rapid eye movement. (b) Child of age 7 years in sleep stage I. (c) Child of age 8 years in sleep stage III. (d) Alpha rhythm of an adult. (e) EEG of an adult with paroxysms. The vertical lines represent the segmentation boundaries. Reproduced with permission from G. Bodenstein and H.M. Praetorius, Feature extraction from the electroencephalogram by adaptive segmentation, *Proceedings of the IEEE*, 65(5):642–652, 1977. ©IEEE.

Figure 8.45 shows examples of detection of transients in two contralateral channels of the EEG of a patient with epilepsy. The EEG signal between seizures (inter-

ictal periods) is expected to exhibit a large number of sharp waves. The length of the arrows shown in the figure was made proportional to the cumulated suprathreshold part of the squared prediction error in order to indicate how pronounced the event was regarded to be by the algorithm.

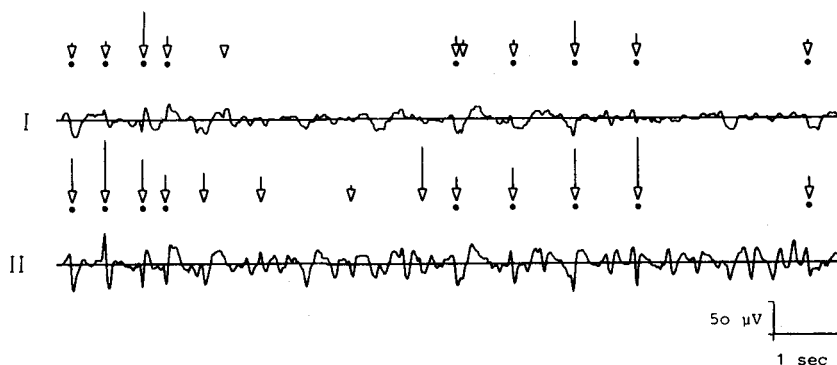


Figure 8.45 Example of detection of transients in the EEG signal of a patient with epilepsy. The signals shown are from contralateral channels between seizures (interictal period). The longer the arrow, the more pronounced the transient detected at the corresponding time instant. Transients detected simultaneously in the two contralateral channels are marked with dots. Reproduced with permission from G. Bodenstein and H.M. Praetorius, Feature extraction from the electroencephalogram by adaptive segmentation, *Proceedings of the IEEE*, 65(5):642–652, 1977. ©IEEE.

The method was further extended to parallel analysis of multichannel EEG signals by Bodenstein et al. [413] and Creutzfeldt et al. [414]. Procedures were proposed for computerized pattern classification and labeling of EEG signals, including clustering of similar segments and state diagrams indicating the sequence of the types of activity found in an EEG record. Figure 8.46 illustrates the record produced by the application of the procedure to two channels of an EEG signal. Typical EEG segments belonging to the four clusters detected in the signal are shown on the left and right sides of the upper portion of the figure. Each signal segment is labeled with the frequencies (FRQ, in Hz) and amplitudes (AMP, in μV) of the resonances detected using an eighth-order AR model. The central column of the upper portion of the figure illustrates the PSDs of the corresponding segments on the left (solid line) and right (dashed line). The middle portion of the figure provides the state diagram, indicating the transitions between the four states (represented by the four clusters of the EEG segments) detected in the two channels of the signal. The states represent: 1, background; 2, eyes open; 3, paroxysm; and 4, epileptiform spike-and-wave complexes. The values on the right of the state diagram give the percentage of the total duration of the signal for which the EEG was in the corresponding states. The bottom portion of the figure illustrates singular events, that is, segments that could not be grouped with any of the four clusters. It was indicated that the segments of most EEG signals could be clustered into at most five states, and that the summarized

record as illustrated in Figure 8.46 could assist clinicians in analyzing lengthy EEG records in an efficient manner.

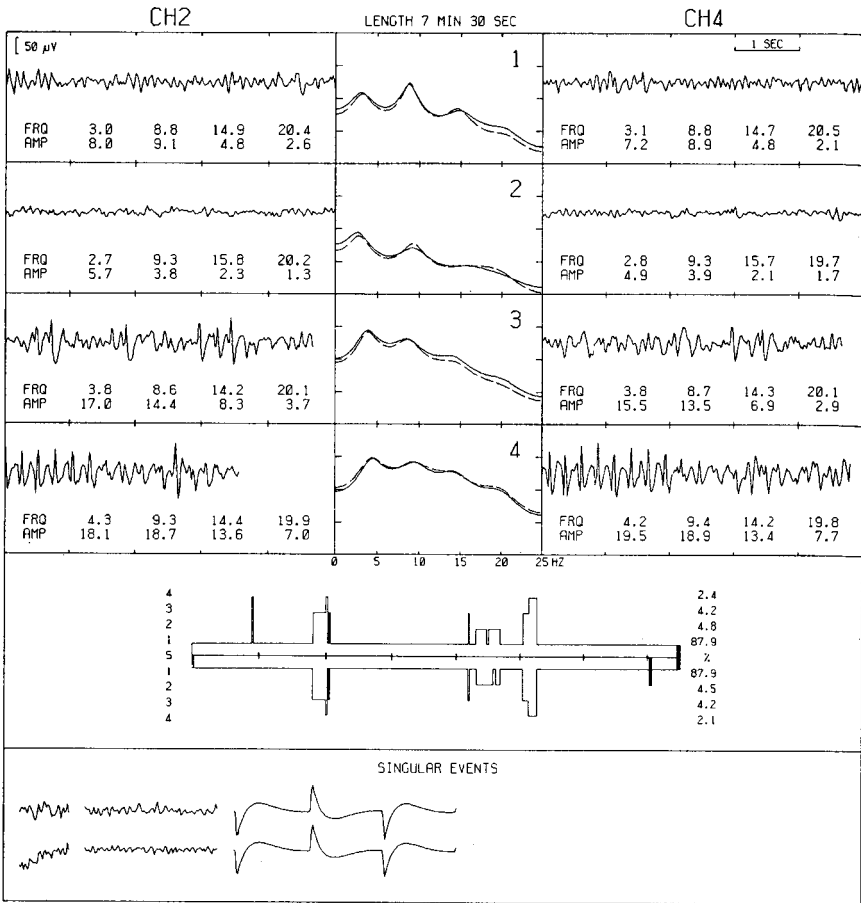


Figure 8.46 Example of application of segmentation and pattern analysis to the EEG signal of a patient with epileptiform activity. Refer to the text for details. Reproduced with permission from G. Bodenstein, W. Schneider, and C.V.D. Malsburg, Computerized EEG pattern classification by adaptive segmentation and probability-density-function classification. Description of the method, *Computers in Biology and Medicine*, 15(5):297–313, 1985. ©Elsevier Science.

8.10 Application: Adaptive Segmentation of PCG Signals

We have noted several times that the PCG signal is nonstationary. Let us now assess the feasibility of adaptive segmentation of PCG signals using the RLSL method (see Section 8.6.2), with no other signal being used as a reference.

Figure 8.47 illustrates the results of segmentation of the PCG signal of a normal subject. The top trace shows the PCG signal over three cardiac cycles; the segment boundaries detected are indicated by the vertical dotted lines as well as by the triangular markers on the time axis. The second trace illustrates a plot of the conversion factor γ_c : The conversion factor drops from unity whenever there is a change in the signal's characteristics, in particular at the boundaries of S1 and S2. A threshold of 0.995 (indicated by the horizontal line overlaid on the second trace) applied to γ_c and a condition imposing a minimum segment length of 50 samples (50 *ms*) were used to obtain the segment boundaries. The third and fourth traces illustrate the ECG and carotid pulse signals of the subject acquired simultaneously with the PCG. The segment boundaries obtained by the RLSL method agree well with the readily noticeable S1 and S2 boundaries as well as the QRS and dicrotic notch positions. (See also Sections 1.2.9, 2.3, and 4.10 for related details.)

Figure 8.48 illustrates the results of adaptive segmentation of the PCG signal of a subject with systolic murmur due to aortic stenosis. The results in this case, however, are not as clear or as easy to interpret as in the preceding case. The method has identified the beginning of S1 and S2; furthermore, the split nature of S2 has been identified by an additional segment boundary within each S2. However, the method has not reliably identified the boundaries between the episodes of S1 and systolic murmur illustrated: The condition on the minimum segment length has affected the placement of the segment boundary after the beginning of S1. Use of other conditions on γ_c may provide better segmentation results.

8.11 Application: Time varying Analysis of HRV

The heart rate is controlled by the ANS and the CNS: The vagal and sympathetic activities lead to a decrease or increase, respectively, in the heart rate (see Section 1.2.5). We saw in Sections 2.2.4 and 7.8 how respiration affects heart rate and how Fourier analysis may be extended to analyze HRV. When heart rate data such as beat-to-beat *RR* intervals are collected over long periods of time (several hours), the signal could be expected to be nonstationary.

Bianchi et al. [400] extended AR-modeling techniques for time-variant PSD analysis of HRV data in order to study transient episodes related to ischemic attacks. The prediction error was weighted with a forgetting factor, and a time-varying AR model was derived. The RLS algorithm was used to update the AR-model coefficients at every *RR* interval sample (every cardiac cycle). The AR coefficients were then used to compute a time-varying PSD. The following frequency bands were indicated to be of interest in the analysis of *RR* interval PSDs: VLF band in the range 0 – 0.03 *Hz* related to humoral and thermoregulatory factors; low-frequency (LF)

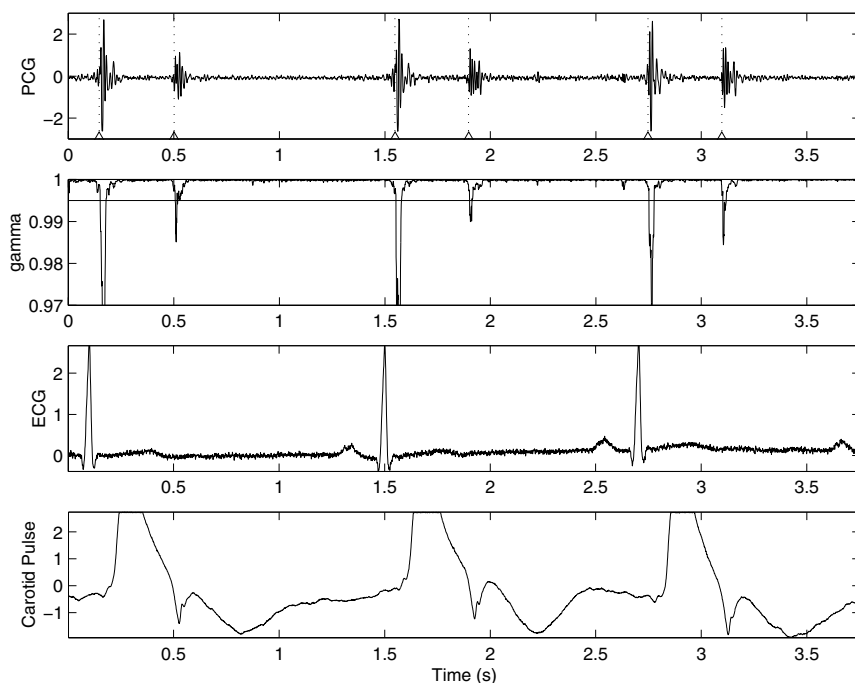


Figure 8.47 Adaptive segmentation of the PCG signal of a normal subject using the RLSL method. Top to bottom: PCG signal (the vertical dotted lines and triangular markers represent the segmentation boundaries); conversion factor γ_c (the horizontal line is the threshold used); ECG; carotid pulse (clipped due to saturation).

band in the range $0.03 - 0.15 \text{ Hz}$ related to sympathetic activity; high-frequency (HF) band in the range $0.18 - 0.4 \text{ Hz}$ related to respiration and vagal activity.

(Note: According to the standards of measurement, physiological interpretation, and clinical use of HRV published by the Task Force of the European Society of Cardiology and the North American Society of Pacing Electrophysiology [466], the frequency bands of HRV are slightly different from those given above and are: VLF, $\leq 0.04 \text{ Hz}$; LF, $0.04 - 0.15 \text{ Hz}$; and HF, $0.15 - 0.4 \text{ Hz}$.)

Figure 8.49 shows an RR interval series including an ischemic episode (delineated by B for beginning and E for ending points, respectively). Figure 8.50 shows the time-varying PSD in the form of a spectrogram. Figure 8.51 shows a segment of RR interval data and a few measures derived from the data.

Some of the important observations made by Bianchi et al. (and illustrated by the spectrogram in Figure 8.50 and the parameters in Figure 8.51) are as follows:

- There is an increase in LF power about $1.5 - 2 \text{ minutes}$ before an ischemic event.
- The RR variance decreases as an episode begins.

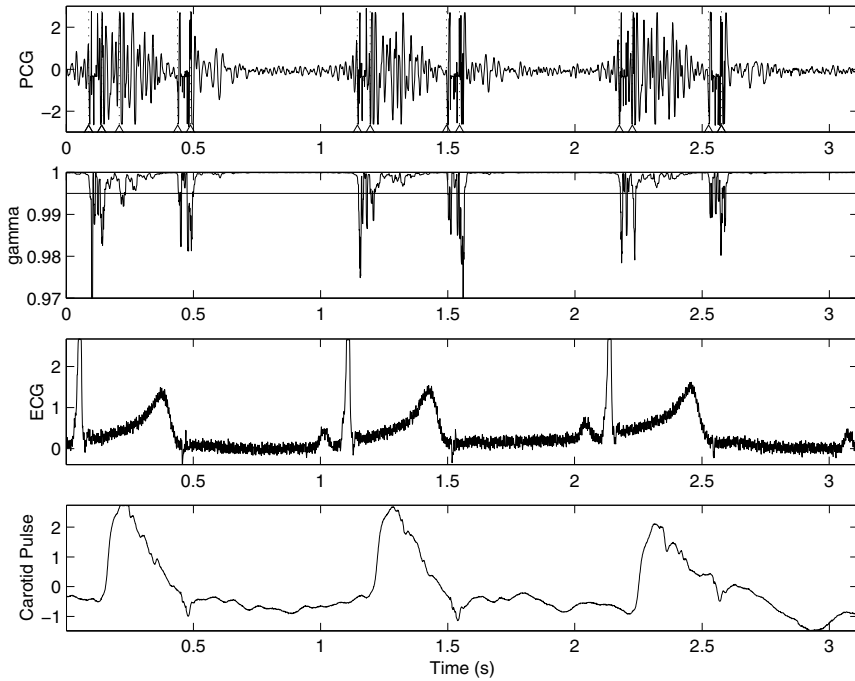


Figure 8.48 Adaptive segmentation of the PCG signal of a subject (female, 11 years) with systolic murmur due to aortic stenosis. Top to bottom: PCG signal (the vertical lines and triangular markers represent the segmentation boundaries); conversion factor γ_c (the horizontal line is the threshold used); ECG; carotid pulse.

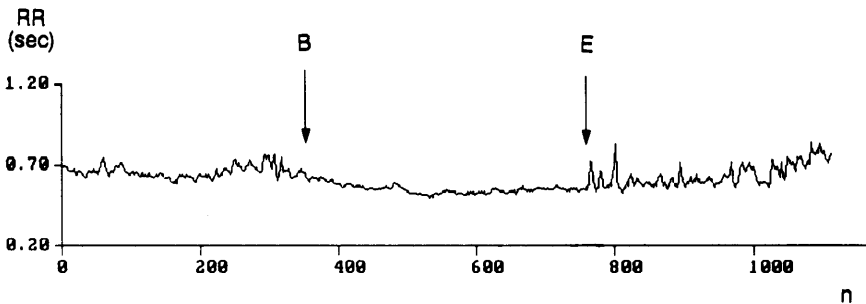


Figure 8.49 *RR* interval series including an ischemic episode. B denotes the beginning and E denotes the end of the episode. Reproduced with permission from A.M. Bianchi, L. Mainardi, E. Petrucci, M.G. Signorini, M. Mainardi, and S. Cerutti, Time-variant power spectrum analysis for the detection of transient episodes in HRV signal, *IEEE Transactions on Biomedical Engineering*, 40(2):136–144, 1993. ©IEEE.

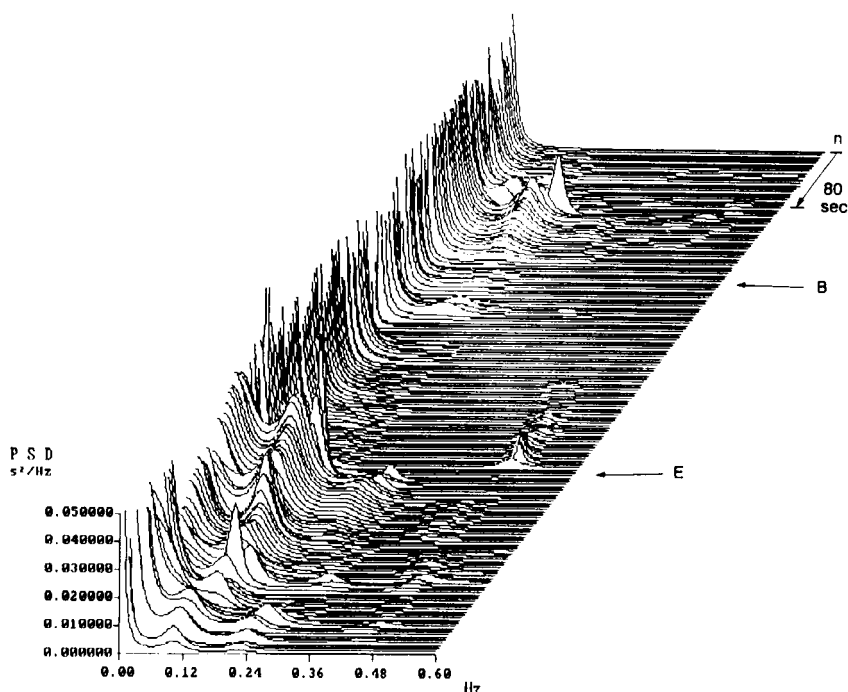


Figure 8.50 Spectrogram of the *RR* interval series in Figure 8.49. Time progresses from the top to the bottom. B denotes the beginning and E denotes the end of an ischemic episode. Reproduced with permission from A.M. Bianchi, L. Mainardi, E. Petrucci, M.G. Signorini, M. Mainardi, and S. Cerutti, Time-variant power spectrum analysis for the detection of transient episodes in HRV signal, *IEEE Transactions on Biomedical Engineering*, 40(2):136–144, 1993. ©IEEE.

- There is a predominant rise in LF power at the end of an ischemic episode.
- A small HF component appears toward the end of an episode.
- Early activation of an LF component precedes tachycardia and ST displacement in the ECG that are generally indicative of the onset of an ischemic episode.
- The results suggest an arousal of the sympathetic system before an acute ischemic attack.

Time-varying AR-modeling techniques have also been applied for the analysis of EEG signals [398]. Time-varying ARMA modeling techniques have been applied to analyze EGG signals [69].

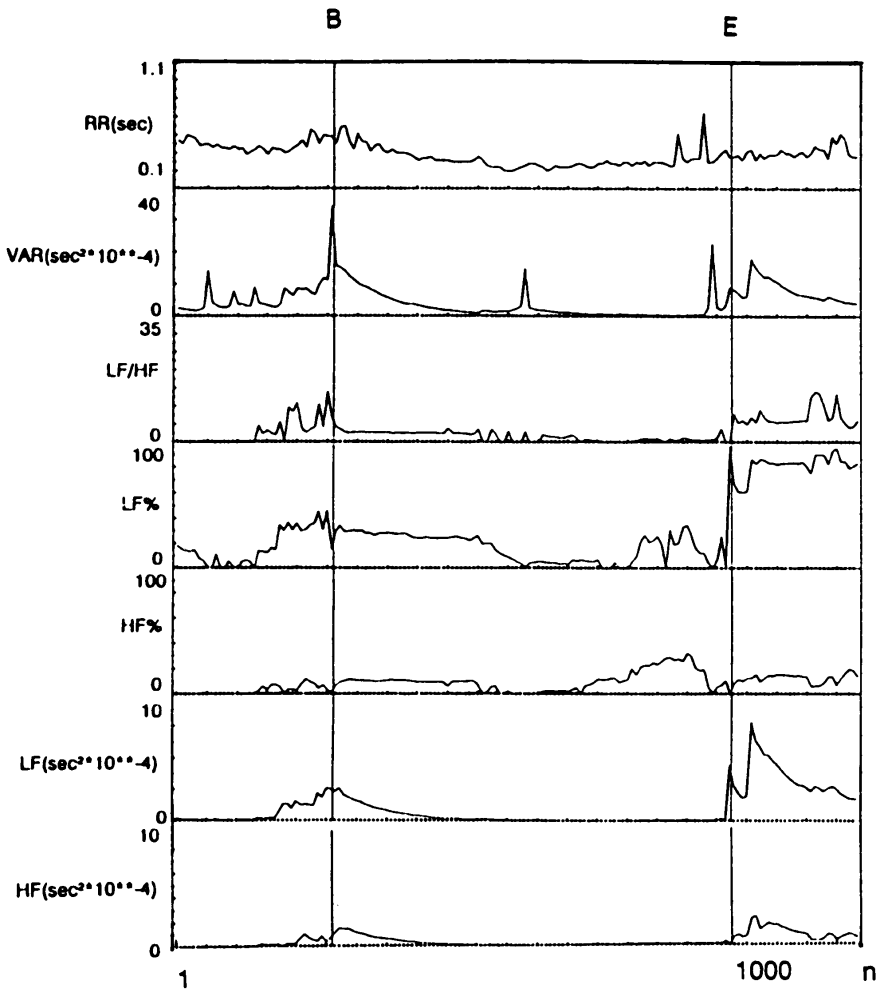


Figure 8.51 Top to bottom: *RR* interval series including an ischemic episode; variance; low-frequency (LF) to high-frequency (HF) power ratio; percentage of LF power; percentage of HF power; LF power; and HF power. B denotes the beginning and E denotes the end of the episode. Reproduced with permission from A.M. Bianchi, L. Mainardi, E. Petrucci, M.G. Signorini, M. Mainardi, and S. Cerutti, Time-variant power spectrum analysis for the detection of transient episodes in HRV signal, *IEEE Transactions on Biomedical Engineering*, 40(2):136–144, 1993. ©IEEE.

8.12 Application: Detection of Epileptic Seizures in EEG Signals

Zhou et al. [467] proposed measures of lacunarity and fluctuation intensity (FI) of DWT coefficients of EEG signals for the detection of epileptic seizures. EEG signals were subjected to DWT-based decomposition with the Daubechies 4 wavelet and five scales, and wavelet coefficients at scales 3, 4, and 5 were selected for further processing. The selection of the scales mentioned above was based on the observation that seizure signals in intracranial EEG signals usually occur in the range of 3 to 29 Hz .

Noting that ictal EEG segments typically display larger fluctuations than interictal segments, the feature FI was defined to measure the intensity of the fluctuations in an EEG signal as

$$FI(s) = \frac{1}{N} \sum_{i=1}^{N-1} |d(i+1) - d(i)|, \quad (8.133)$$

where N is the number of DWT coefficients $d(i)$ for scale s . It was demonstrated that the values of FI during seizures were typically greater than those during other periods.

Lacunarity is a scale-dependent measure of heterogeneity of a signal or image, and quantifies the presence of gaps [275, 468]. If $p(m, l)$ represents the probability of data samples whose amplitude is equal to m , l represents the length of an epoch, and $[A, B]$ is the range of values of the data samples, it follows that

$$\sum_{m=A}^B p(m, l) = 1. \quad (8.134)$$

With

$$M_1(l) = \sum_{m=A}^B m p(m, l) \quad (8.135)$$

and

$$M_2(l) = \sum_{m=A}^B m^2 p(m, l), \quad (8.136)$$

the measure of lacunarity was obtained as

$$\lambda(l) = \frac{M_2(l) - [M_1(l)]^2}{[M_1(l)]^2}. \quad (8.137)$$

Zhou et al. [467] computed lacunarity values of the EEG wavelet coefficients at scales 3, 4, and 5. It was demonstrated that the values of lacunarity during seizures were typically lower than those during other periods.

Including further postprocessing and classification steps, Zhou et al. [467] analyzed a data set of 289.14 h of intracranial EEG data recorded from 21 patients, and obtained a sensitivity of 96.25% in the detection of epileptic seizures with a false detection rate of 0.13/ h and a mean delay time of 13.8 s . See Zhou et al. [467] for

a comparative analysis of their results with those obtained in other recent works on the detection of epileptic seizures in EEG signals.

8.13 Application: Analysis of Crying Sounds of Infants

Várallyay [114] developed techniques to analyze time-varying spectral patterns in crying sounds of infants. It was observed that during a crying episode, the volume, pitch, and tone of the signal vary over time. The varying patterns of the pitch (or fundamental frequency) was referred to as the *cry melody*. Just as the melody of normal speech expresses the mood and intention of a speaker, the melody of crying is expected to relate to the physical and physiological state of the infant. Cry melodies are expected to possess characteristic patterns for various states, such as pain, hunger, discomfort, and boredom. It was also noted that, whereas a person with normal hearing can control the sound or speech produced, a person with hearing impairment is unable to exercise such control due to the lack of auditory feedback. Hirschberg [469, 470] observed high-pitched cry melodies with fundamental frequencies in the range of 1,000 to 2,000 Hz in the case of infants with dysphonia, as compared to the range of 400 to 500 Hz for cries of healthy infants.

The spectral structure of a cry segment is expected to be regular, containing a fundamental frequency and its harmonics. Dynamic variations of the fundamental frequency characterize the cry melody. In order to follow the variations of the fundamental frequency over time, Várallyay [114] divided cry segments into short-time windows of duration 40 ms . The fundamental frequencies of several infant cries that were analyzed were found to vary between 200 and 1,000 Hz , with the most common values being between 300 and 600 Hz . In order to follow variations of the fundamental frequency, three fundamental units of cry melody patterns were defined as falling (-1), flat (0), and rising ($+1$). Cry melodies were coded as sequences of these fundamental patterns.

Figure 8.52 shows the temporal and spectral variations in a cry signal. The top-most trace shows the time-varying amplitude of five cry segments. The second trace shows the spectrogram in which the time-varying patterns of the fundamental frequency and its harmonics (that is, the cry melody) are clearly seen. The two traces at the bottom of the figure show the cry melody as a function of time; the lowest trace has five guidelines superimposed on the cry melody to facilitate its interpretation over predefined frequency ranges.

Várallyay [114] analyzed 2,762 crying sounds recorded from 316 infants. The mean value of the fundamental frequency of the melodies was higher for infants with hearing impairment (425.51 ± 78.10 Hz) than for the control group (408.74 ± 64.77 Hz).

8.14 Application: Adaptive Time frequency Analysis of VAG Signals

Krishnan et al. [273] proposed several methods to derive time-varying parameters from TFDs for the analysis and classification of VAG signals. Specifically, four

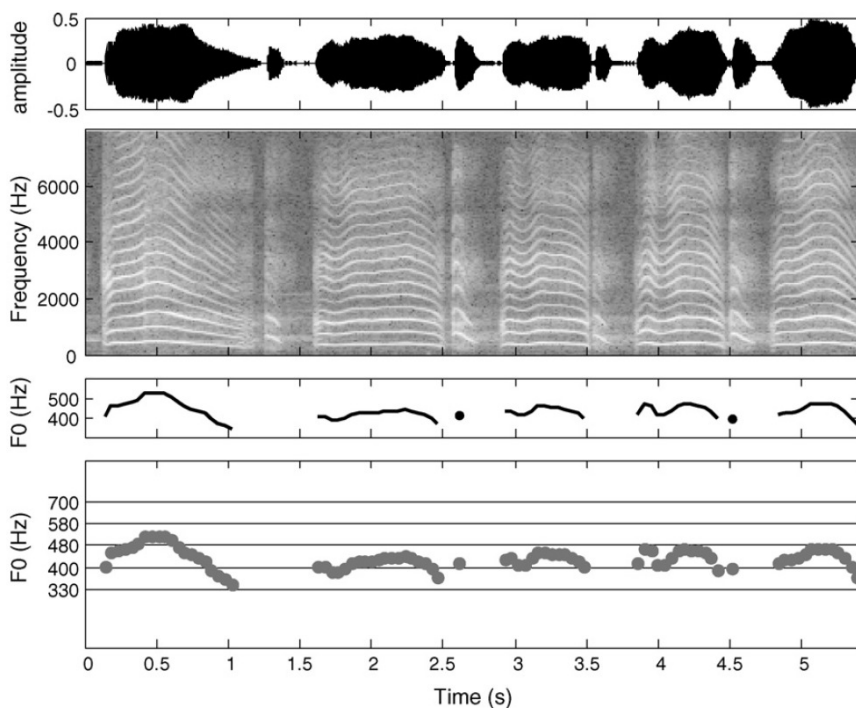


Figure 8.52 Top to bottom: Time-varying amplitude of five cry segments; spectrogram; cry melody as a function of time; the previous item with five guidelines superimposed. Reprinted with permission from G. Várallyay Jr., *The melody of crying*, *International Journal of Pediatric Otorhinolaryngology*, 71(11):1699–1708, 2007, ©Elsevier.

TF features were derived from OMPTFDs (see Section 8.7.5) of VAG signals: the energy parameter (EP), the energy spread parameter (ESP), the frequency parameter (FP), and the frequency spread parameter (FSP). EP was computed as the mean of the OMPTFD $M(t, \omega)$ along each time slice, which gives a representation of energy variation with time as

$$EP(t) = \frac{1}{\omega_m} \sum_{\omega=0}^{\omega_m} M(t, \omega), \quad (8.138)$$

where ω_m is the maximum frequency present in the signal. Signals generated by pathological knees are expected to be highly time-variant (nonstationary) because of differences in cartilage roughness and nonuniformity. Thus, $EP(t)$ of an abnormal VAG signal is expected to show large variations over time. Although $EP(t)$ may be computed directly from the signal, its derivation as above from the OMPTFD includes the benefits of adaptive filtering because only the coherent structures in the signal are retained (see Section 8.7.2).

The parameter ESP, computed as the SD of $M(t, \omega)$ along each time slice, measures the spread of energy over frequency for each time slice as

$$\text{ESP}(t) = \left[\frac{1}{\omega_m} \sum_{\omega=0}^{\omega_m} [M(t, \omega) - \text{EP}(t)]^2 \right]^{\frac{1}{2}}. \quad (8.139)$$

ESP could serve as a feature suitable for the analysis of multicomponent signals. Abnormal VAG signals generated as a result of friction between rough cartilage surfaces may have more components because of the nonuniformity of the surfaces, and hence a higher ESP than normal VAG signals.

The parameter FP, also known as IMF, is given by the first moment of $M(t, \omega)$ along each time slice, as

$$\text{FP}(t) = \frac{\sum_{\omega=0}^{\omega_m} \omega M(t, \omega)}{\sum_{\omega=0}^{\omega_m} M(t, \omega)}. \quad (8.140)$$

FP characterizes the frequency dynamics of the signal. Movement of the knee during VAG signal acquisition may cause linear or nonlinear FM components, with the modulation index depending on the state of lubrication, stiffness, and roughness of the cartilage surfaces [445]. Pathological knees have less lubricated and rougher cartilage surfaces than normal knees, and hence the variations in FP for pathological knees are expected to be different from those for normal knees.

The parameter FSP is given by the second central moment of $M(t, \omega)$ along each time slice as

$$\text{FSP}(t) = \left[\frac{\sum_{\omega=0}^{\omega_m} [\omega - \text{FP}(t)]^2 M(t, \omega)}{\sum_{\omega=0}^{\omega_m} M(t, \omega)} \right]^{\frac{1}{2}}. \quad (8.141)$$

FSP gives the spread of frequency about the mean frequency for each time instant. The spread of frequency at an instant of time could arise as a result of AM components. AM effects may occur in VAG signals and may be dependent on the quality and intensity of the sound produced due to joint vibration. FSP could serve as a good feature to identify noisy knees.

In order to facilitate the derivation of a global decision on each VAG signal, the mean and SD of the arrays of each of the four parameters were calculated. The mean and SD characterize the central tendency and dispersion of the nonstationary parameters.

The VAG signal of a normal subject is shown in Figure 8.53. The signal demonstrates a normal noisy knee with a click and grinding sound heard during auscultation. The OMPTFD of the signal is shown in part (b) of the same figure. The TFD was constructed using only the coherent structures of the signal and the number of TF atoms used was 588. Optimization of the marginals was achieved in two iterations. The click and grinding sound are shown as high-frequency activities in the TFD and have been represented with good TF localization.

Figure 8.54 shows the VAG signal of a subject with chondromalacia patella. Grinding sound was heard during auscultation. The OMPTFD shown in part (b)

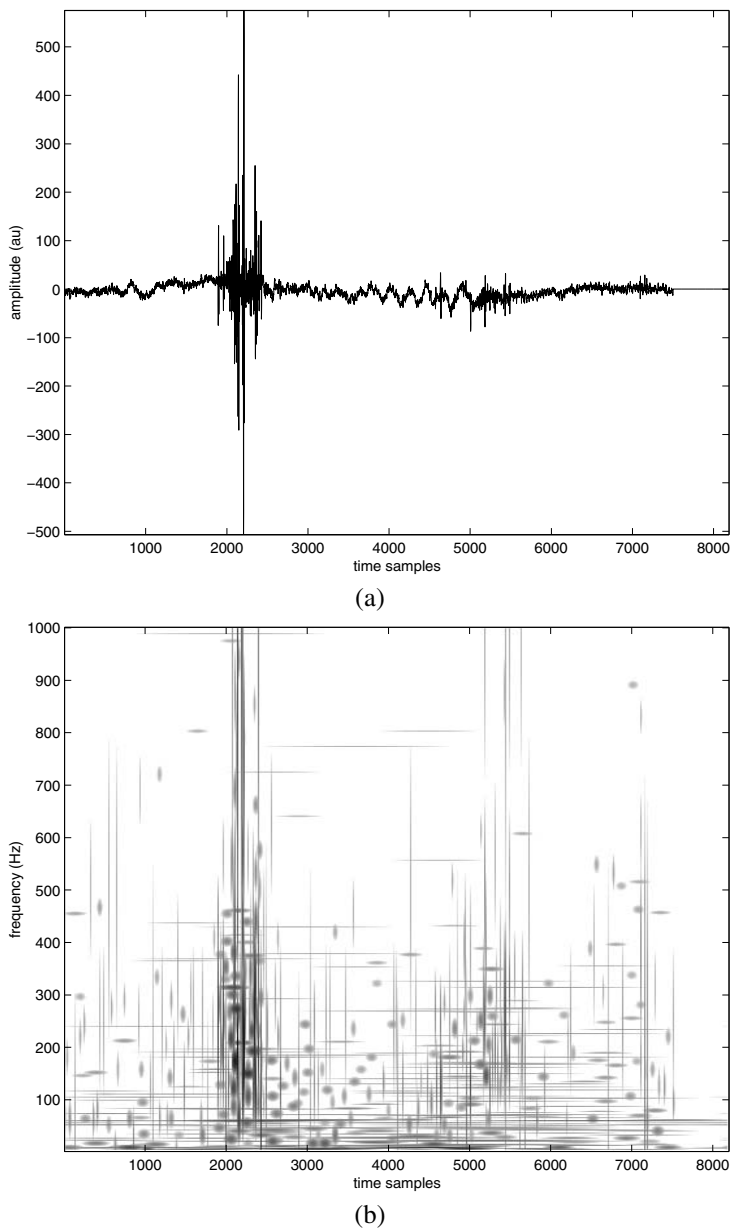


Figure 8.53 (a) VAG signal of a normal subject. A click and grinding sound were heard during auscultation of the knee. au: arbitrary acceleration units. Sampling rate = 2 kHz . (b) OMPTFD of the signal. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, and C.B. Frank, Adaptive time-frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Transactions on Biomedical Engineering*, 47(6):773–783, June 2000. ©IEEE.

of the same figure was constructed using 803 atoms, and the marginals were optimized in two iterations. The OMPTFD indicates two linear FM components (chirps) in the high-frequency range of 500 Hz to 800 Hz . Several tonal components below 400 Hz can also be identified in the TFD.

Figures 8.55 and 8.56 show the EP, ESP, FP, and FSP waveforms derived from the OMPTFD of the normal VAG signal shown in Figure 8.53. Figures 8.57 and 8.58 show the EP, ESP, FP, and FSP waveforms derived from the OMPTFD of the abnormal VAG signal shown in Figure 8.54. It may be readily seen that the parameters of the abnormal VAG signal have larger variations over time as compared to the corresponding parameters of the normal VAG signal. Whereas visual comparative analysis of the TFDs of the normal and abnormal signals could be difficult, several differences are readily observable in the plots of the related parameters.

The EP and ESP features are dependent on the data acquisition protocol and related parameters, in particular, the gain of the amplifier and the power of the VAG signal. To overcome this problem, the values of the CV of EP and ESP were computed and used with the mean and SD of FP and FSP for pattern classification. Using a database consisting of 71 VAG signals, with 51 normal and 20 abnormal signals restricted to chondromalacia patella, an overall accuracy of 77.5% and area under the ROC curve of 0.75 were achieved. The frequency-related parameters were observed to play a major role in the classification task. See Rangayyan and Krishnan [445] for descriptions of methods to detect different types of FM components in TFDs.

It should be noted that, in the analysis of signals using AR coefficients, dominant poles, and other parameters as described in the preceding sections of the present chapter, each signal is divided into a variable number of quasistationary segments. In TF analysis, each signal in its entirety is represented by a small number of features and a global decision is made on each signal, rather than the segment-by-segment decision made in the other approaches mentioned. The TF method does not require labeling of signal segments as normal or abnormal, and it eliminates the need to estimate the joint angle corresponding to the pathology as observed during arthroscopy. See Krishnan et al. [273] and Kim et al. [471] for further discussion on related topics.

8.15 Remarks

We have now reached the stage where we have extended the application of a number of signal processing, modeling, and analysis techniques to nonstationary and multicomponent biomedical signals. Fixed or adaptive segmentation of the signals into quasistationary segments is one approach to facilitate the analysis of such signals using traditional techniques, and we studied several approaches for segmentation. Adaptive segmentation facilitates not only the identification of distinct and separate events at unknown time instants in the given signal, but also the characterization of events of variable duration using the same number of parameters. We have also studied advanced and recently developed techniques such as wavelets and TFDs that permit the analysis of nonstationary and multicomponent biomedical signals without segmentation, and BSS techniques that can help in separating mixed and overlapping

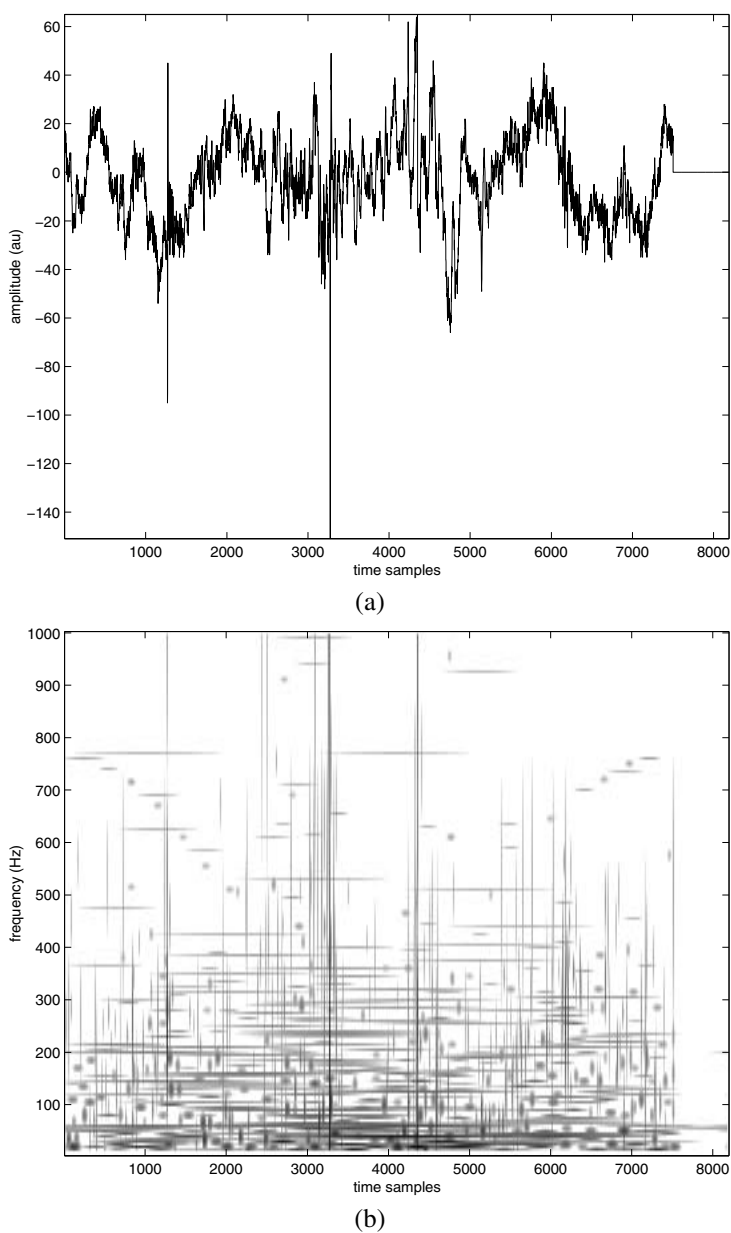
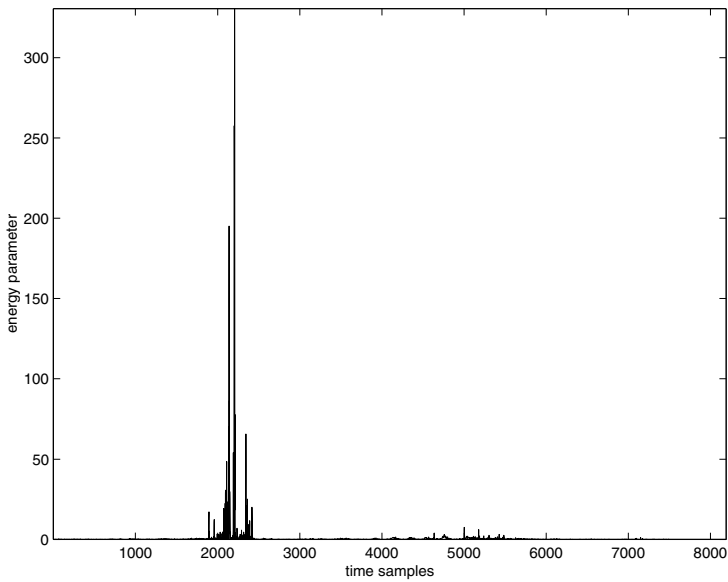
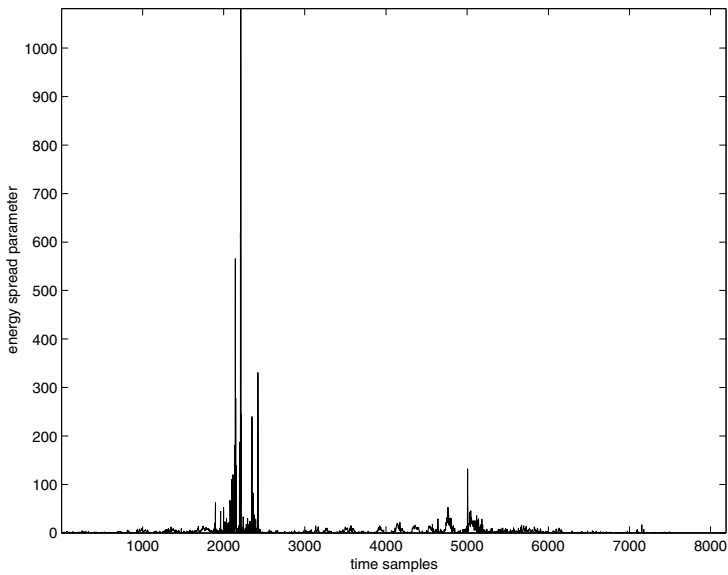


Figure 8.54 (a) VAG signal of a subject with knee-joint pathology. Grinding sound was heard during auscultation of the knee. au: arbitrary acceleration units. Sampling rate = 2 kHz. (b) OMPTFD of the signal. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, and C.B. Frank, Adaptive time-frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Transactions on Biomedical Engineering*, 47(6):773–783, June 2000. ©IEEE.



(a)



(b)

Figure 8.55 (a) EP and (b) ESP obtained from the OMPTFD of the normal VAG signal in Figure 8.53. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, and C.B. Frank, Adaptive time-frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Transactions on Biomedical Engineering*, 47(6):773–783, June 2000. ©IEEE.

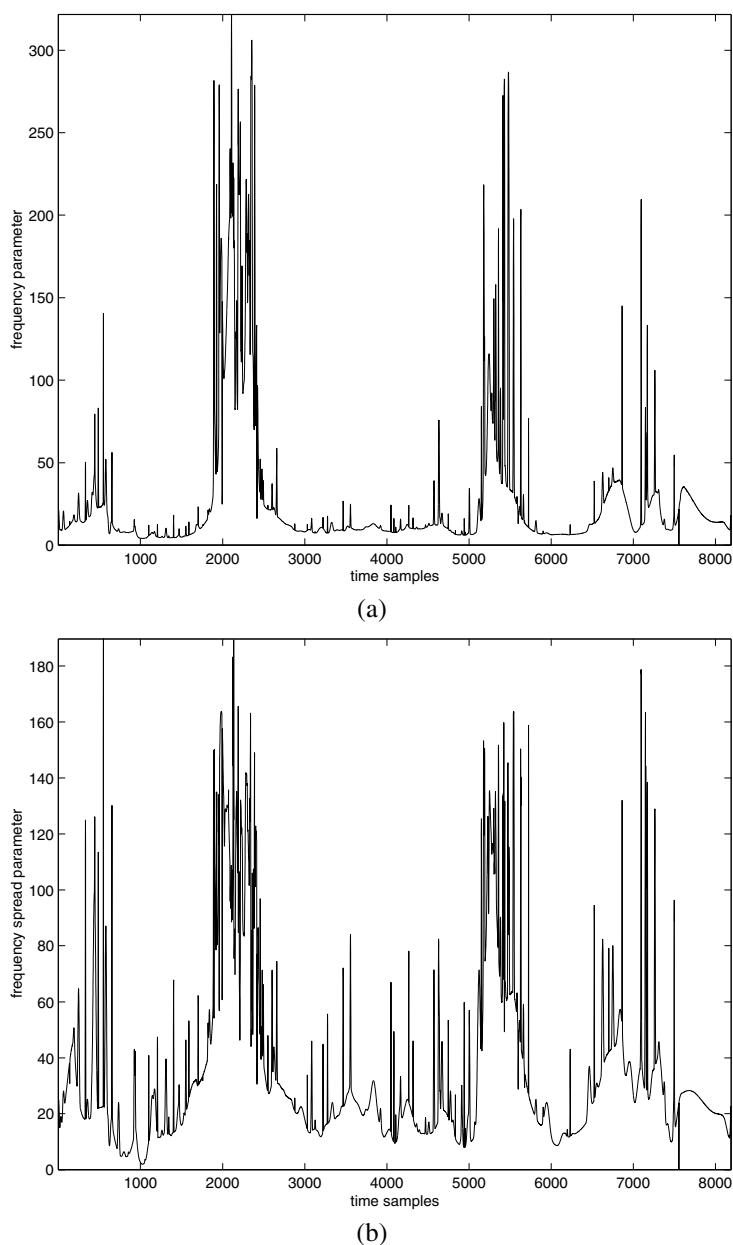


Figure 8.56 (a) FP and (b) FSP obtained from the OMPTFD of the normal VAG signal in Figure 8.53. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, and C.B. Frank, Adaptive time-frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Transactions on Biomedical Engineering*, 47(6):773–783, June 2000. ©IEEE.

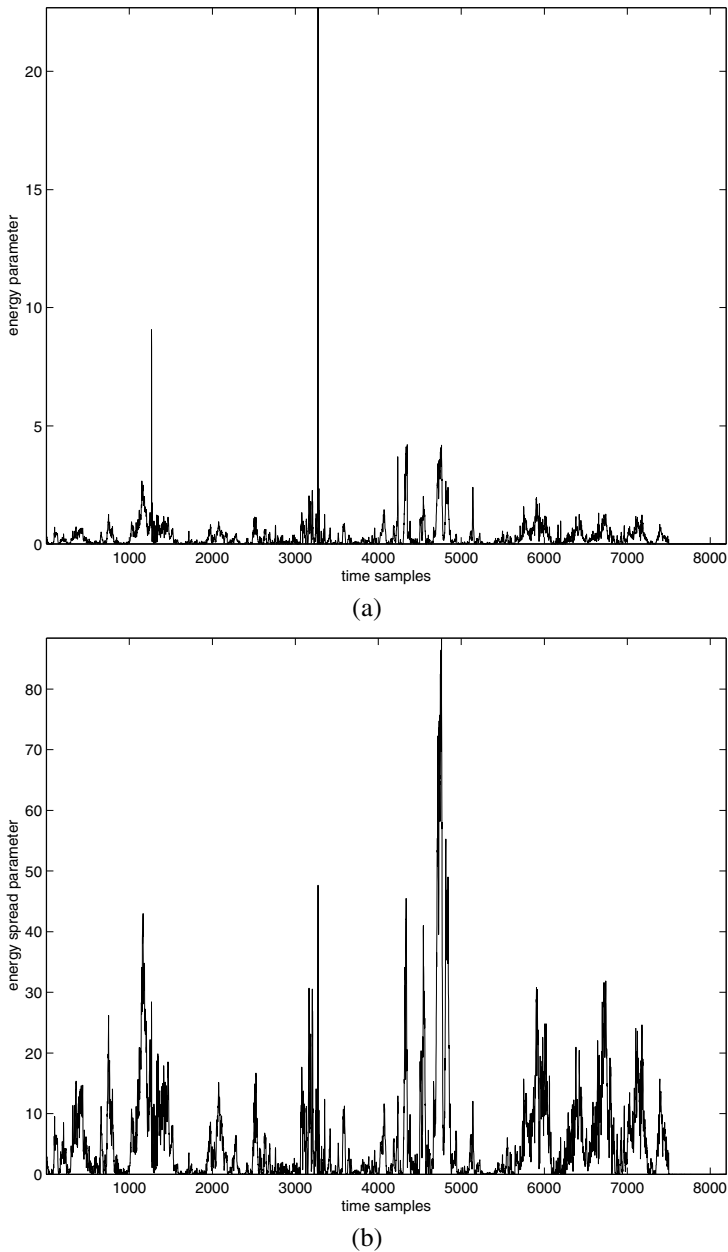


Figure 8.57 (a) EP and (b) ESP obtained from the OMPTFD of the abnormal VAG signal in Figure 8.54. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, and C.B. Frank, Adaptive time-frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Transactions on Biomedical Engineering*, 47(6):773–783, June 2000. ©IEEE.

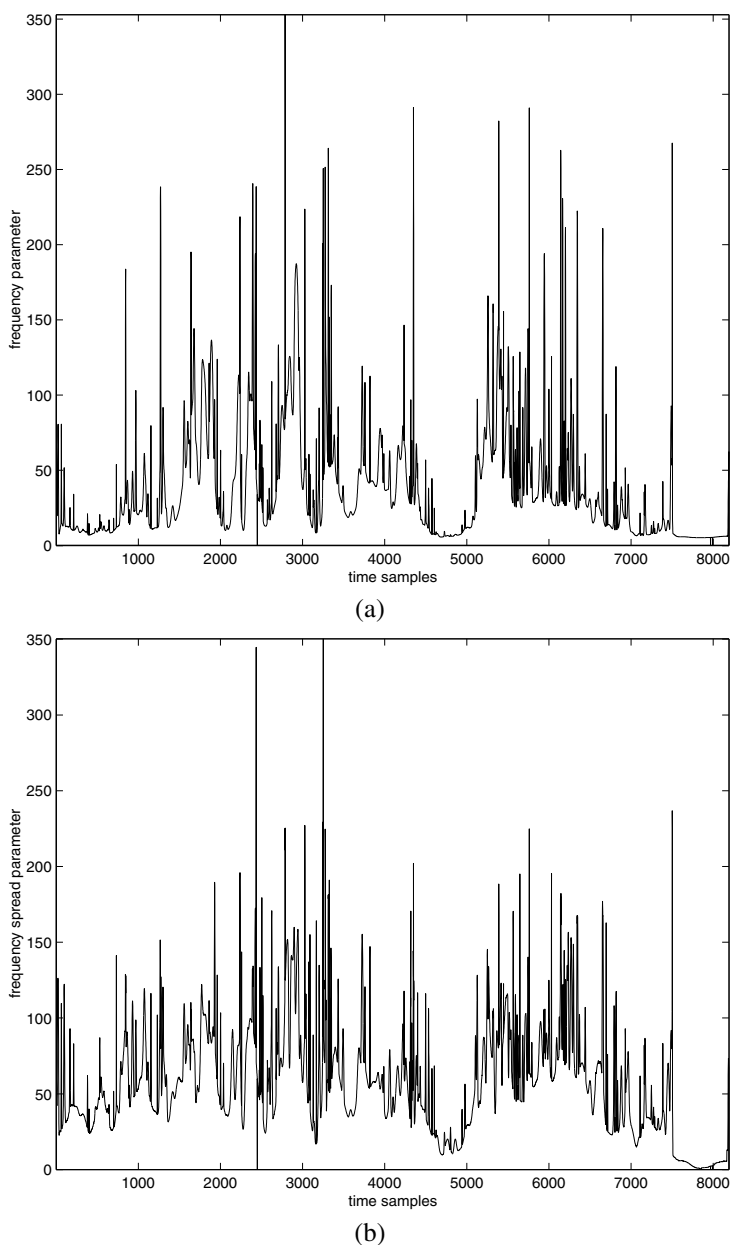


Figure 8.58 (a) FP and (b) FSP obtained from the OMPTFD of the abnormal VAG signal in Figure 8.54. Reproduced with permission from S. Krishnan, R.M. Rangayyan, G.D. Bell, and C.B. Frank, Adaptive time-frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Transactions on Biomedical Engineering*, 47(6):773–783, June 2000. ©IEEE.

components. The results obtained using the techniques studied in the present chapter provide advantages in pattern classification (see Chapter 9) as well as efficient representation and analysis of signals.

8.16 Study Questions and Problems

1. Describe the characteristics of PCG signals that would make them nonstationary. Propose signal processing strategies to break a PCG signal into quasistationary segments.
2. Discuss features of the EEG that make the signal nonstationary. Propose signal processing strategies to detect each type of nonstationarity and to break an EEG signal into quasistationary segments.
3. Investigate features of the EMG that make the signal nonstationary. Propose signal processing strategies to track the time-varying characteristics of the signal. Under what conditions can the signal be partitioned into quasistationary segments? What are the physiological features that you would be able to derive from each segment?
4. Propose an algorithm to perform the segmentation of PCG signals into four parts per cardiac cycle as: (a) the first heart sound; (b) systolic murmur, if present; (c) the second heart sound; and (d) diastolic murmur, if present. If your proposal includes the use of other signals, explain the need and rationale for the use of such signals. Explain the relationship between events in your reference signals and the events of interest in the PCG signal. Provide sketches of typical signals and the results of your methods to illustrate your procedures. Document your procedures using a flowchart or an algorithmic listing. Give at least three nontrivial equations representing important steps in your procedures.

8.17 Laboratory Exercises and Projects

Note: Data files related to the exercises are available at the site

<http://people.ualgary.ca/~ranga/enel563>

1. The speech signal of the word “safety” is given in the file `safety.wav`. You may use the program `safety.m` to read the data. Explore the use of short-time statistics such as *ZCR* and *RMS* values for segmentation of the signal. Study the effect of the duration of the short-time analysis window on the trends in the parameters computed and on segmentation.
2. The files `pec1.dat`, `pec22.dat`, `pec33.dat`, and `pec52.dat` give the PCG, ECG, and carotid pulse signals of two normal subjects and two patients with systolic murmur. You may use the program `plotpec.m` to read the data. Explore the use of short-time *ZCR*, *RMS*, and AR-model coefficients for segmentation of the signals. Evaluate the segment boundaries obtained in relation to the events in the PCG signals as well as the corresponding events in the ECG and carotid pulse channels.

CHAPTER 9

PATTERN CLASSIFICATION AND DIAGNOSTIC DECISION

An important final purpose of biomedical signal analysis is to classify a given signal into one of a few known categories and to arrive at a diagnostic decision regarding the condition of the patient. A physician or medical specialist may achieve this goal via visual or auditory analysis of the signal presented: Comparative analysis of the given signal with others of known diagnoses or established protocols and sets of rules assist in such a decision-making process. The basic knowledge, clinical experience, expertise, and intuition of the physician play significant roles in this process. Some measurements may also be made from the given signal to assist in its analysis, such as the QRS width from an ECG signal plot.

When signal analysis is performed via the application of computer algorithms, the typical result is the extraction of a number of numerical or quantitative features. When the numerical features relate directly to measures of the signal, such as the QRS width and RR interval of an ECG signal, the clinical specialist may be able to use the features in his or her diagnostic logic. Even indirect measures, such as the frequency content of PCG signals and murmurs, may find such direct use. However, when parameters such as AR-model coefficients and spectral statistics are derived, a human analyst is not likely to be able to comprehend and analyze the features. Furthermore, as the number of the computed features increases, the associated diagnos-

tic logic may become too complicated and unwieldy for human analysis. Computer methods would then be desirable to realize the classification and decision process.

At the outset, it should be borne in mind that a biomedical signal forms but one piece of information in arriving at a diagnosis: The classification of a given signal into one of many categories may assist in the diagnostic procedure, but will almost never be the only factor. Regardless, pattern classification based upon signal analysis is an important aspect of biomedical signal analysis, and forms the theme of this chapter. Remaining within the realm of CAD as introduced in Figure 1.53 and Section 1.5, it would be preferable to design methods so as to assist a medical specialist in arriving at a diagnosis rather than to provide a decision.

9.1 Problem Statement

A number of measures and features have been derived from a biomedical signal. Explore methods to classify the signal into one of a few specified categories. Investigate the relevance of the features and the classification methods in arriving at a diagnostic decision about the patient.

Observe that the features may have been derived manually or by computer methods. Note the distinction between classifying the given signal and arriving at a diagnosis regarding the patient: The connection between the two tasks or steps may not always be direct. In other words, a pattern classification method may facilitate the labeling of a given signal as being a member of a particular class; arriving at a diagnosis of the condition of the patient will most likely require the analysis of several items of clinical and other information. Although it is common to work with a prespecified number of pattern classes, many problems do exist where the number of classes is not known *a priori*.

The following sections present a few illustrative case studies. A number of methods for pattern classification, decision making, and evaluation of the results of classification are reviewed and illustrated.

9.2 Illustration of the Problem with Case Studies

9.2.1 Diagnosis of bundle branch block

Bundle-branch block affects the propagation of the excitation pulse through the conduction system of the heart to the ventricles. A block in the left bundle branch results in delayed activation of the left ventricle as compared to the right; a block in the right bundle branch has the opposite effect. Essentially, contraction of the two ventricles becomes asynchronous. The resulting ECG typically displays a wider-than-normal QRS complex (100 – 120 *ms* or more), which could have a jagged or slurred shape as well [33]; see Figure 1.28.

The orientation of the cardiac electromotive forces is affected by bundle-branch block. The initial forces in left bundle-branch block are directed more markedly to the left-posterior, whereas the terminal forces are directed to the superior-left and

posterior parts of the left ventricle [33]. Left bundle-branch block results in the loss of Q waves in leads I, V5, and V6. The following logic assists in the diagnosis of incomplete left bundle-branch block [472]:

IF (QRS duration ≥ 105 ms and ≤ 120 ms) AND
 (QRS amplitude is negative in leads V1 and V2) AND
 (Q or S duration ≥ 80 ms in leads V1 and V2) AND
 (no Q wave is present in any two of leads I, V5, and V6) AND
 (R duration > 60 ms in any two of leads I, aVL, V5, and V6) THEN
the patient has incomplete left bundle-branch block.

Incomplete right bundle-branch block is indicated by the following conditions [472]:

IF (QRS duration ≥ 91 ms and ≤ 120 ms) AND
 (S duration ≥ 40 ms in any two of leads I, aVL, V4, V5, and V6) AND
 in lead V1 or V2 EITHER
 [(R duration > 30 ms) AND (R amplitude > 100 μ V) AND
 (no S wave is present)] OR
 [(R' duration > 30 ms) AND (R' amplitude > 100 μ V) AND
 (no S' wave is present)] THEN
the patient has incomplete right bundle-branch block.

[Note: The first positive deflection of a QRS complex is referred to as the R wave and the second positive deflection (if present) is referred to as the R' wave. Similarly, S and S' indicate the first and second (if present) negative deflections, respectively, after the R wave.]

Note that the logic or decision rules given above may be used either by a human analyst or in a computer algorithm after the durations and amplitudes of the various waves mentioned have been measured or computed. Cardiologists with extensive training and experience may arrive at such decisions via visual analysis of an ECG record without resorting to actual measurements.

9.2.2 Normal or ectopic ECG beat?

PVCs caused by ectopic foci could be precursors of more serious arrhythmia, and hence the detection of such beats is important in cardiac monitoring. As illustrated in Sections 5.4.2 and 5.7 as well as in Figures 5.1 and 5.11, PVCs possess shorter preceding RR intervals than normal beats and display bizarre waveshapes that are markedly different from those of the normal QRS complexes of the same subject. Therefore, a simple rule to detect PVCs or ectopic beats could be as follows:

IF (the RR interval of the beat is less than the normal at the current heart rate) AND
 (the QRS waveshape is markedly different from the normal QRS of the patient)
 THEN *the beat is a PVC.*

As in the preceding case study of bundle-branch block, the logic above may be easily applied for visual analysis of an ECG signal by a physician or a trained observer. Computer implementation of the first part of the rule relating in an objective or quantitative manner to the RR interval is simple. However, implementation of the second condition on waveshape, being qualitative and subjective, is neither direct nor easy. Regardless, we have seen in Chapter 5 how we may characterize waveshape. Figures 5.1 and 5.11 illustrate the application of waveshape analysis to quantify the differences between the shapes of normal QRS complexes and ectopic beats. Figure 5.2 suggests how a 2D feature space may be divided by a simple linear decision boundary to categorize beats as normal or ectopic. We shall study the details of such methods later in this chapter.

9.2.3 Is there an alpha rhythm?

The alpha rhythm appears in an EEG record as an almost-sinusoidal wave (see Figure 1.39); a trained EEG technologist or physician can readily recognize the pattern at a glance from an EEG record plotted at the standard scale. The number of cycles of the wave may be counted over one or two seconds of the plot if an estimate of the dominant frequency of the rhythm is desired.

In computer analysis of EEG signals, the ACF and PSD may be used to detect the presence of the alpha rhythm. We saw in Chapter 4 how these two functions demonstrate peaks at the basic period or dominant frequency of the rhythm, respectively (see Figure 4.9). A peak-detection algorithm may be applied to the ACF, and the presence of a significant peak in the range $75 - 125 \text{ ms}$ for the delay or lag may be used as an indication of the existence of the alpha rhythm. If the PSD is available, the fractional power of the signal in the band $8 - 12 \text{ Hz}$ (see Equation 6.40) may be computed: A high value of the fraction indicates the presence of the alpha rhythm. Note that the logic described above includes the qualifiers “significant” and “high”; experimentation with a number of signals that have been categorized by experts should assist in assigning a numerical value to represent the significance of the features described.

9.2.4 Is a murmur present?

Detection of the presence of a heart murmur is a fairly simple task for a trained physician or cardiologist: In performing auscultation of a patient with a stethoscope, the cardiologist needs to determine the existence of noise-like, high-frequency sounds between the low-frequency S1 and S2. It is necessary to exercise adequate care to reject high-frequency noise from other sources such as breathing, wheezing, and scraping of the stethoscope against the skin or hair. The cardiologist also has to distinguish between innocent physiological murmurs and those due to cardiovascular defects and diseases. Further discrimination between different types of murmurs requires more careful analysis: Figure 5.6 illustrates a decision tree to classify systolic murmurs based upon envelope analysis.

We have seen in Chapters 6 and 7 how we may derive frequency-domain parameters that relate to the presence of murmurs in the PCG signal. Once we have derived such numerical features for a number of signals of known categories of diseases (diagnoses), it becomes possible to design and train classifiers to categorize new signals into one of a few prespecified classes.

The preceding case studies suggest that the classification of patterns in a signal may, in some cases, be based upon thresholds applied to quantitative measurements obtained from the signal; in some other cases, it may be based upon objective measures derived from the signal that attempt to quantify certain notions regarding the characteristics of signals belonging to various categories. Classification may also be based upon the differences between certain measures derived from the signal on hand and those of established examples with known categorization. The succeeding sections of this chapter describe procedures for classification of signals based upon the approaches suggested above.

9.3 Pattern Classification

Pattern recognition or classification may be defined as categorization of input data into identifiable classes via the extraction of significant features or attributes of the data from a background of irrelevant detail [267, 268, 417, 473–476]. In biomedical signal analysis, after quantitative features have been extracted from the given signals, each signal may be represented by a feature vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$, which is also known as a measurement vector or a pattern vector. When the values x_i are real numbers, $\mathbf{x}$ is a point in an n -dimensional Euclidean space: Vectors of similar objects may be expected to form clusters as illustrated in Figure 9.1.

For efficient pattern classification, measurements that could lead to disjoint sets or clusters of feature vectors are desired. This point underlines the importance of appropriate design of the preprocessing and feature extraction procedures. Features or characterizing attributes that are common to all patterns belonging to a particular class are known as *intraset* or *intraclass features*. Discriminant features that represent differences between pattern classes are called *interaset* or *interclass features*.

The pattern classification problem is that of generating optimal decision boundaries or decision procedures to separate the data into pattern classes based on the feature vectors. Figure 9.1 illustrates a simple linear decision function or boundary to separate 2D feature vectors into two classes.

9.4 Supervised Pattern Classification

Problem: You are provided with a number of feature vectors with classes assigned to them. Propose techniques to characterize the boundaries that separate the classes.

Solution: A given set of feature vectors of known categorization is often referred to as a *training set*. The availability of a training set facilitates the development of mathematical functions that can characterize the separation between the classes. The

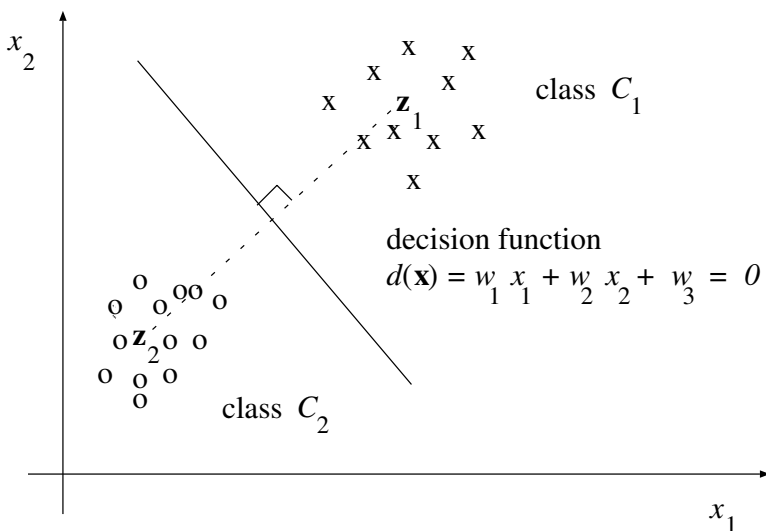


Figure 9.1 Two-dimensional feature vectors of two classes C_1 and C_2 . The prototypes of the two classes are indicated by the vectors $\mathbf{z}_1$ and $\mathbf{z}_2$. The linear decision function shown $d(\mathbf{x})$ (solid line) is the perpendicular bisector of the straight line joining the two class prototypes (dashed line).

functions may then be applied to new feature vectors of unknown classes to classify or recognize them. This approach is known as *supervised pattern classification*. A set of feature vectors of known categorization that is used to evaluate a classifier designed in this manner is referred to as a *test set*. The following sections describe a few methods that can assist in the development of discriminant and decision functions.

9.4.1 Discriminant and decision functions

A general linear discriminant or decision function is of the form

$$d(\mathbf{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_{n+1} = \mathbf{w}^T \mathbf{x}, \quad (9.1)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n, 1)^T$ is the feature vector augmented by an additional entry equal to unity, and $\mathbf{w} = (w_1, w_2, \dots, w_n, w_{n+1})^T$ is a correspondingly augmented weight vector. A two-class pattern classification problem may be stated as

$$d(\mathbf{x}) = \mathbf{w}^T \mathbf{x} \begin{cases} > 0 & \text{if } \mathbf{x} \in C_1, \\ \leq 0 & \text{if } \mathbf{x} \in C_2, \end{cases} \quad (9.2)$$

where C_1 and C_2 represent the two classes. The discriminant function may be interpreted as the boundary separating the classes C_1 and C_2 , as illustrated in Figure 9.1.

In the general case of an M -class pattern classification problem, we will need M weight vectors and M decision functions to perform the following decisions:

$$d_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} \begin{cases} > 0 & \text{if } \mathbf{x} \in C_i, \\ \leq 0 & \text{otherwise,} \end{cases} \quad (9.3)$$

for $i = 1, 2, \dots, M$, where $\mathbf{w}_i = (w_{i1}, w_{i2}, \dots, w_{in}, w_{i,n+1})^T$ is the weight vector for the class C_i .

Three cases arise in solving this problem [473]:

Case 1: Each class is separable from the rest by a single decision surface:

$$\text{if } d_i(\mathbf{x}) > 0, \text{ then } \mathbf{x} \in C_i. \quad (9.4)$$

Case 2: Each class is separable from every other individual class by a distinct decision surface, that is, the classes are pairwise separable. There are $M(M-1)/2$ decision surfaces given by $d_{ij}(\mathbf{x}) = \mathbf{w}_{ij}^T \mathbf{x}$, such that

$$\text{if } d_{ij}(\mathbf{x}) > 0 \forall j \neq i, \text{ then } \mathbf{x} \in C_i. \quad (9.5)$$

[Note: $d_{ij}(\mathbf{x}) = -d_{ji}(\mathbf{x})$.]

Case 3: There exist M decision functions $d_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x}$, $k = 1, 2, \dots, M$, with the property that

$$\text{if } d_i(\mathbf{x}) > d_j(\mathbf{x}) \forall j \neq i, \text{ then } \mathbf{x} \in C_i. \quad (9.6)$$

This is a special instance of Case 2. We may define

$$d_{ij}(\mathbf{x}) = d_i(\mathbf{x}) - d_j(\mathbf{x}) = (\mathbf{w}_i - \mathbf{w}_j)^T \mathbf{x} = \mathbf{w}_{ij}^T \mathbf{x}. \quad (9.7)$$

If the classes are separable under Case 3, they are separable under Case 2; the converse, in general, is not true.

When patterns need to be separated into multiple categories, it may be possible to convert the problem into a series of binary decision problems by considering the separation of each class against the set of remaining classes.

Patterns that may be separated by linear decision functions as above are said to be *linearly separable*. In other situations, an infinite variety of complex decision boundaries may be formulated by using generalized decision functions based upon nonlinear functions of the feature vectors as

$$d(\mathbf{x}) = w_1 f_1(\mathbf{x}) + w_2 f_2(\mathbf{x}) + \dots + w_K f_K(\mathbf{x}) + w_{K+1} \quad (9.8)$$

$$= \sum_{i=1}^{K+1} w_i f_i(\mathbf{x}). \quad (9.9)$$

Here, $\{f_i(\mathbf{x})\}$, $i = 1, 2, \dots, K$, are real, single-valued functions of $\mathbf{x}$; $f_{K+1}(\mathbf{x}) = 1$.

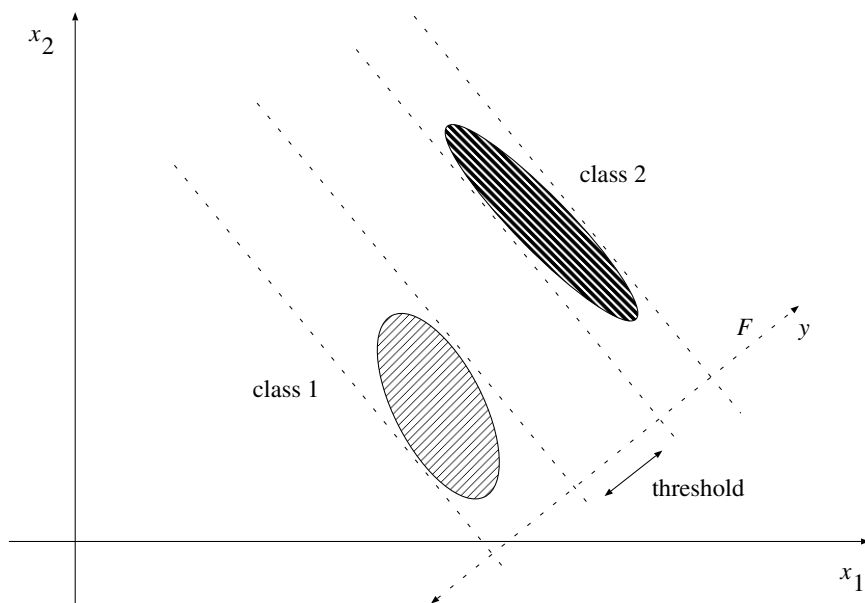


Figure 9.2 Illustration of a classifier based upon FLDA.

Whereas the functions $f_i(\mathbf{x})$ may be nonlinear in the n -dimensional space of $\mathbf{x}$, the decision function may be formulated as a linear function by defining a transformed feature vector $\mathbf{x}^\dagger = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_K(\mathbf{x}), 1]^T$. Then, $d(\mathbf{x}) = \mathbf{w}^T \mathbf{x}^\dagger$, with $\mathbf{w} = [w_1, w_2, \dots, w_K, w_{K+1}]^T$. Once evaluated, $\{f_i(\mathbf{x})\}$ is just a set of numerical values, and $\mathbf{x}^\dagger$ is simply a K -dimensional vector augmented by an entry equal to unity.

See Section 9.11.1 for an illustration of application of linear discriminant analysis to an ECG signal with PVCs.

9.4.2 Fisher linear discriminant analysis

Fisher linear discriminant analysis (FLDA) is a technique that projects multidimensional data on to a 1D space or line which allows for improved discrimination [267]; dimensionality reduction is achieved in the process. In a pattern classification problem with two classes, the projection in FLDA is designed to maximize the distance between the means of the feature vectors for the two classes and minimize their variance within each class.

Figure 9.2 illustrates the concept of a linear classifier based on FLDA. Consider two classes, C_1 and C_2 , in which each sample is represented by the features $\mathbf{x} = [x_1, x_2]^T$. If only the feature x_1 is used for classification, it is equivalent to the situation when all of the data points are projected on to the x_1 axis. Such a projection causes overlapping regions containing samples from both classes; thus, x_1

provides poor separation between the classes C_1 and C_2 . The feature x_2 also shows overlap between the two classes and provides poor separation. However, by inspection of Figure 9.2, it is evident that the classes C_1 and C_2 can be clearly separated by a straight line oriented at about 130° to the abscissa. Hence, a classifier can be designed using projections of the feature vectors or samples provided on to the line labeled as F in Figure 9.2. FLDA provides the weights that define the projection as explained above [267, 282].

Let a given set of feature vectors

$$X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} = \{X_1, X_2\} \quad (9.10)$$

be partitioned into N_1 training samples in subset X_1 , corresponding to class C_1 , and N_2 training samples in subset X_2 , corresponding to class C_2 , with $N_1 + N_2 = N$. The projection of $\mathbf{x}_i$ on to the FLDA discriminant line is

$$y_i = \mathbf{w}^T \mathbf{x}_i. \quad (9.11)$$

The projected value y_i belongs to the subset Y_1 or Y_2 . The weight vector $\mathbf{w}$ defines the direction of the axis of the projected values y_i . The mean of all of the samples belonging to each class before projection is

$$\mathbf{m}_k = \frac{1}{N_k} \sum_{\mathbf{x}_i \in X_k} \mathbf{x}_i, \quad (9.12)$$

where $k = 1, 2$, and $i = 1, 2, \dots, N$. The mean of the projected values in each class is

$$\begin{aligned} \tilde{m}_k &= \frac{1}{N_k} \sum_{y_i \in Y_k} y_i \\ &= \frac{1}{N_k} \sum_{\mathbf{x}_i \in X_k} \mathbf{w}^T \mathbf{x}_i, \\ &= \mathbf{w}^T \mathbf{m}_k, \quad k = 1, 2. \end{aligned} \quad (9.13)$$

The difference between the means of the classes after projection is

$$|\tilde{m}_1 - \tilde{m}_2| = |\mathbf{w}^T \mathbf{m}_1 - \mathbf{w}^T \mathbf{m}_2|. \quad (9.14)$$

The variance or scatter of the projected samples y_i in each class is

$$\begin{aligned} \tilde{s}_k^2 &= \sum_{y_i \in Y_k} (y_i - \tilde{m}_k)^2 \\ &= \sum_{\mathbf{x}_i \in X_k} (\mathbf{w}^T \mathbf{x}_i - \mathbf{w}^T \mathbf{m}_k)^2 \\ &= \sum_{\mathbf{x}_i \in X_k} \mathbf{w}^T (\mathbf{x}_i - \mathbf{m}_k) (\mathbf{x}_i - \mathbf{m}_k)^T \mathbf{w}, \quad k = 1, 2 \end{aligned}$$

$$= \mathbf{w}^T \mathbf{S}_k \mathbf{w}, \quad (9.15)$$

where

$$\mathbf{S}_k = \sum_{\mathbf{x}_i \in X_k} (\mathbf{x}_i - \mathbf{m}_k)(\mathbf{x}_i - \mathbf{m}_k)^T, \quad k = 1, 2. \quad (9.16)$$

$\mathbf{S}_k$ is known as the within-class or intraclass scatter matrix for class k .

The total within-class scatter of the two classes is

$$\tilde{s}_1^2 + \tilde{s}_2^2 = \mathbf{w}^T \{ \mathbf{S}_1 + \mathbf{S}_2 \} \mathbf{w} = \mathbf{w}^T \mathbf{S}_W \mathbf{w}, \quad (9.17)$$

with $\mathbf{S}_W = \mathbf{S}_1 + \mathbf{S}_2$.

The FLDA criterion function is defined as

$$J(\mathbf{w}) = \frac{|\tilde{m}_1 - \tilde{m}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}. \quad (9.18)$$

Optimal separation between the classes is achieved when $J(\mathbf{w})$ is at its maximum. The denominator term of the function $J(\mathbf{w})$ is given by Equation 9.17. The numerator gives the separation between the means of the projected samples for the two classes, and is obtained as follows:

$$\begin{aligned} (\tilde{m}_1 - \tilde{m}_2)^2 &= (\mathbf{w}^T \mathbf{m}_1 - \mathbf{w}^T \mathbf{m}_2)^2 \\ &= \mathbf{w}^T (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T \mathbf{w} \\ &= \mathbf{w}^T \mathbf{S}_B \mathbf{w}. \end{aligned} \quad (9.19)$$

Here, $\mathbf{S}_B = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T$ is the between-class or interclass scatter matrix. The FLDA criterion is given by

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}. \quad (9.20)$$

The weight vector $\mathbf{w}_o$ that maximizes $J(\mathbf{w})$ can be derived by solving a generalized eigenvalue problem. When a new sample is to be classified, its feature vector $\mathbf{x}$ is projected using Equation 9.11 and two-category classification is performed as

$$\mathbf{x} \in \begin{cases} C_1 & \text{if } y = \mathbf{w}_o^T \mathbf{x} < T_1, \\ C_2 & \text{otherwise,} \end{cases} \quad (9.21)$$

where T_1 is a threshold.

Illustration of application: Rangayyan and Wu [264] applied a number of statistical measures for screening of VAG signals (see Section 5.12.1 for a description of the data set used). The parameters used included FF computed for the full duration of each swing cycle and for the first and second halves (labeled as FF_1 and FF_2), corresponding to extension and flexion, as well as the entropy (H), skewness (S), and kurtosis (K). The features could not perform screening with high accuracy

on their own: the six features FF, FF_1, FF_2, S, K , and H individually provided A_z values of 0.72, 0.73, 0.68, 0.70, 0.61, and 0.60, respectively. Figure 9.3 shows the scatter plot of the feature vector $[FF_1, FF_2]$ for the set of 89 VAG signals. The straight line shown in the figure represents the decision function obtained via FLDA. The equation for the decision boundary is

$$0.0058 FF_1 - 0.0010 FF_2 - 0.0192 = 0. \quad (9.22)$$

Due to substantial overlap of the samples in the two categories, FLDA could correctly classify only 19/38 abnormal signals and 40/51 normal signals, yielding an average classification accuracy of 66%. The use of FLDA with all of the six features listed above and the LOO method for cross-validation did not provide any better results, with $A_z = 0.72$. The results obtained and the illustration in Figure 9.3 indicate that the samples are not linearly separable using the measures derived. Mu et al. [274] obtained much better classification performance with $A_z = 0.95$ using the same features as above but with a genetic algorithm for feature selection and the strict two-surface proximal classifier. See Section 9.8.1 for the results of application of a neural network to the same data set.

See Figure 5.2 and Section 9.1.1 for additional illustrations of application of linear decision functions to the classification of ECG signals.

9.4.3 Distance functions

Consider M pattern classes represented by their prototype patterns $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_M$. The prototype of a class is typically computed as the average of all of the feature vectors belonging to the class. Figure 9.1 illustrates schematically the prototypes $\mathbf{z}_1$ and $\mathbf{z}_2$ of the two classes shown.

The Euclidean distance between an arbitrary pattern vector $\mathbf{x}$ and the i^{th} prototype is given as

$$D_i = \|\mathbf{x} - \mathbf{z}_i\| = \sqrt{(\mathbf{x} - \mathbf{z}_i)^T (\mathbf{x} - \mathbf{z}_i)}. \quad (9.23)$$

A simple rule to classify the pattern vector $\mathbf{x}$ would be to choose that class for which the vector has the smallest distance:

$$\text{if } D_i < D_j \forall j \neq i, \text{ then } \mathbf{x} \in C_i. \quad (9.24)$$

A simple relationship may be established between discriminant functions and distance functions as follows [473]:

$$\begin{aligned} D_i^2 &= \|\mathbf{x} - \mathbf{z}_i\|^2 = (\mathbf{x} - \mathbf{z}_i)^T (\mathbf{x} - \mathbf{z}_i) \\ &= \mathbf{x}^T \mathbf{x} - 2\mathbf{x}^T \mathbf{z}_i + \mathbf{z}_i^T \mathbf{z}_i = \mathbf{x}^T \mathbf{x} - 2(\mathbf{x}^T \mathbf{z}_i - \frac{1}{2} \mathbf{z}_i^T \mathbf{z}_i). \end{aligned} \quad (9.25)$$

Choosing the minimum of D_i^2 is equivalent to choosing the minimum of D_i (as all $D_i > 0$). Furthermore, from the equation given above, it follows that choosing

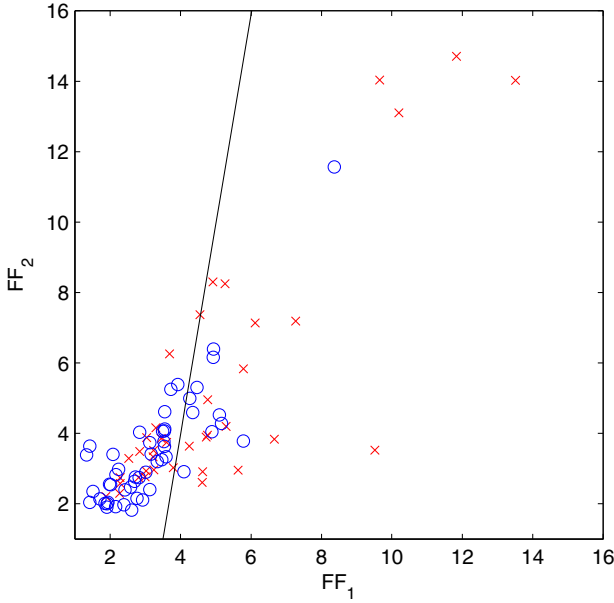


Figure 9.3 Illustration of classification of a data set of 89 VAG signals using FLDA. The circles represent normal samples of the feature vector $[FF_1, FF_2]^T$ and the crosses represent abnormal samples. The straight line is the FLDA decision function, given by $0.0058 FF_1 - 0.0010 FF_2 - 0.0192 = 0$. Figure courtesy of Tingting Mu, University of Liverpool, Liverpool, UK.

the minimum of D_i^2 is equivalent to choosing the maximum of $(\mathbf{x}^T \mathbf{z}_i - \frac{1}{2} \mathbf{z}_i^T \mathbf{z}_i)$. Therefore, we may define the decision function

$$d_i(\mathbf{x}) = (\mathbf{x}^T \mathbf{z}_i - \frac{1}{2} \mathbf{z}_i^T \mathbf{z}_i), \quad i = 1, 2, \dots, M. \quad (9.26)$$

A decision rule may then be stated as follows:

$$\text{if } d_i(\mathbf{x}) > d_j(\mathbf{x}) \quad \forall j \neq i, \text{ then } \mathbf{x} \in C_i. \quad (9.27)$$

This is a linear discriminant function, which becomes obvious from the following representation: If z_{ij} , $j = 1, 2, \dots, n$, are the components of $\mathbf{z}_i$, let $w_{ij} = z_{ij}$, $j = 1, 2, \dots, n$; $w_{i,n+1} = -\frac{1}{2} \mathbf{z}_i^T \mathbf{z}_i$; and $\mathbf{x} = [x_1, x_2, \dots, x_n, 1]^T$. Then, $d_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x}$, $i = 1, 2, \dots, M$, where $\mathbf{w}_i = [w_{i1}, w_{i2}, \dots, w_{i,n+1}]^T$. Therefore, distance functions may be formulated as linear discriminant or decision functions.

9.4.4 The nearest neighbor rule

Suppose that we are provided with a set of N sample patterns $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N\}$ of known classification: Each pattern belongs to one of M classes $\{C_1, C_2, \dots, C_M\}$.

We are then given a new feature vector $\mathbf{x}$ whose class needs to be determined. Let us compute a distance measure $D(\mathbf{s}_i, \mathbf{x})$ between the vector $\mathbf{x}$ and each sample pattern. Then, the nearest-neighbor rule states that the vector $\mathbf{x}$ is to be assigned to the class of the sample that is the closest to $\mathbf{x}$:

$$\mathbf{x} \in C_i \text{ if } D(\mathbf{s}_i, \mathbf{x}) = \min\{D(\mathbf{s}_l, \mathbf{x})\}, \quad l = 1, 2, \dots, N. \quad (9.28)$$

A major disadvantage of the above method is that the classification decision is made based upon a single sample vector of known classification. The nearest neighbor may happen to be an outlier that is not representative of its class. It would be more reliable to base the classification upon several samples: We may consider a certain number k of the nearest neighbors of the sample to be classified, and then seek a majority opinion. This leads to the so-called *k-nearest-neighbor* or *k-NN rule*: Determine the k nearest neighbors of $\mathbf{x}$, and use the majority of equal classifications in this group as the classification of $\mathbf{x}$.

9.5 Unsupervised Pattern Classification

Problem: *We are given a set of feature vectors with no categorization or classes attached to them. No prior training information is available. How may we group the vectors into multiple categories?*

Solution: The design of distance functions and decision boundaries requires a training set of feature vectors of known classes. The functions so designed may then be applied to a new set of feature vectors or samples to perform pattern classification. Such a procedure is known as *supervised* pattern classification due to the initial training step. In some situations a training step may not be possible, and we may be required to classify a given set of feature vectors into either a prespecified or unknown number of categories. Such a problem is labeled as *unsupervised* pattern classification and may be solved by cluster-seeking methods.

9.5.1 Cluster seeking methods

Given a set of feature vectors, we may examine them for the formation of inherent groups or clusters. This is a simple task in the case of 2D vectors, where we may plot them, visually identify groups, and label each group with a pattern class; see Figures 9.1 and 9.2. Allowance may have to be made to assign the same class to multiple disjoint groups. Such an approach may be used even when the number of classes is not known at the outset. When the vectors have a dimension higher than three, visual analysis will not be feasible. It then becomes necessary to define criteria to group the given vectors on the basis of similarity, dissimilarity, or distance measures. A few examples of such measures are as follows [473]:

- Euclidean distance

$$D_E^2 = \|\mathbf{x} - \mathbf{z}\|^2 = (\mathbf{x} - \mathbf{z})^T(\mathbf{x} - \mathbf{z}) = \sum_{i=1}^n (x_i - z_i)^2. \quad (9.29)$$

Here, $\mathbf{x}$ and $\mathbf{z}$ are two feature vectors; the latter could be a class prototype, if available. A small value of D_E indicates greater similarity between the two vectors than a large value of D_E .

▪ Mahalanobis distance

$$D_M^2 = (\mathbf{x} - \mathbf{m})^T \mathbf{C}^{-1} (\mathbf{x} - \mathbf{m}), \quad (9.30)$$

where $\mathbf{x}$ is a feature vector being compared to a pattern class for which $\mathbf{m}$ is the class mean vector and $\mathbf{C}$ is the covariance matrix. Inclusion of the (inverse of the) covariance of the distribution of the class in the distance measure allows consideration of the scatter of the constituent samples of the class: A large scatter of the population diminishes the distance measure as compared to a tighter cluster. A small value of D_M indicates a higher potential membership of the vector $\mathbf{x}$ in the class than a large value of D_M .

▪ Normalized dot product (cosine of the angle between the vectors $\mathbf{x}$ and $\mathbf{z}$)

$$D_d = \frac{\mathbf{x}^T \mathbf{z}}{\|\mathbf{x}\| \|\mathbf{z}\|}. \quad (9.31)$$

A large dot product value indicates a greater degree of similarity between the two vectors than a small value.

The covariance matrix is defined as

$$\mathbf{C} = E[(\mathbf{y} - \mathbf{m})(\mathbf{y} - \mathbf{m})^T], \quad (9.32)$$

where the expectation operation is performed over all feature vectors $\mathbf{y}$ that belong to the class. The covariance matrix provides the covariance of all possible pairs of the features in the feature vector over all samples belonging to the given class. The elements along the main diagonal of the covariance matrix provide the variance of the individual features that make up the feature vector. The covariance matrix represents the scatter of the features that belong to the given class. The mean and covariance need to be updated as more samples are added to a given class in a clustering procedure.

When the Mahalanobis distance needs to be calculated between a sample vector and a number of classes represented by their mean and covariance matrices, a pooled covariance matrix may be used if the numbers of members in the various classes are unequal and low [474]. For example, if the covariance matrices of two classes are $\mathbf{C}_1$ and $\mathbf{C}_2$, and the numbers of members in the two classes are N_1 and N_2 , the pooled covariance matrix is given by

$$\mathbf{C} = \frac{(N_1 - 1)\mathbf{C}_1 + (N_2 - 1)\mathbf{C}_2}{N_1 + N_2 - 2}. \quad (9.33)$$

Various performance indices may be designed to measure the success of a clustering procedure [473]. A measure of the tightness of a cluster is the sum of the squared

errors performance index:

$$J = \sum_{j=1}^{N_c} \sum_{\mathbf{x} \in S_j} \|\mathbf{x} - \mathbf{m}_j\|^2, \quad (9.34)$$

where N_c is the number of cluster domains, S_j is the set of samples in the j^{th} cluster,

$$\mathbf{m}_j = \frac{1}{N_j} \sum_{\mathbf{x} \in S_j} \mathbf{x} \quad (9.35)$$

is the sample mean vector of S_j , and N_j is the number of samples in S_j .

A few other examples of performance indices are:

- Average of the squared distances between the samples in a cluster domain.
- Intracluster variance.
- Average of the squared distances between the samples in different cluster domains.
- Intercluster distances.
- Scatter matrices.
- Covariance matrices.

A simple cluster-seeking algorithm [473]: Suppose we have N sample patterns $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$.

1. Let the first cluster center $\mathbf{z}_1$ be equal to any one of the samples, for example, $\mathbf{z}_1 = \mathbf{x}_1$.
2. Choose a nonnegative threshold θ .
3. Compute the distance D_{21} between $\mathbf{x}_2$ and $\mathbf{z}_1$. If $D_{21} < \theta$, assign $\mathbf{x}_2$ to the domain (class) of cluster center $\mathbf{z}_1$; otherwise, start a new cluster with its center as $\mathbf{z}_2 = \mathbf{x}_2$. For the subsequent steps, let us assume that a new cluster with center $\mathbf{z}_2$ has been established.
4. Compute the distances D_{31} and D_{32} from the next sample $\mathbf{x}_3$ to $\mathbf{z}_1$ and $\mathbf{z}_2$, respectively. If D_{31} and D_{32} are both greater than θ , start a new cluster with its center as $\mathbf{z}_3 = \mathbf{x}_3$; otherwise, assign $\mathbf{x}_3$ to the domain of the closer cluster.
5. Continue to apply steps 3 and 4 by computing and checking the distance from *every* new (unclassified) pattern vector to *every* established cluster center and applying the cluster-assignment or cluster-creation rule.
6. Stop when every given pattern vector has been assigned to a cluster.

Note that the procedure does not require knowledge of the number of classes *a priori*. Note also that the procedure does not assign a real-world class to each cluster: it merely groups the given vectors into disjoint clusters. A subsequent step is required to label each cluster with a class related to the actual problem. Multiple clusters may relate to the same real-world class and may have to be merged.

A major disadvantage of the simple cluster-seeking algorithm is that the results depend upon

- the first cluster center chosen for each domain or class,
- the order in which the sample patterns are considered,
- the value of the threshold θ , and
- the geometrical properties (distributions) of the data (or the feature-vector space).

The maximin-distance clustering algorithm [473]: This method is similar to the previous “simple” algorithm, but first identifies the cluster regions that are the farthest apart, as follows.

1. Let $\mathbf{x}_1$ be the first cluster center $\mathbf{z}_1$.
2. Determine the farthest sample from $\mathbf{x}_1$, and call it cluster center $\mathbf{z}_2$.
3. Compute the distance from each remaining sample to $\mathbf{z}_1$ and to $\mathbf{z}_2$. For every pair of these computations, save the minimum distance, and select the maximum of the minimum distances. If this “maximin” distance is an appreciable fraction of the distance between the cluster centers $\mathbf{z}_1$ and $\mathbf{z}_2$, label the corresponding sample as a new cluster center $\mathbf{z}_3$; otherwise stop forming new clusters and go to Step 5.
4. If a new cluster center was formed in Step 3, repeat Step 3 using a “typical” or the average distance between the established cluster centers for comparison.
5. Assign each remaining sample to the domain of its nearest cluster center.

The term “maximin” refers to the combined use of maximum and minimum distances between the given vectors and the centers of the clusters already formed.

The K -means algorithm [473]: The preceding “simple” and “maximin” algorithms are intuitive procedures. The K -means algorithm is based on iterative minimization of a performance index that is defined as the sum of the squared distances from all points in a cluster domain to the cluster center, as follows.

1. Choose K initial cluster centers $\mathbf{z}_1(1), \mathbf{z}_2(1), \dots, \mathbf{z}_K(1)$. The index in parentheses represents the iteration number.
2. At the k^{th} iterative step, distribute the samples $\{\mathbf{x}\}$ among the K cluster domains, using the relation

$$\mathbf{x} \in S_j(k) \text{ if } \|\mathbf{x} - \mathbf{z}_j(k)\| < \|\mathbf{x} - \mathbf{z}_i(k)\| \forall i = 1, 2, \dots, K, i \neq j, \quad (9.36)$$

where $S_j(k)$ denotes the set of samples whose cluster center is $\mathbf{z}_j(k)$.

3. From the results of Step 2, compute the new cluster centers $\mathbf{z}_j(k+1)$, $j = 1, 2, \dots, K$, such that the sum of the squared distances from all points in $S_j(k)$ to the new cluster center is minimized. In other words, the new cluster center $\mathbf{z}_j(k+1)$ is computed so that the performance index

$$J_j = \sum_{\mathbf{x} \in S_j(k)} \|\mathbf{x} - \mathbf{z}_j(k+1)\|^2, \quad j = 1, 2, \dots, K, \quad (9.37)$$

is minimized. The $\mathbf{z}_j(k+1)$ that minimizes this performance index is simply the sample mean of $S_j(k)$. Therefore, the new cluster center is given by

$$\mathbf{z}_j(k+1) = \frac{1}{N_j} \sum_{\mathbf{x} \in S_j(k)} \mathbf{x}, \quad j = 1, 2, \dots, K, \quad (9.38)$$

where N_j is the number of samples in $S_j(k)$. The name “ K -means” is derived from the manner in which cluster centers are sequentially updated.

4. If $\mathbf{z}_j(k+1) = \mathbf{z}_j(k)$ for $j = 1, 2, \dots, K$, the algorithm has converged: Terminate the procedure. Otherwise, go to Step 2.

The behavior of the K -means algorithm is influenced by:

- the number of cluster centers specified,
- the choice of the initial cluster centers,
- the order in which the sample patterns are considered, and
- the geometrical properties (distributions) of the data (or the feature-vector space).

See Section 9.11.3 for an illustration of application of the K -means method to an ECG signal with PVCs.

9.6 Probabilistic Models and Statistical Decision

Problem: *Pattern classification methods such as discriminant functions are dependent upon the set of training samples provided. Their success, when applied to new cases, will depend upon the accuracy of representation of the various pattern classes by the training samples. How can we design pattern classification techniques that are independent of specific training samples and optimal in a broad sense?*

Solution: Probability functions and probabilistic models may be developed to represent the occurrence and statistical attributes of classes of patterns. Such functions may be based upon large collections of data, historical records, or mathematical models of pattern generation. In the absence of information as above, a training step with samples of known categorization is required to estimate the required model

parameters. It is common practice to assume a Gaussian PDF to represent the distribution of the features for each class, and to estimate the required mean and variance parameters from the training sets. When PDFs are available to characterize pattern classes and their features (see Sections 5.12.2 and 5.12.3), optimal decision functions may be designed based upon statistical functions and decision theory. The following sections describe a few methods that fall into this category.

9.6.1 Likelihood functions and statistical decision

Let $P(C_i)$ be the probability of occurrence of class C_i , $i = 1, 2, \dots, M$; this is known as the *a priori*, *prior*, or unconditional probability. The *a posteriori* or *posterior* probability that an observed sample pattern $\mathbf{x}$ comes from C_i is expressed as $P(C_i|\mathbf{x})$. If a classifier decides that $\mathbf{x}$ comes from C_j when it actually came from C_i , then the classifier is said to incur a *loss* L_{ij} , with $L_{ii} = 0$ or a fixed operational cost and $L_{ij} > L_{ii} \forall j \neq i$.

Since $\mathbf{x}$ may belong to any of M classes under consideration, the expected loss, known as the *conditional average risk* or *loss*, in assigning $\mathbf{x}$ to C_j is [473]

$$R_j(\mathbf{x}) = \sum_{i=1}^M L_{ij} P(C_i|\mathbf{x}). \quad (9.39)$$

A classifier could compute $R_j(\mathbf{x})$, $j = 1, 2, \dots, M$, for each sample $\mathbf{x}$ and then assign $\mathbf{x}$ to the class with the smallest conditional loss. Such a classifier will minimize the total expected loss over all decisions, and is called the *Bayes classifier*. From a statistical point of view, the Bayes classifier represents an optimal classifier.

According to Bayes formula, we have [417, 473]

$$P(C_i|\mathbf{x}) = \frac{P(C_i) p(\mathbf{x}|C_i)}{p(\mathbf{x})}, \quad (9.40)$$

where $p(\mathbf{x}|C_i)$ is called the *likelihood function* of class C_i or the *state-conditional PDF* of $\mathbf{x}$, and $p(\mathbf{x})$ is the PDF of $\mathbf{x}$ regardless of class membership (unconditional). [Note: $P(y)$ is used to represent the probability of occurrence of an event y ; $p(y)$ is used to represent the PDF of a random variable y . Probabilities and PDFs involving a multidimensional feature vector are multivariate functions with dimension equal to that of the feature vector.] Bayes formula shows how observing the sample $\mathbf{x}$ changes the *a priori* probability $P(C_i)$ to the *a posteriori* probability $P(C_i|\mathbf{x})$. In other words, Bayes formula provides a mechanism to update the *a priori* probability $P(C_i)$ to the *a posteriori* probability $P(C_i|\mathbf{x})$ due to the observation of the sample $\mathbf{x}$. Then, we can express the expected loss as [473]

$$R_j(\mathbf{x}) = \frac{1}{p(\mathbf{x})} \sum_{i=1}^M L_{ij} p(\mathbf{x}|C_i) P(C_i). \quad (9.41)$$

As $\frac{1}{p(\mathbf{x})}$ is common for all j , we could modify $R_j(\mathbf{x})$ to

$$r_j(\mathbf{x}) = \sum_{i=1}^M L_{ij} p(\mathbf{x}|C_i) P(C_i). \quad (9.42)$$

In a two-class case with $M = 2$, we obtain the following expressions [473]:

$$r_1(\mathbf{x}) = L_{11} p(\mathbf{x}|C_1) P(C_1) + L_{21} p(\mathbf{x}|C_2) P(C_2). \quad (9.43)$$

$$r_2(\mathbf{x}) = L_{12} p(\mathbf{x}|C_1) P(C_1) + L_{22} p(\mathbf{x}|C_2) P(C_2). \quad (9.44)$$

$$\mathbf{x} \in C_1 \text{ if } r_1(\mathbf{x}) < r_2(\mathbf{x}), \quad (9.45)$$

that is,

$$\begin{aligned} \mathbf{x} \in C_1 \text{ if } & L_{11} p(\mathbf{x}|C_1) P(C_1) + L_{21} p(\mathbf{x}|C_2) P(C_2) \\ & < L_{12} p(\mathbf{x}|C_1) P(C_1) + L_{22} p(\mathbf{x}|C_2) P(C_2), \end{aligned} \quad (9.46)$$

or equivalently,

$$\mathbf{x} \in C_1 \text{ if } (L_{21} - L_{22}) p(\mathbf{x}|C_2) P(C_2) < (L_{12} - L_{11}) p(\mathbf{x}|C_1) P(C_1). \quad (9.47)$$

This expression may be rewritten as [473]

$$\mathbf{x} \in C_1 \text{ if } \frac{p(\mathbf{x}|C_1)}{p(\mathbf{x}|C_2)} > \frac{P(C_2)}{P(C_1)} \frac{(L_{21} - L_{22})}{(L_{12} - L_{11})}. \quad (9.48)$$

The LHS of the inequality above, which is a ratio of two likelihood functions, is often referred to as the *likelihood ratio*:

$$l_{12}(\mathbf{x}) = \frac{p(\mathbf{x}|C_1)}{p(\mathbf{x}|C_2)}. \quad (9.49)$$

Then, Bayes decision rule for $M = 2$ is [473]:

1. Assign $\mathbf{x}$ to class C_1 if $l_{12}(\mathbf{x}) > \theta_{12}$, where θ_{12} is a threshold given by $\theta_{12} = \frac{P(C_2)}{P(C_1)} \frac{(L_{21} - L_{22})}{(L_{12} - L_{11})}$.
2. Assign $\mathbf{x}$ to class C_2 if $l_{12}(\mathbf{x}) < \theta_{12}$.
3. Make an arbitrary or heuristic decision if $l_{12}(\mathbf{x}) = \theta_{12}$.

The rule may be generalized to the M -class case as [473]:

$$\mathbf{x} \in C_i \text{ if } \sum_{k=1}^M L_{ki} p(\mathbf{x}|C_k) P(C_k) < \sum_{q=1}^M L_{qj} p(\mathbf{x}|C_q) P(C_q), \quad (9.50)$$

$j = 1, 2, \dots, M, j \neq i$.

In most pattern classification problems, the loss is nil for correct decisions. The loss could be assumed to be equal to a certain quantity for all erroneous decisions. Then, $L_{ij} = 1 - \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise} \end{cases} \quad (9.51)$$

and

$$\begin{aligned} r_j(\mathbf{x}) &= \sum_{i=1}^M (1 - \delta_{ij}) p(\mathbf{x}|C_i) P(C_i) \\ &= p(\mathbf{x}) - p(\mathbf{x}|C_j) P(C_j), \end{aligned} \quad (9.52)$$

since

$$\sum_{i=1}^M p(\mathbf{x}|C_i) P(C_i) = p(\mathbf{x}). \quad (9.53)$$

The Bayes classifier will assign a pattern $\mathbf{x}$ to class C_i if

$$p(\mathbf{x}) - p(\mathbf{x}|C_i)P(C_i) < p(\mathbf{x}) - p(\mathbf{x}|C_j)P(C_j), \quad j = 1, 2, \dots, M, \quad j \neq i, \quad (9.54)$$

that is,

$$\mathbf{x} \in C_i \quad \text{if} \quad p(\mathbf{x}|C_i)P(C_i) > p(\mathbf{x}|C_j)P(C_j), \quad j = 1, 2, \dots, M, \quad j \neq i. \quad (9.55)$$

This is nothing more than using the decision functions

$$d_i(\mathbf{x}) = p(\mathbf{x}|C_i) P(C_i), \quad i = 1, 2, \dots, M, \quad (9.56)$$

where a pattern $\mathbf{x}$ is assigned to class C_i if $d_i(\mathbf{x}) > d_j(\mathbf{x}) \forall j \neq i$ for that pattern. Using Bayes formula, we get

$$d_i(\mathbf{x}) = P(C_i|\mathbf{x}) p(\mathbf{x}), \quad i = 1, 2, \dots, M. \quad (9.57)$$

Since $p(\mathbf{x})$ does not depend upon the class index i , this can be reduced to

$$d_i(\mathbf{x}) = P(C_i|\mathbf{x}), \quad i = 1, 2, \dots, M. \quad (9.58)$$

The different decision functions given above provide alternative yet equivalent approaches, depending upon whether $p(\mathbf{x}|C_i)$ or $P(C_i|\mathbf{x})$ is used (or available). Estimation of $p(\mathbf{x}|C_i)$ would require a training set for each class C_i . It is common to assume a Gaussian distribution and estimate its mean and variance using the training set.

9.6.2 Bayes classifier for normal patterns

The univariate normal or Gaussian PDF for a single random variable x is given by

$$p(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{1}{2} \left(\frac{x - m}{\sigma} \right)^2 \right], \quad (9.59)$$

which is completely specified by two parameters: the mean

$$m = E[x] = \int_{-\infty}^{\infty} x p(x) dx, \quad (9.60)$$

and the variance

$$\sigma^2 = E[(x - m)^2] = \int_{-\infty}^{\infty} (x - m)^2 p(x) dx. \quad (9.61)$$

In the case of M pattern classes and pattern vectors $\mathbf{x}$ of dimension n governed by multivariate normal PDFs, we have

$$p(\mathbf{x}|C_i) = \frac{1}{(2\pi)^{n/2} |\mathbf{C}_i|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \mathbf{m}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \mathbf{m}_i) \right], \quad (9.62)$$

$i = 1, 2, \dots, M$, where each PDF is completely specified by its mean vector $\mathbf{m}_i$ and its $n \times n$ covariance matrix $\mathbf{C}_i$, with

$$\mathbf{m}_i = E_i[\mathbf{x}], \quad (9.63)$$

and

$$\mathbf{C}_i = E_i[(\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^T]. \quad (9.64)$$

Here, $E_i[\]$ denotes the expectation operator over the patterns belonging to class C_i .

Normal distributions occur frequently in nature and have the advantage of analytical tractability. A multivariate normal PDF reduces to a product of univariate normal PDFs when the elements of $\mathbf{x}$ are mutually independent (then the covariance matrix is a diagonal matrix).

Earlier, we formulated the decision functions

$$d_i(\mathbf{x}) = p(\mathbf{x}|C_i) P(C_i), \quad i = 1, 2, \dots, M. \quad (9.65)$$

Given the exponential in the normal PDF, it is convenient to use

$$d_i(\mathbf{x}) = \ln [p(\mathbf{x}|C_i) P(C_i)] = \ln p(\mathbf{x}|C_i) + \ln P(C_i), \quad (9.66)$$

which is equivalent in terms of classification performance as the natural logarithm $\ln$ is a monotonically increasing function. Then [473],

$$d_i(\mathbf{x}) = \ln P(C_i) - \frac{n}{2} \ln 2\pi - \frac{1}{2} \ln |\mathbf{C}_i| - \frac{1}{2} [(\mathbf{x} - \mathbf{m}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \mathbf{m}_i)], \quad (9.67)$$

$i = 1, 2, \dots, M$. The second term does not depend upon i ; therefore, we can simplify $d_i(\mathbf{x})$ to

$$d_i(\mathbf{x}) = \ln P(C_i) - \frac{1}{2} \ln |\mathbf{C}_i| - \frac{1}{2} [(\mathbf{x} - \mathbf{m}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \mathbf{m}_i)], \quad i = 1, 2, \dots, M. \quad (9.68)$$

The decision functions above are hyperquadrics; hence, the best that a Bayes classifier for normal patterns can do is to place a general second-order (quadratic) decision surface between each pair of pattern classes. In the case of true normal distributions of patterns, the decision functions as above are optimal on an average basis: They minimize the expected loss with the simplified loss function $L_{ij} = 1 - \delta_{ij}$ [473].

If all the covariance matrices are equal, that is, $\mathbf{C}_i = \mathbf{C}$, $i = 1, 2, \dots, M$, we get

$$d_i(\mathbf{x}) = \ln P(C_i) + \mathbf{x}^T \mathbf{C}^{-1} \mathbf{m}_i - \frac{1}{2} \mathbf{m}_i^T \mathbf{C}^{-1} \mathbf{m}_i, \quad i = 1, 2, \dots, M, \quad (9.69)$$

after omitting terms independent of i . The Bayesian classifier is now represented by a set of linear decision functions.

Before one may apply the decision functions as above, it would be appropriate to verify the Gaussian nature of the PDFs of the variables on hand by conducting statistical tests [6, 268]. Furthermore, it would be necessary to derive or estimate the mean vector and covariance matrix for each class; sample statistics computed from a training set may serve this purpose.

See Section 9.11.2 for an illustration of application of the Bayesian method to an ECG signal with PVCs.

9.7 Logistic Regression Analysis

Logistic classification is a statistical technique based on a logistic regression model that estimates the probability of occurrence of an event [477–479]. The technique is designed for problems where patterns are to be classified into one of two classes. When the response variable is binary, theoretical and empirical considerations indicate that the response function is often curvilinear. The typical response function is shaped as a forward or backward tilted “S” and is known as a sigmoidal function. The function has asymptotes at 0 and 1.

In logistic pattern classification, an event is defined as the membership of a pattern vector in one of the two classes. The method computes a variable that depends upon the given parameters and is constrained to the range $[0, 1]$ so that it may be interpreted as a probability. The probability of the pattern vector belonging to the second class is simply the difference between unity and the estimated value.

For the case of a single feature or parameter, the logistic regression model is given as

$$P(\text{event}) = \frac{\exp(b_0 + b_1 x)}{1 + \exp(b_0 + b_1 x)}, \quad (9.70)$$

or equivalently,

$$P(\text{event}) = \frac{1}{1 + \exp[-(b_0 + b_1 x)]}, \quad (9.71)$$

where b_0 and b_1 are coefficients estimated from the data, and x is the independent (feature) variable. The relationship between the independent variable and the estimated probability is nonlinear; it follows an S-shaped curve that resembles the integral of a Gaussian function. In the case of an n -dimensional feature vector $\mathbf{x}$, the model can be written as

$$P(\text{event}) = \frac{1}{1 + \exp(-z)}, \quad (9.72)$$

where z is the linear combination

$$z = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_n x_n = \langle \mathbf{b}, \mathbf{x} \rangle, \quad (9.73)$$

that is, z is the dot product of the augmented feature vector $\mathbf{x}$ with a coefficient or weight vector $\mathbf{b}$.

In linear regression, the coefficients of the model are estimated using the method of least squares; the selected regression coefficients are those that result in the smallest sum of squared distances between the observed and the predicted values of the dependent variable. In logistic regression, the parameters of the model are estimated using the maximum likelihood method [268, 477]; the coefficients that make the observed results “most likely” are selected. Since the logistic regression model is nonlinear, an iterative algorithm is necessary for estimation of the coefficients [478, 479]. A training set is required to design a classifier based upon logistic regression.

9.8 Neural Networks

In many practical problems, we may have no knowledge of the prior probabilities of patterns belonging to one class or another. No general classification rules may exist for the patterns on hand. Clinical knowledge may not yield symbolic knowledge bases that could be used to classify patterns that demonstrate exceptional behavior. In such situations, conventional pattern classification methods as described in the preceding sections may not be well suited for classification of pattern vectors. Artificial neural networks (ANNs), with the properties of experience-based learning and fault tolerance, should be effective in solving such classification problems [267, 475, 476, 480–483].

Figure 9.4 illustrates a two-layer perceptron with one hidden layer and one output layer for pattern classification. The network learns the similarities among patterns directly from their instances in the training set that is provided initially. Classification rules are inferred from the training data without prior knowledge of the pattern class distributions in the data. Training of an ANN classifier is typically achieved by the *back-propagation* algorithm [267, 475, 476, 480–483]. The actual output of the ANN

y_k is calculated as

$$y_k = f \left(\sum_{j=1}^J w_{jk}^{\#} x_j^{\#} - \theta_k^{\#} \right), \quad k = 1, 2, \dots, K, \quad (9.74)$$

where

$$x_j^{\#} = f \left(\sum_{i=1}^I w_{ij} x_i - \theta_j \right), \quad j = 1, 2, \dots, J, \quad (9.75)$$

and

$$f(\beta) = \frac{1}{1 + \exp(-\beta)}. \quad (9.76)$$

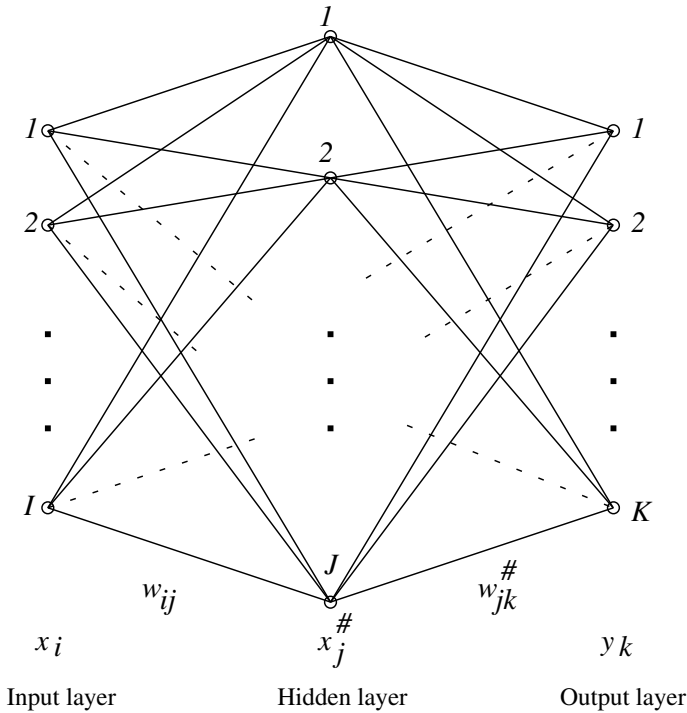


Figure 9.4 A two-layer perceptron.

In the equations given above, θ_j and $\theta_k^{\#}$ are node offsets; w_{ij} and $w_{jk}^{\#}$ are node weights; x_i are the elements of the pattern vectors (input parameters); and I , J , and K are the numbers of nodes in the input, hidden, and output layers, respectively. The weights and offsets are updated by

$$w_{jk}^{\#}(n+1) = w_{jk}^{\#}(n) + \eta[y_k(1-y_k)(d_k - y_k)]x_j^{\#} + \alpha[w_{jk}^{\#}(n) - w_{jk}^{\#}(n-1)], \quad (9.77)$$

$$\theta_k^\#(n+1) = \theta_k^\#(n) + \eta[y_k(1-y_k)(d_k - y_k)](-1) + \alpha[\theta_k^\#(n) - \theta_k^\#(n-1)], \quad (9.78)$$

$$\begin{aligned} w_{ij}(n+1) &= w_{ij}(n) \\ &+ \eta \left[x_j^\#(1 - x_j^\#) \sum_{k=1}^K \{y_k(1 - y_k)(d_k - y_k)w_{jk}^\#\} \right] x_i \\ &+ \alpha[w_{ij}(n) - w_{ij}(n-1)], \end{aligned} \quad (9.79)$$

and

$$\begin{aligned} \theta_j(n+1) &= \theta_j(n) \\ &+ \eta \left[x_j^\#(1 - x_j^\#) \sum_{k=1}^K \{y_k(1 - y_k)(d_k - y_k)w_{jk}^\#\} \right] (-1) \\ &+ \alpha[\theta_j(n) - \theta_j(n-1)], \end{aligned} \quad (9.80)$$

where d_k are the desired outputs, α is a momentum term, η is a gain term, and n refers to the iteration number. Equations 9.77 and 9.78 represent the back-propagation steps, with $y_k(1 - y_k)x_j^\#$ being the sensitivity of y_k to $w_{jk}^\#$, that is, $\frac{\partial y_k}{\partial w_{jk}^\#}$.

The training algorithm is repeated until the errors between the desired outputs and the actual outputs for the training data are smaller than a predetermined threshold value. Shen et al. [483] present a LOO approach to determine the most suitable values for the parameters J , η , and α .

9.8.1 ANNs with radial basis functions

An ANN with RBF [484] or RBF network (RBFN) with a feed-forward hidden layer (see Figure 9.5) applies a nonlinear transformation from the input space to a high-dimensional hidden space, and then produces responses through a linear output transformation. Cover [485] showed that it is possible to transform a given a set of samples that is not linearly separable into one that is linearly separable by applying a nonlinear transformation to project it into a higher-dimensional space. This approach forms the motivation for the use of nonlinear kernel-based methods in pattern classification and machine learning applications.

Consider a set of N labeled input feature vectors, $\mathbf{x}_n$, $n = 1, 2, \dots, N$, characterizing the N signals in a study, each of which is an $M \times 1$ vector. Let y_n be the desired output or classification response for the n^{th} signal, represented by its feature vector $\mathbf{x}_n$. With reference to the RBFN shown in Figure 9.5, we have the output of the network as

$$\widetilde{y}_n = \sum_{j=1}^J w_j \phi(\mathbf{x}_n, \mathbf{c}_j) + w_0, \quad (9.81)$$

where $\widetilde{y}_n$ indicates an estimate of y_n , the RBF ϕ is defined as

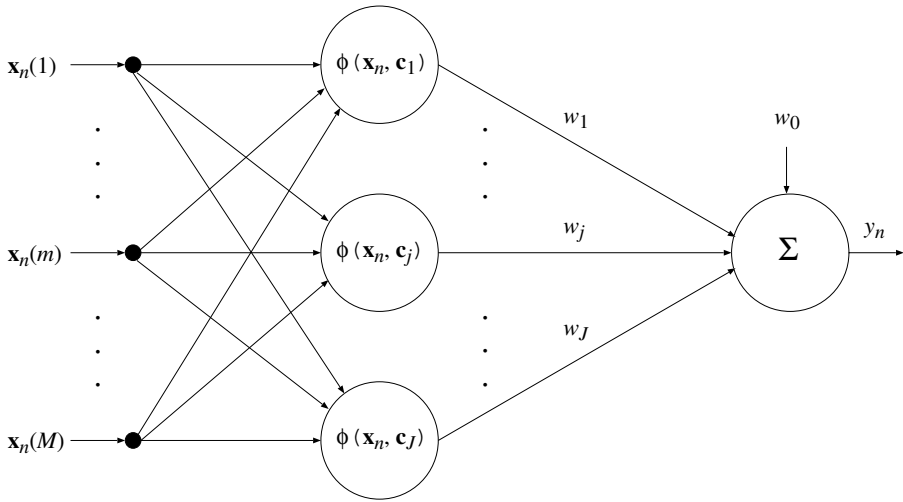


Figure 9.5 Schematic representation of an RBF network. The inputs to the network, $\mathbf{x}_n(1), \mathbf{x}_n(2), \dots, \mathbf{x}_n(M)$, are the M components of the feature vector $\mathbf{x}_n$ of a signal to be classified. The functions shown as ϕ are the RBFs. The hidden layer has J neurons.

$$\phi(\mathbf{x}_n, \mathbf{c}_j) = \exp \left(-\log_e(2) \frac{\|\mathbf{x}_n - \mathbf{c}_j\|^2}{\sigma^2} \right), \quad (9.82)$$

w_j is the weight and $\mathbf{c}_j$ is the center vector for the j^{th} neuron in the hidden layer, J is the number of neurons in the hidden layer, w_0 is the bias, and σ is the spread parameter that determines the width of the area in the input space to which each hidden neuron responds.

The major challenge in the design of an RBFN is the selection of the centers. The selection of the centers in a random fashion commonly leads to a relatively large network with high computation complexity. Rangayyan and Wu [264] applied the orthogonal least-squares (OLS) method [486], a systematic method for center selection which can reduce the size of the RBFN, for screening of VAG signals.

According to Equation 9.81, the mapping performed by the RBFN can be viewed as a regression model, expressed in matrix form as

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} = \begin{bmatrix} 1 & \phi(\mathbf{x}_1, \mathbf{c}_1) & \cdots & \phi(\mathbf{x}_1, \mathbf{c}_J) \\ 1 & \phi(\mathbf{x}_2, \mathbf{c}_1) & \cdots & \phi(\mathbf{x}_2, \mathbf{c}_J) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \phi(\mathbf{x}_N, \mathbf{c}_1) & \cdots & \phi(\mathbf{x}_N, \mathbf{c}_J) \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_J \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_N \end{bmatrix}, \quad (9.83)$$

which is equivalent to

$$\mathbf{y} = \Phi \mathbf{w} + \mathbf{e}, \quad (9.84)$$

where Φ is the $N \times (J+1)$ regression matrix with the RBFs; $\mathbf{y}$ represents the vectorial form of the corresponding values y_n for $n = 1, 2, \dots, N$; $\mathbf{w} = [w_0, w_1, \dots, w_J]^T$; and $\mathbf{e}$ is the approximation error.

The centers of the RBFN are chosen from the set of input feature vectors (a total of N candidates). The task of the OLS method is to perform a systematic selection of fewer than N centers so that the network size can be reduced with minimal degradation of performance during the learning procedure. From Equation 9.83, we can see that there is a one-to-one correspondence between the centers of the RBFN and the coefficients in the regression matrix Φ . At each step of the OLS regression, a new center can be selected in such a manner that the incremental variance of the desired output is maximized. Suppose that there are $Q < N$ centers selected. The OLS solution yielding the weights is given by [486]

$$\tilde{\mathbf{w}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y} = \Phi^+ \mathbf{y}, \quad (9.85)$$

where Φ^+ represents the pseudoinverse of the regression matrix Φ . The output of the RBFN is then expressed as

$$\tilde{\mathbf{y}} = \Phi \tilde{\mathbf{w}} = [\phi_1, \phi_2, \dots, \phi_Q] \tilde{\mathbf{w}}, \quad (9.86)$$

where $\tilde{\mathbf{y}}$ denotes the portion of $\mathbf{y}$ that is within the vectorial space spanned by the columns ϕ_q of the regression matrix Φ .

By using Gram–Schmidt orthogonalization [484], the regression matrix can be decomposed as

$$\Phi = \mathbf{B} \mathbf{A} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_Q] \begin{bmatrix} 1 & a_{11} & a_{12} & \cdots & a_{1Q} \\ 0 & 1 & a_{23} & \cdots & a_{2Q} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}, \quad (9.87)$$

where $\mathbf{A}$ is a $Q \times Q$ upper-triangular matrix with 1s on the main diagonal, and $\mathbf{B}$ is an $N \times Q$ matrix with mutually orthogonal columns $\mathbf{b}_q$ such that

$$\mathbf{B}^T \mathbf{B} = \mathbf{H} = \text{diag}[h_1, h_2, \dots, h_Q], \quad (9.88)$$

where the $Q \times Q$ matrix $\mathbf{H}$ is a diagonal matrix with elements h_k given by

$$h_k = \mathbf{b}_k^T \mathbf{b}_k = \sum_{q=1}^N b_{kq}^2. \quad (9.89)$$

By substituting Equation 9.87 into Equation 9.84, we obtain

$$\mathbf{y} = \mathbf{B} \mathbf{A} \mathbf{w} + \mathbf{e} = \mathbf{B} \mathbf{g} + \mathbf{e}, \quad (9.90)$$

where $\mathbf{g} = \mathbf{A} \mathbf{w}$. In Equation 9.90, the desired output vector $\mathbf{y}$ is expressed as a linear combination of the mutually orthogonal columns of the matrix $\mathbf{B}$. The OLS solution for the coordinate vector $\mathbf{g}$ is given by

$$\tilde{\mathbf{g}} = (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T \mathbf{y} = \mathbf{B}^+ \mathbf{z} = \mathbf{H}^{-1} \mathbf{B}^T \mathbf{y}. \quad (9.91)$$

The k^{th} component of the vector $\tilde{\mathbf{g}}$ is given by

$$g_k = \frac{\mathbf{b}_k^T \mathbf{y}}{\mathbf{b}_k^T \mathbf{b}_k}. \quad (9.92)$$

Because Gram–Schmidt orthogonalization ensures the orthogonality between the approximation error $\mathbf{e}$ and $\mathbf{B}\mathbf{g}$ in Equation 9.90, we have

$$\mathbf{y}^T \mathbf{y} = \mathbf{g}^T \mathbf{B}^T \mathbf{B} \mathbf{g} + \mathbf{e}^T \mathbf{e} = \mathbf{g}^T \mathbf{H} \mathbf{g} + \mathbf{e}^T \mathbf{e} = \sum_{k=1}^Q h_k g_k^2 + \mathbf{e}^T \mathbf{e}. \quad (9.93)$$

Because there is a one-to-one correspondence between the elements of the regression vector $\mathbf{g}$ and the RBF centers $\mathbf{c}_j$, each term in the summation above reflects the contribution of each of the RBF centers. We can, therefore, define an error reduction ratio (ϵ) with respect to the j^{th} RBF center as [486]

$$\epsilon_j = \frac{h_j g_j^2}{\mathbf{y}^T \mathbf{y}}. \quad (9.94)$$

The error reduction ratio offers an effective criterion for the selection of RBF centers in a regression model. At each step of the regression, an RBF center is selected so as to maximize the error reduction ratio toward a tolerance value.

The input layer of the RBFN used by Rangayyan and Wu [264] contained $M = 5$ nodes to accept the set of five features (FF_1, FF_2, S, K, H) extracted from each VAG signal (see Sections 5.12 and 9.4.2). The spread parameter σ was varied over the range $[1, 6]$, and the number of hidden nodes J was varied over the range $[1, 30]$. The resulting output values were used to derive ROC curves and the associated A_z values. The highest A_z value obtained was 0.82 using the RBFN classifier with $\sigma = 6$ and $J = 23$ hidden nodes using $N = 89$ VAG signals. In another work, Rangayyan and Wu [265] used an RBFN with the turns count computed for the first and second halves of each VAG signal, and obtained better classification performance with $A_z = 0.92$.

Regardless of the good results they provide and their popularity, RBFNs pose problems related to the derivation of optimal network parameters and generalization from a training set to a test set.

9.9 Measures of Diagnostic Accuracy and Cost

Pattern recognition or classification decisions that are made in the context of medical diagnosis have implications that go beyond statistical measures of accuracy and validity. We need to provide a clinical or diagnostic interpretation of statistical or rule-based decisions made with signal pattern vectors.

Consider the simple situation of *screening*, which represents the use of a test to detect the presence or absence of a specific disease in a certain study population. The decision to be made is binary: A given subject has or does not have the disease in question. Let us represent by A the event that a subject has the particular pathology, and by N the event that the subject does not have the disease. Let the prior probabilities $P(A)$ and $P(N)$ represent the fractions of subjects with the disease and the subjects without the disease, respectively, in the test population. Let T^+ represent a positive screening test result (indicative of the presence of the disease) and T^- a negative result. The following possibilities arise [487]:

- A *true positive* (TP) is the situation when the test is positive for a subject with the disease (also known as a *hit*). The true-positive fraction (TPF) or *sensitivity* S^+ is given as $P(T^+|A)$ or

$$S^+ = \frac{\text{number of TP decisions}}{\text{number of subjects with the disease}}. \quad (9.95)$$

The sensitivity of a test represents its capability to detect the presence of the disease of concern.

- A *true negative* (TN) represents the case when the test is negative for a subject who does not have the disease. The true-negative fraction (TNF) or *specificity* S^- is given as $P(T^-|N)$ or

$$S^- = \frac{\text{number of TN decisions}}{\text{number of subjects without the disease}}. \quad (9.96)$$

The specificity of a test indicates its accuracy in identifying the absence of the disease of concern.

- A *false negative* (FN) is said to occur when the test is negative for a subject who has the disease of concern; that is, the test has missed the case. The probability of this error, known as the false-negative fraction (FNF), is $P(T^-|A)$.
- A *false positive* (FP) is defined as the case where the result of the test is positive when the individual being tested does not have the disease. The probability of this type of error or false alarm, known as the false-positive fraction (FPF), is $P(T^+|N)$.

Table 9.1 summarizes the classification possibilities. Note that

- $FNF + TPF = 1$,
- $FPF + TNF = 1$,
- $S^- = 1 - FPF = TNF$, and
- $S^+ = 1 - FNF = TPF$.

A summary measure of accuracy may be defined as [487]

$$\text{accuracy} = S^+ P(A) + S^- P(N), \quad (9.97)$$

where $P(A)$ is the fraction of the study population that actually has the disease (that is, the prevalence of the disease) and $P(N)$ is the fraction of the study population that is actually free of the disease.

If the prior probabilities are not available, the accuracy of classification can be estimated as

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (9.98)$$

In the same way, we have

$$TPF = \frac{TP}{TP + FN} \quad (9.99)$$

and

$$TNF = \frac{TN}{TN + FP}. \quad (9.100)$$

Actual Group	Predicted Group	
	Without the Disease	With the Disease
Without the Disease	$S^- = TNF$	FPF
With the Disease	FNF	$S^+ = TPF$

Table 9.1 Schematic representation of a classification matrix. S^- denotes the specificity (true-negative fraction or TNF), FPF denotes the false-positive fraction, FNF denotes the false-negative fraction, and S^+ denotes the sensitivity (true-positive fraction or TPF).

The efficiency of a test may also be indicated by its predictive values. The *positive predictive value* PPV of a test, defined as

$$PPV = 100 \frac{TP}{TP + FP}, \quad (9.101)$$

represents the percentage of the cases labeled as positive by the test that are actually positive. The *negative predictive value* NPV , defined as

$$NPV = 100 \frac{TN}{TN + FN}, \quad (9.102)$$

represents the percentage of cases labeled as negative by the test that are actually negative.

When a new test or method of diagnosis is being developed and tested, it is necessary to use another previously established method as a reference to confirm the

presence or absence of the disease. Such a reference method is often called the *gold standard*, and its results are referred to as the *ground truth*. When computer-based methods need to be tested, it is common practice to use the diagnosis or classification provided by an expert in the field as the gold standard. Results of biopsy, other established laboratory or investigative procedures, or long-term clinical follow-up in the case of normal subjects may also serve this purpose. The term “actual group” in Table 9.1 indicates the result of the gold standard, and the term “predicted group” refers to the result of the test conducted.

Health-care professionals (and the general public) would be interested in knowing the probability that a subject with a positive test result actually has the disease: This is given by the conditional probability $P(A|T^+)$. The question could be answered by using Bayes theorem [268] as

$$P(A|T^+) = \frac{P(A) P(T^+|A)}{P(A)P(T^+|A) + P(N)P(T^+|N)}. \quad (9.103)$$

Note that $P(T^+|A) = S^+$ and $P(T^+|N) = 1 - S^-$. In order to determine the posterior probability as above, the sensitivity and specificity of the test and the prior probabilities of negative cases and positive cases (the rate of prevalence of the disease) should be known.

A cost matrix may be defined, as in Table 9.2, to reflect the overall cost effectiveness of a test or method of diagnosis. The cost of conducting the test and arriving at a TN decision is indicated by C_N ; this is the cost of subjecting a normal person to the test for the purposes of screening for a disease. The cost of the test when a TP is found is shown as C_A ; this might include the costs of further tests, treatment, and follow-up, which are secondary to the test itself but part of the screening and health-care program. The value C^+ indicates the cost of an FP result; this represents the cost of erroneously subjecting an individual without the disease to further tests or therapy. Whereas it may be easy to identify the costs of clinical tests or treatment procedures, it is difficult to quantify the traumatic and psychological effects of an FP result and the consequent procedures on a normal subject. The cost C^- is the cost of an FN result: The presence of the disease in a patient is not diagnosed, the condition worsens with time, the patient faces more complications of the disease, and the health-care system or the patient has to bear the costs of further tests and delayed therapy.

A loss factor due to misclassification may be defined as

$$L = FPF \times C^+ + FNF \times C^-. \quad (9.104)$$

The total cost of the screening program may be computed as

$$C_S = TPF \times C_A + TNF \times C_N + FPF \times C^+ + FNF \times C^-. \quad (9.105)$$

Metz [487] provides more details on the computation of the costs of diagnostic tests.

Actual Group	Predicted Group	
	Without the Disease	With the Disease
Without the Disease	C_N	C^+
With the Disease	C^-	C_A

Table 9.2 Schematic representation of the cost matrix of a diagnostic method.

9.9.1 Receiver operating characteristics

Measures of overall correct classification of patterns as percentages provide limited indications of the accuracy of a diagnostic method. The provision of separate percentage correct classification for each category, such as sensitivity and specificity, can facilitate improved analysis. However, these measures do not indicate the dependence of the results upon the decision threshold. Furthermore, the effect of the rate of incidence or prevalence of the particular disease is not considered.

From another perspective, it is desirable to have a screening or diagnostic test that is both highly sensitive and highly specific. In reality, however, such a test is usually not achievable. Most tests are based on clinical measurements that can assume limited ranges of a variable (or a few variables) with an inherent trade-off between sensitivity and specificity. The relationship between sensitivity and specificity is illustrated by the ROC curve, which facilitates improved analysis of the classification accuracy of a diagnostic method [338, 487–489].

Consider the situation illustrated in Figure 9.6. For a given diagnostic test with the decision variable z , we have predetermined state-conditional PDFs of the decision variable z for actually negative cases indicated as $p(z|N)$ and for actually positive indicated as $p(z|A)$. As indicated in Figure 9.6, the two PDFs will almost always overlap, given that no method can be perfect. The user or operator needs to determine a decision threshold (indicated by the vertical line) so as to strike a compromise between sensitivity and specificity. Lowering the decision threshold will increase TPF at the cost of increased FPF . (Note: TNF and FNF may be derived easily from FPF and TPF , respectively.)

An ROC curve is a graph that plots (FPF , TPF) points obtained for a range of decision threshold or cut points of the decision method (see Figure 9.7). The cut point could correspond to the threshold of the probability of prediction. By varying the decision threshold, we get different decision fractions, within the range $[0, 1]$. An ROC curve describes the inherent detection (diagnostic or discriminant) characteristics of a test or method: A receiver (user) may choose to operate at any point along the curve. The ROC curve is independent of the prevalence of the disease or disorder being investigated as it is based upon normalized decision fractions. As all cases may be simply labeled as negative or all may be labeled as positive, an ROC curve has to pass through the points $(0, 0)$ and $(1, 1)$.

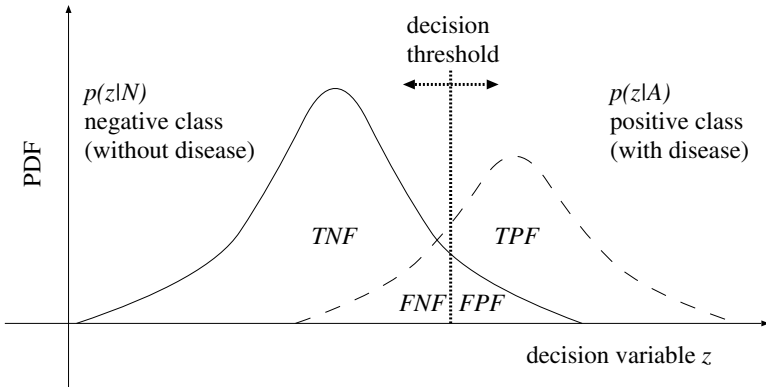


Figure 9.6 State-conditional PDFs of a diagnostic decision variable z for negative and positive cases. The vertical line represents the decision threshold. Based on a similar figure in T.M. Cabral and R.M. Rangayyan, *Fractal Analysis of Breast Masses in Mammograms*, Morgan & Claypool, 2012.

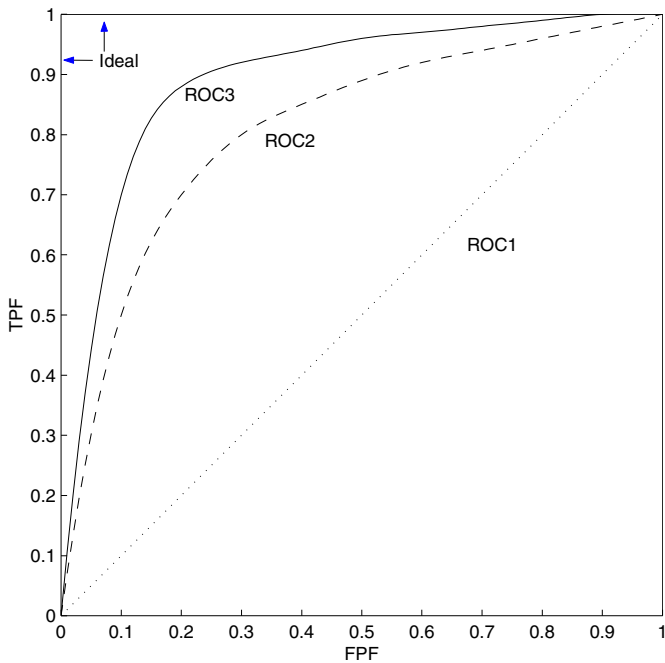


Figure 9.7 Examples of receiver operating characteristic curves.

In a diagnostic situation where a human operator or specialist is required to provide the diagnostic decision, ROC analysis is usually conducted by requiring the specialist to rank each case as one of five possibilities [487]:

1. definitely or almost definitely negative,
2. probably negative,
3. possibly positive,
4. probably positive,
5. definitely or almost definitely positive.

Item 3 above may be replaced by “indeterminate” if appropriate. Various values of TPF and FPF are then calculated by varying the decision threshold from level 5 to level 1 according to the decision items listed above. The resulting (FPF, TPF) points are then plotted to form an ROC curve. The maximum likelihood estimation method [490] is commonly used to fit a binormal curve to data as above.

A summary measure of effectiveness of a test is given by the area under the ROC curve, traditionally labeled as A_z . It is clear from Figure 9.7 that A_z is limited to the range $[0, 1]$. A test that gives a larger area under the ROC curve indicates a better method than one with a smaller area: in Figure 9.7, the method corresponding to ROC3 is better than the method corresponding to ROC2; both are better than the method represented by ROC1 with $A_z = 0.5$. An ideal method will have an ROC curve that follows the vertical line from $(0, 0)$ to $(0, 1)$ and then the horizontal line from $(0, 1)$ to $(1, 1)$, with $A_z = 1$: The method has $TPF = 1$ with $FPF = 0$, which is ideal. (*Note:* This would require the PDFs represented in Figure 9.6 not to overlap.)

9.9.2 McNemar's test of symmetry

Problem: Suppose we have two methods to perform a certain diagnostic test. How may we compare the classification performance of one against that of the other?

Solution: Measures of overall classification accuracies such as a percentage of correct classification or the area under the ROC curve provide simple measures to compare two or more diagnostic methods. If more details are required as to how the classifications of groups of cases vary from one method to another, McNemar's test of symmetry [491, 492] would be an appropriate tool.

McNemar's test is based on the construction of contingency tables that compare the results of two classification methods. The rows of a contingency table represent the outcomes of one of the methods used as the reference, possibly a gold standard (labeled as Method A in Table 9.3); the columns represent the outcomes of the other method, which is usually a new method (Method B) to be evaluated against the gold standard. The entries in the table are counts that correspond to particular diagnostic categories, which in Table 9.3 are labeled as normal, indeterminate, and abnormal. A separate contingency table should be prepared for each true category of the patterns;

for example, normal and abnormal. (The class “indeterminate” may not be applicable as a true category.) The true category of each case may have to be determined by a third method (for example, biopsy or surgery).

Method A	Method B			Total
	Normal	Indeterminate	Abnormal	
Normal	a (1)	b (2)	c (3)	$R1$
Indeterminate	d (4)	e (5)	f (6)	$R2$
Abnormal	g (7)	h (8)	i (9)	$R3$
Total	$C1$	$C2$	$C3$	N

Table 9.3 Schematic representation of a contingency table for McNemar’s test of asymmetry [448,493].

In Table 9.3, the variables $a, b, c, d, e, f, g, h,$ and i denote the counts in each cell, and the numbers in parentheses denote the cell number. The variables $C1, C2,$ and $C3$ denote the total numbers of counts in the corresponding columns; $R1, R2,$ and $R3$ denote the total numbers of counts in the corresponding rows. The total number of cases in the true category represented by the table is $N = C1 + C2 + C3 = R1 + R2 + R3$.

Each cell in a contingency table represents a paired outcome. For example, in evaluating the diagnostic efficiency of Method B versus Method A, cell number 3 will contain the number of samples that were classified as normal by Method A but as abnormal by Method B. The row totals ($R1, R2,$ and $R3$) and the column totals ($C1, C2,$ and $C3$) may be used to determine the sensitivity and specificity of the methods.

High values along the main diagonal (a, e, i) of a contingency table (see Table 9.3) indicate no change or difference in diagnostic performance with Method B as compared to Method A. In a contingency table for truly abnormal cases, a high value in the upper-right portion (cell number 3) will indicate an improvement in diagnosis (higher sensitivity) with Method B as compared to Method A. In evaluating a contingency table for truly normal cases, Method B will have a higher specificity than Method A if a large value is found in cell 7. McNemar’s method may be used to perform detailed statistical analysis of improvement in performance based upon contingency tables if large numbers of cases are available in each category [491,492].

Illustration of application: Krishnan et al. [493] proposed two methods for auditory display and sonification of processed VAG signals. Table 9.4 shows the contingency table for 18 abnormal VAG signals comparing their classification by listening to direct playback of the signal data or after a sonification technique based on the *IMF* and envelope of a given VAG signal. The table shows substantial improvement in sensitivity with the sonification method (15/18) as compared to direct

playback (4/18). Ten abnormal signals that were called normal with direct playback were correctly classified as abnormal with the sonification method.

Direct Playback	Sonification			Total
	Normal	Indeterminate	Abnormal	
Normal	2	0	10	12
Indeterminate	1	0	1	2
Abnormal	0	0	4	4
Total	3	0	15	18

Table 9.4 Contingency table for a method of sonification of VAG signals versus direct playback with 18 abnormal signals [448,493].

Table 9.5 shows the contingency table for 19 normal VAG signals comparing their classification by listening to direct playback of the signal data or after sonification. The results indicate poorer specificity (8/19) of the results of sonification as compared to direct playback (13/19). The results indicate that the higher sensitivity of the sonification method was gained at the expense of a decrease in specificity. See Krishnan et al. [493] for further details.

Direct Playback	Sonification			Total
	Normal	Indeterminate	Abnormal	
Normal	8	2	3	13
Indeterminate	0	0	1	1
Abnormal	0	0	5	5
Total	8	2	9	19

Table 9.5 Contingency table for a method of sonification of VAG signals versus direct playback with 19 normal signals [448,493].

9.10 Reliability of Features, Classifiers, and Decisions

In most practical applications of biomedical signal analysis, the researcher is presented with the problem of designing a system for pattern classification and decision

making using a small number of training samples (signals), with no knowledge of the distributions of the features or parameters computed from the signals. The size of the training set, relative to the number of features used in the pattern classification system, determines the accuracy and reliability of the decisions made [494,495]. One should not increase the number of features to be used without a simultaneous increase in the number of training samples, as the two quantities together affect the bias and variance of the classifier. On the other hand, when the training set has a fixed number of samples, the addition of more features beyond a certain limit will lead to poorer performance of the classifier: this is known as the “curse of dimensionality.” It is desirable to be able to analyze the bias and variance of a classification rule while isolating the effects of the functional form of the distributions of the features used.

Raudys and Jain [495] give a rule-of-thumb table for the number of training samples required in relation to the number of features used in order to remain within certain limits of classification errors for five pattern classification methods. When the available features are ordered in terms of their individual classification performance, the optimal number of features to be used with a certain classification method and training set may be determined by obtaining unbiased estimates of the classification accuracy with the number of features increased one at a time in order. A point is reached when the performance deteriorates, which will indicate the optimal number of features to be used. This method, however, cannot take into account the joint performance of various combinations of features: Exhaustive combinations of all features may have to be evaluated to take this aspect into consideration. Software packages such as the Statistical Package for the Social Sciences (SPSS) [478,479] provide programs to facilitate feature evaluation and selection as well as the estimation of classification accuracies.

Durand et al. [331] reported on the design and evaluation of several pattern classification systems for the assessment of bioprosthetic heart valves based upon 18 features computed from PCG spectra (see Section 6.5). Based upon the rule of thumb that the number of training samples should be five or more times the number of features used, and with the number of training samples limited to data from 20 normal and 28 degenerated valves, exhaustive combinations of the 18 features taken 2, 3, 4, 5, and 6 at a time were used to design and evaluate pattern classification systems. The Bayes method was seen to provide the best performance (98% correct classification) with six features; as many as 511 combinations of the 18 features taken six at a time provided correct classification between 90% and 98%. The nearest-neighbor algorithm with the Mahalanobis distance provided 94% correct classification with only three features and did not perform any better with more features.

9.10.1 Separability of features

Normalized distance between PDFs: Consider a feature x that has the means m_1 and m_2 and SD values σ_1 and σ_2 for the two classes C_1 and C_2 . Assume that the PDFs $p(x|C_1)$ and $p(x|C_2)$ overlap; then, the area of overlap is related to the error of classification. If the SD values σ_1 and σ_2 remain constant, the overlap between the

PDFs decreases as $|m_1 - m_2|$ increases. If the means remain constant, the overlap increases as σ_1 and σ_2 increase (the dispersion of the features increases). These observations are captured by the normalized distance between the means, defined as [496]

$$d_n = \frac{|m_1 - m_2|}{\sigma_1 + \sigma_2}. \quad (9.106)$$

The measure d_n provides an indicator of the statistical separability of the PDFs. A limitation of d_n is that $d_n = 0$ if $m_1 = m_2$ regardless of σ_1 and σ_2 .

See Section 5.12.3 for a discussion on the *KLD*.

Divergence: Let us rewrite the likelihood ratio in Equation 9.49 as

$$l_{ij}(\mathbf{x}) = \frac{p(\mathbf{x}|C_i)}{p(\mathbf{x}|C_j)}. \quad (9.107)$$

Applying the logarithm, we get

$$l'_{ij}(\mathbf{x}) = \ln[l_{ij}(\mathbf{x})] = \ln[p(\mathbf{x}|C_i)] - \ln[p(\mathbf{x}|C_j)]. \quad (9.108)$$

The divergence D_{ij} between the PDFs $p(\mathbf{x}|C_i)$ and $p(\mathbf{x}|C_j)$ is defined as [496]

$$D_{ij} = E[l'_{ij}(\mathbf{x})|C_i] + E[l'_{ji}(\mathbf{x})|C_j], \quad (9.109)$$

where

$$E[l'_{ij}(\mathbf{x})|C_i] = \int_{\mathbf{x}} l'_{ij}(\mathbf{x}) p(\mathbf{x}|C_i) d\mathbf{x}. \quad (9.110)$$

Divergence has the following properties [496]:

- $D_{ij} > 0$;
- $D_{ii} = 0$;
- $D_{ij} = D_{ji}$; and
- if the individual features $x_1, x_2, \dots, x_n$ are statistically independent, then $D_{ij}(x_1, x_2, \dots, x_n) = \sum_{k=1}^n D_{ij}(x_k)$.

It follows that adding more features that are statistically independent of one another will increase divergence and statistical separability.

In the case of multivariate Gaussian PDFs, we have [496]

$$\begin{aligned} D_{ij} &= \frac{1}{2} \text{Tr}[(\mathbf{C}_i - \mathbf{C}_j)(\mathbf{C}_j^{-1} - \mathbf{C}_i^{-1})] \\ &+ \frac{1}{2} \text{Tr}[(\mathbf{C}_i^{-1} + \mathbf{C}_j^{-1})(\mathbf{m}_i - \mathbf{m}_j)(\mathbf{m}_i - \mathbf{m}_j)^T]. \end{aligned} \quad (9.111)$$

The second term in the equation given above is similar to the normalized distance d_n as defined in Equation 9.106 and is zero for PDFs with identical means; however, due to the first term, $D_{ij} \neq 0$ unless the covariance matrices are identical.

In the case of the existence of multiple classes $C_i, i = 1, 2, \dots, m$, the pairwise divergence values may be averaged to obtain a single measure across all of the m classes as [496]

$$D_{\text{av}} = \sum_{i=1}^m \sum_{j=1}^m p(C_i) p(C_j) D_{ij}. \quad (9.112)$$

A limitation of both d_n and D_{ij} is that they increase without an upper bound as the separation between the means increases. On the other hand, the error of classification is limited to the range $[0, 100]\%$ or $[0, 1]$.

Jeffries–Matusita distance: The Jeffries–Matusita (JM) distance provides an improved measure of the separation between PDFs as compared to the normalized distance and divergence. The JM distance between the PDFs $p(\mathbf{x}|C_i)$ and $p(\mathbf{x}|C_j)$ is defined as [496]

$$J_{ij} = \left\{ \int_{\mathbf{x}} \left[\sqrt{p(\mathbf{x}|C_i)} - \sqrt{p(\mathbf{x}|C_j)} \right]^2 d\mathbf{x} \right\}^{1/2}. \quad (9.113)$$

In the case of multivariate Gaussian PDFs, we have [496]

$$J_{ij} = \sqrt{2[1 - \exp(-\alpha)]}, \quad (9.114)$$

where

$$\begin{aligned} \alpha = & \frac{1}{8} (\mathbf{m}_i - \mathbf{m}_j)^T \left(\frac{\mathbf{C}_i + \mathbf{C}_j}{2} \right)^{-1} (\mathbf{m}_i - \mathbf{m}_j) \\ & + \frac{1}{2} \ln \left[\frac{|(\mathbf{C}_i + \mathbf{C}_j)|/2}{(|\mathbf{C}_i||\mathbf{C}_j|)^{1/2}} \right], \end{aligned} \quad (9.115)$$

where $|\mathbf{C}_i|$ is the determinant of $\mathbf{C}_i$.

An advantage of the JM distance is that it is limited to the range $[0, \sqrt{2}]$. $J_{ij} = 0$ when the means of the PDFs are equal and the covariance matrices are zero matrices. Pairwise JM distances may be averaged over multiple classes similar to the averaging of divergence as in Equation 9.112. The JM distance determines the upper and lower bounds on the error of classification [496].

It should be observed that divergence and JM distance are defined for a given feature vector $\mathbf{x}$. The measures would have to be computed for all combinations of features in order to select the best feature set for a particular pattern classification problem.

See Section 5.12.3 for a description of the *KLD* between PDFs.

The *t*-test and the *p*-value: The *t*-test is used to assess whether the means of a feature, x , for two groups or classes are statistically different from each other [10, 282, 497, 498]. The test is performed by assessing the difference between the means of the two groups of the features relative to their spread or variability. If it is assumed that the variance, σ^2 , is the same for the two groups, the *t*-statistic, t_s , is computed as

$$t_s = \frac{\mu_1 - \mu_2}{\sigma_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad (9.116)$$

where μ_1 and n_1 are the mean and size of the first group, μ_2 and n_2 are the mean and size of the second group, and σ_p is the pooled SD , defined as

$$\sigma_p = \sqrt{\frac{(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2}{n_1 + n_2 - 2}}. \quad (9.117)$$

If the variance is not the same for the two groups, t_s is computed as

$$t_s = \frac{\mu_1 - \mu_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}, \quad (9.118)$$

where σ_1^2 is the variance of the first group and σ_2^2 is the variance of the second group. The number of degrees of freedom is equal to $n_1 + n_2 - 2$.

The significance level is usually set at 0.05; this means that, five times out of 100, the difference between the means is concluded to be statistically significant when it is not actually so. This is the probability of being incorrect if the null hypothesis (which assumes that $\mu_1 = \mu_2$) is rejected. Given the degrees of freedom and significance level, the T -critical value, T_c , can be determined using the t -distribution look-up table. When t_s , computed with either Equation 9.118 or Equation 9.116, is larger than T_c , the null hypothesis is rejected.

The t -distribution curve shown in Figure 9.8 illustrates the significance level and p -value in terms of the area under the t -distribution curve. The significance level is the sum of the area under the curve to the right of $+T_c$ and the area to the left of $-T_c$ (all of the areas labeled as A and B in Figure 9.8). The p -value is the sum of the area under the curve to the right of $+t_s$ and the area to the left of $-t_s$ (all of the areas labeled as B in Figure 9.8).

The p -value is a statistical measure of the probability that the results observed in a study could have occurred by chance. A small p -value is a rejection of the null hypothesis in favor of the alternative hypothesis (which assumes that $\mu_1 \neq \mu_2$) because it indicates how unlikely it is that a test statistic as extreme as or more extreme than that given by the data will be observed from the population if the null hypothesis were true. For example, a p -value of 0.01 indicates that there is a one out of a hundred chance that the result occurred by chance. A test resulting in a p -value less than 0.05 is considered to indicate that the difference between the means is statistically significant. A p -value less than 0.01 is taken to indicate that the difference between the means is statistically highly significant.

9.10.2 Feature selection

In a practical application of pattern classification, several parameters may be required to discriminate between multiple classes. Because most features lead to limited discrimination between classes due to the overlap in the ranges of their values for the various classes, it is common to use several features. However, various costs are associated with the measurement, derivation, and computation of each feature. It would be advantageous to be able to assess the contribution of each feature to the

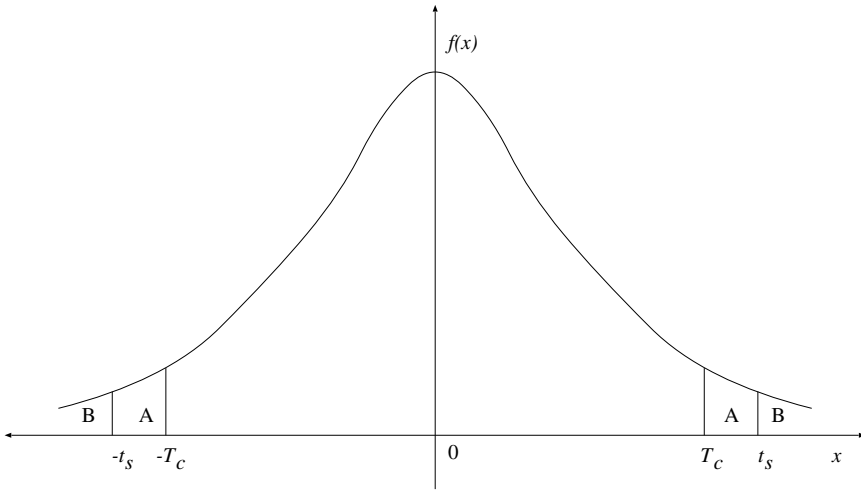


Figure 9.8 The t -distribution curve. Reproduced with permission from T.M. Cabral and R.M. Rangayyan, *Fractal Analysis of Breast Masses in Mammograms*, Morgan & Claypool, 2012. ©Morgan & Claypool.

task of discriminating between the classes of interest, and to select the feature set that provides the best separation between classes and the lowest classification error. The notion of statistical separability of features between classes is useful in addressing these concerns [496]. Various parameters related to the separation of features between classes may be used to select features, such as A_z [338,487,488] and the p -value [498].

Popular methods for feature selection [8,267,293,499] include sequential forward selection, sequential backward selection, and stepwise regression. In sequential forward selection, starting with an empty set, features are sequentially included in the set until the inclusion of additional features does not improve the classification performance. To begin, the best single feature in terms of separation of the classes of interest, such as the p -value, is determined and included in the set of selected features. The selected feature is then used in combination with each remaining feature, in turn, to create pairs. The classification performance of each pair is evaluated and the best pair is selected for the next iteration. The process of including more and more features and evaluation of performance is continued until a prespecified number of features is selected or the classification performance does not improve much with the inclusion of more features.

Sequential backward selection starts with the full set of features and eliminates features until the removal of more features deteriorates the classification performance. To begin with, the feature whose elimination leads to the largest separation of classes is identified and removed. The process is continued by removing one feature at a time until a prespecified number of features remain or any further removal of features causes deterioration in the classification performance.

Stepwise regression [293, 500] may be seen as a variant of forward selection. At each stage, a new feature is added and the obtained model is checked for any possible elimination of the selected features without substantially increasing an error measure. The process is continued until the error is minimized or the improvement achieved is below a limit. The method starts with an initial model and then evaluates the discriminating power of incrementally larger and smaller models at each iteration [293, 500]. At each iteration, the p -values of an F -statistic are computed to evaluate the models with and without a potential feature. The feature is included if sufficient evidence is found to support its inclusion; otherwise, it is removed.

Advanced methods for feature selection include genetic algorithms and genetic programming [274, 501]. Instead of selecting a subset of a given set of features, one could also take the approach of reducing the dimension of the given feature vector via PCA (see Section 8.8.1). PCA helps in removing redundancy in the given features. It should be noted that the result of PCA does not include any of the given features directly but provides transformed values derived from the given feature vector. The transform itself is derived by using statistical measures of a population of feature vectors.

In general, it is expected that selecting a reduced set of features as above will lead to higher classification accuracy than the case with the full set of features. A smaller set of features will also facilitate the design and implementation of more efficient classifiers at lower cost than a larger set. See Banik et al. [293, 502], Nandi et al. [501], and Mu et al. [274, 503] for additional discussions on feature selection and examples of application.

9.10.3 The training and test steps

In the situation when the number of available sample vectors with known classification is limited, questions arise as to how many of the samples may be used to design or train a classifier, with the understanding that the classifier so designed needs to be tested using an independent set of samples of known classification as well. When a sufficiently large number of samples are available, they may be randomly split into two approximately equal sets, one for use as the training set and the other to be used as the test set. The random splitting procedure may be repeated a number of times to generate several classifiers. Finally, one of the classifiers so designed may be selected based upon its performance in both the training and test steps.

When the available number of labeled samples is small, the bootstrap method [504, 505] may be used. In this procedure, a training set is created by drawing at random a large number of samples from the available pool of labeled samples with replacement; that is, a given sample may be drawn and included a number of times in the training set. Similarly, a test set is created by random drawing of samples with replacement. The resampling procedure may be repeated a number of times to obtain multiple estimates of various measures of classification performance.

In the procedure known as k -fold cross-validation, the original labeled data set is partitioned randomly into k subsets of equal size with no overlap. For example, in 10-fold cross-validation with $k = 10$, the given set of samples is randomly divided

into 10 subsets with no overlap. Out of the k subsets, one subset is set aside to test the classifier being designed, and the remaining $(k - 1)$ subsets are used together as the training data. The training and testing process, also known as cross-validation, is repeated k times, with each of the k subsets used only once as the test data. The k results from the k -fold cross-validation process may be averaged to obtain an overall measure of performance of the features used and classification method chosen. In k -fold cross-validation, all data samples are used for both training and testing, and each sample is used only once for testing; this cannot be guaranteed in the random resampling procedure described in the preceding paragraphs.

The leave-one-out method: The LOO method [268] is suitable for the estimation of the classification accuracy of a pattern classification technique, particularly when the number of available samples is small. In this method, one of the available samples is excluded, the classifier is designed with the remaining samples, and then the classifier is applied to the excluded sample. The validity of the classification so performed is noted. This procedure is repeated with each available sample: if N training samples are available, N classifiers are designed and tested. The training and test sets for any one classifier so designed and tested are independent. However, whereas the training set for each classifier has $N - 1$ samples, the test set has only one sample. This procedure is equivalent to k -fold cross-validation with $k = N$. In the final analysis, every sample will have served $(N - 1)$ times as a training sample but only once as a test sample. An average classification accuracy is then computed using all of the test results.

Let us consider a simple case in which the covariances of the sample sets of two classes are equal. Assume that two sample sets, $S_1 = \{\mathbf{x}_1^{(1)}, \mathbf{x}_2^{(1)}, \dots, \mathbf{x}_{N_1}^{(1)}\}$ from class C_1 and $S_2 = \{\mathbf{x}_1^{(2)}, \mathbf{x}_2^{(2)}, \dots, \mathbf{x}_{N_2}^{(2)}\}$ from class C_2 , are given. Here, N_1 and N_2 are the numbers of samples in the sets S_1 and S_2 , respectively. Assume also that the prior probabilities of the two classes are equal to each other. Then, according to the Bayes classifier and assuming $\mathbf{x}$ to be governed by a multivariate Gaussian PDF, a sample $\mathbf{x}$ is assigned to class C_1 if

$$(\mathbf{x} - \mathbf{m}_1)^T(\mathbf{x} - \mathbf{m}_1) - (\mathbf{x} - \mathbf{m}_2)^T(\mathbf{x} - \mathbf{m}_2) > \theta, \quad (9.119)$$

where θ is a threshold, and the sample mean $\tilde{\mathbf{m}}_i$ is given by

$$\tilde{\mathbf{m}}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} \mathbf{x}_j^{(i)}. \quad (9.120)$$

In the LOO method, one sample $\mathbf{x}_k^{(i)}$ is excluded from the training set and then used as the test sample. The mean estimate for class C_i without $\mathbf{x}_k^{(i)}$, labeled as $\tilde{\mathbf{m}}_{ik}$, may be computed as

$$\tilde{\mathbf{m}}_{ik} = \frac{1}{N_i - 1} \left[\sum_{j=1}^{N_i} \mathbf{x}_j^{(i)} - \mathbf{x}_k^{(i)} \right], \quad (9.121)$$

which leads to

$$\mathbf{x}_k^{(i)} - \tilde{\mathbf{m}}_{ik} = \frac{N_i}{N_i - 1}(\mathbf{x}_k^{(i)} - \tilde{\mathbf{m}}_i). \quad (9.122)$$

Then, testing a sample $\mathbf{x}_k^{(1)}$ from C_1 can be carried out as

$$\begin{aligned} & (\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_{1k})^T(\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_{1k}) - (\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_2)^T(\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_2) \\ &= \left(\frac{N_1}{N_1 - 1} \right)^2 (\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_1)^T(\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_1) - (\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_2)^T(\mathbf{x}_k^{(1)} - \tilde{\mathbf{m}}_2) > \theta. \end{aligned} \quad (9.123)$$

Note that when $\mathbf{x}_k^{(1)}$ is tested, only $\tilde{\mathbf{m}}_1$ is changed and $\tilde{\mathbf{m}}_2$ is not changed. Likewise, when a sample $\mathbf{x}_k^{(2)}$ from C_2 is tested, the decision rule is

$$\begin{aligned} & (\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_1)^T(\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_1) - (\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_{2k})^T(\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_{2k}) \\ &= (\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_1)^T(\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_1) - \left(\frac{N_2}{N_2 - 1} \right)^2 (\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_2)^T(\mathbf{x}_k^{(2)} - \tilde{\mathbf{m}}_2) < \theta. \end{aligned} \quad (9.124)$$

The LOO method provides the least biased (practically unbiased) estimate of the classification accuracy of a given classification method for a given training set; it is useful when the number of samples available with known classification is small. The LOO approach may be applied on a sample (signal or feature vector) basis or on a patient basis if multiple signals are included in the data set for each patient.

9.11 Application: Normal versus Ectopic ECG Beats

We have seen the distinctions between normal and ectopic (PVC) beats in the ECG in several different contexts (see Sections 1.2.5, 5.4.2, 5.7, and 9.2.2, as well as Figures 5.1 and 5.11). We shall now see how we can put together several of the topics we have studied so far for the purpose of detecting PVCs in an ECG signal.

9.11.1 Classification with a linear discriminant function

Training step: Figure 9.9 shows the ECG signal of a patient with several ectopic beats, including episodes of bigeminy (alternating normal beats and PVCs). The beats in the portion of the signal in Figure 9.9 were manually labeled as normals ('o' marks) or PVCs ('x' marks) and were used to train a pattern classification system. The training set includes 121 normal beats and 39 PVCs.

The following procedure was applied to the signal to detect each beat, compute features, and develop a pattern classification rule:

1. The signal was filtered with a Butterworth lowpass filter of order 8 and cut-off frequency 70 Hz to remove noise (see Section 3.6.1); the sampling rate is 200 Hz.
2. The Pan–Tompkins algorithm was applied to detect each beat (see Section 4.3.2).

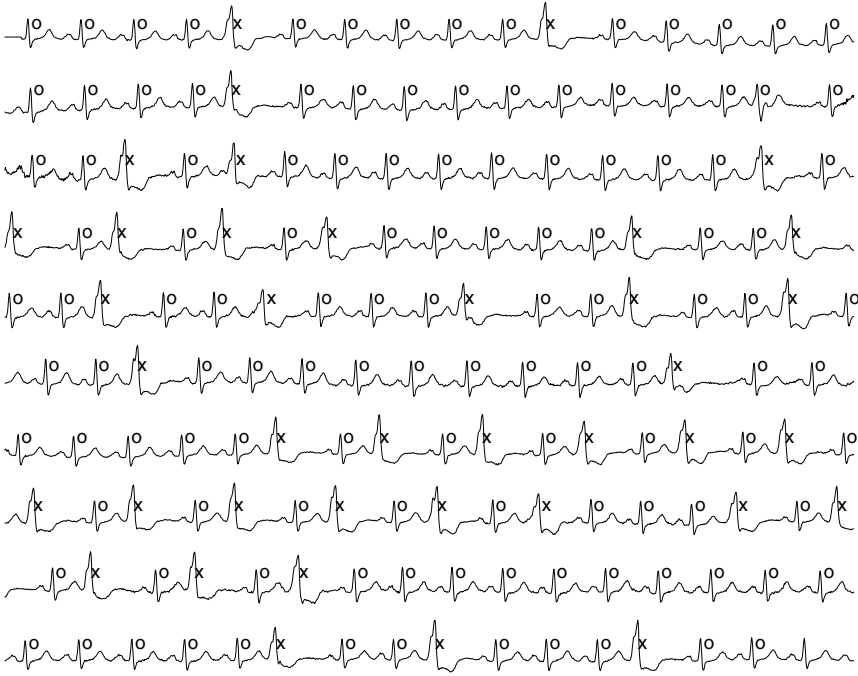


Figure 9.9 The ECG signal of a patient (male, 65 years) with PVCs (training set). Each strip is of duration 10 s; the signal continues from top to bottom. The second half of the seventh strip and the first half of the eighth strip illustrate an episode of bigeminy. Each beat was manually labeled as normal ('o') or PVC ('x'). The last beat was not processed.

3. The QRS – T portion of each beat was segmented by selecting the interval from the sample 160 ms before the peak of the Pan–Tompkins output to the sample 240 ms after the peak (see Figure 5.11).
4. The RR interval and form factor FF were computed for each beat (see Sections 5.6.4 and 5.7 and Figure 5.11). Figure 9.10 illustrates the feature vector plot for the training set.
5. The prototype (mean) feature vectors were computed for the normal and PVC groups in the training set. The prototype vectors are $[RR, FF] = [0.66, 1.58]$ and $[RR, FF] = [0.45, 2.74]$ for the normal and PVC classes, respectively.
6. The equations of the straight line joining the two prototype vectors and its normal bisector were determined; the latter is a linear decision function (see Section 9.4.1 and Figure 9.1). Figure 9.10 illustrates the two lines.

7. The equation of the linear decision function was obtained as $RR - 5.56FF + 11.44 = 0$. The decision rule was derived as

$$\text{if } RR - 5.56FF + 11.44 \begin{cases} > 0 & \text{normal beat,} \\ \leq 0 & \text{PVC.} \end{cases} \quad (9.125)$$

All of the beats in the training set were correctly classified by the decision rule in Equation 9.125.

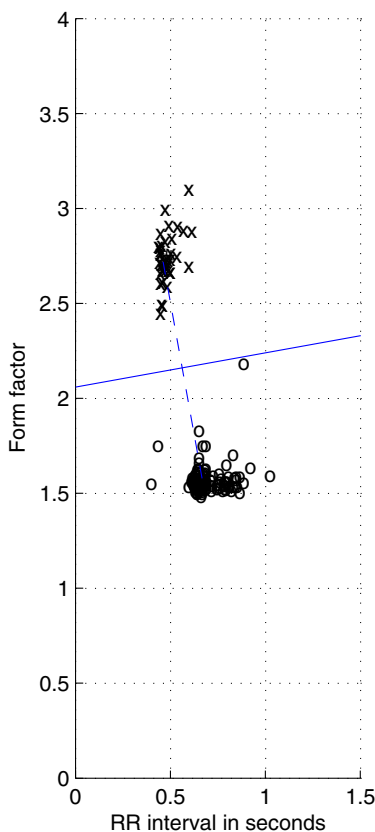


Figure 9.10 The $[RR, FF]$ feature-vector space corresponding to the ECG in Figure 9.9 (training set). Normal: 'o', PVC: 'x'. The straight line joining the two prototype vectors (dashed) and its normal bisector (solid) are also shown; the latter is a linear decision function.

Observe from Figure 9.10 that a simple threshold on FF alone can effectively separate the PVCs from the normals in the training set. A viable classification rule to detect PVCs may also be stated in a manner similar to that in Section 9.2.2. The example given here is intended to serve as a simple illustration of the design of a 2D linear decision function.

Test step: Figure 9.11 illustrates an ECG segment immediately following that in Figure 9.9. The same procedure as described above was applied to detect the beats in the signal in Figure 9.11 and to compute their features, which were used as the test set. The decision rule in Equation 9.125 was applied to the feature vectors and the beats in the signal were automatically classified as normal or PVC. Figure 9.12 illustrates the feature-vector space of the beats in the test set, along with the decision boundary given by Equation 9.125. Figure 9.11 shows the automatically applied labels for each beat: All of the 37 PVCs were correctly classified, and only one of the 120 normal beats was misclassified as a PVC (that is, there was one FP).

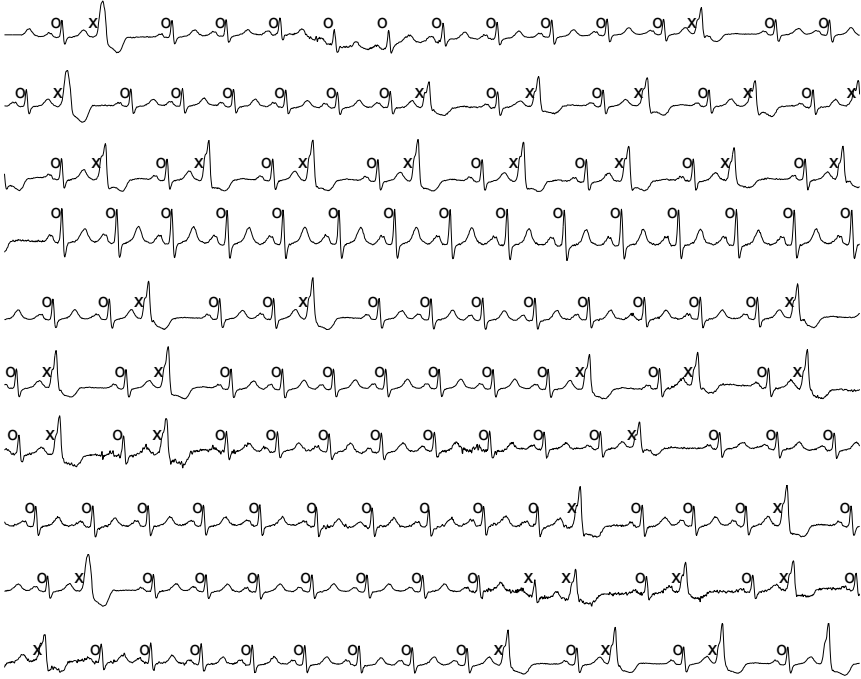


Figure 9.11 The ECG signal of a patient with PVCs (test set); this portion immediately follows that in Figure 9.9. Each strip is of duration 10 s; the signal continues from top to bottom. Each beat was automatically labeled as normal ('o') or PVC ('x') by the decision rule stated in Equation 9.125. The 10th beat in the 9th strip with $[RR, FF] = [0.66, 2.42]$ was misclassified. The last beat was not processed.

It should be observed that a PVC has, by definition, an RR interval that is less than that for a normal beat (at the same heart rate). However, the heart rate of a subject will vary over time, and the reference RR interval to determine the prematurity of PVCs needs to be updated periodically. A decision rule as in Equation 9.125 cannot be applied on a continuing basis even to the same subject. Note that the proposed method can be extended for the identification of sinus beats (originating from

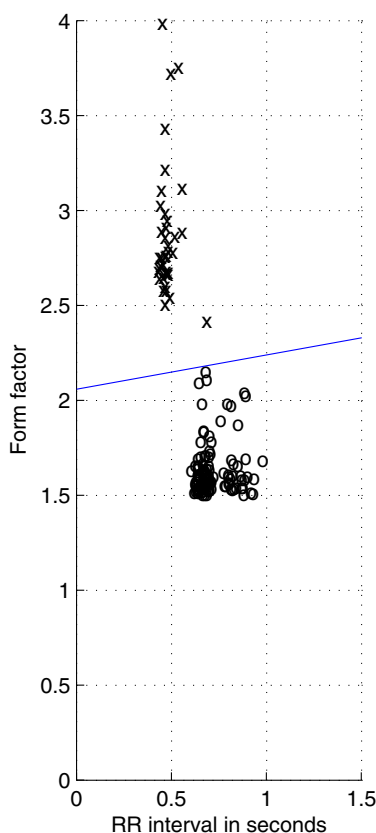


Figure 9.12 $[RR, FF]$ feature-vector space corresponding to the ECG in Figure 9.11 (test set). Normal: 'o', PVC: 'x'. The straight line is the linear decision function given in Equation 9.125. The 'x' mark closest to the decision boundary with $[RR, FF] = [0.66, 2.42]$ corresponds to an FP classification.

the SA node) that meet the prematurity condition due to sinus arrhythmia but are, nevertheless, normal in waveshape.

The FF values will depend upon the waveshape of each ECG beat, which will vary from one ECG lead to another. Therefore, the same decision rule based upon waveshape cannot be applied to all ECG leads of even the same subject. Furthermore, a given subject may have PVCs originating from various ectopic foci resulting in widely different waveshapes even in the same ECG lead. A shape factor to be used for pattern classification must be capable of maintaining different values for (a) PVCs of various waveshapes as one group and (b) normal beats as another group.

The preceding illustration is intended to serve as a simple example of the design of a pattern classification system; in practice, more complex decision rules based upon more than two features are required. Furthermore, it should be observed that

a pattern classification procedure as described above provides beat-by-beat labeling; the overall diagnosis of the patient's condition requires many other items of clinical information and the expertise of a cardiologist.

9.11.2 Application of the Bayes classifier

In another classification experiment with the same ECG signal as in the preceding section, the waveform for each beat was segmented using the Pan–Tompkins algorithm. Each signal was normalized by subtracting its mean and dividing by the maximum value of the result. The area under each segmented and normalized QRS–T wave, referred to as $QRSTA$, was computed as follows (see Section 5.4.3). The baseline value of each beat was obtained as the average of its starting and ending sample values. The baseline was subtracted and the signal was rectified, that is, its absolute value was obtained. $QRSTA$ was computed as the sum of all of the rectified values multiplied by the sampling interval. The feature vector $[QRSTA, FF]^T$ was computed for each beat.

The mean and SD of each feature were computed using the samples in the training set with 123 normal beats and 39 PVCs. Each feature value was normalized by dividing by its SD . Using the training set only, the 1D conditional and posterior probability functions were estimated. Figures 9.13 and 9.14 show the estimated functions using equal prior probabilities for the normal and PVC classes. In a real ECG signal, the number of PVCs could be expected to be far lower than the number of normal beats. Figures 9.15 and 9.16 show the estimated functions using prior probabilities of 0.999 and 0.001 for the normal and PVC classes, respectively. The four figures listed above show how the features vary in their probabilities across the two classes and also how the assumed prior probabilities can affect the posterior probabilities derived.

The 2D scatter of the feature vectors $[QRSTA, FF]^T$ for the training set is shown in Figure 9.17. Assuming the prior probabilities for the two classes to be equal, the Bayesian classifier was derived (see Section 9.6.1). Ellipses are shown for each cluster (normal or PVC) to show the boundaries of the 2D Gaussian functions estimated at σ , 2σ , and 3σ in Figure 9.17; the black contour shown is the Bayesian decision boundary, which misclassifies only one normal beat as a PVC.

Figure 9.18 demonstrates the application of the Bayesian decision function shown in Figure 9.17 to the test set of ECG signals with 183 normal beats and 53 PVCs. It is seen that the classifier correctly recognizes all PVCs but fails to identify a small number of normal beats. However, inspection of the two scatter plots indicates that the samples are linearly separable; a linear classifier without the need to assume prior probabilities could possibly lead to better results.

9.11.3 Classification using the K means method

The K -means clustering method (see Section 9.5.1) was also applied to the ECG signal described in the preceding sections. Figures 9.19 to 9.22 show the evolution

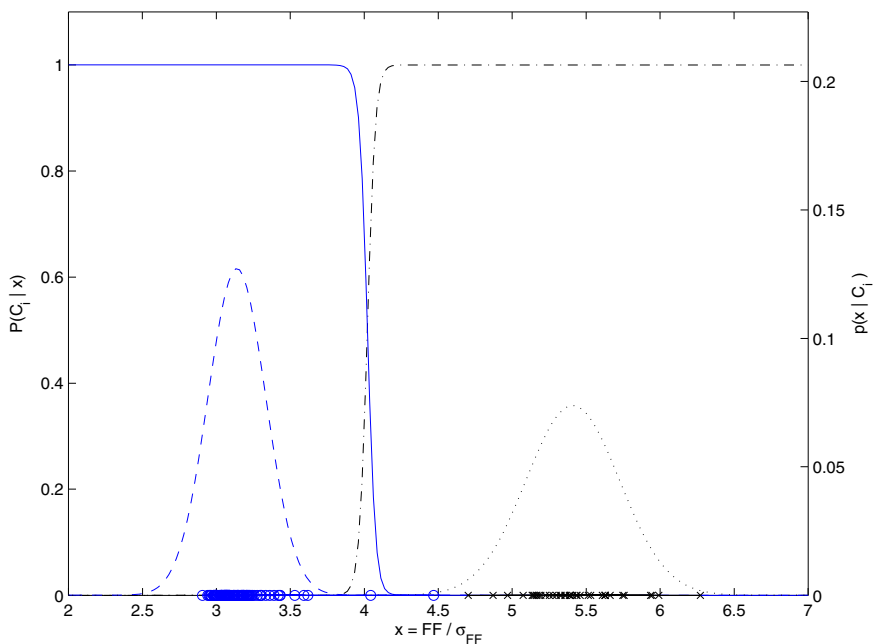


Figure 9.13 Conditional and posterior probability functions estimated for the feature FF for the training set of normal ECG signals ('o') and PVCs ('x'). The prior probabilities using the two classes were assumed to be equal. See also Figure 9.15. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

of class means and the separating boundary over four iterations. It is seen that, after the fourth and final iteration, all of the samples are correctly classified.

9.12 Application: Detection of Knee joint Cartilage Pathology

Moussavi et al. [94], Krishnan et al. [95], and Rangayyan et al. [96] proposed a series of adaptive segmentation, modeling, and pattern classification techniques for the detection of knee-joint cartilage pathology using VAG signals (see Sections 1.2.14, 5.12, 6.6, 8.2.3, 8.6, and 8.14). In consideration of the fact that VAG signals are non-stationary, each VAG signal was first divided into locally stationary segments using the RLS or the RLSL algorithm (see Sections 8.6.1 and 8.6.2). Each segment was considered as a separate signal and modeled by the forward–backward linear prediction or the Burg-lattice method (see Section 8.6.2). The model coefficients or poles were used as parameters for pattern classification.

A striking difference that may be appreciated visually and aurally between normal and abnormal VAG signals is that abnormal signals are much more variable in amplitude across a swing cycle than normal signals. However, this difference is

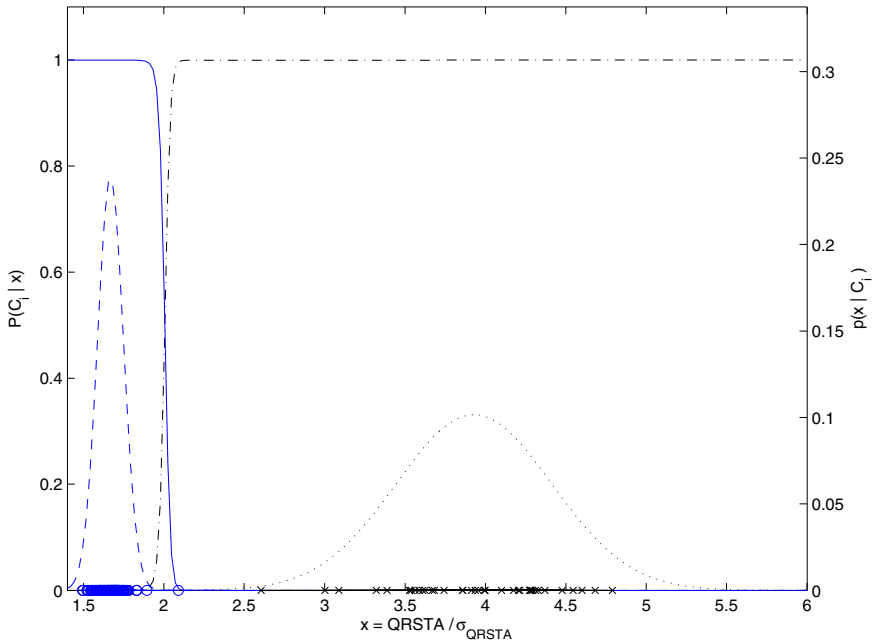


Figure 9.14 Conditional and posterior probability functions estimated for the feature *QRSTA* using the training set of normal ECG signals ('o') and PVCs ('x'). The prior probabilities for the two classes were assumed to be equal. See also Figure 9.16. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

lost in the process of dividing the signals into segments and considering each segment as a separate signal. To overcome this problem, the means (time averages) of the segments of each subject's signal were computed, and then the variance of the means was computed across the various segments of the same signal. The variance of the means represents the above-mentioned difference, and it was used as one of the discriminant features. (The *MS* or *RMS* values of VAG segments may be more suitable for this purpose than their mean values.)

In addition to quantitative parameters derived from VAG signal analysis, clinical parameters (to be described in the following paragraphs) related to the subjects were also investigated for possible discriminant capabilities. At the outset, as shown in Figure 9.23, knee joints of the subjects in the study were categorized into two groups: normal and abnormal. The normal group was divided into two subgroups: normal-silent and normal-noisy. If no sound was heard during auscultation, a normal knee was considered to be normal-silent; otherwise, it was considered to be normal-noisy. All knees in the abnormal group used were examined by arthroscopy (see Section 8.2.3 and Figure 8.2) and divided into two groups: arthroscopically normal and arthroscopically abnormal.

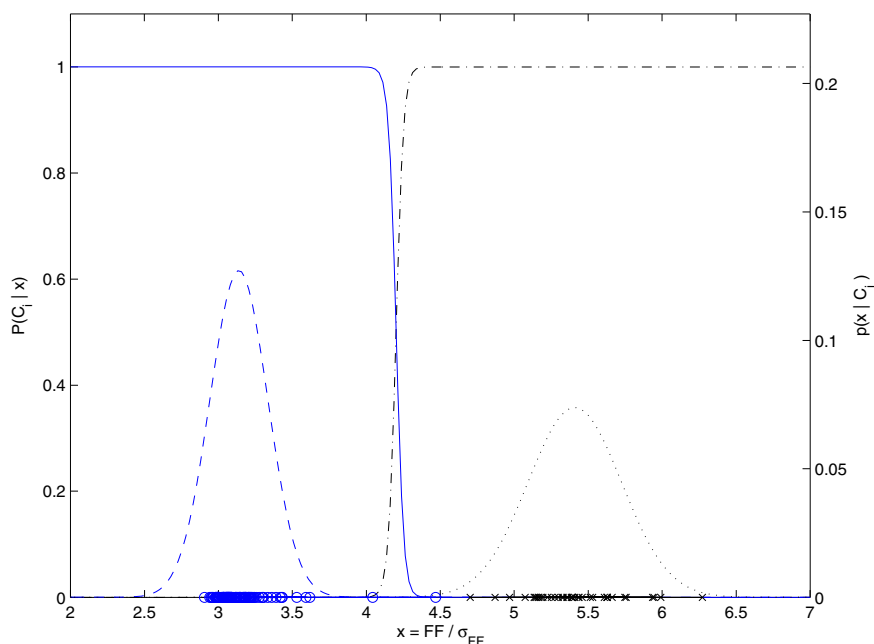


Figure 9.15 Conditional and posterior probability functions estimated for the feature FF using the training set of normal ECG signals ('o') and PVCs ('x'). The prior probabilities of 0.999 and 0.001 were assumed for the normal and PVC classes, respectively. See also Figure 9.13. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

Labeling of VAG signal segments was achieved by comparing the auscultation and arthroscopy results of each patient with the corresponding segmented VAG and joint angle signals. Localization of the pathology was performed during arthroscopy and the joint angle ranges where the affected areas could come into contact with other joint surfaces were estimated. These results were then compared with the auscultation reports to determine whether the joint angle(s) at which pathology existed corresponded to the joint angle(s) at which sound was heard. For example, if it was found from the arthroscopy report of a patient that the abnormal parts of the patient's knee could cause contact in the range $30^\circ - 90^\circ$, VAG signal segments of the subject corresponding to the angle range of $30^\circ - 90^\circ$ were labeled as arthroscopically abnormal; the rest of the segments of the signal were labeled as arthroscopically normal.

Categorization into four groups as above was done based upon the presumptions that normal-noisy and arthroscopically abnormal signals might be distinguishable in their characteristics, and that normal-silent and arthroscopically normal knees would also be distinguishable. The possibilities of arthroscopically normal knees being associated with sounds, normal-noisy knees not having any associated pathology, and

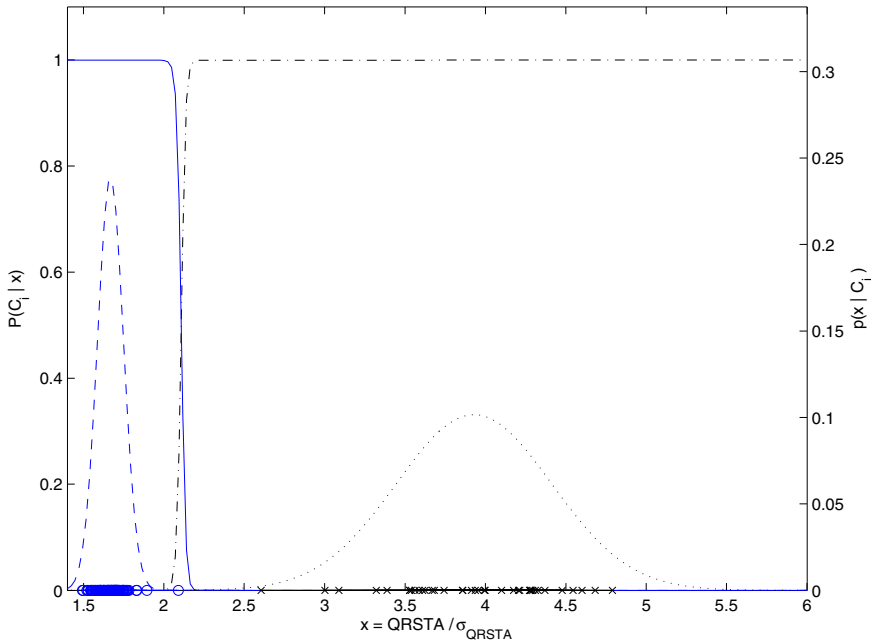


Figure 9.16 Conditional and posterior probability functions estimated for the feature *QRSTA* using the training set of normal ECG signals (‘o’) and PVCs (‘x’). The prior probabilities of 0.999 and 0.001 were assumed for the normal and PVC classes, respectively. See also Figure 9.14. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

normal-silent knees having undetermined pathologies were also admitted. Krishnan et al. [95] further subdivided the arthroscopically normal and arthroscopically abnormal categories into silent and noisy categories, thereby having a total of six categories; this is not shown in Figure 9.23.

Based on clinical reports and auscultation of knee joints, the following clinical parameters were chosen as features (in addition to AR-model parameters) for classification:

Sound: The sound heard by auscultation during flexion and extension movement of the knee joint was coded as

- 0— silent,
- 1— click,
- 2— pop,
- 3— grinding, or
- 4— a mixture of the above-mentioned sounds or other sounds.

Each segment of the VAG signals was labeled with one of the above codes.

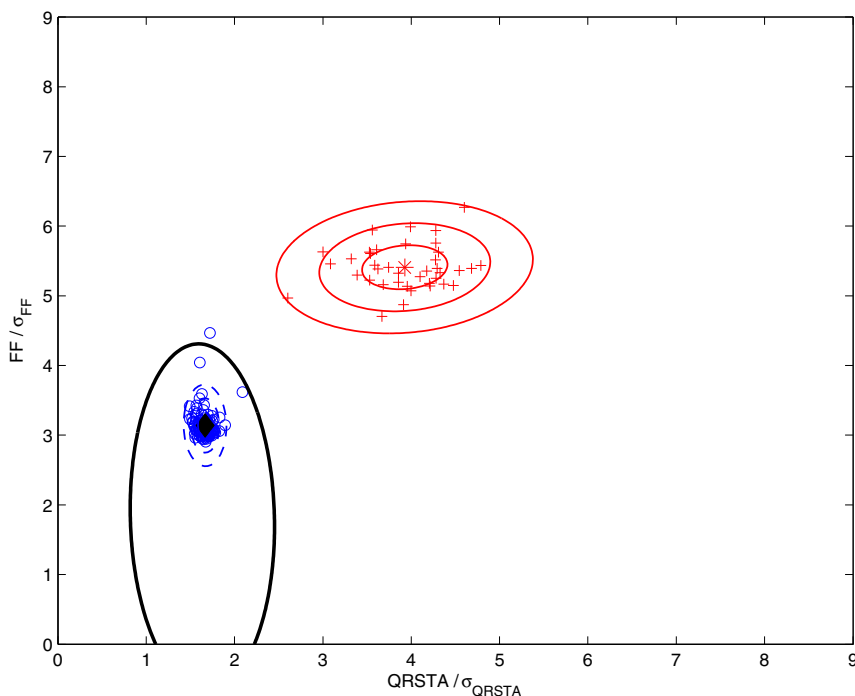


Figure 9.17 2D scatter plot of the feature vectors $[QRSTA, FF]^T$ for the training set of ECG signals. The circles represent normal beats and the plus marks represent PVCs; the diamond and star symbols represent the corresponding average or prototype vectors. The ellipses show the boundaries of the 2D Gaussian functions estimated at σ , 2σ , and 3σ for each class. The partial elliptical contour is the Bayesian decision boundary. See also Figure 9.18. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

Activity level: The activity level of each subject was coded as

- 1— exercising once per week or less,
- 2— exercising two or three times per week, or
- 3— exercising more than three times per week.

Age: The age of the subject in years.

Gender: The gender of the subject, which was coded as

- 0— female, or
- 1— male.

Among the parameters mentioned above, gender may not be a discriminant parameter; however, it is customary to record gender in clinical analysis. Note that

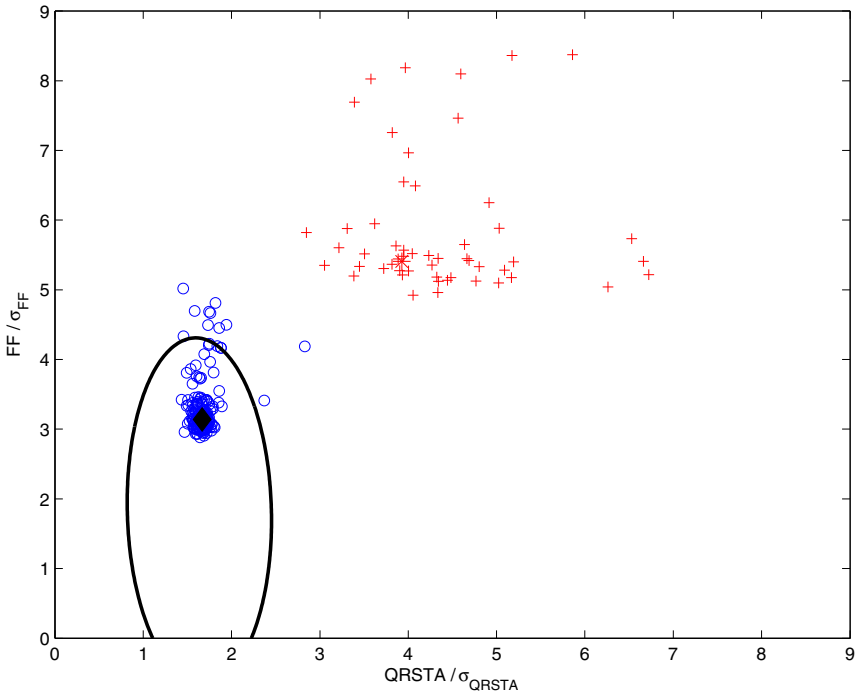


Figure 9.18 2D scatter plot of the feature vectors $[QRSTA, FF]^T$ for the test set of ECG signals. The circles represent normal beats and the plus marks represent PVCs; the diamond and star symbols represent the corresponding average or prototype vectors. The partial elliptical contour is the Bayesian decision boundary derived from the training set shown in Figure 9.17. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

among the four parameters listed above, only the first one can vary between the different segments of a given subject's VAG signal.

Moussavi et al. [94] compared the performance of various sets of features in the classification of VAG signals into two groups and four groups (see Figure 9.23) with random selections of cases. Using a set of 540 segments obtained from 20 normal subjects and 16 subjects with cartilage pathology, different numbers of segments were randomly selected for use in the training step of designing a discriminant function, and finally the selection which provided the best result was chosen for the final classification system. Two-group classification accuracies in the range 77 – 91% and four-group classification accuracies in the range 65 – 88% were obtained.

By combining the steps of classification into two groups and four groups, a two-step method was proposed by Moussavi et al. [94]; a block diagram of this method is illustrated in Figure 9.24. The algorithm first uses training sets to design classifiers for two and four groups. The resulting discriminant functions are used as Classifier 1 (two groups) and Classifier 2 (four groups), respectively. An unknown signal, which

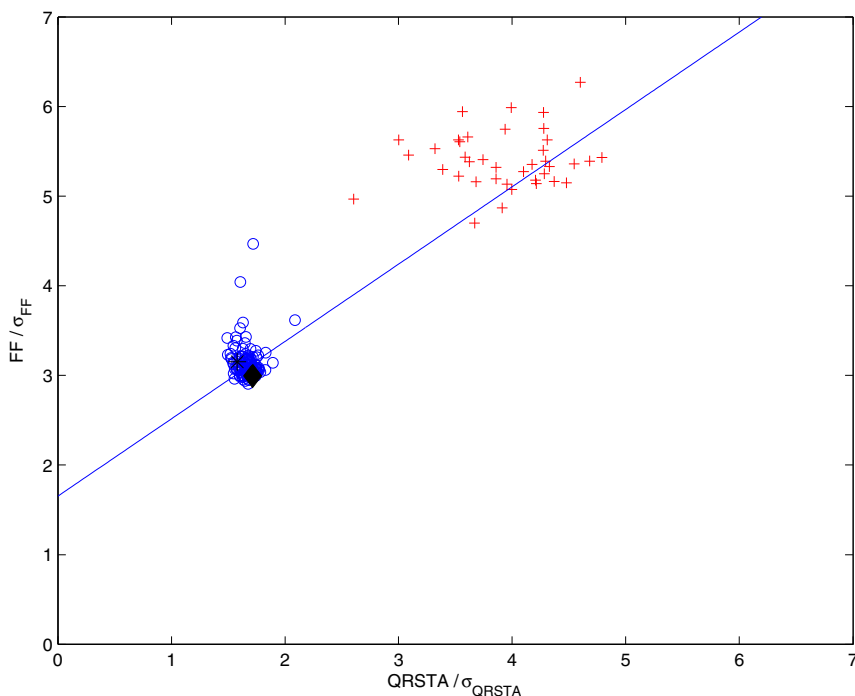


Figure 9.19 2D scatter plot of the feature vectors $[QRSTA, FF]^T$ for the training set of ECG signals with the results of the K -means method after one iteration. The circles represent normal beats and the plus marks represent PVCs; the diamond and star symbols represent the corresponding mean vectors. The straight line is the separating boundary. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

has been adaptively divided into segments, enters Classifier 1. If segments spanning more than 90% of the duration of the signal are classified as being normal, the signal (or subject) is considered to be normal. On the other hand, if more than 90% of the duration of the signal is classified as being abnormal, the signal (or subject) is considered to be abnormal. If more than 10% but less than 90% of the signal duration is classified as abnormal, the signal goes to Classifier 2, which classifies the signal into four groups (see Figure 9.23). In the second step, if more than 10% of the duration of the signal is classified as being arthroscopically abnormal, the signal is considered to be abnormal; otherwise it is considered to be normal. At this stage, information on the numbers of segments belonging to the four categories shown in Figure 9.23 is available, but the final decision is on the normality of the whole signal (subject or knee joint).

The two-step diagnosis method was trained with 262 segments obtained from 10 normal subjects and eight subjects with cartilage pathology, and it was tested with 278 segments obtained from a different set of 10 normal subjects and eight subjects

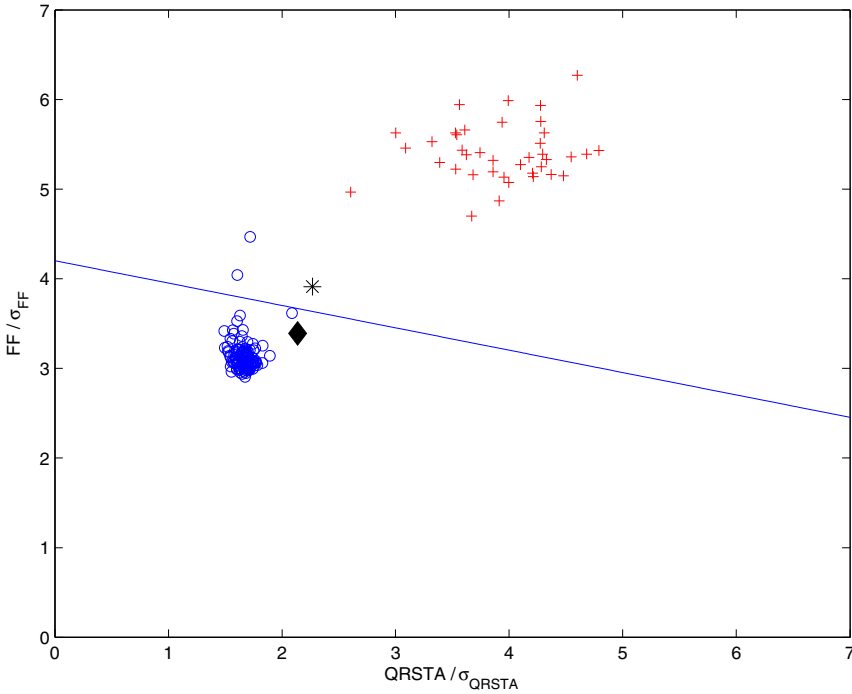


Figure 9.20 2D scatter plot of the feature vectors $[QRSTA, FF]^T$ for the training set of ECG signals with the results of the K -means method after two iterations. The circles represent normal beats and the plus marks represent PVCs; the diamond and star symbols represent the corresponding mean vectors. The straight line is the separating boundary. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

with cartilage pathology but without any restriction on the kind of abnormality. Except for one normal signal which was indicated as being abnormal over 12% of its duration, all of the signals were correctly classified. The results also showed that all of the abnormal signals including signals associated with chondromalacia grades I to IV (see Section 8.2.3 and Figure 8.2) were classified correctly. Based upon this result, it was indicated that the method has the ability to detect chondromalacia patella at its early stages as well as advanced stages. Krishnan et al. [95] and Rangayyan et al. [96] reported on further work along these directions.

9.13 Remarks

The subject of pattern classification is a vast area by itself. The topics presented in this chapter provide a brief introduction to the subject with several illustrations of application to biomedical signals.

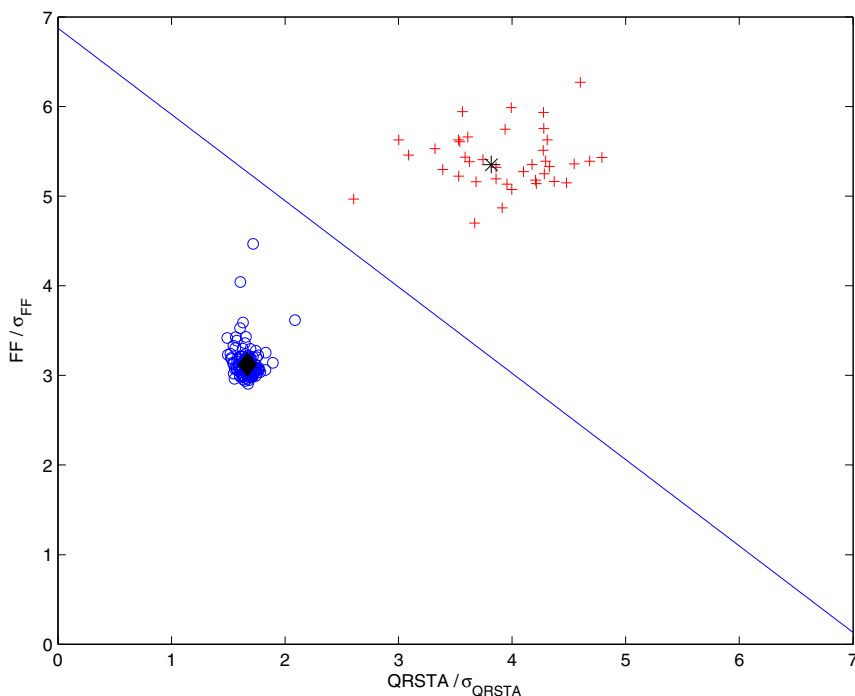


Figure 9.21 2D scatter plot of the feature vectors $[QRSTA, FF]^T$ for the training set of ECG signals with the results of the K -means method after three iterations. The circles represent normal beats and the plus marks represent PVCs; the diamond and star symbols represent the corresponding mean vectors. The straight line is the separating boundary. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

We have now seen how biomedical signals may be processed and analyzed to extract quantitative features that may be used to classify the signals as well as to design diagnostic decision functions. Practical development of such techniques is usually hampered by a number of limitations related to the extent of discriminant information present in the signals selected for analysis, as well as the limitations of the features designed and computed. Artifacts inherent in the signal or caused by the signal acquisition systems impose further limitations.

A pattern classification system that is designed with limited data and information about the chosen signals and features will provide results that should be interpreted with due care. Above all, it should be borne in mind that the final diagnostic decision requires far more information than that provided by biomedical signals and their analysis: this aspect is best left to the physician or health-care specialist in the spirit of computer-aided diagnosis.

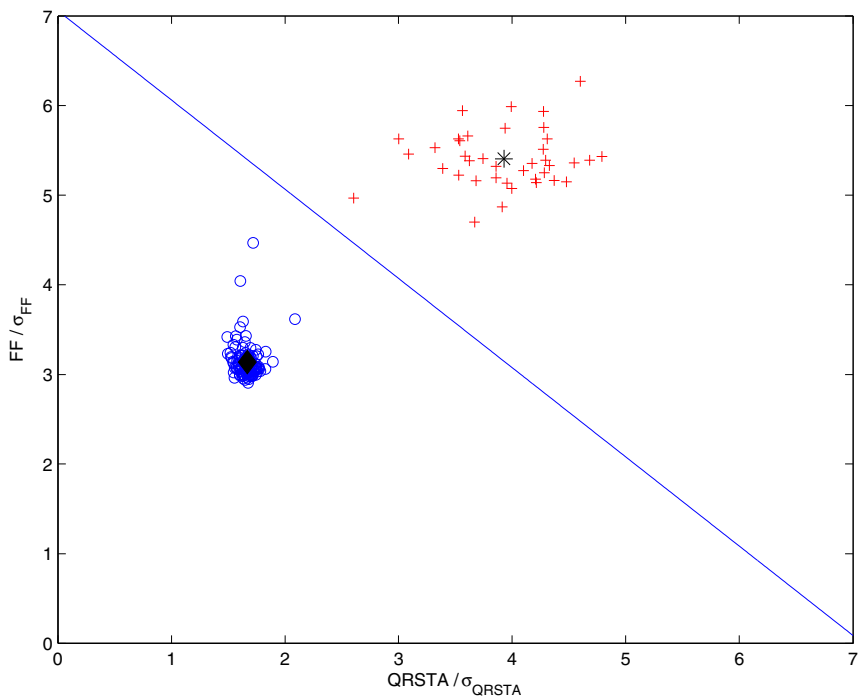


Figure 9.22 2D scatter plot of the feature vectors $[QRSTA, FF]^T$ for the training set of ECG signals with the results of the K -means method after the fourth and final iteration. The circles represent normal beats and the plus marks represent PVCs; the diamond and star symbols represent the corresponding mean vectors. The straight line is the separating boundary. Figure courtesy of Fábio José Ayres, University of Calgary, Calgary, Alberta, Canada.

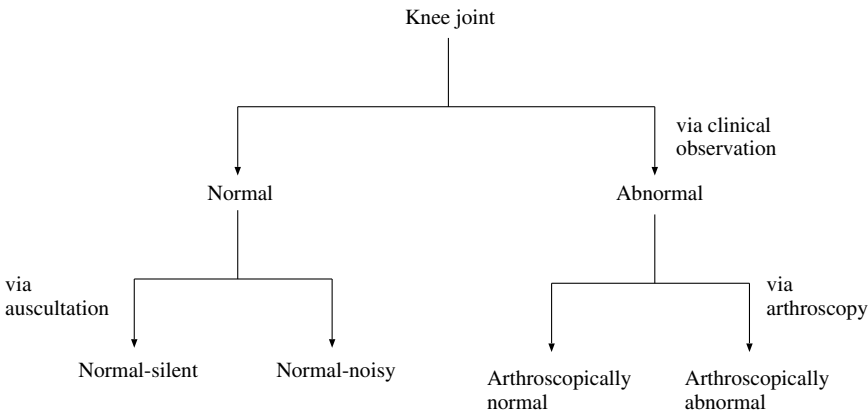


Figure 9.23 Categorization of knee joints based upon auscultation and arthroscopy.

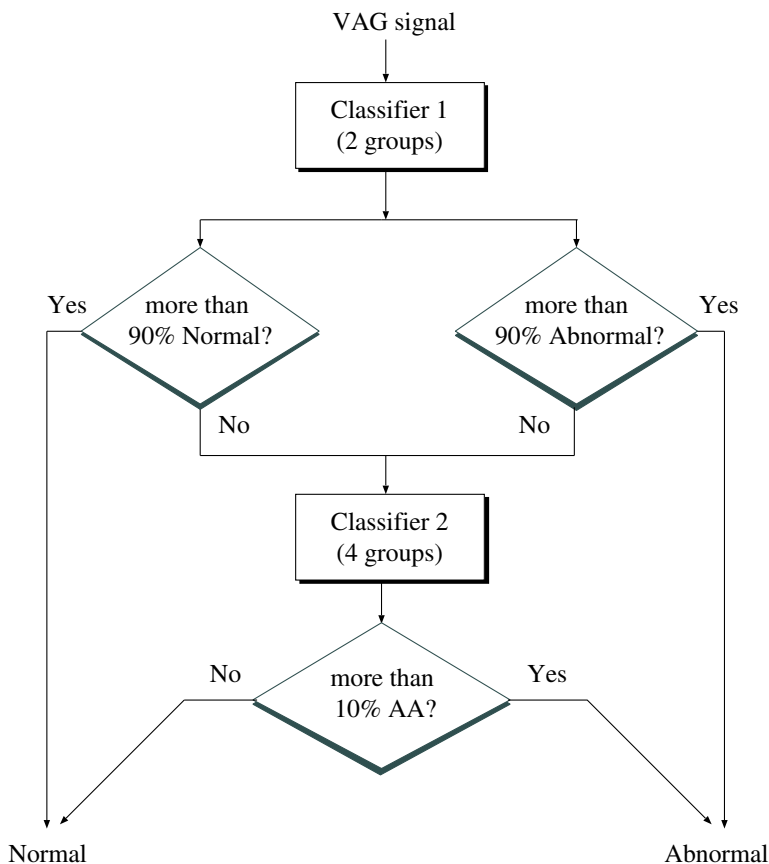


Figure 9.24 A two-step classification method for the diagnosis of cartilage pathology. AA, arthroscopically abnormal. See also Figure 9.23. Reproduced with permission from Z.M.K. Moussavi, R.M. Rangayyan, G.D. Bell, C.B. Frank, K.O. Ladly, and Y.T. Zhang, Screening of vibroarthrographic signals via adaptive segmentation and linear prediction modeling, *IEEE Transactions on Biomedical Engineering*, 43(1):15–23, 1996. ©IEEE.

9.14 Study Questions and Problems

1. The prototype vectors of two classes of signals are specified as Class 1: $[1, 0.5]$, and Class 2: $[3, 3]$. A new sample vector is given as $[2, 1]$. Plot the feature-vector space and describe your observations. Give the equations for two measures of similarity or dissimilarity, compute the measures for the sample vector, and classify the sample as Class 1 or Class 2 using each measure.
2. In a three-class pattern classification problem, the three decision boundaries are $d_1(\mathbf{x}) = -x_1 + x_2$, $d_2(\mathbf{x}) = x_1 + x_2 - 5$, and $d_3(\mathbf{x}) = -x_2 + 1$. Draw the decision boundaries on a sheet of graph paper. Classify the sample pattern vector $\mathbf{x} = [6, 5]$ using the decision functions.

3. Two pattern class prototype vectors are given to you as $\mathbf{z}_1 = [3, 4]$ and $\mathbf{z}_2 = [10, 2]$. Classify the sample pattern vector $\mathbf{x} = [4, 5]$ using (a) the normalized dot product, and (b) the Euclidean distance. Plot the feature-vector space and describe your observations.
4. A researcher makes two measurements per sample on a set of 10 normal and 10 abnormal samples. The set of feature vectors for the normal samples is $\{[2, 6], [22, 20], [10, 14], [10, 10], [24, 24], [8, 10], [8, 8], [6, 10], [8, 12], [6, 12]\}$. The set of feature vectors for the abnormal samples is $\{[4, 10], [24, 16], [16, 18], [18, 20], [14, 20], [20, 22], [18, 16], [20, 20], [18, 18], [20, 18]\}$. Plot the scatter diagram of the samples in both classes in the feature-vector space. Draw a linear decision function to classify the samples with the least error of misclassification. Write the decision function as a mathematical rule.
- How many (if any) samples are misclassified by your decision function? Mark the misclassified samples on the plot.
- Two new observation sample vectors are provided to you as $\mathbf{x}_1 = [12, 15]$ and $\mathbf{x}_2 = [14, 15]$. Classify the samples using your decision rule.
- Now, classify the samples $\mathbf{x}_1$ and $\mathbf{x}_2$ using the k -nearest-neighbor method, with $k = 7$. Measure distances graphically on your graph paper plot and mark the neighbors used in this decision process for each sample.
- Comment upon the results — whether the two methods resulted in the same classification result or not — and provide reasons.
5. A researcher makes measurements of RR intervals (in seconds) and FF for a number of ECG beats including (i) normal beats, (ii) PVCs, and (iii) normal beats with a compensatory pause (NBCP). The values (training set) are given in Table 9.6.

Normal Beats		PVCs		NBCPs	
RR	FF	RR	FF	RR	FF
0.700	1.5	0.600	5.5	0.800	1.2
0.720	1.0	0.580	6.1	0.805	1.1
0.710	1.2	0.560	6.4	0.810	1.6
0.705	1.3	0.570	5.9	0.815	1.3
0.725	1.4	0.610	6.3	0.790	1.4

Table 9.6 Training set of $[RR, FF]$ feature vectors.

- (a) Plot the $[RR, FF]$ feature-vector points for the three classes of beats. (b) Compute the prototype vectors for each class as the class means. Indicate the prototypes on the plot. (c) Derive linear discriminant functions (or decision functions) as the perpendicular bisectors of the straight lines joining the prototypes. State the decision rule(s) for each type of beat. (d) Three new beats are observed to have the parameters listed in Table 9.7. Classify each beat using the decision functions derived in part (c).
6. For the training data given in the preceding problem, compute the mean and covariance matrices of the feature vectors for each class, as well as the pooled covariance matrix. Design a classifier based upon the Mahalanobis distance using the pooled covariance matrix.

Beat No.	RR	FF
1	0.650	5.5
2	0.680	1.9
3	0.820	1.8

Table 9.7 Test set of $[RR, FF]$ feature vectors.

7. You have won a contract to design a software package for CAD of cardiovascular diseases using the heart sound signal (PCG) as the main source of information. The main task is to identify the presence of murmurs in systole and/or diastole. You may use other signals for reference.

Propose a signal processing system to (i) acquire the required signals; (ii) preprocess them as required; (iii) extract at least two features for classification; and (iv) classify the PCG signals as: class 1, normal (no murmurs); class 2, systolic murmur; class 3, diastolic murmur; or class 4, systolic and diastolic murmur.

Provide a block diagram of the complete procedure. Explain the reason behind the application of each step and state the expected results or benefits. Provide algorithmic details and/or mathematical definitions for at least two major steps in your procedure.

Draw a schematic plot of the feature-vector space and indicate where samples from the four classes listed above would fall. Propose a framework of decision rules to classify an incoming signal as belonging to one of the four classes.

9.15 Laboratory Exercises and Projects

Note: Data files related to the exercises are available at the site

<http://people.ucalgary.ca/~ranga/enel563>

1. The data file `ecgpvc.dat` contains the ECG signal of a patient with PVCs (see Figures 9.9 and 9.11). Refer to the file `ecgpvc.m` for details. Use the first 40% of the signal as training data to develop a PVC detection system (see Section 9.11). Develop code to segment the QRS – T portion of each beat using the Pan–Tompkins method (see Section 4.3.2), and compute the RR interval, QRS width (see Figure 4.6), and FF for each beat (see Section 5.6.4). Design linear discriminant functions using (i) RR and QRS width and (ii) QRS width and FF as the features; see Figure 9.10. Analyze the results in terms of TPF and FPF .

Code the decision function into your program as a classification rule. Test the pattern classifier program with the remaining 60% of the signal as the test signal. Compute the test-stage classification accuracy in terms of TPF and FPF .

2. Repeat the previous exercise replacing the linear discriminant function with the k -nearest-neighbor method, with $k = 1, 3, 5$, and 7. Evaluate the method with feature sets composed as (a) RR and QRS width, (b) QRS width and FF , and (c) RR , FF , and QRS width.

Compare the performances of the three classifiers and provide reasons for any differences between them.

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